

MECHANICS AND WAVES AND OSCILLATIONS

J.C. Upadhyaya

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PREFACE

Present book entitled, “Mechanics and Waves and Oscillations” has been written according to the new unified syllabus prescribed for B.Sc., (I) Physics students of all universities. The prescribed course has been divided into four units : first two units deal with mechanics and the rest two units, waves and oscillations. Hence the book has been divided into two parts : Part I–Mechanics and Part II–Waves and Oscillations. In fact, the course on mechanics forms the foundation of entire physics and this is why this course is taught first of all to the students in any physics syllabus. Theory of oscillations emerges from the ideas of mechanics and the waves are generated by oscillations. The learned experts of the curriculum committee have followed the right sequence of course chapters on mechanics, oscillations and waves.

Part-I of the book has been divided into six chapters First chapter deals with the vectors, the mathematical tool, needed necessarily to study the concepts of physics, Second chapter discusses the basic principles of mechanics. This chapter begins with the well known Newton’s laws of motion and their application to variable mass system. In the middle of the chapter, the fundamental laws of conservation of energy, linear and angular momentum are discussed in reasonably details with proper examples. As an application to these basic laws, finally the chapter deals with the collision problem and scattering. Mechanics of rigid bodies is discussed in chapter third where conceptually the climax of classical ideas of mechanics is reached. Fourth and fifth chapters describe respectively mechanics of continuous media and central forces. Newtonian ideas of space and time fail at speeds comparable to the speed of light. Revolutionary ideas regarding space and time and the fundamental equation $E = mc^2$ of mass-energy equivalence result from the Einstein’s postulates of special theory of relativity. This theory with its consequences is discussed in the sixth chapter i.e., the last chapter of mechanics. Second part of the book is divided in the remaining six chapters, where first three chapters are devoted to the theory of vibrations and the rest three chapters to the waves, propagating in a material medium.

In order to grasp the basic concepts of physics and to understand the complicity of subject matter it is necessary that the students have sufficient practice to solve the related problems. *To stress this point, the subject committee of the unified syllabus has made it necessary that the problems, based on each chapter of the book, are to be solved by the students.* This is why we have given a sufficient number of selected informative and modern solved problems inside each chapter followed by a good number of unsolved problems at the end of the chapter. The unsolved problems have been systematically and methodically so arranged that after solving some simpler problems in the beginning an average student is encouraged to tackle relatively difficult problems and a good student feels pleasure and intellectually satisfied after solving some difficult problems given in the last. Regarding the solved problems, we advise to students. *“Always try to solve a problem in writing before looking to its solution. This will help you in locating your difficulties and consequently in removing them.”*

In view of the latest pattern of university examination papers, theoretical questions have been systematically arranged at the end of each chapter of the book in the two heads : (i) Long Answer Questions and (ii) Short Answer Questions.

In writing this book, we have tried to present the subject matter in a simple language so that an average student feels interesting in following the text. We have also kept in mind that after studying this book, the base of knowledge of the students becomes so strong that they may not feel difficulty in understanding the subject matter in higher classes.

Though best effort has been made not to have any mistake inside the book, even then some mistakes may have inadvertently crept in. The author shall feel highly grateful to the readers who will point out and inform regarding any mistake. Further any suggestion towards the improvement of the subject matter will be thankfully received

This book is dedicated to my family members for their constant support and assistance in so many ways during the preparation of the book.

Date : October 06, 2009

—J.C. Upadhyaya

CONTENTS

Chapter	Pages
1. Vector Analysis	1–70
1.1. Scalars and vectors	1
1.2. Scalars and vector fields	2
1.3. Derivatives of a vector with respect to a scalar	4
1.4. Partial derivatives	5
1.5. Gradient of scalar field	9
1.6. Divergence	16
1.7. Curl	22
1.8. Vector integration	32
1.9. Equations of motion in vector form for constant acceleration	33
1.10. Line, surface and volume integrals	37
1.11. Stokes's theorem	42
1.12. Gauss's theorem	43
1.13. Green's theorem	45
Questions and Problems	
Appendix to Chapter-1	
Introduction to Vector Algebra	
A1. Different types of vectors	56
A2. Addition and subtraction of vectors	57
A3. Rectangular components of a vector in a plane	58
A4. Right handed and left handed coordinate systems	59
A5. Components of a vector	59
A6. Position vector	60
A7. Product of two vectors	61
A8. Triple products	64
A9. Finite and infinite rotations : angular velocity	65
A10. Circular motion	67
2. Mechanics of Particles	71—155
Part A : Laws of Motion and Variable Mass System	71–89
2.1 Mechanics	71
2.2 Motion and rest	71
2.3 Frame of reference	71
2.4 Newton's laws of motion	72
2.5 Impulse of a force	74
2.6 Newton's second law of motion as applied to a system of particles	74
2.7 Motion of variable mass system.	77
2.8 Motion of a rocket	80
Part B : Conservation of Energy	89—99
2.9 Conservation of energy	89
2.9.1. Work	89
2.9.2. Kinetic energy: Work - energy theorem	90
2.9.3. Conservative force and conservative force field	91
2.9.4. Potential energy	91
2.9.5. Conservation of mechanical energy	92
2.10 Non-conservative forces– General law of conservation of energy	93
2.11 Examples of conservation of mechanical energy– Conservative systems	94
Part C : Conservation of Linear Momentum	99—109
2.12. Conservation of linear momentum	99
2.13. Examples of conservation of linear momentum	101

2.14. Centre of mass	103
2.15. Centre of mass frame of reference and laboratory frame of reference	106
Part D : Conservation of Angular Momentum	109—118
2.16. Angular momentum and torque— Conservation of angular momentum	109
2.17. Conservation of angular momentum for a system of particles	110
2.18. Examples of conservation of angular momentum	112
2.19. Spin and orbital angular momentum	113
Part E : Collisions and Scattering	118—142
2.20. Collision of two particles (in 3-dimension)	118
2.21. Elastic collision in one dimension	119
2.22. Inelastic collision in one dimension	120
2.23. Elastic collision in two dimensions (<i>study in L-frame</i>)	121
2.24. Elastic collision in two dimensions (<i>study in C-frame</i>)	122
2.25. Elastic oblique collision	124
2.26. Impact parameter, scattering angle and scattering cross-section	136
2.27. Rutherford scattering cross-section	138
2.28. Head-on collision of an α -particle with the nucleus	140
Questions and Problems	
3. Mechanics of Rigid Bodies	156—209
3.1. Definition of rigid body	156
3.2. Rotational kinematic relations	156
3.3. Moment of inertia about an axis	158
3.4. Angular momentum, torque and angular impulse	160
3.5. Kinetic energy of a rotating body about an axis	163
3.6. Motion of a body rolling down an inclined plane	164
3.7. Translatory and rotatory motions: a comparison	165
3.8. Laws of rotational motion	166
3.9. Determination of moment of inertia about an axis	167
3.10. Calculation of moment of inertia of some rigid bodies of regular shape	168
3.11. Angular momentum and inertia tensor	179
3.12. Principal axes and principal moments of inertia	182
3.13. Torque on a rigid body: Equation of motion of a rotating body	184
3.14. Euler's equations of motion for a rigid body	185
3.15. Rotational kinetic energy of a rigid body	187
3.16. Force-free motion of a symmetrical top	188
3.17. Precession of a symmetrical top under the action of external torque	190
3.18. Gyroscope	192
3.19. Gyrostat	193
3.20. Precession of the equinoxes	195
Questions and Problems	
4. Mechanics of Continuous Media	210—247
4.1. Elasticity and isotropic materials	210
4.2. Stress, strain and Hooke's law	210
4.3. Elastic constants of isotropic solids	212
4.4. Shearing stress equivalent to an equal linear tensile stress and equal compressive stress at right angles to each other	214
4.5. Relations between various elastic constants	215
(i) Relation connecting Y , K and σ	215
(ii) Relation connecting Y , η and σ	216
(iii) Relations connecting Y , K and η	216
(iv) Relation connecting K , η and σ	217
4.6. Work done per unit volume in deforming a body – Internal energy due to strain	218
4.7. Beams	222
4.7.1. Types of support	222
4.7.2. Classification of beams	223

4.7.3. Types of load	223
4.7.4. Shearing force, bending moment and sign convention	224
4.7.5. Relation between shearing force and bending moment	225
4.8. Types of bending	226
4.9. Bending of a beam—some definitions	232
4.10. Bending moment and deduction of the expression	232
4.11. Cantilever with end load	234
4.12. Transverse vibrations of a loaded cantilever	236
4.13. Beam supported at its ends and loaded in the middle	236
4.14. Determination of Young's modulus by bending of a beam	238
Questions and Problems	
5. Central Forces	248—316
5.1. Central force	248
5.2. Conservative nature of central forces	250
5.3. Conservative force as a negative gradient of potential energy—curl of a conservative force	252
5.4. Areal velocity under central force	254
5.5. Newton's law of universal gravitation	257
5.6. Gravitational field and potential	257
5.7. Gravitational potential and field due to a thin spherical shell	259
5.8. Gravitational potential and field due to a solid sphere	262
5.9. Gravitational potential and field due to a thick spherical shell	264
5.10. Velocity of escape	265
5.11. Motion of a system of two particles under central force	270
5.12. Centre of mass and relative coordinate system	272
5.13. Equation of motion under central force—Equation of orbit	273
5.14. Motion under inverse square law force and equation of motion	275
5.15. Kepler's laws of planetary motion	279
5.16. Mathematical explanation of Kepler's laws and their deduction	280
5.17. Deduction of Newton's law of universal gravitation from Kepler's laws	285
5.18. Inertial frames of reference	292
5.19. Non-inertial frames and fictitious forces	294
5.20. Centripetal acceleration and centrifugal force	295
5.21. Satellite in circular orbit	296
5.22. Rotating frame of reference and Coriolis force	299
5.23. Motion relative to the earth	301
5.24. Foucault's pendulum	305
Questions and Problems	
6. Special Theory of Relativity	317—366
6.1. Frames of reference—Inertial frames	317
6.2. Galilean transformations	317
6.3. Principle of relativity—Hypothesis of Galilean invariance	319
6.4. Deduction of the law of conservation of linear momentum using Galilean invariance	320
6.5. Transformation of force from one inertial system to another	321
6.6. Absolute frame of reference	322
6.7. Michelson-Morley experiments	322
6.8. Shortcomings of Galilean transformations	325
6.9. Postulates of special theory of relativity	326
6.10. Lorentz transformations	327
6.11. Consequences of Lorentz transformation	329
(1) Length contraction	329
(2) Simultaneity	330
(3) Time dilation	331
(4) Addition of velocities	333
6.12. Distinctive features of Galilean and Lorentz transformations	334
6.13. Conservation of momentum at relativistic speeds—Variation of mass with velocity	342

6.14. Relativistic energy: Mass–energy relation ($E = mc^2$)	344
6.15. Examples of mass–energy conversion	346
6.16. Relation between momentum and energy and conservation laws	346
6.17. Transformation of momentum and energy	347
6.18. Particles with zero rest mass	348
6.19. Concept of four-vector formulation–four dimensional space	354
Questions and Problems	
7. Fundamentals of Vibrations	367—417
7.1. Oscillatory motion	367
7.2. Simple harmonic motion and harmonic oscillator	367
7.3. Differential equation of S.H.M. and its solution	368
7.4. Energy of harmonic oscillator	371
7.5. Linear oscillations of spring–mass system	382
7.6. Angular oscillations	387
7.6.1. Compound pendulum – Determination of g	387
7.6.2. Torsional pendulum	391
7.7. Additional examples of harmonic oscillator	396
(1) Simple pendulum	396
(2) Charge oscillations in L–C circuit	397
(3) Oscillations of two particles connected by a spring–diatomic molecule	398
7.8. Principle of superposition	401
7.9. Superposition of two S.H.M.'s of same frequency along a line	402
7.10. Combination of two mutually perpendicular harmonic vibrations of same frequency	403
7.11. Composition of two rectangular S.H.M.'s of nearly equal frequencies	404
7.12. Composition of two rectangular S.H.M.'s of frequencies in the ratio 2 : 1.	404
7.13. Lissajous figures	405
7.14. Superposition of two S.H.M.'s of different frequencies - modulation and beats	408
Questions and Problems	
8. Damped and forced Oscillations	418—451
8.1. Introduction	418
8.2. Damped harmonic oscillator	418
8.3. Energy and power dissipation of damped harmonic oscillator–relaxation time	421
8.4. Quality factor Q	422
8.5. LCR circuit	422
8.6. Forced (or driven) harmonic oscillator	428
8.7. Resonance (amplitude) and sharpness of resonance–half width of resonance curve	432
8.9. Quality factor and effect of damping	433
8.10. Velocity resonance	435
8.11. Power absorption and resonance width	435
8.12. Driven LCR circuit	437
8.13. Mechanical and electrical analogues	440
8.14. Free, damped and forced vibrations—a comparison	441
8.15. Transient state and steady state	441
Questions and Problems	
9. Complex Vibrations	452—466
9.1. Periodic function	452
9.2. Fourier's theorem	452
9.3. Limitations of Fourier's theorem	454
9.4. Application of Fourier's theorem	454
9.4.1. Fourier analysis of square wave	454
9.4.2. Fourier analysis of saw-tooth wave	459
9.4.3. Fourier analysis of triangular wave	460
9.5. Importance of Fourier's theorem	463
Questions and Problems	

10. Vibrations of Bars	467—490
10.1. Introduction	467
10.2. Longitudinal vibrations in a solid bar— establishment of wave equation and velocity of longitudinal waves	467
10.3. Wave equation for longitudinal vibrations and its general solution	468
10.4. Stationary longitudinal waves in bars	469
10.5. Longitudinal vibrations in a bar, free at both ends (Free-free bar)	470
10.6. Longitudinal vibrations in a bar, fixed at both ends (Fixed-fixed bar)	472
10.7. Longitudinal vibrations in a bar fixed at the mid point	474
10.8. Longitudinal vibrations in a bar fixed at one end and free at the other (Fixed-free bar)	476
10.9. Transverse vibrations of a bar— Differential wave equation	477
10.10. General solution of the wave equation for transverse vibrations	478
10.11. Transverse vibrations in a bar fixed at one end and free at one the other (Clamped free bar)	480
10.12. Transverse birations in a bar free at both ends (Free-ree bar)	481
10.13. Velocity of transverse waves in a bar	483
10.14. Comparision of transverse vibrations of bar and string	484
10.15. Comparision of logitudinal and transverse vibrations in a bar	484
10.16. Tunning fork	484
Questions and Problems	
11. Vibrating Strings	491—514
11.1. Wave motion: Transverse and longitudinal waves	491
11.2. Wave pulse on a string and equation of a wave	491
11.3. Transverse wave propagation along a stretched string: Velocity of transverse waves	494
11.4. General solution of the wave equation	495
11.5. Modes of vibration of stretched string clamped (fixed) at both ends (<i>Overtones and harmonics</i>)	496
11.6. Plucked string	499
11.7. Laws of transverse vibration of strings	500
11.8. Energy of vibrating string: Energy density	501
11.9. Energy transport and rate of energy flow	502
11.10. Transverse impedance	504
Questions and Problems	
12. Ultrasonics	515—524
12.1. Introduction	515
12.2. Production of ultrasonic waves	515
(1) Magnetostriction method	515
(2) Piezo-electric method	516
12.3. Detection of ultrasonics	518
12.4. Determination of wavelength and velocity of ultrasonic waves	519
12.5. Applications of ultrasonic waves	520
Questions and Problems	
* University Examination Papers	i—ix

1

Vector Analysis

1-1. Scalars and Vectors

Measurable physical quantities can be divided mainly into two classes :

(1) Scalars : *Those physical quantities, which have magnitude only but no particular direction in space, are called scalars.* Examples of scalar quantities are mass, volume, density, temperature, electric potential, charge etc.

The complete representation of a scalar quantity requires (i) a *unit of the same nature*, (ii) a *numerical value*, stating how many times the unit is contained in the quantity. For example, the mass of a body is 50 gm, we mean that the unit of measurement is a gramme and 50 such units when brought together, will comprise the mass of the body.

(2) Vectors : *Those physical quantities, which have directions as well as magnitudes, are called vectors. Moreover, a vector quantity must satisfy the parallelogram law of addition of vectors.* Familiar examples of vectors are displacement, velocity, acceleration, force, momentum, magnetic and electric field intensities etc.

The complete specification of a vector quantity requires (i) a *unit of the same kind*, disregarding direction, (ii) a *numerical value*, and (iii) a *statement of direction*. For example, the velocity of a motor car is 60 km/hour due east.

Representation of a vector : A scalar quantity is represented in light face italics or by ordinary letters such as A . For the representation of a vector quantity we write it by a letter bearing an arrow over its top (e.g., $\vec{A}$), while in printing it is also represented in bold face type letters (e.g., $\mathbf{A}$). The magnitude of a vector is represented in italics. For example, A is the magnitude of a vector $\vec{A}$. The magnitude of a vector is also called mod $\vec{A}$ or $|\vec{A}|$.

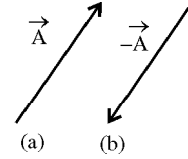


Fig. 1.1. Representation of a vector.

Graphically, a vector is represented by an arrow whose length is equal to the magnitude of the vector [fig. 1.1 (a)]. For this purpose, a convenient scale is chosen and the line, drawn for the magnitude of the vector, bears the arrow-head in the direction of the vector in space. Fig. 1.1(b) shows a vector $-\vec{A}$ which is antiparallel to $\vec{A}$ and having the same magnitude ($|\vec{A}|$).

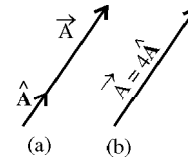


Fig. 1.2. Unit vector.

Unit vector : A vector, whose magnitude is unity, is called a unit vector. A unit vector in the direction of $\vec{A}$ is written as $\hat{A}$ and is given by [fig. 1.2 (a)] :

$$\hat{A} = \frac{\vec{A}}{A} \text{ or } \vec{A} = A \hat{A} \quad \dots(1)$$

The unit vector $\hat{A}$ is read as 'A hat' or 'A caret', Now, if $|\vec{A}| = 4$, then $\vec{A} = 4 \hat{A}$ [fig. 1.2 (b)].

Generally, the unit vectors along the three rectangular axes i.e., X, Y and Z-axes of the cartesian co-ordinate system are denoted by $\hat{i}$, $\hat{j}$ and $\hat{k}$ respectively. If A_x , A_y and A_z are the magnitudes of components

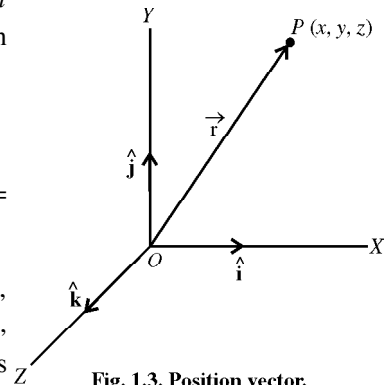


Fig. 1.3. Position vector.

2 | Mechanics and Waves and Oscillations

of a vector $\vec{A}$ along X, Y and Z-axes respectively, then the vector $\vec{A}$ is expressed in such a co-ordinate system as

$$\vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k} \quad \dots(2)$$

If x, y, z are the co-ordinates of a point P in a cartesian co-ordinate system, then the position vector $\vec{r}$ of the point P [fig. 1.3] is expressed as

$$\vec{r} = x \hat{i} + y \hat{j} + z \hat{k}$$

Elementary ideas regarding properties of vectors are given at the end of this chapter as **Appendix-Indroduction to Vector Algebra**.

1.2. Scalar and Vector Fields

If a physical quantity is expressed as a continuous function of the position of a point in a region of space, then such a function is called the *function of position* or *point function* and the region in which it specifies the physical quantity is called the *field*. According to the nature of the physical quantity, there are two main kinds of the field :

(1) Scalar field and (2) Vector field.

(1) Scalar field : If a scalar quantity is given as a continuous function $\Phi(x, y, z)$ of position in a region, then we speak of a **scalar field**. The examples of scalar field are (i) distribution of temperature in a medium, (ii) distribution of electrostatic or gravitational potential etc. The field can be mapped graphically by a series of surfaces - such as isothermal or equipotential surfaces - upon each of which the function $\Phi(x, y, z)$ has a definite constant value. Such surfaces are called **level surfaces**. Thus, the surface $\Phi(x, y, z) = \text{constant}$ is a level surface.

For example, if one end of a metallic rod is heated and the other end is kept cold, then the temperature is different at different points of the rod and hence the distribution of temperature (θ) in the material of the rod is an example of scalar field. In the steady condition, each transverse section of the rod, the value of the temperature (θ) is the same. In **fig. 1.4** at S_1 section the temperature is θ_1 and at S_2 section the temperature is θ_2 . Such surfaces are called **isothermal surfaces**. In **fig. 1.4** isothermal surfaces S_1 and S_2 are **plane surfaces**. These isothermal surfaces may be of any shape instead plane surfaces, but the condition is that at every point of a surface the temperature is the same. In **fig. 1.5** the contours of θ_1, θ_2 and θ_3 temperatures are isothermal surfaces. Similarly in **fig 1.6**, due to a point charge Q , the surfaces of equal potential, called as **equipotential surfaces**, are S_1 and S_2 . These equipotential surfaces are spherical and the values of potential at these

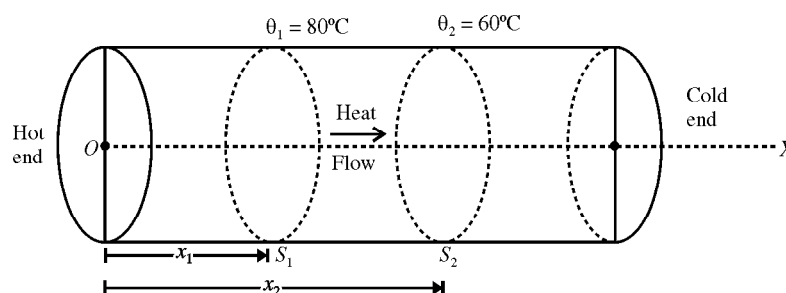


Fig. 1.4. Distribution of temperature in a metallic rod.

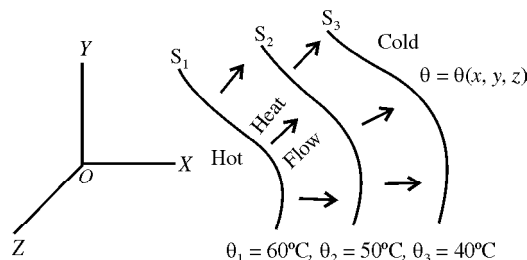


Fig. 1.5. Isothermal surfaces.

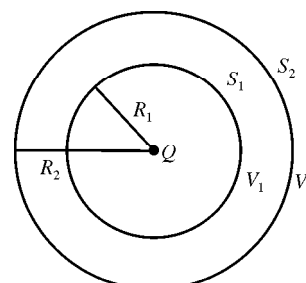


Fig. 1.6. Equipotential surfaces due to a point charge.

surfaces are V_1 and V_2 respectively. Thus, a scalar field can be mapped by equipotential surfaces.

(2) **Vector field** : If a vector quantity is given as a continuous vector function $\vec{F}(x, y, z)$ of position, then we speak of a **vector field**. Examples of a vector field are (i) the distribution of velocity in a fluid, (ii) the distribution of electric or magnetic field intensity etc. *At any given point of the field, the function $\vec{F}(x, y, z)$ is specified by a vector of definite magnitude and direction, both of which generally change continuously from point to point throughout the field region.*

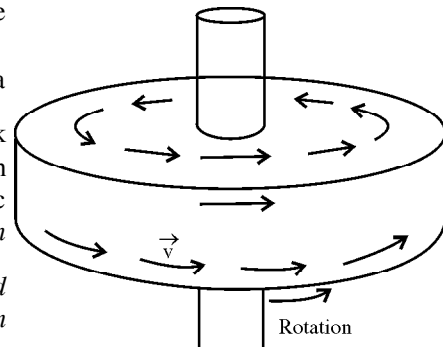


Fig. 1.7. Rotating rigid body.

For example, the velocity $\vec{v}$ of a particle of rotating rigid body will be a vector quantity whose magnitude and direction change from point to point. Hence, the velocity $\vec{v}(x, y, z)$ of the particles of the rotating rigid body represents a vector field [fig. 1.7].

Some other examples of vector field :

(i) **Stream-line motion of a liquid** : A simple example of a vector field is the stream-line flow of a liquid in a narrow tube. At different points of the tube, the velocity $\vec{v}$ of the particles of the liquid is different, the velocity will be less where the tube is wide and greater where the tube is narrow [fig. 1.8]. This velocity of the flowing liquid in the tube can be represented by vectors at different points. Here the velocity $\vec{v}(x, y, z)$ at different points inside the tube represents a vector field.

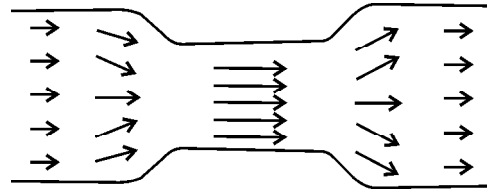


Fig. 1.8. Stream-line flow of a liquid.

In stream-line motion of a viscous liquid in a capillary tube, the velocity of different layers of the liquid is different and hence the velocity distribution of a flowing liquid on a transverse section of the capillary tube is an example of vector field.

(ii) Due to a point charge Q , the electrostatic field $\vec{E}$ is an example of vector field. Clearly, the direction of the electric field $\vec{E}$ at any point P is perpendicular to the equipotential surface S [fig. 1.9]. In between the two plates of a parallel plate condenser, the electric field $\vec{E}$, having direction from positive plate to negative plate, will be uniform. Similarly, the distribution of gravitational field and of magnetic field are examples of vector fields.

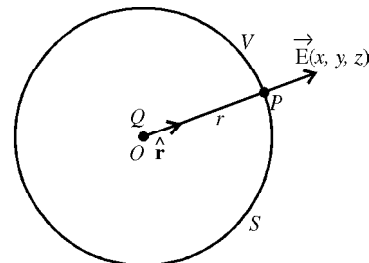


Fig. 1.9. Equipotential surface and electrostatic field due to a point charge.

Flux lines : In a vector field, the vector quantity $\vec{F}(x, y, z)$ at a point (x, y, z) is represented by definite magnitude and direction. Taking any arbitrary point, proceed an infinitesimal distance in the direction of the vector at that point so as to arrive at a closely neighbourhood point. Now, if we proceed from there in a similar way we trace out a curved line. The tangent at any point of this curved line gives the direction of the vector at that point and we call it a **vector line, line of flow or flux line**. For the representation of the magnitude of the vector at any point on a flux line, we draw a very small surface perpendicular thereto and select a number of points per unit area upon this surface,

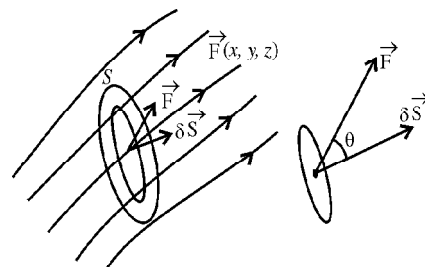


Fig. 1.10. Flux lines.

4 | Mechanics and Waves and Oscillations

which are numerically proportional to the magnitude of the vector. Now, flux line may be drawn through each of these points. Thus the vector field has been mapped out by flux lines. The direction of the lines is that of the vector function and their density, i.e., *the number of lines crossing per unit area perpendicular to their direction, is a measure of the magnitude of the vector*. Thus, if we consider a small area δS perpendicular to vector function $\vec{F}$, then the number of flux lines passing through that area will be $\delta\phi = F\delta S$. If the vector area is $\vec{\delta S} (= \delta S \hat{n})$ and θ is the angle between $\vec{F}$ and $\vec{\delta S}$, then the number of flux lines will be $\vec{F} \cdot \vec{\delta S} = F\delta S \cos \theta$ [fig. 1.10]. Now if we want to know the flux lines ϕ through any surface, then

$$\phi = \int_S \vec{F} \cdot d\vec{S}$$

In the above discussion, the scalar or vector point functions are single valued at every point of the field.

1.3. Derivative of a Vector with respect to a Scalar

If a vector $\vec{A}(u)$ or briefly written as $\vec{A}$ is a function of scalar variable u , then it may be differentiated with respect to the scalar u . The derivative of $\vec{A}$ with respect to u is defined as

$$\frac{d\vec{A}}{du} = \lim_{\delta u \rightarrow 0} \frac{\vec{A}(u + \delta u) - \vec{A}(u)}{\delta u} = \frac{\delta \vec{A}}{\delta u} \quad \dots(1)$$

where as the scalar variable changes from u to $u + \delta u$, the vector changes from $\vec{A}$ to $\vec{A} + \delta \vec{A}$ so that $\vec{A}(u + \delta u) - \vec{A}(u) = \delta \vec{A}$.

If $\vec{A}(u) = A_1(u)\hat{i} + A_2(u)\hat{j} + A_3(u)\hat{k}$, then

$$\frac{d\vec{A}}{du} = \frac{dA_1}{du}\hat{i} + \frac{dA_2}{du}\hat{j} + \frac{dA_3}{du}\hat{k} \quad \dots(2)$$

If the position vector $\vec{r}$ of a particle, given by $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, changes with time (as scalar variable), then the derivative of $\vec{r}$ with respect to time gives the velocity of the particle i.e.,

$$\vec{v} = \frac{d\vec{r}}{dt} = \frac{dx}{dt}\hat{i} + \frac{dy}{dt}\hat{j} + \frac{dz}{dt}\hat{k} = v_x\hat{i} + v_y\hat{j} + v_z\hat{k} \quad \dots(3)$$

The time derivative of velocity $\vec{v}$ is called acceleration $\vec{a}$ i.e.,

$$\vec{a} = \frac{d\vec{v}}{dt} = \frac{dv_x}{dt}\hat{i} + \frac{dv_y}{dt}\hat{j} + \frac{dv_z}{dt}\hat{k} \quad \dots(4)$$

If $\vec{A}(u)$ and $\vec{B}(u)$ are vector functions of scalar variable u and $\Phi(u)$ is a scalar function, then the usual rules of differentiation can be extended to these functions. Thus

$$\frac{d}{du}(\vec{A} + \vec{B}) = \frac{d\vec{A}}{du} + \frac{d\vec{B}}{du} \quad \dots(5)$$

$$\frac{d}{du}(\vec{A} \cdot \vec{B}) = \vec{A} \cdot \frac{d\vec{B}}{du} + \vec{B} \cdot \frac{d\vec{A}}{du} \quad \dots(6)$$

$$\frac{d}{du}(\vec{A} \times \vec{B}) = \vec{A} \times \frac{d\vec{B}}{du} + \frac{d\vec{A}}{du} \times \vec{B} \quad \dots(7)$$

$$\frac{d}{du}(\Phi \vec{A}) = \Phi \frac{d\vec{A}}{du} + \frac{d\Phi}{du} \vec{A} \quad \dots(8)$$

The order of factors does not change the value of the derivative of scalar product in (6) but the order of factors is to be strictly maintained in case of the derivative of vector product in (7).

1.4. Partial Derivatives

Let $F(x, y, z)$ be a function of three variables x, y, z . If we assume y and z as constants and change only the variable x , then the derivative of F , so obtained, is called the **partial derivative of F with respect to x** and it is represented by $\partial F / \partial x$. Thus

$$\frac{\partial F}{\partial x} = \lim_{\delta x \rightarrow 0} \frac{F(x + \delta x, y, z) - F(x, y, z)}{\delta x} \quad \dots(1)$$

Similarly, if we assume x and z as constants and change y , then the **partial derivative of F with respect to y** will be

$$\frac{\partial F}{\partial y} = \lim_{\delta y \rightarrow 0} \frac{F(x, y + \delta y, z) - F(x, y, z)}{\delta y} \quad \dots(2)$$

Further, if we assume x and y constants, then the **partial derivative of F with respect to z** will be

$$\frac{\partial F}{\partial z} = \lim_{\delta z \rightarrow 0} \frac{F(x, y, z + \delta z) - F(x, y, z)}{\delta z} \quad \dots(3)$$

Thus $\partial F / \partial x, \partial F / \partial y$ and $\partial F / \partial z$ are the **partial derivatives of the function $F(x, y, z)$ with respect to x, y and z respectively**. The meaning of a partial derivative e.g., $\partial F / \partial x$ is $(\partial F / \partial x)_{y, z = \text{constant}}$ and it is sometimes written as $(\partial F / \partial x)_{y, z}$.

Second order partial derivatives : Partial derivatives $\partial F / \partial x, \partial F / \partial y$ and $\partial F / \partial z$ can be differentiated partially with respect to x and y . Thus, the partial derivatives of $\partial F / \partial x$ with respect to x and y will be represented respectively as

$$\frac{\partial}{\partial x} \left(\frac{\partial F}{\partial x} \right) = \frac{\partial^2 F}{\partial x^2} \text{ and } \frac{\partial}{\partial y} \left(\frac{\partial F}{\partial x} \right) = \frac{\partial^2 F}{\partial y \partial x} \quad \dots(4)$$

Similarly, the partial derivatives of $\partial F / \partial y$ with respect to x and y can be written respectively as

$$\frac{\partial^2 F}{\partial x \partial y} \text{ and } \frac{\partial^2 F}{\partial y^2} \quad \dots(5)$$

$$\text{Clearly, } \frac{\partial^2 F}{\partial y \partial x} = \frac{\partial^2 F}{\partial x \partial y} \quad \dots(6)$$

For example, suppose $F(x, y, z) = x^3 y z$; hence

$$\frac{\partial F}{\partial x} = 3x^2 y z, \frac{\partial F}{\partial y} = x^3 z, \frac{\partial F}{\partial z} = x^3 y, \frac{\partial^2 F}{\partial x^2} = 6xy z, \frac{\partial^2 F}{\partial x \partial y} = 3x^2 z, \frac{\partial^2 F}{\partial y \partial x} = 3x^2 z \text{ etc.}$$

Ex. 1. Let the position vector of moving particle be represented by

$$\vec{r} = \vec{r}(t) = t^2 \hat{i} - 2t \hat{j} + (t^2 + 2t) \hat{k}, \text{ where } t \text{ is time.}$$

(a) Show that the particle goes through the point $(4, -4, 8)$ m. At what time does it pass through the point ?

6 | Mechanics and Waves and Oscillations

(b) Find the velocity vector and speed of the particle at any time t . Also find these quantities at the time when it passes through the point $(4, -4, 8)$.

Sol. $\vec{r} = t^2 \hat{i} - 2t \hat{j} + (t^2 + 2t) \hat{k}$, ... (i)

(a) If the particle passes through the point $(4, -4, 8)$ m, at the time t , then the position vector is

$$\vec{r} = 4\hat{i} - 4\hat{j} + 8\hat{k} \text{ m} \quad \dots \text{(ii)}$$

Comparing the X-components of (i) and (ii) we obtain

$$t^2 = 4 \text{ or } t = 2 \text{ sec.}$$

From (i) at $t = 2$ sec., $\vec{r} = 4\hat{i} - 4\hat{j} + 8\hat{k}$ m. This is exactly eq. (ii). Hence, the particle passes through the point $(4, -4, 8)$ m at $t = 2$ sec.

(b) Velocity $\vec{v} = \frac{d\vec{r}}{dt} = 2t\hat{i} - 2\hat{j} + 2(t+1)\hat{k} \text{ m/sec.}$

$$\text{Speed, } v = |\vec{v}| = 2\sqrt{t^2 + 1 + (t+1)^2} = 2\sqrt{2}\sqrt{t^2 + t + 1} \text{ m/sec.}$$

At $t = 2$ sec $\vec{v} = 4\hat{i} - 2\hat{j} + 6\hat{k} \text{ m/sec}$; $v = \sqrt{4^2 + 2^2 + 6^2} = 2\sqrt{14} \text{ m/sec.}$

Ex. 2. Show that the radial and transverse components of the velocity of a particle $\vec{v} = d\vec{r}/dt$ are $\frac{dr}{dt} \hat{r}$ and $r \frac{d\hat{r}}{dt}$ respectively.

Sol. If $\hat{r}$ is a unit vector along $\vec{r}$, then

$$\vec{r} = r\hat{r}, \text{ where } r \text{ is the magnitude or length of vector } \vec{r}.$$

Hence the velocity of the particle is

$$\vec{v} = \frac{d\vec{r}}{dt} = \frac{d}{dt}(r\hat{r}) = \frac{dr}{dt}\hat{r} + r\frac{d\hat{r}}{dt} \quad \dots \text{(i)}$$

Now, for unit vector $\hat{r}$, $\hat{r} \cdot \hat{r} = 1$

Differentiating it with respect to t ,

$$\hat{r} \cdot \frac{d\hat{r}}{dt} + \frac{d\hat{r}}{dt} \cdot \hat{r} = 0 \text{ or } \hat{r} \cdot \frac{d\hat{r}}{dt} = 0 \quad \dots \text{(ii)}$$

Thus $\hat{r}$ and $d\hat{r}/dt$ are perpendicular to each other. But $\hat{r}$ is a unit vector along $\vec{r}$, hence $d\hat{r}/dt$ is a vector perpendicular to $\vec{r}$. Therefore, from eq. (i), the velocity $\vec{v} = d\vec{r}/dt$ can be resolved in two components :

(1) a component $\frac{dr}{dt} \hat{r}$ along $\vec{r}$, having magnitude dr/dt ; it is called the **radial component**.

(2) a component $r \frac{d\hat{r}}{dt}$ perpendicular to $\vec{r}$, having magnitude $r \left| \frac{d\hat{r}}{dt} \right|$; it is called **transverse component**.

Ex. 3 If $\vec{A}$ is vector of constant magnitude, show that $\vec{A} \cdot \frac{d\vec{A}}{dt} = 0$ (A. N. U. 2003)

Sol. The magnitude of the vector $\vec{A}$ is constant i.e.,

$$|\vec{A}| = A = \text{constant}$$

Hence $\vec{A} \cdot \vec{A} = A^2 = \text{constant}$

Differentiating,

$$\vec{A} \cdot d\vec{A} + d\vec{A} \cdot \vec{A} = 0 \text{ or } 2\vec{A} \cdot d\vec{A} = 0 \text{ or } \vec{A} \cdot d\vec{A} = 0.$$

Example 4 : Radial and Transverse Components of Velocity and Acceleration of a Particle Moving in a Plane. Show that the radial and transverse velocities of a particle moving in a plane are respectively $\frac{dr}{dt}$ and $r \frac{d\theta}{dt}$ and accelerations in the same directions are respectively,

$$\frac{d^2r}{dt^2} - r \left(\frac{d\theta}{dt} \right)^2 \text{ and } 2 \frac{dr}{dt} \frac{d\theta}{dt} + r \frac{d^2\theta}{dt^2}.$$

Sol. : Let $\hat{i}$ and $\hat{j}$ be mutually perpendicular vectors in the plane of the paper. Also let $\hat{r}$ and $\hat{p}$ be the unit vectors along and perpendicular to $\vec{r}$.

Now, as shown in **fig. 1.11**,

$$\vec{r} = x\hat{i} + y\hat{j}$$

or $\vec{r} = r \cos \theta \hat{i} + r \sin \theta \hat{j}$

$$\therefore \hat{r} = \frac{\vec{r}}{r} = \cos \theta \hat{i} + \sin \theta \hat{j} \quad \dots(i)$$

and $\hat{p} = \frac{\vec{p}}{p} = -\cos(90 - \theta) \hat{i} + \sin(90 - \theta) \hat{j}$

or $\hat{p} = -\sin \theta \hat{i} + \cos \theta \hat{j} \quad \dots(ii)$

Now,
$$\begin{aligned} \vec{v} &= \frac{d\vec{r}}{dt} = \frac{d}{dt}(r \cos \theta \hat{i} + r \sin \theta \hat{j}) \\ &= \left(\frac{dr}{dt} \cos \theta - r \sin \theta \frac{d\theta}{dt} \right) \hat{i} + \left(\frac{dr}{dt} \sin \theta + r \cos \theta \frac{d\theta}{dt} \right) \hat{j} \\ &= \frac{dr}{dt} (\cos \theta \hat{i} + \sin \theta \hat{j}) + r \frac{d\theta}{dt} (-\sin \theta \hat{i} + \cos \theta \hat{j}) \\ &= \frac{dr}{dt} \hat{r} + r \frac{d\theta}{dt} \hat{p} \end{aligned}$$

[from (i) and (ii)] $\dots(iii)$

$\therefore$ **Radial component of velocity** $= dr/dt$ $\dots(iv)$

and **Transverse component of velocity** $= r d\theta/dt$ $\dots(v)$

Again from (i) $\frac{d\hat{r}}{dt} = (-\sin \theta \hat{i} + \cos \theta \hat{j}) \frac{d\theta}{dt} = \hat{p} \frac{d\theta}{dt}$ $\dots(vi)$

and from (ii) $\frac{d\hat{p}}{dt} = (-\cos \theta \hat{i} - \sin \theta \hat{j}) \frac{d\theta}{dt} = -\hat{r} \frac{d\theta}{dt}$ $\dots(vii)$

Now, acceleration $\vec{a} = \frac{d\vec{v}}{dt} = \frac{d}{dt} \left(\frac{dr}{dt} \hat{r} + r \frac{d\theta}{dt} \hat{p} \right)$

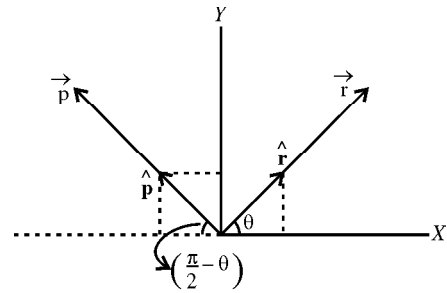


Fig. 1.11. Radial and transverse unit vectors in a plane.

$$\begin{aligned}
&= \frac{d^2 r}{dt^2} \hat{\mathbf{r}} + \frac{dr}{dt} \frac{d\hat{\mathbf{r}}}{dt} + \frac{dr}{dt} \frac{d\theta}{dt} \hat{\mathbf{p}} + r \frac{d^2 \theta}{dt^2} \hat{\mathbf{p}} + r \frac{d\theta}{dt} \frac{d\hat{\mathbf{p}}}{dt} \\
&= \frac{d^2 r}{dt^2} \hat{\mathbf{r}} + 2 \frac{dr}{dt} \frac{d\theta}{dt} \hat{\mathbf{p}} + r \frac{d^2 \theta}{dt^2} \hat{\mathbf{p}} - r \left(\frac{d\theta}{dt} \right)^2 \hat{\mathbf{r}} \quad [\text{from (vi) and (vii)}] \\
&= \left\{ \frac{d^2 r}{dt^2} - r \left(\frac{d\theta}{dt} \right)^2 \right\} \hat{\mathbf{r}} + \left(2 \frac{dr}{dt} \frac{d\theta}{dt} + r \frac{d^2 \theta}{dt^2} \right) \hat{\mathbf{p}} \quad \dots(7)
\end{aligned}$$

Therefore, the **radial acceleration** = $\frac{d^2 r}{dt^2} - r \left(\frac{d\theta}{dt} \right)^2$... (8)

transverse acceleration = $2 \frac{dr}{dt} \frac{d\theta}{dt} + r \frac{d^2 \theta}{dt^2} = \frac{1}{r} \frac{d}{dt} \left(r^2 \frac{d\theta}{dt} \right)$... (9)

Ex. 4. If $\phi = xy + yz + zx$, find $\frac{\partial \phi}{\partial x}$, $\frac{\partial \phi}{\partial y}$ and $\frac{\partial \phi}{\partial z}$. Find also the values of these quantities at the point (3, -1, 2).

Sol. $\phi = xy + yz + zx$
 $\therefore \frac{\partial \phi}{\partial x} = y + z; \frac{\partial \phi}{\partial y} = x + z; \frac{\partial \phi}{\partial z} = x + y.$

At the point (3, -1, 2), $\frac{\partial \phi}{\partial x} = -1 + 2 = 1$; $\frac{\partial \phi}{\partial y} = 3 + 2 = 5$; $\frac{\partial \phi}{\partial z} = 3 - 1 = 2$.

Ex. 5 If $\vec{A} = xz \hat{\mathbf{i}} + (2x^2 - y) \hat{\mathbf{j}} - yz^2 \hat{\mathbf{k}}$, obtain the values of $\frac{\partial A_x}{\partial x}$, $\frac{\partial A_y}{\partial y}$ and $\frac{\partial A_z}{\partial z}$ at the point (1, -1, 1).

Sol. $\vec{A} = xz \hat{\mathbf{i}} + (2x^2 - y) \hat{\mathbf{j}} - yz^2 \hat{\mathbf{k}}$
 $\therefore A_x = xz, A_y = 2x^2 - y$ and $A_z = -yz^2$
Hence, $\frac{\partial A_x}{\partial x} = z, \frac{\partial A_y}{\partial y} = -1$ and $\frac{\partial A_z}{\partial z} = -2yz$

At the point (1, -1, 1), $\frac{\partial A_x}{\partial x} = 1, \frac{\partial A_y}{\partial y} = -1$ and $\frac{\partial A_z}{\partial z} = -2$.

Ex. 6. If $\Phi(x, y, z) = x^2 yz$ and $\vec{A} = 3x^2 y \hat{\mathbf{i}} + yz^2 \hat{\mathbf{j}} - xz \hat{\mathbf{k}}$, find the value of $\frac{\partial}{\partial z} (\Phi \vec{A})$ and $\frac{\partial^2}{\partial y \partial z} (\Phi \vec{A})$ at the point (1, -2, -1).

Sol. Given, $\Phi \vec{A} = (x^2 yz) (3x^2 y \hat{\mathbf{i}} + yz^2 \hat{\mathbf{j}} - xz \hat{\mathbf{k}}) = 3x^4 y^2 z \hat{\mathbf{i}} + x^2 y^2 z^3 \hat{\mathbf{j}} - x^3 yz^2 \hat{\mathbf{k}}$
 $\therefore \frac{\partial}{\partial z} (\Phi \vec{A}) = \frac{\partial}{\partial z} (3x^4 y^2 z \hat{\mathbf{i}} + x^2 y^2 z^3 \hat{\mathbf{j}} - x^3 yz^2 \hat{\mathbf{k}}) = 3x^4 y^2 \hat{\mathbf{i}} + 3x^2 y^2 z^2 \hat{\mathbf{j}} - 2x^3 yz \hat{\mathbf{k}}$
and $\frac{\partial^2}{\partial y \partial z} (\Phi \vec{A}) = \frac{\partial}{\partial y} (3x^4 y^2 \hat{\mathbf{i}} + 3x^2 y^2 z^2 \hat{\mathbf{j}} - 2x^3 yz \hat{\mathbf{k}}) = 6x^4 y \hat{\mathbf{i}} + 6x^2 yz^2 \hat{\mathbf{j}} - 2x^3 z \hat{\mathbf{k}}$
At the point (1, -2, -1), $x = 1, y = -2, z = -1$, and the required values at the point are
 $\frac{\partial}{\partial z} (\Phi \vec{A}) = 12 \hat{\mathbf{i}} + 4 \hat{\mathbf{j}} - 4 \hat{\mathbf{k}}$ and $\frac{\partial^2}{\partial y \partial z} (\Phi \vec{A}) = -12 \hat{\mathbf{i}} - 12 \hat{\mathbf{j}} + 2 \hat{\mathbf{k}}.$

1.5. Gradient of Scalar Field

If $\Phi(x, y, z)$ be a scalar function of position in space (i.e., of the co-ordinate of x, y, z), then its partial derivatives along the three orthogonal axes are $\partial\Phi/\partial x, \partial\Phi/\partial y$ and $\partial\Phi/\partial z$. The **gradient of the scalar function** Φ is defined by

$$\text{grad } \Phi = \hat{\mathbf{i}} \frac{\partial\Phi}{\partial x} + \hat{\mathbf{j}} \frac{\partial\Phi}{\partial y} + \hat{\mathbf{k}} \frac{\partial\Phi}{\partial z} \quad \dots(1)$$

In Vector Algebra, vector differential operator $\vec{\nabla}$ (read as ‘del’ or ‘nebla’) plays an important role. The del operator is defined by

$$\vec{\nabla} = \hat{\mathbf{i}} \frac{\partial}{\partial x} + \hat{\mathbf{j}} \frac{\partial}{\partial y} + \hat{\mathbf{k}} \frac{\partial}{\partial z} \quad \dots(2)$$

Operating with $\vec{\nabla}$ on the scalar function Φ , we have

$$\vec{\nabla} \Phi = \hat{\mathbf{i}} \frac{\partial\Phi}{\partial x} + \hat{\mathbf{j}} \frac{\partial\Phi}{\partial y} + \hat{\mathbf{k}} \frac{\partial\Phi}{\partial z} \quad \dots(3)$$

This is the same expression as defined for the gradient of the scalar function Φ or $\text{grad } \Phi$. **The del operator is a vector operator and when operated on a scalar, it converts the scalar into a vector.** This vector ($\vec{\nabla} \Phi$) is called the **gradient of the scalar**. Thus

$$\vec{\nabla} \Phi = \hat{\mathbf{i}} \frac{\partial\Phi}{\partial x} + \hat{\mathbf{j}} \frac{\partial\Phi}{\partial y} + \hat{\mathbf{k}} \frac{\partial\Phi}{\partial z} = \text{grad } \Phi \quad \dots(4)$$

For example, the scalar quantities, such as temperature and electric potential, can be represented from point to point in space by a scalar function $\Phi(x, y, z)$ and we say that a scalar field is present in the space. The scalar field can be mapped out by level surfaces, upon each of which $\Phi(x, y, z)$ has a constant value.

Consider two surfaces S_1 and S_2 very close to each other on which the values of scalar function are Φ and $\Phi + d\Phi$ respectively. Suppose P and Q are points on the two surfaces [fig. 1.12], whose distances from the origin O are $\vec{r}$ and $\vec{r} + d\vec{r}$ respectively so that vector distance $\vec{PQ} = d\vec{r}$. If dn represents the distance (PN) along the normal from the point P (on surface S_1) to the surface S_2 , we may write,

$$dn = dr \cos \theta = \hat{\mathbf{n}} d\vec{r} \quad \dots(5)$$

where $\hat{\mathbf{n}}$ is the unit normal to the surface S_1 at P .

The rate of increase of Φ at P in the direction of $\hat{\mathbf{n}}$ is greatest and is equal to $\partial\Phi/\partial n$ [fig. 1.13].

Therefore,

$$d\Phi = \frac{\partial\Phi}{\partial n} dn = \frac{\partial\Phi}{\partial n} \hat{\mathbf{n}} d\vec{r}$$

$$\text{But } \vec{\nabla} \Phi d\vec{r} = \left(\hat{\mathbf{i}} \frac{\partial\Phi}{\partial x} + \hat{\mathbf{j}} \frac{\partial\Phi}{\partial y} + \hat{\mathbf{k}} \frac{\partial\Phi}{\partial z} \right) (\hat{\mathbf{i}} dx + \hat{\mathbf{j}} dy + \hat{\mathbf{k}} dz)$$

$$= \frac{\partial\Phi}{\partial x} dx + \frac{\partial\Phi}{\partial y} dy + \frac{\partial\Phi}{\partial z} dz = d\Phi$$

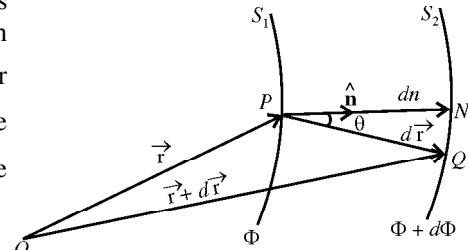


Fig. 1.12. Surfaces of constant scalar function $\Phi(x, y, z)$

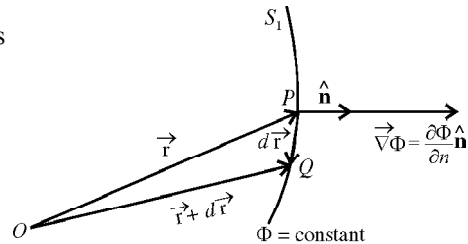


Fig. 1.13. Vector $\vec{\nabla} \Phi$ at a point P , being normal to $\Phi = \text{constant}$ surface.

$$\therefore d\Phi = \frac{\partial\Phi}{\partial n} \hat{n} d\vec{r} = \vec{\nabla}\Phi d\vec{r} \quad \dots(6)$$

$$\text{Therefore, } \vec{\nabla}\Phi = \left(\frac{\partial\Phi}{\partial n}\right)\hat{n}. \quad \dots(7)$$

Hence, the magnitude of the vector $\vec{\nabla}\Phi$ is equal to the maximum rate of change Φ with respect to the space variables and has the direction of that change.

In particular, if $d\vec{r}$ lies on the surface S_1 , we have from (6)

$$d\Phi = \vec{\nabla}\Phi d\vec{r} = 0 \quad \dots(8)$$

which means that the vector $\vec{\nabla}\Phi$ is normal to the surface $\Phi = \text{constant}$.

If S_1 and S_2 surfaces are plane, then the gradient of scalar field Φ [fig. 1.14] is

$$\vec{\nabla}\Phi = \frac{\partial\Phi}{\partial x} \hat{i} = \frac{d\Phi}{dx} \hat{i} \text{ or } |\vec{\nabla}\Phi| = \frac{d\Phi}{dx} \quad \dots(9)$$

where the vector $\vec{\nabla}\Phi$ is perpendicular to the surface $\Phi = \text{constant}$ i.e., the vector $\vec{\nabla}\Phi$ is along X-axis and $d\Phi/dx$ is equal to the maximum rate of its change in that direction.

Physical Significance of Gradient

The gradient of scalar field is of great importance in different fields of physics. For example, if V is the electrostatic potential at a distance r from a point charge Q , then equipotential surface will be spherical [fig. 1.15]. The electric field $\vec{E}$ at the point P distant r from Q is related to the electrostatic potential V as

$$\vec{E} = -\vec{\nabla}V = -\text{grad } V \quad \dots(10)$$

where the potential $V(x, y, z)$ and field $\vec{E}(x, y, z)$ represent the scalar and vector fields respectively at the point P .

If $\hat{r}$ is unit vector along $\vec{OP}$, then

$$\vec{E} = -\vec{\nabla}V = -\frac{\partial V}{\partial r} \hat{r} = -\frac{dV}{dr} \hat{r} \quad \dots(11)$$

where $\vec{E}$ is along the unit vector $\hat{r}$ and obviously $\hat{r}$ is perpendicular to the equipotential surface. Here, the negative gradient of the electric potential (scalar field) is the electric field intensity $\vec{E}$ (vector field); the field intensity $\vec{E}$ at the point P is a vector which is along the direction of greatest rate of decrease of potential (i.e., along $\hat{r}$) and has the value equal to the maximum rate of change of potential (dV/dr).

Similarly, the gravitational field intensity $\vec{E}$ is related with gravitational potential V as

$$\vec{E} = -\vec{\nabla}V.$$

Now, if q charge is present at the point P , then the potential energy will be $U(= qV)$ and the force on that charge q will be

$$\vec{F} = -\vec{\nabla}U = -\text{grad } U = -\frac{\partial U}{\partial r} \hat{r} = -\frac{dU}{dr} \hat{r} \quad \dots(12)$$

In a parallel plate condenser its plates are at V_1 and V_2 potential ($V_1 > V_2$), then

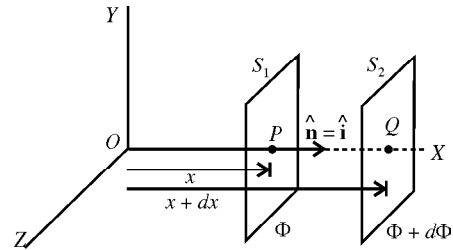


Fig. 1.14.

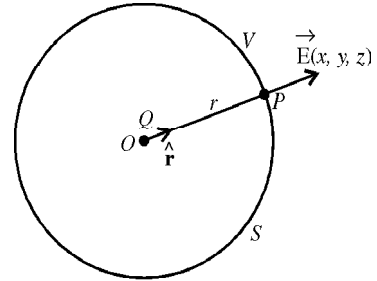


Fig. 1.15.

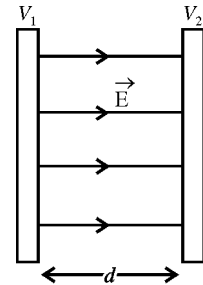


Fig. 1.16.

the electric field intensity $\vec{E}$ will be uniform and in the direction perpendicular to the plates [fig. 1.16].

$$\vec{E} = -\frac{\partial V}{\partial x} \hat{i} = -\frac{V_2 - V_1}{d} \hat{i} \text{ or } E = \frac{V_1 - V_2}{d} \quad \dots(13)$$

Here, the uniform electric field intensity $\vec{E}$ is obtained by the gradient of the potential along X-axis and its value is obtained by eq. (10).

In a **potentiometer**, the potential decreases on the wire from point A to point B and the value of potential gradient is V/l , where V is the potential difference between the ends of potentiometer wire and l is the length of the wire [fig. 1.17].

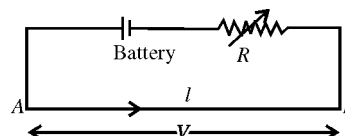


Fig. 1.17.

In **heat conductivity experiment** by Searle's apparatus, if θ_1 and θ_2 are the temperatures of S_1 and S_2 isothermal surfaces respectively, situated correspondingly at x_1 and x_2 , then the thermal gradient is given by

$$\frac{d\theta}{dx} = \frac{\theta_2 - \theta_1}{x_2 - x_1} = -\frac{\theta_1 - \theta_2}{x_2 - x_1} \quad \dots(14)$$

which is negative and heat flow is in the direction of decrease of temperature.

Ex. 1. In conduction of heat in a metallic rod under steady state, an isothermal surface is defined as $\theta = 4x - 5y + 3z$. Find the temperature field and show that the temperature field is uniform.

Sol. Let the temperature field be $\vec{E}$. Then $\vec{E} = -\text{grad } \theta = -\left(\hat{i} \frac{\partial \theta}{\partial x} + \hat{j} \frac{\partial \theta}{\partial y} + \hat{k} \frac{\partial \theta}{\partial z}\right)$

$$\begin{aligned} \text{or} \quad \vec{E} &= -\left[\hat{i} \frac{\partial}{\partial x}(4x - 5y + 3z) + \hat{j} \frac{\partial}{\partial y}(4x - 5y + 3z) + \hat{k} \frac{\partial}{\partial z}(4x - 5y + 3z)\right] \\ &= -[\hat{i}(4) + \hat{j}(-5) + \hat{k}(3)] = -4\hat{i} + 5\hat{j} - 3\hat{k}. \end{aligned}$$

As this field is independent of x, y, z , hence it is a uniform field.

Ex. 2. If $V = 4x^2 + 3y^2 - 9z^2$ represents the electrostatic potential at a point, find the electric field intensity at a point $(-1, 2, 3)$.

Sol. Electric field intensity $\vec{E}$ is given by

$$\begin{aligned} \vec{E} &= -\text{grad } V = -\left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z}\right)(4x^2 + 3y^2 - 9z^2) \\ &= -[8x\hat{i} + 6y\hat{j} - 18z\hat{k}] = -8x\hat{i} - 6y\hat{j} + 18z\hat{k} \end{aligned}$$

$$\text{At point } (-1, 2, 3), \quad \vec{E} = -8 \times -1\hat{i} - 6 \times 2\hat{j} + 18 \times 3\hat{k} = 8\hat{i} - 12\hat{j} + 54\hat{k}.$$

Ex. 3 Find $\text{grad } \phi$ if $\phi = (x^2 + y^2 + z^2)^{1/2}$. (O.U. 2003)

Sol. Here, $\phi = (x^2 + y^2 + z^2)^{1/2}$

$$\therefore \text{grad } \phi = \vec{\nabla} \phi = \vec{\nabla} (x^2 + y^2 + z^2)^{1/2}$$

$$\begin{aligned} \therefore \vec{\nabla} \phi &= \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z}\right)(x^2 + y^2 + z^2)^{1/2} \\ &= \frac{\hat{i}}{2}(x^2 + y^2 + z^2)^{-1/2}(2x) + \frac{\hat{j}}{2}(x^2 + y^2 + z^2)^{-1/2}(2y) + \frac{\hat{k}}{2}(x^2 + y^2 + z^2)^{-1/2}(2z) \\ &= (x\hat{i} + y\hat{j} + z\hat{k})(x^2 + y^2 + z^2)^{-1/2} \end{aligned}$$

But $x\hat{i} + y\hat{j} + z\hat{k} = \vec{r}$ is the position vector and magnitude of $\vec{r}$ is $r = (x^2 + y^2 + z^2)^{1/2}$. Hence

$$\text{grad } \phi = \vec{r} r^{-1} = \frac{\vec{r}}{r} = \hat{r}$$

Ex. 4. The potential function in an electric field is represented by $V = C(x^2 + y^2 + z^2)$, where C is a constant. Show that the electric field is radial.

Sol. Here, $V = C(x^2 + y^2 + z^2)$

$$\therefore \text{Electric field } \vec{E} = -\vec{\nabla} V = -\left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z}\right) C(x^2 + y^2 + z^2) = -2C(x\hat{i} + y\hat{j} + z\hat{k})$$

$$\text{But, } x\hat{i} + y\hat{j} + z\hat{k} = \vec{r}, \quad \therefore \vec{E} = -2C\vec{r}$$

As a field along or opposite to $\vec{r}$ is radial, hence $\vec{E}$ is **radial field**.

Ex. 5. If $\vec{r}$ is the position vector of a point, deduce the value of $\text{grad } (1/r)$.

Sol. Position vector $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$; its magnitude $r = (x^2 + y^2 + z^2)^{1/2}$

$$\begin{aligned} \text{Hence, } \text{grad } \left(\frac{1}{r}\right) &= \text{grad } (x^2 + y^2 + z^2)^{-1/2} = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z}\right) (x^2 + y^2 + z^2)^{-1/2} \\ &= \left[\hat{i} \frac{\partial}{\partial x} (x^2 + y^2 + z^2)^{-1/2} + \hat{j} \frac{\partial}{\partial y} (x^2 + y^2 + z^2)^{-1/2} + \hat{k} \frac{\partial}{\partial z} (x^2 + y^2 + z^2)^{-1/2}\right] \end{aligned}$$

$$\text{Now, } \frac{\partial}{\partial x} (x^2 + y^2 + z^2)^{-1/2} = \left(-\frac{1}{2}\right) (x^2 + y^2 + z^2)^{-3/2} 2x = \frac{-x}{(x^2 + y^2 + z^2)^{3/2}}$$

$$\text{Similarly, } \frac{\partial}{\partial y} (x^2 + y^2 + z^2)^{-1/2} = \frac{-y}{(x^2 + y^2 + z^2)^{3/2}} \text{ and } \frac{\partial}{\partial z} (x^2 + y^2 + z^2)^{-1/2} = \frac{-z}{(x^2 + y^2 + z^2)^{3/2}}$$

$$\begin{aligned} \therefore \text{grad } \left(\frac{1}{r}\right) &= \left[\hat{i} \left\{ \frac{-x}{(x^2 + y^2 + z^2)^{3/2}} \right\} + \hat{j} \left\{ \frac{-y}{(x^2 + y^2 + z^2)^{3/2}} \right\} + \hat{k} \left\{ \frac{-z}{(x^2 + y^2 + z^2)^{3/2}} \right\}\right] \\ &= \frac{-(x\hat{i} + y\hat{j} + z\hat{k})}{(x^2 + y^2 + z^2)^{3/2}} = \frac{-\vec{r}}{r^3}. \end{aligned}$$

Ex. 6. If the electric potential at any point due to a charge q is given by $V = q/r$, then show that the electric intensity $\vec{E}$ is given by

$$\vec{E} = \frac{q}{r^3} \vec{r}$$

Sol. We know that $\vec{E} = -\text{grad } V$

$$\text{Here, } V = \frac{q}{r}, \quad \therefore \vec{E} = -\text{grad } \left(\frac{q}{r}\right) = -q \text{grad } \left(\frac{1}{r}\right)$$

$$\text{But } \text{grad } \left(\frac{1}{r}\right) = \left(\frac{-\vec{r}}{r^3}\right), \quad (\text{Prove as above})$$

$$\text{Therefore, } \vec{E} = -q \left(\frac{-\vec{r}}{r^3}\right) = \frac{q}{r^3} \vec{r}.$$

Ex. 7. If $\vec{r} = \hat{i}x + \hat{j}y + \hat{k}z$, prove that $\text{grad } r^n = nr^{n-2} \vec{r}$.

Sol. $\text{grad } r^n = \vec{\nabla} r^n = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) (x^2 + y^2 + z^2)^{n/2} \quad [\because r = (x^2 + y^2 + z^2)^{1/2}]$

$$= \hat{i} \left(\frac{n}{2} \right) (x^2 + y^2 + z^2)^{(n/2)-1} 2x + \hat{j} \left(\frac{n}{2} \right) (x^2 + y^2 + z^2)^{(n/2)-1} 2y$$

$$+ \hat{k} \left(\frac{n}{2} \right) (x^2 + y^2 + z^2)^{(n/2)-1} 2z$$

$$= n (\hat{i}x + \hat{j}y + \hat{k}z) (x^2 + y^2 + z^2)^{(n-2)/2} = n \vec{r} r^{n-2}.$$

Ex. 8. The scalar potential ϕ in a field is defined by $\phi = 6x - 7xy + 9yz - 10z$. Calculate the gradient of the field. **(O. U. 2003)**

Sol. Here,

$$\phi = 6x - 7xy + 9yz - 10z$$

$\therefore$ $\text{grad } \phi = \vec{\nabla} \phi = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) (6x - 7xy + 9yz - 10z)$

$$= \hat{i} \frac{\partial}{\partial x} (6x - 7xy + 9yz - 10z) + \hat{j} \frac{\partial}{\partial y} (6x - 7xy + 9yz - 10z)$$

$$+ \hat{k} \frac{\partial}{\partial z} (6x - 7xy + 9yz - 10z).$$

$$= \hat{i}(6 - 7y) + \hat{j}(-7x + 9z) + \hat{k}(9y - 10)$$

$$= (6 - 7y)\hat{i} + (-7x + 9z)\hat{j} + (9y - 10)\hat{k}.$$

Ex. 9. If $\phi = x^2y - xz^3$, find $\vec{\nabla} \phi$. **(O.U. 2005)**

Sol. $\vec{\nabla} \phi = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) (x^2y - xz^3) = \hat{i}(2xy - z^3) + \hat{j}(x^2) + \hat{k}(-3xz^2)$

$$= (2xy - z^3)\hat{i} + x^2\hat{j} - 3xz^2\hat{k}.$$

Ex. 10. If $S = S(x, y, z) = x^2 - x^2y + xy^2z^2$, find $\text{grad } S$ at the point $(2, -1, -3)$.

Sol. $\text{grad } S = \vec{\nabla} S$, where $\vec{\nabla} = \hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z}$

$\therefore$ $\text{grad } S = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) (x^2 - x^2y + xy^2z^2)$

$$= \hat{i} \frac{\partial}{\partial x} (x^2 - x^2y + xy^2z^2) + \hat{j} \frac{\partial}{\partial y} (x^2 - x^2y + xy^2z^2) + \hat{k} \frac{\partial}{\partial z} (x^2 - x^2y + xy^2z^2)$$

$$= \hat{i}(2x - 2xy + y^2z^2) + \hat{j}(-x^2 + 2xyz^2) + \hat{k}(2zxy^2)$$

Here, $x = 2$, $y = -1$ and $z = -3$, hence

$$\text{grad } S = \hat{i}[2 \times 2 - 2 \times 2 \times (-1) + (-1)^2 \times (-3)^2] + \hat{j}[-(2)^2 - 2 \times 2 \times (-1) \times (-3)^2]$$

$$+ \hat{k}[2 \times (-3) \times 2 \times (-1)^2]$$

$$= 17\hat{i} - 40\hat{j} - 12\hat{k}.$$

Ex. 11. If ϕ and Ψ are differential scalar functions of x , y and z , then prove that

$$\text{grad } (\phi + \Psi) = \text{grad } \phi + \text{grad } \Psi \quad \text{or} \quad \vec{\nabla} (\phi + \Psi) = \vec{\nabla} \phi + \vec{\nabla} \Psi$$

Sol. $\text{grad } (\phi + \Psi) = \vec{\nabla} (\phi + \Psi) = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) (\phi + \Psi)$

$\therefore$ $\vec{\nabla} (\phi + \Psi) = \hat{i} \frac{\partial}{\partial x} (\phi + \Psi) + \hat{j} \frac{\partial}{\partial y} (\phi + \Psi) + \hat{k} \frac{\partial}{\partial z} (\phi + \Psi)$

$$\begin{aligned}
&= \hat{\mathbf{i}} \frac{\partial \phi}{\partial x} + \hat{\mathbf{i}} \frac{\partial \psi}{\partial x} + \hat{\mathbf{j}} \frac{\partial \phi}{\partial y} + \hat{\mathbf{j}} \frac{\partial \psi}{\partial y} + \hat{\mathbf{k}} \frac{\partial \phi}{\partial z} + \hat{\mathbf{k}} \frac{\partial \psi}{\partial z} \\
&= \left(\hat{\mathbf{i}} \frac{\partial \phi}{\partial x} + \hat{\mathbf{j}} \frac{\partial \phi}{\partial y} + \hat{\mathbf{k}} \frac{\partial \phi}{\partial z} \right) + \left(\hat{\mathbf{i}} \frac{\partial \psi}{\partial x} + \hat{\mathbf{j}} \frac{\partial \psi}{\partial y} + \hat{\mathbf{k}} \frac{\partial \psi}{\partial z} \right) \\
&= \vec{\nabla} \phi + \vec{\nabla} \psi = \text{grad } \phi + \text{grad } \psi.
\end{aligned}$$

Ex. 12. If ϕ and ψ are differential functions of x, y, z , prove that $\text{grad } (\phi\psi) = \phi \vec{\nabla} \psi + \psi \vec{\nabla} \phi$.

Sol.
$$\begin{aligned}
\text{grad } (\phi\psi) &= \left(\hat{\mathbf{i}} \frac{\partial}{\partial x} + \hat{\mathbf{j}} \frac{\partial}{\partial y} + \hat{\mathbf{k}} \frac{\partial}{\partial z} \right) (\phi\psi) \\
&= \hat{\mathbf{i}} \left(\phi \frac{\partial \psi}{\partial x} + \psi \frac{\partial \phi}{\partial x} \right) + \hat{\mathbf{j}} \left(\phi \frac{\partial \psi}{\partial y} + \psi \frac{\partial \phi}{\partial y} \right) + \hat{\mathbf{k}} \left(\phi \frac{\partial \psi}{\partial z} + \psi \frac{\partial \phi}{\partial z} \right) \\
&= \phi \left(\hat{\mathbf{i}} \frac{\partial \psi}{\partial x} + \hat{\mathbf{j}} \frac{\partial \psi}{\partial y} + \hat{\mathbf{k}} \frac{\partial \psi}{\partial z} \right) + \psi \left(\hat{\mathbf{i}} \frac{\partial \phi}{\partial x} + \hat{\mathbf{j}} \frac{\partial \phi}{\partial y} + \hat{\mathbf{k}} \frac{\partial \phi}{\partial z} \right) = \phi \vec{\nabla} \psi + \psi \vec{\nabla} \phi.
\end{aligned}$$

Ex. 13. If $\vec{\mathbf{r}} = x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z\hat{\mathbf{k}}$, prove that $\text{grad } f(r) = \hat{\mathbf{r}} \frac{\partial f(r)}{\partial r}$ and $\text{grad } f(r) \times \vec{\mathbf{r}} = 0$.

Sol.
$$\text{grad } f(r) = \hat{\mathbf{i}} \frac{\partial f(r)}{\partial x} + \hat{\mathbf{j}} \frac{\partial f(r)}{\partial y} + \hat{\mathbf{k}} \frac{\partial f(r)}{\partial z}$$

But,
$$\frac{\partial f(r)}{\partial x} = \frac{\partial f(r)}{\partial r} \frac{\partial r}{\partial x} \text{ and } r = \sqrt{x^2 + y^2 + z^2}; \text{ so that } \frac{\partial r}{\partial x} = \frac{1}{2} \frac{2x}{\sqrt{x^2 + y^2 + z^2}} = \frac{x}{r}.$$

$$\therefore \frac{\partial f(r)}{\partial x} = \frac{x}{r} \frac{\partial f(r)}{\partial r}; \text{ similarly } \frac{\partial f(r)}{\partial y} = \frac{y}{r} \frac{\partial f(r)}{\partial r} \text{ and } \frac{\partial f(r)}{\partial z} = \frac{z}{r} \frac{\partial f(r)}{\partial r}$$

$$\therefore \text{grad } f(r) = \frac{x\hat{\mathbf{i}}}{r} \frac{\partial f(r)}{\partial r} + \frac{y\hat{\mathbf{j}}}{r} \frac{\partial f(r)}{\partial r} + \frac{z\hat{\mathbf{k}}}{r} \frac{\partial f(r)}{\partial r} = (x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z\hat{\mathbf{k}}) \frac{1}{r} \frac{\partial f(r)}{\partial r} = \frac{\vec{\mathbf{r}}}{r} \frac{\partial f(r)}{\partial r} = \hat{\mathbf{r}} \frac{\partial f(r)}{\partial r}.$$

Thus,
$$\text{grad } f(r) \times \hat{\mathbf{r}} = \frac{\partial f(r)}{\partial r} \hat{\mathbf{r}} \times \hat{\mathbf{r}} = \mathbf{0}. \quad [\because \text{For any vector } \vec{\mathbf{A}} \times \vec{\mathbf{A}} = \mathbf{0}.]$$

Ex. 14. Obtain a unit vector normal to the surface $xyz = K$ (constant) at the point $(1, -1, -2)$.

Sol. The direction of $\text{grad } \phi$ is normal to the surface $\phi = \text{constant}$. Hence for surface, $\phi = xyz = \text{constant}$, we have

$$\vec{\nabla} \phi = \left(\hat{\mathbf{i}} \frac{\partial}{\partial x} + \hat{\mathbf{j}} \frac{\partial}{\partial y} + \hat{\mathbf{k}} \frac{\partial}{\partial z} \right) (xyz) = yz\hat{\mathbf{i}} + zx\hat{\mathbf{j}} + xy\hat{\mathbf{k}}$$

Its value at $(1, -1, -2)$,
$$\vec{\nabla} \phi = (-1)(-2)\hat{\mathbf{i}} + (-2)(1)\hat{\mathbf{j}} + 1(-1)\hat{\mathbf{k}} = 2\hat{\mathbf{i}} - 2\hat{\mathbf{j}} - \hat{\mathbf{k}}$$

$$\therefore \text{Unit normal vector } \hat{\mathbf{n}} = \frac{\vec{\nabla} \phi}{|\vec{\nabla} \phi|} = \frac{2\hat{\mathbf{i}} - 2\hat{\mathbf{j}} - \hat{\mathbf{k}}}{\sqrt{2^2 + (-2)^2 + (-1)^2}} = \frac{2\hat{\mathbf{i}} - 2\hat{\mathbf{j}} - \hat{\mathbf{k}}}{3} = \frac{2}{3}\hat{\mathbf{i}} - \frac{2}{3}\hat{\mathbf{j}} - \frac{1}{3}\hat{\mathbf{k}}.$$

Another unit normal vector to the surface is
$$-\left(\frac{2}{3}\hat{\mathbf{i}} - \frac{2}{3}\hat{\mathbf{j}} - \frac{1}{3}\hat{\mathbf{k}} \right).$$

Ex. 15. The temperature at a point (x, y, z) is given by $T = x^2 - y^2 + xyz + 270$, find in what direction and with what rate, the temperature is increasing maximum at the point $(-1, 2, 3)$.

Sol. Here,
$$\vec{\nabla} T = (2x + yz)\hat{\mathbf{i}} + (-2y + xz)\hat{\mathbf{j}} + xy\hat{\mathbf{k}}$$

At the point $(-1, 2, 3)$,
$$\vec{\nabla} T = 4\hat{\mathbf{i}} - 7\hat{\mathbf{j}} - 2\hat{\mathbf{k}}.$$

Hence, the temperature increase at the maximum rate along the vector $(4\hat{i} - 7\hat{j} - 2\hat{k})$ and the maximum rate is

$$\frac{\partial T}{\partial r} = |\vec{\nabla} T| = \sqrt{4^2 + 7^2 + 2^2} = \sqrt{69}.$$

Ex. 16. If $\vec{a}$ is a constant quantity, then show that $\text{grad}(\vec{a} \cdot \vec{r}) = \vec{a}$.

Sol. Suppose $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ and $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$.

$\therefore \vec{a} \cdot \vec{r} = a_1x + a_2y + a_3z$; where a_1, a_2, a_3 are constants.

$$\begin{aligned} \text{Hence, } \text{grad}(\vec{a} \cdot \vec{r}) &= \hat{i} \frac{\partial}{\partial x}(a_1x + a_2y + a_3z) + \hat{j} \frac{\partial}{\partial y}(a_1x + a_2y + a_3z) + \hat{k} \frac{\partial}{\partial z}(a_1x + a_2y + a_3z) \\ &= \hat{i}(a_1 + 0 + 0) + \hat{j}(0 + a_2 + 0) + \hat{k}(0 + 0 + a_3) = a_1\hat{i} + a_2\hat{j} + a_3\hat{k} = \vec{a}. \end{aligned}$$

Ex. 17. The electric field at any point (x, y, z) is given by

$$\vec{E} = [2x\hat{i} + 2y\hat{j} + 2z\hat{k}].$$

Derive an equation for the equipotential surface.

$$\text{Sol. } \vec{E} = -\vec{\nabla} V = -\left[\hat{i} \frac{\partial V}{\partial x} + \hat{j} \frac{\partial V}{\partial y} + \hat{k} \frac{\partial V}{\partial z}\right] \quad \dots(i)$$

$$\text{Given, } \vec{E} = [2x\hat{i} + 2y\hat{j} + 2z\hat{k}] \quad \dots(ii)$$

Comparing eqs. (i) and (ii) we obtain

$$-\frac{\partial V}{\partial x} = 2x, \quad -\frac{\partial V}{\partial y} = 2y \quad \text{and} \quad -\frac{\partial V}{\partial z} = 2z.$$

$$\text{Now, } dV = \left(\frac{\partial V}{\partial x} dx + \frac{\partial V}{\partial y} dy + \frac{\partial V}{\partial z} dz\right)$$

$$\therefore dV = -2x dx - 2y dy - 2z dz$$

$$\text{Integrating, } V = \int [-2x dx - 2y dy - 2z dz]$$

$$V = -(x^2 + y^2 + z^2) + C, \text{ where } C \text{ is a constant.} \quad \dots(iii)$$

For constant potential V , the surface represented by eq. (iii) is an equipotential surface.

Ex. 18. Show that the $\vec{\nabla} \Phi$ is a vector perpendicular to the surface $\Phi(x, y, z) = C$, where C is constant.

Sol. Let S be a surface on which the scalar function $\Phi(x, y, z) = C$, constant. Consider $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ being the position vector to any point P on the surface.

$$\begin{aligned} \text{Now, } d\Phi &= \frac{\partial \Phi}{\partial x} dx + \frac{\partial \Phi}{\partial y} dy + \frac{\partial \Phi}{\partial z} dz \\ &= \left(\hat{i} \frac{\partial \Phi}{\partial x} + \hat{j} \frac{\partial \Phi}{\partial y} + \hat{k} \frac{\partial \Phi}{\partial z}\right) (dx + dy + dz) \\ &= \vec{\nabla} \Phi \cdot d\vec{r} \end{aligned}$$

If $d\vec{r}$ lies on the surface S (for which Φ is constant and $d\Phi$ is zero) close to P , then

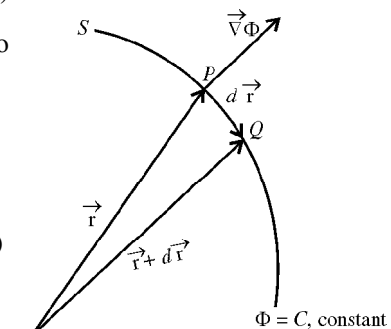


Fig. 1.18.

$$d\Phi = \vec{\nabla} \Phi \cdot d\vec{r} = 0$$

This means that the vector $\vec{\nabla} \Phi$ is normal to $d\vec{r}$ and hence to the surface $\Phi = C$, constant.

1.6. Divergence

If $\vec{A}(x, y, z)$ is a vector field, the scalar product of the vector operator $\vec{\nabla}$ and $\vec{A}$ is a scalar and is called the **divergence of $\vec{A}$** , i.e.,

$$\text{div } \vec{A} = \vec{\nabla} \cdot \vec{A} = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \cdot (\hat{i} A_x + \hat{j} A_y + \hat{k} A_z)$$

$$\text{or} \quad \text{div } \vec{A} = \vec{\nabla} \cdot \vec{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} \quad \dots(1)$$

On the basis of the explanation, given below, we can define $\text{div } \vec{A}$ as follows : **The divergence of a vector field at a point is a scalar quantity which represents the flux of that vector field diverging per unit volume from that point.**

Further, we see in the following discussion that the quantity $\vec{\nabla} \cdot \vec{A}$ at a point is the amount of flux per unit volume diverging from that point. Hence, the name divergence originated from the interpretation of $\vec{\nabla} \cdot \vec{A}$.

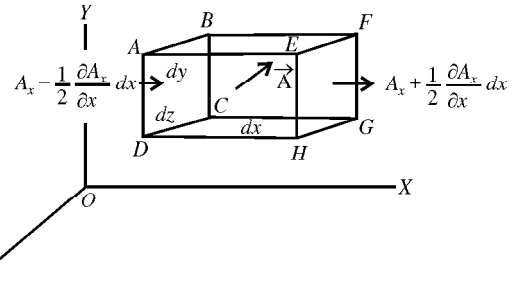


Fig. 1.19. Divergence of a vector field.

Physical Interpretation and Significance of $\text{div } \vec{A}$

In order to understand the meaning and physical significance of $\text{div } \vec{A}$, we consider an example of the flow of a liquid. Here, the velocity $\vec{v}$ of the liquid represents the vector field $\vec{A}(x, y, z)$. Consider the flow of the liquid in a space. In that space, let $\vec{A}$ be the value of a vector field at the mid point of a small rectangular parallelepiped with sides dx, dy and dz [fig. 1.19]. The axial component of the vector $\vec{A}$ are A_x, A_y, A_z , and in general, the value of the vector $\vec{A}$ is different at different points of the field.

On the face $ABCD$ the value of the X -component of the vector $\vec{A}$ is $A_x - \frac{1}{2} \frac{\partial A_x}{\partial x} dx$.

As the face is infinitesimally small, this value can be taken as uniform over the face $ABCD$.

Similarly, for the face $EFGH$, the X -component is $A_x + \frac{1}{2} \frac{\partial A_x}{\partial x} dx$.

The **flux** through a face is defined as the product of the area of the face and the normal component of the vector upon it.

$$\therefore \text{Flux entering the face } ABCD = \left(A_x - \frac{1}{2} \frac{\partial A_x}{\partial x} dx \right) dy dz$$

$$\text{and} \quad \text{flux leaving the face } EFGH = \left(A_x + \frac{1}{2} \frac{\partial A_x}{\partial x} dx \right) dy dz$$

Hence, the excess of flux leaving the parallelepiped over that entering it in the X -direction is

$$\left(A_x + \frac{1}{2} \frac{\partial A_x}{\partial x} dx \right) dy dz - \left(A_x - \frac{1}{2} \frac{\partial A_x}{\partial x} dx \right) dy dz = \frac{\partial A_x}{\partial x} dx dy dz$$

In the hydromechanical case, this represents the **net volume** of the fluid passing per second in the X -direction. Similarly, contributions parallel to Y and Z directions are $\frac{\partial A_y}{\partial x} dx dy dz$ and $\frac{\partial A_z}{\partial x} dx dy dz$.

Hence the **net flux diverging from the element** is

$$d\phi = \left(\frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} \right) dx dy dz = (\vec{\nabla} \cdot \vec{A}) dx dy dz = (\text{div } \vec{A}) dV \quad \dots(2)$$

This quantity $d\phi$ tells us the net volume of the liquid diverging from $dV = dx dy dz$.

$$\text{Hence,} \quad \frac{d\phi}{dV} = \text{div } \vec{A} = \vec{\nabla} \cdot \vec{A} \quad \dots(3)$$

Thus $\vec{\nabla} \cdot \vec{A}$ (or $\text{div } \vec{A}$) at a point gives the amount of flux diverging per unit volume from that point. If $\vec{A}$ is denoting the velocity of the moving fluid, then $\vec{\nabla} \cdot \vec{A}$ tells the rate at which the fluid is diverging from the point per unit volume. Vector function $\vec{A}$ can be the volume of a flowing gas and hence we can use the word fluid instead liquid.

Equation of Continuity

If in a fluid, the divergence of the field (velocity) at a point is positive, then either the fluid is *expanding* and the *density at that point is decreasing* or that point is the **source of fluid**. If the divergence at a point is negative, then either the liquid is *contracting* and its *density is increasing at that point* or that point is **negative source** or **sink**.

If at the point (x, y, z) , the velocity of the fluid is $\vec{v}$ and density is ρ , then the vector $\rho \vec{v} = \vec{A}$ represents the mass of the fluid passing through unit area per second and the net mass going out of the volume dV per second will be $\text{div}(\rho \vec{v}) dV$. Due to this, the density of the fluid in that volume dV will be less and if $d\rho/dt$ is the rate of change of density with time, then $(\partial\rho/\partial t)dV$ will be the decrease in the mass of the fluid per second in the volume dV . Hence,

$$\text{div}(\rho \vec{v}) dV = -\frac{\partial \rho}{\partial t} dV \quad \text{i.e.,} \quad \frac{\partial \rho}{\partial t} + \text{div}(\rho \vec{v}) = 0 \quad \dots(4)$$

This equation (4) is called the **equation of continuity**. Here, in general ρ is a function of x, y, z and time t i.e., $\rho = \rho(x, y, z, t)$ and the fluid in general is compressible.

Thus, $\text{div } \vec{A}$ or $\vec{\nabla} \cdot \vec{A}$ at a point is the amount of flux diverging per unit volume from that point and hence the divergence of a vector field is an important concept which has physical importance in the different fields of physics. We have seen its importance in **hydrodynamics**. If $\vec{J}$ represents the current density, at a point in the charge flow, then $\text{div } \vec{J}$ gives the rate of flow of charge per unit volume at that point so that the charge density is decreasing there (for positive $\text{div } \vec{J}$) i.e.,

$$\text{div } \vec{J} = -\frac{\partial \rho}{\partial t} \quad \text{or} \quad \frac{\partial \rho}{\partial t} + \text{div } \vec{J} = 0 \quad \dots(5)$$

This is known as *equation of continuity in charge flow*.

In Quantum Mechanics, the *divergence of the probability current density* and the *time derivative of the position probability density* is represented by an equation of continuity similar to eq. (5).

Solenoidal Vector

If the flux, entering any element of field space is exactly balanced by that leaving it, the quantity $\text{div } \vec{A} = 0$. Then, there is no source or sink, nor its density is changing, i.e., the fluid is incompressible. If the fluxes entering and leaving an element are equal, the lines of flow of the vector $\vec{A}$ should form either

18 | Mechanics and Waves and Oscillations

closed curves (e.g., in case of magnetic field of a current) or extend to infinity. A vector which satisfies this condition is called **solenoidal**.

For example, one of the Maxwell's electromagnetic field equation is

$$\text{div } \vec{B} = 0 \quad \text{or} \quad \vec{\nabla} \cdot \vec{B} = 0 \quad \dots(6)$$

where magnetic induction $\vec{B}$ is a **solenoidal vector**.

Ex. 1. If $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, find the value of $\text{div } \vec{r}$.

Sol.

$$\begin{aligned} \text{div } \vec{r} &= \vec{\nabla} \cdot \vec{r} = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) (x\hat{i} + y\hat{j} + z\hat{k}) \\ &= \frac{\partial x}{\partial x} + \frac{\partial y}{\partial y} + \frac{\partial z}{\partial z} = 1 + 1 + 1 = 3. \end{aligned}$$

Ex. 2. If $\vec{A} = x^2y\hat{i} - 2y^3z^2\hat{j} + xy^2z\hat{k}$, find the divergence of this vector at the point (1, 1, 1).

Sol.

$$\begin{aligned} \text{div } \vec{A} &= \vec{\nabla} \cdot \vec{A} = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \cdot (x^2y\hat{i} - 2y^3z^2\hat{j} + xy^2z\hat{k}) \\ &= \frac{\partial}{\partial x}(x^2y) + \frac{\partial}{\partial y}(-2y^3z^2) + \frac{\partial}{\partial z}(xy^2z) = 2xy - 6y^2z^2 + xy^2 \end{aligned}$$

At the point (1, 1, 1), $\vec{\nabla} \cdot \vec{A} = 2(1)(1) - 6(1)^2(1)^2 + (1)(1)^2 = -3$. (S. K. U. 2003)

Ex. 3 If $\vec{A} = \hat{i}y + \hat{j}(x^2 + y^2) + \hat{k}(yz + zx)$, then find $\text{div } \vec{A}$ at (1, -2, 3).

Sol.

$$\begin{aligned} \text{div } \vec{A} &= \vec{\nabla} \cdot \vec{A} \\ &= \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) [\hat{i}y + \hat{j}(x^2 + y^2) + \hat{k}(yz + zx)] \\ &= \frac{\partial}{\partial x}(y) + \frac{\partial}{\partial y}(x^2 + y^2) + \frac{\partial}{\partial z}(yz + zx) = 0 + 2y + y + x = x + 3y \end{aligned}$$

Hence at point (1, -2, 3), the value of $\text{div } \vec{A}$ is given by

$$\text{div } \vec{A} = 1 + 3 \times -2 = 1 - 6 = -5.$$

Ex. 4. (a) If ϕ is a scalar function, prove that $\text{div grad } \phi = \nabla^2 \phi$.

(b) Prove that $\text{div grad } (1/r) = 0$.

Sol. (a)

$$\begin{aligned} \text{div (grad } \phi) &= \vec{\nabla} \cdot (\vec{\nabla} \phi) = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \cdot \left(\hat{i} \frac{\partial \phi}{\partial x} + \hat{j} \frac{\partial \phi}{\partial y} + \hat{k} \frac{\partial \phi}{\partial z} \right) \\ &= \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2} = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) \phi = \nabla^2 \phi. \end{aligned}$$

where $\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$ is called Laplacian operator.

(b) $\text{div grad } \left(\frac{1}{r} \right) = \nabla^2 \left(\frac{1}{r} \right) = \frac{\partial^2}{\partial x^2} \left(\frac{1}{r} \right) + \frac{\partial^2}{\partial y^2} \left(\frac{1}{r} \right) + \frac{\partial^2}{\partial z^2} \left(\frac{1}{r} \right)$

But

$$\frac{\partial^2}{\partial x^2} \left(\frac{1}{r} \right) = \frac{\partial}{\partial x} \left[\frac{\partial}{\partial x} \left(\frac{1}{r} \right) \right] = \frac{\partial}{\partial x} \left(-\frac{1}{r^2} \frac{\partial r}{\partial x} \right)$$

$$\therefore r^2 = x^2 + y^2 + z^2, \therefore 2r \frac{\partial r}{\partial x} = 2x \text{ or } \frac{\partial r}{\partial x} = \frac{x}{r}$$

$$\begin{aligned} \text{Hence, } \frac{\partial^2}{\partial x^2} \left(\frac{1}{r} \right) &= -\frac{\partial}{\partial x} \left(\frac{x}{r^3} \right) = -\frac{1}{r^3} - \frac{\partial}{\partial x} \left(\frac{1}{r^3} \right) = -\frac{1}{r^3} + \frac{3x}{r^4} \frac{\partial r}{\partial x} \\ &= -\frac{1}{r^3} + \frac{3x}{r^4} \frac{x}{r} = -\frac{1}{r^3} + \frac{3x^2}{r^5} \end{aligned}$$

$$\text{Similarly, } \frac{\partial^2}{\partial y^2} \left(\frac{1}{r} \right) = -\frac{1}{r^3} + \frac{3y^2}{r^5} \text{ and } \frac{\partial^2}{\partial z^2} \left(\frac{1}{r} \right) = -\frac{1}{r^3} + \frac{3z^2}{r^5}$$

$$\therefore \operatorname{div} \operatorname{grad} \left(\frac{1}{r} \right) = -\frac{3}{r^3} + \frac{3}{r^5} (x^2 + y^2 + z^2) = -\frac{3}{r^3} + \frac{3}{r^3} = 0.$$

Ex. 5. If $\vec{A}$ and $\vec{B}$ are vector fields, then show that

$$\operatorname{div}(\vec{A} + \vec{B}) = \operatorname{div} \vec{A} + \operatorname{div} \vec{B}.$$

$$\begin{aligned} \text{Sol. } \operatorname{div}(\vec{A} + \vec{B}) &= \vec{\nabla} \cdot (\vec{A} + \vec{B}) = \left[\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right] [(A_x + B_x) \hat{i} + (A_y + B_y) \hat{j} + (A_z + B_z) \hat{k}] \\ &= \frac{\partial}{\partial x} (A_x + B_x) + \frac{\partial}{\partial y} (A_y + B_y) + \frac{\partial}{\partial z} (A_z + B_z) \\ &= \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} + \frac{\partial B_x}{\partial x} + \frac{\partial B_y}{\partial y} + \frac{\partial B_z}{\partial z}, = \vec{\nabla} \cdot \vec{A} + \vec{\nabla} \cdot \vec{B} \\ &= \operatorname{div} \vec{A} + \operatorname{div} \vec{B}. \end{aligned}$$

Ex. 6. If r is the position vector of a point, then show that

$$(i) \operatorname{div} \vec{r} = 3 \text{ and } (ii) \vec{\nabla}(\vec{r} \cdot \vec{A}) = \vec{A}. \quad (\text{S. V. U. 2002; O. U. 1997; K. U. 1989})$$

Sol. (i) The position vector $\vec{r} = \hat{i}x + \hat{j}y + \hat{k}z$

$$\begin{aligned} \therefore \operatorname{div} \vec{r} &= \vec{\nabla} \cdot \vec{r} = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) (\hat{i}x + \hat{j}y + \hat{k}z) \\ &= \frac{\partial x}{\partial x} + \frac{\partial y}{\partial y} + \frac{\partial z}{\partial z} = 1 + 1 + 1 = 3. \end{aligned}$$

$$(ii) \quad \vec{r} \cdot \vec{A} = (\hat{i}x + \hat{j}y + \hat{k}z) \cdot (\hat{i}A_x + \hat{j}A_y + \hat{k}A_z) = xA_x + yA_y + zA_z$$

$$\therefore \vec{\nabla}(\vec{r} \cdot \vec{A}) = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) (xA_x + yA_y + zA_z) = \hat{i}A_x + \hat{j}A_y + \hat{k}A_z = \vec{A}.$$

Ex. 7. If $\vec{r} = \hat{i}x + \hat{j}y + \hat{k}z$, find,

$$(i) \operatorname{div} (r^3 \vec{r}) \quad (ii) \operatorname{div} (r^n \vec{r}) \quad (iii) \operatorname{div} (r^{-3} \vec{r}).$$

Sol. (i) $\operatorname{div} (r^3 \vec{r})$

$$\begin{aligned} &= \vec{\nabla} \cdot (r^3 \vec{r}) = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \{ r^3 (\hat{i}x + \hat{j}y + \hat{k}z) \} = \frac{\partial}{\partial x} (xr^3) + \frac{\partial}{\partial y} (yr^3) + \frac{\partial}{\partial z} (zr^3) \\ &= \left\{ r^3 + x \frac{\partial}{\partial x} (x^2 + y^2 + z^2)^{3/2} \right\} + \left\{ r^3 + y \frac{\partial}{\partial y} (x^2 + y^2 + z^2)^{3/2} \right\} + \left\{ r^3 + z \frac{\partial}{\partial z} (x^2 + y^2 + z^2)^{3/2} \right\} \\ &= \left\{ r^3 + x \cdot \frac{3}{2} (x^2 + y^2 + z^2)^{1/2} \cdot 2x \right\} + \dots + \dots \\ &= 3r^3 + 3(x^2 + y^2 + z^2)^{1/2} (x^2 + y^2 + z^2) = 3r^3 + 3r^3 = 6r^3. \end{aligned}$$

$$\begin{aligned}
\text{(ii) } \operatorname{div} \vec{r} &= \vec{\nabla} \cdot (r^n \vec{r}) = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \cdot \{r^n (\hat{i}x + \hat{j}y + \hat{k}z)\} = \frac{\partial}{\partial x} (xr^n) + \frac{\partial}{\partial y} (yr^n) + \frac{\partial}{\partial z} (zr^n) \\
\therefore r &= (x^2 + y^2 + z^2)^{1/2}, \quad \therefore r^n = (x^2 + y^2 + z^2)^{n/2} \\
\therefore \operatorname{div} (r^n \vec{r}) &= \left(r^n + x \frac{\partial}{\partial x} r^n \right) + \left(r^n + y \frac{\partial}{\partial y} r^n \right) + \left(r^n + z \frac{\partial}{\partial z} r^n \right) \\
&= 3r^n + x \frac{\partial}{\partial x} (x^2 + y^2 + z^2)^{n/2} + y \frac{\partial}{\partial y} (x^2 + y^2 + z^2)^{n/2} + z \frac{\partial}{\partial z} (x^2 + y^2 + z^2)^{n/2} \\
&= 3r^n + x \frac{n}{2} (x^2 + y^2 + z^2)^{n/2-1} 2x + y \frac{n}{2} (x^2 + y^2 + z^2)^{n/2-1} 2y + z \frac{n}{2} (x^2 + y^2 + z^2)^{n/2-1} 2z \\
&= 3r^n + n(x^2 + y^2 + z^2) (x^2 + y^2 + z^2)^{n/2-1} = 3r^n + n(x^2 + y^2 + z^2)^{n/2} \\
&= 3r^n + nr^n = (3 + n)r^n.
\end{aligned}$$

$$\begin{aligned}
\text{(iii) } \operatorname{div} (r^{-3} \vec{r}) &= \vec{\nabla} \cdot \frac{\vec{r}}{r^3} = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \cdot \left(\frac{\hat{i}x + \hat{j}y + \hat{k}z}{r^3} \right) = \frac{\partial}{\partial x} \left(\frac{x}{r^3} \right) + \frac{\partial}{\partial y} \left(\frac{y}{r^3} \right) + \frac{\partial}{\partial z} \left(\frac{z}{r^3} \right) \\
&= \frac{1}{r^3} - x \frac{3}{r^4} \frac{\partial r}{\partial x} + \frac{1}{r^3} - y \frac{3}{r^4} \frac{\partial r}{\partial y} + \frac{1}{r^3} - z \frac{3}{r^4} \frac{\partial r}{\partial z} \\
\text{But, } r^2 &= x^2 + y^2 + z^2, \quad \therefore \frac{\partial r}{\partial x} = \frac{x}{r}, \quad \frac{\partial r}{\partial y} = \frac{y}{r}, \quad \frac{\partial r}{\partial z} = \frac{z}{r} \\
\therefore \operatorname{div} \left(\frac{\vec{r}}{r^3} \right) &= \frac{1}{r^3} - \frac{3x^2}{r^5} + \frac{1}{r^3} - \frac{3y^2}{r^5} + \frac{1}{r^3} - \frac{3z^2}{r^5} = \frac{3}{r^3} - \frac{3}{r^5} (x^2 + y^2 + z^2) = \frac{3}{r^3} - \frac{3}{r^5} r^2 = \frac{3}{r^3} - \frac{3}{r^3} = 0.
\end{aligned}$$

Ex. 8. Show that divergence of a vector field obeying inverse square law is zero i.e., such a field is solenoidal.

Sol. Vector field obeying inverse square law is represented as

$$\vec{F} \propto \frac{1}{r^2} \hat{r} \quad \text{or} \quad \vec{F} = \frac{K}{r^2} \hat{r} = \frac{K}{r^3} \vec{r} = \frac{K}{r^3} (\hat{i}x + \hat{j}y + \hat{k}z)$$

where $\vec{r} = \hat{i}x + \hat{j}y + \hat{k}z$ and K is a constant.

$$\begin{aligned}
\therefore \operatorname{div} \vec{F} &= \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z} = \frac{\partial}{\partial x} \left(\frac{Kx}{r^3} \right) + \frac{\partial}{\partial y} \left(\frac{Ky}{r^3} \right) + \frac{\partial}{\partial z} \left(\frac{Kz}{r^3} \right) \\
&= \left\{ \frac{K}{r^3} + Kx \left(\frac{-3}{r^4} \right) \frac{\partial r}{\partial x} \right\} + \left\{ \frac{K}{r^3} + Ky \left(\frac{-3}{r^4} \right) \frac{\partial r}{\partial y} \right\} + \left\{ \frac{K}{r^3} + Kz \left(\frac{-3}{r^4} \right) \frac{\partial r}{\partial z} \right\} \\
&= \frac{3K}{r^3} - \frac{3K}{r^4} \left\{ x \frac{\partial r}{\partial x} + y \frac{\partial r}{\partial y} + z \frac{\partial r}{\partial z} \right\}
\end{aligned}$$

$$\text{But } r^2 = x^2 + y^2 + z^2$$

$$\therefore 2r \frac{\partial r}{\partial x} = 2x \quad \text{or} \quad \frac{\partial r}{\partial x} = \frac{x}{r}. \quad \text{Similarly } \frac{\partial r}{\partial y} = \frac{y}{r} \quad \text{and} \quad \frac{\partial r}{\partial z} = \frac{z}{r}.$$

$$\text{Thus, } \operatorname{div} \vec{F} = \frac{3K}{r^3} - \frac{3K}{r^4} \left[x \frac{x}{r} + y \frac{y}{r} + z \frac{z}{r} \right] = \frac{3K}{r^3} - \frac{3K}{r^5} (x^2 + y^2 + z^2) = \frac{3K}{r^3} - \frac{3K}{r^5} (r^2) = 0.$$

Hence field $\vec{F}$ is solenoidal.

Ex. 9. Prove that : $\operatorname{div} \operatorname{grad} r^n = n(n+1)r^{n-2}$.

$$\text{Sol. } \operatorname{div} \operatorname{grad} r^n = \vec{\nabla} \cdot \vec{\nabla} r^n$$

$$\text{Here, } \vec{\nabla} r^n = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) r^n = nr^{n-1} \left(\frac{\partial r}{\partial x} \hat{i} + \frac{\partial r}{\partial y} \hat{j} + \frac{\partial r}{\partial z} \hat{k} \right).$$

But, $r^2 = x^2 + y^2 + z^2$, $\therefore 2r \frac{\partial r}{\partial x} = 2x$ or $\frac{\partial r}{\partial x} = \frac{x}{r}$. Similarly, $\frac{\partial r}{\partial y} = \frac{y}{r}$, $\frac{\partial r}{\partial z} = \frac{z}{r}$

Hence, $\vec{\nabla} r^n = nr^{n-1} \left(\frac{x}{r} \hat{i} + \frac{y}{r} \hat{j} + \frac{z}{r} \hat{k} \right) = nr^{n-2} (x \hat{i} + y \hat{j} + z \hat{k})$

$$\begin{aligned} \therefore \operatorname{div} \operatorname{grad} r^n &= \vec{\nabla} \cdot \vec{\nabla} r^n = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) nr^{n-2} (x \hat{i} + y \hat{j} + z \hat{k}) \\ &= \frac{\partial}{\partial x} (nr^{n-2} x) + \frac{\partial}{\partial y} (nr^{n-2} y) + \frac{\partial}{\partial z} (nr^{n-2} z) = n(n-2)r^{n-3} \frac{\partial r}{\partial x} x + nr^{n-2} \frac{\partial x}{\partial x} + \dots \\ &= nr^{n-2} + n(n-2)r^{n-3} \frac{x^2}{r} + nr^{n-2} + n(n-2)r^{n-3} \frac{y^2}{r} + nr^{n-2} + n(n-2)r^{n-3} \frac{z^2}{r} \\ &= 3nr^{n-2} + n(n-2)r^{n-3} \left(\frac{x^2 + y^2 + z^2}{r} \right) = 3nr^{n-2} + n(n-2)r^{n-2} \\ &= 3nr^{n-2} + n^2 r^{n-2} - 2nr^{n-2} = nr^{n-2} + n^2 r^{n-2} = n(n+1)r^{n-2} \quad [\because r^2 = x^2 + y^2 + z^2] \end{aligned}$$

Ex. 10. If $r = (x^2 + y^2 + z^2)^{1/2}$, prove that $\operatorname{div} \operatorname{grad} f(r) = \vec{\nabla} \cdot \vec{\nabla} f(r) = \frac{\partial^2 f(r)}{\partial r^2} + \frac{2}{r} \frac{\partial f(r)}{\partial r}$. Hence prove that $\operatorname{div} \operatorname{grad} r^n = n(n+1)r^{n-2}$.

Sol. We have proved that $\operatorname{grad} f(r) = \frac{\vec{r}}{r} \frac{\partial f(r)}{\partial r}$ Hence,

$$\vec{\nabla} \cdot \vec{\nabla} f(r) = \vec{\nabla} \cdot \frac{\vec{r}}{r} \frac{\partial f(r)}{\partial r} = \frac{\partial}{\partial x} \left(\frac{x}{r} \frac{\partial f(r)}{\partial r} \right) + \frac{\partial}{\partial y} \left(\frac{y}{r} \frac{\partial f(r)}{\partial r} \right) + \frac{\partial}{\partial z} \left(\frac{z}{r} \frac{\partial f(r)}{\partial r} \right)$$

Its first term will be

$$\begin{aligned} \frac{\partial}{\partial x} \left(\frac{x}{r} \frac{\partial f(r)}{\partial r} \right) &= \frac{1}{r} \frac{\partial f(r)}{\partial r} + x \frac{\partial}{\partial x} \left(\frac{1}{r} \frac{\partial f(r)}{\partial r} \right) = \frac{1}{r} \frac{\partial f(r)}{\partial r} + x \left(-\frac{1}{r^2} \right) \frac{\partial r}{\partial x} \frac{\partial f(r)}{\partial r} + \frac{x}{r} \frac{\partial}{\partial x} \left(\frac{\partial f(r)}{\partial r} \right) \\ &= \frac{1}{r} \frac{\partial f(r)}{\partial r} - \frac{x}{r^2} \frac{x}{r} \frac{\partial f(r)}{\partial r} + \frac{\partial}{\partial r} \left(\frac{\partial f(r)}{\partial r} \right) \frac{\partial r}{\partial x} \frac{x}{r} = \frac{1}{r} \frac{\partial f(r)}{\partial r} - \frac{x^2}{r^3} \frac{\partial f(r)}{\partial r} + \frac{\partial^2 f(r)}{\partial r^2} \frac{x}{r} \\ &= \frac{1}{r} \frac{\partial f(r)}{\partial r} - \frac{x^2}{r^3} \frac{\partial f(r)}{\partial r} + \frac{x^2}{r^2} \frac{\partial^2 f(r)}{\partial r^2} \end{aligned}$$

Similarly finding other terms and adding them, we get

$$\begin{aligned} \vec{\nabla} \cdot \vec{\nabla} f(r) &= \left(\frac{1}{r} \frac{\partial f(r)}{\partial r} - \frac{x^2}{r^3} \frac{\partial f(r)}{\partial r} + \frac{x^2}{r^2} \frac{\partial^2 f(r)}{\partial r^2} \right) + \left(\frac{1}{r} \frac{\partial f(r)}{\partial r} - \frac{y^2}{r^3} \frac{\partial f(r)}{\partial r} + \frac{y^2}{r^2} \frac{\partial^2 f(r)}{\partial r^2} \right) \\ &\quad + \left(\frac{1}{r} \frac{\partial f(r)}{\partial r} - \frac{z^2}{r^3} \frac{\partial f(r)}{\partial r} + \frac{z^2}{r^2} \frac{\partial^2 f(r)}{\partial r^2} \right) \\ &= \frac{3}{r} \frac{\partial f(r)}{\partial r} - \left(\frac{x^2 + y^2 + z^2}{r^3} \right) \frac{\partial f(r)}{\partial r} + \left(\frac{x^2 + y^2 + z^2}{r^2} \right) \frac{\partial^2 f(r)}{\partial r^2} = \frac{\partial^2 f(r)}{\partial r^2} + \frac{2}{r} \frac{\partial f(r)}{\partial r} \\ &\quad [\because r^2 = x^2 + y^2 + z^2] \end{aligned}$$

If $f(r) = r^n$, then $\vec{\nabla} \cdot \vec{\nabla} f(r) = \nabla^2 r^n = n(n-1)r^{n-2} + \frac{2}{r} nr^{n-1}$

or $\operatorname{div} \operatorname{grad} r^n = n(n+1)r^{n-2}$.

Ex. 11. If $\vec{A} = (x + 3y)\hat{i} + (y - z)\hat{j} + (x + cz)\hat{k}$ is a solenoidal vector, find the value of c .

Sol. Here $A_x = x + 3y$, $A_y = y - z$, $A_z = x + cz$; $\therefore \frac{\partial A_x}{\partial x} = 1$, $\frac{\partial A_y}{\partial y} = 1$, $\frac{\partial A_z}{\partial z} = c$.

Hence $\vec{\nabla} \cdot \vec{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} = 1 + 1 + c = 2 + c$.

But $\vec{A}$ is a solenoidal vector, hence $\vec{\nabla} \cdot \vec{A} = 0$ i.e., $2 + c = 0$ or $c = -2$.

Ex. 12. Prove that : $\text{div}(\psi \vec{A}) = (\text{grad } \psi) \cdot \vec{A} + \psi \text{div } \vec{A}$.

Sol. Let $\vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$, $\therefore \psi \vec{A} = \psi A_x \hat{i} + \psi A_y \hat{j} + \psi A_z \hat{k}$

$$\begin{aligned} \text{div}(\psi \vec{A}) &= \vec{\nabla} \cdot (\psi \vec{A}) = \frac{\partial}{\partial x}(\psi A_x) + \frac{\partial}{\partial y}(\psi A_y) + \frac{\partial}{\partial z}(\psi A_z) \\ &= \psi \frac{\partial A_x}{\partial x} + \frac{\partial \psi}{\partial x} A_x + \psi \frac{\partial A_y}{\partial y} + \frac{\partial \psi}{\partial y} A_y + \psi \frac{\partial A_z}{\partial z} + \frac{\partial \psi}{\partial z} A_z \\ &= \psi \left(\frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} \right) + \left(\frac{\partial \psi}{\partial x} A_x + \frac{\partial \psi}{\partial y} A_y + \frac{\partial \psi}{\partial z} A_z \right) \\ &= \psi (\vec{\nabla} \cdot \vec{A}) + \left\{ \hat{i} \frac{\partial \psi}{\partial x} + \hat{j} \frac{\partial \psi}{\partial y} + \hat{k} \frac{\partial \psi}{\partial z} \right\} \cdot (A_x \hat{i} + A_y \hat{j} + A_z \hat{k}) = \psi \text{div } \vec{A} + (\text{grad } \psi) \cdot \vec{A}. \end{aligned}$$

Ex. 13. Prove that : $\vec{A}(\vec{\nabla} \cdot \vec{r}) - (\vec{A} \cdot \vec{\nabla})\vec{r} = 2\vec{A}$.

Sol. $\vec{A}(\vec{\nabla} \cdot \vec{r}) = \vec{A} \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \cdot (x\hat{i} + y\hat{j} + z\hat{k}) = \vec{A} \left(\frac{\partial x}{\partial x} + \frac{\partial y}{\partial y} + \frac{\partial z}{\partial z} \right) = \vec{A}(1 + 1 + 1) = 3\vec{A}$.

and $(\vec{A} \cdot \vec{\nabla})\vec{r} = \left(A_x \frac{\partial}{\partial x} + A_y \frac{\partial}{\partial y} + A_z \frac{\partial}{\partial z} \right) (x\hat{i} + y\hat{j} + z\hat{k}) = (A_x \hat{i} + A_y \hat{j} + A_z \hat{k}) = \vec{A}$.

Hence, $\vec{A}(\vec{\nabla} \cdot \vec{r}) - (\vec{A} \cdot \vec{\nabla})\vec{r} = 3\vec{A} - \vec{A} = 2\vec{A}$.

Ex. 14. If $\vec{a}$ is a constant vector and $\vec{r}$ is the position vector, prove that $\text{div}(\vec{a} \times \vec{r}) = 0$.

Sol. Here, $\vec{a} = \hat{i}a_x + \hat{j}a_y + \hat{k}a_z = \text{constant}$ and $\vec{r} = \hat{i}x + \hat{j}y + \hat{k}z$

$$\begin{aligned} \therefore \text{div}(\vec{a} \times \vec{r}) &= \vec{\nabla} \cdot (\vec{a} \times \vec{r}) = \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ a_x & a_y & a_z \\ x & y & z \end{vmatrix} = \frac{\partial}{\partial x}(a_y z - a_z y) + \frac{\partial}{\partial y}(a_z x - a_x z) + \frac{\partial}{\partial z}(a_x y - a_y x) \\ &= \left[0 \cdot z + a_y \frac{\partial z}{\partial x} - 0 \cdot y - a_z \frac{\partial y}{\partial x} \right] + \left[0 \cdot x + a_z \frac{\partial x}{\partial y} - 0 \cdot z - a_x \frac{\partial z}{\partial y} \right] + \left[0 \cdot y + a_x \frac{\partial y}{\partial z} - 0 \cdot x - a_y \frac{\partial x}{\partial z} \right] = 0. \end{aligned}$$

1.7. Curl

If $\vec{A}(x, y, z)$ is a vector field, the cross product of the operator $\vec{\nabla}$ and $\vec{A}$ is a vector and is called curl or rotation of $\vec{A}$. Thus,

$$\text{curl } \vec{A} = \vec{\nabla} \times \vec{A} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix}$$

$$= \hat{i} \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) + \hat{j} \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) + \hat{k} \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) \quad \dots(1)$$

The curl of a vector field $\vec{A}$ at a point is related to the line integral $\left(\oint \vec{A} \cdot d\vec{l} \right)$, taken around the boundary of an infinitesimal small plane area situated around that point and curl $\vec{A}$ is defined as “**Curl of a vector field ($\vec{A}$) at a point is a vector (curl $\vec{A}$) whose magnitude is equal to the maximum value of the line integral of that vector ($\oint \vec{A} \cdot d\vec{l}$) per unit area along the boundary of an infinitesimal small area around that point and the direction is along that infinitesimal small area.**” This definition will emerge from the following discussion :

The concept of curl is very useful in Vector Calculus and has significant and wide applications in hydrodynamics and electro-magnetism.

Physical Interpretation and Significance of curl $\vec{A}$

In order to understand the physical significance of curl of a vector field $\vec{A}$, we consider the **circulation of a fluid** around an infinitesimal small area in XY -plane. In hydrodynamics, the line integral of a vector field $\vec{A}$ along a closed curve (as the velocity $\vec{v}$ of a liquid in whirlpools and eddies) is called the **circulation** of $\vec{A}$ along that path. For convenience, we take this path as infinitesimal rectangle in XY -plane and calculate **the line integral of this vector field $\vec{A}$ around the infinitesimal rectangle.**

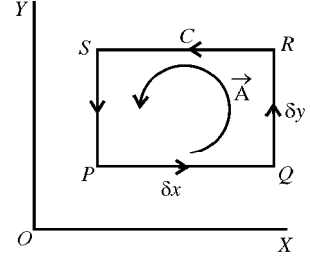


Fig. 1.20. Curl or circulation of a vector field.

In fig 1.20, we consider an infinitesimal rectangle $PQRS$ with sides δx and δy in XY -plane. Let the X - and Y - components of $\vec{A}$ at the point P be A_x and A_y respectively. The average value of the X -component of $\vec{A}$ over

the path PQ can be taken $A_x + \frac{\partial A_x}{\partial x} \cdot \frac{\delta x}{2}$ and over the path RS as $A_x + \frac{\partial A_x}{\partial x} \cdot \frac{\delta x}{2} + \frac{\partial A_x}{\partial y} \cdot \delta y$.

$$\begin{aligned} \text{Therefore, } \int_P^Q \vec{A} \cdot d\vec{l} + \int_R^S \vec{A} \cdot d\vec{l} &= \left(A_x + \frac{\partial A_x}{\partial x} \cdot \frac{\delta x}{2} \right) \delta x - \left(A_x + \frac{\partial A_x}{\partial x} \cdot \frac{\delta x}{2} + \frac{\partial A_x}{\partial y} \cdot \delta y \right) \delta x \\ &= - \frac{\partial A_x}{\partial y} \cdot \delta x \delta y \end{aligned} \quad \dots(2)$$

because the direction of the path RS is towards the negative direction of X .

$$\text{Similarly, } \int_Q^R \vec{A} \cdot d\vec{l} + \int_S^P \vec{A} \cdot d\vec{l} = \frac{\partial A_y}{\partial x} \delta x \delta y \quad \dots(3)$$

By adding (2) and (3), we get the line integral

$$\oint_{PQRS} \vec{A} \cdot d\vec{l} = \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) \delta x \delta y \quad \dots(4)$$

But $\left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right)$ is the Z -component of curl $\vec{A}$ or $\vec{\nabla} \times \vec{A}$ and the infinitesimal area $\delta x \delta y$ may be regarded as the magnitude to the vector area $\delta S_z \hat{k}$. Thus,

$$\oint_{PQRS} \vec{A} \cdot d\vec{l} = (\text{curl } \vec{A})_z dS_z = (\text{curl } \vec{A}) \cdot \delta S_z \hat{k} \quad \dots(5)$$

This equation is correct for any infinitesimal area of magnitude δS_z at P , having boundary C , i.e.,

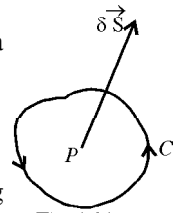


Fig. 1.21.

$$\oint_C \vec{A} \cdot d\vec{l} = (\text{curl } \vec{A}) \cdot \delta S_z \hat{k} \quad \dots(6)$$

Now, if we consider any infinitesimal plane area $\delta \vec{S} = \delta S_x \hat{i} + \delta S_y \hat{j} + \delta S_z \hat{k}$ at P oriented in any direction with a boundary C [fig. 1.21], then evidently,

$$\oint_C \vec{A} \cdot d\vec{l} = (\text{curl } \vec{A}) \cdot \delta \vec{S} \quad \dots(7)$$

The maximum value of $\oint_C \vec{A} \cdot d\vec{l} = |\text{curl } \vec{A}| \delta S$.

Thus, the magnitude of $\text{curl } \vec{A}$ at a point is the maximum value of the line integral of $\vec{A}$ along a boundary C $\left(\oint_C \vec{A} \cdot d\vec{l}\right)$ per unit area; this boundary C is that of an infinitesimal plane area (δS) at the point under consideration.

From (7)

$$\oint_C \vec{A} \cdot d\vec{l} = (\text{curl } \vec{A})_x \delta S_x \hat{i} + (\text{curl } \vec{A})_y \delta S_y \hat{j} + (\text{curl } \vec{A})_z \delta S_z \hat{k} \quad \dots(8)$$

where $(\text{curl } \vec{A})_x = \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}\right)$, $(\text{curl } \vec{A})_y = \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right)$ and $(\text{curl } \vec{A})_z = \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right)$.

These are the cartesian components of $\text{curl } \vec{A}$ and

$$\text{curl } \vec{A} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ A_x & A_y & A_z \end{vmatrix} = \vec{\nabla} \times \vec{A} \quad \dots(9)$$

In order to understand the **physical picture of circulation of moving fluid** (e.g., water) **at a point**, we put a paddle-wheel at that point [fig 1.22]. If the velocity of the fluid over one part of the circumference of the wheel is more than that at the other, the wheel will rotate and if we divide the circulation $\oint \vec{v} \cdot d\vec{l}$ by the area δS of the wheel, then we will obtain the value of the component of $\text{curl } \vec{v}$ along the axis of the wheel.

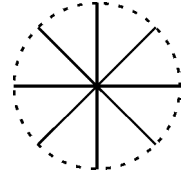


Fig. 1.22. Rotation of paddle wheel

In a circulating fluid, the value of $\text{curl } \vec{v}$ at a point is a measure of angular velocity of the fluid close to that point.

Rotation of a Rigid Body

A simple example of curl of a vector is a rotating rigid body about an axis. Let a rigid body be rotating about an axis AB with an angular velocity $\vec{\omega}$. Any particle P of the body will move on a circle of radius $r \sin \theta$; the position vector of P is $\vec{r}$ with respect to the origin O on the axis AB [fig. 1.23].

Therefore, the value of the linear velocity of the body will be $\omega r \sin \theta$ which can be represented in vector form as

$$\vec{v} = \vec{\omega} \times \vec{r} \quad \dots(10)$$

In general, this equation represents velocity of any particle of the rotating rigid body.

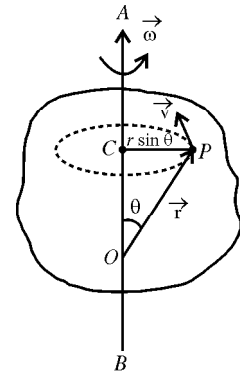


Fig. 1.23. Rotating rigid body.

$$\text{Now, we calculate } \vec{\nabla} \times \vec{v} = \vec{\nabla} \times (\vec{\omega} \times \vec{r}) \quad \dots(11)$$

For this, we use formula for vector triple product

$$\vec{A} \times (\vec{B} \times \vec{C}) = (\vec{A} \cdot \vec{C})\vec{B} - (\vec{A} \cdot \vec{B})\vec{C} \quad \dots(12)$$

Remember that $\vec{\nabla}$ is not ordinary vector, but it has the properties of both vector and differential operator. Hence, it is always written before those variable quantities which can be differentiated. Therefore, using (12), we have

$$\vec{\nabla} \times (\vec{\omega} \times \vec{r}) = (\vec{\nabla} \cdot \vec{r})\vec{\omega} - (\vec{\omega} \cdot \vec{\nabla})\vec{r} \quad \dots(13)$$

$$\text{But,} \quad \vec{\omega}(\vec{\nabla} \cdot \vec{r}) = \vec{\omega} \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) (x\hat{i} + y\hat{j} + z\hat{k}) = \vec{\omega} \left(\frac{\partial x}{\partial x} + \frac{\partial y}{\partial y} + \frac{\partial z}{\partial z} \right) = 3\vec{\omega}$$

$$\text{and} \quad (\vec{\omega} \cdot \vec{\nabla})\vec{r} = \left(\omega_x \frac{\partial}{\partial x} + \omega_y \frac{\partial}{\partial y} + \omega_z \frac{\partial}{\partial z} \right) (x\hat{i} + y\hat{j} + z\hat{k}) = \hat{i}\omega_x + \hat{j}\omega_y + \hat{k}\omega_z = \vec{\omega}$$

Substituting these values in (13) and using (11), we have

$$\vec{\nabla} \times \vec{v} = \vec{\nabla} \times (\vec{\omega} \times \vec{r}) = 3\vec{\omega} - \vec{\omega} = 2\vec{\omega} \text{ i.e., } \text{curl } \vec{v} = 2\vec{\omega} \text{ or } \vec{\omega} = \frac{1}{2} \text{curl } \vec{v} \quad \dots(14)$$

This result gives us again a clue as the name curl $\vec{v}$ (or rotation $\vec{v}$ or rot $\vec{v}$ as it is sometimes called). Hence *if a rigid body is rotating about an axis, then the curl of the linear velocity at a point is equal to two times its angular velocity*. In the case of circular motion of a fluid, an infinitesimal volume around a point at an instant can be understood as a rigid body so that the curl of the velocity of the fluid at that point is equal to two times its angular velocity, while the instantaneous axis of rotation is along curl $\vec{v}$.

If in the motion of a body we have the curl of the body, then particle of the body are moving with an instantaneous angular velocity. Such kind of motion is called **rotational** or **vortical**.

Irrotational Vector–Conservative vector field

If the curl of a vector field $\vec{A}$ is zero i.e.,

$$\text{curl } \vec{A} = \vec{\nabla} \times \vec{A} = 0 \quad \dots(15)$$

Then $\vec{A}$ is called **irrotational vector** or **lamellar vector**. We give two important examples of this vector : (1) **gravitational force** and (2) **electrostatic force**. Both kinds of forces can be represented by the following expression :

$$\vec{F} = \frac{k\hat{r}}{r^2} = \frac{k\vec{r}}{r^3} \quad \dots(16)$$

where $\vec{r} = r\hat{r}$ is the vector distance from the centre of force; for gravitational force, $k = -Gm_1m_2$ and for electrostatic forces, $k = q_1q_2/4\pi\epsilon_0$.

$$\text{Here,} \quad \vec{\nabla} \times \vec{F} = \vec{\nabla} \times \frac{k\vec{r}}{r^3} = k \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ x/r^3 & y/r^3 & z/r^3 \end{vmatrix} = 0 \quad \dots(17)$$

where $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, and $|\vec{r}| = r = (x^2 + y^2 + z^2)^{1/2}$ and $\frac{\partial}{\partial y} \left(\frac{z}{r^3} \right) - \frac{\partial}{\partial z} \left(\frac{y}{r^3} \right) = -\frac{3zy}{r^5} + \frac{3yz}{r^5} = 0$ etc.

If $\vec{F} = f(r)\hat{r}$, which represents a central force with $f(r)$ as a scalar function of r , then it can be proved that

$$\vec{\nabla} \times \vec{F} = \vec{\nabla} \times f(r)\hat{r} = 0 \quad \dots(18)$$

Hence, *a central force is an example of irrotational vector*.

Now, if a vector field can be represented as the gradient of a scalar function ϕ i.e., $\vec{A} = \text{grad } \phi$, then that vector field $\vec{A}$ will be irrotational, because

$$\begin{aligned}\text{curl } \vec{A} &= \text{curl grad } \phi = \vec{\nabla} \times \vec{\nabla} \phi = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ \partial\phi/\partial x & \partial\phi/\partial y & \partial\phi/\partial z \end{vmatrix} \\ &= \hat{i} \left[\frac{\partial^2 \phi}{\partial y \partial z} - \frac{\partial^2 \phi}{\partial z \partial y} \right] + \hat{j} \left[\frac{\partial^2 \phi}{\partial z \partial x} - \frac{\partial^2 \phi}{\partial x \partial z} \right] + \hat{k} \left[\frac{\partial^2 \phi}{\partial x \partial y} - \frac{\partial^2 \phi}{\partial y \partial x} \right] = 0 \quad \dots(19)\end{aligned}$$

Such an irrotational vector field, which can be represented by the gradient of a scalar function is called conservative force field and the curl of conservative force field is always zero.

Examples of conservative force field are (1) gravitational field and (2) electrostatic field, which can be represented as the negative gradient of the potential V i.e., for a conservative field $\vec{E}$,

$$\vec{E} = -\vec{\nabla} V \text{ and } \text{curl } \vec{E} = -\vec{\nabla} \times \vec{\nabla} V = 0 \quad \dots(20)$$

as proved above.

Ex. 1. If $\vec{r} = \hat{i}x + \hat{j}y + \hat{k}z$, then prove that :

(i) $\text{curl } \vec{r} = 0$.

(A. N. U. 2008; O. U. 1997)

(ii) $\text{curl } (\vec{r} r^n) = 0$.

$$\begin{aligned}\text{Sol. (i) } \text{curl } \vec{r} &= \vec{\nabla} \times (\hat{i}x + \hat{j}y + \hat{k}z) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ x & y & z \end{vmatrix} \\ &= \hat{i} \left(\frac{\partial z}{\partial y} - \frac{\partial y}{\partial z} \right) + \hat{j} \left(\frac{\partial x}{\partial z} - \frac{\partial z}{\partial x} \right) + \hat{k} \left(\frac{\partial y}{\partial x} - \frac{\partial x}{\partial y} \right) = 0. \\ &\quad [\because x, y, z \text{ are independent variables}]\end{aligned}$$

$$\begin{aligned}\text{(ii) } \text{curl } (\vec{r} r^n) &= \vec{\nabla} \times (\hat{i}x r^n + \hat{j}y r^n + \hat{k}z r^n) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ x r^n & y r^n & z r^n \end{vmatrix} \\ &= \hat{i} \left[\frac{\partial(z r^n)}{\partial y} - \frac{\partial(y r^n)}{\partial z} \right] + \hat{j} \left[\frac{\partial(x r^n)}{\partial z} - \frac{\partial(z r^n)}{\partial x} \right] + \hat{k} \left[\frac{\partial(y r^n)}{\partial x} - \frac{\partial(x r^n)}{\partial y} \right] \\ &= \hat{i} \left[z n r^{n-1} \frac{\partial r}{\partial y} - y n r^{n-1} \frac{\partial r}{\partial z} \right] + \dots \\ &= \hat{i} n r^{n-1} \left[\frac{z y}{r} - \frac{y z}{r} \right] + \dots \quad \left[\because \frac{\partial r}{\partial y} = \frac{\partial \sqrt{x^2 + y^2 + z^2}}{\partial y} = \frac{y}{r} \right] \\ &= n r^{n-2} [\hat{i}(zy - yz) + \hat{j}(xz - zx) + \hat{k}(xy - yx)] = 0.\end{aligned}$$

Ex. 2. Prove that $\text{curl } (\vec{A} + \vec{B}) = \text{curl } \vec{A} + \text{curl } \vec{B}$.

$$\begin{aligned}\text{Sol. } \text{curl } (\vec{A} + \vec{B}) &= \vec{\nabla} \times (\vec{A} + \vec{B}) = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \times [(A_x + B_x)\hat{i} + (A_y + B_y)\hat{j} + (A_z + B_z)\hat{k}] \\ &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ (A_x + B_x) & (A_y + B_y) & (A_z + B_z) \end{vmatrix}\end{aligned}$$

$$\begin{aligned}
&= \hat{\mathbf{i}} \left[\frac{\partial}{\partial y} (A_z + B_z) - \frac{\partial}{\partial z} (A_y + B_y) \right] + \hat{\mathbf{j}} \left[\frac{\partial}{\partial z} (A_x + B_x) - \frac{\partial}{\partial x} (A_z + B_z) \right] \\
&\quad + \hat{\mathbf{k}} \left[\frac{\partial}{\partial x} (A_y + B_y) - \frac{\partial}{\partial y} (A_x + B_x) \right] \\
&= \hat{\mathbf{i}} \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) + \hat{\mathbf{j}} \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) + \hat{\mathbf{k}} \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) \\
&\quad + \hat{\mathbf{i}} \left(\frac{\partial B_z}{\partial y} - \frac{\partial B_y}{\partial z} \right) + \hat{\mathbf{j}} \left(\frac{\partial B_x}{\partial z} - \frac{\partial B_z}{\partial x} \right) + \hat{\mathbf{k}} \left(\frac{\partial B_y}{\partial x} - \frac{\partial B_x}{\partial y} \right) \\
&= \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ A_x & A_y & A_z \end{vmatrix} + \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ B_x & B_y & B_z \end{vmatrix} \\
&= \vec{\nabla} \times \vec{A} + \vec{\nabla} \times \vec{B} = \text{curl } \vec{A} + \text{curl } \vec{B}.
\end{aligned}$$

Ex. 3. Prove that

$$\vec{\nabla} \times (\vec{A} \times \vec{B}) = \vec{A} (\nabla \cdot \vec{B}) - \vec{B} \times (\vec{\nabla} \cdot \vec{A}) \text{ or } \text{curl } (\vec{A} \times \vec{B}) = \vec{A} \text{ div } \vec{B} - \vec{B} \text{ div } \vec{A} \quad (\text{A. U. 1996; K. U. 1997})$$

Sol. In case of vector triple product,

$$\vec{A} \times (\vec{B} \times \vec{C}) = \vec{B} (\vec{A} \cdot \vec{C}) - \vec{C} (\vec{A} \cdot \vec{B}) \quad \dots(i)$$

Replacing $\vec{A}, \vec{B}, \vec{C}$ in (i) by $\vec{\nabla}, \vec{A}, \vec{B}$ respectively, we obtain

$$\vec{\nabla} \times (\vec{A} \times \vec{B}) = \vec{A} (\vec{\nabla} \cdot \vec{B}) - \vec{B} (\vec{\nabla} \cdot \vec{A})$$

$$\text{or } \vec{\nabla} \times (\vec{A} \times \vec{B}) = \vec{\nabla} (\vec{\nabla} \cdot \vec{B}) - (\vec{\nabla} \times \vec{A}) \quad \dots(ii)$$

$$\text{or } \text{curl } (\vec{A} \times \vec{B}) = \vec{A} \text{ div } \vec{B} - \vec{B} \text{ div } \vec{A}. \quad \dots(iii)$$

Ex. 4. Prove that

$$(i) \text{ curl grad } \phi = 0.$$

(K. U. 2005; A. N. U. 2006; A. U. 2003)

$$(ii) \text{ div curl } \vec{A} = 0$$

(K. U. 2007; A. U. 2002)

$$(iii) \text{ curl } (\phi \vec{A}) = \phi \text{ curl } \vec{A} + \text{grad } \phi \times \vec{A}$$

$$(iv) \text{ grad div } \vec{A}$$

(K. U. 1989)

$$= \hat{\mathbf{i}} \frac{\partial}{\partial x} \left(\frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} \right) + \hat{\mathbf{j}} \frac{\partial}{\partial y} \left(\frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} \right) + \hat{\mathbf{k}} \frac{\partial}{\partial z} \left(\frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} \right).$$

$$\text{Sol. (i) curl grad } \phi = \vec{\nabla} \times \vec{\nabla} \phi = \left(\hat{\mathbf{i}} \frac{\partial}{\partial x} + \hat{\mathbf{j}} \frac{\partial}{\partial y} + \hat{\mathbf{k}} \frac{\partial}{\partial z} \right) \times \left(\hat{\mathbf{i}} \frac{\partial \phi}{\partial x} + \hat{\mathbf{j}} \frac{\partial \phi}{\partial y} + \hat{\mathbf{k}} \frac{\partial \phi}{\partial z} \right)$$

$$= \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ \partial\phi/\partial x & \partial\phi/\partial y & \partial\phi/\partial z \end{vmatrix} = \hat{\mathbf{i}} \left(\frac{\partial^2 \phi}{\partial y \partial z} - \frac{\partial^2 \phi}{\partial z \partial y} \right) + \hat{\mathbf{j}} \left(\frac{\partial^2 \phi}{\partial z \partial x} - \frac{\partial^2 \phi}{\partial x \partial z} \right) + \hat{\mathbf{k}} \left(\frac{\partial^2 \phi}{\partial x \partial y} - \frac{\partial^2 \phi}{\partial y \partial x} \right)$$

As $\frac{\partial^2 \phi}{\partial y \partial z} = \frac{\partial^2 \phi}{\partial z \partial y}$ etc, hence **curl grad** $\phi = 0$.

$$\begin{aligned}
 \text{(ii) } \operatorname{div} \operatorname{curl} \vec{A} &= \vec{\nabla} \cdot (\vec{\nabla} \times \vec{A}) = \vec{\nabla} \cdot \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ A_x & A_y & A_z \end{vmatrix} \\
 &= \vec{\nabla} \cdot \left[\hat{i} \left\{ \frac{\partial A_x}{\partial y} - \frac{\partial A_y}{\partial z} \right\} + \hat{j} \left\{ \frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right\} + \hat{k} \left\{ \frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right\} \right] \\
 &= \frac{\partial}{\partial x} \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) + \frac{\partial}{\partial y} \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) + \frac{\partial}{\partial z} \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) \\
 &= \frac{\partial^2 A_z}{\partial x \partial y} - \frac{\partial^2 A_y}{\partial x \partial z} + \frac{\partial^2 A_x}{\partial y \partial z} - \frac{\partial^2 A_z}{\partial y \partial x} + \frac{\partial^2 A_y}{\partial z \partial x} - \frac{\partial^2 A_x}{\partial z \partial y}
 \end{aligned}$$

If $\vec{A}$ is a perfect differential, then $\frac{\partial^2 A_z}{\partial x \partial y} = \frac{\partial^2 A_z}{\partial y \partial x}$ etc. and therefore **div curl** $\vec{A} = 0$.

$$\begin{aligned}
 \text{(iii) } \operatorname{curl} (\phi \vec{A}) &= (\nabla \times (\phi \vec{A})) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ \phi A_x & \phi A_y & \phi A_z \end{vmatrix} \\
 &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ \phi A_x & \phi A_y & \phi A_z \end{vmatrix} \\
 &= \hat{i} \left[\frac{\partial}{\partial y} (\phi A_z) - \frac{\partial}{\partial z} (\phi A_y) \right] + \hat{j} \left[\frac{\partial}{\partial z} (\phi A_x) - \frac{\partial}{\partial x} (\phi A_z) \right] + \hat{k} \left[\frac{\partial}{\partial x} (\phi A_y) - \frac{\partial}{\partial y} (\phi A_x) \right] \\
 &= \left[\phi \frac{\partial A_z}{\partial y} + A_z \frac{\partial \phi}{\partial y} - \phi \frac{\partial A_y}{\partial z} - A_y \frac{\partial \phi}{\partial z} \right] + \hat{j} \left[\phi \frac{\partial A_x}{\partial z} + A_x \frac{\partial \phi}{\partial z} - \phi \frac{\partial A_z}{\partial x} - A_z \frac{\partial \phi}{\partial x} \right] \\
 &\quad + \hat{k} \left[\phi \frac{\partial A_y}{\partial x} + A_y \frac{\partial \phi}{\partial x} - \phi \frac{\partial A_x}{\partial y} - A_x \frac{\partial \phi}{\partial y} \right] \\
 &= \phi \left[\hat{i} \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) + \hat{j} \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) + \hat{k} \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) \right] \\
 &\quad + \left[\hat{i} \left(A_z \frac{\partial \phi}{\partial y} - A_y \frac{\partial \phi}{\partial z} \right) + \hat{j} \left(A_x \frac{\partial \phi}{\partial z} - A_z \frac{\partial \phi}{\partial x} \right) + \hat{k} \left(A_y \frac{\partial \phi}{\partial x} - A_x \frac{\partial \phi}{\partial y} \right) \right] \\
 &= \phi \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ A_x & A_y & A_z \end{vmatrix} + \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial\phi/\partial x & \partial\phi/\partial y & \partial\phi/\partial z \\ A_x & A_y & A_z \end{vmatrix} = \phi \operatorname{curl} \vec{A} + (\operatorname{grad} \phi) \times \vec{A}.
 \end{aligned}$$

$$\begin{aligned}
 \text{(iv) } \operatorname{grad} \operatorname{div} \vec{A} &= \vec{\nabla} (\vec{\nabla} \cdot \vec{A}) = \vec{\nabla} \cdot \left[\left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) (\hat{i} A_x + \hat{j} A_y + \hat{k} A_z) \right] \\
 &= \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \left(\frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} \right)
 \end{aligned}$$

$$\begin{aligned}
&= \hat{\mathbf{i}} \left(\frac{\partial^2 A_x}{\partial x^2} + \frac{\partial^2 A_y}{\partial x \partial y} + \frac{\partial^2 A_z}{\partial x \partial z} \right) + \hat{\mathbf{j}} \left(\frac{\partial^2 A_x}{\partial y \partial x} + \frac{\partial^2 A_y}{\partial y^2} + \frac{\partial^2 A_z}{\partial y \partial z} \right) + \hat{\mathbf{k}} \left(\frac{\partial^2 A_x}{\partial z \partial x} + \frac{\partial^2 A_y}{\partial z \partial y} + \frac{\partial^2 A_z}{\partial z^2} \right) \\
&= \hat{\mathbf{i}} \left(\frac{\partial^2 A_x}{\partial x^2} + \frac{\partial^2 A_y}{\partial x \partial y} + \frac{\partial^2 A_z}{\partial x \partial z} \right) + \hat{\mathbf{j}} \left(\frac{\partial^2 A_x}{\partial y \partial x} + \frac{\partial^2 A_y}{\partial y^2} + \frac{\partial^2 A_z}{\partial y \partial z} \right) + \hat{\mathbf{k}} \left(\frac{\partial^2 A_x}{\partial z \partial x} + \frac{\partial^2 A_y}{\partial z \partial y} + \frac{\partial^2 A_z}{\partial z^2} \right).
\end{aligned}$$

Ex. 5. Prove that $\text{div} (\vec{\mathbf{A}} \times \vec{\mathbf{B}}) = \vec{\mathbf{B}} \cdot \text{curl} \vec{\mathbf{A}} - \vec{\mathbf{A}} \cdot \text{curl} \vec{\mathbf{B}}$.

Sol. Let $\vec{\mathbf{A}} \times \vec{\mathbf{B}} = \vec{\mathbf{C}}$, then in components

$$\begin{aligned}
\vec{\mathbf{C}} &= C_x \hat{\mathbf{i}} + C_y \hat{\mathbf{j}} + C_z \hat{\mathbf{k}} = \vec{\mathbf{A}} \times \vec{\mathbf{B}} \\
&= (A_y B_z - A_z B_y) \hat{\mathbf{i}} + (A_z B_x - A_x B_z) \hat{\mathbf{j}} + (A_x B_y - A_y B_x) \hat{\mathbf{k}}
\end{aligned}$$

Now, $\text{div} (\vec{\mathbf{A}} \times \vec{\mathbf{B}}) = \text{div} \vec{\mathbf{C}} = \vec{\nabla} \cdot \vec{\mathbf{C}}$

$$\begin{aligned}
&= \frac{\partial C_x}{\partial x} + \frac{\partial C_y}{\partial y} + \frac{\partial C_z}{\partial z} \\
&= \frac{\partial}{\partial x} (A_y B_z - A_z B_y) + \frac{\partial}{\partial y} (A_z B_x - A_x B_z) + \frac{\partial}{\partial z} (A_x B_y - A_y B_x) \\
&= B_x \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) + B_y \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) + B_z \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) \\
&\quad - \left[A_x \left(\frac{\partial B_z}{\partial y} - \frac{\partial B_y}{\partial z} \right) + A_y \left(\frac{\partial B_x}{\partial z} - \frac{\partial B_z}{\partial x} \right) + A_z \left(\frac{\partial B_y}{\partial x} - \frac{\partial B_x}{\partial y} \right) \right] \\
&= (B_x \hat{\mathbf{i}} + B_y \hat{\mathbf{j}} + B_z \hat{\mathbf{k}}) \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ A_x & A_y & A_z \end{vmatrix} \\
&\quad - (A_x \hat{\mathbf{i}} + A_y \hat{\mathbf{j}} + A_z \hat{\mathbf{k}}) \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ B_x & B_y & B_z \end{vmatrix} \\
&= \vec{\mathbf{B}} \cdot \text{curl} \vec{\mathbf{A}} - \vec{\mathbf{A}} \cdot \text{curl} \vec{\mathbf{B}}.
\end{aligned}$$

Ex. 6. If $\vec{\mathbf{a}}$ is a constant vector and $\vec{\mathbf{r}}$ is a position vector, then find $\text{curl} (\vec{\mathbf{a}} \times \vec{\mathbf{r}})$. (A. N. U. 1989)

Sol. Here, $\vec{\mathbf{a}} \times \vec{\mathbf{r}} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ a_x & a_y & a_z \\ x & y & z \end{vmatrix} = (a_y z - a_z y) \hat{\mathbf{i}} + (a_z x - a_x z) \hat{\mathbf{j}} + (a_x y - a_y x) \hat{\mathbf{k}}$

$$\begin{aligned}
\text{Hence, } \text{curl} (\vec{\mathbf{a}} \times \vec{\mathbf{r}}) &= \vec{\nabla} \times (\vec{\mathbf{a}} \times \vec{\mathbf{r}}) = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ a_y z - a_z y & a_z x - a_x z & a_x y - a_y x \end{vmatrix} \\
&= \hat{\mathbf{i}} \left[\frac{\partial}{\partial y} (a_x y - a_y x) - \frac{\partial}{\partial z} (a_z x - a_x z) + \dots \right] \\
&= \hat{\mathbf{i}}(a_x + a_x) + \hat{\mathbf{j}}(a_y + a_y) + \hat{\mathbf{k}}(a_z + a_z) = 2(a_x \hat{\mathbf{i}} + a_y \hat{\mathbf{j}} + a_z \hat{\mathbf{k}}) = 2\vec{\mathbf{a}}.
\end{aligned}$$

Ex. 7. If $\vec{\mathbf{v}} = \vec{\omega} \times \vec{\mathbf{r}}$, where $\vec{\omega}$ is a constant vector, prove that $\text{curl} \vec{\mathbf{v}} = 2\vec{\omega}$. (A. N. U. 2003)

Sol. Given $\vec{\mathbf{v}} = \vec{\omega} \times \vec{\mathbf{r}}$

Now, $\vec{\omega} = \omega_x \hat{i} + \omega_y \hat{j} + \omega_z \hat{k}$ and $\vec{r} = x \hat{i} + y \hat{j} + z \hat{k}$

$$\therefore (\vec{\omega} \times \vec{r}) = (\omega_y z - y \omega_z) \hat{i} + (\omega_z x - z \omega_x) \hat{j} + (\omega_x y - x \omega_y) \hat{k}$$

$$\begin{aligned} \text{Now, } \vec{\nabla} \times \vec{v} &= \vec{\nabla} \times (\vec{\omega} \times \vec{r}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ (\omega_y z - y \omega_z) & (\omega_z x - z \omega_x) & (\omega_x y - x \omega_y) \end{vmatrix} \\ &= \hat{i} \left[\frac{\partial}{\partial y} (\omega_x y - x \omega_y) - \frac{\partial}{\partial z} (\omega_z x - z \omega_x) \right] + \hat{j} \left[\frac{\partial}{\partial z} (\omega_y z - y \omega_z) - \frac{\partial}{\partial x} (\omega_x y - x \omega_y) \right] \\ &\quad + \hat{k} \left[\frac{\partial}{\partial x} (\omega_z x - z \omega_x) - \frac{\partial}{\partial y} (\omega_y z - y \omega_z) \right] \\ &= (\omega_x + \omega_x) \hat{i} + (\omega_y + \omega_y) \hat{j} + (\omega_z + \omega_z) \hat{k} \\ &= 2 [\omega_x \hat{i} + \omega_y \hat{j} + \omega_z \hat{k}] = 2 \vec{\omega}. \end{aligned}$$

Thus, $\text{curl } \vec{v} = 2 \vec{\omega}$.

Ex. 8. If $\vec{A}$ and $\vec{B}$ are irrotational vectors, then prove that $\vec{A} \times \vec{B}$ is a solenoidal vector.

Sol. Given, $\vec{\nabla} \times \vec{A} = 0$ and $\vec{\nabla} \times \vec{B} = 0$

We know that $\vec{\nabla} \times (\vec{\nabla} \times \vec{A}) = \vec{\nabla} (\text{curl } \vec{A}) - \vec{A} (\text{curl } \vec{B})$.

$$\therefore \vec{\nabla} (\vec{A} \times \vec{B}) = \vec{B} (\vec{\nabla} \times \vec{A}) - \vec{A} (\vec{\nabla} \times \vec{B}) = 0.$$

Thus $\vec{A} \times \vec{B}$ is solenoidal vector.

Ex. 9. If $\vec{F} = xy^2 \hat{i} + 2x^2 yz \hat{j} - 3yz^2 \hat{k}$, find $\text{div } \vec{F}$, $\text{curl } \vec{F}$ at the point (1, -1, 1).

$$\begin{aligned} \text{Sol. (a) } \text{div } \vec{F} &= \vec{\nabla} \cdot \vec{F} = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) (xy^2 \hat{i} + 2x^2 yz \hat{j} - 3yz^2 \hat{k}) \\ &= \frac{\partial}{\partial x} (xy^2) + \frac{\partial}{\partial y} (2x^2 yz) + \frac{\partial}{\partial z} (-3yz^2) = y^2 + 2x^2 z - 6yz. \end{aligned}$$

At the point (1, -1, 1), $\text{div } \vec{F} = 1 + 2 + 6 = 9$.

$$\begin{aligned} \text{(b) } \text{curl } \vec{F} &= (\vec{\nabla} \times \vec{F}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ xy^2 & 2x^2 yz & -3yz^2 \end{vmatrix} \\ &= \left[\frac{\partial}{\partial y} (-3yz^2) - \frac{\partial}{\partial z} (2x^2 yz) \right] \hat{i} + \left[\frac{\partial}{\partial z} (xy^2) - \frac{\partial}{\partial x} (-3yz^2) \right] \hat{j} \\ &\quad + \left[\frac{\partial}{\partial x} (2x^2 yz) - \frac{\partial}{\partial y} (xy^2) \right] \hat{k} \\ &= -(3z^2 + 2x^2 y) \hat{i} + (4xyz - 2xy) \hat{k} \end{aligned}$$

At the point (1, -1, 1), $\text{curl } \vec{F} = -\hat{i} - 2\hat{k}$.

Ex. 10. Prove that : $\vec{F} = a(x\hat{i} + y\hat{j})$ is a conservative force field.

Sol. Here, for a conservative force field, $\text{curl } \vec{F} = 0$

$$\begin{aligned}
\text{curl } \vec{F} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ ax & ay & 0 \end{vmatrix} \\
&= \hat{i} \left[\frac{\partial(0)}{\partial y} - \frac{\partial(ay)}{\partial z} \right] - \hat{j} \left[\frac{\partial(0)}{\partial x} - \frac{\partial(ax)}{\partial z} \right] + \hat{k} \left[\frac{\partial(ay)}{\partial x} - \frac{\partial(ax)}{\partial y} \right] \\
&= \hat{i}|0-0| - \hat{j}|0-0| + \hat{k}|0-0| = 0
\end{aligned}$$

Therefore, $\vec{F} = a(x\hat{i} + y\hat{j})$ is a conservative force field.

Ex. 11. Show that $\vec{A} = e^{-x}(-yz\hat{i} + z\hat{j} + y\hat{k})$ is irrotational. (O. U. 1995)

Sol. When $\vec{A}$ is an irrotational vector, $\vec{\nabla} \times \vec{A} = 0$

Here, $\vec{A} = -yze^{-x}\hat{i} + ze^{-x}\hat{j} + ye^{-x}\hat{k}$

$$\begin{aligned}
\therefore \vec{\nabla} \times \vec{A} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ -yze^{-x} & ze^{-x} & ye^{-x} \end{vmatrix} \\
&= \hat{i} \left[\frac{\partial}{\partial y}(ye^{-x}) - \frac{\partial}{\partial z}(ze^{-x}) \right] - \hat{j} \left[\frac{\partial}{\partial x}(ye^{-x}) - \frac{\partial}{\partial z}(-yze^{-x}) \right] \\
&\quad + \hat{k} \left[\frac{\partial}{\partial x}(ze^{-x}) - \frac{\partial}{\partial y}(-yze^{-x}) \right] \\
&= \hat{i}(e^{-x} - e^{-x}) - \hat{j}(-ye^{-x} + ye^{-x}) + \hat{k}(-ze^{-x} + ze^{-x}) = 0.
\end{aligned}$$

Hence the vector $\vec{A}$ is irrotational.

Ex. 12. Find the values of constants a, b, c if the vector $\vec{A} = \hat{i}(x + 2y + az) + \hat{j}(bx - 3y - z) + \hat{k}(4x + cy + 2z)$ is irrotational.

Sol. We know that vector $\vec{A}$ is irrotational, when $\vec{\nabla} \times \vec{A} = 0$.

For the given vector $\vec{A}$,

$$\begin{aligned}
\vec{\nabla} \times \vec{A} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ (x+2y+az) & (bx-3y-z) & (4x+cy+2z) \end{vmatrix} \\
&= \hat{i} \left\{ \frac{\partial}{\partial y}(4x+cy+2z) - \frac{\partial}{\partial z}(bx-3y-z) \right\} + \hat{j} \left\{ \frac{\partial}{\partial z}(x+2y+az) - \frac{\partial}{\partial x}(4x+cy+2z) \right\} \\
&\quad + \hat{k} \left\{ \frac{\partial}{\partial x}(bx-3y-z) - \frac{\partial}{\partial y}(x+2y+az) \right\} \\
&= \hat{i}(c+1) + \hat{j}(a-4) + \hat{k}(b-2)
\end{aligned}$$

But this is equal to zero. Then

$$= c - 1, a = 4 \text{ and } b = 2. \text{ or } a = 4, b = 2, c = -1.$$

Ex. 13. Prove the vector identity

$$\vec{\nabla} \times (\vec{\nabla} \times \vec{A}) = \vec{\nabla} (\vec{\nabla} \cdot \vec{A}) - \nabla^2 \vec{A} \text{ or } \text{curl curl } \vec{A} = \text{grad div } \vec{A} - \nabla^2 \vec{A}. \quad (\text{O.U. 2007})$$

Sol. We know that in case of a vector triple product,

$$\vec{A} \times (\vec{B} \times \vec{C}) = \vec{B} (\vec{A} \cdot \vec{C}) - (\vec{A} \cdot \vec{B}) \vec{C}$$

Replacing in the above equation $\vec{A}, \vec{B}$ and $\vec{C}$ by $\vec{\nabla}, \vec{\nabla}$ and $\vec{A}$ respectively, we get

$$\vec{\nabla} \times (\vec{\nabla} \times \vec{A}) = \vec{\nabla} (\vec{\nabla} \cdot \vec{A}) - (\vec{\nabla} \cdot \vec{\nabla}) \vec{A}$$

$$\text{or } \vec{\nabla} \times (\vec{\nabla} \times \vec{A}) = \vec{\nabla} (\vec{\nabla} \cdot \vec{A}) - \nabla^2 \vec{A} \quad [\because \vec{\nabla} \cdot \vec{\nabla} = \nabla^2]$$

This proves the given vector identity. The identity can also be written as curl

$$\text{curl } \vec{A} = \text{grad div } \vec{A} - \nabla^2 \vec{A}.$$

Ex. 14. Show that $\vec{\nabla} (\vec{A} \times \vec{r}) = \vec{r} (\vec{\nabla} \times \vec{A})$, where $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ is the position vector.

$$\text{Sol. } \vec{A} \times \vec{r} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ A_x & A_y & A_z \\ x & y & z \end{vmatrix} = \hat{i}(A_y z - A_z y) + \hat{j}(A_z x - A_x z) + \hat{k}(A_x y - A_y x)$$

$$\begin{aligned} \therefore \vec{\nabla} (\vec{A} \times \vec{r}) &= \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \cdot [\hat{i}(A_y z - A_z y) + \hat{j}(A_z x - A_x z) + \hat{k}(A_x y - A_y x)] \\ &= \frac{\partial}{\partial x} (A_y z - A_z y) + \frac{\partial}{\partial y} (A_z x - A_x z) + \frac{\partial}{\partial z} (A_x y - A_y x) \\ &= z \frac{\partial A_y}{\partial x} - y \frac{\partial A_z}{\partial x} + x \frac{\partial A_z}{\partial y} - z \frac{\partial A_x}{\partial y} + y \frac{\partial A_x}{\partial z} - x \frac{\partial A_y}{\partial z} \\ &= x \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) + y \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) + z \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) \\ &= (\hat{i}x + \hat{j}y + \hat{k}z) \cdot \left[\hat{i} \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) + \hat{j} \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) + \hat{k} \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) \right] = \vec{r} (\vec{\nabla} \times \vec{A}) \end{aligned}$$

$$\text{Thus, } \vec{\nabla} (\vec{A} \times \vec{r}) = \vec{r} (\vec{\nabla} \times \vec{A}).$$

1-8. Vector Integration

Let $\vec{A}(u)$ be a vector function of scalar variable u . If there exists a vector function $\vec{B}(u)$ such that

$\vec{A}(u) = \frac{d}{du} [\vec{B}(u)]$, then the **indefinite integral** of $\vec{A}(u)$ is defined as

$$\int \vec{A}(u) du = \int \frac{d}{du} [\vec{B}(u)] du = \vec{B}(u) + \vec{C} \quad \dots(1)$$

where $\vec{C}$ is an arbitrary constant vector independent of u .

If $\vec{A}(u)$ is expressed as

$$\vec{A}(u) = A_1(u) \hat{i} + A_2(u) \hat{j} + A_3(u) \hat{k} \quad \dots(2)$$

Then the indefinite integral can be expressed as

$$\int \vec{A}(u) du = \hat{i} \int A_1(u) du + \hat{j} \int A_2(u) du + \hat{k} \int A_3(u) du = \vec{B}(u) + \vec{C} \quad \dots(3)$$

The **definite integral** between the limits $u = \alpha$ and $u = \beta$ is defined as

$$\int_{\alpha}^{\beta} \vec{A}(u) du = \int_{\alpha}^{\beta} \frac{d}{du} [\vec{B}(u)] du = \left[\vec{B}(u) + \vec{C} \right]_{\alpha}^{\beta} = \vec{B}(\beta) - \vec{B}(\alpha) \quad \dots(4)$$

As an application of the method of vector integration we will deduce here the equation of motion in vector form.

1.9. Equations of Motion in Vector Form for Constant Acceleration

In Newtonian Mechanics, a particle moves with constant acceleration under constant force. From Newton's second law of motion ($\vec{F} = m \vec{a}$), the acceleration for constant force is given by

$$\vec{a} = \frac{\vec{F}}{m} = \frac{d\vec{v}}{dt} = \text{constant} \quad \dots(1)$$

where $\vec{a} = \frac{d\vec{v}}{dt}$, $\vec{v} = \frac{d\vec{r}}{dt}$ and $\vec{r}$ = position vector of the particle at any time t .

From eq. (1), the integration of the relation $d\vec{v} = \vec{a} dt$ gives

$$\int d\vec{v} = \int \vec{a} dt \text{ or } \vec{v} = \vec{a} t + \vec{C} \quad \dots(2)$$

where $\vec{C}$ is the constant of integration. Let at $t = 0$, $\vec{v} = \vec{u}$, the initial velocity. Then

$$\vec{u} = 0 + \vec{C} \text{ or } \vec{C} = \vec{u} \quad \dots(3)$$

Therefore, eq., (2) assumes the form

$$\vec{v} = \vec{u} + \vec{a} t \quad \dots(4)$$

which is the **first equation of motion** in vector form.

Now, $\vec{v} = \frac{d\vec{r}}{dt} = \vec{u} + \vec{a} t$

Integration of it yields

$$\int d\vec{r} = \int \vec{u} dt + \int \vec{a} t dt \text{ or } \vec{r} = \vec{u} t + \frac{1}{2} \vec{a} t^2 + \vec{C}_1$$

where $\vec{C}_1$ is the constant of integration.

If at $t = 0$, $\vec{r} = \vec{r}_0$ is the initial position vector, then

$$\vec{r}_0 = 0 + \vec{C}_1 \text{ or } \vec{C}_1 = \vec{r}_0$$

Therefore, $\vec{r} = \vec{r}_0 + \vec{u} t + \frac{1}{2} \vec{a} t^2 \quad \dots(5)$

If we write $\vec{r} - \vec{r}_0 = \vec{s}$, the **displacement** of the particle in time t , then

$$\vec{s} = \vec{u} t + \frac{1}{2} \vec{a} t^2 \quad \dots(6)$$

This is the **second equation of motion**.

From eq. (1)

$$\vec{a} = \frac{d\vec{v}}{dt}$$

Scalar product of $\vec{v}$ on both sides of this equation gives

$$\vec{v} \cdot \vec{a} = \vec{v} \cdot \frac{d\vec{v}}{dt} \quad \text{or} \quad \frac{d\vec{s}}{dt} \cdot \vec{a} = \vec{v} \cdot \frac{d\vec{v}}{dt} \quad \left[\because \frac{d\vec{r}}{dt} = \frac{d\vec{s}}{dt} = \vec{v} \right]$$

This can be integrated as

$$\int \vec{a} \cdot d\vec{s} = \int \vec{v} \cdot d\vec{v} \quad \text{or} \quad \vec{a} \cdot \vec{s} = \frac{1}{2} \vec{v} \cdot \vec{v} + \vec{C}_2 \quad \dots(7)$$

where $\vec{C}_2$ is the constant of integration.

Remember that $\vec{a}$ is constant and $d\left(\frac{1}{2} \vec{v} \cdot \vec{v}\right) = \frac{1}{2} \left(d\vec{v} \cdot \vec{v} + \vec{v} \cdot d\vec{v}\right) = \vec{v} \cdot d\vec{v}$.

When $\vec{s} = 0$, $\vec{v} = \vec{u}$ and therefore from eq. (7)

$$0 = \frac{1}{2} \vec{u} \cdot \vec{u} + \vec{C}_2 \quad \text{or} \quad \vec{C}_2 = -\frac{u^2}{2}$$

Hence, eq. (7) becomes

$$\vec{a} \cdot \vec{s} = \frac{v^2}{2} - \frac{u^2}{2} \quad \left[\because \vec{v} \cdot \vec{v} = v^2 \right]$$

$$\text{or} \quad v^2 = u^2 + 2 \vec{a} \cdot \vec{s} \quad \dots(8)$$

which is the **third equation of motion**.

Thus the **three equations of motion in vector form** are :

$$\vec{v} = \vec{u} + \vec{a} t; \quad \vec{s} = \vec{u} t + \frac{1}{2} \vec{a} t^2; \quad v^2 = u^2 + 2 \vec{a} \cdot \vec{s} \quad \dots(9)$$

In general, these equations of motion in vector form represent the **motion of a particle in three dimensions. If the motion is along a straight or in one dimension**, then

$$v = u + at, \quad s = ut + \frac{1}{2} at^2 \quad \text{and} \quad v^2 = u^2 + 2as \quad \dots(10)$$

Writing $\vec{u}$, $\vec{v}$ and $\vec{a}$ in the component form,

$$\vec{u} = u_x \hat{i} + u_y \hat{j} + u_z \hat{k}, \quad \vec{v} = v_x \hat{i} + v_y \hat{j} + v_z \hat{k} \quad \text{and} \quad \vec{a} = a_x \hat{i} + a_y \hat{j} + a_z \hat{k}$$

and if at $t = 0$, the particle is at the origin, i.e., $\vec{r}_0 = 0$, then

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

Thus, when a particle moves with constant acceleration in three dimensions, the form of the equations of motion will be

$$v_x = u_x + a_x t; \quad v_y = u_y + a_y t; \quad v_z = u_z + a_z t \quad \dots(11a)$$

$$x = u_x t + \frac{1}{2} a_x t^2; \quad y = u_y t + \frac{1}{2} a_y t^2; \quad z = u_z t + \frac{1}{2} a_z t^2 \quad \dots(11b)$$

$$\text{and} \quad v_x^2 = u_x^2 + 2a_x x; \quad v_y^2 = u_y^2 + 2a_y y; \quad v_z^2 = u_z^2 + 2a_z z \quad \dots(11c)$$

In case of motion in two dimensions or in a plane, eqs. (11a) and (11b) are sufficient.

Ex. 1. A force $\vec{P} = 25\hat{i} + 6\hat{j}$ newtons acts on a particle of mass 2.5 kg for 5 sec. If the initial position of the particle is $\vec{r}_0 = 6\hat{j} + 8\hat{k}$ metre and the initial velocity $\vec{u} = 2.5\hat{i} + 3\hat{k}$ metre/sec, calculate : (i) the final velocity of the particle, (ii) the final position of the particle and (iii) the work done by the force on the particle.

Sol. (i) Final velocity, $\vec{v} = \vec{u} + \vec{a}t$

Here, $\vec{a} = \frac{\vec{P}}{m} = \frac{25\hat{i} + 6\hat{j}}{2.5}$ m/sec², $\vec{u} = 2.5\hat{i} + 3\hat{k}$ m/sec and $t = 5$ sec.

$$\therefore \vec{v} = 2.5\hat{i} + 3\hat{k} + (25\hat{i} + 6\hat{j})5/2.5 = 2.5\hat{i} + 3\hat{k} + 50\hat{i} + 12\hat{j} = 52.5\hat{i} + 12\hat{j} + 3\hat{k} \text{ m/sec.}$$

$$\begin{aligned} \text{(ii) Final position, } \vec{r} &= \vec{r}_0 + \vec{u}t + \frac{1}{2}\vec{a}t^2 = 6\hat{j} + 8\hat{k} + (2.5\hat{i} + 3\hat{k})5 + \frac{1}{2}\frac{(25\hat{i} + 6\hat{j})}{2.5} \cdot 5^2 \\ &= 6\hat{j} + 8\hat{k} + 12.5\hat{i} + 15\hat{k} + 125\hat{i} + 30\hat{j} = 137.5\hat{i} + 36\hat{j} + 23\hat{k}. \end{aligned}$$

(iii) Work done on the particle

$$\begin{aligned} &= \vec{P} \cdot (\vec{r} - \vec{r}_0) = (25\hat{i} + 6\hat{j}) \cdot (137.5\hat{i} + 30\hat{j} + 15\hat{k}) \\ &= 3437.5 + 180 = 3617.5 \text{ joules.} \end{aligned}$$

Ex. 2. Evaluate the integral $\int_1^2 A(u)du$, if $A(u) = (6u^2 - 1)\hat{i} + (2u - 3)\hat{j} + (6u^2 - 4u)\hat{k}$

$$\begin{aligned} \text{Sol. } \int_1^2 A(u)du &= \int_1^2 [(6u^2 - 1)\hat{i} + (2u - 3)\hat{j} + (6u^2 - 4u)\hat{k}] du \\ &= \hat{i} \int_1^2 (6u^2 - 1)du + \hat{j} \int_1^2 (2u - 3)du + \hat{k} \int_1^2 (6u^2 - 4u)du \\ &= \hat{i} [2u^3 - u]_1^2 + \hat{j} [u^2 - 3u]_1^2 + \hat{k} [2u^3 - 2u^2]_1^2 \\ &= \hat{i} [16 - 2 - (2 - 1)] + \hat{j} [4 - 6 - (1 - 3)] + \hat{k} [16 - 8 - (2 - 2)] \\ &= \hat{i} [14 - 1] + \hat{j} [-2 + 2] + \hat{k} [8] \\ &= 13\hat{i} + 8\hat{k}. \end{aligned}$$

Ex. 3. At the time $t = 0$, a particle of mass m at rest experiences a time dependent force $\vec{F} = \vec{a}t(T - t)$, where $\vec{a}$ is a constant vector and T is the time during which the given force acts.

Calculate the distance covered while the force acted.

$$\text{Sol. } \vec{F} = \vec{a}t(T - t)$$

$$\text{or } m \frac{d\vec{v}}{dt} = \vec{a}t(T - t) \text{ or } d\vec{v} = \frac{\vec{a}}{m}(tT - t^2)dt \quad \dots(i)$$

Hence at any time t the velocity $\vec{v}$ of the particle is obtained by integrating (i) i.e.,

$$\vec{v} = \frac{d\vec{s}}{dt} = \int_0^t \frac{\vec{a}}{m}(tT - t^2)dt = \frac{\vec{a}}{m} \left(\frac{Tt^2}{2} - \frac{t^3}{3} \right)$$

Therefore, the distance covered in time T is

$$\vec{s} = \int_0^T \frac{\vec{a}}{m} \left(\frac{Tt^2}{2} - \frac{t^3}{3} \right) dt = \frac{\vec{a}}{m} \left[\frac{Tt^3}{6} - \frac{t^4}{12} \right]_0^T$$

$$= \frac{\vec{a}}{m} \left[\frac{T^4}{6} - \frac{T^4}{12} \right] = \frac{\vec{a}}{12m} T^4.$$

Ex. 4. If vector $\vec{A}$ is the function of time, determine the value of $\vec{A} \cdot \frac{d^2 \vec{A}}{dt^2} dt$. (O. U. 2008)

Sol. Integrating by parts,

$$\int \vec{A} \times \frac{d^2 \vec{A}}{dt^2} dt = \vec{A} \times \frac{d\vec{A}}{dt} - \int \frac{d\vec{A}}{dt} \times \frac{d\vec{A}}{dt} dt + \vec{C}$$

$$\text{or} \quad \int \vec{A} \times \frac{d^2 \vec{A}}{dt^2} dt = \vec{A} \times \frac{d\vec{A}}{dt} + \vec{C}$$

where $\vec{C}$ is the constant of integration and $\frac{d\vec{A}}{dt} \times \frac{d\vec{A}}{dt} = 0$ (cross product of same vectors).

Ex. 5. A particle is moving with an acceleration, given by

$$\vec{a} = 2e^{-t} \hat{i} + 5 \cos t \hat{j} - 3 \sin t \hat{k}.$$

If the particle is located at $(2, -3, 2)$ at time $t = 0$ and is moving with a velocity given by $4\hat{i} - 3\hat{j} + 2\hat{k}$, find (a) the velocity and (b) the displacement of the particle at any time t ,

$$\text{Sol. (a) Given,} \quad \vec{a} = \frac{d\vec{v}}{dt} = 2e^{-t} \hat{i} + 5 \cos t \hat{j} - 3 \sin t \hat{k}$$

$$\text{Hence} \quad d\vec{v} = (2e^{-t} \hat{i} + 5 \cos t \hat{j} - 3 \sin t \hat{k}) dt$$

$$\begin{aligned} \text{Integrating,} \quad \vec{v} &= \int (2e^{-t} \hat{i} + 5 \cos t \hat{j} - 3 \sin t \hat{k}) dt \\ &= -2e^{-t} \hat{i} + 5 \sin t \hat{j} - 3 \cos t \hat{k} + \vec{C} \end{aligned}$$

$$\therefore \text{At } t = 0, \quad \vec{v}_0 = -2\hat{i} + 3\hat{k} + \vec{C}$$

$$\text{Given :} \quad \vec{v}_0 = 4\hat{i} - 3\hat{j} + 2\hat{k} \text{ at } t = 0. \text{ Hence}$$

$$-2\hat{i} + 3\hat{k} + \vec{C} = 4\hat{i} - 3\hat{j} + 2\hat{k} \text{ or } \vec{C} = 6\hat{i} - 3\hat{j} - \hat{k}$$

$$\text{Hence} \quad \vec{v} = -2e^{-t} \hat{i} + 5 \sin t \hat{j} + 3 \cos t \hat{k} + 6\hat{i} - 3\hat{j} - \hat{k}$$

$$\text{or} \quad \vec{v} = (6 - 2e^{-t}) \hat{i} + (5 \sin t - 3) \hat{j} + (3 \cos t - 1) \hat{k} \quad \dots(i)$$

(b) Putting $\vec{v} = \frac{d\vec{r}}{dt}$ in (i) and integrating, we obtain

$$\vec{r} = \int [(6 - 2e^{-t}) \hat{i} + (5 \sin t - 3) \hat{j} + (3 \cos t - 1) \hat{k}] dt$$

$$\vec{r}_0 = (6t + 2e^{-t}) \hat{i} - (5 \cos t + 3t) \hat{j} + (3 \sin t - t) \hat{k} + \vec{C}$$

$$\text{At } t = 0, \quad \vec{r}_0 = 2\hat{i} - 5\hat{j} + \vec{C}.$$

But the particle is located at $(2, -3, 2)$ at $t = 0$, we have $\vec{r}_0 = 2\hat{i} - 3\hat{j} + 2\hat{k}$. Hence

$$2\hat{i} - 5\hat{j} + \vec{C}_1 = 2\hat{i} - 3\hat{j} + 2\hat{k} \text{ or } \vec{C} = 2\hat{j} + 2\hat{k}$$

Therefore, $\vec{r} = (6t + 2e^{-t})\hat{i} + (2 - 5\cos t - 3t)\hat{j} + (3\sin t - t + 2)\hat{k}$... (ii)

1-10. Line, Surface and Volume Integrals

1-10-1. Line Integral : Let PQ be any curve, drawn in a vector field and $d\vec{l}$ an element of length along this curve at any point B . Let $\vec{A}$ represent the vector at B in a direction making an angle θ with $d\vec{l}$ [fig. 1.24]. In general if $\vec{A}$ varies in magnitude and direction from point to point along the curve, the integral

$$\int_P^Q \vec{A} \cdot d\vec{l} = \int_P^Q A \cos \theta dl \quad \dots(1)$$

is defined as the **line integral** of $\vec{A}$ along the curve PQ .

In terms of cartesian components,

$$\vec{A} = A_x\hat{i} + A_y\hat{j} + A_z\hat{k} \text{ and } d\vec{l} = dx\hat{i} + dy\hat{j} + dz\hat{k}.$$

Hence, $\int \vec{A} \cdot d\vec{l} = \int_P^Q (A_x dx + A_y dy + A_z dz) \quad \dots(2)$

Examples of Line Integral : (1) If a force $\vec{F}$ is acting on a moving particle then the line integral $\int \vec{F} \cdot d\vec{l}$ over the path described by the particle is the work done by the force i.e.,

$$W = \int_P^Q \vec{F} \cdot d\vec{l} \quad \dots(3)$$

(2) If a unit positive charge is moved in an **electric field** $\vec{E}$ from P to Q , then the **line integral** $\int \vec{E} \cdot d\vec{l}$ represents, the **potential difference** (ΔV) between these points i.e.,

$$\Delta V = - \int_P^Q \vec{E} \cdot d\vec{l} \quad \dots(4)$$

(3) For a conservative vector field $\vec{A}$, the value of the line integral along a closed path is zero i.e.,

$$\oint \vec{A} \cdot d\vec{l} = 0 \quad \dots(5)$$

Conservative Vector Field and Exact Differential : If $\text{curl } \vec{A} = 0$ (or $\vec{\nabla} \times \vec{A} = 0$), then we can find a scalar function $\phi(x, y, z)$ for which $\vec{A} = \vec{\nabla} \phi$ (because $\text{curl grad } \phi = \vec{\nabla} \times \vec{\nabla} \phi = 0$). Now if $\vec{A}$ is the gradient of a scalar function ϕ i.e., $\vec{A} = \vec{\nabla} \phi$, then

$$\vec{A} \cdot d\vec{l} = \vec{\nabla} \phi \cdot d\vec{l} = \left(\hat{i} \frac{\partial \phi}{\partial x} + \hat{j} \frac{\partial \phi}{\partial y} + \hat{k} \frac{\partial \phi}{\partial z} \right) (dx\hat{i} + dy\hat{j} + dz\hat{k})$$

or $\vec{A} \cdot d\vec{l} = \frac{\partial \phi}{\partial x} dx + \frac{\partial \phi}{\partial y} dy + \frac{\partial \phi}{\partial z} dz = d\phi \quad \dots(6)$

Hence the line integral is

$$\int_P^Q \vec{A} \cdot d\vec{l} = \int_P^Q d\phi = \phi(Q) - \phi(P) \quad \dots(7)$$

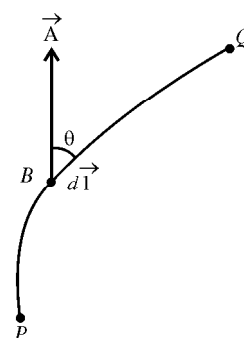


Fig. 1.24.

where $\phi(P)$ and $\phi(Q)$ are the values of the function ϕ at the end points P and Q of the path of integration. Since the value of the integral $\int_P^Q \vec{A} \cdot d\vec{l}$ depends on the end points P and Q , this value is **independent of the path** along which we integrate from P to Q . Such a vector field $\vec{A}$, for which the value of the integral $\int_P^Q \vec{A} \cdot d\vec{l}$ is the same between two given points regardless of the path of integration, is called **conservative vector field**. Examples of conservative vector field are gravitational and electrostatic fields.

Exact Differential : The differential $d\phi$ of a scalar function $\phi(x, y, z)$ in (1) is called an **exact differential**. The value of the integral of such a differential $d\phi$ is independent of the path of integration and depends only on the end points P and Q of the path. Thus $\text{curl } \vec{A} = 0$ is a necessary and sufficient condition for $\vec{A} \cdot d\vec{l}$ to be an exact differential $d\phi$, where $\vec{A}$ is expressed as $\vec{A} = \vec{\nabla} \phi$.

1-10-2. Surface Integral or Flux : Consider any element of infinitesimal area $d\vec{S}$ upon a surface S in a vector field. If $\vec{A}$ be the value of the vector at the middle point of the element, then the integral over the surface [fig. 1.25]

$$\iint_S \vec{A} \cdot d\vec{S} = \iint_S A \cos \theta \, dS \quad \dots(8)$$

is defined as the **surface integral** or **total flux** of vector $\vec{A}$ through the whole surface S .

In cartesian components,

$$\iint_S \vec{A} \cdot d\vec{S} = \iint_S (A_x dS_x + A_y dS_y + A_z dS_z) \quad \dots(9)$$

For example, suppose that $\vec{A} = \vec{v}$ represents the velocity of a moving fluid. If we draw a fixed surface S in the moving liquid, then the surface integral $\iint_S \vec{v} \cdot d\vec{S}$ tells us the **rate of flow of liquid** or the **volume of the liquid flowing** per second through the surface S .

In an electric field $\vec{E}$, the **surface integral** $\iint_S \vec{E} \cdot d\vec{S}$ represents the **electric flux** through the surface S .

1-10-3. Volume Integral : If we consider a closed surface in space in a vector field $\vec{A}$ enclosing a volume V , then

$$\iiint_V \vec{A} \cdot d\vec{V} = \int_x \int_y \int_z (A_x \hat{i} + A_y \hat{j} + A_z \hat{k}) dx \, dy \, dz \quad \dots(8)$$

is defined as the **volume integral** [fig. 1-26].

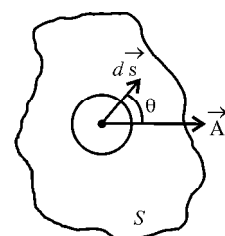


Fig. 1.25. Surface integral.

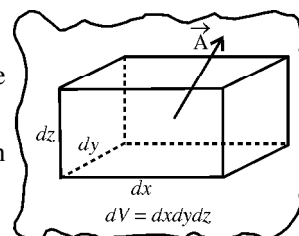


Fig. 1.26 : Volume integral.

Ex. 1. Evaluate the line integral of function $\vec{F} = 6x\hat{i} + 4y\hat{j}$ between the points $(0, 0)$ and $(2, 2)$ in Y - X plane. (O. U. 1988)

Sol. The line integral from P to Q is given by

$$\int_P^Q \vec{F} \cdot d\vec{l} = \int_{0,0}^{2,2} (6x\hat{i} + 4y\hat{j}) (dx\hat{i} + dy\hat{j})$$

where $\vec{F}$ and $d\vec{l}$ are confined to XY -plane.

$$\text{Hence, } \int_P^Q \vec{F} \cdot d\vec{l} = \int_{(0,0)}^{(2,2)} (6x \, dx + 4y \, dy) = \int_0^2 6x \, dx + \int_0^2 4y \, dy$$

$$= [3x^2]_0^2 + [2y^3]_0^2 = 12 + 8 = 20.$$

Ex. 2. Evaluate $\int_C \vec{F} \cdot d\vec{r}$ where $\vec{F} = xy\hat{i} + (x^2 + y^2)\hat{j}$ and C is the arc of the curve, $y = x^2 - 4$, from $(2, 0)$ to $(4, 12)$.

Sol.
$$\int_C \vec{F} \cdot d\vec{r} = \int [xy\hat{i} + (x^2 + y^2)\hat{j}] (dx\hat{i} + dy\hat{j} + dz\hat{k}) = \int [xy dx + (x^2 + y^2) dy]$$

Now, $y = x^2 - 4$ (equation of the curve)

$\therefore dy = 2x dx$

Hence,
$$\begin{aligned} \int_C \vec{F} \cdot d\vec{r} &= \int [x(x^2 - 4)dx + [x^2 + (x^2 - 4)^2]2x dx] \\ &= \int [(x^3 - 4x)dx + (x^2 + x^4 + 16 - 8x^2)2x dx] \\ &= \int [(x^3 - 4x)dx + (2x^5 - 14x^3 + 32x)dx] \\ &= \int_2^4 (2x^5 - 13x^3 + 28x)dx = \left[\frac{2x^6}{6} - \frac{13x^4}{4} + \frac{28x^2}{2} \right]_2^4 \\ &= \left[\frac{x^6}{3} - \frac{13x^4}{4} + 14x^2 \right]_2^4 = \frac{1}{3}(4^6 - 2^6) - \frac{13}{4}(4^4 - 2^4) + 14(4^2 - 2^2) \\ &= 1344 - 780 + 168 = 732. \end{aligned}$$

Ex. 3. Evaluate $\oint_C \vec{A} \cdot d\vec{r}$ around the closed curve C defined by $y = x^2$ and $y^2 = x$ [see **fig.1.27**] with $\vec{A} = (x - y)\hat{i} + (x + y)\hat{j}$.

Sol. The line integral around the closed curve C , confined to XY -plane, is given by

$$\begin{aligned} \oint_C \vec{A} \cdot d\vec{r} &= \oint_C [(x - y)\hat{i} + (x + y)\hat{j}] \cdot [dx\hat{i} + dy\hat{j}] \\ &= \oint_C [(x - y)dx + (x + y)dy] \end{aligned}$$

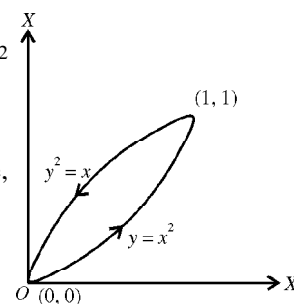


Fig. 1.27.

Therefore, for the given curve,

$$\begin{aligned} \oint_C \vec{A} \cdot d\vec{r} &= \int_0^1 (x - x^2)dx - 2 \int_0^1 (x - x^2)2x dx \quad \text{along } y = x^2 \text{ curve} \\ &\quad + \int_1^0 (y^2 - y)2y dy - \int_1^0 (y^2 - y)dy \quad \text{along } y^2 = x \text{ curve} \end{aligned}$$

Thus,
$$\begin{aligned} \oint_C \vec{A} \cdot d\vec{r} &= \int_0^1 (x - x^2)dx - 2 \int_0^1 (x^2 - x^3)dx - 2 \int_1^0 (y^3 - y^2)dy - \int_1^0 (y^2 - y)dy \\ &= \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^1 - 2 \left[\frac{x^3}{3} - \frac{x^4}{4} \right]_0^1 + 2 \left[\frac{y^4}{4} - \frac{y^3}{3} \right]_1^0 + \left[\frac{y^3}{3} - \frac{y^2}{2} \right]_1^0 \\ &= \left(\frac{1}{2} - \frac{1}{3} \right) + \left(\frac{2}{3} - \frac{2}{4} \right) + 2 \left(-\frac{1}{4} + \frac{1}{3} \right) + \left(-\frac{1}{3} - \frac{1}{2} \right) = \frac{2}{3}. \end{aligned}$$

Ex. 4. (a). If $\vec{F} = (2xy + x^2)\hat{i} + x^2\hat{j} + 2xz\hat{k}$, then show that it is conservative.

(b) Calculate the amount of work done by this force in moving a particle from (0, 1, 2) to (5, 2, 7).

Sol. (a) For a conservative force, $\text{curl } \vec{F} = 0$

$$\begin{aligned}\text{Here, curl } \vec{F} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2xy + z^2 & x^2 & 2xz \end{vmatrix} \\ &= \hat{i} \left[\frac{\partial}{\partial x}(2xz) - \frac{\partial}{\partial z}(x^2) \right] + \hat{j} \left[\frac{\partial}{\partial z}(2xy + z^2) - \frac{\partial}{\partial x}(2xz) \right] + \hat{k} \left[\frac{\partial}{\partial x}(x^2) - \frac{\partial}{\partial y}(2xy + z^2) \right] \\ &= 0 + \hat{j}(2z - 2z) + \hat{k}(2x - 2x) = 0.\end{aligned}$$

Thus, $\vec{F} = (2xy + z^2)\hat{i} + x^2\hat{j} + 2xz\hat{k}$ is a **conservative force**.

$$\begin{aligned}\text{(b) Work done} &= \int_A^B \vec{F} \cdot d\vec{r} = \int_A^B (F_x dx + F_y dy + F_z dz) = \int_A^B [(2xy + z^2) dx + x^2 dy + 2xz dz] \\ &= \int_A^B [(2xy dx + x^2 dy) + (z^2 dx + 2xz dz)] = \int_A^B [d(x^2 y) + d(z^2 x)] \\ &= \int_A^B d(x^2 y + z^2 x) = [x^2 y + z^2 x]_{A(0,1,2)}^{B(5,2,7)} = [5 \times 5 \times 2 + 7 \times 7 \times 5 - 0 - 0] \\ &= 50 + 245 = \mathbf{295 \text{ units.}}\end{aligned}$$

Ex. 5. Given $\vec{A} = (2xy + z^3)\hat{i} + (x^2 + 2y)\hat{j} + (3xz^2 - 2)\hat{k}$. (a) Show that $\vec{\nabla} \times \vec{A} = 0$.

(b) Find a scalar function ϕ such that $\vec{A} = \vec{\nabla} \phi$.

(c) Show that $\int_C \vec{A} \cdot d\vec{r}$ is independent of the path C joining the points (1, -1, 1) and (2, 1, 2)

and find its value.

$$\text{Sol. Given } \vec{A} = (2xy + z^3)\hat{i} + (x^2 + 2y)\hat{j} + (3xz^2 - 2)\hat{k} \quad \dots(i)$$

$$\text{(a) } \vec{\nabla} \times \vec{A} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2xy + z^3 & x^2 + 2y & 3xz^2 - 2 \end{vmatrix} = \mathbf{0} \quad \dots(ii)$$

$$\begin{aligned}\text{(b) When } \vec{A} &= \vec{\nabla} \Phi, \vec{A} \cdot d\vec{r} = \left(\frac{\partial \Phi}{\partial x} \hat{i} + \frac{\partial \Phi}{\partial y} \hat{j} + \frac{\partial \Phi}{\partial z} \hat{k} \right) \cdot (dx \hat{i} + dy \hat{j} + dz \hat{k}) \\ &= \frac{\partial \Phi}{\partial x} dx + \frac{\partial \Phi}{\partial y} dy + \frac{\partial \Phi}{\partial z} dz = d\Phi \quad \dots(iii)\end{aligned}$$

This is an exact differential. In the present case,

$$\begin{aligned}d\Phi = \vec{A} \cdot d\vec{r} &= [(2xy + z^3)\hat{i} + (x^2 + 2y)\hat{j} + (3xz^2 - 2)\hat{k}] \cdot (dx \hat{i} + dy \hat{j} + dz \hat{k}) \\ &= (2xy + z^3)dx + (x^2 + 2y)dy + (3xz^2 - 2)dz \\ &= [(2xy + z^3)dx + x^2 dy + 3xz^2 dz] + 2y dy - 2 dz \\ &= d(x^2 y + xz^3) + d(y^2) + d(-2z)\end{aligned}$$

$$= d(x^2y + xz^3 + y^2 - 2z) \quad \dots(\text{iv})$$

Hence, $\Phi = x^2y + xz^3 + y^2 - 2z \quad \dots(\text{v})$

(c) For the given expression (i) for $\vec{A}$,

$$\vec{\nabla} \times \vec{A} = 0 \text{ and } \vec{A} = \text{grad } \phi \text{ and } \vec{A} \cdot d\vec{r} = \vec{\nabla} \phi \cdot d\vec{r} = d\phi = d(x^2y + xz^3 + y^2 - 2z).$$

This is an exact differential and hence **its integral** $\int_C \vec{A} \cdot d\vec{r} = \int d\phi$ **is independent of the path.**

The value of this integral is

$$\begin{aligned} \int_{(1,-1,1)}^{(2,1,2)} \vec{A} \cdot d\vec{r} &= \int_{(1,-1,1)}^{(2,1,2)} d(x^2y + xz^3 + y^2 - 2z) \\ &= \left[x^2y + xz^3 + y^2 - 2z \right]_{(1,-1,1)}^{(2,1,2)} = \mathbf{18}. \end{aligned}$$

Ex. 6. If $\vec{A} = (2x^2 - 4y)\hat{i} - 7yz\hat{j} - 10xz^2\hat{k}$ evaluate $\int_C \vec{A} \cdot d\vec{r}$ from $(0, 0, 0)$ to $(1, 1, 1)$ along the path C with $x = t$, $y = t^2$ and $z = t^3$. (O. U. 2007)

Sol.
$$\begin{aligned} \int_C \vec{A} \cdot d\vec{r} &= \int_C (2x^2 - 4y)\hat{i} - 7yz\hat{j} - 10xz^2\hat{k} \cdot (dx\hat{i} + dy\hat{j} + dz\hat{k}) \\ &= \int_C [(2x^2 - 4y)dx - 7yzdy - 10xz^2dz] \end{aligned}$$

If $x = t$, $y = t^2$, $z = t^3$, points $(0, 0, 0)$ and $(1, 1, 1)$ correspond to $t = 0$ and $t = 1$ respectively. Then

$$\begin{aligned} \int_C \vec{A} \cdot d\vec{r} &= \int_0^1 [2t^2 - 4t^2]dt - 7t^2t^3d(t^2) - 10t(t^3)^2d(t^3) \\ &= \int_0^1 [6t^2dt - 14t^6dt - 30t^9dt] \\ &= \left[6\frac{t^3}{3} - 14\frac{t^7}{7} + 30\frac{t^{10}}{10} \right]_0^1 \\ &= [2 - 2 + 3] = \mathbf{3}. \end{aligned}$$

Ex. 7. If $\vec{A} = (3x^2 - 6yz)\hat{i} + (2y + 3xz)\hat{j} + (1 - 4xyz^2)\hat{k}$, evaluate $\int_C \vec{A} \cdot d\vec{r}$ from $(0, 0, 0)$ to $(1, 1, 1)$ along the paths C given by :

(a) $x = t$, $y = t^2$, $z = t^3$.

(b) The straight lines from $(0, 0, 0)$ to $(0, 0, 1)$, then to $(0, 1, 1)$ and then to $(1, 1, 1)$.

(c) The straight line joining the two points $(0, 0, 0)$ and $(1, 1, 1)$.

Sol. Here,
$$\begin{aligned} \int_C \vec{A} \cdot d\vec{r} &= \int_C \left\{ (3x^2 - 6yz)\hat{i} + (2y + 3xz)\hat{j} + (1 - 4xyz^2)\hat{k} \right\} \cdot (dx\hat{i} + dy\hat{j} + dz\hat{k}) \\ &= \int_C [(3x^2 - 6yz)dx + (2y + 3xz)dy + (1 - 4xyz^2)dz] \end{aligned}$$

(a) When $x = t$, $y = t^2$, $z = t^3$, points $(0, 0, 0)$ and $(1, 1, 1)$ correspond to $t = 0$ and $t = 1$ respectively.

In such a case,

$$\int_C \vec{A} \cdot d\vec{r} = \int_{t=0}^{t=1} [\{3t^2 - 6(t^2)(t^3)\}dt + \{2t^2 + 3(t)(t^3)\}d(t^2) + \{1 - 4(t)(t^2)(t^3)^2\}d(t^3)]$$

$$\begin{aligned}
&= \int_0^1 [(3t^2 - 6t^5)dt + (4t^3 + 6t^5)dt + (3t^2 - 12t^{11})dt] \\
&= [(t^3 - t^6) + (t^4 + t^6) + (t^3 - t^{12})]_0^1 = 2.
\end{aligned}$$

(b) Along the straight line from (0, 0, 0) to (0, 0, 1), $x = 0$, $y = 0$, $dx = 0$, $dy = 0$ but z varies from 0 to 1. Then the integral over this part of the path is

$$\begin{aligned}
\int_C \vec{A} \cdot d\vec{r} &= \int_C [3x^2 - 6yz]dx + (2y + 3xy)dy + (1 - 4xyz^2)dz \\
&= \int_{z=0}^1 [3(0)^2 - 6(0)(z)]0 + \{2(0) + 3(0)(z)\}0 + \{1 - 4(0)(0)(z^2)\}dz \\
&= \int_0^1 dz = [z]_0^1 = 1. \quad \dots(i)
\end{aligned}$$

Along the straight line from (0, 0, 1) to (0, 1, 1), $x = 0$, $z = 1$, $dx = 0$, $dz = 0$ and y varies from 0 to 1. Hence, the integral over this part of the path is

$$\int_{y=0}^1 [3(0)^2 - 6(y)(1)]0 + \{2y + 3(0)(1)\}dy + \{1 - 4(0)(y)(1)^2\}0 = \int_0^1 2y dy = [y^2]_0^1 = 1. \quad \dots(ii)$$

Along the straight line from (0, 1, 1) to (1, 1, 1), $y = 1$, $z = 1$, $dy = 0$, $dz = 0$ and x varies from 0 to 1. Hence the integral over this straight line is

$$\int_{x=0}^1 [3x^2 - 6(1)(1)]dx + \{2(1) + 3x(1)\}0 + \{1 - 4x(1)(1)^2\}0 = \int_0^1 (3x^2 - 6)dx + [x^3 - 6x]_0^1 = -5. \quad \dots(iii)$$

Thus the total value of the integral along the given paths is

$$\int_C \vec{A} \cdot d\vec{r} = 1 + 1 - 5 = -3.$$

(c) Along the straight line joining (0, 0, 0) and (1, 1, 1) obviously we have $x = t$, $y = t$, $z = t$ and hence. $dx = dy = dz = dt$. Then

$$\begin{aligned}
\int_C \vec{A} \cdot d\vec{r} &= \int_C [(3x^2 - 6yz)dx + (2y + 3xz)dy + (1 - 4xyz^2)dz] \\
&= \int_{t=0}^1 [(3t^2 - 6t^2)dt + (2t + 3t^2)dt + (1 - 4t^4)dt] \\
&= \int_0^1 (2t + 1 - 4t^4)dt = \left[t^2 + t - \frac{4t^5}{5} \right]_0^1 = 2 - \frac{4}{5} = \frac{6}{5}. \quad \dots(iv)
\end{aligned}$$

Observe that the values of the integral between the two points (0, 0, 0) and (1, 1, 1) are different for the two paths.

1.11. Stoke's Theorem

The theorem states that *the line integral of a vector $\vec{A}$ around any closed curve C is equal to the surface integral of $\text{curl } \vec{A}$ taken over any surface S of which the curve is a boundary edge, i.e.,*

$$\oint_C \vec{A} \cdot d\vec{l} = \iint_S \text{curl } \vec{A} \cdot d\vec{S} = \iint_S (\vec{\nabla} \times \vec{A}) \cdot d\vec{S} \quad \dots(1)$$

Proof. If a vector $\vec{A}$ be a function of position, then its line integral along a closed curve C is given by

$$L = \oint_C \vec{A} \cdot d\vec{l}$$

where $d\vec{l}$ is the small element of the path. Now divide the enclosed area into two parts by a line pq so as to form two closed curves C_1 and C_2 [fig. 1.28]. If L_1 and L_2 are the line integrals for the two paths C_1 and C_2 , then their sum ($L_1 + L_2$) will be equal to L for the curve C . The reason lies in the fact that the part pq is traversed from p to q for the curve C_1 and from q to p for the path C_2 so that its contribution is cancelled. Thus, if we divide the area enclosed by the curve C into a number of smaller areas $\delta S_1, \delta S_2, \dots, \delta S_n, \dots$ bounded by curves

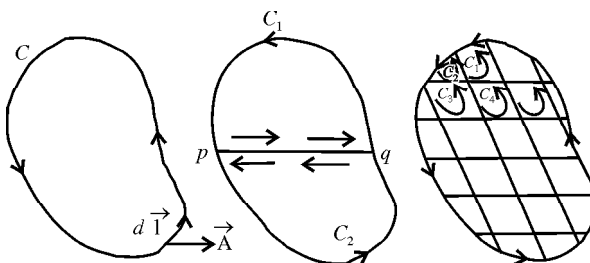


Fig. 1.28. Proving Stoke's theorem.

$C_1, C_2, C_3, \dots, C_n, \dots$, then

$$\oint_C \vec{A} \cdot d\vec{l} = \sum \oint_{C_n} \vec{A} \cdot d\vec{l} \quad \dots(2)$$

The curve C may be divided in such a way that small areas $\delta S_1, \delta S_2, \dots$ may lie in different planes.

If $\hat{n}$ is a unit vector positive outward normal vector upon δS_n , then the vector area corresponding to the small area δS_n is

$$\hat{n} \delta S_n = \vec{\delta S}_n$$

But from the definition of $\text{curl } \vec{A}$ at a point inside δS_n , we have

$$\text{curl } \vec{A} \cdot \vec{\delta S}_n = \oint_{C_n} \vec{A} \cdot d\vec{l}$$

$$\text{Thus, } \oint_C \vec{A} \cdot d\vec{l} = \sum \oint_{C_n} \vec{A} \cdot d\vec{l} = \sum \text{curl } \vec{A} \cdot \vec{\delta S}_n$$

$$\text{i.e., } \oint_C \vec{A} \cdot d\vec{l} = \iint_S \text{curl } \vec{A} \cdot \vec{\delta S} \text{ or } \oint_C \vec{A} \cdot d\vec{l} = \iint_S \text{curl } \vec{A} \cdot \hat{n} dS \quad \dots(3)$$

This is *Stoke's theorem*. It should be noted that the boundary curve or the related surface need not to be plane one.

If the surface to which Stoke's theorem is applied is a closed surface, the length of the boundary

$$\text{edge is zero and then } \iint_S \text{curl } \vec{A} \cdot \vec{\delta S} = 0 \quad \dots(4)$$

In case, $\text{curl } \vec{A}$ or $\vec{\nabla} \times \vec{A} = 0$ everywhere, then according to Stoke's theorem the line integral of $\vec{A}$ around any closed curve vanishes.

1.12. Gauss's Theorem

According to this theorem, the volume integral of the divergence of a vector field $\vec{A}$ taken over any volume V is equal to the surface integral of $\vec{A}$ taken over the closed surface surrounding the volume. That is

$$\iiint_V (\vec{\nabla} \cdot \vec{A}) dV = \iint_S \vec{A} \cdot \vec{\delta S} \quad \dots(1)$$

Proof. In cartesian coordinates,

$$\begin{aligned}\vec{\nabla} \cdot \vec{A} &= \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) (A_x \hat{i} + A_y \hat{j} + A_z \hat{k}) \\ &= \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}, \quad dV = dx \, dy \, dz\end{aligned}$$

and

$$\begin{aligned}\vec{A} \cdot d\vec{S} &= (A_x \hat{i} + A_y \hat{j} + A_z \hat{k}) \cdot (dS_x \hat{i} + dS_y \hat{j} + dS_z \hat{k}) \\ &= A_x dS_x + A_y dS_y + A_z dS_z \\ &= A_x dy \, dz + A_y dx \, dz + A_z dx \, dy.\end{aligned}$$

Hence, Gauss's theorem in cartesian coordinates is

$$\iiint_V \left(\frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} \right) dx \, dy \, dz = \iint_S (A_x dS_x + A_y dS_y + A_z dS_z) \quad \dots(2)$$

Let us consider the first part of the left-hand side integral. As shown along a strip of cross-section $dy \, dz$ extending from $P_1(x_1, y, z)$ to $P_2(x_2, y, z)$, [fig. 1.29] i.e.,

$$\begin{aligned}\iiint_V \frac{\partial A_x}{\partial x} dx \, dy \, dz &= \iint_S [A_x]_{P_1}^{P_2} dy \, dz \\ &= \iint_S [A_x(P_2) - A_x(P_1)] dy \, dz \\ &= \iint_S [A_x(x_2, y, z) - A_x(x_1, y, z)] dy \, dz\end{aligned}$$

Evidently, at P_1 , $A_x \, dy \, dz = -A_x \, dS_x$ and at P_2 , $A_x \, dy \, dz = A_x \, dS_x$ (since at P_1 and P_2 , the X -components of area have opposite directions).

$$\therefore \iiint_V \frac{\partial A_x}{\partial x} dx \, dy \, dz = \iint_S A_x \, dS_x \quad \dots(3)$$

Similarly, $\iiint_V \frac{\partial A_y}{\partial y} dx \, dy \, dz = \iint_S A_y \, dS_y \quad \dots(4)$

and $\iiint_V \frac{\partial A_z}{\partial z} dx \, dy \, dz = \iint_S A_z \, dS_z \quad \dots(5)$

Hence, adding (3), (4) and (5), we have

$$\iiint_V \left(\frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} \right) dx \, dy \, dz = \iint_S (A_x dS_x + A_y dS_y + A_z dS_z)$$

or $\iiint_V (\vec{\nabla} \cdot \vec{A}) dV = \iint_S \vec{A} \cdot d\vec{S}$ or $\iiint_V (\text{div } \vec{A}) dV = \iint_S \vec{A} \cdot d\vec{S} \quad \dots(6)$

This proves the Gauss's theorem of divergence.

If we write $d\vec{S} = \hat{n} \, dS$, where $\hat{n}$ is an outward normal unit vector to the area dS , then Gauss's theorem of divergence can be expressed as

$$\iiint_V (\text{div } \vec{A}) dV = \iint_S \vec{A} \cdot \hat{n} \, dS \quad \dots(7)$$

* The reader should not confuse that we have written for

$$\iint [A_x(P_2) - A_x(P_1)] dy \, dz = \iint (A_x dS_x + A_x dS_x) \quad \text{only } \iint A_x dS_x,$$

because if we take all the stripes then each integral $\iint A_x dS_x$ will account for the half surface integral. Hence $\iint A_x dS_x$ account for the surface integral for X -component.

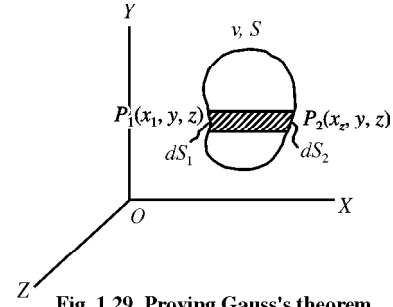


Fig. 1.29. Proving Gauss's theorem.

1.13. Green's Theorem

We can make some important transformations by the use of Gauss's theorem.

According to Gauss's theorem,

$$\iiint_V (\vec{\nabla} \cdot \vec{A}) dV = \iint_S \vec{A} \cdot d\vec{S} \quad \dots(1)$$

Let us consider that the vector field $\vec{A}$ be the product of scalar function ϕ_1 and the gradient of another scalar function ϕ_2 , i.e.,

$$\vec{A} = \phi_1 \vec{\nabla} \phi_2 = \hat{i} \phi_1 \frac{\partial \phi_2}{\partial x} + \hat{j} \phi_1 \frac{\partial \phi_2}{\partial y} + \hat{k} \phi_1 \frac{\partial \phi_2}{\partial z}$$

$$\begin{aligned} \vec{\nabla} \cdot \vec{A} &= \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} = \frac{\partial}{\partial x} \left(\phi_1 \frac{\partial \phi_2}{\partial x} \right) + \frac{\partial}{\partial y} \left(\phi_1 \frac{\partial \phi_2}{\partial y} \right) + \frac{\partial}{\partial z} \left(\phi_1 \frac{\partial \phi_2}{\partial z} \right) \\ &= \phi_1 \left(\frac{\partial^2 \phi_2}{\partial x^2} + \frac{\partial^2 \phi_2}{\partial y^2} + \frac{\partial^2 \phi_2}{\partial z^2} \right) + \frac{\partial \phi_1}{\partial x} \cdot \frac{\partial \phi_2}{\partial x} + \frac{\partial \phi_1}{\partial y} \cdot \frac{\partial \phi_2}{\partial y} + \frac{\partial \phi_1}{\partial z} \cdot \frac{\partial \phi_2}{\partial z} = \phi_1 \nabla^2 \phi_2 + \vec{\nabla} \phi_1 \cdot \vec{\nabla} \phi_2 \end{aligned}$$

where $\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$ is called **Laplacian operator**.

Substituting for $\vec{\nabla} \cdot \vec{A}$ in eq. (1), we get

$$\iiint_V \left(\phi_1 \nabla^2 \phi_2 + \vec{\nabla} \phi_1 \cdot \vec{\nabla} \phi_2 \right) dV = \iint_S (\phi_1 \vec{\nabla} \phi_2) \cdot d\vec{S} \quad \dots(2)$$

We call the transformation represented by eq. (2), as the **first form of Green's theorem**.

If we mutually interchange the scalar functions ϕ_1 and ϕ_2 in eq. (2), we obtain

$$\iiint_V \left(\phi_2 \nabla^2 \phi_1 + \vec{\nabla} \phi_2 \cdot \vec{\nabla} \phi_1 \right) dV = \iint_S (\phi_2 \vec{\nabla} \phi_1) \cdot d\vec{S} \quad \dots(3)$$

Subtracting eq. (3) from eq. (2), we get

$$\iiint_V (\phi_1 \nabla^2 \phi_2 - \phi_2 \nabla^2 \phi_1) dV = \iint_S (\phi_1 \vec{\nabla} \phi_2 - \phi_2 \vec{\nabla} \phi_1) \cdot d\vec{S} \quad \dots(4)$$

Eq. (4) represents the **second form of Green's theorem**.

These transformations (2) and (4), obtained by the use of Gauss's theorem, find important applications in the field of electro-dynamics and hydro-dynamics.

Ex. 1. Using Stoke's theorem, show that

$$\oint_C \vec{r} \cdot d\vec{l} = 0, \text{ where } \vec{r} \text{ is position vector.}$$

Sol. According to Stoke's theorem, $\oint_C \vec{A} \cdot d\vec{l} = \iint_S \text{curl } \vec{A} \cdot d\vec{S}$

Replacing the vector $\vec{A}$ by position vector $\vec{r}$, we obtain

$$\oint_C \vec{r} \cdot d\vec{l} = \iint_S \text{curl } \vec{r} \cdot d\vec{S} = \iint_S 0 \cdot d\vec{S} = 0 \quad [\because \text{curl } \vec{r} = 0]$$

Therefore, $\oint_C \vec{r} \cdot d\vec{l} = 0$.

Ex. 2. Use Stoke's theorem to evaluate

$$\oint_C (y z dx + x z dy + x y dz)$$

where C is the curve $x^2 + y^2 = 1$ and $z = y^2$.

Sol. According to Stoke's theorem,

$$\oint_C \vec{A} \cdot d\vec{l} = \iint_S \text{curl } \vec{A} \cdot d\vec{S}$$

Here, we want to evaluate the integral $\oint_C (yz \, dx + xz \, dy + xy \, dz)$.

We can write the integral as

$$\begin{aligned} \oint_C (yz \, dx + xz \, dy + xy \, dz) &= \oint_C [(yz \, \hat{i} + xz \, \hat{j} + xy \, \hat{k}) \cdot (dx \, \hat{i} + dy \, \hat{j} + dz \, \hat{k})] \\ &= \oint_C \vec{A} \cdot d\vec{l} . \end{aligned}$$

Hence, $\vec{A} = yz \, \hat{i} + xz \, \hat{j} + xy \, \hat{k}$

$$\begin{aligned} \text{Now } \text{curl } \vec{A} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ yz & xz & xy \end{vmatrix} \\ &= \hat{i} \left(\frac{\partial}{\partial y} xy - \frac{\partial}{\partial z} xz \right) + \hat{j} \left(\frac{\partial}{\partial z} yz - \frac{\partial}{\partial x} xy \right) + \hat{k} \left(\frac{\partial}{\partial x} xz - \frac{\partial}{\partial y} yz \right) \\ &= \hat{i}(x - x) + \hat{j}(y - y) + \hat{k}(z - z) = 0 . \end{aligned}$$

Using Stoke's theorem

$$\oint_C \vec{A} \cdot d\vec{l} = \iint_S \text{curl } \vec{A} \cdot d\vec{S} = \iint_S 0 \cdot d\vec{S} = 0$$

Hence the value of the given integral is zero.

Ex. 3. If $\vec{A} = -y \, \hat{i} + x \, \hat{j}$, then calculate (a) $\text{curl } \vec{A}$ (b) the line integral $\int \vec{A} \cdot d\vec{l}$ for the closed curve $x^2 + y^2 = r^2$, $z = 0$. Hence verify Stoke's theorem.

Sol. (a) Here, $A_x = -y$, $A_y = x$, $A_z = 0$; $\therefore \frac{\partial A_x}{\partial y} = -1$, $\frac{\partial A_y}{\partial x} = 1$ and other derivatives are zero.

$$\text{Hence } \text{curl } \vec{A} = \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) \hat{k} = (1 + 1) \hat{k} = 2 \hat{k} .$$

$$(b) \quad \int \vec{A} \cdot d\vec{l} = \int (-y \, \hat{i} + x \, \hat{j}) (dx \, \hat{i} + dy \, \hat{j}) = \int -y \, dx + \int x \, dy .$$

For the circle $x^2 + y^2 = r^2$, $x = r \cos \theta$, $y = r \sin \theta$;

$$\therefore dx = -r \sin \theta \, d\theta, \, dy = r \cos \theta \, d\theta .$$

$$\text{Hence, } \int \vec{A} \cdot d\vec{l} = \int_0^{2\pi} r^2 (\cos^2 \theta + \sin^2 \theta) d\theta = \int_0^{2\pi} r^2 d\theta = 2\pi r^2$$

$$\text{Now, according to Stoke's theorem, } \int \vec{A} \cdot d\vec{l} = \iint_S \text{curl } \vec{A} \cdot d\vec{S}$$

$$\text{Here, } \iint_S \text{curl } \vec{A} \cdot d\vec{S} = \iint_S 2\hat{k} \cdot (dS)\hat{k} = \iint_S 2dS = 2S = 2\pi r^2 .$$

$$\text{Therefore, } \int \vec{A} \cdot d\vec{l} = \iint_S \text{curl } \vec{A} \cdot d\vec{S}$$

Thus, **Stoke's theorem is verified.**

Ex. 4. Verify Stoke's theorem for the vector $\vec{A} = x(\hat{i}x + \hat{j}y)$ integrated round the square, in the plane $z = 0$, whose sides are along the lines $x = 0$, $y = 0$, $x = a$, $y = a$. (O. U. 1995)

Sol. According to Stoke's theorem

$$\oint_C \vec{A} \cdot d\vec{l} = \iint_S \text{curl } \vec{A} \cdot d\vec{S} \quad \dots(i)$$

The sides of the given square can be represented as shown in **fig. 1.30**.

$$\begin{aligned} \text{Here, } \oint_C \vec{A} \cdot d\vec{l} &= \int_{OP} \vec{A} \cdot d\vec{l} + \int_{PQ} \vec{A} \cdot d\vec{l} + \int_{QR} \vec{A} \cdot d\vec{l} \\ &\quad + \int_{RO} \vec{A} \cdot d\vec{l} \end{aligned}$$

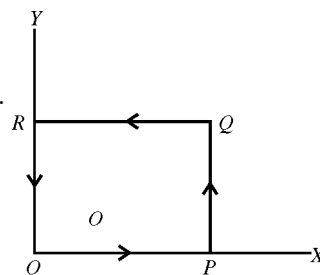


Fig. 1.29.

$$\text{Now, } \oint_{OP} \vec{A} \cdot d\vec{l} = \int_0^a x(\hat{i}x + \hat{j}y) \cdot \hat{i} dx = \int_0^a x^2 dx = \frac{a^3}{3}$$

$$\oint_{PQ} \vec{A} \cdot d\vec{l} = \int_0^a x(\hat{i}x + \hat{j}y) \cdot \hat{j} dy = \int_0^a ay dy = \frac{a^3}{2}$$

$$\oint_{QR} \vec{A} \cdot d\vec{l} = \int_a^0 x(\hat{i}x + \hat{j}y) \cdot \hat{i} dx = -\int_0^a x^2 dx = -\frac{a^3}{3}$$

$$\text{and } \oint_{RO} \vec{A} \cdot d\vec{l} = \int_a^0 x(\hat{i}x + \hat{j}y) \cdot \hat{j} dy = -\int_0^a 0(\hat{i}x + \hat{j}y) \cdot \hat{j} dy = 0$$

$$\text{Thus, } \oint_C \vec{A} \cdot d\vec{l} = \frac{a^3}{3} + \frac{a^3}{2} - \frac{a^3}{3} + 0 = \frac{a^3}{2} \quad \dots(ii)$$

$$\text{Also } \iint_S \text{curl } \vec{A} \cdot d\vec{S} = \iint_S \text{curl } x(\hat{i}x + \hat{j}y) \cdot d\vec{S}$$

$$\begin{aligned} \therefore \iint_S \text{curl } \vec{A} \cdot d\vec{S} &= \int_0^a \int_0^a \hat{k}y \cdot \hat{k} dx dy \\ &= \int_0^a \int_0^a y dx dy = \int_0^a dx \int_0^a y dy \\ &= a \cdot \frac{a^2}{2} = \frac{a^3}{2} \quad \dots(iii) \end{aligned}$$

From eqs. (ii) and (iii) we see that the values of L. H. S. and R. H. S. of eq. (i) are equal. Thus **Stoke's theorem is verified.**

$$\text{But } \text{curl } x(\hat{i}x + \hat{j}y) = \vec{\nabla} \times (\hat{i}x^2 + \hat{j}xy) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 & xy & 0 \end{vmatrix} = \hat{k}y \text{ and } d\vec{S} = \hat{k} dx dy$$

Ex. 5. A closed surface S encloses a volume V . Find the value of $\text{div } \vec{F}$ and $\iint_S \vec{F} \cdot d\vec{S}$ for a vector field $\vec{F} = 3x\hat{i} + 4y\hat{j} + 2z\hat{k}$.

Sol. According to Gauss's divergence theorem, $\iint_S \vec{F} \cdot d\vec{S} = \iiint_V (\text{div } \vec{F}) dV$

$$\therefore \text{div } \vec{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z} = \frac{\partial}{\partial x}(3x) + \frac{\partial}{\partial y}(4y) + \frac{\partial}{\partial z}(2z) = 3 + 4 + 2 = 9$$

$$\text{Hence, } \iint_S \vec{F} \cdot d\vec{S} = \iiint_V (\text{div } \vec{F}) dV = 9V.$$

Ex. 6. Prove that $\iint_S \vec{r} \cdot d\vec{S} = 3V$, where $\vec{r}$ is the position vector at any point on the surface enclosing volume V . (A. N. U. 1989)

Sol. According to Gauss's theorem, $\iint_S \vec{A} \cdot d\vec{S} = \iiint_V (\text{div } \vec{A}) dV$

Here, $\vec{A} = \vec{r}$ and $\text{div } \vec{A} = \text{div } \vec{r} = 3$.

$$\therefore \iint_S \vec{r} \cdot d\vec{S} = \iiint_V (\text{div } \vec{r}) dV = \iiint_V 3 dV = 3V.$$

Ex. 7. Prove that $\iint_S \text{curl } \vec{F} \cdot d\vec{S}$ for any closed surface.

Sol. According to Gauss's theorem, we have

$$\iint_S \text{curl } \vec{F} \cdot d\vec{S} = \iiint_V \vec{\nabla} \cdot (\text{curl } \vec{F}) dV = \iiint_V \text{div curl } \vec{F} dV$$

But div of curl is zero. Therefore

$$\iint_S \text{curl } \vec{F} \cdot d\vec{S} = 0.$$

Ex. 8. If $\hat{n}$ is the unit outward drawn normal to any closed surface of area S , show that

$$\iiint_V (\text{div } \hat{n}) dV = S.$$

Sol. According to Gauss's divergence theorem,

$$\iiint_V (\text{div } \vec{A}) dV = \iint_S \vec{A} \cdot d\vec{S}$$

$$\text{Hence, } \iiint_V (\text{div } \hat{n}) dV = \iint_S \hat{n} \cdot d\vec{S} = \iint_S \hat{n} \cdot \hat{n} dS = \iint_S dS = S \quad [\because d\vec{S} = \hat{n} dS]$$

Ex. 9. Find $\iint_S \vec{F} \cdot \hat{n} dS$, where $\vec{F} = 4xz\hat{i} - y^2\hat{j} + yz\hat{k}$ and S is the surface of a cube bounded by $x = 0$, $x = 1$, $y = 0$, $y = 1$, $z = 0$, $z = 1$. **(K. U. 1988)**

Sol. According to Gauss's theorem, we have

$$\iint_S \vec{F} \cdot \hat{n} dS = \iiint_V \text{div } \vec{F} dV = \iiint_V \left(\frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z} \right) dx dy dz$$

Here, $F_x = 4xz$, $F_y = -y^2$, $F_z = yz$.

$$\begin{aligned} \therefore \iint_S \vec{F} \cdot \hat{n} dS &= \iiint_V \left[\frac{\partial}{\partial x}(4xz) + \frac{\partial}{\partial y}(-y^2) + \frac{\partial}{\partial z}(yz) \right] dx dy dz = \iiint_V (4z - 2y + y) dx dy dz \\ &= \int_0^1 \int_0^1 \int_0^1 (4z - y) dx dy dz = \int_0^1 \int_0^1 [2z^2 - yz]_0^1 dx dy = \int_0^1 \int_0^1 (2 - y) dy dz \\ &= \int_0^1 \left[2y - \frac{y^2}{2} \right]_0^1 dz = \int_0^1 \frac{3}{2} dz = \left[\frac{3}{2} z \right]_0^1 = \frac{3}{2} = 1.5. \end{aligned}$$

Ex. 10. Obtain Gauss's divergence theorem from Green's theorem. **(S. V. U. 2007)**

Sol. According to Gauss's theorem,

$$\iiint_V (\phi_1 \nabla^2 \phi_2 - \phi_2 \nabla^2 \phi_1) dV = \iint_S (\phi_1 \nabla \phi_2 - \phi_2 \nabla \phi_1) \cdot d\vec{S}$$

Let ϕ_1 be a constant scalar function. Then $\nabla \phi_1 = 0$ and the Green's theorem assumes the form

$$\iiint_V (\phi_1 \nabla^2 \phi_2) dV = \iint_S \phi_1 \nabla \phi_2 \cdot d\vec{S} \quad \text{or} \quad \phi_1 \iiint_V \nabla^2 \phi_2 dV = \phi_1 \iint_S \vec{\nabla} \phi_2 \cdot d\vec{S}$$

$$\text{or} \quad \iiint_V \nabla^2 \phi_2 dV = \iint_S \vec{\nabla} \phi_2 \cdot d\vec{S}$$

$$\text{or} \quad \iiint_V \vec{\nabla} \cdot \vec{\nabla} \phi_2 dV = \iint_S \vec{\nabla} \phi_2 \cdot d\vec{S}$$

Suppose $\vec{\nabla} \phi_2 = \vec{A}$. Hence

$$\iiint_V (\vec{\nabla} \cdot \vec{A}) dV = \iint_S \vec{A} \cdot d\vec{S}$$

This is what is known as Gauss's divergence theorem.

QUESTIONS

Long Answer Questions

1. Distinguish between vector field and scalar field. Give one example of each of them. (K. U. 2005)
2. What are partial derivatives ? Obtain an expression for the gradient of a scalar function.
(S. V. U. 1995; S. K. U. 1995)
3. Define gradient of a vector field. Give its physical significance. What are scalar and vector fields ? Give examples.
(S. V. U. 2007)
4. What is meant by gradient of a scalar function ? Prove that $(\nabla \phi) \cdot \hat{n}$ and $\nabla \phi$ is always normal to the surface $\phi = \text{constant}$. Give two examples of gradient of a scalar field.
5. Define divergence of a vector field. Obtain its value in cartesian coordinates and prove that $\text{div} \vec{A} = \vec{\nabla} \cdot \vec{A}$ where $\vec{\nabla}$ is a vector operator.
6. What is meant by curl of a vector function ? Derive an expression for it in cartesian coordinates. Give its physical significance.
(A. N. U. 2008; K. U. 2007)
7. Obtain the expression for $\text{curl} \vec{A}$ and prove that $\vec{A} = \text{curl} \vec{A} \times \vec{r}$ where $\vec{A}$ is a vector. (O. U. 2003)
8. Explain the nature of vector and scalar fields. Derive an expression for curl of a vector field. What is its physical significance ?
(A. N. U. 2006)
9. What is line integral ? What is its physical significance.
(S. K. U. 1994)
10. State and prove Stoke's theorem.
(O. U. 2007; S. K. U. 2007)
11. State and prove Stoke's theorem. Explain how the theorem resembles Gauss's divergence theorem in certain aspects ?
(O. U. 2001; S. V. U. 2003)
12. What is the curl of a vector field ? State and prove Stoke's theorem.
(O. U. 2009; A. N. U. 2007, 05)
13. What are surface and volume integrals ? State and prove Gauss's divergence theorem.
(S. V. U. 2005)
14. State and prove Gauss's divergence theorem. What is its physical significance ?
(A. N. U. 2007; O. U. 2005)
15. Define curl and div of a vector. State and prove Gauss's divergence theorem.
(A. U. 2005; S. V. U. 2003)
16. Define grad, div and curl operators. Explain their physical meanings. State and prove Gauss's law for vector integration.
(S. V. U. 2002)
17. What are scalar and vector products ? Explain the characteristics of a vector product of two vectors. State and prove Gauss's theorem.
(A. U. 2007)
18. What is gradient of scalar field. State and prove Green's theorem.
(K. U. 2005)
19. State and prove Greens's theorem.
(S. K. U. 2005; N. U. 2001)

20. If $\vec{A} = \theta \vec{\nabla} \phi$ where θ and ϕ are the scalar functions of position. Show that

$$\iiint_V (\phi \nabla^2 \theta + \vec{\nabla} \theta \cdot \vec{\nabla} \phi) dV = \iint_S (\theta \vec{\nabla} \phi) \cdot d\vec{S}$$

Short Answer Questions

1. What are scalar and vector fields ? Give examples. (K. U. 2005; A. N. U. 1990)
2. Explain the different types of vector fields with examples. (A. N. U. 2006)
3. Explain gradient of a scalar field. (S. V. U. 2005)
4. Write short notes on (a) the Del operator and (b) Laplacian operator. (A. U. 1990)

[Laplacian Operator is defined by

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

The Laplacian operator is equal to the scalar product of del operator with itself i.e.,

$$\vec{\nabla} \cdot \vec{\nabla} = \hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \cdot \hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} = \nabla^2$$

If Φ is a function of x, y, z , then

$$\nabla^2 \Phi = \frac{\partial^2 \Phi}{\partial x^2} + \frac{\partial^2 \Phi}{\partial y^2} + \frac{\partial^2 \Phi}{\partial z^2}$$

is called Laplace's equation.]

5. Derive an expression for the divergence of a vector field. (A. N. U. 2002)
6. Explain the terms gradient and divergence. (A. N. U. 2003)
7. Explain the meaning of divergence of a vector field and give its physical significance.
8. Define Gradient, Divergence and Curl. Give examples. What is their physical significance ? (A. N. U. 2003)
9. Explain the divergence and curl of a vector field with two examples for each. (O. U. 2003)
10. Explain the terms divergence and curl of a vector field and give their physical significance. (A. U. 2007)
11. Prove that curl of gradient is zero. (A. N. U. 2003, 01)
12. What is line integral ? What is its physical significance ? (S. K. U. 1994)
13. What is the significance of a line integral ? (A. U. 2007)
14. State Stoke's theorem. (A. U. 2007; 2005)
15. State Stoke's, Gauss's and Green's theorems. (S. V. U. 2007; O. U. 2005; N. U. 2001)
16. Obtain Gauss's, divergence theorem from Green's theorem. (S. V. U. 2007)
17. Give physical interpretation of : (a) Gradient of a scalar field; (b) Divergence and curl of a vector field.
18. If $\vec{F} = 3x \hat{i} + 4y \hat{j} + 2z \hat{k}$, find $\text{div } \vec{F}$.
[Ans. 9]
19. Find the value of $\text{div } \vec{E}$, if $\vec{E} = -\text{grad } \phi$.
[Ans. $\text{div } \vec{E} = -\nabla^2 \phi$]
20. Divergence of vector is scalar and the gradient of a scalar is vector. Explain them with examples.
21. What is the meaning of $\vec{\nabla} \cdot \vec{A} = 0$, where $\vec{A}$ is a vector.

[Ans. The meaning of $\vec{\nabla} \cdot \vec{A} = 0$, is the flux lines of vector $\vec{A}$ are closed.]

22. Show that $\iiint_S \vec{r} \cdot d\vec{S} = 3V$.
23. What is a solenoidal vector ? Give example.
24. Prove that curl of a conservative field is zero.
25. Give an example of irrotational field.
26. Explain the meaning of (i) Solenoidal field and (ii) Lamellar field.
27. If $\text{curl } \vec{A} = 0$ and $\text{div } \vec{A} = 0$, state the nature of the vector field.
[Ans. For $\text{curl } \vec{A} = 0$, the field is irrotational and $\vec{A} = \text{grad } \phi$. For $\text{div } \vec{A} = 0$, the field is solenoidal. Hence, $\text{div } \vec{A} = \vec{\nabla} \cdot \vec{\nabla} \phi = \nabla^2 \phi = 0$, which is Laplace equation.]
28. Show that the vector $\vec{A} = \hat{i}yz + \hat{j}xz + \hat{k}xy$ is solenoidal non-rotational.

PROBLEMS

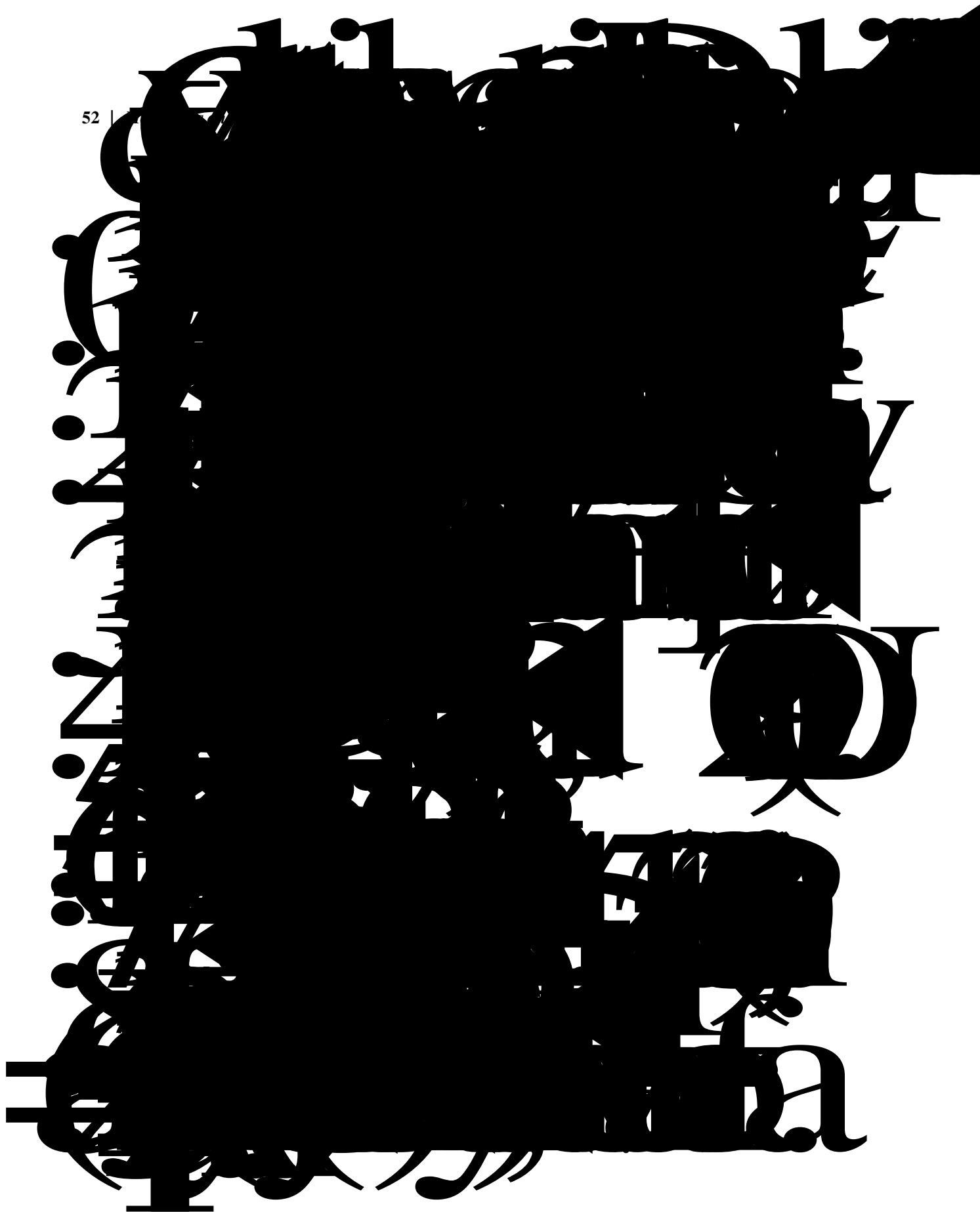
[Derivatives of Vector]

1. The position of a particle at any time t is represented as $\vec{r} = \cos t \hat{i} + \sin t \hat{j} + t \hat{k}$. Show that both speed and the magnitude of the acceleration are constant.
2. The position vector of a particle at any time t is given by $\vec{r} = a_1 \cos \omega t \hat{i} + a_1 \sin \omega t \hat{j} + a_2 t^2 \hat{k}$, where ω , a_1 and a_2 are constants. Show that although the speed of the particle increases with time but the magnitude of the acceleration is always constant.
3. The angular momentum of a particle of mass m is defined by $\vec{L} = \vec{r} \times m(d\vec{r}/dt)$. Show that

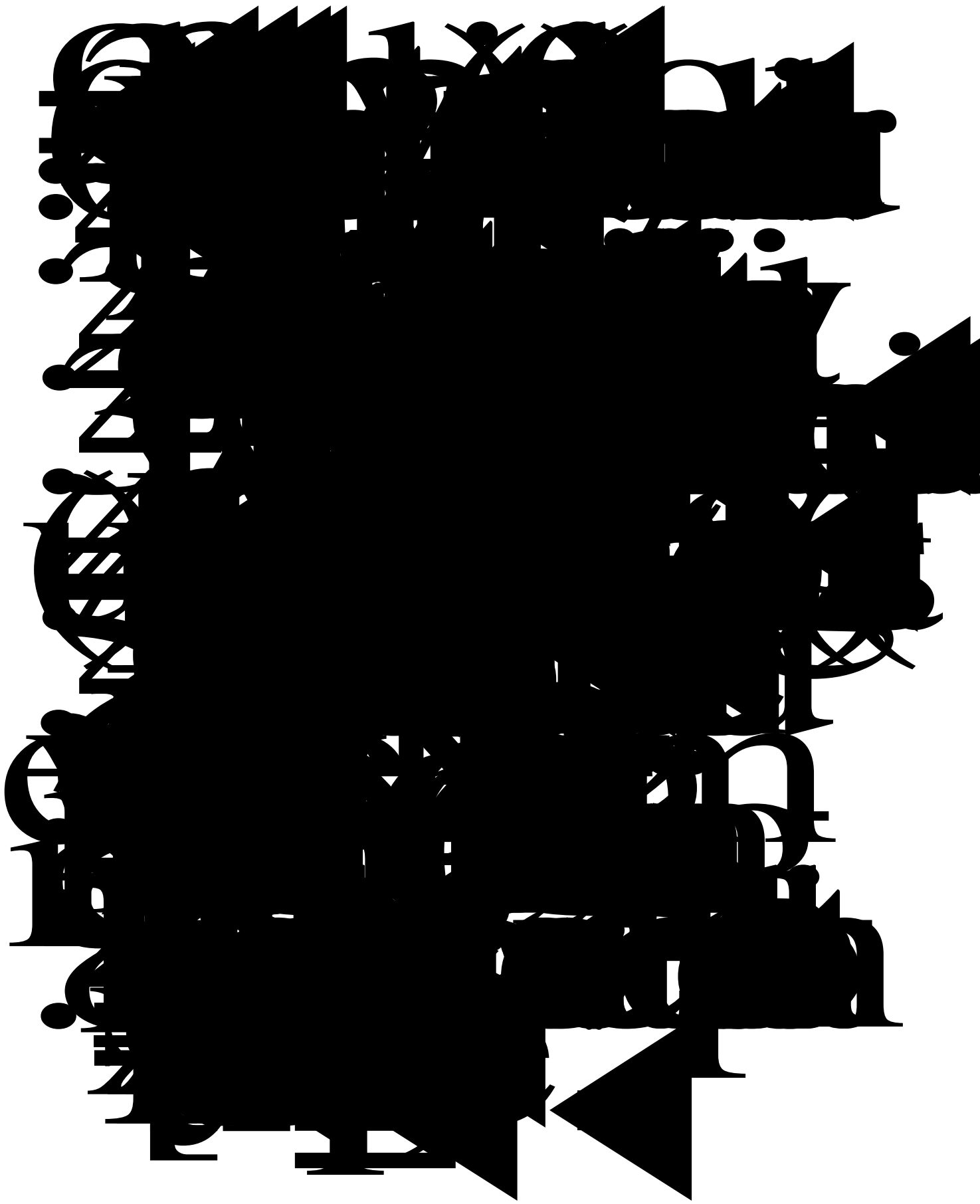
$$\frac{d\vec{L}}{dt} = \vec{r} \times m \frac{d^2 \vec{r}}{dt^2}.$$
4. If $\phi = 5x - 7x^2y + 8y^2 + 16yz - 5z$, find the partial derivatives $\partial\phi/\partial x$, $\partial\phi/\partial y$ and $\partial\phi/\partial z$ and the values of the derivatives at the point $(2, 4, -3)$.
[Ans. $5 - 14xy$, $-7x^2 + 16y + 16z$ and $16y - 5$; 107, -12, 59]
5. Given $\vec{A} = x^2 \hat{i} - y \hat{j} + xz \hat{k}$ and $\vec{B} = y \hat{i} + x \hat{j} - xyz \hat{k}$, find the value of $\frac{\partial^2 (\vec{A} \times \vec{B})}{\partial x \partial y}$ at the point $(1, -1, 2)$.
[Ans. $4 \hat{i} - 8 \hat{j}$]

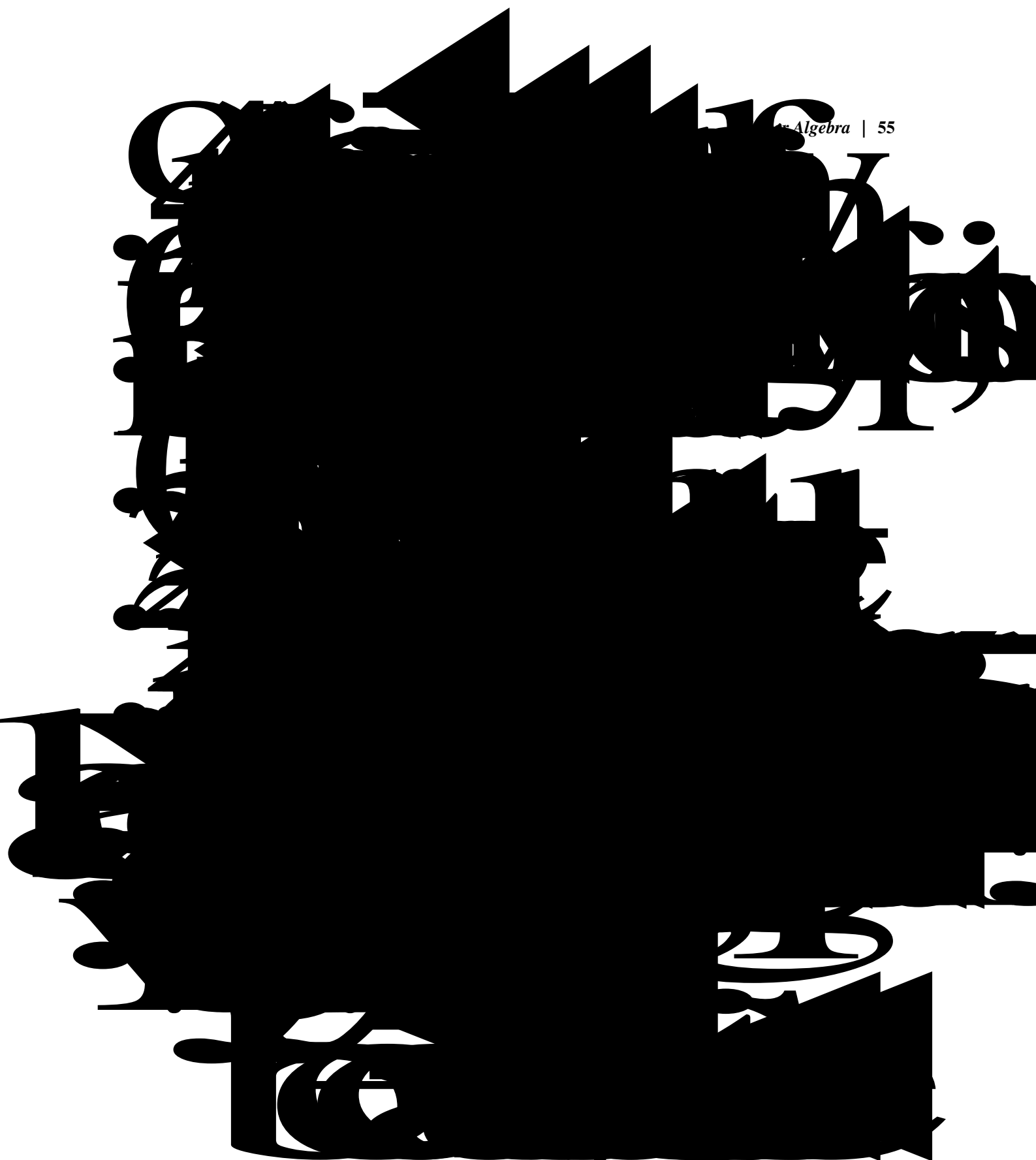
[Gradient of a Scalar Field]

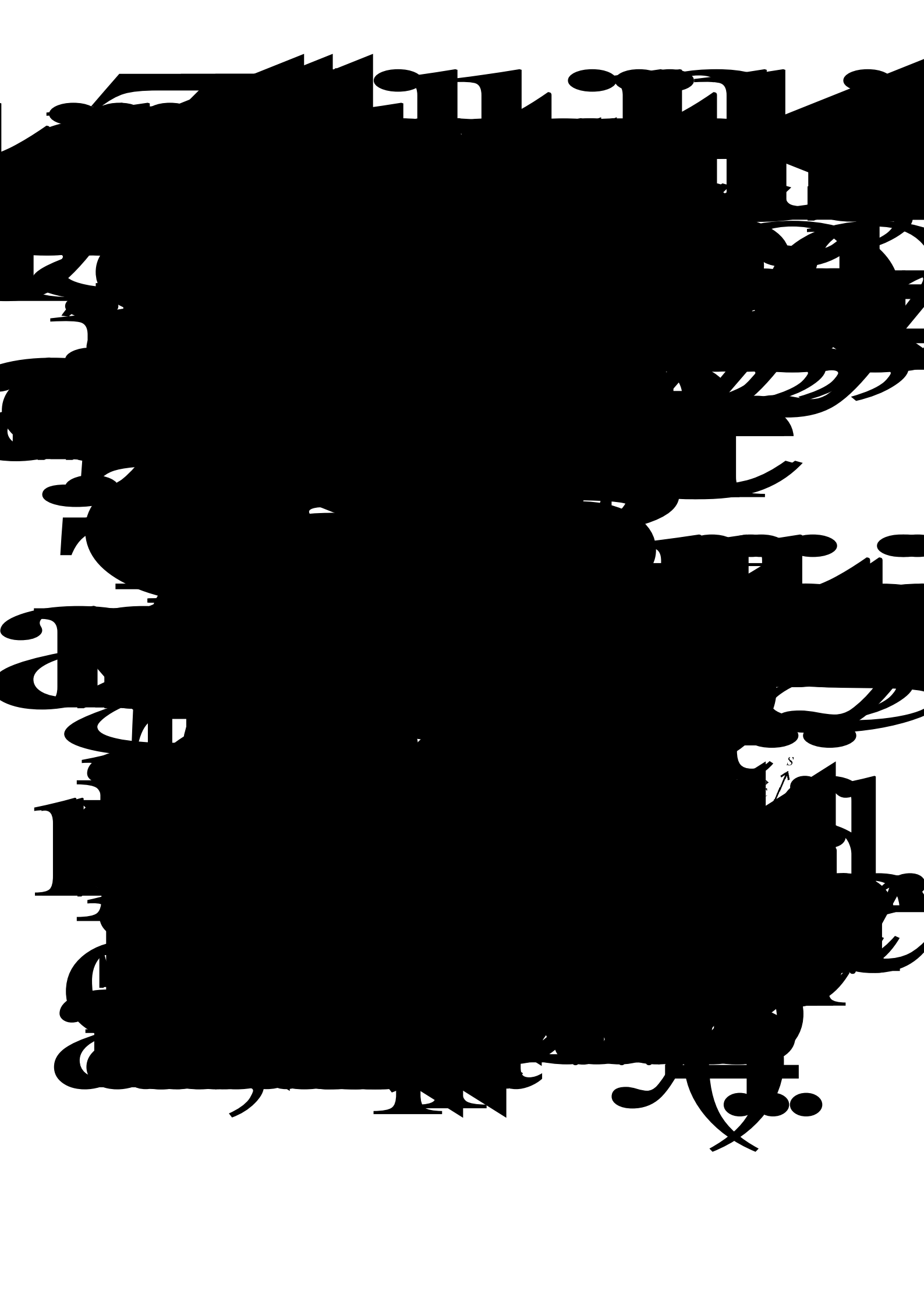
6. If the electrostatic potential at a point is $V = 2x^2 - 3y^2 - z^2$, find the electric field intensity at the point $(1, 1, 1)$.
[Ans. $-4 \hat{i} + 6 \hat{j} + 2 \hat{k}$]
7. A scalar field is given by $V = x^2 + y^2 + z^2$. Find the expression for potential gradient at the point $(1, 1, 1)$.
[Ans. $2x \hat{i} + 2y \hat{j} + 2z \hat{k}$]











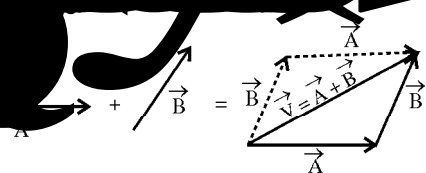


Fig. 2. Addition of two vectors.

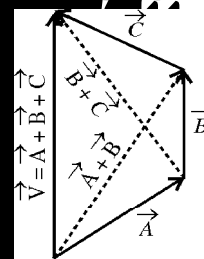


Fig. 3. Addition of several vectors.

components of the vector $\vec{A}$ and are represented by A_x and A_y respectively [fig. 5].
Now, in $\triangle OPQ$

$$\cos a = \frac{OP}{OR}$$

$$\text{and} \quad \sin a = \frac{OQ}{OR}$$

where a is the angle between the vector $\vec{A}$ and X- axis.

$$\begin{aligned} \therefore \quad OP &= OR \cos a \text{ and } OQ = OR \sin a \\ \text{or} \quad A_x &= A \cos a \text{ and } A_y = A \sin a = A \cos b \dots (1) \end{aligned}$$

where $b = \frac{\pi}{2} - a$ is the angle between $\vec{A}$ and Y-axis.

Eq. (1) represents the expressions of the two rectangular components of the vector $\vec{A}$.

If $\hat{i}$ and $\hat{j}$ are unit vectors along X- and Y- axes respectively, then vectorially the two rectangular components are written as

$$\vec{A} = A_x \hat{i} + A_y \hat{j} \dots (2)$$

Now, from the law of vector addition,

$$\vec{A} = \vec{A_x} + \vec{A_y} \text{ or } \vec{A} = \vec{A_x} + \vec{A_y} \dots (3)$$

Hence the sum of the two vector components $\vec{A_x}$ and $\vec{A_y}$ give the vector $\vec{A}$.

$$\text{Thus, } \vec{A} = \vec{A_x} + \vec{A_y} \text{ or } \vec{A} = \vec{A_x} + \vec{A_y} \dots (4)$$

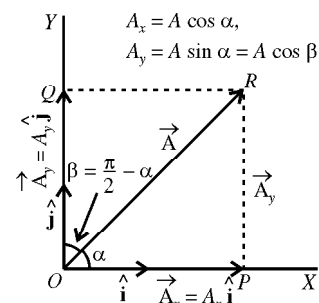


Fig. 5. Resolution of a vector in a plane.

Obviously, $\vec{OR} = \vec{OP} + \vec{PR}$ or $\vec{A} = \vec{A}_x + \vec{A}_y + \vec{A}_z$... (5)

Magnitude of vector, $A = \sqrt{A_x^2 + A_y^2 + A_z^2}$... (6)

Also, $\cos \alpha = \frac{A_x}{A}$, $\cos \beta = \frac{A_y}{A}$, $\cos \gamma = \frac{A_z}{A}$... (7)

These $\cos \alpha$ and $\cos \beta$ are called direction cosines of the vector $\vec{A}$. The vector $\vec{A}$ in eqn. (4) can also be written as

$$\vec{A} = A(\cos \alpha \hat{i} + \cos \beta \hat{j} + \cos \gamma \hat{k})$$

Hence, $\vec{A} = A \hat{r}$... (8)

where $\hat{r} = \cos \alpha \hat{i} + \cos \beta \hat{j} + \cos \gamma \hat{k}$... (9)

Obviously,

A4. Right Handed and Left Handed Co-ordinate Systems

The co-ordinate systems which are generally used are rectangular or **orthogonal**. There may be of two types : (i) right handed [fig. 6(a)], (ii) left handed [fig. 6(b)]. Evidently both co-ordinate systems are the mirror image of each other.

In case of the right handed coordinate system, the thumb, the fore-finger and the middle finger of the right hand give the relative direction of X, Y and Z axes respectively. In such a case, a rotation from X to Y axis will appear clockwise if seen from the origin towards the positive direction of Z-axis. Fundamentally, any one of the two co-ordinate systems may be employed but the right handed system is commonly used.

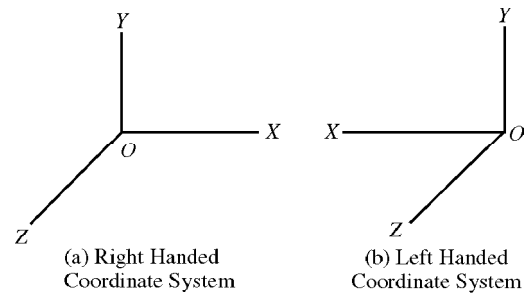


Fig. 6.

A5. Components of a Vector

The components of a vector $\vec{A}$ are any number of vectors whose sum is $\vec{A}$. The components, most frequently used, are those **parallel to X, Y and Z axes** of the right handed cartesian co-ordinate system and they are called **rectangular components** of the vector.

Let the origin O be one extremity of a vector $\vec{A}$. Now draw a rectangular parallelepiped with its three edges lying along the three axes such that $\vec{A}$ is the diagonal through O. If $\vec{A}_x$, $\vec{A}_y$ and $\vec{A}_z$ are the vector intercepts along X, Y and Z axes respectively [fig. 7], then

$$\vec{A} = \vec{A}_x + \vec{A}_y + \vec{A}_z$$

Let $\hat{i}$, $\hat{j}$, $\hat{k}$ represent vectors of unit magnitude along the X, Y and Z axes respectively; these unit vectors along the axes will be frequently used in the later work. Now, if A_x , A_y and A_z are the magnitudes of the vectors $\vec{A}_x$, $\vec{A}_y$ and $\vec{A}_z$ respectively, the

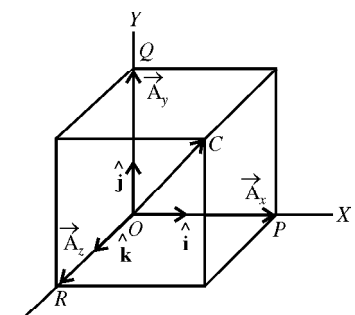


Fig. 7. Components of a vector.

$$\vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$$

Therefore, equation (1) can be written as

$$\mathbf{r} = A_x \hat{\mathbf{i}} + A_y \hat{\mathbf{j}} + A_z \hat{\mathbf{k}} \quad \dots(2)$$

By geometry, the magnitude of the vector $\mathbf{r}$ is

$$OC^2 = OP^2 + OQ^2 + OR^2$$

$$\text{or } A^2 = A_x^2 + A_y^2 + A_z^2 \text{ or } A = |\mathbf{r}| \quad \dots(3)$$

The magnitude of the component vectors are given by

$$A_x = A \cos(\angle X), \quad A_y = A \cos(\angle Y), \quad A_z = A \cos(\angle Z). \quad \dots(4)$$

$$\text{Thus, } \cos(\angle X) = A_x/A, \cos(\angle Y) = A_y/A \text{ and } \cos(\angle Z) = A_z/A \quad \dots(5)$$

With the help of the cosines given in (5) the direction of the vector $\mathbf{r}$ can be found. Therefore, these are called direction cosines.

If we put the values of A_x, A_y, A_z from eq. (4) in eq. (3), we have

$$\cos^2(\angle X) + \cos^2(\angle Y) + \cos^2(\angle Z) = 1 \quad \dots(6)$$

Thus, the sum of the squares of three direction cosines is equal to unity.

The unit vector $\hat{\mathbf{r}}$ along $\mathbf{r}$ can be written as

$$\hat{\mathbf{r}} = \frac{\mathbf{r}}{r} = \frac{A_x \hat{\mathbf{i}} + A_y \hat{\mathbf{j}} + A_z \hat{\mathbf{k}}}{A} \quad [\text{from eq. (2)}]$$

where $A_x/A, A_y/A$ and A_z/A are the direction cosines of $\mathbf{r}$

$$\hat{\mathbf{r}} = \cos(\angle X) \hat{\mathbf{i}} + \cos(\angle Y) \hat{\mathbf{j}} + \cos(\angle Z) \hat{\mathbf{k}} \quad \dots(7)$$

The addition of the two vectors can be written in terms of components as

$$\mathbf{r} = \mathbf{r}_1 + \mathbf{r}_2 = (A_x \hat{\mathbf{i}} + A_y \hat{\mathbf{j}} + A_z \hat{\mathbf{k}}) + (B_x \hat{\mathbf{i}} + B_y \hat{\mathbf{j}} + B_z \hat{\mathbf{k}})$$

$$\text{or } \mathbf{r} = (A_x + B_x) \hat{\mathbf{i}} + (A_y + B_y) \hat{\mathbf{j}} + (A_z + B_z) \hat{\mathbf{k}} \quad \dots(8)$$

$$\text{Its magnitude } = r = |\mathbf{r}| \quad \dots(9)$$

Its direction cosines are $(A_x + B_x)/r, (A_y + B_y)/r$, and $(A_z + B_z)/r$.

A6. Position vector

If the co-ordinate of a point P be (x, y, z) relative to the origin of Cartesian co-ordinate system, fig. 8(a)], then its distance from the origin can be expressed in the vector form as

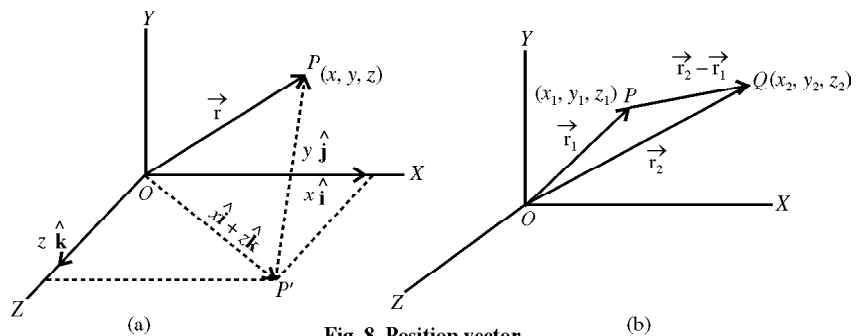


Fig. 8. Position vector.

$$\mathbf{r} = x \hat{\mathbf{i}} + y \hat{\mathbf{j}} + z \hat{\mathbf{k}} \quad \dots(1)$$

We call it ($\vec{r}$) the position vector of the point P [fig. 8(a)]. The direction cosines of $\mathbf{r}$ are obviously $(x/r), (y/r)$ and (z/r) .

If there are two points P and Q, having co-ordinates (x_1, y_1, z_1) and (x_2, y_2, z_2) , then from the origin O the position vectors $\mathbf{r}_1 = \vec{OP}$ and $\mathbf{r}_2 = \vec{OQ}$ will be represented as

$$\vec{r} = x_1\vec{i} + y_1\vec{j} + z_1\vec{k} \text{ and } \vec{s} = x_2\vec{i} + y_2\vec{j} + z_2\vec{k}$$

$$\vec{r} - \vec{s} = (x_2 - x_1)\vec{i} + (y_2 - y_1)\vec{j} + (z_2 - z_1)\vec{k} \quad \dots(2)$$

The relative position vector $\vec{PQ} = \vec{r} - \vec{s}$ is shown in fig. 8(b).

A7. Product of Two Vectors

In vector algebra, the product of two vectors is defined in two different ways :

(1) **Scalar product** : If the multiplication of two vectors gives a scalar, the product is called scalar product.

(2) **Vector product** : If the two vectors are multiplied such that the result of multiplication is a vector, the product is called vector product.

Both types of product are very much useful in Physics. We will discuss in the following each one of them.

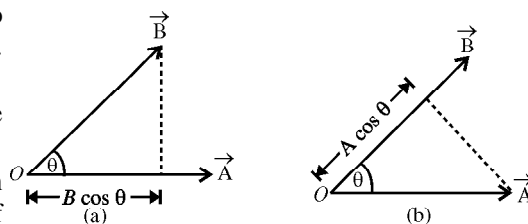


Fig. 9. Scalar product.

(1) Scalar or Dot Product of Two Vector

The scalar product of two vectors $\vec{A}$ and $\vec{B}$ is defined as a scalar quantity $AB \cos q$, where A and B are the magnitudes of the vectors and q is the angle between their directions.

We write the scalar product of two vectors as $\vec{A} \cdot \vec{B}$ which is read $\vec{A}$ dot $\vec{B}$ and therefore it is also called **dot product**. Thus

$$\vec{A} \cdot \vec{B} = AB \cos q \quad \dots(1)$$

The **scalar product $AB \cos q$ of the vectors $\vec{A}$ and $\vec{B}$** may be considered as either the multiplication of the magnitude of $\vec{A}$ and the projection of $\vec{B}$ along $\vec{A}$ [$AB \cos q = A (B \cos q)$], [fig. 9(a)] or the multiplication of the magnitude of $\vec{B}$ and the projection of $\vec{A}$ along $\vec{B}$ [$AB \cos q = B (A \cos q)$], [fig. 9(b)].

It is clear from the definition of the scalar product that the **commutative law** of multiplication holds, i.e.,

$$\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A} = AB \cos q \quad \dots(2)$$

If the vectors $\vec{A}$ and $\vec{B}$ are parallel (pointing in the same direction) to each other, then ($q = 0$)

$$\vec{A} \cdot \vec{B} = AB \cos q = AB$$

and if the vectors are similar, then $\vec{A} \cdot \vec{A} = A^2$... (3)

If this property is applied to the unit vectors $\vec{i}, \vec{j}, \vec{k}$, which are mutually perpendicular to each other, then

$$\vec{i} \cdot \vec{i} = \vec{j} \cdot \vec{j} = \vec{k} \cdot \vec{k} = 1 \quad \dots(4)$$

In case, if the two vectors are perpendicular to each other, i.e.,

$$\cos q = \cos 90^\circ = 0 \text{ and then } \vec{A} \cdot \vec{B} = 0 \quad \dots(5)$$

This is the condition for two orthogonal vectors. If this property is applied to unit vectors $\vec{i}, \vec{j}, \vec{k}$, then

$$\vec{i} \cdot \vec{j} = \vec{j} \cdot \vec{i} = \vec{i} \cdot \vec{k} = \vec{k} \cdot \vec{i} = 0 \quad \dots(6)$$

As a consequence of these results we have

$$\vec{A} \cdot \vec{B} = (A_x\hat{i} + A_y\hat{j} + A_z\hat{k}) \cdot (B_x\hat{i} + B_y\hat{j} + B_z\hat{k}) \text{ or } \vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z \quad \dots(7)$$

We can also write

$$|\vec{A}|^2 = A_x^2 + A_y^2 + A_z^2 = A^2. \quad \dots(8)$$

Examples of Scalar Product are : work and power. If a constant force $\vec{F}$ acts on a body and produces a displacement $\vec{s}$ [fig. 10], then the work W done by the force on the body is defined as

$$W = (\text{component of force along the displacement}) (\text{distance})$$

$$= (F \cos \theta) s = \vec{F} \cdot \vec{s} \quad \dots(9)$$

The rate of doing work is called power. Thus,

$$\text{Power, } P = \frac{dW}{dt} = \frac{d(\vec{F} \cdot \vec{s})}{dt} = \vec{F} \cdot \frac{d\vec{s}}{dt} = \vec{F} \cdot \vec{v} \quad \dots(10)$$

where $\vec{v}$ is the velocity of the body.

(2) Vector or Cross Product

The vector (or cross) product of two vectors $\vec{A}$ and $\vec{B}$ inclined at an angle θ to each other is defined as a vector whose magnitude is $AB \sin \theta$ and direction is that of a unit vector $\hat{n}$ perpendicular to the plane containing $\vec{A}$ and $\vec{B}$ [fig. 11].

Symbolically, the vector product of $\vec{A}$ and $\vec{B}$ is denoted by $\vec{A} \times \vec{B}$ and is read as $\vec{A}$ cross $\vec{B}$. Thus, we have

$$|\vec{A} \times \vec{B}| = AB \sin \theta \quad \dots(11)$$

The direction of the product $\vec{A} \times \vec{B}$ i.e., that of the unit vector $\hat{n}$ is determined by the right hand screw rule. According to this rule, if a right-hand-screw* is turned from the direction of $\vec{A}$ towards that of $\vec{B}$ through the smaller angle, the direction of its advance gives the direction of $\vec{A} \times \vec{B}$ (or that of $\hat{n}$). [fig. 12].

From the definition of vector product it is evident that a change in the **order of the factors in a cross product reverses the sign of the product**, because the sense of rotation of the screw is reversed, i.e.,

$$\vec{A} \times \vec{B} = AB \sin \theta \hat{n} = - \vec{B} \times \vec{A} \quad \dots(12)$$

Hence, the vector product of two vectors is not commutative and the order of the terms should be strictly maintained. But the distributive law still holds, i.e.,

$$\vec{A} \times (\vec{B} + \vec{C}) = \vec{A} \times \vec{B} + \vec{A} \times \vec{C} \quad \dots(13)$$

If two vectors $\vec{A}$ and $\vec{B}$ are parallel or collinear, i.e.,

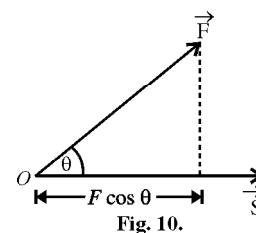


Fig. 10.

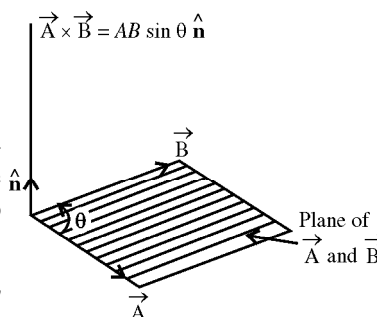


Fig. 11. Vector product.

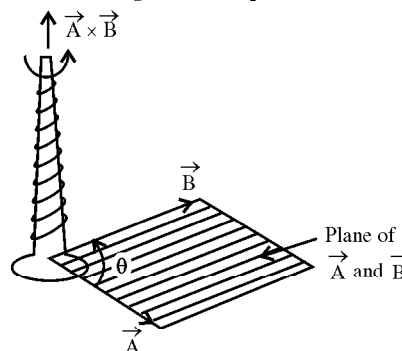


Fig. 12. Motion of right hand screw.

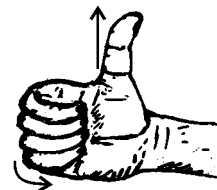


Fig. 13. Right hand screw rule.

* A right hand screw is one that if one's right hand is placed [fig. 13] with the fingers pointing in the direction of rotation, the screw advances in the direction of the thumb.

$\sin q = \sin 0 = 0$ and then from eq. (11), we have

$$\hat{a} \times \hat{a} = 0$$

...(14)

If we apply this property to unit vectors $\hat{i}, \hat{j}, \hat{k}$ [fig. 14], then

$$\hat{i} \times \hat{i} = \hat{j} \times \hat{j} = \hat{k} \times \hat{k} = 0$$

...(15)

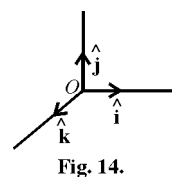


Fig. 14.

In case, if $\vec{a}$ and $\vec{b}$ are perpendicular to each other, then

$$\vec{a} \times \vec{b} = AB \hat{n} \text{ or } \vec{a} \times \vec{b} = AB \sin q \hat{n}$$

Applying this property to unit vectors $\hat{i}, \hat{j}, \hat{k}$, which are mutually orthogonal, we have

$$\hat{i} \times \hat{j} = \hat{k}, \hat{j} \times \hat{k} = \hat{i}, \hat{k} \times \hat{i} = \hat{j}; \hat{j} \times \hat{i} = -\hat{k}, \hat{k} \times \hat{j} = -\hat{i}, \hat{i} \times \hat{k} = -\hat{j} \quad \dots(16)$$

If we write the vectors in terms of their rectangular components, then

$$\vec{a} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k} \text{ and } \vec{b} = B_x \hat{i} + B_y \hat{j} + B_z \hat{k}$$

$$\text{Now, } \vec{a} \times \vec{b} = (A_x \hat{i} + A_y \hat{j} + A_z \hat{k}) \times (B_x \hat{i} + B_y \hat{j} + B_z \hat{k})$$

$$= (A_y B_z - A_z B_y) \hat{i} + (A_z B_x - B_z A_x) \hat{j} + (A_x B_y - A_y B_x) \hat{k}$$

...(17)

This result can be remembered easily, when written in the determinant form, i.e.,

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

...(18)

Examples of Cross Product

(1) **Area** of a parallelogram : If two vectors $\vec{a}$ and $\vec{b}$ are respectively the sides of a parallelogram OPRQ, then from fig 15 area of the parallelogram

$$= OP (QN) = A (B \sin q) = AB \sin q = |\vec{a} \times \vec{b}| \quad \dots(19)$$

Thus, the area of the parallelogram with sides $\vec{a}$ and $\vec{b}$ is $|\vec{a} \times \vec{b}|$. This is twice the area of the triangle OPR with sides $\vec{a}$ and $\vec{b}$ i.e.,

$$\text{Area of triangle (OPR)} = \frac{1}{2} |\vec{a} \times \vec{b}|$$

$$\dots(20)$$

The direction of $\vec{a} \times \vec{b}$ is perpendicular to the plane of the parallelogram [fig. 15]; thus we may think of $\vec{a} \times \vec{b}$ as the vector area of the parallelogram. However, the direction of the vector area depends how we sweep the area i.e., $\vec{a} \times \vec{b}$ or $\vec{b} \times \vec{a}$.

Not only the area of a parallelogram, but any plane area may be considered as a vector whose magnitude is equal to the area and direction normal to it.

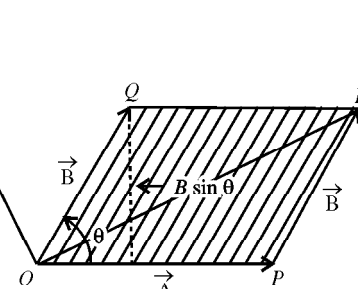


Fig. 15. Area of parallelogram.

(2) **Torque and angular momentum** : Consider a force $\vec{F}$, acting on a point P of a body which can rotate it about a point O. The torque or moment of the force (in magnitude) is equal to the product of the force F and its perpendicular distance $r \sin q$ i.e., $T = F r \sin q$. In vector form, the torque is represented as

$$\vec{\tau} = \vec{r} \times \vec{F} \quad \dots(21)$$

where $\vec{r}$ is the vector distance of the point P relative to O [fig. 16]. The magnitude of the torque is $Fr \sin \theta$ and its direction is perpendicular to the plane containing $\vec{r}$ and $\vec{F}$.

If $\vec{p}$ is the linear momentum of a particle, then its moment of momentum i.e., angular momentum $\vec{J}$ about a point O is defined as

$$\vec{J} = \vec{r} \times \vec{p} \quad \dots(22)$$

where $\vec{r}$ is the distance of the particle from O. The direction of angular momentum is perpendicular to the plane of $\vec{r}$ and $\vec{p}$.

A8. Triple Products

If $\vec{a}$, $\vec{b}$ and $\vec{c}$ are three vectors then the vector product of $\vec{a}$ and $\vec{b}$ is a vector. This product of $\vec{a}$ and $\vec{b}$ can have a scalar as well as a vector product with the vector $\vec{c}$. Thus, there are two types of triple products, namely $(\vec{a} \times \vec{b}) \cdot \vec{c}$ and $(\vec{a} \times \vec{b}) \times \vec{c}$ which have their frequent use in physical processes.

(1) Scalar Triple Product

The product $(\vec{a} \times \vec{b}) \cdot \vec{c}$ is a scalar, say V . Finally, the dot product $(\vec{a} \times \vec{b}) \cdot \vec{c}$ is a scalar, and hence such a triple product is called scalar one. The vector $\vec{a} \times \vec{b}$ is a vector normal to the plane of $\vec{a}$ and $\vec{b}$ with a magnitude $V \cos \theta$ equal to the area of the shaded parallelogram [fig. 17]. Now,

$$(\vec{a} \times \vec{b}) \cdot \vec{c} = V C \cos \theta$$

$$\theta = (\text{angle between } \vec{a} \times \vec{b} \text{ and } \vec{c})$$

$$= (\text{component of } \vec{c} \text{ along } \vec{a} \times \vec{b})$$

$$= (\text{area of the base of the parallelepiped formed by } \vec{a} \text{ and } \vec{b})$$

$$(\text{its perpendicular height})$$

$$= \text{volume of the parallelepiped of edges } \vec{a}, \vec{b}, \vec{c}. \quad \dots(23)$$

Thus, the scalar triple product $(\vec{a} \times \vec{b}) \cdot \vec{c}$ represents the volume of the parallelepiped whose sides are vectors $\vec{a}$ and $\vec{b}$.

Any face of this parallelepiped can be taken as the base and hence its volume can be represented by the following three expressions :

$$(\vec{a} \times \vec{b}) \cdot \vec{c} = (\vec{b} \times \vec{c}) \cdot \vec{a} = (\vec{c} \times \vec{a}) \cdot \vec{b} \quad \dots(24)$$

To obtain the volume with positive sign the cyclic order of $\vec{a}$ and $\vec{b}$ should be maintained. As the order of terms in scalar product is meaningless,

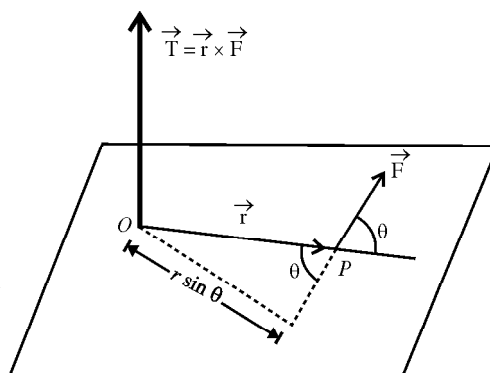


Fig. 16. Torque.

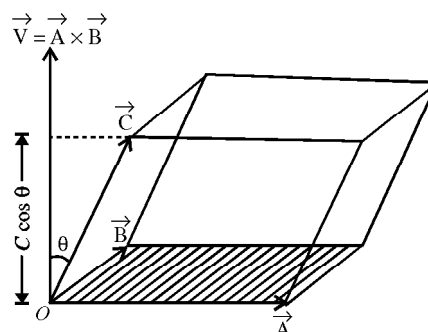


Fig. 17. Scalar triple product.

$$\vec{a} \cdot (\vec{b} \times \vec{c}) = \vec{b} \cdot (\vec{c} \times \vec{a}) = \vec{c} \cdot (\vec{a} \times \vec{b}) \quad \dots(3)$$

Thus the dot and the cross in the scalar triple product may be interchanged without altering the value of the product. So, a scalar triple product is often written as $[\vec{a} \vec{b} \vec{c}]$ without showing the dot or cross.

From relations (2) and (3) we see that three vectors have six identical scalar triple products, which may be written concisely as $[\vec{a} \vec{b} \vec{c}]$ i.e.,

$$[\vec{a} \vec{b} \vec{c}] = \vec{a} \cdot (\vec{b} \times \vec{c}) = (A_x\vec{a} + A_y\vec{a} + A_z\vec{a}) \cdot \vec{a}$$

$$\text{where } \vec{a} = A_x\vec{i} + A_y\vec{j} + A_z\vec{k}, \vec{b} = B_x\vec{i} + B_y\vec{j} + B_z\vec{k} \text{ and } \vec{c} = C_x\vec{i} + C_y\vec{j} + C_z\vec{k}.$$

$$\text{or } [\vec{a} \vec{b} \vec{c}] = \vec{a} \cdot (\vec{b} \times \vec{c}) \quad \dots(4)$$

Coplanar vectors : If three vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ are in the same plane, then the volume of the parallelepiped, having sides $\vec{a}$, $\vec{b}$ and $\vec{c}$, must be zero i.e.,

$$\text{Volume } [\vec{a} \vec{b} \vec{c}] = 0 \text{ or } \vec{a} \cdot (\vec{b} \times \vec{c}) = 0 \quad \dots(5)$$

This is the condition of coplanarity of three vectors.

(2) Vector Triple Product $\vec{a} \times (\vec{b} \times \vec{c})$

The product $\vec{a} \times (\vec{b} \times \vec{c})$ is a vector normal to the plane containing $\vec{b}$ and $\vec{c}$. Therefore, $\vec{a} \times (\vec{b} \times \vec{c})$ is a vector, say $\vec{d}$ normal to the plane containing $\vec{b}$ and $\vec{c}$, i.e., vector $\vec{d}$ lies in the plane of $\vec{a}$ and $\vec{a} \times (\vec{b} \times \vec{c})$ [fig. 18].

Conveniently, we have taken, the vector $\vec{a}$ along X-axis, $\vec{b}$ in the XY-plane and $\vec{c}$ in any direction. Hence $\vec{b} \times \vec{c}$ will be a vector along Z-axis and the triple product $\vec{a} \times (\vec{b} \times \vec{c})$ in the XY-plane or in the plane of $\vec{a}$.

The vector triple product may be expressed as the sum of two terms :

$$\vec{a} \times (\vec{b} \times \vec{c}) = \vec{b}(\vec{a} \cdot \vec{c}) - \vec{c}(\vec{a} \cdot \vec{b}) \quad \dots(6)$$

A9. Finite and Infinitesimal Rotations : Angular Velocity

A finite rotation or angular displacement cannot be represented by a vector because the sum of two such rotations one after the other, is not given by the parallelogram law of addition.

For example, let the point P of the sphere in fig. 19. move to position Q by a 90° rotation about OR axis. Let this rotation be represented by $\vec{\omega}_1$. Further motion from Q to R by 90° may be represented by $\vec{\omega}_2$.

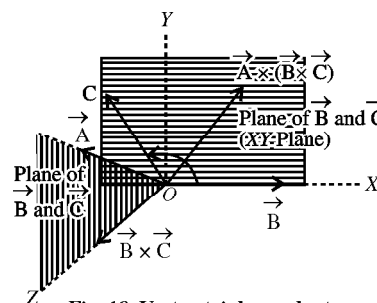


Fig. 18. Vector triple product.

but we observe that to bring the point P to the point R, we can perform a single rotation of 90° about QO axis and hence the total rotation (P to Q and Q to R) is 90° be represented by the vector $\vec{OQ}$. As $\vec{OQ}$ is not the diagonal of the parallelogram formed by $\vec{OP}$ and $\vec{OR}$, we conclude that finite rotations or angular displacements cannot be represented as vector quantities.

We can show, however, that the infinitesimal rotations can be represented by vectors. In fig. 20, we consider very small rotations about axes OR and OP. For this purpose, we move any point A on the sphere first to B about OR axis and then to C about P axis. The total movement AC could be done by a single rotation about an axis OQ, which is in the direction of the resultant of the original vectors along OR and OP. This addition is also commutative. Now, if $\vec{\omega}$ is the infinitesimal rotation occurring in dt time, then the instantaneous angular velocity $\vec{\omega}$ of the rigid body is defined by the relation [fig. 21],

$$\vec{\omega} = \frac{d\theta}{dt} \vec{n} \quad \text{or} \quad \vec{\omega} = \frac{d\vec{\theta}}{dt}$$

where the vector $\vec{\omega}$ is along the axis of infinitesimal rotation.

The angular velocity $\vec{\omega}$ is a vector, having magnitude ω and direction, in which the angular velocity vector $\vec{\omega}$ points, is **given by the right hand screw rule**, i. e., if the fingers of right hand are rotated in the sense of rotation (keeping the thumb perpendicular to fingers), the thumb gives the correct direction of $\vec{\omega}$.

In case, if rigid body rotates with angular velocity $\vec{\omega}$ about an axis AB, then any particle P of the body, having position vector $\vec{r}$ relative to a point O on the axis of rotation, describes a circular path of radius $r \sin \theta$ [fig. 21]. Therefore the magnitude of the linear velocity of the particle is $\omega r \sin \theta$, which can be expressed in vector form as

$$\vec{v} = \vec{\omega} \times \vec{r} \quad \dots(1)$$

Thus, in general eq (2) tells us the linear velocity of any particle of a rigid body

In order to deal with a particle P in circular motion, let the origin O be shifted at the centre of the circle [fig. 22].

Even now, the linear velocity of the particle P is represented by the relation

$$\vec{v} = \omega \times \vec{r} \quad \dots(3)$$

which has the magnitude $v = \omega r$ because $\sin \theta = 1$.

Taking cross product with $\vec{\omega}$ on the both sides of eq. (2), we have

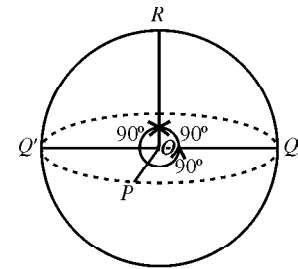


Fig. 19. Finite rotations (Angular displacements)

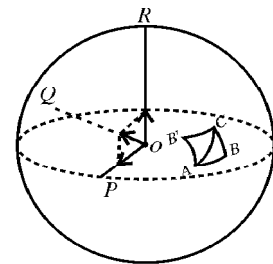


Fig. 20. Infinitesimal rotation (angular displacements)

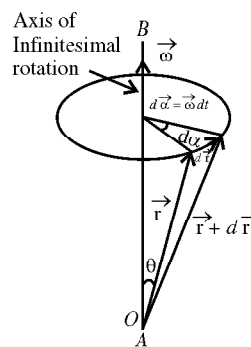


Fig 21(a) Instantaneous angular velocity $\vec{\omega} = \frac{d\theta}{dt}$ for a rotating rigid body.

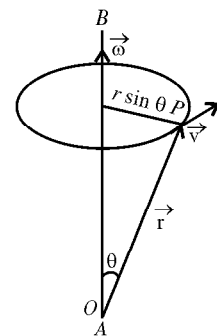


Fig 21(b) Representation of the velocity of a particle of a rotating rigid body.

$$\begin{aligned} \vec{\omega} \times \vec{r} &= \vec{\omega} \times \vec{r} \\ &= \vec{\omega} \times \vec{r} - \vec{r} \times \vec{\omega} = \vec{\omega} \times \vec{r} \quad \dots(4) \\ [\vec{\omega} \times \vec{r} \text{ and } \vec{r} \text{ are perpendicular to each other}] \end{aligned}$$

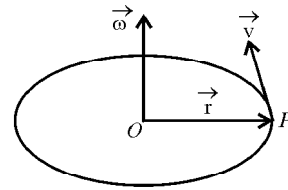


Fig. 22. Circular motion of a particle.

$$|\vec{v}| = r\omega \quad \dots(5)$$

whence angular velocity in circular motion can be calculated.

10. Circular Motion

Let us consider a particle P, moving on the circumference of a circle of constant radius r [fig. 23]. The position vector $\vec{r}$, which is the function of the scalar variable t , is rotating with a constant angular velocity ω . At any time t , the position vector $\vec{r}$ can be expressed in the components form as

$$\begin{aligned} \vec{r} &= r \cos \omega t \hat{i} + r \sin \omega t \hat{j} \\ \text{or} \quad \vec{r} &= r \hat{r} \quad \dots(1) \end{aligned}$$

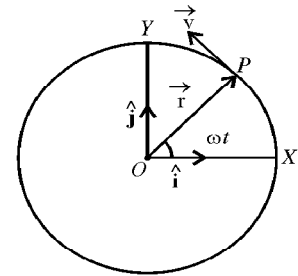


Fig. 23. Discussion of circular motion by vector method.

To find the velocity of the particle P in circular motion, differentiate eq. (1) with respect to time, i.e.,

$$\frac{d\vec{r}}{dt} = \vec{v}$$

$$[\frac{dr}{dt} = 0, \text{ because radius } r \text{ is constant for a circle}]$$

$$\text{or} \quad \vec{v} = \frac{d\vec{r}}{dt} = -r\omega \sin \omega t \hat{i} + r\omega \cos \omega t \hat{j} \quad \dots(2)$$

Obviously, $\hat{r}$ is a unit vector perpendicular to $\vec{v}$ (because $\vec{r} \cdot \vec{v} = 0$) along $\vec{v}$. Hence

$$\vec{v} = -r\omega \hat{\theta} \quad \dots(3)$$

This establishes the important result that the speed of a particle in uniform circular motion is

$$v = r\omega \quad \dots(4)$$

Differentiating eq. (2), we have the acceleration of the particle in circular motion, i.e.,

$$\vec{a} = \frac{d\vec{v}}{dt} = -r\omega^2 \cos \omega t \hat{i} - r\omega^2 \sin \omega t \hat{j}$$

$$\text{But } \vec{r} = r \hat{r}$$

$$\vec{a} = -\omega^2 \vec{r} \quad \dots(5)$$

Hence the magnitude of the acceleration in circular motion is

$$a = r\omega^2 \quad \dots(6)$$

The direction of the acceleration is along $-\hat{r}$, which is directed towards the centre of the circle. Therefore, this acceleration is called centripetal (centre seeking) acceleration. If m is the mass of the particle, then the value of the centripetal force is $m\omega^2 r$. This is also the value of centrifugal force in the frame of the rotating particle.

$$\text{But } v = \omega r, \therefore a = \omega^2 r \quad \dots (7)$$

The radius vector sweeps out 2π angle in one rotation, hence the angle ω will be swept out in 2π rotation, i.e. frequency of revolutions is

$$n = \frac{\omega}{2\pi} \text{ or } \omega = 2\pi n \quad \dots (8)$$

Thus the angular speed ω is related to the frequency n and so it is called angular frequency. If one cycle is completed in time T , then period

$$T = \frac{1}{n} \quad \dots (9)$$

Ex. 1. If $\vec{A} = 3\hat{i} + 2\hat{j} + \hat{k}$ and $\vec{B} = 2\hat{i} - \hat{j} + \hat{k}$, calculate the magnitudes and direction cosines of $\vec{A}$ and $\vec{B}$.

$$\text{Sol. Here, } |\vec{A}| = \sqrt{3^2 + 2^2 + 1^2} = \sqrt{14} \text{ and } |\vec{B}| = \sqrt{2^2 + (-1)^2 + 1^2} = \sqrt{6}$$

$$\therefore \text{Direction cosines of } \vec{A} \text{ are } \frac{3}{\sqrt{14}}, \frac{2}{\sqrt{14}}, \frac{1}{\sqrt{14}}$$

$$\text{and } \text{Direction cosines of } \vec{B} \text{ are } \frac{2}{\sqrt{6}}, \frac{-1}{\sqrt{6}}, \frac{1}{\sqrt{6}}$$

\ Direction cosines of $\vec{A}$ are $\frac{3}{\sqrt{14}}, \frac{2}{\sqrt{14}}, \frac{1}{\sqrt{14}}$ and that of $\vec{B}$ are $\frac{2}{\sqrt{6}}, \frac{-1}{\sqrt{6}}, \frac{1}{\sqrt{6}}$.

x. 2. If $\vec{A} = 4\hat{i} + 6\hat{j} - 3\hat{k}$ and $\vec{B} = 2\hat{i} + 5\hat{j} + 7\hat{k}$, find out the angle between $\vec{A}$ and $\vec{B}$.

$$\text{Sol. Here } |\vec{A}| = \sqrt{4^2 + 6^2 + (-3)^2} = \sqrt{61} \text{ and } |\vec{B}| = \sqrt{2^2 + 5^2 + 7^2} = \sqrt{74}$$

$$\therefore \vec{A} \cdot \vec{B} = 4 \times 2 + 6 \times 5 + (-3) \times 7 = 8 + 30 - 21 = 17$$

$$\text{Now, } \vec{A} \cdot \vec{B} = |\vec{A}| |\vec{B}| \cos \theta$$

$$\text{Here, } 17 = (\sqrt{61}) (\sqrt{74}) \cos \theta \Rightarrow \cos \theta = \frac{17}{\sqrt{61} \sqrt{74}} = \frac{17}{\sqrt{4514}}$$

$$\therefore \cos \theta = \frac{17}{\sqrt{4514}} \Rightarrow \theta = \cos^{-1} \left(\frac{17}{\sqrt{4514}} \right) = 75.43^\circ \text{ or } 75.43^\circ$$

This θ is the angle between the vectors $\vec{A}$ and $\vec{B}$.

Ex. 3. A particle moves from a point $\vec{r}_1 = \hat{i} + \hat{j} + \hat{k}$ to $\vec{r}_2 = 2\hat{i} + 2\hat{j} + \hat{k}$ under the action of a force $\vec{F} = 3\hat{i} + \hat{j} - \hat{k}$. Find the work done.

$$\text{Sol. The displacement of the particle, } \vec{r} = (\vec{r}_2 + 2\hat{k}) - (\vec{r}_1 + \hat{k}) = \hat{i} + \hat{j} + \hat{k}$$

$$\text{Force on the particle, } \vec{F} = 3\hat{i} + \hat{j} - \hat{k}$$

$$\therefore \text{Work done} = \vec{F} \cdot \vec{r} = (3\hat{i} + \hat{j} - \hat{k}) \cdot (\hat{i} + \hat{j} + \hat{k}) = 3 + 1 - 1 = 3$$

Ex. 4. A particle moves from position $3\hat{i} + \hat{j} - 6\hat{k}$ to $14\hat{i} + 1\hat{j} + 9\hat{k}$ metre while a uniform force $4\hat{i} + 1\hat{j} + 3\hat{k}$ newton acts on it. Calculate the work done by the force.

Sol. Force $\vec{F} = 4\hat{i} + 1\hat{j} + 3\hat{k}$ newton.

Displacement $\vec{d} = (14\hat{i} + 13\hat{j} + 9\hat{k}) - (3\hat{i} + 2\hat{j} - 6\hat{k}) = 11\hat{i} + 11\hat{j} + 15\hat{k}$ metre

\ Work done $= \vec{F} \cdot \vec{d} = (4\hat{i} + 1\hat{j} + 3\hat{k}) \cdot (11\hat{i} + 11\hat{j} + 15\hat{k}) = 44 + 11 + 45 = 100$ joules.

Ex. 5. Find a vector having modulus 6 and perpendicular to the vectors $\vec{A} = 2\hat{i} + 2\hat{j} + \hat{k}$ and $\vec{B} = 2\hat{i} - 2\hat{j} + 3\hat{k}$. Find the sine of the angle between $\vec{A}$ and $\vec{B}$.

Sol. $\vec{C} = \vec{A} \times \vec{B}$

Vector $\vec{C}$ is perpendicular to $\vec{A}$ and $\vec{B}$, but its magnitude is $|\vec{C}| = \sqrt{4^2 + 2^2 + 4^2} = 6$.

\ A unit vector along $\vec{C}$ is $\hat{C} = \frac{\vec{C}}{|\vec{C}|} = \frac{4\hat{i} - 2\hat{j} - 4\hat{k}}{6}$

Therefore, a vector, whose magnitude is 6 and is perpendicular to $\vec{A}$ and $\vec{B}$ is

$$\vec{C} = 4\hat{i} - 2\hat{j} - 4\hat{k}$$

But $\sin \theta = \frac{|\vec{A} \times \vec{B}|}{AB} = \frac{6}{\sqrt{2^2 + 2^2 + 1^2} \sqrt{2^2 + 2^2 + 3^2}} = \frac{6}{\sqrt{9} \sqrt{17}} = \frac{2}{\sqrt{17}}$

Ex. 6. Find the area of the parallelogram determined by the vectors $(\hat{i} + 2\hat{j} + 3\hat{k})$ and $(3\hat{i} - 2\hat{j} + \hat{k})$.

Sol. Area of the parallelogram (magnitude) $= |\vec{A} \times \vec{B}| = |(\hat{i} + 2\hat{j} + 3\hat{k}) \times (3\hat{i} - 2\hat{j} + \hat{k})|$
 $= |4\hat{i} - 2\hat{j} - 4\hat{k}| = \sqrt{4^2 + 2^2 + 4^2} = 6$ sq. units.

\ Direction of area $= \frac{4\hat{i} - 2\hat{j} - 4\hat{k}}{6} = \frac{2}{3}(\hat{i} - \hat{j} - \hat{k})$.

Ex. 7. If $\vec{A} = 4\hat{i} - 5\hat{j} + \hat{k}$, $\vec{B} = 2\hat{i} - 10\hat{j} - 7\hat{k}$ and $\vec{C} = 5\hat{i} + 7\hat{j} - 4\hat{k}$, calculate the following:

(i) $\vec{A} \cdot \vec{B}$ (ii) $\vec{A} \cdot \vec{C}$

Sol. (i) $\vec{A} \cdot \vec{B} = (4\hat{i} - 5\hat{j} + \hat{k}) \cdot (2\hat{i} - 10\hat{j} - 7\hat{k}) = 4(2) + (-5)(-10) + (1)(-7) = 8 + 50 - 7 = 51$

(ii) $\vec{A} \cdot \vec{C} = (4\hat{i} - 5\hat{j} + \hat{k}) \cdot (5\hat{i} + 7\hat{j} - 4\hat{k}) = 4(5) + (-5)(7) + (1)(-4) = 20 - 35 - 4 = -19$

Ex. 8. A parallelepiped with one vertex at the origin has three adjacent vertices at the points $(10, -5, 3)$, $(3, -4, 7)$ and $(-5, -6, 3)$ in rectangular co-ordinates (x, y, z) . Calculate its volume.

Sol. The position vectors of the three given vertices may be expressed as $\vec{a} = 10\hat{i} - 5\hat{j} + 3\hat{k}$,

70 | *Mechanics and Waves and Oscillations*

$\vec{d} = 3\vec{a} - 4\vec{b} + \vec{c}$ and $\vec{e} = -5\vec{a} - 6\vec{b} + 3\vec{c}$. Hence the volume of the parallelepiped, whose edges are three vectors, is

$$\vec{a} \cdot (\vec{b} \times \vec{c}) = 6 \quad \Rightarrow \quad \vec{d} \cdot \vec{e} = 306.$$

Ex. 9. A rigid body is spinning with an angular velocity of 2 radians **per second** about an axis parallel to $\vec{a}$ passing through the point $\vec{r}$. **m. Find** the velocity of a particle at the point $\vec{p}$.

Sol. If $\vec{r}$ is the position vector of the particle relative to the given point, then

$$\vec{r} = \vec{p} - \vec{r}$$

Angular velocity of the particle about $\vec{a}$ axis is

$$\vec{\omega} = \omega \vec{a}$$

$$\vec{v} = \vec{\omega} \times \vec{r}$$

$$= \omega \vec{a} \times \vec{r}$$

$$\text{Its magnitude} = \omega r \sin \theta$$

2

Mechanics of Particles

Part A : Laws of Motion and Variable Mass System

2-1. Mechanics

When we look around, we observe the objects in *motion* and at *rest*. *Mechanics is a branch of physics which deals with physical objects in motion and at rest under the influence of external and internal interactions.* The motion of a body is influenced by the bodies, surrounding it. The cause of this influence is the *interaction* between the bodies. In order to analyze and predict the motion of bodies due to different kinds of interaction, some important concepts have been discovered such as momentum, force and energy. We may say that *mechanics is the study of the principles, related to momentum, force and energy which apply to the analysis of all kinds of motion.*

Mechanics was developed since ancient times on the basis of observations on the motion of material objects. Although efforts were made earlier to propose theoretical hypotheses regarding the relationship of force and motion, but it was not until Newton announced his famous laws of motion in 1687. Later the subject of mechanics had been greatly developed by a large number of scientists, specifically by Lagrange and Hamilton, and more recently by Eienstein, Schrodinger, Heisenberg, Dirac etc.

2-2. Motion and Rest

When the position of a body changes with time, we ordinarily say that the body is in the state of motion and if its position does not change with time it is said to be at rest. There may be two kinds of rest and motion—**absolute** and **relative**. A body is said to be in absolute motion when its motion is observed with respect to a point which is stationary in space. But in the universe we do not know any point which remains always stationary. Therefore, absolute motion can only be imagined and practically it has no importance. In the universe every body is in the state of motion; earth, Sun and other stars are moving all the time and thus any body does not remain stationary. Hence absolute rest is also imaginary.

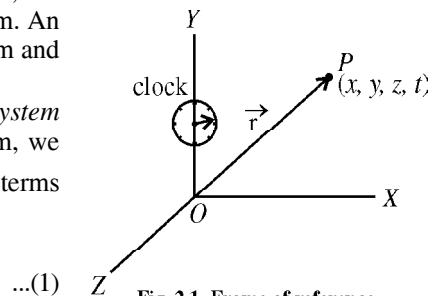
In fact **motion and rest both are relative**. A body will be said to be in the state of motion or rest with respect to another, accordingly if its position changes or does not change relative to the other. For example, a tree on the ground is at rest relative to earth; but the same tree is in the state of motion relative to Sun because firstly, the earth with this tree is constantly rotating round its axis and secondly, it is moving in its orbit round the Sun. Relative to a passenger sitting inside the train other passengers are at rest, but the outside trees, buildings etc. are in the state of motion.

2-3. Frame of Reference

From the above description we see that in order to describe the motion of a body we must know that with respect to what this motion has been measured. *If we imagine a co-ordinate system attached to a rigid body and we describe the position of any particle in space relative to it, then such a co-ordinate system is called **frame of reference**.* For the location of the objects, the position vectors are drawn from the **origin** of the co-ordinate system. An observer uses a frame of reference which is stationary relative to him and generally the position of the observer coincides with the origin.

The simplest frame of reference is a *cartesian co-ordinate system* X, Y, Z. To study the motion of a particle P relative to this system, we draw position vector $\vec{r}$ from the origin O at an instant [fig. 2.1]. In terms of (x,y,z) co-ordinates, the position vector $\vec{r}$ is represented as

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k} \quad \dots(1)$$



To describe fully the motion of the particle, we must know its position at different instants of time, i.e., the position co-ordinate $\vec{r}$ should be expressed as a function of time t . Then the **velocity** and **acceleration** of the particle will be given by

$$\vec{v} = \frac{d\vec{r}}{dt} = \frac{dx}{dt} \hat{i} + \frac{dy}{dt} \hat{j} + \frac{dz}{dt} \hat{k} \quad \dots(2)$$

and

$$\vec{a} = \frac{d\vec{v}}{dt} = \frac{d^2\vec{r}}{dt^2} = \frac{d^2x}{dt^2} \hat{i} + \frac{d^2y}{dt^2} \hat{j} + \frac{d^2z}{dt^2} \hat{k} \quad \dots(3)$$

If a physical phenomenon (e.g., a particle passing through some point P) occurs in space, its position is known as the *point*; the *time* of occurrence and the *point* taken together are called **an event**. In general, for the location of an event in a frame of reference we require its position and time of occurrence and therefore for this purpose four co-ordinates (x, y, z, t) are required. The reference system, employed for this purpose, is called *space-time frame of reference*.

2.4. Newton's Laws of Motion

Sir Isaac Newton expressed his ideas of motion in the form of three laws which are known as the **laws of mechanics**. In fact, mechanics is a study of certain general relations that describe the interactions of material bodies. One general property of a material body is its **inertial mass**. Another new concept useful in describing interactions is **force**. These two concepts, inertial mass and force, were defined in a quantitative manner by Isaac Newton. The definitions of mass and force are contained in his three laws of motion.

(1) Law of inertia : *A body remains at rest or moves with constant velocity, unless not disturbed by some external influence.* The property of a body that it cannot change its state of rest or uniform motion is called **inertia** and the influence under which the velocity of a particle changes is called **force**. The quantitative definitions of *force* and measure of inertia of a body, which we call **mass**, are contained in second and third laws of motion.

(2) Law of force. *The time-rate of change of momentum is proportional to the impressed force, i.e.,*

$$\vec{F} = \frac{d\vec{p}}{dt} \quad \dots(1)$$

Every body possesses the property of inertia or resistance to motion. This inertia is different for different bodies. The measure to this inertia for translation is called the **mass** of a body and is denoted by m . If $\vec{v}$ be the velocity of a body of mass m , then its momentum is defined by

$$\vec{p} = m\vec{v}, \quad \therefore \quad \vec{F} = \frac{d}{dt}(m\vec{v})^*$$

Newton considered that mass of a body remains constant during its motion. Therefore,

$$\vec{F} = m \frac{d\vec{v}}{dt} = m\vec{a} \quad \dots(2)$$

i.e.,

$$\text{Force} = \text{mass} \times \text{acceleration}$$

This is the fundamental law of classical mechanics. Quantitatively, first law is the special case of second law, because if force is not acting on a body, i.e., $\vec{F} = 0$, then

$$\frac{d\vec{v}}{dt} = 0, \quad \therefore \quad \vec{v} = \text{constant} \quad \dots(3)$$

This constant may be zero, which represents the state of rest.

* Common experience tells us that a greater pull or force is required to change a definite amount of velocity in a certain time for a massive body than a lighter body. So Newton considered that mass also be included in the definition of force by defining the momentum as mass times velocity.

Evidently, Newton's second law uses two new quantities (mass and force) neither of which have been rigorously defined. However, his third law in combination with the second law defines uniquely mass and force.

Newton's first and second laws of motion are valid only in inertial frames, defined later.

(3) Law of action and reaction. *To every action there is always equal and opposite reaction.* This means that if 1 and 2 bodies are interacting mutually, then

$$\vec{F}_{12} = -\vec{F}_{21} \quad \dots(4)$$

i.e., *force on 1st body due to 2nd = - force on 2nd body due to 1st.*

$$\text{But } \vec{F}_{12} = \text{Rate of change of momentum of the 1st body} = \frac{d}{dt}(m_1 \vec{v}_1) \quad \dots(5)$$

$$\text{and similarly } \vec{F}_{21} = \frac{d}{dt}(m_2 \vec{v}_2) \quad \dots(6)$$

Now, from (4), (5) and (6), we have

$$m_1 \frac{d\vec{v}_1}{dt} = -m_2 \frac{d\vec{v}_2}{dt}$$

If $a_1 = \left| \frac{d\vec{v}_1}{dt} \right|$ and $a_2 = \left| \frac{d\vec{v}_2}{dt} \right|$ denote for accelerations, then in magnitude

$$m_1 a_1 = m_2 a_2 \text{ or } m_2 = m_1 a_1 / a_2 \quad \dots(7)$$

This means that if an isolated system of two bodies is interacting among themselves, then by measuring their accelerations the ratio of their masses can be determined. If one is the standard body of mass 1 kg (say m_1), then the other (m_2) can be determined. Thus, *Newton's third law defines mass uniquely and hence equation (2) can be used to define the force in a unique way.*

Limitations of the Newton's Laws

Newton's first and second laws hold good only when observed in an unaccelerated frame of reference. These laws appear to be violated if the observer himself is accelerated. If a particle is experiencing no force and its acceleration is zero relative to an observer, then another observer, moving with an acceleration relative to the first one, will see that particle is in acceleration, i.e., a fictitious force is acting on the particle, relative to the second observer. Newton realised the difficulty and he specified that his laws of motion are valid only when the observer is stationed in an inertial frame of reference. Newton defined an **inertial system** as any system which is not accelerating relative to the fixed stars. In an inertial (unaccelerated) frame of reference, force $\vec{F}$ is equal to $m\vec{a}$, but in an accelerated (non-inertial) reference system the true force $\vec{F}$ is not equal to $m\vec{a}$, if $\vec{a}$ is the acceleration observed from the non-inertial system.

According to modern views, Newton's third law of motion is not correct when a force is acting at a distance because the forces or actions cannot move faster than the speed of light. In other words, if the first particle produces any change in the second particle, changed force (or reaction) reaches on the first particle after a finite interval of time. This means simultaneously action is not equal to reaction.* Remember that Newton's third law is still correct for the bodies at rest and for contact forces.

Inertial Frames : Those unaccelerated frames of reference, in which Newton's first law and second law hold good, are called **inertial frames**. In such frames, if a body is not acted by external force, it continues in its state of rest or uniform translatory motion (law of inertia) and hence they are called **inertial frames**. Also, if a frame is an inertial frame, then all those frames, which are moving with constant velocity relative to it, are also inertial.

* A simple example for the gross violation of third law may be quoted here. Let a charged particle move away from a wire carrying a current. The wire exerts a magnetic force on the charged particle, while at the same time the net force on the wire due to the particle is exactly zero. (See Chap. 8, Fundamental Physics by J. Orear, John Wiley & Sons, Inc., New York, 1965).

2.5. Impulse of a Force

In daily life, we see several examples in which a large force is applied on a body for a short interval of time e.g., hitting a cricket-ball by a bat, striking an iron piece by hammer etc. In such cases, **the product of the force and the short interval of time is called the impulse of force.**

If a constant force $\vec{F}$ is acting on a particle for a short interval of time Δt , then the impulse of this force is given by

$$\text{Impulse} = \vec{F} \Delta t \quad \dots(1)$$

If the change in the velocity of the body in the interval Δt is $\Delta \vec{v}$, then

$$\text{Impulse} = \vec{F} \Delta t = m \Delta \vec{v} = \Delta \vec{p} \quad \left(\because \vec{F} = \frac{m \Delta \vec{v}}{\Delta t} \text{ or } \vec{F} \Delta t = m \Delta \vec{v} \right) \quad \dots(2)$$

where $\Delta \vec{p} = m \Delta \vec{v} = m(\vec{v}_2 - \vec{v}_1)$ = change in momentum of the body. If the force $\vec{F}$ varies in the time interval from $t = t_1$ to $t = t_2$, then the impulse transferred will be

$$\int_{t_1}^{t_2} \vec{F} dt = \int_{t_1}^{t_2} \frac{m d\vec{v}}{dt} dt = m \int_{\vec{v}_1}^{\vec{v}_2} d\vec{v} = m(\vec{v}_2 - \vec{v}_1) = \vec{p}_2 - \vec{p}_1 \text{ or } \int_{t_1}^{t_2} \vec{F} dt = m \Delta \vec{v} = \Delta \vec{p} \quad \dots(3)$$

i.e., **impulse transferred = change in momentum.**

2.6. Newton's Second Law of Motion as applied to a System of Particles

If a particle of mass m is moving with velocity $\vec{v}$, its momentum is given by $\vec{p} = m \vec{v}$ and according to Newton's second law of motion, the external force $\vec{F}$ acting on it is equal to the rate of change of momentum, i.e.,

$$\vec{F} = \frac{d\vec{p}}{dt} = \frac{d}{dt}(m \vec{v}) \quad \dots(1)$$

In Newtonian mechanics, m is independent of time or velocity, then

$$\vec{F} = m \frac{d\vec{v}}{dt} = m \frac{d^2 \vec{r}}{dt^2} \quad \dots(2)$$

In a system, there may be one, two or more particles. The discussion for a single particle can be generalized for a system of particles. We can assume a body to be composed of a number of small particles and thus a body can be considered as a system of particles.

Suppose that there is a system of particles. This system may be in the form of a rigid body. There are forces of mutual interaction between every particle and other particles. These forces are called **internal forces**. In the system, the force on i th particle will be given by

$$\vec{F}_i = \vec{F}_i^e + \sum_{j=1}^n \vec{F}_{ij} \quad \dots(1)$$

where $\vec{F}_i^e$ is the external force on the i th particle due to external source of the system; $\vec{F}_{ij}$ is the force on i th particle due to j th particle of the system and the force on the i th particle due to all the other particles of the system ($j = 1$ to n) is represented by the sign of summation in eq. (1) and $\vec{F}_{ii}$, the force on the i th particle due to itself is naturally zero.

According to Newton's second law of motion,

$$\vec{F}_i = \frac{d\vec{p}_i}{dt} = m_i \frac{d\vec{v}_i}{dt} = m_i \frac{d^2 \vec{r}_i}{dt^2} \quad \dots(2)$$

where m_i is the mass of i th particle.

Now, if we take the sum for all the particles of the system, the resultant force $\vec{F}$ with the help of eq. (1) will be obtained, i.e.,

$$\vec{F} = \sum_{i=1}^n \vec{F}_i = \sum_{i=1}^n \vec{F}_i^e + \sum_{i=1}^n \sum_{\substack{j=1 \\ i \neq j}}^n \vec{F}_{ij} \quad \dots(3)$$

But $\sum_i \vec{F}_i = \sum_i m_i \frac{d^2 \vec{r}_i}{dt^2} = \frac{d^2}{dt^2} \sum_i m_i \vec{r}_i$ and $\sum_i \vec{F}_i^e = \vec{F}^e$ is the **total external force**. Also according to Newton's third law of motion, two particles apply equal and opposite forces on each other, i.e.,

$$\vec{F}_{ij} = -\vec{F}_{ji} \quad \text{or} \quad \vec{F}_{ij} + \vec{F}_{ji} = 0 \quad \text{or} \quad \sum_{i=1}^n \sum_{\substack{j=1 \\ i \neq j}}^n \vec{F}_{ij} = 0$$

$$\text{Hence} \quad \vec{F} = \frac{d^2}{dt^2} (\sum_i m_i \vec{r}_i) = \vec{F}^e \quad \text{or} \quad \vec{F}^e = \frac{d^2}{dt^2} (\sum_i m_i \vec{r}_i) \quad \dots(4)$$

Thus the resultant force $\vec{F}$ on the total system of particles is equal to the total external force acting on it and the internal forces in the system mutually cancel the effect of each other. Eq. (4) has been deduced on the basis of Newton's laws of motion.

In an inertial frame of reference, the position vector $\vec{R}$ of centre of mass of a system of particles is defined as*

$$\vec{R} = \frac{\sum m_i \vec{r}_i}{\sum m_i} = \frac{\sum m_i \vec{r}_i}{M} \quad \text{or} \quad \sum m_i \vec{r}_i = M \vec{R}$$

where $M = \sum m_i = m_1 + m_2 + m_3 \dots$ is the total mass of the system of particles.

Thus, from (4), we have

$$\vec{F}^e = \frac{d^2}{dt^2} (M \vec{R}) = M \frac{d^2 \vec{R}}{dt^2} = M \vec{a}_{cm}$$

where $\vec{a}_{cm}$ is acceleration of centre of mass.

This equation is similar to equation of motion for a single particle in accordance with Newton's second law.

Ex. 1. A force of 10 N is applied on an object of mass 5 kg for 5 seconds. Calculate : (i) impulse of the force, (ii) change in momentum of the object, (iii) change in velocity of object, (iv) acceleration of object.

Sol. Here, $m = 5$ kg, $F = 10$ N and $\Delta t = 5$ sec.

(i) Impulse of force $= F \Delta t = 10 \times 5 = \mathbf{50 \text{ newton-sec.}}$

(ii) Now, Change in momentum = Impulse of the force $= \mathbf{50 \text{ kg-ms}^{-1}}$

(iii) Change in velocity of object $\Delta v = \frac{\text{Change in momentum}}{\text{Mass of object}} = \frac{50}{5} = \mathbf{10 \text{ m/sec.}}$

(iv) Acceleration $a = \frac{\Delta v}{\Delta t} = \frac{10}{5} = \mathbf{2 \text{ m/sec.}}$

Ex. 2. Two blocks of masses $m_1 = 1$ kg and $m_2 = 3$ kg are connected by a light string and placed on a frictionless table. When m_1 is pulled by a force F , both

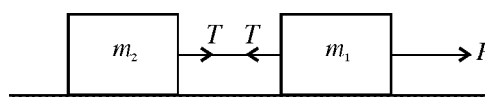


Fig. 2.2.

* For definition of centre of mass, see Art. 2.14.

blocks move with an acceleration $a = 5 \text{ m/s}^2$ [fig. 2.2]. Determine the forces F and tension in the string.

Sol. Let T be the tension in the string. Obviously, the force T is acting on the block m_2 and $F - T$ on the block m_1 . Hence according to Newton's second law of motion,

$$F - T = m_1 a = 1 \times 5 = 5 \text{ newton and } T = m_2 a = 3 \times 5 = 15 \text{ newton}$$

Therefore

$$F = 20 \text{ newton.}$$

Ex. 3. Consider a massless string going over a frictionless pulley. Blocks of masses m_1 and m_2 are attached at the ends of the string. Such a system is called **Atwood machine** [fig. 2.3]. Find the acceleration of the blocks and tension in the string.

Sol. The pulley is frictionless and hence it will not rotate. If m_2 is greater than m_1 and T is the tension in the string, then according to Newton's second law, equations of motion of the two masses are

$$T - m_1 g = m_1 a \quad \dots(i)$$

and

$$m_2 g - T = m_2 a \quad \dots(ii)$$

where a is the acceleration which will be the same for the two masses because the string is continuous.

Adding eqs. (i) and (ii), we obtain

$$(m_2 - m_1) g = (m_1 + m_2) a \text{ or } a = \frac{m_2 - m_1}{m_1 + m_2} g \quad \dots(iii)$$

$$\text{Therefore, } T = m_1 \left(g + \frac{m_2 - m_1}{m_1 + m_2} g \right) = \frac{2m_1 m_2}{m_1 + m_2} g \quad \dots(iv)$$

Ex. 4. A 20 gm bullet travelling at 500 m/s penetrates 10 cms into a wooden block. What is the average force exerted by the block on the bullet ? (O.U. 2002)

Sol. From the third equation of motion

$$v^2 = u^2 + 2as$$

Here $v = 0$, $u = 500 \text{ m/sec}$ and $s = 10 \text{ cm} = 0.10 \text{ m}$

$$\therefore 0 = 500^2 + 2 \times a \times 0.1 \text{ or } a = -\frac{500^2}{2 \times 0.1} = -1.25 \times 10^6 \text{ m/sec}^2$$

Hence average force exerted by the block on the bullet ($m = 20 \text{ gm} = 0.020 \text{ kg}$) is

$$F = -ma = 0.02 \times 1.25 \times 10^6 = 2.5 \times 10^4 \text{ N}$$

Ex. 5. A 1 kg ball drops vertically on the floor with a speed of 25 m/sec. It rebounds with an initial speed of 10 m/sec. (i) What impulse acts on the ball during contact and the ball making contact for 0.002 sec. (ii) What is the average force extended on the floor ? (A.N.U. 1988)

Sol. Impulse $F\Delta t = \text{Change in momentum} = m(v_2 - v_1)$

Taking positive direction up the floor, we have $v_1 = -25 \text{ m/sec}$ and $v_2 = 10 \text{ m/sec}$.

Here, mass of the ball $m = 1 \text{ kg}$ and $\Delta t = 0.002 \text{ sec}$

$$\therefore \text{Impulse } F\Delta t = 1 [10 - (-25)] = 35 \text{ kg-m/sec}$$

$$\text{Hence average force } F\Delta t = \frac{m(v_2 - v_1)}{\Delta t} = \frac{35}{0.002} = 1750 \text{ N.}$$

Ex. 6. A 150 gm ball moving at 12 m/sec hits a bat and rebounds with a velocity of 20 m/sec. If the duration of collision is 0.001 sec., calculate the average force extended on the ball by the bat. (S.K.U. 1989)

Sol. Impulse $F\Delta t = \text{Change in momentum} = m(v_2 - v_1)$

Here, $\Delta t = 0.001 \text{ sec}$, $m = 150 \text{ gm} = 0.15 \text{ kg}$, $v_2 = 20 \text{ m/sec}$ and $v_1 = -12 \text{ m/sec}$.

$\therefore$ Average force on the ball by the bat is

$$F = \frac{m(v_2 - v_1)}{\Delta t} = \frac{0.15(20 + 12)}{0.001} = \frac{4.8}{0.001} = 4800 \text{ N.}$$

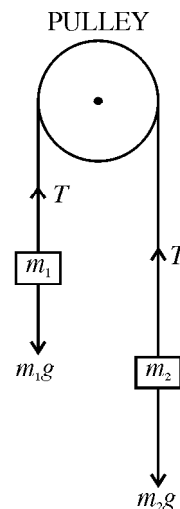


Fig. 2.3 : Atwood machine

Ex. 7. A moving ball strikes another of mass 0.2 kg at rest and exerts a force of 100 N for a time of 10 millisecond. Find the speed acquired by the ball at rest. (N.U. 2003)

Sol. Impulse $F\Delta t$ = Change in momentum = $m(v_2 - v_1)$

Here, $F = 100$ N and $\Delta t = 10 \times 10^{-3}$ sec. Mass of the ball at rest $m = 0.2$ kg

Initial speed of the ball at rest, $v_1 = 0$. Final speed of this ball, $v_2 = v$, say.

$\therefore 100 \times 10 \times 10^{-3} = 0.2 \times v$ or $1 = 0.2v$ or $v = 1/0.2 = 5$ m/sec.

Ex. 8. An empty plastic box, having mass m , is found to move up with an acceleration $g/6$ inside deep water. How much sand should be put inside the box so that it may move down with an acceleration $g/6$.

Sol. Let the force of buoyancy on the box be F . When the box is moving down,

$$F - mg = mg/6 \quad \dots(i)$$

Suppose some sand of mass m' is placed inside the box so that it moves down with an acceleration $g/6$. In this situation

$$(m + m')g - F = (m + m')g/6 \quad \dots(ii)$$

Adding eqs. (i) and (ii), we get

$$m'g = \frac{mg}{6} + (m + m')\frac{g}{6}$$

or

$$m'g - m'\frac{g}{6} = \frac{mg}{6} + \frac{mg}{6}$$

or

$$\frac{5m'g}{6} = \frac{mg}{3} \quad \text{or} \quad m' = \frac{2m}{5}.$$

2.7. Motion of Variable Mass System

So far we have considered the motion of system, having constant mass. There may be certain systems in which mass does not remain constant during motion but changes with time. The mass of the system may increase (taken positive) or decrease (taken negative) with time. Simplest example of the motion of variable mass system is a falling rain drop. When it falls, the moisture may condense over its surface or the water may evaporate so that the mass of the drop changes with time. An interesting example of the motion of a variable mass system is that of a rocket where the burnt fuel in the form of hot gases is ejected, resulting in the change of mass of the rocket with time.

Let a system of mass M be moving with a velocity $\vec{v}$ at any instant t in an inertial frame of reference (say laboratory frame of reference) from which we are observing. At a later time, the mass of the system changes by an amount of ΔM (a positive quantity, in the case of increasing mass and a negative quantity when mass is ejected) to $M + \delta M$. If initially δM be moving with velocity $\vec{u}$ in the laboratory frame of reference, then the initial momentum of the total system (when mass is increasing) is [fig. 2.4 (a)]

$$\vec{p}_i = M\vec{v} + \delta M\vec{u}$$

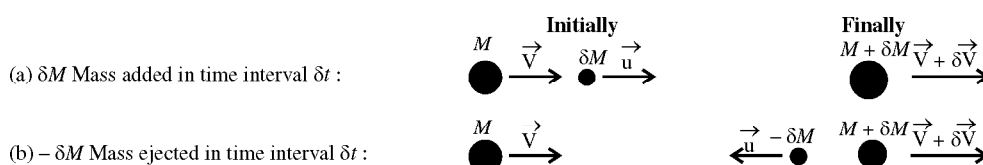


Fig. 2.4. Change of mass of a system in time Δt .

Final momentum of the total system is

$$\vec{P}_f = (M + \delta M) (\vec{V} + \delta \vec{V})$$

Hence the change in momentum is

$$\delta \vec{P} = \vec{P}_f - \vec{P}_i = (M + \delta M) (\vec{V} + \delta \vec{V}) - (M \vec{V} + \delta M \vec{u})$$

$$\text{or, } \delta \vec{P} = (M + \delta M) (\vec{V} + \delta \vec{V}) - \delta M \vec{u} - M \vec{V} \quad \dots(1)$$

When mass of the system is decreasing as in the case of a rocket, the ejected mass is $-\delta M$, moving with velocity $\vec{u}$ in the given frame of reference [fig. 2.4 (b)]. In such a case, the initial and final momenta of the total system are

$$\vec{P}_i = M \vec{V} \text{ and } \vec{P}_f = (M + \delta M) (\vec{V} + \delta \vec{V}) - \delta M \vec{u}$$

$$\text{Hence } \delta \vec{P} = (M + \delta M) (\vec{V} + \delta \vec{V}) - \delta M \vec{u} - M \vec{V} \quad \dots(2)$$

This is the change in the momentum in the time interval δt . Observe that in both cases the equations are the same. In the former case δM (or in the later case $-\delta M$) is a positive quantity.

Now, let an external force $\vec{F}_{ext}$ be acting on the entire system. This external force may be due to gravity and air resistance due to atmosphere (but not the force that propels the rocket).

According to Newton's second law of motion, the rate of change of momentum of a system gives the external force acting on the system i.e.,

$$\vec{F}_{ext} = \frac{d\vec{P}}{dt} = \lim_{\delta t \rightarrow 0} \frac{\delta \vec{P}}{\delta t}$$

$$\begin{aligned} \text{Therefore } \vec{F}_{ext} &= \lim_{\delta t \rightarrow 0} \frac{\delta \vec{P}}{\delta t} = \lim_{\delta t \rightarrow 0} \frac{(M + \delta M)(\vec{V} + \delta \vec{V}) - \delta M \vec{u} - M \vec{V}}{\delta t} \\ &= \lim_{\delta t \rightarrow 0} \left[\frac{M \delta \vec{V}}{\delta t} + (\vec{V} - \vec{u}) \frac{\delta M}{\delta t} + \delta \vec{V} \frac{\delta M}{\delta t} \right] \end{aligned}$$

$$\text{As } \delta t \rightarrow 0, \delta \vec{V} = 0, \frac{\delta \vec{V}}{\delta t} = \frac{d\vec{V}}{dt} \text{ and } \frac{\delta M}{\delta t} = \frac{dM}{dt}.$$

$$\text{Hence } \vec{F}_{ext} = M \frac{d\vec{V}}{dt} + (\vec{V} - \vec{u}) \frac{dM}{dt} \quad \dots(3)$$

This equation expresses the Newton's second law as applied to the system of variable mass. In eq. (3), $dM/dt = +m$, when the mass of the system is increasing with time and $dM/dt = -m$, when the mass of the system is decreasing with time for positive m quantity.

Eq. (3) can be written in the form :

$$\frac{M d\vec{V}}{dt} = \vec{F}_{ext} + (\vec{u} - \vec{V}) \frac{dM}{dt} \quad \dots(4)$$

This equation can easily be used to discuss the motion of a rocket. The quantity $\vec{u} - \vec{V} = \vec{v}_{rel}$

represents the velocity of ejected mass (gases) relative to the main body (rocket). Thus, we can write the force equation of the main body (rocket) as

$$M \frac{d\vec{V}}{dt} = \vec{F}_{ext} + \vec{v}_{rel} \frac{dM}{dt} \quad \dots(5)$$

We can interpret this equation as follows. $M d\vec{V}/dt$ is the mass (instantaneous mass of the rocket) times its acceleration. The first term on the right, $\vec{F}_{ext}$ refers to the external force on the mass M . The second term on the right, $\vec{v}_{rel} dM/dt$, represents the rate at which momentum is being transferred to (or from) the mass M due to leaving (or added) mass. This can be interpreted as the force exerted on the mass M due to the ejection (or addition) of mass. For a rocket this force is called **thrust** or **reaction force**, exerted on the rocket by the expelled gases i.e.,

$$\text{Reaction force or thrust, } \vec{F}_{reaction} = \vec{v}_{rel} \frac{dM}{dt} \quad \dots(6)$$

Thus the equation of motion of a variable mass system, such as rocket, is [from eq. (5)]

$$M \frac{d\vec{V}}{dt} = \vec{F}_{ext} + \vec{F}_{reaction} \quad \dots(7)$$

Special Cases

We may write eq. (3) as

$$\vec{F}_{ext} = \frac{d}{dt}(M \vec{V}) - \vec{u} \frac{dM}{dt} \quad \dots(8)$$

Case I. In case of a system of constant mass, we have the force equation in view of Newton's second law of motion as $\vec{F}_{ext} = M \vec{a}$. Eq. (8) reduces to this form, when $dM/dt = 0$ i.e., $M = \text{constant}$. In such a case,

$$\vec{F}_{ext} = \frac{d}{dt}(M \vec{V}) = M \frac{d\vec{V}}{dt} = M \vec{a} \quad \dots(9)$$

Case II. We consider a case, when $\vec{u} = 0$. In such a case, we are looking at the motion of a variable mass system from a reference frame in which the ejected mass is at rest ($\vec{u} = 0$). Thus from eq. (8), we have

$$\vec{F}_{ext} = \frac{d}{dt}(M \vec{V}) = \vec{V} \frac{dM}{dt} + M \frac{d\vec{V}}{dt} \quad \dots(10)$$

Equation of Motion of Variable Mass System in Absence of External Force

When $\vec{F}_{ext} = 0$, we obtain from eq. (5)

$$M \frac{d\vec{V}}{dt} = \vec{v}_{rel} \frac{dM}{dt} \quad \dots(11)$$

If the ejected mass is moving in opposite direction to the motion of the main body with a speed v relative to it, then for one dimensional motion $v_{rel} = -v$ and then eq. (11) takes the form :

$$M \frac{dV}{dt} = -v \frac{dM}{dt} \quad \dots(12)$$

where the force on the body is the thrust $-v dM/dt$ due to leaving mass. Remember that dM/dt is a negative quantity, when mass is ejected.

2.8. Motion of a Rocket

Rocket works on the *principle of jet propulsion*. From the word ‘jet’ we mean a high velocity stream of fluid issuing out of a nozzle or orifice, as for example a jet of steam, a jet of gas etc. The method of driving a body or machine through the agency of jet is called *jet propulsion*. The principle of jet propulsion depends on the law of conservation of momentum, according to which the momentum of jet, emerging from a body in the backward direction is equal and opposite to the momentum of the body in the forward direction.

In a rocket, the fuel and the oxidiser are burnt in the combustion chamber [fig. 2.5]. The heat of combustion makes the pressure very high inside the chamber so that a jet of hot gases emerges from the tail of the rocket with a very high exhaust velocity and in consequence the rocket moves in the forward direction.

According to the type of fuel used, rockets are classified as (a) liquid fuel rockets and (b) solid fuel rockets. In the former type of rockets, liquid hydrogen or liquid paraffin is employed as fuel and liquid oxygen, hydrogen peroxide (H_2O_2) or nitric acid as oxidiser. In the solid fuel rockets, the fuel (e.g. gunpowder) itself contains the oxidiser and therefore a separate combustion chamber is not needed.

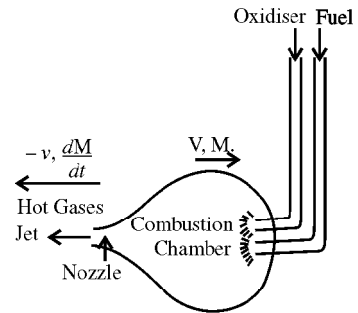


Fig. 2.5 : Rocket.

Rocket Equation and Deduction of the Expression for Final Velocity

Let M be the mass of a rocket at any instant t and V be its velocity in the laboratory frame (an inertial frame). Let us consider for simplicity that the motion of the rocket is confined to one dimension and the direction of its velocity or acceleration dV/dt is in the forward (positive) direction. If a mass m of hot gases emerges out per second with exhaust velocity $-v$ relative to the rocket in the backward direction, then the rate of change of mass of the rocket is $dM/dt = -m$.

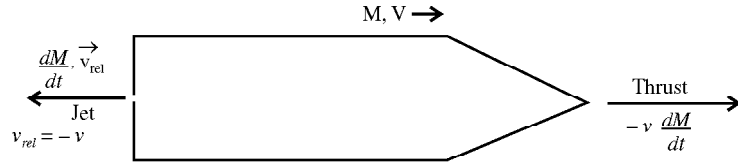


Fig. 2.6 : Motion of rocket.

According to Newton's second law as applied to the motion of a system of variable mass, we have

$$M \frac{d\vec{V}}{dt} = \vec{F}_{ext} + \vec{v}_{rel} \frac{dM}{dt} \quad \dots(1)$$

Thus case of a rocket, for one dimensional motion

$$M \frac{dV}{dt} = F_{ext} - v \frac{dM}{dt} \quad \dots(2)$$

where F_{ext} is the external force on the rocket due to gravity or atmospheric drag and $-v dM/dt$ is the **thrust**. Note that dM/dt is negative and hence thrust on the rocket is positive.

Let us consider the motion of the rocket in a distant space so that no external force acts i.e., $F_{ext} = 0$. In eq. (1), $\vec{v}_{rel}$ is the relative velocity of the gases with magnitude v relative to the rocket in the opposite direction to its motion and is replaced by $-v$ to obtain eq. (2). Hence from eq. (2), we have

$$M \frac{dV}{dt} = -v \frac{dM}{dt} \quad \dots(3)^*$$

where $-v dM/dt$ is the thrust on the rocket due to ejected mass of the gases.

This is the **fundamental equation governing the motion of a rocket under no external influence.**

If the velocity of the ejected gases (v) and rate of burning of fuel (dM/dt) are constants, then MdV/dt will be constant. As M is decreasing with time, *the acceleration of the rocket will increase with the progress of time.*

Let us deduce an expression for the final velocity of the rocket. We assume that the exhaust velocity is constant.

From eq. (3), we obtain

$$dV = -v \frac{dM}{M} \quad \dots(4)$$

Integrating it, we have

$$V = -v \log_e M + C$$

where C is a constant of integration.

If initially mass of the rocket is M_0 and velocity V_0 , then

$$V_0 = -v \log_e M_0 + C \text{ or } C = v \log_e M_0 + V_0$$

Hence

$$V = -v \log_e M + v \log_e M_0 + V_0$$

or

$$V = V_0 + v \log_e \left(\frac{M_0}{M} \right) \quad \dots(5)$$

This is the *expression for the final velocity of the rocket.*

Hence, to achieve a high final velocity of the rocket it is necessary that (i) the value of exhaust velocity v should be large and (ii) its final mass M should be much less than its initial mass M_0 .

The magnitude of exhaust velocity depends on the nozzle design and the amount of heat, produced by fuel. Theoretically, the limit of exhaust velocity is equal to the root mean square velocity of the gas molecules at the temperature of the combustion chamber. Its practical limit is nearly 2 km/sec, because it becomes difficult to attain temperatures more than 3000°C.

Practically the ratio of the initial and final masses M_0/M of the rocket has also an upper limit. In case of a **single stage rocket** (which has only one engine), the final mass M is mainly due to the fuel containers and it cannot be smaller than 10% of M_0 . In the solid fuels rockets, final mass M may be smaller than this; but in such a case this advantage is upset by smaller exhaust velocity v which affects in consequence the final velocity V .

If $M_0/M = 10$, initial velocity $V_0 = 0$ and exhaust velocity $v = 2$ km/sec, then final velocity from eq. (5) is

$$V = 0 + 2 \log_e (10) = 2 \times 2.3 = 4.6 \text{ km/sec.}$$

* Alternatively, the equation of rocket motion can be deduced as follows :

Instantaneous velocity of the rocket = $+V$

Velocity of gases relative to the rocket = $-v$

Velocity of gases in laboratory frame = $-v + V$

Momentum of the burnt fuel, ejected per second = $m (-v + V)$

Hence momentum gained by the rocket per second i.e., force on the rocket = $-m (-v + V) = \frac{dM}{dt}(-v + V)$

But force on the rocket = $\frac{d}{dt}(MV)$

$$\therefore \frac{d}{dt}(MV) = \frac{dM}{dt}(-v + V) \text{ or } \frac{dM}{dt}V + M \frac{dV}{dt} = -v \frac{dM}{dt} + V \frac{dM}{dt}$$

$$\text{Thus } M \frac{dV}{dt} = -v \frac{dM}{dt}$$

This velocity is much less than the escape velocity (11 km/sec) or the orbital velocity near the surface of the earth (8 km/sec). Hence such a large velocity cannot be obtained with the help of a one stage rocket. Employing two or three stage rockets, higher final velocities can easily be obtained.

Multistage Rocket

In order to obtain higher velocities, multistage rockets, made of two or more stages, are employed. In a **two stage rocket**, two rockets are connected in series [fig. 2.7]. First stage rocket is large and heavy and second stage rocket is comparatively small and light. When the rocket is sent in space, the first stage rocket is used first and when its fuel is burnt, it gets detached and is discarded. Now, the second stage rocket starts on with the initial velocity V_0 , imparted to it by the first stage. Hence according to eq. (5), the final velocity will be nearly twice the velocity, achieved by only one stage rocket. It will be slightly less than the double because for first stage the ratio M_0/M will be somewhat smaller. It is due to the fact that in M , the final mass of the first stage, the entire mass of the second stage remains included.

In a **three stage rocket**, three rockets are connected in series. When the fuel of the second stage is exhausted, it is also detached from the rocket and the third stage rocket starts to work acquiring more velocity. Addition of further stages increases further the final velocity.

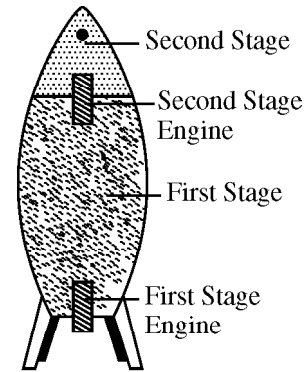


Fig. 2.7 : Two stage rocket.

Effect of Gravity

Near the surface of the earth we cannot neglect the force of gravity. If the rocket is moving vertically in upward direction, then force of gravity ($-Mg$) will be acting in the downward direction. This is the external force and hence $F_{ext} = -Mg$ is to be put in eq. (2).

Hence the form of the equation (2) is obtained to be

$$M \frac{dV}{dt} = -v \frac{dM}{dt} - Mg \quad \dots(6)$$

This equation gives the net force on the rocket at any instant t .

Integrating eq. (6) we get

$$V = -v \log_e M - gt + C \quad \dots(7)$$

When $t = 0$, $M = M_0$ and $V = V_0$, $\therefore C = V_0 + v \log_e M_0$

Hence eq. (7) takes the form

$$V = -v \log_e M - gt + V_0 + v \log_e M_0$$

or

$$V = V_0 + v \log_e \frac{M_0}{M} - gt \quad \dots(8)$$

where t is the time required for burning of the fuel.

It is clear from eq. (7) that if the time t is small, then due to the attraction of the earth the reduction in the value of V is negligible. But if the fuel burns very slowly, the rocket may not be able at all to rise or it may keep hovering without attaining any velocity. Hence it is necessary that the fuel is burnt rapidly in the first part of its journey. But the fuel should not be burnt very rapidly, as it results disadvantageous in two ways : (i) if the rocket acquires a very high velocity before it has moved above the atmosphere, there will be a greater wastage of energy due to air resistance, (ii) due to its larger acceleration, the weight of the passengers may increase very large.

The body of the rocket is made cylindrical and elongated like the shape of the cigar and its nose is sharply pointed so as to reduce the air pressure on its individual parts to the very minimum. Its frame is made of a heat resisting material and its velocity during the first part of its flight through the denser layers of the air is kept sufficiently low.

2.8. Thrust on a Rocket using Bernoulli's Equation

We wish to calculate the thrust on a rocket produced by the escape of exhaust gases from its combustion chamber. Let cross-sectional area of the chamber be A . It is filled with a gas of density ρ at a pressure p . Let there be a small orifice of cross-sectional area A_0 at the bottom of the chamber [fig. 2.8].

If v_0 be the velocity of the gases, escaping through the orifice, then we know that the thrust on the rocket is given by

$$\text{Thrust} = v_0 \frac{dM}{dt} \quad \dots(1)$$

where dM/dt is the rate at which the mass of the gases is ejected from the rocket.

According to Bernoulli's equation

$$\frac{p}{\rho} + \frac{1}{2} \frac{v^2}{g} + h = \frac{p_0}{\rho} + \frac{1}{2} \frac{v_0^2}{g} + h_0$$

$$\text{or} \quad p - p_0 = \frac{1}{2} \rho (v_0^2 - v^2) + \rho g (h_0 - h) \quad \dots(2)$$

where p_0 represents the atmospheric pressure just outside the orifice. For a gas the density is so small that we may neglect the variation in pressure with height in a chamber i.e., neglecting the term $\rho g (h_0 - h)$ in (2), we obtain

$$p - p_0 = \frac{1}{2} \rho (v_0^2 - v^2)$$

$$\text{or} \quad v_0^2 = \frac{2(p - p_0)}{\rho} + v^2 \quad \dots(3)$$

where v is the velocity of the flowing gas inside the chamber and v_0 is the velocity of gas through the orifice. We assume here the flow of gas as steady and incompressible for pressure and exhaust speeds which are not too high.

We assume the continuity of mass flow

$$Av = A_0 v_0 \quad \dots(4)$$

where the density is to be considered constant.

If the orifice is very small so that $A_0 \ll A$, then from (4) $v_0 \gg v$. Hence we can neglect v^2 in comparison to v_0^2 in (3). Hence

$$v_0^2 = \frac{2(p - p_0)}{\rho} \quad \dots(5)$$

Now, the rate at which the mass of the gas is flowing out the orifice is

$$\frac{dM}{dt} = A_0 v_0 \rho \quad \dots(6)$$

Therefore, from (1)

$$\text{Thrust} = v_0 \frac{dM}{dt} = v_0 \times A_0 v_0 \rho = A_0 \rho v_0^2 \quad \dots(7)$$

Substituting the value of v_0^2 from (5), we obtain the expression for the thrust on a rocket according to Bernoulli's equation as

$$v_0 \frac{dM}{dt} = 2A_0 (p - p_0) \quad \dots(8)$$

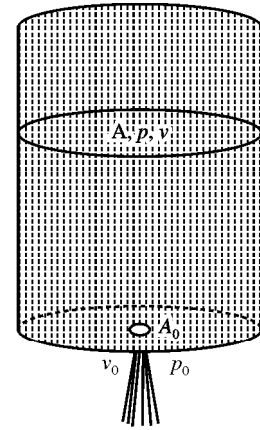


Fig. 2.8 Ejected gases from the orifice of the combustion chamber of the rocket.

Ex. 1. A block of metal weighing 4 kg resting on a frictionless plane. It is struck by a jet releasing water at a rate of 1 kg/s and at a speed of 5 m/s [fig. 2.9]. Calculate the resulting acceleration of the block.

Sol. According to Newton's second law,

$$F = \frac{dp}{dt} = \frac{d}{dt}(mv),$$

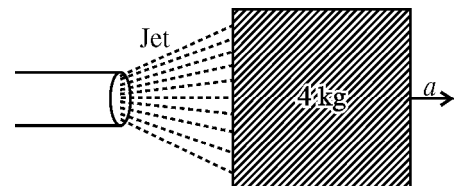


Fig. 2.9.

where m is the mass and v its velocity at any instant.

If v remains constant and m changes with time, then

$$F = v \frac{dm}{dt} \quad \dots(i)$$

This is the situation in the present problem. Here a jet of water strikes a block [fig. 2.9] at a rate of $dm/dt = 1 \text{ kg/s}$ with speed $v = 5 \text{ m/s}$. After striking the block, the velocity of water of the jet becomes zero and hence the momentum imparted to the block per second is $v dm/dt$. In other words, the force on the block due to the jet of water is

$$F = v \frac{dM}{dt} = 5 \times 1 = 5 \text{ N}$$

If this force produces an acceleration a in the block of mass $M = 4 \text{ kg}$, then

$$a = F/M = 5/4 = 1.25 \text{ m/s}^2.$$

Ex. 2. A machine gun is mounted on a trolley which is moving in a horizontal frictionless track. The instantaneous mass of the system (trolley + gun) is 250 kg when the gun is firing bullets, each of mass 10 gram , at the rate of $10 \text{ bullets per second}$ in a direction opposite to the motion of the trolley. Find the acceleration of the trolley and the average thrust of the ejecting bullets on the system. The velocity of bullets relative to the trolley (or gun) is 500 m/s .

Sol. Equation of motion of variable mass system in absence of external force is

$$M \frac{dV}{dt} = -v \frac{dM}{dt}$$

Machine gun (+ trolley), firing bullets, is an example of variable mass system [fig. 2.10].

Here the instantaneous mass of system (trolley + gun) $M = 250 \text{ kg}$

Rate of change of mass $\frac{dM}{dt} = \text{Mass of bullets fired per second}$

$$= -10 \times \frac{10}{1000} = -\frac{1}{10} \text{ kg/sec.}$$

Velocity of bullet relative to trolley = 500 m/sec .

$$\begin{aligned} \therefore \text{Instantaneous acceleration of the trolley } \frac{dV}{dt} &= -\frac{v}{M} \frac{dM}{dt} \\ &= \frac{500}{250} \times \frac{1}{10} = 0.2 \text{ m/sec}^2. \end{aligned}$$

$$\text{Average thrust} = -v \frac{dM}{dt} = 500 \times \frac{1}{10} = 50 \text{ N.}$$

Ex. 3. A rocket burns 0.02 kg of fuel per second ejecting it as a gas with a velocity of $10,000 \text{ m/sec}$. What force does the gas exert on the rocket ? (A.N.U. 2005)

Sol. Thrust or force on the rocket $F_{\text{reaction}} = v \frac{dM}{dt}$

Here, $v = 10,000 \text{ m/sec}$ and $dM/dt = 0.02 \text{ kg/sec}$

$$\therefore F_{\text{reaction}} = 10,000 \times 0.02 = 200 \text{ N}$$

Ex. 4. Neglecting the attraction of the earth show that

$$V = V_0 - v \log_e (1 - mt/M_0)$$

where V is the velocity acquired in t sec, v is the exhaust velocity and m is the constant rate of burning its fuel.

Sol.
$$V = V_0 + v \log_e (M_0/M)$$

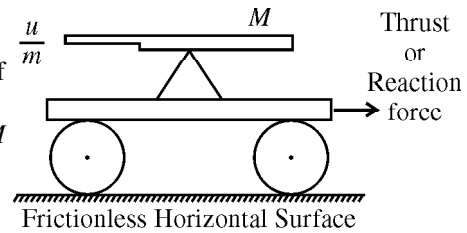


Fig. 2.10.

As the rate of burning of the fuel is m , in t seconds mt mass will come out of the rocket i.e., at this instant mass of the rocket $M = M_0 - mt$,

$$\therefore V = V_0 + v \log_e \left(\frac{M_0}{M_0 - mt} \right) \text{ or } V = V_0 - v \log_e \left(\frac{M_0 - mt}{M_0} \right)$$

or $V = V_0 - v \log_e (1 - mt/M_0)$.

Ex. 5. A rocket having an initial mass M_0 starts from rest. When it attains a velocity v , its mass becomes M . What is the ratio of (M_0/M) when the velocity of exhaust gases is equal to v (numerically).

(O. U. 1988; A.U. 1990)

Sol. Final velocity $V = V_0 + v \log_e \left(\frac{M_0}{M} \right)$

Here, initial velocity of the rocket is zero i.e., $V_0 = 0$ and $V = v$.

$$\therefore v = 0 + v \log_e \left(\frac{M_0}{M} \right) \text{ or } \log_e \left(\frac{M_0}{M} \right) = 1 \text{ or } \frac{M_0}{M} = e^1 = e = 2.717.$$

Ex. 6. A rocket having initial mass 250 kg ejects fuel at the rate of 5 kg/s with a velocity 2 km/sec vertically downward relative to itself. Calculate its velocity 25 second after start, taking initial velocity to be zero neglecting gravity. ($\log_{10} 2 = 0.3$)

Sol. $V = V_0 + v \log_e M_0/M$

Here, initial mass of the rocket $M_0 = 250$ kg

and mass consumed in 25 sec = $5 \times 25 = 125$ kg

$\therefore$ Remaining mass $M = 250 - 125 = 125$ kg

Also, given $V_0 = 0$ and $v = 2$ km/sec.

$$\therefore \text{Velocity after 25 seconds of start, } V = 0 + 2 \times \log_e (250/125) \\ = 2 \times 2.3 \times \log_{10} 2 = 2 \times 2.3 \times 0.3 = \mathbf{1.38 \text{ km/sec.}}$$

Ex. 7. In the case of a rocket motion, show that greater the value of $M_{\text{fuel}}/M_{\text{vehicle}}$, the greater be speed attained by rocket. (A.N.U. 1988)

Sol. $V = V_0 + v \log_e \frac{M_0}{M}$

Here, M_0 = initial mass of rocket = initial mass of the fuel + mass of the vehicle = $M_{\text{fuel}} + M_{\text{vehicle}}$ and $M = M_{\text{vehicle}}$, mass of the vehicle, when all the fuel is burnt away.

Hence speed attained by the rocket

$$V = V_0 + v \log_e \frac{M_{\text{fuel}} + M_{\text{vehicle}}}{M_{\text{vehicle}}} = V_0 + v \log_e \left(1 + \frac{M_{\text{fuel}}}{M_{\text{vehicle}}} \right)$$

Hence greater the value of $M_{\text{fuel}}/M_{\text{vehicle}}$, greater speed will be attained by the rocket.

Ex. 8. A 5000 kg rocket is set for vertical firing. If the exhaust speed is 500 metres/sec, how much gas must be ejected per second to supply the thrust needed (i) to overcome the weight of the rocket, (ii) to give the rocket an initial upward acceleration of 19.6 m/sec^2 ?

Sol. (i) At any instant, the net upward force on the rocket is

$$F = M \frac{dV}{dt} = -v \frac{dM}{dt} - Mg$$

Now, just to overcome the weight of the rocket, the net force $F = 0$. So that

$$\frac{dM}{dt} = -\frac{Mg}{v} = -\frac{5000 \times 9.8}{500} = -98 \text{ kg/sec.}$$

Hence the gas should be ejected at the rate of **98 kg/sec**.

(ii) If a is the upward acceleration of the rocket, then

$$-v \frac{dM}{dt} - Mg = Ma \quad \text{or} \quad \frac{dM}{dt} = -\frac{M}{v}(g+a)$$

or
$$\frac{dM}{dt} = -\frac{5000 \times (19.6 + 9.8)}{500} = -294 \text{ kg/sec.}$$

Thus, to give the rocket an initial upward acceleration of 19.6 m/sec^2 , the gas must be ejected at the rate of **294 kg/sec.**

Ex. 9. A rocket of mass 20 kg has 180 kg of fuel. The exhaust velocity of the fuel is 1.6 km/sec. Calculate the minimum rate of consumption of fuel so that the rocket may rise from the ground. Also calculate the ultimate vertical speed gained by the rocket when the rate of consumption of the fuel is (i) 2 kg/sec. (ii) 20 kg/sec.

Sol.
$$M \frac{dV}{dt} = mv - Mg$$

For lifting of the rocket from the ground it is necessary that at $t = 0$,

$$mv - M_0g > 0 \quad \text{or} \quad m > \frac{M_0g}{v}; \quad \therefore m > \frac{200 \times 9.8}{1.6 \times 1000} \quad \text{or} \quad m > 1.225 \text{ kg/sec,}$$

i.e., minimum rate of consumption of fuel so that the rocket may just rise = **1.225 kg/sec.**

Now,
$$V_{\max} = v \log_e \frac{M_0}{M} - gt$$

If t is the time, in which the rocket gains the ultimate speed, then

$$(i) t = \frac{\text{mass of the fuel}}{m} = \frac{180}{2} = 90 \text{ sec.}; \quad (ii) t = \frac{180}{20} = 9 \text{ sec.}$$

Therefore, in case (i),
$$V_{\max} = 1.6 \log_e \frac{200}{20} - \frac{9.8 \times 90}{1000} \text{ km/sec.}$$

$$= 1.6 \times 2.3 - 0.882 = 3.68 - 0.882$$

$$= \mathbf{2.8 \text{ km/sec.}}$$

For case (ii),
$$V_{\max} = 1.6 \log_e \frac{200}{20} - \frac{9.8 \times 9}{1000} = 3.68 - 0.0882 = \mathbf{3.6 \text{ km/sec.}}$$

Ex. 10. A 6000 kg rocket is set for vertical firing. If the exhaust speed is 1000 m/sec, how much gas must be ejected per second to supply the thrust needed to overcome the weight of the rocket ?

(A.N.U. 2008)

Sol. At any instant t , the net upward force on the rocket is given by

$$F = M \frac{dV}{dt} = -v \frac{dM}{dt} - Mg$$

In order to overcome the weight of the rocket, the net force $F = 0$. Hence

$$0 = -v \frac{dM}{dt} - Mg \quad \text{or} \quad \frac{dM}{dt} = -\frac{Mg}{v}$$

Here, $M = 6000 \text{ kg}$, $v = 1000 \text{ m/sec}$.

$$\therefore \frac{dM}{dt} = -\frac{6000 \times 9.8}{1000} = -58.8 \text{ kg/sec}$$

Hence the gas must be ejected at the rate of **58.8 kg/sec.**

Ex. 11. A rocket of mass 20 kg has 180 kg fuel. The exhaust velocity of the fuel is 1.6 km/s. Calculate the minimum rate of consumption of fuel so that the rocket may rise from the ground.

(A.N.U. 2007)

Sol. Rocket equation is

$$M \frac{dV}{dt} = -v \frac{dM}{dt} - Mg$$

The rocket will rise from the ground, when

$$M \frac{dV}{dt} \text{ or } -v \frac{dM}{dt} - Mg > 0$$

For minimum rate of consumption of the fuel,

$$-v \frac{dM}{dt} - Mg = 0 \text{ or, } \frac{dM}{dt} = -\frac{Mg}{v}$$

Here, $M = 20 \text{ kg} + 180 \text{ kg} = 200 \text{ kg}$, $v = 1.6 \text{ km/sec} = 1600 \text{ m/sec}$.

$$\therefore \frac{dM}{dt} = -\frac{200 \times 9.8}{1600} = -1.225 \text{ kg/sec.}$$

Thus the minimum rate of the consumption of fuel or ejection of gases is **1.225 kg/sec**.

Ex. 12. A rocket of mass 40 kg has a fuel of mass 360 kg inside it. The exhaust velocity of the fuel is 2 km/s. The fuel is burning at the rate of 4 kg/s. Find the final velocity of rocket. (S.K.U. 2007)

Sol. Here, exhaust velocity of the fuel $v = 2 \text{ km/sec}$, mass of the rocket container = 40 kg, mass of fuel = 360 kg

$\therefore$ Total mass $M = 40 + 360 = 400 \text{ kg}$

Rate of burning of the fuel = $dM/dt = 4 \text{ kg/sec}$.

$\therefore$ Total time taken by the fuel for burning, $t = \frac{360 \text{ kg}}{4 \text{ kg/sec}} = 90 \text{ sec}$.

The final velocity of the rocket is given by

$$\begin{aligned} V &= V_0 + v \log_e \left(\frac{M_0}{M} \right) - gt \\ &= 0 + 2 \log_e \left(\frac{400}{40} \right) - \left(\frac{9.8 \times 90}{1000} \right) \text{ km/sec.} \\ &= 2 \times 2.303 \log_{10} 10 - 0.882 = \mathbf{3.724 \text{ km/sec.}} \end{aligned}$$

Ex. 13. An empty rocket weighs 5,000 kg and contains 40,000 kg of fuel. If the exhaust velocity of the fuel 2.0 km/sec., find the maximum velocity gained by the rocket. (Given that $\log_e 10 = 2.3$, $\log_{10} 3 = 0.4771$). (A.N.U. 2002; A.U. 1994; S.V.U. 1988)

Sol. $V = V_0 + v \log_e \frac{M_0}{M}$

Here, $V_0 = 0$, $v = 2 \text{ km/sec}$, $M_0 = 5,000 \text{ kg} + 40,000 \text{ kg} = 45,000 \text{ kg}$ and $M = 5,000 \text{ kg}$

Rocket will gain maximum velocity, when its all the fuel is consumed.

$$\begin{aligned} \therefore V_{\max} &= 0 + 2 \log_e \frac{45,000}{5,000} = 2 \log_e 9 = 2 \log_e (3)^2 \\ &= 2 \times 2 \log_e 3 = 2 \times 2 \times \log_e 10 \times \log_{10} 3 \\ &= 2 \times 2 \times 2.3 \times 0.4771 = \mathbf{3.39 \text{ km/sec.}} \end{aligned}$$

Ex. 14. If the two stage rocket weighs 100 kg and 10 kg separately and contains 800 kg and 10 kg of fuel separately, find the final velocity that can be achieved with an exhaust velocity of 1.5 km/sec. (Given : $\log_e 10 = 2.3$ and $\log_{10} 2 = 0.30$).

Sol. The velocity of a rocket at any time t is given by

$$V = V_0 + v \log_e (M_0/M)$$

In the first case, when the first stage rocket ends, $V_0 = 0$, $v = 1.5 \text{ km/sec}$, $M_0 = 100 + 800 + 10 + 90 = 1000 \text{ kg}$ and $M = 200 \text{ kg}$

$$\begin{aligned} \therefore V_1 &= 0 + 1.5 \log_e (1000/200) = 1.5 \log_e 10 \log_{10} (10/2) \\ &= 1.5 \times 2.3 \times (\log_{10} 10 - \log_{10} 2) \\ &= 1.5 \times 2.3 \times (1 - 0.3) = 1.5 \times 2.3 \times 0.7 = \mathbf{2.415 \text{ km/sec.}} \end{aligned}$$

Now, this is the initial velocity for the second stage rocket and therefore the final velocity of the rocket at the end of second stage will be

$$V = 2.415 + 1.5 \times 2.3 \log_{10} \left(\frac{90+10}{10} \right) = 2.415 + 1.5 \times 2.3 \times \log_{10} 10 \\ = 2.415 + 1.5 \times 2.3 = 2.415 + 3.45 = \mathbf{5.865 \text{ km/sec.}}$$

Ex. 15. A rocket starts vertically upwards with speed u . Prove that its speed v at a height h is given by

$$u^2 - v^2 = \left[2gh / (1 + h/R) \right]$$

where R is the radius of the earth and g is the acceleration due to gravity at the surface of the earth. (S.V.U. 2007)

Sol. If M is the mass of the earth and R is the radius of the earth, then

$$g = \frac{GM}{R^2} \quad \dots(i)$$

If m is the mass of the rocket, then applying law of conservation of energy

$$\frac{1}{2}mu^2 - \frac{GMm}{R} = \frac{1}{2}mv^2 - \frac{GMm}{(R+h)} \quad \dots(ii)$$

where u is the initial speed of the rocket and v at a height h .

$$\text{From (ii), } u^2 - v^2 = 2 \frac{GM}{R} - \frac{2GM}{R+h} = \frac{2GMR}{R^2} \left(1 - \frac{R}{R+h} \right)$$

$$\text{or } u^2 - v^2 = 2gR \left(\frac{h}{R+h} \right) = \frac{2ghR}{R+h}$$

$$\text{or } u^2 - v^2 = \frac{2gh}{1+h/R} \quad \dots(iii)$$

This is the desired relation.

Ex. 16. A machine gun fires 50 gm bullets at a speed of 1000 m/sec. The gun man holding the machine gun in his hands, can exert an average force of 180 newtons against the gun. Determine the maximum number of bullets he can fire in a minute. (K.U. 1990)

$$\text{Sol. Momentum of each bullet} = \left(\frac{50}{1000} \right) \times 1000 = 50 \text{ kg-m/sec.}$$

If n be the maximum number of bullets fired in one minute, then

$$\text{momentum imparted per sec.} = 50 \times \left(\frac{n}{60} \right) = \frac{5n}{6} \text{ kg-m sec}^{-2}.$$

But, force exerted by gun man against the gun = rate at which momentum imparted to the bullets

$$\text{i.e., } 180 \text{ N} = \frac{5n}{6} \text{ N}$$

$$\therefore n = \frac{6 \times 180}{5} = \mathbf{216}.$$

Ex. 17. A spherical rain drop, falling through fog, accumulates mass due to condensation at a rate proportional to its cross-sectional area and velocity. Find the acceleration of the rain drop in terms of its radius (r) and velocity (v). Assume the rain drop starts from rest.

Sol. Let at any instant t the velocity of the rain drop be v and its radius be r . If ρ is the density of the drop, then its cross-sectional area A and mass m are given by

$$A = \pi r^2 \text{ and } m = \frac{4}{3} \pi r^3 \rho \quad \dots(i)$$

According to the problem, the rate at which the mass of the drop increases is given by

$$\frac{dm}{dt} \propto Av \text{ or } \frac{dm}{dt} = C\pi r^2 v \quad \dots(ii)$$

where C is a constant of proportionality.

Motion of a drop, when water is condensing on it with time, is a system of variable mass. When Newton's second law is applied to a system of variable mass, we have

$$m \frac{dv}{dt} = F_{ext} + v_{rel} \frac{dm}{dt} \quad \dots(iii)$$

Here, $F_{ext} = mg$, $v_{rel} = u - v = -v$ ($u = 0$ for small droplets of fog)

$$m \frac{dv}{dt} = mg - v \frac{dm}{dt} \text{ or } \frac{dv}{dt} = g - \frac{v}{m} \frac{dm}{dt} \quad \dots(iv)$$

Substituting for dm/dt from (ii), we obtain the acceleration of the drop ($a = dv/dt$) as

$$a = g - v \frac{(C\pi r^2 v)}{\frac{4}{3}\pi r^3 \rho} \text{ or } a = g - \frac{3C}{4\rho} \frac{v^2}{r}$$

Writing $3C/4\rho = K$, a constant, the desired expression for the acceleration of the rain drop is

$$a = g - \frac{Kv^2}{r} \quad \dots(v)$$

Part B : Conservation of Energy

2.9. Conservation of Energy

Energy is a quantity which can be converted from one form to another but it cannot be created or destroyed. In any process, the total energy, *i.e.*, the sum of all energy present in all different forms, remains the same. This is known as **principle of conservation of energy**. In this chapter we plan to discuss the conservation of energy related to mechanics, *i.e.*, mechanical energy. This involves the ideas of work, kinetic energy and potential energy.

2.9.1. Work

We consider a particle P , moving along a curve AB under the action of a variable force $\vec{F}$. Let at any instant the position vector of the particle be $\vec{r}$. If this force $\vec{F}$ displaces the particle through a distance $d\vec{r}$ [fig. 2.11], then the infinitesimal small amount of work dW done by this force on the particle is defined as

$$dW = \vec{F} \cdot d\vec{r} = F dr \cos \theta$$

If the particle is moved from a point A , to the point B , along the path AB , then the total work done by the force on the particle is given by

$$W = \int_A^B \vec{F} \cdot d\vec{r} \quad \dots(1)$$

If the force $\vec{F}$ remains constant and the displacement $\vec{r}$ is along a straight line, then

$$W = \int_A^B \vec{F} \cdot d\vec{r} = \vec{F} \cdot \vec{r} = F \cos \theta r \quad \dots(2)$$

where θ is the angle between the vectors $\vec{F}$ and $\vec{r}$.

If F_x , F_y and F_z are the rectangular components of the force $\vec{F}$ and $d\vec{r} = dx\hat{i} + dy\hat{j} + dz\hat{k}$, then from eq. (1), we get

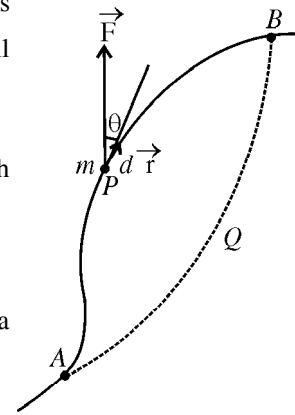


Fig. 2.11 : Work done along a path.

$$W = \int_A^B (F_x \hat{i} + F_y \hat{j} + F_z \hat{k}) (dx \hat{i} + dy \hat{j} + dz \hat{k}) = \int_A^B (F_x dx + F_y dy + F_z dz) \quad \dots(3)$$

The rate of doing work (dW/dt) is defined as **power**, i.e.,

$$P = \frac{dW}{dt}$$

If $\vec{F}$ is the instantaneous force acting on the particle, then

$$P = \frac{dW}{dt} = \vec{F} \cdot \frac{d\vec{r}}{dt} = \vec{F} \cdot \vec{v}$$

which is the expression for the instantaneous power. The average power during a time interval t is obtained by dividing the total work W by time t so that $P_{av} = W/t$.

2.9.2. Kinetic Energy : Work-energy Theorem

The energy of a particle is defined as the capacity of doing work and it is measured by the amount of work which it can do in the position in which it is. The energy due to the speed of a particle is called its kinetic energy. It is generally represented by the letter K .

According to Newton's second law of motion,

$$\vec{F} = m \frac{d\vec{v}}{dt}$$

where $d\vec{v}/dt$ is the acceleration, produced in a particle of mass m by the force $\vec{F}$.

The amount of work done on the particle, when it is taken from the point A to the point B , is given by

$$W = \int_A^B \vec{F} \cdot d\vec{r} = m \int_A^B \frac{d\vec{v}}{dt} \cdot d\vec{r} \quad \dots(4)$$

$$\text{Now, } d\vec{r} = \frac{d\vec{r}}{dt} dt = \vec{v} dt \text{ and so } \frac{d\vec{v}}{dt} \cdot d\vec{r} = \left(\frac{d\vec{v}}{dt} \cdot \vec{v} \right) dt = \frac{d}{dt} \left(\frac{1}{2} \vec{v} \cdot \vec{v} \right) dt = d \left(\frac{v^2}{2} \right).$$

$$\therefore W = \frac{m}{2} \int_A^B d(v^2) = \frac{m}{2} [v^2]_A^B = \frac{m}{2} (v_B^2 - v_A^2)$$

where v_A and v_B are the speeds of the particle at the points A and B respectively.

$$\text{Thus } W = \int_A^B \vec{F} \cdot d\vec{r} = \frac{1}{2} m v_B^2 - \frac{1}{2} m v_A^2 \quad \dots(5)$$

This equation shows that the work done on a free particle depends only upon initial and final speeds of the particle and is *independent of the nature of the force* and the path followed by the particle.

$$\text{The energy } \frac{1}{2} m v^2 \text{ is called the kinetic energy* of the particle, i.e., } K = \frac{1}{2} m v^2. \quad \dots(6)$$

Thus, $W = \text{Kinetic energy } (K_B) \text{ at } B - \text{Kinetic energy } (K_A) \text{ at } A = \text{Change in kinetic energy.}$

Thus, *the work done by a force on a particle is equal to the change in its kinetic energy*. This is known as **work-energy theorem** and has been represented in equation form by eq. (5).

When the work done on the particle by the external force is positive, its kinetic energy increases. In case, if the work done on the particle by the force is negative, there is a decrease in its kinetic energy. Actually, in such a case the particle does work in moving itself against the force and loses its kinetic energy.

* In case, if a particle has initially speed v and finally it acquires zero speed against a force $\vec{F}$, then the work done by the particle against the forces is $W = \int_A^B \vec{F} \cdot d\vec{r} = -m \int_A^B \frac{d\vec{v}}{dt} \cdot d\vec{r} = -\frac{m}{2} \int_v^0 d(v^2) = \frac{1}{2} m v^2$.

Thus, a particle of speed v has the capacity to do $\frac{1}{2} m v^2$ work. This means that $\frac{1}{2} m v^2$ is the kinetic energy of the particle.

2-9-3. Conservative Force and Conservative Force Field

A force is said to be conservative if the work done by it in moving a particle from one point to another point depends only on these points and not on the path followed. This means that if a particle is taken from the point A to the point B through APB or AQB paths and if each time the amount of work done $\left(\int_A^B \vec{F} \cdot d\vec{r} \right)$ by the force is the same, then this force is conservative. Thus, for a conservative force

$$\int_A^B \vec{F} \cdot d\vec{r} = \int_A^B \vec{F} \cdot d\vec{r} \quad \dots(7)$$

(Path APB) (Path AQB)

Paths APB and AQB are any two arbitrary paths, joining the points A and B.

The region in which a particle experiences a conservative force is called a **conservative force field**.

A central force of the form $\vec{F} = F(r) \hat{r}$ is a conservative force. Gravitational, electrostatic and elastic forces are the examples of central force and all these forces are conservative in nature.

2-9-4. Potential Energy

The potential energy of a particle is defined as the energy stored in it due to its position. It is measured by that amount of work which this particle can do, when it moves from its present position to some standard position. It is generally represented by the letter U .

Let us consider a particle in a conservative force field (e.g., gravitational, electrostatic etc.). The conservative force ($\vec{F}$), which is acting on the particle can be just counterbalanced by a suitable applied force $\vec{F}_a$ i.e.,

$$\vec{F} + \vec{F}_a = 0 \quad \text{or} \quad \vec{F}_a = -\vec{F} \quad \dots(8)$$

Now, if the particle is moved very slowly against the conservative force from one position to another without developing the kinetic energy, the work done in moving the particle will appear as its potential energy. We define the difference in potential energy between two positions of the particle as equal to the work done by such an applied force on the particle in moving it from the 1st position to 2nd position, i.e.,

$$U_2 - U_1 = \int_1^2 \vec{F}_a \cdot d\vec{r} \quad \dots(9a)$$

$$\text{But } \vec{F}_a = -\vec{F},$$

$$\therefore U_2 - U_1 = - \int_1^2 \vec{F} \cdot d\vec{r} \quad \dots(9b)$$

Remember that the definition of difference of potential energy $U_2 - U_1$ in eq. (9b) is true only for conservative forces (not for non-conservative forces, described later).

Now, if the work done on the particle by conservative force $\vec{F}$ is represented by W , then

$$U_2 - U_1 = -W = - \int_1^2 \vec{F} \cdot d\vec{r} \quad \dots(10)$$

Hence the difference of potential energy at two points of the field is the negative of the work done by the conservative force. When positive work is done by the conservative force, the potential energy is reduced and for negative work, the potential energy increases.

The potential energy at a point is obtained by assigning zero potential energy to any convenient point and then calculating the difference of potential energies. Now, if we assume the 1st position at ∞ and the potential energy to be **zero** there, then the potential energy of the particle at a point distant $\vec{r}$ is given by

$$U(\vec{r}) = - \int_{\infty}^{\vec{r}} \vec{F} \cdot d\vec{r} \quad \dots(11a)$$

or
$$U(\vec{r}) = \int_{\infty}^{\vec{r}} \vec{F}_a \cdot d\vec{r} \quad \dots(11b)$$

Thus, the potential energy of a particle at a point $\vec{r}$ is defined as the amount of work done by an applied force in moving the particle from infinity to that point.

It is often convenient to choose the reference or standard position to be that at which the force acting on the particle is zero. Thus the force exerted by a spring is zero when the spring has its normal unstretched length; so we generally say that in this position, the potential energy is zero. In case of the gravitational and electrostatic forces, the particle experiences no force of interaction, when it is at infinite distance. Hence in such cases we assign **zero potential energy at ∞** . In case of the earth, the gravitational force (mg) is constant near its surface and hence we find it convenient to take the zero of potential energy at the earth's surface, not at infinity.

Also it is easy to show that for a conservative force $\vec{F}$,

$$\text{Curl } \vec{F} = 0 \text{ or } \vec{\nabla} \times \vec{F} = 0$$

2.9.5. Conservation of Mechanical Energy

Let a particle of mass m move from one point to another under the influence of a conservative force. Then according to eq. (5) and eq. (9b), the amount of work done by the force on the particle will be given by

$$W = \int_1^2 \vec{F} \cdot d\vec{r} = \frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2 = U_1 - U_2 \quad \dots(12)$$

where v_1 and v_2 are the speeds of the particle at the first and second points respectively.

Eq. (12) expresses that in a conservative force field, if the kinetic energy of the particle increases in moving from one point to another, then its potential energy should decrease.

From eq. (12), we get

$$\frac{1}{2}mv_1^2 + U_1 = \frac{1}{2}mv_2^2 + U_2 \quad \dots(13)$$

Writing the kinetic energy $\frac{1}{2}mv^2 = K$, we have

$$K_1 + U_1 = K_2 + U_2. \quad \dots(14)$$

This means that the sum of kinetic and potential energies of a particle remains constant at any point of a conservative force field. This is known as the **law of conservation of mechanical energy**. Thus if a particle moves under the action of such a force so that its mechanical energy (potential energy and kinetic energy) remains constant, this force is said to be a conservative force.

The energy in one form can be converted into some other form. For example, the mechanical energy can be converted in the form of heat and light. However, in any process, the total energy i.e., the sum of all energy present in various forms remains constant. This is known as the *general law of principle of conservation of energy*.

An important note on potential energy : If a particle interacts with another particle or a system of particles and experiences similar to gravitational or electrostatic forces, then such interaction forces are conservative forces. If this one particle system moves in a conservative force field due to an external source, then

$$U_2 - U_1 = - \int_1^2 \vec{F} \cdot d\vec{r} = -W = -(K_2 - K_1) = K_1 - K_2$$

where U_1 and K_1 are the potential and kinetic energies in the first position and U_2 and K_2 in the second position. Hence in a conservative force field,

$$K_1 + U_1 = K_2 + U_2 = E \text{ (total mechanical energy)} \quad \dots(15)$$

i.e., total mechanical energy of the particle remains constant.

In fact, potential energy is not the property of a particle (or body) of its own, but it is the property of the system; this system consists of the particle (or body) and another body due to which a force is acting on the particle. The potential energy of a body above the surface of the earth is shared by the body and

earth both. When the body is allowed to fall, the body and earth move towards each other. However, due to the large mass of the earth, its speed is negligible and practically all the kinetic energy developed is in the share of the body. This is why all the potential energy or kinetic energy is assumed to be associated with the body. In case, if there is a system of two masses whose masses are comparable with each other, the potential energy will be said to be associated with the system of two bodies, not with any one of them.

Hence *potential energy is that energy which is stored in a body or particle due to its position in a force-field or stored in the system due to its configuration*. When the configuration of the system changes, its potential energy changes. Remember that when we say the potential energy of a system, the conservative forces are internal.

2-10. Non-Conservative Forces : General Law of Conservation of Energy

A force is said to be non-conservative, if the work done by that force on a particle that moves between two points depends on the path taken between those points.

Frictional and viscous forces are the examples of non-conservative forces. In case of such forces, if a particle moves from A to B and then from B to A , the work done around the closed path is not zero, but the particle itself loses its kinetic energy along both paths. Thus for non-conservative force,

$$\oint \vec{F} \cdot d\vec{r} = W_{(A \rightarrow B)} + W_{(B \rightarrow A)} \neq 0. \quad \dots(1)$$

For example, if we push a block over a rough table between any two points by various paths, the distance traversed varies and accordingly the work done by the frictional force and hence the work done depends on the path. Hence we cannot define a potential energy like function for non-conservative force.

Difference between Conservative and Non-conservative Forces

(1) When a particle moves under the influence of a conservative force, the work done by it in moving the particle from one point to another depends only on these points and is independent of the path followed, while in case of a non-conservative force, the work done on the particle depends on the length of the path. For example, for gravitational (conservative) force, the work done is independent of the paths between two points but when frictional (non-conservative) force is present the work done has different values for different paths between two points.

(2) The work done under the influence of conservative force is equal to the difference in potential energies ($U_1 - U_2$) in initial and final positions, while for non-conservative forces we can not define potential energy or difference of potential energies ($U_1 - U_2$).

(3) The work done on a particle is zero, when it moves along a closed path under a conservative force. However for a non-conservative force, the work done on the particle along a closed path is non-zero.

(4) The mechanical energy (sum of kinetic and potential energies) of a particle, moving under the influence of a conservative force, remains constant, while it does not remain constant in case of a non-conservative force.

Work Done by Conservative, Non-conservative and External Forces and Work-Energy Theorem

For conservative forces, *the sum of potential and kinetic energies, i.e., mechanical energy is constant, i.e.,*

$$K + U = E, \text{ a constant (mechanical energy)}$$

$$\text{or} \quad \Delta K + \Delta U = 0 = \Delta E \quad (\text{for conservative forces}) \quad \dots(2)$$

But the work done by conservative forces is equal to negative of the increase in potential energy, i.e., $W_c = -\Delta U = -(U_2 - U_1)$.

Now let us suppose that in addition to the conservative forces, a non-conservative force due to friction acts on the particle. If W_{nc} is the work done by the frictional force and W_c by conservative force, then according to work-energy theorem, the work done ($W_{nc} + W_c$) on the particle should be equal to the change (ΔK) in the kinetic energy of the particle. Thus,

$$W_{nc} + W_c = \Delta K = K_2 - K_1 \quad \dots(3)$$

If in addition to frictional and conservative forces, an external force is also acting on the particle, then according to work-energy theorem,

$$W_{ex} + W_{nc} + W_c = \Delta K = K_2 - K_1 \quad \dots(4)$$

Remember that the work done due to all the forces, acting on the particle, is equal to the change in its kinetic energy.

But

$$W_c = -\Delta U = U_1 - U_2,$$

then

$$W_{ex} + W_{nc} = \Delta K + \Delta U = \Delta E$$

or

$$W_{ex} + W_{nc} = E_2 - E_1 \quad \dots(5)$$

Hence if external and frictional forces are acting on the particle, then the work done by them is equal to the change in the mechanical energy.

When external force is not acting, the work done by the non-conservative or frictional force is given by

$$W_{nc} = E_2 - E_1 \quad \dots(6)$$

This equation shows that if a frictional force acts on a particle, the total mechanical energy is not constant, but changes by the amount of work done by the frictional force.

But W_{nc} , which is the work done by friction on the particle, is always negative, hence we see from eq. (6) that the final mechanical energy $E_2 (= K_2 + U_2)$ is less than the initial mechanical energy $E_1 (= K_1 + U_1)$. The lost mechanical energy appears in the form of heat. The heat energy developed is exactly equal to the mechanical energy dissipated. This heat energy, so produced, is equal to the work done by the particle. If Q is the heat energy developed, then evidently $W_{nc} = -Q$ and hence we get

$$\Delta E + Q = 0 \quad \dots(7)$$

This shows that there is no change in the sum of the mechanical energy and heat energy of the system so far conservative and frictional forces are acting on the system. Therefore, the total energy may again be said to be **conserved**.

Similarly, if other non-conservative forces are acting, then the mechanical energy lost is converted into other forms of energy, therefore, *the total energy in general is conserved*, i.e.,

$$\Delta E + Q + \text{change in other forms of energy} = 0. \quad \dots(8)$$

This is the *general law of conservation of energy*.

Thus, the total energy is always conserved while kinetic and potential energies are conserved only when conservative forces act.

In case, non-conservative forces, e.g., friction are not acting on the particle and in addition to the conservative force, an external force is acting, then from eq. (5)

$$W_{ex} = E_2 - E_1 \quad \dots(9)$$

Hence *the work done by external force is equal to the change in mechanical energy of the particle*.

2.11. Examples of Conservation of Mechanical Energy – Conservative Systems

Here we will discuss two important one dimensional conservative systems :

(1) Spring-mass system and

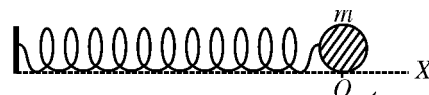
(2) Body-earth system.

In case of the first system spring force (linear restoring force) between the spring and mass and in case of second system, the gravitational force between a body and earth both are the conservative forces.

(1) Spring-mass system : For sufficiently small displacements, a linear restoring force may be produced by a stretched or compressed spring.

Let us consider a massless spring whose one end is fixed to a rigid wall and other end is attached to a particle of mass m . Let the system be placed horizontally on a frictionless table [fig. 2.12]. Let us take the equilibrium position of the particle to be the origin $x = 0$ and the line along the length of the spring as X -axis. Now, if the spring is extended or compressed slowly by the

(a) Normal spring



(b) Stretched spring



(c) Compressed spring

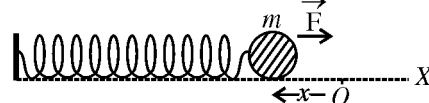


Fig. 2.12.

application of an external force, work will be done on the system (spring + particle) against the restoring force $\vec{F}$ and this work will become the potential energy of the system. If we assume the potential energy of the system in the undisplaced position to be zero, then the potential energy at the displacement x of the particle will be given by

$$U = - \int_0^x \vec{F} \cdot d\vec{r} = - \int_0^x F dx$$

But $F = -Cx$.

$$\therefore U = \int_0^x Cx dx = \frac{1}{2}Cx^2 \quad \dots(1)$$

If a graph between U and x is plotted, it will be **parabola** [fig. 2.13].

If the maximum displacement of the particle is a , then the potential energy developed in the system is

$$U = \frac{1}{2}Ca^2$$

Now, if the particle of mass m be released from the **rest** position at $x = a$, let us calculate the kinetic energy of the particle at any displacement x .

On releasing the particle, it will move under restoring force, which is **conservative** in nature, towards the origin. So the total energy E of the system will remain constant. Initially, total energy E is given by

$$E = K + U = 0 + \frac{1}{2}Ca^2 = \frac{1}{2}Ca^2.$$

If v be the speed of the particle at the displacement x , then

$$\text{kinetic energy of the system } K = \frac{1}{2}mv^2; \text{ potential energy of the system } U = \frac{1}{2}Cx^2$$

$$\text{Hence, total energy } E = \frac{1}{2}mv^2 + \frac{1}{2}Cx^2$$

Using the law of conservation of energy, we have

$$\frac{1}{2}Ca^2 = \frac{1}{2}mv^2 + \frac{1}{2}Cx^2 \quad \dots(2)$$

$$\text{or } \frac{1}{2}mv^2 = \frac{1}{2}C(a^2 - x^2), \text{ whence } v = \pm \sqrt{\frac{C}{m}(a^2 - x^2)} \quad \dots(3)$$

This equation gives the kinetic energy and the velocity of the particle at any displacement x .

When the particle reaches the origin $x = 0$, its kinetic energy and hence the velocity (v_0) will be maximum. Thus,

$$\frac{1}{2}mv_0^2 = \frac{1}{2}Ca^2 \text{ and } v_0 = \pm \sqrt{\frac{C}{m}}a \quad \dots(4)$$

This means that the maximum potential energy $\left(\frac{1}{2}Ca^2\right)$ of the particle with respect to the origin

converts entirely in the form of kinetic energy $\left(\frac{1}{2}mv_0^2\right)$ at $x = 0$.

(2) Body-earth system–Force of gravity : We know that the radius of the earth is nearly 6,400 km. Hence we can have practically the value of acceleration due to gravity (g) to be the same for a height of few kilometers from the earth's surface.

For body-earth system, we take Z -axis of the co-ordinate system vertically up with origin at the earth's surface. We assume the surface of the earth as standard position so that at $z = 0$, $U_{(0)} = 0$. The

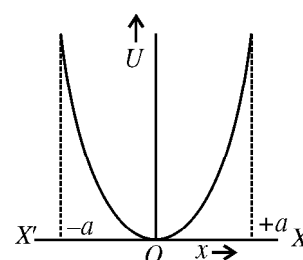


Fig. 2.13.

position of a body under consideration will be taken close to the surface of the earth so that the acceleration due to gravity acting on the body may be taken as constant. If the height of the body above the earth's surface is z , then its potential energy will be

$$U_{(z)} = -\int_0^z \vec{F} \cdot d\vec{r} = -\int_0^z (-mg\hat{k}) \cdot dz\hat{k} \text{ i.e., } U_{(z)} = \int_0^z mg \, dz = mgz \quad \dots(5)$$

where $\vec{F} = m\vec{g} = -mg\hat{k}$ is the force of gravity on the body and $\vec{g} = -g\hat{k}$ is the acceleration due to gravity near the surface of the earth.

Therefore the work done in carrying out the body upto a height z will be mgz . In this position the capacity of doing the work is mgz and hence its potential energy is mgz . This satisfies the relation $F = -\frac{\partial U}{\partial z} = -mg$ for a conservative force.

Let the body be thrown up with initial velocity v_0 . As the force of gravity is a conservative force, hence from the law of conservation of mechanical energy,

Initial mechanical energy (at $z = 0$) = Mechanical energy at z height

$$\text{i.e.,} \quad \frac{1}{2}mv_0^2 + 0 = \frac{1}{2}mv^2 + mgz$$

where v is the speed of the body at a height z .

$$\text{Hence} \quad \frac{1}{2}mv_0^2 - \frac{1}{2}mv^2 = mgz \quad \dots(6)$$

Thus, the loss in the kinetic energy of the body is equal to the work done by the body on the gravitational field and will exist in the field. This is the potential energy of the body or body-earth system. Here the work done on the body by gravitational field is $-mgz$. At the maximum height h , the potential energy will be mgh and kinetic energy zero, i.e.,

$$\frac{1}{2}mv_0^2 = \frac{1}{2}mv^2 + mgz = 0 + mgh \text{ or } \frac{1}{2}mv_0^2 = mgh \quad \dots(7)$$

When the body comes back, gravitational field will do the work on the body and its kinetic energy will increase and on reaching the earth's surface, the kinetic energy of the body will be equal to its initial kinetic energy $\frac{1}{2}mv_0^2$.

Ex. 1. If a force $F = Ax + Bx^2$ acts parallel to X -axis on an object and moves it from $x = 1$ to $x = 2$, calculate the work done.

$$\begin{aligned} \text{Sol. Work done} &= \int_1^2 F_x \, dx = \int_1^2 (Ax + Bx^2) \, dx = \left[\frac{Ax^2}{2} + \frac{Bx^3}{3} \right]_1^2 \\ &= \frac{A}{2}[2^2 - 1^2] + \frac{B}{3}[2^3 - 1^3] = \frac{3}{2}A + \frac{7}{3}B. \end{aligned}$$

Ex. 2. A particle moves along half the circumference of a circle of 1 metre radius. Calculate the work done if the force at any point is inclined at 60° to the tangent at that point and has 5 newtons magnitude.

$$\text{Sol.} \quad \text{Work done} = \int_A^B \vec{F} \cdot d\vec{s} = \int_A^B F \cos \theta \, ds$$

Here, $F = 5$ newtons and $\cos \theta = \cos 60^\circ = \frac{1}{2}$

$$\therefore \text{Work} = \int_A^B 5 \cdot \frac{1}{2} \, ds = \frac{5}{2} (\text{length } s \text{ of the semicircle}) = \frac{5}{2}(\pi r) = \frac{5}{2} \times 3.14 \times 1 = \mathbf{7.85 \text{ joules.}}$$

Ex. 3. A body of mass m is revolving in a circle of radius r under the influence of a centripetal force $F = -k/r^2$, where k is a constant. Determine the total energy of the body.

Sol. Centripetal force, $\frac{mv^2}{r} = \frac{k}{r^2}$, $\therefore$ Kinetic energy $\frac{1}{2}mv^2 = \frac{k}{2r}$

$$\text{Potential energy, } U = -\int_{\infty}^r \vec{F} \cdot d\vec{r} = \int_{\infty}^r \left(\frac{k\hat{r}}{r^2} \right) \cdot d\vec{r} = \int_{\infty}^r \frac{k}{r^2} dr = -\frac{k}{r}$$

Hence Total energy, $E = \text{Kinetic energy} + \text{Potential energy}$

$$= \frac{k}{2r} - \frac{k}{r} = -\frac{k}{2r}.$$

Ex. 4. A body of mass 4 kg slides on a horizontal frictionless table with a speed of 2 m/sec. It is brought to rest in compressing a spring in its path. By how much is the spring compressed if the force constant of the spring is 64 kg/m.

Sol. Kinetic energy of the body $= \frac{1}{2}mv^2 = \frac{1}{2} \times 4 \times 2^2 = 8$ joule.

Let the spring be compressed through a distance x so that the body loses its all kinetic energy and comes to rest. Obviously, the lost kinetic energy will be stored in the form of potential energy $\frac{1}{2}kx^2$ of the system. Thus,

$$\frac{1}{2}kx^2 = \frac{1}{2}mv^2 = 8 \text{ joule or } x^2 = \frac{16}{k} = \frac{16}{64} = \frac{1}{4}, \text{ whence } x = \mathbf{0.5 \text{ metre.}}$$

Ex. 5. A body is allowed to slide on a frictionless track from rest position under gravity. The track ends in a circular loop of diameter D . What is the minimum height of the body in terms of D so that it may complete successfully the loop.

Sol. When the body is at A, the kinetic energy gained by the body is

$$\frac{1}{2}mv^2 = mgh \text{ or } v^2 = 2gh \quad \dots(i)$$

For the least value of h , the value of v should be least in such a way that the body just reaches up to B. In this position, the necessary centripetal force $2mV^2/D$ is provided by the weight of the body [fig. 2.14], i.e.,

$$\frac{2mV^2}{D} = mg$$

or $V^2 = \frac{Dg}{2} \quad \dots(ii)$

Also $\frac{1}{2}mv^2 - \frac{1}{2}mV^2 = mgD \text{ or } v^2 - V^2 = 2gD \quad \dots(iii)$

Putting the values from (i) and (ii), we get

$$2gh - \frac{1}{2}Dg = 2gD \text{ or } h = \frac{5}{4}D.$$

Ex. 6. A mass m is revolving in a vertical circle at the end of a string of r meter length. Calculate the difference in kinetic energies at the top and bottom of the circle. Prove that the tension of the string at the lowest point exceeds the tension at the topmost point by $6mg$.

Sol. As the mass m reaches from position A to the position B, [fig. 2.15] it will lose kinetic energy, but its potential energy will increase by the same amount. Thus,

$$\frac{1}{2}mv_1^2 - \frac{1}{2}mv_2^2 = mg \times 2r \quad \dots(i)$$

At the lowest point $T_1 - mg = mv_1^2/r$

At the top point $T_2 + mg = mv_2^2/r$

where T_1 and T_2 are the tensions of the string at A and B respectively.

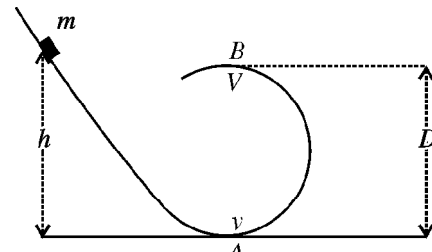


Fig. 2.14.

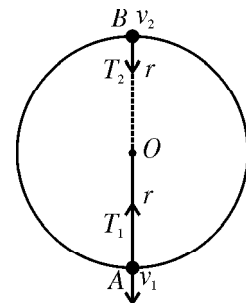


Fig. 2.15.

$$\therefore T_1 - T_2 - 2mg = \frac{mv_1^2}{r} - \frac{mv_2^2}{r} = 4mg \quad [\text{From eq (i)}]$$

or

$$T_1 - T_2 = 6mg.$$

Ex. 7. A particle slides along a track with elevated ends and a flat central part 3 metres long [fig. 2.16]. The curved portions of the track are frictionless while for flat part the coefficient of friction is 0.2. If the particle is released from A (height 1.5 metres above the flat part), find the position where the particle will come to rest.

Sol. Potential energy of the particle at A relative to B = mgh .

When the particle will reach at B along the frictionless track, its potential energy will convert in the form of its kinetic energy $\frac{1}{2}mv^2$. So that

$$\frac{1}{2}mv^2 = mgh \quad \dots(i)$$

Now, the particle, moves a distance, say d , along the flat part BC (for which $\mu = 0.2$) and then comes to rest. Hence the kinetic energy $\frac{1}{2}mv^2$ of the particle will be used against the friction. So that

$$\frac{1}{2}mv^2 = \mu R d = \mu mg d \quad [\because \text{Reaction } R = mg] \quad \dots(ii)$$

From eq. (i) and (ii), we have

$$mgh = \mu mg d \quad \text{or} \quad d = h/\mu = 1.5/0.2 = 7.5 \text{ m.}$$

This distance 7.5 m is greater than the distance $BC = 3$ m. So first, the particle will move frictional part $BC = 3$ m and then it will rise up towards D and will come back to C with the same speed, which it was having previously at C (since elevated part is frictionless). Then the particle will travel the frictional distance $CB = 3$ metres and will rise at B towards A and again on returning it will move a distance **1.5 m** upto O.

Ex. 8. A body of mass m is suspended from a spring. It comes to rest after a downward displacement x_0 . If a linear force is acting, then prove that

(i) the spring (force constant) is mg/x_0 ,

(ii) the gravitational energy lost is mgx_0 .

(iii) the elastic potential energy gained is $\frac{1}{2}mgx_0$.

What happened to the remaining energy ?

Sol. In the equilibrium position, the downward force mg on the spring will be balanced by the restoring force in the spring. So that

$$mg = -\text{Restoring force} = -(-kx_0) = kx_0$$

whence

$$k = mg/x_0 \quad \dots(i)$$

$$\text{Loss in gravitational potential energy} = mgx_0 \quad \dots(ii)$$

$$\text{Elastic potential energy} = \int_0^{x_0} kx \, dx = \frac{1}{2}kx_0^2 = \frac{1}{2}mgx_0 \quad \dots(iii)$$

$$\text{Remaining energy} = mgx_0 - \frac{1}{2}mgx_0 = \frac{1}{2}mgx_0 \quad \dots(iv)$$

This energy is first changed to kinetic energy and finally to heat, when initial oscillations stop.

Ex. 9. A body of mass 1 kg, initially at rest is dropped from a height of 2 metres onto a vertical spring having force constant 490 N/m. Calculate the maximum distance x through which the spring will be compressed.

Sol. Loss in gravitational potential energy = Gain in potential energy by the spring

$$\text{i.e.,} \quad mg(h + x) = \frac{1}{2}kx^2$$

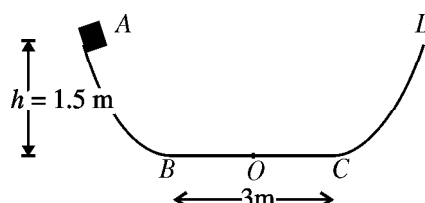


Fig. 2.15.

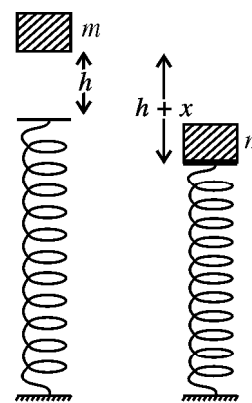


Fig. 2.17.

Here, $m = 1$ kg, $g = 9.8$ m/sec², $h = 2$ m and $k = 490$ N/m

$$\therefore 1 \times 9.8 (2 + x) = \frac{490}{2} x^2 \quad \text{or} \quad 100x^2 = 4x + 8$$

or $25x^2 - x - 2 = 0$

whence $x = \frac{1 \pm \sqrt{1+200}}{50} = \frac{15.177}{50} = 0.3035$ m (Negative value is inadmissible)

Part C : Conservation of Linear Momentum

2.12. Conservation of Linear Momentum

If a force $\vec{F}$ is acting on a particle of mass m , then according to Newton's second law of motion, we have

$$\vec{F} = \frac{d\vec{p}}{dt} = \frac{d}{dt}(m\vec{v}) \quad \dots(1)$$

where $\vec{p} = m\vec{v}$ is the linear momentum of the particle. Momentum is a vector quantity.

If the external force acting on the particle is zero, then

$$\vec{F} = d\vec{p}/dt = 0 \quad \text{or} \quad \vec{p} = m\vec{v} = \text{a constant} \quad \dots(2)$$

Thus, *in absence of an external force, the linear momentum of a particle remains constant.* This is **principle of conservation of linear momentum** for a single particle.

Conservation of linear momentum for a system of two particles : Suppose that in an isolated system there are two particles of masses m_1 and m_2 . No external force is acting on the system. However both particles are exerting interaction forces on each other as in the collision of two particles. We assume that these forces satisfy Newton's third law of motion and we say that these forces are *Newtonian type*.

This is the property of the Newtonian forces that the force $\vec{F}_{21}$ which first particle (1) exerts upon the other (2) is the negative of the force $\vec{F}_{12}$ exerted by second (2) on first (1), i.e.,

$$\vec{F}_{21} = -\vec{F}_{12} \quad \text{or} \quad \vec{F}_{12} + \vec{F}_{21} = 0 \quad \dots(3)$$

If at any instant t , the velocities of the particles are $\vec{v}_1$ and $\vec{v}_2$, then their momenta will be $\vec{p}_1 = m_1\vec{v}_1$ and $\vec{p}_2 = m_2\vec{v}_2$ respectively. Now, according to Newton's second law of motion,

$$\vec{F}_{12} = \frac{d\vec{p}_1}{dt} = \frac{d}{dt}(m_1\vec{v}_1) \quad \text{and} \quad \vec{F}_{21} = \frac{d\vec{p}_2}{dt} = \frac{d}{dt}(m_2\vec{v}_2)$$

Hence eq. (3) assumes the form

$$\frac{d\vec{p}_1}{dt} + \frac{d\vec{p}_2}{dt} = 0 \quad \text{or} \quad \frac{d}{dt}(\vec{p}_1 + \vec{p}_2) = 0$$

i.e., $\vec{p}_1 + \vec{p}_2 = \text{a constant} \quad \text{or} \quad m_1\vec{v}_1 + m_2\vec{v}_2 = \text{a constant} \quad \dots(4)$

Thus, *in absence of external forces on a system of two particles, their total linear momentum is conserved.*

In a collision of two particles, if the velocities of the particles before the collision are $\vec{v}_1$ and $\vec{v}_2$ and after the collision are $\vec{v}_1'$ and $\vec{v}_2'$ respectively, then according to principle of conservation of linear momentum,

$$m_1 \vec{v}_1 + m_2 \vec{v}_2 = m_1 \vec{v}_1' + m_2 \vec{v}_2' \quad \dots(5)$$

This is to be remembered that in an isolated system of two particles, the linear momentum of individual particle may change, but their total linear momentum is always conserved.

Conservation of linear momentum of a system of n particles. Now, let us consider a system of n particles whose masses are $m_1, m_2, \dots, m_n$. The system can be a **rigid body** in which the particles are in fixed positions with respect to one another, or it can be a **collection of particles** in which there may be all kinds of internal motion.

Suppose that the particles of the system are interacting with each other and are also acted by external forces. If at an instant $\vec{p}_1 = m_1 \vec{v}_1, \vec{p}_2 = m_2 \vec{v}_2, \dots, \vec{p}_n = m_n \vec{v}_n$ are the momenta of the particles of masses $m_1, m_2, \dots, m_n$ respectively then the total momentum ($\vec{P}$) of the system is the vector sum of the momenta of individual particles, i.e.,

$$\vec{P} = \vec{p}_1 + \vec{p}_2 + \vec{p}_3 + \dots + \vec{p}_n \quad \dots(6a)$$

$$\text{or} \quad \vec{P} = m_1 \vec{v}_1 + m_2 \vec{v}_2 + \dots + m_n \vec{v}_n \quad \dots(6b)$$

Differentiating it with respect to time t , we have

$$\frac{d\vec{P}}{dt} = \frac{d\vec{p}_1}{dt} + \frac{d\vec{p}_2}{dt} + \dots + \frac{d\vec{p}_n}{dt}$$

$$\text{or} \quad \frac{d\vec{P}}{dt} = \vec{F}_1 + \vec{F}_2 + \dots + \vec{F}_n \quad \dots(7)$$

where $\vec{F}_1, \vec{F}_2, \dots, \vec{F}_n$ represent the forces acting on the particles of masses $m_1, m_2, \dots, m_n$ respectively.

These forces include external and internal forces both. But according to Newton's third law, the internal forces exist in pairs of equal and opposite forces, they balance each other and so do not contribute any thing to the total force. Hence the right hand side sum in eq. (5) represents the resultant force $\vec{F}_{ext}$ only due to the *external forces* acting on all the particles of the system. The internal forces cannot change the total momentum of the system, because being equal and opposite they produce equal and opposite changes in the momentum. Hence, if we want to change the total momentum of the system of the particles, it is necessary to apply external forces on that system. Then the sum of external forces is

$$\vec{F}_{ext} = \frac{d\vec{P}}{dt} = \frac{d}{dt}(\vec{p}_1 + \vec{p}_2 + \dots + \vec{p}_n)$$

If the resultant external force is zero, then

$$\frac{d\vec{P}}{dt} = 0 \quad \text{or} \quad \vec{P} = \text{a constant. i.e., } \vec{P} = \vec{p}_1 + \vec{p}_2 + \vec{p}_3 + \dots + \vec{p}_n = \text{a constant} \quad \dots(8)$$

Thus, *if the resultant external force acting on a system of particles is zero, the total linear momentum of the system remains constant.* This simple but quite general result is called the **law of conservation of linear momentum** for a system of particles. It is to be noted that the momenta of individual particles may change, but their sum, i.e., the total linear momentum remains unaltered in the absence of external forces.

We know that if $\vec{V}$ is the velocity of centre of mass, then the total linear momentum $\vec{P}$ of the system is

$$\vec{P} = M\vec{V}$$

where $M = m_1 + m_2 + \dots$ is the total mass of the system. Hence in absence of external forces, the conservation of linear momentum of a system implies

$$\vec{V} = \text{a constant} \quad \dots(9)$$

Hence, in absence of external forces, the velocity of centre of mass remains constant.

The law of conservation of momentum is a fundamental and exact law of nature. No violation of it has ever been found and it has been thoroughly checked by all kinds of experiments.

2.13. Examples of Conservation of Linear Momentum

According to the principle of conservation of linear momentum, in absence of external forces the total linear momentum of a system is conserved. We are giving below some examples as an application of this principle.

(1) Firing of bullet from a gun : In the beginning when the bullet and the gun are in the rest position, total linear momentum of the system (gun + bullet) is zero. As the bullet is fired, the bullet after coming out of the gun moves with a velocity $\vec{v}$ (say) so that the bullet acquires a forward momentum $m\vec{v}$, where m is the mass of the bullet. In view of the principle of conservation of linear momentum, the total linear momentum of the system, gun + bullet, must remain zero even after the firing of the bullet. Hence the gun should acquire an equal backward momentum, resulting a backward push on the man. If M is the mass of the gun and $\vec{V}$ its velocity, then from the law of conservation of linear momentum,

$$m\vec{v} + M\vec{V} = 0 \quad \text{or} \quad \vec{V} = -m\vec{v}/M \quad \dots(1)$$

Therefore, the gun recoils back with a velocity $\vec{V} = -m\vec{v}/M$ in a direction opposite to the velocity $\vec{v}$ of the bullet. As $m \ll M$, $V \ll v$, i.e., the velocity of the gun is much smaller in comparison to the velocity of the bullet.

(2) Disintegration of a nucleus or blasting of a bomb : Radioactive decay of a nucleus is another example of recoil velocity. Let a nucleus emit an α -particle. If m and M are the masses of the α -particle and the remaining nucleus and their velocities are $\vec{v}$ and $\vec{V}$ respectively after the decay, then the total linear momentum will be $m\vec{v} + M\vec{V}$. If initially the composite nucleus is at rest, then according to the law of conservation of linear momentum,

$$m\vec{v} + M\vec{V} = 0$$

$$\text{Hence the recoil velocity of the nucleus } \vec{V} = -\frac{m\vec{v}}{M} \quad \dots(2)$$

Similarly if a body or a bomb, which is at rest initially, explodes into two (or more) parts, then from the law of conservation of linear momentum [fig. 2.18].

$$m_1\vec{v}_1 + m_2\vec{v}_2 = 0 \quad \dots(3)$$

where the body breaks into two parts after the explosion.

If initially the velocity of the body or bomb of mass m is $\vec{v}$, then from the law of conservation of momentum

$$m\vec{v} = m_1\vec{v}_1 + m_2\vec{v}_2 \quad \dots(4)$$

(3) Motion of two masses connected by a spring : Suppose that two bodies of masses m_1 and m_2 are connected by a spring and are at rest on a horizontal frictionless table. When the masses are pulled outwardly, the spring extends in length and force of tension develops in the spring. Now, if the bodies are left free, then there is no external force on the system. As the total system is at rest in the beginning (at the instant just released), its total linear momentum must remain zero afterwards also. Thus the two masses will move in the opposite directions because only then the total momentum is zero.

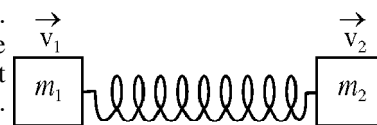


Fig. 2.19.

If m_1 and m_2 are the masses of the two bodies and their respective velocities are $\vec{v}_1$ and $\vec{v}_2$, then according to the principle of conservation of linear momentum, [fig. 2.19]

$$m_1 \vec{v}_1 + m_2 \vec{v}_2 = 0 \text{ or } \vec{v}_2 = -\frac{m_1 \vec{v}_1}{m_2} \quad \dots(5)$$

Obviously the velocities of the two bodies are in opposite direction.

If the magnitudes of the velocities are v_1, v_2 , then

$$\frac{v_2}{v_1} = \frac{m_1}{m_2} \quad \dots(6)$$

This equation will be correct at all instants after the release of the two bodies. The kinetic energy of the bodies will be $K_1 = \frac{1}{2} m_1 v_1^2$ and $K_2 = \frac{1}{2} m_2 v_2^2$. Hence the ratio of their kinetic energies is given by

$$\frac{K_2}{K_1} = \frac{\frac{1}{2} m_2 v_2^2}{\frac{1}{2} m_1 v_1^2} = \frac{m_2^2 v_2^2}{m_1^2 v_1^2} \frac{m_1}{m_2}$$

But $m_1 v_1 = m_2 v_2$ from eq. (6), hence

$$\frac{K_2}{K_1} = \frac{m_1}{m_2} \quad \dots(7)$$

From this equation, it is clear that if the mass m_1 has smaller mass, then its kinetic energy will be greater than that of greater mass m_2 .

Fraction f_1 of the kinetic energy (K_1) of m_1 with respect to the total kinetic energy ($K_1 + K_2$) of the system is given by

$$f_1 = \frac{K_1}{K_1 + K_2} = \frac{\frac{1}{2} m_1 v_1^2}{\frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2} = \frac{\frac{1}{2} m_1 v_1^2}{\frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_1^2 v_1^2 / m_2} = \frac{1}{1 + \frac{m_1}{m_2}} \quad [\because m_1 v_1 = m_2 v_2]$$

$$\text{Thus, } f_1 = \frac{m_2}{m_1 + m_2} \quad \dots(8a)$$

Similarly fraction of kinetic energy of the second mass is

$$f_2 = \frac{m_1}{m_1 + m_2} \quad \dots(8b)$$

Hence if $m_1 < m_2$, fraction of kinetic energy of m_1 will be greater when compared to that of m_2 .

Clearly, the kinetic energy of the moving spring-mass system changes with time, but the division of this energy in the two bodies will not depend on time and will remain constant with lower mass, having greater energy.

Eqs. (7) and (8) are valid for a body falling in the gravitational field of the earth. If m_1 is the mass of the body and m_2 is the mass of the earth, then in their centre of mass frame of reference, the body acquires approximately the total energy ($f_1 = 1$), while the earth acquires extremely small energy ($f_2 \approx 0$). This is the cause the total energy of a body falling in the gravitational field of the earth is assumed to be conserved and the kinetic energy of the earth is not considered in the calculation.

An important example of conservation of linear momentum is the process of collision of two particles which will be discussed later on.

Ex. 1. An electron is accelerated by a potential difference of 20 volts. What is its momentum ?

$$\begin{aligned} \text{Sol. } p &= mv = \sqrt{2m(\frac{1}{2}mv^2)} = \sqrt{2m(eV)} \quad [\because \frac{1}{2}mv^2 = \text{change} \times \text{potential difference} = eV] \\ &= \sqrt{2 \times 0.9 \times 10^{-30} \times 1.6 \times 10^{-19} \times 20} = 2.4 \times 10^{-24} \text{ kg-m/sec.} \end{aligned}$$

Ex. 2. A nucleus, initially at rest, decays radioactively by emitting an electron of momentum 1.73 MeV/c and at right angles to the direction of electron a neutrino with momentum 1 MeV/c. In what direction does the nucleus recoil ? What is its momentum in MeV/c and in S.I. units. If the mass of the residual nucleus is 3.9×10^{-25} kg, calculate its kinetic energy in electron volts.

Sol. Initial momentum = 0

Assuming the direction of the electron as X-axis, we have

momentum of the electron = $1.73\hat{i}$ MeV/c, momentum of the neutrino = $1\hat{j}$ MeV/c,

momentum of the nucleus $\vec{p} = mv$ MeV/c.

From the law of conservation of momentum, we have

$$1.73\hat{i} + 1\hat{j} + m\vec{v} = 0 \quad \text{or} \quad m\vec{v} = -(1.73\hat{i} + 1\hat{j}) \text{ MeV/c.}$$

$\therefore$ Its magnitude = $mv = \sqrt{(1.73)^2 + 1^2} = 2 \text{ MeV/c approx.}$

$$= \frac{2 \times 10^6}{3 \times 10^8} \times 1.6 \times 10^{-19} = 10.66 \times 10^{-22} \text{ kg-m/sec.}$$

Direction cosines of $\vec{p}$ are $(-\frac{\sqrt{3}}{2}, -\frac{1}{2}, 0)$, i.e., direction of the recoil of nucleus makes angle of 210° with X-axis.

Now, the speed of the residual nucleus = $\frac{10.66 \times 10^{-22}}{3.9 \times 10^{-25}} \text{ m/sec.}$

$$\begin{aligned} \therefore \text{K.E. of the residual nucleus} &= \frac{1}{2} mv^2 = \frac{1}{2} (3.9 \times 10^{-25}) \left(\frac{10.66 \times 10^{-22}}{3.9 \times 10^{-25}} \right)^2 \text{ J} \\ &= \frac{1}{2} \frac{(3.9 \times 10^{-25})}{1.6 \times 10^{-19}} \left(\frac{10.66 \times 10^{-22}}{3.9} \right)^2 \text{ eV} = 9.1 \text{ electron-volts.} \end{aligned}$$

2.14. Centre of Mass

The centre of mass of a system of particles or body is that point at which the total mass of a body can be assumed to be concentrated. For the regular geometrical shape of the bodies, the centre of mass is that point about which the body is symmetrical. For the objects of irregular shape, it depends on the distribution of mass.

Let us consider a system of N particles, having masses $m_1, m_2, \dots, m_N$, and instantaneous position vectors $\vec{r}_1, \vec{r}_2, \dots, \vec{r}_N$ respectively in a frame of reference XYZ [fig. 2.20]. The **centre of mass** of this system at this instant is the point whose position vector $\vec{R}$ is given by

$$M\vec{R} = m_1\vec{r}_1 + m_2\vec{r}_2 + \dots + m_N\vec{r}_N \quad \dots(1)$$

where M is the total mass of the system, i.e.,

$$M = \sum m_i = m_1 + m_2 + \dots + m_N$$

$$\text{Therefore, } \vec{R} = \frac{m_1\vec{r}_1 + m_2\vec{r}_2 + \dots + m_N\vec{r}_N}{m_1 + m_2 + \dots + m_N} = \frac{\sum_{i=1}^N m_i \vec{r}_i}{\sum_{i=1}^N m_i} \quad \dots(2)$$

The sum $\sum m_i \vec{r}_i$ is called the **first moment of mass for the system**.

If the co-ordinates of centre of mass of the system are (X, Y, Z) , then

$$\vec{R} = X\hat{i} + Y\hat{j} + Z\hat{k} \quad \dots(3)$$

But $\vec{r}_i = x_i\hat{i} + y_i\hat{j} + z_i\hat{k}$ and then comparing (2) and (3), we get

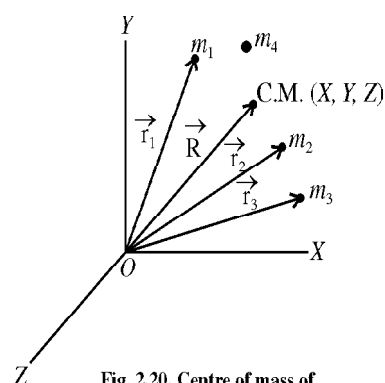


Fig. 2.20. Centre of mass of a system of particles.

$$X = \frac{m_1 x_1 + m_2 x_2 + \dots + m_N x_N}{m_1 + m_2 + \dots + m_N} = \frac{\sum m_i x_i}{\sum m_i}; \quad Y = \frac{m_1 y_1 + m_2 y_2 + \dots + m_N y_N}{m_1 + m_2 + \dots + m_N} = \frac{\sum m_i y_i}{\sum m_i};$$

$$Z = \frac{m_1 z_1 + m_2 z_2 + \dots + m_N z_N}{m_1 + m_2 + \dots + m_N} = \frac{\sum m_i z_i}{\sum m_i} \quad \dots(4)$$

Centre of Mass of a System of Two Particles

If at an instant, the position vectors of two particles are $\vec{r}_1$ and $\vec{r}_2$ in a reference system, then the centre of mass of this two particle-system is given by [fig. 2.21]

$$\vec{R} = \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2}{m_1 + m_2} \quad \dots(5)$$

where m_1 and m_2 are the masses of the two particles.

Hence, the co-ordinates of the centre of mass of two particle system are :

$$X = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2}, \quad Y = \frac{m_1 y_1 + m_2 y_2}{m_1 + m_2}, \quad Z = \frac{m_1 z_1 + m_2 z_2}{m_1 + m_2} \quad \dots(6)$$

From eq. (5),

$$m_1 \vec{r}_1 + m_2 \vec{r}_2 = m_1 \vec{R} + m_2 \vec{R}$$

$$\text{or} \quad m_1 (\vec{r}_1 - \vec{R}) + m_2 (\vec{r}_2 - \vec{R}) = 0$$

$$\text{i.e.,} \quad m_1 \vec{r}_{1C} + m_2 \vec{r}_{2C} = 0 \quad \text{or} \quad m_1 \vec{r}_{1C} = -m_2 \vec{r}_{2C} \quad \dots(7)$$

where $\vec{r}_{1C} = \vec{r}_1 - \vec{R}$ and $\vec{r}_{2C} = \vec{r}_2 - \vec{R}$ are the position vectors of m_1 and m_2 respectively with respect to the centre of mass (C.M.) [fig. 2.21].

From eq. (3), it is clear that the position vectors of m_1 and m_2 are in opposite directions relative to C.M. Hence, the centre of mass exists on the line joining the two particles. From eq. (3) in magnitude,

$$m_1 |\vec{r}_{1C}| = m_2 |\vec{r}_{2C}| \quad \text{or} \quad \frac{|\vec{r}_{1C}|}{|\vec{r}_{2C}|} = \frac{m_2}{m_1} \quad \dots(8)$$

Hence, **the centre of mass divides internally the line joining two particles in inverse ratio of their masses.**

For simplicity, the two particles of masses m_1 and m_2 are taken at a distance d from one-another. If we take the origin O at m_1 and X-axis along the line joining m_1 to m_2 , then the co-ordinates of m_1 and m_2 will be $(0, 0, 0)$ and $(d, 0, 0)$ respectively [fig. 2.22]. The co-ordinates (X, Y, Z) of the centre of mass of this two particle system will be given by

$$X = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2} = \frac{m_2 d}{m_1 + m_2}, \quad Y = \frac{m_1 y_1 + m_2 y_2}{m_1 + m_2} = 0, \quad Z = 0 \quad \dots(9)$$

Thus the co-ordinates of the centre of mass are $\left(\frac{m_2 d}{m_1 + m_2}, 0, 0\right)$. If m_1 and m_2 are at distances d_1 and d_2 respectively from the centre of mass C , then

$$d_1 = \frac{m_2 d}{m_1 + m_2} \quad \text{and} \quad d_2 = d - d_1 = d - \frac{m_2 d}{m_1 + m_2} \quad \text{or} \quad d_2 = \frac{m_1 d}{m_1 + m_2} \quad \dots(10)$$

$$\text{Therefore,} \quad \frac{d_1}{d_2} = \frac{m_2}{m_1} \quad \text{or} \quad \frac{\text{The distance of } m_1 \text{ from C.M.}}{\text{The distance of } m_2 \text{ from C.M.}} = \frac{m_2}{m_1} \quad \dots(11)$$

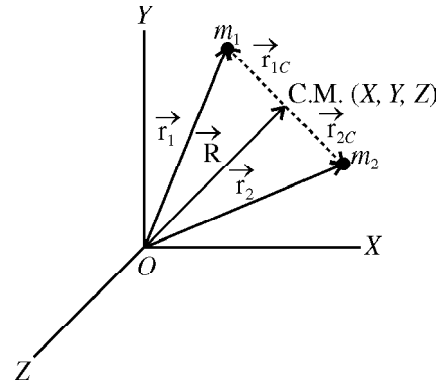


Fig. 2.21. Centre of mass of a system of two particles.

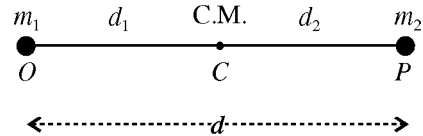


Fig. 2.22.

Centre of Mass of a Rigid Body

A rigid body may be considered as a system of closely packed particles, having continuous distribution of mass. The centre of mass of such a body will be given on replacing the sign of summation (Σ) by the sign of integration in the equation. The centre of mass of (X, Y, Z) a system of particle is given by

$$X = \frac{\sum_i m_i x_i}{\sum_i m_i}, \quad Y = \frac{\sum_i m_i y_i}{\sum_i m_i}, \quad Z = \frac{\sum_i m_i z_i}{\sum_i m_i} \quad \dots(12)$$

If the system of particles is in the form of body of continuous distribution of mass, then we take an infinitesimal small mass dm at (x, y, z) and write the centre of mass (X, Y, Z) of the body as follows:

$$X = \frac{\int x dm}{\int dm} = \frac{1}{M} \int x dm; \quad Y = \frac{\int y dm}{\int dm} = \frac{1}{M} \int y dm; \quad Z = \frac{\int z dm}{\int dm} = \frac{1}{M} \int z dm. \quad \dots(13)$$

where the integration is taken on proper limits.

The position of centre of mass of a symmetrical body having uniform density, can easily be known by symmetry. Hence, the centre of mass of a thin uniform rod is at its mid-point; of a circular disc or sphere is at its centre and so on.

Motion of the Centre of Mass

From eq. (1),

$$M \vec{R} = m_1 \vec{r}_1 + m_2 \vec{r}_2 + \dots + m_N \vec{r}_N$$

Differentiating eq. (1) with respect to time t , we get

$$M \frac{d\vec{R}}{dt} = m_1 \frac{d\vec{r}_1}{dt} + m_2 \frac{d\vec{r}_2}{dt} + \dots + m_N \frac{d\vec{r}_N}{dt}$$

Now, writing $\vec{V}$ for the velocity $d\vec{R}/dt$ of the centre of mass and $\vec{v}_1, \vec{v}_2, \dots$ for the velocities of the particles, we have

$$M \vec{V} = m_1 \vec{v}_1 + m_2 \vec{v}_2 + \dots + m_N \vec{v}_N$$

$$\text{or} \quad M \vec{V} = \sum_i m_i \vec{v}_i = \vec{P} \quad \dots(14)$$

$$\text{or} \quad \vec{V} = \frac{\sum_i m_i \vec{v}_i}{M} = \frac{m_1 \vec{v}_1 + m_2 \vec{v}_2 + \dots + m_N \vec{v}_N}{m_1 + m_2 + \dots + m_N} \quad \dots(15)$$

This equation gives the **velocity of the centre of mass**.

Middle term of eq. (14) is the total linear momentum of the system ($\vec{P}$). Thus, *the total linear momentum of a system is equal to the product of the total mass and velocity of the centre of mass*. In absence of external forces, the total linear momentum ($\sum m_i \vec{v}_i$) remains constant, hence from eq. (14), we have

$$M \vec{V} = \text{a constant} \quad \text{or} \quad \vec{V} = \text{a constant} \quad \dots(16)$$

This is a remarkable property of the centre of mass that its **velocity remains constant in the absence of external forces**. For example, when a radioactive nucleus, moving with constant velocity, decays into a number of particles, the particles move in various directions with different velocities, but the velocity of their centre of mass remains the same. However, if initially the centre of mass is at rest, it will remain at the same position afterwards.

If the eq. (14) is again differentiated with respect to time t , we get

$$\frac{d\vec{P}}{dt} = M \frac{d\vec{V}}{dt} = \sum_N \frac{d}{dt} (m_i \vec{v}_i) = \sum_N \vec{F}_i$$

where $\vec{F}_i$ is the force acting on any particle i of the system. Hence, the total force $\sum_N \vec{F}_i$ on all the particles of the system will be equal to the external force $\vec{F}_{ext}$, because taking all the particles of the system, the forces of their mutual interaction cancel together (Newton's third law), i.e.,

$$\vec{F}_{ext} = \frac{d\vec{P}}{dt} = M \frac{d\vec{V}}{dt} = m \vec{a}_{cm} \quad \dots(17)$$

Thus, the acceleration of the centre of mass is due to only the external forces and is given by Newton's second law of motion. Thus, *the centre of mass of a system of particles moves as if it were a particle of mass equal to the total mass of the system, subjected to the external forces applied to the system.*

Centre of Gravity

The gravitational force on a body is the vector sum of the gravitational forces acting on the individual elements or particles (atoms) of the body. In place of considering all the individual elements of the body, we may say : *"The gravitational force $\vec{F}_G$ acting on a body effectively acts at a single point, called the centre of gravity of the body."* The word effectively implies that the forces on the individual elements are somehow switched off and the force $\vec{F}_G$ is turned on, the net force and net torque on the body will not change.

If $\vec{g}$ is the acceleration due to gravity and it is the same for all elements of the body, then the centre of gravity of the body coincides with the centre of mass. Normally when we consider small bodies on the surface of the earth (or above), the centre of mass and centre of gravity are taken to be the same. However, if a body is very much extended so that $\vec{g}$ is different at its different elements, then the centre of gravity and centre of mass for the body will not coincide. Further the centre of mass is defined independently of any gravitational effect and the concept of centre of mass will remain meaningful where gravitational force may not be considered.

2-15. Centre of Mass Frame of Reference and Laboratory Frame of Reference

No external force acts on an isolated system. Hence the centre of mass of an isolated system moves with constant velocity in an inertial system. *An inertial frame attached with the centre of mass of an isolated system of particles is called the **centre of mass frame of reference** or **C-frame of reference**.* The centre of mass of the system will be at rest with respect to the C-frame. We know that if $\vec{V}$ is the velocity of the centre of mass in an inertial frame, then the total linear momentum of a system of particles is given by

$$\vec{P} = \sum m_i \vec{v}_i = M \vec{V} \quad \dots(1)$$

where $\sum m_i = M$ is the mass of all the particles of the system. But in the C-frame, velocity of centre of mass is zero, i.e., $\vec{V} = 0$. Hence in *the centre of mass frame of reference, the total linear momentum of the system ($\sum m_i \vec{v}_{ic}$) is always zero i.e.,*

$$\sum m_i \vec{v}_{ic} = \sum m_i (\vec{v}_i - \vec{V}) = 0 \quad \dots(2)$$

where $\vec{v}_{ic} = \vec{v}_i - \vec{V}$ = velocity of i th particle relative to the velocity of centre of mass.

For this reason, the C-frame is also called the **zero-momentum frame**. The centre of mass of a system in absence of external forces moves with constant velocity in an inertial frame, so the centre of mass frame of reference for the isolated system is an **inertial frame**. For a system of particles, which is acted by external forces, the C-frame becomes accelerated, i.e., non-inertial and is not of much use.

Centre of mass frame of reference or C-frame is very much useful for the analysis of many experiments, which are difficult to be analyzed in the laboratory frame. If we attach a frame of reference with the laboratory (i.e., the laboratory is at rest in this frame), then such a frame of reference is called **laboratory frame of reference** or **L-frame**.

Suppose in the L-frame, a particle of mass m_1 is moving along X-axis with velocity $\vec{u}$ towards another particle of mass m_2 at rest. The velocity of the centre of mass of this two particle system will be given by

$$\vec{V} = \frac{m_1 \vec{u} + m_2 \times 0}{m_1 + m_2} = \frac{m_1 \vec{u}}{m_1 + m_2} \quad \dots(3)$$

Hence the velocities of the two particles in the C-frame will be

$$\vec{u}_{1C} = \vec{u} - \vec{V} = \vec{u} - \frac{m_1 \vec{u}}{m_1 + m_2} = \frac{m_2 \vec{u}}{m_1 + m_2} \quad \dots(4)$$

and

$$\vec{u}_{2C} = 0 - \vec{V} = -\frac{m_1 \vec{u}}{m_1 + m_2} \quad \dots(5)$$

Also the momenta of the two particles in C-frame will be

$$\vec{p}_{1C} = m_1 \vec{u}_{1C} = \frac{m_1 m_2 \vec{u}}{m_1 + m_2} \quad \text{and} \quad \vec{p}_{2C} = m_2 \vec{u}_{2C} = -\frac{m_1 m_2 \vec{u}}{m_1 + m_2}$$

$$\text{Hence} \quad \vec{p}_{1C} = -\vec{p}_{2C} \quad \text{or} \quad |\vec{p}_{1C}| = |\vec{p}_{2C}| \quad \text{or} \quad m_1 u_{1C} = m_2 u_{2C} \quad \dots(6)$$

Thus in C-frame the magnitude of the linear momenta of the two particles are equal, while in L-frame the linear momenta of the two particles are different, i.e., $m_1 u$ and zero.

Ex. 1. The mass and position vectors of two particles are respectively 6 kg, $(6\hat{i} + 7\hat{j})$ m and 2 kg, $(2\hat{i} + 10\hat{j} - 8\hat{k})$ m. Find the position vector of centre of mass.

Sol. Position vector of the centre of mass of two-particle system is given by

$$\vec{R} = \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2}{m_1 + m_2} = \frac{6(6\hat{i} + 7\hat{j}) + 2(2\hat{i} + 10\hat{j} - 8\hat{k})}{6 + 2} = \frac{40\hat{i} + 62\hat{j} - 16\hat{k}}{8} = 5\hat{i} - 7.75\hat{j} - 2\hat{k} \text{ m.}$$

Ex. 2. If the centre of mass of three particles of masses 10, 20 and 30 gm be at the point (1, 1, 1) m, then where should a fourth particle of 40 gm be placed so that the combined centre of mass be at (0, 0, 0) ?

$$\text{Sol.} \quad X = \frac{m_1 x_1 + m_2 x_2 + m_3 x_3}{m_1 + m_2 + m_3} \quad \text{or} \quad 1 = \frac{0.01 x_1 + 0.02 x_2 + 0.03 x_3}{0.01 + 0.02 + 0.03}$$

$$\text{or} \quad x_1 + 2x_2 + 3x_3 = 6 \quad \dots(i)$$

Suppose 40 gm mass is placed at a point (x_4, y_4, z_4) so that the combined C.M. is at (0, 0, 0). Now,

$$0 = \frac{0.01 x_1 + 0.02 x_2 + 0.03 x_3 + 0.04 x_4}{0.01 + 0.02 + 0.03 + 0.04}$$

$$\text{or} \quad x_1 + 2x_2 + 3x_3 + 4x_4 = 0 \quad \dots(ii)$$

Subtracting eq. (i) from eq. (ii), we have

$$4x_4 = -6 \quad \text{or} \quad x_4 = -3/2$$

In the same way, $y_4 = -3/2$ and $z_4 = -3/2$.

Therefore 40 gm mass should be placed at $(-3/2, -3/2, -3/2)$ m.

Ex. 3. Two particles of masses 100 gm and 300 gm have at a given time position $(2\hat{i} + 5\hat{j} + 13\hat{k})$ m and $(-6\hat{i} + 4\hat{j} - 2\hat{k})$ m respectively, and velocities $(10\hat{i} - 7\hat{j} - 3\hat{k})$ and $(7\hat{i} - 9\hat{j} + 6\hat{k})$ m/sec respectively.

Deduce (i) the instantaneous position of the centre of mass, (ii) the velocity of the second particle in a frame of reference of travelling with the centre of mass.

Sol.

$$(i) \quad \vec{R} = \frac{m_1 r_1 + m_2 r_2}{m_1 + m_2} = \frac{0.1(10\hat{i} - 7\hat{j} - 13\hat{k}) + 0.3(-6\hat{i} + 4\hat{j} + 2\hat{k})}{0.1 + 0.3} = \frac{-16\hat{i} + 17\hat{j} + 7\hat{k}}{4} \text{ m.}$$

$$(ii) \quad \vec{V} = \frac{m_1 v_1 + m_2 v_2}{m_1 + m_2} = \frac{0.1(10\hat{i} - 7\hat{j} - 3\hat{k}) + 0.3(7\hat{i} - 9\hat{j} + 6\hat{k})}{0.1 + 0.3} = \frac{31\hat{i} - 34\hat{j} + 15\hat{k}}{4} \text{ m.}$$

$\therefore$ Velocity of the second particle in the centre of mass frame of reference

$$= \vec{v}_2 - \vec{V} = 7\hat{i} - 9\hat{j} + 6\hat{k} - \frac{31\hat{i} - 34\hat{j} + 15\hat{k}}{4} = \frac{-3\hat{i} - 2\hat{j} + 9\hat{k}}{4} \text{ m/sec.}$$

Ex. 4. The velocities of three particles of masses 20 gm, 30 gm and 50 gm are $10\hat{i}$, $10\hat{j}$ and $10\hat{k}$ cm/sec respectively in an inertial frame. Find the velocity of centre of mass of the system. What are the velocities of the three particles in that frame which is moving with the centre of mass. Assume the axes of the two frames are parallel mutually.

Sol.
$$\vec{V} = \frac{m_1 \vec{v}_1 + m_2 \vec{v}_2 + m_3 \vec{v}_3}{m_1 + m_2 + m_3} = \frac{20(10\hat{i}) + 30(10\hat{j}) + 50(10\hat{k})}{20 + 30 + 50} = 2\hat{i} + 3\hat{j} + 5\hat{k} \text{ cm/sec.}$$

Hence the velocities of the particles in the centre of mass frame of reference are :

$$\vec{v}_{1C} = \vec{v}_1 - \vec{V} = 10\hat{i} - (2\hat{i} + 3\hat{j} + 5\hat{k}) = 8\hat{i} - 3\hat{j} - 5\hat{k} \text{ cm/sec.}$$

$$\vec{v}_{2C} = \vec{v}_2 - \vec{V} = 10\hat{j} - (2\hat{i} + 3\hat{j} + 5\hat{k}) = -2\hat{i} + 7\hat{j} - 5\hat{k} \text{ cm/sec.}$$

$$\vec{v}_{3C} = \vec{v}_3 - \vec{V} = 10\hat{k} - (2\hat{i} + 3\hat{j} + 5\hat{k}) = -2\hat{i} - 3\hat{j} + 5\hat{k} \text{ cm/sec.}$$

Ex. 5. A bomb in flight explodes into two fragments when its velocity is $(10\hat{i} + 2\hat{j})$ m/sec. If the smaller mass M flies with velocity $(20\hat{i} + 50\hat{j})$ m/sec, deduce the velocity of the larger mass $3M$. Deduce also the velocities of the fragments in the centre of mass reference frame, and show these in a diagram.

Sol. Total mass of the bomb $= 3M + M = 4M$

Initial velocity of the bomb $= 10\hat{i} + 2\hat{j}$ m/sec.

$\therefore$ Initial momentum of the system $= 4M(10\hat{i} + 2\hat{j})$

Let $\vec{v}$ be the velocity of the heavier fragment. So that the

final momentum $= M(20\hat{i} + 50\hat{j}) + 3M\vec{v}$

From the law of conservation of momentum, we have

$$M(20\hat{i} + 50\hat{j}) + 3M\vec{v} = 4M(10\hat{i} + 2\hat{j})$$

whence
$$\vec{v} = \frac{1}{3}(20\hat{i} - 42\hat{j}) \text{ m/sec.}$$

There is no external force, hence the centre of mass moves with constant velocity, i.e., $\vec{V} = (10\hat{i} + 2\hat{j})$ m/sec. Further the velocities of the lighter and heavier masses in the centre of mass frame of reference are :

$$20\hat{i} + 50\hat{j} - (10\hat{i} + 2\hat{j}) = (10\hat{i} + 48\hat{j}) \text{ m/sec.}$$

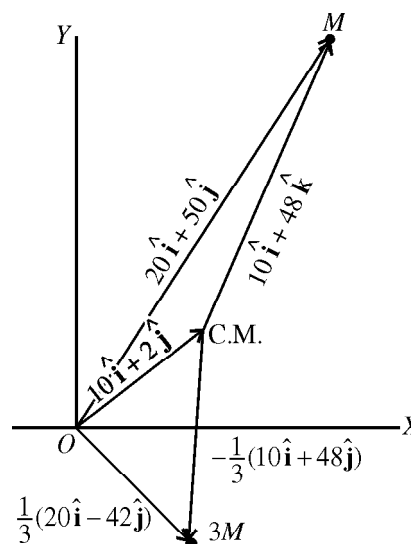


Fig. 2.23.

and $\frac{1}{3}(20\hat{i} - 42\hat{j}) - (10\hat{i} + 2\hat{j}) = \frac{1}{3}(-10\hat{i} - 48\hat{j})$ m/sec. respectively.

Different velocities shown in [fig. 2.23].

Part D : Conservation of Angular Momentum

2.16. Angular Momentum and Torque—Conservation of Angular Momentum

Angular momentum : The angular momentum of a particle is defined as the moment of its linear momentum. It is a vector quantity and denoted by $\vec{L}$.

Suppose that a particle of mass m is moving with velocity $\vec{v}$ in the plane of the paper. Its linear momentum will be $m\vec{v}$. If $\vec{r}$ is the position vector of the particle P from a point O in the plane of the paper, then the moment of the linear momentum, i.e., angular momentum of the particle about O in magnitude will be given by [fig. 2.24]

$$L = mv(ON) = mvr_0$$

or

$$L = mvr \sin \theta = pr \sin \theta \quad \dots(1)$$

where θ is the angle between the vectors $\vec{r}$ and $\vec{p}$ [fig. 2.24].

In vector form, the angular momentum is given by

$$\vec{L} = \vec{r} \times \vec{p} = \vec{r} \times m\vec{v} \quad \dots(2)$$

where $\vec{p} = m\vec{v}$, is the linear momentum in an inertial frame in which the reference point is stationary.

Angular momentum is a vector quantity whose magnitude ($pr \sin \theta$) is equal to the product of linear momentum and its perpendicular distance from the reference point; its direction is perpendicular to both $\vec{r}$ and $\vec{p}$ (or $\vec{v}$). The component of $\vec{L}$ along any line or axis passing through the fixed reference point is often called the angular momentum of the particle about this axis. S.I. units of the angular momentum are kg-m²/sec or J-sec.

If the particle P is in **circular motion** and the reference point O is the centre, [fig. 2.25] then its linear velocity $\vec{v}$ will be given by $\vec{v} = \vec{\omega} \times \vec{r}$,

where $\vec{\omega}$ is the angular velocity of the particle in the perpendicular direction of the plane of the circle. Hence in circular motion, the angular momentum of the particle is given by

$$\vec{L} = \vec{r} \times m\vec{v} = m[\vec{r} \times (\vec{\omega} \times \vec{r})] = m[\vec{\omega}(\vec{r} \cdot \vec{r}) - \vec{r}(\vec{r} \cdot \vec{\omega})] = mr^2\vec{\omega} \quad \dots(3)$$

because $\vec{\omega}$ and $\vec{r}$ are mutually perpendicular so that $\vec{\omega} \cdot \vec{r} = 0$.

Thus, in the circular motion of a particle, its angular momentum (about O) and angular velocity are in the same direction and the angular momentum is given by

$$\begin{aligned} \vec{L} &= mr^2\vec{\omega} \\ \text{Its magnitude} \quad L &= mr^2\omega = mvr \end{aligned} \quad \dots(4)$$

Remember that the moment of inertia of a particle about the axis of rotation (direction of $\vec{\omega}$) is equal to the product of the mass (m) of the particle and square of the perpendicular distance (r^2) from that axis, i.e., $I = mr^2$ and then

$$\vec{L} = I\vec{\omega} \quad \dots(5)$$

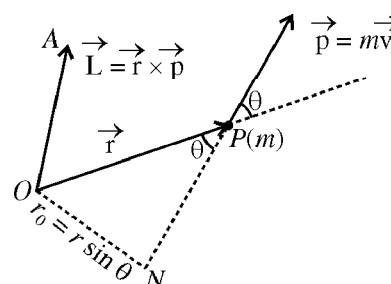


Fig. 2.24. Angular momentum of a particle.

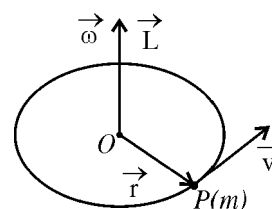


Fig. 2.25. Circular motion.

Torque : Consider a force $\vec{F}$, acting on the point P of a body which can rotate it about a point O . The body may be in the form of a particle at P . The torque or moment of the force (in magnitude) is equal to the product of the magnitude of the force F and its perpendicular distance $r \sin \theta$, i.e., $\tau = Fr \sin \theta$. In vector form, the torque about the point O , is represented as

$$\vec{\tau} = \vec{r} \times \vec{F} \quad \dots(6)$$

where $\vec{r}$ is the vector distance of the point P relative to O [fig. 2.26]. The magnitude of the torque is $Fr \sin \theta$ and its direction is perpendicular to the plane containing $\vec{r}$ and $\vec{F}$.

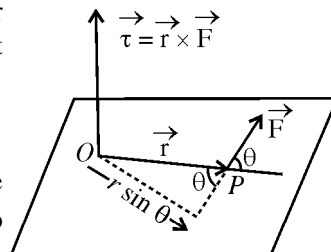


Fig. 2.26 : Torque.

Relation between Angular Momentum and Torque : Differentiating eq. (2) with respect to t , we get

$$\begin{aligned} \frac{d\vec{L}}{dt} &= \frac{d}{dt}(\vec{r} \times \vec{p}) = \frac{d\vec{r}}{dt} \times \vec{p} + \vec{r} \times \frac{d\vec{p}}{dt} \\ \text{or} \quad \frac{d\vec{L}}{dt} &= \vec{r} \times \frac{d\vec{p}}{dt} = \vec{r} \times \vec{F} \end{aligned} \quad \left[\because \frac{d\vec{r}}{dt} \times \vec{p} = \vec{v} \times m\vec{v} = 0 \right]$$

where $\vec{F} = d\vec{p}/dt$ is the force applied on the particle.

The vector product of $\vec{r}$ and $\vec{F}$ is called **torque** or **moment of force** about the reference point and is represented by $\vec{\tau}$. Thus,

$$\vec{\tau} = \frac{d\vec{L}}{dt} = \vec{r} \times \vec{F} \quad \dots(7)$$

Hence, **the torque is equal to the rate of change of angular momentum** ($\vec{L}$). Its unit is newton-metre.

Now, if the torque acting on the particle is zero, then from eq. (7)

$$\vec{\tau} = d\vec{L}/dt = 0 \quad \text{or} \quad \vec{L} = \text{constant} \quad \dots(8)$$

Thus, *in absence of the external torque on a particle, its angular momentum remains constant.* For a rotating particle, this is the **law of conservation of angular momentum**. Remember that

$$\vec{\tau} = \vec{r} \times \vec{F} = 0 \text{ in three cases : (i) } \vec{F} = 0, \text{ (ii) } \vec{r} = 0 \text{ and (iii) } \vec{r} \parallel \vec{F}.$$

2.17. Conservation of Angular Momentum for a System of Particles

If a number of particles are moving independently or attached to one another in the form of a rigid body, then the angular momentum of the system about any point is defined as the vector sum of the angular momenta of the separate particles about the same point.

Now, if $\vec{L}_1, \vec{L}_2, \dots$ are the angular momenta of various particles of a system about a given point, the total angular momentum $\vec{L}$ of the system about the point is given by

$$\vec{L} = \vec{L}_1 + \vec{L}_2 + \dots = (\vec{r}_1 \times m_1 \vec{v}_1) + (\vec{r}_2 \times m_2 \vec{v}_2) + \dots \quad \text{or} \quad \vec{L} = \sum_{n=1}^N (\vec{r}_n \times m \vec{v}_n) \quad \text{or} \quad \sum_{n=1}^N (\vec{r}_n \times \vec{p}_n) \quad \dots(1)$$

where $n = 1, 2, \dots, N$ are the particles in the system.

$$\therefore \text{Torque, } \vec{\tau} = \frac{d\vec{L}}{dt} = \sum_{n=1}^N \vec{r}_n \times \frac{d}{dt}(m_n \vec{v}_n) = \sum_{n=1}^N \vec{r}_n \times \vec{F}_n \quad \dots(2)$$

In this summation, the moments of the internal forces of mutual interaction balance each other, because each pair of equal and opposite collinear forces (action and reaction) will have equal and opposite moments about the reference point so that the addition of these two moments is zero. Hence, *the rate of change of angular momentum of the system about any point is the sum of the torques about that point of all the external forces acting on the system.* Thus $\vec{\tau}$ in eq. (2) represents the external torque on the system.

In a free system left to itself, the external torque $\vec{\tau} = 0$. Then

$$\frac{d\vec{L}}{dt} = 0 \text{ or } \vec{L} = \text{constant. i.e., } \vec{L} = \vec{L}_1 + \vec{L}_2 + \dots = \text{constant} \quad \dots(3)$$

Thus, *the total angular momentum of a system of particles is constant if the resultant external torque acting on the system is zero.* This is **the principle of conservation of angular momentum.**

In the absence of an external torque, total angular momentum ($\vec{L} = \vec{L}_1 + \vec{L}_2 + \dots$) of the system remains constant, although the angular momenta of the particles may mutually interchange.

Angular momentum ($\vec{L}$) and angular momentum about centre of mass ($\vec{L}_{cm}$) : The total angular momentum $\vec{L}$ of a system of particles can be expressed in a convenient and important form by using the velocity of the centre of mass and velocities of the particles relative to the centre of mass. If $\vec{R}$ and $\vec{V}$ are the position vector and velocity of the centre of mass and $\vec{r}_{nC}$ and $\vec{v}_{nC}$ those of a particle of mass m_n relative to the centre of mass, then the position vector and velocity of the particle with respect to the reference point are given by

$$\vec{r}_n = \vec{R} + \vec{r}_{nC} \quad [\because \vec{r}_{nC} = \vec{r}_n - \vec{R}] \text{ and } \vec{v}_n = \vec{V} + \vec{v}_{nC}$$

Hence the total angular momentum for a system of N particles is

$$\begin{aligned} \vec{L} &= \sum_{n=1}^N m_n (\vec{R} + \vec{r}_{nC}) \times (\vec{V} + \vec{v}_{nC}) \\ &= \sum m_n (\vec{R} \times \vec{V}) + \sum m_n (\vec{R} \times \vec{v}_{nC}) + \sum m_n (\vec{r}_{nC} \times \vec{V}) + \sum m_n (\vec{r}_{nC} \times \vec{v}_{nC}) \\ &= \vec{R} \times \vec{V} \sum m_n + \vec{R} \times (\sum m_n \vec{v}_{nC}) + (\sum m_n \vec{r}_{nC}) \times \vec{V} + \sum m_n (\vec{r}_{nC} \times \vec{v}_{nC}) \end{aligned} \quad \dots(4)$$

$$\text{But } \vec{r}_{nC} = \vec{r}_n - \vec{R} \text{ or } m_{nC} \vec{r}_{nC} = m_n \vec{r}_n - m_n \vec{R}$$

$$\text{or } \sum m_{nC} \vec{r}_{nC} = \sum m_n \vec{r}_n - \sum m_n \vec{R} = \sum m_n \vec{r}_n - M \vec{R} \quad \dots(5)$$

where $M = \sum m_n$ = mass of all the particles of the system.

But according to the property of centre of mass, we know that

$$M \vec{R} = m_1 \vec{r}_1 + m_2 \vec{r}_2 + \dots = \sum m_n \vec{r}_n$$

$$\text{Hence from eq. (5), we obtain } \sum m_n \vec{v}_{nC} = 0 \quad \dots(6)$$

$$\text{Also, } \vec{v}_{nC} = \vec{v}_n - \vec{V} \text{ or } m_n \vec{v}_{nC} = m_n \vec{v}_n - m_n \vec{V}$$

$$\text{or } \sum m_n \vec{v}_{nC} = \sum m_n \vec{v}_n - \sum m_n \vec{V} = \sum m_n \vec{v}_n - M \vec{V} \quad \dots(7)$$

The velocity of centre of mass is given by

$$M\vec{V} = m_1\vec{v}_1 + m_2\vec{v}_2 + \dots = \sum m_n\vec{v}_n$$

$$\text{Thus from eq. (7), we have } \sum m_n\vec{v}_{nC} = 0 \quad \dots(8)$$

Hence, from eq. (4), we have

$$\vec{L} = \vec{R} \times M\vec{V} + \sum (\vec{r}_{nC} \times m_n\vec{v}_{nC}) \quad \dots(9)$$

In this equation $\sum (\vec{r}_{nC} \times m_n\vec{v}_{nC})$ represents the angular momentum of the system about the centre of mass, say $\vec{L}_{cm}$, and hence

$$\vec{L} = \vec{L}_{cm} + \vec{R} \times M\vec{V} \quad \dots(10)$$

Now, $M\vec{V} = \vec{P}$ is the total linear momentum of the system. The quantity $\vec{R} \times \vec{P}$ or $\vec{R} \times M\vec{V}$ is the angular momentum of the centre of mass about the reference point. Thus

$$\vec{L} = \vec{L}_{cm} + \vec{R} \times \vec{P} \quad \dots(11)$$

Thus, *the angular momentum of a system of particles about a point or origin (situated in inertial frame) is equal to the sum of angular momentum of the system about the centre of mass and the angular momentum of the centre of mass about the reference point.*

2.18. Examples of Conservation of Angular Momentum

(1) Motion under central force—Motion of a planet or satellite : A central force, acting on a particle, depends only upon the magnitude of its distance from a fixed centre. If $\vec{r}$ is the instantaneous position vector of the particle relative to the fixed centre, then the central force is represented by the relation,

$$\vec{F} = f(r)\hat{r}, \text{ where } f(r) \text{ is a scalar function of distance } r \text{ and } \hat{r} = \vec{r}/r.$$

The torque acting on the particle is

$$\vec{\tau} = d\vec{L}/dt = \vec{r} \times \vec{F} = \vec{r} \times f(r)\hat{r} = f(r)\left[\vec{r} \times \vec{r}/r\right] = 0 \quad [\because \vec{r} \times \vec{r} = 0] \quad \dots(1)$$

$$\text{So that } \vec{L} = \vec{r} \times m\vec{v} = \text{constant} \quad \dots(2)$$

Thus, *the angular momentum of a particle, moving under central force, is conserved.*

We know that a planet moves around Sun under gravitational force and this force is directed towards the centre of Sun. Hence this is a central force and consequently the angular momentum ($\vec{L}$) of the planet is conserved. Similarly, the angular momentum of moon or satellite, moving around the earth, remains constant.

(2) Rotating man with dumb-bells in hands on a turn-table : If a system is rotating with angular velocity ω , then its angular momentum about the axis of rotation is represented as

$$L = I\omega \quad \dots(3)$$

where I is the momentum of inertia of the system about the axis of rotation. Hence for conservation of angular momentum, $I\omega$ should remain constant. This means that if I_1, ω_1 are the moment of inertia and angular velocity of the system about the axis of rotation initially and I_2, ω_2 are the corresponding quantities finally, then in absence of external torque, the conservation of angular momentum of the system demands

$$I_2\omega_2 = I_1\omega_1 \quad \dots(4)$$

Thus, if the moment of inertia of the system decreases i.e., $I_2 < I_1$, then $\omega_2 > \omega_1$ i.e., the system will rotate with more angular velocity.

A simple but well known example for the demonstration of angular momentum is given here. A man stands on a turn table, holding his arms extended [fig. 2.27(a)] with a dumb-bell in each hand. The turn table is free to rotate. If someone sets the man in slow rotation and he then drops his hands to his sides, he will rotate much more rapidly [fig. 2.27(b)]. The total mass of the rotating system remains constant. Hence, a decrease in the distance of the dumb-bells from the axis of rotation results a corresponding decrease in the moment of inertia of the system about the axis of rotation, because the moment of inertia of the system depends upon the distribution of its mass about the axis. Since the angular momentum of the system remains constant, the angular velocity of rotating system is increased due to the decrease in the moment of inertia of the system.

Similarly, a circus acrobat, a diver or a skater performing a pirouette on the toe of one skate takes advantage of this principle.

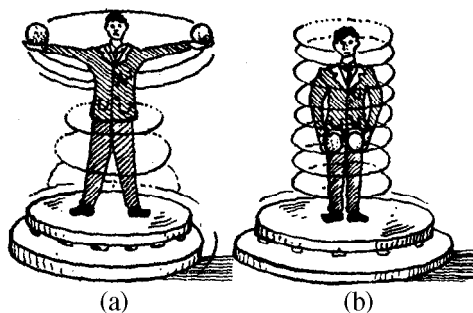


Fig. 2.27.

(3) Scattering of protons or α -particles by a heavy nucleus : Suppose that a proton of charge e and mass m is moving towards a heavy nucleus. The charge on the nucleus is Ze , where Z is the atomic number. Electrostatic force of repulsion will act on the nucleus so that its path will be hyperbola [fig. 2.28]. The perpendicular distance p between the nucleus A and the foot B of the perpendicular drawn from A over the initial direction of the proton is called the **impact parameter**. If the velocity of the proton at infinite distance is v_0 , its initial kinetic energy is $\frac{1}{2}mv_0^2$ and the angular momentum about the nucleus A is mv_0p . If the distance of closest approach of the proton from the nucleus is d , then at this distance kinetic energy = $\frac{1}{2}mv_c^2$, angular momentum = $mv_c d$ and potential energy = Ze^2/d .

The electrostatic force is a central force, hence for the conservation of angular momentum and energy, we have

$$mv_0p = mv_c d \quad \dots(5)$$

$$\text{and} \quad \frac{1}{2}mv_0^2 = \frac{1}{2}mv_c^2 + \frac{1}{4\pi\epsilon_0} \frac{Ze^2}{d} \quad \dots(6)$$

Substituting the value of v_c from eq. (5) in eq. (6), we have

$$\frac{1}{4\pi\epsilon_0} \frac{Ze^2}{d} = \frac{1}{2}mv_0^2 \left\{ 1 - \left(\frac{p}{d} \right)^2 \right\} \quad \dots(7)$$

from which d , the distance of closest approach of the proton, can be calculated.

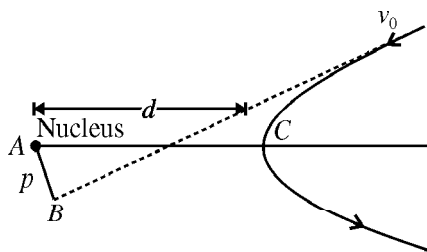


Fig. 2.28 : Scattering of protons by a heavy nucleus.

2.19. Spin and Orbital Angular Momentum

We know that the earth moves in its orbit around the Sun and simultaneously it rotates about its own axis. In the first case the angular momentum of the earth is due to its **orbital motion** and is directed perpendicular to the plane of the orbit. In the second case, the angular momentum is due to the **spinning** of the earth and is along the axis of rotation of the earth. These are called **orbital angular momentum** and **spin angular momentum** respectively. The total angular momentum of the earth about the Sun is the vector sum of these two quantities, whose directions are inclined at $23\frac{1}{2}^\circ$ to each other. Similarly, other planets of the solar system have the orbital and spin angular momenta. The Sun itself is spinning about its axis and hence has spin angular momentum. *The total angular momentum of the solar system is equal to the vector addition of all these angular momenta.* The planets, which are massive and have large radii of orbit, contribute the major portion to the total angular momentum of the system; for example contribution of the planet Jupiter is nearly 60% and that of Saturn nearly 20%. The rotation of the Sun about an axis through its centre (more correctly the centre of mass of the solar system) accounts for only about 2% of the total angular momentum in the solar system.

For the earth, spin angular momentum = $I\omega = \left(\frac{2}{5}mR^2\right)\omega$, ... (1)

where m is the mass of earth, R its radius and $\omega = \frac{2\pi}{24 \times 60 \times 60}$ rad/sec is its angular velocity about the

axis of rotation. Here, $I = \frac{2}{5}mR^2$ is the moment of inertia of the earth about its axis, assuming it in the form of a solid sphere.

Its orbital angular momentum = $mr^2\omega'$... (2)

where r is the distance between the centres of the earth and the Sun and $\omega' = 2\pi/(365 \times 24 \times 60 \times 60)$ radian/sec is its angular velocity about the Sun.

Ex. 1. A 1 kg stone tied at one end of a string is rotated by holding the other end in hand at the rate of 2 revolutions per second. Length of the string is 50 cm. What is the total angular momentum of the stone ?
(A. N. U. 1996)

Sol. Angular momentum in circular motion = $\vec{L} = mr^2\vec{\omega}$

Its magnitude $L = mr^2\omega$

Here, $m = 1$ kg, $n = 2$ rev./sec., $r = 0.5$ m

$$L = mr^2(2\pi n)$$

$$= 1 \times (0.5)^2 \times 2 \times 3.14 \times 2 = \mathbf{3.14 \text{ J-sec.}}$$

The direction of angular momentum is along the angular velocity vector or perpendicular to the plane of circular motion.

Ex. 2. A 500 gm stone is revolved at the end of a 0.4 m long string at the rate of 12.5 rad./s. What is its angular momentum ?
(K.U. 1993)

Sol. As the stone is moving in circular motion, its angular momentum is given by

$$\vec{L} = mr^2\vec{\omega}$$

Its magnitude $L = mr^2\omega$

Here, $m = 0.5$ kg, $r = 0.4$ m and $\omega = 12.5$ rad/sec.

Angular momentum $L = 0.5 \times (0.4)^2 \times 12.5 = \mathbf{1 \text{ joule-second.}}$

Its direction is along angular velocity $\vec{\omega}$ i.e., perpendicular to the plane of circular motion.

Ex. 3. Calculate the angular momentum of a satellite of mass m which moves in a circular orbit of radius r referred to the centre of the orbit. Express the result in terms of G , M , m and r , where M is the mass of earth.

Sol. Angular momentum $\vec{L} = \vec{r} \times m\vec{v}$ or $L = mrv \sin \theta$

Here, for circular orbit $\theta = \pi/2$ and so $\sin \theta = 1$, $\therefore L = mrv$

Its direction is perpendicular to the plane of motion.

Also, $\frac{mv^2}{r} = \frac{GMm}{r^2}$ or $(mvr)^2 = GMm^2r$, $\therefore L = (GMm^2r)^{1/2}$.

Ex. 4. A 100 gm stone is revolved at the end of a 50 cm long string at the rate of 2 revolutions per second. Determine its angular momentum. If after 25 sec it is making only one revolution per second, find the mean torque.

Sol. In case of circular motion,

$$\therefore \vec{L} = mr^2\vec{\omega} = 0.1 \times (0.5)^2 \times (2\pi \times 2) = \mathbf{0.314 \text{ J-sec. } \perp \text{ to the plane of motion.}}$$

Mean torque = $\Delta L/\Delta t = mr^2\Delta\omega/\Delta t = 0.1 \times (0.5)^2 \times 2\pi/25 = \mathbf{6.25 \times 10^{-3} \text{ n/m along } \vec{L}}.$

Ex. 5. Calculate the angular momentum of the spherical earth rotating about its own axis. (Mass of the earth = 6×10^{24} kg and mean radius = 6.4×10^6 m) (A.N.U. 2002)

Sol. Angular momentum of earth rotating about its axis is given by

$$\vec{L} = I \vec{\omega}$$

Here, Moment of inertia of the earth about its own axis is

$$I = \frac{2}{5} MR^2 = \frac{2}{5} \times (6 \times 10^{24}) \times (6.4 \times 10^6)^2 = 9.8 \times 10^{37} \text{ kg-m}^2$$

and angular velocity $\omega = \frac{2\pi}{T} = \frac{2\pi}{24 \times 60 \times 60} = 7.27 \times 10^{-5} \text{ rad.}$

$\therefore L = (9.8 \times 10^{37}) \times (7.27 \times 10^{-5}) = 7.1 \times 10^{33} \text{ kg-m}^2/\text{sec along the axis of rotation.}$

Ex. 6. Find the orbital angular momentum of moon about the earth, given the mass of the moon = 7.36×10^{23} kg, average distance of the moon from the earth = 3.84×10^8 m, period of revolution of moon = 27.3 days. (A.U. 1990)

Sol. Assuming the moon to be moving around earth in a circular orbit, its angular momentum is given by

$$L = mr^2\omega$$

Here, $\omega = \frac{2\pi}{27.3 \times 24 \times 60 \times 60} = 2.663 \times 10^{-6} \text{ rad/sec.}, m = 7.36 \times 10^{23} \text{ kg}, r = 3.84 \times 10^8 \text{ m.}$

$\therefore L = (7.36 \times 10^{23}) \times (3.84 \times 10^8)^2 \times (2.663 \times 10^{-6})$
 $= 2.889 \times 10^{35} \text{ joule-sec. perpendicular to the plane of motion.}$

Ex. 7. The speed of a particle of mass 20 kg, that is moving along a circle of radius 1.5 m, increases at the rate of 0.5 m/sec for every second. Find the torque acting on it. (A.N.U. 2005)

Sol. Torque in circular motion $\vec{\tau} = \vec{r} \times \vec{F}$

Its magnitude $\tau = rF \sin \theta = rF$

because θ between $\vec{r}$ and $\vec{F}$ is $\pi/2$ and $\sin \theta = \sin \pi/2 = 1$.

Here, $r = 1.5 \text{ m}, F = m dv/dt = 20 \times 0.5 = 10 \text{ m/sec}^2$.

$\therefore$ Torque, $\tau = 1.5 \times 10 = 15 \text{ N-m.}$

Its direction is perpendicular to the plane of the $\vec{r}$ and $\vec{F}$ or plane of circular motion.

Ex. 8. Compare the value of the angular momentum of the moon in its orbit about the earth with the value of the angular momentum of rotation of the moon about its own axis.

Sol. Angular momentum of the moon about the earth is $L_1 = mvr = mr^2\omega$

Here, mass of the moon $m = 7.35 \times 10^{22} \text{ kg}$, radius of moon's orbit $r = 3.84 \times 10^8 \text{ m}$ and angular velocity about the earth $\omega = \frac{2\pi}{T} = \frac{2\pi}{2.36 \times 10^6} \text{ rad/sec}$, where $T = 2.36 \times 10^6 \text{ sec}$ is the time of one revolution of the moon round the earth.

$\therefore L_1 = 7.35 \times 10^{22} \times (3.84 \times 10^8)^2 \times \frac{2\pi}{2.36 \times 10^6} = 2.89 \times 10^{34} \text{ J-sec.}$

Now we know that the moon always keeps the same side facing the earth. Therefore, it must make one revolution about its axis in the time it makes one revolution about the earth. Hence, the period (and thus angular frequency) for revolution about its axis is identical to its period of revolution about the earth. Thus the angular momentum L_2 of the moon about its axis is

$$L_2 = I\omega = \frac{2}{5} mR^2\omega, \text{ where } R \text{ is the radius of moon}$$

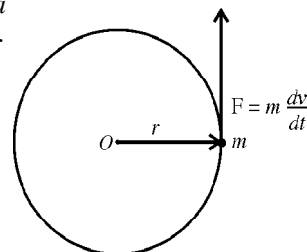


Fig. 2.29.

$$= \frac{2}{5} \times 7.35 \times 10^{22} \times (1.74 \times 10^6)^2 \times \frac{2\pi}{2.36 \times 10^6} = 2.37 \times 10^{29} \text{ J-sec.}, \therefore \frac{L_1}{L_2} = 1.22 \times 10^5.$$

Ex. 9. A particle of mass 20 gm moving in a circle at 4 cm radius with constant speed of 10 cm/sec. What is its angular momentum about (i) the centre of the circle, (ii) a point on the axis of the circle and at 3 cm distance from its centre? Which of these will be in the same direction?

Sol. (i) $L = mvr = 0.02 \times 0.1 \times 0.04 = 8 \times 10^{-5} \text{ J-sec.}$

Its direction is perpendicular to the plane of the circle.

(ii) $L = mvr = 0.02 \times 0.1 \times 0.05 = 10^{-4} \text{ J-sec.}$

Its direction is perpendicular to the plane of the position vector (OA) and instantaneous velocity of the particle, i.e., its direction is OB which is changing with time. In case (i) the direction of L remains the same.

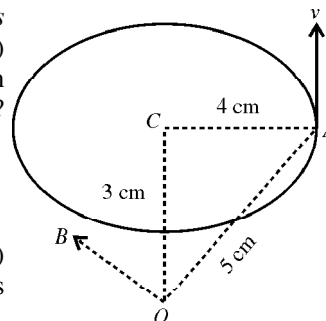


Fig. 2.30.

Ex. 10. A light weight man holds two heavy dumb-bells with outstretched arms while standing on a turn table. A slow push is given to the system so that he rotates at a rate of 1 revolution per minute. Then he pulls the dumb-bells in towards his chest. If the dumb-bells are originally 60 cm from his axis of rotation and are pulled into 10 cm from the axis of rotation, then find the frequency of revolution. Neglect the angular momentum of the man in comparison to that of the dumb-bells.

Sol. If M is the mass of each dumb-bell, then the initial angular momentum of the system,

$$L_1 = 2Mr_1^2\omega_1 = 2Mr_1^2(2\pi n_1) \quad [\because L = mr^2\omega \text{ and } \omega = 2\pi n]$$

where the angular momentum of the man has been left out because of his light weight.

$$\text{Final angular momentum, } L_2 = 2Mr_2^2\omega_2 = 2Mr_2^2(2\pi n_2)$$

$\therefore$ For the conservation of angular momentum $L_1 = L_2$

$$\text{i.e.,} \quad 2Mr_1^2(2\pi n_1) = 2Mr_2^2(2\pi n_2) \text{ or } n_2 = n_1 r_1^2 / r_2^2$$

Here $n_1 = 1 \text{ rev. per minute}$, $r_1 = 0.6 \text{ m}$ and $r_2 = 0.1 \text{ m}$.

$$\therefore n_2 = 1 \times (0.6)^2 / (0.1)^2 = 36$$

Thus, finally the system will rotate at a rate of **36 revolutions per minute**.

Ex. 11. An electron moves about a proton in circular orbit of radius $0.5 \text{ \AA}$. Calculate (i) the orbit angular momentum of the electron about the proton and (ii) the total energy in joules and electron-volts.

Sol. (i) For a circular orbit of the electron, the angular momentum is given by

$$\vec{L} = |\vec{L}| = |\vec{r} \times m\vec{v}| = mvr \quad [\because \vec{r} \text{ and } \vec{v} \text{ are } \perp \text{ to each other for a circular orbit.}]$$

Here the necessary centripetal force for the rotation of the electron in the circular path is provided by the coulombian force between the electron and the proton, i.e.,

$$\frac{mv^2}{r} = \frac{1}{4\pi\epsilon_0} \frac{e^2}{r^2} \text{ or } m^2 v^2 r^2 = \frac{1}{4\pi\epsilon_0} m e^2 r \quad \dots(i)$$

$$\therefore \vec{L} = mvr = \sqrt{me^2 r / 4\pi\epsilon_0} = [9 \times 10^9 \times 9 \times 10^{-31} \times (1.6 \times 10^{-19})^2 \times (0.5 \times 10^{-10})]^{1/2} \\ = 1.02 \times 10^{-34} \text{ J-sec.}$$

$$(ii) \text{ Kinetic energy of the electron, } K = \frac{1}{2} mv^2 = \frac{1}{4\pi\epsilon_0} \frac{e^2}{2r} \quad [\because \text{From relation (i), } mv^2 = \frac{1}{4\pi\epsilon_0} \frac{e^2}{r}]$$

$$\text{Potential energy of the electron, } U = -\frac{1}{4\pi\epsilon_0} \frac{e^2}{r}$$

$$\therefore \text{ Total energy of the electron } = K + U = \frac{1}{4\pi\epsilon_0} \frac{e^2}{2r} - \frac{1}{4\pi\epsilon_0} \frac{e^2}{r} = -\frac{1}{4\pi\epsilon_0} \frac{e^2}{2r}$$

$$\begin{aligned}
&= -9 \times 10^9 \frac{(1.6 \times 10^{-19})^2}{2 \times 0.5 \times 10^{-10}} = -23.04 \times 10^{-19} \text{ joule} \\
&= -\frac{23.04 \times 10^{-19}}{1.6 \times 10^{-19}} = \mathbf{14.4 \text{ eV.}}
\end{aligned}$$

Ex. 12. Determine the impact parameter of a 2 MeV α -particle whose distance of closest approach to a gold nucleus ($Z = 79$) is 2×10^{-11} cm.

Sol. Here, charge on the α -particle = $2e$ and kinetic energy $\frac{1}{2}mv_0^2 = 2 \times 10^6 \times 1.6 \times 10^{-19}$ J

$$\text{But } \frac{1}{4\pi\epsilon_0} \cdot \frac{Ze(2e)}{d} = \frac{1}{2}mv_0^2 \left\{ 1 - \left(\frac{p}{d} \right)^2 \right\}$$

$$\therefore 1 - \left(\frac{p}{d} \right)^2 = \frac{1}{4\pi\epsilon_0} \cdot \frac{2Ze^2}{d} \times \frac{2}{mv_0^2} = 9 \times 10^9 \frac{2 \times 79 \times (1.6 \times 10^{-19})^2}{2 \times 10^{-13} \times 3.2 \times 10^{-13}} = 0.57$$

whence,

$$p = d\sqrt{1-0.57} = 2 \times 10^{-13} \sqrt{0.43} = \mathbf{1.3 \times 10^{-13} \text{ m.}}$$

Ex. 13. A ballet dancer spins about a vertical axis at the rate of one revolution per second with her arms stretched when she folds her arms, her moment of inertia decreases by 60%. Calculate the new rate of revolution. (A.N.U. 2007)

Sol. Let I_1 and I_2 be the moment of inertia of the dancer when the arms are stretched and folded respectively.

As the moment of inertia decreases by 60%, hence

$$I_2 = I_1 - \frac{60}{100} \times I_1 = I_1 - 0.6I_1 = 0.4I_1$$

Applying the law of conservation of angular momentum, we have

$$I_2\omega_2 = I_1\omega_1$$

If n_1 and n_2 are initial and final rate of revolution, then

$$\omega_1 = 2\pi n_1 \text{ and } \omega_2 = 2\pi n_2$$

$$\text{Hence } I_2(2\pi n_2) = I_1 \times (2\pi n_1) \text{ or } I_2 n_2 = I_1 n_1 \text{ or } n_2 = \frac{I_1}{I_2} \times n_1$$

or

$$n_2 = \frac{I_1}{0.4I_1} \times 1 = \mathbf{2.5 \text{ rev./sec.}}$$

Ex. 14. A particle of mass m moves along a curve given by

$$\vec{r} = [\alpha \cos \omega t \hat{i} + \beta \sin \omega t \hat{j}]$$

Find the angular momentum of the particle about the origin.

(O.U. 1990)

$$\text{Sol. Particle velocity } \vec{v} = \frac{d\vec{r}}{dt} = \frac{d}{dt} [\alpha \cos \omega t \hat{i} + \beta \sin \omega t \hat{j}] = \omega [-\alpha \sin \omega t \hat{i} + \beta \cos \omega t \hat{j}]$$

$$\text{Angular momentum } \vec{L} = \vec{r} \times \vec{p} = \vec{r} \times m \vec{v} = m (\vec{r} \times \vec{v})$$

$$= m\omega [\{\alpha \cos \omega t \hat{i} + \beta \sin \omega t \hat{j}\} \times \{-\alpha \sin \omega t \hat{i} + \beta \cos \omega t \hat{j}\}]$$

$$= m\omega [-\alpha^2 \cos \omega t \sin \omega t (\hat{i} \times \hat{i}) + \alpha\beta \cos^2 \omega t (\hat{i} \times \hat{j})$$

$$- \alpha\beta \sin^2 \omega t (\hat{j} \times \hat{i}) + \beta^2 \sin \omega t \cos \omega t (\hat{j} \times \hat{j})]$$

As $\hat{i} \times \hat{i} = 0$, $\hat{j} \times \hat{j} = 0$, $\hat{i} \times \hat{j} = \hat{k}$ and $\hat{j} \times \hat{i} = -\hat{k}$, hence

$$\begin{aligned}\vec{L} &= m\omega [\alpha\beta \cos^2 \omega t \hat{k} + \alpha\beta \sin^2 \omega t \hat{k}] \\ &= m\omega\alpha\beta \hat{k} [\cos^2 \omega t + \sin^2 \omega t] = m\omega\alpha\beta \hat{k}.\end{aligned}$$

Part E : Collisions and Scattering

2-20. Collision of Two Particles (in Three Dimensions)

By collision normally we mean the striking of two bodies or particles. When two particles approach each other, their mutual interaction changes the motion continuously and in consequence to it, their momenta are changed. Such a process is called **collision**. From the physics point of view a collision must not necessarily be a physical contact of the colliding particles. For example, an electron (or proton) approaching an atom is subjected to electrostatic forces and therefore collision takes place.

Let us consider two particles of masses m_1 and m_2 , moving with velocities $\vec{u}_1$ and $\vec{u}_2$ respectively before the collision. If their velocities after the collision are $\vec{v}_1$ and $\vec{v}_2$, then according to the law of conservation of momentum we have

$$m_1 \vec{u}_1 + m_2 \vec{u}_2 = m_1 \vec{v}_1 + m_2 \vec{v}_2 \quad \dots(1)$$

where $\vec{u}_1$, $\vec{u}_2$, $\vec{v}_1$ and $\vec{v}_2$ in general have different directions in space and such a collision is three dimensional.

Here, the terms ‘before’ and ‘after’ the collision mean that the initial (and final) positions are widely separated so that the interaction forces between the particles become effectively zero. Hence the potential energy before and after the collision remains the same.

Two particle collisions are of two types :

(i) Elastic collision : This is the collision in which the kinetic energy of the system is conserved along with the conservation of linear momentum i.e.,

$$\frac{1}{2} m_1 u_1^2 + \frac{1}{2} m_2 u_2^2 = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2 \quad \dots(2)$$

Examples : (1) According to kinetic theory of gases, elastic collision occur between the molecules of a gas.

(2) Collisions between electrons, protons and neutrons are usually elastic.

(ii) Inelastic collision : In an elastic collision, the linear momentum is conserved but the kinetic energies before and after the collision are not equal, i.e.,

$$\frac{1}{2} m_1 u_1^2 + \frac{1}{2} m_2 u_2^2 \neq \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2 \quad \dots(3)$$

Examples : (1) Collisions between gross bodies are always inelastic e.g., collision of two stones.

(2) When two bodies stick together after collision, the collision is completely inelastic e.g., a bullet hitting a target and remaining embedded in it. In case of an inelastic collision between large particles (gross bodies), kinetic energy is converted in some other forms and the lost kinetic energy usually appears in the form of heat in the colliding bodies.

Let us consider the case of inelastic atomic collisions. If the initial kinetic energy of an atom or molecule is sufficiently large, then in collision with some other particle it may absorb some energy. Due to this absorption of energy, its internal structure is changed and we say that the atom has been excited to some higher energy level. The amount of energy used in this process is called *excitation energy**. The final kinetic energy of the system is thus reduced by an amount equal to the excitation energy and then

* The excitation energy has a few definite values for each kind of atom or molecule. Generally, this energy is remitted in the form of light either immediately or after a short interval of time.

$$\frac{1}{2}m_1u_1^2 + \frac{1}{2}m_2u_2^2 = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 + E \quad \dots(4)$$

It is also possible that the atom or molecule may be in the excited state before the collision and during the collision its internal excitation energy may change in the form of kinetic energy, so that we have

$$\frac{1}{2}m_1u_1^2 + \frac{1}{2}m_2u_2^2 + E = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 \quad \dots(5)$$

In general, the collision of two particles is a three dimensional problem. The final velocities $\vec{v}_1$ and $\vec{v}_2$ of the particles cannot be calculated uniquely only by the knowledge of their initial velocities and conservation laws of momentum and energy. The reason lies in the fact that any velocity has three components in three dimensions and hence for the determination of $\vec{v}_1$ and $\vec{v}_2$, six independent equations are required. But equation (1) gives three equations in component form and equations (2), (4) and (5) give only one equation. In case, if after the collision the direction of movement of one particle is also known, then we can calculate $\vec{v}_1$ and $\vec{v}_2$ completely.

2.21. Elastic Collision in One Dimension

Let us consider an elastic head-on collision of two particles of masses m_1 and m_2 , which are moving originally along the same line with velocities u_1 and u_2 respectively [fig. 2.31]. After the collision if their final velocities are v_1 and v_2 , then for the elastic collision from the laws of conservation of momentum and kinetic energy, we have



Fig. 2.31. One dimensional head on collision between two particles.

$$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2 \quad \dots(1)$$

$$\text{and} \quad \frac{1}{2}m_1u_1^2 + \frac{1}{2}m_2u_2^2 = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 \quad \dots(2)$$

From equations. (1) and (2), we have

$$m_1(u_1 - v_1) = m_2(v_2 - u_2) \quad \dots(3)$$

$$\text{and} \quad m_1(u_1^2 - v_1^2) = m_2(v_2^2 - u_2^2) \quad \dots(4)$$

Dividing eq. (4) by (3), we have

$$\frac{u_1^2 - v_1^2}{u_1 - v_1} = \frac{v_2^2 - u_2^2}{v_2 - u_2} \quad \text{or} \quad \frac{(u_1 - v_1)(u_1 + v_1)}{u_1 - v_1} = \frac{(v_2 - u_2)(u_2 + v_2)}{v_2 - u_2}$$

$$\therefore \quad \begin{aligned} u_1 + v_1 &= v_2 + u_2 \\ \text{or} \quad u_1 - u_2 &= v_2 - v_1 \end{aligned} \quad \dots(5)$$

If $u_1 > u_2$, then $v_2 > v_1$.

Thus in an elastic collision between two particles in one dimension, the relative velocity of approach before collision is equal to the relative velocity of separation.

Further, from eq. (3)

$$v_2 - u_2 = \frac{m_1}{m_2}(u_1 - v_1)$$

$$\text{or} \quad v_2 + \frac{m_1v_1}{m_2} = u_2 + \frac{m_1u_1}{m_2} \quad \dots(6)$$

$$\text{Also from (5),} \quad v_2 - v_1 = -u_2 + u_1 \quad \dots(7)$$

Subtracting (7) from (6)

$$\left(\frac{m_1}{m_2} + 1\right)v_1 = 2u_2 + \left(\frac{m_1}{m_2} - 1\right)u_1$$

$$\therefore v_1 = \frac{2u_2}{1 + m_1/m_2} + \frac{\left(\frac{m_1}{m_2} - 1\right)u_1}{1 + m_1/m_2} = \frac{(m_1 - m_2)u_1}{m_1 + m_2} + \frac{2m_2u_2}{m_1 + m_2} \quad \dots(8)$$

Putting the value of v_1 in (7), we get

$$v_2 = \frac{2m_1u_1}{m_1 + m_2} + \frac{(m_2 - m_1)u_2}{m_1 + m_2} \quad \dots(9)$$

Eqs. (8) and (9) give the final velocities of the two colliding particles.

Special Cases

(1) If $m_1 = m_2 = m$, then from eqs. (8) and (9)

$$v_1 = u_2 \text{ and } v_2 = u_1 \quad \dots(10)$$

Thus, in case of **one dimensional elastic collision between two particles of equal masses, the particles exchange their velocities after collision.**

(2) If $u_2 = 0$ i.e., the second particle at rest, then

$$v_2 = \frac{2m_1u_1}{m_1 + m_2} \text{ and } v_1 = \frac{(m_1 - m_2)u_1}{m_1 + m_2} \quad \dots(11)$$

In such a case, we consider three situations :

(a) If $m_1 = m_2 = m$, then

$$v_2 = u_1 \text{ and } v_1 = 0 \quad \dots(12)$$

Thus *after the collision of two particles of equal masses, first particle is stopped and second particle (initially at rest) takes off with the initial velocity of the first particle.*

(b) If $m_2 \gg m_1$, then $v_1 = -u_1$ and $v_2 = 0$

(c) If $m_1 \gg m_2$, then $v_1 = u_1$ and $v_2 = 2u_1$

From (a), we see that in order to slow down the fast neutrons in a nuclear reactor they are made to collide with stationary targets (nuclei) of nearly the same mass as that of the neutrons ($m_1 = m_2$). That is why paraffin (which is rich in hydrogen, having nucleus as proton with nearly the same mass as a neutron) works as a very good moderator.

2-22. Inelastic Collision in One Dimension (*Particles Sticking Together After Collision*)

Consider a perfectly inelastic collision of two particles of masses m_1 and m_2 . Let $\vec{u}_1$ and $\vec{u}_2$ be their respective velocities before collision. If two particles stick together after collision and move with a velocity $\vec{v}$, considering the collision in one dimension, we have

$$\text{momentum of particles before collision} = m_1u_1 + m_2u_2$$

$$\text{momentum of particles after collision} = (m_1 + m_2)v$$

According to the law of conservation of momentum, we have

$$m_1u_1 + m_2u_2 = (m_1 + m_2)v$$

whence

$$v = \frac{m_1u_1 + m_2u_2}{(m_1 + m_2)} \quad \dots(1)$$

Thus, if we know the masses of two particles and their initial velocities before collision, we can obtain their common velocity v after collision.

In case when the second particle is at rest i.e., $u_2 = 0$. From eq. (1), we have

$$v = \frac{m_1u_1}{(m_1 + m_2)} \quad \dots(2)$$

Loss of Kinetic Energy in An Inelastic Collision

$$\text{Kinetic energy before collision, } K_i = \frac{1}{2}m_1u_1^2 \quad \dots(3)$$

$$\text{Kinetic energy after collision, } K_f = \frac{1}{2}(m_1 + m_2)v^2 \quad \dots(4)$$

Dividing eq. (4) by (3), we get

$$\frac{K_f}{K_i} = \frac{\frac{1}{2}(m_1 + m_2)v^2}{\frac{1}{2}m_1u_1^2} \quad \dots(5)$$

Substituting the value of v from eq. (2) in eq. (5), we obtain

$$\frac{K_f}{K_i} = \frac{1}{2}(m_1 + m_2) \left(\frac{m_1u_1}{m_1 + m_2} \right)^2 \times \frac{1}{\frac{1}{2}m_1u_1^2} = \frac{m_1}{(m_1 + m_2)} \quad \dots(6)$$

As $(m_1 + m_2) > m_1$, $K_f/K_i < 1$. Thus the kinetic energy after collision is less than the kinetic energy before collision in an inelastic collision.

$$\begin{aligned} \text{Loss of kinetic energy} &= \frac{1}{2}m_1u_1^2 - \frac{1}{2}(m_1 + m_2)v^2 = \frac{1}{2}m_1u_1^2 - \frac{1}{2}(m_1 + m_2) \left(\frac{m_1u_1}{m_1 + m_2} \right)^2 \\ &= \frac{1}{2}m_1u_1^2 - \frac{m_1^2u_1^2}{2(m_1 + m_2)} = \frac{1}{2}m_1u_1^2 \left[1 - \frac{m_1}{m_1 + m_2} \right] \\ &= \frac{m_1m_2u_1^2}{2(m_1 + m_2)} \quad \dots(7) \end{aligned}$$

which is always positive.

We see from eq. (7) that in an inelastic collision, there is always loss in kinetic energy.

2.23. Elastic Collision in Two Dimensions (Study in L-frame)

Let us consider a particle of mass m_1 , colliding elastically with another particle of mass m_2 , which is initially at rest in the laboratory frame of reference.* Suppose in this frame, the initial velocity of m_1 is $\vec{u}_1$ and after the collision its final velocity be $\vec{v}_1$ in a direction making an angle θ_1 , with its initial path. This angle θ_1 , through which the particle of mass m_1 has been deflected by collision, is called **scattering angle**. Now, if we take the direction of $\vec{u}_1$ along X-axis and the plane containing $\vec{u}_1$ and $\vec{v}_1$ as X-Y plane, then the final velocity $\vec{v}_2$ of m_2 will have no Z-component, because $m_1\vec{u}_1$ and $m_1\vec{v}_1$ have no Z-components. If $\vec{v}_2$

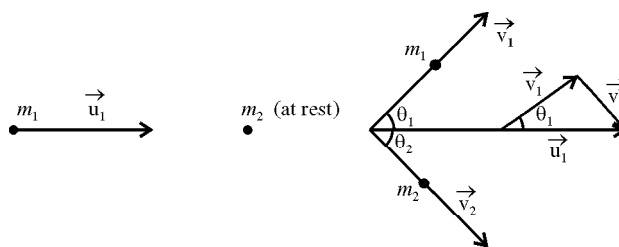


Fig. 2.32 Collision in two dimension in laboratory frame.

makes an angle θ_2 with the initial path of m_1 , we have the following equations for the conservation of momentum and kinetic energy :

$$m_1u_1 = m_1v_1 \cos \theta_1 + m_2v_2 \cos \theta_2 \quad \dots(1)$$

$$m_1v_1 \sin \theta_1 = m_2v_2 \sin \theta_2 \quad \dots(2)$$

$$\frac{1}{2}m_1u_1^2 = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 \quad \dots(3)$$

Here, v_1 , v_2 , θ_1 and θ_2 are the four unknown quantities which cannot be determined with the help

* In case of scattering of α -particles and Compton effect initial velocity of m_2 is zero in the laboratory frame.

of above three equations. In case, if one quantity (say θ_1) is known, the solution of eqs. (1), (2) and (3) will give the values of v_1 , v_2 and θ_2 . Here, we solve the three equations for given θ_1 and $m_1 = m_2$. Hence we obtain eqs. (1) and (2) as follows :

$$u_1 = v_1 \cos \theta_1 + v_2 \cos \theta_2 \quad \dots(4)$$

$$v_1 \sin \theta_1 = v_2 \sin \theta_2 \quad \dots(5)$$

From eq. (4), $u_1 - v_1 \cos \theta_1 = v_2 \cos \theta_2$ or $(u_1 - v_1 \cos \theta_1)^2 = (v_2 \cos \theta_2)^2$

$$\therefore u_1^2 + v_1^2 \cos^2 \theta_1 - 2u_1 v_1 \cos \theta_1 = v_2^2 \cos^2 \theta_2 \quad \dots(6)$$

Squaring both sides of eq. (5), we get

$$v_1^2 \sin^2 \theta_1 = v_2^2 \sin^2 \theta_2 \quad \dots(7)$$

On adding eqs. (6) and (7), we obtain

$$u_1^2 + v_1^2 - 2u_1 v_1 \cos \theta_1 = v_2^2 \quad \dots(8)$$

For $m_1 = m_2$, we have from eq. (3)

$$u_1^2 = v_1^2 + v_2^2 \text{ or } u_1^2 - v_1^2 = v_2^2 \quad \dots(9)$$

If we subtract eq. (9) from eq. (8), we obtain

$$2v_1^2 - 2u_1 v_1 \cos \theta_1 = 0 \text{ or } v_1 - u_1 \cos \theta_1 = 0 \text{ or } v_1 = u_1 \cos \theta_1 \quad \dots(10)$$

Hence we can calculate

$$\text{Now, from eq. (9) and (10), } v_2^2 = u_1^2 - u_1^2 \cos^2 \theta_1 = u_1^2 (1 - \cos^2 \theta_1) = u_1^2 \sin^2 \theta_1$$

$$\therefore v_2 = u_1 \sin \theta_1 \quad \dots(11)$$

Thus we can calculate v_2 .

From eqs. (10) and (11), we obtain on squaring and adding,

$$v_1^2 + v_2^2 = u_1^2 \quad \dots(12)$$

This means that v_1 is perpendicular to v_2 i.e.,

$$\theta_1 + \theta_2 = 90^\circ \quad \dots(13)$$

Thus in a perfectly elastic collision between particles of equal masses, if one particle is initially at rest, the two particles always move off at right angles to each other after collision.

2.24. Elastic Collision in Two Dimensions (Study in C-frame)

Consider a particle of mass m_1 , colliding elastically with another particle of mass m_2 , which is initially at rest in the laboratory frame of reference. Let in this frame, the initial velocity of m_1 be $\vec{u}_1$ and after the collision its final velocity

be $\vec{v}_1$ in a direction making an angle θ_1 , with its initial path [fig. 2.33]. This angle θ_1 , through which the particle of mass m_1 has been deflected by collision, is called **scattering angle**. If we take the direction of $\vec{u}_1$ along X-axis and the plane containing $\vec{u}_1$ and $\vec{v}_1$ as X-Y plane, then the final velocity $\vec{v}_2$ of m_2 will have no

Z-component, because $m_1 \vec{u}_1$ and $m_1 \vec{v}_1$ have no Z-components. Let $\vec{v}_2$ make an angle θ_2 with the initial path of m_1 .

If we consider this problem in laboratory frame, we know that the solution is not possible for all the unknown quantities after the collision. If the above problem is viewed in the centre of mass frame of reference, then its solution is simpler and more informative.

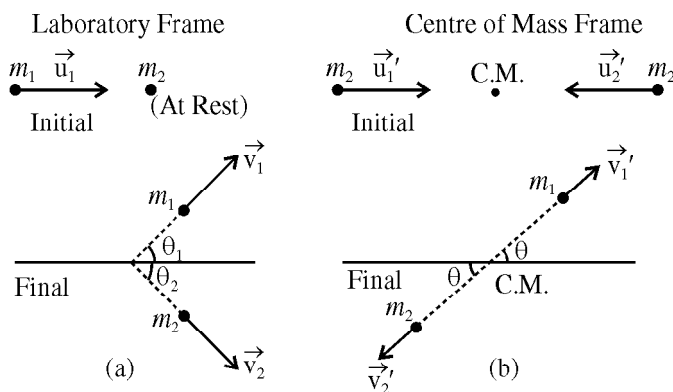


Fig. 2.32.

In the laboratory frame, the velocity $\vec{V}$ of the centre of mass of the system will be given by

$$(m_1 + m_2)\vec{V} = m_1\vec{u}_1 + m_2 \times 0$$

or
$$\vec{V} = \frac{m_1\vec{u}_1}{m_1 + m_2} \quad \dots(1)$$

The centre of mass frame of reference moves with this velocity $\vec{V}$ relative to the laboratory frame i.e., centre of mass (C.M.) is at rest in the centre of mass frame of reference. In this frame, the initial and final velocities of the two particles are given by [fig. 2.33].

$$\vec{u}_1' = \vec{u}_1 - \vec{V}; \quad \vec{u}_2' = -\vec{V} \quad \text{and} \quad \vec{v}_1' = \vec{v}_1 - \vec{V}; \quad \vec{v}_2' = \vec{v}_2 - \vec{V} \quad \dots(2)$$

The centre of mass is at rest in this frame, hence the particles should have equal and opposite momenta, i.e.,

$$m_1\vec{u}_1' = -m_2\vec{u}_2'; \quad m_1\vec{v}_1' = -m_2\vec{v}_2'. \quad \dots(3)$$

This can be proved as follows :

$$m_1\vec{u}_1' = m_1(\vec{u}_1 - \vec{V}) = m_1\left(\vec{u}_1 - \frac{m_1\vec{u}_1}{m_1 + m_2}\right) = \frac{m_1m_2\vec{u}_1}{m_1 + m_2}$$

and
$$-m_2\vec{u}_2' = m_2\vec{V} = m_2\frac{m_1\vec{u}_1}{m_1 + m_2} = \frac{m_1m_2\vec{u}_1}{m_1 + m_2} \quad \text{i.e.,} \quad m_1\vec{u}_1' = -m_2\vec{u}_2'$$

Similarly, we can prove that $m_1\vec{v}_1' = -m_2\vec{v}_2'$.

It means that the final velocities of both particles are making the same angle (θ) with the initial direction [fig. 2.33(b)] and in magnitude

$$m_1u_1' = m_2u_2' \quad \text{and} \quad m_1v_1' = m_2v_2' \quad \dots(4)$$

In this frame, according to the law of conservation of energy, we have

$$\frac{1}{2}m_1u_1'^2 + \frac{1}{2}m_2u_2'^2 = \frac{1}{2}m_1v_1'^2 + \frac{1}{2}m_2v_2'^2 \quad \dots(5)$$

Substituting the values u_2' and v_2' from eq. (4) in eq. (5), we have

$$\frac{1}{2}m_1u_1'^2 + \frac{1}{2}m_2\left(\frac{m_1^2u_1'^2}{m_2^2}\right) = \frac{1}{2}m_1v_1'^2 + \frac{1}{2}m_2\left(\frac{m_1^2v_1'^2}{m_2^2}\right)$$

or
$$\frac{1}{2}m_1u_1'^2\left(1 + \frac{m_1}{m_2}\right) = \frac{1}{2}m_1v_1'^2\left(1 + \frac{m_1}{m_2}\right)$$

whence
$$u_1' = v_1' \quad \dots(6)$$

$\therefore u_2' = v_2' \quad \text{[from eq. (4)]} \quad \dots(7)$

Thus we see that in the centre of mass frame of reference, the magnitudes of the particle velocities are not altered by an elastic collision. Also in this frame the scattering angle θ can have any value. But in the laboratory frame, the scattering angle θ_1 cannot have all values, as shown below :

$$\tan \theta_1 = \frac{v_1 \sin \theta_1}{v_1 \cos \theta_1} = \frac{v_1' \sin \theta}{v_1' \cos \theta + V} \quad \dots(8)$$

[Because from eq. (2), we have $\vec{v}_1 = \vec{v}_1' + \vec{V}$.

In component form, $v_1 \sin \theta_1 = v_1' \sin \theta$

and $v_1 \cos \theta_1 = v_1' \cos \theta + V$, as $\vec{V}$ is along X-axis.]

Now, from eq. (1), we have

$$V = \frac{m_1 u_1}{m_1 + m_2} = \frac{m_1 (u_1' + V)}{m_1 + m_2} \text{ or } (m_1 + m_2) V = m_1 (u_1' + V); \therefore V = \frac{m_1 u_1'}{m_2} = \frac{m_1 v_1'}{m_2} \quad [\because u_1' = v_1']$$

Substituting this value of V in eq. (8), we get

$$\tan \theta_1 = \frac{v_1' \sin \theta}{v_1' \cos \theta + m_1 v_1' / m_2} \text{ or } \tan \theta_1 = \frac{\sin \theta}{\cos \theta + m_1 / m_2} \quad \dots(9)$$

We consider three cases corresponding to $m_1 > m_2$, $m_1 = m_2$ and $m_1 < m_2$.

Case I : if $m_1 > m_2$ or $m_1/m_2 > 1$, the denominator can never be zero, i.e.,

$$\tan \theta_1 < \infty \text{ or } \theta_1 < \pi/2.$$

Thus, we see that if a heavy particle is incident on a light particle initially at rest, the heavy particle will not bounce backward as a result of collision.

Case II : If $m_1 = m_2$, then taking $\cos \theta = -1$, the denominator becomes zero. Hence θ_1 can have values upto $\pi/2$.

Case III : If $m_1 < m_2$ or $m_1/m_2 < 1$, $\tan \theta_1$ can have also negative values and hence in such a case any value of θ_1 is allowed.

2.25. Elastic Oblique Collision

Consider two bodies of masses m_1 and m_2 in spherical form which are moving in different directions

with velocities $\vec{u}_1$ and $\vec{u}_2$ respectively. The two bodies collide with each other obliquely and after collision

they move in different directions with velocities $\vec{v}_1$

and $\vec{v}_2$ respectively [fig. 2.34].

In case, the two bodies move in the same plane before and after collision, we have a *two-dimensional collision*. However, if the bodies collide in such a way that their motions do not confine to the same plane, we have a *three-dimensional collision*.

Here, let us consider a two dimensional oblique collision. We take the line joining the centres of two bodies at the time of collision as X -axis and the line at right angles to it in the plane of the paper as Y -axis.

Let $\vec{u}_1$ and $\vec{u}_2$ make α and β angles and $\vec{v}_1$ and $\vec{v}_2$ make θ and ϕ angles with the X -axis, as shown in [fig. 2.34]

As the velocities of the bodies along Y -axis are not affected by the collision, hence

$$v_1 \sin \theta = u_1 \sin \alpha \quad \dots(1)$$

$$\text{and} \quad v_2 \sin \phi = u_2 \sin \beta \quad \dots(2)$$

For elastic collision, according to Newton's law

$$v_2 \cos \phi - v_1 \cos \theta = u_1 \cos \alpha - u_2 \cos \beta \quad \dots(3)$$

According to the law of conservation of momentum along X -axis, we have

$$m_2 v_2 \cos \phi + m_1 v_1 \cos \theta = m_2 u_2 \cos \beta + m_1 u_1 \cos \alpha \quad \dots(4)$$

Multiply eq. (3) by m_1 and add it to eq. (4) i.e.,

$$(m_1 + m_2) v_2 \cos \phi = (m_2 - m_1) u_2 \cos \beta + 2m_1 u_1 \cos \alpha \quad \dots(5)$$

Dividing both sides of eq. (5) by $(m_1 + m_2)$, we get

$$v_2 \cos \phi = \left(\frac{m_2 - m_1}{m_2 + m_1} \right) u_2 \cos \beta + \frac{2m_1 u_1 \cos \alpha}{m_2 + m_1} \quad \dots(6)$$

If we multiply eq. (3) by m_2 and then subtract eq. (4) from it, we obtain

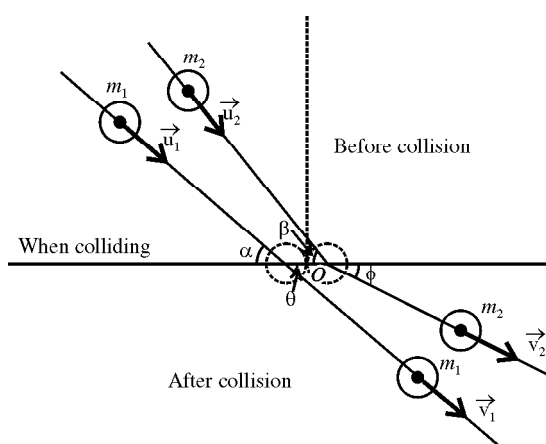


Fig. 2.34 : Elastic oblique collision.

$$v_1 \cos \theta = \left(\frac{m_2 - m_1}{m_2 + m_1} \right) u_1 \cos \alpha + \frac{2m_2 u_2 \cos \beta}{m_2 + m_1} \quad \dots(7)$$

If we square eq. (1) and eq. (7) and then add and take square root, we get the value of final velocity v_1 . Similarly from eq. (2) and eq. (6), we obtain the value of final velocity v_2 . By dividing eq. (1) by eq. (7) and eq. (2) by eq. (6), we obtain $\tan \theta$ and $\tan \phi$. Hence we can obtain the angles θ and ϕ i.e., directions of final velocities $\vec{v}_1$ and $\vec{v}_2$.

Special Cases :

(i) In case the initial velocity of m_2 is zero i.e., $\vec{u}_2 = 0$, then from eq. (2)

$$v_2 \sin \phi = 0 \quad \dots(8)$$

As v_2 is not zero, $\sin \phi = 0$ or $\phi = 0$. This means that **if the mass m_2 were at rest, it will move along the line of impact after collision.**

(ii) If $\vec{u}_2 = 0$ and $m_2 = m_1$, then from eq. (2) $\phi = 0$ and from eq. (7) $\cos \theta = 0$ or $\theta = \pi/2$. Thus, **when a sphere collides obliquely another sphere of the same mass at rest, the directions of motions of the two spheres after collision will be at right angles to each other.**

Ex. 1. In an elastic head-on collision of two particles, assuming the second particle at rest, prove that transference of energy is maximum when their mass ratio is unity. What will happen, if the masses of the two particles are different ?

Sol. Let us consider an elastic head-on collision of two particles of masses m_1 and m_2 . Suppose the mass m_1 be moving with velocity u_1 and m_2 at rest ($u_2 = 0$) initially. After the collision if their final velocities are v_1 and v_2 , then from the laws of conservation of energy and momentum, we have

$$\frac{1}{2} m_1 u_1^2 = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2 \quad \dots(i)$$

$$\text{and} \quad m_1 u_1 = m_1 v_1 + m_2 v_2 \quad \dots(ii)$$

From eqs. (i) and (ii), we have

$$m_1(u_1^2 - v_1^2) = m_2 v_2^2 \quad \dots(iii)$$

$$\text{and} \quad m_1(u_1 - v_1) = m_2 v_2 \quad \dots(iv)$$

Dividing eq. (iii) by eq. (iv), we have

$$u_1 + v_1 = v_2 \quad \dots(v)$$

$$\text{or} \quad v_1 = v_2 - u_1 \quad \dots(vi)$$

Substituting for v_1 from (vi) in eq. (i), we get

$$\frac{1}{2} m_1 u_1^2 = \frac{1}{2} m_1 (v_2 - u_1)^2 + \frac{1}{2} m_2 v_2^2 \quad \text{or} \quad 0 = m_1 v_2^2 - 2m_1 v_2 u_1 + m_2 v_2^2$$

$$\therefore v_2 [v_2(m_1 + m_2) - 2m_1 u_1] = 0$$

whence either $v_2 = 0$, which is impossible or $v_2 = 2m_1 u_1 / (m_1 + m_2)$.

Hence fraction of kinetic energy transferred to the second particle is

$$f = \frac{\frac{1}{2} m_2 v_2^2}{\frac{1}{2} m_1 u_1^2} = \frac{\frac{1}{2} m_2 [2m_1 u_1 / (m_1 + m_2)]^2}{\frac{1}{2} m_1 u_1^2} = \frac{4m_1 m_2}{(m_1 + m_2)^2} = \frac{4m_2 / m_1}{(1 + m_2 / m_1)^2} \quad \dots(vii)$$

When $m_2 / m_1 = 1$, the fraction of transferred kinetic energy $f = \frac{4}{(1+1)^2} = 1$.

Thus, **when the mass ratio is unity, whole of the kinetic energy is transferred to the second particle. In case $m_1 > m_2$ or $m_1 < m_2$, the value of the fraction of transferred kinetic energy will be less than 1.**

Ex. 2. A particle A experiences a perfectly elastic collision with a stationary particle B. Determine their mass ratio if after head-on collision the particles fly apart in opposite directions with equal velocities. (A.N.U. 2001; A.U. 2002)

Sol. According to law of conservation momentum,

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$$

Here, $u_1 = u$, $u_2 = 0$, $v_1 = -v$ and $v_2 = v$ (say)

$$\therefore m_1 u = -m_1 v + m_2 v$$

$$\text{or} \quad v = \left(\frac{m_1}{m_2 - m_1} \right) u \quad \dots(i)$$

According to law of conservation of kinetic energy

$$\frac{1}{2} m_1 u^2 = \frac{1}{2} m_1 v^2 + \frac{1}{2} m_2 v^2$$

$$\text{or} \quad \frac{1}{2} m_1 u^2 = \frac{1}{2} v^2 [m_1 + m_2] \quad \dots(ii)$$

Substituting the value of v from eq. (i) in eq. (ii), we get

$$\frac{1}{2} m_1 u^2 = \frac{1}{2} (m_1 + m_2) \left[\frac{m_1}{(m_2 - m_1)} u \right]^2 \quad \text{or} \quad \frac{1}{2} m_1 u^2 = \frac{1}{2} m_1^2 u^2 \left[\frac{(m_1 + m_2)}{(m_2 - m_1)^2} \right]$$

$$\text{or} \quad 1 = m_1 \left[\frac{(m_1 + m_2)}{(m_2 - m_1)^2} \right] \quad \text{or} \quad (m_2 - m_1)^2 = m_1 (m_1 + m_2)$$

$$\text{or} \quad m_2^2 + m_1^2 - 2m_1 m_2 = m_1^2 + m_1 m_2 \quad \text{or} \quad m_2^2 = 3m_1 m_2 \quad \text{or} \quad m_2 = 3m_1.$$

Therefore, mass ratio $\frac{m_2}{m_1} = 3$.

Ex. 3. A moving particle of mass m collides elastically head-on with a particle of mass $2m$ which is initially at rest. Show that the particle m will lose 8/9th part of its initial kinetic energy after the collision. (A.N.U. 2006)

Sol. We know that in an elastic head-on collision, the final velocity of the first particle is given by

$$v_1 = \frac{2m_2 u_2 + (m_1 - m_2) u_1}{m_1 + m_2}$$

Here, $m_1 = m$, $m_2 = 2m$, $u_1 = u$ (say) and $u_2 = 0$

$$\therefore v_1 = \frac{(m - 2m)u}{m + 2m} = -\frac{u}{3}$$

$$\text{Initial K.E. of the particle of mass } m, = \frac{1}{2} m u^2$$

$$\text{Final K.E. of the particle of mass } m, = \frac{1}{2} m v_1^2 = \frac{1}{2} m \left(-\frac{u}{3} \right)^2 = \frac{1}{2} m \cdot \frac{u^2}{9}$$

$$\begin{aligned} \therefore \text{Loss of kinetic energy of } m \text{ particle} &= \frac{1}{2} m u^2 - \frac{1}{2} m \frac{u^2}{9} \\ &= \frac{1}{2} m u^2 \left(1 - \frac{1}{9} \right) = \frac{8}{9} \times \frac{1}{2} m u^2 \end{aligned}$$

Thus particle m loses 8/9th part of its initial kinetic energy.

Ex. 4. A 4 kg mass moving with a velocity of 20 m/sec collides with another of 12 kg mass moving in opposite direction with a velocity of 40 m/sec. If they stick together during collision, find their common velocity. (S.V.U. 2003; A.U. 2003)

Sol. According to law of conservation of momentum,

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$$

Here, $m_1 = 4$ kg, $m_2 = 12$ kg, $u_1 = 20$ m/sec and $u_2 = -40$ m/sec.

The two masses stick together during the collision. Hence, $v_1 = v_2 = v$ (say), the common velocity of the combined particle.

Thus, $4 \times 20 + 12 \times -40 = (4 + 12)v$

or $80 - 480 = 16v$ or $16v = -400$

$$\therefore \text{Common velocity } v = -\frac{400}{16} = -25 \text{ m/sec.}$$

Thus the combined particle moves with velocity 25 m/sec opposite to the direction of the initial velocity of 4 kg mass.

Ex. 5. Two steel spheres of radii 2 cm and 3 cm move with velocity of 24 cm/sec in opposite directions and collide head-on. If the collision is elastic, calculate the velocities after impact. (A. N.U. 2002)

Sol. If u_1 and u_2 are the initial velocities of masses m_1 and m_2 respectively, then their velocities after an elastic head-on collision are given by

$$v_1 = \frac{(m_1 - m_2)u_1}{m_1 + m_2} + \frac{2m_2 u_2}{m_1 + m_2} \text{ and } v_2 = \frac{2m_1 u_1}{m_1 + m_2} + \frac{(m_2 - m_1)u_2}{m_1 + m_2}$$

Here, $m_1 = \frac{4}{3}\pi r^3 = \frac{4}{3} \times 3.14 \times (0.02)^3$ kg, $m_2 = \frac{4}{3} \times 3.14 \times (0.03)^3$ kg, $u_1 = 0.24$ m/sec and $u_2 = -0.24$ m/sec.

$$\therefore \frac{m_1 - m_2}{m_1 + m_2} = \frac{(0.02)^3 - (0.03)^3}{(0.02)^3 + (0.03)^3} = -\frac{19}{35}, \therefore \frac{m_2 - m_1}{m_1 + m_2} = \frac{19}{35}$$

$$\text{Also, } \frac{2m_1}{m_1 + m_2} = \frac{16}{35} \text{ and } \frac{2m_2}{m_1 + m_2} = \frac{54}{35}$$

$$\text{Hence, } v_1 = \left(-\frac{19}{35}\right) \times 0.24 + \frac{54}{35} \times (-0.24) = -\left(\frac{19+54}{35}\right) \times 0.24 = -\frac{73}{35} \times 0.24 = -0.5006 \text{ m/sec}$$

$$v_2 = \frac{16}{35} \times 0.24 + \frac{19}{35} \times (-0.24) = \left(\frac{16-19}{35}\right) \times 0.24 = -\frac{3}{35} \times 0.24 = -0.0206 \text{ m/sec.}$$

Thus, $v_1 = -50.06$ cm/sec and $v_2 = -2.06$ cm/sec

Negative sign implies that both spheres after collision move in opposite to the direction of the initial velocity of the sphere of 2 cm radius.

Ex. 6. A particle of mass M , moving with a velocity u , makes a head-on collision with a particle of mass m initially at rest so that their final velocities V and v are along the same line. Assuming an elastic collision prove that

$$v = \frac{2u}{1+m/M}.$$

If the particles coalesce on colliding, find the final common velocity and the loss in kinetic energy.

Sol. From the law of conservation of momentum, we have

$$Mu = MV + mv \quad \dots(i)$$

From the law of conservation of energy,

$$\frac{1}{2}Mu^2 = \frac{1}{2}MV^2 + \frac{1}{2}mv^2 \text{ or } u^2 = V^2 + \frac{m}{M}v^2 \quad \dots(ii)$$

Substituting the value of $V^2 = (u - mv/M)^2$ from eq. (i) in eq. (ii), we have

$$u^2 = u^2 + \frac{m^2 v^2}{M^2} - \frac{2mu v}{M} + \frac{m}{M}v^2 \text{ or } 0 = \left(\frac{m}{M} + 1\right)v - 2u, \therefore v = \frac{2u}{1+m/M}.$$

In the second case, $Mu = (m + M)V$ or $V = Mu/(m + M)$

$$\text{Loss of energy} = \frac{1}{2}Mu^2 - \frac{1}{2}(m+M)V^2 = \frac{1}{2}Mu^2 - \frac{1}{2}(m+M)\frac{M^2u^2}{(m+M)^2} = \frac{1}{2}Mu^2 \cdot \frac{m}{m+M}.$$

Ex. 7. A 0.03 kg mass travelling at 0.08 ms^{-1} makes an elastic collision with a 0.05 kg mass at rest. Find the speed of each mass after collision. (A. U. 2001, 1995; S.V.U. 1994)

Sol.
$$v_1 = \frac{(m_1 - m_2)u_1 + 2m_2u_2}{m_1 + m_2} \text{ and } v_2 = \frac{2m_1u_1 + (m_2 - m_1)u_2}{m_1 + m_2}$$

For $u_2 = 0$, $v_1 = \frac{(m_1 - m_2)u_1}{m_1 + m_2}$ and $v_2 = \frac{2m_1u_1}{m_1 + m_2}$

Here, $m_1 = 0.03 \text{ kg}$, $m_2 = 0.05 \text{ kg}$, $u_1 = 0.08 \text{ m/sec}$ and $u_2 = 0$

$$\therefore \frac{m_1 - m_2}{m_1 + m_2} = \frac{0.03 - 0.05}{0.03 + 0.05} = -\frac{0.02}{0.08} = -\frac{1}{4} \text{ and } \frac{2m_1}{m_1 + m_2} = \frac{0.06}{0.08} = \frac{3}{4}$$

Hence
$$v_1 = -\frac{1}{4} \times 0.08 \text{ and } v_2 = \frac{3}{4} \times 0.08$$

or
$$v_1 = -0.02 \text{ m/sec and } v_2 = 0.06 \text{ m/sec.}$$

Ex. 8. Two objects of masses 2 kg and 3 kg moving with velocities 3 m/s and 2 m/s respectively in opposite directions collided with each other and moved with a common velocity. Calculate the loss of kinetic energy. (S.V.U. 2002)

Sol. After the collision, two masses combine together and move with common velocity say v . According to law of conservation of linear momentum,

$$m_1u_1 + m_2u_2 = (m_1 + m_2)v$$

Here, $m_1 = 2 \text{ kg}$, $m_2 = 3 \text{ kg}$, $u_1 = 3 \text{ m/sec}$ and $u_2 = -2 \text{ m/sec}$.

$$\therefore 2 \times 3 + 3 \times -2 = (2 + 3)v \text{ or } 6 - 6 = 5v \text{ or } v = 0$$

Hence after collision, the combined particle comes to rest and hence final kinetic energy is zero.

$$\text{Initial kinetic energy} = \frac{1}{2}m_1u_1^2 + \frac{1}{2}m_2u_2^2 = \frac{1}{2} \times 2 \times 3^2 + \frac{1}{2} \times 3 \times (-2)^2 = 9 + 6 = 15 \text{ joule}$$

$\therefore$ Loss in kinetic energy = **15 J.**

Ex. 9. A body of mass 2 kg collides elastically with another body at rest and then continues to move in the original direction with one-fourth of its original speed. What is the mass of the struck body? (A.N.U. 2002)

Sol. Let m_1 and m_2 be the masses of two bodies and u the initial velocity of m_1 . If v_1 and v_2 be the velocities of m_1 and m_2 after collision, then according to the law of conservation of momentum, we have

$$m_1u = m_1v_1 + m_2v_2$$

Here $m_1 = 2 \text{ kg}$ and $v_1 = u/4$. Therefore,

$$2 \times u = 2 \times u/4 + m_2v_2 \text{ or } m_2v_2 = 3u/2$$

$$\therefore v_2 = 3u/2m_2 \quad \dots(i)$$

Using the law of conservation of kinetic energy, we have

$$\frac{1}{2}m_1u^2 = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2$$

$$\text{i.e., } 2u^2 = 2\left(\frac{u}{4}\right)^2 + m_2v_2^2 \text{ or } u^2 = \frac{u^2}{16} + \frac{m_2}{2}v_2^2 \quad \dots(ii)$$

Substituting the value of v_2 from eq. (i) in eq. (ii), we obtain

$$u^2 = \frac{u^2}{16} + \frac{m_2}{2}\left(\frac{3u}{2m_2}\right)^2 \text{ or } u^2 = \frac{u^2}{16} + \frac{9u^2}{8m_2}$$

or
$$1 = \frac{1}{16} + \frac{9}{8m_2} \quad \text{or} \quad \frac{9}{8m_2} = 1 - \frac{1}{16} = \frac{15}{16}$$

$$\therefore m_2 = \frac{9}{8} \times \frac{16}{15} = 1.2 \text{ kg.}$$

Ex. 10. A 60 kg man is sliding east on the frictionless surface of a frozen pond with a velocity of 0.5 m/sec. He is struck (inelastically) by a 1 kg snow ball whose velocity is 20 m/sec towards the north. What is the man's final velocity ? (A.U. 1989)

Sol. Let $\hat{i}$ and $\hat{j}$ be unit vectors along east and north directions.

Initial linear momentum of the man = $60 \times 0.5 \hat{i}$

Initial linear momentum of the snow ball = $1 \times 20 \hat{j}$

Collision is inelastic. Let the final velocity of the man with snow ball be $\vec{v}$. Conservation of momentum demands

$$60 \times 0.5 \hat{i} + 20 \hat{j} = (60 + 1) \vec{v}$$

Hence
$$\vec{v} = \frac{1}{61} (30 \hat{i} + 20 \hat{j}) = \frac{1}{61} (3 \hat{i} + 2 \hat{j})$$

$$= 0.59 \text{ m/sec at an angle } \theta = \tan^{-1}(2/3) \text{ with the east direction.}$$

Ex. 11. Two particles, having the positions $\vec{r}_1 = (3\hat{i} + 5\hat{j}) \text{ m}$ and $\vec{r}_2 = -(5\hat{i} + 3\hat{j}) \text{ m}$ move with velocities $\vec{v}_1 = (4\hat{i} + 3\hat{j}) \text{ m/sec}$ and $\vec{v}_2 = (a\hat{i} + 7\hat{j}) \text{ m/sec}$. Find :

(i) the value of a , if they collide, (ii) when and where the collision will take place ?

Sol. The two particles will collide after a time t , if they have the same position vectors, i.e.,

$$\vec{r}_1 + \vec{v}_1 t = \vec{r}_2 + \vec{v}_2 t, \quad \therefore (\vec{r}_1 - \vec{r}_2) + (\vec{v}_1 - \vec{v}_2) t = 0 \quad \dots(i)$$

Taking cross product with $(\vec{r}_1 - \vec{r}_2)$, we have

$$(\vec{r}_1 - \vec{r}_2) \times (\vec{v}_1 - \vec{v}_2) = 0 \quad [\because (\vec{r}_1 - \vec{r}_2) \times (\vec{r}_1 - \vec{r}_2) = 0] \quad \dots(ii)$$

$$\therefore (3\hat{i} + 5\hat{j} + 5\hat{i} + 3\hat{j}) \times (4\hat{i} + 3\hat{j} - a\hat{i} - 7\hat{j}) = 0 \quad \text{or} \quad 8(\hat{i} + \hat{j}) \times [(4-a)\hat{i} - 4\hat{j}] = 0$$

or
$$-4\hat{k} - (4-a)\hat{k} = 0 \quad \text{or} \quad 4 + 4 - a = 0 \quad \text{or} \quad a = 8.$$

From eq. (i),
$$t = \frac{(\vec{r}_1 - \vec{r}_2)}{(\vec{v}_1 - \vec{v}_2)} = \frac{-8(\hat{i} + \hat{j})}{-4(\hat{i} + \hat{j})} = 2 \text{ sec.}$$

Place of collision = $\vec{r}_1 + \vec{v}_1 t = 3\hat{i} + 5\hat{j} + (4\hat{i} + 3\hat{j})2 = (11\hat{i} + 11\hat{j}) \text{ m.}$

Ex. 12. Three interacting particles of masses 100, 200 and 400 gm have each a velocity of 20 m/sec magnitude along the positive direction of X-axis, Y-axis and Z-axis respectively. Calculate the velocity of the first particle, when due to forces of interaction the third particle stops moving and the velocity of the second particle is $(10\hat{j} + 5\hat{k}) \text{ m/sec}$.

Sol. Initial momentum = $m_1 \vec{u}_1 + m_2 \vec{u}_2 + m_3 \vec{u}_3 = 2\hat{i} + 4\hat{j} + 8\hat{k} \text{ m/sec.}$

When the third particle stops moving due to their own interactions, then final momentum

$$= m_1 \vec{v}_1 + m_2 \vec{v}_2 + m_3 \vec{v}_3 = 0.1 \vec{v}_1 + 0.2 (10\hat{j} + 5\hat{k})$$

According to the law of conservation of momentum, we have

$$0.1 \vec{v}_1 + 2\hat{j} + \hat{k} = 2\hat{i} + 4\hat{j} + 0.8\hat{k}, \therefore \vec{v}_1 = (20\hat{i} + 20\hat{j} + 70\hat{k}) \text{ m/sec.}$$

[Ballistic pendulum : This pendulum consists of a block of wood, suspended by strings from a rigid support [fig. 2.35] and is used to determine the velocity of a bullet, fired from a gun. By measuring the height to which the block rises, the speed of the bullet can be determined.]

Ex. 13. A bullet of mass m and speed u is allowed to make a completely inelastic collision with a block of much greater mass M [fig. 2.35]. If the block rises to a height h , find the speed of the bullet. Find the loss in energy due to collision.

Sol. If due to inelastic collision the speed of the bullet + block is v , then from the law of conservation of momentum,

$$mu = (m + M) v \quad \dots(i)$$

After the collision the kinetic energy of the bullet + block is given by

$$E = \frac{1}{2}(m + M)v^2$$

Due to this energy the block will rise up and will acquire instantaneous rest position after rising a height h . In this process, the kinetic energy will convert into potential energy, i.e.,

$$\frac{1}{2}(m + M)v^2 = (m + M)gh \text{ or } v = \sqrt{2gh}$$

Substituting the value of v in (i), we get

$$mu = (m + M)\sqrt{2gh},$$

whence

$$u = \frac{m + M}{m}\sqrt{2gh} \quad \dots(ii)$$

$$\text{Loss of energy} = \frac{1}{2}mu^2 - \frac{1}{2}(m + M)v^2 = \frac{1}{2}m\left(\frac{m + M}{m}\right)^2 2gh - \frac{1}{2}(m + M)2gh = (m + M)(M/m)gh.$$

Ex. 14. A 10 gm bullet is fired from a rifle horizontally into a 5 kg block of wood suspended by a string and the bullet gets embedded in the block. The impact causes the block to swing to a height of 5 cm above the initial level. Calculate the velocity of the bullet. (S.V.U. 2007; A.U. 1988)

Sol. Let the velocity of the bullet be u m/sec. Also let v be the velocity of the block with the embedded bullet after impact. Now the system (bullet + block) moves to a height of $h = 5 \text{ cm} = 0.05 \text{ m}$ above its initial level as shown in fig. 2.35.

According to the law of conservation of momentum

$$mu = (M + m)v \text{ or } v = \frac{mu}{(M + m)}$$

Here, $m = 10 \text{ gm} = 0.01 \text{ kg}$ and $M = 5 \text{ kg}$.

$$\therefore v = \frac{0.01u}{(5 + 0.01)} = \frac{0.01u}{5.01} = \frac{u}{501} \quad \dots(i)$$

After the collision the kinetic energy of the system (bullet + block) is $\frac{1}{2}(m + M)v^2$ which will enable to move the system to a height $h = 0.05 \text{ m}$ with potential energy $(m + M)gh$ i.e.,

$$\frac{1}{2}(m + M)v^2 = (m + M)gh \text{ or } v = \sqrt{2gh} = \sqrt{2 \times 9.8 \times 0.05} = \sqrt{0.98} \text{ m/sec} \quad \dots(ii)$$

From (i)

$$u = 501 v = 501 \times \sqrt{0.98} = 4.96 \text{ m/sec.}$$

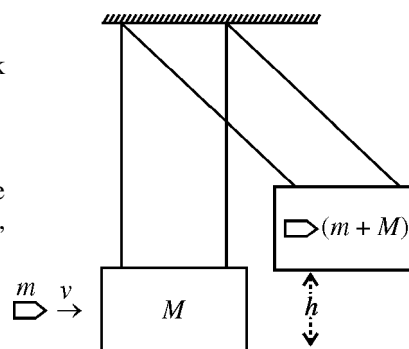


Fig. 2.34.

Ex. 15. A sand bag of mass 10 kg is suspended with a 3 metres long weightless string. A bullet of mass 200 gm is fired with speed 20 m/sec into the bag and stays in the bag. Calculate (i) the speed acquired by the bag, (ii) the maximum displacement of the bag, (iii) energy converted to heat in the collision.

Sol. Mass of the bag $M = 10$ kg; Mass of the bullet $m = 0.2$ kg; Velocity of the bullet $u = 20$ m/sec.; Velocity of the bag $v = ?$

From the law of conservation of momentum,

$$mu = (m + M) v \quad \therefore v = \frac{mu}{m+M} = \frac{0.2 \times 20}{10.2} = \mathbf{0.392 \text{ m/sec.}} \quad \dots(i)$$

After collision, the bag will oscillate like a pendulum. Let the maximum displacement of the bag be x .

Restoring force at this displacement is

$$F = -kx = -mg \sin \theta = -mg \cdot x/l$$

where l is the length of the string.

$$\therefore k = \frac{mg}{l} = \frac{10.2 \times 9.8}{3}$$

At this maximum displacement x , the entire kinetic energy $\left[\frac{1}{2} \times 10.2 \times (0.392)^2\right]$ of the system (bag + bullet) is changed into potential form, i.e., $\int_0^x kx dx = \frac{1}{2} kx^2$.

$$\therefore \frac{1}{2} kx^2 = \frac{1}{2} \times 10.2 \times (0.392)^2$$

$$\text{or } x = \sqrt{\frac{10.2 \times (0.392)^2 \times 3}{10.2 \times 9.8}} = 0.392 \times 0.54 = \mathbf{0.21 \text{ metre.}} \quad \dots(ii)$$

$$\begin{aligned} \text{Energy converted into heat} &= \frac{1}{2} mu^2 - \frac{1}{2} (m+M) v^2 = \frac{1}{2} \times 0.2 \times 20^2 - \frac{1}{2} \times 10.2 \times (0.392)^2 \\ &= 40 - 0.8 = \mathbf{39.2 \text{ joules.}} \quad \dots(iii) \end{aligned}$$

Ex. 16. Two particles of masses m_1 and m_2 stick together on collision. If m_2 is at rest and m_1 is moving along X-axis with velocity $\vec{u}_1$, then calculate the following :

(a) What is the velocity of the combined particle ?

(b) What is the ratio of the kinetic energy after the collision to the initial kinetic energy ?

(c) Describe the motion before and after the collision in the reference frame in which the centre of mass is at rest.

Sol. (a) Let us consider the collision of a particle of mass m_1 , moving with velocity $\vec{u}_1$ with another stationary particle of mass m_2 . If on collision both the particles are attached and have composite velocity $\vec{v}$, then from the law of conservation of momentum we have

$$m_1 \vec{u}_1 = (m_1 + m_2) \vec{v}$$

So that the velocity of the composite particle is

$$\vec{v} = \frac{m_1 \vec{u}_1}{m_1 + m_2} \quad \dots(i)$$

$$(b) \text{ Kinetic energy before the collision } K_i = \frac{1}{2} m_1 u_1^2$$

$$\text{Kinetic energy after the collision } K_f = \frac{1}{2} (m_1 + m_2) v^2$$

$$\therefore \frac{K_f}{K_i} = \frac{\frac{1}{2}(m_1+m_2)v^2}{\frac{1}{2}m_1u_1^2} = \frac{1}{2}(m_1+m_2) \frac{m_1^2u_1^2}{(m_1+m_2)^2} \times \frac{1}{\frac{1}{2}m_1u_1^2} = \frac{m_1}{m_1+m_2} \quad \dots(ii)$$

Evidently $K_f < K_i$ and so the kinetic energy is lost in the collision and this lost energy appears in the internal excitations (heat etc.) of the composite system.

(c) Velocity of the centre of mass

$$\vec{V} = \frac{m_1 \vec{u}_1}{m_1+m_2}$$

In the centre of mass frame of reference, the velocity of the first particle is

$$\vec{u}_1 = \vec{u}_1 - \vec{V} = m_2 \vec{u}_1 / (m_1 + m_2)$$

and that of second particle is

$$\vec{u}_2' = \vec{u}_2 - \vec{V} = -\vec{V} = -m_1 \vec{u}_1 / (m_1 + m_2).$$

On colliding, the particles stick together, the new combined particle has mass $(m_1 + m_2)$ and must be at rest in the centre of mass frame of reference. Relative to the laboratory frame, the new particle has the velocity $\vec{V}$, which is exactly the same as the velocity of the combined particle $\vec{v}$.

Ex. 17. Two objects of masses $m_1 = 200$ gm and $m_2 = 500$ gm possess velocities $\vec{v}_1 = 10\hat{i}$ m/sec. and $\vec{v}_2 = (3\hat{i} + 5\hat{j})$ m/sec just prior to a collision during which they become permanently attached to each other. Calculate (a) the velocity of the centre of mass. (b) the final momentum of the combination in the laboratory frame, (c) the initial and final momenta in the centre of mass frame, (d) what fraction of the initial total kinetic energy is associated with the motion after collision ?

$$\text{Sol. (a)} \quad \vec{V} = \frac{m_1 \vec{v}_1 + m_2 \vec{v}_2}{m_1 + m_2} = \frac{0.2(10\hat{i}) + 0.5(3\hat{i} + 5\hat{j})}{0.2 + 0.5} = \frac{35\hat{i} + 35\hat{j}}{7} \text{ m/sec.}$$

$$(b) \quad \text{Final momentum} = (m_1 + m_2) \vec{V} = 0.7 \vec{V} = 3.5\hat{i} + 2.5\hat{j}.$$

(c) In the centre of mass frame of reference, the centre of mass of the system remains at rest. It means that in the centre of mass frame the initial momentum of the system is zero and hence final momentum of the system will be also zero.

$$(d) \quad K_i = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 = \frac{1}{2}(0.2)(10)^2 + \frac{1}{2}(0.5)[3^2 + 5^2] \\ = 10 + 8.5 = 18.5 \text{ joules.}$$

Now employing the law of conservation of momentum

$$m_1 \vec{u}_1 + m_2 \vec{u}_2 = (m_1 + m_2) \vec{v},$$

where $\vec{v}$ is the final velocity of the combined particle in the laboratory system. Note that $\vec{V}$ is identical to $\vec{v}$.

$$\therefore K_f = \frac{1}{2}(m_1+m_2)v^2 = \frac{1}{2} \times 0.7[5^2 + (25/7)^2] = 13.2 \text{ joules}$$

$$\therefore \frac{K_f}{K_i} = \frac{13.2}{18.5} = 0.72.$$

Ex. 18. A gas molecule having a speed of 30 m/sec. collides elastically with another molecule of the same mass which is initially at rest. After collision, the first molecule moves at an angle of 30° to the initial direction. Find the speed of each molecule after collision. (A.N.U. 1990)

Sol. Since the gas molecule collides elastically with another molecule of the same mass at rest, after collision, the two molecules move at right angles to each other i.e., $\theta_1 = 30^\circ$ and $\theta_2 = 60^\circ$. [fig. 2.36]

Given $u_1 = u = 30$ m/sec. and $u_2 = 0$.

Applying the law of conservation of momentum along the initial direction of motion, we have

$$mu = mv_1 \cos 30^\circ + mv_2 \cos 60^\circ$$

$$\text{or} \quad 30 = v_1 \left(\frac{\sqrt{3}}{2} \right) + v_2 \left(\frac{1}{2} \right)$$

$$\therefore \quad \sqrt{3}v_1 + v_2 = 60 \quad \dots(i)$$

Also applying the law of conservation of momentum perpendicular to the initial direction, we have

$$0 = mv_1 \sin 30^\circ - mv_2 \sin 60^\circ \text{ or } 0 = v_1 \left(\frac{1}{2} \right) - v_2 \left(\frac{\sqrt{3}}{2} \right)$$

$$\therefore \quad v_1 - \sqrt{3} v_2 = 0. \text{ or, } v_1 = \sqrt{3}v_2 \quad \dots(ii)$$

Substituting the value of v_1 in (i), we get

$$3v_2 + v_2 = 60 \text{ or } 4v_2 = 60 \text{ or } v_2 = 15 \quad \dots(iii)$$

$$\therefore \text{ From (i) } v_1 = 15\sqrt{3} \quad \dots(iv)$$

Thus, $v_1 = 15\sqrt{3}$ m/sec. and $v_2 = 15$ m/sec.

Ex. 19. A ball moving at a speed of 2.2 m/sec strikes an identical stationary ball. After collision one ball moves at 1.1 m/sec at 60° angle with the original line of motion. Find the velocity of the other ball. (A.N.U. 2001; S.V.U. 1990)

Sol. Let m be the mass of each ball and u , the initial velocity of the first ball. Also, let v_1 and v_2 be the final velocities of the balls respectively after collision. [fig. 2.32].

Here, $u_1 = u = 2.2$ m/sec. $u_2 = 0$, $\theta_1 = 60^\circ$, $\theta_2 = \theta = ?$, $v_1 = 1.1$ m/sec and $v_2 = ?$

Applying the law of conservation of momentum along original line of motion, we have

$$mu = mv_1 \cos 60^\circ + mv_2 \cos \theta \text{ or } 2.2 = 1.1 \times \frac{1}{2} + v_2 \cos \theta$$

$$\therefore \quad v_2 \cos \theta = 2.2 - 0.55 = 1.65 \quad \dots(i)$$

Also, applying conservation of momentum perpendicular to the original line of motion, we have

$$0 = mv_1 \sin 60^\circ - mv_2 \sin \theta \text{ or } v_2 \sin \theta = 1.1 \times 0.866 [\because \sin 60^\circ = 0.866]$$

$$\therefore \quad v_2 \sin \theta = 0.9526 \quad \dots(ii)$$

Now, squaring and adding eqs. (i) and (ii), we get

$$v_2^2 = (1.65)^2 + (0.9526)^2 \text{ or } v_2 = \sqrt{(1.65)^2 + (0.9526)^2} = 1.9 \text{ m/sec.}$$

Dividing eq. (ii) by eq. (i), we obtain

$$\tan \theta = \frac{0.9526}{1.65} = 0.577, \therefore \theta = \tan^{-1} (0.577) = 30^\circ.$$

Ex. 20. A ball moving with a speed of 18 m/s strikes an identical stationary ball such that after the collision the direction of each ball makes an angle of 30° with the original line of motion. Find the speeds of the two balls after the collision. Is kinetic energy conserved in this collision process?

Sol. If m is the mass of the each ball, then applying the law of conservation of momentum along original direction of motion in the given problem [fig. 2.37].

$$m \times 18 = mv_1 \cos 30^\circ + mv_2 \cos 30^\circ$$

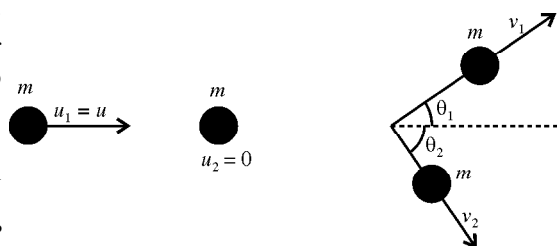


Fig. 2.36.

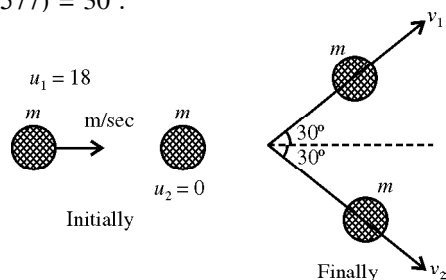


Fig. 2.37.

$$\text{or} \quad 18 = \frac{v_1\sqrt{3}}{2} + \frac{v_2\sqrt{3}}{2} = \frac{\sqrt{3}}{2}(v_1 + v_2)$$

$$\text{or} \quad v_1 + v_2 = \frac{36}{\sqrt{3}} = 12\sqrt{3} \quad \dots(i)$$

The initial momentum of the balls along perpendicular direction = 0

Applying the law of conservation of momentum along the perpendicular direction,

$$0 = mv_1 \sin 30 - mv_2 \sin 30 \text{ or } v_1 - v_2 = 0 \text{ or } v_1 = v_2 \quad \dots(ii)$$

From (i) and (ii), we get

$$v_1 = 6\sqrt{3} \text{ m/sec. and } v_2 = 6\sqrt{3} \text{ m/sec.}$$

$$\text{Kinetic energy before collision, } K_1 = \frac{1}{2}m_1u_1^2 + \frac{1}{2}m_2u_2^2 = \frac{1}{2}m \times 18^2 = 162m$$

$$\begin{aligned} \text{Kinetic energy after collision } K_2 &= \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 \\ &= \frac{1}{2}m(6\sqrt{3})^2 + \frac{1}{2}m(6\sqrt{3})^2 = 108m \end{aligned}$$

Thus $K_1 \neq K_2$. This means that **kinetic energy is not conserved in the collision process.**

Ex. 21. A bomb of mass 40 kg explodes in flight at the instant when its velocity is $(10\hat{i} + 5\hat{j})$ m/sec. It splits into two fragments of mass ratio 1 : 3 and the larger mass is observed to have velocity $v_2 = (10\hat{i} + 50\hat{j} + 10\hat{k})$ m/sec just after explosion.

Find (i) velocity v_1 of smaller fragment just after explosion, and (ii) kinetic energy produced in the explosion.

Sol. (i) According to the law of conservation of momentum,

Initial momentum of the bomb = Final momentum of the fragments

$$m\vec{u} = m_1\vec{v}_1 + m_2\vec{v}_2$$

$$\therefore 40(10\hat{i} + 5\hat{j}) = 10\vec{v}_1 + 30(100\hat{i} + 50\hat{j} + 10\hat{k}) \text{ or } 10\vec{v}_1 + 400\hat{i} + 200\hat{j} - 3000\hat{i} - 1500\hat{j} - 300\hat{k}$$

$$\therefore \vec{v}_1 = -(260\hat{i} + 130\hat{j} + 30\hat{k})$$

$$\text{Its magnitude is } v_1 = \sqrt{\{(260)^2 + (130)^2 + (30)^2\}}$$

$$\text{The magnitude of } \vec{v}_2 \text{ is } v_2 = \sqrt{\{(100)^2 + (50)^2 + (10)^2\}}$$

$$\begin{aligned} \text{(ii) Kinetic energy after explosion} &= \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 \\ &= \frac{1}{2} \times 10 \times [(260)^2 + (130)^2 + (30)^2] + \frac{1}{2} \times 30 \times [(100)^2 + (50)^2 + (10)^2] \\ &= 5 \times 85400 + 15 \times 12600 = 61600 \text{ joule} \end{aligned}$$

$$\text{Kinetic energy before explosion} = \frac{1}{2}mu^2 = \frac{1}{2}40(10^2 + 5^2) = 2500 \text{ joule}$$

Kinetic energy produced in explosion = 61600 joule - 2500 joule = **59100 joule.**

Ex. 22. A shell of mass 2 kg and moving at the velocity of 4 m/sec, suddenly explodes into two equal fragments. The fragments go in directions inclined with the original line of motion with equal velocities. If the explosion imparts 48 joules of translational kinetic energy to the fragment divided equally, find the velocity and direction of each fragment.

Sol. Initial momentum of the shell = $mu = 2 \times 4 = 8 \text{ kg-m/sec}$

After explosion, two fragments of equal masses, 1 kg each, go in direction inclined at angles θ_1 and θ_2 (say) with the original direction of motion with equal velocities, say v (**fig. 2.38**).

According to the law of conservation of momentum along the original direction of motion

$$2u = 1v \cos \theta_1 + 1v \cos \theta_2$$

$$\text{or } 2 \times 4 = v (\cos \theta_1 + \cos \theta_2) \quad \dots(i)$$

According to the law of conservation of momentum along the perpendicular direction

$$2 \times v \sin \theta_1 = 2 \times v \sin \theta_2 = 0$$

$$\text{or } \sin \theta_1 = \sin \theta_2 \text{ or } \theta_1 = \theta_2 = \theta \text{ (say)} \quad \dots(ii)$$

This means that after explosion, two fragments move at the same angles (θ) with the original direction of motion.

Hence from (i), we have

$$8 = 2v \cos \theta \text{ or } v \cos \theta = 4 \quad \dots(iii)$$

$$\text{Initial K.E. of the shell} = \frac{1}{2}mu^2 = \frac{1}{2} \times 2 \times 4^2 = 16 \text{ joules}$$

Energy imparted by the explosion to the two fragments = 48 joules

$\therefore$ Total energy of the two fragments = 48 + 16 = 64 joules

This is divided equally to the fragments. Hence,

Energy of each fragment = 64/2 = 32 joules

$$\text{But kinetic energy of each fragment} = \frac{1}{2} \times 1 \times v^2 = \frac{v^2}{2}$$

$$\frac{v^2}{2} = 32 \text{ joules or } v^2 = 32 \times 2 \text{ or } v = 8 \text{ m/sec.} \quad \dots(iv)$$

Substituting the value of v from (iv) in (iii), we get

$$8 \cos \theta = 4 \text{ or } \cos \theta = \frac{1}{2} \text{ or } \theta = 60^\circ \quad \dots(v)$$

Thus each fragment moves with a velocity of 8 m/sec, in a direction inclined at an angle 60° with the original direction of motion of the shell.

Ex. 23. A body of mass 5 kg, initially at rest explodes and break into three fragments of masses in ratio 1 : 1 : 3. The two pieces of equal mass fly off perpendicularly to each other with a speed of 30 m/sec each. What is the velocity of the heavier fragment ?

Sol. Masses of three fragments = 1 kg, 1 kg and 3 kg.

The two pieces of equal mass, 1 kg, fly off perpendicularly with a speed of 30 m/sec. Let one piece (1 kg) move along X-axis and the other piece (1 kg) along Y-axis (**fig. 2.39**).

Hence their velocities are $30\hat{i}$ and $30\hat{j}$ m/sec and linear momenta are $30\hat{i}$ and $30\hat{j}$ respectively.

Let the velocity of 3 kg fragment be $\vec{v}$. Hence its linear momentum is $3\vec{v}$.

From the law of conservation of momentum, we have
Initial momentum of the body

= Final momentum of fragments

$$0 = 30\hat{i} + 30\hat{j} + 3\vec{v}$$

$$\therefore \vec{v} = -10(\hat{i} + \hat{j})$$

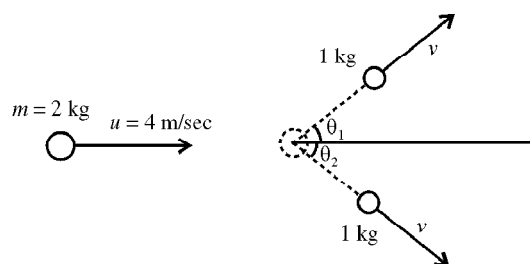


Fig. 2.38.

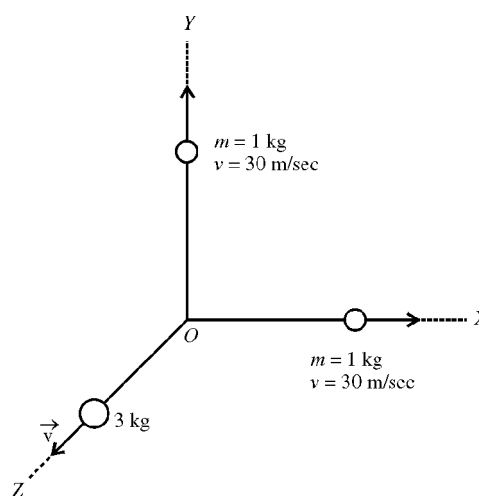


Fig. 2.39.

Magnitude $v = 10\sqrt{2}$ m/sec.

Direction cosine $\cos \theta = -\frac{10}{10\sqrt{2}} = -\frac{1}{\sqrt{2}}$ or $\theta = \cos^{-1}\left(-\frac{1}{\sqrt{2}}\right) = 135^\circ$

Thus the velocity of the third fragment is $10\sqrt{2}$ m/sec.

2.26. Impact Parameter, Scattering Angle and Scattering Cross-Section

The problem of scattering of particles at atomic scale in a central force field is extremely important in modern physics. For example, the scattering of a beam of protons or α -particles by the atomic nuclei of a target is an interesting problem in nuclear physics. Of course, the problem is quantum mechanical, but it can be dealt classically because the procedures for dealing the scattering problem in classical as well as in quantum mechanics are similar. Further a classical study of the scattering problem gives a chance to the beginner to learn the language of the problem.

Let us consider a uniform beam of particles incident on a centre of force. All the particles of the beam have the same mass and energy. The *intensity* of the beam I_0 is the number of particles crossing unit area per unit time normal to the direction of the beam. This I_0 is also called *flux density*. We assume that the force between an incident particle and the particle at the centre of force falls off to zero at large distances.

When a positively charged particle, say an α -particle or proton, approaches a massive nucleus of positive charge $+Ze$ (centre of force), it will experience force of repulsion and the path of the particle will deviate from incident straight line. In fact, the path AB of the α -particle is hyperbola and the nucleus $(+Ze)$ is at its focus [fig. 2.40]. After passing the centre of force, as the particle goes away, the force acting on it will decrease and finally at large distances, the force will become zero. This results again in the straight line motion but in general in a different direction and we say that the particle has been *scattered*.

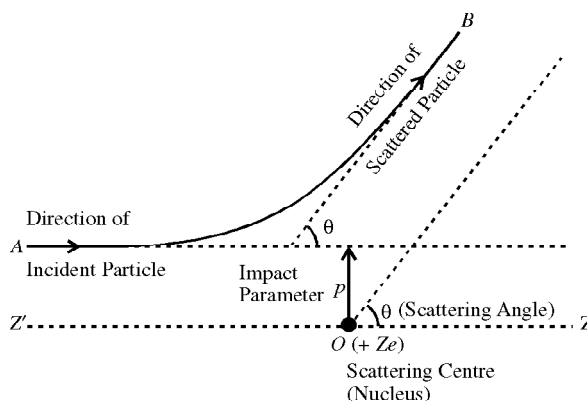


Fig. 2.40 : Impact parameter and scattering angle.

Scattering angle (θ)

The angle between the incident and scattered directions of the particle is called scattering angle and is denoted by θ .

Impact Parameter (p)

If we draw a perpendicular on the direction of the incident particle from the scattering centre, then the length of this perpendicular is known as impact parameter p .

Scattering Cross-section

Consider a uniform beam of particles, moving with a flux of I_0 particles per unit area towards a scattering centre (e.g., an atom of a target). Imagine that the scattering centre presents an area $d\sigma$ perpendicular to the path of the beam such that whatever particles hit $d\sigma$ area are scattered into a solid angle $d\Omega$ [fig. 2.41]. Thus the number of particles, scattered into $d\Omega$ solid angle per second are $I_0 d\sigma$. If $I(\Omega)$ (intensity of the scattered particles) is defined as the number of particles scattered in the direction Ω per unit solid angle per unit time, then the number of scattered particles in the small solid angle $d\Omega$ about Ω direction is given by

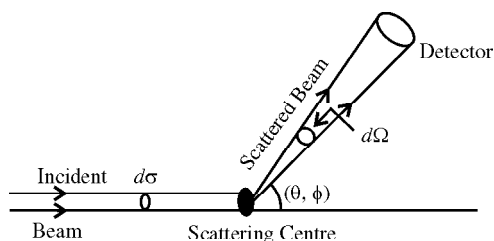


Fig. 2.41 : Differential scattering cross-section.

$$I_0 d\sigma = I(\Omega) d\Omega \text{ or } \frac{d\sigma}{d\Omega} = \frac{I(\Omega)}{I_0} \quad \dots(1)$$

The quantity $\frac{d\sigma}{d\Omega} = \sigma(\Omega)$ is called **differential scattering cross-section** or simply scattering cross-section for scattering in Ω -direction i.e.,

$$\sigma(\Omega) = \frac{d\sigma}{d\Omega} = \frac{I(\Omega)}{I_0} \quad \dots(2)$$

Thus, **the differential scattering cross-section is the ratio of the number of the scattered particles per second per unit solid angle and the flux density of the incident particles.**

Total area presented for scattering by the scattering centre is called **total scattering cross-section** i.e.,

$$\sigma = \int d\sigma = \int \sigma(\Omega) d\Omega = \frac{\int I(\Omega) d\Omega}{I_0} \quad \dots(3)$$

The integral $\int I(\Omega) d\Omega$ gives us the total number of scattered particles per unit time, say N . Hence

$$\sigma = \frac{N}{I_0} \quad \dots(4)$$

Thus, **the ratio of the total number of scattered particles per unit time and the flux density of incident particles is called total scattering cross-section (σ).**

Demensions of scattering cross-section :

$$[\sigma] = \frac{[N]}{[I_0]} = \frac{[\text{Number of scattered particles per unit time}]}{[\text{Number of incident particles per unit time per unit area}]} = \frac{[T^{-1}]}{[T^{-1}L^{-2}]} = [L^2]$$

The unit of area is the unit of σ and the area presented transversely to the incident beam for scattering by the centre of force is the total scattering cross-section; this is why this area of transverse section is called cross-section. In nuclear physics, the diameter of a scattering centre is of the order of 10^{-14} to 10^{-15} m. Hence the cross-section is measured in **barn** or **millibarn**.

$$1 \text{ barn} = 10^{-28} \text{ m}^2$$

As the force is central, hence there must be complete symmetry about the axis of incident beam ($Z'Z$) [fig. 2.42]. Solid angle $d\Omega$ will be given by*

$$d\Omega = 2\pi \sin \theta d\theta \quad \dots(5)$$

where θ is the scattering angle. The incident particles crossing through area $d\sigma = 2\pi p dp$, lying between $p + dp$, are given by

$$I_0 d\sigma = I_0 2\pi p dp \quad \dots(6)$$

$$* \quad d\Omega = \frac{ds}{r^2} = \frac{r \sin \theta d\phi r d\theta}{r^2} = \sin \theta d\theta d\phi$$

For symmetry about XX' axis, $d\phi$ is to be integrated from 0 to 2π . In such a case, $d\Omega = 2\pi \sin \theta d\theta$

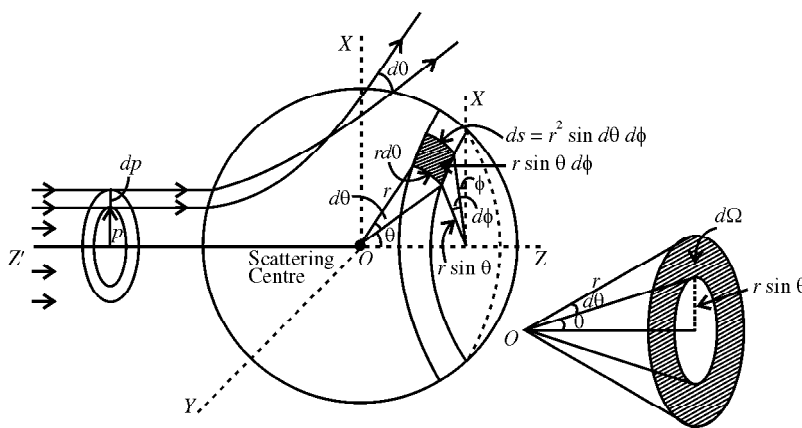


Fig. 2.42 : Scattering.

These particles are scattered into the solid angle $d\Omega = 2\pi \sin \theta d\theta$. But $I(\Omega) = I_0 \sigma(\Omega)$ is the number of particles scattered in the direction Ω per unit solid angle, hence the number of particles scattered into the solid angle $d\Omega$ are

$$I(\Omega) d\Omega = I_0 \sigma(\Omega) d\Omega$$

$$\text{or } I(\Omega) d\Omega = I_0 \sigma(\theta) 2\pi \sin \theta d\theta \quad \dots(7)$$

where $\sigma(\theta)$ represents the differential cross-section for the direction θ . Therefore by using eq. (1) and (6), we get

$$I_0 2\pi p dp = - I_0 \sigma(\theta) 2\pi \sin \theta d\theta$$

Negative sign is introduced, because an increase of p will decrease θ .

$$\text{Thus } \sigma(\theta) = - \frac{p}{\sin \theta} \left[\frac{dp}{d\theta} \right] \quad \dots(8)$$

This gives the **dependence of differential cross-section on the scattering angle θ** .

2.27. Rutherford Scattering Cross-Section

In the Rutherford scattering an α -particle of charge $2e$ and mass m is scattered by a heavy nucleus N [fig. 2.43]. The nucleus is assumed to be at rest during the collision. The charge on the nucleus is Ze , where Z is the atomic number. Suppose the α -particle is moving towards the heavy nucleus with initial velocity v_0 . As the particle approaches the nucleus, the repulsive force ($2Ze^2/4\pi\epsilon_0 r^2$) increases rapidly and the particle changes from a straight line path to a hyperbola APB , having one focus at N as shown in fig. 2.43. The asymptotes AO and BO to the hyperbola give the direction of the incident and scattered particle. The angle COB is the scattering angle θ and NM is the impact parameter p .

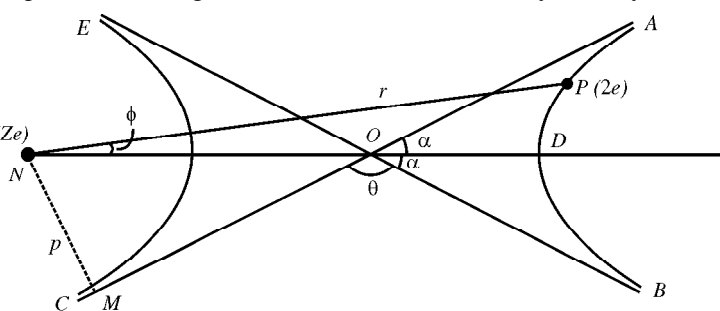


Fig. 2.43 : Scattering in a repulsive force field.

The charged particle is moving in a central force field, hence the equation of its path is given by

$$\frac{l}{r} = 1 + \epsilon \cos \phi \quad \dots(1)$$

where $l = L^2/mK$ and $\epsilon = \sqrt{1 + \frac{2EJ^2}{mK^2}}$

Here the force F is given by

$$F = \frac{2Ze^2}{(4\pi\epsilon_0) r^2} = -\frac{K}{r^2}$$

$$\text{Therefore, } K = -\frac{2Ze^2}{4\pi\epsilon_0} \quad \dots(2)$$

$$\text{Hence } l = \frac{J^2 4\pi\epsilon_0}{2Ze^2 m} \text{ and } \epsilon = \sqrt{1 + \frac{EL^2 (4\pi\epsilon_0)^2}{2Z^2 e^4 m}} \quad \dots(3)$$

As the initial velocity of the particle is v_0 , its total energy is given by

$$E = \frac{1}{2} m v_0^2, \text{ whence } m v_0 = \sqrt{2mE} \quad \dots(4)$$

According to the law of conservation of angular momentum,

$$m v_0 p = m r^2 \omega = L, \text{ whence } m v_0 = \frac{L}{p}$$

Therefore,
$$\frac{L}{p} = \sqrt{2mE} \text{ or } L = p\sqrt{2mE} \quad \dots(5)$$

This gives

$$\epsilon = \sqrt{1 + \frac{E(p\sqrt{2mE})^2 (4\pi\epsilon_0)^2}{2mZ^2 e^4}} \text{ or } \epsilon = \sqrt{1 + \left[\frac{Ep(4\pi\epsilon_0)}{Ze^2} \right]^2} \quad \dots(6)$$

Obviously, $\epsilon > 1$, because $(Ep/Ze^2)^2$ is a positive quantity. Hence eq. (1) represents the path of the charged particle as *hyperbola*.

Since the hyperbolic path must be symmetric about the direction of the periapsis, the scattering angle θ is given by

$$\theta = \pi - 2\alpha \text{ or } \alpha = \frac{\pi}{2} - \frac{\theta}{2} \quad \dots(7)$$

where $\alpha = \angle AOD$.

Further the asymptotic direction is that for which r is infinite (∞) and then $\phi \rightarrow \alpha$. Hence from eq. (1), we have

$$1 + e \cos \alpha = 0 \text{ or } \cos \alpha = -\frac{1}{e} \text{ or } \cos \left(\frac{\pi}{2} - \frac{\theta}{2} \right) = -\frac{1}{e} \text{ or } \sin \frac{\theta}{2} = -\frac{1}{e}$$

Thus
$$\operatorname{cosec} \frac{\theta}{2} = -e$$

Squaring it, we get

$$\operatorname{cosec}^2 \frac{\theta}{2} = e^2 \text{ or } 1 + \cot^2 \frac{\theta}{2} = 1 + \left[\frac{Ep(4\pi\epsilon_0)}{Ze^2} \right]^2 \quad [\text{from (6)}]$$

whence
$$\cot \frac{\theta}{2} = \frac{Ep(4\pi\epsilon_0)}{Ze^2} \quad \dots(8)$$

From which one can find the **scattering angle θ** .

Now from eq. (8), we get an expression for **impact parameter** :

$$p = \frac{Ze^2 \cot \theta/2}{E(4\pi\epsilon_0)} \quad \dots(9)$$

Differentiating eq. (9) we get

$$\frac{dp}{d\theta} = \frac{Ze^2}{2E(4\pi\epsilon_0)} \operatorname{cosec}^2 \frac{\theta}{2} \quad \dots(10)$$

Hence
$$\sigma(\theta) = -\frac{p}{\sin \theta} \left[\frac{dp}{d\theta} \right] = \frac{Ze^2 \cot \frac{\theta}{2}}{E(4\pi\epsilon_0) 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}} \left[\frac{Ze^2}{2E(4\pi\epsilon_0)} \right] \operatorname{cosec}^2 \frac{\theta}{2}$$

or
$$\sigma(\theta) = \frac{1}{4} \left[\frac{Ze^2}{(4\pi\epsilon_0)E} \right] \operatorname{cosec}^4 \frac{\theta}{2} \quad \dots(11)$$

This is the well known expression for the **Rutherford scattering cross-section formula**.

Thus the scattering cross-section or the number of particles scattered per second along the direction θ are proportional to

$$(1) \operatorname{cosec}^4 \frac{\theta}{2},$$

- (2) the square of the charge on the nucleus (Ze),
- (3) the square of the charge on the particle ($2e$), and
- (4) inversely proportional to the square of the initial kinetic energy E .

Thus if N is the number of particles scattered along the angle θ per second, then one can represent

$$N_\theta = C \operatorname{cosec}^4 \theta/2 \quad \dots(12)$$

where C is a constant.

2.28. Head-on Collision of α -Particle with the Nucleus

Let us consider the case when an α -particle moves directly in a straight line towards the nucleus so that the impact parameter $p = 0$ [fig. 2.44]. Let the mass of the α -particle (charge $= +2e$) be m , moving initially with velocity v_0 at a large distance from the nucleus towards it. When it will come in the influence of the positive charge $+Ze$ of the nucleus, due to the force of repulsion its kinetic energy will decrease and convert in the form of potential energy. At a certain distance d from the nucleus the α -particle will stop at P and whole kinetic energy will change in the form of potential energy i.e.,

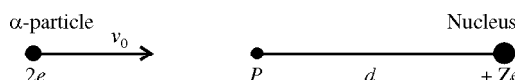


Fig. 2.44 : Distance of closest approach (d).

$$\frac{1}{2}mv_0^2 = \frac{Ze \times 2e}{(4\pi\epsilon_0)d} \quad \dots(1)$$

This distance is called the **distance of closest approach** of the α -particle. Now, due to repulsive force, α -particle will move back and trace its reverse path (i.e., scattering angle $\theta = 180^\circ$).

From eq. (1), we can obtain the value of the distance of closest approach i.e.,

$$d = \frac{4Ze^2}{(4\pi\epsilon_0)mv_0^2} \quad \dots(2)$$

Ex. 1. An α -particle of energy 5 MeV approaches a copper nucleus ($Z = 32$) in the head-on collision. Calculate the distance of nearest approach. (A.N.U. 2007)

Sol. Let d be the distance of nearest approach of the α -particle to the copper nucleus ($Z = 32$). At this distance total kinetic energy $\left(\frac{1}{2}mv_0^2\right)$ of the α -particle will change in the form potential energy i.e.,

$$\frac{1}{2}mv_0^2 = \frac{1}{4\pi\epsilon_0} \frac{(Ze)(2e)}{d}$$

Here, kinetic energy of α -particle, $\frac{1}{2}mv_0^2 = 5 \text{ MeV} = 5 \times 10^6 \times 1.6 \times 10^{-19} \text{ joule}$, $Z = 32$ and $e = 1.6 \times 10^{-19} \text{ coulomb}$.

$$\therefore 5 \times 10^6 \times 1.6 \times 10^{-19} = 9 \times 10^9 \times \frac{32 \times 2 \times (1.6 \times 10^{-19})^2}{d}$$

$$\therefore d = \frac{9 \times 10^9 \times 32 \times 1.6 \times 10^{-19}}{5 \times 10^6} = 1.84 \times 10^{-14} \text{ m.}$$

Ex. 2. A stream of α -particles is bombarded on a mercury nucleus ($Z = 80$) with a velocity $2 \times 10^7 \text{ m/sec}$. If an α -particle is approaching the nucleus in head-on collision, calculate the distance of closest approach. Given mass of α -particle $= 6.4 \times 10^{-27} \text{ kg}$.

Sol. Let the distance of closest approach for an α -particle in head-on collision be d . Then

K.E. of particle = P.E. of α -particle at distance d from nucleus

$$\frac{1}{2}mv_0^2 = \frac{1}{4\pi\epsilon_0} \frac{(Ze)(2e)}{d}$$

$$\begin{aligned}\therefore d &= \frac{1}{4\pi\epsilon_0} \times \frac{4Ze^2}{mv_0^2} \\ &= (9 \times 10^{11}) \times \frac{4 \times 80 \times (1.6 \times 10^{-19})^2}{(6.4 \times 10^{-27}) \times (2 \times 10^7)^2} = \mathbf{2.8 \times 10^{-14} \text{ m.}}\end{aligned}$$

Ex. 3. An α -particle, having a charge $+2e$ is projected directly from infinity with a kinetic energy of 4×10^{-15} J towards a nucleus of charge $+80e$. Find the distance of closest approach of α -particle to the nucleus. (K.U. 2007)

Sol. At the distance of closest approach,

K.E. of α -particle = P.E. of α -particle

$$\frac{1}{2}mv_0^2 = \frac{1}{4\pi\epsilon_0} \cdot \frac{(Ze)(2e)}{d}$$

$\therefore$ Here, $\frac{1}{2}mv_0^2 = 4 \times 10^{-15}$ J, $Z = 80$, $1/4\pi\epsilon_0 = 9 \times 10^9$ S.I. unit and $e = 1.6 \times 10^{-19}$ C.

$$\therefore 4 \times 10^{-15} = 9 \times 10^9 \times \frac{80 \times 2 \times (1.6 \times 10^{-19})^2}{d}$$

whence

$$\begin{aligned}d &= \frac{9 \times 10^9 \times 80 \times 2 \times (1.6 \times 10^{-19})^2}{4 \times 10^{-15}} \\ &= 9 \times 40 \times 2.56 \times 10^{-14} \\ &= \mathbf{9.216 \times 10^{-12} \text{ m.}}\end{aligned}$$

Ex. 4. Alfa particles of energy 4 MeV are scattered back from a gold foil $Z = 79$. Calculate the maximum volume in which the positive charge of the atom is likely to be concentrated. (A.N.U. 1993)

Sol. The α -particle will be scattered back at a distance from gold nucleus where its kinetic energy is converted into potential energy. Let d be the distance of closest approach. Then

K.E. of α -particle = P.E. of α -particle

$$\frac{1}{2}mv_0^2 = \frac{2Ze^2}{4\pi\epsilon_0 d}$$

Here, K.E. = 4 MeV = 4×10^6 eV = $(4 \times 10^6) (1.6 \times 10^{-19})$ joule, $Z = 79$ and $e = 1.6 \times 10^{-19}$ C

$$\therefore (4 \times 10^6) (1.6 \times 10^{-19}) = 9 \times 10^9 \times \frac{2 \times 79 \times (1.6 \times 10^{-19})^2}{d}$$

or

$$\begin{aligned}d &= (9 \times 10^9) \frac{2 \times 79 \times (1.6 \times 10^{-19})^2}{(4 \times 10^6)} \\ &= 5.688 \times 10^{-14} \text{ m}\end{aligned}$$

Maximum volume of positive charge = $\frac{4}{3}\pi d^3 = \frac{4}{3} \times (3.14) \times (5.688 \times 10^{-14})^3 = \mathbf{7.72 \times 10^{-40} \text{ m}^3}$.

Ex. 5. An α -particle with kinetic energy 6.0×10^{-14} joule is scattered at an angle of 60° by Coulomb field of a stationary gold nucleus. Find the impact parameter. (A.U. 1999)

Sol. Impact parameter for α -particles is given by

$$p = \frac{Ze^2 \cot \theta/2}{E(4\pi\epsilon_0)}$$

Here, $E = 6 \times 10^{-14}$ J, $Z = 79$ (for gold), $e = 1.6 \times 10^{-19}$ C, $1/4\pi\epsilon_0 = 9 \times 10^9$ S.I. unit and $\theta = 60^\circ$ so that $\cot \theta/2 = \cot 30^\circ = 1/\tan 30^\circ = \sqrt{3}$.

$$\begin{aligned} \therefore p &= \frac{79 \times (1.6 \times 10^{-19})^2 \times \sqrt{3} \times 9 \times 10^9}{6 \times 10^{-14}} \\ &= \frac{79 \times 2.56 \times 1.732 \times 9 \times 10^{-15}}{6} = 5.25 \times 10^{-13} \text{ m.} \end{aligned}$$

Ex. 6. A 5 MeV α -particle approaches a gold nucleus ($Z = 79$) with an impact parameter of 2.6×10^{-13} m. Through what angle will it be scattered?

Sol. We know that $\cot \frac{\theta}{2} = \frac{Ep(4\pi\epsilon_0)}{Ze^2}$ or $\tan \frac{\theta}{2} = \frac{Ze^2}{Ep(4\pi\epsilon_0)}$

Here, $E = 5 \text{ MeV} = 5 \times 10^6 \times 1.6 \times 10^{-19} \text{ J}$, $p = 2.6 \times 10^{-13} \text{ m}$, $Z = 79$, $e = 1.6 \times 10^{-19} \text{ C}$, and $1/4\pi\epsilon_0 = 9 \times 10^9 \text{ S.I. unit}$.

$$\begin{aligned} \therefore \tan \frac{\theta}{2} &= \frac{79 \times (1.6 \times 10^{-19})^2 \times 9 \times 10^9}{5 \times 10^6 \times 1.6 \times 10^{-19} \times 2.6 \times 10^{-13}} \\ &= \frac{79 \times 2.56 \times 9 \times 10^{-3}}{5 \times 1.6 \times 2.6} = 0.0875 \end{aligned}$$

$\therefore$ For small θ , $\tan \theta = \theta$,

$$\therefore \frac{\theta}{2} = 0.0875 \text{ radian} = 0.0875 \times \frac{180^\circ}{\pi} = 5^\circ$$

Hence $\theta = 10^\circ$.

Ex. 7. In Rutherford's experiment 10^5 particles are scattered at an angle of 2° , calculate the number of α -particles scattered at an angle of 20° .

Sol. The number of particles N_θ scattered per second at an angle θ is given by

$$N_\theta = C \operatorname{cosec}^4 \theta/2$$

$$\text{Therefore, } \frac{N_{(20^\circ)}}{N_{(2^\circ)}} = \frac{\operatorname{cosec}^4 10^\circ}{\operatorname{cosec}^4 1^\circ} = \left(\frac{\sin 1^\circ}{\sin 10^\circ} \right)^4 = \frac{1}{10^4} \quad [\text{for small } \theta, \sin \theta \approx \theta]$$

whence $N_{(20^\circ)} = 10^5/10^4 = 10$ (approximately).

Ex. 8. Determine the differential scattering cross-section and the total scattering cross-section for the scattering of a particle by a rigid elastic sphere.

Sol. Let the radius of the rigid elastic sphere be a (**fig. 2.45**). The impact parameter p for the particle under consideration is

$$p = a \sin \left(\frac{\pi}{2} - \frac{\theta}{2} \right) = a \cos \frac{\theta}{2}$$

$$\text{Hence } \frac{dp}{d\theta} = -\frac{a}{2} \sin \frac{\theta}{2}$$

Differential scattering cross-section

$$\begin{aligned} \sigma(\theta) &= -\frac{p}{\sin \theta} \frac{dp}{d\theta} \\ &= -\frac{a \cos \frac{\theta}{2} \left(-\frac{a}{2} \sin \frac{\theta}{2} \right)}{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}} = \frac{a^2}{4} \end{aligned}$$

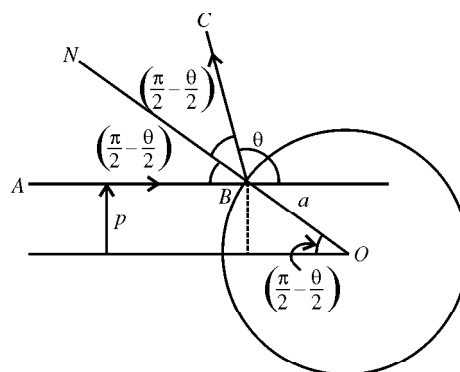


Fig. 2.45.

$$\text{Total scattering cross-section } \sigma = \int \frac{d\sigma}{d\theta} d\Omega = \frac{a^2}{4} \int d\Omega = \frac{a^2}{4} 4\pi = \pi a^2.$$

QUESTIONS

Long Answer Questions

[Newton's Laws of Motion and Motion of a System of Variable Mass -Rocket]

1. State Newton's laws of motion. How do these laws define uniquely the mass and force. Discuss the limitations of Newton's laws.
2. Explain how Newton's law can be applied to a system of particles. (S.V.U. 2002)
3. Formulate Newton's second law for a system of variable mass. Derive the expression for the velocity of rocket at any time. (A.N.U. 2007)
4. Derive the equation of motion of a system of variable mass. Explain the motion of a rocket using the concept of variable mass. (A.N.U. 2006)
5. (a) Show that for a system of variable mass

$$M \frac{d\vec{V}}{dt} = \vec{F}_{external} + \vec{F}_{reaction} \quad (\text{O.U. 2008, 03})$$

- (b) Obtain the velocity of a rocket at any instant of time t . (O.U. 2008, 01)
6. Obtain an expression for the thrust on the rocket using Bernoulli's equation. (S.K.U. 2002)
7. State and prove Bernoulli's theorem. Derive an expression for rocket and thrust based on Bernoulli's theorem. (A.U. 2002)
8. Describe the principle of rocket motion and establish the following expression for the final velocity acquired by the rocket $V = V_0 + v \log_e (M_0/M)$, where the symbols have their usual meaning. On this basis, explain how is two stage rocket superior over a single stage rocket ?
9. Prove that the velocity acquired by a rocket in time t is given as $V = V_0 - v \log_e (1 - mt/M_0)$, where V_0 is the initial velocity of rocket, M_0 is the initial mass of rocket, v is the escape velocity of exhaust gases and m is the rate of consumption of fuel. What will be the value of V , if the attraction due to earth is not neglected ?
10. What is the working principle of a rocket ? Derive the differential equation representing rate of gain of speed by a rocket. How the velocity of the rocket is increased so as to escape from the earth's attracting field ? What is the advantage of two stage rocket in comparison to single stage rocket ?

[Conservation of Energy, Linear and Angular Momenta]

11. State and prove the work-energy theorem. (S.K.U. 2002; A.N.U. 2003, 02)
12. What do you mean by potential energy ? Obtain an expression for potential energy of a particle in a conservative force field. Why this expression is negative ?
13. Prove that the total mechanical energy (= kinetic energy + potential energy) of a system acted upon by a conservative force is always conserved.
14. State and prove the law of conservation of linear momentum for a system of n particles.
15. What is linear momentum ? Prove that if the resultant of external forces acting on a n -particle system, then the total linear momentum of the system is conserved.
16. Explain the centre of mass and the linear momentum of a system. State and prove the law of conservation of linear momentum. (S.K.U. 2003)
17. What is the concept of centre of mass ? Determine the position vector of a system of n particles.
18. Explain the concept of centre of mass. How does it differ from centre of gravity ? Derive formula for its position. On what factors does the position of centre of mass of a body depend ?
19. (a) If the total momentum of a system is conserved, then prove that the centre of mass moves with uniform velocity or remains at rest.
(b) An isolated system of point masses is subjected to the internal forces of interaction only. Show that the centre of mass of the system moves with a constant velocity in an inertial frame.

20. Define angular momentum. State and prove the law of conservation of angular momentum. (S.K.U. 2001)
21. State and explain the law of conservation of angular momentum. State and prove work energy theorem. (S.V.U. 2003)
22. Define angular momentum $\vec{L}$ and torque $\vec{\tau}$. Using Newton's second law in an inertial frame of reference to show that $\vec{\tau} = d\vec{L}/dt$.
23. What is meant by angular momentum of a particle? Show that the rate of change in angular momentum of the particle is equal to the external torque acting on it. Hence derive the law of conservation of angular momentum.
24. Show that the angular momentum $\vec{L}$ of a system of particle can be expressed in the form
- $$\vec{L} = \vec{L}_{cm} + \vec{R} \times \vec{P} \quad (\text{or} \quad \vec{L} = \vec{L}_{cm} + \vec{R} \times M \vec{V})$$
- where $\vec{L}_{cm}$ = angular momentum about the centre of mass, $\vec{R}$ = position vector of centre of mass and $\vec{P} = M \vec{V}$ total linear momentum.

[Collision and Scattering]

25. State and explain the law of conservation of linear momentum. Derive the expression for the final velocities of two bodies in one dimensional collision. (S.V.U. 2002; A.N.U. 1996)
26. (a) Distinguish between elastic and inelastic collisions between two bodies. Discuss elastic collision of two bodies in one dimension. (O.U. 2007)
- (b) For the case of an inelastic collision, obtain an expression for the velocity of the coalesced bodies when one of the bodies was initially at rest. (O.U. 2002)
27. What do you understand by collision? Distinguish between elastic and inelastic collisions. Derive expressions for final velocities of two bodies after a one dimensional elastic collisions between them. (A.U. 2005)
28. What are elastic and inelastic collisions? Give examples. Derive expressions for final velocities of two bodies after a one-dimensional elastic collision between them. (A.U. 2007)
29. (a) What are elastic and inelastic collisions? Give two examples for each. (O.U. 2002)
- (b) Give the theory of elastic collisions in two dimensions. (O.U. 2002)
30. Distinguish between elastic and inelastic collisions. A projectile is a particle in an elastic collision with a target particle of equal mass initially at rest. Show that the two particles move at right angles to each other. (K.U. 1994; A.N.U. 1993)
31. (a) Distinguish between elastic and inelastic collisions. Give examples. (O.U. 2007)
- (b) Discuss the theory of elastic collision in two dimensions in a laboratory frame of reference. (O.U. 2007)
32. Distinguish between elastic and inelastic collisions. Show that the kinetic energy of a system of two particles decreases after an inelastic collision. (S.V.U. 2002)
33. Show that the total kinetic energy decreases in the cases of a completely inelastic collision of a body with another body initially at rest. (O.U. 1990)
34. Distinguish between elastic and inelastic collisions. Obtain expressions for the velocities after collision, when the two spheres collide obliquely. (A.N.U. 1990)
35. What are elastic and inelastic collisions? Give examples. Derive expressions for the final velocities of two bodies after an elastic oblique collision between them? (A.N.U. 2002; S.K.U. 2002)
36. (a) Distinguish between laboratory and centre of mass frame of reference.
- (b) Discuss the phenomenon of oblique and perfectly elastic collision between two bodies in laboratory frame of reference. (O.U. 2003)

37. Explain elastic and inelastic collisions. Deduce the relationship between (a) scattering angles and (b) angle of recoil in the elastic collision between two particles as observed in Lab-frame and C-frame. (K.U. 1997)
38. (a) A particle of mass m_1 collides with another of mass m_2 at rest. After scattering m_1 and m_2 move at angles θ_1 and θ_2 with respect to the original direction of motion of m_1 . How does the collision appear in the centre of mass frame ? (K.U. 1991)
(b) If two particles undergo collision, describe their motion after collision in the centre of mass frame. Also find a relation between the scattering angles in the laboratory frame and the centre of mass frame. (K.U. 1991)
39. Prove that in a collision of a particle of mass m_1 , moving with velocity v_1 in the L-frame (laboratory frame), with a particle of mass m_2 at rest in the L-frame, the angles at which the first particle moves after the collision relative to its initial velocity are given by $\tan \theta = \sin \phi / (\cos \phi + 1/A)$, where $A = m_2/m_1$ and angles θ and ϕ refer to the L and C-frames respectively.
Show also that the maximum value of θ for arbitrary A is given by $\tan \theta = \sqrt{1-A^2}$. Discuss the situation when A is larger than 1 and when it is smaller than 1.
40. In the laboratory frame, a particle of mass m_1 collides with velocity v_1 with another stationary particle of mass m_2 . If the velocity of their centre of mass is v_c , find the velocities of these particles after the collision in the centre of mass frame.
41. Give the theory of inelastic collision in the L and C frames. Deduce the relationship among angles of recoil in the case of elastic collision between two particles as observed in L frame and C-frame. (N.U. 1996; S.K.U. 1995)
42. Explain the terms : (i) impact parameter and (ii) scattering cross-section. The smaller the impact parameter, the larger the scattering angle, prove. (O.U. 2005)
43. Define the terms impact parameter and scattering cross-section. Derive an expression for Rutherford scattering cross-section. (N.U. 2003)
44. Define impact factor parameter. Derive an expression for Rutherford's scattering cross-section. (A.N.U. 2008; A.U. 2001)
45. What is centre of mass. Derive an expression for the Rutherford scattering of α -particles by heavy nucleus. (O.U. 2007; S.V.U. 2002)
46. Derive an expression for Rutherford scattering formula for the collision of α -particles with a heavy nucleus. (A.N.U. 2007)
47. What is Rutherford scattering ? Obtain the equation for the angle of scattering of α -particle in Rutherford scattering. (S.K.U. 2005)
48. Derive the relation between impact parameter and the scattering angle for Rutherford scattering of α -particles. (O.U. 1993)
49. Define the terms 'Impact Parameter' and scattering cross-section. Obtain an expression for the differential scattering cross-section in the case of α -particles striking a thin gold foil and undergoing elastic collision. (A.N.U. 2006)
50. Derive Rutherford's scattering formula. (O.U. 2001; N.U. 2001; A.U. 2001)
51. What is Rutherford scattering ? Obtain an equation for the differential scattering cross-section in the case of α -particle striking a thin gold foil and undergoing elastic collision. (A.U. 1997; S.V.U. 1996)
52. What are elastic and inelastic collisions ? Obtain the equation for the angle of scattering of α -particle in Rutherford scattering. (O.U. 1995)
53. Define the terms impact parameter and scattering cross-section. What is Rutherford scattering ? Obtain an expression for the Rutherford's scattering cross-section and also number of scattered particles per unit area. (O.U. 1995; N.U. 1995)
54. Define scattering cross-section and differential cross-section. Show that scattering cross-section has dimensions of area. (O.U. 2009)

Short Answer Questions

1. Write a note on frame of reference. (A.U. 2007)
2. State Newton's laws of motion. In which frame of reference Newton's first law of motion valid ?
3. What are the limitations of Newton's laws of motion ?
4. What is impulse of a force ?
5. Describe equation of motion for variable mass system. (A.N.U. 2005)
6. Give two examples of variable mass system. (S.V.U. 1997)
7. Describe the motion of variable mass system. (S.V.U. 2007; K.U. 2007)
8. Derive the equation for the motion of Rocket. (S.K.U. 2007)
9. Explain the principle of working of a rocket. (A.N.U. 2008; A.U. 2007)
10. Calculate the thrust on a rocket. (S.V.U. 2004)
11. Explain the principle of multistage rocket. (A.U. 2007)
12. Write a note on multistage rocket. (A.N.U. 2007)
13. Explain the principle of motion of a rocket as a system of variable mass. (S.K.U. 2005)
14. Discuss the motion of rocket under a constant gravitational field. (K.U. 1995; S.K.U. 1994; A.U. 1993; S.V.U. 1996)
15. Discuss the principle of linear conservation of momentum for a system of variable mass and discuss the motion of a rocket. (A.U. 1989; K.U. 1988)
16. Consider the flight of rocket vertically upward and prove that it moves with a non-uniform acceleration.
17. State and explain work-energy theorem. (A.N.U. 2006)
18. Define angular momentum and torque. Derive relation between them. (A.U. 1995)
19. State the principle of conservation of angular momentum. Give an example. (A.U. 1995)
20. State and prove the principle of conservation of linear momentum. (A.U. 1997)
21. State and prove the conservation of angular momentum in case of system of particles. (A.N.U. 1995; A.U. 1991)
22. Obtain the values of the components of angular momentum (L_x , L_y , L_z) of a particle. (O.U. 1990)

[Hint. By definition of angular momentum $\vec{L} = \vec{r} \times \vec{p}$

Expressing position vector $\vec{r}$ and linear momentum in rectangular components i.e., $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ and $\vec{p} = p_x\hat{i} + p_y\hat{j} + p_z\hat{k}$ we have

$$\begin{aligned}\vec{L} = \vec{r} \times \vec{p} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ x & y & z \\ p_x & p_y & p_z \end{vmatrix} \\ &= \hat{i}(yp_z - zp_y) + \hat{j}(zp_x - xp_z) + \hat{k}(xp_y - yp_x) \quad \dots(i)\end{aligned}$$

If L_x , L_y and L_z be the components of $\vec{L}$ along X, Y and Z axes, then

$$\vec{L} = L_x\hat{i} + L_y\hat{j} + L_z\hat{k} \quad \dots(ii)$$

Comparing (i) and (ii), we get

$$L_x = (yp_z - zp_y), L_y = (zp_x - xp_z) \text{ and } L_z = (xp_y - yp_x). \quad \dots(iii)$$

23. State the laws of conservation of linear and angular momentum. (S.K.U. 2007)
24. Define angular momentum. Write the unit of angular momentum.
25. Two bodies of different masses are moving with the same kinetic energy of translation. Which one

has a greater momentum ?

[Ans. Body of greater mass will have the greater momentum.]

$$[\text{Hint. } \frac{1}{2}m_1v_1^2 = \frac{1}{2}m_2v_2^2 \text{ or } \frac{v_1}{v_2} = \sqrt{\frac{m_2}{m_1}} \therefore \frac{p_1}{p_2} = \frac{m_1v_1}{m_2v_2} = \frac{m_1}{m_2} \sqrt{\frac{m_2}{m_1}} = \sqrt{\frac{m_1}{m_2}}]$$

Hence, $m_1 > m_2, p_1 > p_2$

26. The value of initial kinetic energy will be greater or less than final kinetic energy in inelastic collision and why ?
27. The masses of two particles are 1 kg and 4 kg. If the magnitudes of their kinetic energies are equal, then what will be the ratio of their linear momenta ?
[Ans. 1/2]
28. A bullet is fired from a rifle. If the rifle recoils freely, determine whether the kinetic energy of the rifle is greater than, equal to or less than that of the bullet.
[Ans. Less]
29. What is the law of conservation of angular momentum ? What are spin and orbital angular momenta ?
30. Explain the meaning of centre of mass.
31. Find the centre of mass of a two-particle system.
32. What do you mean by the centre of mass frame of reference ?
33. Show that in absence of external forces, the velocity of centre of mass remains constant.
34. Explain the concept of collisions. (S.K.U. 1993)
35. Distinguish between elastic and inelastic collisions. Give examples. (S.K.U. 1993, 94)
36. Distinguish between elastic and inelastic collisions. (A.N.U. 2003, 01; O.U. 2003; A.U. 2002, 01)
37. What are elastic and inelastic collisions ? Mention two examples each. (O.U. 1996)
38. Explain the term 'Impact Parameter'. (K.U. 2007; S.V.U. 2005)
39. Define scattering cross-section. (A.U. 2007)
40. What is impact parameter ? How does the angle of scattering depend on it ? (A.N.U. 2004)
41. What is impact parameter and scattering cross-section ? (A.N.U. 2007; O.U. 2007, 05; S.V.U. 2003)
42. Explain what is meant by scattering cross-section. (S.K.U. 1995; S.V.U. 1994)
43. Write the Rutherford scattering formula and explain the symbols. (S.K.U. 1997)
44. What is Rutherford's scattering cross-section ? Explain its significance. (A.N.U. 1994)
45. Write a short note on Rutherford scattering. (A.U. 2007)
46. Explain the principle of Rutherford scattering. (A.U. 2005)

PROBLEMS

[Laws of Motion]

1. A force produces an acceleration of 8 m/s^2 in body of mass 1 kg and an acceleration of 4 m/s^2 in another body. If both bodies are tied together, find the acceleration produced in the system by this force.
[Ans. 2.67 m/s^2]
2. A uniform rope of length l , resting on a frictionless surface, is pulled at one end by a force F . Determine the tension in the rope at a distance x from this end.
[Ans. $F(l-x)/l$]
3. A rod of length l and mass m is acted by two unequal forces F_1 and F_2 ($< F_1$) [fig. 2.46]. What is the tension in the rod at a distance x from the end of the rod, where F_1 is acting.
[Ans. $[F_1(l-x) + F_2x]/l$]
4. Fig. 2.47 shows a uniform rod of length 60 cm and mass 3 kg. The strings shown in the figure are

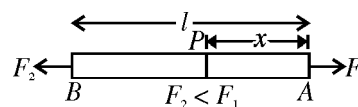


Fig. 2.46.

pulled by constant forces of 10 N and 16 N. Find the force exerted by the 40 cm part of the rod on the 20 cm part. All the surfaces are smooth and the strings and pulleys are light.

[Ans. 12 N]

5. Two blocks of masses m_1 and m_2 are kept in contact on a frictionless table [fig. 2.48]. When the block m_1 is pushed from behind, the two blocks accelerate. If m_1 block exerts a force F on the block m_2 , find the pushing force on m_1 .

[Ans. $F\{1 + (m_1/m_2)\}$]

6. The coefficient of friction between the two blocks of masses m_1 and m_2 is μ [fig. 2.49] and the floor is smooth. Find the maximum horizontal force F , applied on m_2 , so that the two blocks move together.

[Ans. $F = \mu(m_1 + m_2)g$]

7. Two blocks are connected by a string as shown in fig. 2.50. The upper block is suspended by another string. (i) Find the tension in the upper string due to the two blocks suspended in the rest position. (ii) What force F will have to be applied on the upper string to produce an acceleration of 2 m/s^2 in the upward direction in both the blocks? (iii) In this situation what is the tension in the string between the two blocks?

[Ans. (i) 58.8 N, (ii) 70.8 N, (iii) 47.2 N]

8. Fig. 2.51. shows a system of two masses m_1 and m_2 connected by a string. If a constant force $F = m_1g/2$ is applied, find the acceleration of m_1 . Assume the string and pulley to be light and the surface of the table to be smooth.

[Ans. $\frac{m_1g}{2(m_1 + m_2)}$ towards right.]

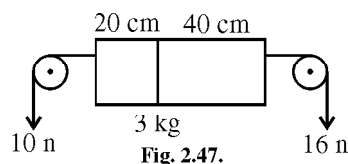


Fig. 2.47.

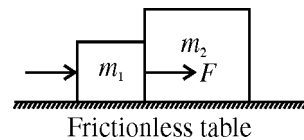


Fig. 2.48.

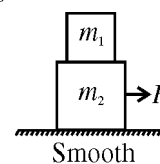


Fig. 2.49.

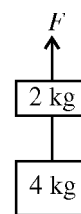


Fig. 2.50.

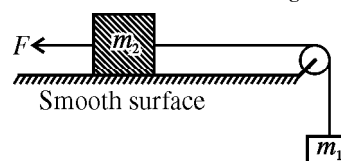


Fig. 2.51.

[Motion of Variable Mass System – Rocket]

9. A rocket burns 0.05 kg of fuel per second and ejects the burnt gases with a velocity of 5000 m/s. Find the reaction force exerted on the rocket by the ejected gas.
[Ans. 250 N]
[Hint. Thrust = Reaction force = $-v \frac{dM}{dt}$, $\frac{dM}{dt} = -0.05 \text{ kg/sec}$.]
10. An empty rocket weighs 6000 kg and contains 44000 kg of fuel. The exhaust velocity of gases is 1 km/s, find the maximum velocity attained by the rocket. (S.K.U. 1993)
[Ans. 19.16 km/sec]
11. A machine gun fires 50 gm bullets at a speed of 1000 m/sec. The gunner holding the machine in his hands, can exert an average force of 180 newtons against the gun. Calculate the number of bullets he can fire per minute.
[Ans. 220 per minute]
12. A rocket having initial mass 240 kg ejects fuel at the rate of 6 kg/sec with a velocity of 2 km/sec vertically downward relative to itself. Calculate its velocity 25 seconds after start, taking initial velocity to be zero and neglecting gravity.
[Ans. 1.96 km/sec]
13. A 6000 kg rocket is set for vertical firing. If the exhaust speed is 1000 m/sec, how much gas must

be ejected each second to supply the thrust needed (a) to overcome the weight of the rocket, (b) to give the rocket an initial upward acceleration of 20 m/sec^2 ?

[Ans. 58.8 kg/sec, 178.8 kg/sec]

14. A rocket starts from rest with an initial mass M_0 and its mass when the fuel is burnt out is M . Neglecting all external forces, show that the ratio of (M_0/M) is equal to e^2 , if the rocket speed is twice the exhaust speed.
15. What will be the ratio of the rocket velocity to the exhaust velocity at the instant that its mass is reduced to $1/20^{\text{th}}$ of its initial mass, its initial velocity being zero.
[Ans. 2.996]
16. A rocket of mass 8000 kg is fired vertically. If exhaust speed is 800 m/s, find the rate of exhaust so that the thrust on the rocket can (i) balance the weight, and (ii) produce an acceleration of $3g$ in it.
[Ans. (i) 98 kg/sec; (ii) 392 kg/sec.] (A.N.U. 1994)
17. A two stage rocket consists of 6000 kg and 500 kg stages and have fuel 40,000 kg and 3,500 kg respectively. If the exhaust velocity is 1.8 km/sec, calculate the speed obtained by last rocket.
[Ans. 6.62 km/sec]
18. If the two stages of a rocket weights 100 kg and 10 kg separately and contain 800 kg and 90 kg fuel separately, what is the final velocity that can be achieved with an exhaust velocity of 1.5 km/sec. (Take $\log_e 10 = 2.3$ and $\log_{10} 2 = 0.30$).
[Ans. 5.9 km/sec] (A.U. 1996; K.U. 1991)
19. If M is the mass of a rocket at any time, then the acceleration of a rocket moving vertically is $\left(\frac{mv}{M} - g\right)$, where m is the mass of the gas ejected per second with v velocity. What is the weight of a body inside the rocket, whose normal weight is W ?
[Ans. Wmv/Mg]
20. If the maximum possible velocity of a rocket be 2 km/sec, calculate the ratio M_0/M for if it is to attain the escape velocity 11.2 km/sec. How long will it take the rocket (starting from rest) to attain this velocity if its rate of change of mass in terms of its initial mass = $1/10$?
[Ans. 9.96 sec]

[Conservation of Energy]

21. Find the work done by a force $F = F_0 + kx$ acting parallel to X -axis on an object which moves along X -axis from x_1 to x_2 .
[Ans. $(x_2 - x_1) \left\{ F_0 + k \frac{x_2 + x_1}{2} \right\}$]
22. Calculate the amount of work done by a force $F = kx^2$ acting on a particle at an angle 60° to displace it from x_1 to x_2 along X -axis.
[Ans. $\frac{k}{6}(x_2^2 - x_1^3)$]
23. If the force acting on a particle at any point (x, y, z) is $(5x\hat{i} + xy\hat{j} + z\hat{k})$, how much work is done, when the particle moves from the point $(5, 2, 1)$ to the point $(5, 3, 2)$?
[Ans. 14 units]
24. A constant force of 5 newton acts on a body of mass 0.5 kg for 2 seconds. If the initial velocity of the body is 4 m/sec in the direction of the force, find (i) the work done by the force, (ii) the final kinetic energy, (iii) the power developed, and (iv) the increase in kinetic energy.
[Ans. (i) 140 joule, (ii) 144 joule, (iii) 70 joule/sec, (iv) 140 joule]
25. Calculate the amount of work done in stretching a spring by 0.1 m, if it is known that the increase in the length of the spring caused by a force of 1 kg-wt is 0.98 metre.
[Ans. 0.5 joule]
26. The scale of a spring balance reads from 0 to 250 kg and is 25 cm long. What is the potential energy of the spring when a 100 kg weight hangs from it ?
[Ans. 49 J]

27. A body of mass 16 kg slides on a horizontal frictionless table with a speed of 1 metre per second. It is brought to rest in compressing a spring in its path. By how much is the spring compressed if the force constant of the spring is 0.25 n/m ?

[Ans. 8 metres]

28. A block of mass m , initially at rest, is dropped from a height h onto a spring of force constant k . Find the maximum distance y through which the spring will be compressed ?

$$\left[\text{Ans. } y = \frac{1}{2} \left\{ \frac{2mg}{k} \pm \sqrt{\left(\frac{2mg}{k} \right)^2 + \frac{8mgh}{k}} \right\} \right]$$

29. A body slides down a curved track which is one quadrant of a circle of radius 1 m. If it starts from rest and there is no friction, find its speed at the bottom of track. ($g = 9.8 \text{ m/sec}^2$)

[Ans. 4.43 m/sec]

30. A pendulum bob of mass M is held out in the horizontal position and then released. If the string is of length l , what is the bob velocity and what is the force on the string when the bob reaches the lowest position ?

[Ans. $v = \sqrt{2gl}$; $T = 3 Mg$]

31. A block weighing 1 kg is released from rest at a point A on a track which is one quadrant of a circle of radius 2 metres [fig. 2.52]. It slides down the track and reaches point B with a speed of 5 m/sec. Now, it slides on a level surface a distance 50 metres upto the point C, where it comes to rest. Calculate the coefficient of friction and the amount of work done against friction as the body slides down the circular arc from A to B.

[Ans. 0.25; 7.1 joule]

[Hint : Work done against friction = $mgh - \frac{1}{2}mv_B^2$, where h is the vertical height of A from B.

$$\text{and } \mu R d = \frac{1}{2}mv_B^2 \text{ or } \mu \times 1 \times 50 = \frac{1}{2} \times 1 \times 5^2.]$$

32. A small block of mass m slides along the frictionless loop-the-loop track shown in fig. 2.53. If it starts from P (having height $6R$ from the bottom), find the tangential and normal components of the force at Q.

[Ans. mg ; $10 mg$]

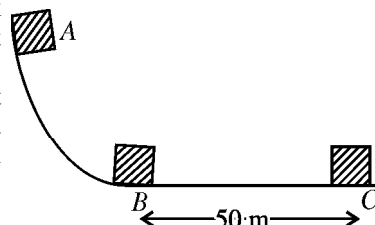


Fig. 2.52.

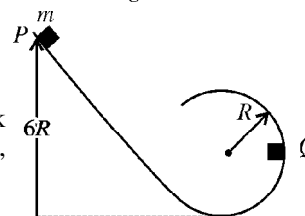


Fig. 2.53.

[Centre of Mass]

33. The distance between the nuclei of CO gas molecule is $1.13 \text{ \AA}$. Determine the position of centre of mass of the molecule relative to the carbon atom. The atomic weight of carbon and oxygen atoms are 12 and 16 respectively.

[Ans. 0.6457]

34. Locate the centre of mass of a system of three particles masses 1 kg, 2 kg and 3 kg placed at the corners of an equilateral triangle of 1 metre side.

[Ans. $(7/12, \sqrt{3}/4, 0)m$, 1 kg as origin]

35. If the centre of mass of three particles of masses 1, 2 and 3 gm be at the point (1, 1, 2) then where should a fourth particle of mass 5 gm be placed so that the combined centre of mass be at (0, 0, 0)?

[Ans. $(-1.2, -1.2, -2.4) \text{ cm}$]

36. Three particles of masses 7, 4 and 10 gm are located at (1, 5, -3), (2, 5, -7) and (3, 3, -1) cm respectively. Find the position vector of their centre of mass.

[Ans. $15/7, 85/21, -59/21 \text{ cm}$]

37. The centre of mass of a system of three particles of masses 4.0 kg, 8.0 kg and 12.0 kg is at (1, 1, 1). Where should another particle of mass 3.0 kg be kept so that centre of mass of entire system may come at (0, 0, 0) ?

[Ans. $(-8, -8, -8)$]

38. The centre of mass of three particles of masses 10, 20 and 30 gm is at $(1, -2, 3)$ cm. Where the particle of mass 40 gm is to be placed so that the new centre of mass is at $(1, 1, 1)$ cm ?
 [Ans. $(1, 5.5, -2)$ cm]
39. In an inertial frame of reference find out the velocity of the centre of mass of a system of three particles of masses 20 g, 30 g and 50 g moving with velocities $(\hat{i} + 2\hat{j})$, $(\hat{j} - 2\hat{k})$ and $(\hat{i} + 3\hat{k})$ respectively. Also find the velocity of each particle in a frame that moves with the centre of mass and has co-ordinate axes parallel to the axes of the first frame.
 [Ans. $\vec{V} = 0.7\hat{i} + 0.7\hat{j} + 0.9\hat{k}$; $0.3\hat{i} + 1.3\hat{j} - 0.9\hat{k}$; $-0.7\hat{i} + 0.3\hat{j} - 2.9\hat{k}$; $0.3\hat{i} - 0.7\hat{j} + 2.1\hat{k}$]
40. Two masses constrained to move in a horizontal plane collide. If $m_1 = 85$ gm, $m_2 = 200$ gm, and initially $\vec{v}_1 = 6.4\hat{i}$ m/sec, $\vec{v}_2 = (-6\hat{i} - 2\hat{j})$ m/sec, find : (i) the velocity of the centre of mass, (ii) the total linear momentum and (iii) the velocities in the reference frame in which the centre of mass is at rest.
 [Ans. (i) $\vec{V} = (-2.8\hat{i} - 1.4\hat{j})$ m/sec; (ii) $(0.796\hat{i} - 0.4\hat{j})$ kg-m/sec; (iii) $(9.2\hat{i} + 1.4\hat{j})$ m/sec; $(3.9\hat{i} - 0.6\hat{j})$ m/sec]

[Angular Momentum and Torque]

41. What is the angular momentum of a particle whose linear momentum is $\vec{p}$ and position vector is $\vec{r}$ about a point whose position vector is $\vec{r}'$?
 [Ans. $(\vec{r} - \vec{r}') \times \vec{p}$]
42. The speed of a body of mass 0.5 kg moving along a circle of radius 0.6 m increases at a constant rate of 0.1 ms^{-2} . Find the torque on the body.
 [Ans. 0.03 N-m]
43. Find the spin angular momentum of the earth about its axis of rotation and orbital angular momentum of the earth about the Sun. Given radius of the earth = 6.4×10^6 m and the orbital radius of the earth round the Sun = 1.5×10^{11} m.
 [Ans. $7.13 \times 10^{33} \text{ kg-m}^2 \text{ sec}^{-1}$, $2.68 \times 10^{40} \text{ kg-m}^2 \text{ sec}^{-1}$]
44. Show that the components of $\vec{L}$ along X, Y and Z axes are given by

$$L_x = yp_z - zp_y; L_y = zp_x - xp_z; L_z = xp_y - yp_x.$$
 If the particle moves only in X-Y plane, show that the resultant angular vector has z-component only.
45. Show that the angular momentum of a particle about a fixed point is equal to the product of its mass and double the area $(\frac{1}{2}r^2\omega)$ described by the rotating line in unit time and has a direction perpendicular to this area.
46. Express total angular momentum in terms of kinetic, potential and total energy of an earth's satellite of mass m in a circular orbit of radius r .
 [Ans. $L = (2mr^2K)^{1/2} = (-mr^2U)^{1/2} = (-2mr^2E)^{1/2}$]
47. A particle of $\frac{1}{2}$ gm mass starts from the point $(1, 0, 2)$ m at a time $t = 0$. If the force acting on it be $2\hat{i} + (1+6t)\hat{j}$ newton, calculate the torque and angular momentum about the point $(2, 2, 4)$ m at time $t = 1$ sec.
 [Ans. $\vec{\tau} = (14\hat{i} - 4\hat{j} + 5\hat{k}) \text{ N-m}$; $\vec{J} = (8\hat{i} - 4\hat{j} - 2\hat{k}) 10^{-3} \text{ kg-m}^2\text{s}^{-1}$]
 [Hint : $\frac{d^2(\vec{r} - \vec{r}_0)}{dt^2} = \frac{\vec{F}}{m} = 2[2\hat{i} + (1+6t)\hat{j}]$ and integrate to get $\vec{v}$ and $\vec{r}$.]
48. A particle of mass m , moving in a circular orbit of radius r , has angular momentum L about the centre. Calculate the kinetic energy of the particle in terms of L , m and r .
 [Ans. $L^2/2mr^2$]

[Hint : For circular motion, $L = mr^2\omega$, $\therefore$ Kinetic energy $\frac{1}{2}mv^2 = \frac{1}{2}m(r\omega)^2 = \frac{m}{2}\left(\frac{L}{mr}\right)^2 = \frac{L^2}{2mr^2}$.]

49. A 500 gm stone is revolved at the end of a 40 cm long string at the rate of 2 rev/sec. What is its angular momentum ? If after 30 sec, it makes only $\frac{1}{2}$ rev/sec; calculate the mean torque applied.
[Ans. 1 joule-sec about the centre of circle; 0.025 N-m]
50. A particle of mass m moves along a path given by $\vec{r} = a\hat{i}\cos\omega t + b\hat{j}\sin\omega t$. Find (a) the torque, (b) the angular momentum about the origin.
[Ans. (a) 0; (b) $2mab\omega\hat{k}$]
51. In the dumb-bell example of page 43, what is the initial and final kinetic energy of the dumb-bell each of mass 20 kg.
[Ans. Initial K.E. = 284 joules, final K.E. = 10,220 joules. The man has to do work in pulling dumb-bells towards him. The kinetic energy of the system will increase by this amount of work.]
52. Determine the angular momentum of a 1 MeV neutron about a nucleus, if the impact parameter is 10^{-12} m.
[Ans. 1.9×10^{-32} kg-m²/sec]
53. The distance of closest approach of a positive particle of charge e to the nucleus of atomic number 80 is 2×10^{-11} cm. Its initial kinetic energy is 2 MeV, obtain the value of the impact parameter for it.
[Ans. 1.69×10^{-13} m]
54. A neutron of energy 1 MeV passes a proton at such a distance that the angular momentum of the neutron relative to the proton approximately equals 10^{-33} J-sec. What is the distance of closest approach ? (Neglect the energy of interaction between the two particles).
[Ans. 4×10^{-14} m]

[Conservation of Linear Momentum and Collisions]

55. A 50 kg man standing on a surface of negligible friction kicks forward a 25 gm stone lying at his feet so that it acquires a speed of 2 metre/sec. What velocity does the man acquire as a result ?
[Ans. -10^3 m/sec.]
56. A shell of mass 0.02 kg is fired by a gun of mass 100 kg. If the muzzle speed of the shell is 80 m/sec, calculate the recoil speed of the gun.
[Ans. 1.6×10^{-2} m/sec]
57. Two bodies of masses 2.0 kg and 3.0 kg are moving with velocities 6 m/s and 4 m/s respectively in mutually perpendicular directions. Find the total linear momentum. If they collide and stick together, find the velocity and direction of motion of the composite body.
[Ans. 16.97 kg-m/s, 3.39 m/s in direction at an angle 135° with the direction of motion of body of mass 2.0 kg]
58. A meteor of 2 kg mass moving with a velocity of 12 km/sec explodes above the atmosphere into two pieces which move along the same path with 15 km/sec and 11 km/sec velocities. Find the mass of each piece and the energy set free by explosion.
[Ans. 0.5 kg and 1.5 kg; 3×10^6 joules]
59. A body of mass 3 kg makes an elastic collision with another body which is at rest. The first body moves in the same direction with its speed reduced by one half. Find the mass of the second body.
[Ans. 1 kg]
60. A 2 kg mass moving with a velocity of 10 m/sec collides with another 6 kg mass moving in opposite direction with a velocity of 20 m/sec. During collision, they stick together. Find their common velocity.
[Ans. 12.5 m/sec. opposite to the direction of initial velocity of 2 kg mass]
61. What percentage of kinetic energy of a moving particle is transferred to a stationary particle, when it strikes the stationary particle of
(a) 100 times its mass and

(b) equal mass ?

[Hint. (a) Here $m_2 = 100 m_1$. Now,

$$v_2 = \left(\frac{2m_1}{m_1 + m_2} \right) u_1 = \left(\frac{2m_1}{m_1 + 100m_1} \right) u_1 = \frac{2u_1}{101}$$

$$\text{Percentage of K.E. transferred} = \frac{\frac{1}{2} m_2 v_2^2}{\frac{1}{2} m_1 u_1^2} \times 100 = \frac{100m_1 \times (2u_1/101)^2}{m_1 u_1^2} \times 100 = 3.992\%$$

(b) When, $m_2 = m_1$,

$$v_2 = \left(\frac{2m_1}{m_1 + m_2} \right) u_1 = \left(\frac{2m_1}{m_1 + m_1} \right) u_1 = u_1$$

$$\text{Percentage of K.E. transferred} = \frac{\frac{1}{2} m_2 v_2^2}{\frac{1}{2} m_1 u_1^2} \times 100 = \frac{m_1 u_1^2}{m_1 u_1^2} \times 100 = 100\%]$$

62. A 5 kg ball and a 10 kg ball approach each other with equal speed of 20 m/sec. What are the speeds after impact if the collision is perfectly elastic ?

[Ans. – 33.3 m/sec, 6.67 m/sec]

63. A particle of mass M moving with a velocity u makes a head-on collision with a particle of mass m initially at rest. If the particles coalesce on colliding and move with a common velocity show that

the loss in kinetic energy is $\frac{1}{2} Mu^2 \left(\frac{m}{m+M} \right)$. (A.U. 1988; O.U. 1988)

64. Originally two particles are at the positions $x_1 = 5$ cm, $y_1 = 0$ and $x_2 = 0$, $y_2 = 10$ cm with $\vec{v}_1 = -4 \times 10^4 \hat{i}$ cm/sec and $\vec{v}_2$ is along the negative direction of Y -axis. (i) What must be the value of $\vec{v}_2$, if they collide ? (ii) What is the velocity of the first particle relative to the second ?

[Ans. (i) -8×10^2 m/sec; (ii) $4 \times 10^2 (2\hat{j} - \hat{i})$ m/sec]

65. A particle of mass m_1 , moving with speed v , makes an elastic head-on collision with another particle of mass m_2 ($m_2 > m_1$). If the particle m_2 has initially equal but opposite momentum to that of m_1 , show that the velocity of the first particle after the collision is $-v$ and if m_2 is at rest in the condition $m_2 \gg m_1$, the loss of kinetic energy of m_1 is $4m_1/m_2$.

66. Two blocks of masses 300 gm and 200 gm are moving toward each other along a horizontal frictionless surface with velocities 50 cm/sec and 100 cm/sec respectively. (a) If the blocks collide and stick together, find their final velocity. (b) Calculate the loss of kinetic energy during the collision. (c) Find the final velocity of each block if the collision is completely elastic.

[Ans. (a) 10 cm/sec; (b) 0.14 joule; (c) -70 cm/sec, 80 cm/sec]

67. A bomb explodes breaking into three pieces and two pieces fly off perpendicular to each other. One piece of mass 1 kg has a velocity of 12 m/s and the second of mass 2 kg has a velocity of 8 m/sec. If the third piece flies off with a velocity of 40 m/s, find its mass.

[Ans. 0.5 kg]

68. A gas molecule having a speed of 300 m/sec collides elastically with another molecule of the same mass which is initially at rest. After collision, the first molecule makes an angle of 30° to the initial direction. Find the speed of each molecule after collision and the angle made with the incident direction by the recoiling target molecule. (A.N.U. 1988)

[Ans. $\theta = 60^\circ$, $v_1 = 150\sqrt{3}$ m/sec and $v_2 = 150$ m/sec]

69. A proton travelling with a speed of 600 m/s interacts with another proton initially at rest. If one proton moves off at an angle of 60° from the initial direction what is the direction of other proton after interaction ? What are the speeds of both the protons. Assume the collision to be perfectly elastic.
[Ans. $\theta = 30^\circ$, $v_1 = 300$ m/s and $v_2 = 519.6$ m/s] (A.U. 1990)
70. A bomb of mass 300 kg explodes in flight at the instant when its velocity is $\vec{v}_0 = (10\hat{i} + 5\hat{j})$ m/sec. It splits in two fragments of mass ratio 1 : 2 and the larger mass is observed to have velocity $\vec{v}_2 = (100\hat{i} + 50\hat{j} + 10\hat{k})$ m/sec just after the explosion. Deduce :
(i) velocity $\vec{v}_1$ of the smaller fragment just after the explosion.
(ii) kinetic energy produced in the explosion.
[Ans. (i) $(-170\hat{i} - 85\hat{j} - 20\hat{k})$ m/sec.; (ii) 0.0409 joule]
71. An α -particle is emitted from a uranium nucleus (U^{238}) in a radioactive decay. The speed and kinetic energy of α -particle are 1.404×10^7 m/sec and 4.212 MeV respectively. Calculate the recoil speed and kinetic energy of the residual nucleus.
[Ans. -2.4×10^5 m/sec., 0.072 MeV]
72. In a radioactive decay of a nucleus, an electron and a neutrino with momentum 1.28×10^{-22} and 6.4×10^{-23} kg-m/sec are emitted at right angles to each other. What is the momentum of the recoiling nucleus ? If the mass of the residual nucleus is 5.8×10^{-26} kg, calculate its recoil kinetic energy.
[Ans. $\approx 1.4 \times 10^{-22}$ kg-m/sec in a direction $\tan^{-1} \frac{1}{2}$ with the electron velocity, 1.1 eV]
73. A stationary body of mass 3 kg explodes into three equal pieces. Two of the pieces fly off at right angles to each other, one with $2\hat{i}$ m/sec and the other with $3\hat{j}$ m/sec. If the explosion takes in 10^{-5} sec, find the average force acting on each piece during the explosion.
[Ans. $2 \times 10^5 \hat{i}$ newton; $3 \times 10^5 \hat{j}$ newton; $-(2\hat{i} + 3\hat{j}) \times 10^5$ newton]
74. A 5 kg object with a speed of 30 m/s strikes a steel plate at an angle of 45° and rebounds at the same speed and same angle. What is the change of the linear momentum in magnitude and direction ?
[Ans. $150\sqrt{2}$ kg-m/sec perpendicular to steel plate]
75. A gun weighing M fires a shell of mass m horizontally and the energy of explosion is such that it can raise the shell to a vertical height h . Show that the speed of the recoil of the gun is $[2m^2 gh/M(m + M)]^{1/2}$.
76. A bullet of mass M is fired with a velocity of 50 m/sec at an angle θ with the horizontal. At the highest point of its trajectory, it collides head-on with a bob of mass $3M$ suspended by a massless string of length $1/3$ m and gets embedded in the bob. After the collision, the string moves through an angle of 120° . Find (i) the angle θ , (ii) the vertical and horizontal co-ordinates of the initial position of the bob with respect to the point of firing of the bullet. [$g = 10$ m/sec²]
[Ans. (i) $\theta = \cos^{-1}(0.8)$; (ii) 45 m, 120 m]
77. A bullet of mass 2 gm, travelling at 500 m/sec, is fired into a ballistic pendulum of mass 1 kg suspended from a cord 1 m long. The bullet penetrates the pendulum and emerges with a velocity of 100 m/sec. Through what vertical height the pendulum rise ?
[Ans. 3.3 cm]
78. A bullet of mass m , moving horizontally with speed v , passes through a pendulum bob of mass M and emerges with a velocity $v/2$. If the length of the pendulum is l , show that the minimum value of v , which will make the pendulum bob to swing through a complete circle is $(2M/m)\sqrt{5gl}$.
79. A rifle bullet of 0.1 kg strikes and embeds itself in a block of 0.9 kg which rests on a horizontal frictionless table and is attached to a spring. The impact compresses the spring through 0.1 metre. If a force of 1 N is required to compress the spring $1/100$ m, calculate the initial velocity of the bullet. What is the velocity of the block just after impact ?
[Ans. 3.33 m/sec; 0.333 m/sec]

80. A steel bob of 1 kg mass is fastened to a cord 1 m and is released when the cord is horizontal [fig. 2.54]. At the bottom of its path, the bob strikes a 5 kg block initially at rest on a frictionless surface. If the collision is elastic, calculate the speed of the bob and speed of the block just after the collision.

[Ans. Bob -2.953 m/sec; block 1.477 m/sec]

81. Two pendulums of masses m_1 and m_2 each of length l are initially situated as shown in [fig. 2.55]. The first pendulum bob from a height h is released and strike the second bob. Assume that the collision is completely inelastic and neglect the mass of string and any frictional effects. How high does the centre of mass rise after the collision?

[Ans. $h [m_1/(m_1 + m_2)]^2$]

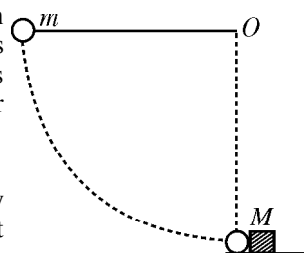


Fig. 2.54.

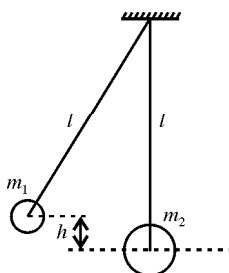


Fig. 2.55.

[Scattering]

82. Show that the distance of closest approach of a proton of energy V electron volts, moving towards a nucleus of atomic number Z is $14.4 (Z/V) \text{ \AA}$.
83. If a particle of charge q and mass m moves with velocity v directly towards a much heavier particle of charge Q , find the distance of closest approach. (Neglect motion of heavier particle)

[Ans. $\frac{1}{4\pi\epsilon_0} \cdot \frac{2Qq}{mv^2}$]

84. An α -particle of velocity 2×10^6 m/sec approaches a gold nucleus ($Z = 79$) in head-on direction. Deduce the nearest distance of approach; assuming Coulomb's law to hold.

[Ans. 2.7×10^{-12} m]

85. An α -particle of energy 10 MeV moves ahead on toward a stationary nucleus of atomic number 80. What is the distance of closest approach?

[Ans. 2.3×10^{-14} m]

86. In Rutherford experiment, 10^3 α -particles are scattered at an angle of 4° , calculate the number of particles scattered at an angle of 14° .

[Ans. 6.7 approx]

87. Determine the differential scattering cross-section for α -particles by Pb ($Z = 82$) nucleus. Calculate the value of impact parameter also.

[Ans. $1.64 \times 10^{-26} \text{ m}^2$; $6.4 \times 10^{-14} \text{ m}$]

88. Show that in the scattering of a charged particle under inverse square repulsive force, the differential scattering cross-section is proportional to $\text{cosec}^4 \theta/2$, where θ is the scattering angle.



Mechanics of Rigid Bodies

3.1. Definition of Rigid Body

A rigid body is defined as a system of particles in which the distances between all pairs of particles remain fixed throughout the motion. Thus, in case of a rigid body if r_{ij} is the distance between any two i th and j th, particles then

$$r_{ij} = \text{constant}$$

In a solid the distances between its constituent particles change to some extent when there is strain due to external forces. If such changes are negligibly small, the solid may be considered as a rigid body. Thus a rigid body will not undergo any change of shape or size due to the application of external forces. However, if on applying external forces, its volume or shape is changed so that the relative positions of the particles of the body are changed, then the body is said to be **deformed**, e.g., rubber.

3.2. Rotational Kinematic Relations

Let a rigid body rotate about a fixed axis. By a fixed axis we mean an axis that is at rest in some inertial frame of reference and does not change direction relative to that frame. **Fig. 3.1** shows a rigid body which rotates about a fixed axis AB . Consider a particle P of the body. Let its position be P_0 at time $t = 0$. Draw a perpendicular P_0O to the axis of rotation. At time t , the particle moves to the position P . If we denote $\angle P_0OP = \theta$, then it is said that the particle P has rotated through an angle θ . In fact, all other particles of the rigid body have also rotated through the same angle θ and we say that the whole body has rotated through an angle θ . This θ is said to be the **angular position** of the rigid body at time t . Hence the rotation of a rigid body is measured by angle θ of OP line from its initial position. Let θ be positive when measured anti-clockwise.

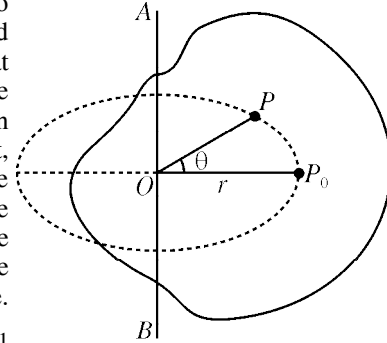


Fig. 3.1.

Angular velocity : If θ_1 is the angular position of P at time t_1 and θ_2 at time t_2 , then the **average angular velocity** in the time interval $t_2 - t_1 = \Delta t$ is defined as

$$\omega_{av} = \frac{\theta_2 - \theta_1}{t_2 - t_1} = \frac{\Delta\theta}{\Delta t} \quad \dots(1)$$

where $\theta_2 - \theta_1 = \Delta\theta$.

The **instantaneous angular velocity** ω is the limit of ω_{av} as Δt approaches zero, i.e., the derivative of θ with respect to t . Thus

$$\omega = \lim_{\Delta t \rightarrow 0} \frac{\Delta\theta}{\Delta t} = \frac{d\theta}{dt} \quad \dots(2)$$

Normally when we say angular velocity, we mean the instantaneous angular velocity. As the body is rigid, all lines like OP rotate the same angle in the same time and hence the angular velocity ω is the characteristic of the entire body.

Angular acceleration : When there is a change in the angular velocity, we say that it has angular acceleration. If ω_1 and ω_2 are the instantaneous angular velocities at times t_1 and t_2 respectively, then the average angular acceleration is defined as

$$\alpha_{av} = \frac{\omega_2 - \omega_1}{t_2 - t_1} = \frac{\Delta\omega}{\Delta t} \quad \dots(3)$$

The **instantaneous angular acceleration** is defined as the limiting value of $\Delta\omega/\Delta t$ as $\Delta t \rightarrow 0$. Thus

$$\alpha = \lim_{\Delta t \rightarrow 0} \frac{\Delta\omega}{\Delta t} = \frac{d\omega}{dt} \quad \dots(4)$$

Normally the term angular acceleration is used for instantaneous angular acceleration. The units of angular velocity and angular acceleration are radian/sec and radian/sec² respectively.

Rotation with constant angular velocity : If the body rotates through equal angles in equal time intervals, we say that it rotates with uniform angular velocity. In such a case,

$$\omega = d\theta/dt = \text{constant and } \theta = \omega t \quad \dots(5)$$

Rotation with constant angular acceleration

Let a rigid body rotate with constant angular acceleration α . From eq. (4), we have

$$\alpha = \frac{d\omega}{dt} \quad \text{or} \quad d\omega = \alpha dt \quad \dots(6)$$

If ω_0 and ω be the angular velocities at times $t = 0$ and $t = t$ respectively, then integrating (6) we obtain

$$\int_{\omega_0}^{\omega} d\omega = \alpha \int_0^t dt \quad \text{or} \quad \omega - \omega_0 = \alpha t$$

$$\text{Thus} \quad \omega = \omega_0 + \alpha t \quad \dots(7)$$

Further from eq. (2)

$$\omega = \frac{d\theta}{dt} \quad \text{or} \quad d\theta = \omega dt = (\omega_0 + \alpha t)dt \quad \dots(8)$$

If θ_0 and θ be the angular positions at $t = 0$ and $t = t$ respectively, then integrating (8), we get

$$\int_{\theta_0}^{\theta} d\theta = \int_0^t (\omega_0 + \alpha t) dt \quad \text{or} \quad \theta - \theta_0 = \omega_0 t + \frac{1}{2} \alpha t^2$$

$$\text{Thus} \quad \theta = \theta_0 + \omega_0 t + \frac{1}{2} \alpha t^2 \quad \dots(9)$$

If $\theta_0 = 0$ at $t = 0$, then

$$\theta = \omega_0 t + \frac{1}{2} \alpha t^2 \quad \dots(10)$$

Substituting the value of $t = (\omega - \omega_0)/\alpha$ from eq. (7) in eq. (10), we get

$$\theta = \omega_0 \left(\frac{\omega - \omega_0}{\alpha} \right) + \frac{1}{2} \alpha \left(\frac{\omega - \omega_0}{\alpha} \right)^2$$

or

$$2\alpha\theta = 2\omega_0(\omega - \omega_0) + (\omega - \omega_0)^2 = \omega^2 - \omega_0^2$$

$$\text{Thus} \quad \omega^2 = \omega_0^2 + 2\alpha\theta \quad \dots(11)$$

Equations. (7), (10) and (11) for rotational motion with constant angular acceleration are similar to equations of motion in straight line motion, i.e.,

$$v = v_0 + at, \quad s = v_0 t + \frac{1}{2} at^2 \quad \text{and} \quad v^2 = v_0^2 + 2as \quad \dots(12)$$

Relation between the linear motion of a particle of a rigid body and its rotation : Let a point P of the rigid body rotate about a fixed axis AB as in **fig. 3.1**. On the rotation of the body, this point P moves in a circle of radius r . If the body or the particle P rotates through an angle θ about the axis AB , then the linear distance moved by the particle along the circle is given by

$$s = r\theta \quad \dots(13)$$

The linear speed along the tangent is

$$v = \frac{ds}{dt} = r \frac{d\theta}{dt} = r\omega \text{ i.e., } v = r\omega \quad \dots(14)$$

and the linear acceleration along the tangent i.e., tangential acceleration is

$$a_t = \frac{dv}{dt} = r \frac{d\omega}{dt} = r\alpha \text{ i.e., } a_t = r\alpha \quad \dots(15)$$

For different particles of the rigid body, the radius r of their circles of rotation has different values, but the angular velocity ω and angular acceleration α are same for all the particles. Therefore, the linear speed and the tangential acceleration of various particles of the body are different.

The **radial (centripetal) acceleration** a_r of a particle moving with speed v in a circle of radius r is given by

$$a_r = \frac{v^2}{r} = \frac{r^2\omega^2}{r} = \omega^2 r \text{ i.e., } a_r = \omega^2 r \quad \dots(16)$$

The resultant acceleration $\vec{a}$ of the particle P can be obtained by adding $\vec{a}_r = a_r \hat{r}$ and $\vec{a}_t = a_t \hat{\theta}$ vectorially, i.e.,

$$\vec{a} = a_r \hat{r} + a_t \hat{\theta} \quad \dots(17)$$

where $\hat{r}$ is unit vector along $\vec{r} = \vec{OP}$ and $\hat{\theta}$ is unit vector perpendicular to $\vec{r}$ [fig. 3.2].

Its magnitude

$$a = \sqrt{a_r^2 + a_t^2} = \sqrt{\left(\frac{v^2}{r}\right)^2 + \left(\frac{dv}{dt}\right)^2} \quad \dots(18)$$

and

$$\tan \phi = \frac{a_t}{a_r} = \left(\frac{dv}{dt}\right) / \left(\frac{v^2}{r}\right) \quad \dots(19)$$

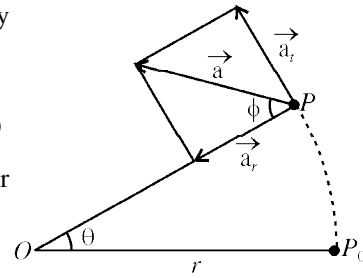


Fig. 3.2. Tangential and radial accelerations.

3.3. Moment of Inertia about an Axis

The *moment of inertia of a mass point about an axis is defined as the product of its mass and the square of its distance from the axis of rotation*. If a particle of mass m be situated at a distance r from the given axis, its moment of inertia I , therefore, is given by

$$I = mr^2 \quad \dots(1)$$

Consider a rigid body of mass m capable of rotating about an axis AB [fig. 3.3]. Assume that the body is composed of a very large number of small particles of masses $m_1, m_2, m_3, \dots$ etc. at distances $r_1, r_2, r_3, \dots$ etc. respectively from the axis AB . Then the moment of inertia of the particle of mass m_1 , about the axis AB is $m_1 r_1^2$, that of the particle of the mass m_2 is $m_2 r_2^2$ and so on. Hence the moment of inertia I of the whole body about the given axis is equal to the sum of the quantities $m_1 r_1^2, m_2 r_2^2, m_3 r_3^2, \dots$ etc. i.e.,

$$I = m_1 r_1^2 + m_2 r_2^2 + m_3 r_3^2 + \dots \quad \dots(2)$$

or,

Hence, *the moment of inertia of a body about an axis is defined as the sum of the products of the masses of different particles, supposed to be constituting the body, and the square of their respective distances from the axis of rotation*. In C.G.S. system, the unit of moment of inertia is **gm-cm²** and in S.I. system it is **kg-m²** and the dimensional formula is $[ML^2]$.

When the body is of continuous homogeneous structure of density ρ , the summation sign may be replaced by integration. Thus

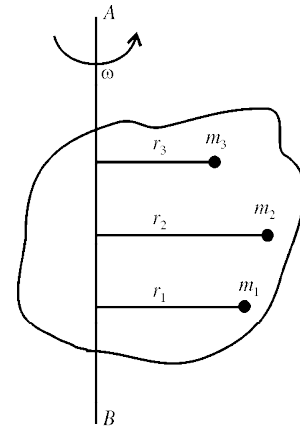


Fig. 3.3. Moment of inertia about an axis.

$$I = \int_{\alpha_1}^{\alpha_2} r^2 dm = \int r^2 \rho dV \quad \dots(3)$$

where $dm = \rho dV$ represents the mass of an infinitesimally small element of the body, situated at a distance r from the axis of rotation. The limits of integration (α_1 and α_2) depend upon the geometrical shape of the body.

Evidently, the moent of inertia of a body about an axis depends upon the following :

- (1) the mass of the body;
- (ii) the distribution of its mass about the given axis.

It is to be noted that the mass (translational inertia) of a body remains constant, but the moment of inertia of a body can be varied, because changing the distance of the mass from the axis of rotation alters the value of moment of inertia.

To have an illustration, let us compare the moment of inertial of a solid circular disc with that of a wheel, having the same mass and the same external radius. In the two cases, the translational inertia (mass) is clearly the same. However, the wheel possesse a considerably large moment of inertia than the disc, the axis of rotation for each has been taken to pass through their respective centres and perpendicular to their planes. The reason obviously lies in the fact that on an average, the particles, constituting the wheel, are situated at a greater distance from the axis of rotation.

It should be noted that if a part whose moment of inertia is I_1 is taken out of a body whose moment of inertia is I , then evidently the moment of inertial of the remaining part will be

$$I_2 = I - I_1 \quad \dots(4)$$

where I , I_1 and I_2 are about the same axis.

Radius of gyration

Whatever may be the shape of a body, it is always possible to find a distance from the axis of rotation at which whole mass of the body could be assumed to be concentrated and even then its moment of inertia about that axis remains unaltered. For this, if the whole mass M of a body be supposed to be concentrated at a distance k from the axis of rotation, then clearly

$$I = Mk^2 = \Sigma mr^2$$

$$\text{or} \quad k = \sqrt{\frac{I}{M}} = \sqrt{\frac{\Sigma mr^2}{M}} \quad \dots(5)$$

This quantity k is called **radius of gyration** of the body about the axis of rotation. Thus, *the radius of gyration of a body, rotating about a given axis, is equal to the distance from the axis of rotation, the square of which, when multiplied by the total mass of the body, will give the moment of inertia of the body about that axis.*

Physical significance of moment of inertia

Newton's first law of motion implies that a body maintains its state of rest or of uniform translatory motion, unless acted upon by some external force. This inertness or inability of a body to change its position of rest or of uniform motion by itself, is called **inertia**. Inertia is a fundamental property of matter and by virtue of which a body opposes an external force tending to change its state of rest or linear motion. Greater the mass of the body, greater is the force required to produce a given linear acceleration in it. Thus, the mass of a body is taken to be a measure of its translational intertia.

Similarly, the state of rotation of a rigid body about an axis remains unaltered, unless an external torque acts upon it. This means that a body, free to rotate about an axis, opposes the external torque, tending to change in its state of rest or of uniform rotation about that axis. This inertia of the body for rotational motion gives an idea about a physical quantity, called its **moment of inertia*** about the axis of rotation and is found to depend not only upon its mass but also on the actual distribution of the mass

* The inertia for rotation is given the name 'moment of inertia' on the analogy of the moment of the couple which this inertia (rotational) opposes.

relative to the axis of rotation. Greater the moment of inertia of a body, greater is the torque required to produce a **desired angular acceleration in it**. Thus in case of rotatory motion, the quantities force and mass of translatory motion are replaced by torque and moment of inertia respectively. This is the physical significance of moment of inertia.

3.4. Angular Momentum, Torque and Angular Impulse

(1) Angular momentum about an axis

The moment of linear momentum of a particle about an axis of rotation is called its angular momentum. It is usually denoted by the letter L .

Consider a rigid body rotating about an axis with an angular velocity ω . Let a particle of mass m of the rigid body be situated at a distance r from the axis of rotation. If it has the linear velocity v , then its linear momentum is mv . Hence, the moment of momentum or angular momentum of the particle about the axis of rotation is given by [fig. 3.4]

$$L = mvr = mr^2\omega \quad [\because v = r\omega] \dots(1)$$

where ω is the angular velocity.

Since the rigid body consists of a number of particles, then the total angular momentum of the body will be the sum of the angular momenta of various particles about the axis of rotation.

Hence angular momentum of the body about the axis of rotation is

$$L = \Sigma mr^2\omega. \dots(2)$$

All the constituent particles of a rotating body have the same angular velocity ω , hence

$$L = \omega \Sigma mr^2 = I\omega \quad [\because \Sigma mr^2 = I] \dots(3)$$

Since the angular velocity is a vector, represented in the form $\vec{\omega}$, then the angular momentum along the axis of rotation can be expressed as

$$\vec{L} = I \vec{\omega} \dots(4)$$

Thus the angular momentum of a rotating rigid body about an axis is the product of moment of inertia and the angular velocity of the body about that axis.

(2) Torque about on axis

If a rigid body is free to rotate about an axis, then by the application of a couple the rotation of the body can be accelerated. The moment of this couple is called **torque** and we say that a torque has been applied on the body.

Let us consider a rigid body be capable to rotate about an axis AB passing through a point O . Suppose m be the mass of a particle P of the body at a distance r from this axis. When the body rotates through a small angle $d\theta$ in a small interval of time dt , P moves a small distance $PP' = ds = rd\theta$ in the same time. If F be the force acting on the particle at P and d^2s/dt^2 its acceleration, then according to Newton's second law of motion,

$$F = m \frac{d^2s}{dt^2}.$$

But $ds = rd\theta$,

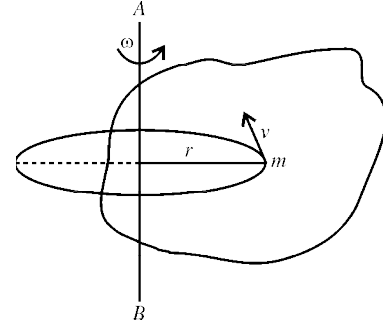


Fig.3.4 Angular momentum of a rigid body about an axis $L = \Sigma mr^2\omega = I\omega$

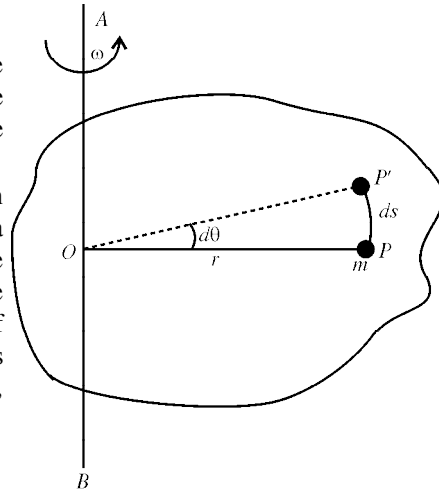


Fig. 3.5. Torque on a rigid body about an axis $\tau = I d\omega/dt = I\alpha$.

$$\therefore \frac{ds}{dt} = r \frac{d\theta}{dt} \text{ and } \frac{d^2s}{dt^2} = r \frac{d^2\theta}{dt^2}.$$

Therefore, the force acting upon the particle at P is given by

$$F = mr \frac{d^2\theta}{dt^2}$$

where $d^2\theta/dt^2$ is the angular acceleration of the particle at P about the axis passing through O . This force is acting on the particle P perpendicular to the line OP ($= r$) in the plane of rotation.

The moment of this force about the axis of rotation is

$$Fr = mr^2 \frac{d^2\theta}{dt^2}.$$

Therefore, the total moment of the forces for the whole body about the axis of rotation i.e., the torque on the body is

$$\tau = \Sigma mr^2 \frac{d^2\theta}{dt^2}.$$

Here, the angular acceleration $d^2\theta/dt^2$ is the same for all particles of the body. Therefore, it may be taken outside of the summation, and hence

$$\tau = \frac{d^2\theta}{dt^2} \Sigma mr^2. \quad \dots(5)$$

But $\Sigma mr^2 = I$, is the moment of inertia of the body about the axis passing through O , therefore

$$\tau = I \frac{d^2\theta}{dt^2} = I \frac{d\omega}{dt} \quad \dots(6)$$

because $\omega = d\theta/dt$ is the angular velocity.

Thus the torque on a rotating rigid body about an axis is equal to the product of moment of inertia and angular acceleration about that axis.

Eq. (6) is called equation of motion for a rotating rigid body about an axis. This is the rotational analogue of Newton's second law of motion i.e., force = mass $\times$ acceleration.

Now, if $d^2\theta/dt^2 = 1$ then $\tau = I$.

Thus, the moment of inertia of a body about an axis is equal to the torque, producing unit angular acceleration in it about that axis.

(3) Relation between torque and angular momentum about an axis

The expression of torque on a rigid body rotating about a fixed axis is given by

$$\tau = I \frac{d\omega}{dt} = I\alpha \quad \dots(7)$$

where I is the moment of inertia about the axis of rotation and $d\omega/dt = \alpha$ is the angular acceleration.

Eq. (7) can be written as

$$\tau = \frac{d}{dt} (I\omega) = \frac{dL}{dt} \quad \dots(8)$$

where $L = I\omega$ is the angular momentum of the body about the given axis.

Thus, the torque acting on a rigid body is equal to the rate of change of angular momentum. Eq. (7) or (8) is called equation of motion of a rotating rigid body about an axis and corresponds to Newton's second law for linear motion i.e.,

$$F = \frac{d}{dt}(mv) = m \frac{dv}{dt} = ma \quad \dots(9)$$

If the external torque $\tau = 0$, then

$$\frac{dL}{dt} = 0 \quad \text{or} \quad L = I\omega = \text{constant}. \quad \dots(10)$$

Thus, *if the external torque on the rotating system is zero, the angular momentum remains constant.* This is known as the **law of conservation of angular momentum** for rotational motion. This law is in complete analogy to the law of conservation of momentum in linear motion, which states that the total linear momentum of an isolated system is constant in absence of external forces.

(4) Angular impulse

The expression for the torque about an axis is given by

$$\tau = I \frac{d\omega}{dt} \quad \dots(11)$$

whence,

$$\tau dt = I \frac{d\omega}{dt} dt.$$

Integrating this expression with respect to time t between the limits $t = 0$ and $t = t$, we get

$$\int_0^t \tau dt = \int_0^t I \frac{d\omega}{dt} dt = \int_{\omega_0}^{\omega} I d\omega \quad \dots(12)$$

where ω_0 and ω are the angular velocities of the body just before and after the time-interval t for which τ is applied.

If the time interval t is very small and torque τ is quite large, then the time integral of the torque, i.e., $\int_0^t \tau dt$ is called the impulse of the torque or angular impulse. Thus, the angular impulse is given by

$$\int_0^t \tau dt = \int_{\omega_0}^{\omega} I d\omega = I\omega - I\omega_0 \quad \dots(13)$$

i.e., angular impulse = change in angular momentum.

Therefore, *the angular impulse is equal to the change in angular momentum of the body.* The concept is analogue of linear impulse ($F dt$) in the translatory motion.

If the torque τ is constant for the time interval t , then

$$\text{angular impulse} = \int_0^t \tau dt = \tau t = I\omega - I\omega_0. \quad \dots(14)$$

(4) Power

If a rigid body is rotating about an axis, under the application of a force F , then for an infinitesimal rotation $d\theta$ the work done is given by

$$dW = Fds = Frd\theta$$

when F is acting along ds and $ds = r d\theta$ (**fig. 3.5**)

But $Fr = \tau$, the torque acting on the body, Hence

$$dW = \tau d\theta$$

Rate of doing work is called power. Hence

$$\text{power } P = \frac{dW}{dt} = \frac{\tau d\theta}{dt} = \tau\omega \quad \dots(15)$$

3.5. Kinetic Energy of a Rotating Body about an Axis

The kinetic energy of a rotating body about an axis depends not only upon its mass and angular velocity but also depends upon the position of the axis and the distribution of mass about that axis.

Suppose a body of mass M is rotating about an axis AB . Assume that the body is composed of a very large number of small particles [fig. 3.6]. Consider a particle P of elementary mass m at a distance r from the axis. If ω is the angular velocity, the linear velocity of the particle P will be $r\omega$.

Therefore, the kinetic energy of rotation of the particle P is $\frac{1}{2}mr^2\omega^2$. Hence the kinetic energy (K.E.) of rotation of the whole body is given by

$$K = \sum \frac{1}{2}mr^2\omega^2 = \frac{1}{2}\omega^2 \sum mr^2$$

since the angular velocity ω is the same for all the constituent particles of the body.

Now, $\sum mr^2 = I$, the moment of inertia about the axis AB ,

$$\therefore \text{K.E. of rotation } K = \frac{1}{2}I\omega^2 \quad \dots(1)$$

If $\omega = 1$, obviously $I = 2 \times K$.

Hence, the moment of inertia of a body may also be defined as twice its kinetic energy of rotation, the angular velocity of the body being unity.

$$\text{But angular momentum } L = I\omega \quad \dots(2)$$

where L is along the axis of rotation,

$$\therefore L = I\sqrt{2K/I} = \sqrt{2KI} \text{ or } K = L^2/2I \quad \dots(3)$$

This is the relation between the rotational kinetic energy and angular momentum about the same axis.

Rolling bodies : Let us calculate the kinetic energy of a rolling body. When a body with circular symmetry (e.g., cylinder, wheel, disc, sphere etc.) rolls on a plane surface, its motion is a combination of translation and rotation. At any instant the axis normal to the plane of the diagram through the point of contact P is the axis of rotation. If the speed of the centre of mass is V (relative to an observer fixed on the surface), the instantaneous angular speed about an axis through P is $\omega = V/R$, where R is the radius of the body. Therefore, at that instant all the particles of the rigid body are moving with the same angular speed* ω about the axis through P and the motion of the body is equivalent to a **pure rotation**. Thus, the total kinetic energy is

$$K = \frac{1}{2}I_P\omega^2,$$

where I_P is the moment of inertia about the axis through P .

From the theorem of parallel axes,

$$I_P = I_{cm} + MR^2 \quad \dots(4)$$

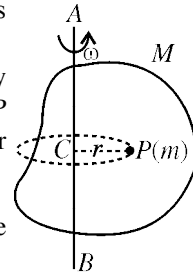
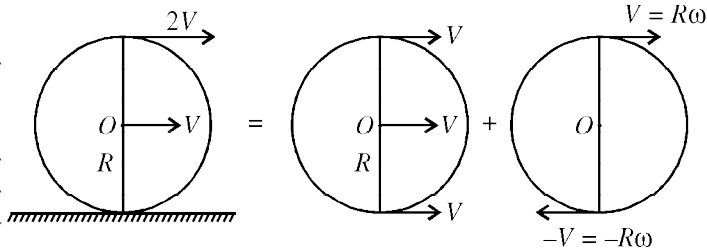


Fig. 3.6.



Rolling Motion = Translational Motion + Rotational Motion

Fig. 3.7.

* It is to be noted that at any instant different particles of the body have different linear speeds. The point P is at rest ($v = 0$) instantaneously, the centre of mass has speed $V = R\omega$ and the highest point of circumference has speed $2R\omega$ relative to P .

where I_{cm} is the moment of inertia of the body of mass M about a parallel axis through O .

$$\therefore K = \frac{1}{2} I_{cm} \omega^2 + \frac{1}{2} MR^2 \omega^2 = \frac{1}{2} I_{cm} \omega^2 + \frac{1}{2} MV^2 \quad \dots(5)$$

If k is the radius of gyration of the body about a parallel axis through O , then $I = Mk^2$ and therefore total kinetic energy

$$K = \frac{1}{2} Mk^2 \frac{V^2}{R^2} + \frac{1}{2} MV^2 = \frac{1}{2} MV^2 \left(\frac{k^2}{R^2} + 1 \right) \quad \dots(6)$$

Alternatively, the motion of the cylinder can be considered as the sum of the translational motion with speed V and pure rotational motion about the centre C with angular speed $\omega = V/R$ [fig. 3.7]. So that, the total kinetic energy is

$$K = \frac{1}{2} MV^2 + \frac{1}{2} I_{cm} \omega^2 = \frac{1}{2} MV^2 \left(1 + \frac{k^2}{R^2} \right) \quad \dots(7)$$

3.6. Motion of a Body Rolling Down an Inclined Plane

If a body rolls down an inclined plane without slipping, it has both translatory and rotatory motions. The body rolls down, hence it suffers vertical descent causing to its loss of potential energy. At the same time the body acquires linear and angular velocities and, therefore, it gains kinetic energy of translation and rotation. If there would have been no loss of energy through friction etc., the loss of potential energy must be equal to the gain in kinetic energy.

Let a body of mass M and radius R be rolling down a plane inclined at an angle θ to the horizontal [fig. 3.8]. If the body acquires an angular speed ω and its centre of mass acquires a linear speed V , after covering a distance s along the inclined plane or a vertical height h ($= s \sin \theta$), then

Loss of potential energy in travelling a vertical height $h = Mgh$.

Gain in
$$\text{K.E.} = \frac{1}{2} I_{cm} \omega^2 + \frac{1}{2} MV^2 = \frac{1}{2} MV^2 \left(\frac{k^2}{R^2} + 1 \right),$$

where k is the radius of gyration about an axis through the centre of mass and parallel to the plane.

According to the law of conservation of energy,

loss in P.E. of the body = gain in K.E. by the body

i.e.,
$$Mgh = \frac{1}{2} MV^2 \left(\frac{k^2}{R^2} + 1 \right)$$

$$\therefore V^2 = \frac{2gh}{1 + k^2/R^2} = \frac{2gs \sin \theta}{1 + k^2/R^2} \quad [\because h = s \sin \theta] \quad \dots(1)$$

If the centre of mass of the body acquires an acceleration a on covering the distance s along the plane after starting from rest, then $V^2 = 2as$. Therefore,

$$2as = \frac{2gs \sin \theta}{1 + k^2/R^2}. \text{ So that, } a = \frac{g \sin \theta}{1 + k^2/R^2} \quad \dots(2)$$

Thus, it is obvious from the above equation that for a given inclination θ , acceleration of the body is inversely proportional to $(1 + k^2/R^2)$. Hence, it follows that lesser the value of k as compared to R the greater is the acceleration of the body rolling down the plane and lesser the time of descent. These effects will be reversed when k is larger. Again, as the expression (2) for acceleration does not contain the mass of the body, the acceleration will be independent of the mass. The magnitude of acceleration and so the time of descent for two bodies of the same mass, size and appearance but of different radius of gyration

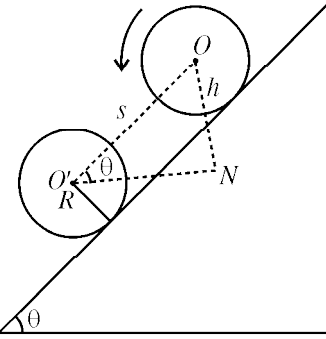


Fig. 3.8.

k (depending upon their internal structure) are different. This difference in times of descent provides an excellent means to distinguish between two spheres of the same mass, size and appearance when one of them is hollow and the other solid.

For identification of the two spheres, one hollow and the other solid but of the same masses (M) and same external radii (R), we allow them to roll down an inclined plane from the same height simultaneously. The moment of inertia of a solid sphere of mass M and radius R about the axis of rotation (diameter) is

$$I = \frac{2}{5}MR^2 = Mk_1^2, \quad \therefore k_1^2 = \frac{2}{5}R^2 \quad \dots(3)$$

where k_1 is the radius of gyration about the axis of rotation.

If the external and internal radii of the hollow sphere are R and R_1 respectively, then its moment of inertia about the axis of rotation is

$$I = \frac{2}{5}M \frac{R^5 - R_1^5}{R^3 - R_1^3} = Mk_2^2$$

where k_2 is the radius of gyration of the hollow sphere about the axis of rotation.

$$\therefore k_2^2 = \frac{2}{5} \frac{R^5 - R_1^5}{R^3 - R_1^3} = \frac{2}{5} \frac{R^5(1 - R_1^5/R^5)}{R^3(1 - R_1^3/R^3)}$$

$$\text{or} \quad k_2^2 = \frac{2}{5}R^2 \frac{1 - (R_1^5/R^5)}{1 - (R_1^3/R^3)} \quad \dots(4)$$

From eq. (4) and eq. (3), we have

$$\frac{k_2^2}{k_1^2} = \frac{1 - (R_1^5/R^5)}{1 - (R_1^3/R^3)} \quad \dots(5)$$

$$\text{But } R > R_1, \text{ hence } \frac{R_1^5}{R^5} < \frac{R_1^3}{R^3} \quad \text{or} \quad 1 - \frac{R_1^5}{R^5} > 1 - \frac{R_1^3}{R^3}$$

$$\text{or} \quad \frac{1 - R_1^5/R^5}{1 - R_1^3/R^3} > 1, \quad \therefore k_2^2 > k_1^2. \quad \dots(6)$$

Thus the value of k^2 for a hollow sphere is greater than that for a solid sphere, although the masses and external radii of the two spheres are the same. Hence it is obvious from the relation (2) that the acceleration of the solid sphere must be greater than that of the hollow one. Therefore, the solid sphere will cover a larger distance on the inclined plane than the hollow sphere in the same interval of time, i.e., if the two spheres are allowed to roll down an inclined plane simultaneously, the solid sphere will reach down the plane first.

3.7. Translatory and Rotatory Motion : A Comparison

When a rigid body moves as a whole along a straight line, its motion is said to be *translatory*. In case of translatory or linear motion, all the particles of the body move in a similar way along parallel straight lines and it makes no difference that we consider the body as a point particle or extended object.

In case of translatory motion, if a body of mass m is moving with velocity $\vec{v}$, its momentum is given by $\vec{p} = m\vec{v}$ and the external force acting on it is given by according to Newton's second law of motion i.e.,

$$\vec{F} = \frac{d\vec{p}}{dt} = \frac{d}{dt}(m\vec{v}) = m\frac{d\vec{v}}{dt} = m\vec{a} \quad \dots(1)$$

where $\vec{a} = d\vec{v}/dt$ is the acceleration. Equation (1) is known as the *fundamental equation of motion for translatory motion of a body*. This equation holds similarly whether the body is in the form of a point

particle or extended object. At any instant of time, each particle of the body moves with the same linear velocity or acceleration.

When a rigid body rotates, it is said to be in **rotatory motion**. In such a motion, we can not assume the body as a point particle and the linear velocity and acceleration of different particles of the body will be different. When a body rotates about an axis, different particles describe different circles of different radii about the axis of rotation.

In case of rotatory motion, the moment of inertia ($I = \Sigma mr^2$) and angular velocity (ω) about an axis are the quantities corresponding to the mass and linear velocity of the body. For a body in rotational motion, the angular momentum is given by

$$L = I\omega$$

which corresponds to linear momentum $p = mv$. If $\alpha = d\omega/dt$ is the angular acceleration and τ the torque acting on the rigid body, then **equation of motion** is

$$\tau = I\alpha$$

which corresponds to equation of motion $F = ma$ for linear motion.

A rigid body may also perform both translational and rotational motion simultaneously. For example, the motion of a rolling cylinder on the floor is composed of linear and rotational motion.

Our study, so far, has revealed that for every physical quantity in translatory motion, we have an analogous quantity appearing in rotatory motion. A glance at the following table is enough to justify the above :

<i>Translatory Motion</i>	<i>Rotatory Motion</i>
(1) Displacement = s	(1) Angular displacement = θ
(2) Velocity = $ds/dt = v$	(2) Angular velocity = $d\theta/dt = \omega$
(3) Acceleration $a = dv/dt$	(3) Angular acceleration $\alpha = d\omega/dt$
(4) Equations of motion :	(4) Equations of rotatory motion :
(a) $v_2 = v_1 + at$	(a) $\omega_2 = \omega_1 + \alpha t$
(b) $s = v_1 t + \frac{1}{2} at^2$	(b) $\theta = \omega_1 t + \frac{1}{2} \alpha t^2$
(c) $v_2^2 = v_1^2 + 2as$	(c) $\omega_2^2 = \omega_1^2 + 2\alpha\theta$
(5) Translational inertia (mass) = m	(5) Moment of inertia = I
(6) Momentum = mv	(6) Angular momentum = $I\omega$
(7) Force	(7) Torque
F = rate of change of linear momentum	τ = rate of change of angular momentum
$= \frac{d}{dt}(mv) = m \frac{dv}{dt}$	$= \frac{d}{dt}(I\omega) = I \frac{d\omega}{dt}$ (about an axis)
$= \text{mass} \times \text{acceleration}$	$= \text{M.I.} \times \text{angular acceleration}$
(8) Impulse = $\int_0^t F dt$	(8) Angular impulse = $\int_0^t \tau dt$
(9) Kinetic energy = $\frac{1}{2} mv^2$	(9) Rotational kinetic energy = $\frac{1}{2} I\omega^2$

3.8. Laws of Rotational Motion

Corresponding to Newton's three laws in the case of linear motion, we have three laws of rotational motion :

- (1) *The state of rotation of a rigid body about an axis remains unaltered unless an external torque*

acts upon it, i.e., according to this law, in the absence of any torque, the angular velocity of the body about any axis will not change; if it is at rest it will remain so and if it is rotating it will continue rotating with the same angular velocity.

(2) The rate of change of angular momentum of a body about a fixed axis is equal to the external torque and takes place in the direction of the torque i.e.

$$\frac{d}{dt} (I \vec{\omega}) = \vec{\tau}$$

(3) To every torque there is an equal and opposite torque. Hence, if a torque is applied by one body upon another, an equal and opposite torque is applied by the latter upon the former, about the same axis of rotation. In other words, a change in the angular momenta of the two bodies about the same axis must be equal and opposite.

3.9. Determination of Moment of Inertia about an Axis

In the case of a continuous, homogeneous body of a definite geometrical shape, the following procedure is adopted for the calculation of moment of inertia of a body about a given axis :

(1) First, an expression is obtained for the moment of inertia of an infinitesimal mass of the body about the given axis. This expression is equal to the product of infinitesimal mass dm and square of its distance x from the axis.

(2) Then, the said expression is integrated over the limits, depending upon the shape of the body, concerned.

Thus, the moment of inertia of a body about a given axis, in general, is given by

$$I = \int x^2 dm \quad \dots(1)$$

where the limits of integration are according to the shape of the body.

(3) Whenever necessary, full use of theorems of perpendicular and parallel axes is made. These theorems are given below :

(i) Theorem of parallel axes. According to this theorem, the moment of inertia of a body about any axis is equal to the sum of moment of inertia about a parallel axis through its centre of mass and the product of the mass of the body and the square of the distance between two axes, i.e.,

$$I = I_{cm} + Ma^2 \quad \dots(2)$$

where $I = M.I.$ of the body about any axis, $I_{cm} = M.I.$ of the body about a parallel axis passing through its centre of mass, $M =$ mass of the body, and $a =$ the distance between the two axes. [fig. 3.9 (a)].

(ii) Theorem of perpendicular axes. This theorem states that the moment of inertia of a plane lamina about an axis perpendicular to its plane is equal to the sum of moments of inertia of the lamina about two axes at right angles to each other, in its own plane, and intersecting each other at the point where the perpendicular axis passes through it.

If I_x and I_y be the moments of inertia of a plane lamina about OX and OY axes, which lie in the plane of the lamina and mutually perpendicular to each other, intersecting at the point O , then the moment of inertia I about an axis, which is passing through O and perpendicular to the plane of the lamina [fig. 3.9 (b)] is given by

$$I = I_x + I_y. \quad \dots(3)$$

where $I_x = \sum my^2$, $I_y = \sum mx^2$ and $I = \sum mr^2$.

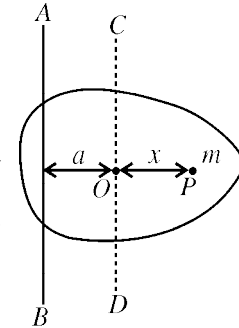


Fig. 3.9(a).

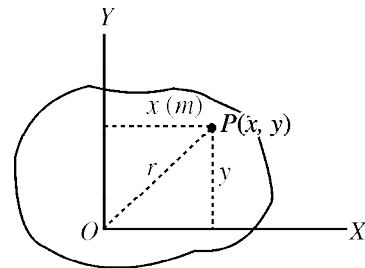


Fig. 3.9(b).

3.10. Calculation of Moment of Inertia of Some Rigid Bodies of Regular Shape*

(1) Moment of inertia of a thin uniform rod : (i) *About an axis passing through the centre of mass and perpendicular to its length.* Let PQ [fig. 3.10] be a thin uniform rod of length l and mass M and AB the axis of rotation passing through the centre of mass O of the rod and perpendicular to its length.

Therefore, mass per unit length = M/l .

Consider a small element of thickness dx at a distance x from O . Hence the mass of the element is $(M/l) dx$ and its $M.I.$ about the axis through O is

$$= \frac{M}{l} dx \cdot x^2.$$

The moment of inertia of the whole rod about AB axis will be the sum of the moments of inertia of all such elements lying between $x = -l/2$ (at P) and $x = l/2$ (at Q) i.e.,

$$\begin{aligned} I &= \int_{-l/2}^{l/2} \frac{M}{l} x^2 \cdot dx = \frac{2M}{l} \int_0^{l/2} x^2 \cdot dx = \frac{2M}{l} \left[\frac{x^3}{3} \right]_0^{l/2} \\ &= \left[\frac{2M}{l} \frac{(l/2)^3}{3} - 0 \right] \end{aligned}$$

$$\text{or} \quad I = \frac{Ml^2}{12}. \quad \dots(1)$$

(ii) *About an axis passing through one end of the rod and perpendicular to its length.* In this case the axis AB passes through the end P of the rod [fig. 3.10].

If I be the moment of inertia of the rod about the axis AB , then according to the theorem of parallel axes

$$I = I_{cm} + Ma^2$$

Here $I_{cm} = Ml^2/12$ and $a = l/2$,

$$\therefore I = \frac{Ml^2}{12} + \frac{Ml^2}{4} \quad \text{or} \quad I = \frac{Ml^2}{3} \quad \dots(2)$$

(2) Moment of inertia of a rectangular lamina : (i) *About an axis perpendicular to its plane and passing through its centre of mass.*

Fig. 3.11 represents a rectangular lamina $ABCD$ of mass M , length l and breadth b . Let YY' be an axis parallel to its side AD and passing through its centre of gravity G . Consider a small strip of width dx , distant x from YY' and also parallel to it. Let σ be the mass per unit area of the lamina, then clearly the mass of the strip is equal to $b \cdot dx \cdot \sigma$.

Therefore, moment of inertia of the strip about $YY' = \sigma b \cdot dx \cdot x^2$.

In order to calculate the moment of inertia of the whole lamina about YY' the above expression is integrated between the limits $x = -l/2$ and $x = +l/2$.

$$\text{Thus, } I_y = \int_{-l/2}^{l/2} \sigma b x^2 dx = 2\sigma b \int_0^{l/2} x^2 dx = 2\sigma b \left[\frac{x^3}{3} \right]_0^{l/2} = 2\sigma b \cdot l^3/24 = Ml^2/12 \quad [\because M = \sigma bl] \quad \dots(3)$$

Similarly, it can be proved that the moment of inertia of the lamina about the axis XX' , an axis parallel to AB or DC and passing through its centre of mass O , will be given by

$$I_x = Mb^2/12. \quad \dots(4)$$

Now, according to the theorem of perpendicular axes, the moment of inertia of the rectangular lamina about an axis perpendicular to its plane and passing through its centre of mass will be expressed as

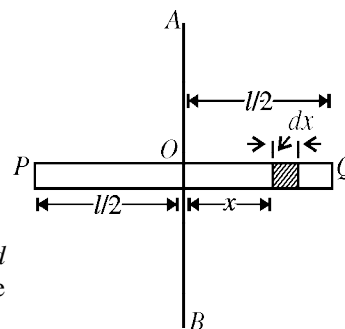


Fig. 3.10.

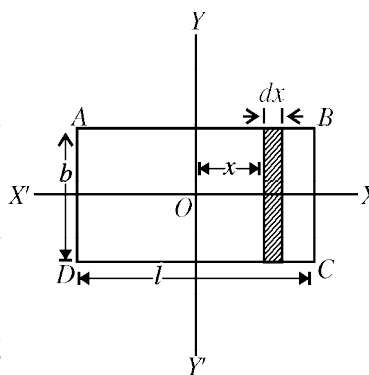


Fig. 3.11.

* Student may leave the calculation part, but they must know the process of calculation and the results.

$$I = I_x + I_y = \frac{Mb^2}{12} + \frac{Ml^2}{12} = M \left(\frac{l^2 + b^2}{12} \right) \quad \dots(5)$$

(3) Moment of inertia of a solid uniform bar of rectangular cross-section : (i) *About an axis perpendicular to its length and passing through its centre of mass.*

Consider a solid uniform bar of length l , breadth b and thickness d , having mass M . Imagine that the bar is composed of a number of thin rectangular laminas, each of them has been placed one above the other as shown in the **fig. 3.12**. Let the mass of the each lamina be m . The length and breadth of each lamina are l and b respectively. The moment of inertia of a lamina about the axis YY' perpendicular to its plane and passing through its centre will be

$$m \frac{l^2 + b^2}{12}.$$

Therefore, the moment of inertia of all the laminas, i.e., of the bar the desired axis (YY') is

$$I_y = \sum m \cdot \frac{l^2 + b^2}{12} = M \frac{l^2 + b^2}{12} \quad [\because \sum m = M] \quad \dots(6)$$

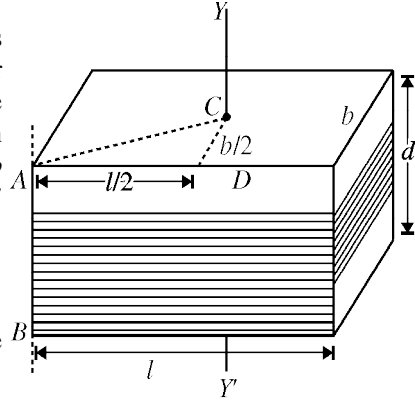


Fig. 3.12.

(ii) *About an axis perpendicular to the length of bar and passing through one of its corners.* Applying the theorem of parallel axes, the moment of inertia of the bar about an axis AB through one of its corners and perpendicular to its length is given by

$$I = I_{cm} + Ma^2$$

Here $I_{cm} = I_y = M \cdot (l^2 + b^2)/12$ and $a^2 = AC^2 = AD^2 = (l/2)^2 + (b/2)^2 = (l^2 + b^2)/4$

$$\therefore I = M \frac{l^2 + b^2}{12} + M \frac{l^2 + b^2}{4} = M \left[\frac{l^2 + b^2 + 3l^2 + 3b^2}{12} \right] \text{ or } I = M \left[\frac{l^2 + b^2}{3} \right] \quad \dots(7)$$

(4) Moment of inertia of a hoop or thin circular ring : (i) *About an axis through its centre and perpendicular to its plane.* Let M be the mass and R the radius of the circular ring or hoop [**fig. 3.13**].

Consider a particle of mass m of the ring. Its moment of inertia about an axis through O and perpendicular to its plane is mR^2 .

Therefore, the moment of inertia I of the entire ring about the given XX' axis will be given by

$$I = \sum mR^2 = R^2 \sum m \text{ or } I = MR^2 \quad [\because \sum m = M] \quad \dots(8)$$

(ii) *About a diameter.* Obviously, the moment of inertia of a hoop is the same about any diameter. Let I be the moment of inertia about a diameter XX' then it will be the moment of inertia about a perpendicular diameter YY' [**fig. 3.13**]. By the application of the theorem of perpendicular axes

$$I_z = I_x + I_y$$

Here $I_z = MR^2$ and $I_x = I_y = I \therefore MR^2 = I + I$ or $I = MR^2/2$. $\dots(9)$

(5) Moment of inertia of a circular disc : (i) *About an axis through its centre and perpendicular to its plane.* Let M be the mass of the disc and R , its radius [**fig. 3.14**], so that its mass per unit area = $M/\pi R^2$.

Consider a concentric elementary ring of radius x and thickness dx .

Then, Area of the ring = circumference $\times$ thickness = $2\pi x \cdot dx$

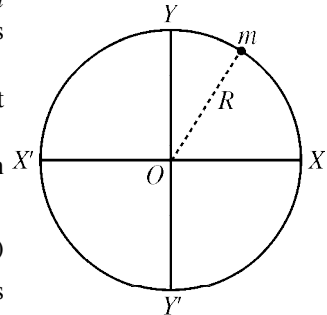


Fig. 3.13.

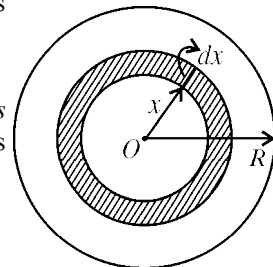


Fig. 3.14.

$$\text{Mass of the ring} = \frac{M}{\pi R^2} 2\pi x dx = \frac{2M \cdot x dx}{R^2}$$

Moment of inertia of this ring about an axis through O and perpendicular to its plane (i.e., plane of the disc)

$$= \frac{2M \cdot x dx}{R^2} \cdot x^2 = \frac{2Mx^3}{R^2} \cdot dx$$

Therefore, the moment of inertia of the whole disc about this axis is

$$I = \frac{2M}{R^2} \int_0^R x^3 dx = \frac{2M}{R^2} \left[\frac{x^4}{4} \right]_0^R = \frac{2M}{R^2} \times \frac{R^4}{4}$$

$$I = \frac{MR^2}{2} \quad \dots(10)$$

(ii) *About a diameter.* Clearly the moment of inertia of the disc is the same about any diameter. Let I be the moment of inertia about any diameter XX' , then it will be the moment of inertia about a perpendicular diameter YY' [fig. 3.15], By the application of the theorem of perpendicular axes,

$$I_z = I_x + I_y$$

Here $I_z = MR^2/2$ and $I_x = I_y = I$,

$$\therefore \frac{MR^2}{2} = I + I \text{ or } I = \frac{MR^2}{4} \quad \dots(11)$$

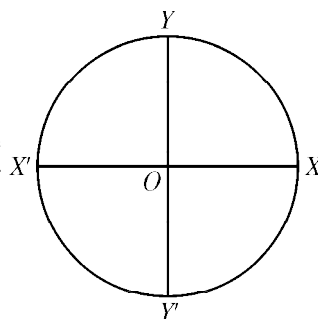


Fig. 3.15.

(6) Moment of inertia of a solid cylinder : (i) *About its own axis or axis of cylindrical symmetry.*

A solid cylinder is just a thick circular disc, or it can be supposed to be composed of a large number of circular discs placed one over the other. Let the mass of each such disc be m and its radius equal to the radius R of the cylinder. Then, the moment of inertia of each disc about the axis of cylinder is $mR^2/2$. Thus the moment of inertia of the cylinder about its own axis is

$$I = \sum m \frac{R^2}{2} = \frac{R^2}{2} \sum m = \frac{MR^2}{2} \quad [\because \sum m = M, \text{ the mass of the cylinder.}] \quad \dots(12)$$

(ii) *About an axis perpendicular to its geometrical axis and passing through its centre of mass.* Let M be the mass of the cylinder, R its radius and l its length, then its mass per unit length = M/l .

Let YY' be the axis of rotation through its centre and perpendicular to its own (geometrical) axis XX' . Consider a thin disc D of thickness dx at a distance x from the centre of the cylinder O [fig. 3.16].

Mass of the disc = $(M/l)dx$

Moment of inertia of this disc about any diameter = $(M/l)dx \times R^2/4$.

By the application of the principle of the parallel axes,

Moment of inertia of this disc about a parallel axis YY' is

$$\frac{M}{l} \cdot dx \cdot \frac{R^2}{4} + \frac{M}{l} \cdot dx \cdot x^2.$$

The moment of inertia of the cylinder about the axis YY' is obtained by integrating the above expression between the limits $x = -l/2$ and $x = l/2$, i.e.,

$$I = \frac{M}{l} \int_{-l/2}^{l/2} \left[\frac{R^2}{4} + x^2 \right] dx = \frac{2M}{l} \left[\frac{R^2 x}{4} + \frac{x^3}{3} \right]_0^{l/2}$$

$$= \frac{M}{l} \left[\frac{R^2}{4} \cdot \frac{l}{2} + \frac{l}{3} \left(\frac{l}{2} \right)^3 \right]$$

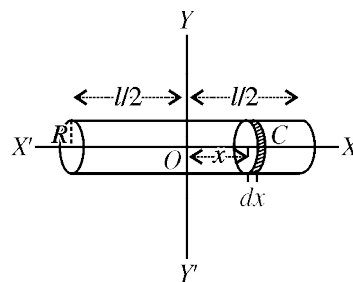


Fig. 3.16.

$$= \frac{M}{l} \left[\frac{R^2 l}{4} + \frac{l^3}{12} \right] = M \left(\frac{R^2}{4} + \frac{l^2}{12} \right) \quad \dots(13)$$

(7) Moment of inertia of a sphere : (i) *About a diameter.* Let **fig. 3.17** represent the section of a sphere of mass M and radius R , through its centre O . Let XX' be the diameter about which the moment of inertia is required to be found. If ρ be the density of the sphere, then its mass

$$M = \frac{4}{3} \pi R^3 \rho.$$

Let us consider a very thin circular disc of thickness dx at a distance x from the centre O . Then radius of the thin disc

$$y = \sqrt{R^2 - x^2}$$

Volume of this disc = $\pi(R^2 - x^2) dx$ and mass = $\pi(R^2 - x^2)dx\rho$

Moment of inertia of the disc about XX'

$$\begin{aligned} &= \frac{\text{its mass} \times (\text{radius})^2}{2} \\ &= \frac{\pi\rho(R^2 - x^2) dx \times (R^2 - x^2)}{2} \\ &= \frac{1}{2} \pi\rho(R^4 - 2R^2x^2 + x^4) dx. \end{aligned}$$

Hence the moment of inertia of the sphere about XX' , can easily be obtained by integrating the above expression between the limits $x = -R$ to $x = R$. Then,

$$\begin{aligned} I &= \int_{-R}^R \frac{\pi\rho}{2} (R^4 - 2R^2x^2 + x^4) dx = 2 \times \frac{\pi\rho}{2} \int_0^R (R^4 - 2R^2x^2 + x^4) dx \\ &= \pi\rho \left[R^4x - 2R^2 \cdot \frac{x^3}{3} + \frac{x^5}{5} \right]_0^R = \pi\rho \left[R^5 - \frac{2R^5}{3} + \frac{R^5}{5} \right] \\ &= \pi\rho \cdot \frac{8R^5}{15} = \frac{2}{5} \times \frac{4}{3} \pi R^3 \rho \times R^2 \quad \left[\because M = \frac{4}{3} \pi R^3 \rho \right] \end{aligned}$$

or

$$I = \frac{2}{5} MR^2 \quad \dots(14)$$

(ii) *About a tangent.* Any tangent to the sphere at any point is parallel to one of its diameters and is at a distance equal to R from the centre. Now, by the application of the theorem of parallel axes, the moment of inertia of the sphere about the tangent is

$$I = \frac{2}{5} MR^2 + MR^2 = \frac{7}{5} MR^2 \quad \dots(15)$$

(8) Moment of inertia of a spherical shell : (i) *About a diameter.* Let **fig. 3.18** represent the section of a shell of mass M and radius R through its centre O . Then,

the surface area of the shell = $4\pi R^2$

$\therefore$ Mass per unit area = $M/4\pi R^2$.

Let us consider a thin circular ring lying between two parallel planes AB and CD perpendicular to XX' . AB and CD planes lie at distances x and $x + dx$ respectively from the centre O of the shell. The radius of the circular ring is $y = \sqrt{R^2 - x^2}$ and the width AC (not dx).

Join OA and OC and let $\angle YOA = \theta$ and $\angle AOC = d\theta$, then

$$x = R \sin \theta \text{ and } AC = R d\theta$$

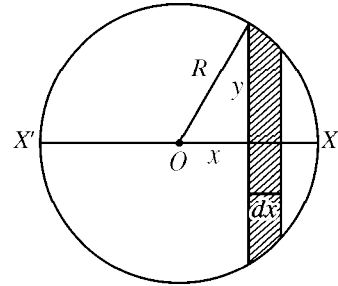


Fig. 3.17.

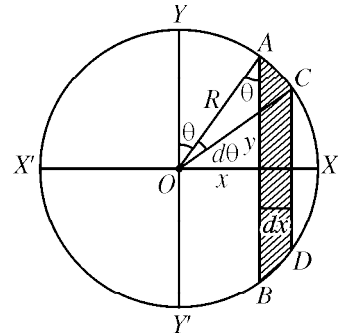


Fig. 3.18.

Differentiating $x = R \sin \theta$, we get

$$dx = R \cos \theta d\theta$$

$$\text{Area of the ring} = 2\pi y \cdot AC = 2\pi R \cos \theta \cdot R d\theta.$$

$$\text{Its mass} = 2\pi R \cos \theta \cdot R d\theta \times \frac{M}{4\pi R^2} = \frac{M}{2R} dx$$

$\therefore$ The moment of inertia of the thin ring about the axis XX'

$$= \text{mass} \times (\text{radius})^2 = \frac{M}{2R} dx \times y^2$$

$$= \frac{M}{2R} (R^2 - x^2) dx \quad [\because R^2 = x^2 + y^2]$$

Hence, the moment of inertia of the whole shell about the diameter XX' is given by

$$I = \int_{-R}^R \frac{M}{2R} (R^2 - x^2) dx = \frac{M}{2R} \int_0^R (R^2 - x^2) dx = \frac{M}{R} \left[R^2 x - \frac{x^3}{3} \right]_0^R = \frac{M}{R} \left[R^3 - \frac{R^3}{3} \right]$$

$$\text{or} \quad I = \frac{2}{3} MR^2 \quad \dots(16)$$

(ii) *About a tangent.* Any tangent of the spherical shell is at a distance R from a parallel diameter. Since moment of inertia of the shell about any diameter is the same, hence applying the theorem of parallel axes, the moment of inertia of the shell about a tangent is given by

$$I = \frac{2}{3} MR^2 + MR^2 \quad \text{or} \quad I = \frac{5}{3} MR^2 \quad \dots(17)$$

Ex. 1. A grindstone starts from rest and has constant angular acceleration of 3 rad/sec². Find (a) the angular displacement, (b) the angular speed of the grindstone 2 sec later.

Sol. (a) Angular displacement, $\theta = \omega_0 t + \frac{1}{2} \alpha t^2$

Here, $\omega_0 = 0$, $\alpha = 3 \text{ rad/sec}^2$ and $t = 2 \text{ sec}$.

$$\therefore \quad \theta = 0 + \frac{1}{2} \times 3 \times 2^2 = \mathbf{6 \text{ radian}}$$

(b) $\omega = \omega_0 + \alpha t$

$$\therefore \quad \text{At } t = 2 \text{ sec, angular speed, } \omega = 0 + 3 \times 2 = \mathbf{6 \text{ rad/sec.}}$$

Ex. 2. A circular metal hoop of mass 1 kg and radius 0.2 metre makes 10 revolutions per sec about its centre. The axis of rotation being normal to the plane of the hoop.

(a) What is the moment of inertia about this axis ?

(b) What is the angular momentum about the same axis ?

(c) Calculate the torque, which will increase the angular momentum by 25% in 10 sec.

(d) If this torque is applied by a force acting tangentially on the rim of the hoop, find the value of this force.

Sol. (a) $I = MR^2 = 1 \times (0.2)^2 = \mathbf{0.04 \text{ kg-m}^2}.$

(b) Here, $\omega = 2\pi n = 2 \times 3.14 \times 10 = 63 \text{ rad/sec}^2$

$$\therefore \quad J = I\omega = 0.04 \times 63 = \mathbf{2.5 \text{ N-m.}}$$

(c) In 10 seconds the angular momentum increases by 25%.

$$\therefore \quad \text{In 1 sec the angular momentum will increase} = \frac{2.5}{10} \times \frac{25}{100} = 0.0625 \text{ N-m}$$

$$\therefore \quad \text{Torque} = \text{Rate of change of angular momentum} = \mathbf{0.0625 \text{ N-m.}}$$

(d) If F is the force, then $F \cdot R = 0.0625 = 0.0625/0.2 = \mathbf{0.312 \text{ newton.}}$

Ex. 3. Determine the moment of inertia of a uniform disc of mass 0.2 kg and radius 0.05 m about an axis along its diameter. (A.U. 2002)

Sol. Moment of inertia of a circular disc about a diameter is given by

$$I = \frac{MR^2}{4} = \frac{(0.2) \times (0.05)^2}{4} = 1.25 \times 10^{-4} \text{ kg-m}^2.$$

Ex. 4. The moment of inertia of a thin circular disc about an axis through its centre and perpendicular to its plane is $\left(\frac{1}{2}\right)mr^2$. Calculate the moment of inertia of the disc :

- (i) about an axis tangential to the disc at a point P .
- (ii) about an axis perpendicular to the disc and passing through P . (O.U. 2003)

Sol. Here, $I = \frac{1}{2}mr^2$

We can obtain the moment of inertia of the disc about a diameter by applying the theorem of perpendicular axes i.e.,

$$I_x + I_y = I \text{ or } 2I_x = I \quad [\because I_x = I_y]$$

$$\therefore I_x = \frac{I}{2} = \frac{1}{2} \left(\frac{1}{2}mr^2 \right) = \frac{1}{4}mr^2$$

(i) The moment of inertia of the disc about a tangent in its plane can be obtained by applying the theorem of parallel axes. Thus

$$I = I_x + mr^2 = \frac{1}{4}mr^2 + mr^2 = \frac{5}{4}mr^2.$$

(ii) The moment of inertia of the disc about a tangent, perpendicular to the plane of the disc, is given by

$$I = \frac{1}{2}mr^2 + mr^2 = \frac{3}{2}mr^2.$$

Ex. 5. The radius of a solid sphere is 5 cm. Find the radius of gyration about its (i) diameter and (ii) tangent.

Sol. (i) M.I. about a diameter : $I = \frac{2}{5}MR^2 = Mk_1^2$

$$\therefore k_1 = R\sqrt{\frac{2}{5}} = 5\sqrt{\frac{2}{5}} \text{ cm} = 3.16 \text{ cm}.$$

(ii) M.I. about a tangent : $I = \frac{7}{5}MR^2 = Mk_2^2$

$$\therefore k_2 = R\sqrt{\frac{7}{5}} = 5\sqrt{\frac{7}{5}} = 5.92 \text{ cm}.$$

Ex. 6. Four spheres each of diameter $2a$ and mass m are placed with their centres on the four corners of a square of side b . Calculate the moment of inertia of the system about any side of the square.

Sol. The four spheres of same mass m and radius a are placed at the corners of the square $PQRS$ [fig. 3.19].

M.I. of P and Q spheres about axis PQ

$$= \frac{2}{5}ma^2 + \frac{2}{5}ma^2 = \frac{4}{5}ma^2$$

...(i)

Similarly, M.I. of R and S spheres about axis $RS = \frac{4}{5}ma^2$.

Since the axis PQ is parallel to SR are at a distance b , therefore the moment of inertia of S and R spheres about the axis PQ

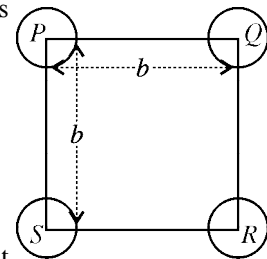


Fig. 3.19.

$$= \frac{4}{5}ma^2 + 2mb^2 \quad \dots(ii)$$

Therefore, total M.I. of the system about the axis PQ is the sum of (i) and (ii), i.e.,

$$= \frac{4}{5}ma^2 + \frac{4}{5}ma^2 + 2mb^2 = \frac{8}{5}ma^2 + 2mb^2.$$

Ex. 7. A uniform thin bar of mass 6 kg and length 2.4 m is bent to make an equilateral hexagon. Calculate its moment of inertia about an axis passing through the centre of mass and perpendicular to the plane of hexagon.

Sol. M.I. of a regular hexagon of side a about an axis passing through the centre of mass and perpendicular to the plane of hexagon is given by

$$I = 6 \left(\frac{ma^2}{12} + m \cdot OP^2 \right) = 6 \left(\frac{ma^2}{12} + m \cdot \frac{3a^2}{4} \right) = 5ma^2$$

[$\because OP = a \cos 30^\circ = a(\sqrt{3}/2)$]

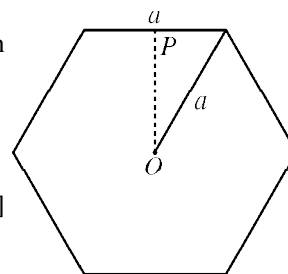


Fig. 3.20.

where m is the mass of side a .

Here, $m = \frac{6}{6} = 1 \text{ kg}$ and $a = \frac{2.4}{6} = 0.4 \text{ m}$.

$\therefore I = 5 \times 1 \times (0.4)^2 = 0.8 \text{ kg-m}^2.$

Ex. 8. A grindstone has a moment of inertia of 50 S.I. units. A constant couple is applied and the grindstone is found to have speed of 150 revolutions per minute, 10 sec after starting from rest. Find the couple.

Sol. The speed of revolutions per sec = $\frac{150}{60} = \frac{5}{2}$.

$\therefore$ Angular velocity acquired $\omega = 2\pi \times \text{number of revolutions per sec} = 2\pi \times \frac{5}{2} = 5\pi \text{ radians/sec.}$

This angular velocity is attained in 10 sec starting from rest, therefore the angular acceleration $d\omega/dt$ is given by

$$\frac{d\omega}{dt} = \frac{5\pi}{10} = \frac{\pi}{2} \text{ rad/sec}^2.$$

$\therefore$ Couple = $I \times d\omega/dt = 50 \times \pi/2 = 78.5 \text{ newton-metre.}$

Ex. 9. The moment of inertia of earth is I . What will be the moment of inertia if the mass is doubled and radius halved ? (S.V.U. 2002)

Sol. The earth may be considered as a solid sphere of mass m and radius R . Hence

$$I = \frac{2}{5}MR^2$$

If mass of the earth is doubled and radius is halved, then

$$I' = \frac{2}{5}(2M)\left(\frac{R}{2}\right)^2 = \frac{MR^2}{5}$$

i.e., $I' = \frac{1}{2} \left[\frac{2}{5}MR^2 \right] = \frac{I}{2}$

Thus, the **moment of inertia of the earth will be halved.**

Ex. 10. The moment of inertia of a solid sphere of radius R is I about any diameter. What will be its moment of inertia when it is converted into a uniform rod of length $2R$ about its perpendicular bisector ? (S.V.U. 2002)

Sol. The moment of inertia of solid sphere about any diameter is given by

$$I = \frac{2}{5}MR^2 \quad \dots(i)$$

The moment of inertia of rod about its perpendicular bisector is given by

$$I' = \frac{Ml^2}{12} = \frac{M(2R)^2}{12} = \frac{MR^2}{3} \quad \dots(ii)$$

Dividing (ii) by (i), we get

$$\frac{I'}{I} = \frac{MR^2/3}{2MR^2/5} = \frac{5}{6} \quad \dots(iii)$$

or $I' = \frac{5}{6} \times I = \frac{5}{6} \times \text{moment of inertia of solid sphere}.$

Ex. 11. A couple of 20 N-m is applied to a fly-wheel of mass 10 kg and radius of gyration 0.5 m. Find the resultant angular acceleration. (O.U. 2008)

Sol. [Flywheel : A flywheel is a large heavy wheel with a long cylindrical axle passing through its centre. Its shape is such that its most of the mass is concentrated at the rim. Its centre of gravity lies on its axis of rotation so that it can come to rest in any desired position.]

Torque $\tau = M.I. \times \text{angular acceleration} = I\alpha$

Here, torque $\tau = 20 \text{ N-m}$ and $I = Mk^2 = 10 \times (0.5)^2 = 2.5 \text{ kg-m}^2$

$$\therefore \alpha = \frac{\tau}{I} = \frac{20}{2.5} = 8 \text{ radians/sec}^2.$$

Ex. 12. A car engine develops 75 kW power when rotating at a speed of 1000 rev/min. What torque does it deliver ? (O.U. 2002)

Sol. In case of a rotating rigid body with angular velocity ω , the power is given by

$P = \tau\omega$ or $\tau = P/\omega$, where τ is the torque.

Here, $P = 75 \text{ kW} = 75 \times 10^3 \text{ watt}$ and $\omega = 2\pi n = 2\pi \times \frac{1000}{60} = \frac{100\pi}{3} \text{ rad/sec}$

$$\therefore \tau = \frac{P}{\omega} = \frac{75 \times 10^3 \times 3}{100 \times 3.14} = \frac{75 \times 30}{3.14} = 716.3 \text{ N-m.}$$

Ex. 13. Given the mass of the earth as $6.0 \times 10^{24} \text{ kg}$ and radius 6400 km. Find the angular momentum and rotational K.E. about its own axis assuming it to a sphere of uniform density. (K.U. 2007)

Sol. Let the earth be a sphere of radius R and mass m .

Hence moment of inertia of the earth about its axis (diameter) is given by

$$I = \frac{2}{5}MR^2 = \frac{2}{5} \times 6.40 \times 10^{24} \times (6400 \times 1000)^3 = 9.83 \times 10^{37} \text{ kg-m}$$

Angular velocity, $\omega = \frac{2\pi}{24 \text{ hours}} = \frac{2 \times 3.14}{24 \times 60 \times 60} = 7.27 \times 10^{-5} \text{ rad/sec.}$

$\therefore$ Angular momentum, $L = I\omega = 9.83 \times 10^{37} \times 7.27 \times 10^{-5} = 7.15 \times 10^{33} \text{ kg-m}^2/\text{sec.}$

Kinetic energy, $E = \frac{1}{2}I\omega^2 = \frac{1}{2} \times 9.83 \times 10^{37} \times (7.27 \times 10^{-5})^2 = 2.6 \times 10^{24} \text{ joule.}$

Ex. 14. A sphere of mass 2.5 kg and diameter 1 m rolls without slipping with a constant velocity of 2 m/s. Calculate its total kinetic energy. (K.U. 2000; A.N.U. 1997)

Sol. Total K.E. = $\frac{1}{2}Mv^2 \left(1 + \frac{k^2}{R^2} \right)$

For a sphere, $I = Mk^2 = \frac{2}{5}MR^2$, $\therefore k^2 = \frac{2}{5}R^2$ or $\frac{k^2}{R^2} = \frac{2}{5}$

$$\therefore \text{Total K. E.} = \frac{1}{2}Mv^2 \left(1 + \frac{2}{5} \right) = \frac{7}{10}Mv^2$$

Here, $M = 2.5 \text{ kg}$ and $v = 2 \text{ m/sec}$

$$\therefore \text{Total kinetic energy} = \frac{7}{10} \times 2.5 \times (2)^2 = 7 \text{ joule.}$$

Ex. 15. A sphere of mass 100 gm and diameter 0.01 m rotates without slipping with a velocity of 0.05 m/sec. Calculate its K.E. of rotations. (O.U. 2007)

$$\begin{aligned} \text{Sol. K.E. of rotation} &= \frac{1}{2} I \omega^2 \\ &= \frac{1}{2} \left(\frac{2}{5} MR^2 \right) \left(\frac{v}{R} \right)^2 = \frac{1}{5} Mv^2 \quad [\because v = R\omega] \\ \therefore &= \frac{1}{5} \times (100 \times 10^{-3}) \times (0.05)^2 = 5 \times 10^{-5} \text{ joule.} \end{aligned}$$

Ex. 16. The kinetic energy of a metal disc rotating at a constant speed of 5 revolutions per second is 100 joule. Find the angular momentum of the disc. (A.U. 2001)

$$\begin{aligned} \text{Sol. Kinetic energy of rotation } E &= \frac{1}{2} I \omega^2 \\ \text{Angular momentum } L = I\omega &= \frac{I\omega^2}{\omega} = \frac{\frac{1}{2} I \omega^2}{\frac{1}{2} \omega} = \frac{2E}{\omega} \end{aligned}$$

Here, $E = 100 \text{ joule}$ and $\omega = 2\pi \times 5 = 10 \times 3.14 \text{ rad/sec}$

$$\therefore L = \frac{2 \times 100}{10 \times 3.14} = 6.366 \text{ kg-m}^2/\text{sec.}$$

Ex. 17. What is the angular momentum of a particle whose rotational kinetic energy is 18 joules, if the angular momentum vector coincides with the axis of rotation and its moment of inertia about the axis is 0.01 kg-m^2 .

Sol. If the angular momentum vector coincides with the axis of rotation, then the angular momentum is given by

$$\begin{aligned} L &= I\omega \\ \text{K.E. of rotation, } E &= \frac{1}{2} I \omega^2 = L^2/2I \\ \therefore L &= \sqrt{2EI} = \sqrt{2 \times 18 \times 0.01} = 0.6 \text{ kg-m}^2/\text{sec.} \end{aligned}$$

Ex. 18. The kinetic energy of a metal disc, rotating at a constant speed of 5 revolutions per second, is 100 joules. Find the angular momentum of the disc. (K.U. 2001, 2000; A.U. 1995)

Sol. Given, $\omega = 2\pi n = 2 \times 3.14 \times 5 = 31.4 \text{ rad/sec}$.

$$\text{Kinetic energy } E = \frac{1}{2} I \omega^2, \therefore I = \frac{2E}{\omega^2} = \frac{2 \times 100}{(31.4)^2} = 0.2028 \text{ kg-m}^2$$

Here, $E = 100 \text{ joule}$, and $I = 0.2028 \text{ kg-m}^2$

$$\therefore L = I\omega = 0.2028 \times 31.4 = 6.368 \text{ kg-m}^2/\text{sec.}$$

Ex. 19. A flywheel in the form of a solid circular disc of mass 5000 kg and radius 1 metre is rotating, making 120 revolutions per minute. Compute the kinetic energy and the angular impulse, if the flywheel is brought to rest in 2 seconds; friction is to be neglected.

Sol. M.I. of the flywheel about its axis

$$= \frac{MR^2}{2} = \frac{5000 \times 1^2}{2} = 2500 \text{ kg-m}^2$$

Angular speed, $\omega = 2\pi n$, where $n = \text{No. of revolutions/sec}$.

$$= 2\pi \times 120/60 = 4\pi \text{ rad/sec.}$$

$$\therefore \text{K.E.} = \frac{1}{2} I \omega^2 = \frac{1}{2} \times 2500 \times (4\pi)^2 = \mathbf{1.97 \times 10^5 \text{ joules.}}$$

$$\begin{aligned} \text{Angular impulse} &= \text{Change in angular momentum} = I\omega \\ &= 2500 \times 4\pi = \mathbf{3.14 \times 10^4 \text{ S.I. units.}} \end{aligned}$$

Ex. 20. A circular disc of radius 0.1 m and mass 1 kg is rotating at the rate of 10 revolutions a second about an axis at right angles to its plane and passing through its centre. Find the work that must be done to increase the rate of revolutions to 20 per second.

Sol. M.I. of the disc about the given axis is

$$I = \frac{MR^2}{2} = \frac{1 \times (0.1)^2}{2} = 5 \times 10^{-3} \text{ kg-m}^2$$

Initial angular velocity $\omega = 2\pi \times \text{number of revolutions per second} = 2\pi \times 10 \text{ rad/sec.}$

Final angular velocity $\omega_2 = 2\pi \times 20 \text{ rad/sec.}$

The work done in increasing the rate of revolutions = Final K.E. – Initial K.E.

$$\begin{aligned} &= \frac{1}{2} I \omega_2^2 - \frac{1}{2} I \omega_1^2 = \frac{1}{2} \times 5 \times 10^{-3} [(40\pi)^2 - (20\pi)^2] \\ &= \frac{1}{2} \times 5 \times 10^{-3} \times 1200 \times \pi^2 = \mathbf{29.5 \text{ joules.}} \end{aligned}$$

Ex. 21. A sphere, a disc, a spherical shell, a ring and a cylinder are allowed to roll down simultaneously on an inclined plane from the same height without slipping. Prove that the sphere reaches down first, then the disc and the cylinder, next the shell and in the last the ring.

Sol. Let M and R be the mass and radius of each body, rolling down an inclined plane at an angle θ with the horizontal. Since the different bodies have different radius of gyration (k), they will be accelerated by different amount and hence will require different time intervals in travelling a certain distance on an inclined plane.

(i) Solid sphere : The moment of inertia of a solid sphere about the axis of rotation (or diameter) is given by

$$I = \frac{2}{5} MR^2 = Mk^2, \text{ where } k \text{ is the radius of gyration.}$$

$$\therefore k^2/R^2 = 2/5$$

$$\therefore \text{Acceleration } a_1 = \frac{g \sin \theta}{1 + (k^2/R^2)} = \frac{g \sin \theta}{1 + 2/5} = \frac{5}{7} g \sin \theta.$$

$$\text{(ii) Disc : } I = \frac{MR^2}{2} = Mk^2 \text{ or } \frac{k^2}{R^2} = \frac{1}{2}, \therefore a_2 = \frac{g \sin \theta}{1 + \frac{1}{2}} = \frac{2}{3} g \sin \theta$$

$$\text{(iii) Spherical shell : } I = \frac{1}{2} MR^2 = Mk^2 \text{ or } \frac{k^2}{R^2} = \frac{2}{3}, \therefore a_3 = \frac{g \sin \theta}{1 + \frac{2}{3}} = \frac{3}{5} g \sin \theta$$

$$\text{(iv) Ring : } I = \frac{1}{2} MR^2 = Mk^2 \text{ or } \frac{k^2}{R^2} = \frac{1}{2}, \therefore a_4 = \frac{g \sin \theta}{1 + 1} = \frac{1}{2} g \sin \theta$$

$$\text{(v) Cylinder : } I = \frac{1}{2} MR^2 = Mk^2 \text{ or } \frac{k^2}{R^2} = \frac{1}{2}, \therefore a_5 = \frac{g \sin \theta}{1 + \frac{1}{2}} = \frac{2}{3} g \sin \theta = a_2$$

Obviously, $a_1 > a_2 (= a_5) > a_3 > a_4$, hence, if a sphere, a disc, a cylinder, a spherical shell and a ring are allowed to roll down an inclined plane simultaneously from the same height without slipping, the sphere will reach down first, then the disc and cylinder, then the shell and in the last ring.

Ex. 22. A solid disc (a) rolls (b) slides from rest down an inclined plane. Neglect friction and compare the velocities in both cases when the disc reaches the bottom of incline.

$$\text{Sol. Here } I = MR^2/2 = Mk^2, \therefore k^2 = R^2/2$$

The acceleration of the disc, when it rolls down over a plane inclined to an angle θ to the horizontal, is given by

$$a = \frac{g \sin \theta}{1 + k^2/R^2} = \frac{g \sin \theta}{1 + R^2/2R^2} = \frac{2g \sin \theta}{3}.$$

If s be the length of the inclined plane, then the velocity V of the disc at the bottom is given by

$$V^2 = 2 \times \frac{2g \sin \theta}{3} \times s \quad \dots(i)$$

since the disc starts from rest.

In case, the disc slides through a distance s , the acceleration a' will be given by

$$a' = g \sin \theta$$

Hence in second case the velocity V' is given by

$$V'^2 = 2g \sin \theta \times s \quad \dots(ii)$$

Dividing (ii) by (i) and taking square root, we get

$$\frac{V}{V'} = \sqrt{\frac{2}{3}}$$

This is the required comparison of velocities in two cases.

Ex. 23. A uniform sphere of mass 2 kg and radius 10 cm is released from rest on an inclined plane which makes 30° angle with the horizontal. Deduce its

(i) angular acceleration,

(ii) linear acceleration along the plane,

(iii) kinetic energy as it travels 2 metres along the plane.

Sol. Linear acceleration along the plane

$$a = \frac{g \sin \theta}{1 + k^2/R^2} = \frac{9.8 \times 1/2}{1 + 2/5} = 3.5 \text{ m/sec}^2 \quad [\because \text{for sphere } k^2 = \frac{2}{5} R^2]$$

Linear velocity, $V = R\omega$, so $\frac{dV}{dt} = R \frac{d\omega}{dt} = a$, linear acceleration.

$\therefore$ Angular acceleration $= a/R = 3.5/0.1 = 35 \text{ rad/sec}^2$.

Gain in K.E. = Loss in P.E. in vertical descent $h (= s \sin \theta)$

$$= Mgs \sin \theta = 2 \times 9.8 \times 2 \times \frac{1}{2} = 19.6 \text{ joules.}$$

Ex. 24. A ball starts from rest and rolls down a 30° inclined plane. How much time would it take to cover 14 metre ? (A.N.U. 1990)

Sol.
$$a = \frac{g \sin \theta}{1 + k^2/R^2}$$

Here, $\frac{k^2}{R^2} = \frac{2}{5}$, $\therefore a = \frac{9.8 \times \sin 30^\circ}{1 + 2/5} = \frac{9.8 \times 0.5}{1.4} = 3.5 \text{ m/sec}^2 \quad [\because I = Mk^2 = \frac{2}{5} MR^2]$

Now, from the equation of motion $s = \frac{1}{2}at^2$

$$\therefore t = \sqrt{\frac{2s}{a}} = \sqrt{\frac{2 \times 14}{3.5}} = 2.8 \text{ sec.}$$

Ex. 25. In a diatomic molecule, two atoms of masses m_1 and m_2 are situated at a distance r . Determine the moment of inertia of the molecule about that axis which is perpendicular to the line joining the two atoms and passes through its centre of mass.

Sol. Moment of inertia of a diatomic molecule : A diatomic molecule, in general, consists of two atoms, separated by some distance. Let the masses of these atoms be m_1 and m_2 , separated by a distance r [fig. 3.2]. We know that the mass of each atom is concentrated on the nucleus, hence r must be the inter-

nuclear distance. If r_1 is the distance of the centre of mass C of the molecule from m_1 , then according to the property of the centre of mass, we have

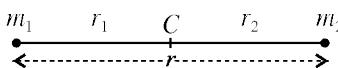


Fig. 3.21.

$$(m_1 + m_2)r_1 = m_2r \quad (\text{Taking } m_1 \text{ as the origin})$$

or
$$r_1 = \frac{m_2r}{m_1 + m_2}, \quad \therefore r_2 - r_1 = \frac{m_1r}{m_1 + m_2}$$

Hence the moment of inertia of the system about an axis passing through C and perpendicular to the line joining the masses is

$$I = m_1r_1^2 + m_2r_2^2 = m_1\left(\frac{m_2r}{m_1 + m_2}\right)^2 + m_2\left(\frac{m_1r}{m_1 + m_2}\right)^2 \quad \text{or} \quad I = \frac{m_1m_2}{m_1 + m_2}r^2 = \mu r^2$$

where $\mu = \frac{m_1m_2}{m_1 + m_2}$ is the **reduced mass** of the system.

As the axis of rotation is perpendicular to the line joining the two particles, the angular momentum will be along the direction of the rotational axis.

Ex. 26. The internuclear distance between the two protons of hydrogen molecule is $0.74 \text{ \AA}$. Calculate the moment of inertia of the molecule about an axis passing through its centre of mass and perpendicular to the line joining the two atoms. ($m_p = 1.7 \times 10^{-27} \text{ kg}$.)

Sol.
$$I = \frac{m_1m_2}{m_1 + m_2}r^2 = \frac{1.7 \times 10^{-27} \times 1.7 \times 10^{-27}}{2 \times 1.7 \times 10^{-27}} \times (0.74 \times 10^{-10})^2 = 4.6 \times 10^{-48} \text{ kg-m}^2.$$

Ex. 27. The interatomic distance in a nitrogen diatomic is $1.3 \text{ \AA}$. Mass of each N atom is $14 u$, where $1 u = 1.67 \times 10^{-27} \text{ kg}$. The average rotational kinetic energy of a nitrogen molecule in air at room temperature is nearly $4 \times 10^{-21} \text{ J}$. Calculate the moment of inertia of the molecule about its centre of mass and rotational frequency. (A.U. 1993)

Sol. For a diatomic molecule,

$$I = \frac{m_1m_2}{m_1 + m_2}r^2$$

For a diatomic molecules of similar atoms, $m_1 = m_2 = m$. Hence

$$I = \frac{m \cdot m}{m + m}r^2 = \frac{m^2}{2m} = \frac{1}{2}mr^2$$

Here, $m = 14 \text{ unit} = 14 \times 1.67 \times 10^{-27} \text{ kg}$ and $r = 1.3 \text{ \AA} = 1.3 \times 10^{-10} \text{ m}$

$$\therefore I = \frac{1}{2}mr^2 = \frac{14}{2} \times 1.67 \times 10^{-27} \times (1.3 \times 10^{-10})^2 = 1.975 \times 10^{-46} \text{ kg-m}^2$$

Kinetic energy of rotation, $E = \frac{1}{2}I\omega^2$,

$$\therefore \omega = \sqrt{\frac{2E}{I}} = \sqrt{\frac{2 \times 4 \times 10^{-21}}{1.975 \times 10^{-46}}} = 6.36 \times 10^{12} \text{ rad/sec.}$$

3.11. Angular Momentum and Inertia Tensor

Let us consider a rigid body rotating with an instantaneous angular velocity $\vec{\omega}$ about an axis AB , passing through a fixed point O (in an inertial frame). The direction of the angular velocity $\vec{\omega}$ will be

along the instantaneous axis of rotation. Take the fixed point O as the origin [fig. 3.22].

Let at any instant $\vec{r}$ be the position vector of a particle P of the body. If the length of the perpendicular drawn from P on AB is $PC = r_0$, then C will be the centre of the circle, described by the point P .

Speed of the particle $P = \omega r_0 = \omega r \sin POC = \omega r \sin \theta$

$\therefore$ Velocity of the particle P , $\vec{v} = \vec{\omega} \times \vec{r}$... (1)

The direction of this velocity $\vec{v}$ at any instant will be perpendicular to the plane of position vector $\vec{r}$ and angular velocity vector $\vec{\omega}$.

The angular momentum of the particle P about the point O is given by

$$\vec{L}_p = \vec{r} \times m \vec{v}$$

whose direction is perpendicular to $\vec{r}$ and $\vec{v}$. Evidently this direction will not coincide in general to the angular velocity vector $\vec{\omega}$.

Hence the angular momentum $\vec{L}$ of the entire body about the point O will be obtained by summing over $\vec{L}_p$ for all the particles of the body, i.e.,

$$\vec{L} = \sum m \vec{r} \times \vec{v} = \sum m \vec{r} \times (\vec{\omega} \times \vec{r})$$

whose direction, in general, will not be along $\vec{\omega}$.

$$\text{Now, } \vec{L} = \sum m \{ (\vec{r} \cdot \vec{r}) \vec{\omega} - (\vec{r} \cdot \vec{\omega}) \vec{r} \} \quad \dots (2)$$

$$\text{or } \vec{L} = \sum \{ m r^2 \vec{\omega} - (\omega r \cos \theta) \vec{r} \} \quad \dots (3)$$

where $\theta = \angle POC$, is the angle between the vectors $\vec{r}$ and $\vec{\omega}$.

$$\text{Hence } \vec{L} = \sum m r^2 [\vec{\omega} - (\omega \cos \theta / r^2) \vec{r}] \quad \dots (4)$$

From this equation, it is clear that for the entire body, in general, the angular momentum $\vec{L}$ and angular velocity $\vec{\omega}$ are not in the same direction.*

We write $\vec{r}$, $\vec{\omega}$ and $\vec{L}$ in the cartesian components :

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}, \quad \vec{\omega} = \omega_x\hat{i} + \omega_y\hat{j} + \omega_z\hat{k} \quad \text{and} \quad \vec{L} = J_x\hat{i} + J_y\hat{j} + J_z\hat{k}.$$

* For example, for a particle of mass m , the angular momentum will be given by $\vec{J}_p = m r^2 [\vec{\omega} - (\omega \cos \theta / r^2) \vec{r}]$

Obviously $\vec{J}_p$ and $\vec{\omega}$ are not in the same direction.

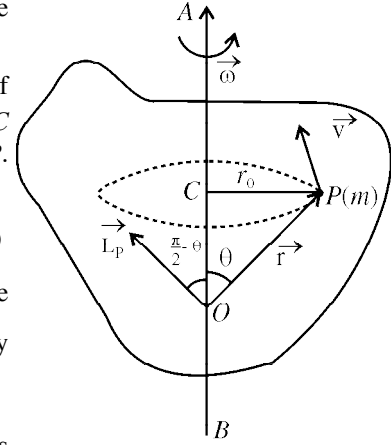


Fig. 3.22.

where the co-ordinate system XYZ with origin O is fixed in the body.

From eq. (2), we have

$$\vec{L} = \Sigma m[(\omega_x \hat{i} + \omega_y \hat{j} + \omega_z \hat{k}) r^2 - \{(x\hat{i} + y\hat{j} + z\hat{k}) (\omega_x \hat{i} + \omega_y \hat{j} + \omega_z \hat{k})\} (x\hat{i} + y\hat{j} + z\hat{k})] \quad \dots(5)$$

where $\vec{\omega} = \omega_x \hat{i} + \omega_y \hat{j} + \omega_z \hat{k}$, $r = x\hat{i} + y\hat{j} + z\hat{k}$ and $r^2 = x^2 + y^2 + z^2$.

If L_x, L_y, L_z are the components of $\vec{L}$ along X, Y, Z axes respectively, then

$$L_x = \Sigma m [(r^2 \omega_x - (x\omega_x + y\omega_y + z\omega_z) x)] \quad \dots(6a)$$

or

$$L_x = \omega_x \Sigma m (r^2 - x^2) - \omega_y \Sigma m xy - \omega_z \Sigma m xz$$

$$\text{Similarly, } L_y = -\omega_x \Sigma m xy + \omega_y \Sigma m (r^2 - y^2) - \omega_z \Sigma m yz \quad \dots(6b)$$

and

$$L_z = -\omega_x \Sigma m xz - \omega_y \Sigma m yz + \omega_z \Sigma m (r^2 - z^2) \quad \dots(6c)$$

Eqs. (6) can be written as

$$L_x = I_{xx}\omega_x + I_{xy}\omega_y + I_{xz}\omega_z; L_y = I_{yx}\omega_x + I_{yy}\omega_y + I_{yz}\omega_z; L_z = I_{zx}\omega_x + I_{zy}\omega_y + I_{zz}\omega_z \quad \dots(7)$$

where

$$I_{xx} = \Sigma m (r^2 - x^2) = \Sigma m (y^2 + z^2); I_{yy} = \Sigma m (r^2 - y^2) = \Sigma m (x^2 + z^2);$$

$$I_{zz} = \Sigma m (r^2 - z^2) = \Sigma m (x^2 + y^2).$$

$$I_{xy} = -\Sigma m xy = I_{yx}; I_{xz} = -\Sigma m xz = I_{zx}; I_{yz} = -\Sigma m yz = I_{zy} \quad \dots(8)$$

The quantity I_{xx} is called **moment of inertia of the body about X-axis**. Similarly I_{yy} and I_{zz} define the moments of inertia of the body about Y - and Z -axes respectively. The quantities $I_{xy}, I_{xz}, \dots$ etc. which involve the sums of products of co-ordinates, are called **products of inertia**.

Any component of the angular momentum vector may be written as

$$L_\alpha = \Sigma_\beta I_{\alpha\beta} \omega_\beta \quad \dots(9)$$

where $\alpha, \beta = x, y, z$.

Eq. (7) may be written in the matrix notation as

$$\begin{pmatrix} L_x \\ L_y \\ L_z \end{pmatrix} = \begin{pmatrix} I_{xx} & I_{xy} & I_{xz} \\ I_{yx} & I_{yy} & I_{yz} \\ I_{zx} & I_{zy} & I_{zz} \end{pmatrix} \begin{pmatrix} \omega_x \\ \omega_y \\ \omega_z \end{pmatrix} \quad \dots(10a)$$

or

$$\vec{L} = \mathbf{I} \vec{\omega} \quad \dots(10b)$$

The nine elements $I_{xx}, I_{xy}, \dots, I_{zz}$ of the (3×3) matrix may be regarded as components of a single entity $\mathbf{I}$. This entity $\mathbf{I}$ is called **inertia tensor**. Since $I_{xy} = I_{yx}$ etc., $\mathbf{I}$ is a **symmetric tensor**. If we denote x, y, z by x_1, x_2, x_3 respectively, then in general any element of the inertial tensor is given by

$$I_{\alpha\beta} = I_{\beta\alpha} = \sum_{i=1}^N m_i [\delta_{\alpha\beta} r_i^2 - x_{i\alpha} x_{i\beta}] \quad \dots(11)$$

where $\alpha, \beta = 1, 2, 3$.

In eqs. (7) and (8), the matrix elements are in the suitable form, if the rigid body is composed of discrete particles. In case of a continuous body, the summation sign is replaced by mass or volume integration. Thus the diagonal element I_{xx} is

$$I_{xx} = \int (r^2 - x^2) dm = \int \rho(\vec{r}) (r^2 - x^2) dV = \int \rho(\vec{r}) (y^2 + z^2) dV \quad \dots(12a)$$

where dm is the mass of an infinitesimal volume element dV and $\rho(\vec{r})$ is the mass density at $\vec{r}$ within dV .

The off-diagonal element I_{xy} is

$$I_{xy} = - \int xy \, dm = - \int \rho(\vec{r}) xy \, dV \quad \dots(12b)$$

In general, the matrix element $I_{\alpha\beta}$ can be written as

$$I_{\alpha\beta} = \int \rho(\vec{r}) (r^2 \delta_{\alpha\beta} - x_\alpha x_\beta) \, dV \quad \dots(13)$$

Moment of Inertia and Angular Momentum about an Axis

Now, we consider the magnitude (L_0) of the component of $\vec{L}$ along the axis of rotation, i.e., $\vec{\omega}$.

The angular momentum $\vec{L}$ of the rigid body is given by [from eq. (3)]

$$\vec{L} = \Sigma [m r^2 \vec{\omega} - (\omega r \cos \theta) \vec{r}] \quad \dots(14)$$

In eq. (14), we see that the first quantity in the right hand side is along $\vec{\omega}$ and in the second quantity, component of $\vec{r}$ along $\vec{\omega}$ will be $r \cos \theta$ [fig. 3.22].

Hence the magnitude of the component (L_0) of the angular momentum $\vec{L}$ along the axis of rotation will be given by

$$L_0 = \Sigma m \{ r^2 \omega - (\omega r \cos \theta) r \cos \theta \} = \Sigma m r^2 \omega (1 - \cos^2 \theta) = \Sigma m r^2 \omega \sin^2 \theta = \omega \Sigma m r_0^2 \quad \dots(15)$$

where $r_0 = r \sin \theta$ is the distance PC and ω is the same for all particles of the body.

In eq. (15) the sum $\Sigma m r_0^2$ is the **moment of inertia of the body about the axis of rotation** i.e.,

$$I = \Sigma m r_0^2 = m_1 r_1^2 + m_2 r_2^2 + \dots \quad \dots(16)$$

Substituting for $\Sigma m r_0^2 = I$ in (14), the angular momentum about the axis of rotation (the component of angular momentum $\vec{L}$ along the axis OA) is

$$\vec{L}_0 = I \vec{\omega} \quad \text{or} \quad L_0 = I \omega \quad \dots(17)$$

Hence **the angular momentum of a rigid body about an axis is equal to the product of moment of inertia about that axis and angular velocity.**

If we have a body with axial symmetry and the body is allowed to rotate about the axis of symmetry, then the component of $\vec{L}_p$ perpendicular to OA will be cancelled by an equal component of the angular momentum of another particle on the opposite side of the dotted circle. Therefore in such a case *the total angular momentum $\vec{L}$ of the body will be along the axis of rotation, i.e.,*

$$\vec{L} = I \vec{\omega} \quad \text{or} \quad L = I \omega. \quad \dots(18)$$

For example, if a sphere rotates about its diameter, $\vec{L}$ and $\vec{\omega}$ are always parallel.

3.12. Principal Axes and Principal Moments of Inertia

From eq. (2) or eq. (10) [Art. 3.11], it is obvious that in general $\vec{L}$ and $\vec{\omega}$ are not in the same direction. However, it can be proved that whatever the shape of a body may be, there are three mutually perpendicular axes, such that the angular momentum of the body is along the angular velocity, when the body is rotated about that axis. The moments of inertia about these axes are denoted by I_1 , I_2 and I_3 and are called **principal moments of inertia** with corresponding axes as **principal axes**. Thus if a body is

rotating about a principal axis, then $\vec{L}$ and $\vec{\omega}$ are in the same direction i.e.,

$$\vec{L} = I \vec{\omega} \quad \dots(1)$$

where I is scalar, being moment of inertia ($I = \Sigma mr_0^2$) about this axis.

In eq. (1) the angular momentum $\vec{L}$ and angular velocity $\vec{\omega}$ are in the same direction along the principal axis. Therefore each will have three components along X, Y, Z axes of an arbitrary co-ordinate system fixed in the body. Thus $\vec{L} = L_x \hat{i} + L_y \hat{j} + L_z \hat{k} = I(\omega_x \hat{i} + \omega_y \hat{j} + \omega_z \hat{k})$, where $\hat{i}, \hat{j}$ and $\hat{k}$ are unit vectors along $X-, Y-, Z-$ axes respectively. Hence

$$L_x = I\omega_x, L_y = I\omega_y, L_z = I\omega_z \quad \dots(2)$$

Equating the expressions of L_x, L_y and L_z in eqs. (7) [Art. 3.11] and (2), we get

$$\begin{aligned} I_{xx}\omega_x + I_{xy}\omega_y + I_{xz}\omega_z &= I\omega_x; \quad I_{xy}\omega_x + I_{yy}\omega_y + I_{yz}\omega_z = I\omega_y; \quad I_{xz}\omega_x + I_{yz}\omega_y + I_{zz}\omega_z = I\omega_z \\ \text{or} \quad (I_{xx} - I)\omega_x + I_{xy}\omega_y + I_{xz}\omega_z &= 0 \\ I_{xy}\omega_x + (I_{yy} - I)\omega_y + I_{yz}\omega_z &= 0 \\ I_{xz}\omega_x + I_{yz}\omega_y + (I_{zz} - I)\omega_z &= 0 \end{aligned} \quad \dots(3)$$

$$\text{or} \quad \begin{pmatrix} I_{xx} - I & I_{xy} & I_{xz} \\ I_{xy} & I_{yy} - I & I_{yz} \\ I_{xz} & I_{yz} & I_{zz} - I \end{pmatrix} \begin{pmatrix} \omega_x \\ \omega_y \\ \omega_z \end{pmatrix} = 0 \quad \dots(4)$$

For these equations to have nontrivial solutions, the determinant of the coefficients must vanish

$$\text{i.e.,} \quad \begin{vmatrix} I_{xx} - I & I_{xy} & I_{xz} \\ I_{xy} & I_{yy} - I & I_{yz} \\ I_{xz} & I_{yz} & I_{zz} - I \end{vmatrix} = 0 \quad \dots(5)$$

This is called **secular** or **characteristic equation** and its solutions are called **eigen values**. Eq. (5) is cubic in I , the three solutions $I = I_1, I_2, I_3$ are the **three principal moments of inertia**.

The direction of a principal axis is determined by substituting for I equal to one of the three roots, say $I = I_1$, in eq. (3) and determine the ratios $\omega_x : \omega_y : \omega_z$ or $\frac{\omega_y}{\omega_x} = \lambda_1, \frac{\omega_z}{\omega_x} = \lambda_2$ (say).

$$\begin{aligned} \text{Thus} \quad \vec{\omega}_1 &= \omega_x \hat{i} + \lambda_1 \omega_x \hat{j} + \lambda_2 \omega_x \hat{k} \\ \therefore \quad \hat{\omega}_1 &= \frac{\hat{i} + \lambda_1 \hat{j} + \lambda_2 \hat{k}}{\sqrt{1 + \lambda_1^2 + \lambda_2^2}} \end{aligned} \quad \dots(6)$$

This determines the direction of $\vec{\omega}_1$ or **the direction of the principal axis corresponding to I_1** . Similarly if we substitute I_2 or I_3 , we may find the direction of the corresponding principal axis by using eq. (3). The magnitude of angular velocity is arbitrary and we are free to take its any value.

If is to be remembered that X, Y, Z is any general co-ordinate system fixed in the body and the co-ordinate system with principal axes as its axes X_0, Y_0, Z_0 has, in general, inclined axes with the X, Y, Z system. If $\hat{i}_0, \hat{j}_0, \hat{k}_0$ be the unit vectors along principal axes, then the angular momentum $\vec{J}$ along an arbitrary direction can be expressed as

$$\begin{aligned} \vec{L} &= L_1 \hat{i}_0 + L_2 \hat{j}_0 + L_3 \hat{k}_0 \\ \vec{L} &= I_1 \omega_1 \hat{i}_0 + I_2 \omega_2 \hat{j}_0 + I_3 \omega_3 \hat{k}_0 \end{aligned} \quad \dots(7a)$$

$$\text{or} \quad \begin{pmatrix} L_1 \\ L_2 \\ L_3 \end{pmatrix} = \begin{pmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{pmatrix} \begin{pmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{pmatrix} \quad \dots(7b)$$

where $\vec{\omega} = \omega_1 \hat{i}_0 + \omega_2 \hat{j}_0 + \omega_3 \hat{k}_0$. Thus the inertia tensor $\mathbf{I}_0$ is diagonal, when we consider the co-ordinate system fixed in the body along the principal axes, i.e.,

$$\mathbf{I}_0 = \begin{pmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{pmatrix}.$$

Symmetric Bodies : In several cases, a body has some regular shape and the principal axes may be found by determining the axis of symmetry of the body. The axis of symmetry is one of the principal axis. For example, in case of a circular solid cylinder, its axis of cylindrical symmetry is one of the principal axes. Suppose the axis of cylindrical symmetry is the Z-axis, then the contribution from (x, y, z) is cancelled by that from $(-x, -y, z)$ because $I_{xz} = -\sum mxz = 0$ and $I_{yz} = -\sum myz = 0$ and $J_z = I_{zz}\omega_z = I_3\omega_z$. Thus Z-axis (or axis of cylindrical symmetry) is the principal axis. Obviously the other two principal axes are in the X-Y plane with $I_1 = I_2$. In general, a rigid body is said to be symmetric, if two of its principal moments of inertia are equal. It may happen that all the three principal moments of inertia are equal, i.e., $I_1 = I_2 = I_3$. This is the case for a cube or a sphere with origin at the centre. Such bodies may be called totally symmetric.

Properties of Inertia Tensor

(1) The inertia tensor is given by

$$\mathbf{I} = \begin{pmatrix} I_{xx} & I_{xy} & I_{xz} \\ I_{yx} & I_{yy} & I_{yz} \\ I_{zx} & I_{zy} & I_{zz} \end{pmatrix}$$

where $I_{xy} = I_{yx} = -\sum mxy$, $I_{xz} = I_{zx} = -\sum mxz$, $I_{yz} = I_{zy} = -\sum myz$.

This means, $I_{\alpha\beta} = I_{\beta\alpha}$, where $\alpha, \beta = x, y, z$. Thus inertia tensor $\mathbf{I}$ is a symmetric tensor.

(2) If I_1, I_2 and I_3 are the principal moment of inertia about the corresponding principal axes, then off diagonal elements of the inertia tensor are zero and the inertia tensor $\mathbf{I}_0$ is diagonal i.e.,

$$\mathbf{I}_0 = \begin{pmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{pmatrix}$$

(3) If for a rigid body $I_1 = I_2$, then the body is called symmetric top. Earth is an example of a symmetric top because its equatorial diameter is somewhat greater than its polar diameter. The tensor for such a body is

$$\mathbf{I}_0 = \begin{pmatrix} I_1 & 0 & 0 \\ 0 & I_1 & 0 \\ 0 & 0 & I_3 \end{pmatrix}$$

(4) In case of a solid sphere or cube, all the three principal moments of inertia are equal i.e., $I_1 = I_2 = I_3$. Such a body is called totally symmetric top.

3-13. Torque on a Rigid Body : Equation of Motion of Rotating Body

The angular momentum of a rigid body about a fixed point is given by

$$\vec{L} = \mathbf{I} \vec{\omega} \quad \dots(1)$$

where $\mathbf{I}$ is the inertia tensor and $\vec{\omega}$ is the instantaneous angular velocity.

The torque is defined as the rate of change of angular momentum. Hence the torque acting on the rigid body about the fixed point is given by

$$\vec{\tau} = \frac{d\vec{L}}{dt} = \frac{d}{dt} (I\vec{\omega}) \quad \dots(2)$$

This is similar to the equation of motion for translatory motion ($\vec{F} = d\vec{P}/dt$), given by Newton's second law. Hence equation (2) is called *equation of motion for a rotating rigid body*.

If the axis of rotation of the rigid body happen to be an axis of symmetry of the body, then $\vec{L}$ and $\vec{\omega}$ will have the same direction and I will be the moment of inertia about the axis of rotation. In such a case,

$$\vec{\tau} = \frac{d\vec{L}}{dt} = I \frac{d\vec{\omega}}{dt} \quad \dots(3)$$

This is the equation of motion of a rotating rigid body about its axis of symmetry. Eq. (3) is a more convenient form of the equation of motion of a rotating rigid body but it is not as general as eq. (2).

3.14. Euler's Equations of Motion for a Rigid Body

If a rigid body is rotating under the action of a torque $\vec{\tau}$ with one point fixed, then the equation of rotational motion in an inertial frame of reference is given by

$$\vec{\tau} = \left[\frac{d\vec{L}}{dt} \right]_i \quad \dots(1)$$

where $\vec{L}$ is the angular momentum and its time derivative refers to the inertial co-ordinate system. The axes of such a co-ordinate system are in fixed directions in space. The time derivatives in this co-ordinate system are represented by subscript i because the equation holds in an inertial frame.

The co-ordinate system, which is fixed in the rigid body, is rotating with an instantaneous angular velocity $\vec{\omega}$. We call this frame of reference as *body co-ordinate system*. We know that the time derivative of a vector in an inertial frame of reference is related to the time derivative in a rotating frame of reference by the following operator equation :

$$\left(\frac{d}{dt} \right)_i = \left(\frac{d}{dt} \right)_r + \vec{\omega} \times \quad \dots(2)$$

Hence the time derivatives of angular momentum $\vec{L}$ in the body co-ordinate and inertial co-ordinate systems are related as

$$\left[\frac{d\vec{L}}{dt} \right]_i = \left[\frac{d\vec{L}}{dt} \right]_r + \vec{\omega} \times \vec{L} \quad \dots(3)$$

$$\text{Thus,} \quad \vec{\tau} = \frac{d\vec{L}}{dt} + \vec{\omega} \times \vec{L} \quad \dots(4)$$

where we have dropped the body subscript because we shall represent the physical quantities of right hand side in the body co-ordinate system.

We choose principal axes for body set of axes. If I_1 , I_2 and I_3 are the principal moments of inertia, then

$$\vec{L} = I_1\omega_1\hat{i} + I_2\omega_2\hat{j} + I_3\omega_3\hat{k} \quad \dots(5)$$

where $\vec{\omega} = \omega_1 \hat{i} + \omega_2 \hat{j} + \omega_3 \hat{k}$ is the angular velocity with components ω_1 , ω_2 and ω_3 along the principal axes.

As the principal moments of inertia and body base vectors $\hat{i}$, $\hat{j}$ and $\hat{k}$ are constants in time with respect to the body co-ordinate system, we find that in the body co-ordinate system, using (5) the time derivative of $\vec{L}$ is

$$\frac{d\vec{L}}{dt} = I_1 \dot{\omega}_1 \hat{i} + I_2 \dot{\omega}_2 \hat{j} + I_3 \dot{\omega}_3 \hat{k} \quad \dots(6)$$

where $\dot{\omega}_1 = d\omega_1/dt$, $\dot{\omega}_2 = d\omega_2/dt$ and $\dot{\omega}_3 = d\omega_3/dt$.

Substituting in (4), we obtain

$$\vec{\tau} = I_1 \dot{\omega}_1 \hat{i} + I_2 \dot{\omega}_2 \hat{j} + I_3 \dot{\omega}_3 \hat{k} + (\omega_1 \hat{i} + \omega_2 \hat{j} + \omega_3 \hat{k}) \times (I_1 \omega_1 \hat{i} + I_2 \omega_2 \hat{j} + I_3 \omega_3 \hat{k}) \quad \dots(7)$$

Writing $\vec{\tau} = \tau_1 \hat{i} + \tau_2 \hat{j} + \tau_3 \hat{k}$, we can obtain the x , y , z components of the torque $\vec{\tau}$ as

$$\tau_1 = I_1 \dot{\omega}_1 + (I_3 - I_2) \omega_2 \omega_3 \quad \dots(8a)$$

$$\tau_2 = I_2 \dot{\omega}_2 + (I_1 - I_3) \omega_3 \omega_1 \quad \dots(8b)$$

$$\tau_3 = I_3 \dot{\omega}_3 + (I_2 - I_1) \omega_1 \omega_2 \quad \dots(8c)$$

Eqs. (8) are known as **Euler's equations** for the motion of a rigid body with one point fixed under the action of a torque.

In case a rigid body is rotating about a fixed axis, say principal Z -axis, then

$$\omega_1 = \omega_2 = 0 \text{ and } \omega_3 = \omega$$

Therefore, from eq. (8) we have the equations of motion as

$$\tau_1 = \tau_2 = 0$$

$$\text{and} \quad \tau_3 = I_3 \dot{\omega} \quad \text{or} \quad \tau = I \frac{d\omega}{dt} \quad \dots(9)$$

where we have put $\tau_3 = \tau$ and $I_3 = I$ corresponding to Z -axis.

Instantaneous angular momentum about Z -axis is

$$L_3 = I_3 \omega_3 \quad \text{or} \quad L = I\omega \quad \dots(10)$$

and instantaneous rotational kinetic energy is

$$K = \frac{1}{2} \vec{\omega} \cdot \vec{L} = \frac{1}{2} I \omega^2 \quad \dots(11)$$

Torque-free Motion of a Rigid Body

When a rigid body is not subjected to any net torque, the Euler's equations of motion of the body with one point fixed reduce to

$$I_1 \dot{\omega}_1 = (I_2 - I_3) \omega_2 \omega_3 \quad \dots(12a)$$

$$I_2 \dot{\omega}_2 = (I_3 - I_1) \omega_3 \omega_1 \quad \dots(12b)$$

$$I_3 \dot{\omega}_3 = (I_1 - I_2) \omega_1 \omega_2 \quad \dots(12c)$$

In case the body is not subjected to any net forces or torques, its centre of mass is either at rest or moves with uniform velocity. Obviously we may discuss the rotational motion of the rigid body in a reference system in which the centre of mass is stationary and choose the centre of mass as fixed point and origin for the principal axes in the body. In such a case, we obtain from eq. (12) two integrals of motion, describing the kinetic energy and angular momentum as constant in time.

If we multiply eqs. (12) by ω_1 , ω_2 , ω_3 respectively and then add, we obtain

$$I_1\omega_1\dot{\omega}_1 + I_2\omega_2\dot{\omega}_2 + I_3\omega_3\dot{\omega}_3 = (I_2 - I_3 + I_3 - I_1 + I_1 - I_2) \omega_1\omega_2\omega_3 = 0$$

or
$$\frac{d}{dt} \left(\frac{1}{2} I_1 \omega_1^2 + \frac{1}{2} I_2 \omega_2^2 + \frac{1}{2} I_3 \omega_3^2 \right) = 0$$

or
$$\frac{1}{2} I_1 \omega_1^2 + \frac{1}{2} I_2 \omega_2^2 + \frac{1}{2} I_3 \omega_3^2 = \frac{1}{2} \vec{\omega} \cdot \vec{L} = \text{constant} \quad \dots(13)$$

which is the **principle of conservation of total rotational kinetic energy** in absence of external torque.

As
$$\vec{\tau} = \frac{d\vec{L}}{dt} = 0, \vec{L} = I_1\omega_1\hat{i} + I_2\omega_2\hat{j} + I_3\omega_3\hat{k} = \text{constant} \quad \dots(14)$$

describes another constant of motion, representing the **principle of conservation of angular momentum**.

3.15. Rotational Kinetic Energy of a Rigid Body

Let a rigid body be rotating about an axis passing through a fixed point in the body with an angular velocity $\vec{\omega}$. The velocity $\vec{v}$ of a particle of mass m of the body is given by

$$\vec{v} = \vec{\omega} \times \vec{r}$$

The kinetic energy of this particle is given by

$$K = \frac{1}{2} m v^2 = \frac{1}{2} m \vec{v} \cdot \vec{v} \quad \dots(1)$$

Total kinetic energy of the entire body is given by

$$\begin{aligned} K &= \sum \frac{1}{2} m v^2 = \sum \frac{1}{2} m \vec{v} \cdot \vec{v} = \frac{1}{2} \sum \vec{v} \cdot m \vec{v} = \frac{1}{2} \sum (\vec{\omega} \times \vec{r}) \cdot m \vec{v} \\ &= \frac{1}{2} \sum \vec{\omega} \cdot (\vec{r} \times m \vec{v}) \quad [\because (\vec{A} \times \vec{B}) \cdot \vec{C} = \vec{A} \cdot (\vec{B} \times \vec{C})] \\ &= \frac{1}{2} \vec{\omega} \cdot \sum (\vec{r} \times m \vec{v}) \quad [\text{because } \vec{\omega} \text{ is the same for all particles}] \end{aligned}$$

But $\vec{L} = \sum (\vec{r} \times m \vec{v})$ about the fixed point, hence

$$K = \frac{1}{2} \vec{\omega} \cdot \vec{L} \quad \dots(2)$$

This is the expression for **rotational kinetic energy** of a rigid body. This is a scalar and can be written in the form

$$\begin{aligned} K &= \frac{1}{2} \vec{\omega} \cdot \vec{L} = \frac{1}{2} \omega_x L_x + \frac{1}{2} \omega_y L_y + \frac{1}{2} \omega_z L_z \\ &= \frac{1}{2} \omega_x (I_{xx} \omega_x + I_{xy} \omega_y + I_{xz} \omega_z) + \frac{1}{2} \omega_y (I_{xy} \omega_x + I_{yy} \omega_y + I_{yz} \omega_z) \\ &\quad + \frac{1}{2} \omega_z (I_{xz} \omega_x + I_{yz} \omega_y + I_{zz} \omega_z) \\ &= \frac{1}{2} I_{xx} \omega_x^2 + \frac{1}{2} I_{yy} \omega_y^2 + \frac{1}{2} I_{zz} \omega_z^2 + I_{xy} \omega_x \omega_y + I_{xz} \omega_x \omega_z + I_{yz} \omega_y \omega_z \quad \dots(3) \end{aligned}$$

The kinetic energy may be written in the compact form as

$$K = \frac{1}{2} \sum_{\alpha, \beta=1}^3 I_{\alpha\beta} \omega_\alpha \omega_\beta \quad \dots(4)$$

In case, I_1 , I_2 , I_3 are the principal moments of inertia, the angular momentum $\vec{L}$ and angular velocity $\vec{\omega}$ are related by

$$L_1 = I_1\omega_1, L_2 = I_2\omega_2, L_3 = I_3\omega_3$$

where L_1, L_2, L_3 are the components of $\vec{L}$ along three principal axes.

Hence the kinetic energy in a co-ordinate system of principal axes is given by

$$K = \frac{1}{2} \vec{\omega} \cdot \vec{L} = \frac{1}{2} \omega_1 L_1 + \frac{1}{2} \omega_2 L_2 + \frac{1}{2} \omega_3 L_3 = \frac{1}{2} \sum_{\alpha=1}^3 I_{\alpha} \omega_{\alpha}^2 \quad \dots(5)$$

We may express the kinetic energy of rotation in the usual form, used in elementary mechanics, as

$$K = \frac{1}{2} I \omega^2 \quad \dots(6)$$

where I is the moment of inertia about the axis of rotation. The proof is as follows :

We know that the component (L_0) of angular momentum $\vec{L}$ along the axis of rotation, i.e., along $\vec{\omega}$ is given by

$$L_0 = I\omega$$

where $I = \Sigma mr_0^2$ is the moment of inertia about the axis of rotation.

Hence the kinetic energy about the axis of rotation is given by

$$K = \frac{1}{2} \vec{\omega} \cdot \vec{L} = \frac{1}{2} \omega L_0 \quad [\because L \cos \theta = L_0]$$

i.e.,

$$K = \frac{1}{2} I \omega^2 \quad \dots(7)$$

3.16. Force-Free Motion of a Symmetrical Top

Now, we use Euler's equation to discuss the force-free motion of a symmetrical rigid body, that is a symmetrical top. We define symmetrical top as a **rigid body** for which $I_1 = I_2$ and the third principal axis (Z-axis) is the axis of symmetry of the body. In such a case Euler's equations for torque free motion of a rigid body [eqs. (12), Art. 3.13] take the form :

$$I_1 \dot{\omega}_1 = (I_1 - I_3) \omega_2 \omega_3 \quad \dots(1a)$$

$$I_1 \dot{\omega}_2 = (I_3 - I_1) \omega_3 \omega_1 \quad \dots(1b)$$

$$\dot{\omega}_3 = 0 \quad \dots(1c)$$

On integrating the last equation, we obtain

$$\omega_3 = \text{constant} \quad \dots(2)$$

Thus the component of the angular velocity (ω) along the symmetry axis is constant with time for force-free motion of the rigid body.

If we now put

$$\Omega = \frac{I_3 - I_1}{I_1} \omega_3 = \text{constant}, \quad \dots(3)$$

then the first two equations of (1) can be written as

$$\dot{\omega}_1 = -\Omega \omega_2 \quad \dots(4a)$$

$$\dot{\omega}_2 = \Omega \omega_1 \quad \dots(4b)$$

Differentiating eq. (4a), we get

$$\ddot{\omega}_1 = -\Omega \dot{\omega}_2 \quad \dots(5)$$

Substituting for $\dot{\omega}_2$ from (4b), we obtain

$$\ddot{\omega}_1 = -\Omega^2 \omega_1 \text{ or } \ddot{\omega}_1 + \Omega^2 \omega_1 = 0 \quad \dots(6)$$

This is well known differential equation, used to deal simple harmonic motion, and its solution is

$$\omega_1 = \omega_c \sin \Omega t \quad \dots(7)$$

where ω_c is some constant and we have chosen initial conditions $\omega_1 = 0$ at $t = 0$ so that the phase constant is zero.

Substituting for ω_1 from (7) in (4a), we get ...(8)

$$\omega_c \Omega \cos \Omega t = \Omega \omega_2 \text{ or } \omega_2 = \omega_c \cos \Omega t$$

Squaring (7) and (8), we get

$$\omega_1^2 + \omega_2^2 = \omega_c^2 \quad \dots(9)$$

This is the equation of a circle with radius ω_c . Obviously ω_1 and ω_2 are perpendicular to each other.

Let us take ω_1 along X-axis and ω_2 along Y-axis. If $\hat{i}$ and $\hat{j}$ are unit vectors along the two principal axes other than the symmetric axis, then the vector

$$\vec{\omega}' = \omega_1 \hat{i} + \omega_2 \hat{j}$$

or

$$\vec{\omega}' = (\omega_c \sin \Omega t) \hat{i} + (\omega_c \cos \Omega t) \hat{j} \quad \dots(10)$$

has a constant magnitude $|\vec{\omega}'| = \omega_c$ and rotates about the third principal axis (symmetry axis) with an angular frequency

$$\Omega = \frac{I_3 - I_1}{I_1} \omega_3$$

as shown in **fig. 3-23**. Hence the total angular velocity $\vec{\omega}$ is given by

$$\vec{\omega} = \omega_1 \hat{i} + \omega_2 \hat{j} + \omega_3 \hat{k} = \vec{\omega}' + \omega_3 \hat{k} \quad \dots(11)$$

with $|\vec{\omega}| = \sqrt{\omega_c^2 + \omega_3^2} = \text{constant}$.

Thus the angular velocity vector $\vec{\omega}$ has constant magnitude and precesses about the Z-axis (axis of symmetry) with constant angular frequency Ω . The precessional motion is shown in **fig. 3-23**.

We observe that $\vec{\omega}$ **vector moves on the surface of a cone about the axis of symmetry** with the constant angular frequency Ω . This precessional motion is with respect to the principal (body) axes, fixed in the body, which are themselves rotating in space with the large frequency ω (relative to Ω). It is to be seen from eq. (3) [$\Omega = (I_3 - I_1) \omega_3 / I_1$] that as I_1 is closer to I_3 , the precessional frequency Ω will be slower compared to the rotation frequency ω . The constant ω_c and ω_3 can be determined, if the constants of the motion, K the kinetic energy and the magnitude of the angular momentum L , are known. Both K and L can be expressed in terms of ω_c and ω_3 as

$$K = \frac{1}{2} I_1 \omega_c^2 + \frac{1}{2} \omega_3^2 \quad \dots(12)$$

and

$$L^2 = I_1^2 \omega_c^2 + I_3^2 \omega_3^2 \quad \dots(13)$$

which can be solved to obtain the values of ω_c and ω_3 .

Frequency and period of precession of angular velocity vector $\vec{\omega}$ (or the axis of rotation) about the axis of symmetry (Z-axis) are given by

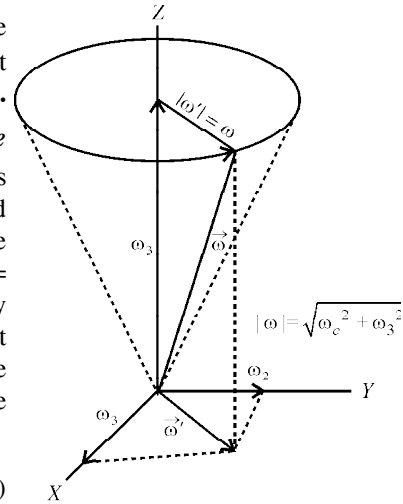


Fig. 3.23. Force-free motion of a symmetrical top ($I_1 = I_2$)

$$f = \frac{\Omega}{2\pi} = \frac{\omega_3}{2\pi} \frac{I_3 - I_1}{I_1} \quad \dots(14)$$

and

$$T = \frac{1}{f} = \frac{2\pi}{\omega_3} \frac{I_1}{I_3 - I_1} \quad \dots(15)$$

Rotation of the earth : One very important example of the application of the theory discussed above, is the case of rotating earth. The rotational motion of the earth may be considered as that of a free body because the external torques of the earth are very weak and therefore one would expect the precession of axis of rotation around the axis of symmetry. The earth is symmetrical about the polar axis and flattened slightly at the poles (shape of oblate spheroid). Consequently I_3 is little greater than I_1 and hence the precessional angular frequency is

$$\Omega = \frac{I_3 - I_1}{I_1} \omega_3 = \frac{\omega_3}{306} \quad \dots(16)$$

Since ω_3 is practically equal to the magnitude of ω , i.e., $\omega_3 \approx \omega$, the angular frequency of the earth,

$$\frac{2\pi}{\omega_3} \approx \frac{2\pi}{\omega} = 1 \text{ day} \quad \dots(17)$$

Hence the **precessional period of the earth** is given by

$$T = \frac{2\pi}{\Omega} = \frac{2\pi}{\omega_3} \cdot \frac{I_1}{I_3 - I_1} = \frac{2\pi}{\omega_3} \times 306 = 306 \text{ days} \quad \dots(18)$$

If some circumstance disturbed the axis of rotation from the figure axis of the earth, one would therefore expect the axis of rotation to precess around the figure axis, i.e., around the North Pole once every 306 days. However, the measured value is 440 days. The disagreement is not due to lack of knowledge of I_1 or I_3 but due to the fact that the earth is not a perfect rigid body and does not possess the shape of oblate spheroid. In fact, the shape of earth resembles a flattened pear. Careful measurements tell us that the earth's rotation axis precesses about the North Pole in a circle of radius 10 m with a period of 440 days. As the latitude depends on the position of the axis of rotation, a measurable change in latitude results.

3-17. Precession of a Symmetric Top under the Action of External Torque

Consider a symmetrical top of mass M spinning rapidly about its symmetry axis with its tip at a point O , fixed in an inertial frame of reference. O is taken as origin of the inertial system. If the axis of the top makes an angle θ with the vertical (Z -axis), from our observation we find that the axis of the spinning top moves about the vertical axis OZ , sweeping out a cone as shown in **fig. 3.24 (a)**. The rotation of the axis of rotation of the spinning top, produced by a torque, is called **precession**. The rate at which the rotation axis moves about the vertical (OZ) axis is called the angular velocity of precession Ω .

Explanation : If the top were not spinning, it will immediately fall down on the ground, when left with its axis at an angle θ to the vertical. When the top

is spinning with angular velocity $\vec{\omega}$ about its symmetry

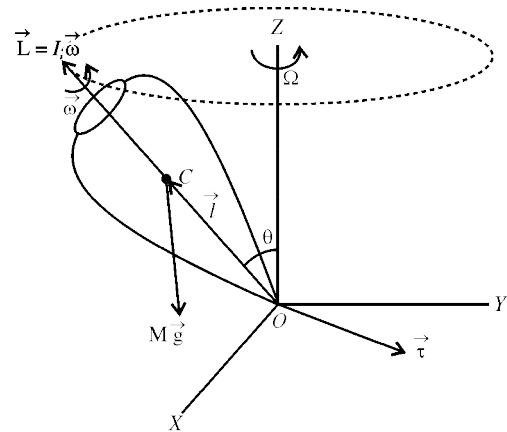


Fig. 3.24 (a). Precession of a spinning top.

axis, it has an angular momentum $\vec{L} = I\vec{\omega}$ directed along its axis. Here I is the moment of inertia of the top about its axis of symmetry. Due to the weight $M\vec{g}$ ($\vec{g}$ = acceleration due to gravity) of the top, there is a torque about O , given by [fig. 3.24(a)]

$$\vec{\tau} = \vec{l} \times M\vec{g} \quad \dots(1)$$

where $\vec{l}$ is the position vector of the centre of mass C with respect to O . The direction of $\vec{\tau}$ is perpendicular to both $\vec{l}$ and $M\vec{g}$ in the horizontal XY plane and its magnitude is $\tau = l mg \sin \theta$.

The equation of motion for a rotating rigid body is

$$\vec{\tau} = \frac{d\vec{L}}{dt} \quad \dots(2)$$

Hence the change in $\vec{L}$ in time dt is

$$d\vec{L} = \vec{\tau} dt \quad \dots(3)$$

Thus the change $d\vec{L}$ in the angular momentum $\vec{L}$ of the spinning top is in the direction of the torque $\vec{\tau}$. But torque $\vec{\tau}$ is perpendicular to position vector $\vec{l}$ and hence perpendicular to the angular momentum vector $\vec{L}$ (as $\vec{l}$ and $\vec{L}$ are along the same direction). Hence $d\vec{L}$ is perpendicular to $\vec{L}$. This means that the angular momentum $\vec{L}$ changes in direction but not in magnitude. Since $\vec{L}$ points along the axis of the top, we find that the axis moves to the right in fig. 3.24 (b) (This situation is similar to the case of circular motion under a centripetal force, where the force is perpendicular to velocity and the velocity changes in direction but not in magnitude). This explains the precession of the axis of the top.

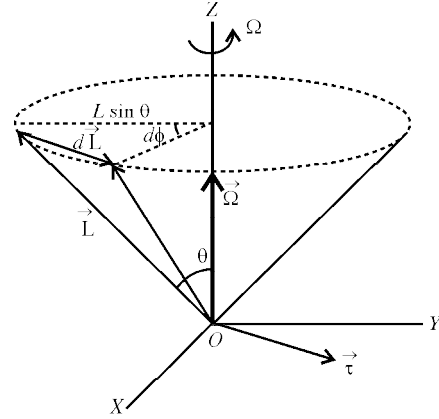


Fig. 3.24 (b). Torque giving rise angular velocity of precession.

The end of the vector $\vec{L}$ (or axis of spinning top) describe a circle around Z -axis of radius $L \sin \theta$ [fig. 3.24(b)]. Therefore, the magnitude of $d\vec{L}$ is

$$dL = L \sin \theta d\phi \quad \dots(4)$$

The angular velocity of precession is defined as the rate at which the axis of the spinning top rotates around Z -axis i.e.,

$$\Omega = \frac{d\phi}{dt} \quad \dots(5)$$

which is represented by a vector $\vec{\Omega}$ along OZ .

$$\text{Hence} \quad dL = L \sin \theta \Omega dt \quad [\because d\phi = \Omega dt] \quad \dots(6)$$

$$\text{But from (3)} \quad |d\vec{L}| = dL = \tau dt \quad \dots(7)$$

From eq. (6) and (7),

$$L \sin \theta \Omega dt = \tau dt \text{ or } \tau = \Omega L \sin \theta \quad \dots(8)$$

But the magnitude of the torque acting on the system in $\tau = Mgl \sin \theta$. Hence

$$\Omega L \sin \theta = Mgl \sin \theta \quad \dots(9)$$

$$\text{or} \quad \Omega = \frac{Mgl}{L} = \frac{Mgl}{I\omega} \quad \dots(10)^*$$

Thus the rate of precession is inversely proportional to the angular momentum of the spinning top. This means that smaller the angular momentum, larger the precessional angular velocity. This is why the precession becomes faster and faster as the spinning top slows down due to friction. Also, we see that the rate of precession is independent of angle θ , between the axis of the spinning top and vertical.

From eqs. (8) and (9), $\tau = \Omega L \sin \theta = Mgl \sin \theta$. Hence in vector form we can express the torque as

$$\vec{\tau} = \vec{\Omega} \times \vec{L} = \vec{l} \times M\vec{g} \quad \dots(11)$$

In general, the angle θ , inclination of the axis of the spinning top with the vertical, does not remain constant, but oscillates between two fixed values θ_1 and θ_2 . The end of the vector $\vec{L}$ oscillates (at the same time $\vec{L}$ precesses around Z-axis) between two circles C_1 and C_2 [fig. 3.25]. This oscillatory motion of axis of the spinning top is called **nutation**.

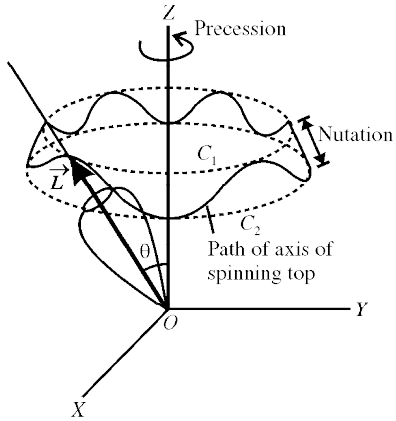


Fig. 3.25. Precession and nutation of the axis of spinning top.

3.18. Gyroscope

A gyroscope is a heavy symmetrical body (top) in the form of a heavy disc or flywheel, rotation rotating at a very high speed about its axis of symmetry and this axis is also free to turn about some fixed point under the action of torque.

A gyroscope is shown in fig. 3.26, where D is a heavy and large disc, rotating with a high angular velocity ω about its axis OA , supported on a pivot at the origin O . Let the weight of the flywheel (disc + axle) $M\vec{g}$ be acting at its centre of mass C , at a distance l from the origin O [fig. 3.26]. Due to the weight $M\vec{g}$ of the fly wheel, there is a torque about O given by

$$\vec{\tau} = \vec{l} \times M\vec{g} \quad \dots(1)$$

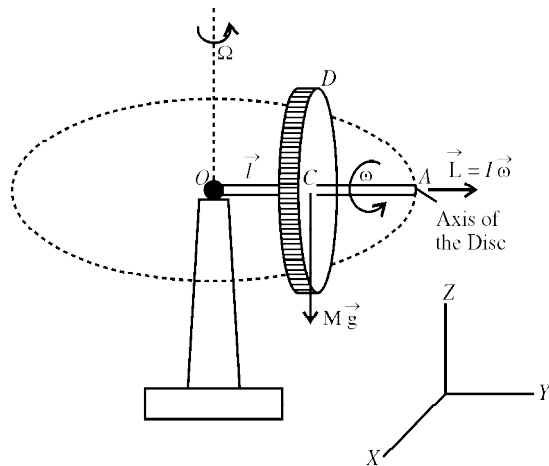


Fig. 3.26. Gyroscope.

* This result is valid if the spin angular velocity ω is very large in comparison to precessional angular velocity Ω . Since there is also angular momentum due to precessional motion, so that the total angular momentum $\vec{L}$ is not exactly along the axis of the top; but if $\Omega \ll \omega$, we can ignore the angular momentum due to precessional motion.

where $\vec{r}$ is the position vector of the centre of mass C from the origin O . The magnitude of the torque is $\tau = Mgl$. The direction of the torque is perpendicular to the plane of $\vec{r}$ and $M\vec{g}$ vectors and is in the horizontal (XY) plane, perpendicular to the axis of the disc.

If I is the moment of inertia of the fly wheel about its axis, then the angular momentum of the system is $\vec{L} = I\vec{\omega}$ along its axis. As the torque $\vec{\tau} = \vec{r} \times M\vec{g}$ is acting on the system, it will change the angular momentum by $d\vec{L}$ in time dt according to the relation

$$d\vec{L} = \vec{\tau} dt \quad \dots(2)$$

because $\vec{\tau} = d\vec{L}/dt$. As the direction of $\vec{\tau}$ is perpendicular to the axis of rotation of the fly wheel or the vector $\vec{L} = I\vec{\omega}$. Hence from (2) the change $d\vec{L}$ is also perpendicular to $\vec{L}$ and the angular momentum changes in direction but not in magnitude. Since the angular momentum points along the axis of the fly wheel, the axis changes in anti-clockwise direction as seen from above. This rotation of the axis of the rotating fly wheel is termed as the **precession**. The rate at which the axis of the disc rotates defines the angular velocity precession (Ω) i.e.,

$$\Omega = \frac{d\phi}{dt} \quad \dots(3)$$

where $d\phi$ is the angle of rotation in time dt .

The end of the vector $\vec{L}$ (or axis of the rotating disc) describes a circle around the vertical direction (Z -axis) of radius L (magnitude of $\vec{L}$) [fig. 3.27] so that

$$dL = Ld\phi = L\Omega dt \quad \dots(4)$$

because from (3) $d\phi = \Omega dt$. But from (2) $dL = \tau dt$. Hence

$$dL = L\Omega dt = \tau dt \text{ i.e., } \tau = L\Omega \quad \dots(5)$$

But $\tau = Mgl$ is the magnitude of the torque [from (1)]. Therefore,

$$L\Omega = Mgl \quad \dots(6)$$

$$\text{whence } \Omega = \frac{Mgl}{L} = \frac{Mgl}{I\omega} \quad \dots(7)$$

Thus the angular velocity of precession is inversely proportional to the angular momentum of the spinning fly wheel of the gyroscope.

If we represent the angular velocity of precession by $\vec{\Omega}$ vector, then the relation (6) can be put in the vector form

$$\vec{\Omega} \times \vec{L} = \vec{r} \times M\vec{g} \quad \dots(8)$$

Remember that the definition of gyroscope is applicable to a spinning top and hence spinning top is also a gyroscope.

3.19. Gyrostat

A gyrostat is a heavy disc or flywheel, rotating at a high speed about its axle passing through its centre of mass and mounted so that the disc and axle are both free to turn as a whole about any axis

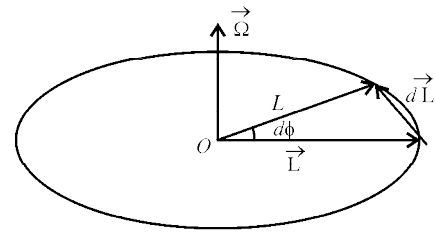


Fig. 3.27. Precession.

perpendicular to its own axle. This is a type of gyroscope.

A gyrostatis shown in **fig. 3.28**. This consists of a heavy circular disc or wheel having its axle AB mounted in gimbals. This disc spins rapidly around the principal axis AB and is arranged so that the total torque about O is zero. In absence of external torque, from equation of motion

$$\vec{\tau} = \frac{d\vec{L}}{dt} = 0 \quad \dots(1)$$

we have angular momentum

$$\vec{L} = I\vec{\omega} = \text{constant} \quad \dots(2)$$

where I is the moment of inertia about the principal axis AB and $\vec{\omega}$ angular velocity.

This angular momentum $\vec{L}$ is along AB or X -axis. If we neglect friction, the spin axis AB can move freely around both the horizontal Y - and the vertical Z axes. If we move the gyroscope around the room, AB always will point in the same direction. Let us set the gyroscope axis so that AB is horizontal and points in the East-West direction (position 1 of **fig. 3.29**). Due of the Earth's rotation we see that AB gradually tilts down (or up). After six hours it is in a vertical position (position 4 of **fig. 3.29**). This apparent rotation of AB is in fact because of the rotation of the earth. Obviously our laboratory moves from position 1 to 4, the orientation of AB remains fixed in space. This experiment proves that the earth rotates about its NS -axis.

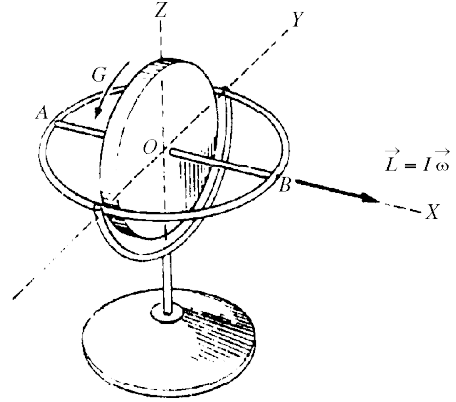


Fig. 3.28. Gyrostatis with spin axis AB .

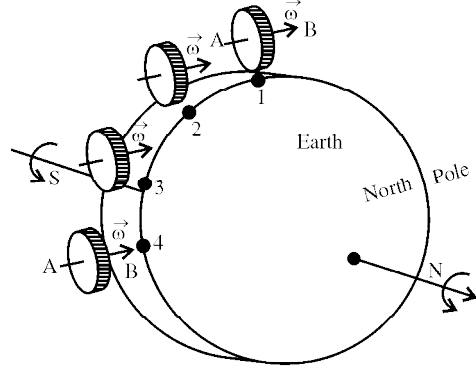


Fig. 3.29. The axis of rotation AB of a gyrostatis under no torque remains fixed in space, and therefore rotates relative to the earth.

Now, if a torque $\vec{\tau}$ is acting on the gyrostatis, its angular momentum from the relation $\vec{\tau} = d\vec{L}/dt$ changes in time dt , given by

$$d\vec{L} = \vec{\tau} dt \quad \dots(3)$$

The change in the angular momentum of the gyrostatis is in the direction of the torque. If the direction of the instantaneous torque is along Y -axis, then the disc with axle precesses about Z -axis such that

$$\vec{\Omega} \times \vec{L} = \vec{\tau} \text{ or } \Omega L = \tau \quad \dots(4)$$

where $\vec{\Omega}$ is the angular velocity of precession.

As torque $\vec{\tau}$ is perpendicular to the angular momentum $\vec{L}$, the change $d\vec{L}$ is also perpendicular to $\vec{L}$ and the angular momentum changes in direction but not in magnitude. This means that the axis of the rotating disc changes direction but the magnitude of the angular momentum $L = I\omega$ remains constant. From (4) the angular velocity of precession is given by

$$\Omega = \frac{\tau}{L} = \frac{\tau}{I\omega} \quad \dots(5)$$

Thus, for a given torque, the precessional rate is inversely proportional (i) to the moment of inertia of the wheel about its axle, and (ii) to the angular velocity of the wheel. Hence, *larger the moment of inertia of the wheel and higher its angular velocity, the smaller will be the rate of precession*. Consequently, a heavier wheel, rotating fast, will make for greater stability of its axis of rotation. This tendency to preserve its axis of rotation endows a gyrostator with a stability of direction, which is made use in ships, aeroplanes etc.

3.20. Precession of the Equinoxes

The equatorial plane (celestial equator) of the earth and the plane of its orbit around Sun (ecliptic) are inclined to each other at an angle 23.5° . Two points P_1 and P_2 at which the ecliptic intersects the equatorial plane are called **equinoxes**. The Sun appears to cross the celestial equator from south to north at the point P_1 (called **vernal equinox**) and from north to South at the point P_2 (called **autumnal equinox**). In the northern hemisphere the vernal equinox occurs on or about March 21 and the autumnal equinox on or about Sept. 23.

We know that the earth deviates slightly from a sphere. It bulges out at the equator and can be approximated by an oblate spheroid of revolution. The rotating earth about its axis through the north and south poles acts as a giant gyroscope. The gravitational attraction due to the Sun and the moon on the equatorial bulge of the earth develops a net torque. Due to this torque, *the earth's axis precesses in space about the normal to the plane of ecliptic in the east-west direction [fig. 3.30]*. This torque is very weak and hence the precession of earth's axis is very slow. This precessional motion of the earth's axis results in an equal change in direction of the line of equinoxes.

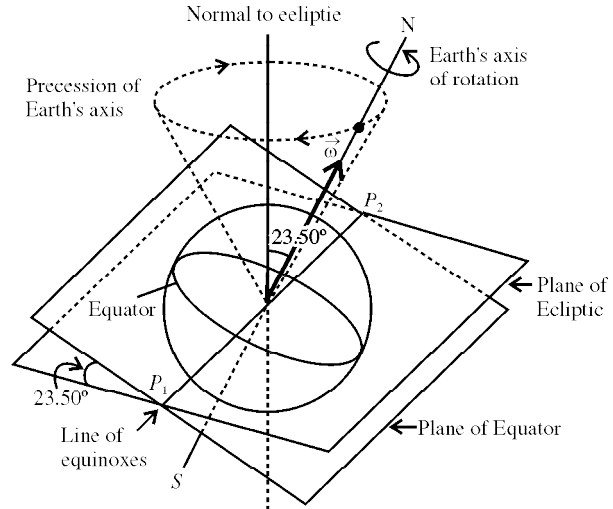


Fig. 3.30 Precession of earth's axis of rotation about normal to ecliptic

This slow motion of the line of equinoxes about the normal to the plane of ecliptic is called the **precession of equinoxes**. The equinoxes move round the ecliptic with a large period of 27,725 years, giving a precessional angular velocity of 7.19×10^{-12} rad/sec. Due to this precessional motion, as time passes, different stars become the polar star. Now the north star (Polaris) is the polar star; in 3000 B.C., Thuban was the polar star and in 14,000 A.D., Vega will be the polar star.

We can calculate the magnitude of the torque which is to be exerted on the earth in order to produce the observed rate of precession of equinoxes.

We use the relation

$$\vec{\tau} = \vec{\Omega} \times \vec{L} \text{ or } \tau = \Omega L \sin \phi$$

for the calculation of the torque.

Here, $\phi = 23.5^\circ$ and angular velocity of precession of the earth, $r = 7.19 \times 10^{-12}$ rad/sec. Angular momentum of the earth is

$$\begin{aligned} L = I\omega &= \frac{2}{5}MR^2\omega = \frac{2}{5} \times (5.98 \times 10^{24} \text{ kg}) \times (6.38 \times 10^6 \text{ m})^2 \times (7.29 \times 10^{-5} \text{ rad/sec}) \\ &= 9.72 \times 10^{37} \text{ kg-m}^2/\text{sec}. \end{aligned}$$

$$\text{Hence } \tau = 7.19 \times 10^{-12} \times 9.72 \times 10^{37} \times 0.399 = 2.79 \times 10^{26} \text{ N-m}$$

Thus, $\tau = 2.79 \times 10^{26}$ N-m is the torque about the centre of mass exerted by the Sun and the Moon on the earth. However this torque is small in planetary terms.

3.21. Applications of Gyroscope and Gyrostat

The property of a rotating gyrostat to tend to preserve its axis of rotation, is used in various technical applications, some of which are discussed below :

(1) Cotrolling of rolling of ships. To avoid the rolling of a ship, a gyro-stabiliser is used. This consists of a motor driven gyroscope spinning about a vertical axis in such a way that the axis may be tilted forward or backward but not sideways. When the ship rolls, say to the right, an automatic device tilts the axis forward, and hence producing an opposing torque. In case, if the ship rolls to the left, the axis of the gyroscope is tilted backwards, thus again opposing the roll. Such kind of arrangement is also used in aeroplanes.

(2) Rifling of barrels of guns and rifles. This is an interesting example of the directional stability of a rapidly rotating body. Spiral grooves are cut in the barrels of guns and rifles, so that before the bullet emerges out, it is forced to move along these grooves, thereby acquiring a spin about an axis along the direction of motion. This gives a much directional stability and ensures its almost uniform linear motion through air.

(3) Riding of Bicycle. When a person rides a bicycle without holding its handle, for taking a turn to one side, he tilts to that side. By doing so, the front wheel, rotating as a gyrostat, experiences couple about the horizontal line of advance; thereby, the axis of the wheel turns about the vertical, and hence the wheel moves to the desired side.

(4) Gyro-compass. The most important application of the gyrostatic effect is in gyrostatic compass. This is used in ships, aeroplanes and submarines to find direction. It consists of a heavy disc, supported freely in frictionless gimbals and driven with the help of an electric motor. It is set revolving with its horizontal axis in the geographical meridian and due to the conservation of its angular momentum and persistence of its direction, the axis continues in the direction of meridian, despite any change in the direction of the supporting frame and hence of the ship. Magnetic disturbances do not affect it and this is an additional advantage over magnetic compass.

(5) Other applications. Automatic pilot, the steering of torpedoes, the artificial horizon and the turn and bank indicators etc. in aeroplanes are comparatively more recent applications of the gyrostatic principles.

Ex. 1. A 12 cm long weightless stick with masses 5 gm and 10 gm attached to its two ends is rotating about an axis passing through its centre of mass and inclined at 30° to its length. If it makes two revolutions per second, find (a) the angular velocity vector; (b) the linear velocity of each mass and its angular momentum vector about the centre of mass; (c) the total angular momentum vector of the system about the centre of mass; (d) the angular momentum about the axis of rotation and (e) the moment of inertia about this axis.

Sol. If the distance of centre of mass from 5 gm mass is x , then

$$0.005 x = 0.01 (0.12 - x) \quad \text{or} \quad x = 0.08 \text{ m}$$

In **fig. 3.31** the centre of mass of the system has been represented by O . About the axis of rotation, A and B masses will describe circles of radius AC and BD respectively. Here, $AC = AO \sin 30^\circ = 0.08 \times \frac{1}{2} = 0.04 \text{ m}$ and $BD = 0.04 \sin 30^\circ = 0.02 \text{ m}$.

(a) Angular velocity $\omega = 2\pi n = 2\pi \times 2 = 12.5 \text{ rad/sec}$ along the axis of rotation.

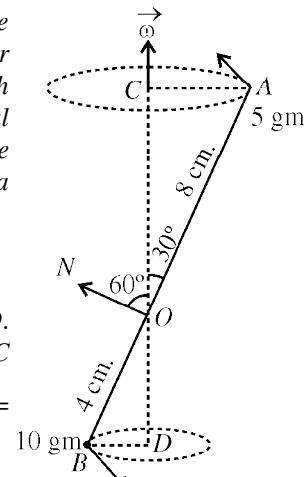


Fig. 3.31.

(b) Linear velocity of 5 gm $v = \omega \times AC = 12.5 \times 0.04 = 0.5$ m/sec.

This velocity is perpendicular to OA , hence the angular momentum of mass A about $O = m \vec{r} \times \vec{v}$.

Its magnitude $= mrv = 0.05 \times OA \times 0.5 = 0.005 \times 0.08 \times 0.5 = 2 \times 10^{-4}$ kg-m²/sec.

Its direction is perpendicular to $\vec{v}$ and $\vec{OA}$, i.e., it is along a line perpendicular to AB , making an angle 60° with the axis of rotation.

Similarly, linear velocity of 10 gm mass $= 12.5 \times 0.02 = 0.25$ m/sec.

Its angular momentum about O has the magnitude $0.01 \times 0.04 \times 0.25 = 10^{-4}$ kg-m²/sec and direction perpendicular to AB , inclined at an angle 60° with the axis of rotation. At any instant both the angular momenta are along the same direction.

(c) Total angular momentum $= 2 \times 10^{-4} + 10^{-4} = 3 \times 10^{-4}$ kg-m²/sec along a direction perpendicular to AB .

(d) The component of total angular momentum along the direction of axis of rotation is the angular momentum about this axis, i.e.,

$$= 3 \times 10^{-4} \cos 60^\circ = 3 \times 10^{-4} \times \frac{1}{2} = 1.5 \times 10^{-4} \text{ kg-m}^2/\text{sec}.$$

(e) Moment of inertia about the axis of rotation

$$= \Sigma mr^2 = 0.005 \times (0.04)^2 + 0.01 \times (0.02)^2 = 1.2 \times 10^{-5} \text{ kg-m}^2.$$

The angular momentum about the axis of rotation (d) can also be expressed as $L = I\omega = 1.2 \times 10^{-5} \times 12.5 = 1.5 \times 10^{-4}$ kg-m²/sec the same result as in (d).

Ex. 2. Two point masses, each of mass m , are connected by a massless rigid rod of length $2a$, forming a dumb-bell. This dumb-bell is rotating with constant angular velocity ω about an axis which makes an angle θ with the rod. Find the magnitudes and the directions of the angular momentum and the torque, applied to the system. How are the results affected, if an identical dumb-bell is symmetrically fixed with the first dumb-bell and the system is rotated with an angular velocity ω about the same axis.

Sol. Let the dumb-bell [fig. 3.32] rotate with an angular velocity $\vec{\omega}$ about the axis AOB in an inertial co-ordinate system. The angular momentum $\vec{L}$ of the two masses is

$$\vec{L} = \vec{L}_1 + \vec{L}_2 = m \vec{r}_1 \times (\vec{\omega} \times \vec{r}_1) + m \vec{r}_2 \times (\vec{\omega} \times \vec{r}_2)$$

As shown in fig. 3.32, $\vec{L}_1$ and $\vec{L}_2$ point in the same direction, but the total angular momentum $\vec{L}$ of the system is not along $\vec{\omega}$. The magnitude of $\vec{L}$ is

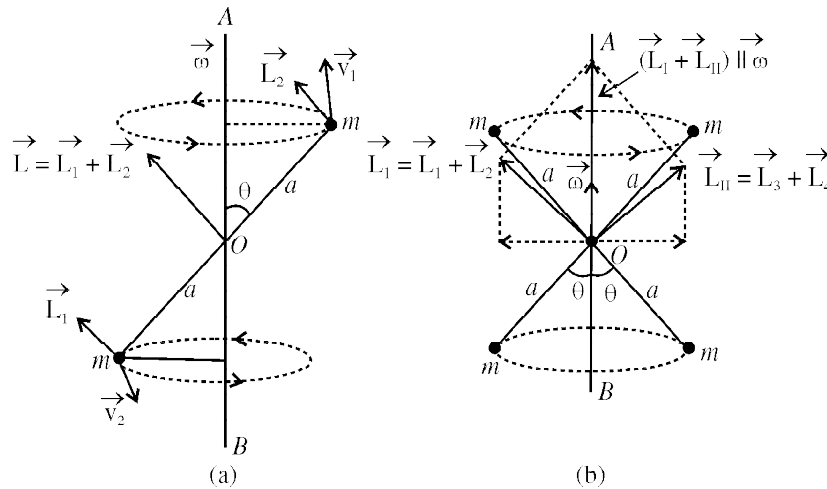


Fig. 3.32.

$$L = ma^2\omega \sin \theta + ma^2\omega \sin \theta = 2ma^2\omega \sin \theta = I\omega \sin \theta$$

where $I = 2ma^2$ is the moment of inertia of the dumb-bell about an axis perpendicular to the length of the connecting rod.

Since $\vec{L}$, which is continuously changing direction, is not constant and hence to maintain the motion, a torque $\vec{\tau}$ is acting on the system. The torque $\vec{\tau}$ is given by

$$\vec{\tau} = \frac{d\vec{L}}{dt}$$

where $\vec{\tau}$ is a vector in the direction in which the tip of the vector $\vec{L}$ is moving. Obviously,

$$\frac{d\vec{L}}{dt} = \vec{\omega} \times \vec{L}$$

in analogy to the relation, $\frac{d\vec{r}}{dt} = \vec{\omega} \times \vec{r}$.

Hence the magnitude of the applied torque is

$$T = \left| \frac{d\vec{L}}{dt} \right| = \omega J \sin (90 - \theta) = \omega J \cos \theta = 2ma^2\omega^2 \sin \theta \cos \theta$$

and its instantaneous direction is perpendicular to the plane of $\vec{\omega}$ and $\vec{L}$. **Fig. 3.32(b)** shows the case when the two identical dumb-bells, given in the problem, are moving symmetrically. In this case the perpendicular components of the angular momenta of the two dumb-bells will cancel and the components parallel to $\vec{\omega}$ add up so that $\vec{L}$ and $\vec{\omega}$ are in the same direction. If there are no resisting forces to the rotating system, the system, once rotated, will rotate indefinitely and in this case there is no need of the external torque.

Ex. 3. Calculate the inertia tensor for the system of four point masses 1 gm, 2 gm, 3 gm and 4 gm, located at the points (1, 0, 0), (1, 1, 0), (1, 1, 1) and (1, 1, -1) cm.

Sol.
$$I_{xx} = \sum_{i=1}^4 m_i (y_i^2 + z_i^2) = 1 \times 0 + 2 \times 1 + 3 \times 2 + 4 \times 2 = 16 \text{ gm-cm}^2$$

Similarly,
$$I_{yy} = \sum_i m_i (x_i^2 + z_i^2) = 17 \text{ gm-cm}^2 \quad \text{and} \quad I_{zz} = \sum_i m_i (x_i^2 + y_i^2) = 19 \text{ gm-cm}^2$$

Also
$$I_{xy} = I_{yx} = -\sum_{i=1}^4 m_i x_i y_i = -[0 + 2 \times 1 \times 1 + 3 \times 1 \times 1 + 4 \times 1 \times 1] = -9 \text{ gm-cm}^2$$

Similarly,
$$I_{xz} = I_{zx} = -\sum_i m_i x_i z_i = 1 \text{ gm-cm}^2 \quad \text{and} \quad I_{yz} = I_{zy} = -\sum_i m_i y_i z_i = 1 \text{ gm-cm}^2$$

Thus the inertia tensor is
$$I = \begin{pmatrix} 16 & -9 & 1 \\ -9 & 17 & 1 \\ 1 & 1 & 19 \end{pmatrix}.$$

Ex. 4. Find the moments of inertia and the products of inertia of a uniform plate of side a and mass M about X , Y and Z axes as shown in **fig. 3.33** and hence write down the moment of inertia matrix.

Sol. Suppose that the surface area density of the material of the plate of side a is σ . Hence if M be the mass of the plate, then

$$\sigma = \frac{M}{a^2} \text{ or } M = \sigma a^2 \quad \dots(i)$$

Consider an element of the plate of length dx and width dy at the point $P(x, y)$ of the plate.

Area of the element = $dx dy$

Mass of the element = $\sigma dx dy$

M.I. of this element about X-axis = $\sigma dx dy y^2$

Hence, the moment of inertia of the full plate about X-axis is given by

$$\begin{aligned} I_{xx} &= \int_{x=0}^a \int_{y=0}^a \sigma dx dy y^2 \\ &= \int_{x=0}^a \sigma dx \left[\frac{y^3}{3} \right]_0^a = \frac{\sigma a^3}{3} \int_0^a dx \\ &= \frac{\sigma a^4}{3} = \frac{1}{3} M a^2 \end{aligned}$$

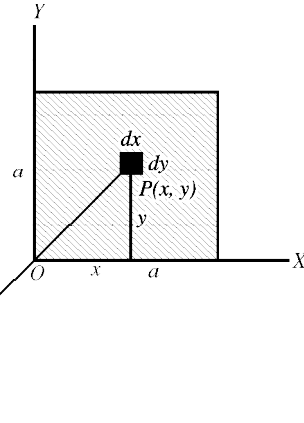


Fig. 3.33.

$$[\because M = \sigma a^2] \dots(ii)$$

Similarly,
$$I_{yy} = \int_{x=0}^a \int_{y=0}^a \sigma dx dy x^2 = \frac{1}{3} M a^2$$

Also,
$$I_{zz} = \int_{x=0}^a \int_{y=0}^a \sigma (x^2 + y^2) dx dy = \frac{1}{3} M a^2 + \frac{1}{3} M a^2 = \frac{2}{3} M a^2 \quad \dots(iii)$$

Product of inertia
$$I_{xy} = I_{yx} = - \int_{x=0}^a \int_{y=0}^a xy \sigma dx dy = - \frac{1}{4} \sigma a^4 = - \frac{1}{4} M a^2$$

As $z = 0$, hence

$$I_{xz} = I_{zx} = I_{yz} = I_{zy} = 0$$

Thus the inertia tensor is

$$I = \begin{pmatrix} \frac{1}{3} M a^2 & -\frac{1}{4} M a^2 & 0 \\ -\frac{1}{4} M a^2 & \frac{1}{3} M a^2 & 0 \\ 0 & 0 & \frac{2}{3} M a^2 \end{pmatrix} = \frac{M a^2}{12} \begin{pmatrix} 4 & -3 & 0 \\ -3 & 4 & 0 \\ 0 & 0 & 8 \end{pmatrix}.$$

Ex. 5. Consider a rectangular parallelepiped of uniform density ρ , mass M with sides a , b and c . For origin O at one corner, find the moments and products of inertia of the parallelepiped by taking the co-ordinate axes along the edges. If $a = b = c$ (case of a cube), determine the inertia tensor.

Sol.

$$\begin{aligned} I_{xx} &= \int \rho (y^2 + z^2) dV \\ &= \int_0^c \int_0^b \int_0^a \rho (y^2 + z^2) dx dy dz \quad (\text{see fig. 3.34}) \\ &= \rho a \int_0^c \int_0^b (y^2 + z^2) dy dz = \rho a \int_0^c \left[\frac{b^3}{3} + z^2 b \right] dz \\ &= \rho a \left[\frac{b^3 c}{3} + \frac{c^3 b}{3} \right] = \frac{\rho a b c}{3} (b^2 + c^2) = \frac{M}{3} (b^2 + c^2), \text{ where } M = \rho a b c. \end{aligned}$$

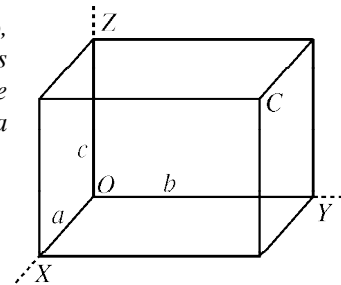


Fig. 3.34.

Similarly, $I_{yy} = \frac{M}{3}(c^2 + a^2)$, $I_{zz} = \frac{M}{3}(a^2 + c^2)$

Also $I_{xy} = -\int_0^c \int_0^b \int_0^a \rho xy dx dy dz = -\rho c \int_0^b \int_0^a xy dx dy = -\rho c \frac{a^2}{2} \frac{b^2}{2} = -\frac{M}{4} ab = I_{yx}$

Similarly, $I_{xz} = I_{zx} = -\frac{1}{4} Mac$ and $I_{yz} = I_{zy} = -\frac{1}{4} Mbc$

For a cube $a = b = c$, then

$$I_{xx} = \frac{2}{3} Ma^2 = I_{yy} = I_{zz}$$

and

$$I_{xy} = -\frac{1}{4} Ma^2 = I_{yx} = I_{xz} = I_{zx} = I_{yz} = I_{zy}$$

Therefore, $I = \frac{Ma^2}{12} \begin{pmatrix} 8 & -3 & -3 \\ -3 & 8 & -3 \\ -3 & -3 & 8 \end{pmatrix}$.

Ex. 6. Consider a homogeneous cube of density ρ , mass M and side a . Taking origin O at one corner and axes along the edges of the cube, determine the inertia tensor, the principal axes and their associated moments of inertia.

Sol. The inertia tensor I for the cube under consideration [fig. 3.35] can be evaluated as above, i.e.,

$$I = \frac{Ma^2}{12} \begin{pmatrix} 8 & -3 & -3 \\ -3 & 8 & -3 \\ -3 & -3 & 8 \end{pmatrix} = \begin{pmatrix} 8\lambda & -3\lambda & -3\lambda \\ -3\lambda & 8\lambda & -3\lambda \\ -3\lambda & -3\lambda & 8\lambda \end{pmatrix} \text{ with } \lambda = \frac{Ma^2}{12}.$$

To find the principal moments of inertia, we solve the secular equation

$$\begin{pmatrix} 8\lambda - I & -3\lambda & -3\lambda \\ -3\lambda & 8\lambda - I & -3\lambda \\ -3\lambda & -3\lambda & 8\lambda - I \end{pmatrix} = 0$$

Subtract the second row from first, we obtain

$$\begin{pmatrix} 11\lambda - I & -(11\lambda - I) & 0 \\ -3\lambda & 8\lambda - I & -3\lambda \\ -3\lambda & -3\lambda & 8\lambda - I \end{pmatrix} = 0$$

or $(11\lambda - I) \begin{pmatrix} 1 & -1 & 0 \\ -3\lambda & 8\lambda - I & -3\lambda \\ -3\lambda & -3\lambda & 8\lambda - I \end{pmatrix} = 0$

or $(11\lambda - I) [(8\lambda - I)^2 - (3\lambda)^2 - 1\{(3\lambda)^2 + 3\lambda(8\lambda - I)\}] = 0$

or $(11\lambda - I) (22\lambda^2 - 13\lambda I + I^2) = 0$

or $(11\lambda - I) (11\lambda - I) (2\lambda - I) = 0$

Therefore, $I_1 = I_2 = 11\lambda$ and $I_3 = 2\lambda$ or $I_1 = I_2 = \frac{11}{12} Ma^2$ and $I_3 = \frac{1}{6} Ma^2$

which are the expressions for the principal moments of inertia.

Since the two roots are identical, $I_1 = I_2$, the principal axis associated with the root I_3 is the axis of symmetry. The moment of inertia tensor, when considered principal axes, is

$$I = \frac{Ma^2}{12} \begin{pmatrix} 11 & 0 & 0 \\ 0 & 11 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

Now we evaluate the directions of the principal axes associated with the three roots. First we

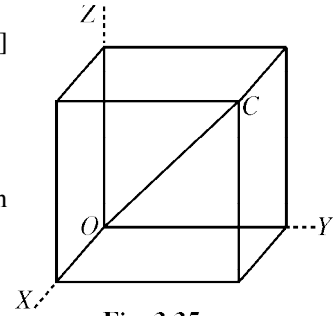


Fig. 3.35.

evaluate the direction of the principal axis associated with the root I_3 . For this, we substitute $I = I_3 = Ma^2/6 = 2\lambda$ in (4) [Art. 3.12]

$$(8\lambda - 2\lambda)\omega_x - 3\lambda\omega_y - 3\lambda\omega_z = 0$$

$$-3\lambda\omega_x + (8\lambda - 2\lambda)\omega_y - 3\lambda\omega_z = 0$$

$$-3\lambda\omega_x - 3\lambda\omega_y + (8\lambda - 2\lambda)\omega_z = 0$$

or $2\omega_x - \omega_y - \omega_z = 0, -\omega_x + 2\omega_y - 2\omega_z = 0, -\omega_x - \omega_y + 2\omega_z = 0$

From which, we obtain

$$\frac{\omega_x}{\omega_y} = 1, \frac{\omega_y}{\omega_z} = 1, \frac{\omega_z}{\omega_x} = 1, \text{ i.e., } \omega_x : \omega_y : \omega_z = 1 : 1 : 1$$

If we take unit vector $\hat{\omega}$ along ω corresponding to $I_3 = Ma^2/6$, then

$$\hat{\omega} = \frac{1}{\sqrt{3}}(\hat{i} + \hat{j} + \hat{k})$$

which is obviously along the diagonal OC of the cube $[\vec{r} = a(\hat{i} + \hat{j} + \hat{k})]$. As $I_1 = I_2$, axes associated with I_1 and I_2 have any mutually perpendicular directions in a plane perpendicular to this diagonal.

Ex. 7. A top is made by passing a light and small pin through the centre of a uniform thin disc of mass 20 gm and radius 2 cm. Find the precession velocity for such a top rotating at 30 rad/sec. Given moment of inertia of a disc about its diameter and perpendicular to the plane are $(MR^2/4)$ and $(MR^2/2)$ respectively.

Sol. Given $I_1 = I_2 = MR^2/4$ and $I_3 = MR^2/2$

Angular velocity of precession,

$$\Omega = \frac{I_3 - I_1}{I_1} \omega = \frac{(MR^2/2) - (MR^2/4)}{(MR^2/4)} \times 30 = 30 \text{ rad/sec.}$$

Ex. 8. A top of mass 1 kg and rotational inertia $5 \times 10^{-4} \text{ kg-m}^2$ is spinning at 30 rotations per second about an axis making an angle of 30° with the vertical. Its centre of mass is 4 cm from the origin. Find its angular speed of precession. (A.N.U. 2007)

Sol. Precessional angular speed of a spinning top is given by

$$\Omega = \frac{\tau}{L} = \frac{Mgl}{I\omega}$$

Here, $M = 1 \text{ kg}$, $g = 9.8 \text{ m/sec}^2$, $l = 4 \text{ cm} = 0.04 \text{ m}$, $I = 5 \times 10^{-4} \text{ kg-m}^2$ and

$$\omega = 2\pi v = 2 \times 3.14 \times 30 = 188.4 \text{ rad/sec}^2.$$

$$\therefore \Omega = \frac{1 \times 9.8 \times 0.04}{5 \times 10^{-4} \times 188.4} = 4.16 \text{ rad/sec.}$$

Ex. 9. A top is spinning 30 rev/sec about an axis making an angle of 30° with the vertical. Its mass is 0.5 kg and its rotational inertia is $5 \times 10^{-4} \text{ kg-m}^2$. The centre of mass is 4 cm from the pivot point. If the spin is clockwise as seen from the above, what is the magnitude and direction of the angular velocity of precession? (A.N.U. 1994)

Sol. The angular velocity of precession is given by

$$\Omega = \frac{Mgl}{I\omega}$$

Here, $r = 0.04 \text{ m}$, $m = 0.5 \text{ kg}$, $l = 5 \times 10^{-4} \text{ kg-m}^2$ and $\omega = 2\pi v = 2\pi \times 30 = 188.4 \text{ rad/sec}$.

$$\therefore \Omega = \frac{0.5 \times 9.8 \times 0.04}{5 \times 10^{-4} \times 188.4} = 2.08 \text{ rad/sec.}$$

From vector relation $\vec{\Omega} \times \vec{L} = \vec{l} \times M \vec{g}$, the direction of angular velocity of precession is **anticlockwise** as seen from above.

Ex. 10. A top is made up of a thin disc of radius 0.12 m and mass 0.2 kg. It is pierced in the centre and a pin of negligible mass is mounted normal to its plane. The pivot under the disc is 0.03 m long. The top is made to spin with its axis making an angle $\theta = 20^\circ$ with the vertical and has a precessional speed of 2 rad per second. Find the angular speed with which the top spins. **(A.U. 1990)**

Sol. Precessional angular velocity of a top is given by

$$\Omega = \frac{Mgl}{I\omega}, \therefore \omega = \frac{Mgl}{I\Omega}$$

where ω is the angular speed of spinning top.

Here, l = distance of the centre of mass from the pivot (origin) = 0.03 m,

Mg = weight of the top = 0.2×9.8 N,

$$I = \text{M.I. of top} = \text{M.I. of disc} = \frac{1}{2}mR^2 = \frac{1}{2} \times 0.2 \times (0.12)^2 = 1.44 \times 10^{-3} \text{ kg-m}^2$$

and $\Omega = 2$ rad/sec.

$$\text{Therefore, } \omega = \frac{0.2 \times 9.8 \times 0.03}{1.44 \times 10^{-3} \times 2} = \mathbf{20.4 \text{ rad/sec.}}$$

11. A gyroscope wheel weighing 1.0 kg and having a radius of gyration 30 cm, is spinning about its axis at a rate of 240 rev./min. The axis is held horizontal and is supported by a point at one end which is 10 cm. from the centre of mass of the wheel. Calculate the angular velocity of the precession of the wheel.

Sol. Angular velocity of precession of a gyroscope, shown in **fig. 3.36**, is given by

$$\Omega = \frac{Mgl}{I\omega}$$

Here, $M = 1$ kg, $l = 10$ cm = 0.1 m, $I = Mk^2 = 1.0 \times (0.3)^2$

$$= 0.09 \text{ kg-m}^2 \text{ and } \omega = 240 \text{ rev/min} = 2\pi \times \frac{240}{60} = 8\pi = 8 \times 3.14 \\ = 25.1 \text{ rad/sec.}$$

$$\therefore \Omega = \frac{1 \times 9.8 \times 0.1}{0.09 \times 25.12} = \mathbf{0.433 \text{ rad/sec.}}$$

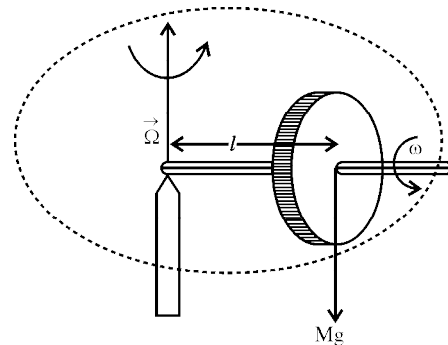


Fig. 3.36.

QUESTIONS

Long Answer Questions

- What is a rigid body ? Deduce the three equations of motion of rotating rigid body about an axis.
- Distinguish between pure translation and pure rotational motion. Deduce the equations of motion for a rotating body. **(S.V.U. 2003)**
- (a) Distinguish between inertia and moment of inertia of a rigid body. Obtain the relation $\tau = I\alpha$.
(b) In the case of a rotating rigid body when the torque τ ceases to act on the body, what can you conclude about the angular momentum of the body ? **(O.U. 2003)**
- Define torque and angular momentum. Obtain the relation between them. Obtain an expression for the total kinetic energy of a rolling body on a plane surface as a sum of translational and rotational kinetic energies. **(S.K.U. 1990)**

5. Show that $\tau = I\alpha$ for a rigid body undergoing rotational motion. τ is the torque acting on the body, I its moment of inertia about the axis of rotation and α the angular acceleration. (K.U. 1991)
6. (a) What is rotational inertia ? Derive an expression for the kinetic energy of rotation of a rigid body.
(b) What is the effect of 'g' of the rotation of the earth ? (O.U. 2001)
[Hint : Because on the rotation of the earth (with angular velocity ω), centrifugal force acts on a body on its surface, reducing its weight (mg). As the mass of the body is constant, g will decrease. The value of g decreases, when we move from N or S pole towards equator due to the rotation of the earth.]
7. What is meant by rigid body ? Obtain an equation of motion for a rigid body rotating about the axis of symmetry. (N.U. 2001)
8. Distinguish between pure translational and pure rotational motion. Deduce the equation of motion for a rotating body. Obtain an expression for the speed of centre of mass of rolling body from an inclined plane. (K.U. 2005)
9. Distinguish between pure translational and pure rotational motion. Deduce the equation of motion for a rotating body. Obtain an expression for the speed of centre of mass of a rolling body from an inclined plane. (K.U. 2005)
10. Find expressions for the final velocity and acceleration of the centre of mass of a body rolling down an inclined plane without slipping. (K.U. 1988; N.U. 1990)
11. A rigid body is rotating about an axis through the origin. Deduce relations connecting the components of total angular momentum with the components of the angular velocity.
12. What is inertia tensor ? What are moments of inertia and products of inertia ? Prove that the inertia tensor is symmetric.
13. What do you mean by principal axes and principal moments of inertia coefficients ?
14. (a) Obtain Euler's equations of motion for a rotating rigid body. (O.U. 2005)
(b) Using these equations, explain conservation of energy and angular momentum. (O.U. 2005)
15. Show that the angular momentum $\vec{L}$ of a rotating rigid body is given by $\vec{L} = \mathbf{I} \vec{\omega}$, where $\vec{\omega}$ is the angular velocity. Show that $\mathbf{I}$ is the tensor of second rank.
16. Show that the kinetic energy of a rotating rigid body can be expressed as $K = \frac{1}{2} \vec{L} \cdot \vec{\omega}$. Hence show that if I is the moment of inertia about the axis of rotation, prove that the kinetic energy can be expressed as $K = \frac{1}{2} I \omega^2$.
17. Show that the kinetic energy of a rotating rigid body in a co-ordinate system of principal axes is given by $K = \frac{1}{2} (I_1 \omega_1^2 + I_2 \omega_2^2 + I_3 \omega_3^2)$
18. Deduce the fundamental equation of motion of a rigid body about fixed axis and show that it is rotational analogue to $\vec{F} = d\vec{P}/dt$ in the case of translation motion.
19. Mention the important properties of inertia tensor. Derive the relationship between angular momentum and inertia. (A.U. 2007)
20. (a) What is rigid body ? Derive the Euler's equations of rotational motion for a rigid body fixed at one end. (O.U. 2007)
(b) Using Euler's equations prove the principle of conservation of energy. (O.U. 2007)
21. Derive Euler's equations of a rotating rigid body about a fixed point. (O.U. 2009, S.V.U. 2003; A.N.U. 2005)
22. Derive Euler's equation of rotational motion for a rigid body fixed at one end. Deduce the law of conservation of energy from Euler's equations. (A.N.U. 2007)
23. Obtain Euler's equation of rotational motions for a rigid body and apply them to verify the conservation laws of energy and angular momentum. (S.V.U. 2005)
24. Define the terms rigid body, moment of inertia and symmetric top. Explain the precession of a top. (S.K.U. 2003; N.U. 2002)
25. Discuss the torque free motion of a symmetric top. (N.U. 2003)

26. Define torque and angular momentum. Explain the law of conservation of angular momentum with examples. Write a short note on gyroscope. (A.U. 2001)
27. What is a symmetric top ? Calculate the precessional velocity of a top and arrive at the relation $\vec{\tau} = \vec{\omega}_p \times \vec{L}$. (O.U. 2002)
28. What is precessional motion ? How does the angular momentum of a spinning top vary as it precesses about the vertical ? (A.N.U. 2006)
29. Derive Euler's equations of motion for a rigid body. Explain the working of a gyroscope. (A.U. 2007)
30. What is precessional motion ? Find angular velocity of precession of a spinning top. Show that the rate of precession is independent of mass but depends only on the distribution of mass. (S.K.U. 2007)
31. What is symmetric top and precessional motion ? Calculate the precessional velocity of the top. (A.N.U. 2008, 07)
32. What is gyroscope ? Briefly explain the working. (A.U.V. 2007)
33. Describe a gyroscope. Explain what is meant by nutation. Give its applications. (S.V.U. 2007)
34. What is a gyroscope ? Explain the working of gyroscope. Explain briefly about precession of equinoxes. (S.K.U. 2005)

Short Answer Questions

1. Define a rigid body. What do you understand by the moment of inertia of rigid body rotating about an axis. Give its physical significance. (A.U. 2007)
2. Obtain the equation of motion for a rotating body. (S.V.U. 2007; A.U. 2005)
3. What is the significance of moment of inertia ? Express angular momentum and torque in term of moment of inertia. (S.V.U. 2002)
4. Define torque and deduce the equation of motion for a body rotating about the axis of symmetry. (S.K.U. 1993)
5. Show that $\tau = I\alpha$ for a rigid body undergoing rotational motion. (K.U. 1991)
6. Obtain an expression for the kinetic energy of a rotating rigid body. (O.U. 2009)
7. Explain about moment of inertia tensor and mention some of its properties. (K.U. 1997)
8. A body of mass m slides down on a smooth inclined plane, having angle θ with the horizontal, and reaches the bottom with a speed v . If the same mass were in the form of a ring which rolls down on an inclined plane (having the same angle θ with the horizontal), what will be the speed of the ring at the bottom ?
[Ans. $v/\sqrt{2}$.]
9. If two circular discs of the same mass and thickness are made from metals having different densities, which disc will have smaller moment of inertia about its central axis ?
[Ans. The disc of more density]
10. A ballet-dancer stretches her hands out for slowing down. On what principle does it depend ?
11. Write Euler's equations for a rigid rotating body and explain the symbols. (S.K.U. 1997, 95)
12. State and prove Euler's equation. (A.N.U. 2003, 2008, 2001)
13. Deduce the fundamental equations for translation and rotation. (A.N.U. 2003)
14. Using Euler's equations verify the law of conservation of angular momentum. (A.N.U. 2007)
15. What is precessional motion ? Obtain an expression for the precessional angular velocity of a spinning top. (O.U. 2002)
16. Show that the precession becomes faster as the spin of the top slows down. (O.U. 2002)
17. What is precessional motion ? How does the angular momentum of a spinning top vary as it precesses about the vertical. (A.N.U. 2006)
18. Derive expression for rate of precession of symmetrical top. (A.N.U. 2005)
19. Derive an expression for the precessional velocity of symmetrical top. (S.K.U. 2007)

20. What is precessional motion ? How does the angular momentum of a spinning top vary on it precesses about the vertical ? (A.N.U. 2008)
21. Explain the principle of gyroscope. (A.U. 2007; A.N.U. 2004)
22. Explain the working of a gyroscope. (O.U. 2008; A.U. 2005)
23. Explain the working of gyroscope and its applications. (A.N.U. 1997)
24. What is a gyroscope ? What are the uses of it ? (A.N.U. 2001)

PROBLEMS

[Rotational Motion about a Fixed Axis]

1. Find the radius of gyration of a disc of mass 100 gm and radius 5 cm about an axis passing through its centre of mass and perpendicular to its plane.
[Ans. 0.035 m.]
2. A uniform solid sphere of mass 1 kg and radius 0.2 metre makes 4 revolutions per second about a diameter. How much torque is required to increase its angular momentum about this axis by 10% in 20 seconds ?
[Ans. 0.002 N-m.]
3. A uniform thin bar of mass 3 kg and length 0.9 metre is bent to make an equilateral triangle. Calculate the moment of inertia about an axis passing through the centre of mass and perpendicular to the plane of the triangle.
[Ans. 0.045 kg-m².]
4. Show that the radius of gyration of a circular plate of radius a about an axis perpendicular to its plane through the circumference is given by $k = a\sqrt{3}/2$.
5. Obtain expressions for rotational kinetic energy of a rigid body in terms of (i) angular momentum and angular velocity, (ii) angular velocity and moment of inertia.
[Ans. K.E. = $\frac{1}{2} L\omega = \frac{1}{2} I\omega^2$.]
6. If the diameter of earth suddenly contracts to its half, how is the length of a day affected ?
[Ans. 6 hrs]
[Hint : From the law of conservation of angular momentum, $I_1\omega_1 = I_2\omega_2$, or $I_1(2\pi/T_1) = I_2(2\pi/T_2)$.
Hence $T_2 = T_1 \left(\frac{I_2}{I_1} \right)$, where $I_1 = \frac{2}{5} MR_1^2$ and $I_2 = \frac{2}{5} MR_2^2$ (M = mass of earth)
As $R_2 = \frac{1}{2} R_1$, $\therefore \frac{I_2}{I_1} = \left(\frac{R_2}{R_1} \right)^2 = \frac{1}{4}$ and $T_2 = T_1 \times \frac{1}{4} = \frac{24}{4} = 6$ hrs. ($\because T_1 = 24$ hrs.)]
7. A sphere of mass 500 gm and diameter 4 cm rolls without slipping with a velocity of 10 cm/sec. Calculate its total energy.
[Ans. 3.5×10^{-3} joules.]
8. Calculate the angular momentum of a body whose rotational kinetic energy is 10 joules, if the angular momentum vector coincides with the axis of rotation and its moment of inertia about this axis is 8 gm – cm².
[Ans. 4×10^{-3} kg-m²/sec.]
9. A metre stick is held vertically with one end on the ground and is then allowed to fall. Calculate the speed of the other end when it touches the ground, assuming that the end on the ground does not slip.
[Ans. 5.42 m/sec.]
[Hint : $\frac{1}{2} I\omega^2 = Mg \times 0.5$.]

10. A solid sphere of mass 50 gm and diameter 2 cm rolls without slipping with a uniform velocity of 5 cm per second. Calculate its total energy. (O.U. 1990)

[Ans. 8.75×10^{-5} joule.]

11. A thin uniform rod of length l is held vertically with one end on the floor. Find the angular velocity of the other end when the rod falls down. Assume that the end of the rod on the floor does not slip.

[Ans. $\omega = \sqrt{3g/l}$.] (K.U. 1990)

[Hint. $Mg \frac{l}{2} = \frac{1}{2} I \omega^2, I = \frac{Ml^2}{3}$.]

12. A sphere of mass 100 gm and diameter 0.01 m rotates without slipping with a velocity of 0.05 m/sec. Calculate its K.E. of rotation. (A.N.U. 1996)

[Ans. 5×10^{-5} joule.]

13. A disc of mass 2 kg rolls without slipping over a horizontal plane with a velocity 4 m/sec. Find the K.E. of the disc. (A.U. 1990)

[Ans. 24 joule]

14. A flywheel when slowed down from 60 rpm to 30 rpm loses 100 J of energy. What is its moment of inertia ? What is the change in the angular momentum ? (A.U. 1990)

[Ans. $(200/3\pi^2) \text{ kg-m}^2$ and $(200/3\pi) \text{ kg-m}^2/\text{sec}$.]

15. Determine moment of inertia of a uniform disc of mass 0.2 kg and radius 0.05 m about an axis passing through its diameter. (A.U. 2001)

[Ans. $1.25 \times 10^{-4} \text{ kg-m}^2$]

16. An automobile develops 100 HP power when rotating at a speed of 1000 rpm. Calculate the torque acting. [1 H.P. = 746 watt] (O.U. 2007)

[Ans. 712.7 N-m.]

[Hints : $\tau = \frac{P}{\omega} = \frac{100 \times 746}{2\pi(1000/60)} = 712.7 \text{ N-m}$.]

17. Calculate the angular momentum of the spherical earth rotating about its own axis. (Mass of the earth = 6×10^{24} kg and mean radius = 6.4×10^6 m) (A.N.U. 2002)

[Ans. $7.13 \times 10^{33} \text{ kg-m}^2/\text{sec}$.]

18. A circular disc of mass 100 kg and radius 50 cm is mounted axially and is rotating. Calculate the K.E. when it executes 25 rev/min. (S.K.U. 1995)

[Ans. 3.85 joule.]

19. A solid sphere of mass 100 gm and radius 2.5 cm rolls without sliding with a uniform velocity of 10 cm per second along a straight line on a smooth horizontal table. Calculate its total energy.

[Ans. 7×10^{-4} joule.]

20. The kinetic energy of a body rotating at a constant speed of 300 rev/min is 100 joule. Find its angular momentum.

[Ans. $6.368 \text{ kg-m}^2/\text{sec}$.]

21. A circular disc of 50 kg and radius 100 cm is mounted axially and made to rotate. Calculate the K.E. it possesses when executing 100 revolutions per minute. (A.U. 1989)

[Ans. 1369.37 joule.]

22. A symmetrical body is rotating about its axis of symmetry, its moment of inertia about the axis of rotation being 1 kg-m^2 and its rate of revolution is 2 rev/sec. What is its angular momentum ?

[Ans. $12.57 \text{ kg-m}^2/\text{sec}$.]

23. Calculate the angular momentum of a body whose rotational kinetic energy is 80 joules if the angular momentum vector coincides with the axis of rotation and its moment of inertia is $8 \times 10^{-7} \text{ kg-m}^2$.
(O.U. 1988)

[Ans. $1.13 \times 10^{-2} \text{ kg-m}^2/\text{sec}$.]

24. A flywheel of mass 500 kg and diameter 1 metre revolves about an axle. Its frequency of revolution is increased by 18 in 5 seconds. Calculate the torque applied. (Assume fly wheel in form of a disc.)

[Ans. 1414 N – m.]

25. A solid spherical ball rolls down a plane inclined at 30° to the horizontal. Calculate the distance travelled in 4 seconds after starting from rest. ($g = 9.8 \text{ m/sec}^2$.)

[Ans. 28 metres.]

26. Calculate the ratio of rotational to translational kinetic energy for a solid cylinder, rolling down a plane without slipping.

[Ans. 1 : 2.]

[Inertia Tensor and Precession]

27. Two particles of masses 2 gm and 4 gm are attached to the two ends of a 6 cm long stick of negligible weight. Calculate the moment of inertia of the system about an axis passing through its centre of mass and perpendicular to the stick. If the stick makes two revolutions per second about this axis, find the linear velocity vector for each particle and hence find the magnitude and direction of the total angular momentum about the centre of mass. Is it in the same direction as that of the angular velocity ?

[Ans. $4.8 \times 10^{-6} \text{ kg-m}^2$, 0.25 and 0.5 m/sec; $6 \times 10^{-5} \text{ kg-m}^2/\text{sec}$ in the direction of angular velocity.]

28. (a) Consider a homogeneous cube of density ρ , mass M and side a . For origin O at one corner and axes along the edges of the cube, determine the inertia tensor..

$$\left[\text{Ans. } I = \frac{Ma^2}{12} \begin{pmatrix} 8 & -3 & -3 \\ -3 & 8 & -1 \\ -3 & -3 & 8 \end{pmatrix} \right]$$

- (b) Determine the inertia tensor, if the origin O is the centre of mass in the above problem and the axes are parallel to the edges. Could you correlate the inertia tensors in the two cases ?

$$\left[\text{Ans. } I = \frac{Ma^2}{6} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}; \text{yes.} \right]$$

$$[\text{Hint : } I_{xx} = I_{xx}^c + M(Y^2 + Z^2) = \frac{Ma^2}{6} + M\left[\left(\frac{a}{2}\right)^2 + \left(\frac{a}{2}\right)^2\right] = \frac{Ma^2}{6} + \frac{Ma^2}{2} = \frac{2}{3}Ma^2]$$

$$\text{and } I_{xy} = I_{xy}^c - MXY = \frac{Ma^2}{4}.]$$

29. Prove that the principal moments of inertia for a system consisting of two particles of masses m_1 and m_2 connected by a massless rigid rod of length a are $I_1 = I_2 = m_1m_2a^2/(m_1 + m_2)$ and $I_3 = 0$.

30. Four point masses, each equal to m are placed at $(a, 0, 0)$, $(0, a, 0)$, $(0, 0, a)$ and (a, a, a) . Find the inertia tensor for this system.

$$[\text{Ans. } I = ma^2 \begin{pmatrix} 4 & -1 & -1 \\ -1 & 4 & -1 \\ -1 & -1 & 4 \end{pmatrix}.]$$

31. Find the inertia tensor for a configuration of four points each of mass m located at $(a, 0, 0)$, $(0, b, 0)$, $(0, 0, c)$ and (a, b, c) .

$$[\text{Ans. } I = m \begin{bmatrix} 2(b^2 + c^2) & -ab & -ca \\ -ab & 2(c^2 + a^2) & -bc \\ -ca & -bc & 2(a^2 + b^2) \end{bmatrix}]$$

32. A rigid body consists of three particles of masses 2, 1, 4 gm located at (1, -1, 1), (2, 0, 2), (-1, 1, 0) cm respectively. Find the angular momentum of the body if it is rotated about the origin with angular velocity $\vec{\omega} = 3\hat{i} - 2\hat{j} + 4\hat{k}$. Find also the principal moments of inertia and directions of the principal axes of the system.

[Ans. $\vec{L} = -6\hat{j} + 42\hat{k}$; $I_1 = 18$, $I_2 = 13 - \sqrt{73}$, $I_3 = 13 + \sqrt{73}$ gm-cm²; $\hat{j} + \hat{k}$, $\frac{1}{6}(1+\sqrt{73})\hat{i} - \hat{j} + \hat{k}$, $\frac{1}{6}(1-\sqrt{73})\hat{i} - \hat{j} + \hat{k}$.]

33. (a) Find the moments of inertia and products of inertia of a uniform square plate of side a and mass M about the X , Y , Z axes, shown as in **fig 3.37**.

- (b) In the above problem, find also the principal moments of inertia and the directions of the principal axes for the plate.

[Ans. (a) $I_{xx} = I_{yy} = \frac{1}{3}Ma^2$, $I_{zz} = \frac{1}{12}Ma^2$, $I_{xy} = I_{yx} = -\frac{1}{4}Ma^2$,

$$I_{xz} = I_{zx} = I_{yz} = I_{zy} = 0.$$

(b) $I_1 = \frac{1}{12}Ma^2$, $I_2 = \frac{7}{2}Ma^2$, $I_3 = \frac{2}{3}Ma^2$; $\hat{i} + \hat{j}$, $\hat{i} - \hat{j}$, $\hat{k}$.]

34. Find the principal moments of inertia at the centre of a uniform rectangular plate of sides a and b .

[Ans. $I_1 = \frac{1}{12}Ma^2$, $I_2 = \frac{1}{12}Mb^2$, $I_3 = \frac{1}{12}M(a^2 + b^2)$.]

35. Find the moments of inertia and products of inertia of the uniform solid sphere $x^2 + y^2 + z^2 = a^2$ in the first octant.

[Ans. $I_{xx} = I_{yy} = I_{zz} = \frac{2}{5}Ma^2$, $I_{xy} = I_{yz} = I_{zx} = -2Ma^2/5\pi$]

36. Determine the principal moments of inertia of a uniform cylinder of radius a and height h .

[Ans. $I_1 = I_2 = \frac{1}{12}M(3a^2 + h^2)$, $I_3 = \frac{1}{2}Ma^2$.]

37. If T be the kinetic energy, $\vec{G}$ be the external torque about the instantaneous axis of rotation and $\vec{\omega}$ the angular velocity, then prove that $\frac{dT}{dt} = \vec{G} \cdot \vec{\omega}$.

[Hint : According to Euler's equations,

$$G_1 = I_1\dot{\omega}_1 + (I_3 - I_2)\omega_2\omega_3 \text{ or } I_1\dot{\omega}_1 = G_1 + (I_2 - I_3)\omega_2\omega_3 \text{ etc.}$$

$$T = \frac{1}{2}I_1\omega_1^2 + \frac{1}{2}I_2\omega_2^2 + \frac{1}{2}I_3\omega_3^2$$

$$\therefore \frac{dT}{dt} = \omega_1(I_1\dot{\omega}_1) + \omega_2(I_2\dot{\omega}_2) + \omega_3(I_3\dot{\omega}_3)$$

$$= \omega_1[G_1 + (I_2 - I_3)\omega_2\omega_3] + \omega_2[G_2 + (I_3 - I_1)\omega_2\omega_3] + \omega_3[G_3 + (I_1 - I_2)\omega_2\omega_3]$$

$$= \omega_1G_1 + \omega_2G_2 + \omega_3G_3 = \vec{G} \cdot \vec{\omega}.]$$

38. From Euler's equations of motion for a rigid body, having no external torque about a fixed point, prove that

$$T = \frac{1}{2}I_1\omega_1^2 + \frac{1}{2}I_2\omega_2^2 + \frac{1}{2}I_3\omega_3^2 = \text{constant and } \vec{L} = I_1\omega_1\hat{i} + I_2\omega_2\hat{j} + I_3\omega_3\hat{k} = \text{constant}$$

where the terms have standard meaning.

39. A gyroscope wheel of mass 2 kg and radius of gyration 0.30 m is spinning about its axle at a rate of 240 revolutions per minute. The axle is held horizontal and is supported by a pivot at one end

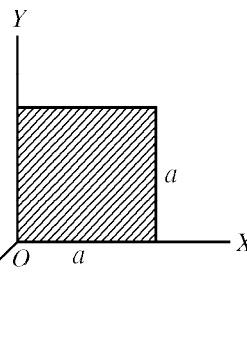


Fig. 3.37.

which is 0.10 m from the centre of mass of the wheel. Calculate the precessional speed of the wheel.

[Ans. 0.43 rad/sec.]

40. A right circular cone of mass 2 kg is pivoted at the apex. The cone is rotating about its longitudinal axis with frequency of 80 rev/sec. Find the precessional frequency if the diameter of the base is 10 cm, the height is 40 cm and the angle of the cone is θ . Assume that the moment of inertia of the cone is $(3/10)MR^2$ and its centre of mass is $(h/4)$ from the base.

[Ans. 7.8 rad/sec.]

[Hints : $\Omega = \frac{Mgl}{I\omega}$; Here, $l = h - h/4 = 3h/4$

$$\Omega = \frac{Mg(3/4)h}{(3/10)MR^2\omega} = \frac{10gh}{4R^2\omega} = 10 \times 9.8 \times 0.4 \times (0.05)^2 \times 2\pi \times 80 = 7.8 \text{ rad/sec.}]$$

Mechanics of Continuous Media

4.1. Elasticity and Isotropic Materials

A body is said to be **rigid**, if the relative positions of its constituent particles remains unchanged, when external forces act upon it. Actually, no body is perfectly rigid and every body can be deformed more or less, by the suitable application of the forces. These applied forces are called **deforming forces**. When there are **small deformations** in a body due to the applied forces, it regains its original shape or size after the removal of the forces. This property of a material body is called **elasticity** i.e., **Elasticity** is that property of matter by virtue of which a body regains the original shape or size on the removal of the deforming forces.

Elastic limit : If the deforming forces are applied on a body upto a certain limit, and the body regains its original shape or size after the removal of the forces, then such a limit of elasticity is called **elastic limit**.

For example, if an iron wire is clamped at one end in the roof and by attaching a hanger on the other end, the wire is loaded, it will extend in length due to the applied weight (force). If the applied weight is not much, the wire will acquire its original length on the removal of the weight. However if the wire is loaded so much that the elastic limit is crossed, then wire will not regain its original length and in fact, it will be permanently extended in length or it will break. Different materials have different limits of elasticity.

It must be noted here that the deformation in the shape or size of a body is the result of equal and opposite forces or couples in equilibrium.

Perfectly elastic and plastic bodies : If the body recovers completely its original condition as soon as the deforming forces are removed, it is said to be **perfectly elastic**, while if it completely retains its modified form, it is said to be **perfectly plastic**. There are, however, no perfectly elastic or plastic bodies and actual bodies lie between the two extremes. The nearest approach to a perfectly elastic body is a quartz fibre and to a perfectly plastic body is putty.

Homogeneous and Isotropic Materials

In this chapter, we plan to study the elastic property of the **homogeneous** and **isotropic** materials. A material is said to be **homogeneous and isotropic**, if its composition is uniform and its properties are the same at all points and in all directions inside it. In these materials, not only elastic property is the same in all directions, but other properties like thermal expansion, heat conduction, refraction etc. are also identical. Nearly all the fluids i.e., liquids and gases are the examples of homogeneous and isotropic materials. Solid metals, like copper, iron, gold etc. in the form of wire or body are considered to be isotropic in view of elasticity.

Solids, for example wood and crystals, are not isotropic. Such materials with different properties in different directions are called **non-isotropic** or **anisotropic**. For example, in cubic crystals, the elastic property is anisotropic.

4.2. Stress, Strain and Hooke's Law

Stress : When a system of external forces act on a body, relative displacements of its various particles take place and consequently the body is deformed. For such deformation, the body offers resistance and internal forces of reaction are developed inside the body. These internal forces, so developed,

are equal in magnitude and opposite to the deforming forces and equilibrium is established. The forces of internal reaction tend to restore the body to its original form. On account of the restoring forces, the body recovers its shape or size, when the deforming forces are removed.

*The restoring force per unit area, set up inside the body, is called **stress** and it is measured by the magnitude of the deforming force acting on unit area of the body within the elastic limit. Thus, if a force F is acting on a area, A , then*

$$\text{stress} = \frac{F}{A}$$

Since stress is the force per unit area, its dimensional formula is $ML^{-1}T^{-2}$ and S.I. unit is N/m^2 . There are two types of stress :

(1) Normal stress : If the stress is normal to the surface, it is called the **normal stress**. The stress is always normal in the case of a change in length of the wire or in the case of the change in the volume of a body. When there is the change in length, the stress is said to be **tensile**. The normal stress is of two types : compressive and expansive (i.e., tensile) accordingly as a decrease or increase in volume takes place.

(2) Tangential or Shearing stress : When the stress is tangential to the surface, it is called a **shearing** or **tangential stress**. Such a stress is produced, when one face of a body is kept fixed and a tangential deforming force is applied on the other face. If the force producing the stress is inclined at an angle normal to the surface, the stress will have the **normal** and **tangential** components.

Strain : When a system of forces or couples acting upon a body causes relative displacements of its various parts, a change in length, volume or shape is produced. The body is then said to be **strained**.

*The change produced in the dimension of a body by a system of forces or couples, in equilibrium, is called **strain** and it is measured by the ratio of change in length to the original length (**longitudinal strain**) or change in volume to the original volume (**volume strain**). If there is a change in shape, the strain, called as **shearing strain**, is measured by the angle through which a line originally perpendicular to the fixed face is turned.*

There are three types of strain :

(1) Longitudinal strain : *If the length of a wire is changed due to deforming forces, then the change per unit length is called **longitudinal strain**.* Let L be the original length of a wire and l be the change in its length due to applied forces. Then,

$$\text{longitudinal strain, } \alpha = \frac{l}{L}$$

(2) Volume strain : *If the volume of a material is changed by the deforming forces, then the change in per unit volume is called **volume strain**.* Let V be original volume of a material and v be the change in volume by the application of deforming forces, then

$$\text{volume strain, } \gamma = \frac{v}{V}$$

(3) Shearing strain : *If one face of a body is fixed and the deforming force on another face is so applied that its shape changes, then the strain is called **shearing strain** and it is measured by the angle (θ) in between a line perpendicular to the fixed face in the initial position and final position of the line on the application of deforming force.*

Thus, the strain is just a ratio or an angle, it is a dimensionless quantity and has no units.

Hooke's Law

This is the basic law of elasticity and was established by Robert Hooke in 1678. This law states that within elastic limit, the stress is proportional to the corresponding strain, i.e.,

$$\text{Stress} \propto \text{Strain} \text{ or } \text{Stress} = E \times \text{Strain i.e., } E = \frac{\text{Stress}}{\text{Strain}}$$

The constant of proportionality E is called **modulus of elasticity**. Its value is independent of the magnitude of the stress or the strain and depends only on the nature of the material and its other conditions. Modulus of elasticity has the same dimensional formula as that of stress, i.e., $ML^{-1}T^{-2}$, since strain is a dimensionless quantity.

Corresponding to three types of strain, we have three moduli of elasticity, namely, (i) **Young's modulus**, corresponding to longitudinal strain, (ii) **Bulk modulus**, corresponding to volume strain, and (iii) **Modulus of rigidity**, corresponding to shearing strain.

4.3. Elastic Constants of Isotropic Solids

(1) Young's modulus : Within the elastic limit, the ratio of longitudinal stress to the corresponding strain, produced, is called **Young's modulus** for the material of a body and is denoted by the letter ' Y '.

Let L be the original length of a wire of cross-section A and it elongates l by the application of a force F . [fig. 4.1] Then in equilibrium,

$$\text{stress} = F/A \text{ and strain} = l/L$$

$$\therefore \text{Young's modulus, } Y = \frac{F/A}{l/L} = \frac{F \cdot L}{A \cdot l} \quad \dots(1)$$

If the radius of a wire is r , then area of cross section of the wire is $A = \pi r^2$. Now, if M kg mass is suspended at the lower end of the wire, then $F = Mg$ and

$$Y = \frac{MgL}{\pi r^2 l} \quad \dots(2)$$

We see that by the application of more force (Mg), if the change in length (l) of the wire is less, the value of Young's modulus (Y) of the material of the wire will be more.

Steel (or iron) is more elastic than rubber : When same force is applied on a steel wire and a rubber piece, rubber is extended much more than steel and ordinarily it is said that rubber is more elastic than steel. However it is not true in the language of physics. If we consider two wires of steel and rubber of the same length (L) and same radius (r) and load each of them by same weight (Mg), then the change (l_r) in length of the rubber wire will be more than that (l_s) for steel wire.

$$\text{Hence} \quad Y_r = \frac{MgL}{\pi r^2 l_r} \text{ and } Y_s = \frac{MgL}{\pi r^2 l_s}$$

$$\therefore \quad \frac{Y_s}{Y_r} = \frac{l_r}{l_s} \text{ i.e., } Y_s > Y_r$$

Hence the value of the Young's modulus or coefficient of elasticity of steel (or iron) is more than that of rubber i.e., **steel (or iron) is more elastic than rubber**.

Force exerted on the two clamped ends of a rod on cooling or heating : If a rod of length L and area of cross-section A is clamped at both the ends and is cooled, then its length should decrease. However the ends are fixed and hence the length even on cooling remains the same. Clearly the clamped ends exert force of tension on the rod i.e., if the rod would have not been clamped, the length L of the rod might have decreased to $L - l$. If the rod of length L is cooled by $\Delta\theta^\circ\text{C}$, then from the definition of coefficient of linear expansion, we have

$$\alpha = \frac{l}{L\Delta\theta} \quad [\because L = L_0 (1 + \alpha\theta)] \text{ or } \Delta L = L_0 \alpha\Delta\theta \text{ or } l = L\alpha\Delta\theta \quad \dots(1)$$

$$\text{Hence} \quad \text{strain} = \frac{l}{L} = \alpha\Delta\theta$$

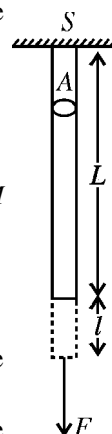


Fig. 4.1.

$$\therefore \text{Young's modulus of rod, } Y = \frac{\text{Stress}}{\text{Strain}} = \frac{\text{Force}}{\text{Area} \times \alpha \Delta \theta} = \frac{\text{Force of tension}}{A \alpha \Delta \theta}$$

$$\text{Hence force of tension at the clamped ends of the rod} = YA\alpha\Delta\theta \quad \dots(2)$$

Similarly, when a rod clamped at both ends, is heated, corresponding to the elongation in its length there is extra force exerted by the clamped (fixed) ends on the rod and the length of the rod remains the same. If the rod is heated by $\Delta\theta^\circ\text{C}$, then the force exerted by the clamped ends on the rod is

$$F = YA\alpha\Delta\theta \quad \dots(3)$$

(2) Bulk modulus : When a solid or a fluid (liquid or gas) is subjected to change in pressure, its volume changes, but the shape remains unchanged. The force per unit area, applied normally and uniformly to the surface of the body (or pressure) gives the **stress** and the change per unit volume, the **strain**. Now, *within the limit of elasticity the ratio of uniform and normal stress on the surface of a body to the volumetric strain is called bulk modulus of the material of the body and it is denoted by the letter 'K'.*

Let V be the original volume of a body and by the application of a pressure p , let the volume be changed to $(V - v)$. Then

$$\text{stress} = p \text{ and volume strain} = v/V$$

$$\therefore \text{bulk modulus, } K = \frac{p}{-v/V} = -\frac{pV}{v} \quad \dots(4)$$

The negative sign indicates here, that by the increase of pressure a decrease in volume occurs.

The reciprocal of bulk modulus of a substance is called its **compressibility**, i.e.,

$$\text{compressibility} = \frac{1}{K} = -\frac{v/V}{p} \quad \dots(5)$$

(3) Modulus of rigidity or Shear modulus : *Within the elastic limit, the ratio of tangential stress to the shearing strain is called modulus of rigidity of the material of a body and is denoted by the letter ' η '.*

In this case, the shape of a body changes, but its volume remains unchanged. Consider a cube of side L , whose lower edge CD is fixed. A tangential force F is applied in the direction shown over the upper face of area $A (= L^2)$. A force of reaction of the same magnitude F acts on the lower fixed face in the opposite direction, as shown in **fig. 4.2**. These two equal and opposite forces, with the lines of action different, constitute a couple. Due to this couple, face AD of the cube is displaced to the position $A'D$ and face BC to the position $B'C$ through an angle θ , known as the angle of shear or shearing strain. The rotation of the cube by the above mentioned couple is prevented by an opposite couple of equal moment, provided by the constraints.

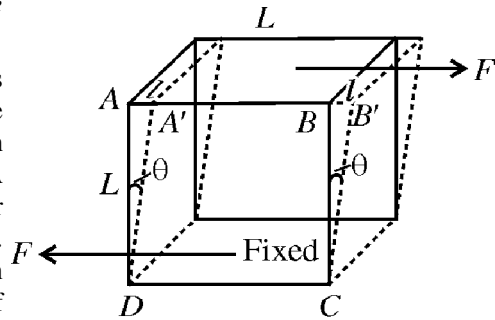


Fig. 4.2. Shearing strain.

Thus, shearing strain $= \theta = \tan \theta = l/L$, if θ is small where $AA' = BB' = l$ and $AD = BC = L$.

$$\text{Shearing stress} = \frac{\text{Tangential force}}{\text{Area of the face}} = \frac{F}{A}$$

$$\text{Modulus of rigidity, } \eta = \frac{F/A}{\theta} = \frac{F}{A \cdot \theta} = \frac{F \cdot L}{Al} \quad \dots(6)$$

This is a relation exactly similar to the one for Young's modulus, but here in place of linear stress,

there is a tangential stress and l , a displacement at right angles L occurs instead along it.

In S.I. system, the units for Y , K and η are newton/m².

(4) **Poisson's ratio** : When a wire is stretched along its length, it is elongated along with a contraction in its diameter. Thus, the length of the wire increases in the direction of the forces, whereas a contraction occurs in the perpendicular direction. This fact is not only true for wire, but for all other bodies under strain.

The ratio of change in dimension to the initial dimension, perpendicular to the direction of the forces, is called **lateral strain**. *Within elastic limit, the lateral strain, is proportional to the longitudinal strain, i.e., the ratio of lateral strain and longitudinal strain is a constant for the material of a body. This constant is known as Poisson's ratio.*

If β and α are the lateral and longitudinal strains respectively, then

$$\text{Poisson's ratio, } \sigma = \frac{\beta}{\alpha}$$

Consider a wire of length L and diameter D . Let the length of the wire be increased by l under the action of two equal and opposite forces and let a corresponding decrease in diameter be d . [fig 4.3]. Then,

Longitudinal strain, $\alpha = l/L$ and Lateral strain, $\beta = d/D$.

$$\therefore \text{Poisson's ratio, } \sigma = \frac{\beta}{\alpha} = \frac{d/D}{l/L} = \frac{L}{l} \cdot \frac{d}{D} \quad \dots(7)$$

Expressing in differential form,

$$\sigma = \frac{L}{D} \frac{dD}{dL} \quad \dots(8)$$

Poisson's ratio is a pure number and has no units and dimensions since it is the ratio of two strains.

4.4. Shearing Stress Equivalent to an Equal Linear Tensile Stress and an Equal Compressive Stress at Right Angles to Each Other

Consider the front view of a cube $ABCD$ of side L [fig. 4.4]. The base DC of the cube is fixed and a tangential force F is applied on the upper face AB .

Now, shearing stress = F/L^2 .

If there is only the force F acting on the upper face AB , then the cube should move as a whole in the direction of the force. But the cube is fixed at the lower face DC , hence an equal and opposite force acts on the face DC . These two forces, therefore, constitute a couple which tends to rotate the cube in the clockwise direction.

The cube, however, does not rotate, hence the face DC which is fixed exerts an equal and opposite couple by exerting forces (say F') on the faces AD and CB shown in the figure.

Now, moment of clockwise couple = $F \times AD$.

and the moment of the anti-clockwise couple = $F' \times AB$.

But as the cube is in equilibrium,

$$\therefore F \times AD = F' \times AB \text{ or } F = F' \quad [\because AB = AD = L]$$

Thus, a tangential force F applied to the face AB gives rise to equal tangential forces acting along all the faces in suitable directions, as shown in [fig. 4.4].

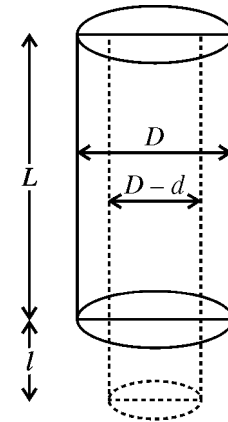


Fig. 4.3. Longitudinal and lateral strains.

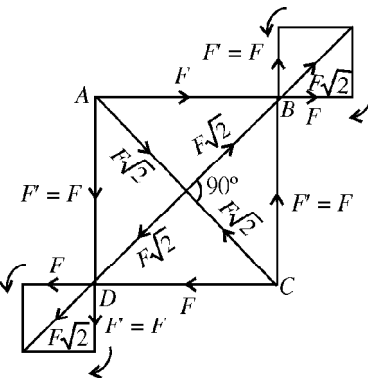


Fig. 4.4. Shearing stress.

The forces acting along the faces AB and CB give rise to a resultant force $F\sqrt{2}$ along DB and forces acting along the face AD and CD also give rise to a resultant force $F\sqrt{2}$ along BD . Hence, an outward pull is exerted along the diagonal BD at the corners B and D , resulting in its extension.

Similarly, at each of the corners A and C a resultant force $F\sqrt{2}$ acts inwards along the diagonal AC , resulting in its contraction.

Thus, a tangential force F acting on one of the face of the cube is equivalent to a force of extension $F\sqrt{2}$ acting outwards along the diagonal BD and a force of compression $F\sqrt{2}$ acting inwards along the diagonal AC .

If a plane is drawn perpendicular to the plane of the paper through the diagonals AC or BD in the cube, its area will be

$$= L \times L\sqrt{2} = L^2\sqrt{2}.$$

Hence, linear tensile stress along $BD = \frac{F\sqrt{2}}{L^2\sqrt{2}} = \frac{F}{L^2}$

Similarly, linear compressive stress along $AC = \frac{F\sqrt{2}}{L^2\sqrt{2}} = \frac{F}{L^2}$

Evidently, F/L^2 is the shearing stress over the face AB of the cube. Thus, a **shearing stress** is equivalent to an equal linear tensile stress and an equal compressive stress at right angles to each other.

4.5. Relations Connecting Various Elastic Constants

The elastic constants are dependent to each other, since any change in the size and shape of a body may be obtained by first changing the size of the body only and then by changing the shape only. Thus, expressions can be derived, showing inter-relations connecting them.

(i) **Relation connecting Y , K and σ [$Y = 3K(1 - 2\sigma)$]** : Let $ABCDEFGH$ represent a cube of unit side. [Fig. 4.5]. Let us consider a force F , which acts normally and uniformly on each of its six faces in the outward direction. Since each force F is acting on unit area, these forces are representing stresses.

If α is the increase per unit length per unit tension along the direction of the force then the elongation produced in each of the edges, namely AB , BF and BC , will be $F\alpha$. If β is the contraction produced per unit length per unit stress perpendicular to the edges, then contraction produced perpendicular to each of the edge, namely AB , BF and BC , will be $F\beta$. Thus the side AB (similarly BF and BC) will increase by $F\alpha$ but simultaneously contract by $F\beta + F\beta$ due to the two other pairs of tensile stresses.

Hence, the sides of the cube become

$$AB = 1 + F\alpha - F\beta - F\beta = 1 + F(\alpha - 2\beta)$$

$$BF = 1 + F\alpha - F\beta - F\beta = 1 + F(\alpha - 2\beta)$$

$$BC = 1 + F\alpha - F\beta - F\beta = 1 + F(\alpha - 2\beta).$$

Hence, the final volume of the cube is

$$AB \times BF \times BC = [1 + F(\alpha - 2\beta)]^3 = 1 + 3F(\alpha - 2\beta) \quad [\text{according to Binomial expansion}]$$

The terms containing higher powers of α and β have been neglected since α and β are very small quantities.

As the initial length of a side of the cube is 1, its initial volume = 1

$\therefore$ Change in volume = Final volume – Initial volume

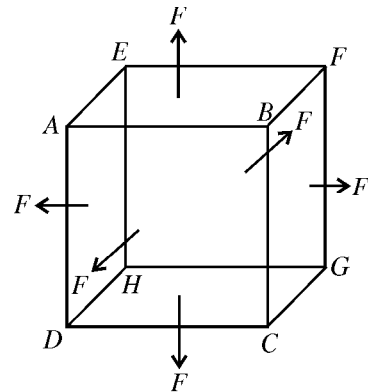


Fig. 4.5.

$$= 1 + 3F(\alpha - 2\beta) - 1 = 3F(\alpha - 2\beta).$$

$$\text{Volume strain} = \frac{3F(\alpha - 2\beta)}{1} = 3F(\alpha - 2\beta)$$

$$\therefore \text{Bulk modulus, } K = \frac{\text{Normal stress}}{\text{Volumetric strain}} = \frac{F}{3F(\alpha - 2\beta)}$$

$$\text{or } K = \frac{1}{3(\alpha - 2\beta)} \text{ or } K = \frac{1/\alpha}{3(1 - 2\beta/\alpha)}.$$

But α is the increase per unit length per unit stress, hence clearly

Stress = 1 (as the area of any edge of the cube is unity)

and Linear strain = $\alpha/1 = \alpha$, $\therefore Y = 1/\alpha$

Also $\beta/\alpha = \sigma = \text{Poisson's ratio}.$

$$\text{Therefore, } K = \frac{Y}{3(1 - 2\sigma)} \text{ or } Y = 3K(1 - 2\sigma) \quad \dots(1)$$

(ii) Relation connecting Y , η and σ [$\eta = Y/[2(1 + \sigma)]$]:

Let $ABCD$ represent the front face of a cube of side L . A tangential force F is applied on its upper face AB and the bottom face DC is fixed. As a result of this force the cube is sheared to $A'B'CD$ through an angle θ [fig. 4.6]. Then, shearing strain is

$$\theta = \frac{AA'}{AD} = \frac{BB'}{BC} = \frac{l}{L},$$

where the displacement $AA' = BB' = l$.

Shearing stress,

$$T = \frac{F}{\text{Area of the upper face of the cube}} = \frac{F}{L^2}$$

$$\therefore \text{Coefficient of rigidity, } \eta = \frac{T}{\theta}.$$

Fig. 4.6.

But a shearing stress along AB is equivalent to a tensile stress along DB and an equal compressive stress along AC at right angles to each other.

Let α and β be the longitudinal and lateral strains per unit stress respectively. Then extension along diagonal DB due to tensile stress = $DB \cdot T \cdot \alpha$

and extension along diagonal DB due to compression stress along $AC = $DB \cdot T \cdot \beta$$

$$\text{Total extension along } DB = DB \cdot T \cdot (\alpha + \beta) = L\sqrt{2} T (\alpha + \beta).$$

Let us draw a perpendicular BN on DB' . Then increase in the length of diagonal DB is nearly equal to NB' . As θ is very small, therefore $\angle AB'C$ is nearly 90° and $\angle BB'N = 45^\circ$.

$$\text{Thus, } NB' = BB' \cos 45^\circ = \frac{BB'}{\sqrt{2}} = \frac{l}{\sqrt{2}}; \therefore L\sqrt{2} T (\alpha + \beta) = \frac{l}{\sqrt{2}} \text{ or } T \frac{L}{l} = \frac{1}{2(\alpha + \beta)}.$$

$$\text{But } T \cdot \frac{L}{l} = \frac{T}{l/L} = \frac{T}{\theta} = \eta; \therefore \eta = \frac{1}{2(\alpha + \beta)} = \frac{1}{2\alpha(1 + \beta/\alpha)}$$

$$\text{But } \beta/\alpha = \sigma \text{ and } Y = \frac{\text{Stress}}{\text{Longitudinal strain}} = \frac{1}{\alpha} [\because \alpha \text{ is the longitudinal strain per unit stress}]$$

$$\text{Therefore, } \eta = \frac{Y}{2(1 + \sigma)} \quad \dots(2)$$

(iii) Relation connecting Y , K and η [$Y = 9K\eta/(\eta + 3K)$]: From relation (1),

$$1 - 2\sigma = Y/3K$$

From relation (2), $2 + 2\sigma = Y/\eta$

Adding the above two equations, we get

$$3 = \frac{Y}{\eta} + \frac{Y}{3K} = Y \left(\frac{1}{\eta} + \frac{1}{3K} \right) = Y \left(\frac{3K + \eta}{3\eta K} \right)$$

Thus

$$Y = \frac{9\eta K}{3K + \eta} \quad \dots(3a)$$

This may be written as

$$\frac{9}{Y} = \frac{3K + \eta}{\eta K} \text{ or } \frac{9}{Y} = \frac{3}{\eta} + \frac{1}{K} \quad \dots(3b)$$

(iv) Relation connecting K , η and σ [$\sigma = (3K - 2\eta)/(2\eta + 6K)$]:

The relation between K , Y and σ for an isotropic solid is [eq. (1)]

$$K = \frac{Y}{3(1 - 2\sigma)} \quad \dots(4)$$

whence

$$1 - 2\sigma = \frac{Y}{3K} \text{ or } 2\sigma - 1 = -\frac{Y}{3K} \text{ or } 2\sigma = 1 - \frac{Y}{3K}$$

$$\therefore \sigma = \frac{1}{2} \left(1 - \frac{Y}{3K} \right) \quad \dots(5)$$

This relation connects the Poisson's ratio (σ) of a material with its Young's modulus (Y) and bulk modulus (K).

η , Y and σ of an isotropic solid are related to [eq. (2)]

$$\eta = \frac{Y}{2(1 + \sigma)} \quad \dots(6)$$

whence,

$$1 + \sigma = \frac{Y}{2\eta}$$

$$\therefore \sigma = \frac{Y}{2\eta} - 1 \quad \dots(7)$$

This relation expresses Poisson's ratio (σ) of a material in terms of its Young's modulus (Y) and modulus of rigidity (η).

From relations (4) and (6), we have

$$Y = 3K(1 - 2\sigma) \text{ and } Y = 2\eta(1 + \sigma) \quad \dots(8)$$

Equating the two values of Y , we have

$$3K(1 - 2\sigma) = 2\eta(1 + \sigma) \text{ or } 3K - 6K\sigma = 2\eta + 2\eta\sigma$$

$$\text{or } \sigma(2\eta + 6K) = 3K - 2\eta \text{ i.e., } \sigma = \frac{3K - 2\eta}{2\eta + 6K} \quad \dots(9)$$

This is the relation between Poisson's ratio of a material (σ) in terms of its bulk modulus (K) and modulus of rigidity (η).

Limiting values of σ

Equating the values of Y in (4), we have

$$3K(1 - 2\sigma) = 2\eta(1 + \sigma) \quad \dots(10)$$

where K and η are essentially positive quantities. Therefore,

(i) if Poisson's ratio is a positive quantity, then both right hand side and left and side of above equation must be positive. This is possible when

$$1 - 2\sigma > 0 \text{ or } \sigma < \frac{1}{2} \text{ or } 0.5. \quad \dots(11)$$

(ii) if σ is a negative quantity, left hand side expression of above equation will be positive. Then, the right hand side expression must also be positive. Hence

$$1 + \sigma > 0 \text{ or } \sigma > -1 \quad \dots(12)$$

Thus, the theoretical limiting values of σ are -1 and 0.5 , i.e.,

$$-1 < \sigma < 0.5 \quad \dots(13)$$

4.6. Work Done Per Unit Volume in Deforming a Body – Internal Energy due to Strain

In order to deform a body, work has to be done. This energy, so spent is stored up in the body, and is called **elastic potential energy** or **internal energy** of the strained body or simply **strain energy**.

We will calculate below, the amount of work done, in three principal cases of strain.

(i) **Longitudinal strain** : When a wire of length L and cross-section A is stretched by a force F , acting along its length, then in equilibrium

$$Y = \frac{F}{A} \cdot \frac{L}{l} \text{ (where } l \text{ is the increase in length.)}; \therefore F = \frac{YAl}{L}$$

For a further small increase in length dl , the small work done, is given by

$$dW = Fdl = \frac{YAl}{L} dl$$

Hence, the total work done in increasing the length of the wire by an amount l is

$$W = \int_0^l \frac{YA}{L} l \, dl = \frac{YAl^2}{2L} = \frac{Fl}{2} \quad [\because \frac{YAl}{L} = F] \quad \dots(1)$$

This amount of work done is stored in the elongated wire as elastic potential energy or internal potential energy.

The volume of the wire = AL .

$$\therefore \text{Work done per unit volume} = \frac{YAl^2}{2L \times AL} = \frac{1}{2} \frac{Yl^2}{L^2} = \frac{1}{2} \cdot \frac{Yl}{L} \cdot \frac{l}{L} \quad \dots(2)$$

$$\text{But } \frac{Yl}{L} = \frac{F}{A} = \text{Stress and } \frac{l}{L} = \text{Strain.}$$

$$\therefore \text{Strain energy per unit volume} = \frac{1}{2} \times \text{Stress} \times \text{Strain.} \quad \dots(3)$$

(ii) **Volume strain** : Consider a body of volume V compressed by an amount v by the application of a uniform pressure p acting normal to its whole surface. Then by the definition of bulk modulus, we have

$$K = \frac{p}{v/V} = \frac{pV}{v} \text{ or } p = K \frac{v}{V}$$

If the surface area of the body be A , then the compressive force acting on the body is

$$F = pA = KAv/V$$

The small work done for a movement dx in the direction of F , that is normal to the surface, is given by

$$dW = Fdx = KAv dx/V.$$

But $Adx = dv$ is the infinitesimal change in volume.

Hence $dW = \frac{K}{V} v dv$

$\therefore$ Total work done for a change of volume v is

$$W = \int_0^v \frac{K}{V} v dv = \frac{K}{V} \frac{v^2}{2} \quad \dots(4)$$

Hence, the work done or strain energy per unit volume

$$= \frac{W}{V} = \frac{1}{2} \frac{K}{V^2} v^2 = \frac{1}{2} \cdot \frac{Kv}{V} \cdot \frac{v}{V} \quad \dots(5)$$

$$= \frac{1}{2} \times \text{Stress} \times \text{Strain}. \quad \dots(6)$$

(iii) **Shearing strain** : Consider the front view of a cube $ABCD$ of side L . [fig. 4.7]. A tangential force F is applied on the upper face. The lower edge of the cube is fixed. As a result of the force F , the upper plane of the cube is displaced through $AA' = BB' = l$. Now, for a further displacement dl of the plane, the amount of work done, is given by

$$dW = F \cdot dl$$

Thus, the total amount of work done for the displacement l is

$$W = \int_0^l F \cdot dl$$

But $\eta = \frac{F}{A\theta} = \frac{F}{L^2} \times \frac{L}{l}$ or $F = \eta \cdot L \cdot l$ [$\because A = L \times L$ and $\theta = l/L$]

$$\therefore W = \int_0^l \eta \cdot L \cdot l \cdot dl = \frac{1}{2} \eta \cdot L \cdot l^2 \quad \dots(7)$$

Volume of the cube $= L^3$

$$\therefore \text{Work done per unit volume} = \frac{1}{2} \cdot \frac{\eta \cdot L \cdot l^2}{L^3} = \frac{1}{2} \cdot \frac{\eta l}{L} \cdot \frac{l}{L} \quad \dots(8)$$

Here $\eta/L = \text{Stress}$ and $l/L = \theta = \text{Strain}$.

$$\therefore \text{Work done or Strain energy per unit volume} = \frac{1}{2} \times \text{Stress} \times \text{Strain}. \quad \dots(9)$$

Ex. 1. A wire of 3.0 m long and 0.625 sq. cm in cross section is found to stretch by 0.3 cm under a tension of 1200 kg wt. What is Young's modulus of the material of the wire ? (A.N.U. 2003; A.U. 2000)

Sol. Young's modulus $Y = \frac{\text{stress}}{\text{strain}} = \frac{F/A}{l/L} = \frac{FL}{lA}$

Here, $F = 1200 \text{ kg wt} = 1200 \times 9.8 \text{ newton}$, $L = 3 \text{ m}$, $l = 0.3 \text{ cm} = 0.3 \times 10^{-2} \text{ m}$, and $A = 0.625 \text{ cm}^2 = 0.625 \times 10^{-4} \text{ m}^2$.

$$\therefore Y = \frac{1200 \times 9.8 \times 3}{(0.3 \times 10^{-2}) \times (0.625 \times 10^{-4})} = 1.88 \times 10^{11} \text{ N/m}^2.$$

Ex. 2. A steel wire 6 m long and a copper wire 4 m long, each of cross section 0.25 cm^2 are fastened end to end and stretched with a tension of 500 N. What is the change in length to each wire ? (O. U. 2002)

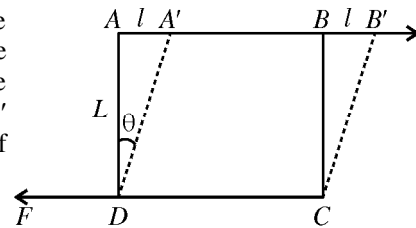


Fig. 4.7.

Sol. $Y = \frac{F \cdot L}{A \cdot l}, \therefore l = \frac{F \cdot L}{A \cdot Y}$

Here, for steel wire, $F = 500 \text{ N}$, $L = 6 \text{ m}$, $A = 0.25 \text{ cm}^2 = 0.25 \times 10^{-4} \text{ m}^2$ and $Y = 2 \times 10^{11} \text{ N/m}^2$.

$\therefore$ Change in length of the steel wire, $l = \frac{500 \times 6}{0.25 \times 10^{-4} \times 2 \times 10^{11}} = 6 \times 10^{-4} \text{ m}$.

and change in length of the copper wire, $l = \frac{500 \times 4}{0.25 \times 10^{-4} \times 1.25 \times 10^{11}} = 6.4 \times 10^{-4} \text{ m}$.

Ex. 3. A uniform steel wire of density $7,800 \text{ kg/m}^3$ is 2.5 m long and mass $15.6 \times 10^{-3} \text{ kg}$. It extends by 1.25 mm when loaded by 8 kg . Calculate the value of Young's modulus for steel. (A.N.U. 2001)

Sol. Given, mass of the wire $m = 15.6 \times 10^{-3} \text{ kg}$; density of the wire $\sigma = 7800 \text{ kg/m}^3$

But, volume of the wire, $V = AL$, Area of cross-section and $L = \text{length}$

$\therefore A = \frac{V}{L} = \frac{15.6 \times 10^{-3}}{7800 \times 2.5} = 8 \times 10^{-7} \text{ m}^2$

$\therefore Y = \frac{FL}{Al} = \frac{8 \times 9.8 \times 2.5}{8 \times 10^{-7} \times 1.25 \times 10^{-3}} = 1.96 \times 10^{11} \text{ N/m}^2$

Ex. 4. An iron wire of radius 1 mm is fixed at its two ends in its normal state at 30°C temperature. When the temperature becomes 20°C , determine the tension in the rod ($Y = 2.2 \times 10^{11} \text{ N/m}^2$ and $\alpha = 11 \times 10^{-6} / ^\circ\text{C}$ for iron).

Sol. Let the tension developed in the wire be T , when it is cooled from 30°C to 20°C . Hence

Stress $= T/\pi (10^{-3})^2$

Now, $\alpha = \frac{L_2 - L_1}{L_1(\theta_2 - \theta_1)}$ or Strain $= \frac{L_2 - L_1}{L_1} = \alpha(\theta_2 - \theta_1) = 11 \times 10^{-6} \times (30 - 20) = 11 \times 10^{-5}$.

$\therefore Y = \frac{\text{Stress}}{\text{Strain}} = \frac{T}{\pi \times (10^{-3})^2} \times \frac{1}{11 \times 10^{-5}} = 2.2 \times 10^{11}$; $\therefore T = \pi \times 2.2 \times 11 = 72.5 \text{ newton}$.

Ex. 5. Calculate Y for rubber if a rubber tube 0.4 metre long, whose external and internal diameters are 0.01 m and 0.004 m respectively, extends 0.0006 m when stretched by a force of 5 kg-wt .

Sol. If r_1 and r_2 be the internal and external radii of the tube, then the area of cross-section of the tube is

$A = \pi (r_2^2 - r_1^2) = 3.14 [(0.005)^2 - (0.002)^2] = 3.14 \times 0.000021 = 6.6 \times 10^{-5} \text{ m}^2$

Here, $F = 5 \text{ kg-wt} = 5 \times 9.8 = 49 \text{ newton}$, $L = 0.4 \text{ m}$ and $l = 0.006 \text{ m}$

$\therefore Y = \frac{FL}{Al} = \frac{49 \times 0.4}{6.6 \times 10^{-5} \times 0.006} = 4.9 \times 10^8 \text{ N/m}^2$

Ex. 6. A force of 10^6 N/m^2 is required for breaking a material. If the density of the material is $3 \times 10^3 \text{ kg/m}^3$, then what should be the length of the wire made of material so that it breaks by its own weight?

Sol. If ρ is the density of the wire, then the weight (force) of the wire of length L and area of cross-section A is

$F = mg = \text{volume} \times \text{density} \times g = (Al) \rho g = \text{Stress} = \frac{F}{A} = \frac{AL\rho g}{A} = L\rho g$

This will be breaking stress $= 10^6 \text{ N/m}^2$, when

$$L\rho g = 10^6$$

whence,
$$L = \frac{10^6}{\rho g} = \frac{10^6}{3 \times 10^3 \times 9.8} = \mathbf{34 \text{ m}}$$

Ex. 7. Calculate the Poisson's ratio for the material. Given, $Y = 12.25 \times 10^{10} \text{ N/m}^2$ and $\eta = 4.55 \times 10^{10} \text{ N/m}^2$. (O.U. 2008)

Sol. Poisson's ratio $\sigma = \frac{Y}{2\eta} - 1 = \frac{12.25 \times 10^{10}}{2 \times 4.55 \times 10^{10}} - 1 = 1.346 - 1 = \mathbf{0.346}$

Ex. 8. Calculate Poisson's ratio for silver. Given its Young's modulus $= 7.25 \times 10^{10} \text{ N/m}^2$ and bulk modulus $= 11 \times 10^{10} \text{ N/m}^2$. (K.U. 2001; N.U. 1996)

Sol. $K = \frac{Y}{3(1-2\sigma)}$ or $Y = 3K(1-2\sigma)$ or $1-2\sigma = \frac{Y}{3K}$

whence
$$\sigma = \frac{1}{2} \left(1 - \frac{Y}{3K} \right)$$

Here, $Y = 7.25 \times 10^{10} \text{ N/m}^2$ and $K = 11 \times 10^{10} \text{ N/m}^2$

$\therefore \sigma = \frac{1}{2} \left[1 - \frac{7.25 \times 10^{10}}{3 \times 11 \times 10^{10}} \right] = \mathbf{0.39}$.

Ex. 9. The modulus of rigidity and Poisson's ratio of material of a wire are $2.87 \times 10^{10} \text{ N/m}^2$ and 0.379 respectively. Find the value of Young's modulus of the material of the wire. (A.N.U. 2005)

Sol. $Y = 2\eta(1+\sigma)$

Here, $\eta = 2.87 \times 10^{10} \text{ N/m}^2$ and $\sigma = 0.379$

$\therefore Y = 2 \times 2.87 \times 10^{10} \times (1 + 0.379) = 2 \times 2.87 \times 10^{10} \times 1.379 = \mathbf{7.915 \times 10^{10} \text{ N/m}^2}$.

Ex. 10. The Young's modulus for steel is $Y = 2 \times 10^{11} \text{ N/m}^2$ and its rigidity modulus $\eta = 8 \times 10^{10} \text{ N/m}^2$. Find the Poisson's ratio and its bulk modulus. (A.N.U. 2008; 07, S.V.U. 2002)

Sol. $\sigma = \frac{Y}{2\eta} - 1$ and $K = \frac{Y}{3(1-2\sigma)}$

Here, $Y = 2 \times 10^{11} \text{ N/m}^2$ and $\eta = 8 \times 10^{10} \text{ N/m}^2$

$\therefore \sigma = \frac{2 \times 10^{11}}{2 \times 8 \times 10^{10}} - 1 = 1.25 - 1 = \mathbf{0.25}$

Hence,
$$K = \frac{2 \times 10^{11}}{3(1-2 \times 0.25)} = \mathbf{1.33 \times 10^{11} \text{ N/m}^2}$$
.

Ex. 11. Determine the Poisson's ratio of rubber if due to pulling a rubber thread, its change in volume is negligible in comparison to its change in length.

Sol. If the rubber thread has the length l and radius r , then volume of the thread is

$$V = \pi r^2 l = \text{constant (given)}$$

Now,
$$\delta V = \pi (2r \delta r l + r^2 \delta l) = 0$$

or
$$2l \delta r + r \delta l = 0 \quad \text{or} \quad -\frac{\delta r}{r} = \frac{\delta l}{2l}$$

Hence, Poisson's ratio, $\sigma = \frac{\text{Lateral strain}}{\text{Longitudinal strain}} = \frac{-\delta r/r}{\delta l/l} = \frac{1}{2} = \mathbf{0.5}$.

Ex. 12. A wire of length 5 m and diameter 1 mm is stretched by 0.3 mm by applying a load of 10.0 kgf. Calculate the internal energy of the strained wire. Take $g = 9.8 \text{ m/s}^2$.

Sol. Strain energy or internal energy of the strained wire $= \frac{1}{2} Fl$.

Here, $F = 10 \text{ kgf} = 10 \times 9.8 \text{ N}$, $l = 0.3 \text{ mm} = 0.3 \times 10^{-3} \text{ m}$.

$\therefore$ Internal energy of the strained wire $= \frac{1}{2} \times (10 \times 9.8) \times (0.3 \times 10^{-3}) = 1.47 \times 10^{-2} \text{ J}$.

4.7. Beams

A beam is a rod or bar of uniform cross-section of a homogeneous, isotropic elastic material, whose length is large in comparison to its thickness or width. The beam is supported at one or more points and is usually loaded perpendicular to its axis. The distance between the supports is called *span*.

When a beam is loaded, its every cross-section faces bending and shear. The load finally gets transferred to supports. The system of applied forces and reactions keep the beam in equilibrium.

When a horizontal beam is loaded with vertical loads, it bends due to the action of the loads. The shearing stress and bending moment are developed inside the beam to resist bending. The amount of bending in the beam depends upon the magnitude and type of the loads, length of the beam, elasticity of the beam and dimensions of the beam.

4.7.1. Types of Support

Beams can have following types of supports :

(1) Simple support or knife edge support :

When a beam rests on a simple or knife edge support, the reaction at the support is at right angles to the beam. **Fig. 4.8** shows two simple supports A and B with normal reactions R_1 and R_2 .

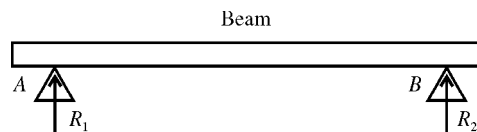


Fig. 4.8. Simple supports.

(2) Roller support :

When a beam rests on a roller support, the reaction at the support is at right angles to the beam [fig. 4.9].

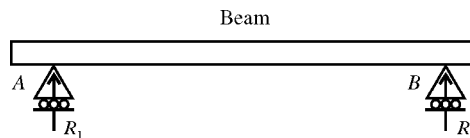


Fig. 4.9. Roller supports.

(3) **Hinged (or pinned) support :** A hinged support keeps the end of the beam stationary. The reaction may have both horizontal and vertical components. [fig. 4.10]

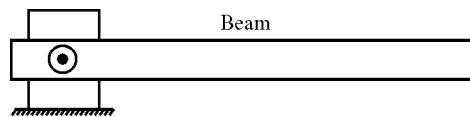


Fig. 4.10. Hinged support.

(4) **Fixed or built-in support :** When a beam is fixed at one end, the support is called fixed or built-in support. In such a case, the reaction is accompanied by a couple. [fig. 4.11]

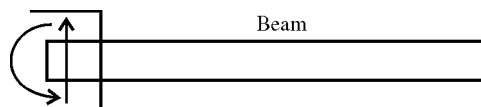


Fig. 4.11. Built-in support.

4.7.2. Classification of Beams

According to the type of supports, the beams are classified as follows :

(1) **Simply supported beam** : A beam is called simply supported, if both of its ends rest on two simple supports or one of its ends is hinged and the other is roller-supported. It has one span [fig. 4.12].

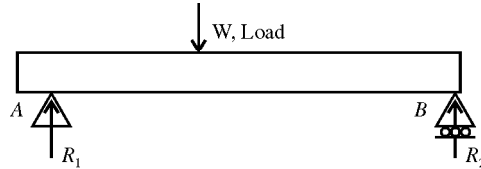


Fig. 4.12. Simply supported beam.

(2) **Cantilever beam** : If a beam is fixed at one end in the wall and free at the other, it is called a cantilever beam [fig. 4.13].

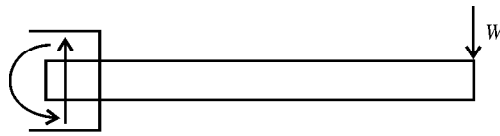


Fig. 4.13. Cantilever.

(3) **Overhanging beam** : If a beam has its one end portion or both end portions, extended in the form of cantilever, it is known as overhanging beam [fig. 4.14].

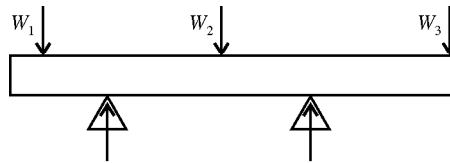


Fig. 4.14. Overhanging beam.

(4) **Rigidly fixed beam** : If both ends of a beam are rigidly fixed or built-in the walls, we call such a beam as rigidly fixed beam [fig. 4.15].

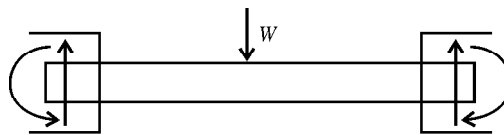


Fig. 4.15. Rigidly fixed beam.

(5) **Continuous beam** : If a beam is supported on more than two supports, then such a beam is called continuous beam [Fig. 4.16].

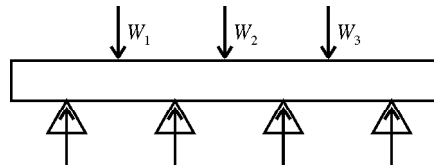


Fig. 4.16. Continuous beam.

4.7.3. Types of Load

A beam may be subjected to the following types of loads :

(1) **Point load or concentrated load** : When a load acts on a beam at a point on a beam, it is called a point load or concentrated load [fig. 4.17]. A point load is represented by an arrow. Practically it is not

possible to apply a load at a point, because it has some contact area on some length of the beam. If the contact length is very small in comparison to the length of the beam, then the applied load is considered to be a point load. For example, in the laboratory experiment of bending of a beam, a hanger with kilogramme weights, suspended at the middle of span, acts as a point load.

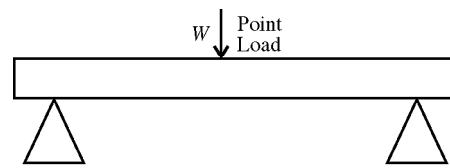


Fig. 4.17.

(2) Uniformly distributed load : When a load is spread over a beam in such a way that the load on per unit length of the beam is the same, such a load is known as uniformly distributed [fig. 4.18]. For example, in a tank containing water, the water load acting on the floor of the tank is known as uniformly distributed load.

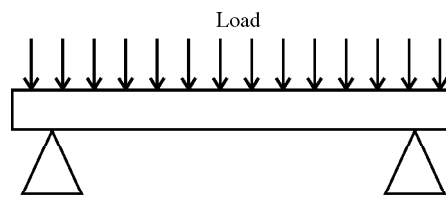


Fig. 4.18. Uniformly distributed load.

(3) Uniformly varying load : If a load is spread over a beam in such a way that the load on per unit length of the beam increases or decreases uniformly along the length of the beam, then such a load is called uniformly varying load [Fig. 4.19].

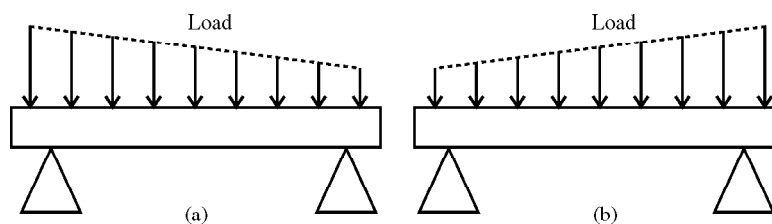


Fig. 4.19. Uniformly varying load.

4-7-4. Shearing Force, Bending Moment and Sign Convention

Shearing Force (F) : When a beam, resting on supports, horizontally is subjected to transverse loads, due to internal forces developed in the beam a shear force acts vertically on the section of a beam. The shearing force at a cross-section of a beam is the force which tries to shear off the section. It is defined as the unbalanced vertical force either to the right or to the left of the section of a portion of the beam.

Bending moment (M) : A bending moment at the cross-section of a beam is the algebraic sum of moments of all the forces about the section acting either to the left or to the right of the section.

Sign Convention : Consider a beam, resting on supports, horizontally. The beam is subjected to transverse loads. In equilibrium, let us take a portion $IJKL$ of the beam. Now, draw a transverse plane PP' , which divides this portion into two portions $IPP'L$ and $PJKP'$ [fig. 4-20 (a)]. Due to internal forces developed due to the loading right hand portion $PJKP'$ experiences a shearing force F and a bending moment M at the section PP' because of the left hand portion [fig. 4-20 (b)]. The left hand portion will also experience a shearing force F and a bending moment M in the opposite sense at the section PP' due to the right hand portion, since the entire beam is in equilibrium. *The direction of a shearing is considered to be positive, if it causes clockwise rotation of the member on which it acts. The upward shearing force F is positive on the left section PP' of the right hand portion $PJKP'$ and the downward shearing force F is positive on the right section PP' of the left hand portion $IPP'L$. The bending moment M is considered*

to be positive when it tends to rotate (i) the left section of the right hand portion of a beam in the clockwise direction, and (ii) the right section of the left hand portion of a beam in the anticlockwise direction **fig. 4.20 (b)** shows the positive shearing force F and positive bending moment M (i) on the left end face PP' of right portion and (ii) on the right end face PP' of the left portion.

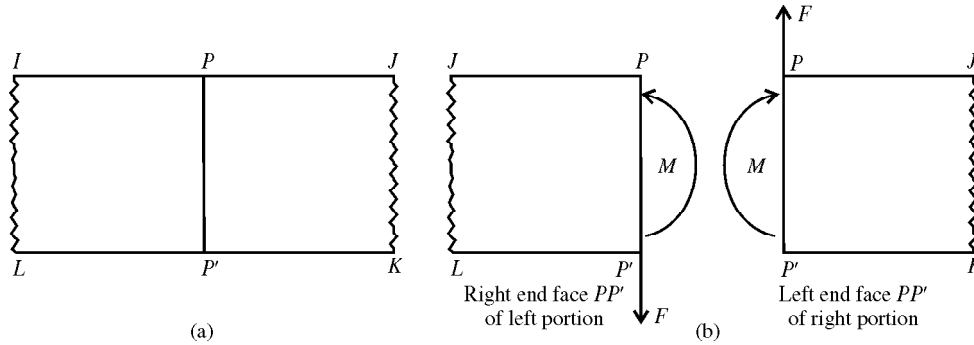


Fig. 4.20. Sign convention for shearing force F and bending moment M ; the values of F and M are positive in the direction as shown in (b).

Remember that choice of sign convention is arbitrary but we have discussed above the sign convention in the majority of scientific applications.

4.7.5. Relation Between Shearing Force and Bending Moment

In the presence of vertical loads on a beam, in equilibrium a shearing force F and a bending moment M acts on the transverse section of a beam due to internal forces. In order to obtain a relation between shearing force F and bending moment M at a section of the beam, let us consider the equilibrium of a small element of length δx of the beam [**fig. 4.21**]. Let F be the upward shearing force and M be the clockwise bending moment on the left face of the element. Then the corresponding quantities on the right face of the element are

$$F + \frac{dF}{dx} \delta x \text{ and } M + \frac{dM}{dx} \delta x$$

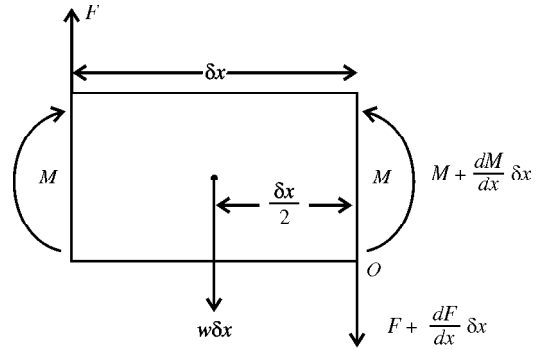


Fig. 4.21. Equilibrium of an element of a beam.

but in the opposite sense. Further the weight of the element $w\delta x$ acts vertically through the mid point of the element, where w is the weight of the beam per unit length.

For the vertical equilibrium of the element.

$$F - w\delta x - \left(F + \frac{dF}{dx} \delta x \right) = 0$$

$$\text{or} \quad w = -\frac{dF}{dx} \quad \dots(1)$$

For no rotation, taking moment about O , we have

$$M + \frac{dM}{dx} \delta x - F\delta x - w\delta x \cdot \frac{\delta x}{2} - M = 0$$

Neglecting second order quantities, we get

$$F = \frac{dM}{dx} \quad \dots(2)$$

Thus the rate of change of bending moment is equal to the shearing force at the section. Substituting for F from (2) in (1), we get

$$w = -\frac{d^2 M}{dx^2} \quad \dots(3)$$

4.8. Types of Bending

Let us consider the neutral axis of a beam. When a beam is bent, the neutral axis takes the form of an arc.* If the value of the radius of curvature R of the neutral axis is the same for all of its point, then the bending is called **uniform bending**. In case, the value of R is different for different parts of the neutral axis, the bending is termed as **non-uniform bending**.

In case of uniform bending of a beam AB [fig. 4.22 (a)], the bending moment is constant and shearing force is zero ($F = dM/dx$) in the span CD . For non-uniform bending [fig. 4.22 (b)], the bending moment changes in the central span as we move along the beam and the shearing force is present at a transverse section.

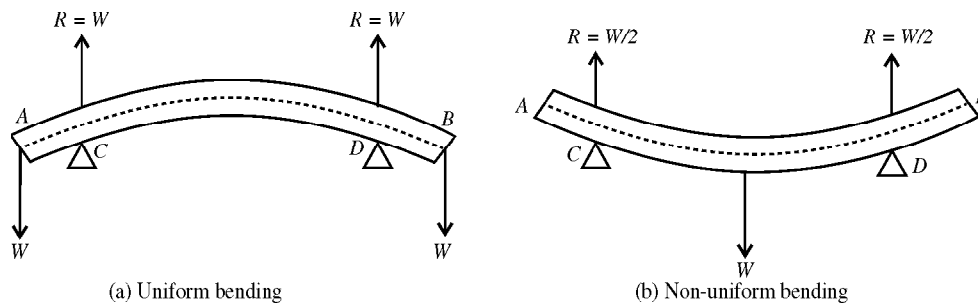


Fig. 4.22.

Uniformly bent beam : Suppose that AB is a uniform beam of length $l + 2a$ [fig. 4.23], resting on supports C and D and carrying a load W at each end. Considering the beam to be light and the symmetry of loading, the reactions at each support is $R = W$.

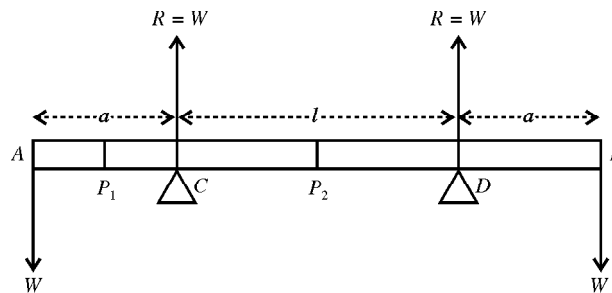


Fig. 4.23. Uniformly bent beam.

Let A be the origin of co-ordinates. Then at a transverse section at P_1 , at a distance x from A for $0 < x < a$ [fig. 4.24], we have for equilibrium

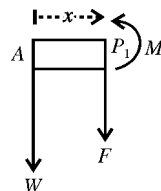


Fig. 4.24.

* Regarding definition of neutral axis sec. Art. 4.9.

$$F + W = 0 \text{ or } F = -W \quad \dots(1)$$

$$\text{Taking moment about A, } M - Fx = 0 \text{ or } M = -Wx \quad \dots(2)$$

At P_2 , at a distance x from A for $a \leq x \leq (a + l)$, we have [fig. 4.25] in equilibrium

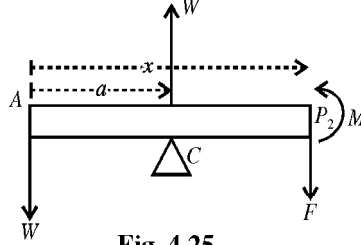


Fig. 4.25.

$$F - W + W = 0 \text{ or } F = 0 \quad \dots(3)$$

and taking moment about C,

$$M - F(x - a) + Wa = 0 \text{ or } M = -Wa = \text{constant} \quad \dots(4)$$

The formula $F = \frac{dM}{dx}$ is also satisfied for the relations (1), (2) and (3), (4).

From relations (3) and (4) we find for the central span CD , the bending moment is constant and shearing force is zero. This means that **the central span of the beam is uniformly bent**.

We know that the bending moment at a section of a beam is given by [Art. 4.10]

$$M = \frac{YI_g}{R} \quad \dots(5)$$

where Y is the Young's modulus of the material of the beam, R is the radius of curvature of neutral axis at the section and I_g is the geometrical moment of inertia of the sectional area

From (4) and (5), for a uniformly bent beam

$$\frac{YI_g}{R} = -Wa \text{ or, } R = -\frac{YI_g}{Wa} \quad \dots(6)$$

This means that the value of radius of curvature is constant in the uniformly bent beam.

Non-uniformly bent beam : Let AB be a uniform beam, resting on two knife edges C and D . It has been loaded by the weight W in the middle of the central span CD of length l of the beam. The beam is considered to be of negligible weight (fig. 4.26).

Let C be the origin of co-ordinates. Then at a transverse section at P_1 , at a distance x from C for $0 < x < l/2$ [fig. 4.27], we have for equilibrium,

$$F - W/2 = 0 \text{ or } F = W/2 \quad \dots(1)$$

For no rotation, taking moment about C ,

$$M - Fx = 0 \text{ or } M = \frac{Wx}{2} \quad \dots(2)$$

Similarly at a point P_2 , for equilibrium of the portion $\frac{l}{2} < x < l$ [fig. 4.28], we have

$$F + W - W/2 = 0 \text{ or } F = -\frac{W}{2} \quad \dots(3)$$

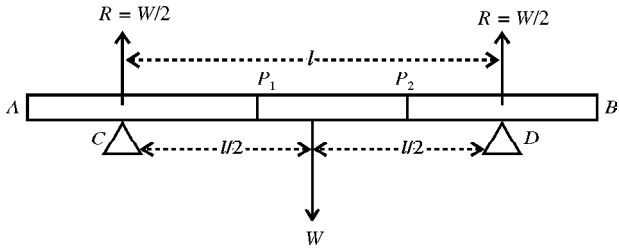


Fig. 4.26. Non-uniformly bent beam.

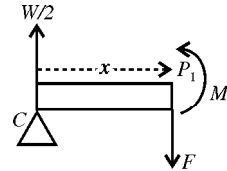


Fig. 4.27.

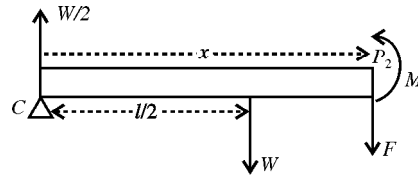


Fig. 4.28.

and
$$M - Fx - W\frac{l}{2} = 0 \text{ or } M = -\frac{W}{2}x + \frac{Wl}{2} \quad \dots(4)$$

Thus the shearing force at a section of the beam is not zero and the bending moment varies as we move along in the central span of the beam. This means that **the bending of the given beam is non-uniform**. Obviously the radius of curvature R ($M = YI_g/R$ or $R = YI_g/M$) is different at different parts of the beam.

Ex. 1. A simply supported beam of span $2a$ carries a point load W in the middle. Find the reactions at the supports and shearing force and bending at a section of the beam. Draw shearing force and bending moment diagrams. The beam has negligible weight in comparison to load.

Sol. In view of the symmetry of the load and reactions, $R_A = R_B = R$ (say) and $R + R = W$ or $R = \frac{W}{2}$. Thus the reaction on the beam due to the supports are each $W/2$ and act vertically upwards as shown in **fig. 4.29**.

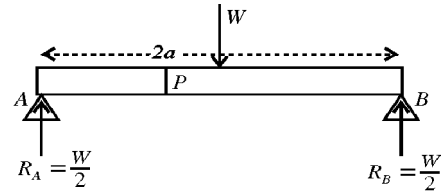


Fig. 4.29.

First consider of a shearing force F and the bending moment M at a section P in the beam at a distance x from the end A , where $0 < x < a$. [**fig. 4.30**]. The unknowns F and M are assumed to act in the positive sense on the right hand face P of the segment AP according to the sign convention. Considering the equilibrium of the element AP , we have

$$F - \frac{W}{2} = 0 \text{ and } M - Fx = 0 \text{ (Take moment about A)}$$

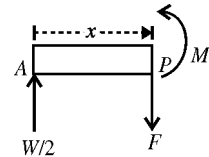


Fig. 4.30.

$$\therefore F = \frac{W}{2} \text{ and } M = Fx \text{ or } M = \frac{W}{2}x \quad \dots(1)$$

Thus the shearing force is $W/2$ downward at the right section P of the segment AP of the beam and the bending moment is $(W/2)x$ in the anticlockwise direction.

Similarly, if P is a section at a distance x from A such that $a < x < 2a$ [**fig. 4.31**], the conditions for equilibrium are

$$F + W - \frac{W}{2} = 0 \text{ and } M - Fx - Wa = 0$$

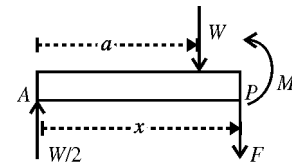


Fig. 4.31.

$$\therefore F = -\frac{W}{2} \quad \dots(2a)$$

$$\text{and } M = Fx + Wa \text{ or } M = -\frac{W}{2}x + Wa \text{ or } M = W\left(a - \frac{x}{2}\right) \quad \dots(2b)$$

In each case (1) or (2), we see that

$$\frac{dM}{dx} = F \text{ and } \frac{d^2M}{dx^2} = 0$$

This is in accordance to the formulae $F = \frac{dM}{dx}$ and $-w = \frac{d^2M}{dx^2}$, because the weight of the beam

is negligible. In view of the relations (1) and (2), shearing force and bending moment diagrams are drawn in **fig. 4.32**.

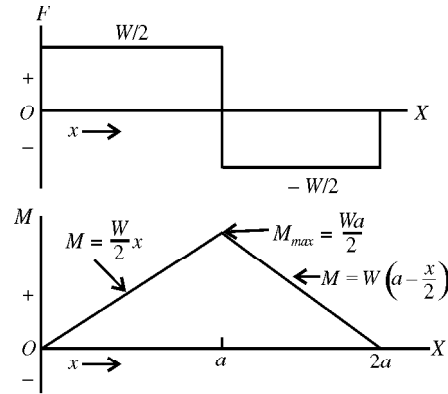


Fig. 4.32. Shearing force and bending moment diagrams for the given simply supported beam

Ex. 2. Consider a cantilever beam of length l , fixed at one end into a wall and loaded at the other end by weight W . Assume that the weight of the beam is negligible. Find the reaction and bending moment at the support. Find also the shearing force and bending moment at a section. Draw shearing force and bending moment diagrams.

Sol. Let R be the vertical reaction and Z be the bending moment at the fixed end [**fig. 4.33**].

Considering the equilibrium of the beam as a whole, we have

$$W - R = 0 \quad \text{and} \quad Z + Wl = 0$$

or

$$R = W \quad \text{and} \quad Z = -Wl$$

Thus the bending moment is anti-clockwise.

Now consider the equilibrium of the portion AP of the beam, where P is a section at a distance x from A such that $0 < x < l$. [**fig. 4.34**] The forces and bending moments are shown in **fig 4.34**. The conditions for equilibrium require

$$F - W = 0 \quad \text{or} \quad F = W$$

and moments about A ,

$$M + Wl - Fx = 0 \quad \text{or} \quad M = -W(l - x).$$

Observe that the formulae $F = W$ and $M = -W(l - x)$ are satisfied and $\frac{d^2M}{dx^2} = 0$, because the weight of the beam is negligible.

Shearing force and bending moment diagrams are shown in **fig. 4.35**.

Ex. 3. Repeat Ex. 2 considering weight per unit length w of the beam.

Sol. Let R be the vertical reaction and Z be the bending moment at the fixed support. Further, given that the weight per unit length of the beam is w [**fig. 4.36**].

Considering the equilibrium of the beam as a whole, we have

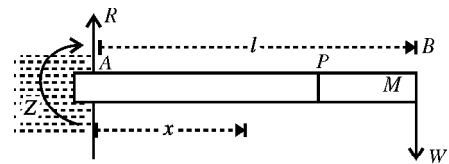


Fig. 4.33. Cantilever.

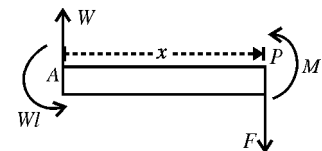


Fig. 4.34.

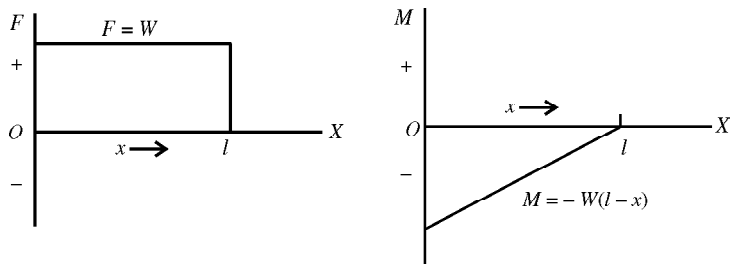


Fig. 4.35. Shearing force and bending moment diagrams for cantilever with negligible weight of the beam

$$R = W + wl \text{ and } Z + Wl + wl \cdot \frac{l}{2} = 0 \text{ or } Z = -Wl - \frac{1}{2} wl^2$$

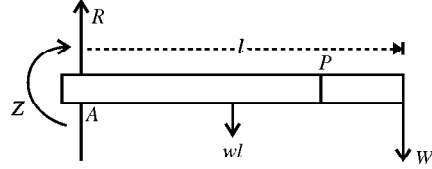


Fig. 4.36.

This bending moment with magnitude $Wl + \frac{1}{2} wl^2$ acts in the anticlockwise direction.

Conditions for equilibrium for $0 < x < l$ [fig. 4.37] are

$$F - (W + wl) + wx = 0 \text{ or } F = W + w(l - x) \quad \dots(1)$$

Also for no rotation taking moment about A

$$M - Fx - wx \cdot \frac{x}{2} + Wl + \frac{1}{2} wl^2 = 0$$

or

$$M = [W + w(l - x)]x + \frac{1}{2} wx^2 - Wl - \frac{1}{2} wl^2$$

or

$$M = -W(l - x) - \frac{1}{2} w(l - x)^2 \quad \dots(2)$$

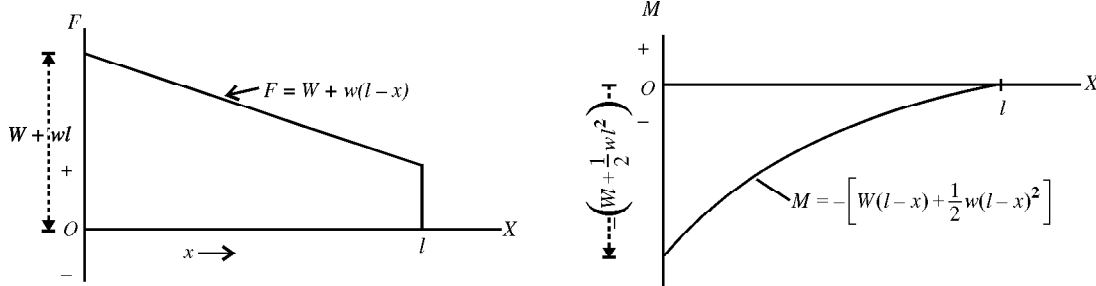


Fig. 4.38. Shearing force and bending moment diagrams for a cantilever with weight per unit length (w) of the beam.

Now, from (2)

$$\frac{dM}{dx} = W + w(l - x) \quad \dots(3)$$

Using (1), we see that the formula $F = dM/dx$ is satisfied.

Further, differentiation of (3) gives

$$\frac{d^2M}{dx^2} = -w \quad \dots(4)$$

which satisfies the desired relation.

Shearing force and bending moment diagrams are drawn in fig. 4.38 in view of the relation (1) and (2).

Ex. 4. A simply supported beam of span 8 m carries loads of 6 kN and 8 kilo newton at a distance of 4 m and 6m from left end as shown in fig. 4.39(a). Compute the reactions at the ends of the beam. Also find the shearing force and bending moment at a section P distant x from A. The weight of the beam is negligible.

Sol. For static equilibrium of the entire beam,

$$R_1 + R_2 = 6 + 10 = 16 \text{ kN} \quad \dots(i)$$

Taking moment about A, $R_2 \times 8 - 6 \times 4 - 10 \times 6 = 0$ or $8 R_2 = 84$

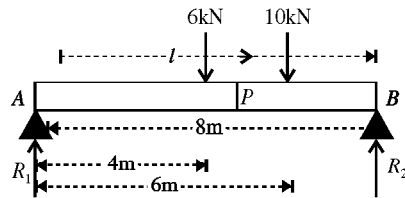


Fig. 4.39 (a).

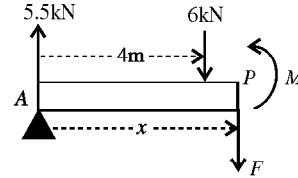


Fig. 4.39 (b).

$$\therefore R_2 = \frac{84}{8} = 10.5 \text{ kN} \quad \dots(ii)$$

$$\text{Hence } R_1 = 16 - R_2 = 16 - 10.5 = 5.5 \text{ kN} \quad \dots(iii)$$

Considering the equilibrium of the portion AP of the beam, we have

$$F + 6 - 5.5 = 0$$

$$\text{or } F = -0.5 \text{ kN} \quad \dots(iv)$$

Taking A as the origin and for no rotation, taking moment about A,

$$M - Fx - 6 \times 4 = 0 \text{ or } M = 0.5x + 24 \text{ kN-m} \quad \dots(v)$$

Observe that the formula $F = dM/dx$ is satisfied by (iv) and (v).

Ex. 5. A beam of span 8 m, overhanging on both sides is loaded as shown in **fig. 4.41**. Find the reactions at both ends. Find also the shearing force and bending moment at the section P at a distance x from A. Neglect the weight of the beam.

Sol. For static equilibrium [fig. 4.40],

$$R_1 + R_2 = 400 + 1000 + 500 = 1900 \text{ N}$$

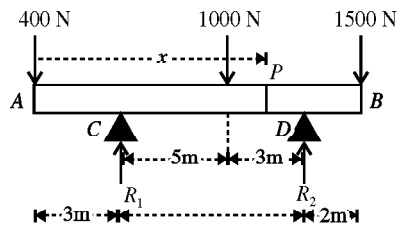


Fig. 4.40.

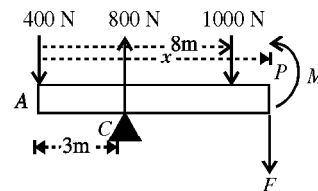


Fig. 4.41.

Taking moment of the forces about C,

$$R_2 \times 8 + 400 \times 3 - 1000 \times 5 - 500 \times 10 = 0$$

$$\text{or, } 8R_2 = 5000 + 5000 - 1200 = 8800 \text{ N}$$

$$\therefore R_2 = 1100 \text{ N}$$

$$\text{Hence } R_1 = 1900 - R_2 = 1900 - 1100 = 800 \text{ N}$$

Considering the equilibrium of the AP portion of the beam [fig. 4.41],

$$F + 1000 - 800 + 400 = 0 \quad \text{or} \quad F = -600 \text{ kN}$$

Taking moment about A (for no rotation)

$$M - Fx - 1000 \times 8 + 800 \times 3 = 0$$

$$\text{or } M = -600x + 5600 \text{ kN-m}$$

Observe that the relation $F = dM/dx$ is satisfied.

4.9. Bending of a Beam : Some Definitions

Beam : A rod (or bar) of uniform rectangular or circular cross-section, whose length is very large in comparison to its thickness or radius is called a beam.

A beam may be considered as made up of a number of thin plane (or cylindrical) layers, collected parallel (or coaxial) to each other. Also, each layer is regarded as to consist of a number of parallel longitudinal fibres, which are called the **longitudinal filaments**.

Neutral surface : When a beam is clamped horizontally at one end and loaded at the other, it undergoes bending. The filaments (or fibres) of outward side are lengthened and subjected to tension, while those of the inner side are shortened and compressed. In between these two portions, there is a layer or surface in which the filaments are neither elongated nor shortened. Such a surface is called neutral surface. In **fig. 4.42**, $EFGH$ is the neutral surface.

Plane of bending : As shown in **fig. 4.42**, the beam has a plane of symmetry $ABCD$ parallel to its length. This plane intersects every transverse section in a straight line, which is an axis of symmetry for that section.

If the bending is uniform, all longitudinal filaments are bent into circular arcs in planes parallel to the plane of symmetry $ABCD$. Then, plane $ABCD$ may be called the **plane of bending**.

The centres of curvature of the longitudinal filaments lie on a straight line perpendicular to the plane of bending. This straight line may be called as the axis of bending. In the **fig. 4.42**, line OO' is the axis of bending.

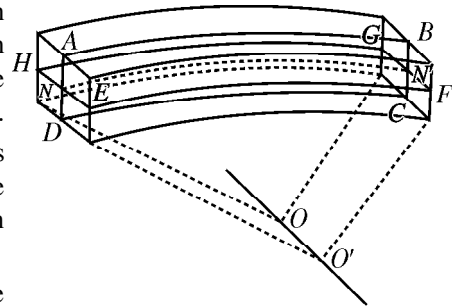


Fig. 4.42.

Neutral axis : The line of intersection of the plane of bending with the neutral surface (both are perpendicular to each other) is called the neutral axis. In **fig. 4.42**, line NN' is the neutral axis.

4.10. Bending Moment and Deduction of the Expression

Consider a beam $ABCD$ fixed at the end AD in the wall. The other end BC of the beam is loaded by a load W **fig. 4.43**. The weight of the beam has been neglected in comparison to load W . Take a transverse section PP' of the beam. This section divides the beam into two portions $APP'D$ and $PBCP'$.

Let us consider the equilibrium of the portion $PBCP'$ of the beam. Since the weight of the beam is negligible, the shearing force F on the left face PP' of the beam must be equal to the load W because of static equilibrium (**fig. 4.43**). These two equal and opposite forces ($F = W$, W) constitute a couple. This clockwise couple which bends the beam, is called bending couple and the moment of this couple is called the **bending moment**.

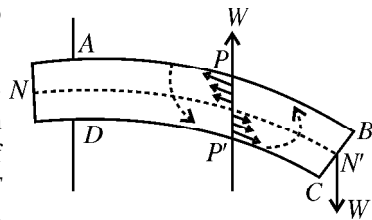


Fig. 4.43.

But the beam is in equilibrium, therefore, there must also be an equal and opposite couple to balance it. The only other forces which act on $PBCP'$ are obviously the forces due to stretching and compression of the filaments of the beam passing across PP' . The filaments in the upper portion of the beam above the neutral surface are elongated and thus, they must be in tension, the portion of the filaments to the left of PP' exerts a pulling force on the portion to the right of PP' ; similarly a portion of the filaments in the lower half of the beam, to the left of PP' , exerts a pushing force on the portion to the right of PP' , because the filaments below the neutral surface are shortened. The distribution of these forces at the section PP' has been shown in **fig. 4.43** and their resultant is a couple which balances the bending couple.

The moment of this balancing couple (due to tensile and compressive stresses) is called the **moment of the resistance to bending**. Since in the position of equilibrium it is equal and opposite to the bending couple, it is referred as **internal bending moment or simply bending moment of the beam**.

Expression for bending moment : Let us consider a small portion of the beam bounded by two transverse sections AB and CD close to each other **fig. 4.44 (a)**. After bending as in **fig. 4.44 (b)**, AC is elongated to $A'C'$ and BD is shortened to $B'D'$. Line EF represents the neutral surface, which is neither stretched nor shortened.

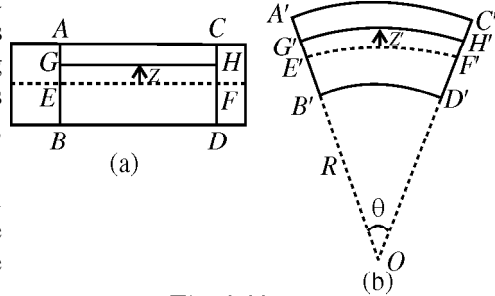


Fig. 4.44.

Let the small portion, under consideration, be bent in the form of a circular arc, subtending an angle θ at the centre of curvature O . Let R be the radius of curvature of the neutral surface $E'F'$.

Consider a filament GH distant z from EF in the unbent position of the beam. After bending its position is shown by $G'H'$ [**fig. 4.44(b)**].

$$\begin{aligned} \text{Now,} & \quad G'H' = (R + z)\theta \\ \text{Before bending} & \quad GH = EF \\ \text{and} & \quad E'F' = EF \text{ (for neutral surface)} \\ \therefore & \quad GH = E'F' \\ \text{But } E'F' = R\theta, \therefore & \quad GH = R\theta \end{aligned}$$

$$\text{Change in the length of the filament} = G'H' - GH = (R + z)\theta - R\theta = z\theta.$$

$$\text{Strain} = \frac{\text{Change in length}}{\text{Initial length}} = \frac{G'H' - GH}{GH} = \frac{z\theta}{R\theta} = \frac{z}{R} \quad \dots(1)$$

If $PQRS$ [**fig. 4.45**] represents a section of the beam at right angles to its length and the plane of bending, then, clearly, the forces acting on the filaments are perpendicular to this section. The line MN lies on the neutral surface.

Let the breadth of the section be $PQ = b$, and its depth, $QR = d$.

The forces, producing elongations and contractions in filaments, act perpendicular to the upper and lower halves, $PQMN$ and $MNRS$ respectively, of the section $PQRS$ and their directions being opposite to each other.

Now, consider a small area δa of the section $PQRS$ about a point A , distant z from the neutral surface. The strain produced in a filament passing through this area will be z/R , as shown above.

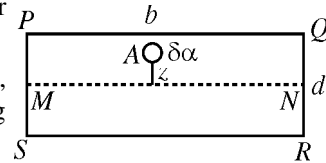


Fig. 4.45.

$$\text{Now,} \quad Y = \text{stress/strain}, \quad \therefore \text{stress} = Y \times \text{strain}.$$

$$\text{Hence,} \quad \text{Stress about the point } A = Y \times z/R,$$

where Y is Young's modulus for the material of the beam.

$$\text{Force on the area } \delta a = Y \cdot \frac{z}{R} \cdot \delta a$$

$$\text{Moment of this force about the line } MN = Y \cdot \frac{z}{R} \cdot \delta a \cdot z = \frac{Y \delta a z^2}{R}$$

But the moments acting on both the upper and the lower halves of the section are in the same direction, therefore, the total amount of the moments acting on the filaments in the section $PQRS$ is given by

$$\Sigma \frac{Y \delta a z^2}{R} = \frac{Y}{R} \Sigma \delta a z^2 = \frac{YI}{R} \quad \dots(2)$$

where $I = \Sigma \delta a z^2$ and is called the *second moment (or geometrical moment of inertia) of the sectional area* about MN and therefore, equal to aK^2 , where a is the whole area of surface $PQRS$ and K , its radius of gyration about MN .

This quantity YI/R is the restoring couple or the **bending moment** of the beam, being equal and opposite to the moment of bending couple due to the load in the position of equilibrium. Thus,

$$\text{Bending moment} = \frac{YI}{R} \quad \dots(3)$$

The quantity YI is known as **flexural rigidity**.

In case of a beam of rectangular cross-section of breadth b and depth d , the area of cross-section is $b \cdot d$ and (radius of gyration)² is equal to $d^2/12$. Hence,

$$I = \text{area of cross-section} \times d^2/12 = b \cdot d \times d^2/12 = bd^3/12^* \quad \dots(4)$$

Similarly, in the case of cylindrical rod of radius r the area of cross-section is πr^2 and the square of radius of gyration of the cross-section about an axis through the centre of the section and perpendicular to the plane of bending is $r^2/4$. Thus

$$I = \pi r^2 \times r^2/4 = \pi r^4/4. \quad \dots(5)$$

In case, the rod is hollow, having r_1 and r_2 as the internal and external radii, then

$$I = \frac{\pi}{4} (r_2^4 - r_1^4). \quad \dots(6)$$

4.11. Cantilever with End Load

A horizontal beam, fixed at one end and loaded at the free end, is called a **cantilever**.

Let a beam of length l be fixed at its one end A and loaded with a load W at B (Fig. 4.47). Let the end B be deflected to the position B' under the action of the load W^{**} . Suppose the axis of X lies horizontally in the direction of the unbent beam and the axis of Y , vertically downwards. Consider a section of the beam, as at P , at a distance x from the end A . Then the moment of the bending couple due to the load W

$$= W(l - x).$$

The cantilever is in equilibrium in the bent position, hence the bending couple must be balanced at the section under consideration by the restoring couple or bending moment.

The bending moment $= YI/R$,

where Y is the Young's modulus of the material of the beam, I geometrical moment of inertia of the section and R the radius of curvature of the neutral axis at the section.

$$\text{Thus,} \quad \frac{YI}{R} = W(l - x) \quad \dots(1)$$

* We want to know geometrical moment of inertia about the axis MN (Fig. 4.46). Take a layer of thickness dz at a distance z from MN . Area of the section of this layer $= b \cdot dz$. Its geometrical moment of inertia about $MN = bdz \cdot z^2$.

$\therefore$ Geometrical M.I. of the total sectional area of the beam about MN is

$$I = \int_{-d/2}^{d/2} bdz \cdot z^2 = 2 \int_0^{d/2} bdz \cdot z^2 = 2b \left[\frac{z^3}{3} \right]_0^{d/2} = \frac{2b}{3} \left(\frac{d}{2} \right)^3 = \frac{bd^3}{12}.$$

$$\text{Similarly, for a circular section,} \quad I = \int_0^r 2\pi x dx x^2 = \frac{\pi r^4}{4}.$$

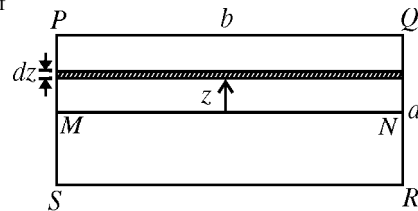


Fig. 4.46.

** We assume that the weight of the beam is negligible in comparison to the load W . Ofcourse, the depression $\delta = BB'$ is much smaller than the length of the beam, but it is shown relatively much more for clarity.

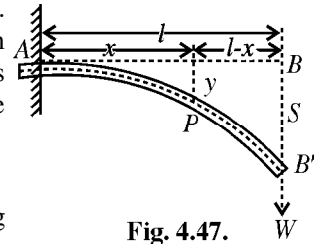


Fig. 4.47.

The radius of curvature R is given by the equation

$$R = \frac{\{1 + (dy/dx)^2\}^{3/2}}{d^2y/dx^2} \quad \dots(2)$$

where y is the depression of the beam at a distance x from the fixed end.

Since y is small, $(dy/dx)^2$ can be neglected in comparison to 1. Hence, we get

$$R = \frac{1}{d^2y/dx^2} \quad \dots(3)$$

Substituting for R in equation (1) above, we have

$$YI \cdot \frac{d^2y}{dx^2} = W(l - x) \quad \dots(4)$$

or

$$\frac{d^2y}{dx^2} = \frac{W}{YI}(l - x) \quad \dots(5)$$

This equation may now be solved to get the value of the depression of the free end *i.e.*, at $x = l$.

Integrating equation (5), we get

$$\frac{dy}{dx} = \frac{W}{YI} \left(lx - \frac{x^2}{2} \right) + C_1 \quad \dots(6)$$

where C_1 is constant of integration, which can be known from the conditions of the problem. One such condition is that at $x = 0$, $dy/dx = 0$.

Applying this conditions in equation (6), we have $C_1 = 0$

and thus
$$\frac{dy}{dx} = \frac{W}{YI} \left(lx - \frac{x^2}{2} \right) \quad \dots(7a)$$

It is to be pointed out that at P , the slope of the curve $\tan \theta = dy/dx$, θ being the angle between X -axis and tangent at P of the bent portion APB' .

Thus
$$\frac{dy}{dx} = \tan \theta = \frac{W}{YI} \left(lx - \frac{x^2}{2} \right) \quad \dots(7b)$$

Integrating once again,

$$y = \frac{W}{YI} \left(\frac{lx^2}{2} - \frac{x^3}{6} \right) + C_2 \quad \dots(8)$$

where C_2 is another constant of integration.

But at $x = 0$, $y = 0$, hence substituting these values in eq. (8), we get $C_2 = 0$

Thus, **the expression for the depression y at a distance x from the fixed end is**

$$y = \frac{W}{YI} \left(\frac{lx^2}{2} - \frac{x^3}{6} \right) \quad \dots(9)$$

At **the free end**, $x = l$ and let $y = \delta$, then the expression for depression is

$$\delta = \frac{W}{YI} \left(\frac{l^2}{2} - \frac{l^2}{6} \right) \text{ or } \delta = \frac{Wl^3}{3YI} \quad \dots(10)$$

For a beam of rectangular cross-section of breadth b and depth d , $I = bd^3/12$ and hence

$$\delta = \frac{4Wl^3}{Ybd^3} \quad \dots(11)$$

For a beam of circular cross-section of radius r , $I = \pi r^4/4$

$$\therefore \delta = \frac{4Wl^2}{3\pi r^4 Y} \quad \dots(12)$$

4.12. Transverse Vibrations of a Loaded Cantilever

Consider a rod of length l , clamped at one end and loaded at its free end with mass M . In the equilibrium position, the deflection of the free end will be given by [eq. (10) above],

$$W = Mg = \frac{3YI}{l^3} \delta \quad \dots(1)$$

Let the mass M be slightly depressed and then released. Now, the cantilever will execute transverse vibrations. If y be the displacement of the free end of the cantilever at any time from the equilibrium position. Then a restoring force F will act opposite to W .

$$\text{Therefore,} \quad W - F = \frac{3YI}{l^3} (\delta + y) \quad \dots(2)$$

Subtracting eq. (13) from eq. (14), we get

$$F = - \frac{3YI}{l^3} y \quad \dots(3)$$

If d^2y/dt^2 be the acceleration produced in the mass M at that instant at the free end, then the force F is Md^2y/dt^2 .

Therefore, the equation of motion is

$$M \frac{d^2y}{dt^2} + \frac{3YI}{l^3} y = 0 \quad \text{or} \quad \frac{d^2y}{dt^2} + \frac{3YI}{Ml^3} \cdot y = 0 \quad \dots(4)$$

This equation represents a S.H.M., whose periodic time is given by

$$T = 2\pi \sqrt{\frac{Ml^3}{3YI}} \quad \dots(5)$$

4.13. Beam Supported at its Ends and Loaded in the Middle

Consider a beam supported on two knife edges K_1 and K_2 at a distance l apart in a horizontal plane. Let it be loaded by a load W at its middle point O . The reaction of each knife edge is obviously equal to $W/2$ acting upwards because of static equilibrium ($R + R = W$ or $R = W/2$). The beam bends as shown in the **fig. 4.48** and the depression is maximum in the middle.

The middle part of the beam is almost horizontal. Hence the beam may be considered as equivalent to two inverted cantilevers OA and OB . The length of each cantilever will be $l/2$, clamped at the end O and carrying an upward load $W/2$ at the other. Therefore, the depression of O below the knife edges is the elevation of the loaded end of such a cantilever from the lowest Y position O (**fig. 4.49**).

Let us consider a vertical section at P , distant x from O and consider the part PB . The moment of the deflecting couple is

$$= \frac{W}{2} \cdot PB = \frac{W}{2} \cdot \left(\frac{l}{2} - x \right)$$

In the equilibrium position this deflection couple must be balanced at the section under consideration by the restoring couple or bending moment, established inside the beam, *i.e.*,

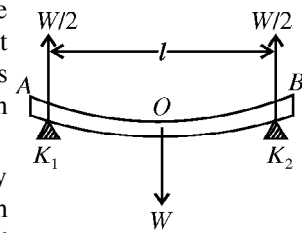


Fig. 4.48.

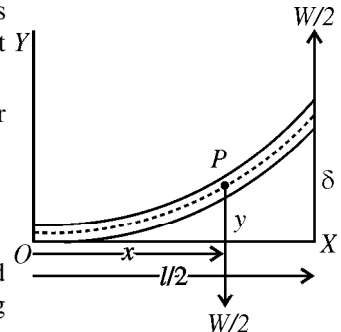


Fig. 4.49.

$$\frac{W}{2} \left(\frac{l}{2} - x \right) = \frac{YI}{R} \quad \dots(1)$$

If y is the elevation of the section P above X -axis, then radius of curvature of the neutral axis at this section is given by

$$R = \frac{1}{d^2 y / dx^2} \quad \dots(2)$$

Hence eq. (1) is

$$\frac{W}{2} \left(\frac{l}{2} - x \right) = \frac{YI d^2 y}{dx^2} \quad \text{or} \quad \frac{d^2 y}{dx^2} = \frac{W}{2YI} \left(\frac{l}{2} - x \right)$$

Integrating this expression, we have

$$\frac{dy}{dx} = \frac{W}{2YI} \left(\frac{l}{2} x - \frac{x^2}{2} \right) + C_1 \quad \dots(3)$$

where C_1 is the constant of integration.

At O , $x = 0$ and $dy/dx = 0$; hence $0 = 0 + C_1$ or $C_1 = 0$,

$$\therefore \frac{dy}{dx} = \frac{W}{2YI} \left(\frac{l}{2} x - \frac{x^2}{2} \right) \quad \dots(4)$$

Integrating further, we have

$$y = \frac{W}{2YI} \left(\frac{l}{2} \frac{x^2}{2} - \frac{x^3}{6} \right) + C_2$$

where C_2 is another constant of integration.

Again at O , $x = 0$ and $y = 0$, $\therefore C_2 = 0$

$$\therefore y = \frac{W}{2YI} \left(\frac{l}{2} \frac{x^2}{2} - \frac{x^3}{6} \right) \quad \dots(5)$$

Now, at the free end, $x = l/2$ and $y = \delta$ (say). Therefore

$$\delta = \frac{W}{2YI} \left(\frac{l}{2} \frac{l^2}{8} - \frac{l^3}{48} \right) \quad \text{or} \quad \delta = \frac{Wl^3}{48YI} \quad \dots(6)$$

If b and d be the breadth and thickness of the beam respectively, then $I = bd^3/12$. Then

$$\delta = \frac{Wl^3}{4Ybd^3} \quad \dots(7)$$

If the cross-section of the beam be a circle of radius r , then

$$\delta = \frac{Wl^3}{12Y\pi r^4} \quad \dots(8)$$

This is the same as the depression at the middle point O . If M be the mass with which the beam has been loaded in the middle, then eq. (7) is

$$\delta = \frac{Mgl^3}{4Ybd^3} \quad \dots(9)$$

The use of this equation is made in the construction of **girders**. The depression of a beam at the middle is proportional directly as the cube of its length and inversely as the width and cube of the depth

or thickness. Thus, in order that the girder may not bend appreciably, its span, *i.e.*, length is made small and its breadth and depth large. At the same time Young's modulus for the material of the beam must be large.

Shape of girders 'I' : When a girder is supported at its ends, its filaments above the neutral surface (or in the upper half), are compressed and those below the neutral surface (or in the lower half) are elongated. The compression or extension of the filaments is maximum at the upper or lower faces of the girder respectively, hence the stresses are maximum at these faces and they decrease, as we move towards the neutral surface from either side. Consequently, the upper and lower faces of the beam should be much stronger than its middle portions. In other words, the middle portion may be made of much smaller breadth than the upper and the lower faces. This is done by manufacturing the girders with their section in the form of **I** so that breadth at the upper and lower faces may be sufficiently larger than that of the inner parts. This saves quite a good amount of the material without sacrificing the strength of the girder.

Stiffness of a beam : The stiffness of a beam is defined as the ratio of maximum depression of the beam to its span. It is usually represented by the symbol $1/n$. For steel girders of large span, n must lie between 1000 and 2000 and for shorter span, between 500 and 700. For timber beams, n should be not less than 360.

4.14. Determination of Young's Modulus by Bending of a Beam

As shown above the depression of a beam of rectangular cross-section, supported at the ends and loaded in the middle, is given by

$$\delta = \frac{Mgl^3}{4Ybd^3} \text{ or } Y = \frac{Mgl^3}{4bd^3\delta}$$

Thus, to determine the Young's modulus of the material of a beam, the beam is supported horizontally and symmetrically on two knife edges, K_1 and K_2 , fixed at a distance l apart, as shown in the **fig.4.51**. A hanger, supported by a knife edge is then placed on the middle of the beam and the load is applied by placing the weight on it. The depression δ of the beam, so produced, is determined by a micrometer screw in equilibrium. At the moment, when micrometer screw just touches the beam, a current starts flowing in the galvanometer G , from the battery, causing to deflect the needle of the former and the reading is noted. A series of loads are applied and observations are taken for load increasing and decreasing. A graph is then plotted between the load of mass M as abscissa [**fig. 4.50**] and the corresponding mean depression as the ordinate. The slope of the straight line, so obtained, given the values δ/M .

$$\text{Now, } Y = \frac{Mgl^3}{4bd^3\delta} = \frac{gl^3}{4bd^3} \cdot \frac{M}{\delta}$$

Thus, knowing the values of l , b and d , Young's modulus of the material of the beam may be determined.

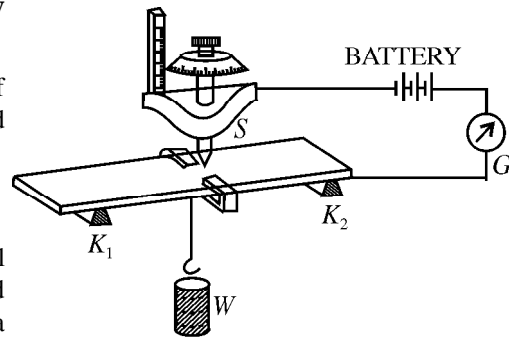


Fig. 4.50.

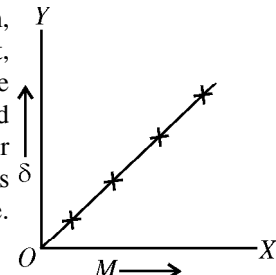


Fig. 4.51.

Ex. 1. The depression at the free end of a cantilever of length 1.5 m is 10 cm when a certain load is suspended at this end. What is the depression at a distance 0.5 m from the fixed end?

Sol. Here, $l = 1.5$ m, $\delta = 10$ cm = 0.1 m, $x = 0.5$ m, $y = ?$

$$\text{Depression at the free end } \delta = \frac{Wl^3}{3YI}, \quad \therefore \frac{W}{YI} = \frac{3\delta}{l^3}$$

Depression at a distance x from the fixed end

$$y = \frac{Wx^2}{2YI} \left(l - \frac{x}{3} \right) = \frac{W}{YI} \frac{x^2(3l-x)}{2 \times 3} = \left(\frac{3l-x}{2l^3} \right) x^2 \delta$$

$$\therefore y = \frac{(3 \times 1.5) - 0.5}{2 \times (1.5)^3} \times (0.5)^2 \times 0.1 = \mathbf{0.0148 \text{ m.}}$$

Ex. 2. A steel wire of 1 mm radius is bent to form a circle of 10 cm radius. What is the bending moment and the maximum stress, if $Y = 2 \times 10^{11} \text{ N-m}^2$.

Sol. Bending moment = YI/R

Here, $I = \pi r^4/4 = 3.14 \times (10^{-3})^4/4$.

$$\therefore \text{Bending moment} = \frac{2 \times 10^{11} \times 3.14 \times (10^{-3})^4}{4 \times 0.1} = \mathbf{1.57 \text{ N-m.}}$$

$$\text{Stress} = Yz/R$$

For maximum stress the value of z should be maximum, i.e., $z = r$.

$$\therefore \text{Maximum stress} = \frac{Yr}{R} = \frac{2 \times 10^{11} \times (10^{-3})}{0.1} = \mathbf{2 \times 10^9 \text{ N-m}^{-2}}.$$

Ex. 3. A cantilever of length l and uniform cross-section shows a depression of 2 cm at the loaded end. What will be the depression at distance $l/4$, $l/2$ and $3l/4$ from the fixed end ?

$$\text{Sol.} \quad y = \frac{Mgx^2}{2YI} \left(l - \frac{x}{3} \right)$$

When $x = l$, $y = 0.02 \text{ cm}$,

$$\therefore 0.02 = \frac{Mgl^2}{2YI} \left(l - \frac{l}{3} \right) = \frac{Mgl^3}{3YI}$$

$$\begin{aligned} \text{When } x = l/4, \quad y_1 &= \frac{Mg}{2YI} \left(\frac{l}{4} \right)^2 \left(l - \frac{l}{12} \right) = \frac{Mgl^2}{32YI} \cdot \frac{11}{12} = \frac{11}{128} \cdot \frac{Mgl^3}{3YI} \\ &= \frac{11}{128} \times 0.02 = \mathbf{1.72 \times 10^{-3} \text{ m.}} \end{aligned}$$

$$\begin{aligned} \text{When } x = l/2, \quad y_2 &= \frac{Mg}{2YI} \left(\frac{l}{4} \right)^2 \left(l - \frac{l}{6} \right) = \frac{Mgl^2}{8YI} \cdot \frac{5}{6} = \frac{5}{16} \cdot \frac{Mgl^3}{3YI} \\ &= \frac{5}{16} \times 0.02 = \mathbf{6.2 \times 10^{-3} \text{ m.}} \end{aligned}$$

$$\begin{aligned} \text{When } x = \frac{3l}{4}, \quad y_3 &= \frac{Mg}{2YI} \left(\frac{3l}{4} \right)^2 \left(l - \frac{l}{4} \right) = \frac{Mgl^2}{2YI} \cdot \frac{27}{64} = \frac{Mgl^3}{3YI} \cdot \frac{81}{128} \\ &= \frac{81}{128} \times 0.02 = \mathbf{1.27 \times 10^{-2} \text{ m.}} \end{aligned}$$

Ex. 4. A 10 cm wide and 0.2 mm thick metal sheet is bent to form a cylinder of 10 cm length and 50 cm radius. If the Young's modulus of the metal is $1.5 \times 10^{11} \text{ N-m}^{-2}$ calculate (i) the stress and strain on the convex surface and (ii) the bending moment.

$$\text{Sol. (i)} \quad \text{Strain} = \frac{z}{R} = \frac{2 \times 10^{-4}}{2} \times \frac{1}{0.5} = \mathbf{2 \times 10^{-4}} \quad [\because \text{for convex surface } z = \text{half thickness of the sheet}]$$

$$\text{Stress} = Y \times \text{strain} = 1.5 \times 10^{11} \times 2 \times 10^{-4} = 3.0 \times 10^7 \text{ N-m}^{-2}$$

$$(ii) \quad \text{Bending moment} = \frac{YI}{R} = \frac{Y}{R} \cdot \frac{bd^3}{12} = \frac{1.5 \times 10^{11}}{0.5} \times \frac{0.1 \times (2 \times 10^{-4})^3}{12} = 2 \times 10^{-2} \text{ N-m.}$$

Ex. 5. A light metal rod of length 60 cm and of radius 1 cm is clamped at one end loaded at the free end, with 5.5 kg. Calculate the depression of the free end, assuming $Y = 9 \times 10^{11}$ dynes/sq. cm. and $g = 980 \text{ cm/sec}^2$. **(L.A.S.)**

Sol. At the free end, $\delta = \frac{Wl^3}{3YI}$

For a circular cross-section of radius r , $I = \pi r^4/4$

$$\therefore \delta = \frac{4Wl^3}{3Y\pi r^4}$$

Here, $W = 5.5 \times 9.8 \text{ N}$, $l = 0.6 \text{ m}$, $Y = 9 \times 10^{10} \text{ N/m}^2$ and $r = 0.01 \text{ m}$.

$$\text{Therefore, } \delta = \frac{4 \times 5.5 \times 9.8 \times (0.6)^3}{3 \times 9 \times 10^{10} \times 3.14 \times (0.01)^4} = 5.495 \times 10^{-3} \text{ m.}$$

Ex. 6. A vertical steel rod of circular cross-section of radius 1 cm is rigidly fixed in the earth. Its upper end is 3 metre from the earth. A thick string which can stand a maximum tension of 2 kg-wt. is tied at the upper end of the rod and pulled horizontally. Find how much the top will be deflected before the string snaps. (Y for steel $= 2 \times 10^{11}$ S.I. units; $g = 10 \text{ m/sec}^2$).

Sol. The lower end of a vertical rod is fixed and the upper end is pulled by a horizontal force, hence this is the case of a cantilever. [fig. 4.42]

The deflection of the upper end, is given by

$$\delta = \frac{Wl^3}{3YI}$$

Here, the horizontal force $W = 2 \times 10 \text{ N/m}^2$; length of the rod $l = 3 \text{ m}$; value of $Y = 2 \times 10^{11} \text{ N/m}^2$.

$$\text{For circular cross-section } I = \frac{\pi r^4}{4} = \frac{\pi \times (0.01)^4}{4} = \frac{\pi \times 10^{-8}}{4}$$

$$\therefore \delta = \frac{2 \times 10 \times (3)^3 \times 4}{3 \times 2 \times 10^{11} \times 3.14 \times 10^{-8}} = 0.1146 \text{ m.}$$

Ex. 7. Compare loads required to produce equal depression for two beams, made of the same material and having the same length and weight with the only difference that while one has circular cross-section, the cross-section of the other is square.

Sol. The depression δ at the middle point of a beam of length l of circular cross-section of radius r , is given by

$$\delta = \frac{W_1 l^3}{12Y\pi r^4}, \text{ where } W_1 \text{ is the weight loaded.} \quad \dots(i)$$

If in case of rectangular cross-section beam the weight loaded be W_2 , then the depression δ' , is given by

$$\delta' = \frac{W_2 l^3}{4bd^3Y}$$

since the two beams have the same length and are made of same material.

But for square cross-section $b = d$

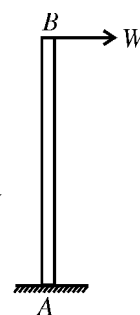


Fig. 4.51.

$$\therefore \delta' = \frac{W_2 l^3}{4b^4 Y} \quad \dots(ii)$$

The two beams have equal weights and made of the same material, therefore their masses and hence volume must be equal, i.e., $\pi r^2 l = b^2 l$ or $b^2 = \pi r^2$

Substituting the value of b^2 in (ii), we get

$$\delta' = \frac{W_2 l^3}{4\pi^2 r^4 Y} \quad \dots(iii)$$

But depression is the same in both cases, i.e., $\delta = \delta'$. Therefore equating (i) and (iii), we get

$$\frac{W_1 l^3}{12Y\pi r^4} = \frac{W_2 l^3}{4\pi^2 r^4 Y} \text{ or } \frac{W_1}{W_2} = \frac{3}{\pi}, \text{ which is the required ratio.}$$

Ex. 8. A cantilever of length l is loaded at a point x_1 , calculate the depression of the tip of the cantilever. If the same load is shifted to the tip, what is the depression of the point x_1 ?

Sol. The expression for the depression y at a distance x from the fixed end, is given by

$$y = \frac{W}{YI} \left(\frac{lx^2}{2} - \frac{x^3}{6} \right) \quad \dots(i)$$

$$\text{Also at } x, \quad \frac{dy}{dx} = \tan \theta = \frac{W}{YI} \left(lx - \frac{x^2}{2} \right) \quad \dots(ii)$$

In the first case, the cantilever is loaded at $OP = x_1$. [fig. 4.52] Hence the beam will not bend beyond x_1 by the load W , but the portion $P'Y$ will be tangent at P' and the total depress $\delta = XY$, will be given by

$$\delta = XX' + X'Y = PP' + P'X' \tan \theta$$

$$\text{Hence } PP' = \frac{Wx_1^3}{3YI}, \quad P'X' = l - x_1 \text{ and } \tan \theta$$

$$= \frac{W}{YI} \left(x_1^2 - \frac{x_1^2}{2} \right) = \frac{Wx_1^2}{2YI}$$

$$\therefore \delta = \frac{Wx_1^3}{3YI} + (l - x_1) \frac{Wx_1^2}{3YI} = \frac{W}{YI} \left(\frac{lx_1^2}{2} - \frac{x_1^3}{6} \right) = \frac{Wx_1^2}{2YI} \left(l - \frac{x_1}{3} \right) \quad \dots(iii)$$

This is the depression of the tip of the cantilever.

In the second case, the same load is shifted to the tip, then the depression at any point x_1 , is given by

$$\delta' = \frac{W}{YI} \left(\frac{lx_1^2}{2} - \frac{x_1^3}{6} \right) = \frac{Wx_1^2}{2YI} \left(l - \frac{x_1}{3} \right) \quad \dots(iv)$$

We see that in two cases the depression is the same.

Ex. 9. A steel strip is clamped horizontally from one end. On applying a 500 gm load at the free end the bending in equilibrium state is 5 cm. Calculate, (i) the potential energy in the strip, (ii) the frequency of vibration if the load is disturbed from equilibrium (neglect mass of the strip itself).

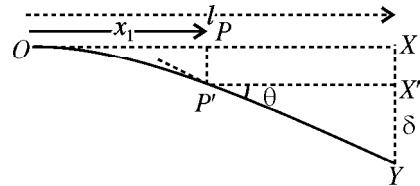


Fig. 4.52.

Sol. (i) The bending is 5 cm by the load 500 gm load = 0.05×9.81 newton, hence force per unit bending or force constant is

$$C = \frac{0.5 \times 9.81}{0.05} = 98.1 \text{ N-m}^{-1}$$

$$\therefore \text{Potential energy} = \int_0^x Cx \, dx = \frac{1}{2} Cx^2 = \frac{1}{2} \times 9.81 \times (0.05)^2 = 0.12 \text{ joules.}$$

$$(ii) \quad \text{Frequency, } n = \frac{1}{2\pi} \sqrt{\frac{C}{M}} = \frac{1}{2\pi} \sqrt{\frac{9.81}{0.5}} = 2.23 \text{ sec}^{-1} \text{ (nearly).}$$

Ex. 10. A solid cylindrical rod of radius 6 mm bends by 8 mm under a certain load. Deduce the bending when all other things remaining the same the beam is replaced by a hollow cylindrical one with external radius 10 mm, internal 8 mm.

Sol. If δ and δ' are the depressions of the two beams, then

$$\delta = \frac{Mgl^3}{3YI} \text{ and } \delta' = \frac{Mgl^3}{3YI'}; \delta' = \frac{I}{I'} \delta$$

$$\text{For solid beam, } I = \pi r^4/4 \text{ and for hollow beam, } I' = \frac{\pi(r_2^4 - r_1^4)}{4},$$

$$\therefore \delta = \frac{6^4}{10^4 - 8^4} \times 8 \times 10^{-3} = 1.76 \times 10^{-3} \text{ m.}$$

Ex. 11. A rod of width 2 cm and thickness 3 mm supported horizontally on two knife-edges 80 cm apart is loaded with weights of 10 gm, hung from its ends, which project 15 cm beyond the knife edges. If the centre of the lath is thereby elevated 2 mm, calculate Young's modulus for the material.

Sol. Let the beam be placed on two knife edges P and Q . It takes the shape as shown in the figure, when loaded at both ends by 0.01 g newton. [fig. 4.53]

$$\therefore \text{Couple acting at } P = 0.01 \text{ g} \times 0.15 \text{ n-m} = \frac{YI}{R}.$$

$$\begin{aligned} \text{Here, } I &= \frac{bd^3}{12} = \frac{0.02 \times (3 \times 10^{-3})^3}{12} \\ &= 4.5 \times 10^{-10}. \end{aligned}$$

Consider the lath is bent into a circle of radius of curvature R , then from the geometry of the circle.,

$$PO \cdot OQ = (2R - OO')OO'$$

$$\text{or } 2 \times 10^{-3} \times 2R = 0.4 \times 0.4$$

$$\text{whence } R = 40 \text{ m}$$

$$\text{Now, } \frac{YI}{R} = \frac{Y \times 4.5 \times 10^{-10}}{10}$$

$$\frac{Y \times 4.5 \times 10^{-10}}{10} = 0.01 \times 9.81 \times 0.15$$

$$\begin{aligned} \text{or } Y &= \frac{4.0 \times 0.01 \times 9.81 \times 0.15}{4.5 \times 10^{-10}} \\ &= 1.308 \times 10^{10} \text{ n-m}^{-2}. \end{aligned}$$

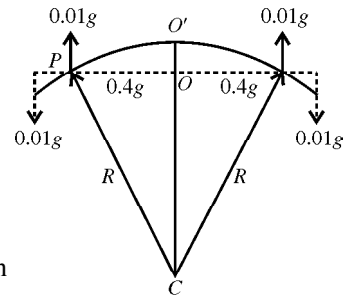


Fig. 4.53.

$$[\because 2R \gg OO']$$

QUESTIONS

Long Answer Questions

1. Explain the meaning of stress and strain. State Hooke's law and hence define Young's modulus, bulk modulus, modulus of rigidity and Poisson's ratio of an isotropic and homogeneous material.
2. Show that the shearing stress is equivalent to an equal linear tensile stress and an equal compressive stress at right angles to each other. Prove that $Y = 3K(1 - 2\sigma)$.
3. Define stress and strain. Derive the relations between the elastic moduli. (S.V.U. 2002)
4. Define 'Young's modulus', 'Bulk modulus' and 'Modulus of rigidity'. Derive the relation between them. (S.V.U. 2005; O.U. 2009)
5. Define the elastic moduli. Derive the relation between them. (K.U. 2007)
6. Explain the elastic constant of isotropic solids. Obtain a relation between them. (A.U. 2007)
7. Define elastic constants of isotropic solids and obtain the relationships between them. (S.V.U. 2007)
8. Define three moduli of elasticity. Derive the relations among the elastic constants Y , K , η and σ . (K.U. 2005; S.V.U. 2003; A.N.U. 2002; S.K.U. 2002, 01)
9. Define the three moduli of elasticity and obtain the relation between them. Discuss the limit of Poisson's ratio (σ). (K.U. 1996)
10. Prove the following relations :

$$Y = 3K(1 - 2\sigma) = 2\eta(1 + \sigma) = \frac{9K\eta}{3K + \eta} \quad \text{and} \quad \sigma = \frac{3K - 2\eta}{6K + 2\eta}. \quad (\text{O.U. 1995})$$

11. (a) Show that the potential energy per unit volume of a strained wire is

$$U = \frac{1}{2} \times \text{Stress} \times \text{Strain}.$$

- (b) Find the work done in stretching a wire.
- (c) Show that the theoretical limiting values of Poisson's ratio are -1 and 0.5 .
12. What do you understand by shearing force and bending moment ? Explain. Show that at a section of a beam shearing force is equal to the rate of change of bending moment.
13. What is a beam ? Explain the meaning of the following terms :
(i) Neutral layer, (ii) Plane of bending, (iii) Bending moment.
14. What do you understand by terms neutral surface, plane of bending, neutral axis and bending moment ? Derive an expression for the moment of couple required to bend uniform metallic bar into an arc of a circle of small curvature.
15. What is bending moment ? Find an expression for the depression at a point distant x from the fixed end of a cantilever.
16. Explain bending moment. Derive an expression for the bending moment for a beam of rectangular cross-section. (A.U. 2007)
17. What is bending moment ? Deduce an expression for it if the beam is of : (i) rectangular, (ii) circular, cross-section.
18. What is a cantilever ? Derive an expression for the depression of the mid point of a beam, supported at the ends and loaded at the middle. (O.U. 2008; K.U. 2005)
19. Show that the depression at the free end of a light cantilever, when the end is loaded is proportional to the cube of its length.
20. Show that loads, required to produce equal depressions for two beams made of the same material and having the same length and weight with the only difference that they are with circular and rectangular cross-sections respectively, are in the ratio $3 : \pi$.

21. What are shearing force and bending moment. Show that the shearing force and bending moment at a section of a beam are related as $F = dM/dx$.
22. What are uniform and non-uniform bending of beams ? Discuss the examples of uniformly bent beam and non-uniformly bent beam.
23. A simply supported beam of span l carries a point load W in the middle. Find the reactions at the supports. Determine also shearing force and bending moment at a section of the beam. Draw shearing force and bending moment diagrams. Assume the weight of the beam to be negligible.

Short Answer Questions

1. Define stress, strain and modulus of elasticity.
2. State Hooke's law and define different moduli of elasticity. (O.U. 2001)
3. Define Young's, bulk, rigidity moduli and Poisson's ratio. Write a note on elastic constants of isotropic solid. (S.K.U. 2007)
4. Define the four elastic constants of isotropic solids. (A.N.U. 2007)
5. Explain briefly about elastic constants. (S.K.U. 2005)
6. Give a brief account of classification of beams. (A.U. 2005)
7. What are shearing force and bending moment ?
8. Explain sign conventions in bending of beams. (A.U. 2007, 05)
9. Explain the uniform and non-uniform bending beams. (K.U. 2005)
10. Classify different types of beams. (O.U. 2005)
11. What are different types of loads ?
12. What are point load and distributed load ?
13. What is cantilever ? (A.U. 2007)
14. Explain steel is more elastic than rubber.
15. Why girders for supporting roofs are formed in the shape of **I**.
16. A cantilever of uniform section is more likely to break near its fixed end.
17. What are the theoretical limits of Poisson's ratio for an isotropic and homogeneous body ?
[Ans. $-1 < \sigma < 0.5$.]
18. What is Poisson's ratio ? Give its limiting values.
19. Obtain in terms of bending moment M , an expression for longitudinal stress in a beam at a distance z from the neutral axis.
20. Compare loads required to produce equal depressions for two beams of same material, length and weight when one has a circular cross-section and other has square cross-section.

PROBLEMS**[Elastic Constants and Their Inter-relations]**

1. A wire 3.0 m long and 0.625 sq. cm in cross-section is found to stretch by 0.3 cm under a tension of 1200 kg. What is Young's modulus of the material of the wire ?
[Ans. 1.88×10^{11} N/m².]
2. One end of a wire 2 m long and 0.2 cm² in cross-section is fixed in a ceiling and a load of 4.8 kg is attached to the free end. Find the extension of the wire. Young's modulus of steel = 2.0×10^{11} N/m². Take $g = 10$ m/s².
[Ans. 2.4×10^{-5} m.]
3. A steel wire of 2 mm diameter is just stretched between two fixed points at a temperature of 20°C. Determine the tension when the temperature falls to 10°C. Coefficient of linear expansion of steel is 0.000011 per °C and Young's modulus for steel is 2.1×10^{11} newton/m².
[Ans. 72.5 N.]

4. Young's modulus of a metal is $2 \times 10^{11} \text{ N/m}^2$ and Poisson's ratio is 0.25. Calculate the bulk modulus.
[Ans. $1.33 \times 10^{11} \text{ N/m}^2$.]
5. A steel wire of length 2.0 m and cross-section $1 \times 10^{-6} \text{ m}^2$ is held between rigid supports with a tension 200 newton. If the middle point of the wire is pulled 5 mm sideways, calculate the change in tension. Also calculate the change in tension if temperature changes by 5°C . [For steel $Y = 2.2 \times 10^{11} \text{ N-m}^2$ and $\alpha = 8.8 \times 10^{-6}/^\circ\text{C}$]
[Ans. 2.75 newton; 8.8 newton.]
6. Two wires A and B are made of same material. The length of the wire A is half the length of wire B and diameter of wire A is the double the diameter of wire B. If same weight is suspended from each of them, compare the increase in their lengths.
[Ans. $l_a/l_b = 1/8$.]
7. The breaking stress of a material is 10^9 N/m^2 and its density is $3 \times 10^3 \text{ kg/m}^3$. Calculate the length of its wire which breaks due to its own weight when suspended.
[Ans. $3.4 \times 10^4 \text{ m}$.]
8. The Young's modulus and bulk modulus for steel are respectively $2 \times 10^{11} \text{ N/m}^2$ and $11 \times 10^{10} \text{ N/m}^2$. Calculate the Poisson's ratio.
[Ans. $\sigma = 0.197$.]
9. Find the formula for the work done in stretching a wire and apply it to find the elastic energy stored up in wire, originally 5 metres long and 1 mm in diameter, which has been stretched by 3/10 mm due to a load of 10 kg. [Take $g = 9.8 \text{ ms}^{-2}$.]
[Ans. 0.0147 joules.]
10. A steel wire of length 2 m is stretched through 2 mm. The cross-sectional area of the wire is 4 mm. Calculate the elastic potential energy stored in the stretched wire. Y for steel = $2 \times 10^{11} \text{ N/m}^2$.
[Ans. 0.8 joule.]
11. A wire 4 metres long, 0.3 mm in diameter is stretched by a force of 8000 gm wt. If the extension in the length amounts to 1.5 mm, calculate the energy stored in the wire.
[Ans. $5.88 \times 10^{-3} \text{ J}$.]
12. A wire of length 2 m and area of cross-section 10^{-6} m^2 is stretched to elongate by 10^{-4} m . Calculate the internal energy of the wire. ($Y = 2 \times 10^{11} \text{ N/m}^2$)
[Ans. $5 \times 10^{-4} \text{ J}$.]

[Bending of Beam]

13. A cantilever of length 50 cm is depressed by 15 mm at the loaded end. Calculate the depression at distance 30 cm from the fixed end.
[Ans. 6.48 mm.]
14. A cylindrical rod of length 1 m and radius 1 cm is uniformly bent into a circular arc of radius 10 m. Calculate the bending moment. (Young's modulus for the material of the rod = $9.1 \times 10^{10} \text{ N-m}^{-2}$)
[Ans. 7.14 N-m.]
15. A steel rod of length 50 cm, width 2 cm and thickness 1 cm is bent in the form of an arc of radius 200 cm. Calculate (i) the stress and strain on the convex surface of the rod, and (ii) the bending moment. ($Y = 2 \times 10^{11} \text{ N-m}^{-2}$.)
[Ans. (i) $2 \times 10^9 \text{ N-m}^{-2}$, 0.01; (ii) 167 N-m.]
16. A 10 cm wide and 0.2 mm thick metal sheet is bent to form a cylinder of 80 cm length and 50 cm

radius. If the Young's modulus of the metal is $1.5 \times 10^{11} \text{ N-m}^{-2}$, calculate (i) the stress and strain on convex surface; (ii) bending moment.

[Ans. (i) $3 \times 10^7 \text{ N-m}^{-2}$, 2×10^{-4} ; (ii) 0.02 N-m.]

17. A rod of rectangular section having thickness and breadth each of 0.5 cm is bent in the form of an arc having radius of curvature 1000 cm. If the Young's modulus of the material of the rod is 10^{11} N-m^{-2} , calculate (i) the stress and strain on convex surface. (ii) bending moment.

[Ans. (i) $2.5 \times 10^7 \text{ N-m}^{-2}$, 2.5×10^{-4} ; (ii) 0.52 N-m.]

18. A steel wire of 1.00 mm radius is bent in the form of a circular arc of radius 50 cm Calculate : (i) the bending moment, (ii) the maximum stress. ($Y = 2 \times 10^{11} \text{ N-m}^{-2}$)

[Ans. (i) 0.314 N-m; (ii) $4 \times 10^8 \text{ N-m}^{-2}$.]

19. The end of a given strip cantilever depresses 10 mm under a certain load. Calculate the depression under same load for another cantilever of same material 2 times in length, 2 times in width and 3 times in thickness (vertical).

[Ans. 1.48 mm.]

20. Show that the longitudinal stress at a distance z from the neutral surface of a beam bent to a radius of curvature R is Yz/R . Where is the tensile stress maximum in a beam, supported at two ends and loaded at the middle ?

[Ans. The lower surface in the middle region.]

21. One end of a 100 cm long rod is fixed horizontally. When a load of 500 gm is suspended at the other end, it is depressed through 5 mm, find the depression at 40 cm distance from the fixed end when a 5 kg load is suspended at (i) the free end, (ii) 40 cm from the fixed end.

[Ans. (i) 1.04 cm; (ii) 0.3 cm.]

22. For the same mass per unit length, show that a beam of square section is stiffer than one of circular section, the deflections being in the ratio 3 : π .

23. A beam one metre long, 4 cm wide, 5 mm thick is supported on two knife-edges 80 cm apart. The depression produced by a 2 kg load hung from the centre of the knife edges is 5 mm. Find Young's modulus of the material of the beam.

[Ans. $9.6 \times 10^{10} \text{ N-m}^{-2}$.]

24. A cylindrical rod of iron AB , values 0.5 cm and length 25 cm is clamped at A so that its axis is horizontal. Find the deflections of the free end of the rod when a force of 3 kg-wt is applied : (i) at B , (ii) at the point midway between A and B . ($Y = 21 \times 10^{11} \text{ dyne/cm}^2$).

[Ans. (i) 0.467 cm; (ii) 0.145 cm.]

25. A cantilever of uniform cross-section and length l shows a depression δ at the loaded end. What will be the depression at a distance $l/2$ from the fixed end ?

[Ans. $\frac{5}{16} \delta$.]

26. A uniform light rod 120 cm long is clamped horizontally at one end. A weight of 100 gm is attached to the free end. Calculate the depression of a free point 90 cm distance from the clamped end. The diameter of the rod is 2 cm. ($Y = 1.013 \times 10^{11} \text{ dyne/cm}^2$, $g = 980 \text{ cm/sec}^2$.)

[Ans. 0.4492 cm.]

27. A cantilever of length 1.0 m depresses by 5 mm when a load of 0.5 kgf is suspended at its free end. Now a load of 5 kgf is suspended (i) at the free end; (ii) at distance 40 cm from the fixed end. When will be the depression at distance 40 cm from the fixed end ?

[Ans. (i) 1.04 cm; (ii) 0.32 cm.]

28. A simply supported beam at its ends, having of span 7 m, carries point loads of 4 kN and 6 kN at the distances of 3 m and 5 m from left end. Calculate the reactions at the supports.

[Ans. 4 kN and 6 kN.]

29. For a simply supported beam AB as shown in fig. 4.44, find the reactions at the supports and draw shear force and bending moment diagrams. Take origin at A and measure x along AB .

[Ans. $R_A = 2.5$ kN, $R_B = 2.5$ kN; for $0 < x < 2$ m, $F = 2.5$ kN and $M = 2.5x$ kN and for $2\text{ m} < x < 4$ m, $F = -2.5$ kN and $M = (10 - 2.5x)$ kN-m; shear force and bending moment diagram is shown in fig. 4.45]

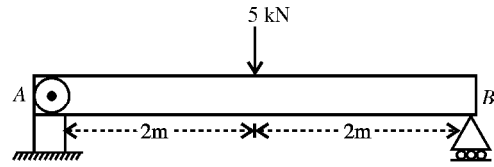


Fig. 4.54.

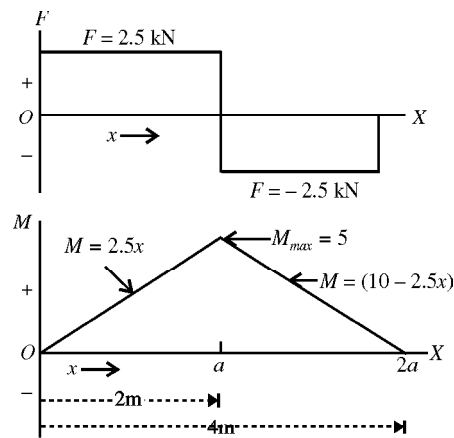


Fig. 4.55.

5

Central Forces

5.1. Central Force

When a force acts on a particle in such a way that it is always directed towards or away from a fixed point and its magnitude depends only upon the distance (r) from the point, then such a force is called a **central force**. Thus, a central force is represented by

$$\vec{F} = F(r) \hat{r} = F(r) \frac{\vec{r}}{r} \quad \dots(1)$$

where $F(r)$ is any function of distance r and $\hat{r} = \vec{r}/r$ is a unit vector along $\vec{r}$ from the fixed centre. The force $\vec{F}$ is attractive or repulsive, if $F(r) < 0$ or $F(r) > 0$ respectively (**fig. 5.1**).

A central force, frequently occurring in practice, is represented by

$$F(r) = -\frac{K}{r^n} \quad \dots(2)$$

where K and n are constants. This is known as the **n th power law of force**.

Now, if $K > 0$, the force is attractive, and if $K < 0$, the force is repulsive. Important cases corresponding to $K \gtrless 0$ are the following :

- (i) When $n = 2$, K may be + ve or - ve,

$$F(r) = -\frac{K}{r^2} \quad \dots(3)$$

This is the **inverse square law force**. In vector form, this force can be represented as

$$\vec{F} = -\frac{K}{r^2} \hat{r} \quad \dots(4)$$

- (ii) When $n = -1$, $K > 0$,

$$F(r) = -Kr \quad \dots(5)$$

This represents a linear restoring force as in case of simple harmonic motion. In one dimension, say along X -axis, this force has the form :

$$F_{(x)} = -Kx \quad \dots(6)$$

In vector form a linear restoring force will be represented as

$$\vec{F} = -Kx \hat{x} \quad \dots(7)$$

where x is the instantaneous displacement of a particle (P) from a fixed point O along X -axis and the $\vec{F} = -Kx \hat{x}$ is the linear restoring force acting on it. (**fig. 5.2**)

Examples of Central Forces

Important examples of central forces are gravitational force between two masses, electrostatic force between two charges and elastic force in spring-mass system. A brief discussion about these forces is given below :

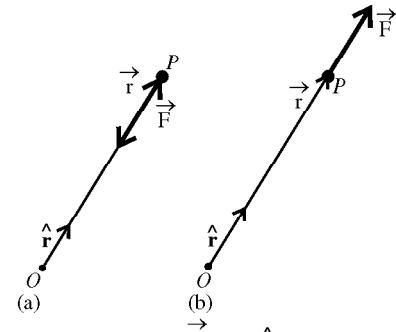


Fig. 5-1. Central force $\vec{F} = F(r) \hat{r}$: (a) $F(r) < 0$, force is attractive, directed towards the centre O , (b) $F(r) > 0$, force is repulsive directed away the centre O .

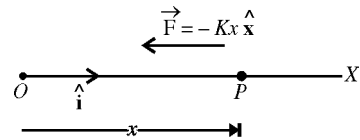


Fig. 5-2. Linear restoring force.

(1) Gravitational force of attraction between two masses : According to Newton's law of universal gravitation, the force of gravitational attraction between any two masses

m_1 and m_2 , separated by a distance $\vec{r}$, is given by (fig. 5.3)

$$\vec{F} = -\frac{Gm_1m_2}{r^2} \hat{r} \quad \dots(8)$$

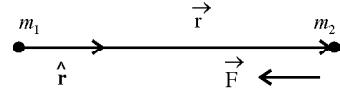


Fig. 5.3. Gravitational force of attraction between two masses.

where $\vec{r}$ is the position vector from m_1 to m_2 , $\hat{r} = \vec{r}/r$ and G the constant of gravitation. In S.I. system, $G = 6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$.

Gravitational force is an important example of central (inverse square law) force with $F(r) = -Gm_1m_2/r^2$ or $F(r) = -K/r^2$ ($K = Gm_1m_2$). The earth moves around Sun under a central force.

(2) Electrostatic force between two charges : According to Coulomb's law, the electrostatic force between two charges q_1 and q_2 , separated by a distance $\vec{r}$, is given by

$$\vec{F} = \frac{1}{4\pi\epsilon_0} \frac{q_1q_2}{r^2} \hat{r} \quad \dots(9)$$

where $\vec{r}$ is the position vector from q_1 to q_2 , $\hat{r} = \vec{r}/r$ and $1/4\pi\epsilon_0$ is the constant, known as the permittivity of the free space, with value $1/4\pi\epsilon_0 = 9 \times 10^9$ S.I. unit.

Electrostatic force is another important example of central (inverse square law) force with $F(r) = q_1q_2/4\pi\epsilon_0r^2$ or $F(r) = -K/r^2$ ($K = -q_1q_2/4\pi\epsilon_0$). In hydrogen atom, an electron moves around proton (nucleus) under central force.

(3) Elastic force in spring-mass system : Consider a spring whose one end is fixed to a rigid support and the other end is attached to a mass m . Let this spring-mass system is placed on a horizontal frictionless table or suspended freely (fig. 5.4). Now, if the spring be extended or compressed by a distance x , the elastic force acting on the mass m , being in a direction opposite to the displacement, is given by

$$F = -kx \text{ or } \vec{F} = -k\vec{x} \quad \dots(10)$$

where k is called **force constant** of the spring.

This elastic force is a linear restoring force and is also an important example of central force.

Important Features of Central Force

(1) The general form of the central force is

$$\vec{F} = F(r) \hat{r}$$

where $F(r)$ is a scalar function of distance r . The force $\vec{F}$ is attractive or repulsive according to $F(r) < 0$ or $F(r) > 0$. Most important examples of central forces are the inverse square law force (gravitational and electrostatic forces) and linear restoring force.

(2) Central force is a conservative force in nature.

(3) When a particle is moving under central force, the torque acting on it is zero ($\vec{\tau} = d\vec{L}/dt = 0$) i.e., the angular momentum of the particle is conserved.

(4) For a particle moving under central force, the areal velocity of the particle is constant.

(5) When a particle is moving under central force, the motion of the particle takes place in a plane.

Theorem 1. When a particle moves under the action of a central force, its angular momentum is conserved.

Proof : A central force is represented by

$$\vec{F} = F(r) \hat{r} = F(r) \frac{\vec{r}}{r} \quad \dots(i)$$

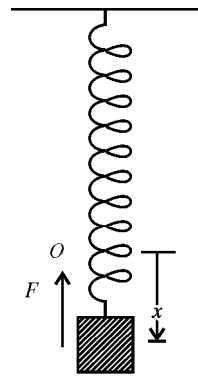


Fig. 5.4. Elastic force in spring-mass system.

where $F(r)$ is a scalar function of distance r and $\hat{r} = \vec{r}/r$.

When a particle is moving under the action of the central force, given by (i), the torque acting on it is given by

$$\vec{\tau} = \frac{d\vec{L}}{dt} = \vec{r} \times \vec{F} = \vec{r} \times F(r) \frac{\vec{r}}{r} = \frac{F(r)}{r} \vec{r} \times \vec{r} = 0 \quad [\because \vec{r} \times \vec{r} = 0] \dots(ii)$$

where $\vec{L}$ is the angular momentum about the origin.

Integration of (ii) gives

$$\vec{L} = \text{constant (vector)} \dots(iii)$$

Thus, when a particle moves under the action of a central force, its angular momentum ($\vec{L}$) is conserved, i.e., $\vec{L}$ remains the same in the magnitude and direction.

Theorem 2. When a particle moves under the influence of a central force, the motion takes place in a plane.

Proof : A central force is given by

$$\vec{F} = F(r) \hat{r} = F(r) \frac{\vec{r}}{r} \dots(i)$$

where $F(r)$ is a scalar function of distance.

$$\text{Torque on the moving particle, } \vec{\tau} = \frac{d\vec{L}}{dt} = \vec{r} \times \vec{F} = \vec{r} \times F(r) \frac{\vec{r}}{r} = 0 \dots(ii)$$

where $\vec{L} = \vec{r} \times \vec{p}$ is the angular momentum of the moving particle about the origin. Here $\vec{r}$ is the position vector $\vec{p} = m\vec{v}$ is the linear momentum [fig. 5.5].

Integration of (ii) gives

$$\vec{L} = \text{constant (vector)} \dots(iii)$$

But

$$\vec{L} = \vec{r} \times \vec{p} \dots(iv)$$

where $\vec{r}$ is the position vector and $\vec{p}$ is the linear momentum [fig. 5.5].

Taking dot product with $\vec{r}$ of both sides of eq. (iv), we have

$$\vec{r} \cdot \vec{L} = \vec{r} \cdot (\vec{r} \times \vec{p}) = (\vec{r} \times \vec{r}) \cdot \vec{p} = 0$$

because in a scalar triple product the position of dot and cross may be interchanged and $\vec{r} \times \vec{r} = 0$.

Therefore, $\vec{r}$ is always perpendicular to the constant vector $\vec{L}$ [fig. 5.5], i.e., the motion of a particle takes place in a plane for a central force.

5.2. Conservative Nature of Central Forces

Conservative force : A force is said to be conservative if the work done by it in moving a particle from one point to another point depends only on these points and not on the path followed. This means that if a particle is taken from the point A to the point B through APB or AQB paths and if each time the

amount of work done $\left(\int_A^B \vec{F} \cdot d\vec{r} \right)$ by the force is the same, then this force is conservative. Thus, for a conservative force,

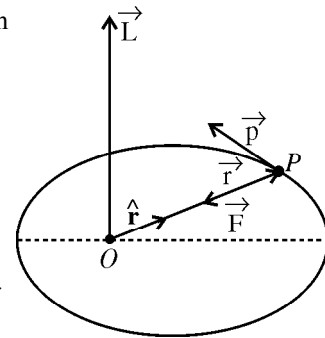


Fig. 5.5.

$$\int_A^B \vec{F} \cdot d\vec{r} = \int_A^B \vec{F} \cdot d\vec{r} \quad \dots(1)$$

(Path APB) (Path AQB)

Paths APB and AQB are any two arbitrary paths, joining the points A and B .

The region in which a particle experiences a conservative force is called a **conservative force field**.

Central Force is a Conservative Force

Central force is a conservative force, if a force acts on a particle in such a way that it is always directed towards or away from a point and its magnitude depends only upon the distance r from the point, then this force is called a **central force**.

In **fig. 5.6**, a central force has been shown to act away from (or toward) the centre. In the central force field* any two arbitrary paths AOB and AQB have been drawn between two points A and B . Taking centre O , two dashed sectors of circles have been drawn with radii r and $r + dr$ respectively. Now take two points P and Q on APB and AQB paths respectively, lying between the two sectors [**fig. 5.6**]. Let $\vec{F}_1$ and $\vec{F}_2$ be the central forces at P and Q and let $d\vec{r}_1$ and $d\vec{r}_2$ be the displacements of the particle between two sectors along two paths respectively. If θ_1 and θ_2 are the angles between $\vec{F}_1$, $d\vec{r}_1$ and $\vec{F}_2$, $d\vec{r}_2$ respectively, then

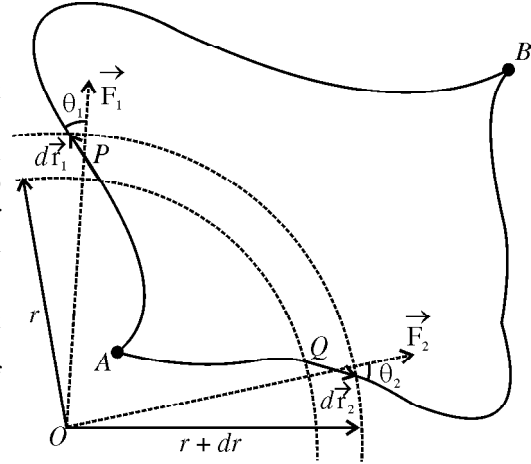


Fig. 5.6. Central force is conservative.

$$\vec{F}_1 \cdot d\vec{r}_1 = F_1 dr_1 \cos \theta_1 \quad \text{and} \quad \vec{F}_2 \cdot d\vec{r}_2 = F_2 dr_2 \cos \theta_2$$

As $\vec{F}_1$ and $\vec{F}_2$ lie at equal distances from the centre O , their magnitudes F_1 and F_2 are **equal**. Also, the separation of the circles measured along the direction of $\vec{F}_1$ is equal to the separation measured along $\vec{F}_2$, hence the projections of $d\vec{r}_1$ and $d\vec{r}_2$ on $\vec{F}_1$ and $\vec{F}_2$ respectively are equal, i.e., $dr_1 \cos \theta_1 = dr_2 \cos \theta_2$. Thus

$$F_1 dr_1 \cos \theta_1 = F_2 dr_2 \cos \theta_2 \quad \text{or} \quad \vec{F}_1 \cdot d\vec{r}_1 = \vec{F}_2 \cdot d\vec{r}_2 \quad \dots(2)$$

Similarly, the same argument can be given for every pair of segments of the paths APB and AQB so that

$$\int_A^B \vec{F}_1 \cdot d\vec{r}_1 = \int_A^B \vec{F}_2 \cdot d\vec{r}_2 \quad \dots(3)$$

(Path APB) (Path AQB)

This means that the work done $W = \int_A^B \vec{F} \cdot d\vec{r}$ by a central force is independent of the path. Hence the central force is conservative in nature.

For conservative forces, the work done around a closed path is always zero : The amount of work done, when a particle reaches from the point A to the point B in a conservative force field, is

$$W_{(A \rightarrow B)} = \int_A^B \vec{F} \cdot d\vec{r} = - \int_B^A \vec{F} \cdot d\vec{r} = - W_{(B \rightarrow A)} \quad \dots(4)$$

* The region in which the force acting on a particle is directed away or towards a centre is called central force field.

Hence, if the particle moves in the same conservative force field from the point B to the point A , then the amount of work done is

$$W_{(B \rightarrow A)} = -W_{(A \rightarrow B)}$$

Thus, the total amount of work done on the particle by a conservative force, when it moves from A to B and then from B to A i.e., along a closed path, is

$$\oint \vec{F} \cdot d\vec{r} = W_{(A \rightarrow B)} + W_{(B \rightarrow A)} = W_{(A \rightarrow B)} - W_{(A \rightarrow B)} = 0 \quad \dots(5)$$

where $\oint$ sign represents the integral over the closed path.

The points A and B are any two arbitrary points in the conservative force field. Hence we see that *for conservative forces the work done around any closed path is zero*. In fact, this is a very important characteristic of conservative forces. Hence we can define a conservative force as follows :

A force is said to be conservative, if the work done on a particle is zero, when the particle moves under such a force around a closed path.

A constant force is an example of conservative force : In a constant and uniform force field, i.e., for $\vec{F} = \text{constant}$, the work done around a closed path is given by

$$\oint \vec{F} \cdot d\vec{r} = \vec{F} \cdot \oint d\vec{r} = 0$$

because for a closed curve $\oint d\vec{r} = 0$. Thus a uniform constant force field is always conservative.

5.3. Conservative Force as a Negative Gradient of Potential Energy ($\vec{F} = -\vec{\nabla} U$) – Curl of a Conservative Force

Potential energy of a particle in the field of a conservative force $\vec{F}$ at a point $\vec{r}$ is defined as.

$$U(\vec{r}) = - \int_{\infty}^{\vec{r}} \vec{F} \cdot d\vec{r} \quad \dots(1)$$

Expressing $\vec{F}$ and $d\vec{r}$ in rectangular components, we have

$$U(\vec{r}) = \int (F_x \hat{i} + F_y \hat{j} + F_z \hat{k}) \cdot (dx \hat{i} + dy \hat{j} + dz \hat{k})$$

$$\text{or} \quad U(\vec{r}) = - \int F_x dx - \int F_y dy - \int F_z dz \quad \dots(2)$$

Differentiating this equation partially with respect to x, y, z , we get

$$F_x = -\frac{\partial U}{\partial x}, \quad F_y = -\frac{\partial U}{\partial y}, \quad F_z = -\frac{\partial U}{\partial z}$$

$$\text{Hence the force} \quad \vec{F} = \hat{i} F_x + \hat{j} F_y + \hat{k} F_z = -\hat{i} \frac{\partial U}{\partial x} - \hat{j} \frac{\partial U}{\partial y} - \hat{k} \frac{\partial U}{\partial z}$$

$$\text{or} \quad \vec{F} = - \left(\hat{i} \frac{\partial U}{\partial x} + \hat{j} \frac{\partial U}{\partial y} + \hat{k} \frac{\partial U}{\partial z} \right) = -\text{grad } U \quad \text{or} \quad -\vec{\nabla} U \quad \dots(3)$$

Thus, a *conservative force is the negative gradient of potential energy U* .

$$\text{Only for one dimension,} \quad F = -\frac{\partial U}{\partial x} \quad \text{or} \quad U = - \int F dx \quad \dots(4)$$

For a conservative force, curl $\vec{F} = 0$: A conservative force can be represented as $\vec{F} = -\vec{\nabla} U$.

$$\therefore \quad \text{curl } \vec{F} = \vec{\nabla} \times \vec{F} = -\vec{\nabla} \times \vec{\nabla} U = - \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_x & F_y & F_z \end{vmatrix} = - \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial U}{\partial x} & \frac{\partial U}{\partial y} & \frac{\partial U}{\partial z} \end{vmatrix}$$

$$= - \left[\hat{i} \left(\frac{\partial^2 U}{\partial y \partial z} - \frac{\partial^2 U}{\partial z \partial y} \right) + \hat{j} \left(\frac{\partial^2 U}{\partial z \partial x} - \frac{\partial^2 U}{\partial x \partial z} \right) + \hat{k} \left(\frac{\partial^2 U}{\partial x \partial y} - \frac{\partial^2 U}{\partial y \partial x} \right) \right] \quad \dots(5)$$

But U is taken as a perfect differential, i.e., $\frac{\partial^2 U}{\partial y \partial z} = \frac{\partial^2 U}{\partial z \partial y}$ etc.

$$\text{Hence, from (5) } \text{curl } \vec{F} = 0 \quad \dots(6)$$

Thus, the curl of a conservative force is zero.

Theorem 1. Prove that a central force is a conservative force.

Proof : For a central force, $\vec{F} = F(r)\hat{r}$...(i)

where $F(r)$ is a scalar function of distance r and $\hat{r} = \vec{r}/r$.

$$\begin{aligned} \text{Now, } \text{curl } \vec{F} &= \vec{\nabla} \times \hat{r} F(r)^* = \frac{1}{r^2 \sin \theta} \begin{vmatrix} \hat{r} & r\hat{\theta} & r\sin\theta\hat{\phi} \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ F(r) & 0 & 0 \end{vmatrix} \\ &= \frac{1}{r^2 \sin \theta} \left[r\hat{\theta} \frac{\partial F(r)}{\partial \phi} - r\sin\theta\hat{\phi} \frac{\partial F(r)}{\partial \theta} \right] \end{aligned} \quad \dots(ii)$$

Since $F(r)$ is the function of distance r only and is independent of θ and ϕ ,

$$\text{curl } \vec{F} = 0. \quad \dots(iii)$$

Thus a central force is always conservative.

As special cases, $F(r) = -k/r^n$, $F(r) = -k/r^2$, $F(r) = -kr$, all are the examples of central force and they are conservative forces.

Theorem 2. Prove that a linear restoring force is a conservative force.

Proof : A force is said to be a linear restoring force, if it is proportional to the displacement of the particle from a fixed point and is directed opposite to the displacement. This means that such a force is the linear function of the displacement, having direction opposite to it.

Let a particle move along X-axis in the influence of linear restoring force $\vec{F}$ [fig. 5.7]. If the fixed point is the

origin O and at any time t the displacement is $\vec{x} = x\hat{i}$ (where $\hat{i}$ is the unit vector along X-axis), then according to the definition,

$$\vec{F} = -kx\hat{i} \quad \text{or} \quad F = -kx \quad \dots(iv)$$

where k is some positive constant, called the **force constant** or **spring factor**. The negative sign indicates that the restoring force acts opposite to the displacement from the origin at $x = 0$.

For sufficiently small displacements, a linear restoring force may be produced by a stretched or compressed spring.

If $\vec{F} = -kx\hat{i}$ is a conservative force, then for it $\text{curl } \vec{F} = 0$, i.e.,

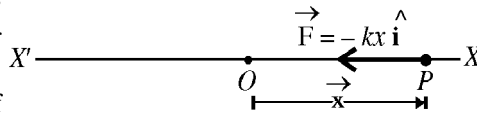


Fig. 5.7.

* In spherical co-ordinates, curl of any vector $\vec{A}$ is given by $\text{curl } \vec{A} = \vec{\nabla} \times \vec{A} = \frac{1}{r^2 \sin \theta} \begin{vmatrix} \hat{r} & r\hat{\theta} & r\sin\theta\hat{\phi} \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ A_r & rA_\theta & r\sin\theta A_\phi \end{vmatrix}$.

$$\text{curl } \vec{F} = -\vec{\nabla} \times kx\hat{i} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ kx & 0 & 0 \end{vmatrix} = 0 \quad \dots(v)$$

Thus a linear restoring force is a conservative force.

5.4. Areal Velocity Under Central Force

Consider a particle of mass m , moving under a central force. If $\vec{r}$ is the instantaneous position vector and $\vec{v}$ the instantaneous velocity of the particle, then its angular momentum is given by $\vec{L} = \vec{r} \times m\vec{v}$. As the particle is moving under central force, its angular momentum remains constant i.e.,

$$\vec{L} = \vec{r} \times m\vec{v} = \text{constant} \quad \dots(1)$$

Let O be the centre of force. If the position vector $\vec{r}$ of the particle P changes from $\vec{r}$ to $\vec{r} + \Delta\vec{r}$ in time Δt , then the vector area $\Delta\vec{S}$ swept by the radius vector in this time is

$$\Delta\vec{S} = \frac{1}{2} \vec{r} \times \Delta\vec{r} \quad \dots(2)$$

This area is swept in Δt time, therefore dividing both the sides of this equation by Δt and taking limit as $\Delta t \rightarrow 0$, then

$$\frac{d\vec{S}}{dt} = \frac{1}{2} \vec{r} \times \vec{v} = \frac{\vec{L}}{2m} \quad \dots(3)$$

This expression gives the areal velocity ($d\vec{S}/dt$) of the particle. But $\vec{L}$ is constant for a particle, moving under central force, so from relation (3) we see that the areal velocity remains constant, when the particle moves in the influence of a central force.

Areal velocity in planetary or satellite motion : The planets move in elliptical orbits about the Sun at a focus.* The gravitational force on the planet is directed towards the centre of the Sun. In other words, this force is a central force. Consequently, the angular momentum ($\vec{L} = \vec{r} \times m\vec{v}$) of the planet will be constant and hence the orbit of motion will be **plane**.

For planetary motion, the constancy of $\vec{L}$ means that the areal velocity $d\vec{S}/dt = (\vec{L}/2m)$ is constant. Thus, *the rate of sweeping out of the area by a radial vector from Sun to the planet is a constant*—this is known as **Kepler's second law of planetary motion**. Similarly, artificial satellites move in elliptical orbits around the earth and the angular momentum is conserved.**

* As more correctly speaking, the focus of the ellipse should be taken at the centre of mass of the Sun and the planet. As the Sun is much heavier than the planet, the centre of mass is very near to the centre of the Sun.

** In case if the artificial satellite comes close to the earth its speed decreases due to the friction in the earth's atmosphere and its angular momentum decreases. But the angular momentum of the earth increases in this process and hence the total angular momentum of the system (earth and satellite) remains constant.

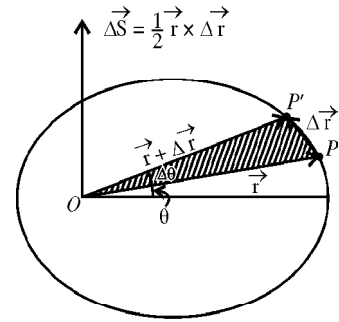


Fig. 5.8. Areal velocity under central force.

In planetary motion, for the conservation of angular momentum ($\vec{L} = \vec{r} \times m\vec{v}$), the planet must move faster at the point of closest approach to the Sun than at the farthest point. The reason lies in the fact that at these points the angular momentum is mvr , (because at these positions $\vec{r}$ is perpendicular to $\vec{v}$) and for its constancy, the shorter r is associated with larger v . If r_1, r_2 are the distances of closest and farthest points of the planet respectively and v_1, v_2 are the corresponding velocities, then

$$mv_1r_1 = mv_2r_2 \text{ or } v_1r_1 = v_2r_2 \quad \dots(4)$$

Ex. 1. Show that the force $F = (y^2 - x^2)\hat{i} + 2xy\hat{j}$ is conservative.

Sol. If the force is conservative, $\text{curl } \vec{F} = 0$

Here, $\vec{F} = (y^2 - x^2)\hat{i} + 2xy\hat{j}$

$$\begin{aligned} \therefore \text{Curl } \vec{F} &= \vec{\nabla} \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ (y^2 - x^2) & 2xy & 0 \end{vmatrix} \\ &= \hat{i} \left[\frac{\partial}{\partial y} 0 - \frac{\partial}{\partial z} (2xy) \right] + \hat{j} \left[\frac{\partial}{\partial z} (y^2 - x^2) - \frac{\partial}{\partial x} 0 \right] + \hat{k} \left[\frac{\partial}{\partial x} (2xy) - \frac{\partial}{\partial y} (y^2 - x^2) \right] \\ &= \hat{k} [2y - 2y] = 0. \end{aligned}$$

Hence the given force is conservative.

Ex. 2. What is the value of 'a' if the force given by the expression $\vec{F} = (2xy + z^2)\hat{i} + x^2\hat{j} + axz\hat{k}$ is to be conservative ? (O.U. 2001)

Sol. For a conservative force, $\text{curl } \vec{F} = 0$.

Here, $\text{curl } \vec{F} = \vec{\nabla} \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ (2xy + z^2) & x^2 & axz \end{vmatrix} = 0$.

$$\text{Therefore, } \hat{i} \left[\frac{\partial}{\partial y} (axz) - \frac{\partial}{\partial z} (x^2) \right] - \hat{j} \left[\frac{\partial}{\partial x} (axz) - \frac{\partial}{\partial z} (2xy + z^2) \right] + \hat{k} \left[\frac{\partial}{\partial x} (x^2) - \frac{\partial}{\partial y} (2xy + z^2) \right] = 0$$

$$\text{or, } -\hat{j} [(az - 2z) + 0] = 0 \quad \text{or} \quad az - 2z = 0 \quad \text{or} \quad a = 2.$$

Ex. 3. (a). If $\vec{F} = (2xy + x^2)\hat{i} + x^2\hat{j} + 2xz\hat{k}$, then show that it is conservative.

(b) Calculate the amount of work done by the above force in moving a particle from (0, 1, 2) to (5, 2, 7).

Sol. (a) For a conservative force, $\text{curl } \vec{F} = 0$.

Here, $\text{curl } \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2xy + x^2 & x^2 & 2xz \end{vmatrix}$

$$= \hat{i} \left[\frac{\partial}{\partial x}(2xz) - \frac{\partial}{\partial z}(x^2) \right] + \hat{j} \left[\frac{\partial}{\partial z}(2xy + z^2) - \frac{\partial}{\partial x}(2xz) \right] + \hat{k} \left[\frac{\partial}{\partial x}(x^2) - \frac{\partial}{\partial y}(2xy + z^2) \right]$$

$$= 0 + \hat{j}(2z - 2z) + \hat{k}(2x - 2x) = 0.$$

Thus, $\vec{F} = (2xy + x^2)\hat{i} + x^2\hat{j} + 2xz\hat{k}$ is a conservative force.

$$\begin{aligned} \text{(b) Work done} &= \int_A^B \vec{F} \cdot d\vec{r} = \int_A^B (F_x dx + F_y dy + F_z dz) = \int_A^B [(2xy + z^2) dx + x^2 dy + 2xz dz] \\ &= \int_A^B [(2xy dx + x^2 dy) + (z^2 dx + 2xz dz)] = \int_A^B [d(x^2 y) + d(z^2 x)] \\ &= \int_A^B d(x^2 y + z^2 x) = [x^2 y + z^2 x]_{A(0,1,2)}^{B(5,2,7)} = [5 \times 5 \times 2 + 7 \times 7 \times 5 - 0 - 0] \\ &= 50 + 245 = \mathbf{295 \text{ units.}} \end{aligned}$$

Ex. 4. At an instant, the position vector of a particle, moving with constant angular velocity, is given by

$$\vec{r} = r \cos \theta \hat{i} + r \sin \theta \hat{j} \text{ with } \theta = \omega t$$

Prove that the force acting on the particle is conservative.

Sol. $\vec{r} = r \cos \theta \hat{i} + r \sin \theta \hat{j}$

Unit vector along $\vec{r}$ is $\hat{r} = \frac{\vec{r}}{r} = \cos \theta \hat{i} + \sin \theta \hat{j}$

Now, velocity $\vec{v} = \frac{d\vec{r}}{dt} = -r \sin \theta \frac{d\theta}{dt} \hat{i} + r \cos \theta \frac{d\theta}{dt} \hat{j} = -r \omega \sin \theta \hat{i} + r \omega \cos \theta \hat{j}$

where $\frac{d\theta}{dt} = \omega$, because $\theta = \omega t$.

Hence, acceleration, $\vec{a} = \frac{d\vec{v}}{dt} = -r \omega \cos \theta \frac{d\theta}{dt} \hat{i} - r \omega \sin \theta \frac{d\theta}{dt} \hat{j}$

or $\vec{a} = -\omega^2 r (\cos \theta \hat{i} + \sin \theta \hat{j}) = -\omega^2 \vec{r} = -\omega^2 r \hat{r} = -\omega^2 \vec{r}$

Thus the force, $\vec{F} = m \vec{a} = -m\omega^2 \vec{r}$

But $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$

$\therefore \vec{F} = -m\omega^2 (x\hat{i} + y\hat{j} + z\hat{k})$

$\therefore \text{Curl } \vec{F} = \vec{\nabla} \times \vec{F} = \vec{\nabla} \times [-m\omega^2 (x\hat{i} + y\hat{j} + z\hat{k})] = (-m\omega^2) [\vec{\nabla} \times (x\hat{i} + y\hat{j} + z\hat{k})]$

$$= (-m\omega^2) \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x & y & z \end{vmatrix} = 0.$$

Hence the force is conservative.

Ex. 5. Show that the force $\vec{F} = \frac{k}{r^n} \hat{r}$ is a conservative force, where k and n are constants.

Sol. $\text{curl } \vec{F} = \vec{\nabla} \times \frac{k}{r^n} \hat{r} = \vec{\nabla} \times \frac{k \vec{r}}{r^{n+1}} \quad \left[\because \hat{r} = \frac{\vec{r}}{r} \right]$

But $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$; $r = (x^2 + y^2 + z^2)^{1/2}$ and hence, $r^{n+1} = (x^2 + y^2 + z^2)^{\frac{n+1}{2}}$

Now, $\text{curl } \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{kx}{(x^2 + y^2 + z^2)^{\frac{n+1}{2}}} & \frac{ky}{(x^2 + y^2 + z^2)^{\frac{n+1}{2}}} & \frac{kz}{(x^2 + y^2 + z^2)^{\frac{n+1}{2}}} \end{vmatrix}$

$$= \hat{i} \left[\frac{kz \times 2y}{(x^2 + y^2 + z^2)^{\frac{n-1}{2}}} - \frac{ky \times 2z}{(x^2 + y^2 + z^2)^{\frac{n-1}{2}}} \right] + \dots$$

$$= 0$$

Hence, the given force is conservative.

5.5. Newton's Law of Universal Gravitation

In the universe, any particle (which has mass) attracts every other particle with a force, known as **gravitational force**. According to **Newton's law of universal gravitation**, the force of attraction between any two particles is directly proportional to the product of their masses and inversely proportional to the square of distance between the two particles. If two particles of masses m_1 and m_2 are separated by a distance r , then in view of this law the gravitational force between the two particles is given by

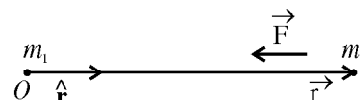


Fig. 5.9.

$$\vec{F} = -G \frac{m_1 m_2}{r^2} \hat{r} = -G \frac{m_1 m_2}{r^3} \vec{r} \quad \dots(1)$$

where $\vec{r}$ is the position vector from m_1 to m_2 , $\hat{r} = \vec{r}/r$ [fig. 5.9] and G the constant of gravitation. The negative sign signifies the force between the two gravitating particles is that of attraction.

Magnitude of the force, $F = Gm_1 m_2 / r^2$.

Value of $G = 6.67 \times 10^{-8}$ dyne-cm²/gm² (C.G.S. units) = 6.67×10^{-11} newton-m²/kg² (S.I. units).

The extreme importance of the gravitational force may be seen by the following examples :

- We are bound to live on earth due to the force of gravitation. Moreover, the gravitational forces keep the atmosphere (containing life providing agency oxygen) to retain on the earth.
- The earth and other planets are bound to move around the Sun (which is the most important source of light and heat for us) only due to gravitational forces. Remember that the life is possible on the earth due to the presence of the Sun.

Similarly, a number of examples may be seen in everyday life showing the importance of gravitational forces.

As Newton's law of gravitation is valid for large inter-planetary distances and small distances on earth, this is why this law is said to be **universal**. This law does not hold true for intermolecular distances of the order of 10^{-7} cm and very large orbital velocities. The gravitational force, which is an inverse square law force ($-K/r^2$), is an important example of central force.

5.6. Gravitational Field and Potential

Gravitational Field : The region around a particle or body in which any other particle experiences

a gravitational force is called **gravitational field**. The intensity of gravitational field at a point due to a mass is defined as the force experienced by unit mass, placed at that point. It is, however, supposed that the introduction of the unit mass at the point does not change the configuration of the field. If M be the mass of a gravitating particle, the intensity of gravitational field at a distance $\vec{r}$ [fig. 5.10] from it is given by

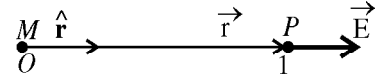


Fig. 5.10.

$$\vec{E} = -\frac{GM \times 1}{r^2} \hat{r} \quad \text{or} \quad \vec{E} = -\frac{GM}{r^2} \hat{r} \quad \dots(1)$$

where $\hat{r} = \vec{r}/r$ is a unit vector along $\vec{r}$. Here the negative sign indicates that the direction of force is opposite to $\hat{r}$ i.e., towards the mass M .

If we place a mass m at a distance $\vec{r}$ from M , then the gravitational force on m is given by

$$\vec{F} = -\frac{GMm}{r^2} \hat{r} \quad \dots(2)$$

From eqs. (1) and (2), we get

$$\vec{F} = m \vec{E} \quad \dots(3)$$

$$\text{or} \quad \vec{E} = \frac{\vec{F}}{m} \quad \dots(4)$$

Thus, the intensity of gravitational field is equal to the force per unit mass. Normally the intensity of field is said to be **field** only. Sometimes intensity of gravitational field is said to be **gravitational attraction**.

Gravitational potential : The gravitational potential at a point in a gravitational field of a body is the amount of work, required to be done on a unit mass in bringing it from infinity (i.e., the state of zero potential) to that point. In other words, the potential at a point is equal to the **potential energy per unit mass**. Generally gravitational potential is represented by V and its value at infinity is taken to be zero. Hence the gravitational potential at a distance $\vec{r}$ from a mass M is given by

$$V = \frac{U(r)}{m} = -\int_{\infty}^{\vec{r}} \vec{E} \cdot d\vec{r} \quad \dots(5)$$

where $U(r)$ is the potential energy of a mass m placed at $\vec{r}$ and $\vec{E}$ is the intensity of the field at that distance.

Remember that the gravitational force is a conservative force and the force on unit mass is $\vec{E}$ and

the definition (5) is according to the definition of potential energy.
$$\left(U(r) = -\int_{\infty}^{\vec{r}} \vec{F} \cdot d\vec{r} \right)$$

Gravitational potential due to a mass M

Eq. (5) is used to find the gravitational potential at a distance $\vec{r}$ due to a point mass M . The intensity of the field $\vec{E}$ at a distance $\vec{r}$ from mass M is given by [fig. 5.11]

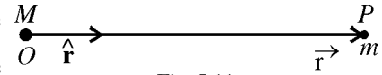


Fig. 5.11.

$$\vec{E} = -\frac{GM}{r^2} \hat{r}$$

Hence

$$V = -\int_{\infty}^{\vec{r}} \vec{E} \cdot d\vec{r} = \int_{\infty}^{\vec{r}} \frac{GM}{r^2} \hat{r} \cdot d\vec{r} = \int_{\infty}^{\vec{r}} \frac{GM}{r^2} dr$$

or
$$V = -\frac{GM}{r} \quad \dots(6)$$

Gravitational Potential Energy of a System of Two Particles

If a mass m is placed at a distance $\vec{r}$ from M , then the **gravitational potential energy** of the system of M and m masses will be

$$U = -\int_{\infty}^{\vec{r}} \vec{F} \cdot d\vec{r} = -\int_{\infty}^{\vec{r}} -\frac{GMm}{r^2} \hat{r} \cdot d\vec{r} = -\frac{GMm}{r} \quad \dots(7)$$

Therefore, from (6) and (7)

$$U = mV \quad \text{or} \quad V = U/m \quad \dots(8)$$

This is the *relation between the gravitational potential energy and gravitational potential*.

The **potential difference between two points** at distances $\vec{r}_1$ and $\vec{r}_2$ in a gravitational field is given by

$$V_2 - V_1 = -\int_{\vec{r}_1}^{\vec{r}_2} \vec{E} \cdot d\vec{r} \quad \dots(9)$$

Relation between the Intensity of Gravitational Field ($\vec{E}$) and Gravitational Potential (V).

From the definition of gravitational potential,

$$V = -\int_{\infty}^{\vec{r}} \vec{E} \cdot d\vec{r} = \int_{\infty}^{\vec{r}} (E_x dx + E_y dy + E_z dz) \quad \dots(10)$$

where $\vec{E} = E_x \hat{i} + E_y \hat{j} + E_z \hat{k}$ and $d\vec{r} = dx \hat{i} + dy \hat{j} + dz \hat{k}$; E_x, E_y, E_z are the components of $\vec{E}$ along X, Y, Z axes respectively. Hence,

$$V = -\int E_x dx - \int E_y dy - \int E_z dz \quad \dots(11)$$

Differentiating V partially with respect to x, y and z , we get

$$\frac{\partial V}{\partial x} = -E_x, \quad \frac{\partial V}{\partial y} = -E_y, \quad \frac{\partial V}{\partial z} = -E_z,$$

i.e.,

$$E_x = -\frac{\partial V}{\partial x}, \quad E_y = -\frac{\partial V}{\partial y}, \quad E_z = -\frac{\partial V}{\partial z}$$

Hence

$$\vec{E} = E_x \hat{i} + E_y \hat{j} + E_z \hat{k} = -\hat{i} \frac{\partial V}{\partial x} - \hat{j} \frac{\partial V}{\partial y} - \hat{k} \frac{\partial V}{\partial z}$$

or

$$\vec{E} = -\vec{\nabla} V = -\text{grad } V \quad \dots(12)$$

For one dimension, $E = -\frac{dV}{dx}$ or $E = -\frac{dV}{dr}$... (13)

5.7. Gravitational Potential and Field due to a Thin Spherical Shell

The point, at which the gravitational potential is to be considered may lie (i) outside the shell, (ii) on the surface of the shell or (iii) inside the shell.

(i) Point outside the shell : Let P be a point [fig. 5.12] at which the potential is to be determined and O the centre of a spherical shell of radius R . The point P lies at a distance r from the centre O of the shell.

Draw two planes CD and EF , perpendicular to OP and close to each other. Thus these planes cut out a slice of the shell in the form of a ring $CDFE$ of radius EG and width CE .

If $\angle EOB = \theta$ and $\angle COE = d\theta$, then

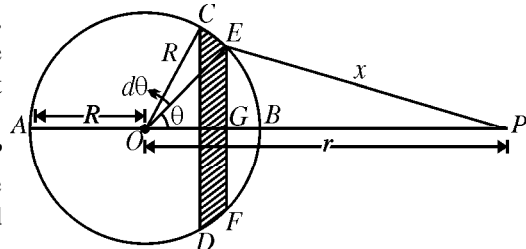


Fig. 5.12.

radius of the ring $EG = OE \sin \theta = R \sin \theta$
 and width of the ring $CE = OE d\theta = R d\theta$
 $\therefore$ Area of the ring = Circumference $\times$ Its width [fig. 5.13]
 $= 2\pi R \sin \theta \times R d\theta$
 $\therefore$ Mass of the ring $= 2\pi R \sin \theta \times R d\theta \times \sigma = 2\pi R^2 \sigma \sin \theta d\theta$
 where σ is the mass per unit area of the surface of the shell.

Since every point of the ring is at the same distance x from the point P , the small potential dV at P due to the ring is given by

$$dV = -G \cdot \frac{\text{Mass of the ring}}{x} = -\frac{G \cdot 2\pi R^2 \sigma \sin \theta d\theta}{x}$$

In triangle OEP , $EP^2 = OE^2 + OP^2 - 2 OE \cdot OP \cdot \cos \theta$
 or $x^2 = R^2 + r^2 - 2Rr \cos \theta$

Differentiate x with respect to θ , $2x dx = 2Rr \sin \theta d\theta$ or $\sin \theta d\theta = x dx / Rr$

$$\therefore dV = \frac{G \cdot 2\pi R^2 \sigma x dx}{Rrx} = -\frac{G \cdot 2\pi R \sigma}{r} dx$$

Integrating this equation for the whole shell between the limits $PB = r - R$, and $PA = r + R$, we get the potential V due to the whole shell at P , i.e.,

$$V = \int_{r-R}^{r+R} -G \cdot \frac{2\pi R \sigma}{r} dx = -G \cdot \frac{2\pi R \sigma}{r} [x]_{r-R}^{r+R} = -G \cdot \frac{2\pi R \sigma}{r} 2R = -G \cdot \frac{4\pi R^2 \sigma}{r}$$

If M is the mass of the shell, then $M = 4\pi R^2 \sigma$. Hence

$$V = -\frac{GM}{r} \quad \dots(1)$$

Thus, the potential at a point outside a thin spherical shell is the same, as if its whole mass were concentrated at its centre.

Field : The intensity of the gravitational field at the point P is given by

$$E = -\frac{dV}{dr} = -\frac{d}{dr} \left(-\frac{GM}{r} \right) = -\frac{GM}{r^2} \quad \dots(2)$$

This is the same as due to a mass M placed at O .

Force on a point mass m : If a mass m is placed at the point P , then the potential energy will be given by

$$U = mV = -\frac{GMm}{r} \quad \dots(3)$$

and the force acting on the mass due to the shell is given by

$$F = -\frac{dU}{dr} = -\frac{GMm}{r^2} \quad \dots(4)$$

Thus, we see that the spherical shell acts at points outside as if all its mass M were concentrated at the centre of the shell.

(ii) Point on the surface of the shell : If we consider the point P to be situated at B , i.e., on the surface of the shell, then we can get the potential there by integrating the above expression for dV , between the limits $x = 0$ and $x = 2R$. Hence, the potential at the point P lying on the surface of the shell, is given by

$$V = \int_0^{2R} -G \cdot \frac{2\pi R \sigma}{r} dx = -G \cdot \frac{2\pi R \sigma}{r} [x]_0^{2R} = -G \cdot \frac{4\pi R^2 \sigma}{r}$$

Here $r = R$ and $4\pi R^2 \sigma = M$.

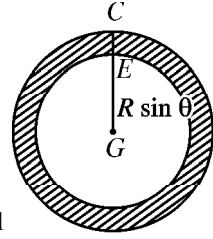


Fig. 5.13.

$$\therefore V = -\frac{GM}{R}. \quad \dots(5)$$

Field : The gravitational field at the point P , when it lies on the surface of the shell, is

$$E = -\frac{dV}{dr} = -\frac{d}{dr}\left(-\frac{GM}{r}\right) = -\frac{GM}{r^2}$$

$$\text{Here, } r = R, \quad \therefore E = -\frac{GM}{R^2}. \quad \dots(6)$$

Thus, the gravitational attraction of a shell at a point on its surface is the same as would be if its whole mass were concentrated at its centre.

(iii) Point inside the shell : Let the point P lie inside the shell at a distance r from O [fig. 5.14]. Proceeding exactly in the same way as before, to find the potential at P due to the ring $CDFE$, we get

$$dV = -\frac{G \cdot 2\pi R \sigma}{r} dx$$

To obtain the potential V at P due to the whole shell, therefore, we integrate the expression for dV between the limits $BP = R - r$ and $AP = R + r$. Thus

$$\begin{aligned} V &= \int_{R-r}^{R+r} -G \cdot \frac{2\pi R \sigma}{r} dx = -G \cdot \frac{2\pi R \sigma}{r} [x]_{R-r}^{R+r} \\ &= -G \cdot \frac{2\pi R \sigma}{r} 2r = -G \cdot \frac{4\pi R^2 \sigma}{R} \end{aligned}$$

$$\text{or } V = -\frac{GM}{R} \quad [\because M = 4\pi R^2 \sigma] \quad \dots(7)$$

This value is the same as the potential at the surface of the shell.

Thus the potential at any point inside a spherical shell is constant and equal to the value of the potential on its surface.

Field : The gravitational field at P is given by

$$E = -\frac{dV}{dr} = -\frac{d}{dr}\left(-\frac{GM}{R}\right) = 0$$

Thus, the intensity of the field at any point inside the spherical shell is zero.

Force on a point mass m : The potential energy, when a mass m is placed at any point P , is given by

$$U = mV = -\frac{GMm}{R} \quad (\text{for } r < R) \quad \dots(9)$$

and the force acting on the mass is given by

$$F = -\frac{dU}{dR} = 0 \quad (\text{for } r < R) \quad \dots(10)$$

Thus, if a mass is placed at any point inside a thin spherical shell, then the force acting on it is always zero. This is a very special property of an inverse square law force. Outside the shell, the force varies as $1/r^2$.

A graph between V and r drawn on the basis of the above results is shown in Fig. 5.15 (a). This shows that the potential is constant at all the points inside the shell, but

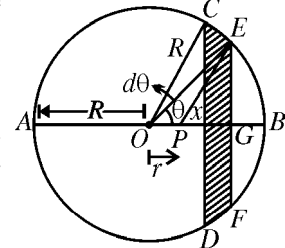


Fig. 5.14.

$$[\because \frac{GM}{R} \text{ is a constant}] \quad \dots(8)$$

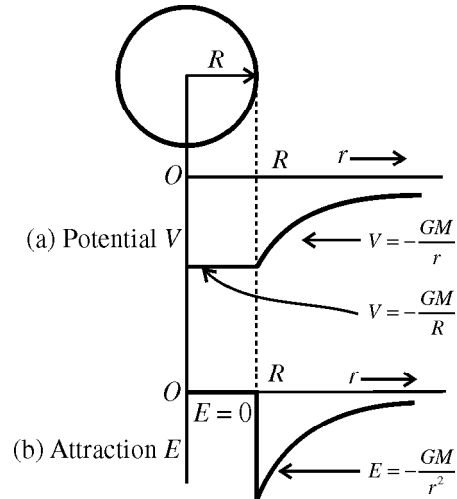


Fig. 5.15.

* U - r graph [i.e., the graph between the potential energy (U) and the distance of the mass from the centre (r)] and F - r graph (i.e., the graph between the force on the mass m and its distance from the centre) are exactly of the same shape as V - r and E - r graphs respectively.

decreases when $r > R$. Fig. 5.15 (b) represents the variation of the intensity of the field with distance r^* .

5.8. Gravitational Potential and Field due to a Solid Sphere

(i) **Point outside the sphere :** Consider a point P outside the sphere at a distance r from its centre O . We have to determine the potential at P .

Imagine the sphere to be divided into a large number of thin spherical shells concentric to the sphere, of masses $m_1, m_2, \dots$ etc. [fig. 5.16].

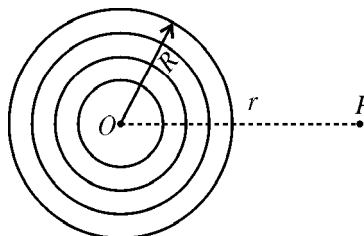


Fig. 5.16.

Potential at P due to each individual shell will be

$$= -\frac{G \times \text{Mass of the shell}}{r}$$

$\therefore$ Potential at P due to all spherical shells of masses $m_1, m_2, m_3, \dots$ etc. is

$$\begin{aligned} V &= -\frac{Gm_1}{r} - \frac{Gm_2}{r} - \frac{Gm_3}{r} - \dots \\ &= -\frac{G}{r}(m_1 + m_2 + m_3 + \dots). \end{aligned}$$

Now, clearly $m_1 + m_2 + \dots = M$, the mass of the sphere,

$$\therefore V = -\frac{GM}{r} \quad (\text{for } r > R) \quad \dots(1)$$

Thus, in the case of a solid sphere, the potential at a point lying outside sphere is the same as if its whole mass were concentrated at the centre.

When the point P lies on the **surface of the sphere**, obviously the potential will be

$$V = -\frac{GM}{R} \quad [\because r = R, \text{ the radius of the sphere}]$$

Intensity of the field : The intensity of the gravitational field at the point r outside the sphere is given by

$$E = -\frac{dV}{dr} = -\frac{d}{dr}\left(-\frac{GM}{r}\right) = -\frac{GM}{r^2} \quad (\text{for } r > R) \quad \dots(2)$$

The field at a point P , when it lies on the **surface of the sphere** is

$$E = -\frac{GM}{R^2} \quad [\because \text{Here } r = R]$$

Force on a mass m : If a point mass m is placed at a distance $r (> R)$, then its potential energy due to the solid sphere is given by

$$U = mV = -\frac{GMm}{r} \quad \dots(3)$$

and the force, acting on the mass m , is

$$F = -\frac{dU}{dr} = -\frac{GMm}{r^2} \quad \dots(4)$$

(ii) **Point inside the sphere :** Let the point P lie inside the sphere at a distance r from its centre O [fig. 5.17]. The radius of the sphere is R and the density of its material ρ .

With O as centre and radius $OP = r$, we draw a sphere. Then the point P lies on the surface of the solid sphere of radius r and inside the spherical shell of internal radius r and external radius R .

Volume of the inner solid sphere $= \frac{4}{3}\pi r^3$; $\therefore$ Mass of the sphere $= \frac{4}{3}\pi r^3 \rho$, where ρ is the density of the sphere.

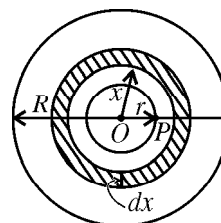


Fig. 5.17.

Hence potential V_1 at P due to the inner solid sphere

$$V_1 = -\frac{G \cdot \frac{4}{3}\pi r^3 \rho}{r} \quad \text{or} \quad V_1 = -\frac{4}{3}\pi r^2 \rho G \quad \dots(5)$$

To determine the potential at P due to the outer spherical shell let us imagine it to be divided into a number of thin concentric shells and consider one such shell of radius x and infinitesimally small thickness dx .

Now, the surface area of the spherical shell $= 4\pi x^2$; volume of the shell $= 4\pi x^2 dx$

$\therefore$ Mass of the shell $= 4\pi x^2 dx \rho$

Remembering that the potential at any point within a spherical shell is the same as on the surface,

we have potential at P due to the shell $= -\frac{G 4\pi x^2 dx \rho}{x} = -4\pi \rho G \cdot x dx$. The potential V_2 at the point P due to shell of internal radius r and external radius R is obtained by integrating this expression between the limits $x = r$ and $x = R$. Thus,

$$V_2 = \int_r^R -4\pi \rho G \cdot x dx = -4\pi \rho G \left[\frac{x^2}{2} \right]_r^R = -4\pi \rho G \left(\frac{R^2}{2} - \frac{r^2}{2} \right) = -2\pi G (R^2 - r^2) \rho \quad \dots(6)$$

Hence the total potential at the point P is

$$\begin{aligned} V &= V_1 + V_2 = -\frac{4}{3}\pi r^2 \rho G - 2\pi \rho G (R^2 - r^2) = -2\pi \rho G \left(\frac{2}{3}r^2 + R^2 - r^2 \right) = -\frac{2}{3}\pi \rho G (3R^2 - r^2) \\ &= -\frac{4}{3}\pi R^3 \rho G \frac{(3R^2 - r^2)}{2R^3} \end{aligned}$$

But $\frac{4}{3}\pi R^3 \rho = M$, the mass of the sphere,

$$\therefore V = -GM \left(\frac{3R^2 - r^2}{2R^3} \right) \quad \dots(7)$$

Field : The **gravitational attraction** or the intensity of the field at the point P is given by

$$\begin{aligned} E &= -\frac{dV}{dr} = -\frac{d}{dr} \left[-GM \left(\frac{3R^3 - r^2}{2R^3} \right) \right] \\ &= -\frac{GM}{2R^3} \cdot 2r = -\frac{GM}{R^3} r \quad \dots(8) \end{aligned}$$

$$\therefore E \propto r. \quad \dots(9)$$

Thus, the intensity of the gravitational field at a point inside a solid sphere is proportional to its distance from the centre and is zero at its centre.

Force on mass m : The potential energy, when a mass m is placed at any point r inside the sphere, is given by

$$U = mV = -GMm \left(\frac{3R^3 - r^2}{2R^3} \right) \quad (\text{for } r < R) \quad \dots(10)$$

and the force acting on the mass is given by

$$F = -\frac{dU}{dr} = -\frac{GMm}{R^3} r \quad (\text{for } r < R) \quad \dots(11)$$

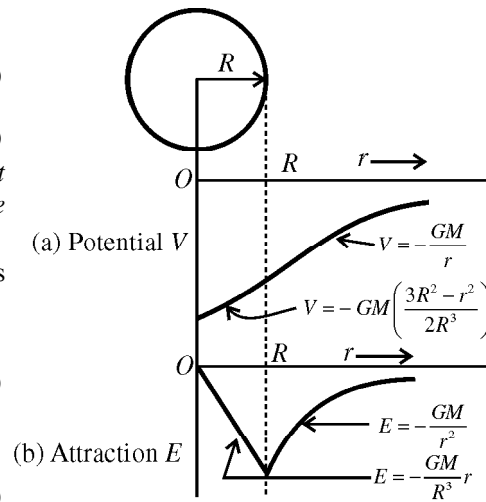


Fig. 5.18.

Thus, the force on a point mass m , placed inside a sphere, is proportional to its distance from the centre and is zero at the centre.

The graph drawn between V and r [drawn on the basis of eq. (1) and (7)] shows that in case of a solid sphere the potential continuously increases upto the centre of the sphere [fig. 5.18(a)]. The graph between E and r represents that the maximum value of intensity occurs at the surface of the sphere and its value decreases on proceeding towards the centre, becoming ultimately zero at the centre [fig. 5.18(b)].

5.9. Gravitational Field and Potential due to a Thick Spherical Shell

(i) **Point outside :** Consider a thick spherical shell of inner radius R_1 and outer radius R_2 with mass M . Similar to the case of sphere, we consider the thick spherical shell, composed of a number of thin spherical shells of masses $m_1, m_2, \dots$ etc. [fig. 5.19]. So that the gravitational potential at a point P at distance r , outside of the spherical shell, is given by

$$V = -\frac{Gm_1}{r} - \frac{Gm_2}{r} - \dots = -\frac{G}{r} (m_1 + m_2 + \dots)$$

or
$$V = -\frac{GM}{r} \quad \dots(1)$$

where $m_1 + m_2 + \dots = M$ is the mass of the thick spherical shell.

Intensity of field :
$$E = -\frac{dV}{dr} = -\frac{GM}{r^2} \quad \dots(2)$$

(ii) **Point on the surface :** At $r=R_2$, $V = -\frac{GM}{R_2}$ and $E = -\frac{GM}{R_2^2}$... (3)

(iii) **Point inside the spherical shell :** Let us consider the thick spherical shell to be divided into a number of concentric thin spherical shells. If we take a thin spherical shell of radius x and thickness dx , then similar to the case of solid sphere, the gravitational potential dV at a point P at r inside the shell is given by [fig. 5.20]

$$dV = -\frac{G4\pi x^2 dx \rho}{x} = -4\pi G \rho x dx$$

where ρ is the density of the material of the shell.

The potential V at the point P due to the thick spherical shell is obtained by integrating from R_1 to R_2 , i.e.,

$$V = \int_{R_1}^{R_2} -4\pi G \rho x dx = -2\pi G (R_2^2 - R_1^2) \rho \quad \dots(4)$$

But $\rho = \frac{M}{\left[\frac{4\pi}{3}(R_2^3 - R_1^3)\right]}$, $\therefore V = -\frac{3}{2}GM \left(\frac{R_2^2 - R_1^2}{R_2^3 - R_1^3}\right)$... (5)

Intensity of field $E = -\frac{dV}{dr} = 0$... (6)

(iv) **Point between two surfaces :** The point P lies at a distance r from the centre O between the two spherical surfaces of radii R_1 and R_2 . [fig. 5.21] Point P is on the surface of the shell of radii R_1 and r , due to which the potential is V_1 (say) and P is inside the spherical shell of radii r and R_2 , due to which the potential is V_2 (say). So that

$$V_1 = -\frac{G \times \frac{4\pi}{3}(r^3 - R_1^3)\rho}{r} \text{ and } V_2 = -2\pi G (R_2^2 - r^2) \rho$$

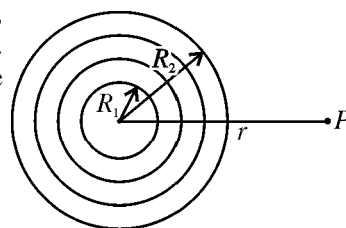


Fig. 5.19.

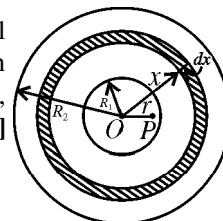


Fig. 5.20.

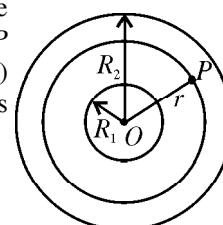


Fig. 5.21.

$$\begin{aligned} \therefore V &= V_1 + V_2 = -\frac{4}{3}\pi G\rho \left[\frac{(r^3 - R_1^3)}{r} + \frac{3}{2}(R_2^2 - r^2) \right] \\ \text{or } V &= -\frac{2}{3}\pi GP \left(3R_2^2 - \frac{2R_1^2}{r} - r^2 \right) \\ \text{or } V &= -\frac{GM}{2} \frac{(3R_2^2 - r^2 - 2R_1^2/r)}{(R_2^3 - R_1^3)} \quad \dots(7) \end{aligned}$$

$$\text{Hence, } E = -\frac{dV}{dr} = -\frac{GM}{r^2} \left(\frac{r^3 - R_1^3}{R_2^3 - R_1^3} \right) \quad \dots(8)$$

5.10. Velocity of Escape

It is a thing of common experience that when a body is projected in the upward direction, it returns back due to the gravitational pull of the earth on it. Now, if the body is projected upwards with such a velocity which will just take the body beyond the gravitational field of the earth, then it will never come back. This velocity of the body is called the **velocity of escape**.

Let us consider a body of mass m lying at a distance r ($> R$) from the centre of the earth (or planet or Sun). The force with which the earth attracts the body is given by

$$F = -\frac{GMm}{r^2}$$

where M is the mass of the earth.

If this mass be moved through a distance dr away from the earth, then the small work done on the body is

$$dW = \frac{GMm}{r^2} dr.$$

In order that the body may not return on the earth, sufficient work must be done on it to impart the body so much kinetic energy that it moves to infinity. Therefore the work done in moving the body from the surface of the earth to infinity is

$$W = \int_R^\infty \frac{GMm}{r^2} dr = \left[-\frac{GMm}{R} \right]_R^\infty = \frac{GMm}{R} \quad \dots(1)$$

Evidently, a body of K.E. greater than this (GMm/R) would escape completely from the earth. Hence the minimum velocity v of projection of the body to escape into space is given by the relation,

$$\frac{1}{2}mv^2 = \frac{GMm}{R} \quad \text{or} \quad v = \sqrt{\frac{2GM}{R}} \quad \dots(2a)$$

$$\text{But } g = GM/R^2, \quad \therefore v = \sqrt{2gR} \quad \dots(2b)$$

Expressions (2a) and (2b) are for the **escape velocity** of a body from earth.

If at the surface of the earth, we take $g = 9.8 \text{ m/sec}^2$ and its radius $R = 6.4 \times 10^6 \text{ m}$, then escape velocity

$$v = \sqrt{2 \times 9.8 \times 6.4 \times 10^6} = 11.12 \times 10^3 \text{ m/sec} = 11.2 \text{ km/sec}. \quad \dots(3)$$

Hence, if a body is projected in the upward direction with a velocity 11.2 km/sec or more than this, it will never return to the earth. Escape velocity is the same for all bodies of different masses.

If the body is situated initially at a distance r ($> R$) from the centre of earth, then for escaping of the body, the kinetic energy needed, will be

* If a body of mass m is situated on the surface of the earth where the value of acceleration due to gravity is g , then $mg = GMm/R^2$ or $g = GM/R^2$.

$$\frac{1}{2}mv^2 = \frac{GMm}{r}$$

$$\text{So that the escape velocity } v = \sqrt{\frac{2GM}{r}} = \sqrt{\frac{2gR^2}{r}} \quad \dots(4)$$

It may be noted that a particle, having velocity equal to or more than escape velocity, will escape even if it has not been projected vertically upwards.

From the expression for escape velocity $v = \sqrt{2GM/R}$, it is clear that if the particle is projected from the surface of some other planet or satellite (i.e., moon), the escape velocity will depend on its **mass** and **radius**. For moon

$$v = \sqrt{\frac{2 \times 6.6 \times 10^{-11} \times 7.34 \times 10^{22}}{1.74 \times 10^6}} = 2.3722 \text{ km/sec.} \quad \dots(5)$$

which is nearly $\frac{1}{5}$ th the value at the earth's surface.

At ordinary temperatures, the root mean square (R.M.S.) velocity for hydrogen is nearly 2 km/sec and for O_2 and CO_2 , it is nearly 0.5 km/sec. This is much below the escape velocity near the surface of the earth, hence the atmosphere is retained on earth's surface. But there may be some leakage of hydrogen (obtained from the dissociation of the vapour of water) because some of the gas molecules have much higher velocity than the mean value. The molecules of oxygen, obtained from the dissociation of water vapour, will be left in earth's atmosphere. Probably, due to this cause, *an excess of oxygen is found in the earth's atmosphere*, while the proportion of hydrogen is much higher in the rest of the universe. The escape velocity for the moon is nearly 2 km/sec, hence the molecules of hydrogen can easily escape out of the moon's atmosphere. Further, the molecules of other gases, having high velocity, will also leak from its atmosphere. *This explains the rarity of atmosphere at moon's surface.*

Thus, the gaseous atmosphere cannot be retained on smaller planets or satellites, which have escape velocity much lower.

Ex. 1. Two bodies of masses M_1 and M_2 are placed distant d apart. Show that at the position, where the gravitational field due to them is zero, the potential is given by

$$V = -\frac{G}{d}(M_1 + M_2 + 2\sqrt{M_1M_2}).$$

Sol. Let the gravitational field be zero at a point distant r from the mass M_1 . Then, the distance of the same point from the mass $M_2 = d - r$. At the point, under consideration, the field due to M_1 and M_2 is zero, i.e.,

$$-\frac{GM_1}{r^2} = -\frac{GM_2}{(d-r)^2} \text{ or } \frac{d-r}{r} = \sqrt{\frac{M_2}{M_1}}$$

Adding 1 on both sides, we have

$$\frac{d}{r} = \frac{\sqrt{M_2} + \sqrt{M_1}}{\sqrt{M_1}} \text{ or } r = \frac{d\sqrt{M_1}}{\sqrt{M_1} + \sqrt{M_2}}; \therefore d - r = d - \frac{d\sqrt{M_1}}{\sqrt{M_1} + \sqrt{M_2}} = \frac{d\sqrt{M_2}}{\sqrt{M_1} + \sqrt{M_2}}.$$

Therefore, potential at the point is

$$\begin{aligned} V &= -\frac{GM_1}{r} - \frac{GM_2}{(d-r)} = -\frac{GM_1}{d\sqrt{M_1}}(\sqrt{M_1} + \sqrt{M_2}) - \frac{GM_2}{d\sqrt{M_2}}(\sqrt{M_1} + \sqrt{M_2}) \\ &= -\frac{G}{d}(M_1 + \sqrt{M_1M_2} + M_2 + \sqrt{M_1M_2}) = -\frac{G}{d}(M_1 + M_2 + 2\sqrt{M_1M_2}). \end{aligned}$$

Ex. 2. Why do we call sometimes g as the intensity of gravity? Show that the earth's surface potential is equal to gR , where R is the radius of the earth.

Sol. If the earth is assumed to be a uniform sphere of radius R , then intensity of gravitational field at any point at its surface is given by

$$E = -\frac{GM}{R^2}, \text{ where } M \text{ is the mass of the earth.}$$

But the force with which the earth attracts a body of mass m lying at its surface is mg , hence

$$mg = -\frac{GMm}{R^2} \text{ or } g = -\frac{GM}{R^2} = E.$$

Thus the gravitational intensity at any point on the earth's surface is g and consequently g is termed as intensity of gravity.

$$\text{Potential at any point on earth's surface} = -\frac{GM}{R} = -\frac{GM}{R^2} R = gR.$$

Ex. 3. Find the gravitational potential at the surface of the earth from the following data :

Radius of earth $R = 6.37 \times 10^6$ m; mean density of the earth $\rho = 5500$ kg/m³ and $G = 6.66 \times 10^{-11}$ S.I. unit.

$$\begin{aligned} \text{Sol. } V &= -\frac{GM}{R} = -\frac{G}{R} \cdot \frac{4}{3} \pi R^3 \rho = -\frac{4}{3} G \pi R^2 \rho \\ &= -\frac{4}{3} \times 6.66 \times 10^{-11} \times 3.142 \times (6.37 \times 10^6)^2 \times 5500 = -6.2 \times 10^7 \text{ J/kg.} \end{aligned}$$

Ex. 4. Assuming the radius of earth $= 6.37 \times 10^6$ m, mean density $= 5.5 \times 10^3$ kg/m³, $G = 6.66 \times 10^{-11}$ nm²/kg², calculate the gravitational potential at the surface of earth. Hence find the velocity to project a body of mass 5.0 kg from the earth's surface so as to reach a height equal to the radius of earth above the earth's surface.

$$\text{Sol. As above, } V = -\frac{GM}{R} = -6.2 \times 10^7 \text{ J/kg}$$

Suppose a mass m is projected from the surface at P of the earth so that it reaches at a height $h (= R)$ at Q [fig 5.22]. From the law of conservation of energy,

Total mechanical energy at P = Total mechanical energy at Q

$$-\frac{GMm}{R} + \frac{1}{2}mv^2 = -\frac{GMm}{R+R} + 0 \quad (\text{At } Q, \text{ K.E.} = 0)$$

$$\text{Hence, } \frac{1}{2}mv^2 = \frac{GMm}{R} - \frac{GMm}{2R} = \frac{GMm}{2R} \text{ or } v^2 = \frac{GM}{R}$$

$$\therefore v = \sqrt{\frac{GM}{R}} = \sqrt{6.2 \times 10^7} = 7.9 \times 10^3 \text{ m/s.}$$

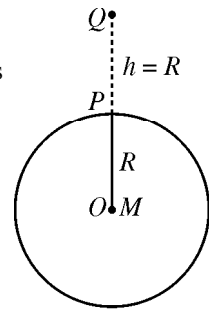


Fig. 5.22.

Ex. 5. If a mass of 50 kg is raised to a height $2R$ from the earth's surface, calculate the change in potential energy. ($g = 9.8$ m/sec²; $R = 6.4 \times 10^6$ m)

Sol. The potential energy of a mass m distance r from a solid sphere of mass M is given by

$$U(r) = \text{Potential} \times m = -\frac{GMm}{r} \quad \left[\because V = -\frac{GM}{r} \right] \quad (\text{referring zero potential energy at infinite distance}).$$

But $g = GM/R^2$, where R is the radius of earth,

$$\therefore U(r) = -\frac{gR^2m}{r}$$

Difference of potential energy of mass m between the points at r_1 and r_2 is

$$U(r_2) - U(r_1) = -\frac{gR^2m}{r_2} + \frac{gR^2m}{r_1}$$

Here $r_1 = R$ and $r_2 = 2R + R = 3R$

$$\therefore U(r_2) - U(r_1) = -\frac{gRm}{3} + gRm = \frac{2}{3}gRm$$

$$= \frac{2}{3} \times 9.8 \times 50 \times 6.4 \times 10^6 = 2.1 \times 10^9 \text{ joules.}$$

Ex. 6. What is the potential energy of a mass of 1 kg on the surface of the earth, referred to zero potential energy at infinite distance ? Calculate also its potential energy at a distance of 10^5 km from the centre of the earth.

Sol. In case of earth (or solid sphere) the potential energy is given by

$$U(r) = -\frac{GMm}{r}$$

(i) For the surface of the earth, $r = R$

$$\therefore U(r) = \frac{GMm}{R} = -\frac{(6.67 \times 10^{-11}) \times 5.98 \times 10^{24} \times 1}{6.37 \times 10^6} = -6.37 \times 10^7 \text{ J.}$$

(ii) $r = 10^5 \text{ km} = 10^8 \text{ m}$,

$$U(r) = -\frac{(6.67 \times 10^{-11}) \times 5.98 \times 10^{24} \times 1}{10^8} = -3.98 \times 10^6 \text{ J.}$$

Ex. 7. What is the velocity of projection of a body so that it reaches to a height equal to twice the radius of the earth ? (A.N.U. 2001)

Sol. The potential energy, when the body is at the surface of the earth, is given by

$$U_1 = -G \frac{Mm}{R} \quad \dots(i)$$

where R = radius of the earth.

Potential energy at height $h = 2R$ is

$$U_2 = -\frac{GMm}{R+h} = -\frac{GMm}{R+2R} = -\frac{GMm}{3R}$$

Hence the increase in potential energy of the body is

$$U_2 - U_1 = -\frac{GMm}{3R} - \left(-\frac{GMm}{R}\right) = \frac{2}{3} \frac{GMm}{R}$$

This increase in potential energy is provided by the kinetic energy of projection of the body i.e.,

$$\frac{1}{2}mv^2 = \frac{2}{3} \frac{GMm}{R}$$

$$v = \sqrt{\frac{4GM}{3R}} = \sqrt{\frac{4 \times 6.67 \times 10^{-11} \times 5.98 \times 10^{24}}{3 \times 6}} = 9133 \text{ m/sec.}$$

Ex. 8. If a mass m is raised to a height equal to the radius of the earth, what is the work done ? (A.N.U. 2003)

Sol. Potential energy at the surface of the earth, $U_1 = -\frac{GMm}{R}$

Potential energy at a height $h = R$, $U_2 = -\frac{GMm}{R+R} = -\frac{GMm}{2R}$

$$\text{Work done} = U_2 - U_1 = -\frac{GMm}{2R} + \frac{GMm}{R} = \frac{GMm}{2R} = \frac{gRm}{2} \quad \left[\because g = \frac{GM}{R^2} \right]$$

Ex. 9. Estimate the mass of the Sun assuming the orbit of the earth round the Sun to be a circle. The distance between the Sun and the earth is $1.49 \times 10^{11} \text{ m}$ and $G = 6.67 \times 10^{-11} \text{ N-mkg}^{-2}$.

(A.N.U. 2007, 01; S.V.U. 2003)

Sol. If m and M are the masses of earth and Sun respectively and the earth is revolving about the Sun in a circular path of radius r with angular velocity ω , then

$$\frac{GMm}{r^3} = m\omega^2 r \quad \text{or} \quad M = \frac{\omega^2 r^2}{G}$$

Here, angular velocity ω of earth around the Sun is given by $\omega = \frac{2\pi}{365 \times 24 \times 60 \times 60}$ rad/sec,
 $r = 1.49 \times 10^{11}$ m and $G = 6.66 \times 10^{-11}$ S.I. unit

$$\therefore M = \left(\frac{2\pi}{365 \times 24 \times 60 \times 60} \right)^2 \times \frac{(1.49 \times 10^{11})^2}{6.66 \times 10^{-11}} = 1.972 \times 10^{30} \text{ kg.}$$

Ex. 10. A mass m is taken from the surface of a sphere of radius a and mass M to its centre. Calculate the necessary work done.

Sol. Work done = $U_2 - U_1$ = Potential energy at the centre – Potential energy at the surface

$$= m(V_2 - V_1) = m \left[-\frac{3}{2} \frac{GM}{a} + \frac{GM}{a} \right] = -\frac{GMm}{2a}.$$

Ex. 11. A body from the surface of the earth is projected with velocity v_0 . Prove that the velocity at a height h is given by

$$v^2 = v_0^2 - \frac{2gh}{1+(h/R)}.$$

Sol. From the law of conservation of energy,

Mechanical energy at earth's surface = Mechanical energy at height h

$$\frac{1}{2}mv_0^2 - \frac{GMm}{R} = \frac{1}{2}mv^2 - \frac{GMm}{h+R}$$

$$\text{or} \quad v^2 = v_0^2 + \frac{2GM}{h+R} - \frac{2GM}{R} = v_0^2 - \frac{2GMh}{R(h+R)}$$

$$\text{or} \quad v^2 = v_0^2 - \frac{2GMh}{R^2(1+h/R)} = v_0^2 - \frac{2gh}{1+(h/R)} \quad \left[\because g = \frac{GM}{R^2} \right]$$

Ex. 12. If a straight tunnel is made between two places of the earth, then find the component of gravitational force along the tunnel at a distance x from the middle point of the tunnel.

Sol. If O is the centre of the earth and C is the middle point of the tunnel (fig. 5.23), then

$$\cos OPC = \frac{CP}{OP} = \frac{x}{r}, \text{ where } r = PO.$$

The gravitational force at P along $PO = \frac{GMm}{R^3} r$

Its component along tunnel = $\frac{GMm}{R^3} r \cos OPC$

$$= \frac{GMmr}{R^3} \cdot \frac{x}{r} = \frac{mgx}{R} \quad \left[\because g = \frac{GM}{R^2} \right]$$

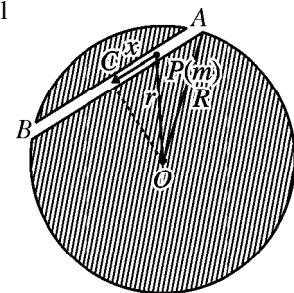


Fig. 5.23.

Ex. 13. For the earth-Sun system, calculate (i) the kinetic energy and (ii) the work which will have to be done in doubling the radius of the orbit of the earth.

Sol. The gravitational potential at the earth due to the Sun = $-GM_s/r$

$$\therefore \text{Potential energy of the earth} = -\frac{GM_s M_e}{r}$$

$$\text{K.E. of the earth} = \frac{1}{2} M_e v^2.$$

In case of earth, the necessary centripetal force for its rotation is provided by gravitational attraction, i.e.,

$$\frac{M_e v^2}{r} = \frac{GM_s M_e}{r^2}$$

$$\therefore \frac{1}{2} M_e v^2 = \frac{1}{2} \frac{GM_s M_e}{r}$$

$$\therefore \text{Total energy of the earth} = -\frac{GM_s M_e}{r} + \frac{1}{2} \frac{GM_s M_e}{r} = -\frac{GM_s M_e}{2r}.$$

Evidently, negative sign indicates that the earth is bound with the earth with this energy, i.e.,

$$\begin{aligned} \text{The binding energy} &= \frac{GM_s M_e}{2r} = \frac{1}{2} M_e v^2 = \frac{1}{2} M_e \times r^2 \omega^2 \\ &= \frac{1}{2} \times 5.98 \times 10^{24} \times \left(\frac{1.5 \times 10^{11} \times 2 \times 3.14}{365 \times 24 \times 60 \times 60} \right)^2 = \mathbf{2.7 \times 10^{33} \text{ joules.}} \\ [\because r &= 1.5 \times 10^{11} \text{ m and } \omega = \frac{2\pi}{365 \times 24 \times 60 \times 60} \text{ rad/sec.}] \end{aligned}$$

When the radius of the earth's orbit is doubled, then the final binding energy will be

$$= \frac{GM_s M_e}{2(2r)} = \frac{1}{2} \frac{GM_s M_e}{2r} = \frac{2.7}{2} \times 10^{33} = \mathbf{1.35 \times 10^{33} \text{ joules.}}$$

Therefore the amount of work done in doubling the radius of the orbit

$$= 2.7 \times 10^{33} - 1.35 \times 10^{33} = \mathbf{1.35 \times 10^{33} \text{ joules.}}$$

5.11. Motion of a System of Two Particles Under Central Force

We plan to study the motion of two-particle system, interacting mutually through central force or inverse square law force. We will see that when central force is acting between two bodies, this two-body problem can always be reduced to one-body problem. Let us consider two particles of masses m_1 and m_2 , whose instantaneous position vectors in an inertial frame with origin O_i are $\vec{r}_1$ and $\vec{r}_2$

[fig. 5.24]. Hence the vector distance of m_1 from m_2 is $\vec{r} = \vec{r}_1 - \vec{r}_2$. If m_1 and m_2 interact via central force* collinear with vector $\vec{r}$, then the forces acting on the two particles are given by

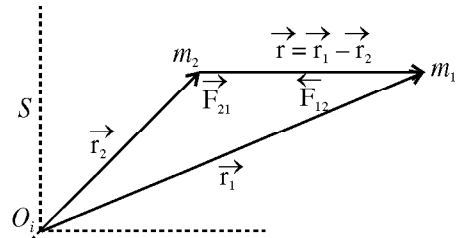


Fig. 5.24.

$$\vec{F}_{12} = m_1 \frac{d^2 \vec{r}_1}{dt^2} \quad \text{and} \quad \vec{F}_{21} = m_2 \frac{d^2 \vec{r}_2}{dt^2} \quad \dots(1)$$

But according to Newton's third law of motion,

$$\vec{F}_{12} = -\vec{F}_{21} = \vec{F} \text{ (say), } \therefore \frac{d^2 \vec{r}_1}{dt^2} = \frac{\vec{F}}{m_1} \quad \text{and} \quad \frac{d^2 \vec{r}_2}{dt^2} = -\frac{\vec{F}}{m_2}$$

* This central force may be electrostatic or gravitational.

Hence
$$\frac{d^2}{dt^2}(\vec{r}_1 - \vec{r}_2) = \vec{F} \left(\frac{1}{m_1} + \frac{1}{m_2} \right)$$

If $\hat{r}$ is the unit vector along $\vec{r} = \vec{r}_1 - \vec{r}_2$ then $\vec{F} = F\hat{r}$, where F is negative for attractive force and positive for repulsive force. For central force, F is any function of distance r . Now,

$$\frac{d^2 \vec{r}}{dt^2} = \left(\frac{1}{m_1} + \frac{1}{m_2} \right) F\hat{r}. \quad \dots(2)$$

If we put $\frac{1}{\mu} = \frac{1}{m_1} + \frac{1}{m_2}$ in this equation, then

$$\frac{d^2 \vec{r}}{dt^2} = \frac{\vec{F}}{\mu} = \frac{F\hat{r}}{\mu} \quad \dots(3)$$

or
$$\vec{F} = \mu \frac{d^2 \vec{r}}{dt^2} \quad \dots(4)$$

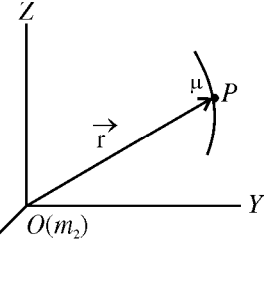


Fig. 5.25.

This is the **equation of motion for a two-particle system, reduced in the form of a one-body problem**, because it is similar to the equation of motion of a single particle of mass μ at vector distance $\vec{r}$ from one of the two particles, considered as centre of force O^* [fig. 5.25]. Thus, the original two-body problem, involving the calculation of $\vec{r}_1$ and $\vec{r}_2$ has been reduced to the calculation of a single vector $\vec{r}$ as function of time. The quantity

$\mu = \frac{m_1 m_2}{m_1 + m_2}$ is therefore called the **reduced mass** of the two particles, having masses m_1 and m_2 . The reduced mass μ has a value less than m_1 or m_2 .

If $m_2 \gg m_1$, then

$$\mu = \frac{m_1 m_2}{m_1 + m_2} = \frac{m_1}{1 + m_1/m_2} \approx m_1 \text{ (say } m) \quad \dots(5)$$

In such a case, we have to consider only a one-body problem. When we do not need much accuracy, this approximation will be sufficient.

In **hydrogen atom**, an electron of mass m revolves round a proton of much heavier mass M . The **reduced mass of the hydrogen atom** will be given by

$$\mu = \frac{mM}{m + M} = m \left(1 + \frac{m}{M} \right)^{-1} = m \left(1 - \frac{m}{M} \right) \quad \dots(6)$$

But
$$\frac{\text{Mass of the electron}}{\text{Mass of the proton}} = \frac{m}{M} = \frac{1}{1836}$$

$$\therefore \mu = m \left(1 - \frac{1}{1836} \right) \quad \dots(7)$$

Hence in case of hydrogen atom, if the energy, period etc. are calculated by assuming that an electron moves round a fixed proton, there will occur some error. To get the correct results, m should be replaced by the reduced mass μ in the expressions of energy, period etc. Actually, in case of hydrogen atom, this error will be very small, because

* Remember that the distance of m_1 from m_2 is $\vec{r}$ and also the distance (of reduced mass) from the origin O is $\vec{r}$ i.e., the path of m_1 with respect to m_2 will be same as that of μ mass relative to O .

$$m \left(1 - \frac{1}{1836} \right) \cong m \text{ nearly} \quad \dots(8)$$

Positronium is a temporary combination of a positron and an electron similar to a hydrogen atom. A positron is a particle which has mass equal to the electron mass but has positive charge. The reduced mass of the positronium is given by

$$\mu' = \frac{mm}{m+m} = \frac{m}{2} \quad \dots(9)$$

($\because m = \text{mass of the electron} = \text{mass of the positron}$)

Thus, the reduced mass of the positronium particle is one-half the mass of the electron.

5.12. Centre of Mass and Relative Co-ordinate System

When reducing two-body problem to one-body problem, we have taken the position vectors $\vec{r}_1$ and $\vec{r}_2$ of masses m_1 and m_2 respectively in an inertial frame. The relative position vector of m_1 relative to m_2 is $\vec{r} = \vec{r}_1 - \vec{r}_2$. If a frame of reference is attached to m_2 , shown in **fig. 5.25** by frame of reference XYZ with origin O , then this frame of reference (XYZ) is a relative co-ordinate system (with respect to the frames attached to O_i).

As a force is acting on m_2 , this m_2 or O is in the state of acceleration in the inertial frame S , hence $OXYZ$ frame will be a non-inertial frame. Looking at the equation,

$$\vec{F} = \mu \frac{d^2 \vec{r}}{dt^2}, \quad \dots(1)$$

one may forget this fact. *Actual paths of both particles will depend on the law of force, initial positions and initial velocities.* When studying planetary motion, we will consider m_2 (or mass of Sun M) as origin O and the relative vector $\vec{r}$ ($= \vec{r}_1 - \vec{r}_2$) of m_1 (or mass of planet m) with respect to m_2 (or M) will be taken as the position vector $\vec{r}$ ($= \vec{r}_1 - \vec{r}_2$) of the reduced mass μ .

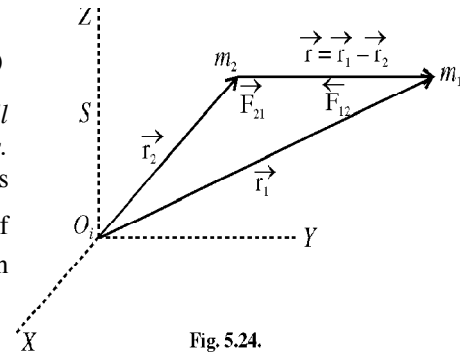


Fig. 5.24.

In **fig. 5.26**, the position vector $\vec{R}$ of centre of mass of m_1 and m_2 masses, situated at distances $\vec{r}_1$ and $\vec{r}_2$ respectively from O_i , is given by

$$\vec{R} = \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2}{m_1 + m_2} \quad \dots(2)$$

$$\text{Also,} \quad \vec{r} = \vec{r}_1 - \vec{r}_2 \quad \dots(3)$$

Solving eqs. (1) and (2), we get

$$\vec{r}_1 = \vec{R} + \frac{m_2}{m_1 + m_2} \vec{r} \quad \text{and} \quad \vec{r}_2 = \vec{R} - \frac{m_1}{m_1 + m_2} \vec{r} \quad \dots(4)$$

$$\text{But} \quad \mu = \frac{m_1 m_2}{m_1 + m_2},$$

$$\text{hence} \quad \vec{r}_1 = \vec{R} + \frac{\mu}{m_1} \vec{r} \quad \text{and} \quad \vec{r}_2 = \vec{R} - \frac{\mu}{m_2} \vec{r} \quad \dots(5)$$

In eq. (5) we have obtained the position $\vec{r}_1$ and $\vec{r}_2$ of the two particles in the inertial frame (S) in terms of the position vector of the centre of mass and relative vector $\vec{r}$.

Differentiating eq. (2) with respect to t , we get the velocity of the centre of mass (C.M.), i.e.,

$$\vec{V} = \frac{d\vec{R}}{dt} = \frac{m_1 d\vec{r}_1/dt + m_2 d\vec{r}_2/dt}{m_1 + m_2} \quad \text{or} \quad \vec{V} = \frac{m_1 \vec{v}_1 + m_2 \vec{v}_2}{m_1 + m_2} \quad \dots(6)$$

Also,

$$\vec{F}_{12} = m_1 \frac{d^2 \vec{r}_1}{dt^2} \quad \text{and} \quad \vec{F}_{21} = m_2 \frac{d^2 \vec{r}_2}{dt^2}$$

As the two masses m_1 and m_2 are interacting mutually, Newton's third law of motion is applicable. so that

$$\vec{F}_{12} + \vec{F}_{21} = 0 \quad \text{or} \quad m_1 \frac{d^2 \vec{r}_1}{dt^2} + m_2 \frac{d^2 \vec{r}_2}{dt^2} = 0 \quad \text{or} \quad \frac{d}{dt} \left(m_1 \frac{d\vec{r}_1}{dt} + m_2 \frac{d\vec{r}_2}{dt} \right) = 0$$

Integrating $m_1 \frac{d\vec{r}_1}{dt} + m_2 \frac{d\vec{r}_2}{dt} = \text{a constant}$ or $m_1 \vec{v}_1 + m_2 \vec{v}_2 = \text{a constant} \quad \dots(7)$

Hence from eq. (6), the velocity of centre of mass is given by

$$\vec{V} = \frac{d\vec{R}}{dt} = \text{a constant} \quad \dots(8)$$

Thus **the centre of mass of two-body system moves with constant velocity**. If we attach an inertial frame to the centre of mass (C.M.), then such a frame of reference is called **centre of mass-frame of reference** (or **C.M. frame of reference**). In this centre of mass-frame of reference, the velocity of centre of mass is zero (i.e., $\vec{V} = 0$). (It is usually convenient to study the two-body problem in C.M. frame). The position vectors of the two particles in the C.M. frame will be obtained as [by putting $\vec{R} = 0$ in eq. (5)]:

$$\vec{r}_1 = \frac{\mu}{m_1} \vec{r} \quad \text{and} \quad \vec{r}_2 = -\frac{\mu}{m_2} \vec{r} \quad \dots(9)$$

It will be proper to discuss about a specific case. If the force $F = \mu\omega^2 r$, then the equivalent particle of mass μ will move on a circle of radius r with respect to O (or m_2) [fig. 5.27]. However in the inertial frame, attached to the centre of mass, both the particles will move in circular orbits of radius r_1 and r_2 with the same angular velocity [fig. 5.28], where the distance between m_1 and m_2 is r and $r = r_1 + r_2$.

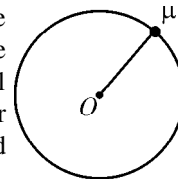


Fig. 5.27.

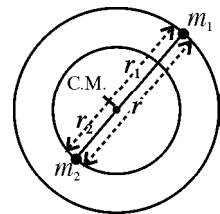


Fig. 5.28.

5.13. Equation of Motion Under Central Force – Equation of Orbit

Consider a particle of mass m , moving under the action of a central force, given by

$$\vec{F} = F(r) \hat{r} \quad \dots(1)$$

where $F(r)$ is the magnitude of the central force and is the scalar function of distance r . Here unit vector

$\hat{r} = \vec{r}/r$ and $\vec{r}$ is the position vector of the particle. This central force, acting on the particle is directed always towards or away a point (centre) and is radial. If $F(r) < 0$ or $F(r) > 0$, the central force is attractive or repulsive respectively.

According to Newton's second law of motion, the force $\vec{F}$ can be expressed as

$$\vec{F} = m \vec{a} \quad \dots(2)$$

where $\vec{a} = d^2 \vec{r} / dt^2$ is the acceleration of the particle. When we are dealing the motion of a two particle system, m is to be replaced by reduced mass μ . In polar co-ordinates (r, θ) , the acceleration is expressed as

$$\vec{a} = (\ddot{r} - r\dot{\theta}^2)\hat{r} + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\hat{\theta} \quad \dots(3)$$

where $(\ddot{r} - r\dot{\theta}^2)$ is the radial acceleration (along $\hat{r}$) and $2\dot{r}\dot{\theta} + r\ddot{\theta}$ is the transverse acceleration (along $\hat{\theta}$).

Here, $\dot{r} = \frac{dr}{dt}$, $\dot{\theta} = \frac{d\theta}{dt}$, $\ddot{r} = \frac{d^2r}{dt^2}$ and $\ddot{\theta} = \frac{d^2\theta}{dt^2}$.

$$\text{Hence, force, } \vec{F} = m \vec{a} = m (\ddot{r} - r\dot{\theta}^2)\hat{r} + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\hat{\theta} \quad \dots(4)$$

From eqs. (1) and (4),

$$m(\ddot{r} - r\dot{\theta}^2)\hat{r} + m(r\ddot{\theta} + 2\dot{r}\dot{\theta})\hat{\theta} = F(r)\hat{r}. \quad \dots(5)$$

On equating the coefficients of $\hat{r}$ and $\hat{\theta}$ of eq. (5)

$$m(\ddot{r} - r\dot{\theta}^2) = F(r) \quad (\because \dot{\theta} = d\theta/dt = \omega, \text{ angular velocity}) \quad \dots(6)$$

$$\text{and} \quad r\ddot{\theta} + 2\dot{r}\dot{\theta} = 0 \quad \dots(7)$$

where eq. (6) is the equation for radial motion.

Eq. (7) can be written as

$$2\dot{r}\dot{\theta} + r\ddot{\theta} = \frac{1}{r} \frac{d}{dt}(r^2\dot{\theta}) = 0$$

$$\text{whence} \quad r^2\dot{\theta} = r^2\omega = h, \text{ a constant} \quad \dots(8)$$

From eq. (8), $\omega = h/r^2$. Hence eq. (6) can be written as

$$m \left[\ddot{r} - r \left(\frac{h}{r^2} \right)^2 \right] = F(r)$$

$$\text{or} \quad m\ddot{r} - \frac{mh^2}{r^3} = F(r) \quad \dots(9)$$

This is the equation of motion of a particle subjected to a central force $\vec{F} = F(r)\hat{r}$.

We can put this equation in a form which is suitable to carry out to study the nature of orbits.

$$\text{Now, } \dot{r} = \frac{dr}{dt} = \frac{dr}{d\theta} \cdot \frac{d\theta}{dt} \cdot \omega = \frac{h}{r^2} \frac{dr}{d\theta} \quad (\because r^2\omega = h)$$

$$\text{Therefore } \ddot{r} = \frac{d^2r}{dt^2} = \frac{d}{dt} \left(\frac{dr}{dt} \right) = \frac{d}{d\theta} \left(\frac{h}{r^2} \frac{dr}{d\theta} \right) \frac{d\theta}{dt} = \omega \frac{d}{d\theta} \left(\frac{h}{r^2} \frac{dr}{d\theta} \right) = \frac{h}{r^2} \frac{d}{d\theta} \left(\frac{h}{r^2} \frac{dr}{d\theta} \right)$$

$$\text{Let } u = \frac{1}{r}, \text{ hence } \frac{du}{d\theta} = -\frac{1}{r^2} \frac{dr}{d\theta} \text{ or } \frac{1}{r^2} \frac{dr}{d\theta} = -\frac{du}{d\theta}$$

$$\therefore \ddot{r} = h^2 u^2 \frac{d}{d\theta} \left(-\frac{du}{d\theta} \right) = -h^2 u^2 \frac{d^2 u}{d\theta^2}$$

Hence eq. (9) takes the form

$$-mh^2u^2 \frac{d^2u}{d\theta^2} - mh^2u^3 = \frac{F(1/u)}{m}$$

$$\therefore \frac{d^2u}{d\theta^2} + u = -\frac{F(1/u)}{h^2u^2} \quad \dots(10)$$

This is the **differential equation of orbit of a particle under the influence of a central force.**

Now, **if the central force is attractive** (e.g., gravitational force), we can write

$$F(r) = -f(r) \text{ or } F(1/u) = -f(1/u)$$

where $f(r)$ or $f(1/u)$ is the magnitude of the force and a positive quantity. In such a case,

$$-\frac{F(1/u)}{m} = \frac{f(1/u)}{m} = p, \text{ force per unit mass.}$$

Hence, from eq. (10), we obtain

$$\frac{d^2u}{d\theta^2} + u = \frac{p}{h^2u^2} \quad \dots(11)$$

This is the **differential equation of the orbit of a particle moving under an attractive central force p per unit mass.**

Equation of Orbit Under Inverse Square Law Force

In case, the central force is an inverse square law force of the type

$$F(r) = -\frac{K}{r^2} \text{ or } F(1/u) = -Ku^2,$$

Then the equation of the orbit assumes the form [from eq. (10)],

$$\frac{d^2u}{d\theta^2} + u = \frac{K}{mh^2} \quad \dots(12)$$

Equation of Orbit Under Gravitational Force between Two Masses

Specifically if the central force is attractive gravitational force between two masses m_1 and m_2 , then

$$\vec{F} = -\frac{Gm_1m_2}{r^2} \hat{r}$$

So that
$$F(r) = -\frac{Gm_1m_2}{r^2} \text{ or } F(1/u) = -Gm_1m_2u^2 \quad \dots(13)$$

In such a case, m is to be replaced by the reduced mass $\mu = m_1m_2/(m_1 + m_2)$ in order to discuss two-body problem in the form of one-body problem ($\vec{F} = \mu \vec{a}$).

Hence, from eq. (10), we obtain

$$\frac{d^2u}{d\theta^2} + u = \frac{Gm_1m_2}{\mu h^2} \quad \dots(14)$$

This is the **differential equation of orbit for a two-particle system moving under gravitational force between two masses.**

5.14. Motion Under Inverse Square Law Force and Equation of Motion

According to the inverse square law force, the force, acting between two particles at a distance r apart, is given by

$$F(r) = -\frac{K}{r^2} \quad \dots(1)$$

where K is a constant. If one particle is at the origin and the other at the position $\vec{r}$, then the force acting on the second particle is given by

$$\vec{F} = -\frac{K}{r^2} \hat{r} \quad \dots(2)$$

Examples of such forces are gravitational and electrostatic forces. For gravitational force between two point masses m_1 and m_2 ; $K = Gm_1m_2$ and for the electrostatic force between point charges q_1 and q_2 ;

$K = -\frac{1}{4\pi\epsilon_0} q_1q_2$ (in S.I. system), when they are situated in air or vacuum, i.e.,

$$\vec{F} = -\frac{Gm_1m_2}{r^2} \hat{r} \quad ; \quad \vec{F} = \frac{1}{4\pi\epsilon_0} \frac{q_1q_2}{r^2} \hat{r} \quad \dots(3)$$

Newton's law of Gravitation ; Coulomb's law

The gravitational force is always attractive. The electrostatic (coulomb) force is attractive if the charges q_1, q_2 have opposite signs and repulsive if q_1, q_2 have the same signs.

The inverse square law force is usually the force between two particles. We know that the motion of a system of two particles under central force can always be reduced to one-body problem. Hence the **equation of motion** for a system of two particles of masses m_1 and m_2 under inverse square law force is given by [using eq. (2)]

$$\mu \frac{d^2 \vec{r}}{dt^2} = -\frac{K}{r^2} \hat{r} \quad \dots(4)$$

where $\mu = m_1m_2/(m_1 + m_2)$ is the reduced mass of the system.

Motion of Two-Body System Under Inverse Square Law of Force of Gravitation

If two bodies of masses m_1 and m_2 are moving under mutual gravitational force, the equation of motion for equivalent one-body problem will be obtained by putting $-\frac{Gm_1m_2}{r^2} \hat{r}$ in place of $\vec{F}$ in $\vec{F} = \frac{\mu d^2 \vec{r}}{dt^2}$, i.e.,

$$\mu \frac{d^2 \vec{r}}{dt^2} = -\frac{Gm_1m_2}{r^2} \hat{r} \quad \dots(5)$$

$$\text{or} \quad \frac{d^2 \vec{r}}{dt^2} = -\frac{G(m_1 + m_2)}{r^2} \hat{r} \quad \dots(6)$$

where $\vec{r}$ is the vector distance of m_1 from m_2 and $\mu = \frac{m_1m_2}{m_1 + m_2}$ is the reduced mass of the two bodies.

Thus the two-body problem under gravitational force has been reduced to one-body problem, where we have to study about the motion of a body of mass μ . In order to obtain the orbit in two-body problem, we have to solve only one-body problem.

In the circular path of a satellite, the centrifugal force on μ mass is balanced by gravitational force, i.e.,

$$\mu\omega^2 r = \frac{GMm}{r^2} \quad \text{or} \quad \omega^2 r = \frac{G(M + m)}{r^2} \quad \dots(7)$$

where $m_1 = m$ is the mass of the satellite and $m_2 = M$ is the mass of the earth.

$$\text{Thus,} \quad \text{period } T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{r^3}{G(M + m)}} \quad \dots(8)$$

If the mass m is negligible in comparison to M , then

$$T = 2\pi\sqrt{\frac{r^3}{GM}} \quad \dots(9)$$

Thus when a satellite is moving around the earth and its mass is not negligible in comparison to the mass of the earth, we will get correct results by replacing $(M + m)$ for M . This correction may be negligible for a satellite of small mass. However this correction may not be negligible for moon, revolving around earth, and heavy planets around Sun.

Theorem 1. Prove that the angular momentum of two-particle system under inverse square law force is conserved.

Proof : If the angular momentum of a system is $\vec{L}$, the torque $\vec{\tau}$ acting on it is given by

$$\vec{\tau} = \frac{d\vec{L}}{dt} = \vec{r} \times \vec{F} \quad \dots(i)$$

$$\text{Here,} \quad \vec{F} = -\frac{K}{r^2} \hat{r} = -\frac{K}{r^3} \vec{r} \quad \left[\because \hat{r} = \frac{\vec{r}}{r} \right]$$

$$\text{Hence} \quad \frac{d\vec{L}}{dt} = \vec{r} \times \left(-\frac{K}{r^3} \vec{r} \right) = -\frac{K}{r^3} \vec{r} \times \vec{r} = 0$$

$$\text{On integration,} \quad \vec{L} = \text{a constant} \quad \dots(ii)$$

Thus *the angular momentum of a system under inverse square law of force is conserved*. In two-body problem, the value of angular momentum is $\mu r^2 \omega = \text{constant}$. This can be proved as follows :

Remember that $\vec{F} = \mu \frac{d^2 \vec{r}}{dt^2}$, hence in equivalent one-body problem angular momentum $\vec{L}$ ($= \vec{r} \times \mu \vec{v}$) will be conserved. Thus

$$\vec{L} = \vec{r} \times \mu \vec{v} = \text{constant} \quad \dots(iii)$$

$$\text{But} \quad \vec{v} = \dot{r} \hat{r} + r\omega \hat{\theta} \quad (\text{in plane polar co-ordinates})$$

where $\hat{r}$ and $\hat{\theta}$ are unit vectors, mutually perpendicular and $\dot{r} = dr/dt$ and $\omega = d\theta/dt$.

$$\text{In magnitude,} \quad v = \sqrt{\dot{r}^2 + r^2 \omega^2}$$

$$\text{Hence} \quad \vec{L} = \vec{r} \times \mu (\dot{r} \hat{r} + r\omega \hat{\theta}) = \mu \dot{r} \vec{r} \times \hat{r} + \mu r \omega \vec{r} \times \hat{\theta} \quad \dots(iv)$$

$$\text{But} \quad \vec{r} \times \hat{r} = \vec{r} \times \vec{r}/r = 0 \quad \text{and} \quad \vec{r} \times \hat{\theta} = r \hat{r} \times \hat{\theta} = r \hat{z},$$

where $\hat{z}$ is a unit vector, in direction perpendicular to the plane of $\hat{r}$ and $\hat{\theta}$.

$$\text{Hence} \quad \vec{L} = \mu r^2 \omega \hat{z} = \text{constant} \quad \dots(v)$$

i.e., in magnitude, $L = \mu r^2 \omega = \text{constant}$.

Theorem 2. Prove that the total energy of a system of two particles is conserved under inverse square force.

Proof : The equation of motion of a system under inverse square law of force is given by

$$\mu \frac{d^2 \vec{r}}{dt^2} = -\frac{K}{r^2} \hat{r} \quad \dots(i)$$

This force is radial and we know that the radial acceleration in plane polar co-ordinates is

$$\ddot{r} - r\dot{\theta}^2 = \frac{d^2r}{dt^2} - r\omega^2.$$

Therefore,
$$\mu \frac{d^2 \vec{r}}{dt^2} = \mu \left(\frac{d^2r}{dt^2} - r\omega^2 \right) \hat{r} = -\frac{K}{r^2} \hat{r}$$

Thus,
$$\mu \frac{d^2r}{dt^2} - \mu r\omega^2 = -\frac{K}{r^2} \quad \dots(ii)$$

But $\omega = \frac{L}{\mu r^2}$ (as proved above) and $F = -\frac{K}{r^2} = -\frac{dU}{dr}$

$$\therefore \mu \frac{d^2r}{dt^2} - \frac{L^2}{\mu r^3} = -\frac{dU}{dr} \quad \text{or} \quad \mu \frac{d^2r}{dt^2} = -\frac{d}{dr} \left(\frac{1}{2} \frac{L^2}{\mu r^2} + U \right)$$

Multiplying by $\frac{dr}{dt}$ on both sides,

$$\mu \frac{d^2r}{dt^2} \frac{dr}{dt} = -\frac{d}{dr} \left(\frac{1}{2} \frac{L^2}{\mu r^2} + U \right) \frac{dr}{dt} \quad \text{or} \quad \frac{d}{dt} \left[\frac{1}{2} \mu \left(\frac{dr}{dt} \right)^2 \right] = -\frac{d}{dt} \left(\frac{1}{2} \mu r^2 \omega^2 + U \right) \quad [\because L = \mu r^2 \omega]$$

or
$$\frac{d}{dt} \left[\frac{1}{2} \mu \left(\frac{dr}{dt} \right)^2 + \frac{1}{2} \mu r^2 \omega^2 + U \right] = 0 \quad \dots(iii)$$

On integration
$$\frac{1}{2} \mu (\dot{r}^2 + r^2 \omega^2) + U = \text{a constant, say } E \quad \dots(iv)$$

But $\dot{r}^2 + r^2 \omega^2 = v^2$, hence
$$\frac{1}{2} \mu v^2 + U(r) = E, \text{ a constant} \quad \dots(v)$$

where E is the total energy (kinetic energy + potential energy) of the system.

Thus *the total energy of a system under inverse square force remains constant.*

The gravitational force between celestial bodies, like Sun and earth, is an inverse-square-force. We plan to study about the motion of celestial bodies under this inverse-square-force.

Ex. 1. Calculate the reduced mass of hydrogen molecule (H_2). The mass of hydrogen atom is 1.7×10^{-27} kg.

Sol. A molecule of hydrogen consists of two hydrogen atoms, of same mass m . The reduced mass of hydrogen molecule is given by

$$\mu = \frac{m \times m}{m + m} = \frac{m}{2}$$

Here, $m = 1.7 \times 10^{-27}$ kg, $\therefore \mu = \frac{1}{2} \times 1.7 \times 10^{-27}$ kg = **8.5×10^{-28} kg.**

Ex. 2. The maximum and minimum distances of a comet from the Sun are 1.4×10^{12} m and 7×10^{10} m respectively. If its velocity nearest to the Sun is 6×10^4 m/sec, what is its velocity when farthest.

Sol. From the law of conservation of angular momentum for the comet,

$$\mu v_1 r_1 = \mu v_2 r_2 \therefore v_1 = \frac{v_2 r_2}{r_1} = \frac{6 \times 10^4 \times 7 \times 10^{10}}{1.4 \times 10^{12}} = 3000 \text{ m/sec.}$$

Ex. 3. Two masses m_1 and m_2 are initially at rest at infinite distance apart. They approach each other due to gravitational attraction. Find their speed of approach at the instant they are at d distance apart.

Sol. Initially, potential energy $U_1 = 0$, kinetic energy $K_1 = 0$, so that initial total energy = 0

$$\text{Reduced mass of the system } \mu = \frac{m_1 m_2}{m_1 + m_2}$$

For two-body problem, one can use equivalent one-body problem in which one mass μ is considered to be moving. If v is the relative velocity then the total final energy at d separation is

$$= \frac{1}{2} \mu v^2 - \frac{G m_1 m_2}{d}$$

From the law of conservation of mechanical energy,

$$\frac{1}{2} \mu v^2 - \frac{G m_1 m_2}{d} = 0 \quad \text{or} \quad v^2 = \frac{2 G m_1 m_2}{\mu d} \quad \text{or} \quad v = \sqrt{\frac{2 G (m_1 + m_2)}{d}}.$$

5.15. Kepler's Laws of Planetary Motion (Motion of Celestial Bodies)

Planets move around the Sun under gravitational force in different orbits. Earth is a planet among the nine planets of the Sun namely, (i) Mercury, (ii) Venus, (iii) Earth, (iv) Mars, (v) Jupiter, (vi) Saturn, (vii) Uranus, (viii) Neptune and (ix) Pluto. Kepler's experimental work regarding the motion of celestial bodies is of extreme importance. After taking observations for a period of more than 20 years on planets and doing a lot of calculational work, Kepler announced three laws related to planetary motion in 1619. This achievement of Kepler that the orbits of planets around the Sun are elliptical, is one of the greatest discoveries in the human history. Kepler's three laws are the following :

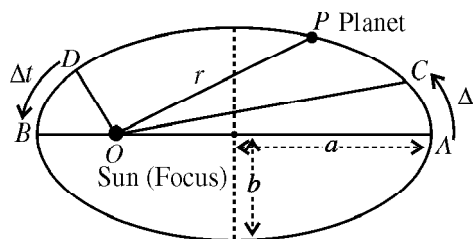


Fig. 5.29.

(1) The law of elliptical orbits : Every planet moves in an elliptical orbit around the Sun, the Sun being at one of the foci.

According to this law, the path of a planet around Sun is in the form of an ellipse and the Sun is at one of its foci. Different planets of the Sun move in different orbits about it.

(2) The law of areas : The radius vector, drawn from the Sun to a planet, sweeps out equal areas in equal times, i.e., the areal velocity of the radius vector is constant.

In [fig. 5.29], the line drawn from the Sun O upto the planet traverses equal areas OAC and ODB in the same time Δt so that the areal speed of the radius vector remains constant. Obviously at different points of the ellipse the linear speed of the planet is different. When the planet is at more distance from the Sun, it moves small distance AC in time Δt and when the planet is close to the Sun, it crosses more distance DB in the same time Δt . This is why the linear speed of the planet is less for greater distance and it is more for smaller distance.

(3) The harmonic law : The square of the period of revolution of the planet around the Sun is proportional to the cube of the semi-major axis of the ellipse.

If the period of revolution of the planet about the Sun is T and the semi-major axis of the ellipse is a , then from Kepler's third law,

$$T^2 \propto a^3 \quad \text{or} \quad T^2 = k a^3 \quad \dots(1)$$

Thus according to this law, the period of revolution of the planet at a more distance relative to the

closer planet will be more. If for one planet, the period of revolution is T_1 and semi-major axis is a_1 and the corresponding quantities are T_2 and a_2 for another planet, then

$$T_1^2 = ka_1^3 \quad \text{and} \quad T_2^2 = ka_2^3$$

So that
$$\frac{T_1^2}{T_2^2} = \frac{a_1^3}{a_2^3} \quad \text{or} \quad \frac{T_1}{T_2} = \left(\frac{a_1}{a_2}\right)^{\frac{3}{2}} \quad \dots(2)$$

From Kepler's laws we get some knowledge regarding the planetary motion on the basis of experimental observation, but this was Newton who gave a theoretical explanation of planetary motion. According to Newton, gravitational force acts between the Sun and a planet. Newton studied thoroughly the Kepler's laws of planetary motion. He announced his famous law of universal gravitation with the help of the three Kepler's laws and his laws of motion. According to Newton's law of universal gravitation, if two masses m_1 and m_2 are placed at a distance r , then the force of gravitational attraction is given by

$$\vec{F} = -\frac{Gm_1m_2}{r^2}\hat{r} \quad \dots(3)$$

where G is called gravitational constant.

In the next section, starting from this equation, we shall study about the the motion of celestial bodies and present theoretical explanation of Kepler's three laws of planetary motion.

Note. If the planetary orbit around Sun is assumed to be circle of radius a , then Kepler's third law will give

$$T^2 \propto a^3 \quad \text{or} \quad T^2 = ka^3 \quad \dots(4)$$

If T_1 and T_2 are the period of revolution of two planets, then

$$T_1^2 = ka_1^3 \quad \text{and} \quad T_2^2 = ka_2^3 \quad \dots(5)$$

where a_1 and a_2 are radii of the circular orbits of the two planets around the Sun.

Therefore,
$$\frac{T_1^2}{T_2^2} = \frac{a_1^3}{a_2^3} \quad \text{or} \quad \frac{T_1}{T_2} = \left(\frac{a_1}{a_2}\right)^{3/2} \quad \dots(6)$$

5-16. Mathematical Explanation of Kepler's Laws and their Deduction

If we study the motion of the system of Sun and a planet, like earth, under gravitational force, it is a two-body problem which can always be reduced to one-body problem. Suppose that the masses of the Sun and the planet are M and m respectively, then the equation of equivalent one-body problem will be

$$\mu \frac{d^2 \vec{r}}{dt^2} = -\frac{GMm}{r^2}\hat{r} \quad \dots(1)$$

where $\vec{r}$ is the vector distance from M to m .

Deduction of Kepler's second law—Conservation of angular momentum and areal velocity : As the gravitational force on the planet due to Sun is a central force, total angular momentum of system is conserved. Proof of this statement is as follows :

Here, force between Sun and planet
$$\vec{F} = -\frac{GMm}{r^2}\hat{r}$$

$$\begin{aligned} \text{Torque, } \vec{\tau} &= \frac{d\vec{L}}{dt} = \vec{r} \times \vec{F} = \vec{r} \times \left(-\frac{GMm}{r^2}\right)\hat{r} \\ &= \vec{r} \times \left(-\frac{GMm}{r^3}\right)\vec{r} = 0 \end{aligned} \quad \left[\because \hat{r} = \frac{\vec{r}}{r} \text{ and } \vec{r} \times \vec{r} = 0 \right] \quad \dots(2)$$

Hence $\vec{L} = \text{a constant}$... (3)

Thus for Sun-planet system similar to one-body problem, we have

$$\vec{L} = \vec{r} \times \mu \vec{v} = \text{a constant} \quad \dots (4)$$

Hence the **angular momentum of Sun-planet system under gravitational force is conserved**. As $\vec{L}$ is constant, its direction will be perpendicular to the plane of $\vec{r}$ and $\vec{v}$. This means that the **motion of the Sun-planet system is in a plane under gravitational force**.

In plane polar co-ordinates, the position vector of the mass μ (or m) from origin O (or M) will be given by $\vec{r} = r\hat{r}$ [fig. 5.30] and its velocity will be

$$\vec{v} = \dot{r}\hat{r} + r\omega\hat{\theta}$$

where $\hat{r}$ and $\hat{\theta}$ are radial and transverse unit vectors respectively and $\dot{r} = \frac{dr}{dt}$

and $\omega = \dot{\theta} = \frac{d\theta}{dt}$.

Hence, $\vec{L} = \vec{r} \times \mu (\dot{r}\hat{r} + r\omega\hat{\theta})$ i.e., $\vec{L} = \mu r\omega (\vec{r} \times \hat{\theta})$ [$\because \vec{r} \times \hat{r} = 0$]

If we take a unit vector $\hat{z}$ (along $\vec{L}$) perpendicular to the plane of motion, then on taking $\hat{r} \times \hat{\theta} = \hat{z}$ and hence $\vec{r} \times \hat{\theta} = r\hat{r} \times \hat{\theta} = r\hat{z}$, we have

$$\vec{L} = \mu r^2 \omega \hat{z} = \text{a constant} \quad \dots (5a)$$

In magnitude, $L = \mu r^2 \omega = \text{a constant} \quad \dots (5b)$

In motion when the position vector $\vec{r}$ changes to $\vec{r} + \Delta \vec{r}$, the area $\Delta \vec{S}$ crossed by the radial vector [fig. 5.31] is

$$\Delta \vec{S} = \frac{1}{2} \vec{r} \times \Delta \vec{r} \quad \dots (6)$$

This area is crossed in Δt time, hence on dividing by Δt on both sides of eq. (6) and on taking the limit $\Delta t \rightarrow 0$, we obtain the areal velocity as

$$\frac{d\vec{S}}{dt} = \frac{1}{2} \vec{r} \times \frac{d\vec{r}}{dt} = \frac{1}{2} \vec{r} \times \vec{v} \quad \dots (7)$$

But from eq. (4) $\vec{r} \times \vec{v} = \vec{L}/\mu$, the areal velocity ($d\vec{S}/dt$) of the planet around the Sun is given by

$$\frac{d\vec{S}}{dt} = \frac{\vec{L}}{2\mu} = \text{a constant} \quad \dots (8)$$

In magnitude, $\frac{dS}{dt} = \frac{L}{2\mu} = \frac{r^2 \omega}{2} \quad (\because L = \mu r^2 \omega) \quad \dots (9)$

Thus, the rate of crossed area by the line, drawn from the Sun upto the planet, i.e., areal velocity of the radius vector remains constant. This is the **Kepler's second law of planetary motion**.

Deduction of Kepler's first law : The differential equation of orbit of a two-particle system, moving under mutual gravitational force, is given by

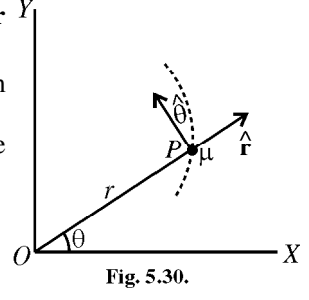


Fig. 5.30.

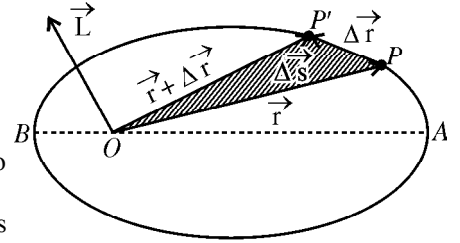


Fig. 5.31.

$$\frac{d^2u}{d\theta^2} + u = \frac{GMm}{\mu h^2} \quad \dots(10)$$

where M and m are the masses of Sun and planet respectively and $\mu = Mm/(M + m)$ is the reduced mass of the Sun-planet system.

In eq. (10), we have used $u = 1/r$ and $r^2\omega = h$. But, the angular momentum of the system is $L = \mu r^2\omega$. Hence $h = r^2\omega = L/\mu$.

Hence, eq. (10) takes the form :

$$\frac{d^2u}{d\theta^2} + u = \frac{\mu GMm}{L^2} \quad \dots(11)$$

This is the equation of orbit for planetary motion around the Sun.

In eq. (11), right hand side has constant quantity and hence there will be a simple solution of this equation. Equation (11) can be written as

$$\frac{d^2}{d\theta^2} \left(u - \frac{\mu GMm}{L^2} \right) + \left(u - \frac{\mu GMm}{L^2} \right) = 0$$

$$\text{or} \quad \frac{d^2u'}{d\theta^2} + u' = 0 \quad \dots(12)$$

where $u' = u - \frac{\mu GMm}{L^2}$.

The solution of eq. (12) is (similar to the equation of simple harmonic motion)

$$u' = A \cos(\theta + \phi) \quad \text{or} \quad u - \frac{\mu GMm}{L^2} = A \cos(\theta + \phi) \quad \dots(13)$$

where A is a constant.

Putting $u = \frac{1}{r}$ in eq. (13), we get

$$\frac{1}{r} - \frac{\mu GMm}{L^2} = A \cos(\theta + \phi) \quad \text{or} \quad \frac{L^2/\mu GMm}{r} = 1 + \frac{L^2 A}{\mu GMm} \cos(\theta + \phi)$$

$$\text{or} \quad \frac{l}{r} = 1 + e \cos(\theta + \phi) \quad \dots(14)$$

where, $l = L^2/\mu GMm$ and $e = L^2 A/\mu GMm$.

Thus, the path of the planet will be ellipse, but in the condition $e < 1$ or $J^2 A/\mu GMm < 1$, this establishes **Kepler's first law**, According to which a planet moves around the Sun in an elliptical orbit and the Sun is at one focus of the orbit.

Now, we want to express e in terms of total energy E of the system which is constant for motion.

$$\text{Total energy, } E = \frac{1}{2} \mu v^2 - \frac{GMm}{r}.$$

At maximum and minimum distances of the planet from the origin, the velocity vector is perpendicular to $\vec{r}$. This means that at these points $v = r \, d\theta/dt = r\omega = L/\mu r$ and therefore

$$E = \frac{L^2}{2\mu r^2} - \frac{GMm}{r}.$$

If the minimum and maximum distances of the planet from the origin are r_{min} and r_{max} respectively, then

$$E = \frac{L^2}{2\mu} \left(\frac{1}{r_{min}} \right)^2 - \frac{GMm}{r_{min}} = \frac{L^2}{2\mu} \left(\frac{1}{r_{max}} \right)^2 - \frac{GMm}{r_{max}} \quad \dots(15)$$

Usually, we take $\phi = \pi$, so that from eq. (14)

$$\frac{l}{r} = 1 - e \cos \theta \quad \dots(16)$$

$$\text{From eq. (16), } \frac{l}{r_{max}} = 1 - e \text{ and } \frac{l}{r_{min}} = 1 + e \quad \dots(17)$$

where r_{min} and r_{max} are the minimum and maximum distances of the planet from the origin, corresponding to $\theta = \pi$ and $\theta = 0$ or $\cos \theta = -1$ and $\cos \theta = 1$ respectively.

$$\text{From eq. (17), } e = \frac{r_{max} - r_{min}}{r_{max} + r_{min}}. \quad \dots(18)$$

On substituting from eqs. (17) in eq. (15) and using $e = L^2 A / \mu G M m$ and $l = L^2 / \mu G M m$, we get

$$e = \left(1 + \frac{2EJ^2}{\mu G^2 m^2 M^2} \right)^{\frac{1}{2}}; \quad E = -\frac{\mu G^2 m^2 M^2}{2L^2} (1 - e^2) \quad \dots(18)$$

Thus, if $e < 1$, the total energy E is negative and so the planet is bound and the orbit is closed in the form of an **ellipse**. The orbit will be **circle** for $e = 0$. So that for circular orbit

$$E = -\mu G m^2 M^2 / 2L^2$$

In case $e = 1$ and $e > 1$, total energy E will be positive and the path will be **parabola** and **hyperbola** respectively. Remember that in solving the two-body problem of the Sun and a planet (e.g., earth), we have neglected the effect of other planets.

Deduction of Kepler's third law : According to the property of an ellipse,

$$\text{latus rectum, } l = \frac{b^2}{a} \quad \dots(19)$$

$$\text{and area of ellipse} = \pi ab \quad \dots(20)$$

where a and b are the semi-major and semi-minor axes respectively.

But $l = L^2 / \mu G m$, hence

$$\frac{b^2}{a} = \frac{L^2}{\mu G M m}, \text{ whence } b^2 = \frac{L^2 a}{\mu G M m} \quad \dots(21)$$

If T is the period of revolution of the planet around the Sun, then

$$T = \frac{\text{Area of ellipse}}{\text{Areal velocity}} = \frac{\pi ab}{L/2\mu} = \frac{2\pi\mu ab}{L}$$

because from eq. (9), areal velocity = $J/2\mu$.

$$\text{Hence } T^2 = \frac{4\pi^2 \mu^2 a^2 b^2}{L^2} = \frac{4\pi^2 \mu^2 a^2}{L^2} \cdot \frac{L^2 a}{\mu G M m} \quad [\text{using eq. (21)}]$$

$$\text{or } T^2 = \frac{4\pi^2 \mu a^3}{G M m} \quad \dots(22)$$

But $\mu = mM/(M + m)$, hence

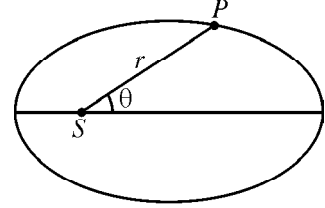


Fig. 5.32.

$$T^2 = \frac{4\pi^2 a^3}{G(M+m)} \quad \text{or} \quad T^2 = ka^3 \quad \dots(23)$$

Since $k = \frac{G\pi^2}{G(M+m)}$ remains constant,

$$T^2 \propto a^3. \quad \dots(24)$$

This is **Kepler's third law**, according to which *the square of period of revolution of a planet around the Sun is directly proportional to the cube of semi-major axis of the ellipse.*

In eq. (24), the constant of proportionality (k) depends on the mass of the planet, hence this constant will be different for different planets. Thus we obtain a more correct statement of the third law, i.e., *the square of the periods of various planets are proportional to the cube of their respective semi-major axes with different proportionality constants, because for each planet, proportionality constant is different.* If we neglect the mass (m) of a planet compared to the mass of the Sun (M), we obtain the same proportionality constant for all the planets i.e.,

$$T^2 = \frac{4\pi^2}{GM} a^3 \quad \dots(25)$$

This is an approximate relation and in fact, this is Kepler's third law of planetary motion. For example, $m/M = 0.1\%$ for Jupiter and $m/M = 3 \times 10^{-4}\%$ for earth. This means that proportionality constant in (25) is different for different planets. However, in case of the Bohr model of atom, all revolving electrons are identical in mass and hence μ and k are the same for all orbits in an atom. This means that proportionality constant is the same for different electrons in different orbits. Therefore **Kepler's third law** is very much true for the electron orbits in the Bohr atom.

Theorem 1. (a) If the minimum and maximum distances of a planet from the Sun are r_1 and r_2 respectively, then

$$\frac{r_2}{r_1} = \frac{1+e}{1-e}.$$

(b) If the maximum and minimum velocities of a planet around Sun are v_1 and v_2 respectively, then

$$\frac{r_2}{r_1} = \frac{v_1}{v_2} \quad \text{and} \quad e = \frac{v_1 - v_2}{v_1 + v_2} = \frac{r_2 - r_1}{r_2 + r_1}.$$

Proof : (a) Let the equation of elliptical orbit of a planet be

$$\frac{l}{r} = 1 + e \cos \theta$$

We see that at $\theta = 0$, the value of r is minimum (r_1) and at $\theta = \pi$, the value of r is maximum (r_2), i.e.,

$$\frac{l}{r_1} = 1 + e \quad \text{and} \quad \frac{l}{r_2} = 1 - e.$$

Dividing the two equations,

$$\frac{r_2}{r_1} = \frac{1+e}{1-e}.$$

(b) At the maximum distance r_2 (point A) and at the minimum distance r_1 (point B), the velocity vector $\vec{v}$ is perpendicular to $\vec{r}$. Therefore, in the expression of the velocity $\vec{v} = \dot{r}\hat{r} + r\dot{\theta}\hat{\theta}$, radial component will be not present, i.e., $\dot{r} = 0$.

Hence at minimum and maximum distances, $\vec{v} = r\omega\hat{\theta}$ and in value $v = r\omega$.

If at r_1 and r_2 , the velocities are v_1 and v_2 respectively then from the law of conservation of angular momentum,

$$\mu r_1^2 \omega_1 = \mu r_2^2 \omega_2 \quad \text{or} \quad r_1^2 \omega_1 = r_2^2 \omega_2$$

But $v_1 = r_1 \omega_1$ and $v_2 = r_2 \omega_2$, hence

$$r_1 v_1 = r_2 v_2 \quad \text{or} \quad \frac{r_2}{r_1} = \frac{v_1}{v_2} \quad \dots(i)$$

As r_2 is maximum and r_1 is minimum, hence the value of the velocity at minimum distance r_1 will be maximum and at maximum distance r_2 it will be minimum. But $\frac{r_2}{r_1} = \frac{1+e}{1-e}$, hence

$$\frac{r_2}{r_1} = \frac{1+e}{1-e} = \frac{v_1}{v_2}; \quad \therefore e = \frac{v_1 - v_2}{v_1 + v_2} = \frac{r_2 - r_1}{r_2 + r_1} \quad \dots(ii)$$

Theorem 2. Prove that in an elliptical orbit, total energy depends only on the semi-major axis of the orbit.

Proof : The apsidal (turning) distances r_1 and r_2 are known as **perihelion** and **aphelion** and are given by

$$\frac{l}{r_1} = 1 + e \quad \text{or} \quad r_1 = \frac{l}{1+e} \quad \dots(i)$$

and $\frac{l}{r_2} = 1 - e \quad \text{or} \quad r_2 = \frac{l}{1-e} \quad \dots(ii)$

The semi-major axis (a) of the ellipse is one-half the sum of these two apsidal distances i.e.,

$$a = \frac{r_2 + r_1}{2} = \frac{1}{2} \left[\frac{l}{1+e} + \frac{l}{1-e} \right] = \frac{l}{1-e^2} \quad \dots(iia)$$

But $l = L^2/\mu K$, ($K = GMm$), then

$$a = -\frac{L^2}{\mu K} \cdot \frac{\mu K^2}{2EL^2} = -\frac{K}{2E} \left[\text{from eq. (25), (Art 5.16), } 1 - e^2 = -\frac{2EL^2}{\mu K^2} \right] \quad \dots(iiib)$$

or $E = -\frac{K}{2a} \quad \dots(iiic)$

Thus in case of an elliptical orbit, the total energy depends solely on the major axis.

5.17. Deduction of Newton's Law of Universal Gravitation from Kepler's Laws

Newton deduced his famous law of universal gravitation from Kepler's laws with the help of his laws of motion.

Let **fig. 5.33** represent a planet P moving in an elliptical orbit around the Sun, the Sun being at one of the foci S of the ellipse. Let P be the position of the planet at any time t and Q at the time $(t + dt)$. Then the area swept out by the radius vector SP in time dt is equal to the area ΔSPQ i.e., area swept out = $\frac{1}{2} SP \times PQ$ (as PQ is infinitesimal small and is considered perpendicular to SP).

Here, $SP = r$ and $PQ = r d\theta$

Hence, area swept out = $\frac{1}{2} r^2 d\theta$.

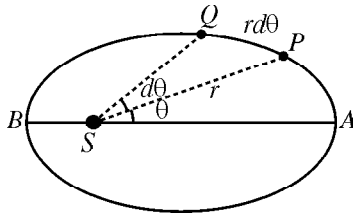


Fig. 5.33.

Therefore, the areal velocity of the radius vector = $\frac{1}{2} r^2 \frac{d\theta}{dt}$.

According to Kepler's second law, this must be a constant. Let it be equal to $h/2$. Then

$$r^2 \frac{d\theta}{dt} = h. \quad \dots(1)$$

The planet is moving on a curved path, hence it is obvious from Newton's first law of motion that a force is acting upon it along the radius PS . Consequently an acceleration is acting on the planet in the direction of the force. Resolving this acceleration into its two components, along and perpendicular to the radius vector, we have

(i) component a_r , along the radius vector, i.e., the radial acceleration of the planet is given by

$$a_r = \frac{d^2 r}{dt^2} - r \left(\frac{d\theta}{dt} \right)^2 \quad \dots(2)$$

and (ii) component a_t , perpendicular to the radius vector, i.e., the transverse acceleration of the planet is given by

$$a_t = \frac{1}{r} \frac{d}{dt} \left(r^2 \frac{d\theta}{dt} \right)$$

But from eq. (1), $r^2 \frac{d\theta}{dt} = \text{a constant}$.

Hence $a_t = 0$.

Thus, the planet has no transverse acceleration and only the radial acceleration is acting on it, i.e., *the force on the planet is directed towards the Sun*.

From eq. (1), we have

$$\frac{d\theta}{dt} = \frac{h}{r^2} = hu^2, \quad \dots(3)$$

where $u = 1/r$.

$$\text{Now,} \quad \frac{dr}{dt} = \frac{d}{dt} \left(\frac{1}{u} \right) = -\frac{1}{u^2} \cdot \frac{du}{dt} = -\frac{1}{u^2} \frac{du}{d\theta} \cdot \frac{d\theta}{dt} = -\frac{1}{u^2} \cdot \frac{du}{d\theta} \cdot hu^2 = -h \frac{du}{d\theta}$$

$$\therefore \quad \frac{d^2 r}{dt^2} = -h \frac{d^2 u}{d\theta^2} \frac{d\theta}{dt} = -h^2 u^2 \cdot \frac{d^2 u}{d\theta^2} \quad \dots(4)$$

Substituting the values of $d\theta/dt$ and $d^2 r/dt^2$ from eq. (3) and (4) in eq. (2), we get

$$a_r = -h^2 u^2 \left(u + \frac{d^2 u}{d\theta^2} \right) \quad \dots(5)$$

Let the equation of the elliptical orbit be

$$\frac{l}{r} = 1 + e \cos \theta \quad \text{or} \quad lu = 1 + e \cos \theta \quad \dots(6)$$

where l is the latus rectum and e the eccentricity of the ellipse.

Differentiating eq. (6) twice with respect to θ , we have

$$l \frac{d^2 u}{d\theta^2} = -e \cos \theta \quad \dots(7)$$

Adding the eq. (6) and eq. (7), we get

$$l \left(u + \frac{d^2 u}{d\theta^2} \right) = 1, \quad \therefore \quad u + \frac{d^2 u}{d\theta^2} = \frac{1}{l}.$$

Substituting this value of $u + \frac{d^2u}{d\theta^2}$ in eq. (5), we get

$$a_r = -\frac{h^2 u^2}{l} = -\frac{h^2}{l} \cdot \frac{1}{r^2}$$

Let $h^2/l = k$ (a constant).

$$\text{Then} \quad a_r = -\frac{k}{r^2} \quad \dots(8)$$

$$\text{or} \quad a_r \propto -\frac{1}{r^2} \quad \dots(9)$$

Thus, the acceleration and hence the force acting on the planet is inversely proportional to the square of its distance from the Sun. Negative sign indicates that the force is one of attraction.

Let a and b be the semi-major and semi-minor axes of the elliptical orbit of the planet. The periodic time t of the planet, i.e., the time for one revolution round the Sun is given by

$$T = \frac{\text{area of the ellipse}}{\text{areal velocity}} = \frac{\pi ab}{\frac{1}{2} r^2 d\theta/dt} = \frac{2\pi ab}{h}$$

$$\therefore T^2 = \frac{4\pi^2 a^2 b^2}{h^2} \quad \dots(10)$$

But $b^2/a = l$, the latus rectum of the ellipse.

$$\therefore T^2 = \frac{4\pi^2 l}{h^2} a^3 = \frac{4\pi^2}{k} a^3 \quad \left[\because k = \frac{h^2}{l} \right] \quad \dots(11)$$

Now, in accordance with the Kepler's third law $T^2 \propto a^3$ for every planet. Thus $4\pi^2/k$ is a constant and therefore k is constant for every planet. In other words, k is independent of the nature of the planet.

Newton compared the motion of a falling body at the earth's surface with the motion of the moon round the earth. If α be the acceleration of the moon towards the earth, then

$$\alpha = \frac{v^2}{R} = \left(\frac{2\pi R}{T} \right)^2 \times \frac{1}{R} = \frac{4\pi^2 R}{T^2}, \quad \left[\because v = R\omega = \frac{2\pi R}{T} \right] \quad \dots(12)$$

where R is the distance from the centre of the earth to the centre of the moon and T the time of one revolution of the moon round the earth.

But $R = 3.8 \times 10^8$ m and $T = 27.3$ days $= 2.36 \times 10^6$ sec.

$$\therefore \alpha = \frac{4 \times (3.14)^2 \times 3.8 \times 10^8}{(2.36 \times 10^6)^2} = 2.7 \times 10^{-3} \text{ cm/sec}^2. \quad \dots(13)$$

The ratio of the acceleration α of the moon towards the centre of the earth to the acceleration g of a falling body at the earth's surface is

$$\frac{\alpha}{g} = \frac{7.7 \times 10^{-3}}{9.81} = \frac{1}{60^2} \text{ approx.} \quad \dots(14)$$

But the distance between the centre of the earth and the moon is 60 times the radius of the earth. Hence the attractive force of the earth is inversely proportional to the square of the distance of the body from the centre, i.e., the law of attraction of falling bodies at the surface of the earth or of the moon towards the earth is the same as that of the attraction of the planets towards the Sun.

Newton realised from this that the attraction of the earth for bodies of its surface or of the Sun for planets is not a special property of them, but it is a common property for all bodies, whether large or small,

i.e., the **attraction is universal and mutual**.

If m and M be the masses of a planet and the Sun respectively, then from eq. (8) the attractive force of the Sun on the planet is given by

$$F = -\frac{km}{r^2} \quad \dots(15)$$

But the attraction is mutual, the attractive force of the planet on the Sun is similarly given by

$$F' = \frac{KM}{r^2} \quad \dots(16)$$

where we have assumed positive direction from the Sun to the planet.

According to Newton's third law of motion, action is equal and opposite to reaction, hence

$$F = -F' \quad \text{or} \quad \frac{km}{r^2} = \frac{KM}{r^2}$$

$$\text{or} \quad \frac{k}{M} = \frac{K}{m} = G, \text{ (say),} \quad \therefore \quad k = MG$$

Substituting the value of k in eq. (15), we get

$$F = -\frac{GMm}{r^2} \quad \dots(17)$$

Thus, *the force of attraction between the Sun and the planet is directly proportional to the product of their masses.*

From the above deductions, Newton enunciated his famous **law of universal gravitation**, which can be written for any two masses m and M in the vector form

$$\vec{F} = -\frac{GMm}{r^2} \hat{r} \quad \dots(18)$$

where $\hat{r} = \vec{r}/r$ is a unit vector from one particle to the other.

Ex. 1. *The mean distance of mars from Sun is 1.524 times the distance of the earth from Sun. Compute the period of revolution of mars around Sun.* (A.N.U. 2008, 05; A.U. 2003)

Sol. T_1 and T_2 are the periods of revolutions of mars and earth around Sun, then according to Kepler's third law,

$$\frac{T_1}{T_2} = \left(\frac{a_1}{a_2}\right)^{3/2}$$

where a_1 and a_2 are the respective distances of the mars and earth from the Sun.

Here, $\frac{a_1}{a_2} = 1.524$ and for earth, $T_2 = 1$ year.

$$\text{Hence} \quad T_1 = \left(\frac{a_1}{a_2}\right)^{3/2} \times T_2 = (1.524)^{3/2} \times 1 \text{ year} = \mathbf{1.88 \text{ year.}}$$

Ex. 2. *The jupiter's period of revolution round the Sun is 12 times that of the earth. Assuming the planetary orbitals to be circular, find how many times the distance between the jupiter and Sun exceeds that between the earth and the Sun.* (K.U. 2003)

Sol. According to Kepler's law,

$$\frac{T_1}{T_2} = \left(\frac{a_1}{a_2}\right)^{3/2} \quad \therefore \quad \frac{a_1}{a_2} = \left(\frac{T_1}{T_2}\right)^{2/3}$$

Here,
$$\frac{\text{Period of revolution of jupiter about Sun}}{\text{Period of revolution of earth about Sun}} = \frac{T_1}{T_2} = 12$$

Hence,
$$\frac{\text{Jupiter's distance from Sun}}{\text{Earth's distance from Sun}} = \frac{a_1}{a_2} = \left(\frac{T_1}{T_2}\right)^{2/3} = (12)^{2/3} = \mathbf{5.24}.$$

Ex. 3. If the earth be one-half of its present distance from the Sun, what will be the number of days in a year ? **(K.U. 2002)**

Sol. If T is the period of revolution of a planet, say earth around the Sun, and a is the distance of the planet from the Sun, then according to Kepler's third law,

$$T^2 = Ka^3 \quad \dots(i)$$

If the distance is reduced to one-half i.e., $a/2$, then the period of revolution, say T' of earth around Sun will be given by

$$T'^2 = K(a/2)^3 \quad \dots(ii)$$

Hence
$$\frac{T'^2}{T^2} = \frac{(a/2)^3}{a^3} \text{ or } \frac{T'^2}{T^2} = \frac{1}{8} \text{ or } T' = \frac{T}{\sqrt{8}}$$

Here, $T=1$ year = 365 days for earth.

$$\therefore T' = \frac{365}{\sqrt{8}} = \mathbf{129 \text{ days.}}$$

Ex. 4. The earth moves in an approximately circular orbit with radius 1.5×10^{11} m. Mars circles around Sun in its orbit in 687 days. What is the radius of mar's orbit around Sun. **(K.U. 1994)**

Sol.
$$\frac{T_1^2}{T_2^2} = \frac{a_1^3}{a_2^3} \text{ or } \frac{a_1}{a_2} = \left(\frac{T_1}{T_2}\right)^{2/3}$$

For mars, $T_1 = 687$ days, for earth $T_2 = 365$ days, a_1 =radius of mar's orbit = ? and a_2 = radius of earth's orbit = 1.5×10^{11} m.

$$\therefore a_1 = \left(\frac{T_1}{T_2}\right)^{2/3} \times a_2 = \left(\frac{687}{365}\right)^{2/3} \times 1.5 \times 10^{11} = \mathbf{2.286 \times 10^{11} \text{ m.}}$$

Ex. 5. The semi-major axes of the orbits of mercury and mars are respectively 0.387 and 1.524 in astronomical units. If the period of mercury is 0.241 year, what is the period of mars. **(A.U. 1988)**

Sol. According to Kepler's third law.

$$\frac{T_1^2}{T_2^2} = \frac{a_1^3}{a_2^3} \text{ or } \frac{T_1}{T_2} = \left(\frac{a_1}{a_2}\right)^{3/2}$$

Here, semi-major axis of orbit of mercury $a_1 = 0.387$ astronomical units, semi-major axis of orbit of mars $a_2 = 1.524$ astronomical units, period of mercury $T_1 = 0.241$ year and period of mars, $T_2 = ?$

$$\therefore T_2 = \left(\frac{a_2}{a_1}\right)^{3/2} \times T_1 = \left(\frac{1.524}{0.387}\right)^{3/2} \times 0.241 \text{ year} = \mathbf{1.9 \text{ years.}}$$

Ex. 6. The period of revolution of planet Neptune is 164.8 years. How many times is it farther from the Sun than the earth-Sun distance ? **(O.U. 2002)**

Sol.
$$\frac{T_1^2}{T_2^2} = \frac{a_1^3}{a_2^3} \text{ or } \frac{a_1}{a_2} = \left(\frac{T_1}{T_2}\right)^{2/3}$$

Here, period of revolution of neptune about Sun $T_1 = 164.8$ years, and period of revolution of earth around Sun $T_2 = 1$ year.

$$\therefore \frac{\text{Distance of neptune from Sun}}{\text{Distance of earth from Sun}} = \frac{a_1}{a_2} = \left(\frac{164.8}{1}\right)^{2/3} = \mathbf{30.06}.$$

Ex. 7. The maximum and minimum distances of a comet from the Sun are 1.6×10^{12} m and 8×10^{10} m respectively. If the speed of the comet at the nearest point is 6×10^4 m/sec., calculate the speed at the farthest point. **(A.N.U. 2007)**

Sol. If r_1 and r_2 are the maximum and minimum distances of the comet from the Sun and corresponding velocities of the comet are v_1 and v_2 respectively, then from the law of conservation of angular momentum, we have

$$\mu r_1 v_1 = \mu r_2 v_2 \text{ or } r_1 v_1 = r_2 v_2$$

Here, $r_1 = 1.6 \times 10^{12}$ m, $r_2 = 8 \times 10^{10}$ m, $v_1 = ?$ and $v_2 = 6 \times 10^4$ m/sec.

$$\therefore v_1 = \frac{r_2 \times v_2}{r_1} = \frac{8 \times 10^{10} \times 6 \times 10^4}{1.6 \times 10^{12}} = \mathbf{3000 \text{ m/sec.}}$$

Ex. 8. The eccentricity of earth's orbit is 0.0167. In the earth's orbit, find the ratio of maximum and minimum velocities.

Sol. If the maximum and minimum velocities of a planet or earth around Sun are v_1 and v_2 respectively and e is the eccentricity of the orbit, then

$$\frac{v_1}{v_2} = \frac{1+e}{1-e}$$

Hence
$$\frac{v_1}{v_2} = \frac{1+0.0167}{1-0.0167} = \mathbf{1.03}.$$

Ex. 9. The greatest and least velocities of a certain planet in its orbital round the Sun are 30 km/sec. and 29.2 km/sec. respectively. Find the eccentricity of the orbit.

Sol. If v_1 and v_2 are the maximum and minimum velocities of the planet around the Sun, then the eccentricity of the planetary orbit is given by

$$e = \frac{v_1 - v_2}{v_1 + v_2} = \frac{30 - 29.2}{30 + 29.2} = \frac{0.8}{59.2} = \mathbf{0.0135}.$$

Ex. 10. Assume that the moon revolves round the earth in an orbit of radius 3.84×10^5 km. If the mass of earth is 6×10^{24} kg and the periodic time of moon is 27.17 days, find the mass of the moon.

Sol. If the masses of the moon and the earth are m and M respectively, then for equivalent one-body problem

$$\mu \omega^2 r = \frac{GMm}{r^2} \quad \dots(i)$$

where $\mu = \frac{mM}{m+M}$ and r is the distance between m and M .

If $T = 2\pi/\omega$ is the period of revolution, then $\omega = 2\pi/T$. Hence from (i)

$$\frac{mM}{m+M} \cdot \left(\frac{2\pi}{T}\right)^2 r = \frac{GMm}{r^2} \quad \text{or} \quad m + M = \frac{4\pi^2 r^3}{GT^2}$$

whence
$$m = \frac{4\pi^2 r^3}{GT^2} - M = \frac{4\pi^2 \times (3.84 \times 10^8)^3}{6.67 \times 10^{-11} \times (27.3 \times 24 \times 60 \times 60)^2} - 6 \times 10^{24}$$

$$= 6.01775 \times 10^{24} - 6 \times 10^{24} = \mathbf{1.775 \times 10^{22} \text{ kg.}}$$

Ex. 11. A planet moves around the Sun in an elliptical orbit. Its minimum and maximum distances from the Sun are r_1 and r_2 respectively, then prove that the total energy of system is $E = -\frac{GMm}{(r_1 + r_2)}$ and its angular momentum is $L = \sqrt{\frac{2GMm}{(r_1 + r_2)}}$, where M is the mass of the Sun.

Sol. From theorem 2, $a = \frac{r_1 + r_2}{2}$ and $a = -\frac{GMm}{2E}$

Hence
$$E = -\frac{GMm}{2a} = -\frac{GMm}{r_1 + r_2} \quad \dots(i)$$

Total energy of the system at maximum and minimum distances remains the same and constant. Hence

$$E = \frac{1}{2} \mu v_1^2 - \frac{GMm}{r_1} = \frac{1}{2} \mu v_2^2 - \frac{GMm}{r_2}$$

But
$$L = \mu r_1 v_1 = \mu r_2 v_2 \quad [\text{see Theorem 1}]$$

$$\therefore E = \frac{L^2}{2\mu r_1^2} - \frac{GMm}{r_1} = -\frac{GMm}{r_1 + r_2}$$

or
$$\frac{L^2}{2\mu r_1^2} = \frac{GMm}{r_1} - \frac{GMm}{r_1 + r_2} = \frac{GMmr_2}{r_1(r_1 + r_2)}$$

Hence,
$$L = \sqrt{\frac{2GMmr_1r_2}{r_1 + r_2}} \quad \dots(ii)$$

Ex. 12. A particle of mass m moves under the action of central force whose potential is $V(r) = Kmr^3$ ($K > 0$), then

(i) For what kinetic energy and angular momentum will the orbit be a circle of radius R about the origin ?

(ii) Calculate the period of circular motion.

Sol.
$$V(r) = Kmr^3$$

Therefore,
$$F = -\frac{\partial V}{\partial r} = -3Kmr^2$$

(i) For circular motion,
$$F = \frac{mv^2}{r} = -3Kmr^2$$

Kinetic energy
$$\frac{1}{2}mv^2 = -\frac{3}{2}Kmr^3$$

Angular momentum
$$= mvr = mr(-3Kr^3)^{1/2} \quad [\text{because } v^2 = -3Kr^3]$$

(ii) Since $v = r\omega = \frac{2\pi r}{T}$, therefore from eq. $v^2 = -3Kr^3$, we have

$$\left[\frac{2\pi r}{T} \right]^2 = -3Kr^3 \quad \text{or} \quad \frac{4\pi^2}{T^2} = -3Kr, \text{ whence } T = \frac{2\pi}{\sqrt{-3Kr}}.$$

Ex. 13. A particle of mass m is observed to move in a spiral orbit given by the equation $r = C\theta$, where C is a constant. Is it moving in a central force field? If it is so, find the force law.

Sol. The differential equation of the orbit is given by

$$\frac{d^2u}{d\theta^2} + u = -\frac{mf(1/u)}{L^2u^2} \quad \dots(i)$$

Since $u = 1/r$ and $r = C\theta$ (given), hence $u = \frac{1}{C\theta}$...(ii)

Differentiating it, we get

$$\frac{du}{d\theta} = -\frac{1}{C\theta^2} \quad \text{and} \quad \frac{d^2u}{d\theta^2} = \frac{2}{C\theta^3} = 2C^2u^3 \quad \dots(iii)$$

Substituting in eq. (i), we get $2C^2u^3 + u = -\frac{m}{L^2u^2}f\left(\frac{1}{u}\right)$

Therefore, $f\left(\frac{1}{u}\right) = -\frac{L^2}{m}(u^3 + 2C^2u^5)$ or $f(r) = -\frac{L^2}{m}\left[\frac{1}{r^3} + \frac{2C^2}{r^5}\right]$...(iv)

Thus the particle is moving under a central force field and the force law is given by eq. (iv).

Ex. 14. Find the time period of revolution of phobos revolving round the planet mars in circular orbit at a distance of 9500 km from its centre. Take the radius of the mars = 3400 km and the acceleration due to gravity on the surface of mars = 3.72 m/sec². (S.V.U. 1990)

Sol. Phobos is the satellite of mars. If M is the mass of mars and m that of a particle at the surface of the mars, then the force of gravitational attraction between the particle and mars is given by

$$\frac{GMm}{R^2} = mg \quad \dots(i)$$

where R is the radius of mars and g is the acceleration due to gravity at the surface of the mars.

Hence from (i), $M = \frac{gR^2}{G} = \frac{3.72 \times (3400 \times 10^3)^2}{6.67 \times 10^{-11}} = 6.448 \times 10^{23} \text{ kg.}$

If the satellite phobos of mass m' moves in a circular orbit of radius r around the planet mars, then the gravitational force on phobos is $\frac{GMm'}{r^2}$ gives the centripetal force ($m'\omega^2r$) for rotation in circular orbit i.e.,

$$\frac{GMm'}{r^2} = m'\omega^2r \quad \text{or} \quad \omega^2 = \frac{GM}{r^3} \quad (\omega = \text{angular velocity of phobos around mars})$$

whence $\omega = \sqrt{\frac{GM}{r^3}} \therefore \omega = \sqrt{\frac{6.67 \times 10^{-11} \times 6.448 \times 10^{23}}{(9500 \times 10^3)^2}} = 2.24 \times 10^{-4} \text{ rad/sec}$

Period of revolution of phobos, $T = \frac{2\pi}{\omega} = \frac{2 \times 3.14}{2.24 \times 10^{-4}} = 2.805 \times 10^4 \text{ sec.}$

5-18. Inertial Frames of Reference

Those unaccelerated frames of reference, in which Newton's first and second laws hold, are called **inertial frames**. In such frames, if a body is not acted by external force, it continues in its state of rest or uniform translatory motion (*law of inertia*) and hence they are called **inertial frames**. Thus, in an inertial frame, if a body is not experiencing any external force its acceleration $\vec{a}$ is given by

$$\vec{a} = \frac{d\vec{v}}{dt} = \frac{d^2\vec{r}}{dt^2} = 0 \quad [\because \vec{F} = m\vec{a} = 0] \quad \dots(1)$$

The law of inertia was first stated by Galileo, therefore the inertial frames are also named as *Galilean frames of reference*.

The position $\vec{r}$, velocity $\vec{v}$ and the acceleration of a particle depend on the frame chosen. Let us see how can we connect the position, velocity and acceleration of a particle measured in two different frames in constant relative motion. Consider an inertial frame S and another frame S' , which is moving with constant velocity $\vec{v}_0$ relative to S . Initially (at $t = 0$) if the positions of the origins of the two frames coincide, then in the two frames the position vectors of any particle P at any instant t can be related by the following expressions [fig. 5.34] :

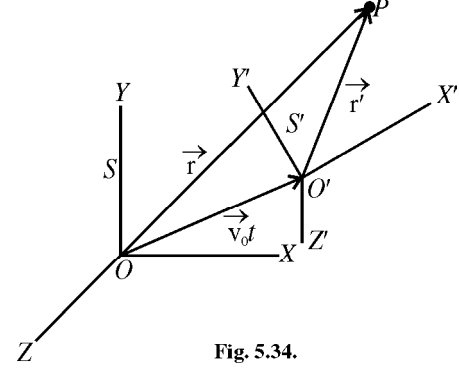


Fig. 5.34.

$$\vec{r} = \vec{OO'} + \vec{r'} \quad \text{or} \quad \vec{r} = \vec{v}_0 t + \vec{r'}$$

or

$$\vec{r'} = \vec{r} - \vec{v} t \quad \dots(2)$$

This relation connects the position vector $\vec{r'}$ of the particle in frame S' to the position vector $\vec{r}$ of the particle in S .

In Newtonian mechanics, two observers O and O' in constant relative motion will measure the same time, i.e., $t = t'$, when the particle is at P . Differentiating equation (2) with respect to time,

$$\frac{d\vec{r'}}{dt} = \frac{d\vec{r}}{dt} - \vec{v}_0 \quad \text{or} \quad \frac{d\vec{r'}}{dt'} = \frac{d\vec{r}}{dt} - \vec{v}_0 \quad [\because \vec{v}_0 = \text{constant}]$$

$$\vec{v'} = \vec{v} - \vec{v}_0 \quad \dots(3)$$

where $\vec{v}$ is the velocity of the particle with respect to S , $\vec{v'}$ is the velocity of the particle with respect to S' and $\vec{v}_0$ the velocity of S' with respect to S . Eq. (3) connects the velocities of the particle P in S' and S frames.

Differentiating eq. (3) with respect to time, we obtain

$$\frac{d\vec{v'}}{dt} = \frac{d\vec{v}}{dt} \quad \text{or} \quad \vec{a'} = \vec{a} \quad \dots(4)$$

Thus, the acceleration of the particle as observed in frame S' is the same as observed in frame S , which is moving with constant velocity relative to S .

According to Newton's second law, the force on the particle P in the frame S is

$$\vec{F} = m\vec{a}$$

and the force on the particle P in the frame S' is

$$\vec{F'} = m\vec{a'}$$

But $\vec{a'} = \vec{a}$, hence

$$\vec{F'} = m\vec{a'} = m\vec{a} = \vec{F} \quad \dots(5)$$

where m is assumed to be the same (independent of velocity) in two frames.

Thus the force on the particle, as measured in S' , using Newton's second law, is the same as measured

in S , using the same law. This means that if S is an inertial frame, then S' is also an inertial frame, because Newton's second law holds good in both the frames.

Further if in absence of external force ($\vec{F} = 0$) in frame S , the acceleration of the particle $\vec{a}$ is zero, then from eq. (5) in frame S' the acceleration $\vec{a}'$ is also zero, i.e., no force ($\vec{F}' = m\vec{a}' = m\vec{a} = 0$) is acting on the particle in the frame S' . This means that if law of inertia (i.e., first law) is valid in frame S , then it is also valid in frame S' .

Thus, we conclude that *if a frame is an inertial frame, then all those frames, which are moving with constant velocity relative to it, are also inertial.*

5.19. Non-inertial Frames and Fictitious Forces

Those frames of reference in which Newton's law of inertia does not hold, are called **non-inertial frames**. All the accelerated and rotating frames are the non-inertial frames of reference. According to Newton's second law, the applied force $\vec{F}_i$ on a body of mass m is given by

$$\vec{F}_i = m\vec{a}_i \quad \dots(1)$$

where $\vec{a}_i$ is the acceleration, observed in an inertial frame. This equation, as it stands, is not valid for accelerated frames because due to its own acceleration, the accelerated observer will observe an acceleration $\vec{a}_n$ different to $\vec{a}_i$ and hence

$$\vec{F}_i \neq m\vec{a}_n \quad \dots(2)$$

If no external force is acting on a particle, even then in the accelerated frame it will appear that a force is acting on it. Suppose that S is an *inertial frame* and another frame S' is moving with an acceleration $\vec{a}_0$ relative to S . [fig. 5.35.] The acceleration of a particle P , on which no external force is acting, will be zero in the frame S ; but in frame S' the observer will find that an acceleration $-\vec{a}_0$ is acting on it. Thus, in frame S' , the observed force on the particle is $-m\vec{a}_0$, where m is the mass of the particle.

Such a force, which does not really act on the particle but appears due to the acceleration of the frame, is called a **fictitious** or **pseudo force**. Hence the fictitious force on the particle P is

$$\vec{F}_0 = -m\vec{a}_0 \quad \dots(3)$$

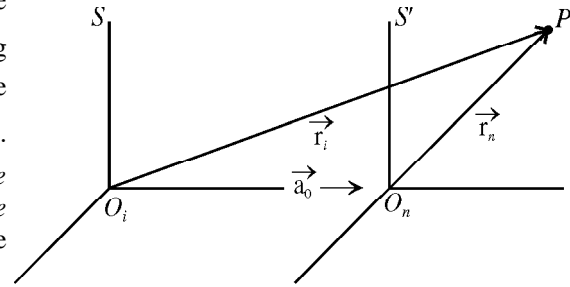


Fig. 5.35.

Here, accelerated frame S' is non-inertial, because in this frame Newton's law of inertia is not valid, i.e., in absence of any external force, the observer of S' frame is observing a force ($-m\vec{a}_0$) on the particle.

Suppose frame S' coincides at $t = 0$ with the inertial frame S . Then at any time t , the position vectors of a particle $\vec{r}_i$ and $\vec{r}_n$ in the inertial and non-inertial frames respectively are connected as [fig 5.35]

$$\vec{r}_i = \vec{r}_n + \frac{1}{2}\vec{a}_0 t^2 \quad \dots(4)$$

where $\vec{a}_0$ is the acceleration of the frame S' with respect to S .

Double differentiation with respect to time gives

$$\frac{d^2 \vec{r}_i}{dt^2} = \frac{d^2 \vec{r}_n}{dt^2} + \vec{a}_0$$

$$\text{or} \quad \vec{a}_i = \vec{a}_n + \vec{a}_0 \quad \dots(5)$$

$$\text{or} \quad \vec{a}_i - \vec{a}_0 = \vec{a}_n$$

$$\text{Thus} \quad m\vec{a}_i - m\vec{a}_0 = m\vec{a}_n$$

$$\text{or} \quad \vec{F}_0 + \vec{F}_i = \vec{F}_n \quad [\text{from eq. (1) and (3)}] \quad \dots(6)$$

This formula gives the observed force $\vec{F}_n$ in the accelerated system.

For example, suppose that a box is falling in the gravitational field of the earth with an acceleration g . Now, if we consider about a particle, falling freely inside the box, the fictitious force on the particle is mg upward, because the acceleration of the frame (box) is g downward or $-g$ (we have taken $+ve$ in the upward direction). As the real force on the particle due to the attraction of the earth $-mg$, the force observed by the observer inside the box is

$$F_n = F_i + F_0 = -mg + mg = 0$$

where we have assumed earth as an inertial frame (stationary).

If the particle has no initial velocity relative to the box, it will seem to remain suspended in mid-air at the same place inside the falling box.

Now, suppose that the box is moved with an acceleration $a_0 = g$ in the upward direction relative to the ground. In such a case, the real force (F_i) and fictitious force (F_0) on the particle are given by

$$F_i = -mg, \quad F_0 = -ma_0 = -mg$$

Hence the total force in the accelerated frame (box) is

$$F_n = F_i + F_0 = -mg - mg = -2mg$$

This means that the observer, stationed in the box having an acceleration g upward, will measure a force $2mg$ downward on the particle.

5.20. Centripetal Acceleration and Centrifugal Force (Circular Motion)

Let us consider a particle P of mass m , moving on the circumference of a circle of constant radius r in X - Y plane in an inertial frame S (XYZ). The position vector $\vec{r}$, which is the function of the scalar variable t , is rotating with a constant angular velocity ω . At any time t , the position vector $\vec{r}$ can be expressed in the components form as [fig. 5.36]

$$\vec{r} = x\hat{i} + y\hat{j}$$

$$\text{or} \quad \vec{r} = r \cos \omega t \hat{i} + r \sin \omega t \hat{j}$$

$$\text{or} \quad \vec{r} = r (\cos \omega t \hat{i} + \sin \omega t \hat{j}) \quad \dots(1)$$

Velocity of the particle P is obtained as

$$\vec{v} = \frac{d\vec{r}}{dt} = r(-\omega \cos \omega t \hat{i} + \omega \sin \omega t \hat{j})$$

[$\because dr/dt = 0$, because radius r is constant for a circle]

$$\text{or} \quad \vec{v} = \omega r (-\sin \omega t \hat{i} + \cos \omega t \hat{j}) \quad \dots(2)$$

$$\text{Its magnitude,} \quad v = \sqrt{\omega^2 r^2 (\sin^2 \omega t + \cos^2 \omega t)} = \omega r \quad \dots(3)$$

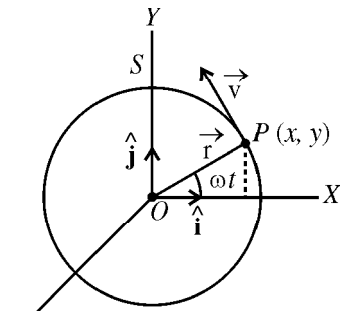


Fig. 5.36.

Differentiating eq. (2), we have the acceleration of the particle in circular motion, i.e.,

$$\vec{a} = \frac{d\vec{v}}{dt} = \omega^2 r (-\cos \omega t \hat{i} - \sin \omega t \hat{j}) = -\omega^2 r (\cos \omega t \hat{i} + \sin \omega t \hat{j})$$

But $\vec{r} = r (\cos \omega t \hat{i} + \sin \omega t \hat{j})$.

Therefore, $\vec{a} = -\omega^2 \vec{r}$... (4)

The direction of this acceleration is directed towards the centre of the circle and hence this acceleration is called **centripetal** (*centre seeking*) **acceleration**. If m is the mass of the particle, then according to Newton's second law, the force acting on the particle in the inertial frame (XYZ) is

$$\vec{F}_i = -m\omega^2 \vec{r} \quad \dots (5)$$

This is called **centripetal** (*centre seeking*) **force**, which is provided by some actual force, needed to rotate the particle on a circular path. Its magnitude is

$$F_i = m\omega^2 r = mv^2/r \quad [\because v = r\omega] \quad \dots (6)$$

For example, if a mass tied to a string, is rotating in a circle, then the centripetal force, necessary for rotation on the circular path, is provided by the tension of the string, acting on the mass towards the centre of the circle.

Now, let us consider a frame S' , which is rotating with an angular velocity ω with respect to the frame S so that the particle is at rest in the rotating frame [fig. 5.37].

Therefore in the rotating (non-inertial) frame S' , the acceleration or force on the particle is zero, i.e., $\vec{a}_n = 0$ or $\vec{F}_n = 0$.

Using the relation $\vec{F}_i + \vec{F}_0 = \vec{F}_n$ and putting $\vec{F}_i = -m\omega^2 \vec{r}$ and $\vec{F}_n = 0$, we obtain

$$-m\omega^2 \vec{r} + \vec{F}_0 = 0 \quad \text{or} \quad \vec{F}_0 = m\omega^2 \vec{r} \quad \dots (7)$$

This fictitious force $\vec{F}_0$ is called **centrifugal force** and is obviously directed away from the centre along the radius vector ($\vec{r}$). In the rotating (non-inertial) frame, in order to produce zero acceleration of the particle, the centrifugal force is balanced by some actual force (e.g., inward tension in a rotating string with a mass).

5.21. Satellite in Circular Orbit

A body, which revolves constantly round a comparatively much larger body, is said to be a **satellite**. We know that the earth rotates round the Sun and moon round the earth in specified orbits. The earth is the satellite of the Sun and the moon that of the earth. These are called **natural satellites**. Each one of these satellites is attracted by its primary by the gravitational force. Scientists have also been able to place man-made satellites. They are called **artificial satellites** and are usually sent in space with the help of high velocity rockets.

In the circular orbit of a satellite around the earth, the gravitational attraction is balanced by centrifugal force, [fig. 5.38] i.e.,

$$GMm/r^2 = m\omega^2 r \quad \dots (1)$$

where M = mass of the earth, m = mass of the satellite, r = distance between the centres of earth and satellite and ω = angular speed of satellite.

$$\text{From (1)} \quad \omega^2 = GM/r^3 \quad \dots (2)$$

Period of revolution of satellite : If T is the *time of revolution*, then

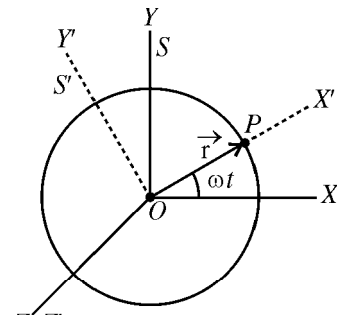


Fig. 5.37.

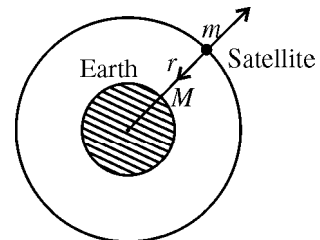


Fig. 5.38.

$$\left(\frac{2\pi}{T}\right)^2 = \frac{GM}{r^3} \quad \text{or} \quad T^2 = \frac{4\pi^2}{GM} r^3$$

$$\text{But } g = \frac{GM}{R^2}, \text{ then } T^2 = \frac{4\pi^2}{gR^2} \cdot r^3 \quad \text{or} \quad T = \frac{2\pi r}{R} \sqrt{\frac{r}{g}} \quad \dots(3)$$

If the satellite revolves very close to earth, then $r = R$

$$T = 2\pi \sqrt{\frac{R}{g}} \quad \dots(4)$$

Taking $R = 6.4 \times 10^6$ m and $g = 9.8$ m/sec², the period of revolution of the satellite near the surface of the earth is obtained to be

$$T = 2 \times 3.14 \sqrt{6.4 \times 10^6 / 9.8} \text{ sec} = 84 \text{ minutes (approx.)}$$

Velocity of satellite : If v be the velocity of the satellite, then

$$v = r\omega = r\sqrt{GM/r^3} = \sqrt{GM/r} = \sqrt{gR^2/r} = R\sqrt{g/r}$$

$$\text{i.e., } v = \sqrt{\frac{GM}{r}} \quad \text{or} \quad v = R\sqrt{\frac{g}{r}} \quad \dots(5)$$

When the satellite is moving around close to the surface of the earth, then $r = R$.

$$\text{In such a case } v = \sqrt{GM/R} \quad \text{or} \quad v = \sqrt{gR} \quad \dots(6)$$

Taking $R = 6.4 \times 10^6$ m and $g = 9.8$ m/sec²,

$$v = \sqrt{9.8 \times 6.4 \times 10^6} \text{ m/sec} = 7.92 \text{ or } 8 \text{ km/sec approx.} \quad \dots(7)$$

Hence if we want to have an artificial satellite close to the surface of the earth, we have to give the orbital velocity about 8 km/sec.

Angular momentum of the satellite :

$$L = mvr = mr\sqrt{\frac{GM}{r}} = \sqrt{GMm^2r} = \sqrt{gR^2m^2r} \quad \dots(8)$$

which is constant because the satellite is moving under central (gravitational) force.

Geostationary satellite

If a satellite with period of revolution $T = 24$ hours is put in a circular orbit with its plane coinciding with equatorial plane, moving from west to east, then to an observer on the earth the satellite will be seen stationary. Such a satellite is called **geostationary satellite**.

Let h be the height (altitude) of a geostationary satellite from the surface of the earth. Then, the distance of the satellite from the centre of the earth is $r = R + h$. Hence the period of revolution of such a satellite is given by [from eq. (3)]

$$T = \frac{2\pi r}{R} \sqrt{\frac{r}{g}} = 2\pi \sqrt{\frac{r^3}{gR^2}}$$

But $T = 24$ hours $= 24 \times 60 \times 60$ sec. and $R = 6.4 \times 10^6$ m and $g = 9.8$ m/sec. Hence

$$2 \times 3.14 \times \left[\frac{r^3}{9.8 \times (6.4 \times 10^6)^2} \right]^{1/2} = 24 \times 60 \times 60$$

whence $r = 4.2 \times 10^7$ m

$\therefore$ Height (altitude) of the geostationary satellite

$$h = r - R = 4.2 \times 10^7 - 6.4 \times 10^6 = 3.56 \times 10^7 \text{ m (see Ex. 5).}$$

Ex. 1. Calculate the fictitious force and the total force acting on a freely falling body of mass 20 kg with reference to a frame moving with a downward acceleration of 6 m/sec².

Sol. Assuming earth as inertial frame and positive direction upward

$$F_i = -mg = -20 \times 9.8 = -196$$

Here, $a_0 = -6$ m/sec² and $g = -9.8$ m/sec².

Fictitious force, $F_0 = -ma_0 = -20 \times -6 = 120$ N

i.e., fictitious force of amount **120 newtons** is acting in the upward direction.

Total force in the accelerated frame

$$F_n = ma_n = F_i + F_0 = -196 + 120 = -76 \text{ N.}$$

i.e., Total force of amount **76 newtons** is acting downward in the accelerated frame.

Ex. 2. Calculate the effective weight of an astronaut ordinarily weighing 60 kg when his rocket moves vertically (i) upward and (ii) downward with 6g acceleration.

Sol. True force, $F_i = -mg$

Force in accelerated frame or effective weight

$$F_n = F_i + F_0 = -mg - ma_0 = -m(g + a_0)$$

In case (i) $m = 60$ kg and $a_0 = 6g$

$\therefore$ effective weight = $-60(g + 6g) = -420g = \mathbf{420 \text{ kg-weight downward.}}$

In case (ii) $a_0 = -6g$ and hence

effective weight = $-60(g - 6g) = 300g = \mathbf{300 \text{ kg-weight upward.}}$

Ex. 3. A 1 kg stone at the end of a 2 metres long string makes 5 revolutions per second. Calculate the force on the stone as measured in an inertial frame and in a frame which is rotating with the string.

Sol. In an inertial frame, the necessary force to rotate the stone in the circular path is centripetal force $-m\omega^2 r$, i.e.,

$$\begin{aligned} \text{Force on the stone in inertial frame} &= -m\omega^2 r = -1(2 \times 3.14 \times 5)^2 \times 2 = -1974 \text{ newtons} \\ &= \mathbf{1974 \text{ newtons towards the centre (centripetal)}} \end{aligned}$$

This force is supplied by the tension in the string.

In case of the non-inertial frame, rotating with the stone, the acceleration of the stone is **zero**. Thus, the total force in this frame is **zero**. In order to have equilibrium of the stone in the rotating frame, the inward force of tension on the stone is balanced by the centrifugal force (fictitious force), amounting $m\omega^2 r = \mathbf{1974 \text{ newtons.}}$

Ex. 4. An artificial satellite is revolving round the earth at a distance of 620 km. Calculate the minimum velocity and the period of revolution. Radius of earth is 6380 km and acceleration due to earth's gravity at the surface of the earth is 9.8 metres/sec².

Sol. Radius of earth's satellite orbit, $r = \text{Radius of earth} + \text{Distance of satellite from earth's surface.}$
 $= 6380 + 620 = 7000 \text{ km} = 7 \times 10^6 \text{ m/sec.}$

Radius of earth, $R = 6380 \times 10^3 \text{ m}; g = 9.8 \text{ m/sec}^2.$

$$\therefore \text{Period of revolution, } T = \frac{2\pi r}{R} \sqrt{\frac{r}{g}} = \frac{2\pi \times 7 \times 10^6}{6380 \times 10^3} \sqrt{\frac{7 \times 10^6}{9.8}} = \mathbf{5775 \text{ sec.}}$$

$$\text{and} \quad \text{Orbital velocity, } v = R \sqrt{\frac{g}{r}} = 6380 \times 10^3 \sqrt{\frac{9.8}{7 \times 10^6}} = \mathbf{7.55 \times 10^6 \text{ m/sec.}}$$

Ex. 5. Calculate the height of geostationary (equatorial) satellite which is always seen to be over the same point of earth's surface. ($G = 6.66 \times 10^{-11}$ S.I. units; $M = 5.98 \times 10^{24}$ kg.) **(K.U. 2005)**

Sol. Let the height of the equatorial satellite be h . So that the equatorial satellite seems over the same point of earth's surface, i.e., the angular velocity of satellite is the same as that of the earth itself.

Here, angular velocity of satellite = $\frac{2\pi}{24 \times 60 \times 60} = 7.3 \times 10^{-5} \text{ sec}^{-1}$.

$$\text{Also, } \frac{GMm}{r^2} = m\omega^2 r \text{ or } r^3 = \frac{GM}{\omega^2} = \frac{6.66 \times 10^{-11} \times 5.98 \times 10^{24}}{(7.3 \times 10^{-5})^2} = 75 \times 10^{27} \text{ or } r = 4.2 \times 10^7 \text{ m}$$

$$\therefore h = r - R = 4.2 \times 10^7 - 6.4 \times 10^6 = \mathbf{3.56 \times 10^7 \text{ m.}}$$

Ex. 6. A satellite with kinetic energy E_k is revolving round the earth in circular orbit. How much more kinetic energy is to be given to it so that it may just escape into outer space ? (A.N.U. 2003)

Sol. If M and m be the masses of the earth and the satellite respectively, then the gravitational force between the necessary centripetal force for rotation i.e.,

$$\frac{GMm}{r^2} = \frac{mv^2}{r}, \quad r = \text{distance of satellite from the centre of earth}$$

$$\therefore \frac{1}{2}mv^2 = \frac{GMm}{2r} = E_k, \text{ kinetic energy of satellite}$$

Total energy of satellite = P.E. + K.E.

$$= -\frac{GMm}{r} + \frac{GMm}{2r} = -\frac{GMm}{2r}$$

This means that the satellite is bound with the earth with $-GMm/2r$ energy.

Hence the satellite is to be given $GMm/2r = E_k$ more kinetic energy so that it may escape out of the gravitational field of the earth.

5.22. Rotating Frame of Reference and Coriolis Force

The earth itself rotates about its axis in 24 hours. Therefore, the frame fixed on the earth will also rotate with it and so it will be a **non-inertial frame**. Suppose that a frame (X_r, Y_r, Z_r) is rotating with an angular velocity $\vec{\omega}$ relative to an inertial frame $S(X_i, Y_i, Z_i)$. For simplicity, we assume that both of the frames have common origin O and common Z -axis [fig. 5.39]. In case of earth, the common origin O may be thought as the centre of the earth, Z -axis as coinciding with its rotational axis and the frame S' as rotating with earth relative to the non-rotating frame S .*

The position vector of a particle P in both frames will be the same**, i.e., $\vec{r}_i = \vec{r}_r = \vec{R}$, because the origins are coincident.

Now, if the particle P is stationary in the frame S , the observer in the rotating frame S' will see that the particle is moving oppositely with linear velocity $-\vec{\omega} \times \vec{R}$. Thus, if the velocity of the particle in the frame S is

$(d\vec{R}/dt)_i$, then its velocity $(d\vec{R}/dt)_r$ in the rotating frame will be given by

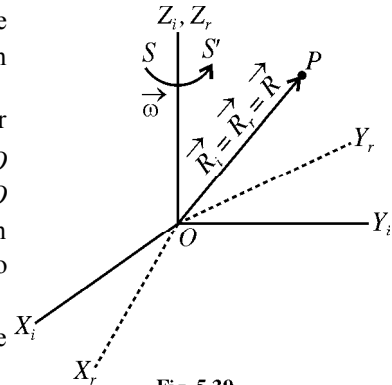


Fig. 5.39.

* Truly speaking, this frame is rotating about Sun. But the assumption of S to be an inertial frame will not add any perceptible error in our measurements on earth.

** The components may have different values, but the direction and magnitude of the position vector will be the same in two frames.

$$\left(\frac{d\vec{R}}{dt}\right)_r = \left(\frac{d\vec{R}}{dt}\right)_i - \vec{\omega} \times \vec{R} \quad \text{or} \quad \left(\frac{d\vec{R}}{dt}\right)_i = \left(\frac{d\vec{R}}{dt}\right)_r + \vec{\omega} \times \vec{R} \quad \dots(1)$$

Actually this equation holds for all vectors and relates the time derivatives of a vector in the frames S and S' . Therefore, relation (1) may be written in the form of operator equation :

$$\left(\frac{d}{dt}\right)_i = \left(\frac{d}{dt}\right)_r + \vec{\omega} \times \quad \dots(2)$$

Writing $d\vec{R}/dt = \vec{v}$ for the velocity of the particle, we have

$$\vec{v}_i = \vec{v}_r + \vec{\omega} \times \vec{R} \quad \dots(3)$$

This relation connects the **velocity** in inertial frame and rotating frame.

Now, if we operate eq. (2) on velocity vector $\vec{v}_i$, we have

$$\left(\frac{d\vec{v}_i}{dt}\right)_i = \left(\frac{d\vec{v}_i}{dt}\right)_r + \vec{\omega} \times \vec{v}_i$$

Substituting the value of $\vec{v}_i$ in the right hand side of this relation, from eq. (3), we get

$$\begin{aligned} \left(\frac{d\vec{v}_i}{dt}\right)_i &= \left[\frac{d}{dt}(\vec{v}_r + \vec{\omega} \times \vec{R})\right]_r + \vec{\omega} \times (\vec{v}_r + \vec{\omega} \times \vec{R}) \\ &= \left(\frac{d\vec{v}_r}{dt}\right)_r + \frac{d\vec{\omega}}{dt} \times \vec{R} + \vec{\omega} \times \left(\frac{d\vec{R}}{dt}\right)_r + \vec{\omega} \times \vec{v}_r + \vec{\omega} \times (\vec{\omega} \times \vec{R}) \end{aligned}$$

Writing the acceleration $d\vec{v}/dt = \vec{a}$, we have

$$\vec{a}_i = \vec{a}_r + 2\vec{\omega} \times \vec{v}_r + \vec{\omega} \times (\vec{\omega} \times \vec{R}) + \frac{d\vec{\omega}}{dt} \times \vec{R} \quad \dots(4)$$

This equation tells us the relation between the acceleration in the inertial and relation frames.

If the angular velocity $d\vec{\omega}/dt = 0$. In such a case, $\vec{\omega}$ is constant as in the case of earth, then

$$\vec{a}_i = \vec{a}_r + 2\vec{\omega} \times \vec{v}_r + \vec{\omega} \times (\vec{\omega} \times \vec{R}) \quad \dots(5)$$

If m is the mass of the particle, then force ($\vec{F}_r$) in the rotating frame is

$$m\vec{a}_r = m\vec{a}_i - 2m\vec{\omega} \times \vec{v}_r - m\vec{\omega} \times (\vec{\omega} \times \vec{R}) \quad \dots(6)$$

or

$$\vec{F}_r = m\vec{a}_i - 2m\vec{\omega} \times \vec{v}_r - m\vec{\omega} \times (\vec{\omega} \times \vec{R}) \quad \dots(7)$$

But the force in the rotating frame,

$$\vec{F}_r = \vec{F}_i + \vec{F}_0 \quad \dots(8)$$

Comparing eqs. (7) and (8), we get

$$\therefore \text{Fictitious force, } \vec{F}_0 = -2m \vec{\omega} \times \vec{v}_r - m \vec{\omega} \times (\vec{\omega} \times \vec{R}) \quad \dots(9)$$

where $-2m \vec{\omega} \times \vec{v}_r$ is called **Coriolis force*** and $-m \vec{\omega} \times (\vec{\omega} \times \vec{R})$, the **centrifugal force**. Both the Coriolis force and centrifugal force are the fictitious forces, resulting from the rotational motion of the frame S' relative to the inertial frame S . These forces are not due to any specific action applied to the particle.

If the particle is at rest (i.e., $\vec{v}_r = 0$) in the rotating frame, then centrifugal force is only the fictitious force

acting on the particle. On the other hand, if the particle is moving with velocity $\vec{v}_r \neq 0$ in the rotating frame, then in addition to centrifugal force, Coriolis force acts on the particle.

5.23. Motion Relative to the Earth

In the rotating frame with the earth, the acceleration of a particle is given by

$$\vec{a}_r = \vec{a}_i - \vec{\omega} \times (\vec{\omega} \times \vec{R}) - 2 \vec{\omega} \times \vec{v}_r \quad \dots(1)$$

where $\vec{a}_i$ is the acceleration in the non-rotating or inertial frame,

$-\vec{\omega} \times (\vec{\omega} \times \vec{R})$ is the **centrifugal acceleration** and $-2 \vec{\omega} \times \vec{v}_r$ is the **Coriolis acceleration**.

An interesting application of eq. (1) is the study of the motion of a body relative to a frame (S') rotating with the earth. The angular velocity of the earth is $\vec{\omega}$ along its axis of rotation. Now consider a particle P close to the earth's surface [fig. 5.40]. If we call $\vec{g}_0$ as the acceleration due to gravity with respect to an inertial or non-rotating frame, then $\vec{a}_i = \vec{g}_0$ and hence the acceleration as measured by an observer rotating with the earth is [from eq. (1)]

$$\vec{a}_r = \vec{g}_0 - \vec{\omega} \times (\vec{\omega} \times \vec{R}) - 2 \vec{\omega} \times \vec{v}_r \quad \dots(2)$$

Hence the acceleration of a body relative to the rotating earth depends on its velocity relative to the earth and on the position vector $\vec{R}$ of the body.

Now we shall discuss the effect of centrifugal and Coriolis accelerations separately.

(a) Effect of centrifugal force : If the particle P is at rest on the earth's surface, then Coriolis term is zero and the acceleration $\vec{a}_r$ measured relative to the earth is called **effective acceleration due to gravity** $\vec{g}$. Thus

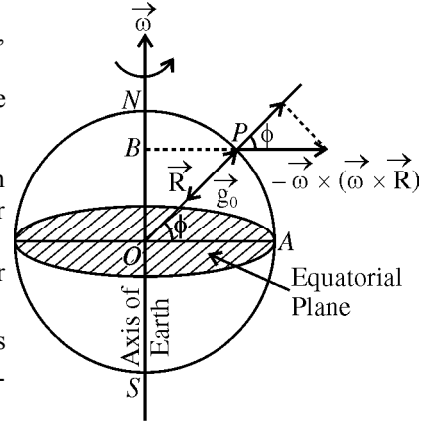


Fig. 5.40.

* The name Coriolis force comes from the name of the scientist G. G. de Coriolis, who described this effect in the rotational motion.

$$\vec{g} = \vec{g}_0 - \vec{\omega} \times (\vec{\omega} \times \vec{R}) \quad \dots(3)$$

Obviously the centrifugal acceleration $-\vec{\omega} \times (\vec{\omega} \times \vec{R})$ acts in the outward direction along P and its magnitude is $\omega^2 R \cos \phi^*$, where ϕ is the latitude at P .

Assuming earth to be spherical, we may consider $\vec{g}_0$ to be pointing towards the centre of the earth along the radial direction. Due to the second term in (3), the direction of $\vec{g}$, called the vertical, deviates slightly from the radial direction and is determined by a plumb line. For practical purposes, the vertical may be assumed to coincide with the radial direction. The magnitude of $\vec{g}$ is slightly less than the value of $\vec{g}_0$ and can be expressed as

$$g = g_0 - \omega^2 R \cos^2 \phi \quad \dots(4)$$

The second term in eq. (4) is very small (about 0.3%) compared to g_0 and accounts for most of the observed variations of acceleration due to gravity with latitude at earth's surface.

(b) Effect of Coriolis force : Coriolis force $(-2m \vec{\omega} \times \vec{v}_r)$ is a fictitious force which acts on a particle only if it is in motion with respect to the rotating frame. Hence, in the rotating frame if a particle moves with velocity $\vec{v}_r$, then it always experiences a force $(-2m \vec{\omega} \times \vec{v}_r)$ **perpendicular** to its path opposite to

the direction of vector product $\vec{\omega} \times \vec{v}_r$. The effect of Coriolis force is not considerable at small (particle) speeds. Let us discuss the effect of Coriolis force, when the particle is moving in a horizontal plane or falling freely on the earth.

(i) Particle moving in a horizontal plane : The Coriolis force causes a moving particle in a horizontal plane in the northern hemisphere to deflect towards the right of its path. In the southern hemisphere, the deflection is towards the left of the path. Let a particle of mass m be projected with velocity $\vec{v}$ in a horizontal plane at a point P on earth's surface with latitude ϕ . Consider a frame XYZ fixed on the earth at P so that X -axis is vertical and YZ is horizontal plane [fig. 5.42]. Now,

$$\vec{v} = v_y \hat{j} + v_z \hat{k} \quad \text{and} \quad \vec{\omega} = \omega \sin \phi \hat{i} + \omega \cos \phi \hat{k}$$

The Coriolis force acting on the particle is

$$\begin{aligned} \vec{F} &= -2m (\vec{\omega} \times \vec{v}) \\ &= -2m (\omega \sin \phi \hat{i} + \omega \cos \phi \hat{k}) \times (v_y \hat{j} + v_z \hat{k}) \end{aligned}$$

* Consider the plane $ONPA$ and take unit vector $\hat{i}$ and $\hat{k}$ along OA and ON

[fig. 5.41]. So that $\vec{\omega} = \omega \hat{k}$ and $\vec{R} = R \cos \phi \hat{i} + R \sin \phi \hat{k}$.

$$\begin{aligned} \text{Now } \vec{\omega} \times (\vec{\omega} \times \vec{R}) &= (\vec{\omega} \cdot \vec{R}) \vec{\omega} - (\vec{\omega} \cdot \vec{\omega}) \vec{R} \\ &= \omega^2 R \sin \phi \hat{k} - \omega^2 R [\hat{i} \cos \phi + \hat{k} \sin \phi] \\ &= -\omega^2 R \cos \phi \hat{i} \end{aligned}$$

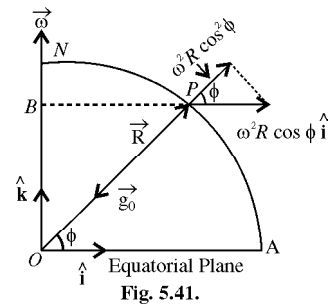


Fig. 5.41.

$$= 2m\omega v_y \cos \phi \hat{i} - 2m\omega v_z \sin \phi \hat{j} + 2m\omega v_y \sin \phi \hat{k} \quad \dots(5)$$

If $\vec{F}_H$ and $h - \frac{1}{2}gT^2$ are the horizontal and vertical components of the Coriolis force, then

$$\vec{F}_H = 2m\omega v_z \sin \phi \hat{j} - 2m\omega v_y \sin \phi \hat{k} \quad \dots(6)$$

Its magnitude

$$F_H = 2m\omega v \sin \phi \quad [\because \sqrt{v_y^2 + v_z^2} = v] \quad \dots(7)$$

$$\text{Also} \quad \vec{F}_V = 2m\omega v_y \cos \phi \hat{i} \quad \dots(8)$$

$$\text{Its magnitude} \quad F_V = 2m\omega v_y \cos \phi \quad \dots(9)$$

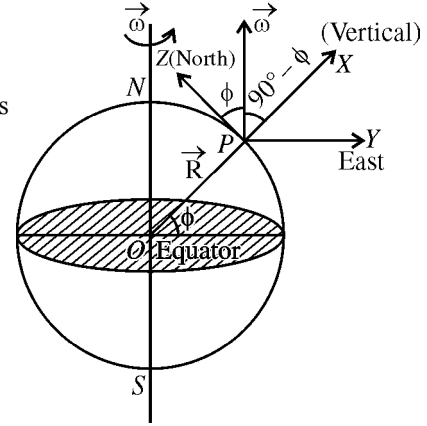


Fig. 5.42.

Obviously the horizontal component of the Coriolis force tends to deviate the path of the particle toward the right in the northern hemisphere. For example, if the particle is projected towards north, due to the Coriolis force it is deflected toward east.

The magnitude of horizontal Coriolis force is $2m\omega v \sin \phi$ and is zero at $\phi = 0$, i.e., at equator. The effect of Coriolis force is appreciable due to its horizontal component because in the vertical direction, its effect is masked by the large gravitational force.

(ii) Freely falling body at earth's surface : One of the important effect of Coriolis force is that a freely falling body deviates from its true vertical path. This deviation is always towards east direction in either of the hemisphere of the earth. Due to the horizontal component of earth's angular velocity, Coriolis force starts to act on freely falling body in the horizontal direction and deviates it from the true vertical direction. We deduce below an expression for this deviation.

Let us consider the free fall of a body from a height h on the surface of the earth at a latitude ϕ . The earth is rotating about its axis with an angular velocity $\vec{\omega}$. At the point P of the earth, take X -axis vertically, Y -axis along east and Z -axis along north. [fig. 5.42]. If $\hat{i}, \hat{j}, \hat{k}$, are unit vectors along these axes, then the angular velocity $\vec{\omega}$ can be represented as

$$\vec{\omega} = \omega \cos \left[\frac{\pi}{2} - \phi \right] \hat{i} + \omega \cos \phi \hat{k} \quad \text{or} \quad \vec{\omega} = \omega (\sin \phi \hat{i} + \cos \phi \hat{k}) \quad \dots(10)$$

As the effective value of the acceleration due to gravity g is the combined effect of the centrifugal acceleration and acceleration in the inertial frame, then substituting $\vec{a}_i - \vec{\omega} \times (\vec{\omega} \times \vec{R}) = -\hat{i}g$ in the equation we get

$$\begin{aligned} \vec{a}_i &= \vec{a}_r + 2\vec{\omega} \times \vec{v}_r + \vec{\omega} \times (\vec{\omega} \times \vec{R}), \\ -\hat{i}g &= \vec{a}_r + 2\vec{\omega} \times \vec{v}_r \end{aligned} \quad \dots(11)$$

Here the velocity of the body $\vec{v}_r$ is almost along X -axis with negligible Y and Z components and hence we have

$$\vec{v}_r = \hat{i} \frac{dx}{dt} \quad \dots(12)$$

Writing $\vec{a}_r$ in component form, we get from eq. (11)

$$-\hat{i}g = \hat{i}\frac{d^2x}{dt^2} + \hat{j}\frac{d^2y}{dt^2} + \hat{k}\frac{d^2z}{dt^2} + 2\omega(\sin\phi\hat{i} + \cos\phi\hat{k}) \times \hat{i}\frac{dx}{dt}$$

or

$$-\hat{i}g = \hat{i}\frac{d^2x}{dt^2} + \hat{j}\left[\frac{d^2y}{dt^2} + 2\omega\frac{dx}{dt}\cos\phi\right] + \hat{k}\frac{d^2z}{dt^2}$$

Now, the three component equations are :

$$\frac{d^2x}{dt^2} = -g \quad \dots(13a) \quad \frac{d^2y}{dt^2} = -2\omega\frac{dx}{dt}\cos\phi \quad \dots(13b) \quad \frac{d^2z}{dt^2} = 0 \quad \dots(13c)$$

Integrating eq. (13a), we get

$$\frac{dx}{dt} = -gt + C$$

But initially at $t = 0$, $dx/dt = 0$, therefore $C = 0$.

$$\text{Thus} \quad \frac{dx}{dt} = -gt. \quad \dots(14)$$

Integrating it further, we get

$$x = -\frac{1}{2}gt^2 + C$$

Initially at $t = 0$, the distance of the stone from the earth $x = h$. This gives $h = C$, and hence

$$x = -\frac{1}{2}gt^2 + h$$

Finally, when at $t = T$, the stone touches and ground $x = 0$. Therefore

$$h - \frac{1}{2}gT^2 = 0 \text{ or } T = \sqrt{2h/g} \quad \dots(15)$$

$$\text{Now,} \quad \frac{d^2y}{dt^2} = -2\omega\frac{dx}{dt}\cos\phi \text{ or } \frac{d^2y}{dt^2} = 2\omega gt \cos\phi \quad [\text{using eq. (14)}]$$

Integrating it, we get

$$\frac{dy}{dt} = \omega gt^2 \cos\phi \text{ and hence } y = \frac{\omega gt^3}{3} \cos\phi \quad \dots(16)$$

The constants of integration in (16) have been taken zero, because initially the stone has no displacement and velocity in the Y -direction is zero.

Finally, at $t = T$, the maximum horizontal displacement will be given by

$$Y = \frac{\omega g T^3}{3} \cos\phi \quad \dots(17)$$

Substituting the value of T from eq. (15), we get

$$Y = \frac{\omega g}{3} \left[\frac{2h}{g} \right]^{3/2} \cos\phi \text{ or } Y = \left[\frac{8}{9g} \right]^{1/2} h^{3/2} \omega \cos\phi \quad \dots(18)$$

Thus a freely falling body at latitude ϕ is displaced horizontally due east by Coriolis force by an amount, given by eq. (18). **At the equator** ($\phi = 0$), the easterly deflection is given by

$$Y = \left[\frac{8}{9g} \right]^{1/2} h^{3/2} \omega \quad \dots(19)$$

Some Other Effects of Coriolis Force

(1) Formation of cyclones :

When a low pressure zone is created at a place on earth, the wind will flow radially from high pressure regions to the central low pressure region. However the Coriolis force deviates the air molecules toward the right of their path in the northern hemisphere, resulting in anticlock-wise motion of the air particles and formation of the cyclone [fig. 5.43 (a)]. The cyclone also moves along with the centre of low pressure region. In the southern hemisphere, the formation of cyclone is clockwise [fig. 5.43(b)].

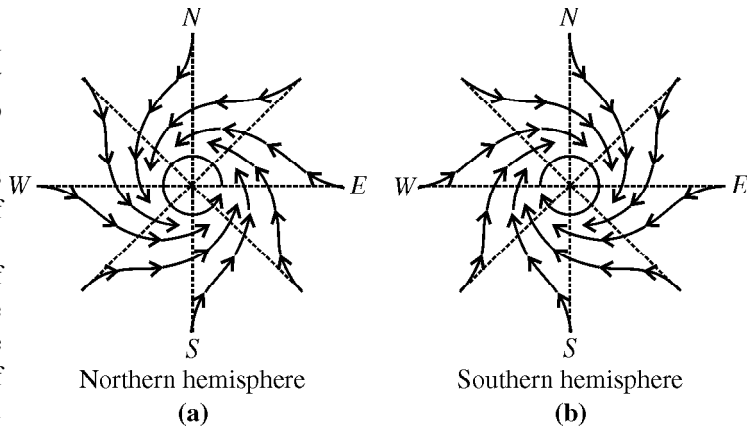


Fig. 5.43 : Formation of cyclones.

(2) In the **northern hemisphere**, flowing water in a river is acted by Coriolis force towards right of its path so that the right bank of the river is eroded more in comparison to its left bank. In the southern hemisphere, the effect is more on the left bank.

(3) **Rotation of plane of oscillation of Foucault's pendulum** : Foucault's pendulum is similar to a simple pendulum, with a heavy bob and long strong suspension wire. If the pendulum is oscillated, Coriolis force acts toward the right of its path which results in the rotation of the plane of oscillation slowly in the clockwise direction in the northern hemisphere. This experimental fact demonstrates that the earth rotates about its axis.

5.24. Foucault's Pendulum

Experimentally this fact, that earth rotates about its axis and hence the frame attached to it is not an inertial frame, is demonstrated by Foucault's pendulum. The experiment was first performed by Foucault in 1851. Foucault's pendulum is similar to a simple pendulum in which a very heavy bob is suspended by means of a long strong wire [fig. 5.44]. Thus if it is once vibrated, it vibrates for considerably large time with a large period (≈ 17 sec.). Foucault took in his experiment a bob of 28 kg mass and a wire of 70 metres length. The upper end of the wire is attached to a rigid support in such a way that the pendulum may vibrate with equal freedom in any direction.

When Foucault's pendulum is allowed to oscillate in the northern hemisphere, its plane of oscillation rotates from east to west (or clockwise as seen from above). It is not possible that the point of support might have turned the plane of vibration. The only possibility is this that the floor, i.e., the earth under the pendulum may be rotating. If such a pendulum is imagined to

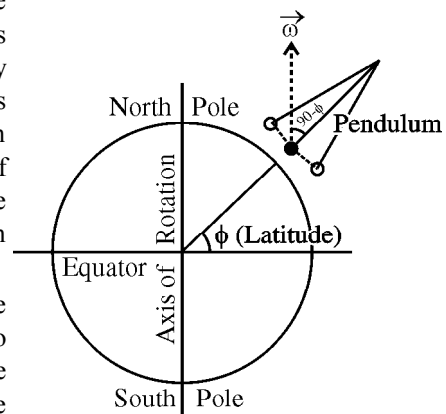


Fig. 5.44. Foucault's pendulum.

vibrate at the north pole of the earth, the plane of oscillation will remain fixed in an inertial frame or solar reference frame. But the earth under the pendulum is rotating once every 24 hours, therefore an observer on the earth will see that the plane of oscillation is turning from east to west, opposite to earth's

rotation. At any other latitude ϕ , the angular velocity $\vec{\omega}$ can be resolved into two components : vertical component $\omega \sin \phi$ and horizontal component $\omega \cos \phi$ in the north direction. Evidently the later component will not have any perceptible effect on the pendulum. The vertical component will make the plane of its oscillation rotate with an angular velocity $\omega \sin \phi$. Hence the **time of one rotation of the plane of oscillation** will be given by

$$T = \frac{2\pi}{\omega \sin \phi} \quad \dots(1)$$

As the earth rotates once in 24 hours with an angular velocity ω , i.e., $2\pi/\omega = 24$ hours, hence

$$T = \frac{24 \text{ hours}}{\sin \phi} \quad \dots(2)$$

This rotation will be clockwise in the northern hemisphere and opposite to it in the southern hemisphere. In fact, this rotation, as seen by the observer on the earth, is due to Coriolis force. When the pendulum moves along a horizontal line, Coriolis force acts in the perpendicular direction to its velocity and in consequence the plane of oscillation turns. Thus, *the rotation of the plane of oscillation of Foucault's pendulum demonstrates that the earth rotates about its axis.*

Ex. 1. A bullet is fired horizontally in the north direction with a velocity of 500 m/sec at 30° N latitude. Calculate the horizontal component of Coriolis acceleration and the consequent deflection of the bullet as it hits a target 250 metres away. Also determine the vertical displacement of the bullet due to gravity. If the mass of the bullet is 100 gm, find the Coriolis force.

Sol. If we take X-axis vertically, Z-axis towards north and Y-axis along east, then the velocity of the bullet is

$$\vec{v} = 500 \hat{k} \text{ m/sec.}$$

and angular velocity $\vec{\omega} = \omega (\hat{k} \cos 30^\circ + \hat{i} \sin 30^\circ)$

since the angular velocity vector $\vec{\omega}$ of the earth is directed parallel to its axis and is inclined at 30° to the horizontal [fig. 5.45].

$$\text{Here, } \omega = \frac{2\pi}{24 \times 60 \times 60} = 7.2 \times 10^{-5} \text{ rad.sec.}$$

$$\therefore \text{ Coriolis acceleration} = -\vec{\omega} \times \vec{v}$$

$$= -2\omega (\hat{k} \cos 30^\circ + \hat{i} \sin 30^\circ) \times 500 \hat{k}$$

$$= 2 \times 7.2 \times 10^{-5} \times 10^2 \times \frac{1}{2} \hat{j} = 0.036 \text{ m/sec}^2 \text{ towards east.}$$

$$\text{Time of journey} = \frac{250}{500} = \frac{1}{2} \text{ sec.}$$

$\therefore$ Deflection of the bullet due to Coriolis acceleration

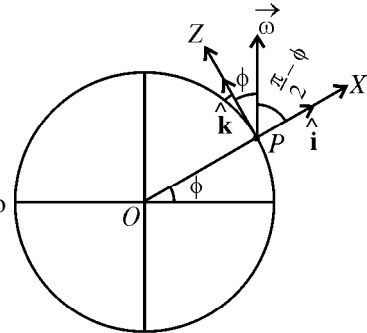


Fig. 5.45.

$$= \frac{1}{2}at^2 = \frac{1}{2}0.036 \times \left(\frac{1}{2}\right)^2 = 4.5 \times 10^{-3} \text{ m towards east.}$$

$$\text{Vertical displacement of the bullet due to gravity} = \frac{1}{2}gt^2 = \frac{1}{2} \times 9.8 \times \left(\frac{1}{2}\right)^2 = 1.23 \text{ m}$$

$$\text{Coriolis force} = -2m\vec{\omega} \times \vec{v} = 0.1 \times 0.036 \hat{i} = 3.6 \times 10^{-3} \text{ newton towards east.}$$

Ex. 2. Calculate the magnitude and direction of the coriolis force on an ice block of mass 5×10^8 kg near the North Pole moving with a speed of 0.02 m/sec. Given the angular velocity of rotation of the earth 0.727×10^{-4} rad/sec. (O.U. 2009)

$$\text{Sol. Coriolis force} = -2m\vec{\omega} \times \vec{v}_r \quad \dots(i)$$

$$\text{Force in magnitude} = 2m\omega v_r \sin \phi \quad \dots(ii)$$

where m is the mass of the body, ω the angular velocity of rotation of the earth, v_r the speed of the body in the rotating frame and ϕ the latitude.

$$\text{Here, } m = 5 \times 10^8 \text{ kg, } \omega = 0.727 \times 10^{-4} \text{ rad/sec, } v_r = 0.02 \text{ m/sec and } \sin \phi = \sin \frac{\pi}{2} = 1 \quad (\because \phi = \pi/2)$$

$$\therefore \text{ Magnitude of force } F = 2 \times 5 \times 10^8 \times 0.727 \times 10^{-4} \times 0.02 \times 1 = 1.454 \times 10^8 \text{ N}$$

The ice block is moving near the The North Pole. Hence its direction of velocity may be taken in horizontal plane. In view of eq. (i), the direction of the force will be **in the horizontal plane towards the right of its path of motion.**

Ex. 3. Prove that the observed acceleration due to gravity g_ϕ at the latitude ϕ is related to its real value g by the relation

$$g_\phi^2 = (g \cos \phi - \omega^2 R \cos \phi)^2 + (g \sin \phi)^2.$$

Sol. If a particle is at rest at ϕ latitude, then it is not acted by Coriolis force.

$$\text{Therefore, } \vec{a}_i = \vec{a}_r + \vec{\omega} \times (\vec{\omega} \times \vec{R})$$

If we take Z-axis along the axis of the earth and X-axis perpendicular to it, then

$$\vec{a}_r = \vec{a}_i - \vec{\omega} \times (\vec{\omega} \times \vec{R})$$

or

$$g_\phi = -g (\cos \phi + \hat{k} \sin \phi) - \omega \times [\omega \hat{k} \times R (\hat{i} \cos \phi + \hat{k} \sin \phi)]$$

$$= -g (\hat{i} \cos \phi + \hat{k} \sin \phi) - \omega^2 R \cos \phi \hat{i}$$

$$= -\hat{i} (g \cos \phi - \omega^2 R \cos \phi) - \hat{k} g \sin \phi$$

$$\therefore g_\phi^2 = g_\phi^2 \quad g_\phi = (g \cos \phi - \omega^2 R \cos \phi)^2 + (g \sin \phi)^2.$$

Ex. 4. Prove that the plane of oscillation of Foucault's pendulum rotates 15° sin ϕ per hour, when ϕ is the latitude of the place.

$$\text{Sol. Period of one rotation of plane of oscillation } T = \frac{24 \text{ hours}}{\sin \phi}$$

If the plane of oscillation be rotating with an angular velocity α per hour, then

$$T = \frac{2\pi}{\alpha} = \frac{360^\circ}{\alpha} \text{ hour, } \therefore \frac{360^\circ}{\alpha} = \frac{24}{\sin \phi}$$

whence $\alpha = 15^\circ \sin \phi$ per hour.

Ex. 5. In how much time will the plane of oscillation of a Foucault's pendulum rotate through 90° at 30° latitude ?

Sol. Period of one rotation, (i.e., 360°) of the plane of oscillation of Foucault's pendulum is

$$T = \frac{24 \text{ hours}}{\sin \phi}$$

Hence the plane of oscillation of the pendulum will rotate through 90° in time

$$= \frac{24 \text{ hours}}{\sin \phi} \times \frac{90^\circ}{360^\circ} = \frac{6 \text{ hours}}{\sin 30^\circ} = 12 \text{ hours.}$$

QUESTIONS

Long Answer Questions

1. What is a central force ? Show that the angular momentum of a particle under central force is conserved and its direction is perpendicular to the plane of the orbit.
2. What is a central force ? Show that if a particle moves under a central force
 - (i) the angular momentum is conserved.
 - (ii) the areal velocity remains constant. (O.U. 1990)
3. Explain what do you understand by central forces ? Give examples. When a particle moves under the action of a central force, prove that its angular momentum is conserved and the motion takes place in a plane.
4. What are central forces ? Give some examples. Show that the central forces are conservative. (O.U. 2008, 07; A.N.U. 2008, 07, 05; A.U. 2007)
5. What is a central force ? Explain its conservative nature. Show that the curl of a central force is zero. (O.U. 2003)
6. Prove that central forces are conservative. Prove that conservative force is equal to the negative gradient of potential energy. If the earth be one-half of its present distance from Sun, what will be the number of days in a year ? (S.K.U. 2007)
7. Define central forces with examples. Explain the term conservative force and show that central forces are conservative. (A.U. 1988)
8. What is central force ? Show that central force is conservative force. State Kepler's laws of planetary motion. (A. U. 2005)
9. Define gravitational field and gravitational potential. Derive the relation between them. (S.K.U. 2002)
10. Define the gravitational potential and gravitational field intensity. Obtain the expressions for gravitational potential at a point outside the shell and inside the shell. (A.N.U. 2001)
11. Deduce the expression for the gravitational potential at a point due to point mass.
12. Deduce expressions for gravitational potential due to a uniform solid sphere at a point : (i) inside (ii) on the surface, and (iii) outside the sphere. Also find the gravitational field at these points, and hence prove that the gravitational potential at the centre of sphere is $3/2$ times that on its surface.
13. What do you understand by the reduced mass ? How is the two-body problem reduced to a single body problem ?
14. Define central force. Explain the conservative nature of central forces. Deduce equation of motion of a particle. (S.K.U. 1989)
15. What is central force ? Give two examples. Derive the equation of motion of a particle under the action of a central force. (A.N.U. 1996; K.U. 1997)

16. Obtain the differential equation $\frac{d^2u}{d\theta^2} + u = \frac{p}{h^2u^2}$ of the orbit of a particle moving under an attractive central force p per unit mass. (O.U. 2007)
17. What do you mean by inverse square force ? Show that the equation of orbit of a planet around Sun under inverse square force is elliptical.
18. What do you mean by an inverse square force ? Establish the equation of motion of a particle moving under an inverse square force. Show that the orbit is elliptical if the force is attractive. Find the point of revolution of the particle.
19. What is central force ? Obtain the equation of motion of a particle in an inverse square law force field. Hence derive Kepler's laws of planetary motion. (A.N.U. 1997; A.U. 1991)
20. State Kepler's laws of planetary motion and prove second law of motion. (A.U. 1996)
21. State Kepler's second law and show that it is a consequence of the law of conservation of angular momentum. (S.V.U. 1997; K.U. 1989)
22. Deduce Kepler's laws of planetary motion assuming the force between Sun and planet to be of inverse square law nature. (K.U. 1995)
23. State and prove Kepler's laws of planetary motion. (A.U. 2007; S.K.U. 2002; S.V.U. 1997)
24. State Newton's law of universal gravitation. Based on this law derive Kepler's third law. (A.N.U. 2004; A.U. 1997)
25. State Kepler's law of planetary motion. Deduce these laws using Newton's inverse square law of gravitation. (N.U. 1994)
26. Write the equation for the eccentricity of a planet which determines the shape of the orbit of a planet round the Sun. Discuss the different cases. (S.V.U. 1990)
27. State Kepler's laws of planetary motions. Derive the third law of planetary motion. (S.K.U. 2005)
28. State Kepler's laws of planetary motion. Explain the orbital motion of satellites based on Kepler's laws. (A.U. 2003; N.U. 2001)
29. What are Kepler's laws of planetary motion ? Show how the universal law of gravitation of Newton has been derived from these laws ?
30. What are inertial and non-inertial frames ? What is fictitious force ? Discuss centrifugal force.
31. What are fictitious forces ? How are these related to non-inertial frames ? What are the effects of Coriolis force due to earth's rotation ? Explain.
32. What is Coriolis force ? Under what conditions does it come into play ? Discuss in general terms the effect of Coriolis force produced as a result of earth's motion.
33. Find out horizontal and vertical components of Coriolis force due to the rotation of the earth on particle of mass m , projected horizontally with velocity $\vec{v}$.
34. What is Coriolis force ? Show that the total Coriolis force acting on a body of mass m in a rotating frame is $-2m\vec{\omega} \times \vec{v}_r$, where $\vec{\omega}$ is the angular velocity of rotating frame and $\vec{v}_r$ is the velocity of the body in rotating frame.

Short Answer Questions

1. What is central force ? Give two examples. (O.U. 2007; S.V.U. 2005; S.K.U. 2003)
2. Give two examples of central forces. (A.U. 2005)
3. Define central forces and give three examples. (S.V.U. 2007)
4. Show that central forces are conservative in nature. (K.U. 2007; O.U. 2005; S.K.U. 2005)
5. Show that central forces are independent of path. (S.V.U. 2002)

6. What are central forces ? What are their characteristics ? (A.N.U. 2003, O.U. 2009)
7. Prove that the centre of two particles subjected to their mutual interaction alone (central force) moves with a constant velocity. (A.N.U. 2008)
8. Show that the conservation of angular momentum of a system applied to planetary motion leads to law of conservation of areal velocity. (A.U. 1995)
9. Show that for a particle moving under the influence of a central force, the areal velocity is constant. (A.N.U. 2004)
10. Earth moves in an orbit around Sun under its gravitational force. Show that its orbit lies in a plane and that the vector joining the earth and the Sun sweeps out equal areas in equal times.
11. Define field strength and potential. What is the relation between them ? (S.V.U. 2002)
12. Define gravitational field and give two examples. (A.N.U. 2005)
13. Show that the angular momentum of a particle is conserved when moving under central force.
14. Show that the areal velocity of a particle moving under an inverse square force always remains constant.
15. Show that the angular momentum and total energy of a body moving under an inverse square force remains conserved.
16. The radius vector drawn from the Sun to a planet sweeps out equal areas in equal times.
17. Show that the angular momentum of a particle about a fixed point is equal to the product of its mass and double the area ($\frac{1}{2}r^2\omega$) described by the rotating line in unit time and has a direction perpendicular to this area.
18. Explain Kepler's law of planetary motion. (A.N.U. 2007; A.U. 2007; S.K.U. 2003)
19. Explain briefly the motion of a satellite. (A.N.U. 2008, 06)
20. Show that Kepler's second law on planetary motion is an application of the principle of conservation of angular momentum. (O.U. 2007)
21. What is the value of gravitational force on a mass m at the centre of a solid sphere ?
[Ans. Zero]
22. If a body of mass m be projected vertically upward from earth's surface (radius of earth = R) to reach a height of $10R$, how much kinetic energy be given to the body ?
[Ans. $\frac{10}{11}mgR$]
23. Show that the intensity of the gravitational field can be expressed as $\vec{E} = -\text{grad } V$, where V is the gravitational potential ?
24. Distinguish between inertial and non-inertial frames. (O.U. 2008)
25. Differentiate between real and fictitious forces.
26. What is Coriolis force ?
27. Prove that a frame of reference fixed with the earth is non-inertial frame.
28. Discuss the effect of centrifugal and Coriolis forces due to earth's rotation.
29. What is Coriolis force ? Obtain the effect of Coriolis force over a mass freely falling from a height H .
30. How is it demonstrated that the earth rotates about its own axis by Foucault's pendulum ?
31. Discuss some geographical effects of Coriolis force due to rotation of the earth.

PROBLEMS

[Conservative Forces]

1. (i) Show that the force $\vec{F} = (2xy + yz^2)\hat{i} + (x^2 + xz^2)\hat{j} + xyz\hat{k}$ is conservative.
(ii) Show that the force $\vec{F} = yz\hat{i} + zx\hat{j} + xy\hat{k}$ is conservative.

(iii) The position of a moving particle at any instant is given by $\vec{r} = A \cos \theta \hat{i} + A \sin \theta \hat{j}$. Show that the force acting on it is a conservative one. Also calculate the total energy of the particle.

(iv) If $\vec{F} = \cos y \hat{i} - x \sin y \hat{j} - \cos z \hat{k}$, prove that the force-field is conservative.

2. You are given a force $\vec{F} = (y^2 - x^2) \hat{i} + 3xy \hat{j}$. Compute the line integral from the point (0, 0) to the point (x_0, y_0) along the path made up of two straight sections (0, 0) to $(x_0, 0)$ and $(x_0, 0)$ to (x_0, y_0) . Compare with the result you get by taking other two sides of the rectangle as the path of integration. Is the force conservative ?

[Ans. Non-conservative.]

3. Yukawa potential energy is given by $U = -\frac{r_0}{r} U_0 e^{-r/r_0}$. What is the force corresponding to this energy ? Is it a conservative force ?

[Ans. $\vec{F} = -\frac{r_0 U_0}{r^3} \left(1 + \frac{r}{r_0}\right) e^{-r/r_0} \hat{r}$; Yes.]

[Gravitational Potential and Field]

4. Two lead spheres of 20 cm and 2 cm in diameter are placed with their centres 100 cm apart. Calculate the force of attraction between the spheres. Given the radius of the earth as 6.37×10^6 m and its mean density $5,530 \text{ kg/m}^3$ (sp. gravity of lead = 11.5). (Meerut 1968)

[Ans. 16×10^{-11} newton.]

5. The force of gravitational attraction between two masses m_1 and m_2 at a distance r is given by $\vec{F} = -\frac{Gm_1 m_2}{r^2} \hat{r}$. What is the potential energy at this separation. Calculate the necessary work done if the distance between them is increased from $r = r_0$ to $r = r_0 + l$.

[Ans. $-\frac{GMm}{r}$; $W = \frac{Gm_1 m_2 l}{r_0(r_0 + l)}$.]

[Hint : $W = -\frac{Gm_1 m_2}{r_0 + l} + \frac{Gm_1 m_2}{r_0}$.]

6. Two masses of 1 kg and 4 kg are placed at a distance of 12 cm. A third mass of 1 kg is placed on the line in between the two masses so that it experiences no resultant force. What is the position ? Calculate the binding energy of the system and energy needed in removing the third mass alone.

[Ans. 4 cm from 1 kg mass; 7.2×10^{-11} joules ; 5×10^{-11} joules.]

7. Calculate the energy required to bring a body of a mass 1 kg from the centre of earth to its surface. Given : $G = 6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$, $R = 6.4 \times 10^6$ m, $\rho = 5.5 \times 10^3 \text{ kg/m}^3$.

[Ans. 2×10^7 J]

8. What is the potential energy of a mass of 1 kg at a distance of 10^6 km from the centre of earth ? (zero potential energy at ∞).

[Ans. -3.98×10^5 J]

9. Calculate the amount of work done to move 1 kg mass from the surface of the earth to a point 10^5 km from the centre of the earth.

[Ans. 5.83×10^7 joules]

10. If for the case of the moon $R_m = 1.7 \times 10^6$ m and $M_m = 7.3 \times 10^{22}$ kg, find (i) the gravitational acceleration at the surface of the moon, and (ii) the escape velocity from the moon.

[Ans. (i) 1.70 m/sec^2 ; (ii) $2.4 \times 10^3 \text{ m/sec}$.]

312 | *Mechanics and Waves and Oscillations*

11. Deduce an expression for the gravitational potential due to a sphere at an external point. Hence calculate the amount of work required to send a body of mass m from earth's surface to a height (i) $R/2$, (ii) $10R$, and (iii) $1000R$, where R is the radius of the earth. Express the result in m , R and g .
[Ans. (i) $\frac{1}{3} gRm$; (ii) $(10/11)gRm$; (iii) $(1000/1001)gRm$]
12. A mass m is raised to a height nR , where R is the radius of the earth and n is an integer. Calculate the change in its potential energy.
[Ans. $\frac{n}{n+1} mgR$]
13. With what velocity should a body of mass m be projected vertically upwards so that it may be able to reach a height nR above earth's surface ?
[Ans. $\sqrt{\frac{2n}{n+1}} gR$]
14. (a) A meteorite is initially at rest at a distance equal to six times the radius of the earth from the centre of the earth. Calculate the speed with which it strikes with the earth.
[Ans. 10.2 km/sec]
(b) What should be the velocity given to a body to attain the height of one-half the radius of earth ?
[Ans. 6.47 km/sec]
15. If a body is to be projected vertically upwards from earth's surface to reach a height of $10R$, how much velocity should be given ?
[Ans. 10.6×10^3 m/s]
16. A rocket is fixed with 90% of escape velocity. Find the maximum height reached by the rocket.
[Ans. $h = 4.2R$, R = radius of earth.]
17. Compare the gravitational potentials at the centre and at the surface of a solid sphere.
[Ans. 3 : 2]
18. Compare the gravitational potentials on the surfaces of two planets of same density, but of diameters in the ratio 1 : 2.
[Ans. 1 : 4]
19. A uniform spherical shell is of mass 5 kg and radius 10 cm. Calculate the gravitational potential at the surface and intensity of gravitational field at the centre of shell.
[Ans. 3.33×10^{-9} J/kg; Zero]
20. The weight of a man on earth's surface is 60 kg. If the radius of earth is doubled, how much would he weigh (i) if the mass of the earth remains constant, (ii) if the average density of the earth remains constant ?
[Ans. (i) 15 kg; (ii) 120 kg]
21. A man can jump vertically 1.5 metres on the earth's surface. Calculate the maximum radius of a sphere of the same mean density as the earth from whose gravitational field he could escape by jumping. (radius of earth = 6.41×10^6 m).
[Ans. 3.103 km]
22. Prove that the least velocity with which a particle must be projected from the surface of a planet radius R and mean density ρ , in order that it may escape completely is $v = R\sqrt{8\pi\rho G/3}$.

[Reduced Mass]

23. Two masses 10 gm and 90 gm are connected by a weightless rod of length 10 cm. Calculate the reduced mass of the system.
[Ans. 0.09 kg]

24. If m is the mass of hydrogen atom, find the reduced mass of H_2 , HD and D_2 , where D stands for deuterium with mass $2m$.

[Ans. $m/2$, $2m/3$, m].

25. The distance between two protons of hydrogen molecule is 0.7×10^{-8} cm. Calculate the moment of inertia of the hydrogen molecule about an axis passing through the mid point on the internuclear axis and perpendicular to its length. (Mass of proton = 1.6×10^{-27} kg-m².)

[Ans. 3.92×10^{-48} kg-m²]

26. Initially a small mass m and relatively heavy mass M are at infinite separation. If mass M is considered to be fixed, what will be the velocity of m relative to M ? If both bodies are allowed to move freely, what will be the relative velocity?

[Ans. $\sqrt{\frac{2GM}{d}}$; $\sqrt{\frac{2G(M+m)}{d}}$]

[Hint : In the first condition, $\frac{1}{2}mv^2 = \frac{2GMm}{d}$ and in the second condition $\frac{1}{2}\mu v^2 = \frac{GMm}{d}$.]

[Kepler's Laws]

27. The periodic time of venus is 224.7 days and that of earth is 365.26 days. Find the ratio of the major axes of the orbits of venus and earth. (A.N.U. 1989)

[Ans. 0.7231]

28. The maximum and minimum velocities of a satellite are v_{max} and v_{min} respectively. Prove that the eccentricity of the orbit of the satellite is

$$e = \frac{v_{max} - v_{min}}{v_{max} + v_{min}}.$$

29. The eccentricity of the earth's orbit is 0.0125. Calculate the ratio of the maximum and minimum speeds of the earth.

[Ans. 1.025]

30. A satellite has its largest and smallest orbital speeds v_{max} and v_{min} respectively. If the satellite has a period T , then show that it moves in an elliptical orbit having major axis with length $\frac{T}{2\pi} \sqrt{v_{max}v_{min}}$.

31. A satellite revolves round a planet in elliptical orbit. Its maximum and minimum distances from the planet are 1.5×10^7 m and 0.5×10^7 m respectively. If the speed at the satellite at the farthest point be 5×10^3 m/s, calculate the speed at nearest point. (A.N.U. 1996)

[Ans. 1.5×10^4 m/sec]

32. The maximum and minimum distances of comet from the Sun are 1.5×10^{12} m and 8×10^{10} m respectively. If the speed of the comet at the farthest point is 3200 m/sec., find its speed at the nearest point. (O.U. 1990)

[Ans. 6×10^4 m/sec]

33. If the mass of the Sun is assumed to be (i) infinite, (ii) finite, find the equation of motion of the planet and the period of revolution in both the conditions. Suppose that the planet of mass m is moving at a distance r from the Sun of mass M .

[Ans. (i) $\ddot{\mathbf{r}} = -\frac{GM}{r^2} \hat{\mathbf{r}}$, $T = 2\pi\sqrt{\frac{r^3}{GM}}$; (ii) $\ddot{\mathbf{r}} = -\frac{G(M+m)}{r^2} \hat{\mathbf{r}}$, $T = 2\pi\sqrt{\frac{r^3}{G(M+m)}}$]

34. A planet of mass m is revolving around Sun in an elliptical orbit. Its minimum and maximum distances from the Sun are r_1 and r_2 respectively. Show that the period of revolution is

$$T = \pi \sqrt{\frac{(r_1 + r_2)^3}{2gR^2}}, \text{ where } g \text{ is acceleration due to gravity.}$$

314 | Mechanics and Waves and Oscillations

35. A particle of mass m describes an elliptical orbit about a centre of attractive force at one of its focus given by $-k/r^2$, where k is a constant. Show that the speed v of the particle at any point of the orbit is given by

$$v^2 = \frac{k}{m} \left[\frac{2}{r} - \frac{1}{a} \right], \text{ where } a \text{ is the semi-major axis.}$$

$$[\text{Hint : } E = T \text{ (K.E.)} + V = \frac{1}{2}mv^2 - \frac{k}{r}]$$

36. If a planet were suddenly stopped in its orbit, supposed to be circular, prove that it will fall into the Sun in a time which is $\sqrt{2}/8$ times of its period.

37. A particle of mass m moves in a central force field defined by $F = -\frac{k}{r^4}$. Show that if E is the total energy supplied to the particle, then its speed is given by

$$v = \sqrt{\frac{k}{mr^2} + \frac{2E}{m}}.$$

38. If the orbit described under a central force be given by $r = a(1 + \cos \theta)$ with centre at the origin, find the law of force.

$$[\text{Ans. } f(r) \propto 1/r^4]$$

39. A particle describes a conic $r = \frac{p}{1 + e \cos \theta}$, where p and e involve constant quantities. Show that the force under which particle is moving is a central force. Deduce the force law.

$$[\text{Ans. } F = -\frac{k}{r^2}]$$

40. A double star consists of two stars of masses m and $2m$ at a distance d apart. They rotate about their centre of mass. Find the period of revolution.

$$[\text{Ans. } 2\pi d \sqrt{\frac{d}{3Gm}}]$$

41. A double star is moving about its centre of mass under gravitational force. If the total mass of this star is M and period of revolution T , then show that the distance d between them is given by

$$d = \left[\frac{GM}{(2\pi/T)^2} \right]^{1/3}.$$

[Non-inertial Frames and Satellite]

42. What is the fictitious force and total force acting on a freely falling body of 3 kg mass with reference to a frame moving with a downward acceleration of 4 m/sec² ?

$$[\text{Ans. 12 newtons upward; 17.4 newtons downward}]$$

43. A rocket is moving upward with acceleration of 3g. Calculate the effective weight of a man sitting in it, if his actual weight is 75 kg.

$$[\text{Ans. 300 kg-wt}]$$

44. A 2 kg stone at the end of 0.5 metre long string makes 4 revolutions in 3 seconds. Calculate the forces on the stone as measured in an inertial frame and in a frame which is rotating with the string.

$$[\text{Ans. } F_i = -70.1 \text{ N; Total force } F_n = 0, F_0 = 70.1 \text{ N}]$$

45. Find the altitude of geo-stationary satellite of radius $R = 6400$ km.

(K.U. 2005)

$$[\text{Ans. } 3.56 \times 10^7 \text{ m}]$$

46. The first artificial satellite was circling round the earth at a distance of 900 km. Taking the radius

of the earth equal to 6371 km and mass 5.983×10^{24} kg and G equal to 6.61×10^{-11} S.I. units, find the velocity of satellite and its period of revolution.

[Ans. Velocity = 7.40 km/sec.; Time period = 1 hr 42 min 52 sec]

47. A satellite moves in a circular orbit round the earth at height $R_e/2$ from earth's surface where R_e is the radius of the earth. Calculate its period of revolution. ($R_e = 6.38 \times 10^6$ m)

[Ans. 9.3×10^3 sec]

48. A small satellite revolves round a planet in an orbit just above the planet's surface. Taking mean density of the planet as 8000 kg/m^3 and G as 6.66×10^{-11} S.I. units, find the time period of satellite.

[Ans. 4206 sec]

49. An astronaut, in an artificial satellite orbiting the earth, is in a state of weightlessness. Explain why ? If the satellite is orbiting the earth at an altitude of 100 km, find its period of revolution. Assume the orbit to be circular. (Mass of the earth = 5.98×10^{24} kg; mean radius of the earth = 6371 km; $G = 6.668 \times 10^{-24}$ S.I. units). (I.A.S.)

[Ans. 5,177 sec]

50. An object fixed with respect to the surface of a planet identical in mass and radius of the earth experiences zero gravitational attraction at the equator. What is the length of a day on the planet ? ($g = 9.8 \text{ m/sec}^2$, $R = 6.4 \times 10^6$ m)

[Ans. 84.5 minutes]

51. An artificial satellite is moving in a circular orbit around the earth with a speed equal to half the magnitude of escape velocity from the earth. Determine the height of the satellite above the earth's surface. If the satellite stopped suddenly in its orbit and allowed to fall freely onto the earth, find the speed with which it hits the surface of the earth.

[Ans. $h = R_e$; $v = \sqrt{GM/R_e}$]

52. A satellite of mass M_s is orbiting the earth in a circular orbit of radius R . It starts losing energy slowly at a constant rate C due to friction. If M_e and R_e denote the mass and radius of the earth respectively, show that the satellite falls on the earth at time

$$t = \frac{GM_s M_e}{2C} \left(\frac{1}{R_e} - \frac{1}{R} \right).$$

53. A satellite of mass m revolves around the earth in a circular orbit of radius r . Obtain expressions for the orbital velocity and time period. (O.U. 2002)

[Ans. $v = R\sqrt{\frac{g}{r}}$, $T = \frac{2\pi r}{R}\sqrt{\frac{r}{g}}$ (R = radius of earth)]

[Coriolis Force]

54. Find the horizontal component of Coriolis force acting on a body of mass 0.1 kg moving northward with a horizontal velocity of 100 m/sec at 30° N latitude on the earth.

[Ans. 7.2×10^{-4} newtons towards east]

55. A person in a jet plane is flying along the equator due east with a speed of 450 m/sec. What is his Coriolis acceleration ?

[Ans. $6.56 \times 10^{-3} \text{ m/sec}^2$]

56. A stone is dropped with zero initial velocity from the top of a 100 m high tower at the equator. Calculate horizontal displacement of the stone due to earth's rotation.

(Take $g = 10 \text{ m/sec}^2$ and $\omega = 7.2 \times 10^{-5} \text{ rad./sec}$)

[Ans. 2.15 cm]

57. Calculate the eastward deviation of a stone falling freely from a height 50 metres above ground at latitudes (i) 60° N, (ii) 45° N and (iii) at equator. ($\omega = 7.3 \times 10^{-5} \text{ rad./sec}$, $g = 9.8 \text{ m/sec}^2$)

[Ans. (i) 0.388 cm; (ii) 0.549 cm; (iii) 0.776 cm]

58. 'A' suggests that if a stone is thrown vertically upwards (from a plane in the northern hemisphere) so as to attain a maximum height ' h ' on its return to the earth, it will have no horizontal displacement. He argues that during the return journey the stone experiences exactly equal and opposite Coriolis force as it does during the onwards journey and, therefore, the net displacement on the ground will be zero. 'B' disagrees to this view. What is your answer to this riddle? Support your answer with proper derivation.

[Ans. The net displacement will be double of the displacement at maximum height towards west.]

59. A body is thrown vertically upward with a velocity u . Prove that it will fall back on a point displaced to the west by a distance equal to

$$\frac{4}{3} \left(\frac{8h^3}{g} \right)^{1/2} \omega \cos \phi,$$

where ϕ is the latitude and $h = u^2/2g$.

60. A body of mass 10 gm is at rest in an inertial frame. Describe its motion in a frame rotating with an angular speed of 10 rad/sec. The particle is situated at a distance of 5 cm from the axis of rotation. Determine the Coriolis and centrifugal forces.

[Ans. Centrifugal force = 0.05 N directed outwardly; Coriolis force = 0.1 N directed inwardly. The addition of the two fictitious forces is 0.05 N inwardly, i.e., this explains the observed circular motion and it seems that a force (centripetal) 0.05 N is acting towards the axis of rotation.]

61. What is the fictitious acceleration of the Sun relative to a frame rotating with the earth. (Distance between the Sun and the earth = 1.5×10^{11} m)

[Ans. 7.8×10^2 m/sec² towards earth]

62. The acceleration due to gravity measured in an earth bound co-ordinate system is represented by g . Due to earth's rotation, g differs from the true acceleration due to gravity g_0 . Assuming that the earth is a sphere of radius R and it is rotating with angular speed ω , find g as a function of latitude ϕ .

[Ans. $g = g_0 [1 - (2x - x^2) \cos^2 \lambda]^{1/2}$, where $x = \omega^2 R/g_0$]

[Hint : In the rotating frame with the earth the total fictitious acceleration on the Sun

$$= -2\vec{\omega} \times \vec{v}_r - \vec{\omega} \times (\vec{\omega} \times \vec{R}) = 2\vec{\omega} \times (\vec{\omega} \times \vec{R}) - \vec{\omega} \times (\vec{\omega} \times \vec{R}) = \vec{\omega} \times (\vec{\omega} \times \vec{R}) = -\omega^2 \vec{R}.]$$

63. In how much time will the plane of oscillation of a Foucault's pendulum turn through 180° at 30° latitude?

[Ans. 24 hours]

64. A Foucault's pendulum is oscillating at any time in the east-west direction. Calculate the time in which its plane of oscillation will start pointing 18° north of east, if the latitude of the place be 45° .

[Ans. 32.2 hours]



6

Special Theory of Relativity

6.1. Frames of Reference – Inertial Frames

In order to describe the motion of a body we must know with respect to what this motion has been measured. If we attach a coordinate system to a rigid body and describe the motion of a particle relative to it, then such a coordinate system is called **frame of reference**. The simplest frame of reference is a cartesian coordinate system XYZ with origin O , as shown in **fig. 6.1**. For the location of a particle, the position vector is drawn from the origin O of the coordinate system. In order to know the position of the particle at different times, an observer O may be stationed at the origin with a clock to measure the time t . **The position and time recorded together constitute an event, represented by four coordinates (x, y, z, t) .** For example, if a particle passes through the point P in space with coordinates x, y, z at instant of time t , then the coordinates (x, y, z, t) represent the concerned event of occurrence at P .

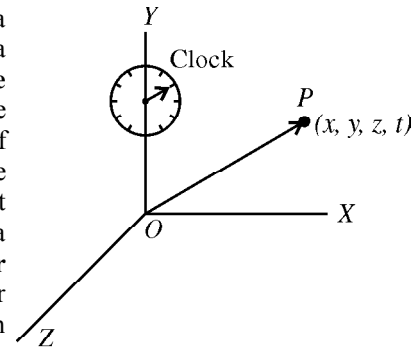


Fig. 6.1. Frame of reference.

Inertial frames of reference : *Those unaccelerated frames of reference, in which Newton's first and second laws hold, are called inertial frames.* In such frames, if a body is not acted by external force, it continues in its state of rest or uniform translatory motion (*law of inertia*) and hence they are called **inertial frames**. Thus, in an inertial frame, a body not experiencing any force ($\vec{F} = 0$) appears unaccelerated ($\vec{a} = 0$) because from Newton's second law

$$\vec{F} = m \vec{a} = 0 \text{ or } \vec{a} = \frac{d^2 \vec{r}}{dt^2} = 0 \quad \dots(1)$$

The law of inertia (Newton's first law of motion) was first stated by Galileo, therefore the inertial frames are also named as *Galilean frames of reference*. Further all those frames, which are moving with constant velocity relative to an inertial frame are also inertial.

6.2. Galilean Transformations

At any instant, the coordinates of a point or particle in space will be different in different coordinate systems. The equations which provide the relationship between the coordinates of a point in two reference systems are called **transformation equations**.

Consider two inertial frames S and S' . The frame S' is moving with constant velocity $\vec{v}$ relative to the frame S [fig. 6.2].

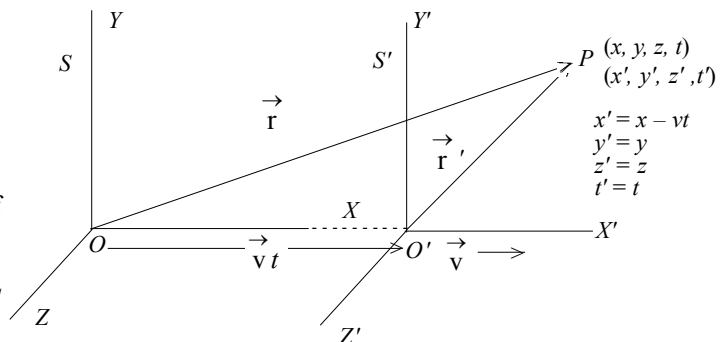


Fig. 6.2 : Representation of Galilean transformations.

For convenience, if we assume (i) that the origins of the two frames coincide at $t = 0$, (ii) that the coordinate axes of the second frame are parallel to that of the first and (iii) that the velocity of the second frame relative to the first is v along X -axis, then the position vectors of a particle at any instant t in the two frames are related by the equation

$$\vec{r}' = \vec{r} - \vec{v}t \quad \dots(1)$$

In the component form, the coordinates are related by the equations

$$x' = x - vt; \quad y' = y; \quad z' = z \quad \dots(2)$$

Eqs. (1) or (2) expresses the transformation of coordinates from one inertial frame to another and they are referred as **Galilean transformations**.

The form of eq. (1) or (2) depends, of course, on the relative motion of two frames of reference, but it also depends implicitly upon certain assumptions regarding the nature of time and space. In the first place, it has been assumed that the time t is independent of any particular frame of reference i.e., if t and t' be the times recorded by the observers O and O' of an event occurring at P , then $t' = t$. If we add the equation $t' = t$, then the **Galilean transformation equations** are expressed as

$$x' = x - vt; \quad y' = y; \quad z' = z; \quad t' = t \quad \dots(3)^*$$

The assumption $t' = t$ implies that the time interval $\Delta t = t_2 - t_1$ between two events as measured in S -frame is equal to the time interval $\Delta t' = t'_2 - t'_1$ of the same events as observed in S' -frame i.e.,

$$t'_2 - t'_1 = t_2 - t_1 \text{ or } \Delta t' = \Delta t \quad \dots(4)$$

We can also consider that frame S is moving with velocity $-v$ along the negative X -axis with respect to S' frame. Then the transformation equations from frame S' to S are

$$x = x' + vt'; \quad y = y'; \quad z = z'; \quad t = t' \quad \dots(5)$$

These are known as **inverse Galilean transformations**.

The other assumption, regarding the nature of the space, is that the distance between two points (or two particles) is independent of any particular frame of reference. Evidently, this assumption is expressed by the form of the transformation eq. (1) or (3). If a rod has length L in the frame S with the end coordinates x_1 and x_2 , then $L = x_2 - x_1$.

If at the same time the end coordinates of the rod in S' are x'_1 and x'_2 , then $L' = x'_2 - x'_1$. But for any time t , from eq. (3), we have

$$x'_2 - x'_1 = x_2 - x_1$$

Therefore, $L' = L \quad \dots(6)$

Thus the length or distance between two points is invariant under Galilean transformations.

Differentiating eq.(1) with respect to time, we get

$$\frac{d\vec{r}}{dt} = \vec{v} + \frac{d\vec{r}'}{dt} = \vec{v} + \frac{d\vec{r}'}{dt'} \quad [\because t = t']$$

$$\text{or} \quad \vec{u} = \vec{v} + \vec{u}' \quad \dots(7)$$

where $\vec{u}$ and $\vec{u}'$ are the observed velocities in S and S' frames respectively.

Eq. (7) transforms the velocity of a particle from one frame to another and is known as **Galilean (or classical) law of addition of velocities**.

Again differentiating eq. (7) with respect to time t , we have

* To this set by equations, the name 'Galilean Transformations' was given by Albert Einstein to distinguish it from the Lorentz Transformations.

$$\frac{d\vec{u}}{dt} = 0 + \frac{d\vec{u}'}{dt} = \frac{d\vec{u}'}{dt'} \quad [\because t = t']$$

$$\text{or} \quad \vec{a} = \vec{a'} \quad \dots(8)$$

Hence according to Galilean transformations, the accelerations of a particle relative to S and S' frames are equal.

This is to be remembered that the Galilean transformations are based basically on two assumptions :

- (1) ***There exists an universal time t which is the same in all reference systems.***
- (2) ***The distance between two points in various inertial systems is the same.***

Thus if any two events occur simultaneously for any observer, then they must occur simultaneously for all observers. In other words, *the time interval between two given events must be identical for all systems of reference.*

6.3. Principle of Relativity—Hypothesis of Galilean Invariance

Absolute velocity of a body has no meaning. The velocity has a meaning only when it is measured relative to some other body or frame of reference. If two bodies are moving with uniform relative velocity, it is impossible to decide which of them is at rest or which of them is moving. This is known as ***principle of relativity***. However, acceleration has an absolute meaning. For example, if we are sitting in a windowless accelerated aircraft, we can perform an experiment and measure its acceleration. But if the aircraft is moving with uniform velocity, we cannot measure its velocity. Of course, we measure its velocity relative to a body outside. Thus the principle of relativity can be alternatively stated as follows :

It is impossible to perform an experiment which will measure the state of uniform velocity of a system by observations, confined to that system.

The motion of a body itself has no meaning unless, we do not know with respect to which this motion has been measured. This led Newton to think about the absolute space and it represents an absolute frame with respect to which every motion should be measured. However, in view of this principle of relativity, we can not perform an experiment which will measure the uniform velocity of a reference system relative to the absolute frame by observations confined to that system.

In the unaccelerated windowless ship all experiments performed inside it will appear the same whether this ship is stationary or in uniform motion. Newton stated the principle of relativity as follows :

The motions of bodies included in a given space are the same among themselves whether that space is at rest or moving uniformly forward in a straight line.

Study of the physical laws involves the measurements of accelerations, forces etc among bodies. The principle of relativity can be stated in an elegant form as follows :

The basic laws of physics are identical in all inertial systems which move with uniform velocity with respect to one another.

This principle is called ***Galilean or Newtonian principle of relativity*** and sometimes it is named as ***hypothesis of Galilean invariance***. In fact, the principle of relativity is a fundamental postulate and is entirely consistent with the theory of special relativity. If any two inertial systems, moving with constant relative velocity, are connected by Galilean transformations, the principle of relativity is modified as :

The basic laws of physics are invariant in form in two reference systems, connected by Galilean transformations.

This statement is somewhat special than the principle of relativity in the sense that it means the assumptions that the time and the space intervals are independent of the frame of reference. We shall see later in the theory of special relativity that the Galilean transformations are not correct, but the appropriate exact transformation equations are the Lorentz transformation equations for connecting any two frames in uniform relative

motion. Thus, the principle of relativity may be stated as :

The basic laws of physics are invariant in form in two inertial frames, connected by Lorentz transformations.

6.4. Deduction of the Law of Conservation of Linear Momentum Using Galilean Invariance

If a body of mass m is moving with a velocity $\vec{v}$, its momentum $\vec{p}$ is given by the equation $\vec{p} = m \vec{v}$.

Suppose that two particles of masses m_1 and m_2 collide with each other. If the velocities of these particles before the collision are $\vec{u}_1, \vec{u}_2$ and after the collision are $\vec{v}_1, \vec{v}_2$ respectively, then according to the law of conservation of linear momentum we have

$$m_1 \vec{u}_1 + m_2 \vec{u}_2 = m_1 \vec{v}_1 + m_2 \vec{v}_2 \quad \dots(1)$$

The collision may be of two types—*elastic* and *inelastic*. In an elastic collision all the kinetic energy of the incoming particles reappears after the collision as the kinetic energy of the same particles. In the usual inelastic collision, part of the kinetic energy of incoming particles appears after the collision as some form of internal excitation energy (such as heat) of one or more particles and rest part appears in the form of their kinetic energy. It is important to note that the law of conservation of momentum is applicable even in an inelastic collision.

Here, we will establish the law of conservation of linear momentum, based on the assumptions of Galilean invariance and conservation of energy.

Suppose the initial and the final velocities of the two colliding particles are $\vec{u}_1, \vec{u}_2$ and $\vec{v}_1, \vec{v}_2$ respectively in a frame S . In general, if the collision is inelastic, then according to the law of conservation of energy in the frame S , we have

$$\frac{1}{2} m_1 u_1^2 + \frac{1}{2} m_2 u_2^2 = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2 + E \quad \dots(2)$$

where E is the part of kinetic energy of incoming particles is lost in the collision in the form of heat, light etc. Here, we have assumed that the masses m_1, m_2 are not altered in the collision.

If another frame S' is moving with constant velocity $\vec{v}$ relative to S , then according to the Galilean transformation the velocities in the frame S' are

$$\vec{u}'_1 = \vec{u}_1 - \vec{v}, \vec{u}'_2 = \vec{u}_2 - \vec{v}; \vec{v}'_1 = \vec{v}_1 - \vec{v}, \vec{v}'_2 = \vec{v}_2 - \vec{v} \quad \dots(3)$$

Now, according to the hypothesis of Galilean invariance, if the law of conservation of energy holds in the frame S , it should also hold in the frame S' , moving with a constant velocity relative to S . Therefore, in frame S' the law of conservation of energy tells us that

$$\frac{1}{2} m_1 u_1'^2 + \frac{1}{2} m_2 u_2'^2 = \frac{1}{2} m_1 v_1'^2 + \frac{1}{2} m_2 v_2'^2 + E^* \quad \dots(4)$$

If the law of conservation of energy is to be invariant under Galilean transformations, then substituting the values of $\vec{u}'_1, \vec{u}'_2, \vec{v}'_1, \vec{v}'_2$ from eq. (3) in eq. (4) we have

$$\frac{1}{2} m_1 (\vec{u}_1 - \vec{v})^2 + \frac{1}{2} m_2 (\vec{u}_2 - \vec{v})^2 = \frac{1}{2} m_1 (\vec{v}_1 - \vec{v})^2 + \frac{1}{2} m_2 (\vec{v}_2 - \vec{v})^2 + E$$

* We have assumed that the lost energy E is the same in both frames. This is in agreement with experiment.

But $(\vec{u}_1 - \vec{v})^2 = (\vec{u}_1 - \vec{v}) \cdot (\vec{u}_1 - \vec{v}) = u_1^2 + v^2 - 2\vec{u}_1 \cdot \vec{v}$ etc.

$$\therefore \frac{1}{2} m_1 (u_1^2 + v^2 - 2\vec{u}_1 \cdot \vec{v}) + \frac{1}{2} m_2 (u_2^2 + v^2 - 2\vec{u}_2 \cdot \vec{v}) = \frac{1}{2} m_1 (v_1^2 + v^2 - 2\vec{v}_1 \cdot \vec{v}) + \frac{1}{2} m_2 (v_2^2 + v^2 - 2\vec{v}_2 \cdot \vec{v}) + E$$

$$\text{or } \frac{1}{2} m_1 u_1^2 - m_1 (\vec{u}_1 \cdot \vec{v}) + \frac{1}{2} m_2 u_2^2 - m_2 (\vec{u}_2 \cdot \vec{v}) = \frac{1}{2} m_1 v_1^2 - m_1 (\vec{v}_1 \cdot \vec{v}) + \frac{1}{2} m_2 v_2^2 - m_2 (\vec{v}_2 \cdot \vec{v}) + E \quad \dots(5)$$

Subtracting eq. (2) from eq. (5), we have

$$m_1 \vec{u}_1 \cdot \vec{v} + m_2 \vec{u}_2 \cdot \vec{v} = m_1 \vec{v}_1 \cdot \vec{v} + m_2 \vec{v}_2 \cdot \vec{v}$$

$$\text{or } (m_1 \vec{u}_1 + m_2 \vec{u}_2) \cdot \vec{v} = (m_1 \vec{v}_1 + m_2 \vec{v}_2) \cdot \vec{v}$$

$$\text{Thus, } m_1 \vec{u}_1 + m_2 \vec{u}_2 = m_1 \vec{v}_1 + m_2 \vec{v}_2 \quad \dots(6)$$

This is the law of conservation of linear momentum.

6.5. Transformation of Force from One Inertial System to Another

Suppose that the force $\vec{F}$ on a particle of mass m in the frame S is represented by Newton's second law

$$\vec{F} = m \vec{a} \quad \dots(1)$$

But according to the postulate that the laws of physics are the same in the frame S and in another frame S' , which is in uniform motion relative to S , we have

$$\vec{F}' = m' \vec{a}' \quad \dots(2)$$

in the frame S' .

We have shown in Art. 6.2 that the acceleration of the particle is the same in two inertial frames, connected by Galilean transformations, i.e.,

$$\vec{a}' = \vec{a} \quad \dots(3)$$

where in the deduction basically we have assumed the invariance of space and time separately.

In Newtonian mechanics, the mass is independent of velocity and hence

$$m = m' \quad \dots(4)$$

Thus,

$$\vec{F} = m \vec{a} = m' \vec{a}' = \vec{F}' \quad \dots(5)$$

This means that if the relation $\vec{F} = m \vec{a}$ (Newton's second law) is used to define the force, then in all inertial systems, the force $\vec{F}$ will have the same magnitude and direction, independent of the relative velocities

of the reference frames. This means that if $\vec{F} = m \vec{a}$ in S -frame, then $\vec{F}' = m' \vec{a}'$ with $m = m'$ in S' -frames and

$\vec{F} = \vec{F}'$. Thus Newton's second law is invariant under Galilean transformations. As Newton's first law ($\vec{F} = 0$) can be deduced from second law and third law involves forces. We may also say that *Newton's laws of motion are the same in all inertial frames or invariant under Galilean transformations.*

Since, in mechanics, the conservation principles *e.g.*, those for energy, linear momentum and angular momentum—all can be shown to be the consequence of Newton's laws of motion, it follows that *the laws of mechanics are the same in all inertial frames under Galilean transformations.*

6.6 Absolute Frame of Reference

According to principle of relativity, the basic laws of physics are the same in all inertial frames. We know that Newton's laws of motion (or laws of mechanics) are the same in all inertial frames under Galilean transformation. In the field of electrodynamics, Maxwell's equations play the same role as Newton's laws of motion in mechanics. If the principle of relativity is extended to electrodynamics, Maxwell's equations should remain the same in two inertial frames with uniform constant relative motion. According to Maxwell's electromagnetic theory, the speed of light in vacuum is $c (= 3 \times 10^8 \text{ m/sec})$ in all directions irrespective of the motion of the source or observer. Hence according to the principle of relativity, the speed of light must be the same in all inertial systems, independent of the motion of the source or observer.

Earlier Galilean transformation were considered to be the correct transformations in two inertial systems in constant relative motion. However, if Galilean transformations are used, the speed of light will depend on the speed of a moving observer. Let us consider an inertial frame (say S frame) in which the speed of light is c in all directions. If any observer (say S' frame) is moving with velocity v towards the source of light, then according to Galilean transformations he will measure the speed of light as $c + v$ and if he is moving away from the source of light with velocity v , he will measure the speed of light as $c - v$. This means that the moving observer towards or away the source, will measure the speed of light c_o , given by according to Galilean transformations

$$c_o = c \pm v \quad \dots(1)$$

Hence different observers moving with different uniform velocities will measure different speeds of light. Therefore an inertial frame of reference, in which the speed of light is c in all directions, must be a special or preferred frame of reference. Such a special frame reference is called **absolute frame of reference**. Any frame of reference which is in a state of motion relative to this absolute frame should be less suitable. If such an absolute frame is known, it seems possible that the **absolute motion** of the bodies may be measured. Michelson-Morley experiments were performed to detect the influence of the motion of the earth with respect to so called absolute frame. However negative results were obtained from these experiments and the idea of constancy of speed of light in vacuum was established. Further the idea of the existence of an absolute frame of reference was discarded. This also implies that the principle of relativity, according to which the speed of light is constant in all inertial frames, is correct and the Galilean transformations are incorrect. In fact, the correct transformation equations are Lorentz transformations which will be deduced later.

6.7. The Michelson-Morley Experiments

Michelson-Morley experiment was designed to determine the motion of the earth relative to a frame of reference in which the speed of light is c in all the directions. This frame is the privileged or absolute frame of reference. The earth itself cannot be the absolute frame of reference because it is moving around the sun in its orbit with a speed $v = 3 \times 10^4 \text{ m/sec}$ and therefore at any time if the earth is identified with the motion of the preferred frame, it will have a speed of $6 \times 10^4 \text{ m/sec}$ after 6 months relative to the absolute (inertial) frame of reference. In any case, the earth has a speed at least $v = 3 \times 10^4 \text{ m/sec}$ relative to an inertial frame at some instant or the other during 6 months of the year. Assuming that the absolute frame is situated at the centre of the sun and the earth is moving with a speed $3 \times 10^4 \text{ m/sec}$ relative to it. Therefore, an observer on earth will measure a speed $(c - v)$ for light moving along the direction of its motion and $(c + v)$ for light moving in opposite direction. We shall see how this assumption (*the change of velocity of light due to the motion of the observer*) was contradicted by the famous experiments of Michelson and Morley, conducted in the year 1880.

In principle, in the experiment of Michelson and Morley, a ray of light starts from a monochromatic source of light S and falls on a half-silvered glass plate P , where a part of light is reflected towards the mirror M_1 and part of it is transmitted towards the mirror M_2 [fig 6.3(a)]. PM_1 and PM_2 are mutually perpendicular directions and they are nearly equal. Let $PM_1 = PM_2 = l$. The two rays, reflected from the mirrors M_1 and M_2 unite again at P and interference fringes are obtained. These fringes may be seen through the telescope T . The entire

apparatus is floated on mercury, contained in a large vessel so that the interferometer may be rotated in any desired direction.

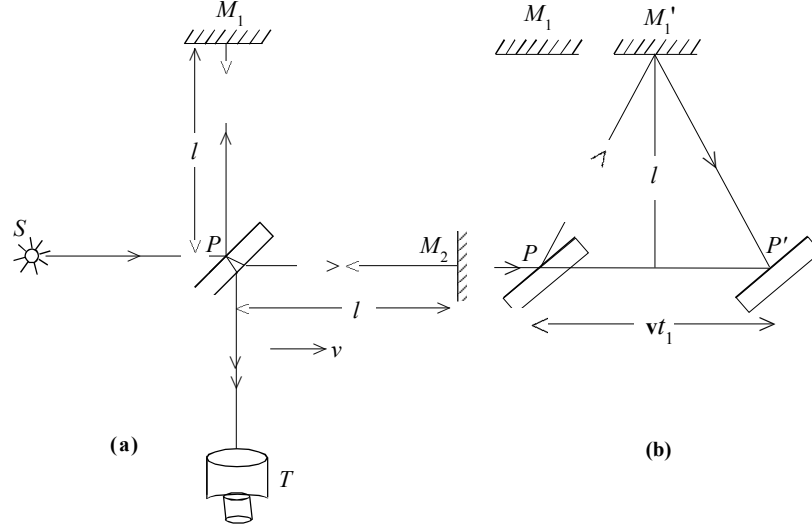


Fig. 6.3 : Michelson-Morley Experiment

In the experiment the instrument is first set up in such a way that arm PM_2 is parallel to the motion of the earth in space. Therefore, the apparatus moves in the direction PM_2 with velocity v relative to the absolute frame. After reflection at P , the ray moves towards the mirror M_1 . In the time, the ray reaches the mirror, it moves to the position M_1' so that the reflection occurs at this position M_1' . If t_1 is the time taken by the ray in traversing the path $PM_1'P'$, in this duration plate P will move a distance $PP' = vt_1$. Hence from **fig 6.3 (b)**, we have

$$PM_1' = \sqrt{PN^2 + M_1'N^2} = \sqrt{(vt_1/2)^2 + l^2}$$

Hence, distance $PM_1'P' = 2PM_1' = 2\sqrt{l^2 + v^2t_1^2/4}$.

This distance $PM_1'P'$ has been traversed by the ray in the absolute frame, in which the velocity of light is c in all the directions. Hence

$$ct_1 = 2\sqrt{l^2 + v^2t_1^2/4} \quad \text{or} \quad c^2t_1^2 = 4l^2 + v^2t_1^2 \quad \text{or} \quad t_1^2 = \frac{4l^2}{c^2 - v^2}$$

Thus

$$t_1 = \frac{2l}{\sqrt{c^2 - v^2}} = \frac{2l}{c(1 - v^2/c^2)^{1/2}} = \frac{2l}{c} \left(1 - \frac{v^2}{c^2}\right)^{-1/2}$$

$$= \frac{2l}{c} \left(1 + \frac{v^2}{2c^2}\right) \quad \left[\text{by Binomial theorem for } \frac{v^2}{c^2} \ll 1 \right] \dots (1)$$

The transmitted light ray at P is travelling with velocity $(c - v)$ relative to the mirror M_2 , because the mirror is moving with the velocity of the earth (v). This ray after reflection at M_2 will travel with a velocity $(c + v)$ relative to the plate P , because now the plate is moving opposite to the ray. If t_2 be the time taken by the transmitted ray to travel the distance l to the mirror M_2 and back, then

$$t_2 = \frac{l}{c-v} + \frac{l}{c+v} \quad (\text{where } PM_2 = l) = \frac{2lc}{c^2 - v^2} = \frac{2l}{c} \left(1 - \frac{v^2}{c^2}\right)^{-1}$$

$$= \frac{2l}{c} \left(1 + \frac{v^2}{c^2}\right) \quad [\text{using Binomial series for } v^2/c^2 \ll 1] \quad \dots(2)$$

Hence the difference in times taken to traverse by the two paths is

$$\Delta t = t_2 - t_1 = \frac{2l}{c} \left(1 + \frac{v^2}{c^2}\right) - \frac{2l}{c} \left(1 + \frac{v^2}{2c^2}\right) = \frac{2l}{c} \cdot \frac{v^2}{2c^2} = \frac{lv^2}{c^3} \quad \dots(3)$$

Therefore, the difference in the distance travelled by the two rays of light i.e.,

$$\text{the path difference} = c\Delta t = lv^2/c^2.$$

Finally, the whole apparatus is turned through 90° so that the other arm PM_1 becomes coincident with the earth's velocity (v) in space [fig. 6.4]. This causes the difference of path in the opposite direction and hence the displacement of fringes should correspond to the path difference $\Delta S = 2lv^2/c^2$. We know that if the path difference between two interfering rays changes by λ , there is a shift of one fringe on the screen. Hence the path difference $\Delta S = 2lv^2/c^2$ will correspond to the fringe shift, given by

$$\Delta N = \frac{\Delta S}{\lambda} = \frac{2lv^2}{c^2\lambda} \quad \dots(4)$$

In experiment, distance l was taken nearly 11 metres. Hence the expected fringe-shift on the screen for $\lambda = 5800 \text{ \AA}$ (yellow colour) will be

$$\Delta N = \frac{2lv^2}{c^2\lambda} = \frac{2 \times 11 \times (3 \times 10^4)^2}{(3 \times 10^8)^2 \times 5800 \times 10^{-10}} \approx 0.4$$

Thus the shift for yellow colour is expected to be 0.4 of a fringe-width. The expected shift of 0.4 of a fringe could be measured easily in the experiment. However, no shift of fringes was observed. The experiment was repeated several times, but such displacement was never seen.

Such results, which obtained against the expected results, were called as **null or negative results of Michelson-Murley experiments**.

One may argue that accidentally the absolute frame has the same velocity $3 \times 10^4 \text{ m/sec}$ with respect to sun as does the earth. Therefore, at this instant, the earth is at rest in the absolute reference system. However, Michelson and Morley repeated their experiment six months later and a four times magnified effect could be expected; but again nothing was observed [fig. 6.5]. The experiments since then have been repeated several times under different circumstances and always the same negative result was obtained.

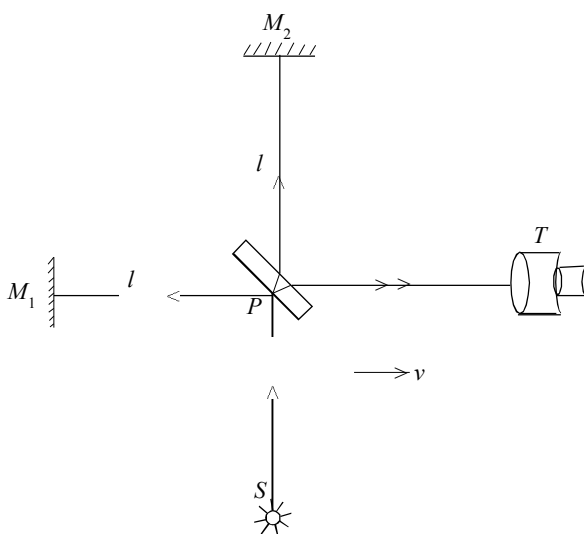


Fig. 6.4 : Michelson-Morley apparatus rotated by $\pi/2$.

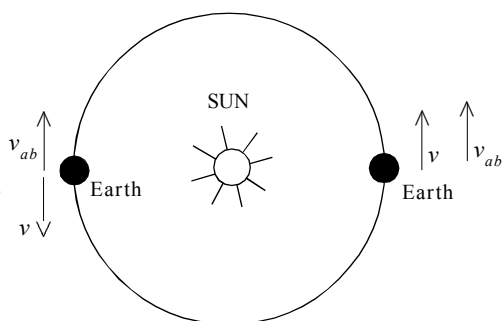


Fig. 6.5

The negative results of Michelson-Morley suggest that the value of v relative to the absolute frame should be zero. In other words, ***the speed of light in vacuum must be the same (c) in all inertial frames***. It does not depend upon the motion of the observer or source as expected on the basis of Galilean transformations. This is the **physical significance of the negative results of Michel-Morley experiments**. Further these negative results show the validity of the principle of relativity in the fields of electrodynamics and optics. In fact, *the principle of relativity is a fundamental truth applicable to all areas of physics*.

6.8. Shortcomings of Galilean Transformations

1. The laws of mechanics are invariant under Galilean transformations. However laws of electrodynamics (Maxwell's field equations) are not invariant under these transformations.

2. In an inertial frame, if the speed of light is c , then according to Galilean transformations, the speed of light will be measured as $c' = c \pm v$ in another inertial frame S' , which is moving towards or away from S with velocity v . However, speed of light c , being the same in all inertial frames, is an experimental fact.

3. In Galilean transformations, the informations could be transmitted instantaneously so that $t' = t$ in any inertial frame. However, there is finite speed of transmitting a information and this speed can not be more than speed of light.

In view of the above, the Galilean transformations need to be modified and the correct transformations are the Lorentz transformations. The later transformations will be deduced in the next article taking into consideration the two postulates of the Einstein's special theory of relativity.

Ex. 1. If the origin of two frames S and S' coincide at $t = t' = t_0$ and S' moves with a velocity $\vec{v}$ having its axis parallel to S , find the Galilean transformations.

Sol. Here the Galilean transformations are given by

$$\vec{r} = \vec{v} (t' - t_0) + \vec{r}'$$

The axes of S and S' are parallel, hence along the three orthogonal axes, the form of the Galilean transformations is

$$\begin{aligned} x &= v_x (t' - t_0) + x' & x' &= x - v_x (t - t_0) \\ y &= v_y (t' - t_0) + y' & y' &= y - v_y (t - t_0) \\ z &= v_z (t' - t_0) + z' & z' &= z - v_z (t - t_0) \end{aligned} \quad \text{or} \quad \begin{aligned} & & & \\ & & & \\ & & & \end{aligned}$$

$$\text{and} \quad t = t' \quad \text{and} \quad t' = t.$$

Ex. 2. A man can row a boat with a velocity V in still water. How much time does he take to cross a stream of width d and return to the same point if velocity of the stream is v ? How much time does he take a distance d upstream and return? Calculate in first order (v^2/V^2) the difference between the two times.

Sol. According to the problem, the man should row the boat perpendicular to the direction of flow so as to return at the same point. If the velocity V of the boat makes an angle θ with the direction of the flow, then it can be resolved into two components :

(i) $V \cos \theta$, and (ii) $V \sin \theta$.

Now, the component $V \cos \theta$ should balance the velocity of the stream i.e.,

$$V \cos \theta = v \quad \text{or} \quad \cos \theta = v/V.$$

If t be the time required in crossing and returning, then

$$t = \frac{d}{V \sin \theta} + \frac{d}{V \sin \theta} = \frac{2d}{V \sin \theta} = \frac{2d}{V \sqrt{1 - \cos^2 \theta}} = \frac{2d}{V \sqrt{1 - v^2/V^2}} = \frac{2d}{\sqrt{V^2 - v^2}}$$

In the second case, time required is

$$t' = \frac{d}{V-v} + \frac{d}{V+v} = \frac{2dV}{V^2 - v^2}$$

Now,

$$t = \frac{2d}{V(1 - v^2/V^2)^{1/2}} = \frac{2d}{V} \left(1 - \frac{v^2}{V^2}\right)^{-1/2} = \frac{2d}{V} \left(1 + \frac{v^2}{2V^2}\right)$$

and

$$t' = \frac{2dV}{V^2 - v^2} = \frac{2d}{V(1 - v^2/V^2)} = \frac{2d}{V} \left(1 - \frac{v^2}{V^2}\right)^{-1} = \frac{2d}{V} \left(1 + \frac{v^2}{V^2}\right)$$

where we have used Binomial expansion and retained the terms in first order in v^2/V^2 .

Hence,

$$t' - t = \frac{2d}{V} \frac{v^2}{2V^2} = \frac{dv^2}{V^3}.$$

Ex. 3. Calculate the expected fringe shifts in the Michelson-Morley experiment, if the distance of each plate is 2 m and the wavelength of monochromatic radiation is (a) 6000 Å and (b) 4000 Å. (A.U. 1990)

Sol. Expected fring-shift $\Delta N = \frac{2lv^2}{c^2\lambda}$

Here, $l = 2\text{m}$, $v = 3 \times 10^4 \text{ m/sec.}$, $c = 3 \times 10^8 \text{ m/sec.}$

(a) For $\lambda = 6000 \text{ Å} = 6000 \times 10^{-10} \text{ m}$,

$$\Delta N = \frac{2 \times 2 \times (3 \times 10^4)^2}{6000 \times 10^{-10} \times (3 \times 10^8)^2} = \mathbf{0.067 \text{ of a fringe.}}$$

(b) For $\lambda = 4000 \text{ Å} = 4000 \times 10^{-10} \text{ m}$,

$$\Delta N = \frac{2 \times 2 \times (3 \times 10^4)^2}{4000 \times 10^{-10} \times (3 \times 10^8)^2} = \mathbf{0.1 \text{ of a fringe.}}$$

Ex. 4. What will be the fringe shift in Michelson-Morley experiment if the effective length of each path is 6 m and light wavelength of 6000 Å is used. Earth's velocity is $3 \times 10^4 \text{ m-s}^{-1}$ and $c = 3 \times 10^8 \text{ m-s}^{-1}$. (A.N.U. 2002)

Sol. Fringe shift $\Delta N = \frac{2lv^2}{\lambda c^2} = \frac{2 \times 6 \times (3 \times 10^4)^2}{(6000 \times 10^{-10}) \times (3 \times 10^8)^2} = \mathbf{0.201 \text{ of a fringe.}}$

6.9. Postulates of Special Theory of Relativity

We have seen that an extension of the principle of relativity to the fields of electrodynamics and optics demands the constancy of speed of light c in vacuum in all inertial systems. This fact is verified by the negative results of Michelson-Morley experiments. In 1905, Einstein propounded the special theory of relativity which makes a conceptual change in the ideas of space and time. According to Newtonian mechanics, time and space are invariant for all observers, while both are variable at high speeds for various observers in special theory of relativity.

The two fundamental postulates of the special theory of relativity are the following :

(1) *All the laws of physics have the same form in all inertial systems, moving with constant velocity relative to one another.* This postulate is just the principle of relativity.

(2) *The speed of light is constant in vacuum in every inertial system.* This postulate is an experimental fact and asserts that the speed of light does not depend on the direction of propagation in vacuum and the relative velocity of the source and the observer. In fact, the second postulate is contained in the first because it predicts the speed of light c to be constant of nature.

The name special theory of relativity comes from the fact that this theory permits the independence of the physical laws of those coordinate systems which are moving with constant velocity relative to one another. Later, Einstein propounded his *general theory of relativity* which allows for the independence of the physical laws of all coordinate systems, having any general relative motion.

These two postulates of special theory of relativity look to be very simple, but they have revolutionised the physics with far reaching consequences. First we deduce transformation equations, connecting any two inertial systems moving with constant relative velocity. The transformation should be such that they are applicable to both Newtonian mechanics and electromagnetism. Such transformations were deduced by Einstein in 1905 and are known as Lorentz transformations because Lorentz deduced them first in his theory of electromagnetism.

6.10. Lorentz Transformations

Suppose that S and S' be the two inertial frames of reference. S' is moving along positive direction of X -axis with velocity v relative to the frame S [fig. 6.6]. Let t and t' be the times recorded in two frames. For our convenience, we will assume that the origins O and O' of the two co-ordinate systems coincide at $t = t' = 0$.

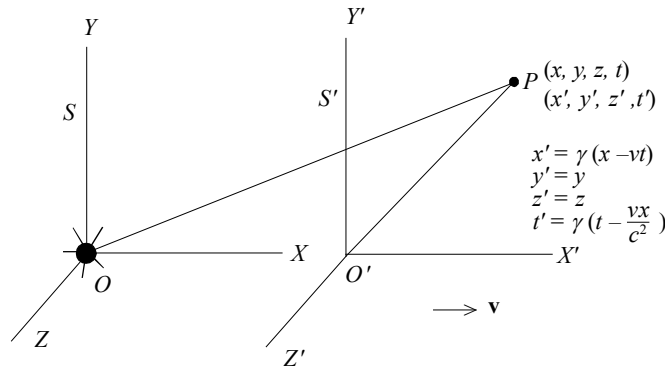


Fig. 6.6 : Representation of Lorentz transformations

Now suppose that a source of light is situated at the origin O in the frame S , from which a wavefront of light is emitted at $t = 0$. When the light reaches at the point P , let the positions and times, measured by the observers O and O' , be (x, y, z, t) and (x', y', z', t') respectively. If the velocity of light is c , then the time measured by the light signal in traversing the distance OP in frame S is

$$t = \frac{OP}{c} = \frac{(x^2 + y^2 + z^2)^{1/2}}{c} \quad \text{or} \quad x^2 + y^2 + z^2 = c^2 t^2. \quad \dots(1)$$

This equation represents the equation of wavefront in frame S . According to the special theory of relativity, the velocity of light will be c in the second frame S' . Hence, in frame S' , the time required by the light signal in travelling the distance $O'P$ is given by

$$t' = \frac{O'P}{c} = \frac{(x'^2 + y'^2 + z'^2)^{1/2}}{c} \quad \text{or} \quad x'^2 + y'^2 + z'^2 = c^2 t'^2 \quad \dots(2)$$

which is the equation of the wavefront in frame S' .

Now transformation equations relating x, y, z, t and x', y', z', t' should be such that eq. (2) transforms to eq. (1). The Galilean transformations connect the measurements in the two frames according to the following equations :

$$x' = x - vt; \quad y' = y; \quad z' = z; \quad t' = t$$

Substituting these values in eq. (2), we get

$$(x - vt)^2 + y^2 + z^2 - c^2 t^2 = 0$$

or $x^2 - 2xvt + v^2 t^2 + y^2 + z^2 - c^2 t^2 = 0$... (3)

This equation is certainly not in agreement with eq. (1) because it contains an extra term $(-2xvt + v^2 t^2)$. Thus the Galilean transformation fails. Further $t \neq t'$, (because $t = OP/c$ and $t' = O'P/c$) which does not agree with the Galilean transformation equations. If the principle of the constancy of the speed of light is valid in all frames, there should exist some transformation which reduces to the Galilean one for $v/c \rightarrow 0$ and which transform

$$x'^2 + y'^2 + z'^2 - c^2 t'^2 = 0 \text{ into } x^2 + y^2 + z^2 - c^2 t^2 = 0.$$

When we look at the eq. (1) and eq. (3), we find that the terms of y and z are in agreement. Hence we can say $y' = y$ and $z' = z$. The extra term $(-2xvt + v^2 t^2)$ indicates that transformations in x and t should be modified so that this extra term is cancelled.

We note that for the observer O , the distance $OO' = vt$ and therefore when $x' = 0$ (point O'), $x = vt$. This suggests the transformation $x' = \alpha(x - vt)$ because only then for $x' = 0$, $x = vt$. Since t' is different from t and may be depending on x , so that in general we may also assume that $t' = \alpha'(t + fx)$. Here α , α' and f are constants, to be determined (for Galilean transformations $\alpha = \alpha' = 1$ and $f = 0$). Now substitution for x' , y' , z' and t' in (2), we have

$$\alpha^2(x - vt)^2 + y^2 + z^2 = c^2 \alpha'^2(t + fx)^2$$

or $\alpha^2(x^2 - 2xvt + v^2 t^2) + y^2 + z^2 = c^2 \alpha'^2(t^2 + 2fxt + f^2 x^2)$

or $x^2(\alpha^2 - f^2 \alpha'^2 c^2) - 2xt(\alpha^2 v + fc^2 \alpha'^2) + y^2 + z^2 = \left(\alpha'^2 - \frac{\alpha^2 v^2}{c^2} \right) c^2 t^2$... (4)

This result obtained from applying transformations from S' to S , must be identical to eq. (1). Therefore,

$$\alpha^2 - f^2 \alpha'^2 c^2 = 1 \quad \dots(i); \quad \alpha^2 v + fc^2 \alpha'^2 = 0 \quad \dots(ii); \quad \alpha'^2 - \frac{\alpha^2 v^2}{c^2} = 1 \quad \dots(iii)$$

Substituting the value of $f = -\frac{\alpha^2 v}{\alpha'^2 c^2}$ from eq. (ii) in eq. (i), we get

$$\alpha^2 - \frac{\alpha^4 v^2}{\alpha'^2 c^2} = 1 \quad \text{or} \quad 1 - \frac{\alpha^2 v^2}{\alpha'^2 c^2} = \frac{1}{\alpha^2}$$

But from (iii)

$$\alpha'^2 = 1 + \alpha^2 v^2 / c^2. \text{ Hence}$$

$$1 - \frac{\alpha^2 v^2 / c^2}{1 + \alpha^2 v^2 / c^2} = \frac{1}{\alpha^2} \quad \text{or} \quad \frac{1}{1 + \alpha^2 v^2 / c^2} = \frac{1}{\alpha^2} \quad \text{or} \quad \alpha^2 = 1 + \frac{\alpha^2 v^2}{c^2}$$

or $\alpha^2 - \frac{\alpha^2 v^2}{c^2} = 1 \quad \text{or} \quad \alpha^2 = \frac{1}{1 - v^2 / c^2} \quad \text{or} \quad \alpha = \frac{1}{\sqrt{1 - v^2 / c^2}}$

Therefore, $\alpha'^2 = 1 + \frac{v^2 / c^2}{1 - v^2 / c^2} = \frac{1}{1 - v^2 / c^2} \quad \text{or} \quad \alpha' = \frac{1}{\sqrt{1 - v^2 / c^2}} = \alpha$

Thus from (ii)

$$f = -v/c^2$$

Therefore,

$$x' = \alpha(x - vt) = \frac{1}{\sqrt{1 - v^2 / c^2}} (x - vt)$$

and

$$t' = \alpha'(t + \beta x) = \frac{1}{\sqrt{1 - v^2/c^2}} \left(t - \frac{vx}{c^2} \right)$$

Thus, the new transformation equations, which are in agreement with the invariance of velocity of light c , are

$$x' = \frac{x - vt}{\sqrt{1 - v^2/c^2}}; \quad y' = y, \quad z' = z; \quad t' = \frac{t - vx/c^2}{\sqrt{1 - v^2/c^2}} \quad \dots(5)$$

These equations are called **Lorentz transformations**, because they were first obtained by Dutch physicist H. Lorentz.

We note that when $v \ll c$ i.e. $v/c \rightarrow 0$, we get the Galilean transformations from the Lorentz transformations i.e.,

$$x' = x - vt, \quad y' = y, \quad z' = z, \quad t' = t \quad \dots(6)$$

In most of the cases, which we encounter on earth, c is a velocity very large compared with the great majority of velocities i.e. $v \ll c$ so that the results of Lorentz transformations do not differ to any great extent from those of the Galilean transformations; but from a theoretical point of view the Lorentz transformations represent a most profound conceptual change specially in relation to space and time.

For convenience, sometimes we put $\beta = v/c$ and

$$1/\sqrt{1 - v^2/c^2} = 1/\sqrt{1 - \beta^2} = \gamma. \quad \dots(7)$$

Hence the Lorentz transformations are written as

$$x' = \gamma(x - vt); \quad y' = y; \quad z' = z; \quad t' = \gamma \left(t - \frac{vx}{c^2} \right) \quad \dots(8)$$

In the derivation of these equations, we assumed that frame S' is moving in positive X -direction with velocity v relative to the frame S . But if we say that frame S is moving with $-v$ velocity relative to S' along negative X -direction, then the transformations are

$$x = \gamma(x' + vt'); \quad y = y'; \quad z = z'; \quad t = \gamma \left(t' + \frac{vx'}{c^2} \right) \quad \dots(9)$$

These are known as **inverse Lorentz transformations**.

From practical point of view at low speeds, there is no difference between the Lorentzian and Galilean transformations and we use the later in most of the problems which we encounter. However, when we have to deal with very fast particles having velocities comparable to c , such as electrons in the atoms, cosmic ray particles, we must use the Lorentz transformations.

6.11. Consequences of Lorentz Transformations

Now, let us discuss the consequences of Lorentz transformations regarding the lengths of the bodies and the time intervals between given events.

(1) Length contraction : In order to measure the length of an object in motion, relative to an observer, the positions of the two end points must be recorded simultaneously. Consider a frame S relative to which a rod is moving with velocity v along the X -axis. Let

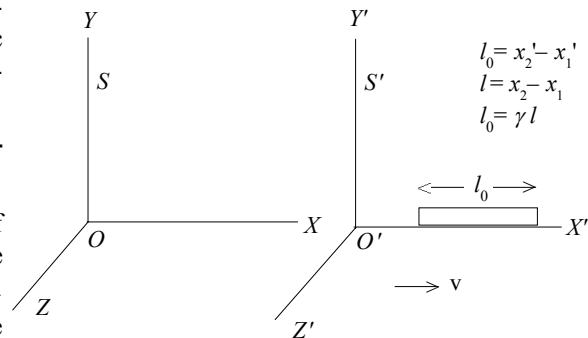


Fig. 6.7 : Contraction of moving rod

us associate a frame S' with the rod so that the rod is at rest in S' [fig. 6.7]. If in this frame, the x -coordinates of the ends of the rod are x_1' and x_2' , then

$$l_0 = x_2' - x_1'$$

This length l_0 has been measured by a stationary observer relative to the rod and is called the **proper length** of the rod. It is not necessary that the observer O' should measure the positions of the end points of the rod simultaneously, because the rod is at rest relative to him.

If the X -coordinates of the end points of that rod in frame S are measured to be x_1 and x_2 at the same time t , then in this frame the observed length of the rod is

$$l = x_2 - x_1$$

According to Lorentz transformations

$$x_1' = \gamma (x_1 - vt) \text{ and } x_2' = \gamma (x_2 - vt)$$

$$\therefore x_2' - x_1' = \gamma (x_2 - x_1) \text{ or } l_0 = \gamma l$$

$$\text{Thus } l = l_0 \sqrt{1 - v^2/c^2} \quad \dots(1)$$

As the factor $\sqrt{1 - v^2/c^2}$ is smaller than unity, we have $l < l_0$. This means that the length of the rod (l), as measured by an observer relative to which the rod is in motion, is smaller than its proper length. Such a contraction of length in the direction of motion relative to an observer is called **Lorentz-Fitzgerald contraction**. However, there will be no change in length in the perpendicular direction of motion.

If the rod is at rest in the frame S , then its proper length is

$$l_0 = x_2 - x_1$$

Now the observer of S' at time t' measures the end coordinates of the rod as x_2' and x_1' , then according to Lorentz transformations, we have

$$x_1 = \gamma (x_1' + vt') \text{ and } x_2 = \gamma (x_2' + vt')$$

$$\text{Hence, } x_2 - x_1 = \gamma (x_2' - x_1') \text{ or } l_0 = \gamma l \text{ or } l = l_0 \sqrt{1 - v^2/c^2} \quad \dots(2)$$

This means that a rod at rest in S appears to be contracted to the observer O' . Thus, *a length is contracted, if there is relative motion between it and the observer.*

(2) Simultaneity : If two events occur at the same time in a frame, they are said to be *simultaneous*. Suppose that S' frame is moving relative to S along positive direction of X -axis with velocity v . Let two events occur simultaneously in frame S at the points P_1 and P_2 with the coordinates (x_1, y_1, z_1, t_1) and (x_2, y_2, z_2, t_2) respectively as measured by the observer O of the S -frame [Fig. 6.8]. As the two events are simultaneous in frame S , we have $t_2 = t_1$. If t_1' and t_2' are the corresponding times of the same two events are measured by O' of frame S' , then the use of Lorentz transformation equations gives

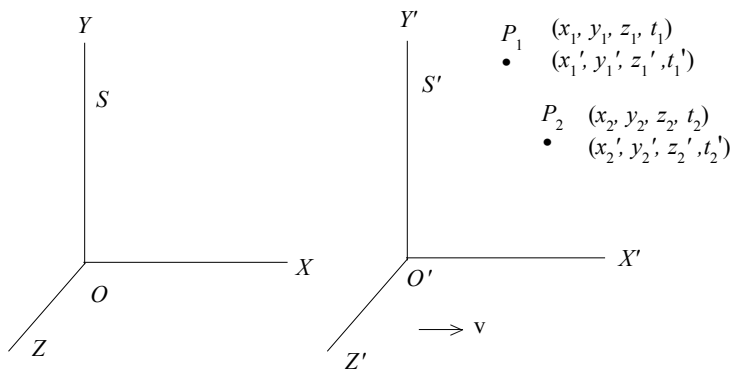


Fig. 6.8 : Representation of two events in two inertial frames

$$t_1' = \frac{t_1 - vx_1/c^2}{\sqrt{1 - v^2/c^2}} \quad \text{and} \quad t_2' = \frac{t_2 - vx_2/c^2}{\sqrt{1 - v^2/c^2}}$$

Therefore,

$$t_2' - t_1' = \frac{t_2 - t_1}{\sqrt{1 - v^2/c^2}} - \frac{(v/c^2)(x_2 - x_1)}{\sqrt{1 - v^2/c^2}}$$

As $t_2 = t_1$ or $t_2 - t_1 = 0$, we have

$$t_2' - t_1' = - \frac{(v/c^2)(x_2 - x_1)}{\sqrt{1 - v^2/c^2}} \quad \dots(3)$$

We observe that $t_2' - t_1' \neq 0$. This means that the two events at two different points P_1 and P_2 which are simultaneous for O in frame S are not simultaneous for the observer O' of the frame S' , moving with speed v along X -axis relative to S . Thus the simultaneity is not absolute, but relative.

(3) Time Dilation : Let a frame S' be moving along X -axis with velocity v relative to S . Now, if a clock being at rest in the frame S' , measures the time t_1' and t_2' of two events occurring at a fixed position x' at P in this frame (**fig. 6.9**), then the interval of time between these events is

$$\Delta t' = t_2' - t_1' = \Delta t_0 \quad (\text{say})$$

Now, according to Lorentz transformations, we have

$$t_1 = \gamma(t_1' + vx_1'/c^2) \quad \text{and} \quad t_2 = \gamma(t_2' + vx_2'/c^2)$$

$$\text{Therefore } t_2 - t_1 = \gamma(t_2' - t_1') \quad \text{or} \quad \Delta t = \gamma \Delta t_0$$

Thus

$$\Delta t = \frac{\Delta t_0}{\sqrt{1 - v^2/c^2}} \quad \dots(4)$$

$$\text{As } 1/\sqrt{1 - v^2/c^2} > 1, \quad \Delta t > \Delta t_0.$$

Thus, the time interval, measured in the frame S is larger than the time interval in the frame S' , in which the two events are occurring at a certain point x' . This effect is called **time dilation** (lengthening of time interval). This means *to stationary observer the moving clock will appear to go slow*. In consequence we may say that the two events appear to take a longer time when they occur in a body in motion relative to the observer than when the body is at rest relative to the observer. *The time interval between any two events which has been measured relative to a frame in which the same two events take place at a certain point, is minimum and is called proper time interval.* It is represented by Δt_0 or $\Delta \tau$, given by

$$\Delta \tau = \Delta t_0 = \Delta t \sqrt{1 - v^2/c^2} \quad \dots(5)$$

For example, an observer sitting in a moving rocket, takes the readings of the needle of his clock at 9 a.m. and 10 a.m., then this $\Delta t = 1$ hour is the proper interval of time.

Now, if $\Delta \tau$ is the decay half life of mesons of radioactive matter as measured in the frame S' in which the particle is at rest, then

$$\Delta t = \frac{\Delta \tau}{\sqrt{1 - v^2/c^2}} \quad \dots(6)$$

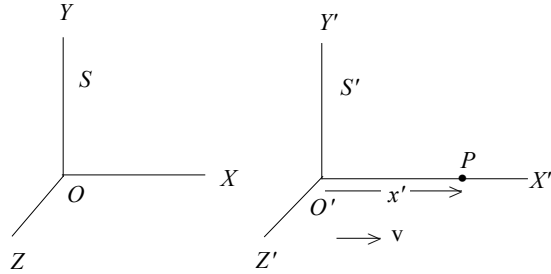


Fig. 6.9 : Two events occur in frames S' at a fixed position x'

is the decay half life observed in a frame S in which the particles are moving with velocity v .

Verification of Time Dilation : (1) A muon or μ -meson is an elementary particle, whose mean life is 2.2×10^{-6} sec in a frame in which it is at rest. When cosmic rays fall on the upper part of the atmosphere, π -mesons are produced and they decay with a very short life to μ -mesons. The velocity of the μ -mesons should be less than the speed of light c , so the expected distance, travelled by them, is less than $2.2 \times 10^{-6}c = (2.2 \times 10^{-6} \times 3 \times 10^8)$ or 0.66 km. Thus, in absence of any dilation effect, the μ -mesons should not travel a distance greater than 0.66 km and hence there should be no possibility for these μ -mesons, created at the top of the atmosphere, to reach the earth's surface (a distance near about 10 km). But experimentally on the earth's surface an appreciable number of μ -mesons can be detected after travelling about 10 km of the atmosphere [fig. 6.10]. This becomes possible only when we consider the life-time of a fast μ -meson to be much longer than the measured life-time of a μ -meson at rest. The μ -mesons of cosmic rays, detected on the earth's surface, have an energy of the order of corresponding to $v = 0.999c$ or $\gamma = 30$ so that their average life according to the special theory of relativity is

$$\Delta t = \Delta \tau / \sqrt{1 - v^2/c^2} = \gamma \Delta \tau = 30 \times 2.2 \times 10^{-6} \approx 6.6 \times 10^{-5} \text{ sec.}$$

Hence the distance travelled by a μ -meson is nearly $6.6 \times 10^{-5} \times 0.999c$ or 20 km.

Thus there is sufficient possibility that some μ -mesons reach on the earth's surface.

(2) It is known that π^+ meson is positively charged particle, having mass 273 times that of an electron. π^+ meson decays into a μ^+ meson (mass = $215 m_e$) and a neutrino with a mean life about 2.5×10^{-8} sec in a frame in which it is at rest. If the velocity of the π^+ mesons in the laboratory frame be $0.9c$, then the expected life time in this frame is

$$\Delta t = \frac{2.5 \times 10^{-8}}{[1 - (0.9c/c)^2]^{1/2}} = \frac{2.5 \times 10^{-8}}{(1 - 0.81)^{1/2}} = 5.7 \times 10^{-8} \text{ sec.}$$

Therefore, on an average, before decaying π^+ meson will travel a distance twice (distance = $5.7 \times 10^{-8} \times 0.9c$) as we would expect non-relativistically (distance = $2.5 \times 10^{-8} \times 0.9c$). If the velocity of pions is close to the velocity of light (c), expected life time in the laboratory frame may be many times than the proper value. Experiments have been done for the life time of π^+ mesons and the results are in good agreement with the predicted time dilation for the appropriate velocity.

The π^+ mesons, produced at a target, are passed through a slit and the muons, generated in the decay, are observed by a detector [fig. 6.11]. Therefore number of muons, detected per second, is proportional to the flux (number of particles arriving per second) of pions and hence the detector shows the reading proportional to the flux of pion beam. If Δt is the mean life time, then the number of particles, left unchanged, after time t is given by

$$N = N_0 e^{-t/\Delta t} = N_0 e^{-x/v\Delta t} \quad \dots(7)$$

where N_0 is the initial number of particles and $t = x/v$; v is the velocity of the particles and x the distance, travelled by the pions in time t . The original flux of pions is obtained by reducing their velocity to zero by allowing them to enter a solid block and measuring the muons emitted. Hence from eq. (7) the value of mean life

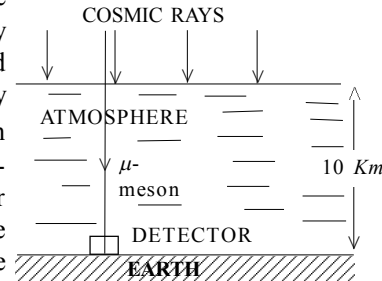


Fig. 6.10

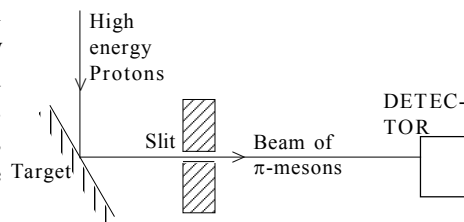


Fig. 6.11

time (Δt) of moving pions can be calculated. To determine the proper mean life time $\Delta\tau$, pions are brought to rest by introducing a solid at the beginning of the path. The time required between the entrance into the stopping solid and the emission of muons is determined by electronic device. This gives the proper mean life $\Delta\tau$ of π^+ mesons. The two values of mean life time of pions (Δt and $\Delta\tau$) satisfy the time dilation relation

$$\Delta t = \Delta\tau \sqrt{1 - v^2/c^2}$$

Thus it will be seen that for the pions of larger velocity, life-time (Δt) will be dilated accordingly.

Twin Paradox in Special Relativity : Consider two twins A and B , each 20 years of age. Twin A remains at rest at the origin O and twin B takes a round trip space voyage to a star with velocity $v = 0.99c$ relative to A . The star is 10 light years away from O . We want to determine the age of A and B as B finishes his journey.

According to A , the time taken by B in the round trip is

$$t_1 = \frac{20 \text{ light years}}{0.99c} = \frac{20 \text{ years} \times c}{0.99c} = 20.2 \text{ years.}$$

Thus according to A his own age, as B completes the journey, is $20 + 20.2 = 40.2$ years.

According to B , the time of journey (proper time interval) is given by

$$t_2 = 20 \text{ years} \times \sqrt{1 - v^2/c^2} = 20 \times \sqrt{(1 - 0.99)^2} = 2.8 \text{ years.}$$

Hence according to B , his own age after the journey is $20 + 2.8 = 22.8$ years.

These two statements are different. After the space journey, will one of the two twins look younger than the other? This is twin paradox in the theory of special relativity.

(4) Addition of Velocities : Let the coordinates of a particle in frame S be (x, y, z, t) and in frame S' (x', y', z', t'), then the components of its velocity in two frames can be written as

$$u_x = dx/dt, \quad u_y = dy/dt, \quad u_z = dz/dt \quad \text{in } S$$

and

$$u'_x = dx'/dt', \quad u'_y = dy'/dt', \quad u'_z = dz'/dt' \quad \text{in } S'.$$

According to the inverse Lorentz transformations

$$x = \gamma(x' + vt'), \quad y = y', \quad z = z', \quad t = \gamma(t' + vx'/c^2)$$

Therefore,

$$dx = \gamma(dx' + vdt'), \quad dy = dy', \quad dz = dz' \text{ and } dt = \gamma(dt' + vdx'/c^2)$$

$\therefore$

$$u_x = \frac{dx}{dt} = \frac{\gamma(dx' + vdt')}{\gamma\left(dt' + \frac{vdx'}{c^2}\right)} = \frac{\frac{dx'}{dt'} + v}{1 + \frac{v}{c^2} \frac{dx'}{dt'}}$$

or

$$u_x = \frac{u'_x + v}{1 + vu'_x/c^2} \quad \dots(8a)$$

Also,

$$u_y = \frac{dy}{dt} = \frac{dy'}{\gamma\left(dt' + \frac{vdx'}{c^2}\right)} = \frac{dy'/dt'}{\gamma\left(1 + \frac{v}{c^2} \frac{dx'}{dt'}\right)}$$

or

$$u_y = \frac{u'_y}{\gamma(1 + vu'_x/c^2)} \quad \dots(8b)$$

Similarly,
$$u_z = \frac{u_z}{\gamma(1 + vu'_x/c^2)} \quad \dots(8c)$$

This is the **relativistic law of addition of velocities** while in classical mechanics $u_x = u'_x + v$, $u'_y = u_y$ and $u'_z = u_z$. We get the later (Galilean) equations, when v is much less than the speed of light c .

If we take the Lorentz transformations, we can prove that

$$u'_x = \frac{u_x - v}{1 - vu_x/c^2}; \quad u'_y = \frac{u_y}{\gamma(1 - vu_x/c^2)}; \quad u'_z = \frac{u_z}{\gamma(1 - vu_x/c^2)} \quad \dots(9)$$

In case a particle (as photon) is moving with a velocity c in the frame S' and S' is moving with velocity c relative to S along positive X -axis direction, then from eq. (8 a) we have

$$u_x = \frac{c + c}{1 + cc/c^2} = c \quad [\text{because } u'_x = c, v = c] \dots(10)$$

Thus the speed of photon in the second frame is also c i.e., the velocity of light is the same for all inertial frames whatever their relative speeds may be. This result is in accordance with the Michelson-Morley experiment. In fact, the Lorentz transformations and hence the law of addition of velocities have been deduced by assuming the velocity of light constant for all inertial observers.

6.12 Distinctive Features of Galilean and Lorentz Transformations

Let an inertial frame S' move with velocity v along X -axis relative to another inertial frame S .

Galilean Transformation (Galilean Relativity)	Lorentz Transformations (Special Relativity)
1. Galilean transformations from S -frame to S' frame are $x' = x - vt, y' = y, z' = z, t' = t$	1. Lorentz transformation from S -frame to S' -frame are $x' = \gamma(x - vt), y' = y, z' = z, t' = \gamma\left(t - \frac{vx}{c^2}\right)$
2. Length of a rod is the same in S and S' frames under Galilean transformation i.e., $l' = l$	2. Length of a stationary rod in S' -frame is observed as contracted in S -frame in view of Lorentz transformations: $l = l'\sqrt{1 - v^2/c^2}$ where c is the speed of light.
3. Time interval of two events in S and S' -frame is the same i.e., $\Delta t' = \Delta t$	3. Time interval of two events occurring at a point in S' -frame is observed to be dilated in S -frame i.e., $\Delta t = \Delta t' / \sqrt{1 - v^2/c^2}$
4. Galilean law of addition of velocities is $u'_x = u_x - v, u'_y = u_y, u'_z = u_z$	4. Relativistic law of addition of velocities is $u'_x = \frac{u_x - v}{1 - vu_x/c^2},$ $u'_y = \frac{u_y}{\gamma(1 - vu_x/c^2)}, u'_z = \frac{u_z}{\gamma(1 - vu_x/c^2)}$
5. Signals and informations can be transmitted with infinite speed.	5. Signals and informations can be transmitted with finite speed, maximum speed being the speed of light in vacuum.

[Lorentz Transformations]**Ex. 1.** Prove that $x^2 + y^2 + z^2 - c^2 t^2$ is invariant under Lorentz transformation. (S.K.U. 1995; N.U. 1993)**Sol.** Lorentz transformation from S -frame to S' -frame are

$$x' = \gamma (x - vt), y' = y, z' = z \text{ and } t' = \gamma (t - vx/c^2)$$

where

$$\gamma = 1/\sqrt{1 - v^2/c^2}$$

The invariance of $x^2 + y^2 + z^2 - c^2 t^2$ under Lorentz transformation means

$$x'^2 + y'^2 + z'^2 - c^2 t'^2 = x^2 + y^2 + z^2 - c^2 t^2$$

Using Lorentz transformation

$$\begin{aligned} x'^2 + y'^2 + z'^2 - c^2 t'^2 &= \gamma^2 (x - vt)^2 + y^2 + z^2 - c^2 \gamma^2 (t - vx/c^2)^2 \\ &= \gamma^2 x^2 + \gamma^2 v^2 t^2 - 2\gamma^2 vxt + y^2 + z^2 - c^2 \gamma^2 t^2 - c^2 \gamma^2 v^2 x^2/c^4 - 2\gamma^2 vxt \\ &= x^2 \gamma^2 (1 - v^2/c^2) + y^2 + z^2 - t^2 \gamma^2 (c^2 - v^2) \\ &= x^2 + y^2 + z^2 - c^2 t^2 \end{aligned}$$

$$[\because \gamma^2 (1 - v^2/c^2) = (1 - v^2/c^2)/(1 - v^2/c^2) = 1 \text{ and } \gamma^2 (c^2 - v^2) = \frac{(c^2 - v^2)}{(1 - v^2/c^2)} = c^2]$$

Hence $x^2 + y^2 + z^2 - c^2 t^2$ is invariant under Lorentz transformations.**[Length Contraction]****Ex. 2.** If a rod travels with a speed $v = 0.6c$ along its length, calculate the percentage of contraction. (S.V.U. 2007; O.U. 1995)**Sol.** Let a rod be at rest with length l_0 in frame S' . If it is moving with speed v along its length (say X -axis) relative to frame S , then the length, measured in S -frame, will be given by

$$l = l_0 \left[1 - \frac{v^2}{c^2} \right]^{1/2} = l_0 \left[1 - \frac{(0.6c)^2}{c^2} \right]^{1/2} = l_0 [1 - 0.36]^{1/2} = l_0 [0.64]^{1/2} = 0.8 l_0$$

$$\text{Hence \% of contraction} = \frac{l_0 - l}{l_0} \times 100 = \frac{l_0 - 0.8l_0}{l_0} \times 100 = 0.2 \times 100 = 20\%.$$

Ex. 3. Calculate the velocity of the rod when its length will appear 90% of its proper length.

(A.U. 2001; A.U. 1994)

$$\text{Sol.} \quad l = l_0 \sqrt{1 - \frac{v^2}{c^2}} \quad \dots(i)$$

$$\text{Here,} \quad \frac{l}{l_0} = \frac{90}{100} = 0.9$$

$$\text{From (i),} \quad \frac{l}{l_0} = \sqrt{1 - \frac{v^2}{c^2}} \text{ or } \frac{l^2}{l_0^2} = 1 - \frac{v^2}{c^2} \text{ or } \frac{v^2}{c^2} = 1 - \frac{l^2}{l_0^2} \text{ or } v = c \sqrt{1 - \frac{l^2}{l_0^2}}$$

$$\therefore v = 3 \times 10^8 \sqrt{1 - (0.9)^2} = 3 \times 10^8 \times \sqrt{0.19} = 1.308 \times 10^8 \text{ m/sec.}$$

Ex. 4. Find the velocity with which a body should travel so that its length becomes half of the rest length? (S.V.U. 2002)

$$\text{Sol.} \quad l = l_0 \sqrt{1 - \frac{v^2}{c^2}} \text{ or } v = c \sqrt{1 - \frac{l^2}{l_0^2}}$$

$$\text{Here, } \frac{l}{l_0} = \frac{1}{2}, \quad \therefore v = 3 \times 10^8 \sqrt{1 - \left(\frac{1}{2}\right)^2} = 3 \times 10^8 \times \frac{\sqrt{3}}{2} = 2.6 \times 10^8 \text{ m/sec.}$$

Ex. 5. Find the proper length of the rod which appears to have a length of 1 m moving with a velocity of $0.5c$ (c = velocity of light = 3×10^8 m/s). (O.U. 2001)

Sol. If l_0 be the proper length of a rod measured by a stationary observer relative to the rod, then the length of the rod (l), as measured by an observer relative to which the rod is in motion, is given by

$$l = l_0 \sqrt{1 - v^2/c^2}, \therefore l_0 = \frac{l}{\sqrt{1 - v^2/c^2}}$$

Here, $l = 1$ m and $v/c = 0.5$ $c/c = 0.5$.

$$\therefore l_0 = \frac{l}{\sqrt{1 - (0.5)^2}} = \frac{l}{\sqrt{0.75}} = 1.15 \text{ m}$$

Ex. 6. A rod of 1 m length is moving with a velocity of 0.6×10^8 m/s with respect to a stationary observer. Find the length of the rod (in m) along its direction of motion as seen by the observer. (A.N.U. 2006)

Sol. $l = l_0 \sqrt{1 - v^2/c^2}$

Here, $l_0 = 1$ m, $v = 0.6 \times 10^8$ m/sec. and $c = 3 \times 10^8$ m/sec.

$$\therefore l = 1 \sqrt{1 - \left(\frac{0.6 \times 10^8}{3 \times 10^8}\right)^2} = \sqrt{1 - \left(\frac{1}{5}\right)^2} = \sqrt{\frac{24}{25}} = 0.98 \text{ m}$$

Ex. 7. A rod has length 1 m. When the rod in a satellite moving with a velocity that is one half of the velocity of light relative to laboratory, what is the length of the rod as determined by an observer (a) in the satellite, and (b) in the laboratory. (O.U. 2001)

Sol. (a) Since the observer in the satellite is at rest relative to the rod, the length of the rod as measured by him will be **1 metre**.

(b) Since the rod in the satellite is moving relative to the laboratory,

$$l = l_0 \sqrt{1 - \frac{v^2}{c^2}} = 1 \sqrt{1 - \left(\frac{0.5c}{c}\right)^2} = 1 \times (0.75)^{1/2} = 0.866 \text{ m.}$$

Ex. 8. A rocket ship is 100 metre long on the ground. When it is in flight, its length is 99 metres to an observer on the ground. What is its speed? (O.U. 2009; A.N.U. 1995)

Sol. $l = l_0 \sqrt{1 - v^2/c^2}, \therefore v = c \sqrt{1 - \frac{l^2}{l_0^2}}$

Here, $l_0 = 100$ m, $l = 99$ m and $c = 3 \times 10^8$ m/sec

$$\therefore v = 3 \times 10^8 \sqrt{1 - \left(\frac{99}{100}\right)^2} = 3 \times 10^8 \sqrt{1 - 0.9801} = 4.23 \times 10^8 \text{ m/sec.}$$

Ex. 9. Calculate the percentage of contraction in the length of a rod moving with a speed of $0.8c$ (where c is the velocity of light) in a direction at an angle of 60° with its own axis. (O.U. March 2007)

Sol. Let the rod be at rest in frame S' with its proper length l_0 . In this frame, its components are $l_{x0} = l_0 \cos 60^\circ = l_0/2$ and $l_{y0} = l_0 \sin 60^\circ = l_0 \sqrt{3}/2$

Let l be the length in S -frame. If l_x and l_y are rectangular components in S -frame, then

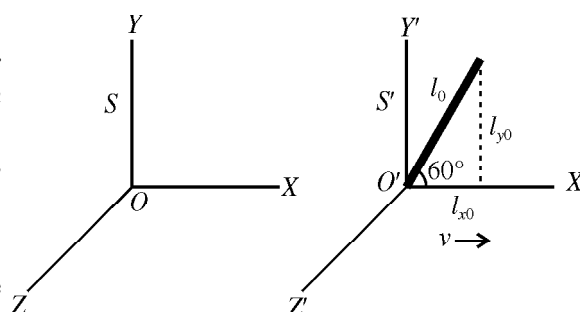


Fig. 6.12.

$$l_x = l_{x0} \sqrt{1 - \frac{v^2}{c^2}} = l_{x0} \sqrt{1 - (0.8)^2} = 0.6 l_{x0} = 0.6 \times l_0/2 = 0.3 l_0 \text{ and } l_y = l_{y0} = l_0\sqrt{3}/2$$

because the length will not contract along Y -direction.

$$\therefore l = \sqrt{l_x^2 + l_y^2} = \sqrt{(0.3l_0)^2 + (l_0\sqrt{3}/2)^2} = l_0 \sqrt{(0.3)^2 + (0.866)^2} = 0.916 l_0$$

$$\therefore \% \text{ contraction} = \frac{l_0 - l}{l_0} \times 100 = \frac{l_0 - 0.916l_0}{l_0} \times 100 = 0.084 \times 100 = \mathbf{8.4\%}$$

Ex. 10. Determine the length and the orientation of a rod of length 10 metres in a frame of reference which is moving with $0.6c$ velocity in a direction making 30° angle with the rod.

Sol. The proper length of the rod along the direction of the moving frame $l_{x_0} = 10 \cos 30^\circ$. Therefore, the length measured in the moving frame is

$$\begin{aligned} l_x &= l_{x_0} \sqrt{1 - v^2/c^2} = 10 \cos 30^\circ \sqrt{1 - (0.6c)^2/c^2} \\ &= 10 \cos 30^\circ \sqrt{1 - 0.36} = 8 \cos 30^\circ = 4\sqrt{3} \text{ m.} \end{aligned}$$

Since the length does not contract perpendicular to the direction of motion of the moving frame,

$$l_y = l_{y_0} = 10 \sin 30^\circ = 5 \text{ m.}$$

Hence the length of the rod observed in the moving frame, is given by

$$l = \sqrt{l_x^2 + l_y^2} = \sqrt{48 + 25} = \sqrt{73} = \mathbf{8.54 \text{ m.}}$$

If the rod makes an angle θ with X -axis of the moving frame, then

$$\tan \theta = \frac{l_y}{l_x} = \frac{5}{4\sqrt{3}} = 0.72 \text{ or } \theta = \tan^{-1}(0.72) = \mathbf{35.8^\circ}.$$

Ex. 11. Show that if V_0 is the rest volume of a cube of side l_0 then $V_0(1 - \beta^2)^{1/2}$ is the volume viewed from a reference frame moving with uniform velocity v in a direction parallel to an edge of the cube.

Sol. One side of the cube = l_0 ; Rest volume of the cube $V_0 = l_0^3$.

Let the edges of the cube lie parallel to X, Y, Z axes. Consider that one of its edges lie along the X -axis of the moving frame.

Now, the edge along X -axis as observed by an observer in moving frame is

$$l = l_0 \sqrt{1 - \beta^2}, \text{ where } \beta^2 = v^2/c^2.$$

But there is no contraction along Y and Z directions, hence the volume of the cube as observed by the observer in the moving frame is

$$l_0 \sqrt{1 - \beta^2} \cdot l_0 \cdot l_0 = l_0^3 \sqrt{1 - \beta^2} = V_0 \sqrt{1 - \beta^2}.$$

Ex. 12. A vector in a coordinate system is represented by $4\hat{i} + 3\hat{j}$. How far is the vector represented by an observer in the system S if it moves with velocity of 1.8×10^8 m/s along X -axis relative to the coordinate system?

Sol. The projection of the vector $4\hat{i} + 3\hat{j}$ along X -axis is 4 and projection along Y -axis is 3. Since the frame moves along X -axis, only X -component (4) will contract.

Thus, $l_x = l_{x0} \sqrt{1 - \frac{v^2}{c^2}} = 4 \sqrt{1 - \left(\frac{1.8}{3.0}\right)^2} = 4 \sqrt{1 - \left(\frac{3}{5}\right)^2} = 4 \sqrt{1 - \left(\frac{9}{25}\right)} = 4 \times \frac{4}{5} = 3.2$

Also, $l_y = l_{y0} = 3$

Hence the vector in S -frame is $3.2\hat{i} + 3\hat{j}$.

[Time Dilation]

Ex. 13. The mean life of a meson is 2×10^{-8} sec. Calculate the mean life of a meson moving with a velocity of $0.8c$. (A.N.U. 1994)

Sol.
$$\Delta t = \frac{\Delta \tau}{\sqrt{1 - v^2/c^2}}$$

Here, mean life of the meson $\Delta \tau = 2 \times 10^{-8}$ sec. and $v = 0.8c$

$$\therefore \text{Mean life of the moving meson, } \Delta t = \frac{2 \times 10^{-8}}{\sqrt{1 - (0.8c/c)^2}}$$

$$= \frac{2 \times 10^{-8}}{\sqrt{1 - 0.64}} = \frac{2 \times 10^{-8}}{0.6} = 3.3 \times 10^{-8} \text{ sec.}$$

Ex. 14. A meson has a speed $0.8c$ relative to the ground. Find how far the meson travels relative to the ground, if its speed remains constant and the time of its flight, relative to the system, in which it is at rest, is 2×10^{-8} sec.

Sol. Time of flight of meson relative to the earth is

$$\Delta t = \frac{\Delta \tau}{\sqrt{1 - v^2/c^2}}$$

Proper time of flight of meson $\Delta \tau = 2 \times 10^{-8}$ sec. Hence

$$\Delta t = \frac{2 \times 10^{-8}}{\sqrt{1 - (0.8c/c)^2}} = \frac{2 \times 10^{-8}}{0.6} = 3.33 \times 10^{-8} \text{ sec.}$$

Distance travelled by the meson relative to the ground

$$= v \Delta t = 0.8 \times 3 \times 10^8 \times 3.33 \times 10^{-8} = 8 \text{ m.}$$

Ex. 15. A stationary μ -meson decays in 2.2×10^{-8} sec. What will be length of its path if it is moving towards the earth with velocity $0.99c$. (K.U. 1994)

Sol. Time of flight relative to the earth is

$$\Delta t = \frac{\Delta \tau}{\sqrt{1 - v^2/c^2}} = \frac{2.2 \times 10^{-8}}{\sqrt{1 - (0.99)^2}} = \frac{2.2 \times 10^{-8}}{\sqrt{1 - 0.9801}} = 15.6 \times 10^{-8} \text{ sec.}$$

$\therefore$ Length of the path $= v \Delta t = 0.99 \times 3 \times 10^8 \times 15.6 \times 10^{-8} = 46.33 \text{ m}$

Ex. 16. A beam of μ meson travels with a speed $v = 0.6c$. Their mean life time as observed in the laboratory is found to be 2.9×10^{-6} sec. What is the mean life time of μ -mesons when they decay at rest. (O.U. 1995)

Sol.
$$\Delta t = \frac{\Delta \tau}{\sqrt{1 - v^2/c^2}} \text{ or } \Delta t = \Delta \tau \sqrt{1 - \frac{v^2}{c^2}}$$

Here, $\Delta t = 2.9 \times 10^{-6}$ sec. and $v = 0.6c$

$$\therefore \Delta t = 2.9 \times 10^{-6} \times \sqrt{1 - (0.6c/c)^2} = 2.9 \times 10^{-6} \sqrt{1 - 0.36} = 2.9 \times 0.8 \times 10^{-6} = \mathbf{2.32 \times 10^{-6} \text{ sec.}}$$

Ex. 17. (a) What is the mean life of a burst of π^+ mesons travelling with $v = 0.73c$, if the proper mean life time is 2.3×10^{-8} sec ?

(b) What is the distance travelled at $v = 0.73c$ during one mean life ?

(c) What distance will be travelled without relativistic effect ?

Sol. (a)
$$\Delta t = \frac{\Delta \tau}{\sqrt{1 - v^2/c^2}} = \frac{2.3 \times 10^{-8}}{\sqrt{1 - (0.73)^2}} = \frac{2.3 \times 10^{-8}}{\sqrt{0.4671}} = \mathbf{3.6 \times 10^{-8} \text{ sec.}}$$

(b) Distance travelled by the burst of π^+ meson

$$= v \Delta t = 0.73 \times 3 \times 10^8 \times 3.6 \times 10^{-8} = \mathbf{7.9 \text{ m.}}$$

(c) Distance travelled without relativistic effect

$$= v \Delta \tau = 0.73 \times 3 \times 10^8 \times 2.3 \times 10^{-8} \approx \mathbf{5.5 \text{ m.}}$$

Ex. 18. A beam of particles of half life 2×10^{-6} sec travels in the laboratory with speed 0.96 times the speed of light. How much distance does the beam travel before the flux falls to $\frac{1}{2}$ times the initial flux ?

Sol. If Δt is the observed half life, then

$$\Delta t = \frac{\Delta \tau}{\sqrt{1 - v^2/c^2}} = \frac{2 \times 10^{-6}}{\sqrt{1 - (0.96)^2}} = \frac{2 \times 10^{-6}}{0.28} \text{ sec.}$$

Evidently, in this observed half life the flux will fall to $\frac{1}{2}$ times the initial flux. In this time the distance travelled by the beam is

$$= v \Delta t = 0.96 \times 3 \times 10^8 \times \frac{2 \times 10^{-6}}{0.28} = \mathbf{2,000 \text{ m.}}$$

Ex. 19. A clock keeps correct time. With what speed should it be moved relative to an observer so that it may seem to lose 2 minutes in 24 hours ?

Sol. Let the clock be at rest in a frame S . In this frame, the clock should show $\Delta \tau = 23$ hours 58 minutes. The observer, relative to which frame S is moving with speed v , will measure the time $\Delta t = 24$ hours = 24×60 minutes in his clock. Now

$$\Delta t = \frac{\Delta \tau}{\sqrt{1 - v^2/c^2}}$$

or
$$1 - \frac{v^2}{c^2} = \left[\frac{\Delta \tau}{\Delta t} \right]^2 = \left[\frac{23 \text{ hrs } 58 \text{ mts}}{24 \text{ hrs}} \right]^2 = \left[1 - \frac{2}{24 \times 60} \right]^2$$

$$= 1 - \frac{2 \times 2}{24 \times 60} = 1 - \frac{1}{360} \text{ approx.}$$

Hence
$$v^2 = \frac{c^2}{360} = \frac{(3 \times 10^8)^2}{360} \text{ or } v = \frac{3 \times 10^8}{\sqrt{360}} = \mathbf{1.58 \times 10^7 \text{ m/s.}}$$

Ex. 21. A clock showing correct time when at rest and loses 2 hours per day when it is moving. What is its speed ? (A.N.U. 1989; S.K.U. 1989)

Sol. $\Delta t = \frac{\Delta \tau}{\sqrt{1 - v^2/c^2}}$ or $1 - \frac{v^2}{c^2} = \left(\frac{\Delta \tau}{\Delta t}\right)^2$ or $\frac{v^2}{c^2} = 1 - \left(\frac{\Delta \tau}{\Delta t}\right)^2$ or $v = c \sqrt{1 - \left(\frac{\Delta \tau}{\Delta t}\right)^2}$

Here, $\Delta t = 24$ hours and $\Delta \tau = 24 - 2 = 22$ hours

$\therefore v = 3 \times 10^8 \sqrt{1 - \left(\frac{22}{24}\right)^2} = 1.2 \times 10^8 \text{ m/sec.}$

Ex. 21. A clock showing correct time when at rest, loses one hour in a day when it is moving. What is its velocity ? (A.U. 1995)

Sol. $\Delta t = \frac{\Delta \tau}{\sqrt{1 - v^2/c^2}}$ or $v = c \sqrt{1 - \left(\frac{\Delta \tau}{\Delta t}\right)^2}$

Here, $c = 3 \times 10^8$ m/sec, $\Delta t = 24$ hours and $\Delta \tau = 24 - 1 = 23$ hours

$\therefore v = 3 \times 10^8 \sqrt{1 - (23/24)^2} = 3 \times 10^8 \times 0.286 = 8.58 \times 10^7 \text{ m/sec.}$

[Addition of Velocities]

Ex. 22. In the laboratory one particle A has velocity $v_x = +2 \times 10^8$ m/sec and another particle B has velocity $v_x = -2 \times 10^8$ m/sec. Calculate the velocity of A relative to B.

Sol. If we assume that the particle B, which is moving with velocity -2×10^8 m/sec relative to the laboratory is at rest in frame S [fig. 6.13]. Clearly relative to this frame, the frame S' of the laboratory is moving with 2×10^8 m/sec. Hence the velocity of the particle A relative to the frame S (i.e. relative to the particle B) is

$$u_x = \frac{u'_x + v}{1 + \frac{vu'_x}{c^2}} = \frac{2 \times 10^8 + 2 \times 10^8}{1 + \frac{(2 \times 10^8)^2}{(3 \times 10^8)^2}} = \frac{4 \times 10^8}{1 + 4/9} = 2.77 \times 10^8 \text{ m/sec.}$$

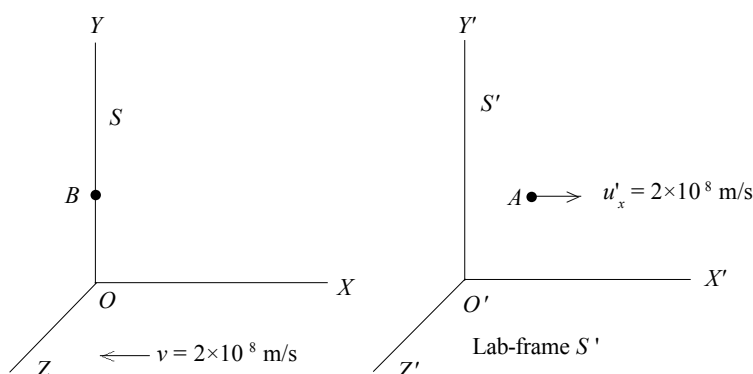


Fig. 6.13 : Particle A has velocity $u'_x = 2 \times 10^8$ m/s in S'-frame (Lab-frame). Particle B is at rest in S-frame which is moving with $v = 2 \times 10^8$ m/s along -ve X-axis relative to S' i.e., B is moving opposite to A with 2×10^8 m/s in the Lab-frame.

Ex. 23. A spaceship moving away from the earth with velocity $0.5c$ fires a rocket whose velocity relative to the space is $0.5c$ (a) away from the earth, (b) towards the earth. Calculate the velocity of the rocket as observed from the earth in two cases.

Solution : Let the velocity of the rocket as observed from the earth be u .

(a) In the 1st case,
$$u = \frac{u' + v}{1 + \frac{u'v}{c^2}}$$

Here, $u' = 0.5c$, $v = 0.5c$. Therefore

$$u = \frac{0.5c + 0.5c}{1 + \frac{0.5c \times 0.5c}{c^2}} = \frac{c}{1.25} = 0.8c.$$

(b) In the 2nd case, $u' = -0.5c$, Hence

$$u = \frac{-0.5c + 0.5c}{1 - \frac{0.5c \times 0.5c}{c^2}} = 0.$$

Ex. 24. A particle has a velocity 6×10^7 m/sec in the X - Y plane at an angle of 60° with X -axis in the system S . Determine the magnitude and direction of its velocity in system S' , when S' has a velocity 3×10^7 m/sec along the positive X -axis.

Solution : The x and y components of the velocity in S' frame are given by

$$u_{x'} = \frac{u_x - v}{1 - u_x v / c^2}, \quad u_{y'} = \frac{u_y \sqrt{1 - v^2 / c^2}}{(1 - u_x v / c^2)}$$

Here, $u_x = 6 \times 10^7 \cos 60^\circ = 3 \times 10^7$ m/sec. $u_y = 6 \times 10^7 \sin 60^\circ = 3\sqrt{3} \times 10^7$ m/sec and $v = 3 \times 10^7$ m/sec.

Therefore,
$$u_{x'} = \frac{3 \times 10^7 - 3 \times 10^7}{1 - \frac{3 \times 10^7 \times 3 \times 10^7}{(3 \times 10^8)^2}} = 0$$

Also,
$$u_{y'} = \sqrt{1 - \left[\frac{3 \times 10^7}{3 \times 10^8} \right]^2} \times \frac{3\sqrt{3} \times 10^7}{\left[1 - \frac{3 \times 10^7 \times 3 \times 10^7}{(3 \times 10^8)^2} \right]}$$

$$= \sqrt{\frac{99}{100}} \times \frac{3\sqrt{3} \times 10^7 \times 100}{99} = \sqrt{\frac{3}{11}} \times 10^8 = 5.2 \times 10^7 \text{ m/sec.}$$

Hence the velocity in S' frame is

$$\vec{u}' = 0\hat{i} + 5.2 \times 10^7 \hat{j} \quad \text{or} \quad \vec{u} = 5.2 \times 10^7 \hat{j}$$

This means that the particle will appear to an observer in S' to be moving along the Y -axis with velocity 5.2×10^7 m/sec.

6.13. Conservation of Momentum at Relativistic Speeds –Variation of Mass with Velocity

We assume that the law of conservation of momentum is valid even at relativistic speeds. Below, first we shall show that with the Newtonian definition of momentum ($\mathbf{p} = m_0 \mathbf{v}$), the momentum is not conserved in collision problem at relativistic speeds and then we shall give the necessary modification so that the fundamental law of conservation of momentum retains its validity at any speed.

Consider the collision of two particles A and B of equal masses m_0 in X - Y plane [fig. 6.14]. Let the two particles initially have equal and opposite velocities in a frame S . Therefore, before the collision, x and y components of the velocity for the particle A are $-v_x$ and $-v_y$ and for B are v_x and v_y . Now, if they collide elastically, their X -components of velocity will not change due to collision but the Y -components will have opposite sign.

The changes of the momenta of the particles A and B along Y -direction due to collision will be given by

$$\Delta p_a = m_0 v_y - (-m_0 v_y) = 2m_0 v_y$$

and $\Delta p_b = -m_0 v_y - (m_0 v_y) = -2m_0 v_y$

Hence the total change in the momentum along Y -direction after the collision is

$$\Delta p_a + \Delta p_b = 2m_0 v_y - 2m_0 v_y = 0$$

Thus the Newtonian momentum is conserved along Y -axis. This is also evident from fig.6.14 that the velocities of the particles along X -axis are not changed by the collision and therefore in X -direction the change in momentum is also zero.

Next, we see the collision of the same particles from a frame S' which is moving with velocity v_x along X -axis relative to S (fig. 6.15). Now, according to Lorentz transformations, x - and y -components of the velocities of the particles in frame S' will be seen as :

(1) For particle A , having $-v_x$ and $-v_y$ velocity components in S frame

$$v'_x(A) = \frac{-v_x - v_x}{1 + v_x^2/c^2} = -\frac{2v_x}{1 + v_x^2/c^2} \quad \dots(1a)$$

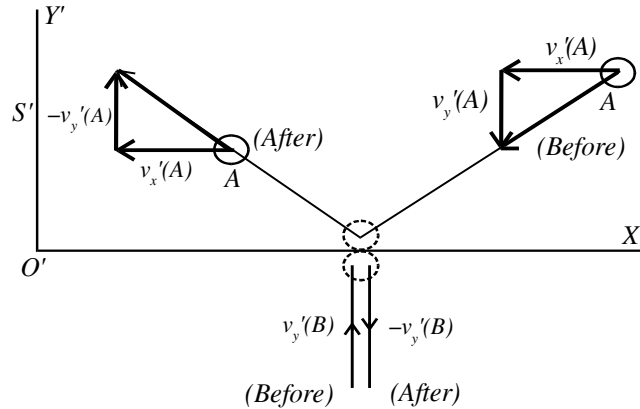


Fig. 6.15 : Collision in S' -frame

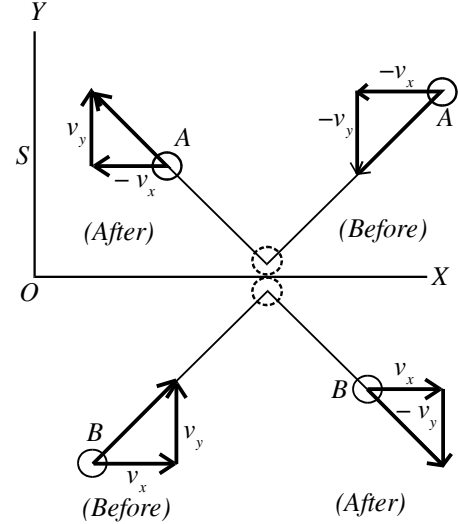


Fig. 6.14 : Collision in S -frame

[because $u'_x = \frac{u_x - v}{1 - vu_x/c^2}$ and here $u_x = -v_x$ in S and $v = v_x$].

$$\text{and} \quad v'_y(A) = \frac{-v_y}{\gamma(1 + v_x^2/c^2)} \quad \dots(1b)$$

(2) For particle B , having $+v_x$ and $+v_y$ velocity components in frame S ,

$$v'_x(B) = \frac{v_x - v_x}{1 - v_x^2/c^2} = 0 \quad \text{and} \quad v'_y(B) = \frac{v_y}{\gamma(1 - v_x^2/c^2)} \quad \dots(2)$$

By the collision the x -components of the velocities of the particles in frame S' are not changed [fig. 6.15] so that the change in momentum of the system along X -axis is zero. But after the collision the signs of $v'_y(A)$ and $v'_y(B)$ are changed and hence according to Newtonian mechanics, the change in y -components of the momenta of the particles will be given by

$$\Delta p'_a = \frac{2m_0 v_y}{\gamma(1 + v_x^2/c^2)} \quad \text{and} \quad \Delta p'_b = -\frac{2m_0 v_y}{\gamma(1 - v_x^2/c^2)} \quad \text{i.e.,} \quad \Delta p'_a + \Delta p'_b \neq 0 \quad \dots(3)$$

Thus in frame S' , there occurs a change in total momentum of the system. Hence, *if the masses of the particles are assumed to be constant at relativistic speeds then the law of conservation of momentum is not valid in all inertial frames.*

For the validity of the conservation of momentum, one has to assume that the mass of a particle depends upon its velocity relative to the frame of reference, chosen for description. Now, if in frame S' the masses of the particles A and B have become m_1 and m_2 respectively, then for the conservation of the momentum, we have

$$\Delta p'_a + \Delta p'_b = \frac{2m_1 v_y}{\gamma(1 + v_x^2/c^2)} - \frac{2m_2 v_y}{\gamma(1 - v_x^2/c^2)} = 0 \quad \text{i.e.,} \quad \frac{m_1}{m_2} = \frac{1 + v_x^2/c^2}{1 - v_x^2/c^2} \quad \dots(4)$$

From eq. (1a), $(v'_x)^2 = \frac{4v_x^2}{(1 + v_x^2/c^2)^2}$. This can be written as

$$1 - \frac{(v'_x)^2}{c^2} = 1 - \frac{4v_x^2/c^2}{(1 + v_x^2/c^2)^2} = \frac{(1 - v_x^2/c^2)^2}{(1 + v_x^2/c^2)^2} \quad \dots(5)$$

From eqs. (4) and (5): $\frac{m_1}{m_2} = \frac{1}{\sqrt{1 - v_x^2/c^2}}$ or $m_1 = \frac{m_2}{\sqrt{1 - v_x^2/c^2}}$

If we consider that the particle B is at rest in frame S' so that $m_2 = m_0$ and A is moving with velocity $v'_x = v$ in this frame, then

$$m_1 = \frac{m_0}{\sqrt{1 - v^2/c^2}} \quad \dots(6)$$

When $v=0$, $m_1=m_0$, this is called the **rest mass** or **proper mass** of the particle. We have considered in our description the particles of equal masses i.e., correctly of equal rest masses. Thus, the mass of a particle in a frame, in which it has the speed v , will be given by

$$m = \frac{m_0}{\sqrt{1 - v^2/c^2}} \quad \dots(7)$$

This relation represents the **variation of mass with velocity**. This result is an important consequence of special theory of relativity and has been verified by several experiments in connection with high energy particles. In 1908, Bucherer showed from his experiment that e/m , ratio of charge and mass, for fast moving

electrons was smaller than that for slow moving electrons, because the mass of electrons with higher speed is more and the charge e remains constant.

Thus the relativistic momentum is given by

$$\vec{p} = m \vec{v} = \frac{m_0 \vec{v}}{\sqrt{1 - v^2/c^2}} \quad \dots(8)$$

With this definition of momentum, our law of conservation of momentum will hold even at relativistic velocities for the two-body collision of identical particles. This may be shown that even if particle B has a different mass from particle A , the above definition will hold in order to conserve the momentum. For $v/c \ll 1$, the expression of momentum reduces to the classical definition $\vec{p} = m_0 \vec{v}$.

Ultimate speed of particles : Now, it can be proved theoretically that no material particle can attain the speed of light c , i.e., c is the ultimate speed of particles. From eq. (7), if a graph is plotted between the mass m of a particle and its speed v [fig. 6.16], then in the condition $v \ll c$, $m = m_0$ which is equal to the classical value, but when the value of v becomes greater than $c/2$, the mass of the particle m increases and this departure from the classical value cannot be neglected. When the particle speed v is approaching the speed of light c , its mass tends to be infinite.

Hence the force, applied to accelerate the particle, increases its mass while speed remains nearly constant. This shows that by accelerating a material particle, it cannot attain the speed equal to or greater than the speed of light. **Thus the ultimate speed of any material particle is always less than the speed of light c .**

6.14 Relativistic Energy : Mass-Energy Relation ($E = mc^2$)

Suppose a force $F = \frac{d}{dt}(mv)$ be acting on a particle of mass m so that its kinetic energy increases. The gain in kinetic energy will be equal to the work done on the particle. If the force displaces the particle through a distance dr along its line of action, then the infinitesimal gain in the kinetic energy is

$$dE_k = F dr = \frac{d(mv)}{dt} dr = v d(mv) \quad [\text{because } v = \frac{dr}{dt}]$$

If the particle starts from rest ($v = 0$) and acquires velocity v under the action of the force, then the gain in the kinetic energy by the particle will be given by

$$E_k = \int dE_k = \int_0^v v d(mv)$$

Integrating this equation by parts, we obtain

$$\begin{aligned} E_k &= mvv \Big|_0^v - \int_0^v mvdv = mv^2 - \int_0^v \frac{m_0 v dv}{\sqrt{1 - v^2/c^2}} * \\ &= \frac{m_0 v^2}{\sqrt{1 - v^2/c^2}} + m_0 c^2 \sqrt{1 - \frac{v^2}{c^2}} - m_0 c^2 \end{aligned}$$

* For integrating put $1 - v^2/c^2 = \alpha$ and hence $v dv = -c^2 d\alpha/2$.

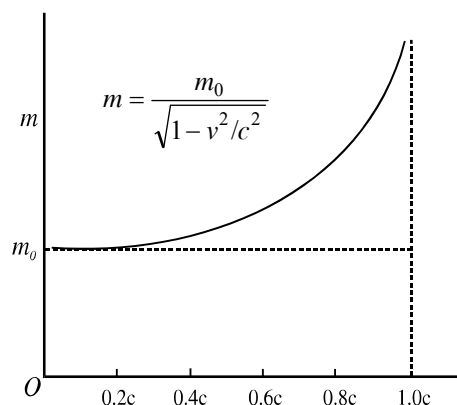


Fig. 6.16 : Variation of mass with velocity

$$= \frac{m_0 c^2}{\sqrt{1 - v^2/c^2}} - m_0 c^2 = mc^2 - m_0 c^2$$

$$\text{Thus} \quad E_k = (m - m_0)c^2 = \frac{m_0 c^2}{\sqrt{1 - v^2/c^2}} - m_0 c^2 \quad \dots(1)$$

where $m = m_0 / \sqrt{1 - v^2/c^2}$.

Eq. (1) is the expression for **relativistic kinetic energy**. It shows that *the gain in kinetic energy corresponds to an increase in mass*.

The quantity $m_0 c^2$, occurring in relation (1), is due to the rest mass of the particle and is called the **rest energy** or **proper energy** E_0 of the particle i.e., $E_0 = m_0 c^2$. Thus, the **total energy** of the particle, when it is moving with velocity v , is

$$\begin{aligned} E &= \text{kinetic energy } (E_k) + \text{Rest energy } (E_0) \\ &= (m - m_0)c^2 + m_0 c^2 = mc^2 \end{aligned}$$

$$\text{Thus} \quad E = \frac{m_0 c^2}{\sqrt{1 - v^2/c^2}} = mc^2 \quad \dots(2)$$

This energy E is called the **relativistic energy** (total energy) of a particle, having relativistic mass m . Thus, there exists a very close relation between mass and energy, unknown in classical physics. This is well known **Einstein's mass-energy relation**.

The relativistic kinetic energy can be expressed as

$$\begin{aligned} E_k &= E - E_0 = (m - m_0)c^2 = m_0 c^2 (1 - v^2/c^2)^{-1/2} - m_0 c^2 \\ E_k &= m_0 c^2 \left[1 + \frac{1}{2} \cdot \frac{v^2}{c^2} + \frac{3}{8} \cdot \frac{v^4}{c^4} + \dots \right] - m_0 c^2 \end{aligned}$$

In the limit $v^2/c^2 \ll 1$, we have

$$E_k = m_0 c^2 \left(1 + \frac{v^2}{2c^2} \right) - m_0 c^2 = \frac{1}{2} m_0 v^2 \quad \dots(3)$$

This relation is the classical result for the kinetic energy.

We see from eq. (1) and (2) that the increase in kinetic energy or total energy ΔE of a particle is associated with a corresponding increase in mass Δm according to the relation,

$$\Delta E = \Delta m c^2 \quad \dots(4)$$

It is known that one kind of energy e.g., kinetic energy can be converted in other forms and hence all forms of energy must be associated with them some mass. According to Einstein, eq. (4) is the most important consequence of the special theory of relativity. He considers that an amount of energy ΔE in any form is equivalent to a mass $\Delta m = \Delta E / c^2$ and conversely, any mass Δm is equivalent to an energy $\Delta E = \Delta m c^2$. This is called the **principle of equivalence of mass and energy**. Thus there is the possibility that mass can be changed into energy and vice-versa. The truth of this fact has been verified by a number of experiments.

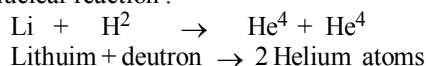
In the language of Einstein, the mass of a body is the measure of the quantity of its energy. This means that a system of inertial mass m is equivalent to an energy $E = mc^2$. Further the rest mass of a body cannot be distinguished from the mass due to the energy possessed by it. Thus, there is the possibility that the rest mass of a body is due to some form of energy and an interchange between rest mass and energy may occur.

6.15. Examples of Mass-Energy Conversion

(1) Electron-positron annihilation : A positron is an elementary particle, having rest mass equal to that of an electron and charge (+ e) equal and opposite to that of an electron. When an electron and a positron come together, they can annihilate with the production of γ -rays. It is found that the energy associated with the γ -rays is equal to the total energy ($2mc^2$) associated with the masses of both particles. Similarly, it has been observed that from γ -rays a pair of electron and positron is produced. These observed phenomena of electron-positron annihilation and pair production confirm most directly the principle of mass-energy equivalence.

(2) Nuclear energy : (i) An enormous amount of destructive energy, obtained from an atom bomb, is a consequence of change of mass into energy. In a U^{235} atom bomb, the nucleus of uranium atom breaks into two parts, having total rest mass slightly less than the original mass of the atom so that the difference of the two masses Δm appears in the form of energy $\Delta E = \Delta mc^2 = \Delta m \times (3 \times 10^8)^2$ joules per atom. A huge amount of energy is released by the disintegration of a large number of U^{235} atoms in an atom bomb.

(ii) Similarly, in some nuclear reactions two light nuclei combine to produce other nuclei, whose total mass is slightly less than the mass of the original nuclei. The difference of these two masses is released in the form of energy. For example, consider the nuclear reaction :



In fact the mass of left hand side atoms is in excess to the mass of right hand side atoms by an amount $\Delta m = 0.02381 \text{ amu}^*$. The calculated energy $\Delta E = \Delta mc^2$ is 22.17 MeV and the observed value 22.2 MeV are in very close agreement.

(iii) The most important source of solar energy is considered the fusion of protons to form helium. The energy release per helium atom, formed from four protons and two electrons, can be calculated from the net change in mass, i.e.,

$$\begin{aligned} \Delta m &= (\text{mass of the 4 protons} + \text{mass of 2 electrons}) - \text{mass of helium atom} \\ &= 4 \times 1.6725 \times 10^{-27} + 2 \times 0.911 \times 10^{-30} - 6.647 \times 10^{-27} = 0.045 \times 10^{-27} \text{ kg.} \end{aligned}$$

Thus the release of energy per atom

$$\Delta E = \Delta mc^2 = 0.045 \times 10^{-27} \times (3 \times 10^8)^2 = 25 \text{ MeV} \quad [1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}]$$

The sun radiates energy at the rate of $3.8 \times 10^{26} \text{ J/sec}$, so its mass is decreasing $3.8 \times 10^{26} / (3 \times 10^8)^2 = 4 \times 10^9 \text{ kg/sec}$.

(3) A number of examples may be quoted, showing the principle of mass-energy equivalence. An internal energy of a body in any form, increases its mass. Thus, a compressed spring is heavier than in the uncompressed state; a hot body loses mass in becoming cold. In such cases, the change of energy is small, hence the change in mass $[\Delta m = \Delta E / (3 \times 10^8)^2]$ is extremely small and is not measurable.

(4) Binding energy : The amount of energy needed to separate the particles (neutrons and protons) of a nucleus to infinite distances is called the *binding energy* of the nucleus. Thus, it is expected that mass of a nucleus to be smaller than the mass of the constituents of the nucleus by an amount $\Delta m = \Delta E/c^2$, where ΔE is the binding energy of the nucleus.

6.15. Relation between Momentum and Energy and Conservation Laws

If a particle of rest mass m_0 moves with a velocity v , its momentum p is given by

$$p = \frac{m_0 v}{\sqrt{1 - v^2/c^2}} \quad m_0 v \gamma \quad \text{or} \quad p^2 = m_0^2 v^2 \gamma^2 \quad \dots(1)$$

* **Atomic mass unit (amu)** is the unit of mass, used in atomic physics and is equal to 1/12th of the rest mass of C^{12} atom.
 $1 \text{ amu} = 1.6604 \times 10^{-27} \text{ kg}$

$$\text{Equivalent energy (1 amu) } \Delta E = 1.6604 \times 10^{-27} \times (3 \times 10^8)^2 = 1.49 \times 10^{10} \text{ J} = \frac{1.49 \times 10^{10}}{1.6 \times 10^{-19}} \text{ eV} = 931 \text{ MeV.}$$

where
$$\gamma^2 = \frac{1}{\sqrt{1 - v^2/c^2}} = \frac{c^2}{c^2 - v^2} \quad \text{or} \quad \gamma^2 v^2 = \gamma^2 c^2 - c^2 \quad \dots(2)$$

Substituting for $\gamma^2 v^2$ in (1) from eq. (2), we have

$$p^2 = m_0^2 \gamma^2 c^2 - m_0^2 c^2 \quad \text{or} \quad p^2 c^2 = m_0^2 \gamma^2 c^4 - m_0^2 c^4$$

or
$$m_0^2 \gamma^2 c^4 = p^2 c^2 + m_0^2 c^4$$

But
$$E = mc^2 = m_0 \gamma c^2, \text{ Therefore,}$$

$$E^2 = p^2 c^2 + m_0^2 c^4 \quad \dots(3)$$

The relation connects the total energy (E) of a particle to its momentum (p). The quantity $m_0^2 c^4$ is constant and hence in relativity a particle has definite energy corresponding to a specific value of its momentum. Thus, the law of conservation of energy will be valid in a collision (even inelastic) where the law of conservation of momentum is assumed to be correct. Thus, according to the law of conservation of momentum in a two particle collision

$$\vec{p}_1 + \vec{p}_2 = \vec{p}_1' + \vec{p}_2' \quad \dots(4)$$

and according to the law of conservation of energy

$$E_1 + E_2 = E_1' + E_2' \quad \dots(5)$$

where E_1, E_2 are the relativistic energies of the particles before the collision and E_1', E_2' after the collision.

From relation (3), we get

$$E^2 - p^2 c^2 = m_0^2 c^4 \quad \dots(6)$$

The quantity $E^2 - p^2 c^2$ is Lorentz invariant. If we transform from one reference frame S to frame S' with $p \rightarrow p'$ and $E \rightarrow E'$, then the invariance of (6) means that

$$E'^2 - p'^2 c^2 = E^2 - p^2 c^2 = m_0^2 c^4$$

This can be proved easily by applying the transformation equations (see Art. 6.17).

We want to emphasize here that the rest mass m_0 of a particle is invariant under Lorentz transformations.

6.17. Transformation of Momentum and Energy

The momentum $\vec{p}$ of a particle, having rest mass m_0 and velocity $\vec{v}$, is given by

$$\vec{p} = m \vec{v} \quad \text{or} \quad \vec{p} = m_0 \vec{v} \gamma \quad \dots(1)$$

where $\gamma = \frac{1}{\sqrt{1 - v^2/c^2}} = \frac{dt}{d\tau}$; $d\tau$ is the proper time interval for the particle and dt is the time interval observed relative to a frame S .

Now,
$$p_x = m_0 v_x \gamma = m_0 \frac{dx}{dt} \frac{dt}{d\tau} \quad \text{or} \quad p_x = m_0 \frac{dx}{d\tau}$$

Similarly,
$$p_y = m_0 \frac{dy}{d\tau} \quad \text{and} \quad p_z = m_0 \frac{dz}{d\tau} \quad \dots(2)$$

and
$$E = \frac{m_0 c^2}{\sqrt{1 - v^2/c^2}} = m_0 c^2 \frac{dt}{d\tau} \quad \text{or} \quad \frac{E}{c^2} = m_0 \frac{dt}{d\tau} \quad \dots(3)$$

The rest mass m_0 and proper interval time $d\tau$ are Lorentz invariants, it follows from eq. (2) and (3) that

p_x, p_y, p_z and E/c^2 must transform from frame S to frame S' under Lorentz transformations exactly as x, y, z, t , transform i.e., similar to the transformations

$$x' = \gamma(x - vt); y' = y; z' = z \text{ and } t' = \gamma(t - vx/c^2) \quad \dots(4)$$

Thus the transformations of energy and momentum are

$$p_x' = \gamma(p_x - vE/c^2); p_y' = p_y; p_z' = p_z; E' = \gamma(E - p_x v) \quad \dots(5)$$

where t has been replaced by E/c^2 .

The inverse transformations are

$$p_x = \gamma(p_x' + vE'/c^2); p_y = p_y'; p_z = p_z'; E = \gamma(E' + p_x' v) \quad \dots(6)$$

Now, if the momentum and energy of a particle are known, we can determine its velocity as

$$v_x = \frac{dx}{dt} = \frac{dx}{d\tau} \frac{d\tau}{dt} = \frac{p_x}{m_0} \frac{m_0 c^2}{E} = \frac{c^2 p_x}{E} \quad \left(p_x = m_0 \frac{dx}{d\tau} \right)$$

$$\text{Thus,} \quad \vec{v} = \frac{c^2 \vec{p}}{E} \quad \text{or} \quad \vec{p} = \frac{\vec{v}}{c^2} E \quad \dots(7)$$

$$\text{Alternatively,} \quad \vec{p} = m \vec{v} = \frac{E}{c^2} \vec{v} \quad (\text{because } E = mc^2)$$

6.18. Particles with Zero Rest Mass

The relativistic energy E of a particle of rest mass m_0 and momentum p is given by

$$E = (p^2 c^2 + m_0^2 c^4)^{1/2} \quad \dots(1)$$

$$\text{When } m_0 = 0, E = pc \text{ or } p = E/c \quad \dots(2)$$

$$\text{But } p = v \frac{E}{c^2}, \text{ hence } \frac{Ev}{c^2} = \frac{E}{c}$$

$$\text{Therefore, } v = c \quad \dots(3)$$

Thus, **a particle of zero rest mass travels with the speed of light**. If m is the equivalent mass of such a particle (as photon) of energy E , then

$$E = mc^2 \text{ or } m = E/c^2 \text{ and } p = mc = E/c \quad \dots(4)$$

In case of a photon of frequency ν , $E = h\nu$, where h is Planck's constant. Therefore, for a **photon** the relativistic mass (m) and momentum (p) will be given by

$$m = \frac{E}{c^2} = \frac{h\nu}{c^2} \text{ and } p = \frac{h\nu}{c} = \frac{h}{\lambda} \quad (\because c = \nu\lambda) \quad \dots(5)$$

Conversely, one can prove that the particles, moving with speed of light, possess zero rest mass.

Ex. 1. At what speed the mass of an object will be double of its value at rest. (A.N.U. March 2007, 03)

Sol. The mass of a body varies with the speed according to the relation

$$m = \frac{m_0}{\sqrt{1 - v^2/c^2}} \text{ whence } 1 - \frac{v^2}{c^2} = \frac{m_0^2}{m^2} \text{ or } \frac{v^2}{c^2} = 1 - \frac{m_0^2}{m^2} \text{ or } v = c \sqrt{1 - \frac{m_0^2}{m^2}}$$

$$\text{Here, } c = 3 \times 10^8 \text{ m/sec and } \frac{m_0}{m} = \frac{m_0}{2m_0} = \frac{1}{2}$$

$$\therefore v = 3 \times 10^8 \sqrt{1 - \left(\frac{1}{2}\right)^2} = 3 \times 10^8 \times \sqrt{\frac{3}{4}} = 2.6 \times 10^8 \text{ m/sec.}$$

Ex. 2. At what speed, the mass of a body will be one and half times its rest mass ? (O.U. 2009)

Sol.
$$m = \frac{m_0}{\sqrt{1-v^2/c^2}} \text{ or } \frac{m}{m_0} = \frac{1}{\sqrt{1-v^2/c^2}}$$

Here, $\frac{m}{m_0} = \frac{3}{2}$ and $c = 3 \times 10^8$ m/sec.

$$\therefore \frac{3}{2} = \frac{1}{\sqrt{1-v^2/c^2}} \text{ or } 1 - \frac{v^2}{c^2} = \left(\frac{2}{3}\right)^2 = \frac{4}{9}$$

$$\therefore \frac{v^2}{c^2} = 1 - \frac{4}{9} = \frac{5}{9} \text{ or } v = c \sqrt{\frac{5}{9}} = 3 \times 10^8 \times \frac{\sqrt{5}}{3} = 10^8 \sqrt{5} = 2.24 \times 10^8 \text{ m/sec.}$$

Ex. 3. If the total energy of a particle is exactly thrice its rest energy, what is the velocity of the particle. (A.N.U. 2003, 02)

Sol. According to Eienstein's mass-energy relation, total energy E of particle of mass m is given by

$$E = mc^2$$

If m_0 be the rest mass, then rest mass energy $= m_0 c^2$

Here, $E = mc^2 = 3m_0 c^2$ or $m = 3m_0$.

But $m = \frac{m_0}{\sqrt{1-v^2/c^2}},$

$$\therefore 3m_0 = \frac{m_0}{\sqrt{1-v^2/c^2}} \text{ or } 1 - \frac{v^2}{c^2} = \left(\frac{1}{3}\right)^2 \text{ or } \frac{v^2}{c^2} = 1 - \frac{1}{9} = \frac{8}{9}$$

or $\frac{v}{c} = \sqrt{\frac{8}{9}} \text{ or } v = c \sqrt{\frac{8}{9}} = 3 \times 10^8 \times \frac{2\sqrt{2}}{3} = 2.8 \times 10^8 \text{ m/sec.}$

Ex. 4. The rest mass of an electron is 9.1×10^{-31} kg. What will be its mass if it were moving with $(4/5)^{\text{th}}$ the speed of light. (A.N.U. 1997)

Sol.
$$m = \frac{m_0}{\sqrt{1-v^2/c^2}}$$

Here, $m_0 = 9.1 \times 10^{-31}$, kg, $v = \frac{4}{5} c = 0.8c$

$$\begin{aligned} \therefore m &= \frac{9.1 \times 10^{-31}}{\sqrt{1-(0.8c/c)^2}} = \frac{9.1 \times 10^{-31}}{\sqrt{1-0.64}} = \frac{9.1 \times 10^{-31}}{\sqrt{0.36}} \\ &= \frac{9.1 \times 10^{-31}}{0.6} = 15.16 \times 10^{-31} \text{ kg.} \end{aligned}$$

Ex. 5. A particle is moving with 90% of the velocity of light. Compare its relativistic mass with its rest mass. (A.N.U. 1996)

Sol. Relativistic mass $m = \frac{m_0}{\sqrt{1-v^2/c^2}}$, m_0 = rest mass

$$\begin{aligned}\therefore \frac{m}{m_0} &= \frac{1}{\sqrt{1-v^2/c^2}} = \frac{1}{\sqrt{1-(0.9)^2}}, \text{ because } v = \frac{90}{100} \times c = 0.9c \\ &= \frac{1}{\sqrt{1-0.81}} = \frac{1}{\sqrt{0.19}} = \mathbf{2.29}\end{aligned}$$

Ex. 6. With what velocity should a particle travel so that its relativistic mass increases by 10% of its rest mass? (K.U. 1995)

Sol. Relativistic mass $m = \frac{m_0}{\sqrt{1-v^2/c^2}}$, m_0 = rest mass

Given, $m = m_0 + \frac{10}{100} \times m_0 = m_0 + 0.1 m_0 = 1.1 m_0$

$$\therefore 1.1 m_0 = \frac{m_0}{\sqrt{1-v^2/c^2}} \text{ or } 1 - \frac{v^2}{c^2} = \frac{1}{(1.1)^2} \text{ or } \frac{v^2}{c^2} = 1 - \frac{1}{(1.1)^2} = 1 - 0.826 = 0.174$$

$$\therefore v = c \sqrt{0.174} = 3 \times 10^8 \times 0.4166 = \mathbf{1.25 \times 10^8 \text{ m/sec.}}$$

Ex. 7. Deduce the fractional increase of mass of a particle for velocity $0.1c$.

Sol. $m = \frac{m_0}{\sqrt{1-v^2/c^2}}$, $\therefore \frac{m}{m_0} = \frac{1}{\sqrt{1-(0.1c/c)^2}} = \frac{1}{\sqrt{1-0.01}} = \frac{1}{\sqrt{1-99}} = 1.005$

Now, fractional increase of mass $= \frac{m-m_0}{m_0} = \frac{m}{m_0} - 1 = 1.005 - 1 = \mathbf{0.005}$

Ex. 8. An electron and a positron practically at rest come together and annihilate each other. Calculate the energy released.

Solution : Rest mass of the electron and positron

$$M_0 = 2m_e = 2 \times 0.9 \times 10^{-30} \text{ kg.}$$

Energy released $M_0 c^2 = 2 \times 0.9 \times 10^{-30} \times 9 \times 10^{16} = \mathbf{1.62 \times 10^{-13} \text{ J.}}$

Ex. 9. An α -particle is accelerated from rest to a velocity equal to 0.9 times that of light. By how much will its mass increase? Take the rest mass of α -particle as $6.65 \times 10^{-27} \text{ kg}$. (K.U. 1990)

Sol. $m = \frac{m_0}{\sqrt{1-v^2/c^2}}$

$$\begin{aligned}\text{Increase in mass} = m - m_0 &= \frac{m_0}{\left(1 - \frac{v^2}{c^2}\right)^{1/2}} - m_0 = m_0 \left[\left(1 - \frac{v^2}{c^2}\right)^{-1/2} - 1 \right] \\ &= \left[1 + \frac{v^2}{2c^2} - 1 \right] = \frac{m_0 v^2}{2c^2} \quad (\text{using Binomial theorem})\end{aligned}$$

Here, $v = 0.9c$ and $m_0 = 6.65 \times 10^{-27} \text{ kg}$

$$\therefore \text{Increase in mass} = \frac{6.65 \times 10^{-27} \times (0.9c)^2}{2 \times c^2} = \mathbf{5.267 \times 10^{-27} \text{ kg}}$$

Ex. 10. Calculate the speed of an electron which has kinetic energy 2 MeV.

Sol. Kinetic energy $= (m - m_0)c^2 = 2 \text{ MeV}$

$$\begin{aligned}\text{Therefore} \quad m &= m_0 + \frac{2\text{MeV}}{c^2} = 0.91 \times 10^{-30} + \frac{2 \times 10^6 \times 1.6 \times 10^{-19}}{(3 \times 10^8)^2} \\ &= 0.91 \times 10^{-30} + 3.45 \times 10^{-30} = 4.36 \times 10^{-30} \text{ kg}.\end{aligned}$$

$$\text{But} \quad m = \frac{m_0}{\sqrt{1 - v^2/c^2}} \quad \text{or} \quad \frac{v^2}{c^2} = 1 - \left(\frac{m_0}{m}\right)^2, \text{ therefore}$$

$$v = c \sqrt{1 - \left(\frac{m_0}{m}\right)^2} = 3 \times 10^8 \sqrt{1 - \left(\frac{0.91 \times 10^{-30}}{4.36 \times 10^{-30}}\right)^2} = \mathbf{2.93 \times 10^8 \text{ m/sec}}$$

Ex. 11. What is the increase in the relativistic mass of a particle of rest mass 1 gm when it is moving with 0.8 c velocity? Hence find its K.E.

$$\text{Sol.} \quad m = \frac{m_0}{\sqrt{1 - v^2/c^2}} = \frac{10^{-3}}{\sqrt{1 - (0.8)^2}} = \frac{10^{-3}}{0.6} = 1.677 \times 10^{-3} \text{ kg}$$

$$\therefore \text{Increase in mass} = 1.667 \times 10^{-3} - 10^{-3} = \mathbf{0.667 \times 10^{-3} \text{ kg}}$$

$$\therefore \text{K.E.} = (m - m_0)c^2 = 0.667 \times 10^{-3} \times (3 \times 10^8)^2 = \mathbf{6 \times 10^{13} \text{ J.}}$$

Ex. 12. Calculate the K.E. of an electron moving with velocity of 0.98 times the velocity of light in the laboratory frame. $m_0 = 9.11 \times 10^{-31} \text{ kg}$. (A.N.U. 1996)

Sol. Kinetic energy = Total energy – rest energy $= mc^2 - m_0c^2$

$$= \frac{m_0c^2}{\sqrt{1 - v^2/c^2}} - m_0c^2 = m_0c^2 \left[\frac{1}{\sqrt{1 - v^2/c^2}} - 1 \right]$$

Here, $m_0 = 9.11 \times 10^{-31} \text{ kg}$ and $v = 0.98 c$

$$\begin{aligned}\therefore \text{Kinetic energy} &= 9.11 \times 10^{-31} \times (3 \times 10^8)^2 \left[\frac{1}{\sqrt{1 - (0.98)^2}} - 1 \right] = 8.199 \times 10^{-14} (5.025 - 1) \text{ joule} \\ &= \frac{9.199 \times 10^{-14} \times 4.025 \text{ eV}}{1.6 \times 10^{-19}} = 2.06 \times 10^6 \text{ eV} = \mathbf{2.06 \text{ MeV}}\end{aligned}$$

Ex. 13. Find the speed that a proton must be given if its mass is to be twice its rest mass of 1.67×10^{-27} kg. What energy must be given to proton to achieve this speed. (A.U. 1993)

Sol.
$$m = \frac{m_0}{\sqrt{1 - v^2/c^2}} \text{ or } 1 - \frac{v^2}{c^2} = \left(\frac{m_0}{m}\right)^2 \text{ or } \frac{v^2}{c^2} = 1 - \left(\frac{m_0}{m}\right)^2 \text{ or } v = c \sqrt{1 - \left(\frac{m_0}{m}\right)^2}$$

Here, $\frac{m_0}{m} = \frac{m_0}{2m_0} = \frac{1}{2}$,

$$\therefore v = c \sqrt{1 - \left(\frac{1}{2}\right)^2} = 3 \times 10^8 \times \frac{\sqrt{3}}{2} = 2.6 \times 10^8 \text{ m/sec.}$$

Kinetic energy $= mc^2 - m_0c^2 = 2m_0c^2 - m_0c^2 = m_0c^2$

$$= 1.67 \times 10^{-27} \times (3 \times 10^8)^2 \text{ joule} = \frac{1.67 \times 10^{-27} \times (3 \times 10^8)^2 \text{ eV}}{1.6 \times 10^{-19}}$$

$$= 9.394 \times 10^8 \text{ eV} = \mathbf{939.4 \text{ MeV}}$$

Ex. 14. Find the energy in electron volts released when a neutron decays into a proton and an electron. ($m_n = 1.6747 \times 10^{-27}$ kg; $m_p = 1.6725 \times 10^{-27}$ kg; $m_e = 0.9 \times 10^{-30}$ kg.)

Sol. : Change of mass in the decaying process can be represented as

$$\Delta m = \text{Mass of the neutron} - (\text{Mass of proton} + \text{Mass of electron})$$

$$= 1.6747 \times 10^{-27} - [1.6725 \times 10^{-27} + 0.9 \times 10^{-30}]$$

$$= 1.6747 \times 10^{-27} - 1.6734 \times 10^{-27} = 0.0013 \times 10^{-27} \text{ kg.}$$

Hence, energy liberated

$$\Delta E = \Delta mc^2 = 0.0013 \times 10^{-27} \times (3 \times 10^8)^2 \text{ J} = 13 \times 9 \times 10^{-15} \text{ J}$$

$$= \frac{13 \times 9 \times 10^{-15}}{1.6 \times 10^{-19}} \text{ eV} = \mathbf{7.3 \times 10^5 \text{ eV}}$$

Ex. 15. The rest mass of a proton is 1.6725×10^{-27} kg. Find its mass and momentum, when it is moving with 2.7×10^8 m/sec velocity. If it collides with a stationary nucleus of mass 2.7×10^{-26} kg and coalesces, find the velocity of the combined particle.

Sol.
$$m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}} = \frac{1.6725 \times 10^{-27}}{\sqrt{1 - \left(\frac{2.7 \times 10^8}{3 \times 10^8}\right)^2}} = \frac{1.6725 \times 10^{-27}}{\sqrt{1 - 0.81}} = 3.84 \times 10^{-27} \text{ kg.}$$

Momentum $p = mv = 3.84 \times 10^{-27} \times 2.7 \times 10^8 = \mathbf{1.04 \times 10^{-18} \text{ kg-m/sec.}}$

If the velocity of the final particle is V , then using the law of conservation of momentum, we have

$$p = (m_0 + M) V$$

[As V will be much lower than c , hence we will consider only rest masses of the particles.]

Therefore,
$$V = \frac{1.04 \times 10^{-18}}{(1.6725 \times 10^{-27} + 2.5 \times 10^{-26})} = \frac{1.04 \times 10^{-18}}{2.66725 \times 10^{-26}} = \mathbf{3.9 \times 10^7 \text{ m/sec.}}$$

Ex. 16. Express for the momentum of a photon in terms of wavelength λ . How much is the rest mass of the photon. Calculate the relativistic mass of the photon of wavelength 5000 Å

Sol. Momentum of the photon

$$p = \frac{E}{c} = \frac{h\nu}{c} = \frac{h}{\lambda} \quad [\text{as } c = \nu\lambda]$$

Also

$$p = \frac{m_0 v}{\sqrt{1 - v^2/c^2}} \quad \text{or} \quad m_0 = \frac{p\sqrt{1 - v^2/c^2}}{v}$$

For photon, $v = c$, therefore, $1 - v^2/c^2 = 1 - 1 = 0$; hence, $m_0 = 0$. Thus the rest mass of photon is **zero**.

$$\text{Relativistic mass } m = \frac{p}{c} = \frac{h}{c\lambda} = \frac{6.62 \times 10^{-34}}{3 \times 10^8 \times 5000 \times 10^{-10}} = 4 \times 10^{-36} \text{ kg}.$$

Ex. 17. Show that the rest mass of a particle of momentum p and kinetic energy T given by

$$m_0 = \frac{p^2 c^2 - T^2}{2Tc^2}$$

Sol. We know that the relativistic energy E is given by

$$E = E_k + E_0 \quad \text{or} \quad E = T + m_0 c^2 \quad \dots(i)$$

Also

$$E^2 = p^2 c^2 + m_0^2 c^4 \quad \dots(ii)$$

Squaring (i) and equating with (ii), we get

$$T^2 + m_0^2 c^4 + 2Tm_0 c^2 = p^2 c^2 + m_0^2 c^4 \quad \text{or} \quad m_0 = \frac{p^2 c^2 - T^2}{2Tc^2}.$$

Ex. 18 Show that if E and p are relativistic energy and momentum in S -frame, then

$$E'^2 - p'^2 c^2 = E^2 - p^2 c^2 = m_0^2 c^4$$

where E' and p' are the corresponding quantities in S' -frame. Prove that $E^2 = p^2 c^2 + m_0^2 c^4$ is invariant under Lorentz transformations.

Sol. : The transformation of energy and momentum from S to S' frame are

$$p'_x = \gamma(p_x - vE/c^2), p'_y = p_y, p'_z = p_z \text{ and } E' = \gamma(E - vp_x)$$

The relation $E^2 = p^2 c^2 + m_0^2 c^4$ is Lorentz invariant means that for a particle in S -frame if $E^2 = p^2 c^2 + m_0^2 c^4$,

then for the same particle in S' -frame, moving with constant velocity relative to S , we have

$$E'^2 = p'^2 c^2 + m_0^2 c^4$$

In other words

$$E'^2 - p'^2 c^2 = E^2 - p^2 c^2 = m_0^2 c^4$$

Applying transformation relations

$$\begin{aligned} E'^2 - p'^2 c^2 &= E'^2 - c^2(p_x'^2 + p_y'^2 + p_z'^2) \\ &= \gamma^2 (E - vp_x)^2 - c^2 \left[\gamma^2 \left(p_x - \frac{vE}{c^2} \right)^2 + c^2 p_y^2 + c^2 p_z^2 \right] \\ &= -p_x^2 \gamma^2 c^2 \left(1 - \frac{v^2}{c^2} \right) + \gamma^2 E^2 \left(1 - \frac{v^2}{c^2} \right) + c^2 p_y^2 + c^2 p_z^2 \end{aligned}$$

$$= E^2 - c^2(p_x^2 + p_y^2 + p_z^2) = E^2 - p^2 c^2$$

Thus $E^2 - p^2 c^2 = E^2 - p^2 c^2 = m_0^2 c^4$.

Ex. 19. Assume that the sun gets energy by fusion of four hydrogen atoms into a helium atom. The rest mass of hydrogen and helium atom are 1.0081 and 4.0039 amu respectively. Calculate the energy released in each fusion reaction. (1 amu = 1.67×10^{-27} kg).

Sol. Loss of mass in each fusion reaction Δm is given by

$$\begin{aligned}\Delta m &= \text{Mass of four hydrogen atom} - \text{mass of 1 helium atom} \\ &= 4 \times 1.0081 - 4.0039 = 0.0285 \text{ amu} \\ &= 0.0285 \times (1.67 \times 10^{-27}) \text{ kg} = 4.76 \times 10^{-29} \text{ kg}\end{aligned}$$

Hence the energy released in one fusion reaction is given by

$$\begin{aligned}E &= mc^2 = (4.76 \times 10^{-29} \text{ kg}) (3 \times 10^8 \text{ m/sec})^2 = 4.28 \times 10^{12} \text{ joule} \\ &= \frac{4.28 \times 10^{-12}}{1/6 \times 10^{-19}} = 2.67 \text{ MeV}.\end{aligned}$$

6.19. Concept of Four Vector Formulation – Four Dimensional Space

In accordance with the two postulates of the special theory of relativity, namely the constancy of the speed of light in vacuum and the invariance of the basic laws of physics in inertial frames, we deduced earlier the Lorentz transformations. These transformations connect the space-time coordinates (x, y, z, t) and (x', y', z', t') of an event in two inertial frames S and S' and are given by

$$x' = \gamma(x - vt), \quad y' = y, \quad z' = z \quad \text{and} \quad t' = \gamma\left(t - \frac{vx}{c^2}\right) \quad \dots(1)$$

where $\gamma = 1/\sqrt{1 - v^2/c^2}$, the frame S' is moving with constant velocity v along X -axis relative to the frame S .

We find that in relativistic mechanics, the space and time coordinates depend on each other. The time coordinate of one inertial system depends on both the space and time coordinates of another system [$t' = \gamma(t - vx/c^2)$]. Therefore, instead of treating the space and time coordinates separately, it is natural to seek the way so that both the coordinates are dealt together similarly. In fact, H. Minkowski was the first to develop a procedure in which the time coordinate is treated similar to the three space coordinates.

Position Vector in Four Dimensional Space

Minkowski considered a four dimensional cartesian space in which the position is specified by three coordinates x, y, z and the time is referred by a fourth coordinate ict . If we write $x_1 = x$, $x_2 = y$, $x_3 = z$ and $x_4 = ict$, then an event is represented by a point with four coordinates (x_1, x_2, x_3, x_4) ($x_4 = ict$) in the four dimensional space. This point in the four dimensional space is called **world point**. The **position vector** of an event or world point is represented as

$$x_\mu = (x_1, x_2, x_3, x_4) = (\vec{r}, ict) \quad \dots(2)$$

where (x_1, x_2, x_3) or (x, y, z) represent the position vector $\vec{r}$ of the point in three dimensional space and

$x_4 = ict$. $\vec{r}$ ($= x, y, z$) is the space part and ict is the time part of the four dimensional position vector x_μ . Of course, the fourth dimension, referring to time is imaginary. The position vector x_μ of an event in four dimensional space is called **position four-vector**.

The four dimensional space in which the position of an event is represented by the position vector $x_\mu = (x_1, x_2, x_3, x_4)$ ($x_4 = ict$) is called **Minkowski** or **world space**. It is also referred as **space-time-continuum** and sometimes briefly as **four-space**. The square of the magnitude (or length) of the position vector in such a four-space has the form

$$s^2 = x_1^2 + x_2^2 + x_3^2 + x_4^2 = \sum_{\mu=1}^4 x_{\mu}^2 \quad \dots(3)$$

where

$$x_4 = ict.$$

Lorentz transformations are designed so that the speed of light remains constant in S and S' inertial frames (S' is moving with constant velocity v relative to S) and this condition is equivalent to require that the square of the length of the position vector in the four-space is the same under the Lorentz transformations i.e.,

$$s^2 = x^2 + y^2 + z^2 - c^2 t^2 = x'^2 + y'^2 + z'^2 - c^2 t'^2$$

or

$$s^2 = x_1'^2 + x_2'^2 + x_3'^2 + x_4'^2 = x_1^2 + x_2^2 + x_3^2 + x_4^2$$

or

$$s^2 = \sum_{\mu=1}^4 x_{\mu}'^2 = \sum_{\mu=1}^4 x_{\mu}^2$$

This means that **the position vector in the four-space is held invariant under the Lorentz transformations.**

The Lorentz transformations, given by (1), from S -frame to S' -frame can be represented as

$$x'_1 = \gamma(x_1 + i\beta x_4), x'_2 = x_2, x'_3 = x_3 \text{ and } x'_4 = \gamma(-i\beta x_1 + x_4) \quad \dots(5)$$

where we have written $x = x_1, y = x_2, z = x_3, t = x_4/ic$ with $\beta = v/c$ and $\gamma = 1/\sqrt{1-v^2/c^2}$.

In matrix notation, the Lorentz transformations [eqs. (5)] from S -frame to S' -frame will be given by

$$\begin{pmatrix} x'_1 \\ x'_2 \\ x'_3 \\ x'_4 \end{pmatrix} = \begin{pmatrix} \gamma & 0 & 0 & i\gamma\beta \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -i\gamma\beta & 0 & 0 & \gamma \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} \quad \dots(6)$$

This is the four-vector formulation for position coordinates in four dimensional space.

Four-Vectors

A vector in four dimensional Minkowski space is called a **four-vector**. Its components transform from one frame to another similar to Lorentz transformations.

A four-vector A_{μ} is a vector in four dimensional space with components A_1, A_2, A_3 and A_4 and is represented as

$$A_{\mu} = (A_1, A_2, A_3, A_4) = (\vec{A}, iA_4) \quad \dots(7)$$

where $\vec{A} (= A_1, A_2, A_3)$ is the space component and $A_4 (= iA_t)$ is the time component. These components transform from S -frame to S' -frame similar to Lorentz transformations i.e.,

$$\begin{pmatrix} A'_1 \\ A'_2 \\ A'_3 \\ A'_4 \end{pmatrix} = \begin{pmatrix} \gamma & 0 & 0 & i\gamma\beta \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -i\gamma\beta & 0 & 0 & \gamma \end{pmatrix} \begin{pmatrix} A_1 \\ A_2 \\ A_3 \\ A_4 \end{pmatrix} \quad \dots(8a)$$

i.e.,

$$A'_1 = \gamma(A_1 + i\beta A_4), A'_2 = A_2, A'_3 = A_3, A'_4 = \gamma(-i\beta A_1 + A_4) \quad \dots(8b)$$

These transformations are governed by the condition

$$\sum_{\mu=1}^4 A_{\mu}'^2 = \sum_{\mu=1}^4 A_{\mu}^2 \quad \dots(9)$$

The square of the magnitude of the four vector is given by

$$\sum_{\mu=1}^4 A_{\mu}^2 = A_1^2 + A_2^2 + A_3^2 + A_4^2 = A_1^2 + A_2^2 + A_3^2 - A_t^2 \quad \dots(10)$$

Position four-vector is an example of four vector and is represented as

$$x_{\mu} = (x_1, x_2, x_3, x_4) = (x, y, z, ict) \quad \dots(11)$$

Other examples of four vectors are velocity four-vector, momentum four-vector, acceleration four-vector etc., but we shall not go here in further details of the four-vector analysis.

Theorem 1. Show that the length of a four position vector remains unchanged under Lorentz transformations.

Proof. Four position vector is given by

$$x_{\mu} = (x_1, x_2, x_3, x_4) \quad (x_4 = ict)$$

Lorentz transformation are

$$x'_1 = \gamma(x_1 + i\beta x_4), x'_2 = x_2, x'_3 = x_3 \text{ and } x'_4 = \gamma(-i\beta x_1 + x_4)$$

where $\beta = v/c$ and $\gamma = 1 / \sqrt{1 - v^2 / c^2}$.

The length of a four position vector remains unchanged under Lorentz transformation means

$$\sum_{\mu=1}^4 x'^2_{\mu} = \sum_{\mu=1}^4 x^2_{\mu}$$

$$\text{or } x'^2_1 + x'^2_2 + x'^2_3 + x'^2_4 = x_1^2 + x_2^2 + x_3^2 + x_4^2$$

In order to prove the above, let us take the expression of left hand side *i.e.*,

$$\begin{aligned} x'^2_1 + x'^2_2 + x'^2_3 + x'^2_4 &= \gamma^2 (x_1 + i\beta x_4)^2 + x_2^2 + x_3^2 + \gamma^2 (-i\beta x_1 + x_4)^2 \\ &= \gamma^2 x_1^2 - \gamma^2 \beta^2 x_4^2 + 2i\beta \gamma^2 x_1 x_4 + x_2^2 + x_3^2 - \gamma^2 \beta^2 x_1^2 + \gamma^2 x_4^2 - 2i\beta \gamma^2 x_1 x_4 \\ &= \gamma^2 x_1^2 (1 - \beta^2) + x_2^2 + x_3^2 + \gamma^2 x_4^2 (1 - \beta^2) \\ &= x_1^2 + x_2^2 + x_3^2 + x_4^2 \quad [\because \gamma^2 (1 - \beta^2) = \frac{(1 - v^2 / c^2)}{(1 - v^2 / c^2)} = 1] \end{aligned}$$

$$\text{This means } \sum_{\mu=1}^4 x'^2_{\mu} = \sum_{\mu=1}^4 x^2_{\mu}$$

Theorem 2. Show that the length of a four vector remains unchanged under Lorentz transformation.

Proof. Four vector is represented as

$$A_{\mu} = (A_1, A_2, A_3, A_4) \quad (A_4 = iA_t)$$

Lorentz transformations for a four-vector from S -frame to S' -frame are

$$A'_1 = \gamma(A_1 + i\beta A_4), A'_2 = A_2, A'_3 = A_3 \text{ and } A'_4 = \gamma(-i\beta A_1 + A_4)$$

where $\beta = v/c$ and $\gamma = 1 / \sqrt{1 - v^2 / c^2}$

The length of a four vector $A_{\mu} = (A_1, A_2, A_3, A_4)$ remains unchanged under Lorentz transformations means

$$\sum_{\mu=1}^4 A'^2_{\mu} = \sum_{\mu=1}^4 A^2_{\mu}$$

$$\text{or } A'^2_1 + A'^2_2 + A'^2_3 + A'^2_4 = A^2_1 + A^2_2 + A^2_3 + A^2_4$$

To prove the above relation, let us take the expression of left hand side *i.e.*,

$$\begin{aligned} A'^2_1 + A'^2_2 + A'^2_3 + A'^2_4 &= \gamma^2 (A_1 + i\beta A_4)^2 + A_2^2 + A_3^2 + \gamma^2 (-i\beta A_1 + A_4)^2 \\ &= \gamma^2 A_1^2 - \gamma^2 \beta^2 A_4^2 + 2i\beta \gamma^2 A_1 A_4 + A_2^2 + A_3^2 - \gamma^2 \beta^2 A_1^2 + \gamma^2 A_4^2 - 2i\beta \gamma^2 A_1 A_4 \\ &= \gamma^2 A_1^2 (1 - \beta^2) + A_2^2 + A_3^2 + \gamma^2 A_4^2 (1 - \beta^2) \\ &= A_1^2 + A_2^2 + A_3^2 + A_4^2 \quad \left[\because \gamma^2 (1 - \beta^2) = \frac{(1 - v^2/c^2)}{(1 - v^2/c^2)} = 1 \right] \end{aligned}$$

This means
$$\sum_{\mu=1}^4 A'^2_{\mu} = \sum_{\mu=1}^4 A^2_{\mu}$$

Ex. 1. Space Time Interval between Events. If two events are having the space and time coordinates (x_1, y_1, z_1, t_1) and (x_2, y_2, z_2, t_2) respectively in a frame S , then the square of space-time interval between them is defined as

$$s_{12}^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2 - c^2 (t_2 - t_1)^2$$

Show that this interval is invariant under Lorentz transformations.

Sol. Let us consider the above given two events relative to a frame S' moving relative to S along the common X -axis with a velocity v . If the space and time coordinates of the two events in S' are (x'_1, y'_1, z'_1, t'_1) and (x'_2, y'_2, z'_2, t'_2) respectively then according to Lorentz transformations

$$x'_1 = \gamma (x_1 - vt_1), y'_1 = y_1, z'_1 = z_1 \text{ and } t'_1 = \gamma (t_1 - vx_1/c^2)$$

$$\text{and } x'_2 = \gamma (x_2 - vt_2), y'_2 = y_2, z'_2 = z_2 \text{ and } t'_2 = \gamma (t_2 - vx_2/c^2)$$

$$\therefore x'_2 - x'_1 = \gamma (x_2 - x_1) - v(t_2 - t_1), y'_2 - y'_1 = y_2 - y_1, z'_2 - z'_1 = z_2 - z_1$$

$$\text{and } t'_2 - t'_1 = \gamma \{ (t_2 - t_1) - v(x_2 - x_1)/c^2 \} \quad \dots(i)$$

$$\begin{aligned} \text{Now } s_{12}'^2 &= (x'_2 - x'_1)^2 + (y'_2 - y'_1)^2 + (z'_2 - z'_1)^2 - c^2 (t'_2 - t'_1)^2 \\ &= \gamma^2 \{ (x_2 - x_1) - v(t_2 - t_1) \}^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2 - c^2 \gamma^2 \{ (t_2 - t_1) - v(x_2 - x_1)/c^2 \}^2 \\ &= (x_2 - x_1)^2 \gamma^2 (1 - v^2/c^2) + (y_2 - y_1)^2 + (z_2 - z_1)^2 - c^2 (t_2 - t_1)^2 \gamma^2 (1 - v^2/c^2) \end{aligned}$$

$$\text{or } s_{12}'^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2 - c^2 (t_2 - t_1)^2 \quad \left(\because \gamma^2 = \frac{1}{1 - v^2/c^2} \right)$$

$$\text{Thus } s_{12}'^2 = s_{12}^2 \dots(ii)$$

Note. If we examine equation (i), we see (a) that the *time interval* between the two events is not equal in S and S' *i.e.*, $t_2 - t_1 \neq t'_2 - t'_1$, (b) that the *space interval* is not the same in two reference systems *i.e.*,

$$(x'_2 - x'_1)^2 + (y'_2 - y'_1)^2 + (z'_2 - z'_1)^2 \neq (x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2$$

In case of classical mechanics, the time interval $(t_2 - t_1 = t'_2 - t'_1)$ and the space interval both are invariant, *i.e.*, independent of frame of reference used for the measurements. Thus *the theory of relativity discards the classical ideas of space and time from the very beginning and tells that space-time interval between two events is invariant.*

Ex. 2. In frame S , two events have the space-time coordinates $(0, 0, 0, 0)$ and $(5c, 0, 0, 3)$, where time coordinate is in seconds. Find the space-time interval between them. Calculate the velocity of a frame in which

- (a) the two events are simultaneous,
- (b) the first event occurs 1 sec earlier than the second,
- (c) the second event occurs 1 sec earlier than the first.

What is the limit for the maximum time interval between these events ?

Sol. $s_{12}^2 = \Delta x^2 + \Delta y^2 + \Delta z^2 - c^2 \Delta t^2 = 25c^2 - 9c^2 = 16c^2$

Therefore, space-time interval $s_{12} = 4c$, which is space-like.

(a) Let the velocity of a frame S' relative to S be v so that the two events are simultaneous (i.e., $\Delta t' = 0$) in the frame S' . Therefore,

$$\Delta t' = \gamma \left(\Delta t - \frac{v \Delta x}{c^2} \right) = 0 \text{ or } \Delta t = \frac{v \Delta x}{c^2}$$

or $3 = \frac{v \times 5c}{c^2}$, whence, $v = 0.6c$

(b) In S' -frame, the first event occurs 1 sec earlier than the second i.e., $\Delta t' = 1$ sec. Therefore

$$\Delta t' = \gamma \left(\Delta t - \frac{v \Delta x}{c^2} \right) = 1 \text{ or } 1 = \frac{1}{\sqrt{1 - v^2/c^2}} \left(3 - \frac{v \times 5c}{c^2} \right)$$

Squaring and arranging the terms, we get the equation

$$13 \frac{v^2}{c^2} - 15 \frac{v}{c} + 4 = 0, \text{ hence } \frac{v}{c} = 0.7c \text{ or } 0.4c.$$

For having $\Delta t' = +1$ sec, $v = 0.4c$ because $v = 0.7c$ will make $\Delta t'$ negative.

(c) If the second event occurs 1 sec earlier than the first in frame S' ; then $\Delta t' = -1$ sec. So that

$$-1 = \frac{1}{\sqrt{1 - v^2/c^2}} \left(3 - \frac{v \times 5c}{c^2} \right).$$

On squaring and arranging the terms we will get the same equation, as obtained above, and for making $\Delta t' = -1$ sec, the first value i.e., $v = 0.7c$ will be admissible.

There is no limit for the time interval of the two events whose space-time interval is space-like; the time interval may have any value between $\Delta t' = -\infty$ to $\Delta t' = +\infty$.

Ex. 3. Prove that the three dimensional volume element $dx dy dz$ is not invariant under Lorentz transformations while the four dimensional volume element $dx dy dz dt$ is invariant.

Solution : In S -frame let the three dimensional volume element be

$$dV = dx dy dz$$

and four dimensional volume element be

$$dV_\mu = dx dy dz dt$$

In S' -frame, let the corresponding volume elements be dV' and dV'_μ . Let the volume element be at rest in frame S' i.e., $dV' = dV_0$. Now, Lorentz transformations are

$$x' = \gamma (x - vt), y' = y, z' = z.$$

If the observer of S -frame observes dx element at the same time, then

$$dx' = \gamma dx, \text{ also } dy' = dy \text{ and } dz' = dz \quad (t \text{ same})$$

Hence

$$dx' dy' dz' = \gamma dx dy dz$$

$$\text{or} \quad dx \, dy \, dz = \frac{dx' \, dy' \, dz'}{\gamma} \quad \text{or} \quad dV = dV_0 \sqrt{1-\beta^2}$$

Therefore, $dV \neq dV_0$ or $dV \neq dV'$

Now, suppose in S' -frame O' observes any two events at x' at dt' interval of time, then $dt' = d\tau$ is the proper interval of time and from $t = \gamma(t' + vx'/c^2)$, we have

$$dt = \gamma dt' = \gamma d\tau \quad (x' \text{ same})$$

$$\text{Therefore,} \quad dx' \, dy' \, dz' \, dt' = \gamma \, dx \, dy \, dz \, \frac{dt}{\gamma} = dx \, dy \, dz \, dt$$

$$\text{or} \quad dV'_\mu = dV_\mu$$

This proves the given statement.

QUESTIONS

Long Answer Questions

- What are Galilean transformations ?
A frame of reference S' is moving with constant velocity relative to another frame. Write down the transformation of x, y, z, t to x', y', z', t' in the Galilean form. At time $t = 0$, both frames are coincident.
Obtain also the transformations of velocity and acceleration.
- Discuss the basic assumptions implied in the Galilean transformations. Use these transformations to show that the distance between two points is invariant in two inertial frames.
- (a) Show that the laws of mechanics are invariant under Galilean transformations.
(b) Establish the law of conservation of linear momentum on the assumption of Galilean invariance.
(c) Discuss the principle of relativity and invariance of speed of light.
- What is absolute frame of reference ? Why were Michelson-Morely experiments performed ? Discuss the Michelson-Morley experiments and the central conclusion of the experiments.
- What was the aim of Michelson-Morely experiment ? Describe the construction of Michelson-Morely experiment with a sketch. Derive the expression for the fringe shift. (S.K.U. 02)
- Describe the arrangement of the apparatus used in the Michelson Morely experiment with a sketch. Derive an experiment for the fringe shift. (A.N.U. 02)
- Describe Michelson-Morely experiment. What is the physical significance of Michelson-Morley experiment ? (S.V.U. 2007, 05, A.U. 2006)
- Describe the Michelson-Morely experiment and explain the physical significance of negative result. (A.N.U. 2008, 07; A.U. June 2007)
- Describe Michelson-Morely experiment and the importance of its result. (K.U. 2007)
- Describe the Michelson-Morely experiment. Explain the result of the experiment. Mention the significance of the experiment. (K.U. 2005)
- Describe the Michelson-Morely experiment and critically comment on its negative result. (O.U. 2005)
- Describe with relevant theory the experiment of Michelson and Morely. Discuss the result. (S.V.U. 2002)
- Derive Lorentz transformation equations from special theory of relativity. Distinguish between Galilean and Lorentz transformations. (S.K.U. 2007)
- State the postulates of special theory of relativity. Derive the Lorentz transformation equations. (A.N.U. March 2007; O.U. Oct. 2007; S.K.U. 2009, 02; S.V.U. 2003; A.U. 2004)

15. Give the postulates of special theory of relativity. Obtain Lorentz transformation equations for the coordinates in the two frames of reference in relative motion. Show that these equations agree with Galilean transformation at low speed. (A.N.U. 2005; K.U. 1990)
16. Derive Lorentz transformation equations. Explain length contraction in relativity. (S.V.U. 2002)
17. (a) Show how Lorentz transformation equations are superior to Galilean transformations.
(b) Prove that when $v \ll c$, Lorentz transformation reduces to Galilean one.
(c) Prove the following statement :
'To the stationary observer, the moving clock appears to go slow.'
18. Write down Lorentz transformation equations and derive relativistic law of addition of velocities ? (A.U. 1997)
19. Show that the apparent length of a rigid body in the direction of its motion and with a uniform velocity v is reduced by the factor $\sqrt{1 - (v^2 / c^2)}$. Deduce the result.
20. What is time dilation in special relativity ? Deduce an expression for time dilation in regard to the interval between two events measured from two different inertial frames.
21. State the postulates of relativity. Explain length contraction and time dilation. (A.U. 1989)
22. (a) Derive the Lorentz space-time transformation formulae.
(b) Deduce the law of addition of relativistic velocities and show that the velocity of a material particle cannot exceed the speed of light. (O.U. 1993)
23. Write down Lorentz transformations. Derive an expression for the variation of mass of a body with its velocity. (K.U. 1997)
24. State the postulates of the special theory of relativity. Derive the mass-energy relation. (N.U. 2003)
25. State the postulates of special theory of relativity and explain them briefly. Explain principle of mass-energy equivalence with examples. (A.U. March 2007)
26. (a) State the fundamental postulates of special theory of a relativity.
(b) Discuss the mass variation according to special theory of relativity and obtain Einstein's mass-energy relation. (O.U. March 2007)
27. Derive Einstein's mass-energy relation. (S.K.U. 2007; K.U. 2007)
28. Obtain an expression for four vector formation for position co-ordinates in four dimensional space. Show that the length of a four vector remains unchanged under Lorentz transformation. (O.U. 2008)

Short Answer Questions

1. Write a note on frames of reference. (A.U. 2002; S.K.U. 2002)
2. What is a frame of reference ? Distinguish between inertial and non-inertial frames. (O.U. 1990)
3. Distinguish between inertial and non-inertial frames of reference. (O.U. 2008)
4. What is the significance of Michelson-Morely experiments. (O.U. 2003, 02; S.K.U. 2001)
5. State and explain the postulates of special theory of relativity. (A.N.U. 2002; O.U. 2003)
6. Explain the postulates of special theory of relativity. (O.U. 2008)
7. Write about Lorentz transformations. (A.N.U. 2001)
8. Write Lorentz-inverse transformation equations. (S.K.U. 1990)
9. Deduce Lorentz transformations. (N.U. 2003)
10. Distinguish between Galilean and Lorentz transformations. (S.V.U. 2007; A.U. 1989; S.V.U. 1990)
11. What is length contraction and time dilation. (A.N.U. 2002)

12. Explain time dilation. (A.U. March 2007; S.V.U. 2003)
13. What is length (Lorentz) contraction ? Write an expression for it. (O.U. 2009; A.N.U. 2001; A.U. 2002, 01)
14. Obtain the equation of Lorentz-Fitzgerald contraction. (K.U. 2005)
15. What do you understand by length contraction ? (S.V.U. 2005; A.U. 2003)
16. Explain time dilation. Give an example. (A.U. 2005; S.K.U. 1996)
17. What is time dilation ? Has it been experimentally verified ? (S.V.U. 1996)
18. Prove that the addition of any velocity to the velocity of light merely reproduces the velocity of light. (A.N.U. 1990)
19. Use Lorentz transformations to obtain the laws of addition of velocities. (A.U. 1988; S.V.U. 1989; K.U. 1988)
20. Show that the particle which travels at the speed of light must have a zero rest mass.
21. Get relation between momentum and energy according to relativity theory. (S.V.U. 2002)
22. Deduce mass-energy equivalence. (A.N.U. 2007; O.U. 2002)
23. Write down the relativistic expression for kinetic energy of a body and show that for smaller speeds it reduces to the classical expression.
24. What is four-space ?
25. Explain the four-vector formalism. (A.U. June 2007)
26. What is a four-vector ? Give one example.

[Problems]

[Galilean Relativity and Michelson-Morley Experiments]

1. The position of a particle P in a coordinate system S is measured as

$$\vec{r} = (6t^2 - 4t)\hat{i} + (-3t^3)\hat{j} + 3\hat{k} \text{ m}$$
 (a) Determine the constant relative velocity of system S' with respect to S if the position of P is measured as

$$\vec{r} = (6t^2 + 3t)\hat{i} + (-3t^3)\hat{j} + 3\hat{k} \text{ m}$$
 [Ans : (a) $-7\hat{i}$]
 (b) Show that the acceleration of the particle is the same in both systems.
 Prove that the Galilean transformation of a position vector is expressed by $\vec{r} = \vec{r}' + \vec{v}t + \vec{R}$, where $\vec{v}$ is the velocity of the frame S' relative to S and $\vec{R}$ the position vector of the origin O' as measured in S at $t'=0$.
2. In a frame moving with a velocity v , show that the times taken by a wave to travel equal distances d starting from a point and back along and perpendicular to the direction of travel differ by an amount dv^2/V^3 , where V is the velocity of the wave. (Assume $v \ll V$).
3. In the Michelson-Morley experiment, what is the expected fringe-shift, if the effective length of each path is 6 metres and light has $6000 \text{ \AA}$ wavelength ? (Speed of earth $v = 3 \times 10^4 \text{ m/sec.}$)
 [Ans : 0.2 of a fringe]
4. Calculate the expected fringes shift in Michelson-Moreley experiment if the distance of each path is 9 metres and light is of wavelength $6000 \text{ \AA}$. Velocity of earth is $3 \times 10^4 \text{ m/s}$.
 [Ans. 0.3 of a fringe-width]

[Lorentz Transformations]

5. An event at $x = 100$ km, $y = 10$ km and $z = 1.0$ km at $t = 2.0 \times 10^{-4}$ sec in a reference frame S . Another frame S' is moving with speed $0.95c$ relative to S along the common $X-X'$ axis, the origins coinciding at $t = t' = 0$. Calculate the coordinates x' , y' , z' and t' of the event in frame S' .
[Ans. $x' = 137.8$ km, $y' = 10$ km, $z' = 1.0$ km and $t' = 3.74 \times 10^{-4}$ sec]
6. With the help of the Lorentz transformation equations, find an expression $(x^2 - c^2 t^2)$ in terms of x' and t' .
[Ans. $(x'^2 - c^2 t'^2)$.]

[Length Contraction]

7. Calculate the velocity and rod while its length will appear 80% of its proper length. ($c = 3 \times 10^8$ m/s).
[Ans. 1.8×10^8 m/sec.] (S.V.U. 2003)
[Hint. $v = c\sqrt{1 - l^2/l_0^2} = 3 \times 10^8 \sqrt{1 - (0.8)^2} = 1.8 \times 10^8$ m/sec.]
8. The length of a rod is 10 metres in a system S . What is its length as seen by an observer in system S' , when S' has a velocity of $0.8c \hat{i}$ relative to S .
[Ans. 6 m]
9. The length of a rod is 5 metres. If the rod is moving with a velocity $0.5c$ relative to earth, find the length of the rod as observed by an observer moving with the rod.
[Ans. 5 m]
10. A Hypothetical train moving with a speed of $0.6c$ passes by the platform of a small station without being slowed down, the observer on the platform note that the length of the train is just equal to the length of the platform which is 200m.
(i) Find the rest length of the train.
(ii) Find the length of the platform as measured by the observer in the train.
[Ans. (i) 250 m, (ii) 160 m]
11. What will be the apparent length of a meter stick measured by an observer at rest, when the stick is moving along its length with a velocity equal to (i) c , (ii) $c/\sqrt{2}$ (iii) $\sqrt{3}c/2$ (iv) $c/2$.
[Ans. (i) 0, (ii) $1/\sqrt{2}$ m, (iii) $1/2$ m, (iv) $\sqrt{3}/2$ m]
12. A rod has a length of 1 m. It is moving in a space-ship moving with a velocity $0.4c$ relative to the earth. Calculate its length as measured by an observer (i) on space-ship; (ii) on earth.
[Ans. (i) 1 m; (ii) 0.92 m]
13. A rod is seen to move with velocity $0.6c$, in a direction inclined at 60° to the rod by an observer, situated in the laboratory and the length of the rod is measured as 0.8 m. What is the true length of the rod and what is the true direction of its motion?
[Ans. 0.85 m ; $\tan\theta = 1.38$]
14. A vector in system S' is represented by $8\hat{i} + 6\hat{j}$. How can the vector be represented in system S while S' is moving with velocity $0.8c \hat{i}$ relative to S .
[Ans. $4.8\hat{i} + 6\hat{j}$]
15. Show that the circle $x^2 + y^2 = r_0^2$ in frame S , appears to be ellipse in a frame S' which is moving with velocity $v \hat{i}$ relative to S .

[Time Dilation]

16. A particle with a mean proper life time of 2×10^{-6} sec. moves through the laboratory with a speed of $0.9c$. Calculate its life time as measured by an observer in the laboratory.
[Ans. 4.58×10^{-6} sec.]
17. What is the velocity of π -mesons whose observed mean life is 2.5×10^{-7} sec. The proper life of these π -mesons is 2.5×10^{-8} sec.
[Ans. 2.98×10^8 m/sec.]
18. Calculate the mean life of π^+ mesons produced in the cosmic-ray showers, which travel with a velocity of $0.75c$, c being the velocity of light. The proper life of π^+ meson is 2.5×10^{-8} sec. (A.U. 1990)
[Ans. 3.78×10^{-8} sec.]
19. The half life of a particular particle as measured in the laboratory comes out to be 4×10^{-8} sec, when its speed is $0.8c$ and 3×10^{-8} sec, when its speed is $0.6c$. Explain this.
[Ans. In both cases, the proper half life comes to be 2.4×10^{-8} sec. This means that a particle has its own life time independent of observers in motion.]
20. A stationary μ meson decays in 2.2×10^{-6} sec. What will be its length of path, if it is moving towards the earth with velocity $0.99c$? In a frame of reference fixed on the meson, what distance will the earth travel before the meson decays?
[Ans : 4.6×10^3 m; 6.5×10^2 m]
21. A beam of pions has velocity $v = 0.6c$. The pion has a half life of 1.8×10^{-8} sec. How long will it take for the pions to decay? How far will they travel in this time?
[Ans. 2.25×10^{-8} sec; 4.05 m]
22. The proper mean life of the μ meson is approximately 2×10^{-6} sec. Suppose that a burst of μ mesons is produced at some height in the atmosphere travels downward at $v = 0.99c$. If 1% of those in the original burst survive to reach the earth, estimate the initial height.
[Ans. 2×10^4 m]
23. Calculate the velocity of a watch when it seems to be slowed down by 1 minute in 1 hour.
[Ans : 1.8×10^7 m/sec.]
24. A clock gives correct time. With what speed should it be moved relative to an observer so that it may seem to lose 5 minutes in 24 hours?
[Ans : 2.5×10^7 m/sec]
25. If a rocket travels with a constant velocity of $0.8c$ from the earth to a star 4 light years distance, what will be (i) the time taken for the trip according to estimates made on the earth, (ii) the time according to a passenger, (iii) the distance from the earth to the star according to a passenger, and (iv) the velocity of the earth and the star as measured by a passenger during the trip?
[Ans : (i) 5 years, (ii) 3 years, (iii) 2.4 light years, (iv) $0.8c$.]

[Addition of Velocities]

26. A nuclear particle was observed to break into two fragments which moved in opposite directions. The velocity of each fragment was found to be $0.8c$ relative to the laboratory. What was the velocity of one fragment relative to the other?
[Ans : $0.98c$]
27. Two electrons, each of velocity $0.8c$, move towards each other. Find the relative velocity of one electron with respect to the other.
[Ans : $0.97c$]

364 | Mechanics and Waves and Oscillations

28. Rockets A and B are observed from the earth to be travelling with velocities $0.8c$ and $0.7c$ in the same direction. What is the velocity of B as seen by an observer in A .

[Ans : $0.227c$]

29. A particle has a velocity $u' = 3\hat{i} + 4\hat{j} + 2\hat{k}$ m/sec in a coordinate system moving with velocity $0.8c$ relative to laboratory frame along +ve direction of X -axis. Find u in the laboratory frame.

[Ans : $(2.4 \times 10^8 \hat{i} + 2.4 \hat{j} + 7.2 \hat{k})$ m/sec]

30. A rocket is sent with a velocity $0.95c$ and a light wave with velocity c along the same path. What is the velocity of the light wave relative to the rocket?

[Ans. c]

31. A photon is observed from a space-ship moving with speed of $0.9c$. What is the speed of the photon with respect to the space-ship observer.

[Ans. c]

32. In frame A , the components of velocity of a particle are $v_x' = v_y' = 0.6c$, $v_z' = 0$. It is seen from a second frame B , which is moving with velocity $-0.8c$ along X -axis. What are the components of velocity as observed from frame B .

[Ans. $v_x = 0.94c$, $v_y = 0.24c$]

[Relativistic Mass and Energy]

33. Find the mass of an electron moving with a velocity of 1×10^8 m/sec. The rest mass of the electron is 9.1×10^{-31} kg. (A.N.U. 1990)

[Ans. 9.639×10^{-31} kg.]

34. A particle of mass m_0 is moving with a velocity $0.9c$. Calculate its relativistic mass and its kinetic energy. (A.N.U. 1988)

[Ans. $2.294 m_0$; $2.294 m_0 c^2$]

35. The total energy of a particle is exactly twice its rest energy. Find its speed. (K.U. 1991; S.V.U. 1997)

[Ans. $0.866c$]

36. A body of specific heat 0.2 is heated to the temperature 100°C . Calculate the percentage by which the mass increases.

[Ans. $9 \times 10^{-11}\%$]

37. Find the velocity of an electron accelerated through a voltage of 10^5 volts. (Assume rest energy of the electron = 0.5 MeV).

[Ans. 1.66×10^8 m/sec.]

38. Calculate the speed of an electron accelerated to potential of one million electron volts.

[Ans. $0.94c$]

39. Show that the momentum of a particle is $m_0 c$ when its velocity is $c/\sqrt{2}$ (where c is speed of light and m_0 is the rest mass of the particle). (O.U. 1989)

$$\text{[Hint. } m = \frac{m_0}{\sqrt{1-v^2/c^2}} = \frac{m_0}{\sqrt{1-(c/\sqrt{2})^2/c^2}} = \frac{m_0}{\sqrt{1/2}} = \sqrt{2} m_0$$

$$\therefore p = \text{mass} \times \text{velocity} = \sqrt{2} m_0 \times \frac{c}{\sqrt{2}} = m_0 c]$$

40. Compute the mass m and speed v of an electron having kinetic energy 1.5 MeV. Given rest mass of electron $m_0 = 9.11 \times 10^{-31}$ kgm, velocity of light in vacuum $c = 3 \times 10^8$ m/sec.

[Ans. $m = 3.58 \times 10^{-30}$ kg; $v = 2.9 \times 10^8$ m/sec.]

[Hints. Kinetic energy $= mc^2 - m_0c^2$, find $m = ?$. Hence get v from the relation $m = m_0 / \sqrt{1 - (v/c)^2}$.]

41. What is the kinetic energy of a proton (rest mass $= 1.67 \times 10^{-27}$ kg) which is moving with velocity 2.7×10^8 m/sec. (Give your answer in eV)

[Ans. 1746 MeV]

42. Calculate the rest mass of a particle of kinetic energy 50 MeV and momentum 130 MeV/c.

[Ans. 2.56×10^{-28} kg]

43. Calculate the amount of work done in accelerating an electron from rest to a velocity 2.4×10^8 m/sec. (Assume the rest energy of the electron $= 0.5$ MeV).

[Ans. 0.335 MeV]

44. When a positron and an electron annihilate, find the wavelength of the two photons emitted?

[Ans. 2.482×10^{-12} m]

[Hint. $E = m_0c^2 + m_0c^2 = 2mf^2 = 2h\nu = \frac{2hc}{\lambda}$; $\lambda = \frac{h}{m_0c} = \frac{6.62 \times 10^{-34}}{9.1 \times 10^{-31} \times 3 \times 10^8} = 2.482 \times 10^{-12}$ m]

45. Calculate the rest energy of an electron in joules and electron-volts ($m_0 = 9.1 \times 10^{-31}$ kg, $c = 3.0 \times 10^8$ m/sec). Also find the speed at which the total relativistic energy becomes 1.25 times the rest energy.

[Ans. 0.51 MeV, 1.8×10^8 m/sec.]

46. Deduce the minimum energy of gamma-ray photons (in MeV) which can cause electron-positron pair production. ($m_0 = 9.1 \times 10^{-31}$ kg).

[Ans. 1.02 MeV]

47. Calculate the momentum of photon of energy 10^{-19} J.

[Ans. 3.3×10^{-28} kg-m/sec.]

48. Find the velocity that an electron must be given so that its momentum is 10 times its rest mass times the speed of light. What is the energy at this speed?

[Ans. $v = 0.995c$; 8.14×10^{-13} J]

49. The velocity of an object is such that its mass increases by 10%.

(i) By what fraction does its length in the direction of motion decrease?

(ii) If its rest energy is W_0 , what is its kinetic energy?

[Ans: (i) 9.1%, (ii) $W_0/10$]

50. Calculate the velocity of a particle when its rest mass energy is double to its kinetic energy.

[Ans: $0.79c$]

51. How many kilowatt-hours of energy would be liberated by the complete conversion of 4 mgm. mass?

[Ans: 10^5 KWH]

52. Calculate the increase in total rest mass, if two particles each of mass 1 gm are moving with 1.5 km/sec velocity in opposite directions collide and come to rest.

[Ans: 2.5×10^{-14} kg]

53. By what factor is the density of an object increased when it is moving with velocity v ?

[Ans: By the factor $[1/(1 - v^2/c^2)]$]

54. The density of gold is 19.3×10^3 kg m^{-3} in a frame S which is at rest. Calculate its density that an observer in a frame S' would determine if the frame S is moving along the X -axis with a speed $0.9c$.

[Ans. 101.6×10^3 kg m^{-3}]

366 | Mechanics and Waves and Oscillations

- 55.** An electron (rest mass 9.1×10^{-31} kg) is moving with speed $0.99c$. Calculate its total energy? Find also the ratio of Newtonian kinetic energy to the relativistic kinetic energy.

[Ans. 5.8×10^{-13} joule; 0.08]

[Hint. $\frac{\text{Newtonian K.E.}}{\text{Relativistic K.E.}} = \frac{\frac{1}{2}m_0v^2}{(m - m_0)c^2}$]

[Four vectors]

- 56.** Show by the direct application of Lorentz transformation that $x^2 + y^2 + z^2 + \omega^2$ is invariant, where $\omega = ict$, $i = \sqrt{-1}$.
- 57.** In a frame S , two events have the space-time coordinates $(0, 0, 0, 0)$ and $(10c, 0, 0, 6)$. Find the space-time interval between them. Calculate the velocity of a frame in which
- (i) the two events are simultaneous,
 - (ii) the first event occurs 8 sec. earlier than the second, and
 - (iii) the second event occurs 8 sec earlier than the first.

[Ans. (i) $0.6c$, (ii) $-0.18c$, (iii) $0.91c$]



Fundamentals of Vibrations

7.1. Oscillatory Motion

Oscillatory or vibratory phenomenon is very common in nature. In physics, familiar examples of this phenomenon are : (i) oscillations of a simple pendulum, (ii) oscillations of spring-mass system, (iii) vibrations of a tuning fork, (iv) vibrations of a guitar string etc. The terms oscillation and vibration have practically the same meaning and both are used for the same purpose. In general usage, an oscillation is possibly slower, but we shall not consider any such distinction. The vibratory behaviour is responsible for all kinds of wave phenomena. In sound waves, particles of the medium and in electromagnetic waves, electric and magnetic field vectors perform oscillatory motions. In fact, the study of oscillatory phenomena is of extreme importance in physics.

The feature that the oscillatory phenomena have in common is periodicity. If a particle moves such that it retraces its path regularly after equal intervals of time, its motion is said to be **periodic**. This interval of time, required to complete one round trip (cycle) of motion is called **period**. Now, if a particle in periodic motion moves back and forth over the same path about an equilibrium position, we call the motion **vibratory** or **oscillatory**.

The time taken for the system to undergo one complete oscillation is the period T of the system. Since the system executes $1/T$ vibrations per unit time, this quantity is called the frequency of vibration (ν), given by

$$\nu = \frac{1}{T}$$

The dimensions of frequency are $(\text{time})^{-1}$. We sometimes express frequency in cycles (or vibrations) per second. One cycle per second (cps) is known as one hertz (Hz). The **hertz** is the SI unit of frequency.

7.2. Simple Harmonic Motion and Harmonic Oscillator

Simple harmonic motion is a particular type of periodic motion and is very common in nature. A system is said to execute **simple harmonic motion** when it vibrates periodically in such a way that at any instant the restoring force acting on it is proportional to its displacement from its position of stable equilibrium in its path and is always directed towards that position.

A system executing simple harmonic motion is called **harmonic oscillator**.

Let us consider a particle of mass m , executing simple harmonic motion about the equilibrium position O with maximum displacement a on either side [fig. 7.1(a)]. At any instant t , if the displacement of the particle is x , then its acceleration will be d^2x/dt^2 and the force (F) will be md^2x/dt^2 . Now, according to the definition of simple harmonic motion,

$$\text{Restoring force } (F) \propto -\text{displacement } (x)$$

$$\text{or} \quad m \frac{d^2x}{dt^2} \propto -x$$

$$\text{or} \quad m \frac{d^2x}{dt^2} = -kx \quad \text{or} \quad F = -kx \quad \dots(1)$$

where k is the force constant. Here, the negative sign indicates that the force on the particle is directed opposite to x increasing.

The potential energy U of the particle at the displacement x is given by

$$U = -\int_0^x F dx$$

$$= \int_0^x kx dx = \frac{1}{2} kx^2 \dots (2)$$

The variation of force (F) and potential energy (U) of the particle, executing simple harmonic motion, with displacement (x) are shown in **fig. 7.1(c)** and **fig. 7.1(b)** respectively. Since U varies as x^2 , the potential energy will be the same at the maximum displacement $x_{\max} = +a$ or $-a$. This maximum displacement a is the **amplitude** of motion.

Characteristics of Simple Harmonic Motion

Simple harmonic motion is characterized with the following properties :

(1) It is a periodic motion with a certain periodic time T .

(2) The restoring force on the particle is proportional to the displacement from the equilibrium position in a direction opposite to it [**fig. 7.1(c)**].

(3) The potential energy of the oscillating particle varies as the square of the displacement [**fig. 7.1(b)**].

(4) The turning points are situated at equal distances from the equilibrium position.

7.3. Differential Equation of Simple Harmonic Motion and its Solution

For simple harmonic oscillations, we have

$$m \frac{d^2 x}{dt^2} = -kx \text{ or } \frac{d^2 x}{dt^2} + \frac{k}{m} x = 0 \quad \dots (1)$$

Let us put $\omega^2 = k/m$. Then

$$\frac{d^2 x}{dt^2} + \omega^2 x = 0 \quad \dots (2)$$

Eq. (2) is called the **differential equation of simple harmonic motion**.

In order to find the displacement of the particle in S.H.M., we solve eq. (2). Multiplying eq. (2) by $2x/dt$, we obtain

$$\frac{2dx}{dt} \cdot \frac{d^2 x}{dt^2} + \omega^2 \cdot 2x \frac{dx}{dt} = 0$$

Integrating it, we obtain

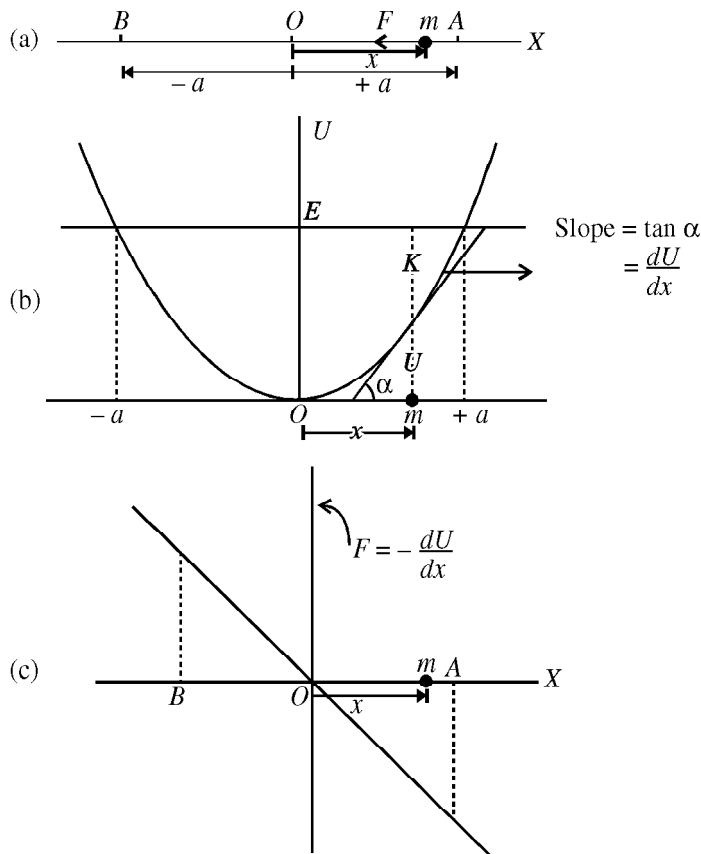


Fig. 7.1 : Simple harmonic motion : (a) Particle displacement between turning points $x = +a$ and $x = -a$, (b) Parabolic potential energy curve, (c) Force versus displacement curve (straight line).

$$\left(\frac{dx}{dt}\right)^2 + \omega^2 x^2 = C, \text{ where } C \text{ is a constant of integration.} \quad \dots(3)$$

The velocity of the oscillating particle is zero at the maximum value of the displacement *i.e.*, at $x = x_{\max} = a$ (say), $dx/dt = 0$ *i.e.*,

$$0 + \omega^2 a^2 = C \text{ or, } C = \omega^2 a^2.$$

Substituting the value of C in eq. (3), we get

$$\left(\frac{dx}{dt}\right)^2 + \omega^2 x^2 = \omega^2 a^2 \text{ or } \left(\frac{dx}{dt}\right)^2 = \omega^2 (a^2 - x^2)$$

$$\text{or } \frac{dx}{dt} = \pm \omega \sqrt{a^2 - x^2} \quad \dots(4)$$

If we take positive root, then eq. (4) can be written as

$$\frac{dx}{\sqrt{a^2 - x^2}} = \omega dt \text{ or } \sin^{-1}\left(\frac{x}{a}\right) = \omega t + \phi$$

$$\text{or } x = a \sin(\omega t + \phi) \quad \dots(5)$$

where ϕ is a constant of integration. If we take negative root, then

$$-\frac{dx}{\sqrt{a^2 - x^2}} = \omega dt \text{ or } \cos^{-1}\left(\frac{x}{a}\right) = \omega t + \phi$$

$$\text{or } x = a \cos(\omega t + \phi) \text{ where } \phi \text{ is a constant of integration.} \quad \dots(6)$$

Eq. (5) and eq. (6), represent sine and cosine forms of the solutions and both are valid solutions of the differential equation (2) of the harmonic oscillator. The two solutions become identical, if the two constants of integration ϕ and ϕ are related as $\phi = \phi - \pi/2$. It is to be seen from eqs. (5) and (6) that in simple harmonic motion, the displacement is represented as a sine or cosine function of time and if a graph is plotted between displacement (x) and time (t), it is called a **sinusoidal graph [fig. 7.2]**. In other words, *the motion represented by sinusoidal graph, is known as simple harmonic motion*. In our later analysis, we shall use the sine form of the solution, given by eq. (5).

Simple harmonic motion is characterized with the following terms :

(i) Displacement and amplitude : At any instant t , the **displacement** of a particle, executing simple harmonic motion, is given by

$$x = a \sin(\omega t + \phi) \quad \dots(7)$$

When $\sin(\omega t + \phi) = 1$ or -1 , the displacement has the maximum value a ($= x_{\max}$) on either side of the equilibrium position $x = 0$. Hence a is the **amplitude** of vibration.

(ii) Period and frequency : Now, let us find the **physical significance** of the constant ω . If the time t in eq. (7) is increased by $2\pi/\omega$, the function becomes

$$x = a \sin[\omega(t + 2\pi/\omega) + \phi] \text{ or } x = a \sin[\omega t + 2\pi + \phi] \text{ or } x = a \sin(\omega t + \phi).$$

That is, the displacement of the particle is the same after a time $2\pi/\omega$. Therefore, $2\pi/\omega$ is the **period** (T) of the motion, *i.e.*,

$$T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{m}{k}} \quad \left(\because \omega^2 = \frac{k}{m}\right) \quad \dots(8)$$

The number of vibrations per second is called the frequency (ν) of the oscillator and is given by

$$\nu = \frac{1}{T} = \frac{\omega}{2\pi} = \frac{1}{2\pi}\sqrt{\frac{m}{k}} \quad \dots(9a)$$

$$\therefore \omega = 2\pi\nu = \frac{2\pi}{T} \quad \dots(9b)$$

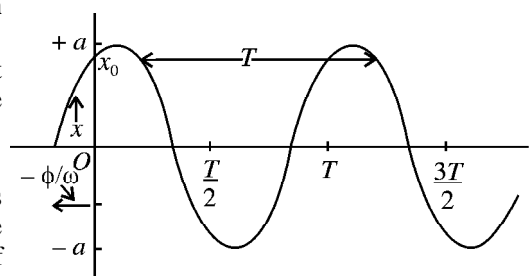


Fig. 7.2 : Simple harmonic motion, $x = a \sin(\omega t + \phi)$, of period T , amplitude a and phase constant ϕ .

The quantity ω is called the **angular frequency**.

(iii) **Velocity and acceleration** : The displacement of the mass executing S.H.M. is given by

$$x = a \sin (\omega t + \phi)$$

Its velocity (v) is given by

$$v = \frac{dx}{dt} = \omega a \cos (\omega t + \phi) \quad \dots(10a)$$

$$\text{or } v = \omega a \sqrt{1 - \sin^2 (\omega t + \phi)}$$

$$= \omega a \sqrt{1 - x^2 / a^2} = \omega \sqrt{a^2 - x^2} \quad \dots(10b)$$

Its acceleration is

$$\frac{dv}{dt} = \frac{d^2x}{dt^2} = -\omega^2 \sin (\omega t + \phi) \quad \dots(11a)$$

$$\text{or } \frac{dv}{dt} = \frac{d^2x}{dt^2} = -\omega^2 x \quad \dots(11b)$$

The variations of displacement (x), velocity (dx/dt) and acceleration (d^2x/dt^2) for $\phi = 0$ with time t are given in **fig. 7.3**. Eqs. (10b) and (11b) give the dependence of velocity and acceleration on the displacement (x). One may observe that the velocity and acceleration lead the displacement in phase by $\pi/2$ and π respectively.

(iv) **Phase** : The phase is an important term, used in the theory of vibrations and waves and hence the concept of phase is to be clearly understood.

The term $(\omega t + \phi)$, occurring in the equation (7) for displacement x , is called the **phase** of vibration. The phase tells us the position of the oscillating particle at any time t . At $t = 0$, the value of the phase is ϕ and therefore ϕ is called **initial phase** or **phase constant**. This gives the information regarding the initial position from where we started to measure the displacement. Suppose at $t = 0$, the particle is at $x = x_0$. So that from eq. (7), we have

$$x_0 = a \sin \phi \text{ or } \phi = \sin^{-1} (x_0/a) \quad \dots(12)$$

One may note that eq. (2) will be satisfied by eq. (7) for any finite value of a and ϕ . The amplitude a and phase constant ϕ are determined by the initial position and

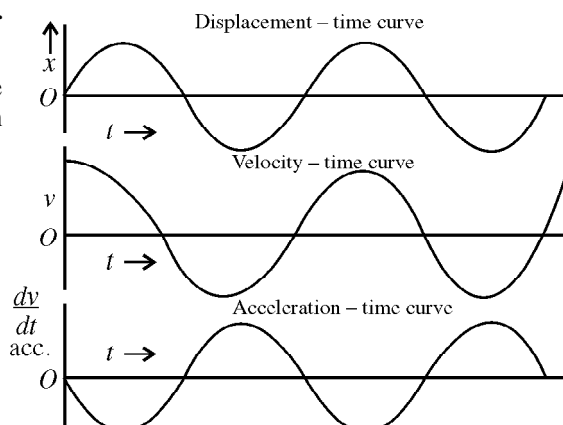


Fig. 7.3 : Variation of displacement (x), velocity (v) and acceleration, with time t .

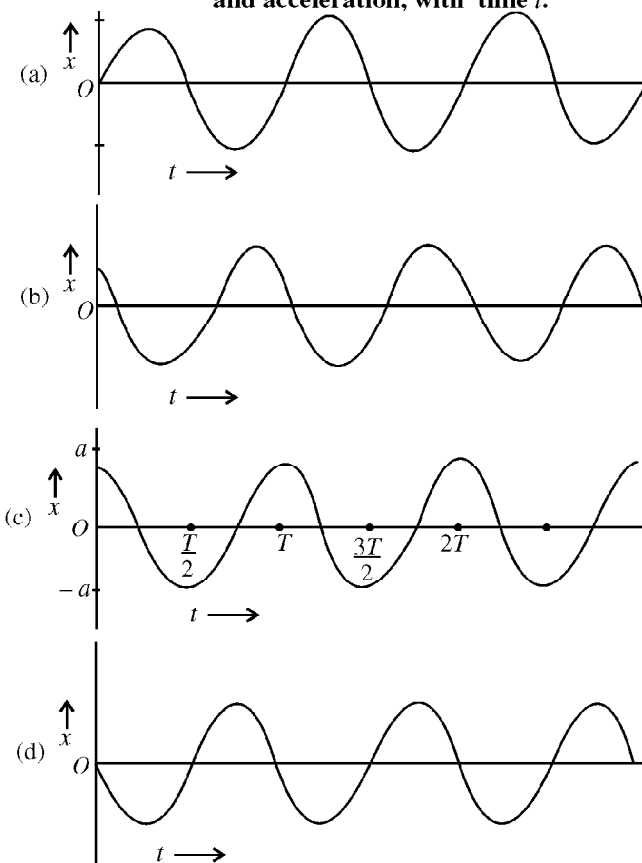


Fig. 7.4 : Displacement-time graphs for S.H.M. represented by $x = a \sin (\omega t + \phi)$: (a) $\phi = 0$, $x_0 = a$, (b) $\phi = \pi/2$, $x_0 = a/\sqrt{2}$, (c) $\phi = \pi/2$, $x_0 = a$ and (d) $\phi = -\pi/2$, $x_0 = 0$.

speed of the particle in simple harmonic motion.*

* Initial position (at $t = 0$), $x_0 = a \sin \phi$ and initial speed, $v_0 = \omega a \cos \phi$

$\therefore a^2 \sin^2 \phi + a^2 \cos^2 \phi = x_0^2 + v_0^2 / \omega^2$ or $a = \sqrt{x_0^2 + v_0^2 / \omega^2}$ and hence $\phi = \sin^{-1} (x_0/a)$.

If we measure the time, when the particle is at the equilibrium position *i.e.*, at $t = 0$, $x = 0$, then from eq. (7), we obtain

$$\sin \phi = 0 \quad \text{or} \quad \phi = 0$$

In this situation, eq. (7) is obtained to be

$$x = a \sin \omega t \quad \dots(13)$$

For $\phi = 0$, we have drawn the displacement-time graph in **fig. 7.4(a)**. For $\phi = \frac{\pi}{4}$, $\frac{\pi}{2}$ and π the graphs are shown in **fig. 7.4(b), (c) and (d)** respectively. In all cases, the graphs, representing SHM's, have exactly the same shape, but there is the shifting of the origin along the time axis. If **fig. 7.4(b)** and **fig. 7.4(c)** represent two separate simple harmonic vibrations, then $\pi/4$ is the phase difference between them and the later is leading in phase. The phase difference for the **fig. 7.4(a)** and **fig. 7.4(d)** is π and the two vibrations are said to be in opposite phase.

Alternative Method of Solving the Differential Equation of Harmonic Oscillator

Differential equation of harmonic oscillator is

$$\frac{d^2x}{dt^2} + \omega^2 x = 0, \quad \text{where } \omega = \sqrt{\frac{k}{m}}.$$

Let the solution of this equation be $x = Ae^{\alpha t}$

Now, substituting the value of $\frac{d^2x}{dt^2}$ and x in the above equation, we have

$$A\alpha^2 e^{\alpha t} + \omega^2 A e^{\alpha t} = 0, \quad \text{or} \quad \alpha^2 + \omega^2 = 0, \quad \text{as in general } A e^{\alpha t} \neq 0.$$

$$\therefore \alpha = \pm \sqrt{-\omega^2} = \pm i\omega, \quad \text{where } i = \sqrt{-1}$$

$$\therefore x = A_1 e^{i\omega t} \quad \text{or} \quad x = A_2 e^{-i\omega t}$$

Hence the general solution of the equation will be given by

$$x = A_1 e^{i\omega t} + A_2 e^{-i\omega t}$$

where A_1 and A_2 are constants, which can be determined from the initial conditions.

$$\begin{aligned} \text{Now,} \quad x &= A_1 (\cos \omega t + i \sin \omega t) + A_2 (\cos \omega t - i \sin \omega t) \\ &= (A_1 + A_2) \cos \omega t + (A_1 i + A_2 i) \sin \omega t. \end{aligned}$$

Let $(A_1 + A_2) = a \sin \phi$ and $(A_1 i + A_2 i) = a \cos \phi$. Then,

$$x = a \sin \phi \cos \omega t + a \cos \phi \sin \omega t$$

or

$$x = a \sin (\omega t + \phi)$$

7.4. Energy of Harmonic Oscillator

A harmonic oscillator possesses two types of energy :

In simple harmonic motion, the oscillating system possesses two types of energy :

(i) **Potential energy**, which is due to its displacement from the mean position.

(ii) **Kinetic energy**, which is due to its velocity.

Hence, at any instant the total energy of the oscillator will be the sum of these two energies. As the system is conservative (*i.e.*, no dissipative or frictional forces are acting), the total mechanical energy $E (= K + U)$ must be conserved.

At the displacement x , the expressions for the potential energy (U), kinetic energy (K) and total energy (E) are given by

$$U = \frac{1}{2} kx^2 = \frac{1}{2} m\omega^2 x^2 \quad [\because \omega^2 = k/m] \quad \dots(1a)$$

or

$$U = \frac{1}{2} m\omega^2 a^2 \sin^2 (\omega t + \phi) \quad [\because x = a \sin (\omega t + \phi)] \quad \dots(1b)$$

$$K = \frac{1}{2}mv^2 = \frac{1}{2}m\omega^2(a^2 - x^2) = \frac{1}{2}k(a^2 - x^2) \quad \dots(2a)$$

$$\text{or} \quad K = \frac{1}{2}m\omega^2[a^2 - a^2 \sin^2(\omega t + \phi)] = \frac{1}{2}m\omega^2 a^2 \cos^2(\omega t + \phi) \quad \dots(2b)$$

$$\begin{aligned} \text{and} \quad E = U + K &= \frac{1}{2}kx^2 + \frac{1}{2}k(a^2 - x^2) = \frac{1}{2}ka^2 \\ &= \frac{1}{2}m\omega^2 a^2 = 2m\pi v^2 a^2 \quad (\because \omega = 2\pi v) \dots(3) \end{aligned}$$

Thus, we see that the total mechanical energy is constant, as we expect and has the value $\frac{1}{2}ka^2$, i.e., the total energy is proportional to the square of the **amplitude**. Hence, if the system is once oscillated the motion will continue for indefinite period without any decrease in amplitude, provided no damping (frictional) forces are acting on the system.

It is clear from the expression $U = \frac{1}{2}kx^2$ that if we plot a graph between the potential energy U and displacement x , we get a parabola having vertex at $x = 0$ [fig. 7.5]. The curve for the total energy E (constant) is the horizontal line. The particle cannot go beyond the points where this line intersects the potential energy curve because U can never be larger than E . These points are **turning points** of the motion and correspond to the maximum displacement (i.e., at $x = \pm a$) the total energy of the oscillator is wholly potential (i.e., $E = U = \frac{1}{2}ka^2$) but the kinetic energy is zero and so the amplitude of motion is $a = \pm\sqrt{2E/k}$. At the equilibrium position, the potential energy is zero, but the kinetic energy has the maximum value $K_{\max} = \frac{1}{2}mv_{\max}^2 = \frac{1}{2}ka^2 = E$ with maximum velocity $v_{\max} = \sqrt{2E/m}$. At other intermediate points the energy is partly kinetic and partly potential but their sum (total energy) is always $\frac{1}{2}ka^2$.

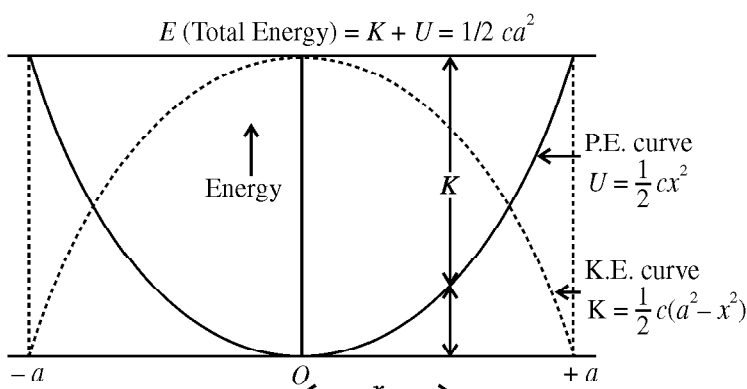


Fig. 7.5 : Variation of potential energy (U), kinetic energy (K) and total energy (E) with displacement (x) in simple harmonic motion.

Average values of kinetic and potential energies : The time average of a quantity $Q(t)$ over a time interval T is given by

$$Q_{av} = \frac{1}{T} \int_0^T Q(t) dt \quad \dots(4)$$

Hence the average kinetic energy for a period T is given by

$$\begin{aligned} K_{av} &= \frac{1}{T} \int_0^T K dt = \frac{1}{T} \int_0^T \frac{1}{2} mv^2 dt = \frac{1}{T} \int_0^T \frac{1}{2} ma^2 \omega^2 \cos^2(\omega t + \phi) dt \quad [\text{from eq. (2)}] \\ &= ma^2 \frac{\omega^2}{2T} \int_0^{T=2\pi/\omega} \frac{1}{2} [1 + \cos(2\omega t + 2\phi)] dx = \frac{1}{4} \frac{ma^2 \omega^2}{2\pi/\omega} \cdot \frac{2\pi}{\omega} = \frac{1}{4} ma^2 \omega^2 = \frac{1}{4} ka^2 \quad \dots(5) \end{aligned}$$

$$\text{Similarly, } U_{av} = \frac{1}{T} \int_0^T U dt = \frac{1}{T} \int_0^T \frac{1}{2} kx^2 dt = \frac{1}{T} \int_0^T \frac{1}{2} ka^2 \sin^2(\omega t + \phi) dt = \frac{1}{4} ka^2 \quad \dots(6)$$

From eqs. (5) and (6), we get

$$K_{av} = U_{av} \quad \dots(7)$$

$$\text{Also,} \quad U_{av} + K_{av} = \frac{1}{2}ka^2 = E = E_{av} \quad \dots(8)$$

because the total energy is constant over the period.

This is an important characteristic of harmonic oscillator that *its average kinetic energy is equal to the average potential energy*.

We shall consider in the next articles some important examples of simple harmonic motion and the superposition of two simple harmonic motions.

Ex. 1. A body is executing S.H.M. and the equation is $x = a \sin (\omega t + \phi)$. If the oscillations are started from the position x_0 with a velocity v_0 , show that the amplitude $a = \sqrt{x_0^2 + (v_0 / \omega)^2}$. **(O.U. March 2008)**

Sol. The displacement of a particle executing S.H.M. is given by

$$x = a \sin (\omega t + \phi) \quad \dots(i)$$

$$\text{Its velocity} \quad v = \frac{dx}{dt} = \omega a \cos (\omega t + \phi) \quad \dots(ii)$$

Given, at $t = 0$, $x = x_0$ and $v = v_0$. Hence from eq. (i) and (ii), we have

$$x_0 = a \sin \phi \quad \dots(iii)$$

$$\text{and} \quad v_0 = \omega a \cos \phi \quad \dots(iv)$$

$$\text{From (iii),} \quad \sin \phi = x_0 / a \quad \dots(v)$$

$$\text{From (iv),} \quad \cos \phi = v_0 / \omega a \quad \dots(vi)$$

Squaring (v) and (vi) and then on adding, we obtain

$$\sin^2 \phi + \cos^2 \phi = \frac{x_0^2}{a^2} + \frac{v_0^2}{\omega^2 a^2} \quad \text{or} \quad 1 = \frac{1}{a^2} \left[x_0^2 + \left(\frac{v_0}{\omega} \right)^2 \right]$$

$$\text{or} \quad a^2 = \left[x_0^2 + (v_0 / \omega)^2 \right], \quad \therefore a = \sqrt{x_0^2 + (v_0 / \omega)^2}.$$

Ex. 2. A particle of mass 5 gm executing simple harmonic motion has amplitude of 8 cm. If it makes 16 vibrations per sec, find its maximum velocity and energy at mean position. **(A.N.U. March 2008)**

Sol. Maximum velocity in S.H.M. is given by

$$v_{\max} = \omega a \quad [\because v = \omega \sqrt{a^2 - x^2}]$$

$$\text{Here,} \quad \omega = 2\pi v = 2 \times 3.14 \times 16 \text{ rad/sec and } a = 8 \text{ cm} = 0.08 \text{ m}$$

$$\therefore v_{\max} = 2 \times 3.14 \times 16 \times 0.08 = \mathbf{8.038 \text{ m/sec.}}$$

$$\text{Energy at mean position} = \frac{1}{2} m v_{\max}^2 = \frac{1}{2} \times 5 \times 10^{-3} \times (8.038)^2 = \mathbf{0.1615 \text{ joule}}$$

Ex. 3. A particle of mass 2 gm moves along the X-axis and is attracted towards origin by a force $8 \times 10^{-3} x$ newton. If it is initially at rest at $x = 10$ cm, find (i) the differential equation of motion, (ii) the position of the particle at any time, (iii) the velocity of the particle at any time, (iv) amplitude and frequency of vibration.

$$\text{Sol. (i)} \quad \text{Force } m \frac{d^2 x}{dt^2} = -8 \times 10^{-3} x \quad \text{or} \quad 2 \times 10^{-3} \frac{d^2 x}{dt^2} = -8 \times 10^{-3} x \quad \text{or} \quad \frac{d^2 x}{dt^2} + 4x = 0$$

This is the differential equation of motion.

$$(ii) \quad \text{Let the solution of the equation be } x = a \sin (2t + \phi) \quad \text{and hence } \frac{dx}{dt} = 2a \cos (2t + \phi)$$

When $t = 0$, $x = 0.1\text{ m}$, $\therefore 0.1 = a \sin \phi$

When $t = 0$, $dx/dt = 0$, $\therefore 0 = 2a \cos \phi$, so that $a = 0$ or $\phi = \pi/2$; but $a \neq 0$, $\therefore \phi = \pi/2$

Therefore $0.1 = a \sin \pi/2$ or $a = 0.1\text{ m}$, $\therefore x = 0.1 \sin (2t + \pi/2) = 0.1 \cos 2t$

This equation gives the position of the particle at any time t .

(iii) Now, $v = dx/dt = -0.2 \sin 2t$.

(iv) Amplitude = **0.1 m**; Frequency, $\nu = \omega/2\pi = 2/2\pi = 1/\pi = \mathbf{0.3\text{ Hz}}$.

Ex. 4. A body having a mass of 4 gm executes S.H.M. The force acting on the body, when displacement is 8 gm, is 24 gm wt. Find the period. If the maximum velocity is 500 cm/sec, find the amplitude and maximum acceleration. ($g = 9.81\text{ m/sec}^2$).

Sol. Force acting on the body at 8 cm

$$\text{displacement} = m\omega^2 x = 4 \times 10^{-3} \times \omega^2 \times 8 \times 10^{-2} = 24 \times 10^{-3} \times 9.81$$

whence

$$\omega = \sqrt{6 \times 981/8} = 27.12\text{ rad/sec.}$$

$$\therefore \text{Period } T = \frac{2\pi}{\omega} = \frac{2 \times 3.14}{27.12} = \mathbf{0.2315\text{ sec.}}$$

The maximum velocity of the body $v = \omega a = 0.5\text{ m/sec.}$

$$\therefore a = \frac{v}{\omega} = \frac{0.5}{27.12} = \mathbf{0.1844\text{ m.}}$$

The maximum acceleration $= \omega^2 a = 735.75 \times 0.1844 = \mathbf{135.67\text{ m/sec}^2}$.

Ex. 5. The maximum acceleration of a linear harmonic oscillator is x and its maximum speed is y . What is its time period? (A.N.U. 2002)

Sol. In case of S.H.M., (Acceleration)_{max} $= \omega^2 a = x$...(i)

(Velocity)_{max} $= \omega a = y$...(ii)

Dividing eq. (i) by eq. (ii), we have

$$\frac{\omega^2 a}{\omega a} = \frac{x}{y} \text{ or } \omega = \frac{x}{y}; \therefore T = \frac{2\pi}{\omega} = \frac{2\pi y}{x}.$$

Ex. 6. The displacement equation of a particle describing simple harmonic motion is $x = 0.01 \sin 50\pi (t + 0.007)$ metre. Calculate the amplitude, time period, maximum velocity and initial phase for the particle. (A.N.U. July 2007)

Sol. Given $x = 0.01 \sin 50\pi (t + 0.007)$...(i)

Equation of S.H.M. $x = a \sin (\omega t + \phi)$...(ii)

Comparing eqs. (i) and (ii),

$$a = \mathbf{0.01} \text{ and } \omega = 50\pi$$

$$\therefore \text{Time period } T = \frac{2\pi}{\omega} = \frac{2\pi}{50\pi} = \mathbf{0.04\text{ sec.}}$$

$$\text{Particle velocity } v = \omega \sqrt{a^2 - x^2}$$

The velocity is maximum (i.e. v_{max}) at $x = 0$ i.e.,

$$v_{\text{max}} = \omega a = 50\pi \times 0.01 = 50 \times 3.14 \times 0.01 = \mathbf{1.67\text{ m/sec.}}$$

Comparison of (i) and (ii) also yields the value of initial phase ϕ i.e.,

$$\therefore \phi = 50\pi \times 0.007 = \mathbf{0.35\pi\text{ radian}}$$

Ex. 7. The displacement of a particle executing S.H.M. is given by

$$X = 0.1 \cos \pi [t + 1/3] \text{ metre.}$$

Calculate the maximum velocity and maximum acceleration of the particle. (K.U. 2007)

Sol. Displacement of the particle in S.H.M. is given by

$$x = 0.1 \cos \pi (t + 1/3) \text{ metre}$$

$$\text{Velocity} \quad v = \frac{dx}{dt} = -0.1 \times \pi \sin \pi (t + 1/3)$$

$$\therefore v_{\max} = 0.1 \times \pi = 0.1 \times 3.14 = \mathbf{0.314 \text{ m/sec.}}$$

$$\text{Acceleration} = \frac{dv}{dt} = -0.1 \times \pi^2 \cos \pi (t + 1/3)$$

$$\text{Maximum acceleration} = 0.1 \times \pi^2 = 0.1 \times (3.14)^2 = \mathbf{0.986 \text{ m/sec}^2}.$$

Ex. 8. A particle under S.H.M. has displacement of 0.4 m at the velocity 0.3 m/s and a displacement 0.3 m at the velocity 0.4 m/s. Calculate amplitude and frequency of the oscillation. (K.U. 1993)

$$\text{Sol. For S.H.M., velocity } v = \omega \sqrt{a^2 - x^2}$$

$$\text{Given, when } x = 0.4 \text{ m, } v = 0.3 \text{ m/sec.}$$

$$\therefore 0.3 = \omega \sqrt{a^2 - (0.4)^2} \quad \dots(i)$$

$$\text{Also, when } a = 0.3, v = 0.4 \text{ m/sec.}$$

$$\therefore 0.4 = \omega \sqrt{a^2 - (0.3)^2} \quad \dots(ii)$$

Squaring (i) and (ii) and dividing

$$\frac{(0.3)^2}{(0.4)^2} = \frac{a^2 - (0.4)^2}{a^2 - (0.3)^2} \text{ or } \frac{9}{16} = \frac{a^2 - 0.16}{a^2 - 0.09}$$

$$\text{or } 9a^2 - 0.81 = 16a^2 - 2.56 \text{ or } 16a^2 - 9a^2 = 2.56 - 0.81$$

$$\text{or } 7a^2 = 1.75 \text{ or } a^2 = 0.25 \text{ or } a = 0.5 \text{ m}$$

Substituting the value of a in eq. (ii), we get

$$0.4 = \omega \sqrt{(0.25 - 0.09)} = 0.4\omega$$

$$\text{whence, } \omega = 1 \text{ or } 2\pi v = 1, \therefore v = \frac{1}{2\pi} = \frac{1}{2 \times 3.14} = \mathbf{0.16 \text{ Hz.}}$$

Ex. 9. A particle performing S.H.M. has a maximum velocity of 0.4 m/s and a maximum acceleration of 0.8 m/sec². Calculate amplitude and the period of the oscillator. (A.N.U. 1994)

$$\text{Sol. Given, } v_{\max} = \omega a = 0.4 \text{ m/sec.}$$

$$\text{and } (\text{acc.})_{\max} = \omega^2 a = 0.8 \text{ m/sec}^2$$

$$\therefore \frac{\omega^2 a}{\omega a} = \frac{0.8}{0.4} \text{ or } \omega = 2 \text{ rad/sec.}$$

$$\text{Period } T = \frac{2\pi}{\omega} = \frac{2 \times 3.14}{2} = \mathbf{3.14 \text{ sec.}}$$

$$\text{Amplitude } a = \frac{v_{\max}}{\omega} = \frac{0.4}{2} = \mathbf{0.2 \text{ m.}}$$

Ex. 10. A particle describing S.H.M. makes 100 complete oscillations per minute and its maximum speed is 5 m/sec. What is the length of its path and its maximum acceleration? Find the velocity when the particle is half way between its mean position and the extreme position. (A.U. April 95)

$$\text{Sol. Frequency } v = 100 \text{ oscillations/min} = \frac{100}{60} = \frac{5}{3} \text{ Hz}$$

$$\therefore \omega = 2\pi v = 2 \times 3.14 \times \frac{5}{3} = 10.47 \text{ rad/sec.}$$

$$\text{Given, } v_{\max} = 5 \text{ m/sec.}$$

$$\text{But } v_{\max} = \omega a, \therefore 5 = 10.47 \times a$$

Hence amplitude $a = \frac{5}{10.47} = 0.48 \text{ m}$

Length of the path of the particle $= 2a = 2 \times 0.48 = \mathbf{0.96 \text{ m}}$

(acceleration) $_{\max} = \omega^2 a = (10.47)^2 \times 0.48 = \mathbf{52.62 \text{ m/sec}^2}$

Now, $v = \omega \sqrt{a^2 - x^2}$

For $x = a/2$, $v = \omega \sqrt{a^2 - (a/2)^2} = \omega \sqrt{3a^2/4} = \omega a \sqrt{3}/2$
 $= \frac{10.47 \times 0.48 \times 1.732}{2} = \mathbf{4.35 \text{ m/sec.}}$

Ex. 11. A particle executes S.H.M. with a period of 0.001 sec and amplitude 0.5 cm. Find its acceleration when it is at 0.2 cm away from its mean position and also obtain its maximum velocity.

(A.N.U. 2002)

Sol. Acceleration $= -\omega^2 x$

Here, $\omega = \frac{2\pi}{T} = \frac{2 \times 3.14}{0.001}$ and $x = 0.2 \text{ cm} = 0.2 \times 10^{-2} \text{ m}$

$\therefore$ Acceleration $= -\left(\frac{2 \times 3.14}{0.001}\right)^2 \times 0.2 \times 10^{-2} = -\mathbf{7.89 \times 10^4 \text{ m/sec}^2}$

Maximum velocity, $v_{\max} = \omega a = \frac{2 \times 3.14}{0.001} \times 0.5 \times 10^{-2} = \mathbf{31.4 \text{ m/sec.}}$

Ex. 12. The displacement equation of a particle describing S.H.M. is $x = 0.5 \cos (10\pi t + \pi/3) \text{ m}$. Calculate (a) amplitude, (b) frequency, (c) phase and (d) displacement after 1 sec. (A.N.U. 2001)

Sol. Standard equation of S.H.M. in cosine form is

$$x = a \cos (\omega t + \phi) \quad \dots(i)$$

Given equation of S.H.M. is $x = 0.5 \cos (10\pi t + \pi/3)$... (ii)

Comparing eqs. (i) and (ii), we have

(a) Amplitude $a = \mathbf{0.5 \text{ m}}$

(b) Angular frequency $\omega = 2\pi\nu = 10\pi$, $\therefore$ Frequency $\nu = \frac{10\pi}{2\pi} = \mathbf{5 \text{ Hz.}}$

(c) Phase (at $t = 1 \text{ sec.}$) $= (\omega t + \phi) = 10\pi \times 1 + \frac{\pi}{3} = \frac{31\pi}{3}$ or $\frac{\pi}{3}$

(d) Displacement x of the particle at time $t = 1 \text{ sec}$ is [using eq. (i)]

$$x = 0.5 \cos \left(10\pi + \frac{\pi}{3} \right) = 0.5 \cos \left(\frac{31\pi}{3} \right) = 0.5 \cos \frac{\pi}{3} = 0.5 \times 0.5 = \mathbf{0.25 \text{ metre.}}$$

Ex. 13. The acceleration of a simple harmonic oscillator is $4\pi^2 \text{ m/s}^2$ at a displacement of 1 m. How many oscillations does it make per second? (S.V.U. 2002)

Sol. In case of a harmonic oscillator, acceleration $= -\omega^2 x$ or $\omega^2 x$ in magnitude

Here, acceleration $= 4\pi^2 \text{ m/sec}^2$

$\therefore 4\pi^2 = \omega^2 \times 1$ or $\omega = 2\pi$

Hence, frequency $\nu = \frac{\omega}{2\pi} = \frac{2\pi}{2\pi} = \mathbf{1 \text{ Hz}}$

Thus, the oscillator makes **1 oscillation per second**.

Ex. 14. The displacement of a linear harmonic oscillator is given by $x = 4 \sin \left(\frac{\pi}{3}t + \frac{\pi}{6} \right)$.

Find the period and velocity at $t = 1$ sec.

(A.N.U. 2003)

Sol. Here, $x = 4 \sin \left(\frac{\pi}{3}t + \frac{\pi}{6} \right)$... (i)

Standard equation of S.H.M. $x = a \sin (\omega t + \phi)$... (ii)

Comparing (i) and (ii), we obtain

$$\omega = \frac{\pi}{3} \text{ or } \frac{2\pi}{T} = \frac{\pi}{3} \text{ or } T = 6 \text{ sec.}$$

Also, velocity $v = \frac{dx}{dt} = 4 \times \frac{\pi}{3} \cos \left(\frac{\pi}{3}t + \frac{\pi}{6} \right)$

$$\therefore \text{At } t = 1 \text{ sec, } v = \frac{4\pi}{3} \cos \left(\frac{\pi}{3} \times 1 + \frac{\pi}{6} \right) = \frac{4\pi}{3} \cos \left(\frac{3\pi}{6} \right) = \frac{4\pi}{3} \cos \left(\frac{\pi}{2} \right) = \frac{4\pi}{3} \times 0 = 0$$

Ex. 15. A body of mass 0.05 kg executes simple harmonic motion. When the displacement from the centre of motion is 0.04 m, the force acting on the body is 18×10^{-3} newton. If the maximum velocity is 4 metre/sec., find the amplitude and maximum acceleration.

Sol. Force, $F = kx = m\omega^2 x$ (in magnitude)

Here, $F = 18 \times 10^{-3}$ N, $m = 0.05$ kg and $x = 0.04$ m

$$\therefore 18 \times 10^{-3} = 0.05 \times \omega^2 \times 0.04 \text{ or } \omega^2 = \frac{18 \times 10^{-3}}{0.05 \times 0.04} = 9 \text{ or } \omega = 3 \text{ rad/sec.}$$

Given, $v_{\max} = 4$ m/sec. $= \omega a$

$$\therefore \text{Amplitude } a = \frac{4}{\omega} = \frac{4}{3} = 1.33 \text{ m.}$$

$$(\text{Acceleration})_{\max} = \omega^2 a = 9 \times \frac{4}{3} = 12 \text{ m/sec}^2.$$

Ex. 16. A simple harmonic wave is represented by

$$y = 10 \sin \left(\frac{2\pi t}{T} + \theta \right)$$

The time period is 30 sec. When $t = 0$, the displacement is 5 cm. Find (a) the phase angle at $t = 7.5$ sec. (b) the phase difference between two points at a time interval of 6 sec. (S.V.U. 1995)

Sol. At $t = 0$, $y = 5$ cm. Hence

$$5 = 10 \sin \theta \text{ or } \sin \theta = \frac{1}{2} \text{ or } \theta = \frac{\pi}{6} \text{ or } 30^\circ$$

Equation of S.H.M. is $y = 10 \sin \left(\frac{2\pi t}{T} + \theta \right)$

$$(a) \text{ Phase angle } \left(\frac{2\pi t}{T} + \theta \right) = \left(\frac{2\pi \times 7.5}{30} + \frac{\pi}{6} \right) = \left(\frac{\pi}{2} + \frac{\pi}{6} \right) = \frac{2\pi}{3} \text{ or } 120^\circ$$

(b) If θ_1 and θ_2 be the phase angles corresponding to times t_1 and t_2 respectively, then

$$\theta_1 = \frac{2\pi t_1}{T} + \theta \text{ and } \theta_2 = \frac{2\pi t_2}{T} + \theta$$

$$\therefore \text{Phase difference} = \theta_2 - \theta_1 = \frac{2\pi}{T}(t_2 - t_1) = \frac{2\pi}{30} \times 6 = \frac{2\pi}{5} \text{ rad or } 72^\circ.$$

Ex. 17. A particle is moving with simple harmonic motion in a straight line, when the distance of the particle from the equilibrium position has the value x_1 and x_2 the corresponding values of velocity

are u_1 and u_2 . Show that the period is $T = 2\pi \left[\frac{x_2^2 - x_1^2}{u_1^2 - u_2^2} \right]^{1/2}$.

Sol. Let the equilibrium position of the particle be O . If P, Q are the two positions of the particle at distances x_1 and x_2 , from O respectively [fig. 7.6], then velocity u_1 of the particle at the position P is given by

$$u_1 = \omega \sqrt{a^2 - x_1^2} \quad \dots(i)$$

The velocity u_2 at position Q is

$$u_2 = \omega \sqrt{a^2 - x_2^2} \quad \dots(ii)$$

Squaring (i) and (ii) and subtracting, we get

$$u_1^2 - u_2^2 = \omega^2(x_2^2 - x_1^2) \quad \text{or} \quad \omega = \sqrt{\frac{u_1^2 - u_2^2}{x_2^2 - x_1^2}}$$

$$\therefore \text{time period } T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{x_2^2 - x_1^2}{u_1^2 - u_2^2}}.$$

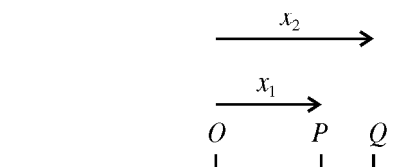


Fig. 7.6.

Ex. 18. A particle moving in simple harmonic motion has speeds 4 cm/sec. and 5 cm/sec. at distances 0.5 cm and 0.3 cm respectively from the mean position. Find the amplitude of oscillation.

(O.U. 2009)

Sol. Speed $v = \omega \sqrt{a^2 - x^2}$

At $x = 0.5$ cm, $v = 4$ cm/sec., $\therefore 4 = \omega \sqrt{a^2 - (0.5)^2} \quad \dots(i)$

At $x = 0.3$ cm, $v = 5$ cm/sec., $\therefore 5 = \omega \sqrt{a^2 - (0.3)^2} \quad \dots(ii)$

Squaring (i) and (ii) and dividing

$$\frac{4^2}{5^2} = \frac{a^2 - (0.5)^2}{a^2 - (0.3)^2} \quad \text{or} \quad \frac{16}{25} = \frac{a^2 - 0.25}{a^2 - 0.09}$$

or $16a^2 - 1.44 = 25a^2 - 6.25$ or $25a^2 - 16a^2 = 6.25 - 1.44$

or $9a^2 = 4.81$ or $a^2 = \frac{4.81}{9}$ or $a = 0.73$ cm.

Ex. 19. If the earth were a homogeneous sphere of radius R , and a straight hole were bored in it through its centre. Show that a particle dropped into the hole will execute a S.H.M. and find out its time period.

Sol. If the mass of the earth is M and R its radius, then the acceleration due to gravity at the surface of the earth given by

$$g = GM/R^2, \text{ where } G \text{ is gravitational constant.}$$

If the density of the earth is ρ , then $M = \frac{4}{3}\pi R^3 \cdot \rho$.

$\therefore g = \frac{4}{3}\pi R \rho \cdot G. \quad \dots(i)$

If g' be the value at depth h below the surface of the earth, then

$$g' = \frac{4}{3}\pi(R-h)\rho \cdot G. \quad \dots(ii)$$

Dividing eq. (ii) by eq. (i), we have $\frac{g'}{g} = \frac{R-h}{R}$ or $g' = \frac{g}{R}(R-h)$.

If we measure the distance from the centre of the earth, putting $R-h = x$, acceleration of the particle

is,
$$\frac{d^2x}{dt^2} = g' = -\frac{g}{R}x$$

Negative sign shows that the acceleration acts towards the centre of the earth, i.e., opposite to x increasing.

$$\therefore \frac{d^2x}{dt^2} + \frac{g}{R}x = 0.$$

Thus, the particle, dropped into the hole, will execute S.H.M. Its periodic time is given by

$$T = 2\pi\sqrt{\frac{R}{g}}.$$

Ex. 20. Two cities on the surface of the earth are joined by a straight smooth underground tunnel of length 640 km. A body is released into the tunnel from one city. How much time will it take to reach the other city? Derive the formula used. $G = 6.67 \times 10^{-11}$ S.I. units and $\rho = 5,520 \text{ kg/m}^3$. Calculate the velocity when the body would be nearest to the centre of the earth in its journey. **(Bombay 1967)**

Sol. Suppose that A and B are the two cities, which are connected by a straight tunnel of length $AB = 640 \text{ km}$.

Let O be the centre of the earth and R its radius. OC is the perpendicular, drawn from the centre O on AB , where C is the mid-point of the tunnel AB . Let a body of mass m be released into the tunnel from its end A and let it be at P, distant x from C at any instant t [fig. 7.7].

If M is the mass of the earth, then on its surface the weight (mg) of the body is given by

$$mg = GMm/R^2.$$

The weight of the body at P, distant $OP (= r)$ from the centre of the earth is

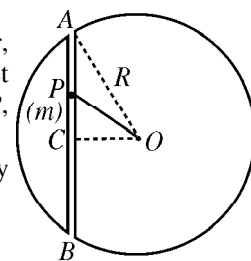


Fig. 7.7.

$$mg' = \frac{GM'm}{r^2} = \frac{G \cdot \frac{4}{3}\pi r^3 \rho m}{r^2} = \frac{G \cdot \frac{4}{3}\pi R^3 \rho mr}{R^3} = \frac{GMm}{R^2} \cdot \frac{r}{R} \quad [\because M' = \frac{4}{3}\pi r^3 \rho \text{ and } M = \frac{4}{3}\pi R^3 \rho]$$

The direction of this force is PO , whose component along PC

$$= \frac{GMm}{R^2} \cdot \frac{r}{R} \cos OPC = \frac{GMm}{R^2} \cdot \frac{r}{R} \cdot \frac{x}{r} = \frac{GMm}{R^3}x$$

Hence the equation of motion of mass m is

$$m \frac{d^2x}{dt^2} + \frac{GMm}{R^3}x = 0 \quad \text{or} \quad \frac{d^2x}{dt^2} + \frac{GM}{R^3}x = 0.$$

The equation represents a S.H.M. of period $T = 2\pi\sqrt{\frac{R^3}{GM}}$ i.e., this is the time in which the body will reach from A to B and again from B to A. Hence, time taken by the body to pass through the tunnel is

$$\frac{T}{2} = \pi\sqrt{\frac{R^3}{GM}} = \pi\sqrt{\frac{R^3}{G \cdot \frac{4}{3}\pi R^3 \rho}} = \sqrt{\frac{3\pi}{4\rho G}} = \sqrt{\frac{3 \times 3.14}{4 \times 5520 \times 6.62 \times 10^{-11}}} = \mathbf{2528 \text{ sec.}}$$

$$\begin{aligned}\text{Velocity at C, } (v) = \omega a &= \sqrt{\frac{GM}{R^3}} \cdot \frac{AB}{2} = \sqrt{G \cdot \frac{4}{3} \pi \rho} \cdot \frac{AB}{2} \\ &= \sqrt{6.67 \times 10^{-11} \times \frac{4}{3} \times 3.14 \times 5520} \cdot \frac{640}{2} \times 10^3 = 3.972 \times 10^2 \text{ m/sec.}\end{aligned}$$

[Energy of Harmonic Oscillator]

Ex. 21. A particle executing S.H.M. has a maximum displacement 'A' and maximum kinetic energy 'E'. Find its displacement at the instant its K.E. is $3E/4$. (O.U. Oct. 2007; A.N.U. 1990)

Sol. Maximum K.E. = $\frac{1}{2} m \omega^2 A^2 = E$

Instantaneous K.E. = $\frac{1}{2} m \omega^2 (A^2 - x^2)$

According to the question

$$\frac{1}{2} m \omega^2 (A^2 - x^2) = \frac{3E}{4} \text{ or } \frac{1}{2} m \omega^2 (A^2 - x^2) = \frac{3}{4} \cdot \frac{1}{2} m \omega^2 A^2$$

or $A^2 - x^2 = \frac{3}{4} A^2$ or $x^2 = A^2 - \frac{3A^2}{4} = \frac{A^2}{4}$

$\therefore x = A/2$.

Ex. 22. The potential and kinetic energies are same for a S.H. oscillator at a displacement of 2 cm. Find its amplitude. (S.V.U. 2002)

Sol. Potential energy $U = \frac{1}{2} kx^2$

Kinetic energy $K = \frac{1}{2} k (a^2 - x^2)$

According to the problem,

$$U = K \text{ i.e., } \frac{1}{2} kx^2 = \frac{1}{2} k (a^2 - x^2) \text{ or } x^2 = a^2 - x^2 \text{ or } a^2 = 2x^2 \text{ or } a = \sqrt{2} x$$

Here, $x = 2 \text{ cm} = 0.02 \text{ m}$, $\therefore a = \sqrt{2} \times 0.02 = 1.414 \times 0.02 \text{ m} = 2.83 \text{ cm}$.

Ex. 23. The total energy of a particle executing S.H.M. is 0.256 J. Time period is 2 seconds. The displacement of the particle at $(T/4)$ sec. is $8\sqrt{2} \text{ cm}$ from the mean position. Calculate the amplitude and mass of the particle. (A.N.U. 1994)

Sol. Equation of S.H.M., $x = a \sin \omega t$

Here, $x = 8\sqrt{2} \text{ cm} = 0.08\sqrt{2} \text{ m}$ and $\omega t = \frac{2\pi}{T} \times \frac{T}{4} = \frac{\pi}{2}$

$\therefore 0.08\sqrt{2} = a \sin \pi/2$ or $a = 0.08\sqrt{2} \text{ m} = 8\sqrt{2} \text{ cm}$.

Total energy in S.H.M. is given by

$$E = \frac{1}{2} m \omega^2 a^2$$

Here, $E = 0.256 \text{ J}$, $\omega = \frac{2\pi}{T} = \frac{2\pi}{2} = 3.14 \text{ rad/sec}$. and $a = 0.08\sqrt{2} = 0.113 \text{ m}$

$\therefore 0.256 = \frac{1}{2} \times m \times (3.14)^2 \times (3.14)^2 \times (0.113)^2$

whence
$$m = \frac{2 \times 0.256}{(3.14)^2 \times (0.113)^2} = \mathbf{4.067 \text{ kg.}}$$

Ex. 24. A particle of mass 5 gm executes S.H.M. has amplitude of 8 cm. If it makes 16 vibrations per sec. Find its maximum velocity and energy at mean position.

Sol. Maximum velocity $v_{\max} = \omega a = 2\pi\nu a$

Here, $\nu = 16 \text{ Hz}$ and $a = 8 \text{ cm} = 0.08 \text{ m}$

$$v_{\max} = 2 \times 3.14 \times 16 \times 0.08 = \mathbf{8.038 \text{ m/sec.}}$$

The energy at mean position is entirely in kinetic form. Hence

$$E = K_{\max} = \frac{1}{2} m v_{\max}^2 = \frac{1}{2} \times 5 \times 10^{-5} \times (8.038)^2 = \mathbf{0.16 \text{ joule.}}$$

Ex. 25. The total energy of a particle executing a S.H.M. of period 2π second is 10.24×10^{-4} joule. The displacement of the particle at $\pi/4$ sec. is $8\sqrt{2}$ cm. Calculate the amplitude of the motion and the mass of the particle.

Sol. The displacement of a particle, executing S.H.M. is given by

$$x = a \sin \pi t = a \sin 2\pi t/T$$

$$[\because \omega = 2\pi/T]$$

$$\therefore 8\sqrt{2} \times 10^{-2} = a \sin \frac{2\pi}{2\pi} \times \frac{\pi}{4} = a \sin \frac{\pi}{4} = \frac{a}{\sqrt{2}}$$

$$\therefore \text{Amplitude of vibration } a = 0.16 \text{ m.}$$

If m be the mass of the particle, then its total energy is given by

$$E = \frac{2m\pi^2 a^2}{T^2} = 10.24 \times 10^{-4} \text{ or } m = \frac{10.24 \times 10^{-4} \times T^2}{2\pi^2 \times a^2}$$

$$\text{Here } T = 2\pi \text{ sec and } a = 0.16 \text{ m, } \therefore m = \frac{10.24 \times 10^{-4} \times 2\pi \times 2\pi}{2 \times \pi^2 \times 0.16 \times 0.16} = \mathbf{0.8 \text{ kg.}}$$

Ex. 26. If the potential energy of a harmonic oscillator in its resting position is 5 joules and the total energy is 9 joules, when the amplitude is 1 metre, what is the force constant? If its mass is 2 kg, what is the period?

Sol. Potential energy in rest position = 5 joules.

At the maximum displacement (equal to amplitude), the potential energy = 9 joules.

$\therefore$ Gain in potential energy = $9 - 5 = 4$ joules.

The gain in energy at the maximum displacement should be equal to $\frac{1}{2} k x_{\max}^2 = \frac{1}{2} k \times (1)^2 = k/2$ joules. Thus,

$$k/2 = 4 \text{ or } k = \mathbf{8 \text{ joules/m.}}$$

$$\therefore \text{Period } T = 2\pi\sqrt{m/k} = 2\pi\sqrt{2/8} = \mathbf{3.14 \text{ sec.}}$$

Ex. 27. What fraction of total energy is kinetic and what fraction is potential when the displacement is one-third of amplitude? (O.U. 1995)

Sol. Given, displacement $x = a/3$.

$$\text{Total energy } E = \frac{1}{2} k a^2, \text{ K.E.} = \frac{1}{2} k (a^2 - x^2) \text{ and P.E.} = \frac{1}{2} k x^2$$

$$\therefore \frac{\text{Kinetic energy}}{\text{Total energy}} = \frac{\frac{1}{2} k (a^2 - x^2)}{\frac{1}{2} k a^2} = \frac{a^2 - (a/3)^2}{a^2} = \frac{a^2 - a^2/9}{a^2} = \mathbf{\frac{8}{9}}$$

$$\text{and} \quad \frac{\text{Potential energy}}{\text{Total energy}} = \frac{\frac{1}{2}kx^2}{\frac{1}{2}ka^2} = \frac{x^2}{a^2} = \frac{(a/3)^2}{a^2} = \frac{1}{9}.$$

Ex. 28. Prove that in simple harmonic motion when the average is taken with respect to the position over one cycle, the average potential energy equals $\frac{1}{6}ka^2$ and the average kinetic energy equals $\frac{1}{3}ka^2$. Explain physically why this result is different to that when the average is taken with respect to time.

$$\begin{aligned} \text{Sol.} \quad U &= \frac{1}{2}kx^2 \quad \text{and} \quad K = \frac{1}{2}k(a^2 - x^2) \\ \therefore U_{av} &= \frac{1}{a} \int_0^a \frac{1}{2}kx^2 dx = \frac{k}{2a} \left(\frac{x^3}{3} \right)_0^a = \frac{1}{6}ka^2 \\ \text{and} \quad K_{av} &= \frac{1}{a} \int_0^a \frac{1}{2}k(a^2 - x^2) dx = \frac{k}{2a} \left[a^2x - \frac{x^3}{3} \right]_0^a = \frac{1}{3}ka^2 \end{aligned}$$

7.5. Linear Oscillations of Spring-Mass System

The system consists of a massless spring, one of its ends is connected to a mass m and the other end is fixed to a rigid support. Here, the mass and spring are on a smooth horizontal surface [fig. 7.8]. If a force is applied on the mass to stretch or compress the spring and then released, the motion of the mass is simple harmonic. If x is the displacement of mass m at any time, then the restoring force exerted by the spring is

$$F_x = -kx, \text{ where } k \text{ is force constant.}$$

$$\text{Also,} \quad \text{force} = m(d^2x/dt^2)$$

Therefore, equation of motion is

$$m \frac{d^2x}{dt^2} = -kx \quad \text{or} \quad \frac{d^2x}{dt^2} + \frac{k}{m}x = 0 \quad \text{or} \quad \frac{d^2x}{dt^2} + \omega^2x = 0 \quad \dots(1)$$

$$\text{where } \omega = \sqrt{k/m}.$$

This equation represents that the motion of spring-mass system is simple harmonic. Its periodic time and frequency is given by

$$T = 2\pi\sqrt{\frac{m}{k}}; \quad n = \frac{1}{2\pi}\sqrt{\frac{k}{m}} \quad \dots(2)$$

The solution of the equation of motion is

$$x = a \sin(\omega t + \phi), \text{ where } \omega = \sqrt{\frac{k}{m}} \quad \dots(3)$$

Now, suppose the spring-mass system is hanging vertically from a rigid support and it is oscillating in earth's gravitational field. [Fig. 7.9].

The question arises : Is there any effect of gravity on the oscillations of the hanging spring-mass arrangement ? For this purpose, first we consider a spring of negligible mass, suspended vertically from a rigid support S [fig. 7.10(a)]. To the free end, now we attach an object of mass m . Due to the weight mg of the object, the spring extends and suppose that the equilibrium position is attained at an extension x_0 of the spring [fig. 7.10 (b)]. At equilibrium, the spring force ($-kx_0$) balances the weight (mg) of the object and the net force is zero i.e.,

$$mg - kx_0 = 0 \quad \dots(4)$$



Fig. 7.8.

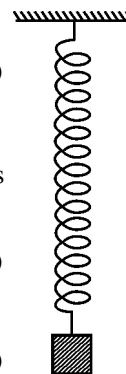


Fig. 7.9.

Now if mass is displaced downwards through distance x from this equilibrium position O , the total spring force will be $-k(x_0 + x)$. Hence the net force is $mg - k(x_0 + x) = -kx$, which tends to restore the object to the equilibrium position O [fig. 7.10(c)]. Thus the restoring force is $F = -kx$ and the equation of motion is

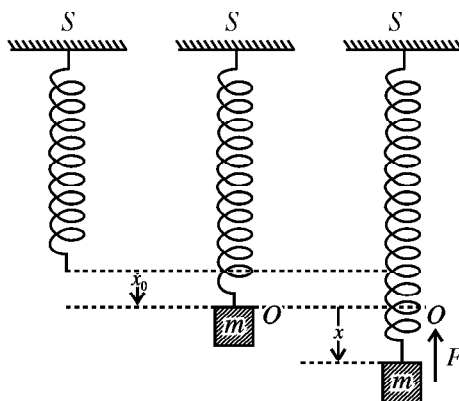


Fig. 7.23 : (a) Spring hanging from a rigid support S ; (b) Equilibrium position of the mass m , attached to the free end of the spring : $mg - kx_0 = 0$; (c) Oscillating mass under the restoring force $F = -kx$.

$$m \frac{d^2x}{dt^2} = -kx \quad \text{or} \quad \frac{d^2x}{dt^2} + \frac{k}{m}x = 0 \quad \dots(5)$$

This is the standard differential equation of simple harmonic motion whose frequency of vibration is given by eq. (2). Thus the gravity has no effect on the force constant or frequency of oscillation. It is to be noted that the oscillations are vertical due to the vertical tension in the spring, caused by the weight of the object.

Ex. 1. A spring is stretched by 8 cm by a force of 10 newton. Find the force constant of the spring. What will be the period of a 4 kg mass suspended by it ? (A.N.U. 2002)

Sol. If a force F stretched a spring by x , then in equilibrium

$$F = kx, \text{ where } k \text{ is force constant of the spring.}$$

Here, $F = 10 \text{ N}$ and $x = 8 \text{ cm} = 0.08 \text{ m}$.

$$\therefore \text{Force constant } k = \frac{F}{x} = \frac{10}{0.08} = 125 \text{ N/m}$$

$$\text{Period } T = 2\pi\sqrt{\frac{m}{k}} = 2 \times 3.14 \sqrt{\frac{4}{125}} = 1.123 \text{ sec.}$$

Ex. 2. A 1.6 kg-wt. extends a spring 8 cm from its unstretched position. The mass is replaced by a body of 50 gm. The mass is pulled and then released, find the period of oscillation.

Sol. $F = kx$, or $k = F/x$

$$\therefore k = \frac{1.6g}{0.08} = \frac{1.6 \times 9.8}{0.08} = 196 \text{ N/m}$$

$$\therefore \text{Period, } T = 2\pi\sqrt{\frac{m}{k}} = 2\pi\sqrt{\frac{50 \times 10^{-3}}{196}} = 0.11 \text{ sec.}$$

Ex. 3. A body of mass 0.2 kg is suspended from a spring, the spring extends by 0.1 m. The velocity of the body is 0.4 m/sec, when the displacement is 0.05 m. Find the time period, frequency and amplitude (A.N.U. 2001)

Sol. $F = mg = kx$ or $k = \frac{mg}{x}$

Here, $m = 0.2$ kg, $g = 9.8$ m/sec.² and $x = 0.1$ m

$$\therefore k = \frac{0.2 \times 9.8}{0.1} = 19.7 \text{ N/m}$$

Period,
$$T = 2\pi\sqrt{\frac{m}{k}} = 2\pi\sqrt{\frac{0.2}{19.6}} = \mathbf{0.634}.$$

Frequency,
$$\nu = \frac{1}{T} = \frac{1}{0.634} = \mathbf{1.577 \text{ Hz}}$$

Also, velocity
$$v = \omega\sqrt{a^2 - x^2}$$

Here, $v = 0.4$ m/sec, $\omega = 2\pi\nu = 2\pi \times 1.577 = 9.897$ rad/sec. and $x = 0.05$ m.

$$\therefore 0.4 = 9.897 \sqrt{a^2 - (0.05)^2} \text{ or, } (0.4)^2 = (9.897)^2 [(a^2 - (0.05)^2)] \text{ or } a^2 = (0.05)^2 + \frac{(0.4)^2}{(9.897)^2}$$

whence
$$a = \mathbf{0.0643 \text{ m.}}$$

Ex. 4. A body of mass 4.9 kg hangs from a spring and oscillates with a period of 0.6 sec. How much will the spring shorten when the body is removed ? (A.N.U. July 2007)

Sol. For spring-mass system period $T = 2\pi\sqrt{\frac{m}{k}}$

Here, $m = 4.9$ kg and $T = 0.6$ sec

$$\therefore 0.6 = 2\pi\sqrt{\frac{4.9}{k}},$$

whence
$$k = \frac{4\pi^2 \times 4.9}{(0.6)^2} = \frac{4 \times (3.14)^2}{(0.6)^2} \times 4.9 = 536.8 \text{ N/m}$$

Due to the weight of the mass let the spring be extended by x_0 i.e., $mg = kx_0$

$$\therefore x_0 = \frac{mg}{k} = \frac{4.9 \times 9.8}{536.8} = \mathbf{0.089 \text{ m} = 8.9 \text{ cm.}}$$

Ex. 5. The length of a weightless spring increases by 2 cm when a weight of 1.0 kg is suspended from it. The weight is pulled down by 10 cm and released. Determine (i) period of oscillation (ii) total energy of oscillation of spring mass system.

Sol.
$$k = \frac{F}{x} = \frac{1.0 \times 9.8}{0.02} = 490 \text{ N/m}$$

$$\therefore \text{Period } T = 2\pi\sqrt{\frac{m}{k}} = 2\pi\sqrt{\frac{1.0}{490}} = \mathbf{0.29 \text{ sec.}}$$

$$\text{Total energy} = \frac{1}{2}ka^2 = \frac{1}{2} \times 490 \times (0.1)^2 = \mathbf{2.45 \text{ joule.}}$$

Ex. 6. A mass of 3 kg is hung from a vertical spring. When a mass of 0.5 kg is gently added the spring is further stretched by 5 cm. If the extra mass is removed and the first mass is set into oscillation, calculate the period of oscillation. (A.N.U. 1997)

Sol. Force constant $k = \frac{F}{x} = \frac{mg}{x} = \frac{0.5 \times 9.8}{5 \times 10^{-2}} = 98 \text{ N/m}$

$$\text{Period } T = 2\pi\sqrt{\frac{m}{k}} = 2 \times 3.14 \sqrt{\frac{3}{98}} = \mathbf{1.1 \text{ sec.}}$$

Ex. 7. The period of oscillation of a spring is 0.1 second. The amplitude of oscillation is 2 cm. A body of mass 0.2 kg is hung from it. Calculate the maximum P.E. that will be stored in it.

(A.N.U. 1996, 93)

Sol. Maximum P.E. = $\frac{1}{2}m\omega^2 a^2 = \frac{1}{2}m \left(\frac{2\pi}{T}\right)^2 a^2 = \frac{2\pi^2 m a^2}{T^2}$ $\left(\because \omega = \frac{2\pi}{T}\right)$

$$= \frac{2 \times (3.14)^2 \times (0.2) \times (0.02)^2}{(0.1)^2} = \mathbf{0.158 \text{ joule.}}$$

Ex. 8. Fig. 7.11 shows a mass M resting on a smooth table between two firm supports A , B and controlled by two massless springs. If the mass M is 30 gm, and the force constants of the two springs are 2 and 1 N/m, deduce (i) the frequency of small oscillations of M , (ii) the energy of oscillations for amplitude 0.5 cm.

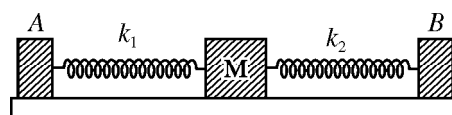


Fig. 7.11.

Sol. If the mass M is displaced through a distance x towards left or right, the restoring forces in the two springs are

$$F_1 = -k_1 x \text{ and } F_2 = -k_2 x$$

acting on the mass along the same direction. So that, the total restoring force on the mass M is $F = -(k_1 + k_2)x$. Thus, the effective force constant of the system is $k = k_1 + k_2 = 2 + 1 = 3$ N/m.

(i) Mass $M = 30$ gm = 0.03 kg

$$\therefore \text{Frequency, } \nu = \frac{1}{2\pi} \sqrt{\frac{k}{M}} = \frac{1}{2\pi} \sqrt{\frac{3}{0.03}} = \frac{5}{\pi} = \frac{5}{3.14} = \mathbf{1.6 \text{ (nearly).}}$$

(ii) Energy $E = \text{Max. P.E.} = \frac{1}{2} k x_{\max}^2 = \frac{1}{2} \times 3 \times (0.5 \times 10^{-2})^2 = \mathbf{3.75 \times 10^{-5} \text{ J.}}$

Ex. 9. (a) Two springs of force constant k_1 and k_2 are connected in series. What will be the effective force constant of the system. (A.N.U. March 2007)

(b) If $M = 30$ gm, $k_1 = 2$ N/m and $k_2 = 1$ N/m, find the effective force constant and period of oscillation.

Sol. (a) When the springs of force constant k_1 and k_2 are connected as shown in **fig. 7.12**, the displacement x of mass M produces the same restoring force F in the two springs. If the two springs are extended through x_1 and x_2 , then

$$F = -kx_1 = -kx_2$$

and therefore

$$x = x_1 + x_2 = -\frac{F}{k_1} - \frac{F}{k_2} = -F \left(\frac{k_1 + k_2}{k_1 k_2} \right), \therefore F = -\frac{k_1 k_2}{k_1 + k_2} x$$

Hence force constant, $k = \frac{k_1 k_2}{k_1 + k_2}$.

(b) Here, $k_1 = 2$ N/m, $k_2 = 1$ N/m and $M = 30$ gm = 0.03 kg

$$\therefore \text{Effective force constant } k = \frac{k_1 k_2}{k_1 + k_2} = \frac{2 \times 1}{2 + 1} = \frac{2}{3} \text{ N/m}$$

$$\text{Period of oscillation } T = 2\pi \sqrt{\frac{M}{k}} = 2\pi \sqrt{\frac{0.03 \times 3}{2}} = \mathbf{1.33 \text{ sec.}}$$

Ex. 10. Spring having mass : Find the period and frequency of oscillation of a body of mass M suspended from a uniform spring of force constant k and mass m is given by $n = \frac{1}{2\pi} \sqrt{\frac{k}{M + m/3}}$. (K.U. 2005)

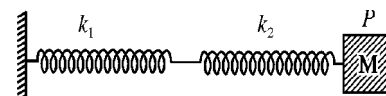


Fig. 7.12.

Sol. If l is the length of the spring, then mass per unit length $= m/l$.

Now, let us consider a length ds of the spring at a distance s (measured along the length) below the fixed upper end [fig. 7.13]. The mass of this small element is $(m/l) ds$.

If the velocity of the lowest point of the spring is v , then evidently the velocity of the length ds at a distance s from rigid support will be $(s/l)v$.

$$\therefore \text{Kinetic energy of the element} = \frac{1}{2} \left(\frac{m}{l} ds \right) \left(\frac{vs}{l} \right)^2$$

$$\therefore \text{K.E. of the whole spring} = \int_0^l \frac{1}{2} \frac{mv^2}{l^3} s^2 ds = \frac{1}{2} \frac{mv^2}{l^3} \cdot \frac{l^3}{3} = \frac{mv^2}{6}$$

$$\text{Kinetic energy of the mass } M \text{ at the lower end of the spring} = \frac{1}{2} Mv^2.$$

$$\text{Total kinetic energy of the system} = \frac{1}{2} Mv^2 + \frac{1}{6} mv^2 = \frac{v^2}{2} \left(M + \frac{m}{3} \right)$$

If x is the displacement of the mass from its mean position, then $v = \frac{dx}{dt}$

and
$$\text{total K.E.} = \left(M + \frac{m}{3} \right) \left(\frac{dx}{dt} \right)^2.$$

$$\text{Potential energy of the spring} = \int_0^x kx dx = \frac{1}{2} kx^2.$$

From the law of conservation of energy

$$\frac{1}{2} \left(M + \frac{m}{3} \right) \left(\frac{dx}{dt} \right)^2 + \frac{1}{2} kx^2 = \text{a constant.}$$

Differentiating it with respect to time and dividing by dx/dt , we have

$$\left(M + \frac{m}{3} \right) \frac{d^2x}{dt^2} + kx = 0 \quad \text{or} \quad \frac{d^2x}{dt^2} + \frac{k}{(M + m/3)} x = 0$$

This is the equation of simple harmonic motion, whose time period and frequency are given by

$$T = 2\pi \sqrt{\frac{(M + m/3)}{k}} \quad \text{and} \quad \nu = \frac{1}{T} = \frac{1}{2\pi} \sqrt{\frac{k}{(M + m/3)}}.$$

Ex. 11. A body of mass 0.095 kg hanging from a spring of mass 0.015 kg oscillates up and down simple harmonically. The force constant of the spring is $10\pi^2$ N/m. Calculate the frequency of oscillations. The spring is now replaced with a heavier one but the same spring constant. How does the frequency change ? (O.U. 2002)

Sol. The frequency of oscillation of a spring, having mass m and loaded with a mass M , is given by

$$\nu = \sqrt{\frac{k}{M + (m/3)}} = \frac{1}{2\pi} \sqrt{\frac{10\pi^2}{0.095 + (0.015/3)}} = \frac{1}{2\pi} \sqrt{\frac{10\pi^2}{0.1}} = \frac{10\pi}{2\pi} = 5 \text{ Hz.}$$

If the spring is replaced with a heavier one, the mass m will increase and hence the frequency ν will decrease.

Ex. 12. In a vertical spring-mass system, the period of the oscillation is 0.26 second when the mass is 0.4 kg and period becomes 0.34 second when an additional mass of 0.3 kg is added. Calculate the mass of the spring and the force constant of the spring. (A.N.U. 1994, S.V.U. 1995)

Sol. When a mass M is suspended from the free end of a spring, having mass m , the period of oscillation is given by

$$T = \sqrt{\frac{M + (m/3)}{k}}$$

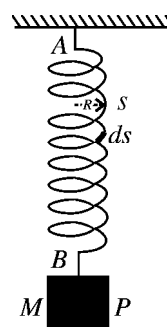


Fig. 7.13.

where k is force constant.

Here, $M = 0.4$ kg, $T = 0.26$ second and when $M = 0.7$ kg, $T = 0.34$ second.

$$\therefore 0.26 = 2\pi \sqrt{\frac{0.4 + (m/3)}{k}} \quad \text{or,} \quad \frac{(0.26)^2}{0.4 + (m/3)} = \frac{4\pi^2}{k} \quad \dots(i)$$

$$\text{Similarly,} \quad 0.34 = 2\pi \sqrt{\frac{0.7 + (m/3)}{k}} \quad \text{or} \quad \frac{(0.34)^2}{0.7 + (m/3)} = \frac{4\pi^2}{k} \quad \dots(ii)$$

From eqs. (i) and (ii), we have

$$\frac{(0.26)^2}{0.4 + (m/3)} = \frac{(0.34)^2}{0.7 + (m/3)} \quad \text{or} \quad \frac{0.7 + (m/3)}{0.4 + (m/3)} = \left(\frac{0.34}{0.26}\right)^2 = 1.71$$

$$m = \mathbf{0.0675 \text{ kg.}}$$

$$\therefore 0.7 + (m/3) = 0.684 + (m/3) \times 1.71$$

$$\text{or} \quad \frac{m}{3} (1.71 - 1) = 0.7 - 0.684 \quad \text{or} \quad m = 3 \times \frac{0.016}{0.71} = \mathbf{0.676 \text{ kg.}}$$

The force constant can be obtained by putting the value of m in (i) i.e.,

$$\frac{(0.26)^2}{0.4 + (0.0676/3)} = \frac{4\pi^2}{k} = \frac{4(3.14)^2}{k} \quad \text{or} \quad \frac{0.0676}{0.4225} = \frac{39.44}{k}$$

$$\therefore k = 39.44 \times \frac{0.4225}{0.0676} = \mathbf{246.5 \text{ N/m.}}$$

7.6. Angular Oscillations —Compound and Torsional Pendulums

In case of spring-mass system, the mass oscillates about an equilibrium position along a straight line. When we observe the oscillations of a pendulum or balance wheel in a watch, the system rotates back and forth about an equilibrium position under the action of restoring torque. In these rotational systems, the restoring torque and moment of inertia replace restoring force and mass of spring-mass system respectively. If the restoring torque in the system is proportional to the angular displacement and directed opposite to it, the system executes **angular simple harmonic vibrations** and a differential equation analogous to the equation of harmonic oscillator can be established. We consider here the restoring torques due to gravity (e.g., in compound pendulum) and due to elasticity (e.g., in torsional pendulum).

7.6.1. Compound Pendulum – Determination of g

A compound pendulum is a rigid body capable of oscillating freely in a vertical plane about a horizontal axis under the action of gravity. Let **fig. 7.14** represent the vertical section of a rigid body through the centre of gravity G . It is capable of oscillating about a horizontal axis through S , the centre of suspension, where the vertical section through G meets the horizontal axis. In equilibrium, the centre of gravity G of the body lies vertically below S . Let the distance $SG = l$. Now if the body is displaced to one side and left free, it oscillates about the equilibrium position. If θ be the angular displacement at any instant t , then the restoring torque about S is given by

$$I \frac{d^2\theta}{dt^2} = - mgl \sin \theta \quad \dots(1)$$

where I is the moment of inertia of the body about the horizontal axis through S and mg is the weight of the body.

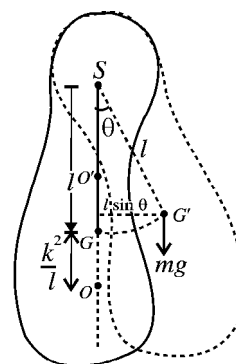


Fig. 7.14 : Compound pendulum.

For sufficiently small angular displacement θ , eq. (1) assumes the form

$$\frac{d^2\theta}{dt^2} + \frac{mgl}{I}\theta = 0$$

or
$$\frac{d^2\theta}{dt^2} + \omega^2\theta = 0 \quad \dots(2)$$

where $\omega = \sqrt{mgl/I}$.

Eq. (2) represents that the compound pendulum is executing a simple harmonic motion whose period of oscillation is given by

$$T = \frac{2\pi}{\omega} \text{ or } T = 2\pi\sqrt{\frac{I}{mgl}} \quad \dots(3)$$

If I_g be the moment of inertia of the rigid body about a parallel axis through its centre of gravity G , then by the application of the theorem of parallel axis, we have

$$I = I_g + ml^2 = mK^2 + ml^2 = m(K^2 + l^2) \quad \dots(4)$$

where K is the radius of gyration of the body about a parallel axis through G .

Hence the expression of periodic time of the pendulum becomes

$$T = 2\pi\sqrt{\frac{m(K^2 + l^2)}{mgl}} \text{ or } T = 2\pi\sqrt{\frac{\frac{K^2}{l} + l}{g}} \quad \dots(5)$$

Here, the periodic time is the same as that of a simple pendulum of length $\left(\frac{K^2}{l} + l\right)$, say L . This length L is called the **length of an equivalent simple**

pendulum. If we take a point O on the line SG produced such that $SO = \frac{K^2}{l} + l$, the length of an equivalent simple pendulum, then the point O is called **centre of oscillation.** (fig. 7.15)

Interchangeability of centres of suspension and oscillation : The distance of the centre of oscillation from the centre of gravity = $K^2/l = l'$ (say).

Thus, the expression for the time period of the compound pendulum can be written as

$$T = 2\pi\sqrt{\frac{l' + l}{g}} \quad \dots(6)$$

If the pendulum is inverted and made to oscillate about the centre of oscillation, then its time period T' will be given by

$$T' = 2\pi\sqrt{\frac{(K^2/l') + l'}{g}}$$

But $K^2/l = l'$ or $K^2 = ll'$

$$\therefore T' = 2\pi\sqrt{\frac{(ll'/l') + l'}{g}} \text{ or } T = 2\pi\sqrt{\frac{l + l'}{g}} \quad \dots(7)$$

From eqs. (6) and (7), we get

$$T = T' \quad \dots(8)$$

Hence, the time periods of oscillation about the centre of suspension and the centre of oscillation are equal, i.e., they are interchangeable.

Condition for maximum and minimum time periods (Variation of T with l) : The period of oscillation of a compound pendulum, is given by

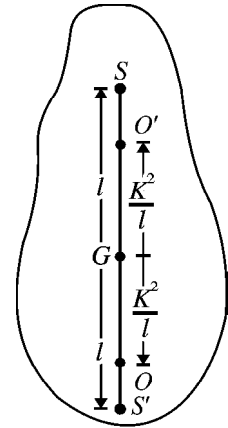


Fig. 7.15.

$$T = 2\pi \sqrt{\frac{(K^2/l) + l}{g}} \quad \dots(9)$$

Obviously, as l is varied, time period T also alters, if other factors remain constant. Relation (16) shows **the variation of time period with length**.

Eq. (9) can be written as

$$T^2 = \frac{4\pi^2}{g} \left(\frac{K^2}{l} + l \right).$$

Differentiating it with respect to l , we get

$$2T \frac{dT}{dl} = \frac{4\pi^2}{g} \left(-\frac{K^2}{l^2} + 1 \right). \quad \dots(10)$$

For minimum and maximum periods $dT/dl = 0$, i.e., $-K^2/l^2 + 1 = 0$ or $l = \pm K$

If we calculate the value of d^2T^2/dl^2 and put $l = K$, the quantity d^2T^2/dl^2 comes out to be positive. Hence for this value of l , time period of compound pendulum has minimum value. Condition for the period to be **minimum** is

$$l = K \quad \dots(11)$$

Thus, *the period of oscillation of the pendulum is minimum, when the distance between the centre of suspension and the centre of gravity is equal to the radius of gyration of the pendulum about a parallel axis through the centre of gravity.*

Now, if we put $l = 0$ in eq. (9), the value of T becomes infinite, i.e.,

$$T = \text{a maximum} \quad \dots(12)$$

In other words, *the period of oscillation of the compound pendulum is maximum, when the axis of suspension passes through centre of gravity of the pendulum.*

Four points, collinear with the C.G., about which the time period is the same : The time period of a compound pendulum, is given by

$$T = 2\pi \sqrt{\frac{K^2 + l^2}{lg}} \quad \dots(13)$$

where K is the radius of gyration of the compound pendulum about an axis passing through its centre of gravity and perpendicular to the plane of the oscillation and l the distance between the centre of suspension and its centre of gravity.

Squaring eq. (13), we get

$$T^2 = 4\pi^2 \left(\frac{K^2 + l^2}{lg} \right) \quad \text{or} \quad l^2 + K^2 = \frac{lgT^2}{4\pi^2} \quad \text{or} \quad l^2 - \frac{gT^2}{4\pi^2} \cdot l + K^2 = 0 \quad \dots(14)$$

This is a quadratic equation in l and gives two values of l . If l_1 and l_2 be two values, then

$$l_1 + l_2 = \frac{gT^2}{4\pi^2} \quad \dots(15)$$

$$\text{and} \quad l_1 l_2 = K^2. \quad \dots(16)$$

Both l_1 and l_2 are positive, because their sum and also product both are positive. From this we conclude that for a particular value of T there are two points distant l_1 and l_2 from the centre of gravity of the pendulum, about which the pendulum may be suspended and yet we have the same time period.

If $l_1 = l$, then $l_2 = K^2/l$ [from eq. (16)]

Now, if two circles are drawn with centre G and radii l and K^2/l [fig. 7.16],

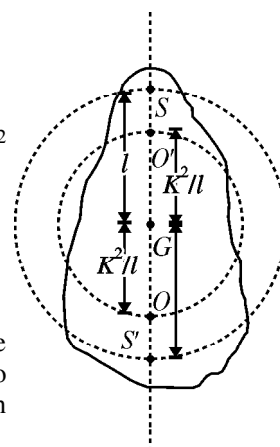


Fig. 7.16 : Compound Pendulum.

the period of oscillation about any point of suspension on either circle is the same. Hence there are infinite number of points in a compound pendulum for which the time period has the same value. If a straight line through G is drawn, it will cut these circle in four points S, O', O, S' . Hence in all there are four points collinear with the C. G. of the pendulum, about which the time periods are equal.

Obviously, if S is the centre of suspension, O is the centre of oscillation or vice-versa; similarly if S' is the centre of suspension, O' is the centre of oscillation or vice-versa. Consider a pair of such points which (a) lie on a straight line passing through G , (b) lie on opposite side of G , and (c) lie at unequal distances from G . In **fig. 7.16**, S and O or S' and O' are such points, then

$$SO = SG + GO = l + \frac{K^2}{l} = \frac{l^2 + K^2}{l}.$$

Similarly, $S'O' = \frac{l^2 + K^2}{l}.$

Substituting in eq. (13), we get

$$T = 2\pi\sqrt{\frac{SO}{g}} = 2\pi\sqrt{\frac{S'O'}{g}} = 2\pi\sqrt{\frac{L}{g}}. \quad \dots(24)$$

where $SO = S'O' = L$ is the length of an equivalent simple pendulum.

Determination of the Value of g by Bar Pendulum

One of the convenient and simple form of a compound pendulum, used to determine the value of g , is the bar pendulum. It is just a uniform metal bar PQ about one metre long, having holes drilled at equal distances along its length symmetrically about its centre of gravity [**fig. 7.17**]. A knife edge can be inserted into any desired hole and is kept on a horizontal support. Now the bar can be made to oscillate in a vertical plane. In an actual experiment, the bar is allowed to oscillate about the horizontal knife edge which is inserted in each hole in turn, starting from P to G . The period of oscillation corresponding to each hole is obtained by noting the time for twenty five oscillations. The bar is then inverted and similarly the periods are obtained about different holes from G to Q .

Now if a graph is plotted between the distance of the centre of suspension from one end of the bar and periodic time, it consists of two curves, symmetrical about a line which cuts X -axis at G' [**fig. 7.18**]. This point corresponds to the centre of gravity G of the bar about which the period is infinite. For a particular value of period T , if we draw a straight line parallel to X -axis, it cuts the two curves at the points A, B, C and D . Obviously if A is the centre of suspension, then C is the corresponding point of oscillation and

AC is the length ($L = l + \frac{K^2}{l} = AG'' + G''C$) of equivalent simple pendulum. Similar is the case for the points B and D and length BD . The mean of the distances AC and BD gives the length L of equivalent simple pendulum. Now one can use the relation $T = 2\pi\sqrt{\frac{L}{g}}$ to determine the value of g .

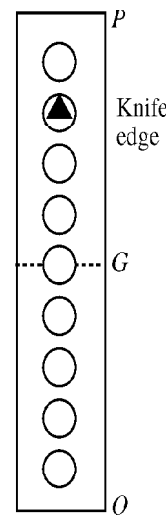


Fig. 7.17 : Bar pendulum.

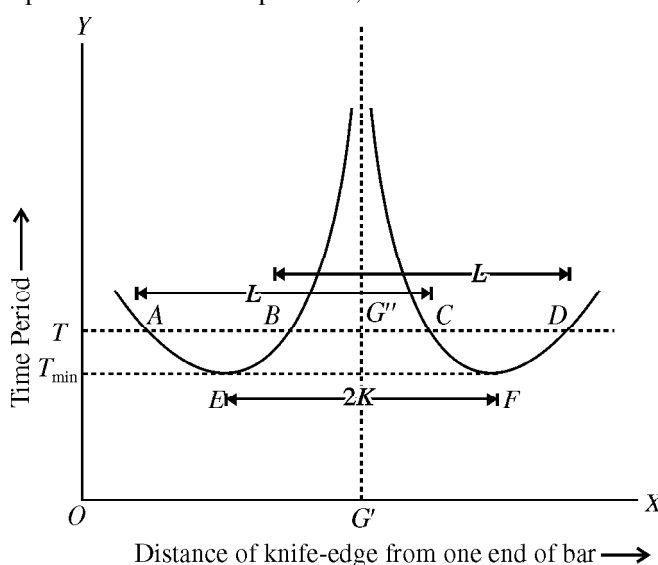


Fig. 7.18 : Graph showing the variation of time period with distance for a bar pendulum.

Points of superiority of a compound pendulum over a simple pendulum : (1) A simple pendulum is an ideal concept only and cannot be realized in actual practice. In the case of a compound pendulum the length of an equivalent simple pendulum can be found easily and accurately. Thus the value of g can be determined accurately with compound pendulum.

(2) The distance between the two positions of knife edges can be measured accurately. In the case of a simple pendulum both the point of suspension and point of oscillation are indefinite and hence the true length of the pendulum can hardly be measured correctly.

(3) The compound pendulum vibrates as a whole and thus there is no possibility of a lag as it may present between the bob and the thread in the case of a simple pendulum.

(4) Due to its large mass and moment of inertia, the compound pendulum keeps on vibrating for a fairly long time and thus the time period can be measured accurately by finding the time for a large number of vibrations. In a simple pendulum the vibrations die out quickly as the mass of the bob is comparatively very small.

7.6.2. Torsional Pendulum

If one end of a fairly thin and long wire is clamped to a rigid support and the other end is attached to the centre of a heavy body (e.g., disc, cylinder or sphere), then this arrangement is called the torsional pendulum [fig. 7.19].

If the disc or sphere be turned in the horizontal plane to twist the wire and then released, it executes torsional vibrations of a definite period about the wire as axis.

Now, if the disc is rotated through an angle θ , then the restoring torque, set up in the wire, will be $-C\theta$, where $C (= \eta\pi^4/2l)$ is the torsional rigidity or couple per unit twist of the wire. Here η is the modulus of rigidity of the material of the wire and the wire has length l with radius r .

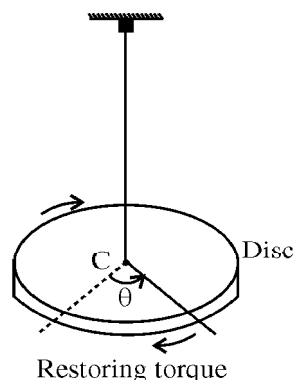


Fig. 7.19 : Torsional pendulum : Restoring torque is opposite to θ increasing.

This torque produces an angular acceleration $d^2\theta/dt^2$ in the disc or sphere. If I be the moment of inertia of the disc ($I = \frac{1}{2}MR^2$)

or sphere ($I = \frac{2}{5}MR^2$) about the wire as axis, then the torque is $I \frac{d^2\theta}{dt^2}$. Hence the equation of motion is

$$I \frac{d^2\theta}{dt^2} + C\theta = 0 \quad \text{or} \quad \frac{d^2\theta}{dt^2} + \frac{C}{I}\theta = 0$$

Putting $\omega^2 = C/I$, we obtain

$$\frac{d^2\theta}{dt^2} + \omega^2\theta = 0 \quad \dots(1)$$

This equation represents a simple harmonic motion of period

$$T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{I}{C}} \quad \dots(2)$$

It may be noted that in the derivation of this formula no approximation is required. Hence the period of torsional oscillations remains the same for large amplitude oscillations, provided that the elastic limit of the suspension wire has not been exceeded.

The solution of eq. (1) is of the form

$$\theta = \theta_0 \sin(\omega t + \phi)$$

where θ_0 = angular amplitude, $\omega = \sqrt{C/I}$ and ϕ = initial phase.

Measurement of Rigidity Modulus (η) by Torsional Pendulum

We know that the expression for the torsional rigidity (C) of the wire used in the torsional pendulum, is

$$C = \frac{\eta \pi r^4}{2l} \quad \dots(3)$$

where η is the rigidity modulus of the material of the wire, r is the radius of the wire and l is its length.

Also the expression for moment of inertial (I) for a disc or cylinder about the axis of symmetry is given by

$$I = \frac{1}{2} MR^2 \quad \dots(4)$$

where M is the mass and R is the radius of the disc or cylinder.

$$\text{In case of sphere,} \quad I = \frac{2}{5} MR^2 \quad \dots(5)$$

where M is the mass and R is the radius of the sphere.

In order to determine the value of rigidity modulus (η) of the wire by a torsional pendulum, the apparatus consists of a long thin specimen wire, whose one end is attached to a rigid support. To the other free end of the wire, a heavy disc (or cylinder or sphere) is fixed at its centre (**fig. 7.19**). The disc is turned in the horizontal plane and then released, it executes torsional oscillations, whose period is given by

$$T = 2\pi \sqrt{\frac{I}{C}} \quad \dots(6)$$

Substituting for I and C from eqs. (3) and (4) we obtain

$$T = 2\pi \sqrt{\frac{MR^2}{2} \cdot \frac{2l}{\eta \pi r^4}} = 2\pi \sqrt{\frac{MR^2 l}{\eta \pi r^4}}$$

$$\text{whence} \quad \eta = \frac{4\pi MR^2 l}{T^2 r^4} \quad \dots(7)$$

Periodic time T is obtained by measuring the time for 20 oscillations of the torsional pendulum. Now, substituting the values of M (mass) and R (radius) of the disc (or cylinder) and also the values of l (length) and r (radius) of the wire in eq. (7), we can determine the value of rigidity modulus of the material of the wire.

Ex. 1. A disc of metal of radius R with its plane vertical can be made to swing about a horizontal axis passing through any one of a series of holes bored along a diameter. Show that the minimum period of oscillation, is given by

$$T = 2\pi \sqrt{\frac{1.414 R}{g}}.$$

Sol. The vibrating disc will act as compound pendulum, whose time period is given by

$$T = 2\pi \sqrt{\frac{(K^2/l) + l}{g}}$$

where K is the radius of gyration of the disc and l the distance between the centre of gravity and the centre of suspension.

The condition for the time period to be minimum is $l = K$.

$$\therefore \quad T = 2\pi \sqrt{\frac{2K}{g}}$$

For disc, $\frac{1}{2}MR^2 = MK^2 \therefore K = \frac{R}{\sqrt{2}}$, where R is the radius of the disc.

$$\therefore T = 2\pi\sqrt{\frac{2 \times R}{\sqrt{2}g}} = 2\pi\sqrt{\frac{1.414R}{g}}.$$

This is the required expression.

Ex. 2. A uniform thin rod of length 120 cm and width 6 cm is swinging in a vertical plane as pendulum about a point A at some distance from one end. If the time of swing is minimum, find the distance of A from the end of the rod.

Sol. Moment of inertia of a uniform thin rod of length l and width b about the axis passing through C.G. and perpendicular to the plane of vibration is given by

$$I = M\left(\frac{l^2 + b^2}{12}\right) = MK^2, \text{ where } K \text{ is the radius of gyration.}$$

Here, $l = 120 \text{ cm} = 1.2 \text{ m}$; $b = 6 \text{ cm} = 0.06 \text{ m}$.

$$\therefore K^2 = \frac{l^2 + b^2}{12} = 0.1203 \text{ m}^2.$$

If the time of swing is minimum, then distance between point of suspension and C.G. = $K = \sqrt{0.1203} \text{ m} = 0.347 \text{ m}$.

Therefore, distance from one end of the rod = $0.60 - 0.347 = 0.253 \text{ m}$.

Ex. 3. A body of mass 200 gm oscillates about a horizontal axis at a distance of 20 cm from its centre of gravity. If the length of the equivalent simple pendulum be 35 cm find its moment of inertia about the axis of suspension.

Sol. The oscillating body acts as compound pendulum, whose time period is given by

$$T = 2\pi\sqrt{\frac{(K^2/l) + l}{g}}.$$

The length $\frac{K^2}{l} + l = L$ is the length of the equivalent simple pendulum.

Here, $l = 0.2 \text{ m}$, $L = 0.35 \text{ m}$, $\therefore 0.35 = \frac{K^2}{0.20} + 0.2$, i.e., $K^2 = 0.03$.

The moment of inertia I of the body about the axis of suspension is

$$I = mK^2 + ml^2 = m(K^2 + l^2) = 0.2 (0.03 + 0.2^2) = 0.014 \text{ kg-m}^2.$$

Ex. 4. A circular disc of radius 20 cm oscillates as a pendulum about a point on its circumference. Calculate the period of oscillation.

Sol. Let the disc of radius 20 cm be suspended from the point P of its circumference. (fig. 7.20)

The moment of inertia of the disc, about an axis passing through its C.G. and perpendicular to its plane will be given by

$$I = MK^2 = \frac{1}{2}M \times (0.2)^2 \text{ or } K^2 = 0.02$$

The distance of centre of oscillation from the point of suspension is

$$L = PO = (K^2/l) + l = 0.3 \text{ m}.$$

$$\therefore \text{Period of oscillation, } T = 2\pi\sqrt{\frac{L}{g}} = 2 \times 3.14 \times \sqrt{\frac{0.3}{9.8}} = 1.098 \text{ sec}.$$

Ex. 5. A compound pendulum is formed by suspending a heavy ring of radius 490.5 cm from a string. What is the length of the string when the period is minimum? Find this period.

Sol. The time period of oscillation of a compound pendulum, is given by

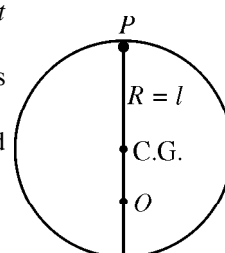


Fig. 7.20.

$$T = 2\pi\sqrt{\frac{(K^2/l) + l}{g}} \quad \dots(i)$$

Here, a heavy ring is suspended with the help of a string. If the weight of the string is negligible with respect to the heavy ring, C.G. of the system will lie at the centre of the ring.

From eq. (i) the time period will be minimum, when $K = l$

But for the ring, $I = MR^2 = MK^2$ or $R = K$.

where R is the radius of the ring (consider all the mass is concentrated at the rim).

$$\therefore R = l$$

i.e., the distance between centre of suspension and centre of ring is R . In other words, for minimum time period, the length of the string must be zero and ring must be suspended to any point of the circumference. Putting in (i) $K = l = R$, we get

$$T_{\min} = \sqrt{\frac{(l^2/l) + l}{g}} = 2\pi\sqrt{\frac{2l}{g}} = 2\pi\sqrt{\frac{2R}{g}}.$$

Here, $R = 490.5$ cm and $g = 981$ cm/sec².

$$\therefore T_{\min} = 2 \times 3.14 \times \sqrt{\frac{2 \times 490.5}{981}} = \mathbf{6.28 \text{ sec.}}$$

Ex. 6. A thin uniform bar of length 120 cm is made to oscillate about an axis through its end. Find the period of oscillation and other points about which it can oscillate with the same period.

Sol. The moment of inertia of a thin bar of length a and mass M , is given by

$$I = \frac{Ma^2}{12} = MK^2, \text{ where } K \text{ is the radius of gyration.}$$

$$\therefore K^2 = \frac{a^2}{12} = \frac{120 \times 120}{12} = 1200.$$

The time period of compound pendulum, is given by

$$T = 2\pi\sqrt{\frac{(K^2/l) + l}{g}} = 2\pi\sqrt{\frac{(1200/60) + 60}{980}} \quad [\because l = SG = 60 \text{ cm}]$$

$$= 2 \times 3.14 \times \sqrt{\frac{2}{7}} = \mathbf{1.796 \text{ sec.}}$$

Let L be the length of equivalent simple pendulum, then

$$T = 2\pi\sqrt{\frac{L}{g}}$$

$$\therefore L = \frac{gT^2}{4\pi^2} = \frac{980}{(3.14)^2} \times \frac{(2 \times 3.14 \times 2)^2}{49} = 80 \text{ cm} = \mathbf{0.8 \text{ m.}}$$

Therefore, there is another point O at a distance of 80 cm, from S or 20 cm from C.G. about which time period will be the same.

If the pendulum is inverted and suspended from S' , time period will be the same due to symmetry. Corresponding to S' , there will be a point O at a distance of 20 cm from C.G.

Hence the bar will have same time periods about four points S, S', O, O' whose distance from S are 0, 120, 80 and 40 cm respectively [fig. 7.21].

Ex. 7. A solid sphere of mass 2 kg and radius 0.05 metre is suspended from a wire. Find the period of oscillation, if the torque required to twist the wire is 4×10^{-3} newton-m/radian.

Sol. Moment of inertia of the sphere about any diameter is

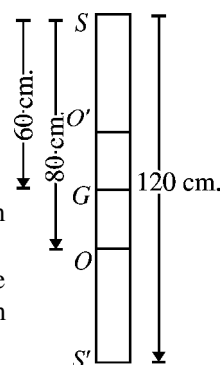


Fig. 7.21.

$$I = \frac{2}{3}MR^2 = \frac{2}{3} \times 2 \times 0.05 \times 0.05 = 0.002 \text{ kg} \times \text{m}^2.$$

$$\therefore \text{Period of oscillation, } T = 2\pi\sqrt{\frac{I}{C}} = 2\pi \times \sqrt{\frac{0.002}{4 \times 10^{-3}}} = \mathbf{4.44 \text{ sec.}}$$

Ex. 8. A body is executing torsional oscillations about a wire of torsional rigidity C . If the total energy of oscillation is E and the moment of inertia of the body about the axis of rotation is I , show that the angular velocity ($d\theta/dt$) at the displacement θ is given by

$$\frac{d\theta}{dt} = 2\pi\sqrt{\frac{C}{I}}\sqrt{\frac{2E}{C} - \theta^2}.$$

Sol. Let C be the couple per unit twist. If the wire is twisted through an angle θ , then the restoring couple is $-C\theta$. Now if the wire is twisted to angle $\theta + d\theta$, then the amount of work done is

$$dW = C\theta d\theta$$

Hence the total amount of work done in twisting the wire through an angle θ is

$$W = \int_0^\theta C\theta d\theta = \frac{1}{2}C\theta^2$$

At the angular displacement θ , this work done becomes the potential energy of the system i.e.,

$$U = \frac{1}{2}C\theta^2.$$

The instantaneous angular displacement θ of a body, executing torsional oscillation is given by

$$\theta = \theta_0 \sin(\omega t + \phi), \text{ where } \omega = \sqrt{\frac{C}{I}}$$

So that
$$\frac{d\theta}{dt} = \theta_0 \omega \cos(\omega t + \phi)^*$$

K.E. at the angular displacement θ is

$$\begin{aligned} K &= \frac{1}{2}I\left(\frac{d\theta}{dt}\right)^2 = \frac{1}{2}I\theta_0^2\omega^2\cos^2(\omega t + \phi) \\ &= \frac{1}{2}I\theta_0^2\omega^2\left(1 - \frac{\theta^2}{\theta_0^2}\right) = \frac{1}{2}I\omega^2(\theta_0^2 - \theta^2) = \frac{1}{2}C(\theta_0^2 - \theta^2) \end{aligned}$$

$$\therefore \text{Total energy, } E = K + U = \frac{1}{2}C(\theta_0^2 - \theta^2) + \frac{1}{2}C\theta^2 = \frac{1}{2}C\theta_0^2$$

$$\therefore \text{Angular amplitude } \theta_0 = \sqrt{\frac{2E}{C}}$$

$$\text{Now, } \frac{d\theta}{dt} = \theta_0 \sqrt{1 - \frac{\theta^2}{\theta_0^2}} = \sqrt{\frac{C}{I}} \sqrt{\theta_0^2 - \theta^2} = \sqrt{\frac{C}{I}} \sqrt{\frac{2E}{C} - \theta^2}.$$

Ex. 9. A sphere of moment of inertia 3355 gm-cm^2 when suspended by a thin wire executes torsional oscillations of period 4.3 seconds . Calculate the maximum angular velocity of the sphere and its average potential energy when the amplitude is $\pi/2$ radians.

Sol. Let the equation for the angular displacement of the sphere be

$$\theta = \theta_0 \sin(\omega t + \phi)$$

$$\therefore \text{Angular velocity } d\theta/dt = \theta_0 \omega \cos(\omega t + \phi)$$

$$\text{Its maximum value} = \theta_0 \omega = 2\pi n \theta_0.$$

Here $\theta_0 = \pi/2$ radians and frequency $n = 1/4.3$ per sec.

$$\therefore \text{Max. angular velocity} = 2 \times 3.14 \times \frac{1}{4.3} \times \frac{3.14}{2} = \mathbf{2.3 \text{ rad/sec.}}$$

* Here, ω is not the actual angular velocity but governing the phase of motion.

$$\begin{aligned} \text{Average potential energy} &= \frac{1}{4} I \omega^2 \theta_0^2 & [\text{In place of } \frac{1}{2} m \omega^2 a^2] \\ &= \frac{1}{4} \times 3355 \times 10^{-7} \times \left(\frac{2\pi}{4.3}\right)^2 \times \left(\frac{\pi}{2}\right)^2 = 4.45 \times 10^{-4} \text{ J.} \end{aligned}$$

7.7. Additional Examples of Harmonic Oscillators

(1) Simple Pendulum

A simple pendulum consists of a small heavy (ideally point) bob of mass m , suspended by means of an inextensible massless string of length l to a rigid support S [fig. 7.22]. If the bob is drawn to one side from its equilibrium position O and then left free, it starts to oscillate in a vertical plane. Let θ be the angular displacement at any time t . Due to the weight mg of the bob, a restoring torque (i.e., moment of force) τ is acting on the system and is given by $\tau = -mgl \sin \theta$.

This is the torque about the point S and bears negative sign because its direction is opposite to θ increasing. If I be the moment of inertia of the bob about the point S and $d^2\theta/dt^2$ be the angular acceleration at the angular displacement θ , then torque can be expressed as $\tau = I \frac{d^2\theta}{dt^2}$

$$\text{Therefore, } I \frac{d^2\theta}{dt^2} = -mgl \sin \theta \quad \dots(1)$$

If the bob is considered a point mass m , then its moment of inertia about S is given by

$$I = ml^2 \quad \dots(2)$$

$$\text{Hence } ml^2 \frac{d^2\theta}{dt^2} = -mgl \sin \theta \quad \text{or} \quad \frac{d^2\theta}{dt^2} + \frac{g}{l} \sin \theta = 0$$

$$\text{or } \frac{d^2\theta}{dt^2} + \omega^2 \sin \theta = 0, \quad \text{where } \omega = \sqrt{\frac{g}{l}}, \quad \dots(3)$$

Taylor's expansion of $\sin \theta$ is

$$\sin \theta = \theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \dots \quad \dots(4)$$

For sufficiently small θ (in radians), we can neglect all terms in the expansion except the first term. In such a case,

$$\sin \theta \approx \theta \quad \dots(5)$$

Thus for small angular amplitude, eq. (3) is obtained to be

$$\frac{d^2\theta}{dt^2} + \omega^2 \theta = 0 \quad \dots(6)$$

Eq. (6) is completely analogous to the differential equation of the harmonic oscillator. The solution of the equation is given by

$$\theta = \theta_m \sin (\omega t + \phi) \quad \dots(7)$$

where θ_m is the maximum angular displacement and ϕ is the phase constant.

The period of oscillation of the simple pendulum is given by

$$T = \frac{2\pi}{\omega} \quad \text{or} \quad T = 2\pi \sqrt{\frac{l}{g}} \quad \dots(8)$$

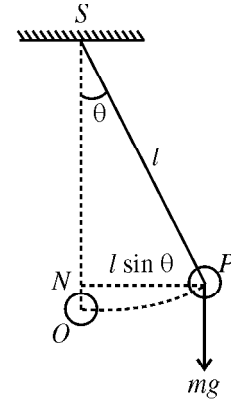


Fig. 7.22 : Simple pendulum.

(2) Charge Oscillations in L - C Circuit

Let us consider a coil of self-inductance L and negligible resistance connected in series with a capacitor of capacitance C [fig. 7.23]. Initially the condenser is charged by a battery to a charge $+Q_0$ on the right plate. Next the connection of the battery is cut off with the help of a key and the condenser is discharged through the inductance L .

Let Q be the charge at any time t on the capacitor C in the circuit. The potential difference across the capacitor C is $V = Q/C$. We take V to be positive when the right plate of the condenser is positively charged and represent the current direction by the arrow in fig. 7.23. Obviously the voltage across the capacitor is the same as the induced voltage $L di/dt$ across the inductance L , where di/dt is the rate of change of the current in the circuit. Thus

$$V = \frac{Q}{C} = L \frac{di}{dt} \quad \dots(1)$$

But $i = -dQ/dt$ (the charge is decreasing on the right plate and the current is flowing as shown),

$$L \frac{d^2 Q}{dt^2} + \frac{Q}{C} = 0 \quad \dots(2)$$

Eq. (2) is analogous to the differential equation of simple harmonic motion of spring-mass system. We can think Q as the charge displacement, L as the charge inertia and $1/C$ as the force constant. Here the electromotive force (Q/C) across the capacitance plays the role of restoring force in the mechanical oscillator.

Eq. (2) can be written as

$$\frac{d^2 Q}{dt^2} + \frac{1}{LC} Q = 0 \text{ or } \frac{d^2 Q}{dt^2} + \omega^2 Q = 0 \quad \dots(3)$$

This is identical to equation of the harmonic oscillator. Thus the charge on the condenser is oscillatory and the frequency of oscillation is given by

$$\nu = \frac{\omega}{2\pi} = \frac{1}{2\pi\sqrt{LC}} \quad \dots(4)$$

The solution of eq. (3) is expressed as

$$Q = Q_0 \sin(\omega t + \phi) \quad \dots(5)$$

Thus we find that the charge on the condenser oscillates between the values $+Q_0$ and $-Q_0$ and its frequency depends on L and C . Differentiating eq. (5) with respect to t , we get the instantaneous current i , i.e.,

$$i = -dQ/dt = -Q_0 \omega \cos(\omega t + \phi)$$

or

$$i = i_0 \sin(\omega t + \phi - \frac{\pi}{2}) \quad \dots(6)$$

where

$$Q_0 \omega = i_0.$$

Thus the current lags in phase by angle $\pi/2$ to the charge.

Practically there is always some resistance in L - C circuit and its effect will be considered later. Now let us calculate the energy stored in the inductance L and capacitance C at any instant t . The current is build up from 0 to i in time t and the energy E_L stored in the inductance L is obtained by integrating the instantaneous power with respect to time i.e.,

$$E_L = \int_0^t V i dt = \int_0^t L \frac{di}{dt} i dt = \int_0^i L i di = \frac{1}{2} L i^2$$

The potential energy stored at the instant t in the capacitor is

$$E_C = \frac{1}{2} \frac{Q^2}{C}$$

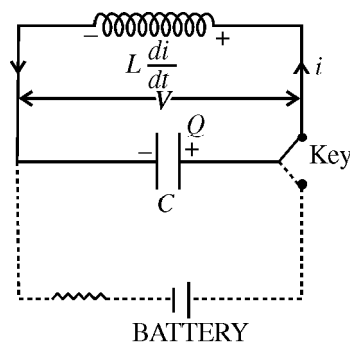


Fig. 7.23 : L - C circuit.

Therefore the total energy in the circuit at any time t is

$$E = E_L + E_C = \frac{1}{2}Li^2 + \frac{1}{2}\frac{Q^2}{C} \quad \dots(7)$$

This equation for total energy is similar to the equation for mechanical oscillator ($E = \frac{1}{2}mv^2 + \frac{1}{2}kx^2$). As Q and i vary with time, the inductance and capacitor exchange energy periodically similar to the energy exchanges in the spring and mass system.

(3) Oscillation of Two Particles Connected by a Spring–Diatomic Molecule

Let us consider two masses m_1 and m_2 connected by a weightless spring. The masses are free to vibrate only along the axis of the spring. Such an oscillating system is called **two body harmonic oscillator**. When due to some reason one of the masses is slightly displaced from its equilibrium position, this produces an extension or compression in the spring. Due to spring action, it exerts a linear restoring force on the masses so that both of the masses vibrate about their mean equilibrium positions and the centre of mass of the system remains stationary.

Suppose that two masses m_1 and m_2 are connected by a massless spring of force constant k . The masses are constrained to move along the axis of the spring [fig. 7.24]. Let r_0 be the normal length of the spring. Let x_1 and x_2 be the coordinates of the two ends of the spring at any instant t . Therefore the instantaneous length of the spring is $r = x_2 - x_1$. The change in length is thus obtained to be

$$x = r - r_0 = (x_2 - x_1) - r_0 \quad \dots(1)$$

For $x > 0$, $x = 0$ and $x < 0$, the spring is extended, normal and compressed respectively. Suppose that at any instant, the spring is increased in length i.e., $x > 0$. The spring is exerting the same force (kx) in magnitude on the two masses but the force $F_1 (= kx)$, acting on m_1 , is opposite to the force $F_2 (= -kx)$ on m_2 i.e.,

$$F_1 = kx \quad \text{and} \quad F_2 = -kx \quad \dots(2)$$

Applying Newton's second law, we obtain the equations of motion of the two masses as follows :

$$m_1 \frac{d^2 x_1}{dt^2} = kx \quad \text{and} \quad m_2 \frac{d^2 x_2}{dt^2} = -kx$$

$$\therefore \frac{d^2 x_1}{dt^2} = \frac{k}{m_1} x \quad \text{and} \quad \frac{d^2 x_2}{dt^2} = -\frac{k}{m_2} x$$

On subtraction

$$\frac{d^2 x_2}{dt^2} - \frac{d^2 x_1}{dt^2} = -\frac{k}{m_2} x - \frac{k}{m_1} x \quad \text{or} \quad \frac{d^2 (x_2 - x_1)}{dt^2} = -\left(\frac{1}{m_1} + \frac{1}{m_2}\right) kx$$

$$\text{or} \quad \frac{d^2 x}{dt^2} = -\frac{k}{\mu} x \quad \text{or} \quad \frac{d^2 x}{dt^2} + \frac{k}{\mu} x = 0 \quad \dots(3)$$

where $\frac{1}{\mu} = \frac{1}{m_1} + \frac{1}{m_2}$ or $\mu = \frac{m_1 m_2}{m_1 + m_2}$ is the **reduced mass** of the system. We have used $\frac{d^2 (x_2 - x_1)}{dt^2}$

$= \frac{d^2 x}{dt^2}$ because r_0 is constant.

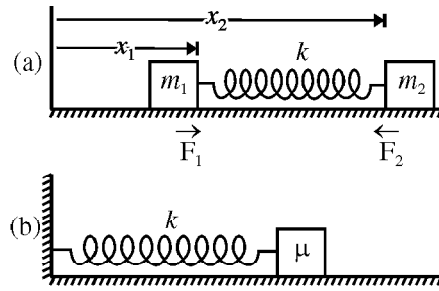


Fig. 7.24 : (a) Two-body oscillator : $F_2 = -kx = -F_1$, $x = (x_2 - x_1) - r_0$, $r_0 =$ unstretched length of the spring of force constant k ; (b) Equivalent single-body oscillator of mass $\mu \left(= \frac{m_1 m_2}{m_1 + m_2} \right)$ attached to a rigid wall by a spring of the same force constant k .

Eq. (3) is identical in form to the equation for simple harmonic oscillator, developed for one-body oscillations. The frequency of oscillation of the system is given by

$$\nu = \frac{1}{2\pi} \sqrt{\frac{k}{\mu}} \quad \dots(4)$$

Thus the system has the same frequency of oscillation as a single object of mass μ , connected by a similar spring of force constant k to a rigid wall. Hence the two-body oscillator [fig. 7.24(a)] is equivalent to one-body oscillator [fig. 7.24(b)]. However in [fig. 7.24(a)] x is the relative displacement of the two masses from their equilibrium positions rather than the displacement of a single mass from its equilibrium position and μ is the reduced mass rather than the mass of single object. The potential energy of the two-body harmonic oscillator, when the length of the spring changes by x , is given by

$$U = \int_0^x kx \, dx = \frac{1}{2} kx^2 = \frac{1}{2} k(r-r_0)^2 \quad \dots(5)$$

Vibrations of a diatomic molecule : A diatomic molecule, *e.g.*, HCl molecule, is an example of two-body system which can oscillate along the axis of symmetry. The atoms in a diatomic molecule are coupled through electromagnetic forces and the bonding between them may be imagined like a spring. Thus we may consider a diatomic molecule as a system of two masses connected by a spring.

In a diatomic molecule, the binding forces are electromagnetic in origin which are composed of attractive and repulsive parts. The equilibrium separation r_0 between the two atoms of the molecule is that at which the attractive and repulsive forces exactly balance each other. If the separation is slightly increased, the two atoms try to come to the equilibrium distance under resultant attractive force (restoring force) and if the separation is slightly decreased, they repel each other to have equilibrium configuration. Thus the vibrations of the molecule may take place, if the system is disturbed by giving some energy.

Ex. 1. A simple pendulum possesses 0.05 joule energy when its length is 2 m and the amplitude of motion of the bob is 4 cm. Calculate its energy (1) when the length remains the same but the amplitude is doubled, (2) the amplitude remains the same but the length is made one-half.

Sol. $E = \frac{1}{2} m\omega^2 a^2$; Here, $E = 0.05 \text{ J}$, $\omega^2 = g/l = g/2$ and $a = 4 \times 10^{-2} \text{ m}$

$$\therefore 0.5 = \frac{1}{2} m(g/2)(4 \times 10^{-2})^2$$

$$(1) \text{ When } a' = 2 \times 4 \times 10^{-2} \text{ m, } E' = \frac{1}{2} m(g/2)(2 \times 4 \times 10^{-2})^2$$

$$\therefore \frac{E'}{0.05} = 4 \text{ or } E' = \mathbf{0.2 \text{ J}}$$

$$(2) \text{ Here } l' = 2/2 = 1 \text{ m,}$$

$$\therefore E'' = \frac{1}{2} m(g/2)(4 \times 10^{-2})^2$$

$$\therefore \frac{E''}{0.05} = 2 \text{ or } E'' = \mathbf{0.1 \text{ J}}$$

Ex. 2. What is the frequency of electrical oscillations in an inductance of 10 millihenry in series with a 2 μf capacitance. If the maximum potential difference across the condenser is 5 volts, find the energy of the oscillating system. (Rewa 2004; Bhopal 04)

$$\text{Sol. } \nu = \frac{1}{2\pi\sqrt{LC}}$$

Here $L = 10 \times 10^{-3} \text{ henry}$ and $C = 2 \times 10^{-6} \text{ farad}$.

$$\therefore \nu = \frac{1}{\sqrt{10 \times 10^{-3} \times 2 \times 10^{-6}}} = \mathbf{1.1 \times 10^3 \text{ Hz.}}$$

$$\text{Energy, } E = \frac{1}{2} CV^2 = \frac{1}{2} \times 2 \times 10^{-6} \times 5^2 = \mathbf{25 \times 10^{-6} \text{ joules.}}$$

Ex. 3. Show that the total energy in the L-C circuit, remains constant with time. What is the expression obtained for the total energy ?

$$\text{Sol.} \quad E = \frac{1}{2} Li^2 + \frac{1}{2} \frac{Q^2}{C} = \frac{1}{2} LQ_0^2 \omega^2 \cos^2 (\omega t + \phi) + \frac{1}{2} \frac{Q_0^2}{C} \sin^2 (\omega t + \phi)$$

$$\text{But} \quad \omega^2 = \frac{1}{LC}, \quad \therefore E = \frac{1}{2} \frac{Q_0^2}{C}.$$

Hence the total energy remains constant with time. The expression for total energy is given above.

Ex. 4. Two masses of 10 gm and 90 gm are connected by a spring of length 10 cm and force constant 10^3 N/m^2 , calculate the frequency of oscillation.

$$\text{Sol.} \quad \nu = \frac{1}{T} = \frac{1}{2\pi} \sqrt{\frac{k}{\mu}}$$

$$\text{Here } k = 10^3 \text{ N/m and } \mu = \frac{m_1 m_2}{m_1 + m_2} = \frac{0.01 \times 0.09}{0.01 + 0.09} = 0.009 \text{ kg}$$

$$\therefore \nu = \frac{1}{2\pi} \sqrt{\frac{10^3}{0.009}} = \mathbf{53 \text{ Hz.}}$$

Ex. 5. A sodium chloride molecule has a natural vibrational frequency of $1.14 \times 10^{13} \text{ Hz}$. Calculate inter-atomic force constant. The masses of sodium and chlorine are 23 a.m.u. and 35 a.m.u. respectively. (1 a.m.u. $1.67 \times 10^{-27} \text{ kg}$). (O.U. 2007; A.N.U. March 2007)

Sol. Frequency of vibration for a diatomic molecule is given by

$$\nu = \frac{1}{2\pi} \sqrt{\frac{k}{\mu}}$$

$$\text{whence for constant, } k = 4\pi^2 \nu^2 \mu$$

$$\text{Here, } \nu = 1.14 \times 10^{13}, \mu = \frac{m_1 m_2}{m_1 + m_2} = \frac{23 \times 1.67 \times 10^{-27} \times 35 \times 1.67 \times 10^{-27}}{23 \times 1.67 \times 10^{-27} + 35 \times 1.67 \times 10^{-27}} = 23.18 \times 10^{-27} \text{ kg}$$

$$\therefore k = 4 \times (3.14)^2 \times (1.14 \times 10^{13})^2 \times 23.18 \times 10^{-27} = \mathbf{118.9 \text{ N/m.}}$$

Ex. 5. Calculate the vibrational frequency of HCl molecule, given the force constant = 839.0 N/m. ($m_H = 1.67 \times 10^{-27} \text{ kg}$ and $m_{Cl} = 35 \times 1.67 \times 10^{-27} \text{ kg}$)

$$\text{Sol. } \nu = \frac{1}{2\pi} \sqrt{\frac{k}{\mu}} = \frac{1}{2\pi} \sqrt{\frac{839.0 \times 36}{35 \times 1.67 \times 10^{-27}}} = \mathbf{11.4 \times 10^{13} \text{ Hz.}}$$

$$\text{where } \mu = \frac{m_H m_{Cl}}{m_H + m_{Cl}} = \frac{1.67 \times 10^{-27} \times 35 \times 1.67 \times 10^{-27}}{1.67 \times 10^{-27} + 35 \times 1.67 \times 10^{-27}} = \frac{35 \times 1.67 \times 10^{-27}}{36}$$

Ex. 6. Consider two discs of masses 25 gm and 50 gm (as shown in fig. 7.25) with massless spring of force constant 10^3 N/m . Calculate the frequency of oscillation of the spring (i) when the system is resting on a table, and (ii) when the table is removed and the system is falling freely.

Sol. In the first case, one disc is resting on the table and therefore the other disc of mass 25 gm will vibrate.

$$\therefore \text{Frequency, } \nu = \frac{1}{2\pi} \sqrt{\frac{k}{m}} = \frac{1}{2\pi} \sqrt{\frac{10^3}{0.025}} = \frac{10^2}{\pi} = \mathbf{31.8 \text{ Hz.}}$$

In the second case, when the system is falling freely, both discs will vibrate and hence

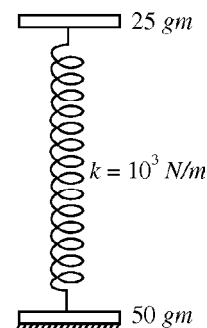


Fig. 7.25

$$v = \frac{1}{2\pi} \sqrt{\frac{k}{\mu}} = \frac{1}{2\pi} \sqrt{\frac{10^3 \times 3}{0.05}} = 39 \text{ Hz.} \left(\because \mu = \frac{m_1 m_2}{m_1 + m_2} = \frac{0.025 \times 0.05}{0.025 + 0.05} = \frac{0.05}{3} \text{ kg} \right)$$

7.8. Principle of Superposition

Let two (or more) simple harmonic motions be acting simultaneously on a particle (or point). At any instant, if x_1 be the displacement of the particle due to one simple harmonic motion and x_2 due to the other, then according to the principle of superposition, the resultant displacement of the particle is the vector sum of the two displacements, i.e.,

$$x = x_1 + x_2 \quad \dots(1)$$

We know that the differential equation for a simple harmonic motion is

$$\frac{d^2 x}{dt^2} + \omega^2 x = 0 \quad \dots(2)$$

This is a **linear homogeneous equation** of second order. Such equation has the characteristic property that the *sum of any two solutions of it is itself a solution*, i.e., if x_1 and x_2 are the solutions of eq. (2), then the sum $x = x_1 + x_2$ of these solutions also satisfies eq. (2). Thus, if x_1 and x_2 are the solutions of eq. (2), then

$$\frac{d^2 x_1}{dt^2} = -\omega^2 x_1 \quad \text{and} \quad \frac{d^2 x_2}{dt^2} = -\omega^2 x_2 \quad \dots(3)$$

The addition of the two equations gives

$$\frac{d^2 (x_1 + x_2)}{dt^2} = -\omega^2 (x_1 + x_2) \quad \text{or} \quad \frac{d^2 x}{dt^2} = -\omega^2 x \quad \dots(4)$$

This means that $x = x_1 + x_2$ satisfies eq. (2) and hence *the differential equation of simple harmonic motion is a linear homogeneous equation and follows the principle of superposition*. In the present work, we shall study only those oscillations which satisfy linear equations and are in accordance with the principle of superposition. It is to be mentioned that the differential equations of damped and driven harmonic oscillators are linear differential equations.

In case of nonlinear equation

$$\frac{d^2 x}{dt^2} = -k_1 x - k_2 x^2 - k_3 x^3 - \dots \dots \dots \quad \dots(5)$$

(e.g., for anharmonic oscillator), the superposition of two solutions does not satisfy the equation itself. In case of simple pendulum, the differential equation has the form

$$\frac{d^2 \theta}{dt^2} + \omega^2 \sin \theta = 0 \quad [\because \omega = \sqrt{g/l}] \quad \dots(6)$$

But $\sin \theta = \theta - \frac{\theta^3}{3!} - \frac{\theta^5}{5!} - \dots$

Hence $\frac{d^2 \theta}{dt^2} = -\omega^2 \theta + \frac{\omega^2}{6} \theta^3 - \frac{\omega^2}{120} \theta^5 - \dots \quad \dots(7)$

Thus the differential equation for the motion of simple pendulum is nonlinear and in such a case, the principle of superposition will be not applicable. However for small θ , θ^3 , θ^5 etc. terms can be neglected and then the equation.

$$\frac{d^2 \theta}{dt^2} = -\omega^2 \theta \quad \text{or} \quad \frac{d^2 \theta}{dt^2} + \omega^2 \theta = 0 \quad \dots(8)$$

is the linear equation, representing simple harmonic motion, and for such small amplitude oscillations of simple pendulum, the principle of superposition is applicable. Large amplitude oscillations of simple pendulum are anharmonic and represented by nonlinear equation (7). In the present chapter, we shall apply the principle of superposition for compounding two simple harmonic motions which are in a straight line

or mutually perpendicular. However this principle is applicable for the superposition of many linear simple harmonic motions.

7.9. Superposition of Two Simple Harmonic Motions of Same Frequency (or Equal Periods) along a Line

Let two simple harmonic motions of same frequency ($\nu = \omega/2\pi$) [equal periods ($= 2\pi/\omega$)] but of different amplitudes and phases be simultaneously acted on a particle in the same straight line. Let the two component vibrations be represented by

$$x_1 = a_1 \sin (\omega t + \phi_1) \quad \dots(1)$$

and

$$x_2 = a_2 \sin (\omega t + \phi_2) \quad \dots(2)$$

where a_1 and a_2 are the amplitudes and ϕ_1 and ϕ_2 the epochs or initial phases of the component vibrations. The phase difference between them is $(\phi_1 - \phi_2)$.

According to the principle of superposition, the resultant displacement of the particle is given by

$$x = x_1 + x_2 \quad \dots(3)$$

Thus,

$$\begin{aligned} x &= a_1 \sin (\omega t + \phi_1) + a_2 \sin (\omega t + \phi_2) \\ &= a_1 [\sin \omega t \cos \phi_1 + \cos \omega t \sin \phi_1] + a_2 [\sin \omega t \cos \phi_2 + \cos \omega t \sin \phi_2] \\ &= \sin \omega t [a_1 \cos \phi_1 + a_2 \cos \phi_2] + \cos \omega t [a_1 \sin \phi_1 + a_2 \sin \phi_2] \end{aligned} \quad \dots(4)$$

Now, suppose

$$a \cos \theta = a_1 \cos \phi_1 + a_2 \cos \phi_2 \quad \dots(5)$$

and

$$a \sin \theta = a_1 \sin \phi_1 + a_2 \sin \phi_2 \quad \dots(6)$$

Then

$$x = a \sin \omega t \cos \theta + a \cos \omega t \sin \theta \quad \text{or} \quad x = a \sin (\omega t + \theta) \quad \dots(7)$$

where a is the resultant amplitude of oscillation, $(\omega t + \theta)$ is its phase.

Eq. (7) represents that **the resultant motion is simple harmonic** whose frequency ($= \omega/2\pi$) is the same as that of any component vibration, but the amplitude a and phase θ are different. Squaring and adding eqns. (5) and (6) we obtain

$$\begin{aligned} a^2 &= a_1^2 + a_2^2 + 2a_1a_2 \cos (\phi_1 - \phi_2) \\ \text{or} \quad a &= \sqrt{a_1^2 + a_2^2 + 2a_1a_2 \cos (\phi_1 - \phi_2)} \end{aligned} \quad \dots(8)$$

which tells us the **amplitude of resultant simple harmonic motion**.

Dividing eq. (6) by eq. (5),

$$\tan \theta = \frac{a_1 \sin \phi_1 + a_2 \sin \phi_2}{a_1 \cos \phi_1 + a_2 \cos \phi_2} \quad \text{or} \quad \theta = \tan^{-1} \frac{a_1 \sin \phi_1 + a_2 \sin \phi_2}{a_1 \cos \phi_1 + a_2 \cos \phi_2} \quad \dots(9)$$

This angle θ gives the **phase of the resultant motion**.

Let us consider some special cases :

(a) If $\phi_1 - \phi_2 = 0$ or $2n\pi$, where n is an integer, the two vibrations are in the **same phase** and the resultant amplitude has the maximum value, *i.e.*,

$$a = \sqrt{a_1^2 + a_2^2 + 2a_1a_2} = a_1 + a_2 \quad \dots(10)$$

(b) If $\phi_1 - \phi_2 = \pi$ or $(2n + 1)\pi$, where n is any integer, the two vibrations are in the **opposite phase** and the resultant amplitude has the minimum value *i.e.*,

$$a = \sqrt{a_1^2 + a_2^2 - 2a_1a_2} = a_1 - a_2 \quad \dots(11)$$

If in addition $a_1 = a_2$, the resultant amplitude $a = 0$, *i.e.*, the particle remains in rest position.

7.10. Combination of Two Mutually Perpendicular Simple Harmonic Vibrations of Same Frequency

Let the two simple harmonic motions of the same frequencies be acting along the axis of X and axis of Y respectively. If their displacements at any time t be x and y , then they can be represented by

$$x = a \sin (\omega t + \phi) \quad \dots(1)$$

$$y = b \sin \omega t \quad \dots(2)$$

where a and b are the amplitudes of the component vibrations having same frequency $\omega/2\pi$ (or period $2\pi/\omega$) and the first vibration is ahead of the second by a phase angle ϕ , i.e., the phase difference between the two vibrations is ϕ .

$$\text{From eq. (1), we have} \quad x = a[\sin \omega t \cos \phi + \cos \omega t \sin \phi] \quad \dots(3)$$

$$\text{But from eq. (2)} \quad \sin \omega t = \frac{y}{b} \text{ and } \cos \omega t = \sqrt{1 - \frac{y^2}{b^2}}$$

Substituting the values of $\sin \omega t$ and $\cos \omega t$ in (3), we get

$$\frac{x}{a} = \frac{y}{b} \cos \phi + \sin \phi \sqrt{1 - \frac{y^2}{b^2}} \quad \text{or} \quad \left(\frac{x}{a} - \frac{y}{b} \cos \phi \right)^2 = \sin^2 \phi \left(1 - \frac{y^2}{b^2} \right)$$

$$\text{Thus} \quad \frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{2xy}{ab} \cos \phi = \sin^2 \phi \quad \dots(4)$$

This expression represents the resultant motion, which is in general an ellipse inclined to the axes of co-ordinates.

Special Cases :

(i) When $\phi = 0$, the relation (4) becomes

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{2xy}{ab} = 0 \quad \text{or} \quad \left(\frac{x}{a} - \frac{y}{b} \right)^2 = 0 \quad \text{or} \quad \pm \left(\frac{x}{a} - \frac{y}{b} \right) = 0. \quad \dots(5)$$

This equation represents a pair of coincident straight line $y = bx/a$, passing through the origin and lying in the first and third quadrants [fig. 7.26(a)].

This motion along a straight line represents the case of *linearly polarized light* in optics.

(ii) When $\phi = \pi/4$, we have $\sin \phi = 1/\sqrt{2}$ and $\cos \phi = 1/\sqrt{2}$. Hence eq. (4) takes the form

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{\sqrt{2}xy}{ab} = \frac{1}{2} \quad \dots(6)$$

This equation represents an oblique ellipse [fig. 7.26(b)].

(iii) When $\phi = \pi/2$, the expression (4) is

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

This equation represents an ellipse, whose axes are coincident with the X - and Y -axes [fig. 7.26(b)]. The semi-major and semi-minor axes of the ellipse are a and b . Further as the time t starts to increase from zero, x begins to decrease from its maximum positive value and y simultaneously begins to go positive from $y = 0$ position. Thus the ellipse is described in the anticlockwise direction [fig. 7.26(c)]. The same ellipse is obtained for $\phi = 3\pi/2$ or $-\pi/2$, but now the motion is in clockwise direction. Thus when the phase difference ϕ between the two

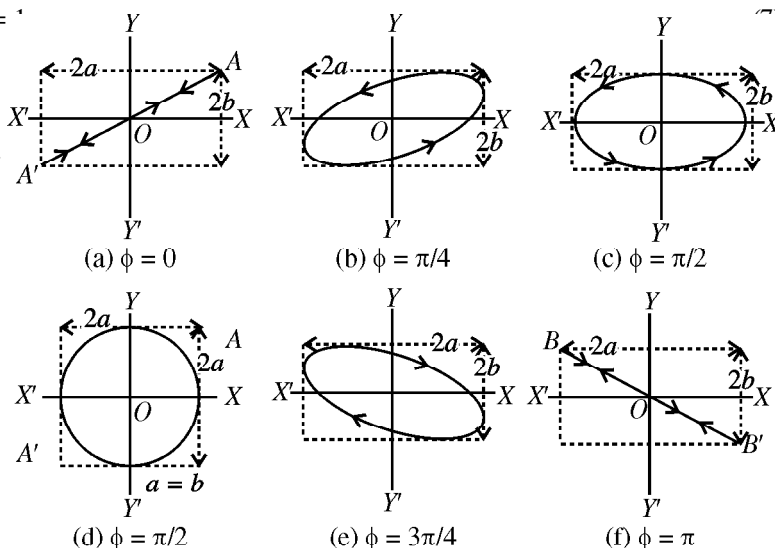


Fig. 7.26.

rectangular simple harmonic motions is $+\pi/2$, the resultant motion is *elliptically polarized*.

Now, if $a = b$, the ellipse degenerates in the circle [fig. 7.26(d)]

$$x^2 + y^2 = a^2 \quad \dots(8)$$

Hence, two simple harmonic motions at right angles to each other of equal amplitudes but with phases differing by $\pi/2$, are equivalent to a uniform circular motion, the radius of the circle being equal to the amplitude of either simple harmonic motion. This case corresponds to circularly polarized light in optics.

Conversely, a uniform circular motion can be resolved into two simple harmonic motions at right angles to each other, their amplitudes being equal, but their phases differing $\pi/2$.

(iv) When $\phi = 3\pi/4$, we have $\sin \phi = 1/\sqrt{2}$ and $\cos \phi = -1/\sqrt{2}$.

Therefore, from eq. (4), we have

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{\sqrt{2}xy}{ab} = \frac{1}{2} \quad \dots(9)$$

This equation represents an oblique ellipse as shown in fig. 7.26(e).

(v) When $\phi = \pi$, eq. (4) is

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{\sqrt{2}xy}{ab} = 0 \quad \text{or} \quad \left(\frac{x}{a} + \frac{y}{b}\right)^2 = 0 \quad \dots(10)$$

which represents a pair of coincident straight lines $y = -bx/a$, passing through the origin and lying in the second and fourth quadrants as BOB' [fig. 7.26(f)].

For any other value of ϕ , except discussed above, the path will be an ellipse, inclined to the co-ordinate axes.

7.11. Composition of Two Rectangular S.H.M.'s of Nearly Equal Frequencies

If the two component vibrations are exactly of equal frequencies (or periods), the elliptical paths, referred to above, remain perfectly steady. But, when the two frequencies differ slightly from each other, there comes about a gradual but progressive change in the relative phase (ϕ) of the two vibrations and consequently the shape of the ellipse slowly undergoes a change. All these elliptic path lie within a rectangle of sides $2a$ and $2b$.

Now, starting with the phase in agreement, i.e., $\phi = 0$, the ellipse coincides with the diagonal $\frac{x}{a} - \frac{y}{b} = 0$ of the rectangle.

As ϕ increases from 0 to $\pi/2$, the ellipse opens out to the form $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, passing through intermediate oblique positions.

When ϕ increases from $\pi/2$ to π , the ellipse closes up again and finally coincides with the other

diagonal $\frac{x}{a} + \frac{y}{b} = 0$ of the rectangle [fig. 7.27].

As the phase difference changes from π to 2π , the reverse process takes place, until the ellipse again coincides with the first diagonal. All these changes have been represented in fig. 7.27.

It should be noted that the frequency of the complete cycle is equal to the difference of the frequencies of the two component simple harmonic vibrations.

7.12. Composition of Two Rectangular S.H.M.'s of Frequencies in the Ratio 2 : 1

Let the two component rectangular simple harmonic vibrations be represented along X-axis and Y-axis as

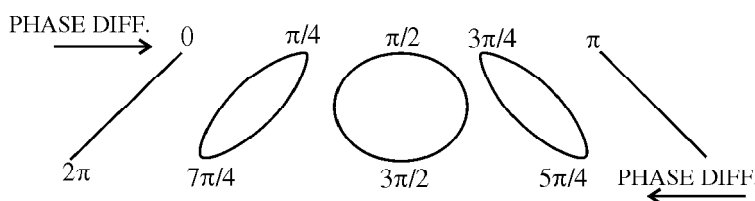
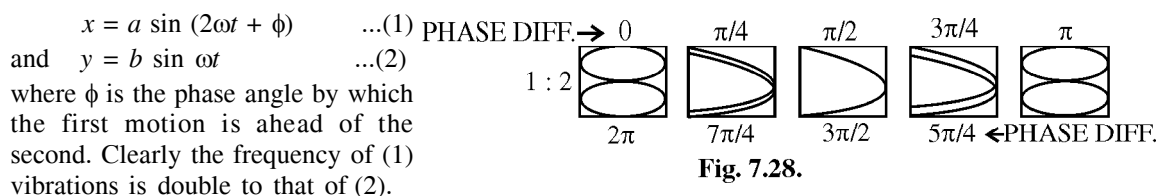


Fig. 7.27.



The resultant motion, which can be obtained by eliminating ωt from the equations (1) and (2), in general will be a curve, having two loops for any value of phase difference and amplitudes.

Here, we will consider only two special cases, given below :

(a) When $\phi = 0$, we get

$$x = a \sin 2\omega t = 2a \sin \omega t \cos \omega t$$

$$\text{or} \quad \frac{x}{a} = 2 \sin \omega t \cos \omega t \quad \text{or} \quad \frac{x}{a} = \frac{2y}{b} \sqrt{1 - \frac{y^2}{b^2}} \quad \text{or} \quad \frac{x^2}{a^2} = \frac{4y^2}{b^2} \left(1 - \frac{y^2}{b^2}\right) = -\frac{4y^2}{b^2} \left(\frac{y^2}{b^2} - 1\right)$$

$$\text{Thus,} \quad \frac{x^2}{a^2} + \frac{4y^2}{b^2} \left(\frac{y^2}{b^2} - 1\right) = 0 \quad \dots(3)$$

This equation represents the figure of '8' [fig. 7.28].

(b) When $\phi = \pi/2$, we have

$$x = a \sin \left(2\omega t + \frac{\pi}{2}\right) = a \cos 2\omega t = a(1 - 2 \sin^2 \omega t)$$

$$\therefore \quad \frac{x}{a} = 1 - \frac{2y^2}{b^2} \quad \left[\because \sin^2 \omega t = \frac{y^2}{b^2} \right]$$

$$\text{or} \quad \frac{2y^2}{b^2} = 1 - \frac{x}{a} = -\frac{x-a}{a}, \quad \therefore y^2 = -\frac{b^2}{2a}(x-a) \quad \dots(4)$$

This is the equation of a parabola with vertex $(a, 0)$.

If the frequencies of the two rectangular vibrations depart slightly from the ratio 2 : 1, the form of the curve slowly varies as ϕ changes [fig. 7.28].

7.13. Lissajous Figures

When a particle oscillates simultaneously under the action of two simple harmonic motions at right angles to each other, the curve traced by the resultant motion of the particle is called **Lissajous figure** after the name of J.A. Lissajous who made an extensive study of such motions. The shape of the curve depends upon the amplitudes, frequencies and phase difference of the component vibrations. We have seen that if the frequencies of the two S.H.Ms. at right angles are equal, the Lissajous figure, in general, is an ellipse and has a definite shape for a particular phase difference [fig. 7.26]. For $\phi = 0$ or π , it is a straight line and for $\phi = \frac{\pi}{2}$ and $a_1 = a_2$, it is a circle.

When the two frequencies of component perpendicular vibrations are in the ratio 2 : 1, the Lissajous figures are relatively complex. It has the shape of figure '8' for $\phi = 0$ or π and that of parabola for $\phi = \frac{\pi}{2}$. If the component frequencies are in the complicated ratio, the Lissajous figures are odd and complicated.

Experimental Demonstration of Lissajous Figures

(1) Optical method : For this purpose, two electrically maintained tuning forks *A* and *B* are set up so that the vibrations of one take place in a vertical plane and those of the other in a horizontal plane. Each fork has a small mirror attached to the side of a prong. Light from a small powerful source, rendered convergent with the help of a convex lens, is allowed to fall on the mirror *M* of the vertical fork *A*. The light, reflected from *M*₁, strikes the mirror *M*₂ of the horizontal fork *B*, from where it is directed towards the screen *S*. The position of the lens is so adjusted that it produces a sharp image on the screens [fig. 7.29].

If now only the vertical fork is vibrated, the spot of light on the screen executes a vertical S.H.M. so rapidly, that only a bright vertical line is observed on the screen. If the horizontal fork alone vibrates, a bright horizontal line is seen. Thus these motions represent two simple harmonic vibrations at right angles to each other. Now, if both the forks vibrate simultaneously, the two rectangular vibrations of the spot of light are compounded and the spot describes the path of the resultant motion on the screen. If the frequencies of the forks bear some simple ratio to each other, a Lissajous figure will be seen on the screen. For example, if the frequencies are in the ratio 2 : 1, a figure of '8' may be obtained. In case if the ratio between the frequencies is approximately two, it will gradually pass through the cyclic of changes of figures as shown in **fig. 7.29**. This method is usually employed for the comparison of the frequencies of two forks.

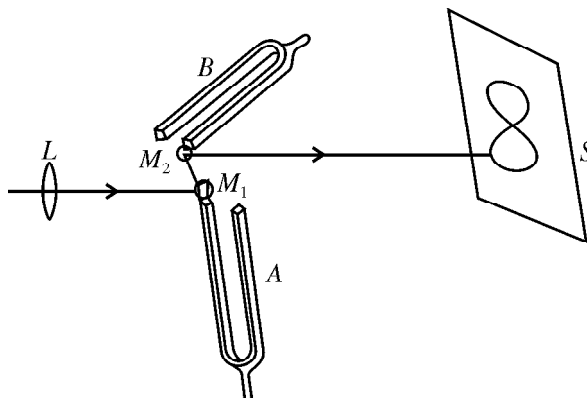


Fig. 7.29.

(2) **By cathode ray oscillograph :** Lissajous figures can be produced easily by means of a cathode ray oscilloscope (CRO). Different alternating sinusoidal voltages are applied at X and Y deflection plates of the CRO and the electron beam traces the resultant effect on the fluorescent screen. When the applied voltages have the same frequency, we can obtain the various curves of **fig. 7.26**, by adjusting the phases and amplitudes. By using the CRO, we can also obtain easily the Lissajous figures for the component frequencies for the ratio 1 : 2 or any other ratio.

Uses of Lissajous Figures

(1) Lissajous figures are used in obtaining beautiful designs for printing in cloth industry.

(2) Lissajous figures find useful applications in the acoustical measurements, *e.g.*, determination of unknown frequency. If the frequencies of vibration of the two rectangular vibrations are not exactly equal or in the ratio 2 : 1, the form of the Lissajous figure undergoes a gradual progressive change. The time, taken by the curve to undergo a complete cycle of changes, enables us to find the frequency of one vibration, if the frequency of the other is known. Let the frequency of one vibration be n and that of the other be slightly different n' , the frequencies being nearly in the ratio 1 : 1. If t be the time in which the cycle undergoes a complete change, then the frequency of the other will be $n' = n \pm 1/t$.

If the frequencies are nearly in the ratio 2 : 1, then $n' = 2n \pm 1/t$.

To find whether n' is greater or less than n or $2n$ (as the case may be) a small piece of wax is attached to the unknown (*e.g.*, the prong of the fork) and again the time of one cycle of changes is determined. If this time increases, the unknown frequency is higher and if it decreases, the unknown frequency is lower than n or $2n$.

(3) One practical application of Lissajous figures is that they enable us to know by inspection the period ratio of its constituent vibrations. It is evident that if the horizontal and vertical lines are drawn on a Lissajous figure and if they cut it m and n times respectively, the required period ratio of the two vibrations is $m : n$. *i.e.*,

$$\frac{\text{Frequency along Y-direction } (v_2)}{\text{Frequency along X-direction } (v_1)} = \frac{m}{n}$$

$$= \frac{\text{Number of intersection points of the Lissajous figure by a parallel line to X-axis}}{\text{Number of intersection points of the figure by a parallel line to Y-axis}}$$

Ex. 1. A simple pendulum is made to oscillate under two simple harmonic motions, given by $x = 5 \cos \omega t$ and $y = 6 \cos (\omega t + \phi)$.

Find the resultant path of the bob, when $\phi = 0$ and $\frac{\pi}{2}$. Draw a plot of the path in each case, indicating the direction of movement of the bob.

Sol. For $\phi = 0$, $x = 5 \cos \omega t$ and $y = 6 \cos \omega t$

$$\therefore \frac{y}{x} = \frac{6}{5} \quad \text{or } y = 1.2x$$

In this case the path of the bob is straight line as shown in **fig. 7.30(a)**.

For $\phi = \frac{\pi}{2}$, $x = 6 \cos \omega t$

and $y = -6 \sin \omega t$

$$\therefore \sin^2 \omega t + \cos^2 \omega t = \frac{y^2}{6^2} + \frac{x^2}{5^2}$$

$$\text{or } \frac{x^2}{25} + \frac{y^2}{36} = 1$$

Now the path of the bob is along the ellipse in clockwise direction as shown in **fig. 7.30(b)**.

Ex. 2. Two vibrations of right angles to one another are described by the equations

$$x = 10 \cos (5\pi t) \quad \text{and} \quad y = 10 \cos (5\pi t + \frac{\pi}{3})$$

Construct the Lissajous figures of the resulting motion.

Sol. $x = 10 \cos (5\pi t)$ and $y = 10 \cos (5\pi t + \frac{\pi}{3})$

Now, $y = 10 \cos 5\pi t \times \cos \frac{\pi}{3} - 10 \sin (5\pi t) \times \sin \frac{\pi}{3}$

$$\text{or } y = \frac{x}{2} - \frac{\sqrt{3}}{2} \sqrt{10^2 - x^2} \quad [\because \sin 5\pi t = \sqrt{1 - (\cos 5\pi t)^2}]$$

$$\text{or } (2y - x)^2 = 3(100 - x^2) \quad \text{or } x^2 + 4y^2 - 4xy = -3x^2 + 300$$

$$\text{or } x^2 + y^2 - xy = 75$$

$$\text{whence, } \frac{x^2}{(5\sqrt{3})^2} + \frac{y^2}{(5\sqrt{3})^2} - \frac{xy}{(5\sqrt{3})(5\sqrt{3})} = 1.$$

Thus the resulting motion is an oblique ellipse.

Ex. 3. Two tuning forks produce a Lissajous's figure. The figure changes in form from a parabola to a figure of eight and again to a parabola, the whole cycle occupying 6 seconds. If the frequency of one fork is 100, find possible frequencies of the other.

Sol. The figure changes from a parabola to a figure of '8' and then again to a parabola. Evidently then the component frequencies are nearly in the ratio 1 : 2 since the frequency of one tuning fork is 100, that of the other may be nearly 200 or 50.

The same figure being repeated in 6 sec., suggests that in 6 seconds the other fork completes one vibration more or less than double or half the vibrations performed by the first fork. Therefore, in one second the other fork will complete $\frac{1}{6}$ vibration more or less than double or half the vibrations made by the first fork. Obviously, therefore, the other fork will in one second, complete $200 \pm \frac{1}{6}$ or $50 \pm \frac{1}{6}$ vibrations.

Hence the possible frequencies of the other fork are $200 \pm \frac{1}{6}$ and $50 \pm \frac{1}{6}$.

Ex. 4. Two tuning forks whose frequencies are approximately in the ratio 2 : 1 are employed to produce Lissajous figures, and it is observed that the figure goes through a cycle of change in 15 sec. On loading slightly the fork of the upper pitch, the figure goes through a cycle of changes in 10 sec. If the fork of lower pitch has a frequency of 300 Hz. calculate the frequency of the other fork before and after loading.

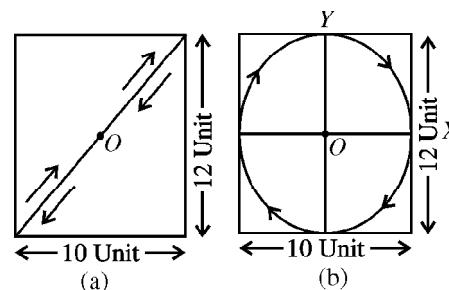


Fig. 7.30.

Sol. Since the frequency of the fork of lower pitch is 200 Hz, that of second may approximately be 600 Hz. Before loading the second fork, the figure goes through a cycle of changes in 15 sec., i.e., the frequency of the second fork may be equal to $600 \pm \frac{1}{15}$.

On loading the second fork, the time of the cycle of changes is reduced to 10 sec. Hence the frequency of the second fork after loading may be $600 \pm \frac{1}{10}$.

But on loading the frequency of the fork must decrease. Therefore, the frequency of the second fork before loading = $600 - \frac{1}{10} = 599 \frac{14}{15}$ Hz. and after loading = $600 - \frac{1}{10} = 599 \frac{9}{10}$ Hz.

Ex. 5. Show that a linear simple harmonic motion can be obtained by the superposition of two equal and opposite circular motions of the same angular frequency.

Sol. Let us consider two equal and opposite circular motions of the same angular frequency ω [fig. 7.31]. At any time t , the position vectors of the rotating point in the two cases are given by

$$\vec{OP}_1 = a \cos \omega t \hat{i} + a \sin \omega t \hat{j}$$

$$\vec{OP}_2 = a \cos \omega t \hat{i} - a \sin \omega t \hat{j}$$

The superposition of two vectors gives

$$\vec{OP}_1 + \vec{OP}_2 = 2a \cos \omega t \hat{i}$$

Thus the resultant displacement due to the two equal and opposite circular motions is along X-axis i.e., straight line and describes a S.H.M. of angular frequency ω .

Ex. 6. In a cathode ray oscilloscope, the electrons are deflected by two mutually perpendicular electric fields according to the displacement-time relations, given by

$$x = 4 \sin (2\pi \nu t + \pi/6) = 4 \sin (2\pi \nu t)$$

Determine the resultant path followed by the electrons.

Sol.
$$\frac{x^2}{4^2} + \frac{y^2}{4^2} - \frac{2xy}{4 \times 4} \cos \frac{\pi}{6} = \sin^2 \frac{\pi}{6}$$

[See eq. (4), Art. 7.10]

or
$$\frac{x^2}{16} + \frac{y^2}{16} - \frac{2xy}{16} \cdot \frac{\sqrt{3}}{2} = \frac{1}{4} \quad \text{or} \quad x^2 + y^2 - \sqrt{3}xy - 4 = 0$$

Thus the resultant path is an ellipse.

Ex. 7. Lissajous figure in the case of two vibrating forks is a parabola, prove that the frequencies are in the ratio 1 : 2.

Sol. Suppose that one fork of frequency ν_1 , is vibrating along X-axis and other of frequency ν_2 along Y-axis. Then if the parabola cuts the X-axis m times and Y-axis n times (fig. 7.32), then

$$\frac{\nu_2}{\nu_1} = \frac{m}{n} = \frac{1}{2} = 1 : 2.$$

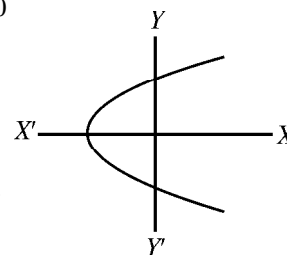


Fig. 7.32.

7.14. Superposition of Two Simple Harmonic Motions of Different Frequencies along the Same Line—Modulation and Beats

In physics, there are a number of phenomena in which the motion is a superposition of two simple harmonic motions, having different angular frequencies ω_1 and ω_2 . For example, when two tuning forks of different frequencies vibrate simultaneously, they emit sinusoidal waves. The oscillatory motion of the eardrum is, in fact, a superposition of two simple harmonic motions due to the two waves. To simplify mathematically, it is assumed that the two harmonic motions have the same amplitude a so that the phase

difference $[(\omega_2 t + \phi_2) - (\omega_1 t + \phi_1)] = (\omega_2 - \omega_1)t + \phi_2 - \phi_1$ between the two vibrations is changing continuously with time. Since the initial phase difference $(\phi_2 - \phi_1)$ does not play any significant role in this case, we assume that the two oscillations have zero initial phase and are represented as

$$x_1 = a \sin \omega_1 t \text{ and } x_2 = a \sin \omega_2 t \quad \dots(1)$$

The resultant motion is obtained as a superposition of the two motions i.e.,

$$x = x_1 + x_2 = a \sin \omega_1 t + a \sin \omega_2 t \quad \dots(2)$$

Modulation

Eq. (2) can be written as

$$x = 2a \cos\left(\frac{\omega_1 - \omega_2}{2}t\right) \sin\left(\frac{\omega_1 + \omega_2}{2}t\right) \quad \dots(3)$$

which represents an oscillatory motion with angular frequency $\frac{\omega_1 + \omega_2}{2}$ and amplitude $2a \cos\left(\frac{\omega_1 - \omega_2}{2}t\right)$.

We define an **average angular frequency** ω_{av} and a **modulated angular frequency** ω_{mod} by

$$\omega_{av} = \frac{\omega_1 + \omega_2}{2} \text{ and } \omega_{mod} = \frac{\omega_1 - \omega_2}{2} \quad \dots(4)$$

and a **modulation amplitude** $a_{mod}(t)$ by

$$a_{mod}(t) = 2a \cos\left(\frac{\omega_1 - \omega_2}{2}t\right) = 2a \cos \omega_{mod} t \quad \dots(5)$$

Therefore the superposed oscillation is given by

$$x = a_{mod}(t) \sin \omega_{av}(t) \quad \dots(6)$$

The superposition of two simple harmonic motions of different frequencies is shown in **fig. 8.8**. The figure also shows the variation of the modulation amplitude with time.

We see from eq. (5) that the modulation amplitude varies with time with a frequency, given by

$$\nu_{mod} = \frac{\omega_{mod}}{2\pi} = \frac{(\omega_1 - \omega_2)/2}{2\pi} = \frac{\nu_1 - \nu_2}{2} \quad \dots(7)$$

In a cycle, the modulation amplitude in eq. (6) is $2a$, 0 , $-2a$ and 0 , as the phase angle $(\omega_{mod} t)$ becomes 0 , $\pi/2$, π and $3\pi/2$ respectively. However the intensity (I) is proportional to the square of the amplitude, it will be maximum two times and also minimum two times in a cycle of the modulation amplitude. For example, in case of two sound waves (of equal amplitudes), the ear will hear the maximum or minimum sound at double the frequency of the modulated amplitude. *This fluctuation in the intensity of sound is called beats*. The frequency of the beats is given by

$$\nu_{beat} = 2\nu_{mod} = \nu_1 - \nu_2 \quad \dots(8)$$

Mathematically, we obtain this result as follows :

$$I \propto a_{mod}^2(t) = 4a^2 \cos^2 \omega_{mod}(t) = 2a^2 (1 + \cos 2\omega_{mod} t) \quad \dots(9)$$

Therefore $a_{mod}^2(t)$ or I oscillates at angular frequency $2\omega_{mod}$ i.e., the frequency of the beats is twice the modulation frequency. However one can hear the beats when $|\nu_1 - \nu_2|$ is less than about 10 Hz.

Ex. 1. Two vibrations along the same line are represented by the equations

$$x_1 = a \sin 20 \pi t \text{ and } x_2 = a \sin 22 \pi t.$$

Determine the beat frequency and draw a graph of the resultant motion over one beat period.

Sol. Here, $\nu_1 = 10$ Hz and $\nu_2 = 11$ Hz. Beat frequency = $|\nu_1 - \nu_2| = 1$ Hz.

$\therefore$ For graph see **fig. 7.33**.

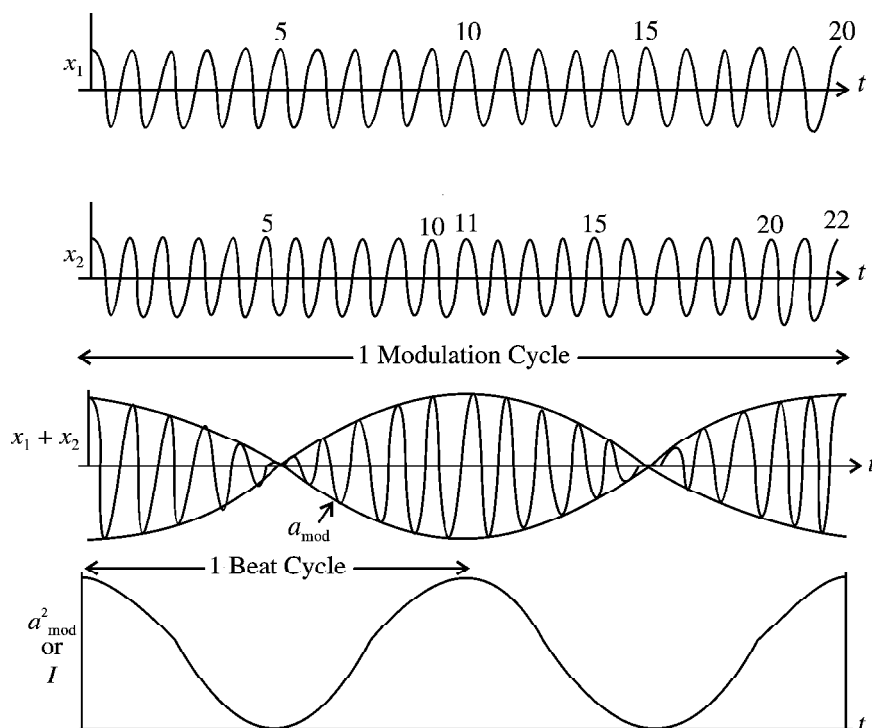


Fig. 7.33 : Modulation and formation of beats.

QUESTIONS

Long Answer Questions

1. What is S.H.M. ? Deduce the equation of motion of SHM and obtain its solution. (O.U. March 2008)
2. What is simple harmonic motion ? Mention its characteristics. Represent displacement, velocity and acceleration in a single diagram. (A.N.U. 2007)
3. What is the simple harmonic motion ? Write its characteristics. Establish the equation of motion of a simple harmonic oscillator and solve it. (S.K.U. 2003)
4. Define simple harmonic motion and explain its characteristics. Derive expressions for the frequency and energy of a simple harmonic oscillator. (A.U. March 2007)
5. (a) Define simple harmonic motion and explain its physical characteristics. (O.U. March 2007, K.U. April 2005)
 - (b) Show that the frequency of the simple harmonic oscillator is independent of the amplitude and its energy is proportional to the square of the amplitude.
6. (a) What are the characteristics of a simple harmonic oscillator ?
 - (b) Obtain an expression for the total energy of a simple harmonic oscillator. Show that the total energy is independent of the instantaneous displacement of the body. (O.U. 2002)
7. Show that the total energy of a simple harmonic oscillator is directly proportional to the square of its amplitude. (O.U. 2002)
8. Derive an expression for the energy of a body executing simple harmonic motion and show that it is directly proportional to the square of the amplitude and inversely proportional to the square of the period.
9. Show that for harmonic oscillator the total mechanical energy remains constant and is proportional to the square of the amplitude. (A.U. 1995)

10. Show that solution of the equation $\frac{d^2x}{dt^2} + \omega^2 x = 0$ is of the form $x = a \sin (\omega t + \phi)$.

Hence prove that for initial displacement x_0 and initial velocity v_0 , the amplitude $a = \sqrt{x_0^2 + v_0^2 / \omega^2}$ and initial phase $\phi = \tan^{-1} (\omega x_0 / v_0)$.

11. Set up the differential equation for the vibration of a spring treating it as a simple harmonic oscillator and obtain its solution. Discuss how the frequency of vibration of the spring is modified, if the mass of spring is taken into consideration. Does the force of gravity has any effect on the frequency of vibration of the spring ? **(A.N.U. 1990)**
12. Derive an expression for the frequency of vibration of a loaded spring taking its mass into consideration. **(S.K.U. 2007)**
13. What is simple harmonic motion ? Obtain the expression for the frequency of vibration of a loaded spring taking its mass into consideration. **(A.N.U. March 2007; S.V.U. 2007)**
14. What are the physical characteristics of simple harmonic motion. Discuss the theory of linear combination of two simple harmonic motions of the same frequencies. **(A.U. 2001)**
15. Discuss the linear combination of two S.H.M.'s, which are (i) with same angular frequency (ii) with slightly different frequencies. Explain how, the resultant phase and amplitudes change ? **(O.U. 2003)**
16. Discuss with mathematical theory the linear combination of simple harmonic vibrations of the same frequency. **(A.U. 2003)**
17. Discuss the effect of combining two S.H.M. in the same direction of the same angular frequency but different amplitudes and phases. How does the amplitude and phase of the resultant vibration get affected if two frequencies are slightly different from each other. **(O.U. 1990)**
18. Find the resultant vibration when two S.H.M. at right angles and differing in phase by 45° , but same frequency are compounded. **(K.U. 1991)**
19. Find the resultant of two S.H.M.'s. of periods in the ratio 1 : 2 and having a phase difference of 45° . **(A.U. 1992)**
20. Define simple harmonic motion. Write any four characteristics of simple harmonic motion. Discuss the theory of linear combination of two simple harmonic motions of the same frequency. **(S.K.U. 2002)**
21. Define simple harmonic motion. Give examples. Discuss the linear combination of two mutually perpendicular simple harmonic vibrations of equal frequencies.
22. Find the resultant of two simple harmonic motions of equal periods when they act at right angles of each other and discuss few cases. **(S.K.U. 2007)**
23. Explain Lissajous figures when two simple harmonic oscillations of the frequency ratio 1 : 1 and phase difference varying between 0 and π , superpose.
24. When two mutually perpendicular simple harmonic motions of amplitudes and time periods in the ratio 1 : 1, but phase difference (i) 90° , (ii) 45° superpose, deduce the expression for the figures obtained.
25. What are Lissajous figures ? Find the resultant of two simple harmonic motions of the same frequency acting along the same line but different phase. **(O.U. 2009; S.V.U. 2003)**
26. Discuss analytically the linear combination of two simple harmonic motions of the same frequency acting at right angles with various phase differences. **(A.U. June 2007)**
27. If two mutually perpendicular simple harmonic oscillations represented by equations $x = a \sin (2\omega t + \phi)$ and $y = b \sin \omega t$, superpose simultaneously on a particle, deduce the equation for the resultant path of the particle. What will be the shape of the path if $\phi = 0^\circ, \pi/2$ and π ?
28. Two perpendicular simple harmonic motions, whose amplitudes are different but the frequencies are in the ratio 2 : 1, are superposed together. Discuss those conditions in which the phase difference is 0, $\pi/2$ and π .

29. What are Lissajou's figures ? Find the resultant of simple harmonic motions of equal period's when they act at right angles to one another. Discuss the different important cases.

(A.N.U. 2008; S.V.U. 2005)

30. What are Lissajou's figures ? Find the resultant of two mutual perpendicular simple harmonic vibrations of same frequency.

(O.U. Oct. 2007)

31. What are Lissajou's figures ? How they can be obtained experimentally ? Give some important uses of Lissajou's figures.

Short Answer Questions

- What are physical characteristics of S.H.M. ? (S.V.U. 2005)
- What is S.H.M. ? What are its characteristics ? (S.K.U. 2005)
- Describe the characteristics of linear harmonic oscillator. (A.N.U. 2003)
- Discuss the physical significance of simple harmonic motion. (A.N.U. 2001)
- Explain how the change of velocity of a S.H. oscillator changes with position and time ? (S.V.U. 2002)
- Derive an expression for the energy of the harmonic oscillator. (A.N.U. 2008; A.U. 2001)
- Describe how the energy of an S.H.O. changes during an oscillation. (S.V.U. 2002)
- At what displacement the potential and kinetic energies of a simple harmonic motion are equal. (O.U. 2001)
- Express the kinetic and potential energies as a function of total energy of harmonic oscillator when its displacement half the amplitude.

[Ans. $K = \frac{3}{4}E, U = \frac{1}{4}E$.]

- Show that the total energy of a simple harmonic oscillator is directly proportional to the square of its amplitude. (K.U. 1995)
- Differentiate between the periodic motion and simple harmonic motion, giving two examples of each.
[Ans. If a moving particle retraces its path regularly after an interval of time, its motion is said to be periodic. Examples : (i) uniform circular motion of a particle, (ii) motion of a planet on elliptical path, (iii) motion of a simple pendulum etc. Simple harmonic motion is a particular type of periodic motion, in which at any instant the restoring force is proportional to displacement and directed opposite to it. Examples : motion of simple pendulum, spring-mass system, bar pendulum etc.]
- A particle of mass 0.1 kg lies in a potential field $V = 50x^2 + 100$ J/kg. What is the force on the particle ? Write the differential equation of motion.

[Ans. Force $F = -10x$ newton; $\frac{d^2x}{dt^2} + 100x = 0$.]

- Find the frequency of oscillation in Q. 12.

[Ans. $5/\pi$.]

- Show that in a compound pendulum, the centre of suspension and centre of oscillation are interchangeable.
- Show that in simple harmonic motion the displacement of the particle is the same after a time $2\pi/\omega$.
- The equation of a particle executing simple harmonic motion is

$$x = (1 \text{ m}) \sin\left[(\pi \text{ s}^{-1})t + \frac{\pi}{3}\right].$$

Write down the amplitude, time period and maximum speed.

[Ans. $a = 1 \text{ m}$, $T = 2\text{ s}$, $v_m = 5\pi \text{ m/s}$.]

- A particle executing S.H.M. comes to rest at the extreme positions. Is the resultant force on the particle zero at these positions ? Explain.

[Ans. No.]

18. Two particles execute simple harmonic motions of the same amplitude and frequency on parallel lines side by side. They cross one another when moving in opposite directions each time their displacement is half their amplitude. What is the phase difference between them ?
[Ans. Phase difference = $2\pi/3$.]
19. In S.H.M. when the displacement is one-half the amplitude, what fractions of the total energy are kinetic and potential ?
[Ans. $K/E = 0.75$ and $U/E = 0.25$.]
20. At what displacement the energy half kinetic and half potential ?
[Ans. $a/\sqrt{2}$.]
21. Given $x = 2.5 \sin \left(\frac{2\pi t}{3} + \frac{\pi}{2} \right)$, then find the values of angular velocity, maximum linear velocity, initial phase and time period.
[Ans. 2.09 radian/sec, 5.22 units, $\pi/2$, 3 sec.]
22. Derive an equation for the frequency of vibration of a spring taking its mass into consideration.
(A.U. 2005; A.N.U. 2005; S.K.U. 2003; S.V.U. 2002)
23. Discuss about the amplitude for linear combination of two simple harmonic oscillators of same frequency.
(A.V.U. 2005)
24. What are Lissajous' figures.
(A.U. March 2007; S.V.U. 2007)
25. Write a short note on Lissajous's figures.
(A.N.U. 2007, A.U. 2005)
26. What are Lissajous figures ? Give an example.
(S.K.U. 1996; S.V.U. 1996)
27. Write a note on the combination of two S.H.M. along a line.
(S.K.U. 1996)
28. What are beats ? How they are produced ?
(A.N.U. 2001; S.V.U. 2003)
29. If the Lissajous figure has the shape of letter '8', what is the ratio of component frequencies ?
[Ans. 2 : 1.]
30. What is the shape of Lissajous figure, when two mutually perpendicular simple harmonic motions of same amplitude and frequency are superposing on each other with a phase difference of $\pi/2$.
[Ans. Circle.]

PROBLEMS

[Simple Harmonic Motion]

1. A S.H.M. is represented by the equation
$$x = 0.12 \cos (12t - \pi/6)$$
where x is in metre and t in second. Find the frequency, time period and maximum velocity.
(A.U. 2002)
[Ans. $\nu = 1.91$ Hz; 0.52 sec; $v_{\max} = 1.44$ m/sec.]
2. A particle executes simple harmonic motion with a period of 0.002 sec and amplitude 10 cm. Find its acceleration when it is at a distance of 4 cm from its mean position. (A.U. 2001, S.V.U. 2003)
[Ans. -3.94×10^5 m/sec²]
3. The displacement equation of a particle describing simple harmonic motion is $x = 0.01 \sin 100 \pi (t + 0.005)$ metre, where x is the displacement of the particle at any instant t . Calculate the amplitude, periodic time, maximum velocity and displacement at the time of start of the motion.
[Ans. $a = 0.01$ m, $T = 0.02$ sec, $v_{\max} = 3.14$ m/sec, $x_0 = 0.01$ m]
4. A particle executes S.H.M. with a period of 0.002 sec and amplitude 10 cm. Find its acceleration when it is 4 cm away from its mean position and also obtain its maximum velocity.
[Ans. -3.9×10^5 m/sec²; 314 m/sec.]
5. The displacement equation for a particle in simple harmonic motion at an instant t is,
$$x = 0.01 \sin 100 \pi (t + 0.005) \text{ m}$$

Calculate (i) amplitude; (ii) time period; (iii) maximum velocity and (iv) displacement at start.

[Ans. (i) 0.01 m; (ii) 0.02 sec.; (iii) 3.14 m/s; (iv) 0.01 m.]

6. A particle makes S.H.M. along a straight line and its velocity when passing through points 3 cm and 4 cm from the centre of its path is 16 cm/sec and 12 cm/sec respectively. Find (a) the amplitude; (b) the time period of motion.

[Ans. 0.05 m; 1.571 sec.]

7. Prove that the velocity of a particle in S.H.M. at a distance $\sqrt{3}/2$ of its amplitude from the centre is half of its velocity at the central position.

8. A test tube weighs 6 gm and has an external diameter of 23 cm is floated in water by placing 10 gm mercury at the bottom of the tube. The tube is depressed by a small amount and then released. Find its time period.

[Ans. 0.453 sec.]

9. The displacement of a particle executing S.H.M. is given by

$$x = 20 \sin\left(\frac{2\pi t}{T} + \phi\right)$$

The period of vibration is 60 seconds. At $t = 0$, the displacement of the particle is 10 cm. Calculate (i) the initial phase, (ii) the phase angle when the displacement is 4 cm and (iii) the phase difference between any two positions of the particle 10 seconds apart.

[Ans. (i) $\pi/6$, (ii) $\sin^{-1}(0.15)$, (iii) $\pi/3$]

10. If a smooth tunnel is bored through the earth along one of its diameters, then calculate the time of one oscillation of the ball. ($R = 6.4 \times 10^3$ km and $g = 9.8$ metres/sec²).

[Ans. 84 minutes.]

11. A particle in S.H.M. has velocities u_1 and u_2 when displacements from the mean positions are x_1 and x_2 respectively. Calculate period, amplitude and maximum speed of the particle.

$$[\text{Ans. } T = 2\pi \sqrt{\frac{x_2^2 - x_1^2}{u_1^2 - u_2^2}}, a = \sqrt{\frac{u_1^2 x_2^2 - u_2^2 x_1^2}{u_1^2 - u_2^2}}, v = \sqrt{\frac{u_1^2 x_2^2 - u_2^2 x_1^2}{x_2^2 - x_1^2}}.]$$

12. A particle of mass m executes simple harmonic motion of amplitude a and frequency n . Find the force constant, the maximum kinetic energy and the average kinetic energy.

[Ans. $4\pi^2 n^2 m$, $2m\pi^2 n^2 a^2$, $m\pi^2 n^2 a^2$.]

13. If the potential energy of a harmonic oscillator in its resting position is 10 joules and the average kinetic energy is 5 joules, what is its total energy at any instant ?

[Ans. 20 joules.]

[Spring-Mass System]

14. A body of mass 0.25 kg hangs from a spring and oscillates. Find the time period and frequency. [Spring force constant = 40 Nm^{-1}]. (A.N.U. 2001)

[Ans. 2.014 sec.]

15. A vertical spring extends by 25 cm, when mass of 5 kg loads it. Calculate the frequency of oscillation, if the spring is loaded with 4 kg and allowed to oscillate. (S.V.U. 2003; O.U. 2003)

[Ans. 1.115 Hz]

16. A body of mass 0.2 kg is hung from a spring of constant 100 N/m. Calculate the value of undamped frequency. (A.N.U. 05)

[Ans. 3.56 Hz]

17. An object of mass 2.5 kg is hanging from a massless spring. Another object of mass 400 gm, hung below the first object, stretches the spring 2 cm farther. If the object of 400 gm is removed and the first object is set into vertical oscillations, determine the period of oscillation.

[Ans. 0.71 sec.]

18. A body of mass 4.9 kg when suspended from a spring, oscillates with a period 0.5 sec. By what amount will the spring be contracted on the removal of the body ?

[Ans. 0.062 m.]

19. One end of a vertical spring is attached to a fixed support and to the lower end a mass m is suspended. If by suspending the mass, the lower end of the spring is stretched through x_0 distance, then find the spring constant and period of oscillation.

[Ans. mg/x_0 ; $2\pi\sqrt{x_0/g}$.]

[Hint : $mg = kx_0$.]

20. (a) Two springs are joined and connected to mass M as shown in **fig. 7.12**. If the springs separately

have force constants k_1 and k_2 , show that the frequency of oscillation is $n = \frac{1}{2\pi} \sqrt{\frac{C_1 C_2}{(C_1 + C_2)m}}$.

- (b) If the two massless springs are attached to a mass M and to fixed supports as shown in **fig. 7.11**,

then show that the frequency of oscillation is given by $n = \frac{1}{2\pi} \sqrt{\frac{C_1 + C_2}{m}}$.

21. In the in **fig. 7.11** M is a mass which is placed between two rigid supports A and B , and is controlled by two massless springs. If the mass of M is 50 gms and the force constants of springs are 3000 and 2000 dynes/cm, deduce :

- (i) the 'frequency' of small oscillations of the mass M .
- (ii) the energy of vibration of amplitude 0.4 cm.
- (iii) the velocity of mass M when it passes through its mean position.

[Ans. (i) 1.59 Hz; (ii) 4×10^{-5} joules; (iii) 0.04 m/sec.]

[Compound Pendulum]

22. A sphere of radius R is suspended by a string from a fixed point, so that the distance from the centre of sphere to the point of suspension is ($l \gg R$). Show that the period of the pendulum is

$$T = 2\pi \sqrt{\frac{I}{g} \left(1 + \frac{R^2}{5l^2} \right)}.$$

If $R = 0.05$ m and $l = 1$ m, what is the length of the equivalent simple pendulum ?

[Ans. 1.001 m.]

23. A body of mass 200 gm oscillates about a horizontal axis at a distance of 20 cm from its centre of gravity. If the length of the equivalent simple pendulum be 35 cm, find its moment of inertia about the axis of suspension.

[Ans. 1.4×10^{-4} kg $\times$ m².]

24. A thin uniform bar of length 1.2 m is made to oscillate about an axis through its end. Find the period of oscillation and other points about which it can oscillate with the same period.

[Ans. 1.796 sec; Points are situated at distance 0.4, 0.8, 1.2 m from one end.]

25. (i) Assuming small amplitude oscillations, what would be the frequency of a simple pendulum of length 2 metres, in an elevator accelerating upward at a rate of 2.0 metres/sec² ?

- (ii) What would be its frequency in free fall ?

[Ans. (i) 0.38 Hz; (ii) Zero.]

26. Prove that a uniform rod of length L hanging as a compound pendulum from a pivot at one end has the same frequency as a simple pendulum of length $2L/3$. for oscillations of small amplitude.

27. A thin rod of length 60 cm is used as a compound pendulum. Find the position of a point such that the period about it is a minimum.

[Ans. 17.3 cm from C.G.]

28. A disc of radius r oscillates as a pendulum about a horizontal axis perpendicular to the plane of the disc through a point $r/2$ from its centre. Calculate the period of oscillation.

[Ans. $T = 2\pi\sqrt{3r/2g}$.]

29. A metal disc of radius R is made to oscillate about an axis passing through a point on its edge. Find (a) the equivalent length of simple pendulum, and (b) the minimum time period of oscillation.

[Ans. $3R/2$; $T = 2\pi\sqrt{3R/2g}$.]

30. Calculate the time of oscillation of a uniform square lamina of side 30 cm about an axis perpendicular to its plane and passing through one of its corners. ($g = 9.8 \text{ m/sec}^2$).

[Ans. $T = \pi\sqrt{13/98}$.]

31. A uniform circular rod with a radius of 2 cm oscillates when suspended from a point on its axis at a distance of 4 cm from one end. If the length of the rod is 1 metre, find the point or points from which, it suspended, the periodic time would remain unaltered.

[Ans. At 31.87 cm, 68.13 cm and 96 cm from the same end.]

32. A uniform rod AB of mass 100 gm and length 120 cm can swing in a vertical plane about A as pendulum. A particle of mass 200 gm is attached to the rod at a distance x from A . Find x such that the period of vibration is a minimum.

[Ans. $x = 2.748 \text{ cm}$.]

[Torsional Pendulum]

33. A solid cylinder of 2 cm radius weighing 200 gm is rigidly connected, with its axis vertical, to the lower end of a fine wire. The period of oscillation of the cylinder under the influence of the torsion of wire, is 2 seconds. Calculate the couple necessary to twist it through four complete turns.

[Ans. $9.9 \times 10^{-3} \text{ N-m}$.]

34. A torsion pendulum consists of a rectangular wood block $8 \text{ cm} \times 12 \text{ cm} \times 3 \text{ cm}$ and with a mass 0.3 kg suspended by means of a wire passing through its centre in such a way that the shortest side is vertical. If the period of torsional oscillations is 2.4 sec, find the torsion constant of the wire.

[Ans. $3.565 \times 10^{-3} \text{ N-m/rad}$.]

35. A solid spherical bob of radius 0.05 m and mass 3.5 kg is suspended from a wire of torsional rigidity $6 \times 10^{-3} \text{ N-m/rad}$ to form a torsion pendulum. Find the period. Find also the period, in case if a hollow sphere of the same mass and same material but with external radius 6 cm were used.

[Ans. 4.8 sec; 6.62 sec.]

36. The balance wheel of a watch vibrates with an angular amplitude of π radians and a period of 0.5 second. Determine (i) the maximum angular speed of the wheel, (ii) the angular speed of the wheel at the displacement $\pi/2$ radians, and (iii) the angular acceleration of the wheel at the displacement $\pi/4$ radians.

[Ans. (i) 39 rad/sec., (ii) 34 rad/sec., (iii) 120 rad/sec².]

[Combination of Two Perpendicular S.H.M.'s]

37. Two forks are used to produce Lissajous figures. It is found that the figure completes its cycle in 10 sec. If the frequency of A (known to be slightly longer than that of B) is 200. Calculate the frequency of B .

[Ans. 199.9 Hz.]

38. Two tuning forks are vibrated perpendicularly and a beam of light is reflected from the mirrors, attached to the prongs of the forks. The elliptical figures are produced on the screen and they change in shape as time progresses. The figures complete a cycle of changes in 10 seconds. If the frequency of one fork is 250 (slightly less than that of the other), determine the frequency of the other.

[Ans. 250.1 Hz.]

39. Two tuning forks P and Q of frequencies close to each other are used to obtain Lissajous figures and it is observed that the figures goes through a cycle of changes in 20 seconds. Now if P is loaded slightly with wax, the figure goes through a cycle of changes in 10 seconds. If the frequency of Q is 300 Hz, what is the frequency of P before and after loading.

[Ans. 299.95 Hz; 299.9 Hz.]

40. Two rectangular simple harmonic vibrations of frequency ratio 2 : 1 are represented as follows :

$$x = a_1 \cos (2\omega t + \phi) \text{ and } y = a_2 \cos \omega t$$

Find the resultant motion, when (i) $\phi = 0$; (ii) $\pi/2$ and (iii) π .

[Ans. (i) Parabola; (ii) Figure of '8'; (iii) Parabola oppositely directed to that of (i).]



Damped and Forced Oscillations

8.1. Introduction

A particle will perform simple harmonic motion with a constant amplitude (or energy) of vibration, provided the restoring force on it is proportional to the displacement ($F = -kx$). In actual practice, we observe that if a physical system, once oscillated, its oscillations die out with the passage of time. For example, if we oscillate a pendulum, the amplitude of vibration does not remain constant but decreases gradually with time and ultimately the pendulum comes to rest. In fact, when a body moves in a medium, frictional forces act on the body in a direction opposite to its movement and the work done against the frictional forces results in the decrease of its mechanical energy. This is why, an oscillating pendulum loses energy and consequently its amplitude of oscillation becomes smaller and smaller as the time elapses.

In the presence of frictional forces, the motion of the oscillator is said to be **damped** by friction and the vibrating system is called **damped harmonic oscillator**. The frictional or damping force on the oscillator arises from the retarding action of the medium. The magnitude of this force is some function of the velocity of the moving body. At ordinary speeds, the damping force (F_d) is found proportional to the velocity (v) in a direction opposite to it i.e., $F_d = -\gamma v$, where γ is a constant called the damping coefficient. It depends on the size of the body.

8.2. Damped Harmonic Oscillator

In the presence of frictional forces, arising due to the viscosity of the medium, the oscillatory system is called **damped harmonic oscillator**. The free oscillations of a physical system in the presence of frictional forces are called **damped harmonic motions**. When damping is taken into account, then a harmonic oscillator is acted by the following forces :

- (1) A restoring force $-kx$ at the displacement x due to the spring action, (K = force constant) and
- (2) A damping force $-\gamma v$, where $v = dx/dt$ is the velocity of the oscillator at the displacement x and the constant γ is the *damping coefficient or frictional force per unit velocity*. This γ is also called **mechanical resistance**.

Thus the resultant force on the oscillating body is given by

$$F = -kx - \gamma \frac{dx}{dt}$$

Using Newton's second law, the equation of motion of the damped harmonic oscillator is given by

$$m \frac{d^2x}{dt^2} = -kx - \gamma \frac{dx}{dt} \quad \dots(1)$$

where m is the mass of the oscillating body.

Dividing by m , eq. (1) can be written as

$$\frac{d^2x}{dt^2} + 2b \frac{dx}{dt} + \omega_0^2 x = 0 \quad \dots(2)$$

where $2b = \gamma/m$ and $\omega_0^2 = k/m$. Here $\omega_0 = \sqrt{k/m}$ is the natural angular frequency of the oscillator in absence of the damping force. The constant $b = \gamma/2m$ is called **damping or decay constant** and the constant $\frac{1}{b} = \frac{2m}{\gamma}$ is termed as **decay modulus**. Also if we put $m/\gamma = \tau = 1/2b$, then τ is called **relaxation time or time constant**.

Equation (2) is called the **differential equation of damped harmonic oscillator**.

Examples of Damped Harmonic Oscillator

(1) A spring-mass system, when displaced from equilibrium, oscillates with decreasing amplitude with time and finally comes to rest.

(2) A pendulum, disturbed from equilibrium position, oscillates with decreasing amplitude with time and ultimately acquires equilibrium position.

(3) Charge oscillations in a *LCR* circuit die away after some time.

(4) Oscillations of a ballistic galvanometer coil with decreasing amplitude.

Solution of the Differential Equation of Damped Harmonic Oscillator

Let the solution of eq. (2) be

$$x = Ae^{\alpha t}$$

Substituting the values of x , dx/dt and d^2x/dt^2 in eq. (2), we have

$$(\alpha^2 + 2b\alpha + \omega_0^2)e^{\alpha t} = 0 \text{ or } \alpha^2 + 2b\alpha + \omega_0^2 = 0$$

$$\therefore \alpha = -b \pm \sqrt{b^2 - \omega_0^2}$$

Hence the solutions of the eq. (2) are

$$x = A_1 e^{(-b + \sqrt{b^2 - \omega_0^2})t} \text{ and } x = A_2 e^{(-b - \sqrt{b^2 - \omega_0^2})t}$$

Thus the most general solution of eq. (2) is

$$x = A_1 e^{(-b + \sqrt{b^2 - \omega_0^2})t} + A_2 e^{(-b - \sqrt{b^2 - \omega_0^2})t}$$

$$\text{i.e., } x = e^{-bt} (A_1 e^{\sqrt{b^2 - \omega_0^2}t} + A_2 e^{-\sqrt{b^2 - \omega_0^2}t}) \quad \dots(4)$$

where A_1 and A_2 are constants, depending upon the initial conditions of motion.

The quantity $\sqrt{b^2 - \omega_0^2}$ is imaginary, real or zero, it depends on the relative values of b and ω_0 . If $b < \omega_0$, then $\sqrt{b^2 - \omega_0^2}$ is imaginary and it is called **underdamped case**. If $b > \omega_0$ or $b = \omega_0$, the situations are known as **overdamped** and **critically damped** respectively.

(i) **Underdamped case** ($b < \omega_0$) i.e., **case of damped harmonic oscillations** : If the damping is so low that $b < \omega_0$, then $\sqrt{b^2 - \omega_0^2} = \sqrt{-1(\omega_0^2 - b^2)} = i\sqrt{\omega_0^2 - b^2} = i\omega$, where $i = \sqrt{-1}$ and $\omega = \sqrt{\omega_0^2 - b^2}$, which is a real quantity. Now, from eq. (4), we have

$$x = e^{-bt} [A_1 e^{i\omega t} + A_2 e^{-i\omega t}] = e^{-bt} [A_1 (\cos \omega t + i \sin \omega t) + A_2 (\cos \omega t - i \sin \omega t)]$$

$$\text{or } x = e^{-bt} [(A_1 + A_2) \cos \omega t + i (A_1 - A_2) \sin \omega t]$$

As x is real quantity, $(A_1 + A_2)$ and $i (A_1 - A_2)$ must be real quantities. Clearly, A_1 and A_2 are complex quantities. If $A_1 + A_2 = a_0 \sin \phi$ and $i (A_1 - A_2) = a_0 \cos \phi$, then

$$x = a_0 e^{-bt} \sin (\omega t + \phi) \quad \dots(5)$$

This equation represents a **damped harmonic motion**. This motion is oscillatory or ballistic whose **periodic time** is given by

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{\sqrt{\omega_0^2 - b^2}} \quad \dots(6)$$

Thus, the effect of damping is to increase the periodic time. If $b = 0$, i.e., no damping, then $T = 2\pi/\omega_0$. But in actual cases, the effect of damping on periodic time is usually negligible except few extreme cases.

The **amplitude** of the damped harmonic oscillations [eq. (5)] is given by

$$a = a_0 e^{-bt} = a_0 e^{-t/2\tau} \quad \dots(7)$$

where a_0 is the amplitude in the absence of damping. In the presence of damping, the amplitude decreases exponentially with time. In **fig 8.1**, time displacement curve is shown for damped harmonic motion. As the maximum value of $\sin (\omega t + \phi)$ is alternately $+1$ and -1 , obviously the time displacement curve of

the vibrating body lies entirely between the curves $a = a_0 e^{-bt}$ and $a = -a_0 e^{-bt}$, shown by the dotted lines.

Logarithmic decrement : The time interval between two successive maximum displacements (i.e., amplitudes) of left hand and right hand sides is $T/2$. If a_n and a_{n+1} are the two successive amplitudes, then

$$a_n = a_0 e^{-bt} \text{ and } a_{n+1} = a_0 e^{-b(t + T/2)}$$

$$\text{Therefore, } \frac{a_n}{a_{n+1}} = \frac{e^{-bt}}{e^{-b(t + T/2)}} = e^{bT/2} \quad \dots(8)$$

The quantity $e^{bT/2}$, say d , is a constant of motion, representing reduction in the amplitude and hence we call it **decrement**.

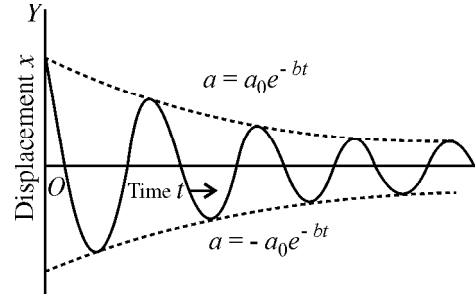


Fig. 8.1. Variation of amplitude with time.

$$\text{Now, } d = e^{bT/2} = \frac{a_n}{a_{n+1}}; \therefore \log_e d = \log_e \frac{a_n}{a_{n+1}} = \frac{bt}{2} = \lambda \text{ (say)} \quad \dots(9)$$

We call λ as the **logarithmic decrement** which is equal to the natural logarithmic of the ratio of two successive amplitudes of vibration.

If a_1 is the first maximum displacement, occurring after $T/4$ time, then

$$a_1 = a_0 e^{-bT/4} \text{ or } a_0 = a_1 e^{bT/4} \quad \dots(10)$$

From which one can determine the amplitude a_0 in absence of damping.

(ii) Overdamped case : If the damping is high i.e., $b > \omega_0$, then $\sqrt{b^2 - \omega_0^2}$, say β , is a real quantity and then from eq. (4), we have

$$x = (A_1 e^{-(b-\beta)t} + A_2 e^{-(b+\beta)t}) \quad \dots(11)$$

As $b > \beta$, both quantities of right hand side decrease exponentially with time and the displacement comes to zero in course of time [fig. 8.2]. In case of high damping the motion is **non-oscillatory**. Such a motion is called **dead beat** or **aperiodic** or **overdamped**. Such a motion is realized practically (i) when a pendulum moves in a very viscous liquid, (ii) in a moving coil galvanometer with a low resistance across its terminals, etc.

(iii) Critically damped case : If $b = \omega_0$, then from eq. (4), we have

$$x = (A_1 + A_2)t e^{-bt} = C t e^{-bt}, \text{ where } C = A_1 + A_2$$

In this equation, there is only one constant, hence it does not provide us the solution of differential equation (2) of second order.

Now, suppose $\sqrt{b^2 - \omega_0^2} = h$, which is a small quantity. Hence from eq. (11), we have

$$x = e^{-bt} [A_1 e^{ht} + A_2 e^{-ht}] = e^{-bt} [A_1(1 + ht + \dots) + A_2(1 - ht + \dots)]$$

Neglecting the small terms, containing h^2 and higher powers of h , we get

$$x = e^{-bt} [(A_1 + A_2) + h(A_1 - A_2)t] \text{ or } x = e^{-bt} (P + Qt) \quad \dots(12)$$

where $P = A_1 + A_2$ and $Q = h(A_1 - A_2)$

$$\text{Also } \frac{dx}{dt} = e^{-bt} Q + e^{-bt} (-b)(P + Qt)$$

If initially at $t = 0$, the displacement of the particle is $x = x_0$ and there the velocity is v_0 , then

$$\begin{aligned} x_0 &= P \text{ and } v_0 = Q - bP \\ &= Q - bx_0 \text{ or } Q = v_0 + bx_0 \end{aligned}$$

$$\therefore x = [x_0 + (v_0 + bx_0)t] e^{-bt} \quad \dots(13)$$

This equation represents that initially the displacement increases due to the factor $[x_0 + (v_0 + bx_0)t]$. But as time elapses, the exponential term becomes relatively more important and the displacement decreases exponentially but quicker than overdamped case [fig. 8.2]. In this case, the oscillatory motion does not just occur. Such a motion is called **critically damped** or **just aperiodic**

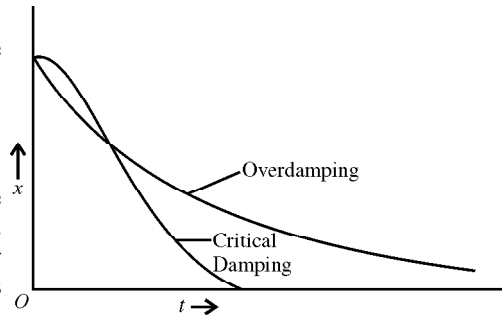


Fig. 8.2. Damped harmonic oscillator—overdamping and critical damping

or **just dead beat**. In a critically damped system, the displacement becomes zero in minimum time. The characteristic of critical damping is used in the measuring instruments, such as ammeters, voltmeters etc, where we want that the pointer or needle settles down to its final reading in a very short time.

8.3. Energy and Power Dissipation of Damped Harmonic Oscillator-Relaxation Time

When a body oscillates in a medium (*e.g.*, oscillations of simple pendulum in air), the frictional or damping forces act on the body in a direction opposite to its movement. In this process work is done by the body in overcoming the frictional forces. This results in a continuous decrease in the mechanical energy of the oscillating body and thereby in diminishing the amplitude of oscillation with time. Finally the vibrating body comes to rest, if left to itself for a sufficiently long time.

The total energy of a damped harmonic oscillator decreases with time. We are interested to find the rate at which the energy is dissipated i.e., the power dissipated, in the weak damping limit ($b \ll \omega_0$).

At any instant t , the displacement of a damped harmonic oscillator is given by

$$x = a_0 e^{-bt} \sin(\omega t + \phi) \quad \dots(1)$$

where $a = a_0 e^{-bt}$ is the amplitude of damped harmonic oscillator at any time t .

In case of harmonic oscillator, we know that the total energy E of the oscillator is proportional to the square of the amplitude. When damping is low, we may consider that around time t , the oscillations are well described over some cycles by a simple harmonic motion, in which the total energy $E(t)$ is given by

$$E = \frac{1}{2} k a^2 = \frac{1}{2} k a_0^2 e^{-2bt} \quad \dots(2)$$

Writing $E_0 = \frac{1}{2} k a_0^2$, the initial value of the energy, we have

$$E = E_0 e^{-2bt} \text{ or } E = E_0 e^{-t/\tau} \quad \dots(3)$$

The energy of the oscillator after one period (T) is

$$E_T = E_0 e^{-2b(t+T)} = E e^{-2bT} \quad \dots(4)$$

Hence the **average power dissipation** is given by

$$P_{av} = \frac{\text{Energy loss in one period}}{\text{Periodic time}} = \frac{E - E_T}{T} = \frac{E - E e^{-2bT}}{T} = \frac{E - E(1 - 2bT)}{T} \quad (\text{as damping is small})$$

$$\text{i.e., } P_{av} = 2bE = \frac{E}{\tau} \quad \dots(5)$$

This dissipation of energy is caused by the damping force $\gamma dx/dt$. Hence if we calculate the negative value of the average rate of doing work by this force, i.e., $\langle \gamma (dx/dt)^2 \rangle$, then we will get exactly the same expression for power dissipation, as obtained above. The proof is as follows :

$$\begin{aligned} \text{Average rate of doing work by the damping force} &= \langle F \frac{dx}{dt} \rangle = -\gamma \langle \left(\frac{dx}{dt} \right)^2 \rangle \\ &= -2bm \left(\frac{1}{2} a_0^2 \omega^2 e^{-2bt} \right) = -2bE = -E/\tau \end{aligned} \quad \dots(6)^*$$

$$\text{For low damping } (b \ll \omega_0), \langle \left(\frac{dx}{dt} \right)^2 \rangle = \langle \omega^2 a_0^2 e^{-2bt} \sin^2(\omega t + \phi) \rangle$$

$$= \frac{1}{2} a_0^2 \omega^2 e^{-2bt} \quad [\text{Time average of } \sin^2(\omega t + \phi) = \frac{1}{2}]$$

where e^{-2bt} has been taken constant for a period.

* For low damping ($b \ll \omega_0$), $\langle (dx/dt)^2 \rangle = \langle a_0^2 \omega^2 e^{-2bt} \sin^2(\omega t + \phi) \rangle = \frac{1}{2} a_0^2 \omega^2 e^{-2bt}$ where time average of $\sin^2(\omega t + \phi)$ is $\frac{1}{2}$ and e^{-2bt} has been taken as constant for a time period.

Hence, **negative of the average rate of doing work** = $\frac{E}{\tau}$ = **power dissipation**.

This loss of energy in damped harmonic oscillations appears in the form of heat in the oscillating system and dissipated in the medium.

Relaxation Time τ

In case of damped harmonic oscillator, the total energy (E) at an instant t can be expressed as

$$E = E_0 e^{-t/\tau} \quad \dots(7)$$

where $\tau = 1/2b$ is relaxation time and b is damping constant.

When $t = \tau$,

$$E = E_0 e^{-1} \text{ or } E = \frac{E_0}{e} \quad \dots(8)$$

Thus the relaxation time is defined as the time in which the total mechanical energy of the damped oscillator decreases to $1/e$ of its initial value.

8.4. Quality Factor Q

When damping is small, the term Q , known as **quality factor**, is very widely used for various oscillating systems. The quality factor Q of an oscillating system is defined as the product of natural angular frequency (ω_0) (in absence of damping) and time constant (τ) of the oscillator i.e.,

$$Q = \omega_0 \tau \quad \dots(1)$$

Evidently Q is a *dimensionless quantity*.

For a lightly damped harmonic oscillator, $\omega \approx \omega_0$. Therefore

$$Q = \omega \tau \quad \dots(2)$$

$$\text{But, } Q = \omega \tau = \frac{2\pi}{T} \cdot \frac{E}{P_{av}} = \frac{2\pi E}{P_{av} T} \quad \left[\because \omega = \frac{2\pi}{T} \text{ and } P_{av} = \frac{E}{\tau} \right]$$

$$\text{or } Q = 2\pi \frac{\text{average energy stored}}{\text{average energy loss per period}} \quad \dots(3)$$

Thus Q -factor of an oscillating system is equal to 2π times the ratio of the average energy stored to the average energy loss per period.

Now, for a mechanical oscillator, $\omega_0 = \sqrt{k/m}$ and relaxation time, $\tau = m/\gamma$, then from (1)

$$Q = \sqrt{\frac{K}{m}} \cdot \frac{m}{\gamma} = \frac{\sqrt{mK}}{\gamma} \quad \dots(4)$$

where γ is damping coefficient.

Thus high value of Q implies low value of the damping for the oscillating system. This Q is a measure of the extent to which an oscillator is free from damping. In case of an **undamped oscillator**, $\gamma = 0$, and hence Q is **infinite**.

The energy of the oscillation ($E = E_0 e^{-t/\tau}$) decreases to e^{-1} of its initial value in the time τ and in this time the oscillator performs $\omega_0 \tau / 2\pi = Q / 2\pi$ oscillations.

8.5. LCR Circuit

Charge oscillation in an LCR circuit is an example of damped harmonic oscillator. Consider a coil of self inductance L and resistance R , connected to a capacitor C [fig. 8.3]. In the beginning, the condenser is charged to a charge q_0 on the right plate by a battery and then it is discharged through the inductance L and capacitance C by switching off the battery circuit and connecting the LCR circuit by a key. At any instant, if q be the charge on the capacitor C , then the potential difference across the capacitor C is

$$V = \frac{q}{C} = L \frac{di}{dt} + Ri$$

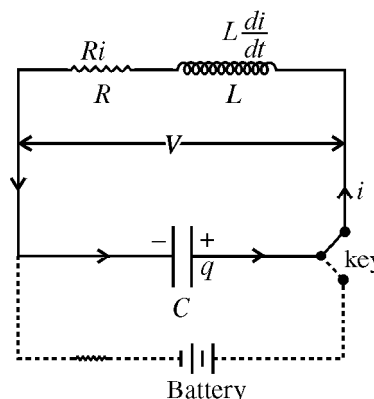


Fig. 8.3. LCR circuit.

$$\text{or } L \frac{d^2 q}{dt^2} + R \frac{dq}{dt} + \frac{q}{C} = 0 \quad \left(\because i = -\frac{dq}{dt} \right) \quad \dots(1)$$

Here L , R , $1/C$ play the role of mass (m), damping (γ) and force constant (K) respectively.

Eq. (1) can be written as

$$\frac{d^2 q}{dt^2} + 2b \frac{dq}{dt} + \omega_0^2 q = 0 \quad \dots(2)$$

where $b = R/2L$ and $\omega_0^2 = 1/LC$.

Eq. (10) is exactly like the differential equation for the damped harmonic oscillator. The discharge will be oscillatory, when

$$\omega_0^2 > b^2 \quad \text{or} \quad \frac{1}{LC} > \frac{R^2}{4L^2}.$$

The solution of eq. (10) is of the form

$$q = q_0 e^{-(R/2L)t} \sin(\omega t + \phi) \quad \dots(11)$$

We note that the charge amplitude, $q_0 e^{-(R/2L)t}$, decays exponentially with time. The decay in charge amplitude is caused by the resistance R , because in absence of R (i.e., $R = 0$), the charge amplitude will remain constant (q_0). Hence in a LCR circuit, resistance is only the dissipative element.

The frequency of the oscillatory discharge is given by

$$\nu = \frac{\omega}{2\pi} = \frac{\sqrt{\omega_0^2 - b^2}}{2\pi} = \frac{1}{2\pi} \sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}} \quad \dots(12)$$

When $R = 0$, we obtain the case of harmonic charge oscillations and the frequency of oscillations, $\nu_0 = 1/2\pi\sqrt{LC}$ for LC circuit. Thus the presence of resistance in the circuit decreases the frequency.

Q-Factor in LCR circuit : For low resistance circuit,

$$Q = \omega_0 \tau = \frac{\omega_0}{2b} = \frac{L\omega_0}{R} \quad \dots(13)$$

For pure LC circuit $R = 0$, hence $Q = \infty$.

Now, if we differentiate eq. (11) with respect to t and place $-dq/dt = i$, then on arranging the differentiated terms, we get

$$i = i_0 e^{-bt} \sin(\omega t + \theta) \quad \dots(14)$$

where i_0 is the maximum value of the circuit.

Cases when $\omega_0^2 \leq b^2$ or $\frac{1}{LC} \leq \frac{R^2}{4L^2}$

In case $\omega_0^2 < b^2$ or $\frac{1}{LC} < \frac{R^2}{4L^2}$, the circuit is overdamped and the discharge will be **non-oscillatory**.

Further, the case, when $\omega_0^2 = b^2$ or $\frac{1}{LC} = \frac{R^2}{4L^2}$ represents **critical damping** or **just aperiodic**.

Ex. 1. Show that the amplitude of a damped harmonic oscillator falls to $1/e$ value in 2τ time, where τ is relaxation time.

Sol.

$$a = a_0 e^{-bt} = a_0 e^{-t/2\tau} \quad \text{or} \quad \frac{a}{a_0} = e^{-t/2\tau}$$

According to the problem, $\frac{a}{a_0} = \frac{1}{e}$

Hence $\frac{1}{e} = e^{-t/2\tau}$ or $e^{-1} = e^{-t/2\tau}$ or $\frac{t}{2\tau} = 1, \therefore t = 2\tau$.

Ex. 2. The amplitude of vibration of a spring-mass system decreases from 10 cm, to 2.5 cm in 200 seconds. Find the damping constant and relaxation time. In how much time does the amplitude falls to $1/e$ times the initial value ?

If the oscillator performs 100 oscillations in the given time, compare the periods with and without damping. What do you conclude ?

Sol. The amplitude at an instant t of a damped oscillator is given by

$$a = a_0 e^{-bt} \text{ or } e^{bt} = \frac{a_0}{a} \text{ or } b = \frac{1}{t} \log_e \frac{a_0}{a}$$

Here, $t = 200$ sec., $a_0 = 10$ cm and $a = 2.5$ cm,

$$\therefore b = \frac{1}{200} \log_e \frac{10}{2.5} = 6.93 \times 10^{-3} \text{ sec}^{-1}; \tau = \frac{1}{2b} = \frac{1}{2 \times 6.93 \times 10^{-3}} = 72 \text{ sec.}$$

$$\text{Now, for } \frac{a}{a_0} = \frac{1}{e}, \quad t = \frac{1}{b} \log_e e = \frac{1}{b} = \frac{1}{6.93 \times 10^{-3}} = 144 \text{ sec.}$$

$$\text{Period, } T = \frac{2\pi}{\sqrt{\omega_0^2 - b^2}} = \frac{200}{100} = 2 \text{ sec.}$$

$$\text{Now, } 2 = \frac{2\pi}{\sqrt{\omega_0^2 - b^2}}, \therefore \omega_0^2 = \pi^2 + b^2 = \pi^2 + (6.93 \times 10^{-3})^2 \text{ or } \omega_0 = 3.1416$$

$$\therefore T_0 = 1.9999 \text{ sec} \approx 2 \text{ sec.}$$

Note that the change in the time period due to damping is negligible.

Ex. 3. A damped vibrating system starting from rest reaches a first amplitude of 500 mm, which reduces to 50 mm after 100 oscillations, each of period 2.3 seconds. Find the damping constant, relaxation time and correction for the first displacement for damping.

Sol. The amplitude of damped vibrations is given by

$$a = a_0 e^{-bt}$$

Time period, $T = 2.3$ sec.

First amplitude of the system will be at $t = T/4$ and the amplitude after 100 oscillations i.e., 201th amplitude will occur at $(100T + T/4)$ time, hence

$$a_1 = a_0 e^{-bT/4} \text{ and } a_{201} = a_0 e^{-b(100T + T/4)},$$

$$\therefore \frac{a_{201}}{a_1} = e^{-100bT} \text{ or } \frac{50}{500} = e^{-100 \times 2.3b} \text{ or } 10^{-1} = e^{-230b} \text{ or } \log_e 10 = 230b,$$

$$\therefore \text{Damping constant, } b = 2.3/230 = 0.01 \text{ sec}^{-1}$$

$$\therefore \text{Relaxation time, } \tau = \frac{1}{2b} = \frac{1}{2 \times 0.01} = 50 \text{ sec.}$$

$$\therefore a_1 = a_0 e^{-bT/4}, \therefore 500 = a_0 e^{-0.01(2.3/4)}$$

$$\text{or } a_0 = 500 e^{0.023/4} \text{ or } \log_{10} a_0 = \log_{10} 500 + \frac{0.023}{4 \times 2.3}$$

$$\text{or } \log_{10} a_0 = 2.699 + 0.0025 = 2.7015, \text{ whence } a_0 = 502.9 \text{ mm.}$$

Thus in absence of damping, the amplitude of motion would have been 502.9 mm.

Hence the correction for the displacement for damping = $502.9 - 500 = 2.9$ mm.

Ex. 4. An underdamped oscillator has its amplitude reduced to $(1/10)^{\text{th}}$ of its initial value after 100 oscillations. If the time period is 2 seconds, calculate (1) the damping constant and (2) the decay modulus. (A.U. 1996)

Sol. The amplitude a of damped oscillator is given by

$$a = a_0 e^{-bt} \quad \dots(i)$$

where a_0 is the amplitude of undamped oscillator.

The first amplitude a_1 is obtained after $T/4$ sec, where T is the periodic time. Hence,

$$a_1 = a_0 e^{-bt(T/4)} \quad \dots(ii)$$

As the successive amplitudes occur at interval of $T/2$, the amplitude after 100 oscillations i.e., 201th amplitude will occur at time $\left(200\frac{T}{2} + \frac{T}{4}\right)$ or $\left(100T + \frac{T}{4}\right)$. Hence

$$a_{201} = a_0 e^{-b(100T + T/4)} \quad \dots(iii)$$

Dividing eq. (iii) by eq. (ii), we get

$$\frac{a_{201}}{a_1} = \frac{e^{-b(100T + T/4)}}{e^{-b(T/4)}} = e^{-100bT}$$

Given, $\frac{a_{201}}{a_1} = \frac{1}{10}$ and $T = 2$ sec.

$$\therefore 10^{-1} = e^{-100 \times b \times 2} \text{ or } \log_e 10 = 200b \text{ or } 2.3 \log_{10} 10 = 200b, \therefore b = \frac{2.3}{200} = 0.0115 \text{sec}^{-1} \quad \dots(i)$$

$$(ii) \text{ Decay modulus} = \frac{1}{b} = \frac{1}{0.0115} = 86.8 \text{ second.} \quad \dots(ii)$$

Ex. 5. The differential equation for a certain system is

$$\frac{d^2x}{dt^2} + 2b\frac{dx}{dt} + \omega_0^2 x = 0$$

If $\omega_0 > b$, find the time in which

- (i) amplitude falls to $1/e$ times the initial value.
- (ii) energy of the system falls to $1/e$ times the initial value,
- (iii) energy falls to $1/e^4$ times the initial value.

Sol. (i) The given differential equation represents the equation of motion of damped harmonic oscillator, whose amplitude of vibration is given by

$$a = a_0 e^{-bt}$$

When amplitude falls to $1/e$ times the initial value a ,

$$a_0 e^{-bt} = a_0/e \text{ or } e^{-bt} = e^{-1} \text{ or } bt = 1, \therefore t = 1/b \text{ sec.}$$

$$(ii) \text{ Energy, } E_t = E_0 e^{-2bt} = E_0/e \therefore e^{-2bt} = e^{-1} \text{ or } 2bt = 1 \text{ or } t = 1/2b \text{ sec.}$$

$$(iii) E_t = E_0 e^{-2bt} = E_0/e^4 \text{ or } 2bt = 4 \text{ or } t = 2/b \text{ sec.}$$

Ex. 6. A damped harmonic oscillator has its first amplitude of 20 cm. It reduces to 2 cm after 100 oscillations, each of period 4.6 sec. Calculate the logarithmic decrement and damping constant. Find the number of oscillations in which the amplitude drops to 50%.

$$\text{Sol. } \lambda = \frac{1}{m} \log_e \frac{a_n}{a_{n+m}} = \frac{1}{100} \log_e \frac{20}{2} = \frac{1}{100} \log_e 10 = \frac{2.3}{100} \log_{10} 10 = 2.3 \times 10^{-2}.$$

$$\lambda = \frac{bT}{2}, \therefore b = \frac{2\lambda}{T} = \frac{2 \times 2.3 \times 10^{-2}}{4.6} = 10^{-2} \text{ sec}^{-1}.$$

$$m = \frac{1}{\lambda} \log_e \frac{a_n}{a_{n+m}} = \frac{1}{2.3 \times 10^{-2}} \times 2.3 \times \log_{10} 2 = 10^2 \times 0.3 = 30.$$

Ex. 7. The frequency of a tuning fork is 300 Hz. If its quality factor Q is 5×10^4 , find the time after which its energy becomes $1/10$ of its initial value. What is the value of relaxation time for the oscillator?

Sol. $E = E_0 e^{-2bt} = E_0 e^{-t/\tau}$. When energy becomes $1/10$ of its initial value,

$$E_0/10 = E_0 e^{-t/\tau} \text{ or } 10 = e^{t/\tau} \text{ or } t/\tau = \log_e 10 = 2.3, \therefore t = 2.3\tau$$

$$\text{But } \tau = \frac{Q}{\omega} = \frac{Q}{2\pi n} = \frac{5 \times 10^4}{2 \times 3.14 \times 300}, \therefore t = \frac{2.3 \times 5 \times 10^4}{2 \times 3.14 \times 300} = 61.1 \text{ sec.}$$

Ex. 8. The amplitude of a second pendulum falls to half initial value in 150 sec. Calculate the Q -factor. (A.N.U. 2000)

Sol. The amplitude of second pendulum in the presence of damping forces will decrease according to the relation :

$$a = a_0 e^{-bt} \text{ or, } \frac{a}{a_0} = e^{-bt}$$

Here, $\frac{a}{a_0} = \frac{1}{2}$ and $t = 150$ sec.

$$\therefore \frac{1}{2} = e^{-150b} \text{ or } e^{150b} = 2 \text{ or } 150b = \log_e 2 = 2.303 \log_{10} 2 \text{ or } 150b = 0.6932$$

$$\therefore b = \frac{0.6932}{150} = 4.62 \times 10^{-3} \text{ sec}^{-1}$$

For second pendulum, $T = 2$ sec, $\therefore \omega = \frac{2\pi}{T} = \frac{2\pi}{2} = \pi = 3.14$ rad/sec.

$$\text{Also } \tau = \frac{1}{2b} = \frac{1}{2 \times 4.62 \times 10^{-3}}$$

$$\text{Hence, } Q = \omega\tau = \frac{3.14}{2 \times 4.62 \times 10^{-3}} = \mathbf{339.8}$$

Ex. 9. The Q -factor of a tuning fork of frequency 512 Hz is given as 8×10^4 . Calculate the time in which the energy is reduced by $\left(\frac{1}{e}\right)^{\text{th}}$ of its initial value. (A.N.U. March 2007)

$$\text{Sol. } Q = \omega\tau, \therefore \tau = \frac{Q}{\omega} = \frac{8 \times 10^4}{2 \times 3.14 \times 512} = 24.88 \text{ sec.}$$

$$\text{Now, } E = E_0 e^{-t/\tau} \text{ or } \frac{E}{E_0} = e^{-t/\tau}, \therefore \frac{1}{e} = e^{-\frac{t}{24.88}} \text{ or } e^{-1} = e^{-\frac{t}{24.88}}$$

$$\therefore \frac{t}{24.88} = 1 \text{ or } t = \mathbf{24.88 \text{ sec}}$$

Ex. 10. The quality factor of a sonometer wire is 2×10^3 . On plucking it makes 240 vibrations per second. Calculate the time in which amplitude decreases to half the initial value. ($\log_{10} 2 = 0.3010$)

$$\text{Sol. Quality factor, } Q = \omega\tau$$

$$\text{Here, } Q = 2 \times 10^3 \text{ and } \omega = 2\pi\nu = 2\pi \times 240 \text{ rad/sec.}$$

$$\therefore 2 \times 10^3 = 2 \times 3.14 \times 240 \times \tau$$

$$\text{whence, } \tau = \frac{2 \times 10^3}{2 \times 3.14 \times 240} = 1.327 \text{ sec.}$$

The amplitude at an instant t is given by

$$a = a_0 e^{-bt} = a_0 e^{-t/2\tau} \quad \left(\because b = \frac{1}{2\tau} \right)$$

$$\text{Given, } \frac{a}{a_0} = \frac{1}{2}. \text{ Hence}$$

$$\frac{a}{a_0} = \frac{1}{2} = e^{-\frac{t}{2\tau}} = e^{-\frac{t}{2 \times 1.327}}$$

$$\therefore e^{t/(2 \times 1.327)} = 2 \text{ or } \frac{t}{2 \times 1.327} = \log_e 2 = 2.3 \times \log_{10} 2$$

$$\text{whence } t = 2 \times 1.327 \times 2.3 \times 0.3010 = \mathbf{1.84 \text{ sec.}}$$

Ex. 11. The Q value of a spring loaded with 0.3 kg is 60. It vibrates with a frequency of 2 Hz. Calculate the force constant and mechanical resistance. (A.N.U. 1997)

Sol. Spring force constant $k = m\omega^2$

Here, $m = 0.3$ kg and $\omega = 2\pi\nu = 2\pi \times 2 = 4\pi$ rad/sec.

$$\therefore k = 0.3 \times (4\pi)^2 = 0.3 \times (4 \times 3.14)^2 = \mathbf{47.37 \text{ N/m.}}$$

Now,
$$Q = \omega\tau = \frac{\omega}{2b} = \frac{\omega m}{\gamma}$$

whence,
$$\gamma = \frac{\omega m}{Q} = \frac{4\pi \times 0.3}{60} \quad (\because Q = 60)$$

$$= \mathbf{0.0628 \text{ kg/sec.}}$$

Ex. 12. A box of 100 gm is attached to one end of a spring whose other end is fixed to a rigid support. When a mass of 900 gm is placed inside the box the system performs 4 vibrations per second and the amplitude falls from 2 cm to 1 cm in 15 seconds. Calculate (i) the force constant, (ii) the relaxation time, (iii) Q of the system and (iv) Q when only 100 gm (empty box) is with the spring. ($\log_e 10 = 2.3$)

Sol. (i) Frequency of vibration of the system, $\nu = 4 \text{ sec}^{-1}$,

$$\therefore \omega_0 = 2\pi\nu = 2 \times 3.14 \times 4 = 25 \text{ rad./sec.} \quad (\text{if the damping is small})$$

But, $\omega_0 = \sqrt{k/m}$ or $c = m\omega_0^2 = 1 \times 25^2 = \mathbf{625 \text{ N-m.}}$

$$(\because m = 100 \text{ gm} + 900 \text{ gm} = 1000 \text{ gm} = 1 \text{ kg})$$

(ii) At any instant t , the amplitude of the system is given by

$$a = a_0 e^{-bt} \text{ or } 0.01 = 0.02 e^{-15b} \text{ or } \frac{1}{2} = e^{-15b} \text{ or } 15b = \log_e 2$$

$$\therefore b = \frac{\log_e 2}{15} = \frac{2.3 \times 0.3}{15} = \mathbf{0.046 \text{ sec}^{-1}}$$

$$\therefore \text{Relaxation time, } \tau = \frac{1}{2b} = \frac{1}{2 \times 0.046} = \mathbf{11 \text{ sec.}}$$

$$(iii) \quad Q = \omega_0 \tau = 25 \times 11 = \mathbf{275.}$$

Damping force $= \gamma dx/dt$, where $\gamma = 2bm = 2 \times 0.046 \times 1$.

(iv) When the box is empty, i.e., the mass is 100 gm = 0.1 kg, the damping force $\gamma dx/dt$ and restoring force kx remain unchanged. Now,

$$\omega_0 = \sqrt{\frac{k}{m'}} = \left(\frac{625}{0.1}\right)^{\frac{1}{2}} = 79 \text{ rad./sec. and } \tau = \frac{1}{2b} = \frac{m'}{\gamma} = \frac{0.1}{2 \times 0.046 \times 1} = 1.1 \text{ sec.}$$

$$\therefore Q = \omega_0 \tau = 79 \times 1.1 = \mathbf{87.}$$

Ex. 13. A condenser of capacity 20 μF is charged and then discharged through an inductance of 10 mH. Calculate the frequency of resulting oscillations. (Resistance of the coil can be neglected).

(A. N.U. 1996)

Sol. Frequency
$$\nu = \frac{1}{2\pi} \sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}}$$

When $R = 0$,
$$\nu = \frac{1}{2\pi\sqrt{LC}}$$

Here, $L = 10 \text{ mH} = 10 \times 10^{-3} \text{ H}$ and $C = 20 \mu\text{F} = 20 \times 10^{-6} \text{ F}$.

$$\therefore \nu = \frac{1}{2 \times 3.14} \sqrt{10 \times 10^{-3} \times 20 \times 10^{-6}} = \mathbf{356.1 \text{ Hz.}}$$

Ex. 14. Calculate the frequency and Q -factor for a circuit with $L = 2 \text{ mH}$, $C = 5 \mu\text{F}$ and $R = 0.2 \Omega$.

(A.N.U. 2002)

Sol. Frequency
$$\nu = \frac{1}{2\pi} \sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}} = \frac{1}{2\pi} \sqrt{\frac{1}{2 \times 10^{-3} \times 5 \times 10^{-6}} - \frac{(0.2)^2}{4 \times (2 \times 10^{-3})^2}}$$

$$= \frac{1}{2\pi\sqrt{2 \times 10^{-3} \times 5 \times 10^{-6}}} = \mathbf{1592 \text{ Hz}}$$

$$\therefore Q = \frac{\omega L}{R} = \frac{2\pi\nu L}{R} = \frac{2 \times 3.14 \times 1592 \times 2 \times 10^{-3}}{0.2} = 99.27$$

Ex. 15. A condenser of 2 microfarad capacity is discharged through a 1 ohm resistance and 2 henry inductance. Show that the system will act as damped harmonic oscillator. Calculate the frequency and quality factor of LC circuit and find the time in which the amplitude of oscillations is reduced to 1%.

Sol.
$$\nu = \frac{1}{2\pi} \sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}}$$

Here, $C = 2 \times 10^{-6}$ farad, $L = 2$ henry and $R = 1$ ohm.

$$\therefore \nu = \frac{1}{2\pi} \sqrt{\frac{1}{2 \times 2 \times 10^{-6}} - \frac{1}{16}} = \frac{10^3}{2\pi \times 2} = 79.5 \text{ Hz.}$$

As resistance is low, so

$$Q = \frac{L\omega}{R} = \frac{2 \times 2\pi}{1} \times \frac{10^3}{2\pi \times 2} = 10^3$$

The amplitude,

$$q = q_0 e^{-R/2L}$$

As amplitude decreases to 1% of its initial value q_0 , we have

$$\frac{q_0}{100} = q_0 e^{-t/4} \text{ or } e^{t/4} = 100 \text{ or } \frac{t}{4} = \log_e 100 = 2.3 \times 2 \text{ or } t = 18.4 \text{ sec.}$$

Ex. 16. In a circuit, $L = 2$ mH and $C = 5$ μ F. If $R = 1$ Ω , 40 Ω , and 100 Ω , discuss the behaviour of the circuit in each case. Find the frequency of oscillation and Q -factor in the relevant cases.

Sol. Here, $L = 2 \times 10^{-3}$ H and $C = 5 \times 10^{-6}$ F

$$\therefore \frac{1}{LC} = \frac{1}{2 \times 10^{-3} \times 5 \times 10^{-6}} = 10^8$$

Case I : $R = 1$ Ω , $\therefore \frac{R^2}{4L^2} = \frac{1}{4 \times (2 \times 10^{-3})^2} = 6.25 \times 10^4$

$$\therefore \frac{1}{LC} > \frac{R^2}{4L^2} \text{ and hence the circuit is oscillatory.}$$

$$\nu = \frac{1}{2\pi} \sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}} = 1.59 \text{ KHz.}$$

and

$$Q = \frac{\omega_0 L}{R} = \frac{2\pi \times 1.59 \times 10^3 \times 2 \times 10^{-3}}{1} = 19.98$$

Case II : $R = 40$ Ω , $\therefore \frac{R^2}{4L^2} = \frac{40 \times 40}{4 \times (2 \times 10^{-3})^2} = 10^8$

Hence $\frac{1}{LC} = \frac{R^2}{4L^2}$ (or $\omega_0^2 = b^2$)

This is the case of **critical damping**.

Case III : $R = 100$ Ω , $\therefore \frac{R^2}{4L^2} = \frac{100^2}{4 \times (2 \times 10^{-3})^2} = 6.25 \times 10^8$

Therefore $\frac{R^2}{4L^2} > \frac{1}{LC}$ (or $b^2 > \omega_0^2$)

This is the case of **overdamping**.

8.6. Forced (or Driven) Harmonic Oscillator

If an external periodic force is applied on a damped harmonic oscillator, then the oscillating system is called **driven** or **forced oscillator** and its oscillations are called **forced (or driven) vibrations**. The applied periodic force is called **driver**. Let the periodic force represented by $F = F_0 \sin pt$, where

F_0 is the maximum value of the force of frequency $p/2\pi$. Hence the differential equation of motion is

$$m \frac{d^2x}{dt^2} = -\gamma \frac{dx}{dt} - kx + F$$

where m is the mass of the oscillating particle, $-\gamma dx/dt$ is the damping force and $-kx$ is the restoring force. Now, the above equation can be written as

$$\frac{d^2x}{dt^2} + \frac{\gamma}{m} \frac{dx}{dt} + \frac{k}{m} x = \frac{F_0}{m} \sin pt$$

or

$$\frac{d^2x}{dt^2} + 2b \frac{dx}{dt} + \omega_0^2 x = f_0 \sin pt \quad \dots(1)$$

where $2b = \gamma/m = 1/\tau$, $\omega_0 = \sqrt{k/m} = 2\pi\nu_0$ is the natural angular frequency of the system in the absence of damping and driven force, and $f_0 = F_0/m$.

Eq. (1) represents the **differential equation of motion for the driven harmonic oscillator**.

If we apply a harmonically oscillating force to a damped harmonic oscillator at a frequency (p), which is, in general, different to the natural frequency of the oscillator, in the beginning we obtain the transient response i.e., in the initial stages, the oscillator tends to vibrate with its natural frequency (ω), while the impressed periodic force tries to impose its own frequency (say p) upon it and hence a sort of tussle takes place between the two. When the transient effects have died away, the oscillator ultimately acquires the steady state and vibrates at the frequency (p) of the applied harmonic force and not at its natural frequency (ω). Such oscillations of the system at the frequency of the applied periodic force are called **forced** or **driven oscillations**. The external applied harmonic force is called the **driver** and the oscillating system itself is called the **driven** or **forced oscillator**. The driving harmonic force delivers periodic impulses to the oscillator so that the loss of energy in doing work against the damping forces is recovered. Consequently in the steady state, the oscillator vibrates with constant amplitude at the frequency of applied harmonic force.

Examples of driven harmonic oscillator :

1. Vibration of a bridge under the rhythm of marching soliders over it.
2. The oscillations of sonometer wire when a vibrating fork is placed on its sounding box.
3. Charge oscillations in an *LCR* circuit fed by alternating voltage.
4. Oscillations of a stretched wire carrying alternating current between the poles of a magnet.

Solution of the Differential Equation of the Driven Harmonic Oscillator–Steady State Solution

Let the steady state solution of the driven harmonic oscillator be

$$x = a \sin (pt - \theta) \quad \dots(2)$$

because the amplitude (a) of the forced oscillations remains constant and the frequency of vibrations is equal to the ($p/2\pi$) of the force. Here θ represents the phase difference between the force and the resultant displacement of the system. Now, from eq. (3), we have

$$\frac{dx}{dt} = pa \cos (pt - \theta) \text{ and } \frac{d^2x}{dt^2} = -p^2 a \sin (pt - \theta)$$

Substituting these values in eq. (1), we get

$$-p^2 a \sin (pt - \theta) + 2bpa \cos (pt - \theta) + \omega_0^2 a \sin (pt - \theta) = f_0 \sin (pt - \theta + \theta) \quad \dots(3)$$

This relation holds for all the values of t . Hence when $(pt - \theta) = 0$ or $2m\pi$, then from eq. (3), we have

$$2bpa = f_0 \sin \theta \quad \dots(4)$$

and when $(pt - \theta) = \pi/2$ or $(2m + 1)\pi/2$, then

$$(\omega_0^2 - p^2)a = f_0 \cos \theta \quad \dots(5)$$

Squaring eqs. (4) and (5) and on adding, we get

$$4b^2 p^2 a^2 + (\omega_0^2 - p^2)^2 a^2 = f_0^2 (\sin^2 \theta + \cos^2 \theta) \text{ or } [4b^2 p^2 + (\omega_0^2 - p^2)^2] a^2 = f_0^2$$

$$\therefore a = \frac{f_0}{\sqrt{(\omega_0^2 - p^2)^2 + 4b^2 p^2}} = \frac{F_0/m}{\sqrt{(\omega_0^2 - p^2)^2 + 4b^2 p^2}} \quad \dots(6)$$

Dividing eq. (4) by eq. (5), we get

$$\tan \theta = \frac{2bp}{\omega_0^2 - p^2} \text{ or } \theta = \tan^{-1} \left(\frac{2bp}{\omega_0^2 - p^2} \right) \quad \dots(7)$$

Substituting these values in eq. (2), we get

$$x = \frac{F_0/m}{\sqrt{(\omega_0^2 - p^2)^2 + 4b^2 p^2}} \sin (pt - \theta) \quad \dots(8)$$

where θ is given by eq. (7)

Thus the amplitude of forced oscillations not only depends on force amplitude (F_0) and mass of the oscillating body (m) but also on the driving frequency (p), natural frequency (ω_0) of the oscillator (in absence of damping) and the damping constant (b). The phase of the displacement lags that of the harmonic force by angle θ which is also depending on p , ω_0 and b . The reader may note that we may sometime use word frequency for angular frequency. Further note that the steady state solution has the frequency of the driving force and its amplitude is constant.

In case of undamped system, $b = 0$ so that from eq. (7), $\theta = 0$ and therefore from eq. (8)

$$x = \frac{F_0/m}{\omega_0^2 - p^2} \sin pt \quad \dots(9)$$

Thus for a simple harmonic oscillator, when a periodic force $F = F_0 \sin pt$ is applied, the driven force and displacement (x) are in the same phase (as $\theta = 0$). Hence we conclude that **phase lag is essentially a consequence of damping**.

Variation of Amplitude (a) and Phase (θ) with the Driving Frequency (p)

We consider below the amplitude (a) and phase lag (θ) of the oscillator at different driving frequencies in the case of low damping.

Case I : Low driving frequency ($p \ll \omega_0$) : When the driving frequency (p) is much less than the natural frequency (ω_0) of the oscillator (in absence of damping), we obtain from eq. (6)

$$a \rightarrow \frac{f_0}{\omega_0^2} = \frac{F_0/m}{k/m} = \frac{F_0}{k} \text{ and } \tan \theta \rightarrow 0 \text{ or } \theta \rightarrow 0 \quad \dots(10)$$

Thus at the low driving frequency, the amplitude does not depend on the mass (m) of the oscillating system or damping but it is controlled by the spring factor (k). In this limit, the periodic force displaces slowly the system against the restoring force of the spring and the displacement is said to be in phase with the driving force.

Case II : Resonant response : Now, if the driving frequency is increased, the value of the amplitude a increases. To find the condition for maximum amplitude (response), we differentiate eq. (6) with respect to p and put $da/dp = 0$ i.e.,

$$\frac{da}{dp} = \left(\frac{F_0}{m} \right) \left(-\frac{1}{2} \right) \frac{2(\omega_0^2 - p^2)(-2p) + 8b^2 p}{[(\omega_0^2 - p^2)^2 + 4b^2 p^2]^{\frac{3}{2}}} = 0 \text{ or } 2b^2 - \omega_0^2 + p^2 = 0$$

$$\text{Hence } p = \sqrt{\omega_0^2 - 2b^2} = p_r \text{ (say)} \quad \dots(11)$$

For this value of p , d^2a/dp^2 is found to be negative. Hence the amplitude a is maximum at $p = p_r$, given by eq. (11).

At $p_r = \sqrt{\omega_0^2 - 2b^2}$, a is maximum, say $a_{\max}$ amplitude, and from eq. (6),

$$a_{\max} = \frac{f_0}{2b\sqrt{\omega_0^2 - b^2}} \quad \dots(12)$$

Thus the amplitude of the oscillating system is maximum at a particular value of the driving frequency (p_r).

*This phenomenon in which the amplitude of the driven oscillator is maximum at a certain driving frequency (p_r) is called **amplitude resonance** and this frequency is called as **amplitude**.*

Resonant frequency : The value of this angular frequency ($p_r = \sqrt{\omega_0^2 - 2b^2}$) is less than the natural frequency of undamped or damped vibrations (ω_0 or $\omega = \sqrt{\omega_0^2 - b^2}$) of the oscillator. However, we are considering low damping so that $p_r \approx \omega_0$ and then

$$a_{\max} = \frac{f_0}{2b\omega_0} = \frac{f_0\tau}{\omega_0} \quad \dots(13)$$

Thus in case of zero damping ($b = 0$), the amplitude will be infinity. But this is not possible as the frictional forces acting on the oscillating system, are never zero.

For low damping, the ratio of the response (amplitude) at resonance to the response at low or zero driving frequency is given by

$$\frac{\text{Response (amplitude) at resonance (for low damping)}}{\text{Response (amplitude) at zero driving frequency}} = \frac{f_0\tau/\omega_0}{f_0/\omega_0} = \omega_0\tau = Q \quad \dots(14)$$

Thus the response at resonance is high for a system with high Q -factor.

At $p = \omega_0$, we obtain from eq. (7)

$$\tan \theta = \infty \text{ or } \theta = \frac{\pi}{2} \text{ and } a = \frac{f_0}{2b\omega_0} \quad \dots(15)$$

Thus at $p = \omega_0$, the amplitude is less than $a_{\max}$ ($= f_0/2b\sqrt{\omega_0^2 - b^2}$).

However for low damping, $p_r \approx \omega_0$ and then amplitude is maximum, given by eq. (15).

Case III : High driving frequency ($p \gg \omega_0$)

: When the driving frequency is much high,

$$\tan \theta \rightarrow -\frac{2b}{p} \rightarrow -0 \text{ or } \theta \rightarrow \pi$$

$$\text{and } a = \frac{f_0}{p^2} \text{ nearly or } a = \frac{F_0}{mp^2} \quad \dots(16)$$

Thus in the high frequency limit, the response decreases rapidly as $1/p^2$ and tends to zero. In this limit, the oscillating mass responds like a free object, shaking rapidly in opposite phase ($\theta = \pi$) to the impressed force of high frequency.

The variation of amplitude a with the frequency p of the driving force is shown in **fig. 8.4(a)**. As the angular frequency is increased from the low or zero frequency limit, the amplitude a increases continuously and becomes maximum at a frequency p_r , which is just before and close to ω_0 . This is **amplitude resonance**. Now, if the driving frequency is further increased, the amplitude (a) decreases regularly and at very high frequencies approaches to very small value. When the damping is greater, the resonant frequency (p_r) is less than ω_0 but in case of low damping the resonant frequency of the force is nearly equal to the natural frequency (ω_0) of the oscillator. It is also clear from these curves that for low damping the height of the peak becomes greater and

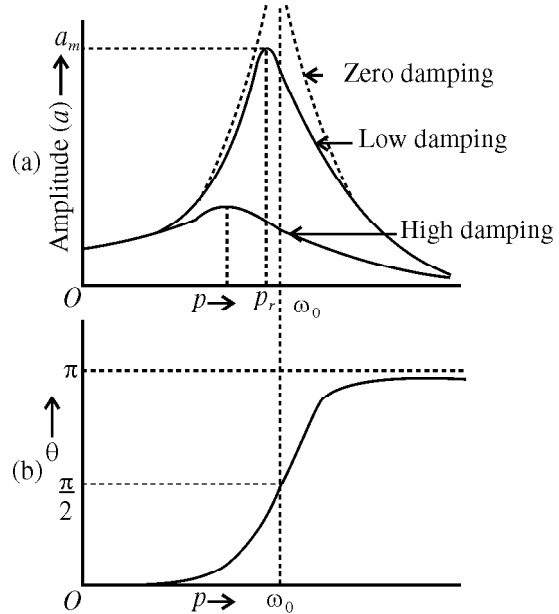


Fig. 8.4 : Forced oscillations : (a) Variation of amplitude (a) of the oscillating system with the driving frequency; (b) Phase lag (θ) of displacement relation to the driving force as a function of driving frequency (p).

when the damping is zero (in the ideal case) the height of the peak (i.e., the maximum amplitude) rises to infinity.

Regarding the phase, we note from our analysis that the displacement x always lags in phase θ relative to the driving force; the value of θ is zero at low frequencies, increases with frequency p and passes $\pi/2$ at $p = \omega_0$ and finally becomes π at high frequencies [fig. 8.4(b)].

Examples of Resonant Vibration

1. When two strings of the same frequency are stretched on the same board and one of them is set into vibrations, it is observed that the other string starts to vibrate.

2. If two pendulums of the same periodic time are suspended from a thin cord, and one of the pendulum is oscillated, the other pendulum begins to oscillate [fig. 8.5].

3. Soldiers passing over a suspension bridge always break steps because if the frequency of their march becomes equal to the natural frequency, then due to resonance the bridge may set into violent vibrations and collapse.

4. In the determination of the speed of sound in air in the laboratory by resonance tube method, a loud sound is heard when the frequency of the vibrating fork becomes equal to the frequency of the air column of the tube.

5. In the driven LCR circuit, the current becomes large when the frequency of the applied alternating voltage becomes equal to the resonant frequency of the circuit.

8.7. Resonance (Amplitude) and Sharpness of Resonance—Half Width of Resonance Curve

In case of driven harmonic oscillator, if we study the variation of amplitude of the oscillating system with the driving frequency, we find that at a particular frequency, called the resonance frequency (p_r), the amplitude is maximum. This phenomenon is called **amplitude resonance**. The amplitude resonance occurs at the frequency, given by

$$p_r = \sqrt{\omega_0^2 - 2b^2} \quad \dots(1)$$

where ω_0 is frequency of undamped oscillator and b is damping constant.

At this frequency, the amplitude of the oscillating system is given by

$$a_{\max} = \frac{f_0}{2b\sqrt{\omega_0^2 - b^2}} = \frac{F_0/m}{2b\sqrt{\omega_0^2 - b^2}} \quad \dots(2)$$

where F_0 is the amplitude of the periodic force $F = F_0 \sin pt$ and m is the mass of the oscillator.

If the damping is small $b \ll \omega_0$, then the resonant frequency is equal to the frequency of the oscillator i.e.,

$$p_r = \omega_0 \text{ and } a_{\max} = \frac{F_0}{2bm\omega_0} = \frac{F_0\tau}{m\omega_0} \quad \dots(3)$$

However at zero or low driving frequency, the amplitude is given by

$$a_0 = \frac{f}{\omega_0^2} = \frac{F_0}{m\omega_0^2} \quad \dots(4)$$

$$\text{Thus, } \frac{a_{\max}}{a_0} = \frac{F_0\tau/m\omega_0}{F_0/m\omega_0^2} = \omega_0\tau = Q$$

$$\text{i.e., } \text{Quality factor, } Q = \frac{a_{\max}}{a_0} = \frac{\text{Amplitude at resonance}}{\text{Amplitude at zero (or low) driven frequency}} \quad \dots(5)$$

Obviously the amplitude (response) will be more for a high Q -oscillating system.

The amplitude of the oscillating system decreases from the maximum value $A_{\max}$ with the change (decrease or increase) of the frequency of the impressed force. In fig 8.4(a), for different values of damping,

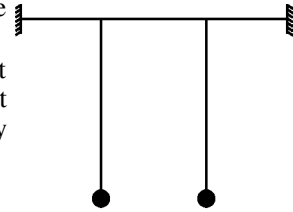


Fig. 8.5.

curves have been drawn between the driven frequency and the amplitude.

Sharpness of resonance : The term **sharpness of resonance** refers to the rate of fall in amplitude with the change of force frequency on each side of the resonant frequency. When the damping is low, the response (amplitude) falls off very rapidly on either side of the resonant frequency and we say that the resonance is **sharp**. On the other hand, for high damping the response falls off very slowly on either side of the resonant frequency and the resonance is said to be **flat**. Thus, the resonance is flat or sharp according to the damping for the oscillating system is large or small. Familiar examples of a flat or sharp resonance are the resonance of an air column and a sonometer wire with a tuning fork respectively. Due to its large damping the air column responds with tuning fork over a fairly wide range in the neighbourhood of resonance. Thus in this case it is actually difficult to get an exact point of resonance and resonance is said to be **flat**. But in case of sonometer wire, the damping is small so that the resonance is **sharp** and hence to have the resonant point, the length of wire is adjusted very precisely.

Half width of amplitude resonance curve : If p_h is the value of angular frequency when the amplitude falls to half the value of p_r at resonance, then Δp , the change in p , is called **half width** of the resonance curve [fig. 8.6], i.e.,

$$\text{Half width, } \Delta p = |p_h - p_r| \quad \dots(6)$$

$$\text{At resonance, } a_{\max} = \frac{f_0}{2b\sqrt{\omega_0^2 - b^2}} = \frac{f_0}{\sqrt{4b^2\omega_0^2 - 4b^4}} \quad \dots(7)$$

At the frequency p_h , the amplitude is $a_{\max}/2$.

$$\therefore \frac{a_{\max}}{2} = \frac{f_0}{\sqrt{(\omega_0^2 - 2b^2 - p_h^2)^2 + 4b^2\omega_0^2 - 4b^4}} \quad [\text{using eq. (6), Art. 8.6}] \quad \dots(8)$$

From (7) and (8), we have

$$(\omega_0^2 - 2b^2 - p_h^2)^2 + 4b^2\omega_0^2 - 4b^4 = 4(4b^2\omega_0^2 - 4b^4)$$

But,

$$p_r^2 = \omega_0^2 - 2b^2$$

$$\therefore (p_r^2 - p_h^2)^2 = 3(4b^2\omega_0^2 - 4b^4) = 3(4b^2p_r^2 + 4b^4) \quad \dots(9)$$

If damping is small, $4b^4$ is negligible and then we have

$$p_r^2 - p_h^2 = \pm\sqrt{3}(2bp_r) \text{ or } p_h^2 = p_r^2 \mp \sqrt{3}(2bp_r)$$

or

$$p_h = p_r \left(1 \mp \sqrt{3} \frac{2b}{p_r} \right)^{1/2} = p_r \left(1 \mp \frac{\sqrt{3}b}{p_r} \right) = p_r \mp \sqrt{3}b$$

$$\therefore \text{Half width, } \Delta p = |p_h - p_r| = \sqrt{3}b \quad \dots(10)$$

This is the expression for half width of the amplitude resonance curve.

$$\text{Full width} = 2\Delta p = 2\sqrt{3}b \quad \dots(11)$$

But,

$$Q = \omega_0\tau = \frac{\omega_0}{2b} \text{ or } b = \frac{\omega_0}{2Q}$$

$$\text{hence Half width, } \Delta p = \sqrt{3} \frac{\omega_0}{2Q} \quad \dots(12)$$

Thus for high Q -oscillating system the half width of the amplitude resonance curve is small.

8.8. Quality Factor and Effect of Damping

The quality factor of an oscillating system with damping is defined as

$$Q = \omega_0\tau = \frac{\omega_0}{2b} \quad \dots(1)$$

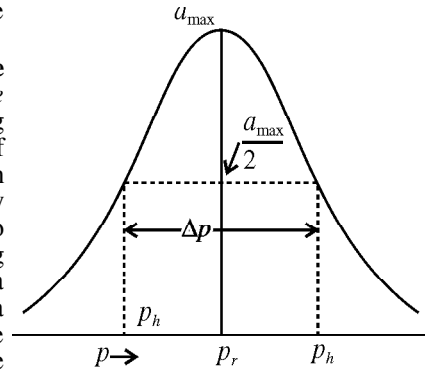


Fig. 8.6. Amplitude resonance curve and half width (Δp)

where τ is relaxation time and b is damping constant.

The quality factor can be interpreted in terms of amplitude of the oscillating system because [eq. (14), Art. 8.6].

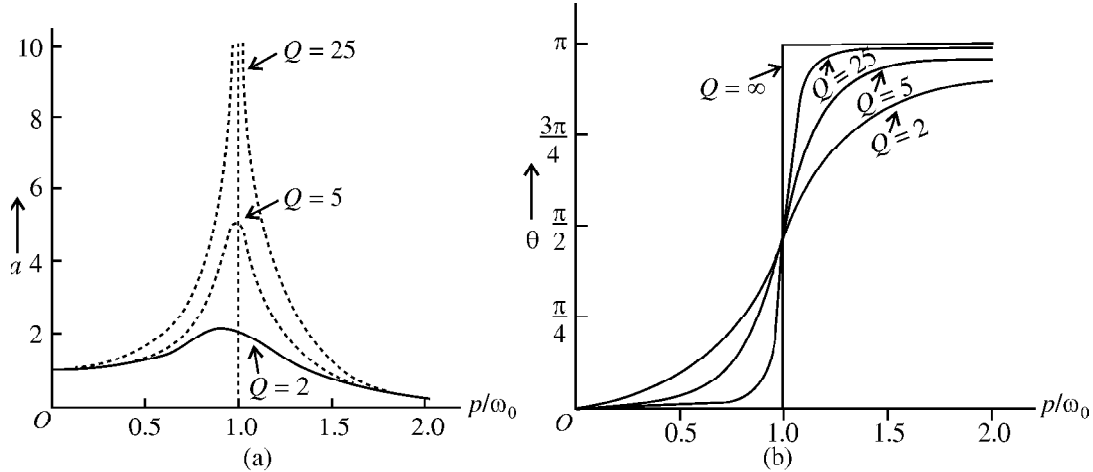


Fig. 8.7 : For different Q (a) Variation of amplitude (a) with driving frequency (ω); (b) Variation of phase lag (θ) with driving frequency (ω).

$$\frac{\text{The amplitude at resonance (for low damping)}}{\text{The at amplitude at low frequency of the driving force } (p \rightarrow 0)} = \frac{f_0 \tau / \omega_0}{f_0 / \omega_0^2} = \omega_0 \tau = Q \quad \dots(2)$$

Thus the Q -factor can be defined as the ratio of amplitude at resonance to the amplitude at low frequencies.

In case of driven oscillator,

$$a = \frac{F_0 / m}{\sqrt{(\omega_0^2 - p^2)^2 + 4b^2 p^2}} \quad \text{and} \quad \theta = \tan^{-1} \left(\frac{2bp}{\omega_0^2 - p^2} \right) \quad \dots(3)$$

Let, $a_0 = \frac{f_0}{\omega_0^2} = \frac{F_0}{m\omega_0^2}$ at low frequencies so that a and θ can be expressed

$$a = a_0 \frac{\omega_0 / p}{[\{(\omega_0 / p) - (p / \omega_0)\}^2 + 1/Q^2]^{\frac{1}{2}}} \quad \text{and} \quad \tan \theta = \frac{1}{(\omega_0 / p) - (p / \omega_0)} \quad \dots(4)$$

Using eq. (4), amplitude (a) and phase lag (θ) versus driving frequency (p) curves are shown in **fig. 8.7** for different values of Q . The amplitude acquires maximum value (a_m) at the amplitude resonant frequency (p_r). Using eq. (11) and (12) of Art. 8.6, we can express a_m and p_r in terms of Q as follows (when damping is considered) :

$$\frac{a_{\max}}{a_0} = \frac{Q}{\sqrt{1 - (1/4Q^2)}} \quad \text{and} \quad \frac{p_r}{\omega_0} = \sqrt{1 - \frac{1}{2Q^2}} \quad \dots(5)$$

For different values of Q , each curve has a maximum value a_m , provided $Q \leq 1/\sqrt{2}$ because clearly from eq. (5), then there are no oscillations. We find that when Q is high (or damping is small), the value of $a_{\max}$ is high and for small Q (or high damping), it is small. The amplitude resonant frequency p_r is less than ω_0 , differing from it more for smaller Q . However we consider, generally, lightly damped or high Q systems, so that $a_{\max}/a_0 = Q$ and $p_r/\omega_0 = 1$ nearly.

8.9. Velocity Resonance

The velocity of the driven oscillator is

$$v = \frac{dx}{dt} = \frac{f_0 p}{\sqrt{(\omega_0^2 - p^2)^2 + 4b^2 p^2}} \cos(pt - \theta) \quad [\text{from eq. (8), Art. 11.2}]$$

or
$$v = v_0 \sin(pt - \theta + \pi/2) \quad \dots(1)$$

where
$$v_0 = \frac{f_0 p}{\sqrt{(\omega_0^2 - p^2)^2 + 4b^2 p^2}} = \frac{f_0}{\sqrt{(p - \omega_0^2/p)^2 + 4b^2}} \text{ is the velocity amplitude}$$

and
$$\theta = \tan^{-1} \left(\frac{2bp}{\omega_0^2 - p^2} \right).$$

Thus the velocity amplitude v_0 varies with the driven frequency p [fig.8.8]. When $p = 0$, $v_0 = 0$ and when $p = \omega_0$, v_0 attains the maximum value. Hence at the frequency $p = \omega_0 = \sqrt{k/m}$ of the impressed force, the velocity has the maximum value and we call it **velocity resonance**. At velocity resonance,

$$\theta = \tan^{-1} \left(\frac{2bp}{0} \right) = \frac{\pi}{2},$$

$\therefore$ Velocity phase constant $= -\theta + \pi/2 = -\pi/2 + \pi/2 = 0$.

This means that at velocity resonance, the velocity of the oscillator is in phase with the applied force. This is the most favourable situation for transfer of energy from the applied force to the oscillator, because the rate of work done on the oscillator by the impressed force is Fv , which is always positive for F and v in phase.

When $p \geq \omega_0$, the velocity amplitude is smaller than at $p = \omega_0$.

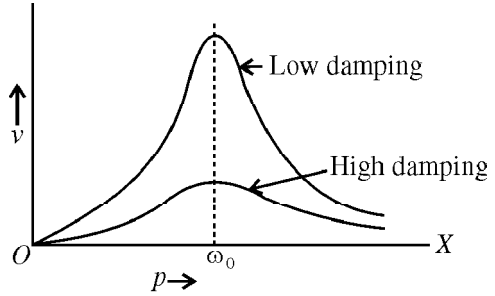


Fig. 8.8. Variation of velocity amplitude with driven frequency (p)

8.10. Power Absorption and Resonance Width

An oscillating physical system loses energy in doing work against damping forces. In a driven harmonic oscillator, this loss of energy is supplied by the applied harmonic force and hence the oscillations of the system are maintained at constant amplitude. One will be interested in knowing the rate at which the energy must be supplied to maintain the vibrations. The rate of supplying the energy or power absorption by the oscillating system at any instant t is given by

$$P = \frac{dW}{dt} = F \frac{dx}{dt} = Fv \quad \dots(1)$$

Now, $F = mf_0 \sin pt$ and $v = \frac{dx}{dt} = \frac{pf_0 \cos(pt - \theta)}{\sqrt{(\omega_0^2 - p^2)^2 + 4b^2 p^2}}$

Hence at any time the energy absorbed by the oscillating system is

$$P = F \frac{dx}{dt} = mf_0 \sin pt \cdot \frac{pf_0 \cos(pt - \theta)}{\sqrt{(\omega_0^2 - p^2)^2 + 4b^2 p^2}} \quad \dots(2)$$

Hence the average power absorbed is

$$P_{av} = \left(F \frac{dx}{dt} \right)_{av} = \frac{mf_0^2 p}{\sqrt{(\omega_0^2 - p^2)^2 + 4b^2 p^2}} [\sin pt \cos(pt - \theta)]_{av} \quad \dots(3)$$

Now the average of $\sin pt \cos(pt - \theta)$ for the period T

$$\begin{aligned} &= \frac{1}{T} \int_0^T \sin pt \cos(pt - \theta) dt = \frac{1}{T} \int_0^T \frac{1}{2} \{ \sin(2pt - \theta) + \sin \theta \} dt = \frac{\sin \theta}{2} \\ &= \frac{1}{2} \frac{2bp}{\sqrt{(\omega_0^2 - p^2)^2 + 4b^2 p^2}} \quad [\text{from eq. (7); Art 8.6}] \end{aligned}$$

$$\therefore P_{av} = \frac{1}{2} m f_0^2 \frac{2b p^2}{(\omega_0^2 - p^2)^2 + 4b^2 p^2} \quad \dots(4)$$

Replacing $\frac{f_0^2 p^2}{(\omega_0^2 - p^2)^2 + 4b^2 p^2}$ by v_0^2 , we have

$$P_{av} = m b v_0^2 = \frac{m v_0^2}{2\tau} \text{ or } \frac{\gamma v_0^2}{2} \quad \dots(5)$$

We see from eq. (4) that when $p = \omega_0$, the absorbed average power is maximum, say P_m , given by

$$P_m = \frac{1}{4} \frac{m f_0^2}{b} = \frac{1}{2} m f_0^2 \tau. \quad \dots(6)$$

Thus the power absorbed is maximum at the frequency of velocity resonance and not at the frequency of amplitude resonance. A graph between P_{av} versus p for a particular value of damping constant b (or $Q = \omega_0 \tau = \omega_0/2b$) is shown in **fig.8.9**.

When steady state is attained, the average power dissipated* should be the same as the average power supplied by the driving force.

Resonance width : If p_h is the value of the driving frequency when the average power falls to half the value at resonance, the half-width of the power resonance curve is given by

$$\Delta p = |p_h - \omega_0| \quad \dots(7)$$

The power drops to its half the maximum value, when

$$\frac{1}{2} P_m = (P_{av})_{p_h}$$

$$\frac{1}{2} \frac{m f_0^2}{4b} = \frac{1}{2} m f_0^2 \frac{2b p_h^2}{(\omega_0^2 - p_h^2)^2 + 4b^2 p_h^2}$$

$$\text{or } 8b^2 p_h^2 = (\omega_0^2 - p_h^2)^2 + 4b^2 p_h^2$$

$$\text{or } \omega_0^2 - p_h^2 = \pm 2b p_h$$

$$\text{or } \omega_0 - p_h = \pm \frac{2b p_h}{\omega_0 + p_h} = \pm \frac{2b}{1 + \omega_0/p_h} \approx \pm b \text{ or } \frac{1}{2\tau} \quad [\text{taking } \omega_0/p_h = 1 \text{ approx.}]$$

$\therefore$ **Half width** of the power versus frequency curve

$$\Delta p = |\omega_0 - p_h| = \frac{1}{2\tau} = \frac{\omega_0}{2Q} \quad [\because Q = \omega_0 \tau] \quad \dots(8)$$

The quantity $2\Delta p$ is called the **band-width** or **full-width** at the half maximum power and is given by

$$2\Delta p = \frac{\omega_0}{Q} = \frac{1}{\tau} \text{ or } 2b \quad \dots(9)$$

Thus the resonance full-width for forced oscillations is equal to the inverse of decay time (τ) for free oscillations.

$$\text{Also, } Q = \omega_0 \tau = \frac{\omega_0}{2\Delta p} = \frac{\text{Frequency at resonance}}{\text{Full width at half maximum power}} \quad \dots(10)$$

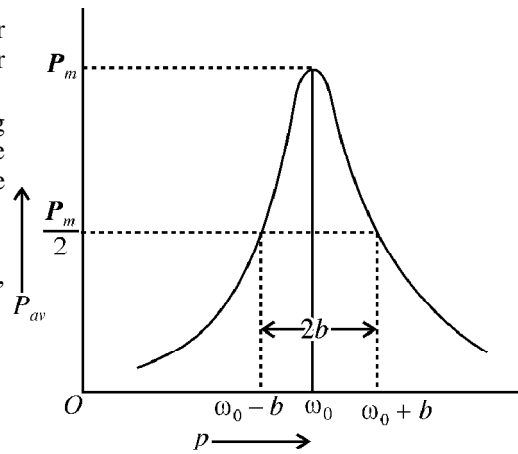


Fig. 8.9 : Average power absorbed (P_{av}) as a function of driving frequency : Full width of the resonance curve $2\Delta p = 2b$ or $1/\tau$; $Q = \omega_0 \tau$.

* Instantaneous power dissipated = (force) $dx/dt = 2bm(dx/dt)^2 = 2bm \cdot \frac{f_0^2 p^2}{(\omega_0^2 - p^2)^2 + 4b^2 p^2} \cos^2(pt - \theta)$

$$\text{Average power dissipated} = \frac{1}{2} m f_0^2 \frac{2b p^2}{(\omega_0^2 - p^2)^2 + 4b^2 p^2} \quad [\because \{\cos^2(pt - \theta)\}_{av} = \frac{1}{2}]$$

Alternatively, $Q = 2\pi \frac{\text{Average energy stored}}{\text{Average energy dissipated per cycle}}$... (11)

At any instant, stored energy of oscillator = K.E + P.E.

$$\begin{aligned} \text{Time averaged stored energy, } E &= \langle \text{K.E.} \rangle + \langle \text{P.E.} \rangle = \frac{1}{2} \left\langle m \left(\frac{dx}{dt} \right)^2 \right\rangle + \left\langle \frac{1}{2} kx^2 \right\rangle \\ &= \frac{1}{2} mp^2 a^2 \langle \cos^2(pt - \theta) \rangle + \frac{1}{2} m\omega_0^2 a^2 \langle \sin^2(pt - \theta) \rangle \\ &= \frac{1}{4} mp^2 a^2 + \frac{1}{4} m\omega_0^2 a^2 = \frac{1}{4} m(p^2 + \omega_0^2) a^2 \end{aligned} \quad \dots (12)$$

At $p \approx \omega_0$, $E = \frac{1}{2} m\omega_0^2 a^2$

Average energy dissipated per sec. = $\left\langle \gamma \frac{dx}{dt} \right\rangle = \gamma p^2 a^2 \langle \cos^2(pt - \theta) \rangle = \frac{m}{2\tau} p^2 a^2$... (13)

Average energy dissipated per cycle (at $p \approx \omega_0$) = $\frac{m\omega_0^2 a^2}{2\tau} \cdot \frac{2\pi}{\omega_0}$... (14)

$$\therefore Q = 2\pi \frac{\frac{1}{2} m\omega_0^2 a^2}{\frac{m\omega_0^2 a^2}{2\tau} \cdot \frac{2\pi}{\omega_0}} = \omega_0 \tau = \frac{\omega_0}{2\Delta p} \quad \dots (15)$$

Sharpness of resonance and quality factor : We may use the term, sharpness of resonance, to refer the rapid rate of fall in power with the frequency on each side of the resonant frequency (ω_0). Thus for small band-width ($2\Delta p$) or high Q system, the resonance, is said to be **sharp**. For low Q system, the band-width will be more and the power falls off slowly. The resonance in such a case is said to be **flat**. Thus Q is a measure of the sharpness of the resonance.

8.11. Driven LCR Circuit

Driven LCR circuit is an example of the driven harmonic oscillator in which the oscillations are sustained by an alternating electromotive force. Consider a circuit containing the inductance L , resistance R and capacity C in series. An alternating E.M.F. $E = E_0 \sin pt$ is applied to this circuit, where E_0 is the peak value [fig.8.10]. If at any instant t , q is the charge on the condenser C , then the potential difference across its plates is q/C . If the current in the circuit at this instant is i , then the self-induced E.M.F. in the inductance is $-L di/dt$. According to Ohm's law, the potential difference between the ends of the resistance is RI . Hence

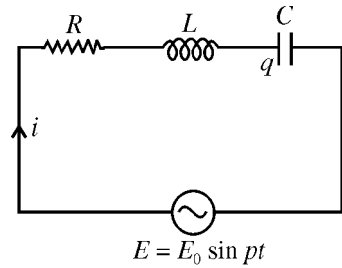


Fig. 8.10.

Total potential difference = $Ri + \frac{q}{C}$ = total E.M.F. = $E - L di/dt$, i.e. $L \frac{di}{dt} + Ri + \frac{q}{C} = E$... (1)

or $\frac{d^2 q}{dt^2} + \frac{R}{L} \frac{dq}{dt} + \frac{q}{LC} = \frac{E_0}{L} \sin pt$. $\left[\because i = \frac{dq}{dt} \right]$... (2)

This equation is similar to the equation of driven harmonic oscillator having R/L in place of $2b$, $1/LC$ for ω_0^2 and E_0/L for f_0 . Hence the steady state solution of the above equation is

$$q = \frac{E_0/L}{\sqrt{\left(\frac{1}{LC} - p^2\right)^2 + \left(\frac{pR}{L}\right)^2}} \sin(pt - \theta) \quad \text{with } \theta = \tan^{-1} \frac{pR/L}{\frac{1}{LC} - p^2} \quad \dots (3)$$

Differentiating q in eq. (3) with respect to t , we can find the current $i = dq/dt$ in the circuit. Practically, we require the knowledge of the current in the circuit*.

$$i = \frac{E_0 p / L}{\sqrt{\left(\frac{1}{LC} - pL\right)^2 + R^2}} \cos(pt - \theta) \text{ or } i = \frac{E_0}{\sqrt{R^2 + \left(pL - \frac{1}{pC}\right)^2}} \sin(pt - \phi) \quad \dots(4a)$$

where
$$\phi = \theta - \frac{\pi}{2} = \tan^{-1} \frac{pL - 1/pC}{R} \left[\because \tan \phi = -\cot \theta = \frac{p^2 - 1/LC}{pR/L} = \frac{pL - 1/pC}{R} \right] \dots(4b)$$

which represents that the current i lags is phase ϕ from the E.M.F. E .

The quantity $\sqrt{R^2 + (pL - 1/pC)^2}$ is called the **impedance** of the circuit and is represented by Z . Its unit is Ohm. Hence

$$i = \frac{E_0}{Z} \sin(pt - \phi) \text{ or } i = i_0 \sin(pt - \phi) \quad \dots(5)$$

where $i_0 = E_0/Z$ is the peak value of the current.

The quantity $(pL - 1/pC)$ is called the **reactance** of the circuit and is denoted by X . Therefore

$$\text{Impedance, } Z = \sqrt{R^2 + X^2} = \sqrt{(\text{Resistance})^2 + (\text{Reactance})^2} \quad \dots(6)$$

Now, the current amplitude $i_0 = \frac{E_0}{Z} = \frac{E_0}{\sqrt{R^2 + X^2}}$ is smaller for the larger value of Z . The expression for

$$Z = \sqrt{R^2 + (pL - 1/pC)^2} \quad \dots(7)$$

depends upon the frequency (p) of the applied alternating E.M.F. We will consider below the important cases, when the driven frequency is increased. We denote $\omega_0 = 1/\sqrt{LC}$ as the undamped frequency of the circuit.

Case I : $p < \omega_0$, the term $-1/pC$ in impedance is **dominant** and is very large and $Z = \sqrt{R^2 + (pL - 1/pC)^2} \approx 1/pC$.

Therefore, the current amplitude is (when $p \rightarrow 0$)

$$i_0 = \frac{E_0}{1/pC} \text{ and } \phi = -\frac{\pi}{2} \quad \dots(8)$$

Thus, for low values of p , the current amplitude has a small value. On increasing the driven frequency (p) gradually, the magnitude of reactance $(pL - 1/pC)$ and hence the impedance Z decreases resulting in the increase of the current amplitude.

Case II : When $p = \omega_0 = 1/\sqrt{LC}$, the reactance

$$X = pL - \frac{1}{pC} = \frac{L}{\sqrt{LC}} - \frac{\sqrt{LC}}{C} = \sqrt{\frac{L}{C}} - \sqrt{\frac{L}{C}} = 0 \quad \dots(9)$$

and so the impedance Z is minimum and is equal to R . In consequence the current amplitude $i_0 = E_0/Z = E_0/R$ becomes maximum. This is the case of **electrical resonance**. If $R = 0$, then the current amplitude at resonance becomes infinite. The **resonant frequency** of the circuit is given by

$$\nu_r = \frac{p}{2\pi} = \frac{F_0}{m} = \frac{1}{2\pi\sqrt{LC}} \quad \dots(10)$$

* **A shorter procedure for solution of eq. (1) :** Let the solution be $i = i_0 \sin(pt - \phi)$. Hence $q = \int i dt = \frac{i_0}{p} \cos(pt - \phi)$

Substituting the values of di/dt , i and q in eq. (1) and proceeding as before, the values of i and ϕ can be calculated.

In such a case, $\phi = 0$ and $i = \frac{E_0}{R} \sin pt$...(11)

Thus at resonance the current and the applied E.M.F. are in the same phase.

Case III : When $p \gg \omega_0$, the term $L\omega$ in the expression for Z becomes dominant and very large and then.

$$i_0 = E_0/pL \text{ and } \phi = \pi/2 \quad \dots(12)$$

The variation of i_0 with p is shown in **fig. 8.11** for different values of R . The smaller value of R , the sharper is the resonance. The variation of phase difference ϕ between the current and the E.M.F. is shown in **fig. 8.12**. The current lags in phase for $p < \omega_0$ and exceeds in phase for $p > \omega_0$.

Power in A.C. circuits : The power of an electrical circuit is defined as the rate of its doing work and at any time it is measured by the product of the current and the potential difference. If at any instant i is the current and E , the E.M.F., then the instantaneous power delivered by the source to the circuit is

$$P = Ei. \quad \dots(13)$$

Now, let us consider the power relations by taking LCR circuit. The instantaneous power delivered by the source to the circuit

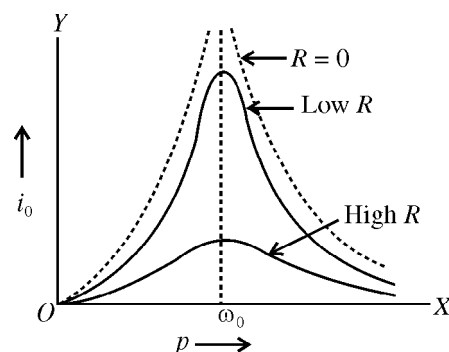


Fig. 8.11.

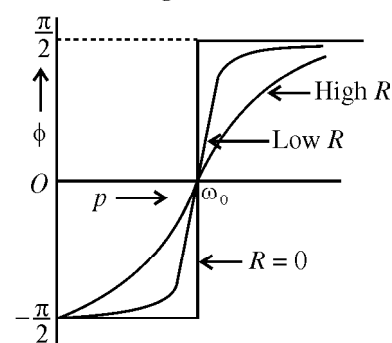


Fig. 8.12.

$$P = Ei = E_0 \sin pt \cdot i_0 \sin (pt - \phi) = \frac{E_0 i_0}{2} [\cos \phi - \cos (2pt - \phi)] \quad \dots(14)$$

where, the average value of $\cos (2pt - \phi)$ for one cycle is zero, hence the average power of an A.C. circuit is given by

$$P_{av} = \frac{E_0 i_0}{2} \cos \phi = \frac{E_0}{\sqrt{2}} \cdot \frac{i_0}{\sqrt{2}} \cos \phi = E_{rms} i_{rms} \cos \phi \quad \dots(15)$$

where, $E_{rms} = \frac{E_0}{\sqrt{2}}$ and $i_{rms} = \frac{i_0}{\sqrt{2}}$.

This $\cos \phi$ in eq. (15) is called the **power factor** of the circuit and it will be given by

$$\cos \phi = \frac{R}{\sqrt{R^2 + (pL - 1/pC)^2}} = \frac{R}{Z} \quad [\text{From eq. (4b)}]$$

At resonance, $\phi = 0$ so the average power delivered to the circuit is maximum and then

$$(P_{av})_{max} = (P_{av})_{res} = \frac{1}{2} E_0 i_0 = \frac{1}{2} \frac{E_0^2}{R} = \frac{1}{2} i_0^2 R \quad \dots(16)$$

But, $i_{rms} = \frac{i_0}{\sqrt{2}} = \frac{1}{\sqrt{2}} \cdot \frac{E_0}{Z} = \frac{E_{rms}}{Z}$

$$\therefore \text{Power in A.C. circuit} = (i_{rms} Z) \cdot i_{rms} R/Z = i_{rms}^2 R \quad \dots(17)$$

Thus in an A.C. circuit, the power appears only in the form of heat.

Analogous to mechanical oscillator, we can show easily that

$$Q = \omega_0 \tau = \frac{L\omega_0}{R} = \frac{\omega_0}{2\Delta p} = \frac{\text{Frequency at resonance}}{\text{Full width at half maximum power}} \quad \dots(18)$$

Q-Factor : Similar to mechanical oscillator, the quality factor is given by

$$Q = \omega_0 \tau = \frac{\omega_0 L}{R}, \text{ where } \omega_0 = \frac{1}{\sqrt{LC}} \quad \dots(19)$$

Full width or the band-width of the power resonance curve is given by

$$2\Delta p = \frac{1}{\tau} = \frac{R}{L} = \frac{\omega_0}{Q} \quad \dots(20)$$

$$\text{Thus } Q = \frac{\omega_0}{2\Delta\omega} = \frac{\text{Frequency at resonance}}{\text{Full width at half maximum power}} \quad \dots(21)$$

The full width $2\Delta\omega$ is small for high Q . Circuits with Q of the order of 10^2 to 10^4 are quite common. The resonant circuit in a radio receiver with Q of the order of 10^2 to 10^3 may select a particular station and discard the others.

8.12. Mechanical and Electrical Analogues

The differential equation for forced oscillations if a mechanical oscillator is

$$m \frac{d^2 x}{dt^2} + \gamma \frac{dx}{dt} + kx = F_0 \sin pt \quad \dots(1)$$

The differential equation for a LCR circuit, driven by an alternating electromotive force is

$$L \frac{d^2 q}{dt^2} + R \frac{dq}{dt} + \frac{q}{C} = E_0 \sin pt \quad \dots(2)$$

Equation (1) and (2) are similar and hence allow analogies between the mechanical and electrical quantities. A look at the equations shows that the mass is analogous to the self-inductance L , the frictional force per unit velocity or mechanical resistance γ to the electrical resistance R , the force constant to the inverse of capacitance C , the force $F (= F_0 \sin pt)$ to the voltage $E (= E_0 \sin pt)$, the displacement x to the charge q and the velocity dx/dt to the current dq/dt .

The velocity (dx/dt) of force oscillator in the steady state is given by

$$v = \frac{f_0}{\sqrt{(p - \omega_0^2 / p)^2 + 4b^2}} \cos(pt - \theta)$$

where

$$f_0 = \frac{F_0}{m}, \quad \omega_0^2 = \frac{k}{m} \quad \text{and} \quad b = \frac{\gamma}{2m}$$

Hence

$$v = \frac{F_0 / m}{\sqrt{(p - k / mp)^2 + (\gamma / m)^2}} \cos(pt - \theta)$$

or

$$v = \frac{F_0}{\sqrt{(pm - k / p)^2 + \gamma^2}} \cos(pt - \theta) \quad \dots(3)$$

The current (dq/dt) in a driven LCR circuit is given by

$$i = \frac{E_0}{\sqrt{(pL - 1 / pC)^2 + R^2}} \cos(pt - \theta) \quad \dots(4)$$

In an AC electrical circuit the quantity $\sqrt{(pL - 1 / pC)^2 + R^2}$ is called impedance and is denoted by Z i.e.,

$$Z = \sqrt{(pL - 1 / pC)^2 + R^2} \quad \dots(5)$$

Equation (3) and (4) are similar, hence the impedance of mechanical oscillator is denoted by Z_m , given by

$$Z_m = \sqrt{(pm - k/p)^2 + \gamma^2} \quad \dots(6)$$

The mechanical impedance is often expressed in mechanical ohm similar to ohm for electrical impedance.

For ready reference, mechanical quantities and the corresponding electrical analogues are given in the following table.

Table : Analogous Mechanical and Electrical Quantities

Mechanical Quantities	Electrical Quantities
1. Mass (m)	1. Inductance (L)
2. Force (F)	2. Voltage (E)
3. Mechanical resistance (γ)	3. Electrical resistance (R)
4. Displacement (x)	4. Charge (q)
5. Velocity ($v = dx/dt$)	5. Current ($i = dq/dt$)
6. Compliance ($1/k$)*	6. Capacitance (C)
7. Mechanical impedance (Z_m)	7. Electrical impedance (Z)

8.13. Free, Damped and Forced Vibrations : A Comparison

Free Vibrations (Undamped harmonic oscillator)	Damped Free Vibrations (Damped harmonic oscillator)	Forced Vibrations (Driven harmonic oscillator)
1. The vibrations of a system free from external force are called free vibrations.	1. The free vibrations of a system in the presence of frictional forces are called damped free vibrations.	1. The vibrations of a system under the action of an external periodic force are called forced vibrations.
2. In case of free vibrations, the energy of vibrating system remains constant.	2. In case of damped free vibrations, the energy of the vibrating system decreases with time because of doing work against damping forces.	2. In case of forced vibrations, the applied harmonic force supplies the energy to the oscillator periodically to compensate for the loss of energy due to damping forces.
3. Such vibrations have constant amplitude and continue indefinitely.	3. The amplitude of such vibrations decreases with time and ultimately the vibrating system comes to rest.	3. Such vibrations with constant amplitude continue as long as the external periodic force acts.
4. The differential equation for undamped harmonic oscillator is $\frac{d^2x}{dt^2} + \omega_0^2 x = 0$ where $\omega_0 = \sqrt{k/m}$	4. The differential equation for damped harmonic oscillator is $\frac{d^2x}{dt^2} + 2b\frac{dx}{dt} + \omega_0^2 x = 0$ where b is damping constant	4. The differential equation for forced oscillator is $\frac{d^2x}{dt^2} + 2b\frac{dx}{dt} + \omega_0^2 x = f_0 \sin pt$ where p is angular frequency of the external periodic force.
5. Period $T = \frac{2\pi}{\omega_0}$	5. Period $T = \frac{2\pi}{\sqrt{\omega_0^2 - b^2}}$	5. The system vibrates with the frequency p of the applied harmonic force in the steady state.

8.14. Transient State and Steady State

In this chapter we have so far studied the steady state behaviour of an oscillating system, acted by a harmonic driving force. In fact, in a practical situation, the driving force is first switched on at some instant on a damped harmonic oscillator. The frequency (p) of driving force is in general different to the natural frequency (ω) of the damped harmonic oscillator. In the beginning, the oscillator tries to vibrate with its natural frequency (ω) and the applied (driven) periodic force tends to impose its own frequency (p) upon it. Hence for some time, the oscillating system is in the **transient state** and the free oscillations

* Inverse of force constant is called compliance.

of frequency ω die out and ultimately the system starts oscillating with a constant amplitude with the frequency (p) of the driven harmonic force. This is called the **steady state**. In the transient state, the oscillations of the system take place under the superposition of two oscillations of frequencies $\omega = \sqrt{\omega_0^2 - b^2}$ and p .

We consider the system to be at rest with no initial displacement at $t = 0$, the time when the driving force is switched on. We are interested in knowing the nature of the transient motion just after the instant $t = 0$.

Mathematically, the problem is to find the general solution of the differential equation of the driven harmonic oscillator :

$$\frac{d^2x}{dt^2} + 2b\frac{dx}{dt} + \omega_0^2x = f_0 \sin pt \quad [f_0 = F_0/m] \quad \dots(1)$$

The steady state solution is not the general solution. According to the theory of differential equations, the general solution $x(t)$ of eq. (1) is the superposition of two solutions $x_1(t)$ and $x_2(t)$, given by

(1) $x_1(t) = a \sin (\omega t - \theta)$, which is the steady state solution (particular solution) of eq. (1).

(2) $x_2(t) = a_0 e^{-bt} \sin (\omega t + \phi)$, which is the general solution of the differential equation,

$$\frac{d^2x}{dt^2} + 2b\frac{dx}{dt} + \omega_0^2x = 0.$$

Thus the general solution of eq. (1) is

$$x = x_1 + x_2 \text{ or } x = a \sin (pt - \theta) + a_0 e^{-bt} \sin (\omega t + \phi) \quad \dots(2)$$

We impose the initial conditions that at $t = 0$, $x = 0$ and $dx/dt = 0$ on the general solution x , given by eq. (2) and its time derivative dx/dt . Now,

$$\frac{dx}{dt} = pa \cos (pt - \theta) + a_0 e^{-bt} [-b \sin (\omega t + \phi) + \omega \cos (\omega t + \phi)] \quad \dots(3)$$

Imposing the, initial conditions on eqs. (2) and (3), we obtain

$$0 = -a \sin \theta + a_0 \sin \phi \text{ and } 0 = pa \cos \theta - a_0 b \sin \phi + a_0 \omega \cos \phi \quad \dots(4)$$

We are interested here in the case in which the *damping is weak and the driving frequency is close to the resonant frequency* i.e., $b \ll \omega$ and $p \approx \omega$. In such a case, the conditions (4) become

$$a_0 \sin \phi = a \sin \theta \text{ and } a_0 \cos \phi = -a \cos \theta \quad \dots(5)$$

$$\text{Dividing } \tan \phi = -\tan \theta \text{ or } \phi = \pi - \theta. \text{ So that } a_0 = a \quad \dots(6)$$

Hence the general solution is

$$x = a \sin (pt - \theta) - a_0 e^{-bt} \sin (\omega t - \theta) \quad \dots(7)$$

Now, first let us consider the case of zero damping ($b = 0$) or ideal oscillator, then from eq. (9) [Art. 8.6],

$$a = \frac{F_0}{m(\omega_0^2 - p^2)}, \theta = 0 \text{ and } \omega = \omega_0 \quad \dots(8)$$

Hence eq. (7) takes the form

$$x = \frac{F_0}{m(\omega_0^2 - p^2)} (\sin pt - \sin \omega_0 t) \text{ or } x = \frac{2F_0}{m(\omega_0^2 - p^2)} \cos \frac{p - \omega_0}{2} t \sin \frac{p + \omega_0}{2} t \quad \dots(9)$$

$$\text{or } x = a_m(t) \sin \omega_a t \quad \dots(10)$$

where
$$a_m(t) = \frac{2F_0}{m(\omega_0^2 - p^2)} \cos \omega_m t \text{ with } \omega_m = \frac{p - \omega_0}{2} \text{ and } \omega_a = \frac{p + \omega_0}{2}$$

This is the case when we were studying the superposition of two simple harmonic motions of two slightly different frequencies and phenomenon of modulation and beats.

Thus the beats occur between the natural and driven oscillations at a frequency $\omega_0 - p$ and in case of zero damping, phenomenon of beat goes on continuously.

Next, we consider the case, when the damping is present i.e., the case of damped harmonic oscillator alongwith the applied harmonic force we see from eq. (7) that at $t = 0$, the displacement is a superposition of two harmonic terms with the driving frequency (p) close to the natural frequency (ω). Thus the initial motion consists of beats between the free and driven oscillations. If there were no damping, these beats would continue forever, as we have seen above. However due to damping the second term of eq. (7) diminishes with time, resulting in weak beats. These are called transient beats and the oscillator is said to be in the **transient state**. After a sufficiently long time compared to the decay time τ , the **steady state solution** becomes dominant [i.e. first term of eq. (7)] and the oscillator executes vibrations of constant amplitude (with no beats) with the frequency of driven harmonic force [fig. 8.13]. In the beginning, due to damping, the oscillator adjusts gradually its phase relative to the driving harmonic force and the relative phase is stabilized in the steady state.

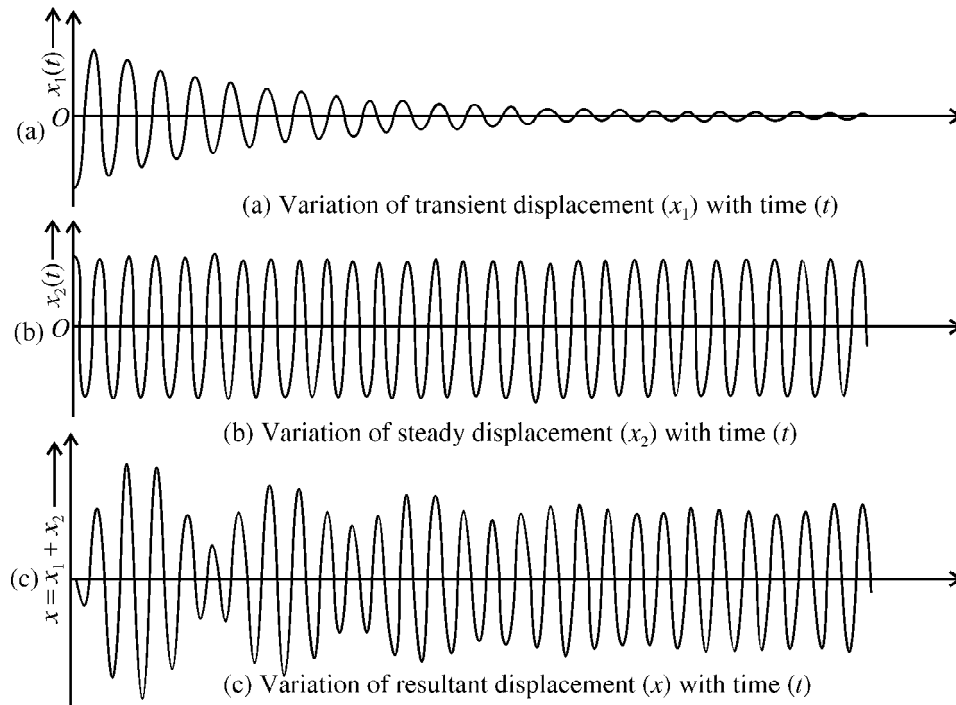


Fig. 11.9 : Variation of displacement of a driven oscillator (for low damping) with time from starting at $t = 0$.

Ex. 1. The amplitude of forced vibrations is given by

$$a = \frac{f}{\sqrt{(\omega_0^2 - p^2)^2 + p^2 / \tau^2}}.$$

If $Q = 40$, calculate the value of $A/A_{\max}$ for $p/\omega_0 = 0.99$.

Sol. $Q = \omega_0 \tau = 50$ or $\tau = 50/\omega_0$

and

$$a = \frac{f}{\sqrt{(\omega_0^2 - p^2)^2 + p^2/\tau^2}} = \frac{f}{\omega_0^2 \sqrt{\left(1 - \frac{p^2}{\omega_0^2}\right)^2 + \frac{1}{2500} \frac{p^2}{\omega_0^2}}}$$

For small damping, at resonance $a_{\max} = \frac{f}{\omega_0^2 \times 1/50} = \frac{50f}{\omega_0^2}$

When $p/\omega_0 = 0.99$, $\frac{a}{a_{\max}} = \frac{1}{50 \sqrt{(1-0.99)^2 + \frac{1}{2500}(0.99)^2}} = 0.71$.

Ex. 2. In case of a forced harmonic oscillator, the amplitude of vibrations increases from 0.02 mm at very low frequencies to a value 5 mm at the frequency 100 Hz. Find (i) Q -factor of the system, (ii) damping constant b and relaxation time and (iii) half width of amplitude resonance curve.

Sol. (i) $Q = \omega_0 \tau = \frac{\omega_0}{2b} = \frac{a_{\max}}{a \text{ at zero or very low frequency}} = \frac{5}{0.02} = 250$.

(ii) Resonant frequency = 100 Hz, $\therefore \omega_0 = 2\pi \times 100$

Hence $\tau = \frac{Q}{\omega_0} = \frac{250}{2\pi \times 100} = 0.4 \text{ sec.}$ and $b = \frac{1}{2\tau} = \frac{1}{2 \times 0.4} = 1.25 \text{ per sec.}$

(iii) Half width $\Delta p = \sqrt{3} \times 1.25 = 2.2 \text{ rad/sec.}$

Ex. 3. If a constant torque of $8 \times 10^{-5} \text{ N-m}$ produces an angular displacement of 2° in a torsion pendulum whose moment of inertia is 10^{-3} kg-m^2 , calculate the frequency of the periodic external torque which will produce resonance.

Sol. Angular displacement = $2^\circ = 2 \times \pi/180$ radian; applied couple = $8 \times 10^{-5} \text{ N-m}$

Restoring couple for unit displacement = $\frac{8 \times 10^{-5}}{2\pi} \times 180 \text{ N-m/rad.}$

At resonance, frequency of the torque = approx. natural frequency

$$= \frac{1}{2\pi} \sqrt{\frac{C}{I}} = \frac{1}{2\pi} \sqrt{\frac{8 \times 10^{-5} \times 180}{2\pi \times 10^{-3}}} = 0.24 \text{ Hz.}$$

Ex. 4. In case of a driven harmonic oscillator, the amplitude of vibrations increases from 0.1 mm at very low frequencies to a maximum value of 5 mm at the frequency 200 Hz. Find Q -factor, damping constant b and relaxation time.

Sol. $Q = \frac{\text{Amplitude at resonance}}{\text{Amplitude at zero or very low frequency}} = \frac{5}{0.1} = 50$.

Here, resonant frequency $\nu_0 = 200 \text{ Hz}$, $\therefore \omega_0 = 2\pi \times 200$

Therefore, $\tau = \frac{Q}{\omega_0} = \frac{50}{2\pi \times 200} = 0.04 \text{ sec.}$ and $b = \frac{1}{2\tau} = 12.6 \text{ per sec.}$

Ex. 5. Find the average energy stored in a 0.1 kg mass, attached to a spring. The mass is oscillating with an amplitude of 5 cm in resonance with a harmonic force of frequency 30 Hz. If the quality factor of the oscillator is 100, calculate the power losses.

Sol. At resonance, $p = \omega_0$, the average energy stored

$$= \frac{1}{2} m \omega_0^2 a^2 = \frac{1}{2} \times 0.1 (2\pi \times 30)^2 \times (5 \times 10^{-2})^2 = 4.44 \text{ J.}$$

Now, $Q = \frac{\text{Average energy stored}}{\text{Average energy dissipated per period}}$

$$\begin{aligned}\text{Period, } T &= 1/30 \text{ sec; } \therefore \text{Average energy dissipated in } 1/30 \text{ sec.} = \frac{2\pi \times 4.44}{100} = \mathbf{0.279 \text{ J.}} \\ \text{Average energy dissipated per sec. or power losses} &= 30 \times 0.279 = \mathbf{8.37 \text{ J.}}\end{aligned}$$

Ex. 6. In case of a forced harmonic oscillator of mass 10 gm the amplitude of vibration is 0.2 mm at very low frequencies. As the frequency is varied, the amplitude value increases to a maximum value of 20 mm for a frequency of 200 Hz and thereafter decreases with further increase in frequency. Find (i) the Q factor of the system (ii) the damping constant and (iii) the half width of the resonance curve.

$$\text{Sol. (i)} \quad Q = \frac{\text{Maximum amplitude at resonance}}{\text{Amplitude at very low frequencies}} = \frac{20 \times 10^{-3}}{0.2 \times 10^{-3}} = \mathbf{100}$$

$$\text{(ii) Resonant angular frequency } \omega_0 = 2\pi \times 200 = 400\pi \text{ sec}^{-1}$$

$$\text{Now,} \quad Q = \omega_0 \tau = \frac{\omega_0}{2b}$$

$$\therefore \text{Damping constant} \quad b = \frac{\omega_0}{2Q} = \frac{400\pi}{2 \times 100} = \frac{400 \times 3.14}{2 \times 100} = \mathbf{6.28 \text{ sec}^{-1}}$$

$$\text{(iii) Half band width} \quad \Delta p = \sqrt{3} b = \sqrt{3} \times 6.28 = 1.732 \times 6.28 = \mathbf{10.88 \text{ rad/sec.}}$$

Ex. 7. An A. C. supply of frequency 10000 and 110 volts is connected across a circuit containing a resistance of 10 ohms, an inductance of 10^{-3} henry and a capacity of $1 \mu\text{F}$. Find the value of the current. What must be the value of the capacity in order that the current may be maximum.

$$\begin{aligned}\text{Sol. Impedance, } Z &= \sqrt{R^2 + (pL - 1/pC)^2} = \sqrt{R^2 + (2\pi\nu L - 1/2\pi\nu C)^2} \\ &= \sqrt{10^2 + \left(2\pi \times 10000 \times 10^{-3} - \frac{1}{2\pi \times 10000 \times 10^{-6}}\right)^2} = 48 \text{ ohms approx.}\end{aligned}$$

$$\therefore \text{Current} = \text{e.m.f.}/Z = 110/48 = \mathbf{2.293 \text{ amp.}}$$

Maximum current will flow when the total resistance is zero and resonance is produced. Hence the maximum current will be when $pL - 1/pC = 0$ i.e., when

$$C = \frac{1}{Lp^2} = \frac{1}{L \times 4\pi^2\nu^2} = \frac{1}{10^{-3} \times 4 \times \pi^2 \times (10^4)^2} = \mathbf{0.253 \times 10^{-6} \text{ farad.}}$$

Ex. 8. An A.C. potential of $1 \times 10^5 \text{ sec}^{-1}$ frequency and 1.0 volt amplitude is applied to a series LCR circuit. If $R = 2.0 \text{ ohm}$, $L = 0.50 \text{ milihenry}$, calculate the value of C to secure resonance. Also calculate the rms value of current and peak potential difference across the condenser.

Sol. Capacity, required to produce resonance, is given by

$$C = \frac{1}{4\pi^2 n^2 L} = \frac{1}{4 \times (3.14)^2 \times (10^5)^2 \times 0.5 \times 10^{-3}} = \mathbf{5 \times 10^{-9} \text{ farad.}}$$

$$I_{\text{rms}} = \frac{E_{\text{rms}}}{R} = \frac{E_0}{R\sqrt{2}} = \frac{1}{2\sqrt{2}} \text{ ampere} \quad [\text{at resonance } Z = R]$$

Peak potential differential across the condenser = peak current $\times$ reactance across condenser

$$= \frac{E_0}{R} \times \frac{1}{pC} = \frac{1}{2} \times \frac{1}{10^5 \times 5 \times 10^{-9}} = \mathbf{10^3 \text{ volts.}}$$

QUESTIONS

Long Answer Questions

1. What are damped oscillations ? Solve the differential equation of a damped harmonic oscillator. Discuss the cases when it is over-damped and critically damped. (A.N.U. 2007)
2. Solve the differential equation of damped harmonic oscillator with all cases and discuss in detail the underdamped case.
3. (a) Derive the equation of motion of damped harmonic oscillator and obtain its solution.
(b) Discuss the conditions under which the oscillations are under-damped, over-damped and critically-damped. (O.U. March 2007)
4. For the following cases represent the variation of displacement against time graphically :
(1) Undamped harmonic oscillator
(2) Overdamped simple harmonic oscillator
(3) Critically damped simple harmonic oscillator.
Mention the corresponding solutions alongside the curves. (O.U. 2002)
5. Find the equation of motion for a damped harmonic oscillator and find its solution. Give two examples of damped harmonic oscillations. (A.N.U. 2006)
6. Form the differential equation for the damped harmonic motion of an oscillator and obtain its solution. Discuss the conditions under which the oscillations are under-damped, over damped and critically damped. (K.U. 2005)
7. Set up the differential equation for forced vibrations and explain the three cases :
(1) $p \ll \omega$, (ii) $p = \omega$ and $p \gg \omega$ where p is angular frequency of the force. (O.U. 2005)
8. (a) What is a damped harmonic oscillator ? Represent graphically the solution of a damped harmonic oscillator.
(b) Explain the terms Logarithmic decrement, Relaxation time and the Q factor of a damped harmonic oscillator. (O.U. 2003)
9. Explain the terms 'damping' and ' Q -factor'. Compare a forced mechanical oscillator with an electrical oscillator. (S.V.U. 2002)
10. What do you understand by the relaxation time, damping constant and quality factor of a damped harmonic oscillator ? Obtain their expressions.
11. Obtain the expressions of average total energy and average power loss for a damped harmonic oscillator.
12. If the damping constant $b \ll \omega_0$, prove that
(a) the average total energy $E = \frac{1}{2} m \omega^2 a_0^2 e^{-2bt}$.
(b) average power dissipation $P = E/\tau$.
(c) average rate of doing work by the damping force $-\gamma dx/dt$ is equal to the average power dissipation (i.e. to E/τ).
13. A particle is oscillating under a damping force. Show that the average power loss (P) is given by $P = E/\tau$, where E is the average energy and τ is the relaxation time. What happens to the dissipated energy ?
14. The amplitude A of forced vibrations in a system is given by

$$A = \frac{F/m}{[(\omega_0^2 - \omega^2)^2 + (\omega/\tau)^2]^{1/2}} \cdot (\text{Proof not required}).$$

Symbols have their usual meanings. Show that in the limit $\omega \ll \omega_0$ the response is independent of the mass, while in the limit $\omega \gg \omega_0$ the response is independent of the spring constant of the system.
15. Give the theory of driven harmonic oscillator. Show that at resonance the displacement always lags behind the driving force in phase by $\pi/2$, while the velocity at resonance is always in phase with the driving force.

16. What are free, damped and forced oscillators ? Obtain the differential equation for the forced oscillator and solve it.
17. What are forced oscillations ? Derive the equation of motion of a driven harmonic oscillator and find its solution. (O.U. 2009)
18. If the frequency of the impressed force on an oscillator is changed gradually, then what changes will take place in the response (amplitude) ? Draw a graph, representing this variation. Prove that the displacement always lags in phase from the driven force. Show by means of a graph the relation between the frequency of the force and this phase difference.
19. What are forced oscillations ? Obtain an expression for the amplitude of forced vibrations and deduce the condition for resonance. (A.U. March 2007, K.U. 2007)
20. (a) Derive differential equation of a driven harmonic oscillator.
(b) Distinguish between velocity resonance and amplitude resonance. (O.U. Oct. 2001)
21. (a) Explain the terms resonance, sharpness of resonance and Q -factor.
(b) Set up the equation of motion for forced vibrations and obtain its solution. (O.U. Oct. 2007)
22. Define quality factor and bandwidth of the sharpness of resonance. Obtain quality factor for a driven harmonic oscillator at resonance. (S.V.U. 2007)
23. (a) Explain forced vibrations. Discuss mathematically the vibrations of a system executing damped S.H.M. when subjected to an external periodic force. Obtain an expression for the amplitude of forced vibrations and discuss the conditions of resonance. (O.U. 1995)
(b) Explain the meaning of sharpness of resonance. (A.U. 1992)
24. Describe the transient and steady states of a forced oscillator.
25. Write down the differential equation for a forced harmonic oscillator and explain the significance of each term in it. Obtain solution of this equation and discuss it. Write the condition of resonance and explain the sharpness of resonance.
26. For a driven harmonic oscillator of natural frequency $\omega/2\pi$ described by

$$m\ddot{x} + r\dot{x} + kx = F \sin pt,$$
 deduce an expression for velocity amplitude x_0 . Show that for the case $p \gg \omega$, $p = \omega$ and $p \ll \omega$, this amplitude is governed by inertia factor, the resistance factor and the spring factor respectively.
27. Explain the power absorption, quality factor and resonance of a forced oscillator.
28. Write down the expression for the amplitude of a forced oscillator. Explain the resonance half power points on the amplitude-frequency curve. Show that the quality factor $Q = \frac{\omega_r}{2\Delta\omega}$, where the symbols have their usual meanings.
29. Prove that for the forced oscillations of a damped oscillator the average power of the applied force is equal to the average power dissipated by the damping force.
30. What do you understand by resonance ? Explain the sharpness of resonance with the help of theoretical results of forced oscillations and deduce the expression for its quality factor.
31. Show that in a driven oscillator, the maximum power is absorbed at the frequency of velocity resonance and not at the frequency of amplitude resonance.
32. Explain the details of the following terms in case of driven harmonic oscillator :
(i) Quality factor, (ii) Sharpness of resonance, and (iii) Velocity resonance.

Short Answer Questions

1. Obtain differential equation of a damped harmonic oscillator. (S.V.U. 2003)
2. Explain about relaxation time and quality factor. (S.K.U. 2007)
3. Form the equation for damped oscillator and obtain its solution. (A.U. March 2007)
4. Write the differential equations for the following cases of damped harmonic oscillator : (i) mechanical, (ii) electrical.

[Hint : (i) $m \frac{d^2x}{dt^2} + \gamma \frac{dx}{dt} + kx = 0$. (ii) $\frac{d^2q}{dt^2} + R \frac{dq}{dt} + \frac{q}{C} = 0$.]

5. Define damping constant and relaxation time.

6. The differential equation for damped harmonic oscillator is $\frac{d^2x}{dt^2} + 2b\frac{dx}{dt} + \omega_0^2x = 0$

If $\omega_0 > b$, find the time in which amplitude falls to $1/e$ times the initial value.

[Ans. $1/b$ sec.]

[Hint : $a = a_0 e^{-bt}$ or $\frac{a_0}{a} = \frac{1}{e} = e^{-bt}$ or $e^{-1} = e^{-bt}$ or $bt = 1$ or $t = 1/b$ sec.]

7. The equation of motion for a damped oscillator is given by

$$\frac{d^2x}{dt^2} + \frac{r}{4} \frac{dx}{dt} + 8x = 0.$$

For what range of values for the damping constant, will motion be under damped.

(O.U. 2005)

8. What is meant by ' Q ' of an oscillator? How is it affected by damping?

(O.U. 2005)

9. The value of quality factor is high for small damping. Explain this statement.

10. Show that the energy of oscillation in a damped harmonic oscillator decreases to e^{-1} of its initial value in the time τ and in this time, the oscillator performs $Q/2\pi$ oscillations.

[Note : This is why τ is sometimes called decay time.]

[Hint : $E = E_0 e^{-t/\tau}$. In time $t = \tau$, $\frac{E}{E_0} = e^{-1}$ or $\frac{E}{E_0} = \frac{1}{e}$

$$Q = \omega_0 \tau \text{ or } \tau = \frac{Q}{\omega_0}, T = \frac{2\pi}{\omega} = \frac{2\pi}{\omega_0} \text{ (for low damping)}$$

$\therefore$ Number of oscillations in τ time $= \frac{\tau}{T} = \frac{Q}{\omega_0} \frac{\omega_0}{2\pi} = \frac{Q}{2\pi}$.]

11. Explain what do you mean by overdamped and critically damped vibrations. (S.V.U. 2007)

12. What is resonance? Explain with two examples. (S.V.U. 2002)

13. Obtain an expression for logarithmic decrement. (O.U. 2008)

14. Discuss the cases of overdamping, underdamping and critical damping. (A.U. June 2007)

15. Explain the phenomenon of resonance. What is Q -factor? (A.U. June 2007, 03)

16. Define quality factor. (A.U. March 2007)

17. Explain Q -factor. Give an expression for Q factor in a mechanical oscillator. (O.U. 2009; A.N.U. 2008)

18. Define Q -factor and mention its importance. (O.U. 2001)

19. Explain how the sharpness of resonance is affected by Q -factor. (A.N.U. March 2007)

20. Explain the terms quality factor, sharpness of resonance and power factor. (K.U. 2007)

21. Define logarithmic decrement to damped harmonic oscillator. Describe the electrical analogue for a simple oscillator. (S.K.U. 2003)

22. Explain the term velocity resonance and amplitude resonance. (S.K.U. 2005)

23. What is resonance? Explain half power points. (K.U. 2005)

24. How are the terms quality factor, band-width and resonance frequency related. Explain using a diagram. (O.U. 05)

25. Show that at resonance for low damping, the phase difference between the force and resultant displacement of the oscillator is $\pi/2$.

26. Explain the electrical analogue of forced oscillator. (O.U. 2002)

27. What are transient and steady states? Explain.

PROBLEMS

[Damped Harmonic Oscillator]

1. The time period of pendulum is 2 sec. If the angular retardation due to air is 0.04 times angular velocity of pendulum and original amplitude is 1° , calculate the amplitude after 10 oscillations.

[Ans. $40'$ approx.]

2. A simple pendulum has a period of 2 sec. and amplitude of 2° . After 10 complete oscillations its amplitude is reduced to 1.5° . Find the relaxation time and damping constant.
[Ans. 34.76 sec; 0.01438 sec^{-1} .]
3. The quality factor of a tuning fork is 6×10^4 and its frequency is 512 Hz. In what time its energy will become $1/10$ of its initial value ?
[Ans. 42.94 sec.]
4. A damped vibrating system, starting from rest reaches the first amplitude of 2 cm which reduces to 0.2 cm in that direction after 50 oscillations, each of period 2.3 sec. Find the correction for the first displacement for damping. ($\log_e 10 = 2.3$)
[Ans. 0.23 mm]
5. In how much time will the amplitude of a damped harmonic oscillator be reduced to half its initial value, if the decay modulus is one second ?
[Ans. 0.69 second]
6. The quality factor of a tuning fork of frequency 256 is given by 4×10^4 . Calculate the time in which the energy is reduced to $(1/e)^{\text{th}}$ of its energy in the absence of damping. How many oscillations does the oscillator make in this time ?
[Ans. 24.88 sec, 3669.28]
7. The oscillations of a tuning fork of frequency 256 Hz die away to $1/e$ times their amplitude in one second. Show that the frequency due to air damping is practically unaffected.
8. Find the Q -factor of a second pendulum if its amplitude falls to one-third of its initial value in 56 oscillations ?
[Ans. 160]
9. The amplitude of an oscillator of frequency 200 Hz falls to $1/10$ of its initial value after 2000 cycles. Calculate (i) its relaxation time, (ii) its quality factor, (iii) time in which its energy falls to $1/10$ of its initial value, (iv) damping constant.
[Ans. (i) 2.174 sec. (ii) $Q = 2730$, (iii) $t = 5$ sec, (iv) $b = 0.23$]
10. The differential equation of an oscillating system is expressed as : $\frac{d^2x}{dt^2} + 2b\frac{dx}{dt} + n^2x = 0$.
If $b \ll n$, find the time in which : (i) the amplitude of oscillation falls to $1/e$ of its initial value, (ii) the energy falls to $1/e$ of its initial value.
[Ans. (i) $(1/b)$ sec. (ii) $(1/2b)$ sec.]
11. The amplitude of an oscillator of frequency 200 hertz reduces to $1/10$ th of its initial amplitude after 2000 oscillations. Calculate the quality factor, relaxation time and damping constant.
[Ans. 2.73×10^3 ; 2.17 sec.]
12. The quality factor of a tuning fork of frequency 384 Hz is 10^4 . How much time will be taken in reducing the energy $1/e$ of its initial value ? How many oscillations take place in this duration ?
[Ans. 4.1 sec.; 1574.47.]
13. The quality factor Q of a sonometer wire is 2×10^2 . On plucking, the wire emits a note of frequency 120 Hz. Find the time in which the emplitude falls to $(1/e^2)$ of the initial value.
[Ans. 10.6 sec.]
14. Give the theory of oscillations in series LCR circuit with small damping. Deduce the frequency and quality factor for a circuit with $L = 2 \text{ mh}$, $C = 5 \text{ } \mu\text{F}$ and $R = 0.2 \text{ ohm}$.
[Ans. $1.59 \times 10^3 \text{ Hz}$; 100.]
15. In a LCR circuit $L = 10 \text{ mh}$, $C = 1 \text{ } \mu\text{F}$ and $R = 0.1 \text{ ohm}$. How long a time does the charge oscillation take to decay half amplitude ? What is the quality factor for the circuit ?
[Ans. 0.14 sec; 1000.]

16. In a LCR circuit, $L = 0.3$ henry and $C = 30 \mu\text{F}$. Find the maximum value of the resistance which will make the series circuit to be oscillatory.

[Ans. 200Ω]

17. A condenser of 3 microfarad capacity is discharged through a 2 ohm resistance and 4 henry inductance. Show that the discharge is oscillatory. Calculate the frequency and Q -factor of the circuit. Find the time in which the charge amplitude becomes 1% of the initial value.

[Ans. The discharge is oscillatory; 18.4 sec.]

[Hint. : If $\frac{1}{LC} > \frac{R^2}{4L^2}$, the discharge is oscillatory.]

[Driven Harmonic Oscillator]

1. Write the equation of motion of a simple harmonic undamped oscillator to which a force $F = F_0 \cos pt$ is applied. Verify that its solution is

$$x = \frac{F_0}{m(\omega_0^2 - p^2)} \cos pt.$$

Discuss the resonance in this case.

2. In a forced harmonic oscillator, the amplitude of vibration increases from 0.01 mm at very low frequencies to a maximum value 5 mm at the frequency 100 c.p.s. Calculate the quality factor, damping constant and half width of the resonance curve.

[Ans. 500; 1.25/sec.; 2.2 rad./sec.]

3. If $Q = 50$, find the values of $A/A_{\max}$, when $p/\omega_0 = 0.98$ and 0.97.

[Ans. 0.45, 0.32.]

4. A mass of 1 kg is suspended from a linear spring with a force constant $C = 10^3$ N/m, and damping coefficient $\gamma = 50 \times 10^{-3}$ N-sec/m. The spring is driven by an external force $F = F_0 \sin pt$, where $F_0 = 2.5$ newton and p is twice the natural frequency of the system. (i) Calculate the amplitude of the resulting motion. (ii) How much is the displacement shifted in phase from the driving force ?

[Ans. (i) 8.3×10^{-4} m; (ii) -179.9° .]

5. For a high Q -system, show that the width of the amplitude resonance curve is nearly $\sqrt{3}b$, where the full-width is measured between those frequencies where $a = a_m/2$.

6. Calculate the average energy stored in a 20 gm mass attached to a spring and vibrating with an amplitude 1 cm in resonance with a periodic force whose frequency is 20 Hz. If the quality factor of the oscillator be 160, how much energy being dissipated per second ?

[Ans. 0.016 J; 0.0125 J.]

7. A condenser of $0.5 \mu\text{F}$ capacity is discharged through a 2 ohm resistance and a 2 henry inductance. Find the frequency of electrical oscillations and the time in which the amplitude of oscillations is reduced to 0.1%.

[Ans. 160 Hz; 14 sec.]

[Hint. : $Q_0/1000 = Q_0 e^{-Rt/2L} = Q_0 e^{-t/2}$ or $e^{t/2} = 1000$.]

8. A torsional pendulum with moment of inertia 900 gm-cm^2 and torsion constant 4×10^{-3} N-m/rad executes resonant vibrations under the influence of an external torque. Calculate the frequency of the applied torque. What should be the Q of the oscillator so that the amplitude of oscillation may reduce to 20%, when the frequency of the applied torque increased by 0.2% ?

[Ans. 1.06 Hz; 187.]

9. A circuit contains a resistance of 2 ohm, an inductance of 0.005 henry and a capacity of $600 \mu\text{F}$ in series with a potential difference of 100 volts at driven frequency 50 Hz. Find the current, the power factor and the power.

[Ans. 26.5 amp (r.m.s value) ahead of the driven voltage by $61^\circ 54'$; $\cos \phi = 0.471$; $p = 1114$ watts.]

10. In an experiment on forced oscillations the frequency of a sinusoidal driving force is changed while its amplitude is kept constant. It is found that the amplitude of vibration is 0.10 mm at very low frequency of the driver and goes upto a maximum of 0.5 mm at driving frequency 200 sec^{-1} . Calculate (i) Q of the system, (ii) relaxation time τ , and (iii) half width of resonance curve.
[Ans. (i) 500; (ii) 0.4 sec; (iii) 2.16 rad/sec.]
11. Calculating the resonating frequency and Q of the circuit containing $0.02 \mu\text{F}$ capacity, 80 mH inductance and 0.25 ohm resistance.
[Ans. $4 \times 10^3 \text{ Hz}$; 8×10^3 .]
12. Find the resonating frequency and Q of a circuit containing $0.04 \mu\text{F}$ capacity, 40 mH inductance and 0.2 ohm resistance. What is the band-width of the power resonance curve ?
[Ans. $\nu_0 = 3.98 \times 10^3 \text{ Hz}$; $Q = 5000$; $\Delta\nu = 5 \text{ rad./sec.}$]
13. A condenser of $0.5 \mu\text{F}$ capacity is discharged through a 2 ohm resistance and a 2 henry inductance. Find the frequency of electrical oscillations and the time in which the amplitude of oscillations is reduced to 0.1% .
[Ans. 160 Hz ; 14 sec.]
14. A resistance of 10 ohms is joined in series with an inductance of 0.5 henry . What capacitance should be put in series with the combination to obtain maximum current ? What will be the potential difference across the resistance, inductance and capacity ? The current is being supplied by a 200 volts and $50 \text{ cycles A.C. mains}$.
[Ans. $20.27 \times 10^{-6} \text{ farad}$; 20 amp ; 200 volts ; 3141.6 volts , 3141.6 volts .]
15. A coil of 3 mH is given. Calculate the capacity which will make the resonance frequency 1000 kilo Hz .
[Ans. 8.85 peko-farad .]
16. LCR circuit has $R = 10 \text{ ohms}$, $L = 200 \text{ micro-henry}$ and $C = 500 \text{ peko-farad}$. Determine the angular frequency separation between the half power points and quality factor for the circuit.
[Ans. $5 \times 10^4 \text{ rad./sec.}$; 63.3 .]



Complex Vibrations

9.1. Periodic Function

Any function which repeats itself regularly over a given interval of time or space is called a periodic function. Mathematically, a function $f(t)$ is said to be periodic if for all time t , we have

$$f(t + T) = f(t)$$

where T is a constant, called the period. For example, sinusoidal functions, $\sin \omega t$ and $\cos \omega t$, are periodic functions of period $T = 2\pi/\omega$ because

$$\sin \omega \left(t + \frac{2\pi}{\omega} \right) = \sin \omega t \text{ and } \cos \omega \left(t + \frac{2\pi}{\omega} \right) = \cos \omega t$$

There may be functions of time t which are periodic but not sinusoidal. A periodic saw-tooth wave form, used in the time base of a cathode ray oscillograph, a square wave generated by an electronic function generator etc. are examples of such periodic functions.

A simple harmonic vibration is the simplest type of vibration. When few such simple harmonic vibrations are superimposed, we obtain a periodic vibration or a periodic function of complex nature. For example, the vibrations represented by saw-tooth wave or square wave is a complex periodic function. Although these functions are quite simple to visualize, but they are very complicated to represent mathematically. Fourier theorem deals with the analysis of such complex periodic functions in terms of simple harmonic components.

9.2. Fourier's Theorem

In 1822, Fourier stated a very important theorem, with the help of which any complex periodic function can be analysed. According to this theorem, *any finite single valued periodic function, whatever, can be expressed as a summation of simple harmonic terms which have frequencies, being integral multiples of that of the given function.* The series, so obtained, is called *Fourier's series*.

Mathematically Fourier's theorem can be represented by the series

$$y = f(\omega t) = A_0 + A_1 \sin \omega t + A_2 \sin 2\omega t + \dots + A_r \sin r \omega t + \dots \\ + B_1 \cos \omega t + B_2 \cos 2\omega t + \dots + B_r \cos r \omega t + \dots \quad \dots(1)$$

or

$$y = f(\omega t) = A_0 + \sum_{r=1}^{\infty} A_r \sin r \omega t + \sum_{r=1}^{\infty} B_r \cos r \omega t \quad \dots(2)$$

where $y = f(\omega t)$ is the displacement of complex periodic vibration of frequency $\omega/2\pi$ at any instant t and r is a positive integer i.e., $r = 1, 2, 3, \dots$. Here the periodic function has been expressed as the summation of sine and cosine components, which have frequencies, multiples of $\omega/2\pi$.

Evaluation of the values of $A_0, A_1, \dots, A_r$ and $B_1, B_2, \dots, B_r$

To find A_0 , multiply equation (1) by dt and integrate for complete vibration, i.e., from 0 to T ($= 2\pi/\omega$), we get

$$\int_0^T y dt = A_0 \int_0^T dt + A_1 \int_0^T \sin \omega t dt + \dots + A_r \int_0^T \sin r \omega t dt \\ + \dots + B_1 \int_0^T \cos \omega t dt + \dots + B_r \int_0^T \cos r \omega t dt + \dots \\ = A_0 T$$

$$\therefore A_0 = \frac{1}{T} \int_0^T y dt \quad \dots(3)^*$$

$$^* \int_0^T \sin r \omega t dt = \left[-\frac{\cos r \omega t}{r \omega} \right]_0^{T=2\pi/\omega} = \frac{-\cos 2\pi r + \cos 0}{r \omega} = 0 \text{ and } \int_0^T \cos r \omega t dt = \left[\frac{\sin r \omega t}{r \omega} \right]_0^{T=2\pi/\omega} = \frac{\sin 2\pi r - \sin 0}{r \omega} = 0$$

To find the value of A_r , multiply equation (1) by $\sin r\omega t \, dt$ and integrate for complete vibration. Then we get

$$\begin{aligned} \int_0^T y \sin r\omega t \, dt &= A_0 \int_0^T \sin r\omega t \, dt + A_1 \int_0^T \sin \omega t \sin r\omega t \, dt + \dots + A_r \int_0^T \sin^2 r\omega t \, dt + \dots \\ &\quad + B_1 \int_0^T \sin r\omega t \cos \omega t \, dt \dots + B_r \int_0^T \sin r\omega t \cos r\omega t \, dt + \dots \\ &= \frac{A_r}{2} \int_0^T (1 - \cos 2r\omega t) \, dt = A_r \times \frac{T}{2} \\ \therefore A_r &= \frac{2}{T} \int_0^T y \sin r\omega t \, dt = \frac{2}{T} \int_0^T y \sin \frac{2\pi r t}{T} \, dt \quad \dots(4) \end{aligned}$$

Similar values of B_r can be found by multiplying eq. (1) by $\cos r\omega t \, dt$ and then on integrating we will get

$$B_r = \frac{2}{T} \int_0^T y \cos r\omega t \, dt = \frac{2}{T} \int_0^T y \cos \frac{2\pi r t}{T} \, dt \quad \dots(5)$$

Thus if we put $r = 1, 2, 3, \dots$ in eq. (4) and (5) we can calculate the coefficients $A_1, A_2, \dots, B_1, B_2, \dots$ of sine and cosine terms. Hence we can analyse $y = f(\omega t)$, i.e., the displacement of complex periodic vibrations in terms of sine and cosine terms.

Significance of the Fourier constant A_0

The constant A_0 of the Fourier's series is evaluated by using the relation (3) i.e.,

$$A_0 = \frac{1}{T} \int_0^T y \, dt = \frac{1}{T} \int_0^T f(\omega t) \, dt$$

Thus A_0 is the average value of the function y or $f(\omega t)$ over the period T and represents the mean displacement for a complete period from the X -axis. The term A_0 denotes the distance of the axis of symmetry of the displacement-time curve of the complex vibration from the axis of coordinates or time axis. Obviously the straight line $y = A_0$ (constant) represents the axis of the complex vibration curve. In a period of complex vibration, if for half the period the function y [or $f(\omega t)$] has positive values and for the other half of the period, y has equal negative values, then the value of A_0 is zero.

For a square wave, when the axis of vibration is at a distance $a/2$ from X -axis, the value of A_0 is $a/2$ [Fig. 9.1 (i)] and when it coincides with X -axis, the value of A_0 is zero [Fig. 9.1 (ii)]; in both figures the displacement y has been measured from X -axis or time axis.

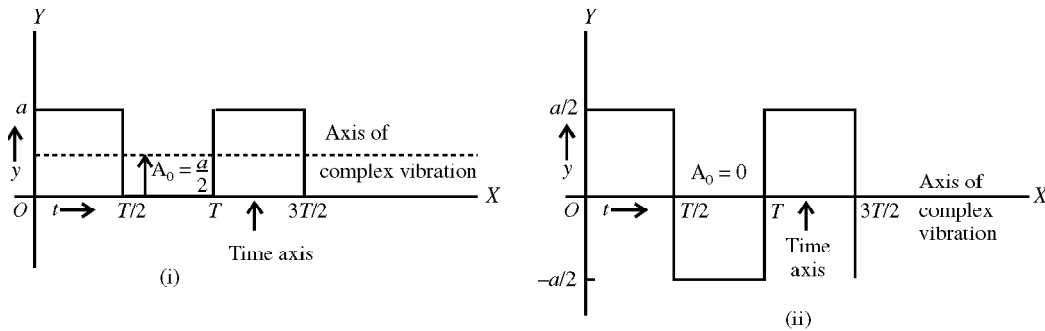


Fig. 9.1 Representing the value of A_0 for a square wave to be (i) $a/2$, when axis of vibration is at a distance of $a/2$ from X -axis and (ii) zero, when both axes coincide.

Fourier series for a periodic function of variable x

In case, if $y = f(x)$ is a periodic function of variable x of period L , then

$$A_0 = \frac{1}{L} \int_0^L f(x) \, dx; A_r = \frac{2}{L} \int_0^L f(x) \sin \frac{2\pi r x}{L} \, dx \text{ and } B_r = \frac{2}{L} \int_0^L f(x) \cos \frac{2\pi r x}{L} \, dx \quad \dots(6)$$

and the series is

$$y = f(x) = A_0 + \sum_{r=1}^{\infty} A_r \sin \frac{2\pi r x}{L} + \sum_{r=1}^{\infty} B_r \cos \frac{2\pi r x}{L} \quad \dots(7)$$

If the periodic function $f(x)$ has a period 2π in the interval $-\pi$ to π , it can be expressed as a Fourier's series of sines and cosines in the form :

$$f(x) = A_0 + \sum_{r=1}^{\infty} A_r \sin rx + \sum_{r=1}^{\infty} B_r \cos rx \quad \dots(8)$$

where the coefficients are evaluated as

$$A_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx; A_r = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin rx dx \text{ and } B_r = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos rx ds \quad \dots(9)$$

9.3. Limitations of Fourier's Theorem

In order to apply Fourier's theorem to practical problems, the theorem is subjected to the following limitations regarding the form of complex vibration:

(1) **The function should be periodic single-valued and continuous.** This means that the displacement should repeat after an interval of time and it must be continuous with only one value at an instant. In other words, the displacement should not have two or more values at the same time. In **fig. 9.2** at the point A, the function is not single-valued but the particle has two displacements at the same instant. In such a case, Fourier's theorem is not applicable.

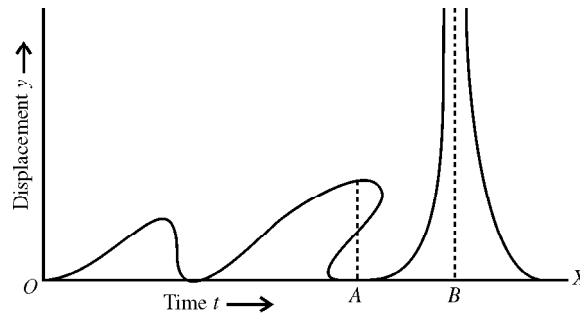


Fig. 9.2. Discussing limitations of Fourier's theorem.

(2) **The function should always be finite.** This means that the displacement during oscillations should always remain finite and it should not be infinite at any time as at B (**fig. 9.2**).

(3) **Moreover, the function should have a finite number of discontinuities and a finite number of maxima and minima in the interval of one oscillation, T .**

9.4. Applications of Fourier's Theorem

The Fourier's analysis is applicable to periodic functions of time as well as those of space. Here, we shall apply the Fourier's theorem to analyse the following complex periodic vibrations :

- (1) Square wave
- (2) Saw-tooth wave
- (3) Triangular wave

These periodic curves cannot be represented by pure sine or cosine curves, but we shall see that they can be built up by compounding a number of sinusoidal curves.

9.4.1 Fourier Analysis of Square Wave

Let us consider the Fourier analysis of a square wave as shown in **fig. 9.3**. We represent the periodic function $y = f(\omega t)$, such as displacement, along Y -axis and time along X -axis. Here, y remains constant from time $t = 0$ to $t = T/2$ and then remains zero for the half period. Hence we may express this function as follows :

$$\begin{aligned} y &= a, & t &= 0 \text{ to } t = T/2 \\ y &= 0, & t &= T/2 \text{ to } t = T. \end{aligned} \quad \dots(1)$$

We have to find the Fourier's series of this function.

According to Fourier's theorem, we have

$$y = A_0 + A_1 \sin \omega t + A_2 \sin 2\omega t + \dots A_r \sin r\omega t + \dots + B_1 \cos \omega t + \dots + B_r \cos r\omega t + \dots \quad \dots(2)$$

$\therefore$

$$\begin{aligned} A_0 &= \frac{1}{T} \int_0^T y dt = \frac{1}{T} \int_0^{T/2} a dt + 0 \\ &= \frac{a}{T} \times \frac{T}{2} = \frac{a}{2} \end{aligned} \quad \dots(3)$$

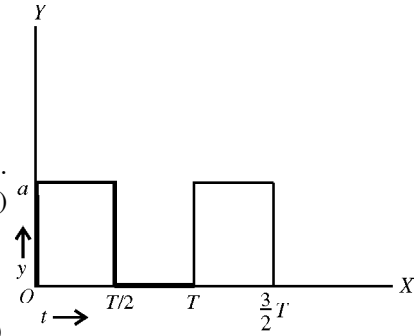


Fig. 9.3. Square wave.

$$\begin{aligned} A_r &= \frac{1}{T} \int_0^{T/2} y \sin r\omega t dt + 0 = \frac{2}{T} \int_0^{T/2} a \sin r\omega t dt \\ &= \frac{2a}{T} \left[-\frac{1}{r\omega} \cos r\omega t \right]_0^{T/2} \quad \left[\because \omega = \frac{2\pi}{T} \right] \\ &= \frac{2a}{T} \times \frac{T}{2\pi r} (-\cos r\pi + 1) = \frac{a}{\pi r} (1 - \cos r\pi) \end{aligned} \quad \dots(4)$$

$$\text{Now, when } r = 0, 2, 4, \dots A_r = 0 \quad \dots(5a)$$

$$\text{and when } r = 1, 3, 5, \dots A_r = \frac{2a}{\pi r} \quad \dots(5b)$$

$$\begin{aligned} B_r &= \frac{2}{T} \int_0^{T/2} a \cos r\omega t dt \\ &= \frac{2}{T} \left[\frac{a}{r\omega} \sin r\omega t \right]_0^{T/2} \quad \left[\text{since } \omega = \frac{2\pi}{T} \right] \\ &= \frac{2}{T} \left[\frac{a}{r\omega} \left(\sin r \times \frac{2\pi}{T} \times \frac{T}{2} - 0 \right) \right] = \frac{2a}{Tr\omega} (\sin r\pi - 1) = 0 \end{aligned} \quad \dots(6)$$

Therefore, substituting the values of A_0 , A_r and B_r in eq. (2) from (3), (5) and (6), we obtain the Fourier's series for the given function as

$$y = \frac{a}{2} + \frac{2a}{\pi} \left[\sin \omega t + \frac{1}{3} \sin 3\omega t + \frac{1}{5} \sin 5\omega t + \dots + \frac{1}{r} \sin r\omega t + \dots \right] \quad \dots(7)$$

We see that the complex periodic function is represented as the superposition of a number of simple harmonic motions (**fig. 9.4**). The frequencies of these component motions are an odd multiple of the fundamental frequency, i.e., in this case only odd harmonics are present. The amplitude of the successive harmonics is proportional to $1, \frac{1}{3}, \frac{1}{5}, \dots$,

$\frac{1}{r}, \dots$. This is the characteristic property of a particular wave form and is independent of the instant for measuring the time. Expression (7) contains only sine terms which depend on the instant from which the time is measured. If the time is

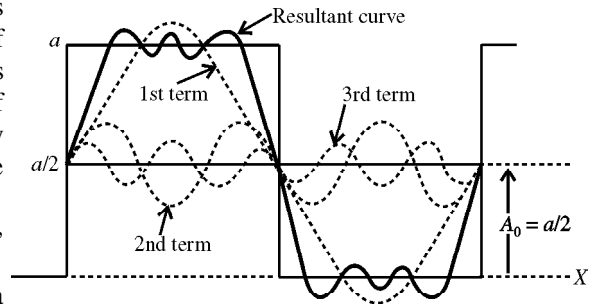


Fig. 9.4. Resultant of first three sine terms and the constant $A_0 = a/2$ of the Fourier's series [eq. 7].

measured $T/4$ second later than in **fig. 9.3**, then the curve will be symmetrical about Y -axis and there will be only cosine terms in the series (see below **Ex. 3**).

This is also clear that A_0 (here equal to $a/2$) is the average value of y . This means that the axis of the curve is at a distance $a/2$ up the time axis. In case, if for equal intervals of time, y has equal positive and negative values, the value A_0 is zero (see below **Ex. 1**) and the axis of the curve is the time axis.

Taking four terms (constant and first three harmonics) of the Fourier's series [eq. (7)], resultant curve is shown in **fig. 9.4** by a thick line. As more higher harmonics are added, the curve more nearly approaches the original, as shown in **fig. 9.3**.

Ex. 1. Find a Fourier's series for the function defined by the equations :

$$y = \begin{cases} +a & \text{for } t = 0 \text{ to } T/2 \\ 0 & \text{for } t = T/2 \\ -a & \text{for } t = T/2 \text{ to } T \end{cases}$$

Sol. The given function, a square wave, is shown in **fig. 9.5**. Let the Fourier's expansion be given by

$$y = A_0 + A_1 \sin \omega t + \dots + A_r \sin r\omega t + \dots + B_1 \cos \omega t + \dots + B_r \cos r\omega t + \dots \quad (1)$$

where $\omega = 2\pi/T$.

$$\begin{aligned} A_0 &= \frac{1}{T} \int_0^T y \, dt \\ &= \frac{1}{T} \int_0^{T/2} (+a) \, dt + 0 \\ &\quad + \frac{1}{T} \int_{T/2}^T (-a) \, dt \\ &= \frac{a}{2} - \frac{a}{2} = 0 \quad \dots(2) \end{aligned}$$

$$A_r = \frac{2}{T} \int_0^T y \sin r\omega t \, dt$$

$$= \frac{2}{T} \int_0^{T/2} a \sin r\omega t \, dt + 0 + \frac{2}{T} \int_{T/2}^T (-a) \sin r\omega t \, dt$$

$$= \frac{2a}{T} \left[-\frac{\cos r\omega t}{r\omega} \right]_0^{T/2} - \frac{2a}{T} \left[-\frac{\cos r\omega t}{r\omega} \right]_{T/2}^T \quad \left[\because \omega = \frac{2\pi}{T} \right]$$

$$= \frac{2a}{Tr\omega} \left[-\cos r \times \frac{2\pi}{T} \times \frac{T}{2} + 1 \right] - \frac{2a}{Tr\omega} \left[-\cos r \times \frac{2\pi}{T} \times T + \cos r \times \frac{2\pi}{T} \times \frac{T}{2} \right]$$

$$= \frac{a}{r\pi} [-\cos r\pi + 1] + \frac{a}{r\pi} [1 - \cos r\pi]$$

$$= \frac{2a}{r\pi} [1 - \cos r\pi]$$

$$= 0, \text{ when } r \text{ is even i.e., } r = 2, 4, 6, \dots \quad \dots(3a)$$

$$= \frac{4a}{r\pi}, \text{ when } r \text{ is odd i.e., } r = 1, 3, 5, \dots \quad \dots(3b)$$

and $B_r = \frac{2}{T} \int_0^{T/2} a \cos r\omega t \, dt + \frac{2}{T} \int_{T/2}^T (-a) \cos r\omega t \, dt$

$$= \frac{2a}{T} \left[\frac{\sin r\omega t}{r\omega} \right]_0^{T/2} - \frac{2a}{T} \left[\frac{\sin r\omega t}{r\omega} \right]_{T/2}^T \quad \left[\text{since } \omega = \frac{2\pi}{T} \right]$$

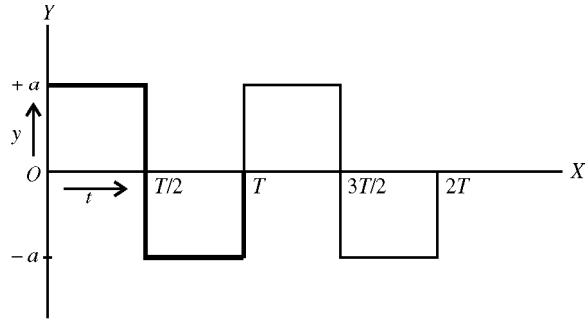


Fig. 9.5. Square wave.

$$\begin{aligned}
&= \frac{a}{r\pi} [\sin r\pi - 0] - \frac{a}{r\pi} [\sin 2r\pi - \sin r\pi] \\
&= \frac{a}{r\pi} [0] - \frac{a}{r\pi} [0] = 0
\end{aligned} \quad \dots(4)$$

Substituting for A_0 , A_r and B_r from (2), (3) and (4) in (1), we get the Fourier's series as

$$\begin{aligned}
y &= \frac{4a}{\pi} \sin \omega t + \frac{4a}{3\pi} \sin 3\omega t + \frac{4a}{5\pi} \sin 5\omega t + \dots + \frac{4a}{r\pi} \sin r\omega t + \dots \\
y &= \frac{4a}{\pi} \left[\sin \omega t + \frac{1}{3} \sin 3\omega t + \frac{1}{5} \sin 5\omega t + \dots + \frac{1}{r} \sin r\omega t + \dots \right]
\end{aligned} \quad \dots(5)$$

Ex. 2. Find the Fourier's series for the function $f(t)$ defined by

$$f(t) = 0 \text{ for } 0 < t < T/2.$$

$$f(t) = 1 \text{ for } T/2 < t < T \text{ and } f(t + T) = f(t).$$

(O.U. March 2007)

Sol. In the given case,

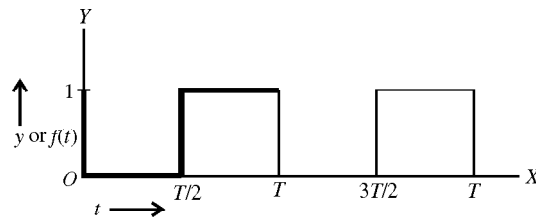


Fig. 9.6. Square wave.

$$\begin{aligned}
&\text{and} \quad y = f(t) = 0 \quad \text{for} \quad 0 < t < T/2 \\
&\text{also} \quad y = f(t) = 1 \quad \text{for} \quad T/2 < t < T \\
&\text{also} \quad f(t + T) = f(t).
\end{aligned} \quad \dots(1)$$

This means that $f(t)$ is a periodic function. The given function, a square wave, is represented in **fig. 9.6**. Let the Fourier's series be given by

$$\begin{aligned}
y &= A_0 + A_1 \sin \omega t + A_2 \sin 2\omega t + \dots \\
&\quad + A_r \sin r\omega t + \dots + B_1 \cos \omega t + \dots + B_r \cos r\omega t + \dots
\end{aligned} \quad \dots(2)$$

Fourier coefficients are evaluated as follows :

$$A_0 = \frac{1}{T} \int_0^T y dt = \frac{1}{T} \int_0^{T/2} 0 \times dt + \frac{1}{T} \int_{T/2}^T 1 \times dt = \frac{1}{T} [t]_{T/2}^T = \frac{1}{2} \quad \dots(3)$$

$$\begin{aligned}
A_r &= \frac{2}{T} \left[\int_0^{T/2} y \sin r\omega t dt + \int_{T/2}^T y \sin r\omega t dt \right] \\
&= 0 + \frac{2}{T} \int_{T/2=\pi/\omega}^{T=2\pi/\omega} \sin r\omega t dt = \frac{2}{2\pi/\omega} \left[-\frac{\cos r\omega t}{r\omega} \right]_{\pi/\omega}^{2\pi/\omega} \\
&= \frac{\omega}{\pi} \times \frac{1}{r\omega} [-\cos 2r\pi + \cos r\pi] \\
&= \frac{1}{r\pi} (\cos r\pi - 1)
\end{aligned} \quad \dots(4)$$

and

$$\begin{aligned}
B_r &= \frac{2}{T} \left[\int_0^{T/2} y \cos r\omega t dt + \int_{T/2}^T y \cos r\omega t dt \right] \\
&= 0 + \frac{2}{T} \int_{T/2=\pi/\omega}^{T=2\pi/\omega} \cos r\omega t dt = \frac{2}{2\pi/\omega} \left[\frac{\sin r\omega t}{r\omega} \right]_{\pi/\omega}^{2\pi/\omega} = 0
\end{aligned} \quad \dots(5)$$

When r is even, $\cos 2r\pi = 1$ and when r is odd i.e., $r = 1, 3, 5, \dots$, $\cos r\pi = -1$. This means from (4)

$$A_r = 0 \quad \text{for even } r$$

$$A_r = -\frac{2}{r\pi} \quad \text{for odd } r \quad \dots(5)$$

Hence from (2), the Fourier's series for the given periodic function is

$$y = f(t) = \frac{1}{2} - \frac{2}{\pi} \sin \omega t - \frac{2}{3\pi} \sin 3\omega t \dots - \frac{2}{r\pi} \sin r\omega t \dots$$

or

$$y = f(t) = \frac{1}{2} - \frac{2}{\pi} \left[\sin \omega t + \frac{1}{3} \sin 3\omega t + \dots + \frac{1}{r} \sin r\omega t + \dots \right] \quad \text{for odd } r \quad \dots(6)$$

Ex. 3. Find the Fourier's coefficients for the function, represented by the following equations :

$$y = 0, \quad -\frac{T}{2} < t < -\frac{T}{4} \quad \dots(i)$$

$$y = a, \quad -\frac{T}{4} < t < +\frac{T}{4} \quad \dots(ii)$$

$$y = 0, \quad \frac{T}{2} < t < \frac{T}{2} \quad \dots(iii)$$

Write the fourier series also.

Sol. The given function is represented in **fig. 9.7**. Let the Fourier series be given by

$$y = A_0 + A_1 \sin \omega t + A_2 \sin 2\omega t + \dots + A_r \sin r\omega t + \dots + B_1 \cos \omega t + \dots + B_r \cos r\omega t + \dots$$

Fourier coefficients for the given function are evaluated as follows :

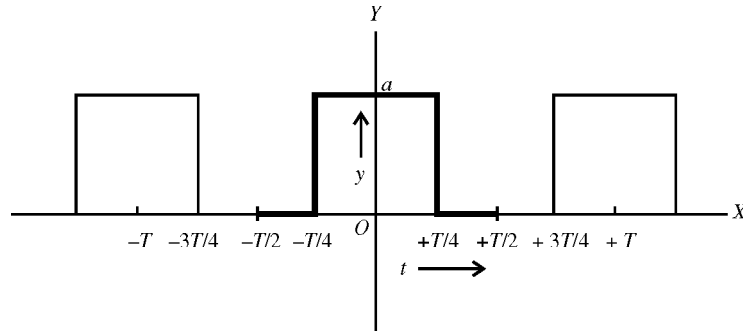


Fig. 9.7. Square wave.

$$A_0 = \frac{1}{T} \int_{-T/2}^{T/2} y dt = \frac{1}{T} \left[\int_{-T/2}^{-T/4} y dt + \int_{-T/4}^{+T/4} y dt + \int_{T/4}^{T/2} y dt \right]$$

$$= \frac{1}{T} \int_{-T/4}^{T/4} a dt = \frac{a}{T} \left[\frac{T}{4} + \frac{T}{4} \right] = \frac{a}{2}$$

$$A_r = \frac{2}{T} \int_{-T/2}^{T/2} y \sin r\omega t dt = \frac{2}{T} \int_{-T/4}^{+T/4} a \sin r\omega t dt = \frac{2a}{T} \left[\frac{-\cos r\omega t}{r\omega} \right]_{-T/4}^{+T/4}$$

$$= \frac{2a}{T} \left[-\frac{\cos \frac{2\pi}{T} rt}{2\pi r/T} \right]_{-T/4}^{+T/4} = \frac{2a}{T} \left[\frac{\cos \frac{r\pi}{2} - \cos r\pi/2}{2\pi r/T} \right] = 0 \quad \left(\because \omega = \frac{2\pi}{T} \right)$$

and

$$B_r = \frac{2}{T} \int_{-T/2}^{T/2} y \cos r\omega t dt = \frac{2}{T} \int_{-T/4}^{T/4} a \cos r\omega t dt$$

$$\begin{aligned}
&= \frac{2}{T} \int_{-T/2}^{T/2} y \cos \omega t \, dt = \frac{2a}{T} \left[\frac{\sin r \omega t}{r \omega} \right]_{-T/4}^{T/4} \\
&= \frac{2a}{T} \times \frac{T}{2\pi r} \times 2 \sin \left(r \times \frac{2\pi}{T} \times \frac{T}{4} \right) = \frac{2a}{r\pi} \sin \left(\frac{r\pi}{2} \right)
\end{aligned}$$

Thus the Fourier coefficients are

$$A_0 = \frac{a}{2}, \quad A_r = 0 \quad \text{and} \quad B_r = \frac{2a}{r\pi} \sin \left(\frac{r\pi}{2} \right)$$

Thus the Fourier series is

$$y = \frac{a}{2} + \frac{2a}{\pi} \left[\cos \omega t + \frac{1}{3} \cos 3\omega t + \frac{1}{5} \cos 5\omega t + \dots \right]$$

9.4.2 Fourier Analysis of Saw-Tooth Wave.

A saw-tooth wave, representing the displacement-time curve of a certain oscillatory system, is represented in fig. 9.8. For a saw-tooth wave, we see that when $t = 0$, $y = a$ and then the displacement decreases linearly with time and when $t = T$, $y = 0$. The curve repeats itself after $t = T$.

Now let us take a point P on the linear portion of the saw-tooth curve with coordinates (t, y) . From the similar triangles AOB and $PP'B$, we have

$$\frac{a}{y} = \frac{T}{T-t} \quad \text{or} \quad y = a \frac{(T-t)}{T}$$

Thus the saw-tooth wave form can be represented as

$$y = \frac{a}{T}(T-t) \quad \text{for } 0 < t < T \dots (1)$$

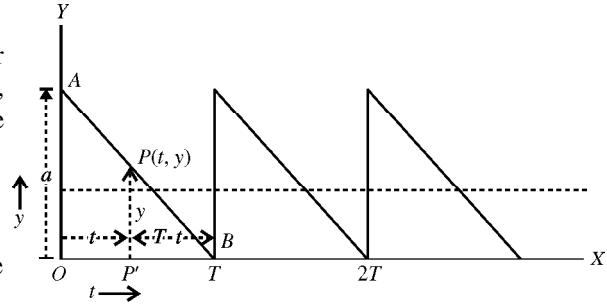


Fig. 9.8. Saw-Tooth wave.

The Fourier series is given by

$$y = A_0 + A_1 \sin \omega t + A_2 \sin 2\omega t + \dots + A_r \sin r\omega t + \dots + B_1 \cos \omega t + \dots + B_r \cos r\omega t + \dots \dots (2)$$

Fourier coefficients of the series for the given wave form are determined as follows :

$$A_0 = \frac{1}{T} \int_0^T y \, dt = \frac{a}{T} \int_0^T \left(1 - \frac{t}{T}\right) dt = \frac{a}{T} \left[t - \frac{t^2}{2T} \right]_0^T = \frac{a}{T} \left[T - \frac{T^2}{2T} \right] = \frac{a}{2} \dots (3)$$

Hence the axis of curve is at a distance $a/2$ up the time axis.

$$\begin{aligned}
A_r &= \frac{2}{T} \int_0^T y \sin r\omega t \, dt = \frac{2a}{T} \int_0^T \left(1 - \frac{t}{T}\right) \sin r\omega t \, dt \\
&= \frac{2a}{T} \int_0^T \sin r\omega t \, dt - \frac{2a}{T^2} \int_0^T t \sin r\omega t \, dt \\
&= \frac{2a}{T} \left\{ -\frac{\cos r\omega t}{r\omega} \right\}_0^T - \frac{2a}{T^2} \left\{ \left(-t \frac{\cos r\omega t}{r\omega} \right)_0^T + \int_0^T \frac{\cos r\omega t}{r\omega} dt \right\} \\
&\quad \text{(Integrating by parts)} \\
&= \frac{2a}{Tr\omega} \{ -\cos r\omega T + \cos(r\omega \times 0) \} + \frac{2a}{T^2} \cdot \frac{T \cos r\omega T}{r\omega} - \frac{2a}{T^2} \left(\frac{\sin r\omega t}{r^2 \omega^2} \right)_0^T
\end{aligned}$$

Now, $-\cos(r\omega T) + \cos 0 = -\cos(r\omega \cdot 2\pi/\omega) + 1 = -\cos 2\pi r + 1 = -1 + 1 = 0$

and $(\sin r\omega t)_0^T = 0 \quad \left[\because T = \frac{2\pi}{\omega} \right]$

$$\therefore A_r = \frac{2a}{T} \cdot \frac{\cos(2\pi r)}{r\omega} = \frac{2a}{Tr\omega} = \frac{2a\omega}{2\pi r\omega} = \frac{a}{r\pi} \quad \dots(4)$$

$$\text{and } B_r = \frac{2}{T} \int_0^T y \cos r\omega t \, dt = \frac{2a}{T} \int_0^T \left(1 - \frac{t}{T}\right) \cos r\omega t \, dt = 0 \quad \dots(5)$$

Hence all the cosine terms of Fourier's series are absent.

Hence, the Fourier's series in this case is

$$y = \frac{a}{2} + \frac{a}{\pi} \left(\sin \omega t + \frac{1}{2} \sin 2\omega t + \frac{1}{3} \sin 3\omega t + \dots + \frac{1}{r} \sin r\omega t + \dots \right) \quad \dots(6)$$

We note that in the series all the harmonics are present, having amplitude in the ratio $1 : \frac{1}{2} : \frac{1}{3} : \dots$ and frequencies in the ratio $1 : 2 : 3 : \dots$. Hence the value of the amplitude decreases for higher harmonics.

The addition of successive terms is represented in **fig 9.9**. We see that the resultant curve tends to resemble the saw-tooth wave form as we include more and more terms inside the bracket.

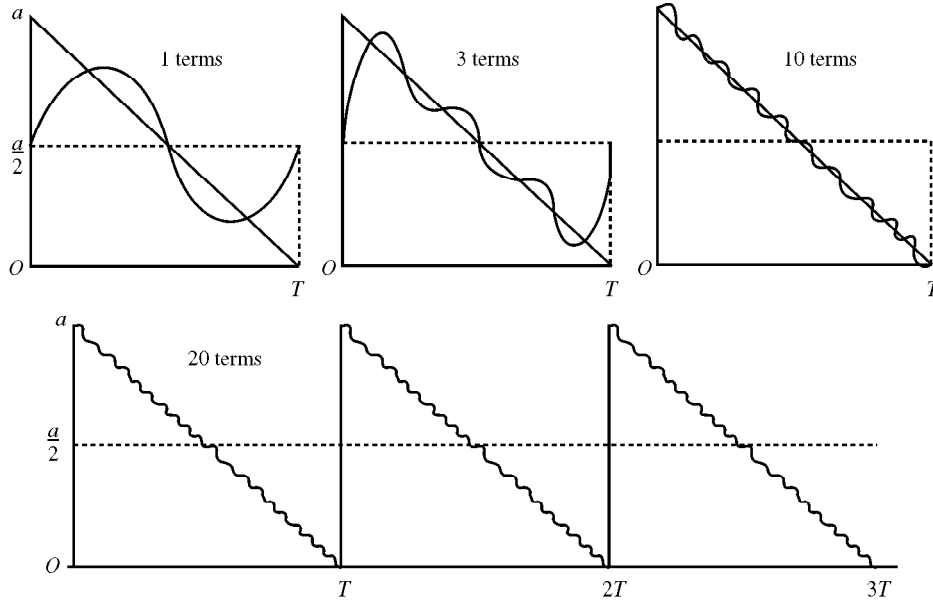


Fig. 9.9.

9.4.3. Fourier Analysis of Triangular wave.

Consider a triangular part OAB of the wave. On the linear part of the wave, let us take a point P with coordinates (t, y) . In similar triangles OAC and OPP' , we have

$$\frac{AC}{OC} = \frac{PP'}{OP'} \quad \text{or} \quad \frac{a}{T/2} = \frac{y}{t} \quad \text{or} \quad y = \frac{2at}{T} \quad \text{for } 0 < t < T/2$$

If we take a point P on AB straight line, then similarly

$$\frac{a}{T - T/2} = \frac{y}{T - t} \quad \text{or} \quad \frac{a}{T/2} = \frac{y}{T - t} \quad \text{or} \quad y = \frac{2a(T - t)}{T} \quad \text{for } T/2 < t < T$$

Hence the displacement at any time t for the triangular wave can be represented as

$$y = \frac{2at}{T} \quad \text{for } 0 < t < T/2$$

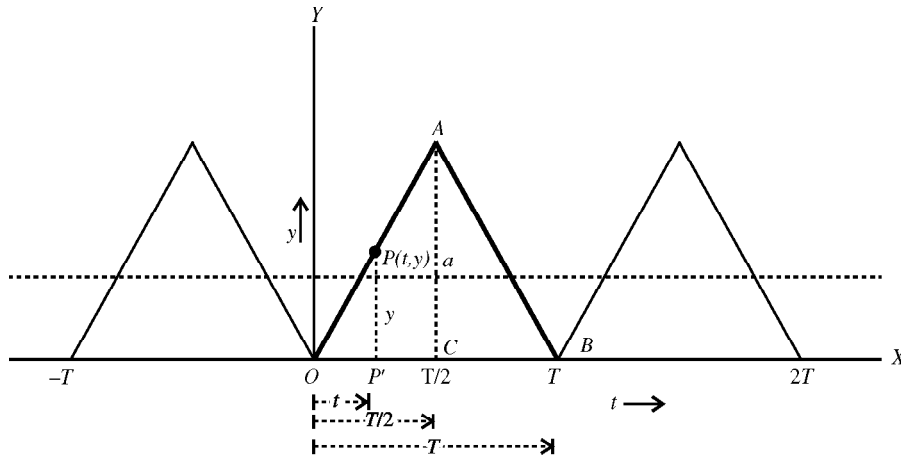


Fig. 9.10. Triangular wave.

and
$$y = \frac{2a(T-t)}{T} \quad \text{for } T/2 < t < T. \quad \dots(1)$$

After $t = T$ the curve repeats itself.

The Fourier series is given by

$$y = f(\omega t) A_0 + A_1 \sin \omega t + A_2 \sin 2\omega t + \dots + A_r \sin r\omega t + \dots + B_1 \cos \omega t + \dots + B_r \cos r\omega t + \dots \dots(2)$$

Now, let us calculate the Fourier coefficients :

$$\begin{aligned} A_0 &= \frac{1}{T} \int_0^T y dt = \frac{1}{T} \left[\int_0^{T/2} \frac{2at}{T} dt + \int_{T/2}^T \frac{2a(T-t)}{T} dt \right] \\ &= \frac{1}{T} \left[\frac{2a}{T} \left(\frac{t^2}{2} \right)_0^{T/2} + \frac{2a}{T} \left\{ Tt - \frac{t^2}{2} \right\}_{T/2}^T \right] = \frac{2a}{T^2} \left[\frac{T^2}{8} + \left(T^2 - \frac{T^2}{2} \right) - \left(\frac{T^2}{2} - \frac{T^2}{8} \right) \right] \\ &= \frac{a}{2} \end{aligned} \quad \dots(3)$$

This $A_0 = a/2$ represents the displacement of the axis of the curve from the time axis.

$$\begin{aligned} A_r &= \frac{2}{T} \int_0^T y \sin r\omega t dt \\ &= \frac{2}{T} \left[\int_0^{T/2} \frac{2at}{T} \sin r\omega t dt + \int_{T/2}^T \frac{2a(T-t)}{T} \sin r\omega t dt \right] \\ &= \frac{4a}{T^2} \left[\int_0^{T/2} t \sin r\omega t dt + T \int_{T/2}^T \sin r\omega t dt - \int_{T/2}^T t \sin r\omega t dt \right] \end{aligned} \quad \dots(4)$$

Now, let us evaluate the value of the integrals in the bracket, taking $T = 2\pi/\omega$.

$$\begin{aligned} I_1 &= \int_0^{T/2=\pi/\omega} t \sin r\omega t dt = \left[-t \frac{\cos r\omega t}{r\omega} \right]_0^{\pi/\omega} - \int_0^{\pi/\omega} -\frac{\cos r\omega t}{r\omega} dt \\ &= -\frac{\pi}{\omega} \frac{\cos r\pi}{r\omega} + \frac{1}{r\omega} \left[\frac{\sin r\omega t}{r\omega} \right]_0^{\pi/\omega} = -\frac{\pi \cos r\pi}{r\omega^2} \end{aligned} \quad \dots(5)$$

$$I_2 = T \int_{T/2}^T \sin r\omega t dt = \frac{2\pi}{\omega} \left[-\frac{\cos r\omega t}{r\omega} \right]_{\pi/\omega}^{2\pi/\omega} = \frac{2\pi(\cos r\pi - 1)}{r\omega^2} \quad \dots(6)$$

and

$$I_3 = \int_{T/2}^T t \sin r\omega t dt = \left[-t \frac{\cos r\omega t}{r\omega} \right]_{\pi/\omega}^{2\pi/\omega} + \int_{\pi/\omega}^{2\pi/\omega} \frac{\cos r\omega t}{r\omega} dt$$

$$= -\frac{2\pi}{\omega \times r\omega} + \frac{\pi \cos r\pi}{\omega r\omega} + \frac{1}{r\pi} \left[\frac{\sin r\omega t}{r\omega} \right]_{\pi/\omega}^{2\pi/\omega} = \frac{\pi(\cos r\pi - 2)}{r\omega^2} \quad \dots(7)$$

Substituting the solves of integrals from (5), (6) and (7) in (4), we have

$$\begin{aligned} A_r &= \frac{4a}{T^2} \left[-\frac{\pi \cos r\pi}{r\omega^2} + \frac{2\pi(\cos r\pi - 1)}{r\omega^2} - \frac{\pi(\cos r\pi - 2)}{r\omega^2} \right] \\ &= \frac{4a\pi}{T^2 r^2 \omega^2} [-\cos r\pi + 2\cos r\pi - 2 - \cos r\pi + 2] \\ &= 0 \end{aligned} \quad \dots(8)$$

$$\begin{aligned} B_r &= \frac{2}{T} \int_0^T y \cos r\omega t \, dt = \frac{2}{T} \left[\int_0^{T/2} \frac{2at}{T} \cos r\omega t \, dt + \int_{T/2}^T \frac{2a(T-t)}{T} \cos r\omega t \, dt \right] \\ &= \frac{4a}{T^2} \left[\int_0^{T/2} t \cos r\omega t \, dt + \int_{T/2}^T T \cos r\omega t \, dt - \int_{T/2}^T t \cos r\omega t \, dt \right] \end{aligned} \quad \dots(9)$$

Now, let us solve the integrals

$$\begin{aligned} I_1 &= \int_0^{T/2=\pi/\omega} t \cos r\omega t \, dt = \left[-t \frac{\sin r\omega t}{r\omega} \right]_0^{\pi/\omega} - \int_0^{\pi/\omega} -\frac{\sin r\omega t}{r\omega} \, dt \\ &= \left[\frac{\cos r\omega t}{(r\omega)^2} \right]_0^{\pi/\omega} = \frac{1}{r^2 \omega^2} (\cos r\pi - 1) \end{aligned} \quad \dots(10)$$

$$I_2 = T \int_{T/2}^T \cos r\omega t \, dt = T \left[-\frac{\sin r\omega t}{r\omega} \right]_{\pi/\omega}^{2\pi/\omega} = 0 \quad \dots(11)$$

$$\begin{aligned} I_3 &= \int_{T/2}^T t \cos r\omega t \, dt = \left[t \frac{\sin r\omega t}{r\omega} \right]_{\pi/\omega}^{2\pi/\omega} - \int_{\pi/\omega}^{2\pi/\omega} \frac{\sin r\omega t}{r\omega} \, dt \\ &= 0 + \left[\frac{\cos r\omega t}{(r\omega)^2} \right]_{\pi/\omega}^{2\pi/\omega} = \frac{1}{(r\omega)^2} (1 - \cos r\pi) \end{aligned} \quad \dots(12)$$

Hence,

$$\begin{aligned} B_r &= \frac{4a}{T^2} \left[\frac{1}{r^2 \omega^2} (\cos r\pi - 1) + 0 - \frac{1}{r^2 \omega^2} (1 - \cos r\pi) \right] \\ &= \frac{4a}{(2\pi/\omega)^2} \cdot \frac{2}{r^2 \omega^2} (\cos r\pi - 1) = \frac{2a}{\pi^2 r^2} (\cos r\pi - 1) \end{aligned} \quad \dots(13)$$

When r is even i.e., $r = 2, 4, 6, \dots$, $\cos r\pi = 1$.

$$\therefore B_r = \frac{2a}{\pi^2 r^2} (1-1) = 0 \quad \dots(14)$$

When r is odd i.e., $r = 1, 3, 5, \dots$, $\cos r\pi = -1$. Then

$$B_r = \frac{2a}{\pi^2 r^2} (-1-1) = -\frac{4a}{\pi^2 r^2} \quad \dots(15)$$

Substituting for A_0 , A_r and B_r from (3), (8), (14) and (15) in (2), we get the Fourier series as

$$\begin{aligned} y = f(\omega t) &= \frac{a}{2} - \frac{4a}{\pi^2 \times 1^2} \cos \omega t - \frac{4a}{\pi^2 \times 3^2} \cos 3\omega t \\ &\quad - \frac{4a}{\pi^2 \times 5^2} \cos 5\omega t \dots - \frac{4a}{\pi^2 \times r^2} \cos r\omega t \dots \end{aligned}$$

$$\text{or} \quad y = \frac{a}{2} - \frac{4a}{\pi^2} \left[\cos \omega t + \frac{1}{3^2} \cos 3\omega t + \frac{1}{5^2} \cos 5\omega t + \dots + \frac{1}{r^2} \cos r\omega t + \dots \right] \quad \dots(16)$$

Thus the triangular wave has the axis $y = a/2$ above the time axis and the component terms are the odd harmonics with frequencies in the ratio $1 : 3 : 5 : \dots$. We may plot the different terms of the series graphically and find their resultant. First term represents the value of the coefficient $A = a/2$ and it gives the displacement of the axis of complex vibration curve from the time axis. 1st, 2nd and 3rd terms in the bracket are represented by dotted curves in **fig. 9.11**. Their resultant is obtained by the superposition of these three curves with constant $A = a/2$ and is shown by thick full line curve (**fig. 9.11**). If a very large number of terms are considered, the resultant will resemble the triangular shape closely (**fig. 9.10**).

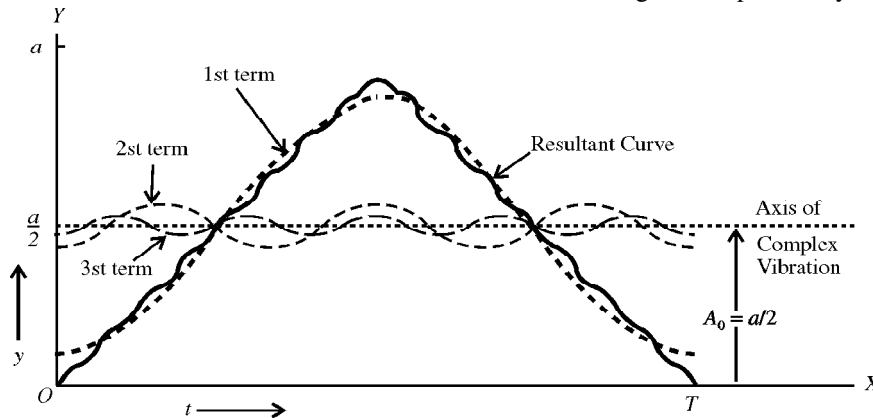


Fig. 9.11. Resultant of first three cosine terms of the series and mean displacement ($a/2$) of the axis of complex vibration.

9.5 Importance of Fourier Theorem

This theorem plays an important role in the analysis of complex periodic vibrations into harmonic components or in the synthesis of such vibrations from harmonic components. Fourier theorem has its applications to audio mechanical and electrical systems where periodic phenomena may occur. Stringed instruments, like piano, guitar, sitar and violin find important role in the world of music. These instruments produce musical sound by transverse vibrations of stretched strings. These vibrations are in general complex and are analysed into harmonic components by applying Fourier's theorem. The harmonic component having the lowest frequency is called fundamental and the other components, differing in frequency to that of fundamental, are called overtones or harmonics. Fourier theory allows us to excite or stop particular harmonic components of the vibrating string. The quality of musical sound depends upon the presence of the number of harmonics and sometimes on the absence of particular harmonic. For example, in piano, the strings are struck approximately $(1/7)$ th of the length so that the seventh harmonic which tends to dissonance with other notes in a chord is eliminated. Fourier theorem is also useful in the analysis of output of rectifier circuits and hence in the designing of suitable filter circuits.

Ex. 1. Find the Founier series expansion for the function $f(x) = 1$ for $-\pi \leq x \leq 0$ and $f(x) = 2$ for $0 \leq x \leq \pi$. (O.U. 2009)

Sol. Let the Founier series be

$$f(x) = A_0 + \sum_{r=1}^{\infty} A_r \sin rx + \sum_{r=1}^{\infty} B_r \cos rx$$

Now,

$$\begin{aligned} A_0 &= \int_{-\pi}^{\pi} f(x) dx = \frac{1}{2\pi} \int_{-\pi}^0 1 \times dx + \frac{1}{2\pi} \int_0^{\pi} 2 \times dx \\ &= \frac{1}{2\pi} [x]_{-\pi}^0 + \frac{1}{2\pi} [2x]_0^{\pi} = \frac{1}{2\pi} \times \pi + \frac{1}{2\pi} \times 2\pi = \frac{1}{2} + 1 = \frac{3}{2} \end{aligned}$$

$$\begin{aligned}
 A_r &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin rx \, dx = \frac{1}{\pi} \left[\int_{-\pi}^0 \sin rx \, dx + \int_0^{\pi} \sin rx \, dx \right] \\
 &= \frac{1}{\pi} \left[\frac{-\cos rx}{r} \right]_{-\pi}^0 + \frac{1}{\pi} \left[\frac{-2 \cos rx}{r} \right]_0^{\pi} = \frac{1}{\pi r} [-1 + \cos r\pi] + \frac{1}{\pi r} [-2 \cos r\pi + 2] \\
 &= \frac{1}{\pi r} (1 - \cos r\pi)
 \end{aligned}$$

When r is even, $A_r = 0$

When r is odd, $A_r = \frac{2}{\pi r}$

$$\begin{aligned}
 B_r &= \frac{1}{\pi r} \int_{-\pi}^{\pi} f(x) \cos rx \, dx = \frac{1}{\pi} \left[\int_{-\pi}^0 \cos rx \, dx + \int_0^{\pi} 2 \cos rx \, dx \right] \\
 &= \frac{1}{\pi r} \left[\frac{\sin rx}{r} \right]_{-\pi}^0 + \frac{2}{\pi} \left[\frac{\sin rx}{r} \right]_0^{\pi} = 0
 \end{aligned}$$

Hence Fournier series is

$$f(x) = \frac{3}{2} + \sum_{r=1}^{\infty} \frac{2}{\pi r} \sin rx, \text{ when } r=1, 3, 5, \dots$$

or
$$f(x) = \frac{3}{2} + \frac{2}{\pi} \left[\sin x + \frac{1}{3} \sin 3x + \frac{1}{5} \sin 5x + \dots + \frac{1}{r} \sin rx + \dots \right]$$

QUESTIONS

Long Answer Questions

1. State and explain Fourier's theorem. Describe how the Fourier coefficients can be evaluated. (S.K.U. 2007)
2. State Fourier's theorem. What are its limitations ? Explain how the constants can be evaluated? (A.N.U. 2008, 03)
3. State the Fourier theorem and evaluate the Fourier coefficients. (A.N.U. March 2005)
4. State and explain Fourier theorem. Explain how you analyse periodic functions using Fourier theorem?
5. Sate Fourier's theorem. Analyse a square wave with the help of Fourier Theroem. (S.V.U. 2002; O.U.2001; A.U.2001)
6. Discuss the application of Fourier's series for the analysis of a square wave. (A.N.U. March 2007)
7. State Fourier's theorem. Obtain Fourier series for sqaure wave. (S.K.U. 2005)
8. State the Fourier's theorem. Analyse a saw-tooth wave using Fourier's theorem. (K.U. 2007)
9. (a) What is Fourier theorem ? Obtain the Fourier coefficients. (O.U. 2008)
(b) Analyse the triangular wave using Fourier theorem. (O.U.2008)
10. State and explain Fourier theorem. How do you use it to analyse a triangular wave. (A.U. June 2007)

Short Answer Questions

1. What is Fourier theorem ?
2. Discuss the conditions for the application of Fourier theorem.
3. Expalin Fourier's theorem. (O.U. 2002)
4. State and explain Fourier theorem. (O.U. 2003; A.N.U. 2002; A.U. 2001)
5. State and explain Fourier theorem. Mention its applications. (A.U. March 2007)
6. What are the limitations of Fourier theorem ?

PROBLEMS

1. Apply Fourier's theorem to analyse the square wave given below (**fig. 9.12**).

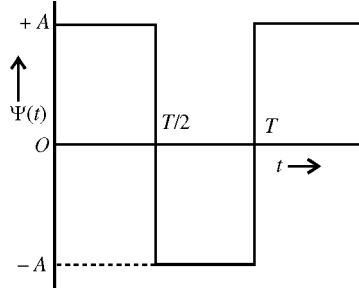


Fig. 9.12

[Ans. $\Psi(t) = \frac{4A}{\pi} (\sin \omega t + \frac{1}{3} \sin 3\omega t + \frac{1}{5} \sin 5\omega t + \dots)$]

[Hint. Mathematically the square wave may be expressed as :

$$\Psi(t) = +A \text{ for } 0 < t < T/2$$

$$\Psi(t) = -A \text{ for } T/2 < t < T$$

2. **Fig. 9.13** gives a periodic function. Obtain a Fourier analysis of this upto the fifth harmonic.

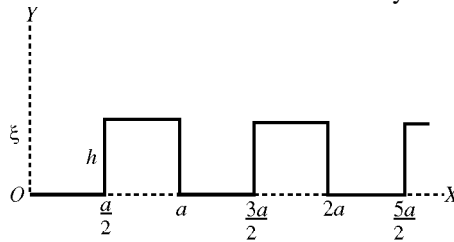


Fig. 9.13.

[Ans. $\xi = \frac{h}{2} - \frac{2h}{\pi} \left[\sin \frac{2\pi x}{a} + \frac{1}{3} \sin \frac{6\pi x}{a} + \frac{1}{5} \sin \frac{10\pi x}{a} + \dots \right]$]

[Hint. The given periodic function can be expressed as :

$$\xi = 0 \text{ for } 0 < x < a/2$$

$$\xi = h \text{ for } a/2 < x < a$$

3. Find the Fourier's series for the saw-tooth function given by

$$y = \frac{a}{T} t \text{ for } 0 < t < T$$

[Ans. $y = \frac{a}{2} - \frac{a}{\pi} \left(\sin \omega t + \frac{1}{2} \sin 2\omega t + \frac{1}{3} \sin 3\omega t + \dots \right)$]

4. State the conditions under which Fourier's theorem may be applicable. Apply it to deduce the harmonic components of a saw-tooth wave of frequency $n/2\pi$ and form given by

$$y = -\frac{a}{2} + a \frac{n}{2\pi} t \text{ for } t = 0 \text{ to } \frac{2\pi}{n}.$$

[Ans. $-\frac{a}{\pi} \left(\sin nt + \frac{1}{2} \sin 2nt + \frac{1}{3} \sin 3nt + \dots \right)$].

5. Find the Fourier series of the periodic function $f(t)$, the period of which is T and the form of which in the first period, i.e., $0 \leq t \leq T$ is given by $f(t) = t(T-t)$.

$$[\text{Ans. } y = \frac{T^2}{6} - \frac{T^2}{\pi^2} \left(\cos \omega t + \frac{1}{2^2} \cos 2\omega t + \frac{1}{3^2} \cos 3\omega t + \dots + \frac{1}{r^2} \cos r\omega t + \dots \right)]$$

6. Find the Fourier coefficients for the function represented by the following equations :

$$y = 0 \text{ for } -\frac{T}{2} < t < \frac{T_1}{2}, y = a \text{ for } -\frac{T_1}{2} < t < +\frac{T_1}{2}, \text{ and } y = 0 \text{ for } \frac{T_1}{2} < t < \frac{T}{2}.$$

$$[\text{Ans. } A_0 = \frac{T_1}{T} a, A_r = 0 \text{ and } B_r = \frac{2a}{r\pi} \left(\frac{r\pi T_1}{T} \right);$$

$$y = \frac{aT_1}{T} + \frac{2a}{\pi} \left[\sin \left(\frac{\pi T_1}{T} \right) \cos \omega t + \frac{1}{2} \sin \left(\frac{2\pi T_1}{T} \right) \cos 2\omega t + \frac{1}{3} \sin \left(\frac{3\pi T_1}{T} \right) \cos 3\omega t + \dots + \frac{1}{r} \sin \left(\frac{r\pi T_1}{T} \right) \cos r\omega t + \dots \right]$$

7. Find the Fourier series for the periodic function shown in **fig. 9.14** given by

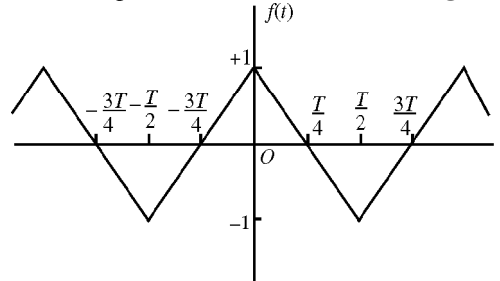


Fig. 9.14.

$$f(t) = \begin{cases} f(t) = 1 + \frac{4t}{T} & \text{for } -\frac{T}{2} < t < 0 \\ f(t) = 1 - \frac{4t}{T} & \text{for } 0 < t < \frac{T}{2} \end{cases}$$

$$[\text{Ans. } f(t) = \frac{8}{\pi^2} \left[\cos \omega t + \frac{1}{3^2} \cos (3\omega t) + \frac{1}{5^2} \cos (5\omega t) + \dots \right]]$$

8. A saw-tooth wave is represented by **fig. 9.15**, which may be expressed as.

$$y = f(t) = A \left(1 - \frac{t}{2T} \right) \text{ from } t = 0 \text{ to } t = T$$

Find the fourier series for this saw-tooth wave.

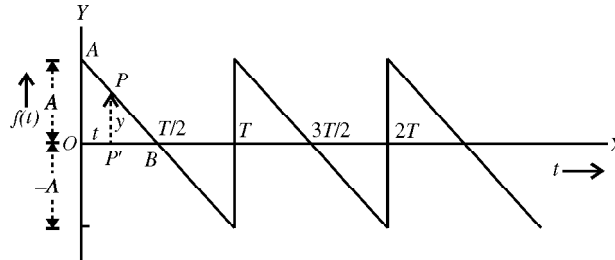


Fig. 9.15.

$$[\text{Ans. } f(t) = \frac{2A}{\pi} \left(\sin \omega t + \frac{1}{2} \sin 2\omega t + \frac{1}{3} \sin 3\omega t + \dots + \frac{1}{r} \sin r\omega t + \dots \right)]$$

$$[\text{Hint. } \frac{PP'}{OA} = \frac{P'B}{OB} \text{ or } \frac{y}{A} = \frac{(T/2) - t}{T/2} \text{ or } y = A \left(1 - \frac{2t}{T} \right)]$$

Vibrations of Bars

10-1. Introduction

A bar or rod is considered to be a body of homogeneous and isotropic material whose length is large in comparison to its diameter. A bar may be made to vibrate *longitudinally*, *transversely* or *torsionally*. The exact mode of vibration depends both on the way of stroking and on the manner the bar is clamped. For the vibration of a bar, it is not needed to be stretched like a wire, because in place of tension, it is the elasticity of the material of the bar which restores its successive portions to their original relative positions.

10-2. Longitudinal Vibrations in a Solid Bar—Establishment of Wave Equation and Velocity of Longitudinal Waves

When a bar is stroked at one end with a hammer or rubbed along its length with a resined cloth, *longitudinal vibrations* are set up in it, moving along its length with a velocity depending upon the elasticity and density of the material of the bar. Cross-sections of the bar vibrate to and fro about their mean positions in the direction of the axis and the elements of the bar are extended or contracted as the longitudinal wave passes through them.

Suppose that longitudinal waves are travelling along an elastic solid bar. Consider two close planes *A* and *B*, perpendicular to the axis of the bar at distances x and $x + \delta x$ respectively from some arbitrary origin [fig. 10.1]. If at any instant *A'* and *B'* are the positions of the planes *A* and *B* respectively, then the displacements at x and at $x + \delta x$ are y ($= AA'$) and $y + \delta y$ ($= BB'$) respectively. Hence the initial length *AB* between the two planes $= \delta x$.

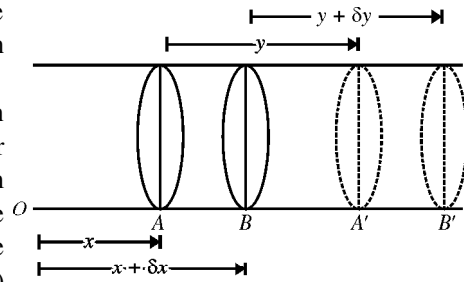


Fig. 10.1. Longitudinal vibrations in a bar.

$$\text{Change in this length} = A'B' - AB = (\delta x + \delta y) - \delta x = \delta y$$

$$[\because A'B' = (x + \delta x) + (y + \delta y) - (x + y) = \delta x + \delta y]$$

$$\therefore \text{Longitudinal strain} = \delta y / \delta x = \partial y / \partial x \quad \dots(1)$$

The change in length of the element *AB* of the bar occurs due to the difference of force acting on the two planes. The force acting on the plane *A* is given by

$$F_a = \text{stress} \times \text{area of cross-section of the bar.}$$

If α is the area of cross-section of the bar, the Young's modulus of its material is given by

$$Y = \frac{\text{stress}}{\text{strain}} \text{ or stress} = Y \frac{\partial y}{\partial x} \text{ or } \frac{F_a}{\alpha} = Y \frac{\partial y}{\partial x}$$

$$\therefore F_a = \alpha Y \frac{\partial y}{\partial x}$$

Similarly the force acting on the plane *B* is given by

$$F_b = F_a + \delta(F_a) = F_a + \frac{\partial}{\partial x}(F_a)\delta x$$

or,
$$F_b = \alpha Y \frac{\partial y}{\partial x} + \alpha Y \frac{\partial^2 y}{\partial x^2} \delta x$$

$\therefore$ Resultant force on the element

$$AB = F_b - F_a = \alpha Y \frac{\partial^2 y}{\partial x^2} \delta x .$$

If ρ is the density of material of the bar, then the mass of the element $AB = \rho \alpha \delta x$. If the acceleration of this small element AB is $\partial^2 y / \partial t^2$, then according to Newton's second law of motion, we have

$$\rho \alpha \delta x \frac{\partial^2 y}{\partial t^2} = \alpha Y \delta x \frac{\partial^2 y}{\partial x^2}$$

or
$$\frac{\partial^2 y}{\partial t^2} = \frac{Y}{\rho} \frac{\partial^2 y}{\partial x^2} \quad \dots(2)$$

We know that the standard differential equation of the wave motion is given by

$$\frac{\partial^2 y}{\partial t^2} = v^2 \frac{\partial^2 y}{\partial x^2} \quad \dots(3)$$

where v is the velocity of the waves.

Hence eq. (2) is the *differential equation for the longitudinal waves in a bar*. The velocity of the longitudinal waves in the bar is obtained by comparing eqs. (2) and (3) i.e.,

$$v^2 = \frac{Y}{\rho} \text{ or } v = \sqrt{\frac{Y}{\rho}} \quad \dots(4)$$

Thus, the *velocity of longitudinal waves in a bar depends upon the density and Young's modulus of its material*.

10.3. Wave Equation for Longitudinal Vibrations and its General Solution

The wave equation for longitudinal vibrations in a bar is given by

$$\frac{\partial^2 y}{\partial t^2} = v^2 \frac{\partial^2 y}{\partial x^2} \quad \dots(1)$$

where $v = \sqrt{\frac{Y}{\rho}}$. This is the one dimensional wave equation for longitudinal waves, travelling along X -axis (axis of the bar). Here y is the longitudinal displacement of a section of the bar and it is the function of x (distance from arbitrary origin) and time t . The general solution of eq. (5) must be a function of the linear combination of t and x i.e.,

$$y = f(t + \alpha x) \quad \dots(2)$$

where α is a constant.

Differentiating eq. (2) with respect to t , we obtain

$$\frac{\partial y}{\partial t} = f'(t + \alpha x)$$

Again differentiating with respect to t ,

$$\frac{\partial^2 y}{\partial t^2} = f''(t + \alpha x) \quad \dots(3)$$

Differentiating eq. (2) with respect to x ,

$$\frac{\partial y}{\partial x} = f'(t + \alpha x)$$

Again differentiating with respect to x ,

$$\frac{\partial^2 y}{\partial x^2} = \alpha^2 f''(t + \alpha x) \quad \dots(4)$$

Substituting the values of $\frac{\partial^2 y}{\partial t^2}$ and $\frac{\partial^2 y}{\partial x^2}$ from eqs. (3) and (4) in eq. (1), we obtain

$$f''(t + \alpha x) = v^2 \alpha^2 f''(t + \alpha x)$$

Whence $\alpha^2 = \frac{1}{v^2}$ or $\alpha = \pm \frac{1}{v}$... (5)

Hence the general solution of the wave equation (1) in view of eq. (2) is given by

$$y = f\left(t - \frac{x}{v}\right) + g\left(t + \frac{x}{v}\right) \quad \dots(10)$$

Clearly the displacement of a particle at a point x in the bar is due to two waves :

- (i) One wave [$y_1 = f(t - x/v)$] is travelling in the positive X -direction
and (ii) the other wave [$y_2 = g(t + x/v)$] is moving along the negative X -direction.

In case of a solid rod, first wave, travelling along the positive X -direction, is the incident wave (say) and the other wave moving along the negative X -direction is the reflected wave.

Simplest and most important case of the general solution of the wave equation is the *sinusoidal solution*, given by

$$y = a \sin \omega \left(t - \frac{x}{v}\right) + a \sin \omega \left(t + \frac{x}{v} + \phi\right) \quad \dots(11)$$

where a is the amplitude of vibration, $\omega/2\pi (= v)$ is the frequency of vibration and ϕ is the phase difference between the two waves. First sine wave [$y_1 = a \sin \omega(t - x/v)$] is the incident wave and the second sine wave [$y_2 = a \sin \{\omega(t + x/v) + \phi\}$] is the reflected wave. By superposition, these two waves may produce *stationary waves*.

10.4. Stationary Longitudinal Waves in Bars

Consider a rod of length l and take O as the origin of X -axis. Let

$$y_1 = a \sin \omega \left(t - \frac{x}{v}\right) \quad \dots(1)$$

represent a plane longitudinal sine wave travelling along the rod with amplitude a .

The wave reflected from the other end ($x = l$) can be represented as

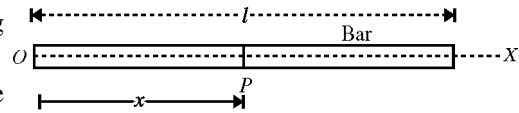


Fig. 10.2.

$$y_2 = a \sin \left[\omega \left(t + \frac{x}{v}\right) + \phi \right] \quad \dots(2)$$

where ϕ is the phase difference.

The incident and reflected waves may give rise to stationary waves so that the displacement at a distance x from the origin is given by

$$y = y_1 + y_2$$

or,
$$y = a \sin \omega \left(t - \frac{x}{v}\right) + a \sin \left[a \sin \omega \left(t + \frac{x}{v}\right) + \phi \right] \quad \dots(3)$$

In fact, this is the general solution of the wave equation in the sinusoidal form. Thus the general solution may represent stationary waves produced in a bar. Eq. (3) can be written as

$$y = 2a \cos \left(\frac{\omega x}{v} + \frac{\phi}{2} \right) \sin \left(\omega t + \frac{\phi}{2} \right)$$

or,
$$y = A \cos \left(kx + \frac{\phi}{2} \right) \sin \left(\omega t + \frac{\phi}{2} \right) \quad \dots(4)$$

where $A = 2a$, is the maximum value of the amplitude and $k = \omega/v = 2\pi/\lambda$ is the wave number corresponding to the wavelength λ .

10.5. Longitudinal Vibrations in a Bar, Free at Both Ends (Free-Free Bar)

The general solution of the wave equation for the longitudinal vibrations in a bar is given by

$$y = A \cos \left(kx + \frac{\phi}{2} \right) \sin \left(\omega t + \frac{\phi}{2} \right) \quad \dots(1)$$

When the standing waves are formed, the free ends of the bar $x=0$ and $x=l$ must be antinodes. Thus, we have the *boundary conditions* (in view of vanishing of $F = \alpha Y dy/dx$ at free ends)

$$(i) \quad \frac{dy}{dx} = 0 \text{ at } x=0 \text{ for all } t \quad \dots(2)$$

$$\text{and } (ii) \quad \frac{dy}{dx} = 0 \text{ at } x=l \text{ for all } t$$

Differentiating eq. (1) with respect to x , we get

$$\frac{dy}{dx} = -A k \sin \left(kx + \frac{\phi}{2} \right) \sin \left(\omega t + \frac{\phi}{2} \right) \quad \dots(3)$$

For boundary condition (i) i.e., at $x=0$, $\frac{dy}{dx} = 0$, from eq. (3), we have

$$0 = \sin \frac{\phi}{2} \text{ or, } \phi = 0 \quad \dots(4)$$

Hence, eq. (1) becomes

$$y = A \cos kx \sin \omega t \quad \dots(5)$$

The second boundary condition (ii) will give by using eq. (3) (for $\phi = 0$)

$$\sin kl = 0 \text{ or, } kl = n\pi, \text{ where } n = 1, 2, 3, \dots \quad \dots(6)$$

$$\text{whence} \quad l = \frac{n\pi}{k} = \frac{n\pi}{2\pi/\lambda} \text{ or } l = \frac{n\lambda}{2} \quad \dots(7)$$

Thus, in case of a bar, free at both ends, standing waves are produced, when the length of the bar is an integral multiple of $\lambda/2$.

We know that for acoustic waves,

$$v = v\lambda \text{ or } v = \frac{v}{\lambda} \quad \dots(8)$$

where v is the frequency of vibration.

$$\text{From eq. (7)} \quad \lambda = \frac{2l}{n} \quad \dots(9)$$

$$\text{Hence,} \quad v = \frac{nv}{2l} \quad \dots(10)$$

where $n = 1, 2, 3, \dots$

If Y is the value of Young's modulus and ρ is the density of the material of the bar, the velocity of the longitudinal waves is given by $v = \sqrt{Y/\rho}$ and hence the frequency of vibration is given by

$$v = \frac{n}{2l} \sqrt{\frac{Y}{\rho}} \quad \dots(11)$$

Thus longitudinal stationary waves are formed in the bar for certain discrete values of frequency [$v = v/2l, v/l, 3v/2l, \dots$, when $n = 1, 2, 3, \dots$ etc.]. At these frequencies, the rod is said to be in resonance and these special frequencies are called **resonant frequencies**.

Harmonics and Overtones

In case of a free-free bar, the standing waves are produced, when the frequency of longitudinal vibrations in the bar is given by

$$\therefore v = \frac{nv}{2l} = \frac{n}{2l} \sqrt{\frac{Y}{\rho}} \quad \dots(12)$$

where

$$n = 1, 2, 3, \dots$$

When $n = 1$, the lowest possible frequency, given by

$$v_1 = \frac{v}{2l} = \frac{1}{2l} \sqrt{\frac{Y}{\rho}} \quad (\text{first harmonic}) \quad \dots(13)$$

is called the **fundamental frequency** of vibration of the bar.

The other frequencies with which standing waves can be formed in the bar are

$$v_2 = \frac{2v}{2l} = \frac{2}{2l} \sqrt{\frac{Y}{\rho}} = 2v_1 \quad (\text{2nd harmonic}) \quad \dots(14)$$

$$v_3 = \frac{3v}{2l} = \frac{3}{2l} \sqrt{\frac{Y}{\rho}} = 3v_1 \quad (\text{3rd harmonic}) \quad \dots(15)$$

etc. Thus possible allowed frequencies of vibration of the bar are an integral multiple of the fundamental frequency. But integral multiples of a frequency are called harmonics, hence the frequencies $v_1, v_2, v_3, \dots$, which are in the ratio $1 : 2 : 3 : \dots$ with the fundamental frequency, represent the **harmonic series**. Fundamental frequency (v_1) is the **first harmonic** and $v_2, v_3, \dots$ are the **2nd harmonic, 3rd harmonic, ...** respectively.

The frequencies with which the bar can vibrate except the fundamental frequency (fundamental tone) are called **overtones**. Thus $v_2 = 2v_1$ is the first overtone, $v_3 = 3v_1$, the second overtone etc.

Normal Modes of Vibration

In a normal mode of vibration of an oscillating system, all its particles vibrate sinusoidally with the same frequency. When the bar vibrates according to eq. (5) with some natural frequency, we say that it is vibrating in a normal mode. For n th normal mode of vibration,

$$k = \frac{n\pi}{l} \quad \text{[(using eq. (6))]} \quad \dots(16a)$$

and

$$\omega = 2\pi v = \frac{n\pi v}{l} \quad \text{[using eq. (10)]} \quad \dots(16b)$$

The frequencies, given by $v = nv/2l$, for $n = 1, 2, 3, \dots$ are called **normal mode frequencies** or simply **normal frequencies** in different modes of vibration of the free-free bar.

Hence in the n th normal mode of vibration of a bar, the equation for displacement at any time t at distance x from the origin is given by [using eq. (5)] :

$$y_n = A_n \cos \frac{n\pi x}{l} \sin \frac{n\pi vt}{l} \quad \dots(17)$$

When $n = 1$, the bar is said to vibrate in *fundamental mode* or *first normal mode* of vibration with frequency given by eq. (13). In such a case, the equation of standing wave will be

$$y_1 = A_1 \cos \frac{\pi x}{l} \sin \frac{\pi vt}{l} \quad \dots(18)$$

At any point x , the amplitude of oscillation of a particle is $A_1 \cos (\pi x/l)$. At $x = 0$ or $x = l$, the value of the amplitude ($\pm A_1$) is maximum. It is zero at $x = l/2$. Thus in the fundamental mode of vibration of the bar, free at both ends, we have antinodes (A) at the ends and one node (N) in the middle [fig. 10.3 (a)].

When $n = 2$, the bar vibrates in *second normal mode* of vibration with frequency given by eq. (14). The equation of standing wave in this mode is

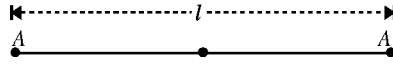
$$y_2 = A_2 \cos \frac{2\pi x}{l} \sin \frac{2\pi vt}{l} \quad \dots(19)$$

Now at $x = 0, l/2$ and l , the value of the amplitude will be maximum and it is zero at $x = l/4$ and $3l/4$.

Thus in the second mode of vibration, in addition to the two antinodes (A) at the ends of the bar, one

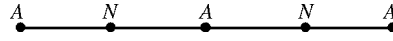
antinode (A) will be formed in the middle ($x = l/2$) and there will be two nodes (N) in between them [fig. 10.3 (b)].

Fundamental Tone
First Harmonic



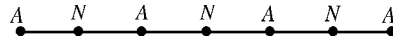
$$(a) \text{ Mode 1 or Fundamental Mode : } \lambda_1 = 2l, v_1 = \frac{v}{2l} = \frac{1}{2l} \sqrt{\frac{Y}{\rho}}$$

2nd Harmonic
First Overtone



$$(b) \text{ Mode 2 : } \lambda_2 = l, v_2 = \frac{v}{l} = \frac{2}{2l} \sqrt{\frac{Y}{\rho}} = 2v_1$$

3rd Harmonic
2nd Overtone



$$(c) \text{ Mode 3 : } \lambda_3 = \frac{2l}{3}, v_3 = \frac{3v}{2l} = \frac{3}{2l} \sqrt{\frac{Y}{\rho}} = 3v_1$$

Fig. 10.3. Normal modes of free-free bar.

In the third mode of vibration ($n = 3$), the equation of standing wave is

$$y_3 = A_3 \cos \frac{3\pi x}{l} \sin \frac{3\pi vt}{l} \quad \dots(20)$$

In this mode, the antinodes (A) will be at $x = 0, l/3, 2l/3$ and l and the nodes (N) at $x = l/6, l/2, 5l/6$ [fig. 10.3 (c)].

Similarly in the n th mode of vibration free-free of the bar there will be $(n + 1)$ antinodes and n nodes.

General Vibration of the Free-Free Bar and the Complete Solution

The general solution of the wave equation, given by eq. (17) i.e.,

$$y_n = A_n \cos \frac{n\pi x}{l} \sin \frac{n\pi vt}{l} \quad \dots(21)$$

represents the n th mode of vibration of a free-free bar with frequency $nv/2l$. As the wave equation is linear, the complete solution or the most general solution is obtained by the linear superposition of all the y_n i.e.,

$$y = \sum_{n=1}^{\infty} A_n \cos \frac{n\pi x}{l} \sin \frac{n\pi vt}{l} \quad \dots(22)$$

This represents the general longitudinal vibration of the bar, resulting from the superposition of all the modes of different frequencies.

10.6. Longitudinal Vibrations in a Bar, Fixed at Both Ends (Fixed-Fixed Bar)

We write the general solution of the wave equation of the longitudinal vibrations in the form :

$$y = A \cos \left(kx + \frac{\phi}{2} \right) \sin \left(\omega t + \frac{\phi}{2} \right) \quad \dots(1)$$

In the case of a bar fixed at both ends, there will be no displacement of particles at the ends of the bar ($x = 0$ and $x = l$). Thus, the boundary conditions are :

(i) $y = 0$ at $x = 0$ for all t , (ii) $y = 0$ at $x = l$ for all t

Eq. (1) is

$$y = A \cos \left(kx + \frac{\phi}{2} \right) \sin \left(\omega t + \frac{\phi}{2} \right) \quad \dots(2)$$

In case of boundary condition (i), we obtain

$$0 = \cos \phi/2 \text{ or } \phi/2 = \pi/2 \text{ or } \phi = \pi \quad \dots(3)$$

Hence eq. (2) takes the form

$$y = A \cos \left(kx + \frac{\pi}{2} \right) \sin \left(\omega t + \frac{\pi}{2} \right) = -A \sin kx \cos \omega t$$

or $y = B \sin kx \cos \omega t$... (4)

where we have put $B = -A$.

For boundary condition (ii),

$$\sin kl = 0 \text{ or } kl = n\pi, \text{ where } n = 1, 2, 3, \dots \quad \dots (5)$$

Hence $l = \frac{n\pi}{k} = \frac{n\pi}{2\pi/\lambda} = \frac{n\lambda}{2}$ or $\lambda = \frac{2l}{n}$... (6)

Now, frequency $\nu = \frac{v}{\lambda} = \frac{nv}{2l} = \frac{n}{2l} \sqrt{\frac{Y}{\rho}}$... (7)

where Y = Young's modulus and ρ = density of the material of the bar.

Thus longitudinal standing waves are formed in the fixed-fixed bar for frequency $\nu_1 = v/2l$, $\nu_2 = v/l$ or $\nu_3 = 3v/2l$ etc. Thus **free-free and fixed-fixed bars have identical frequencies.**

Harmonics and Overtones

The frequencies of harmonic series are obtained from relation (7), given by

$$\nu = \frac{nv}{2l} = \frac{n}{2l} \sqrt{\frac{Y}{\rho}} \quad \dots (8)$$

where eq. (8) for $n = 1, 2, 3, \dots$ represents first, second, third, harmonics respectively i.e.,

$$\nu_1 = \frac{1}{2l} \sqrt{\frac{Y}{\rho}} \quad (\text{fundamental or 1st harmonic})$$

$$\nu_2 = \frac{2v}{2l} = \frac{2}{2l} \sqrt{\frac{Y}{\rho}} = 2\nu_1 \quad (2\text{nd harmonic or 1st overtone})$$

$$\nu_3 = \frac{3v}{2l} = \frac{3}{2l} \sqrt{\frac{Y}{\rho}} = 3\nu_1 \quad (3\text{rd harmonic or 2nd overtone})$$

where ν_1 is the frequency of fundamental tone or first harmonic, ν_2 that of first overtone or second harmonic etc.

Normal Modes

Using eqs (4), (5) and (7) the n th normal mode of vibration of the bar is represented as

$$y_n = B_n \sin \frac{n\pi x}{l} \cos \frac{n\pi \nu t}{l} \quad \dots (9)$$

where $\omega = 2\pi\nu = n\pi\nu/l$ [using eq. (7)]. The frequencies, given by $\nu = nv/2l$ for $n = 1, 2, 3, \dots$ are called the normal frequencies in different modes of vibration of the bar, fixed at both ends.

The equations of standing waves in different modes can be represented as

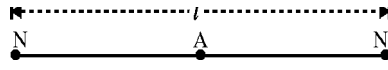
$$y_1 = B_1 \sin \frac{\pi x}{l} \cos \frac{\pi \nu t}{l} \quad (\text{fundamental mode or 1st normal mode})$$

$$y_2 = B_2 \sin \frac{2\pi x}{l} \cos \frac{2\pi \nu t}{l} \quad (2\text{nd normal mode})$$

$$y_3 = B_3 \sin \frac{3\pi x}{l} \cos \frac{3\pi \nu t}{l} \quad (3\text{rd normal mode})$$

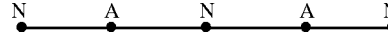
When the rod is vibrating in the fundamental mode, we find that at $x = 0$ and $x = l$, the amplitude is zero and at $x = l/2$ it is maximum ($\pm B_1$). Therefore, reflex in the fundamental mode of vibration of the fixed-fixed bar, we obtain two nodes (N) at the ends and one antinode (A) at the middle [fig. 10.4 (a)]. In the second normal mode, at $x = 0, l/2, l$, three nodes (N) and at $x = l/4, 3l/4$, two antinodes are formed [fig. 10.4 (b)].

Fundamental
First Harmonic



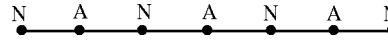
(a) **Mode 1 or Fundamental Mode** : $\lambda_1 = 2l$, $v_1 = \frac{v}{2l} = \frac{1}{2l} \sqrt{\frac{Y}{\rho}}$

2nd Harmonic
1st Overtone



(b) **Mode 2** : $\lambda_2 = l$, $v_2 = \frac{v}{l} = \frac{1}{l} \sqrt{\frac{Y}{\rho}} = 2v_1$

3rd Harmonic
2nd Overtone



(c) **Mode 3** : $\lambda_3 = 2l/3$, $v_3 = \frac{3v}{2l} = \frac{3}{2l} \sqrt{\frac{Y}{\rho}} = 3v_1$

Fig. 10.4. Normal modes of fixed-fixed bar.

In the 3rd mode, there are four nodes (N) and three antinodes (A) in between them at equal distances [fig. 10.4 (c)]. Thus, if the bar is vibrating in the n th normal mode, there will be $(n + 1)$ nodes and n antinodes.

In case of a fixed-fixed bar, the general solution of the wave equation, given by eq. (9), represents the n th mode of vibration. The *complete solution* representing the general vibration, is obtained by the linear superposition of all the modes of different frequencies i.e.,

$$y = \sum_{n=1}^{\infty} B_n \sin \frac{n\pi x}{l} \cos \frac{n\pi v t}{l} \quad \dots(10)$$

10.7. Longitudinal Vibrations in a Bar Fixed at the Mid Point

The general solution of the wave equation, representing stationary waves, for the longitudinal vibrations in a bar, is given by

$$y = A \cos \left(kx + \frac{\phi}{2} \right) \sin \left(\omega t + \frac{\phi}{2} \right) \quad \dots(1)$$

If the bar is fixed (or clamped) at the mid point, the stationary waves will be set up in such a manner that the mid point is always a node and the ends are antinodes. The boundary conditions are :

(i) $\frac{dy}{dx} = 0$ at $x = 0$ and $x = l$ at all t

(ii) $y = 0$ at $x = l/2$ at all t .

Differentiating eq. (1) with respect to x , we get

$$\frac{dy}{dx} = -Ak \sin \left(kx + \frac{\phi}{2} \right) \sin \left(\omega t + \frac{\phi}{2} \right)$$

For boundary condition (i), similar to the case of free-free bar [Art. 10.5], we may obtain the equation of standing waves in the bar as [eq. (5), Art. 10.5]

$$y = A \cos kx \sin \omega t \quad \dots(2)$$

where $k = n\pi/l$ or $\lambda = 2\pi/k = 2l/n$ and $\omega = 2\pi v = 2\pi \frac{v}{\lambda} = \frac{n\pi v}{l} = \frac{n\pi}{l} \sqrt{\frac{Y}{\rho}}$.

If we apply boundary condition (ii), we obtain from eq. (2)

$$0 = \cos \frac{n\pi}{l} \cdot \frac{l}{2} \text{ or } \cos \frac{n\pi}{2} = 0$$

whence

$$\frac{n\pi}{2} = (2r - 1) \frac{\pi}{2}, \text{ where } r = 1, 2, 3, \dots$$

or

$$n = 2r - 1 \text{ i.e., } n = 1, 3, 5, \dots \quad \dots(3)$$

Thus, eq. (2) takes the form

$$y_n = A_n \cos \frac{n\pi x}{l} \sin \frac{n\pi vt}{l} \quad \dots(4)$$

where

$$n = 1, 3, 5, \dots$$

The frequencies of the harmonics ($\nu = \omega/2\pi$) are given by

$$\nu = \frac{n\nu}{2l} = \frac{n}{2l} \sqrt{\frac{Y}{\rho}} \quad \dots(5)$$

Frequency of the fundamental note (or first harmonic) is obtained for $n = 1$ (or $r = 1$) i.e.,

$$\nu_1 = \frac{\nu}{2l} = \frac{1}{2l} \sqrt{\frac{Y}{\rho}} \quad \dots(6a)$$

If we put $n = 3$ (or $r = 2$) and $n = 5$ (or $r = 3$), the frequencies of the first and second overtones (or higher harmonics) are found to be

$$\nu_2 = \frac{3\nu}{2l} = \frac{3}{2l} \sqrt{\frac{Y}{\rho}} = 3\nu_1 \text{ and } \nu_3 = \frac{5\nu}{2l} = \frac{5}{2l} \sqrt{\frac{Y}{\rho}} = 5\nu_1 \quad \dots(6b)$$

Hence the frequencies of the harmonics are in the ratio $1 : 3 : 5 : \dots$. Thus we find that **when stationary waves are formed in a bar clamped at the mid point, only odd harmonics are found and even harmonics are absent.**

Writing $n = 2r - 1$ in eq. (4), we obtain the equation of r th mode of vibration of the bar i.e.,

$$y_r = A_r \cos (2r - 1) \frac{\pi x}{l} \sin (2r - 1) \frac{\pi vt}{l} \quad \dots(7)$$

For $r = 1, 2, 3$, we obtain the first three normal modes of vibration of the bar fixed at the mid point and the corresponding equations of standing waves are obtained from eq. (4) :

$$\begin{aligned} y_1 &= A_1 \cos \frac{\pi x}{l} \sin \frac{\pi vt}{l} && \text{(Fundamental mode)} \\ y_2 &= A_2 \cos \frac{3\pi x}{l} \sin \frac{3\pi vt}{l} && \text{(2nd mode)} \\ y_3 &= A_3 \cos \frac{5\pi x}{l} \sin \frac{5\pi vt}{l} && \text{(3rd mode)} \end{aligned} \quad \dots(8)$$

The first three modes of vibration of a bar fixed at the mid point and free at the two ends are shown in **fig. 10.5**. A mode is always formed at the fixed mid point and antinodes at the two free ends. In the fundamental mode of vibration, a node (N) is formed in the middle at $x = l/2$ and two antinodes (A) at the free ends ($x = 0$ and $x = l$). In the second mode of vibration, three nodes (N) and four antinodes (A) and in the third mode of vibration, 5 nodes (N) and six antinodes (A) are produced. Obviously in the r th mode of vibration of a bar, fixed at the mid point, there will be $(2r - 1)$ nodes and $2r$ antinodes.

Fundamental

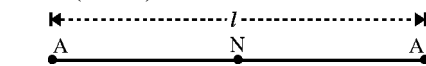
First harmonic

First overtone

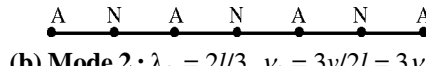
Third harmonic

Second overtone

Fifth harmonic



(a) Mode 1 or fundamental mode : $\lambda_1 = 2l$, $\nu_1 = v/2l$



(b) Mode 2 : $\lambda_2 = 2l/3$, $\nu_2 = 3v/2l = 3\nu_1$



(c) Mode 3 : $\lambda_3 = 2l/5$, $\nu_3 = 5v/2l = 5\nu_1$

Fig. 10.5. Normal modes of a bar fixed at the mid point.

The general solution of the wave equation for a bar fixed at the mid point, given by eq. (7), represents the r th mode of vibration of the bar. Hence the complete solution, representing the general vibration, will be obtained as

$$y = \sum_{n=1}^{\infty} y_n = \sum_{n=1}^{\infty} A_n \cos (2n - 1) \frac{\pi x}{l} \sin (2n - 1) \frac{\pi vt}{l} \quad \dots(9)$$

10.8. Longitudinal Vibrations in a Bar Fixed at One End and Free at the Other (Fixed-Free Bar)

The general solution of the wave equation for the longitudinal vibrations may represent stationary waves and is taken in the form

$$y = A \cos \left(kx + \frac{\phi}{2} \right) \sin \left(\omega t + \frac{\phi}{2} \right) \quad \dots(1)$$

When the standing waves are formed in a bar, fixed at one end and free at the other, a node is formed at the fixed end ($x = 0$) and an antinode on the free end ($x = l$). In such a case, the boundary conditions are :

(i) $y = 0$ at $x = 0$ for all t

$$(ii) \frac{dy}{dx} = 0 \text{ at } x = l \text{ for all } t \quad \dots(2)$$

Using boundary condition (i), we get

$$\cos \phi/2 = 0 \text{ or } \phi/2 = \pi/2 \text{ or } \phi = \pi \quad \dots(3)$$

Hence

$$y = -2a \sin kx \cos \omega t \quad \dots(4)$$

Now

$$\frac{dy}{dx} = -2ak \cos kx \cos \omega t$$

Using boundary condition (ii),

$$\cos kl = 0 \text{ or } kl = (2n-1) \frac{\pi}{2} \quad \dots(5)$$

where $n = 1, 2, 3, \dots$

$$\therefore k = (2n-1) \frac{\pi}{2l} \text{ or } \lambda = \frac{2\pi}{k} = \frac{4l}{2n-1} \quad \dots(6)$$

or

$$l = \frac{(2n-1)\lambda}{4} \quad \dots(7)$$

Thus, in case of a bar fixed at one end only, standing waves are formed when the length of the rod is an odd multiple of $\lambda/4$.

$$\text{Now, frequency, } \nu = \frac{v}{\lambda} = \frac{(2n-1)v}{4l} = \frac{(2n-1)}{4l} \sqrt{\frac{Y}{\rho}} \quad \dots(8)$$

where $n = 1, 2, 3, \dots$

Thus, longitudinal stationary waves are set up in the bar for certain discrete frequencies, corresponding to $n = 1, 2, 3, \dots$

$$\text{For } n = 1, \quad \nu_1 = \frac{v}{4l} = \frac{1}{4l} \sqrt{\frac{Y}{\rho}} \text{ (Fundamental tone/first harmonic)} \quad \dots(9a)$$

$$\text{For } n = 2, \quad \nu_2 = \frac{3v}{4l} = \frac{3}{4l} \sqrt{\frac{Y}{\rho}} = 3\nu_1 \text{ (First overtone/third harmonic)} \quad \dots(9b)$$

$$\text{For } n = 3, \quad \nu_3 = \frac{5v}{4l} = \frac{5}{4l} \sqrt{\frac{Y}{\rho}} = 5\nu_1 \text{ (Second overtone/fifth harmonic)} \quad \dots(9c)$$

We see that the frequencies of fundamental, first overtone, second overtone etc. are in the ratio $1 : 3 : 5 : \dots$ etc. and hence **for a bar fixed at one end, only odd harmonics are found and even harmonics are absent.**

First three normal modes of vibration are obtained for $n = 1, 2, 3$ and the corresponding equations of standing waves are given by eq. (4) i.e.,

$$y = -2a \sin kx \cos \omega t$$

where

$$k = (2n-1) \frac{\pi}{2l} \text{ and } \omega = 2\pi\nu = \frac{(2n-1)\pi v}{2l}$$

Thus,

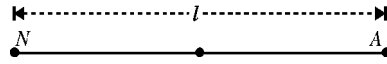
$$y = -2a \sin (2n-1) \frac{\pi x}{2l} \cos \frac{(2n-1)\pi v t}{2l} \quad \dots(10)$$

$$\text{For } n = 1, \quad y_1 = -2a \sin \frac{\pi x}{2l} \cos \frac{\pi vt}{2l} \quad (\text{Fundamental or first mode}) \dots (11a)$$

$$\text{For } n = 2, \quad y_2 = -2a \sin \frac{3\pi x}{2l} \cos \frac{3\pi vt}{2l} \quad (\text{Second mode}) \dots (11b)$$

$$\text{For } n = 3, \quad y_3 = -2a \sin \frac{5\pi x}{2l} \cos \frac{5\pi vt}{2l} \quad (\text{Third mode}) \dots (11c)$$

Fundamental tone
First harmonic



$$\text{(a) Mode 1 or fundamental mode : } \lambda = 4l, v_1 = \frac{v}{4l} = \frac{1}{4l} \sqrt{\frac{Y}{\rho}}$$

First overtone
3rd harmonic



$$\text{(b) Mode 2 : } \lambda_2 = \frac{4l}{3}, v_2 = \frac{3v}{4l} = \frac{3}{4l} \sqrt{\frac{Y}{\rho}} = 3v_1$$

Second overtone
5th harmonic



$$\text{(c) Mode 3 : } \lambda_3 = \frac{4l}{5}, v_3 = \frac{5v}{4l} = \frac{5}{4l} \sqrt{\frac{Y}{\rho}} = 5v_1$$

Fig. 10.6. Normal modes of a bar fixed at one end.

Fig. 10.6 shows first three normal modes of vibration of a bar fixed at one end and free at the other end. In the fundamental mode of vibration, a node is formed at the fixed end and an antinode at the free end at $x = l$. In the second mode of vibration, two nodes are formed at $x = 0$ and $x = 2l/3$ and two antinodes at $x = l/3$ and $x = l$. In the third normal mode, three nodes are obtained at $x = 0, 2l/5$ and $4l/5$ and three antinodes at $x = l/5, 3l/5$ and l . Thus in case of a bar fixed at one end, n th mode of vibration will have n nodes and n antinodes.

The general solution of the wave equation for a fixed free bar, given by eq. (10), represents the n th mode of vibration. The general vibration of the bar is represented by the complete solution of the wave equation and is obtained by the linear superposition of all the modes of different frequencies i.e.,

$$y = \sum_{n=1}^{\infty} (-2a) \sin(2n-1) \frac{\pi x}{2} \cos(2n-1) \frac{\pi vt}{2l} \quad \dots (12)$$

10.9. Transverse Vibrations of a Bar – Differential Wave Equation

Transverse vibrations can be produced in a bar or rod of elastic material. If a bar is clamped at one end and displaced at the free end through a small distance, then on releasing, the bar vibrates transversely. Transverse vibrations can also be produced in a free-free (free at both ends) bar by holding at the two ends and striking a blow near the middle.

When a bar executes transverse vibrations, the strains are elongation and contraction of the filaments of the rod on the two sides of the neutral axis. Hence the forces which come into play are those appearing in the bending of a beam. If we apply a force F perpendicular to the neutral axis, the bending takes place and the bending moment M is given by

$$M = \frac{YI_g}{R} \quad \dots (1)$$

where Y is the Young's modulus of the material of the bar and $I_g = AK^2$ is the geometrical moment of inertia of the section of the bar (A = area of cross-section of the bar and K = geometrical radius of gyration of the area

* In case of a rod of rectangular cross-section of breadth b and thickness (depth) d , $I_g = \frac{bd^3}{12}$ and $A = bd$;

$\therefore I_g = AK^2 = bdK^2 = bd^3/12$, whence $K = d/2\sqrt{3}$.

For a rod of circular cross-section of radius r ; $I_g = \pi r^4/4$ and $I_g = AK^2 = \pi r^2 K^2$; $\therefore \pi r^2 K^2 = \pi r^4/4$ or $K = r/2$.

of cross-section about the neutral plane)*. Here, R is the radius of curvature of the neutral surface, given by

$$R = \frac{\{1 + (\partial y / \partial x)^2\}^{3/2}}{\partial^2 y / \partial x^2}.$$

The X -axis is taken along the neutral axis and y is the displacement at the point x [fig. 10.7].

For small y , dy/dx is also very small and hence neglecting $(dy/dx)^2$ in comparison to 1, we obtain $R = 1/(\partial^2 y / \partial x^2)$. Therefore eq. (1) takes the form

$$M = YAK^2 \frac{\partial^2 y}{\partial x^2} \quad \dots(2)$$

Further the force F is related to the bending moment as

$$F = \frac{\partial M}{\partial x} \quad \dots(3)$$

Let us now consider an element PQ of length δx . The faces of the element, P and Q , are perpendicular to the neutral axis. Let F be the shearing force at the face P . Then the shearing force at the face Q is $F + \delta F = F + \frac{\partial F}{\partial x} \delta x$. Then the unbalanced force on the element is $-\frac{\partial F}{\partial x} \delta x$. If y be the displacement of the element at time t , then according to Newton's second law of motion, this force can be expressed as the product of mass ($m = \rho A \delta x$) and acceleration ($a = d^2 y / dt^2$) i.e.,

$$(\rho A \delta x) \frac{\partial^2 y}{\partial t^2} = -\frac{\partial F}{\partial x} \delta x \quad \dots(4)$$

where ρ is the density of the material of the element (bar) and A is the area of cross-section of the bar.

Eq. (4) gives

$$\rho A \frac{\partial^2 y}{\partial t^2} = -\frac{\partial F}{\partial x} \quad \dots(5)$$

On using eq. (3) and eq. (2) we obtain

$$\rho A \frac{\partial^2 y}{\partial t^2} = -\frac{\partial}{\partial x} \left(\frac{\partial M}{\partial x} \right) = -\frac{\partial^2}{\partial x^2} \left(YAK^2 \frac{\partial^2 y}{\partial x^2} \right)$$

or
$$\rho A \frac{\partial^2 y}{\partial t^2} = -YAK^2 \frac{\partial^4 y}{\partial x^4} \text{ or } \frac{\partial^4 y}{\partial x^4} = \frac{\rho}{YK^2} \cdot \frac{\partial^2 y}{\partial t^2}$$

or
$$\frac{\partial^4 y}{\partial x^4} + \frac{\rho}{YK^2} \cdot \frac{\partial^2 y}{\partial t^2} = 0 \quad \dots(6)$$

This is the **differential wave equation for the transverse waves** in a bar or rod.

10-10. General Solution of the Wave Equation for Transverse Vibrations

The wave equation for transverse vibrations of a bar is given by

$$\frac{\partial^4 y}{\partial x^4} + \frac{\rho}{AK^2} \frac{\partial^2 y}{\partial t^2} = 0 \quad \dots(1)$$

Let the bar be executing simple harmonic transverse vibrations. Suppose that the bar vibrates with an amplitude $a(x)$, where a is a function of x , so that the solution is of the form

$$y = a(x) \sin \omega t \quad \dots(2)$$

where ω is the angular frequency of vibration.

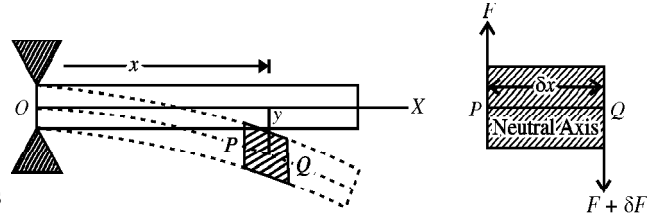


Fig. 10.7. Transverse vibrations of a bar.

Hence,
$$\frac{\partial^2 y}{\partial t^2} = -\omega^2 a(x) \sin \omega t \quad \dots(3a)$$

and
$$\frac{\partial^4 y}{\partial x^4} = \frac{d^4 a(x)}{dx^4} \sin \omega t \quad \dots(3b)$$

Hence, eq. (1) takes the form

$$\frac{d^4 a(x)}{dx^4} - \frac{\rho \omega^2}{YK^2} a(x) = 0$$

or
$$\frac{d^4 a(x)}{dx^4} - \alpha^4 a(x) = 0 \quad \dots(4)$$

where
$$\alpha^4 = \rho \omega^2 / YK^2 \quad \dots(5)$$

To solve eq. (4), we put $\frac{d^4}{dx^4} = D^4$. Then, eq. (4) takes the form

$$D^4 a(x) - \alpha^4 a(x) = 0 \quad \text{or} \quad (D^4 - \alpha^4) a(x) = 0$$

or
$$(D^2 - \alpha^2)(D^2 + \alpha^2) a(x) = 0$$

Therefore, either
$$(D^2 - \alpha^2) a(x) = 0 \quad \text{or} \quad (D^2 + \alpha^2) a(x) = 0$$

i.e., either
$$\frac{d^2 a(x)}{dx^2} - \alpha^2 a(x) = 0 \quad \text{or} \quad \frac{d^2 a(x)}{dx^2} + \alpha^2 a(x) = 0 \quad \dots(6)$$

The solution of
$$\frac{d^2 a(x)}{dx^2} - \alpha^2 a(x) = 0$$
 is
$$a(x) = A \cosh \alpha x + B \sinh \alpha x \quad \dots(7)$$

Also the solution of
$$\frac{d^2 a(x)}{dx^2} + \alpha^2 a(x) = 0$$
 is
$$a(x) = C \cos \alpha x + D \sin \alpha x \quad \dots(8)$$

The general solution is the linear combination of the solutions (7) and (8). Hence

$$a(x) = A \cosh \alpha x + B \sinh \alpha x + C \cos \alpha x + D \sin \alpha x \quad \dots(9)$$

The complete general solution of the wave equation (1) is obtained by putting for $a(x)$ from (9) into eq. (2) i.e.,

$$y = (A \cosh \alpha x + B \sinh \alpha x + C \cos \alpha x + D \sin \alpha x) \sin \omega t \quad \dots(10)$$

where A, B, C and D are arbitrary constants to be determined from the boundary conditions.

Boundary Conditions

(1) At the Fixed (or Clamped) End :

$y = 0$ for all t , because there is no displacement at the fixed end,

$$\frac{dy}{dx} = 0 \quad \text{for all } t, \text{ because there is no slope at the fixed end.}$$

(2) At the Free End :

$$\frac{d^2 y}{dx^2} = 0 \quad \text{for all } t, \text{ because there is no bending moment at the free end i.e., } M = YAK^2 \frac{d^2 y}{dx^2} = 0.$$

$$\frac{d^3 y}{dx^3} = 0 \quad \text{for all } t, \text{ because there is no force at the free end i.e., } F = \frac{dM}{dx} = YAK^2 \frac{d^3 y}{dx^3} = 0.$$

10-11. Transverse Vibrations in a Bar Fixed at One End and Free at the Other (Clamped Free Bar)

The general solution of the wave equation for transverse waves in a bar is given by

$$y = (A \cosh \alpha x + B \sinh \alpha x + C \cos \alpha x + D \sin \alpha x) \sin \omega t \quad \dots(1)$$

where A, B, C, D are constants and $\alpha = (\rho \omega^2 / YK^2)^{1/4}$ (ρ = density of the material of the rod, Y = Young's modulus, K = geometrical radius of gyration of the area of cross-section of the bar about neutral plane and ω = angular frequency).

Let the bar be clamped at the end $x = 0$ and free at the end $x = l$. The *boundary conditions* are :

(i) At $x = 0, y = 0$ and $\frac{dy}{dx} = 0$ for all t .

(ii) At $x = l, \frac{d^2y}{dx^2} = 0$ and $\frac{d^3y}{dx^3} = 0$ for all t .

Differentiating eq. (1) with respect to x and further differentiations yield

$$\frac{dy}{dx} = (A\alpha \sinh \alpha x + B\alpha \cosh \alpha x - C\alpha \sin \alpha x + D\alpha \cos \alpha x) \sin \omega t \quad \dots(2)$$

$$\frac{d^2y}{dx^2} = (A\alpha^2 \cosh \alpha x + B\alpha^2 \sinh \alpha x - C\alpha^2 \cos \alpha x - D\alpha^2 \sin \alpha x) \sin \omega t \quad \dots(3)$$

$$\frac{d^3y}{dx^3} = (A\alpha^3 \sinh \alpha x + B\alpha^3 \cosh \alpha x + C\alpha^3 \sin \alpha x - D\alpha^3 \cos \alpha x) \sin \omega t \quad \dots(4)$$

If we put the condition (i) in eqs. (1) and (2), we obtain

$$A + C = 0 \text{ and } B + D = 0$$

Hence

$$C = -A \text{ and } D = -B$$

... (5)

Putting condition (ii) in eqs. (3) and (4), we get

$$A (\sinh \alpha l + \cos \alpha l) + B (\sinh \alpha l + \sin \alpha l) = 0 \quad \dots(6)$$

$$A (\cosh \alpha l - \sin \alpha l) + B (\cosh \alpha l + \cos \alpha l) = 0 \quad \dots(7)$$

For nonvanishing A and B , the determinant of eqs. (6) and (7) vanishes i.e.,

$$\begin{vmatrix} \cosh \alpha l + \cos \alpha l & \sinh \alpha l + \sin \alpha l \\ \sinh \alpha l - \sin \alpha l & \cosh \alpha l + \cos \alpha l \end{vmatrix} = 0$$

$$\text{or } (\cosh \alpha l + \cos \alpha l)^2 - (\sinh^2 \alpha l - \sin^2 \alpha l) = 0$$

$$\text{or } 2 + 2 \cosh \alpha l \cos \alpha l = 0 \quad [\because \cosh^2 \alpha l - \sinh^2 \alpha l = 1 \text{ and } \cos^2 \alpha l + \sin^2 \alpha l = 1]$$

$$\text{or } (\cosh \alpha l + \cos \alpha l) = -1$$

$$\text{or } \cosh \alpha l = -\sec \alpha l \quad \dots(8)$$

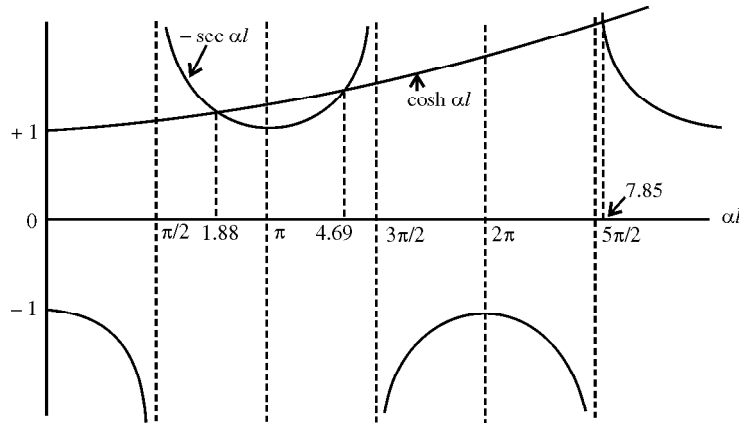


Fig. 10.8. Graphical solution of eq. (8)

The solution of the eq. (8) is obtained by plotting the graph of the two sides of the eq. (8) i.e., $\cosh \alpha l$ and $-\sec \alpha l$ against the angle αl and locating the points of intersection of the two curves [fig. 10.8]. The solutions are obtained to be

$$\alpha l = 1.88, 4.69, 7.85, \dots \quad \dots(9)$$

which are close to $\pi/2, 3\pi/2, 5\pi/2, \dots$, but are exactly equal at higher values.

In eq. (1), the expression for α is

$$\alpha = \left(\frac{\rho \omega^2}{YK^2} \right)^{1/4} \quad \text{or} \quad \alpha^4 = \frac{\rho \omega^2}{YK^2}$$

$$\text{Hence} \quad \omega^2 = \frac{\alpha^4 K^2 Y}{\rho} \quad \text{or} \quad \omega = \alpha^2 K \sqrt{\frac{Y}{\rho}} = \alpha^2 K v \quad \dots(10)$$

where $v = \sqrt{Y/\rho}$ is the velocity of longitudinal waves.

Eq. (10) represents the expression for the *angular frequency* of vibration. Hence the **frequency of vibration** is given by

$$v = \frac{\omega}{2\pi} = \frac{\alpha^2 K v}{2\pi} = \frac{\alpha^2 K}{2\pi} \sqrt{\frac{Y}{\rho}} \quad \dots(11)$$

where $\alpha = \frac{1.88}{l}, \frac{4.69}{l}, \frac{7.85}{l}, \dots$

For a bar of thickness d , $K = d/\sqrt{12}$ and for a bar of radius r , $K = r/2$.

Thus the bar can vibrate in different modes for different values of α , but here upper modes or overtones will not have frequencies in exact integral multiples of the lowest mode (for $\alpha = 1.88/l$). The lowest mode is called *fundamental mode of vibration*. Thus **the frequencies of overtones are not harmonics of fundamental frequency**. First three modes of vibration of a clamped free bar are shown in fig. 10.9. Remember that when upper modes have frequencies in exact integral multiples of the frequency of the fundamental mode, they are called *harmonics*, otherwise they are called *overtones* or *upper partials*.

From eq. (11), the frequency of the *fundamental* is given by

$$v_1 = \frac{(1.88)^2}{l^2} \cdot \frac{K v}{2\pi} = \frac{(1.88)^2}{l^2} \cdot \frac{K}{2\pi} \sqrt{\frac{Y}{\rho}} \quad \dots(12)$$

The frequencies of the **first and second overtones** are given by

$$v_2 = \frac{(4.69)^2}{l^2} \cdot \frac{K v}{2\pi} = \frac{(4.69)^2}{l^2} \cdot \frac{K}{2\pi} \sqrt{\frac{Y}{\rho}} \quad \dots(13)$$

$$\text{and} \quad v_3 = \frac{(7.85)^2}{l^2} \cdot \frac{K v}{2\pi} = \frac{(7.85)^2}{l^2} \cdot \frac{K}{2\pi} \sqrt{\frac{Y}{\rho}} \quad \dots(14)$$

Thus $v_1 : v_2 : v_3 = 1 : 6.27 : 17.55$ and hence the partials do not form a harmonic series. Therefore, the sound emitted by the vibrating bar transversely is *nonmelodius*. This is why in the musical instruments, the vibrating bars or reeds are seldomly used.

10-12. Transverse Vibrations in a Bar Free at Both Ends (Free-Free Bar)

We write the general solution of the wave equation for transverse waves in a bar in the form

$$y = (A \cosh \alpha x + B \sinh \alpha x + C \cos \alpha x + D \sin \alpha x) \sin \omega t \quad \dots(1)$$

where A, B, C, D are constants and $\alpha = (\rho \omega^2 / YK^2)^{1/4}$.

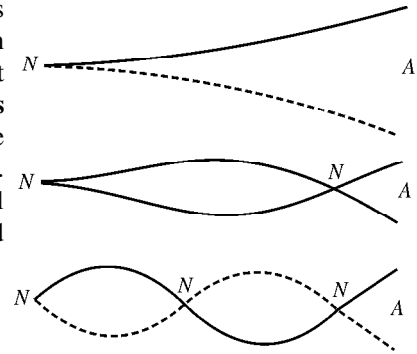


Fig. 10.9. First three modes of vibration of clamped free bar.

When the bar is free at both ends i.e., at $x = 0$ and $x = l$, the *boundary conditions* are :

(i) At $x = 0$, $\frac{d^2 y}{dx^2} = 0$ and $\frac{d^3 y}{dx^3} = 0$ for all t .

(ii) At $x = l$, $\frac{d^2 y}{dx^2} = 0$ and $\frac{d^3 y}{dx^3} = 0$ for all t .

From (1), differentiating y with respect to x two times and three times, we get

$$\frac{d^2 y}{dx^2} = (A\alpha^2 \cosh \alpha x + B\alpha^2 \sinh \alpha x - C\alpha^2 \cos \alpha x - D\alpha^2 \sin \alpha x) \sin \omega t \quad \dots(2)$$

$$\frac{d^3 y}{dx^3} = (A\alpha^3 \sinh \alpha x + B\alpha^3 \cosh \alpha x + C\alpha^3 \sin \alpha x - D\alpha^3 \cos \alpha x) \sin \omega t \quad \dots(3)$$

Using condition (i) in eqs. (2) and (3), we get

$$A - C = 0 \text{ and } B - D = 0$$

or

$$C = A \text{ and } D = B$$

... (4)

Applying condition (ii), in eqs. (2) and (3), we obtain

$$A (\cosh \alpha l - \cos \alpha l) = B (\sin \alpha l - \sinh \alpha l) \quad \dots(5)$$

$$A (\sinh \alpha l + \sin \alpha l) = B (\cos \alpha l - \cosh \alpha l) \quad \dots(6)$$

Dividing eq. (5) by eq. (6),

$$\frac{\cosh \alpha l - \cos \alpha l}{\sinh \alpha l + \sin \alpha l} = \frac{\sin \alpha l - \sinh \alpha l}{\cos \alpha l - \cosh \alpha l}$$

$$(\cosh \alpha l - \cos \alpha l)^2 = \sinh^2 \alpha l - \sin^2 \alpha l$$

Using

$$\cosh^2 \alpha l - \sinh^2 \alpha l = \cos^2 \alpha l + \sin^2 \alpha l = 1, \text{ we get}$$

$$2 - 2 \cosh \alpha l \cos \alpha l = 0$$

or

$$\cosh \alpha l = \sec \alpha l$$

... (7)

The graphical solution of eq. (7) gives

$$\alpha l = 4.73, 7.85, 10.996, \dots$$

... (8)

Now from the relation

$$\alpha = (\rho \omega^2 / YK^2)^{1/4},$$

we get the expressions for angular frequency (ω) and frequency (ν) as :

$$\omega = \alpha^2 K \sqrt{\frac{Y}{\rho}} = \alpha^2 K \nu \text{ and } \nu = \frac{\alpha^2 K}{2\pi} \sqrt{\frac{Y}{\rho}} = \frac{\alpha^2 K \nu}{2\pi} \quad \dots(9)$$

where $\nu = \sqrt{Y/\rho}$ is the velocity of longitudinal waves in the bar.

Hence the frequency of the fundamental mode is

$$\nu_1 = \frac{(4.73)^2}{l^2} \cdot \frac{K}{2\pi} \sqrt{\frac{Y}{\rho}} = \frac{(4.73)^2}{l^2} \cdot \frac{K \nu}{2\pi} \quad \dots(10)$$

The frequencies of the first two overtones are :

$$\nu_2 = \frac{(7.85)^2}{l^2} \cdot \frac{K}{2\pi} \sqrt{\frac{Y}{\rho}} = \frac{(7.85)^2}{l^2} \cdot \frac{K \nu}{2\pi}$$

and $\nu_3 = \frac{(10.996)^2}{l^2} \cdot \frac{K}{2\pi} \sqrt{\frac{Y}{\rho}} = \frac{(10.996)^2}{l^2} \cdot \frac{K \nu}{2\pi} \dots(11)$

Hence the frequencies of the first three modes of vibration of a free-free bar are in the ratio $\nu_1 : \nu_2 : \nu_3 = 1 : 2.76 : 5.40$.

Thus, the partials for the transverse vibrations of a free-face bar do not give a harmonic series. These modes of the free-free bar are shown in **fig. 10.10**.

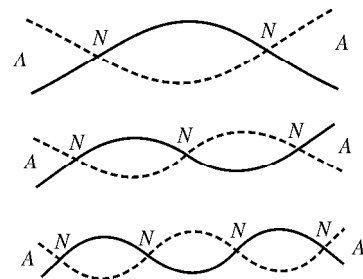


Fig. 10.10. First three modes of a free-free bar.

Comparison of Frequencies for Longitudinal and Transverse Vibrations

We consider the frequencies for longitudinal and transverse vibrations, when a clamped free bar vibrates in fundamental mode.

Frequency of longitudinal vibrations in fundamental mode :

$$v_l = \frac{v}{4l} = \frac{1}{4l} \sqrt{\frac{Y}{\rho}} \quad \dots(12)$$

Frequency of transverse vibrations in fundamental mode :

$$v_t = \frac{(1.88)^2}{l^2} \cdot \frac{K}{2\pi} \sqrt{\frac{Y}{\rho}} \quad \dots(13)$$

$$\text{Hence, } \frac{v_t}{v_l} = \frac{2.24K}{l} \quad \dots(14)$$

Since the value of K ($K = r/2$ for cylindrical bar and $K = d/\sqrt{12}$ for a bar of thickness d), is much less than l , the frequency of transverse vibrations is much less than the frequency of longitudinal vibrations for a bar when in both cases fundamental mode of vibration is considered.

10-13. Velocity of Transverse Waves in a Bar

The differential wave equation for the transverse waves in a rod is given by

$$\frac{\partial^4 y}{\partial x^4} + \left(\frac{\rho}{YK^2} \right) \frac{\partial^2 y}{\partial t^2} = 0 \quad \dots(1)$$

Let the solution of the wave equation be in the plane wave form :

$$y = ae^{i(\omega t - kx)} \quad \dots(2)$$

where ω is the angular frequency and $k = 2\pi/\lambda$ is the wave number.

$$\text{From (2)} \quad \frac{\partial^4 y}{\partial x^4} = k^4 y \text{ and } \frac{\partial^2 y}{\partial t^2} = -\omega^2 y$$

Putting the values of $\partial^4 y/\partial x^4$ and $\partial^2 y/\partial t^2$ in eq. (1), we obtain

$$k^4 y - \left(\frac{\rho}{YK^2} \right) \omega^2 y = 0 \quad \dots(3)$$

The wave velocity (u) is given by

$$u = \frac{\omega}{k} \quad \dots(4)$$

$$\text{Hence } k = \frac{\omega}{u} \quad \dots(5)$$

Substituting $k = \omega/u$ in eq. (3), we get

$$\frac{\omega^4}{u^4} - \left(\frac{\rho}{YK^2} \right) \omega^2 = 0 \text{ or } u^4 = \frac{\omega^2 Y}{\rho} K^2$$

Since $v = \sqrt{Y/\rho}$ represents the velocity of longitudinal waves in the bar, we have

$$u^4 = \omega^2 v^2 K^2 \quad \dots(6)$$

whence

$$u = \sqrt{\omega v K}$$

This is the *expression for the wave velocity* for the transverse vibration of a bar.

From eq. (6), we see that the wave velocity (u) is not constant but a function of angular frequency ω .

Hence the medium is *dispersive* for transverse waves in rods.

From eq. (4), if we put $\omega = ku$, then

$$u = \sqrt{kuvK} \text{ or } \sqrt{u} = \sqrt{kvK}$$

or

$$u = kvK = \frac{2\pi K}{\lambda} \sqrt{\frac{Y}{\rho}} \quad \dots(7)$$

This is another form of the expression for the wave velocity of transverse waves in a bar.

Also from (4) and (7),

$$\frac{\omega}{k} = kvK \text{ or } \omega = k^2 Kv \quad \dots(8)$$

If a number of frequencies are present in the vibrating bar, the group velocity (u_g) is given by

$$u_g = \frac{d\omega}{dk} = 2kKv = \frac{4\pi K}{\lambda} \sqrt{\frac{Y}{\rho}} \quad \dots(9)$$

10-14. Comparison of Transverse Vibrations of Bar and String

(1) When a bar vibrates transversely, this is the elasticity of the material of the bar which is responsible to restore the successive portions of the bar to their original position. However, in case of string it is the tension in the string which is responsible for transverse vibrations.

(2) In case of a string, the velocity of transverse waves is given by $v = \sqrt{Y/\rho}$ and does not depend on the wavelength, but the velocity of the transverse waves in a bar is given by $v = \frac{2\pi K}{\lambda} \sqrt{\frac{Y}{\rho}}$ and hence depends on the wavelength λ .

10-15. Comparison of Longitudinal and Transverse Vibrations in a Bar

S. No.	Longitudinal Vibrations	Transverse Vibrations
1.	Differential wave equation $\frac{\partial^2 y}{\partial t^2} = v^2 \frac{\partial^2 y}{\partial x^2}$ where $v = \sqrt{Y/\rho}$.	1. Differential wave equation $\frac{\partial^2 y}{\partial t^2} = -v^2 K^2 \frac{\partial^4 y}{\partial x^4}$ where $v = \sqrt{Y/\rho}$ and K = geometrical radius of gyration.
2.	Velocity of longitudinal waves in a bar : $v = \sqrt{\frac{Y}{\rho}}$ where Y = young's modulus and ρ = density of material of bar.	2. Velocity of transverse waves in a bar : $u = \sqrt{\omega v K} = \frac{2\pi}{\lambda} Kv = \frac{2\pi K}{\lambda} \sqrt{\frac{Y}{\rho}}$ where ω = angular frequency, v = velocity of longitudinal waves and λ = wavelength.
3.	Velocity of longitudinal waves in a bar is constant and does not depend on frequency.	3. Velocity. of transverse waves in a bar depends upon frequency.
4.	Medium is nondispersive for longitudinal waves in a bar.	4. Medium is dispersive for transverse waves in a bar.
5.	Group velocity for longitudinal waves in a bar is equal to wave velocity.	5. Group velocity for transverse waves in a bar is double of wave velocity.
6.	Overtone are harmonics of fundamental frequency.	6. Overtone are not harmonics of the fundamental frequency.

10-16. Tuning Fork

Tuning forks are very important instruments, used as sources of sound of fixed frequency in physical laboratories. They are also used as standards of measuring small time intervals with high accuracy because due to the fixed frequency of the fork, its period of vibration is very precise.

A tuning fork may be considered as a modified form of a free-free bar, executing transverse vibrations. When such a bar is bent in the shape of big U and added with a stem at the centre, we obtain a **tuning fork**.

Let us consider the transverse vibrations of a free-free bar, vibrating in its fundamental [fig. 10.11]. During the course of vibration, the two free ends of the bar move simultaneously along the same direction. Two nodes (N, N) are formed close to the ends at

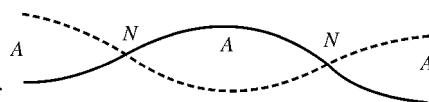


Fig. 10.11. Transverse vibrations of a free-free bar in fundamental mode.

a distance $0.2242l$ from either of the free ends and an antinode (A) is obtained in between the two nodes. As the bar is bent, the two nodes (N, N), are shifted closer and closer [fig. 10.12]. When it takes the final shape of U, the two arms become parallel and in this shape the two nodes (N, N) are closest. Consequently, the amplitude of vibration at the middle antinode becomes very small compared to that at the ends. Now the original mode of vibration is the same (fundamental) and the two arms are seen to move inwards or outwards simultaneously. When a stem is attached at the centre, the system is known to be **tuning fork**. Since the stem has the mass, its addition to the U bar further reduces the amplitude of vibration of the middle antinode and the two nodes (N, N) approach still closer. The outward motion of the arms raises the stem, and their inward motion lowers the stem [fig. 10.13]. In consequence, transverse vibrations of the two arms (prongs of the fork) produces the longitudinal vibrations of the stem. If the stem of the fork is pressed on a surface (resonance box or table), the vibrations of the stem cause small transverse vibrations of the surface.

When a tuning fork is struck mildly, it emits fundamental tone only. Each prong of the fork may be considered as a bar fixed at one end and free at the other end (fixed-free bar). The frequency in the fundamental mode of vibration is given by

$$\nu = \frac{(1.88)^2}{l^2} \cdot \frac{K}{2\pi} \sqrt{\frac{Y}{\rho}} \quad \dots(1)$$

We know that for a bar of rectangular cross-section $K = d/\sqrt{12}$, where d is the thickness of the prong. Hence,

$$\nu = \frac{(1.88)^2}{l^2} \cdot \frac{d}{2\pi\sqrt{12}} \sqrt{\frac{Y}{\rho}} \quad \dots(2)$$

Thus, the frequency of a tuning fork is

- (i) *inversely proportional to the square of its length (l),*
- (ii) *directly proportional to the thickness of the prong (d),*

and (iii) *directly proportional to the velocity of the longitudinal waves ($v = \sqrt{Y/\rho}$) in the material of the fork.*

The width of the prong does not affect the frequency of the fork. This is why *the arms of a fork of low pitch are long and thin, whereas the arms of fork of high pitch are small and thick.*

When the fork is struck vigorously overtones, which have frequencies nearly 6 and 17 times that of fundamental, are produced. However these overtones are feeble and die away much more quickly than the fundamental. Thus we obtain only the *pure fundamental tone* which persists for a long time.

Effect of Temperature : When the temperature increases, the length of the fork and thickness of the prong are increased and the density and Young's modulus of the material are decreased. If α be the coefficient of linear expansion of the material of the fork and β the coefficient of decrease of Young's modulus, then the frequency at temperature t is given by

$$\begin{aligned} \nu_t &= \frac{(1.88)^2}{l_0^2(1+\alpha t)^2} \cdot \frac{d_0(1+\alpha t)}{2\pi\sqrt{12}} \sqrt{\frac{Y_0(1-\beta t)}{\rho_0(1-3\alpha t)}} \\ &= \frac{(1.88)^2}{l_0^2} \cdot \frac{d_0}{2\pi\sqrt{12}} \sqrt{\frac{Y_0}{\rho_0}} \cdot \left(1 + \frac{1}{2}\alpha t\right) \left(1 - \frac{1}{2}\beta t\right) \\ \text{i.e.,} \quad \nu_t &= \nu_0 \left(1 - \frac{\beta - \alpha}{2} t\right) \quad \dots(3) \end{aligned}$$

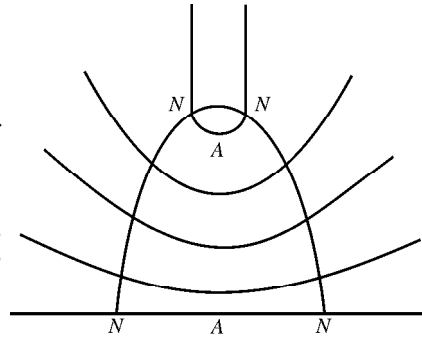


Fig. 10.12. Bending of a bar to form a tuning fork.

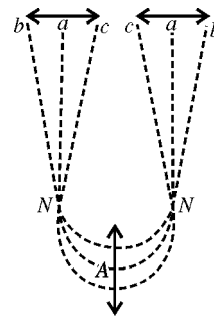


Fig. 10.13. Vibrating tuning fork.

where ν_0 is the fundamental frequency at 0°C .

$$\begin{aligned} \text{For steel, } \alpha &= 12 \times 10^{-6} \text{ K}^{-1} \text{ and } \beta = 240 \times 10^{-6} \text{ K}^{-1} \\ \therefore \nu_t &= \nu_0 (1 - 0.00011t) \end{aligned} \quad \dots(4)$$

Thus, with the rise of temperature the frequency of a tuning fork decreases slowly. This decrease in frequency is about 0.01% for 1°C rise in temperature for steel. Now a days, standard forks (standard of frequency) are usually made of elinvar for which the temperature coefficient is very small and the frequency remains approximately constant.

Ex. 1. What is the wavelength of longitudinal waves of frequency 400 Hz in an alloy whose density is $5,500 \text{ kg/m}^3$ and Young's modulus $8.8 \times 10^{10} \text{ N/m}^2$.

$$\text{Sol. The velocity of sound } v = \sqrt{\frac{Y}{\rho}} = \sqrt{\frac{8.8 \times 10^{10}}{5500}} = 4 \times 10^3 \text{ m/sec}$$

$$\therefore \text{Wavelength } \lambda = v/\nu = 4 \times 10^3/400 = \mathbf{10 \text{ m.}}$$

Ex. 2. A brass rod of length 3 metres is clamped at the centre. It emits a note of frequency 600 Hz, when it vibrates longitudinally in fundamental mode. If the density of the brass is 8300 kg/m^3 , calculate its Young's modulus.

Sol. When a rod of length clamped in the middle, vibrates in fundamental mode, the expression for the frequency of vibration is

$$v = \frac{1}{2l} \sqrt{\frac{Y}{\rho}} \quad \dots(i)$$

where Y is Young's modulus and ρ is the density of the material of the rod.

$$\text{From (i)} \quad Y = 4v^2 l^2 \rho \quad \dots(ii)$$

Here, $v = 600 \text{ Hz}$, $l = 3 \text{ m}$ and $\rho = 8300 \text{ kg/m}^3$. Hence

$$Y = 4 \times (600)^2 \times (3)^2 \times 8300 = \mathbf{10.76 \times 10^{10} \text{ N/m}^2}.$$

Ex. 3. A brass rod 2m in length is damped at its middle point. One end is excited to produce longitudinal vibrations. The frequency of the emitted note is found to be 1000 Hz. Calculate the velocity of longitudinal waves and also Young's modulus of brass assuming the density to be 8500 kg/m^3 . **(O.U. 2009)**

Sol. When a rod is clamped at its middle point, the frequency of longitudinal vibrations in the fundamental note is given by

$$v = \frac{v}{2l}$$

where v is the velocity of longitudinal waves in the rod and l in the length of the rod.

$$\text{Therefore, } v = v \times 2l = 1000 \times 2 \times = \mathbf{4000 \text{ m/sec.}}$$

$$\text{Also } v = \sqrt{\frac{Y}{\rho}},$$

$$\therefore Y = \rho v^2 = 8500 \times (4000)^2 = 1.36 \times 10^{11} \text{ N/m}^2.$$

Ex. 4. Calculate the approximate frequency of the lowest note given by the longitudinal vibrations of a steel bar of length 25 cm., density 8000 kg/m^3 and Young's modulus $2 \times 10^{11} \text{ N/m}^2$.

Sol. The frequency of the lowest note given by the longitudinal vibrations of a free-free bar is given by

$$v = \frac{1}{2l} \sqrt{\frac{Y}{\rho}}$$

Here, $l = 0.25 \text{ m}$, $Y = 2 \times 10^{11} \text{ N/m}^2$ and $\rho = 8000 \text{ kg/m}^3$

$$\therefore v = \frac{1}{2 \times 0.25} \sqrt{\frac{2 \times 10^{11}}{8000}} = \mathbf{10^4 \text{ Hz.}}$$

Ex. 5. A standing wave, produced in a rod, is represented by the equation

$$y = 0.001 \sin(13x) \sin(200t).$$

If the value of Young's modulus of the material of the rod is 10^{11} N/m^2 , calculate the maximum particle velocity and tensile stress at $x = 4 \text{ cm}$.

Sol. Particle velocity, $dy/dt = 0.001 \times 200 \sin(13x) \cos(200t)$

When $\cos(200t) = 1$, the particle velocity will be maximum.

$\therefore$ Maximum particle velocity $= 0.001 \times 200 \sin(13x)$

But at $x = 0.04 \text{ m}$, $13x = 13 \times 0.04 = 0.52 \text{ radian} = 0.52 \times 180/3.14 \text{ degree} = 30^\circ \text{ approx.}$

$$\therefore \sin(13x) = \sin 30^\circ = \frac{1}{2}$$

Hence, Maximum particle velocity $= 0.001 \times 200 \times \frac{1}{2} = 0.1 \text{ m/sec.}$

$$\text{Tensile stress} = Y \frac{dy}{dx} = 10^{11} \times 13 \cos(13x) \sin(200t)$$

Hence at $x = 0.04$, maximum tensile stress

$$= 10^{11} \times 0.001 \times 13 \times \sqrt{3}/2 \quad [\because \cos(13x) = \cos 30^\circ = \sqrt{3}/2]$$

$$= 11.26 \times 10^8 \text{ N/m}^2.$$

Ex. 6. A copper rod of length 0.5 m is clamped at one end and kept free at the other end. If it is made to execute longitudinal vibrations, find the fundamental frequency and frequencies of the first two overtones. (Young's modulus of copper $= 12 \times 10^{10} \text{ Nm}^{-2}$; Density of copper $= 8900 \text{ kg m}^{-3}$)

Sol. When a rod is clamped at one end and kept free at the other end, the frequency of longitudinal vibrations of the rod in the n th mode of vibration is given by

$$v_n = \frac{(2n-1)}{4l} \sqrt{\frac{Y}{\rho}}, \text{ where } n = 1, 2, 3, \dots \text{etc.}$$

Fundamental frequency of the rod corresponds to $n = 1$ i.e.,

$$v_1 = \frac{\sqrt{Y}}{\rho} = \frac{1}{4 \times 0.5} \cdot \sqrt{\frac{12 \times 10^{10}}{8900}} = 1836 \text{ Hz.}$$

For $n = 2$, we obtain the frequency of first overtone i.e.,

$$v_2 = \frac{3}{4l} \sqrt{\frac{Y}{\rho}} = 3v_1 = 3 \times 1836 = 5508 \text{ Hz.}$$

The frequency of second overtone corresponds to $n = 3$ i.e.,

$$v_3 = \frac{5}{4l} \sqrt{\frac{Y}{\rho}} = 5v_1 = 5 \times 1836 = 9180 \text{ Hz.}$$

Ex. 7. The second harmonic of the note emitted from a bar, free at both ends, has the same frequency as the frequency of the third harmonic note emitted from another bar, clamped at one end and free at the other. Both the bars are executing longitudinal vibrations. Determine the ratio of the length of the two bars. Assume that the two rods are made of the same material.

Sol. For a free-free bar, frequency of the second harmonic is given by

$$v_2 = \frac{2}{2l_1} \sqrt{\frac{Y}{\rho}} \quad \dots(i)$$

From a clamped-free bar, the frequency of the third harmonic

$$v_3 = \frac{3}{4l_2} \sqrt{\frac{Y}{\rho}} \quad \dots(ii)$$

According to the question $v_2 = v_3$ i.e.,

$$\frac{2}{2l_1} \sqrt{\frac{Y}{\rho}} = \frac{3}{4l_2} \sqrt{\frac{Y}{\rho}} \text{ or } \frac{l_1}{l_2} = \frac{4}{3}.$$

Ex. 8. A metallic bar of length 1 metre is free at its both ends. The diameter of cross section of the rod is 1 cm. Find the fundamental frequency of the longitudinal vibrations and transverse vibrations of the rod. (Velocity of sound in copper is 4000 m/s.)

Sol. When a free-free bar vibrates longitudinally the fundamental frequency of vibration is given by

$$v_l = \frac{v}{2l} = \frac{4000}{2 \times 1} = \mathbf{2000 \text{ Hz.}}$$

In case of transverse vibrations of a free-free bar, the fundamental frequency of vibration is given by

$$v_t = \frac{(4.73)^2}{l^2} \cdot \frac{v}{2\pi} \quad \left[v = \sqrt{\frac{Y}{\rho}} \right]$$

Here, $K = \frac{r}{2} = \frac{0.01}{2 \times 2} = 0.0025 \text{ m}$, $v = 4000 \text{ m/sec}$, $l = 1 \text{ m}$

$$\therefore v_t = \frac{(4.73)^2 \times 0.0025 \times 4000}{1^2 \times 2 \times 3.14} = \mathbf{35.62 \text{ Hz.}}$$

QUESTIONS

Long Answer Questions

1. What are longitudinal vibrations in a bar ? Deduce an expression for the velocity of longitudinal waves in a solid bar.
2. Set up the differential wave equation when the bar is executing longitudinal vibrations. Give the solution of the longitudinal wave equation.
3. Deduce the wave equation for longitudinal vibrations in a bar in differential form and show that the velocity of the longitudinal waves in the bar is given by

$$v = \sqrt{Y/\rho}.$$

4. Discuss the boundary conditions used in the theoretical treatment of stationary waves in a bar. Using the solution of longitudinal wave equation, explain the modes of vibration of a bar clamped in the middle.
5. Show that the longitudinal vibrations of a free-free bar contain all harmonics while those of a clamped-free bar contain only odd harmonics.
6. Discuss the possible modes of longitudinal vibration of a rod, fixed at one end, the other end being free.
7. Deduce an expression for the fundamental frequency of the longitudinal vibrations of a bar (i) clamped at one end, and (ii) clamped at both ends.
8. Discuss the stationary waves and first three modes of longitudinal vibrations of a free-free bar.
9. Deduce the differential equation for the transverse waves in a thin rod. Hence show that the rod behaves as a dispersive medium.
10. Assuming a plane wave solution of differential wave equation for the transverse vibrations in a bar, show that the velocity of transverse waves is given by

$$u = \sqrt{\omega v K}$$

where ω is the angular frequency of the wave, v the velocity of longitudinal waves and K the geometrical radius of gyration.

11. Assuming a plane wave solution of transverse wave equation in the form $y = a e^{i(\omega t - kx)}$, show that the wave velocity (u) is a function of angular frequency (ω) i.e., the medium is dispersive. Hence show that the group velocity is given by

$$u_g = 2Kk \sqrt{\frac{Y}{\rho}}$$

where symbols have usual meaning.

12. Solve the differential equation in general for simple harmonic transverse vibrations of a bar and discuss the cases for free-free bar.

13. Deduce the expression for normal mode frequencies for transverse vibrations of a bar, clamped at one end and free at the other.
Hence show that the overtones (partials) do not form a harmonic series.
14. For the vibrations of a clamped-free bar in fundamental mode, show that the frequency of transverse vibrations is much less than the frequency of longitudinal vibrations.
15. Explain the vibrations of a tuning fork. Mention the different factors affecting its frequency.
16. Describe the changes in the position of the nodes which occur, when a bar is bent more and more from the straight form to a U-shaped one.
17. Discuss vibrations of a tuning fork. How does the rise of temperature affect the frequency of a tuning fork ?

Short Answer Questions

1. What types of vibrations are possible in a bar ?
2. Show that the expression for tensile stress during longitudinal vibrations of a bar is given by

$$F = Y dy/dx$$
 where terms have usual meaning.
3. Deduce an expression for the velocity of longitudinal sound waves in a bar.
4. Write the equations of standing waves for first three modes of longitudinal vibrations of a free-free bar.
5. Discuss briefly the longitudinal modes of vibration of a bar clamped rigidly at both the ends. (O.U. 2009)
6. Show that only odd harmonics are possible for the longitudinal vibrations of a bar clamped at mid point.
7. Set up the differential wave equation for transverse vibrations of a bar.
8. Deduce an expression for the velocity of transverse waves in a bar.
9. How can you say that the vibrations of a tuning fork can be regarded as the transverse vibrations of a free-free bar, curved at the middle.
10. What is the expression for the frequency of a tuning fork ? Discuss the various factors affecting its frequency.
11. How does the temperature affect the frequency of a tuning fork ?
12. Show that the partials in case of transverse vibrations of a bar do not form a harmonic series.
13. Write the boundary conditions for transverse vibrations of (i) a free-free bar, and (ii) fixed-free bar.
14. What are boundary conditions when a bar vibrates (a) longitudinally and (b) transversely.
(S.K.U. 1990)
15. Compare the transverse vibrations of a bar and those of a string.
16. Compare the longitudinal and transverse vibrations in a bar.

PROBLEMS

1. A metallic rod of length 3 m is clamped at the centre. It emits a note of frequency 600 Hz, when it vibrates longitudinally. If the density of the material of the rod is 8300 kg/m^3 , find the value of Young's modulus.
[Ans. $1.076 \times 10^{11} \text{ N/m}^2$]
2. The density of aluminium is $2.8 \times 10^3 \text{ kg/m}^3$ and its Young's modulus is $7 \times 10^{10} \text{ N/m}^2$. If the frequency of the aluminium rod is 600 Hz, find the velocity of sound and wavelength through the rod.
[Ans. 5000 m/sec, 12m]
3. A metallic rod of length one metre and density 7100 kg/m^3 is clamped at its middle and longitudinal vibrations are set up in it by stroking along its length. If the frequency of the fundamental note produced by it is 2600 Hz, find the velocity of sound in the rod and the Young's modulus of the material of the rod.
[Ans. $v = 5200 \text{ m/s}$ and $Y = 21.39 \times 10^{10} \text{ N/m}^2$]
4. A brass rod, one metre long, is firmly clamped in the middle and one end is stroked by a resined cloth. What is the pitch of the note which you will hear ? (Young's modulus for brass = 10^{11} N/m^2 and density = $9,000 \text{ kg/m}^3$)
[Ans. 1666.6 Hz]
5. A copper rod of length 1.2 m and diameter 1 mm is clamped at one end. Calculate the frequencies of

longitudinal vibrations of the first three harmonics. (For copper, Young's modulus = $12.2 \times 10^{10} \text{ N/m}^2$ and density = 8900 kg m^{-3}).

[Ans. 771.3 Hz, 2314 Hz and 3856 Hz]

6. A certain wave produced in a certain medium is expressed as

$$y = 0.002 \sin(0.15x) \sin(48000t).$$

where x and y are in cm and *in sec*. Calculate the maximum particle velocity and tensile stress at the point $x = 3.5 \text{ cm}$, if Y of the medium = $8 \times 10^{10} \text{ N/m}^2$.

[Ans. 0.48 m/sec; $2.1 \times 10^7 \text{ N/m}^2$]

7. A brass rod of length 1 m is clamped at its mid point. If it is made to vibrate longitudinally, find the fundamental frequency and frequencies of the first two overtones. (Y for brass = $10 \times 10^{10} \text{ N m}^{-2}$ and density of brass = $8.3 \times 10^3 \text{ kg m}^{-3}$)

[Ans. 1735.5 Hz; 5206.5 Hz and 8677.5 Hz]

8. A wooden rod of length 1 m, clamped at its centre, is stroked with a resined cloth. A note of frequency 2048 Hz is heard. Find Young's modulus for the material of rod. (Density of wood = 800 kg/m^3)

[Ans. $1.34 \times 10^{10} \text{ N/m}^2$]

9. In the apparatus of Kundt's tube, a copper rod of one metre length is clamped at a distance of 25 cm from one end. Air is present in the tube in which the velocity of sound is 340 m/sec. Powder in the tube is collected in small heaps with 5 cm distance between two in succession. What is velocity of sound in copper?

[Ans. 3400 m/sec]

10. A steel rod is of 0.002 m radius and has a length of 2 m. What is its fundamental frequency of free-free transverse vibrations? (Young's modulus $Y = 19.5 \times 10^{10} \text{ N/m}^2$ and density $\rho = 7700 \text{ kg m}^{-3}$)

[Ans. 4.47 Hz]

11. A copper rod of length 2 m is free at its both ends. The diameter of the rod is 1 cm. Find the fundamental frequency of longitudinal and transverse vibrations of the rod. (Velocity of sound in copper = 3560 m/sec.)

[Ans. 890 Hz; 7.9 Hz]

12. An aluminium bar, 1.2 m long, has a rectangular cross-section with a thickness of 0.006 m, measured in the direction of transverse vibrations. Calculate the fundamental frequency and two overtones of transverse vibrations of the fixed-free bar. (Velocity of longitudinal waves in the bar = 5150 m/s).

[Ans. 3.48 Hz, 21.82 Hz, 61.07 Hz]

$$\text{[Hint. } v = \frac{(1.88)^2}{l^2} \cdot \frac{Kv}{2\pi} = \frac{(1.88)^2 \times 0.006 \times 5150}{(1.2)^2 \times 2\sqrt{3} \times 2 \times 3.14} = \frac{109.2}{31.3} = 3.48 \text{ Hz.} \quad \left[\because K = \frac{d}{2\sqrt{3}} \right]$$

13. An iron rod of length 2 m, fixed at the mid point, is struck so that it vibrates longitudinally. The fundamental note emitted by the rod is in resonance with the sonometer wire of length 16.2 cm. Keeping the tension in the wire unchanged, a tuning fork of frequency 540 Hz is found in resonance with the same sonometer wire of length 72.6 cm. Calculate the frequency of the iron rod and its Young's modulus. (Density of steel 7700 kg/m^3)

[Ans. 2420 Hz, $1.8 \times 10^{11} \text{ N/m}^2$]

14. A wooden rod of length 1.5 m is held in the middle and set in vibration longitudinally. It gives the same note as an open organ pipe 1.3 m long. Find the speed of sound in the rod. (Speed of sound in air = 338 m/sec.)

[Ans. 388.7 m/sec.]

[Hint. For an open organ pipe, frequency of fundamental $v = v/2l$]

Vibrating Strings

11.1. Wave Motion : Transverse and Longitudinal Waves

The concept of waves is very fundamental in nature. The waves which move in the forward direction are called **progressive** or **travelling waves**. In a travelling wave, there is always a transmission of energy along the direction of propagation of the wave. However in stationary waves, there is no transference of energy from one place to another.

The waves which need material medium to reach from one place to another are called **mechanical waves**. For the propagation of such waves, the medium must have the property of elasticity and inertia. Sound waves, waves on a string etc. are examples of mechanical waves. The waves which can even travel also in vacuum are called **electromagnetic waves**. X-rays, light waves, radio waves etc. are examples of electromagnetic waves.

Mechanical waves are of two types :

(1) **Transverse Waves** : The waves in which each particle of the medium executes oscillatory motion about its mean position in a direction perpendicular to the direction of wave propagation, are called **transverse waves**. For example, the waves produced by the transverse vibration of a stretched string .

(2) **Longitudinal Waves** : The waves in which each particle of the medium executes oscillatory motion about its mean position along the direction of wave propagation, are called **longitudinal waves**. Sound waves, produced in the air, and the waves transmitted along a stretched spring by compressing its one end then releasing it, are the examples of longitudinal waves.

11.2. Wave Pulse on a String and Equation of Wave

Consider a long string with one end fixed to a wall and the other end being held by a person so that the string is tight under tension. Let the person move his hand a little up and down producing a bump in the string near his hand [fig. 11.1]. This disturbance or bump travels along the string towards right with constant speed. **Fig. 11.1** also shows the state of the string at successive instants. In this process, the person does some work on the part of the string close to his hand, providing some energy to that part. This disturbed part exerts elastic force on the part of the string (with inertial mass) just right to it and transfers the energy. The bump thus moves towards right along the string. This process continues and the different parts of the string are successively disturbed, propagating the energy from left to right.

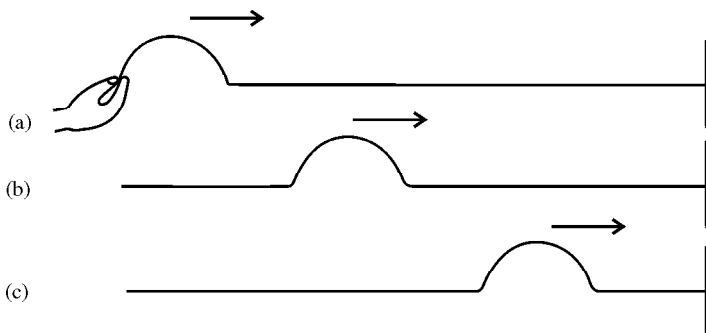


Fig. 11.1. Propagation of wave pulse on a string

If the disturbance is localised only to a small part of space (string) at a time, it is said that a **wave pulse** is passing through that part of space. This wave pulse is formed, when the source (hand) is active only for a short time.

If the source, producing the disturbance, is active for relatively more time, repeating its motion several times, a **wave train** or a **wave packet** is obtained. For example, the person in **fig. 11.1** moves his hand up and down 5 times and then stops, a wave train with 5 loops will advance on the string towards right.

Equation of Travelling Wave

Consider a wave pulse or wave train, travelling on the string towards right with velocity v . If we take the source or the left end of the string as origin ($x = 0$) and X -axis along the string, the vertical displacement y of a particle at the origin is a function of time t (taking $t = 0$, when the particle is at the origin O in the beginning in the undisplaced position) i.e.,

$$y = f(t)$$

where $f(t)$ is a function of time, representing the displacement y of the particle at $x = 0$.

As the disturbance (or wave) is travelling towards right with speed v , this displacement y of the wave will reach a point P , distance x from O , after x/v time. Hence at any instant t the displacement y of the particle at P must be the same as the displacement at the origin x/v seconds earlier i.e.,

$$y = f\left(t - \frac{x}{v}\right) \quad \dots(1)$$

This is the equation for wave, travelling along the positive direction of X -axis with constant velocity v [**fig. 11.2**]. The function f determines the shape and size of the wave. Such a wave is called a **travelling wave** or **progressive wave**. In case of a wave, the displacement y is a function of distance x and time t i.e., $y = y(x, t)$. It is to be noted that the time t and position x must appear in the wave equation in the combination $(t - x/v)$ only. For example,

$$y = a \sin \frac{2\pi}{T} \left(t - \frac{x}{v}\right), y = ae^{-\omega(t - x/v)}$$

etc. are valid wave equations.

If the wave travels in the negative X -direction, its equation will be represented as

$$y = f\left(t + \frac{x}{v}\right) \quad \dots(2)$$

The wave travelling in the positive X -direction may be represented as

$$y = f\left(\frac{vt - x}{v}\right) \text{ or } y = F(x - vt) \quad \dots(3)$$

where F is some other function of x and t .

If the source O vibrates in a *simple harmonic motion*, the equation for displacement y of the particle at the origin will be represented by $y = a \sin \omega t$ and the equation of the wave, produced by such a vibrating source, will be obtained to be

$$y = a \sin \omega \left(t - \frac{x}{v}\right) \quad \dots(4)$$

where a represents the amplitude and ω the angular frequency. If T is the time period of oscillation, then $T = 2\pi/\omega$ and frequency of oscillation $\nu = 1/T = \omega/2\pi$. The wave represented by eq. (4) is called a **sine wave** or **sinusoidal wave** [**fig. 11.3**].

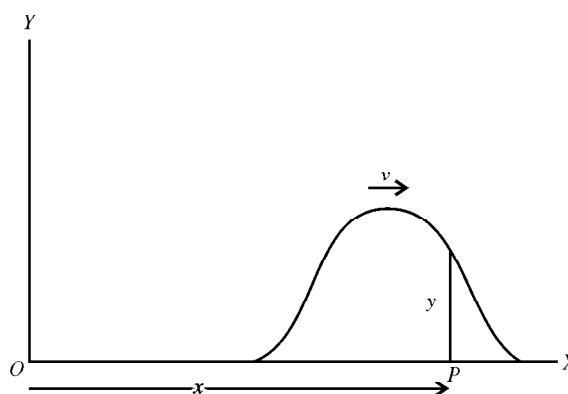


Fig. 11.2. Equation for a travelling wave : $y = f\left(t - \frac{x}{v}\right)$

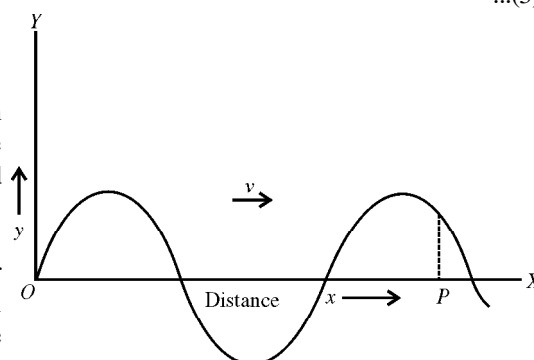


Fig. 11.3. Sine wave.

Various alternative forms of equation of a travelling harmonic wave along positive X -axis can be written as

$$y = a \sin 2\pi v \left(t - \frac{x}{v} \right) \quad \dots(4a)$$

or
$$y = a \sin \frac{2\pi}{T} \left(t - \frac{x}{v} \right)$$

or
$$y = a \sin 2\pi \left(\frac{t}{T} - \frac{x}{\lambda} \right) \quad \dots(4b)$$

or
$$y = a \sin \frac{2\pi v}{\lambda} \left(t - \frac{x}{v} \right)$$

or
$$y = a \sin \frac{2\pi}{\lambda} (vt - x) \quad \dots(4c)$$

The quantity $2\pi/\lambda = 2\pi \bar{v} = k$ is called the **propagation constant**, where $\bar{v} = 1/\lambda$ is the wave number.

$\therefore$
$$y = a \sin (\omega t - kx) \quad \dots(4d)$$

If ϕ is the phase constant, then

$$y = a \sin (\omega t - kx + \phi) \quad \dots(4e)$$

Note that

$$y = a \sin (kx - \omega t) \quad \dots(4f)$$

is also a valid form of equation of a harmonic wave, travelling along positive X -axis, but it has a phase difference of π with the wave in eq. (4d).

Relation between Particle Velocity and Slope of Displacement Curve

The general equation of a wave, travelling along positive direction of X -axis with constant velocity v is given by

$$y = f \left(t - \frac{x}{v} \right) \quad \dots(5)$$

where y is the displacement of a particle on the wave at any time t at a distance x from an arbitrary origin (or source).

Differentiating eq. (5) partially with respect to t , we get the expression for particle velocity i.e.,

$$\frac{\partial y}{\partial t} = f' \left(t - \frac{x}{v} \right) \quad \dots(6)^*$$

Differentiating eq. (1) partially with respect to x , we get the expression for slope of the displacement curve i.e.,

$$\frac{\partial y}{\partial x} = -\frac{1}{v} f' \left(t - \frac{x}{v} \right) \quad \dots(7)$$

From (6) and (7), we obtain

$$\frac{\partial y}{\partial x} = -\frac{1}{v} \frac{\partial y}{\partial t}$$

or
$$\frac{\partial y}{\partial t} = -v \frac{\partial y}{\partial x}$$

$$\text{Particle velocity} = -(\text{Wave velocity}) \times (\text{Slope of displacement curve})$$

* Let $t - x/v = u$. Then

Now,
$$\frac{\partial y}{\partial t} = \frac{\partial f}{\partial u} \cdot \frac{\partial u}{\partial t} = f'(u) \frac{\partial(t - x/v)}{\partial t} = f'(u) = f' \left(t - \frac{x}{v} \right)$$

Also,
$$\frac{\partial y}{\partial x} = \frac{\partial f}{\partial u} \cdot \frac{\partial u}{\partial x} = f'(u) \cdot \frac{\partial(t - x/v)}{\partial x} = f'(u) \left(-\frac{1}{v} \right) = -\frac{1}{v} f' \left(t - \frac{x}{v} \right)$$

11.3. Transverse Wave Propagation along a Stretched String : Velocity of Transverse Waves

In order to study the propagation of transverse waves along a string, first we must define a string. *By a string or wire in acoustics we mean a cord or wire whose length is large compared to its diameter, and which is perfectly uniform and flexible.* The string is stretched with a large tension so that the effects of gravitational forces are negligible.

Consider a string with one end fixed and the other end being pulled by a person with a tension T . Let the person produce a bump or wave pulse by moving his hand up and down. The wave pulse, so produced, travels along the string towards the fixed end. In such a case, the string is thrown into transverse vibration perpendicular to the direction of wave propagation.

Let us take X -axis along the direction of the undisplaced string and Y -axis in the direction of the displacement of the particles of the string at right angles to the X -axis. Now, consider a small element PQ of the string of length equal to δx at a distance x from the origin. Let $P'Q'$

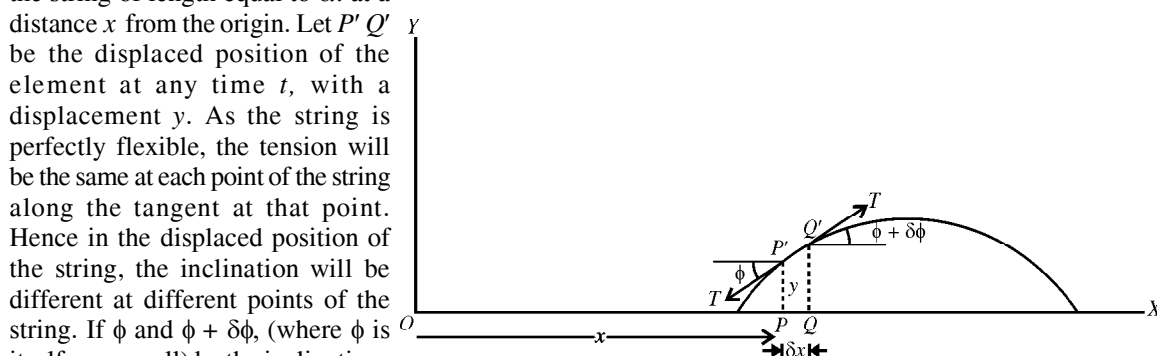


Fig. 11.4. Transverse wave propagation along a string.

be the displaced position of the element at any time t , with a displacement y . As the string is perfectly flexible, the tension will be the same at each point of the string along the tangent at that point. Hence in the displaced position of the string, the inclination will be different at different points of the string. If ϕ and $\phi + \delta\phi$, (where ϕ is itself very small) be the inclinations with X -axis of the tangents at P' and Q' points respectively, then resolving the tension acting at P' and Q' along X and Y axes, we have

$$\Sigma T_x = T \cos(\phi + \delta\phi) - T \cos \phi$$

$$\Sigma T_y = T \sin(\phi + \delta\phi) - T \sin \phi$$

As ϕ and $\delta\phi$ are small, hence

$$\cos(\phi + \delta\phi) \approx \cos \phi, \sin(\phi + \delta\phi) \approx \phi + \delta\phi \text{ and } \sin \phi \approx \phi$$

Thus,

$$\Sigma T_x = 0 \text{ and } \Sigma T_y = T(\phi + \delta\phi) - T\phi = T\delta\phi$$

For small angles,

$$\phi = \tan \phi.$$

Hence, the resultant force along Y -axis is

$$\Sigma T_y = T\delta(\tan \phi) = T\delta \left(\frac{\partial y}{\partial x} \right) = T \frac{\partial}{\partial x} \left(\frac{\partial y}{\partial x} \right) \delta x = T \frac{\partial^2 y}{\partial x^2} \delta x.$$

If m is the mass per unit length of the string, then the mass of small element δx will be $m\delta x$. If the acceleration of this element at the displacement y be $\partial^2 y / \partial t^2$, then according to Newton's second law of motion

$$m\delta x \frac{\partial^2 y}{\partial t^2} = t \frac{\partial^2 y}{\partial x^2} \delta x$$

or

$$\frac{\partial^2 y}{\partial t^2} = \frac{T}{m} \frac{\partial^2 y}{\partial x^2} \quad \dots(1)$$

We know that the standard differential equation of the wave motion is given by

$$\frac{\partial^2 y}{\partial t^2} = v^2 \frac{\partial^2 y}{\partial x^2} \quad \dots(2)$$

* If a string under tension is fixed at both ends and plucked at any point, a wave pulse travels toward either fixed end.

where v is the velocity of the waves.

Hence eq. (1) represents the **differential equation for the transverse waves along a stretched string**. The velocity of the transverse waves along the string is obtained by comparing eqs. (1) and (2) i.e.,

$$v^2 = \frac{T}{m} \text{ or } v = \sqrt{\frac{T}{m}} \quad \dots(3)$$

Thus the velocity of transverse waves along a stretched string depends only on the applied tension T and mass per unit length (or linear mass density) m .

11.4. General Solution of the Wave Equation

The wave equation for transverse waves along a string is given by

$$\frac{\partial^2 y}{\partial t^2} = v^2 \frac{\partial^2 y}{\partial x^2} \quad \dots(1)$$

where v represents the velocity of transverse waves along the stretched string.

Eq. (1) represents the one dimensional wave equation for transverse waves, travelling along X -axis (or string). Here y is the transverse displacement of a particle of the string and it is the function of x and time t . The general solution of eq. (1) must be a function of the linear combination of t and x i.e.,

$$y = f(t + \alpha x) \quad \dots(2)$$

where α is a constant.

Differentiating eq. (2) with respect to t , we obtain

$$\frac{\partial y}{\partial t} = f'(t + \alpha x)$$

Again differentiating with respect to t ,

$$\frac{\partial^2 y}{\partial t^2} = f''(t + \alpha x) \quad \dots(3)$$

Differentiating eq. (3) with respect to x ,

$$\frac{\partial y}{\partial x} = \alpha f'(t + \alpha x)$$

Again differentiating it with respect x ,

$$\frac{\partial^2 y}{\partial x^2} = \alpha^2 f''(t + \alpha x) \quad \dots(4)$$

Substituting the values of $\frac{\partial^2 y}{\partial t^2}$ and $\frac{\partial^2 y}{\partial x^2}$ from eqs. (3) and (4) in eq. (1), we obtain

$$f''(t + \alpha x) = v^2 \alpha^2 f''(t + \alpha x)$$

whence
$$\alpha^2 = \frac{1}{v^2} \text{ or } \alpha = \pm \frac{1}{v} \quad \dots(5)$$

Hence the **general solution of the wave equation** (1) in view of eq. (2) is given by

$$y = f\left(t - \frac{x}{v}\right) + g\left(t + \frac{x}{v}\right). \quad \dots(6)$$

Significance of the General Solution of Wave Equation

When we look at the general solution of the wave equation, represented by eq. (6), we observe that the displacement of a particle at a point x is due to two waves : (i) one wave [$y_1 = f(t - x/v)$] is travelling in the positive X -direction and (ii) the other wave [$y_2 = g(t + x/v)$] is moving along the negative X -direction. In case

of a string, first wave, travelling along the positive X -direction, is the **incident wave** (say) and the other wave, moving along the negative X -direction is the **reflected wave**.

Simplest and most important case of the general solution of the wave equation is the **sinusoidal solution**, given by

$$y = a \sin \omega \left(t - \frac{x}{v} \right) + a \sin \omega \left[\left(t + \frac{x}{v} \right) + \phi \right] \quad \dots(7)$$

where a is the amplitude of vibration, $\omega/2\pi (= \nu)$ is the frequency of vibration and θ is the phase difference between the two waves. First sine wave [$y_1 = a \sin \omega (t - x/v)$] is the incident wave and the second sine wave

$\left[y_2 = a \sin \left\{ \omega \left(t + \frac{x}{v} \right) + \phi \right\} \right]$ is the reflected wave. By superposition, these two oppositely moving waves of the same amplitude, frequency and wavelength may produce **stationary waves**. This is the physical significance of the general solution of the wave equation.

11.5. Modes of Vibration of Stretched String Clamped (Fixed) at Both Ends (Overtones and Harmonics)

Consider a string of length l along X -axis. The string has been clamped (fixed) at $x = 0$ and $x = l$. Now, let a transverse sine wave of angular frequency $\omega (= 2\pi\nu)$ and amplitude a given by

$$y_1 = a \sin \omega (t - x/v) \quad \dots(1)$$

be travelling along the positive X -axis (string) with speed v . This is the incident wave. The wave, reflected from the other end ($x = l$), moving in the negative X -axis, can be represented as

$$y_2 = a \sin [\omega (t + x/v) + \phi] \quad \dots(2)$$

where ϕ is the phase difference between the two waves.

The incident and the reflected waves superpose each other are identical in all respects but differ in phase. Hence they may give rise to standing waves. This is to be remembered that standing or stationary waves are formed by the superposition of two exactly similar waves travelling in opposite directions.

The resultant displacement of the particle of the string at the position x and time t is given by principle of superposition i.e.,

$$y = y_1 + y_2 = a \sin \omega \left(t - \frac{x}{v} \right) + a \sin \left[\omega \left(t + \frac{x}{v} \right) + \phi \right]$$

or,

$$y = 2a \cos \left(\frac{\omega x}{v} + \frac{\phi}{2} \right) \sin \left(\omega t + \frac{\phi}{2} \right)$$

Since $\omega/v = 2\pi \nu/v = 2\pi/\lambda = k$ is the wave number,

$$\therefore y = 2a \cos \left(kx + \frac{\phi}{2} \right) \sin \left(\omega t + \frac{\phi}{2} \right) \quad \dots(3)$$

where λ is the wavelength.

If standing waves are produced, the two ends of the string $x = 0$ and $x = l$ must be nodes because they are kept fixed. Thus, the **boundary conditions** are :

(i) $y = 0$ at $x = 0$ for all t

(ii) $y = 0$ at $x = l$ for all t

If we apply boundary condition (i) to eq. (3), we obtain

$$0 = \cos \phi/2 \text{ or } \phi/2 = \pi/2 \text{ or } \phi = \pi \quad \dots(4)$$

Hence eq. (3) takes the form

$$y = -2a \sin kx \cos \omega t \quad \dots(5)$$

For boundary condition (ii)

$$\sin kl = 0 \text{ or } kl = n\pi, \text{ where } n = 1, 2, 3 \quad \dots(6)$$

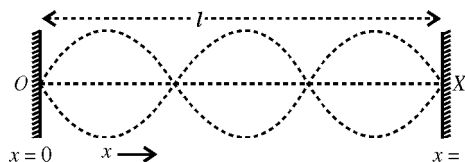


Fig. 11.5. Modes of vibration of a stretched string clamped at both ends.

Hence
$$l = \frac{n\pi}{k} = \frac{n\pi}{2\pi/\lambda} = \frac{n\lambda}{2} \quad \dots(7)$$

Thus in case of a string fixed at both ends standing waves are produced, when the length of the string is an integral multiple of $\lambda/2$.

Harmonics and Overtones

From eq. (7)

$$\lambda = \frac{2l}{n} \quad \dots(8)$$

We know that for a acoustic wave, $v = v\lambda$, where v is the frequency of vibration. Hence

$$v = \frac{v}{\lambda} = \frac{vn}{2l} \quad \dots(9)$$

If T is the tension in the string and m is the mass per unit length, then speed of the transverse waves v along the string is given by

$$v = \sqrt{\frac{T}{m}} \quad \dots(10)$$

Thus, the expression for the frequency of vibration of the string is obtained to be

$$v = \frac{n}{2l} \sqrt{\frac{T}{m}} \quad \dots(11)$$

where $n = 1, 2, 3, \dots$.

For $n = 1$, we obtain the lowest possible frequency, given by

$$v_1 = \frac{1}{2l} \sqrt{\frac{T}{m}} \text{ (fundamental tone or 1st harmonic) } . \quad \dots(12)$$

This is called the **fundamental frequency** of the string. Hence when the string is vibrating with fundamental frequency, the tone emitted by the string is known as **fundamental tone** or **first harmonic**.

The other natural frequencies of the string with which standing waves can be obtained for $n = 2, 3, \dots$ etc. are given by

$$v_2 = \frac{1}{2l} \sqrt{\frac{T}{m}} \text{ (1st overtone or 2nd harmonic) } \quad \dots(13)$$

$$v_3 = \frac{3}{2l} \sqrt{\frac{T}{m}} \text{ (2nd overtone or 3rd harmonic) } \quad \dots(14)$$

etc. Thus, **possible allowed frequencies of vibration of a string clamped at both ends are an integral multiple of the fundamental frequency**. Instead of fundamental frequency the higher frequencies with which the string can vibrate are called **overtones**. In this way, $v_2 = 2v_1$ is the **first overtone** and $v_3 = 3v_1$ is the **second overtone** etc.

An integral multiple of the fundamental frequency is called its **harmonic**. In case of a string, clamped at both ends, the frequencies of all overtones including fundamental are an integral multiple of the frequency of the fundamental i.e., in the ratio 1 : 2 : 3 etc. and hence they form a harmonic series. Thus **for a string fixed at both ends, all the overtones are harmonics of the fundamental frequency and all the harmonics of fundamental frequency are overtones**. Here, fundamental tone is the first harmonic, first overtone is the second harmonic, second overtone is the third harmonic etc.

Distinction between Harmonics and Overtones

The lowest natural frequency of an oscillating system is called its **fundamental tone**. The higher frequencies with which the system can vibrate are called **overtones**. In case of a string fixed at both ends, if v_1 is the frequency of vibration of the fundamental, then $v_2 = 2v_1$, $v_3 = 3v_1$, etc. represent the overtones i.e., first overtone, second overtone, etc.

An integral multiple of the fundamental frequency is called its **harmonic**. In case of a string, clamped at both ends, fundamental tone is the first harmonic, first overtone is the second harmonic, second overtone is the third harmonic etc. There are cases where an overtone may have a frequency which is not exact multiple of the frequency of the fundamental. For example, when a bar, clamped at one end and free at the other, may vibrate transversely with higher frequencies or produce overtones whose frequencies are not an integral multiple of the fundamental. In this case, the overtones are not harmonics of the fundamental frequency because their frequencies (including fundamental) are in the ratio 1 : 6.27 : 17.55 etc.

Normal Modes of Vibration of the String

When all the particles of an oscillating system vibrate sinusoidally with the same frequency, the system is said to vibrate in a normal mode. If the string vibrates according to eq. (5) with some natural frequency, it is said to vibrate in a normal mode. Writing $k = n\pi/l$ from eq. (6) and $\omega = 2\pi\nu = n\pi v/l$ from eq. (9), eq. (5) can be written as

$$y = -2a \sin \frac{n\pi x}{l} \cos \frac{n\pi vt}{l} \quad \dots(15)$$

This equation represents the equation for the displacement of a particle of the string at the position x and time t in the n th normal mode or n th mode. The frequencies, given by $\nu = nv/2l$ (for $n = 1, 2, 3, \dots$) of eq. (9) are called **normal frequencies** of vibration of the vibrating string fixed at both ends.

When $n = 1$, the string is said to vibrate in the **fundamental mode** or **first mode** with frequency $\nu_1 = v/2l$. In such a case, the equation of standing wave is

$$y = -2a \sin \frac{\pi x}{l} \cos \frac{\pi vt}{l} \quad \dots(16)$$

At any point x , the amplitude of vibration of a particle is $2a \sin (\pi x/l)$. The amplitude is zero at $x = 0$ and $x = l$. Its value is maximum at $x = l/2$ for $\sin \pi x/l = 1$. Thus in the fundamental mode of vibration of the string, fixed at both ends, we have two nodes (N) at the ends and one antinode (A) at the middle point of the string [fig. 11.6 (a)].

When $n = 2$, the string vibrates in **second normal mode** with frequency $\nu_2 = v/l = 2\nu_1$. The equation of standing wave in this mode is

$$y = -2a \sin \frac{2\pi x}{l} \cos \frac{2\pi vt}{l} \quad \dots(17)$$

The amplitude $2a \sin 2\pi x/l$ is zero at $x = 0, l/2$ and l and is maximum ($2a$) at $x = l/4$ and $x = 3l/4$. Thus there are three nodes (N) at $x = 0, l/2$ and l and two antinodes (A) at $x = l/4$ and $3l/4$ [fig. 11.6 (b)]. In the second mode, the string will vibrate in two loops. Clearly, at any instant, all the particles of the first loop in between the nodes at $x = 0$ and $x = l/2$ will be in the same phase and simultaneously all the particles of the second loop in between the nodes at $x = l/2$ and $x = l$ will be in the same phase but opposite phase relative to the first loop. For example, the particle at $x = l/4$ will be in the opposite phase relative to the particle at $x = 3l/4$ because at $x = l/4$, $y = -2a \cos 2\pi vt/l$ and at $x = 3l/4$, $y = +2a \cos (2\pi vt/l) = -2a \cos [(2\pi vt/l) + \pi]$.

In the **third mode** of vibration ($n = 3$), the frequency of the vibrating string is $\nu_3 = 3v/2l = 3\nu_1$ and the equation of standing wave is

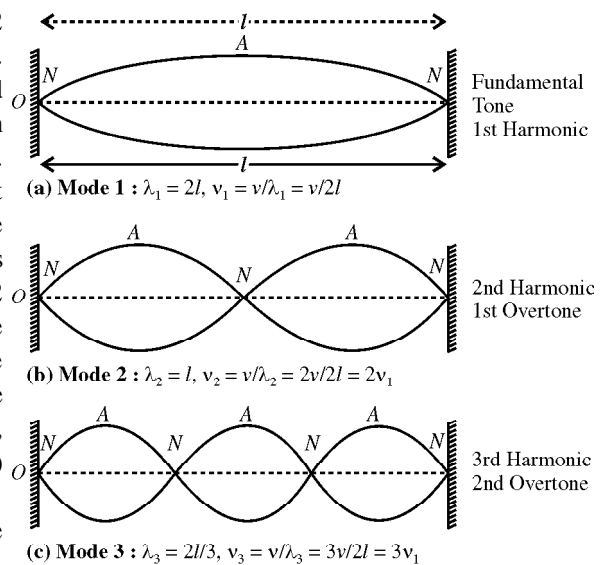


Fig. 11.6. First three normal modes of a string clamped at both ends

$$y = -2a \sin \frac{3\pi x}{l} \cos \frac{3\pi vt}{l} \quad \dots(17)$$

In this mode there will be four nodes at $x = 0, l/3, 2l/3$ and l and three antinodes at $x = l/6, l/2$ and $5l/6$ [fig. 11.6 (c)].

Thus if the string clamped at both ends is vibrating in the n th mode of vibration, the frequency of vibration is n times that of the fundamental ($v_n = nv_1$) and there will be $(n + 1)$ nodes and n antinodes.

When the string of a musical instrument such as sitar is pulled aside (plucked) at some point and released, its shape in vibration may not correspond to any normal mode, discussed above. Actually, the shape of the string is a combination of several normal modes and hence a combination of frequencies are emitted.

11.6. Plucked String

A stretched string is said to be excited by **plucking**, if it is pulled aside at any point through a small distance and then released.

We know that a string fixed at both ends, is capable of vibrating in all possible modes. When the string is vibrating in the n th mode, the equation for displacement is given by

$$y_n = A_n \sin \frac{n\pi x}{l} \cos \frac{n\pi vt}{l} \quad \dots(1)$$

where l is the length of the string, A_n the maximum amplitude of the string in the n th mode and v the speed of waves.

The general vibration of the string, fixed at both ends, is that resulting from the superposition of all the normal modes i.e.,

$$y = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi x}{l} \cos \frac{n\pi vt}{l} \quad \dots(2)$$

Suppose that the string of length l be pulled aside at a point distant a from the end $x = 0$ through a small transverse distance b and then released. The string is said to be *plucked* at $x = a$ and the resulting vibration is that due to *plucking*. The distance b is so small that the tension T in the string is assumed constant. The initial displacement at any point x of the string is given by [fig. 11.7],

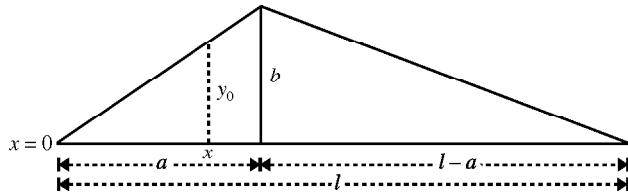


Fig. 11.7. Plucking of a string.

$$y_0 = \frac{b}{a}x \left[\because \frac{y_0}{x} = \frac{b}{a} \right] \text{ from } x = 0 \text{ to } x = a \quad \dots(3)$$

and

$$y_0 = \frac{b}{l-a}(l-x) \left[\because \frac{y_0}{l-x} = \frac{b}{l-a} \right] \text{ from } x = a \text{ to } x = l.$$

Multiply both sides of eq. (1) by $\sin \frac{n\pi x}{l}$ and integrate between the limits $x = 0$ to $x = l$ using proper values of y_0 in each range from (3). Then taking $t = 0$, we have

$$\int_0^a \frac{bx}{a} \sin \frac{n\pi x}{l} dx + \int_a^l \frac{b(l-x)}{(l-a)} \sin \frac{n\pi x}{l} dx = \int_0^l A_n \sin^2 \frac{n\pi x}{l} dx$$

$$\text{whence} \quad A_n = \frac{2bl^2}{n^2 \pi^2 a(l-a)} \sin \frac{n\pi a}{l} \quad \dots(4)$$

Hence the general solution for the plucked string is

$$y = \frac{2bl^2}{\pi^2 a(l-a)} \sum_{n=1}^{\infty} \frac{1}{n^2} \sin \frac{n\pi x}{l} \cos \frac{n\pi vt}{l} \quad \dots(5)$$

Some modes (harmonics) may be absent in (5), because the amplitude of vibration A_n in (4) depends on the distance a of the point of plucking. The n th mode of vibration or n th harmonic will be absent if

$$\sin \frac{n\pi a}{l} = 0 \text{ or } \frac{n\pi a}{l} = m\pi \text{ or } a = \frac{l}{n}m \quad \dots(6)$$

where $m = 1, 2, 3, \dots$ etc.

If the string is plucked at the centre (middle point), then $a = \frac{l}{2}$ and from (6) $\frac{l}{2} = \frac{l}{n}m$ or $n = 2m$ or $n = 2, 4, 6, \dots$. Hence **2nd, 4th, 6th, modes (harmonics) will be absent for a string plucked at the mid point.** All these harmonics require a node at $l = a/2$. In other words, if the string is plucked at the middle point, the string cannot vibrate with frequencies which are even multiples ($n = 2, 4, 6, \dots$) of its fundamental frequency.

If $a = l/3$, $n = 3m$ or $n = 3, 6, 9, \dots$. Hence if a string is plucked at a distance $a = l/3$ from the fixed end, then 3rd, 6th, 9th, ... harmonics will be absent and the remaining modes will be present. All these harmonics need a node at $a = l/3$. Thus, in general **for a plucked string the harmonics (modes), having a node at the point of plucking, are absent.** This is called **Young's law**.

From (4), we see that **the amplitude in the n th mode of vibration is inversely proportional to n^2 .** **Therefore the amplitude of higher harmonics fall off rapidly.** The amplitude of the fundamental mode is four times that of the second mode, nine times that of the third mode etc. i.e., the amplitude of successive modes is falling off rapidly.

It is to be noted that the vibration of the string can be stopped by touching the point of plucking after the plucked string is set into vibration. As the harmonics which have a node at the point of plucking are not excited in accordance with Young's law and the rest of the harmonics which do not have a node at the point of plucking are stopped by damping due to touching, in consequence all vibrations of the string will stop.

When the string is plucked at a single point by a sharp object, the emitted note is very brilliant due to the presence of a large number of modes as in case of the instrument, mandoline. However for a plucked string all the modes having a node at the point of plucking will be absent. In case the string is plucked over a finite length by a round object like finger, a less number of modes are excited so that the sound emitted is softer. The quality of sound given out by a harp is inferior when compared to that given out by a mandoline because the former is plucked by fingers.

11.7. Laws of Transverse Vibrations of Strings

The fundamental frequency of vibration of a string of length l (fixed at both ends), having a linear mass density m and being stretched under tension T , is given by

$$v = \frac{1}{2l} \sqrt{\frac{T}{m}} \quad \dots(1)$$

We can deduce three laws which are called **laws of transverse vibrations of a string** from the above equation.

(1) Law of length : When T and m are constants, $v \propto \frac{1}{l}$ i. e., *the fundamental frequency of vibration of a stretched string is inversely proportional to the length of the string when tension and mass per unit length are kept constants.*

(2) Law of tension : When l and m are constants, $v \propto \sqrt{T}$ i.e., *the frequency of the fundamental mode of a string is directly proportional to the square root of tension where the length and mass per unit length are kept constants.*

(3) Law of mass : When l and T are constants, $v \propto \frac{1}{\sqrt{m}}$ i.e., *the frequency of the fundamental mode of a string is inversely proportional to the square root of mass per unit length of the string when the length and tension are kept constants.*

As mass per unit length $m = \pi r^2 \times l \times \rho = \pi (D^2/4)\rho = \pi D^2 \rho/4$, the frequency of vibration of the string can be written as

$$v = \frac{1}{2l} \sqrt{\frac{4T}{\pi D^2 \rho}} \text{ or } v = \frac{1}{lD} \sqrt{\frac{T}{\pi \rho}} \quad \dots(2)$$

where D is the diameter of the string and ρ the density of its material.

Hence we may write two more laws, deduced from the law of mass.

(a) Law of diameter : When l , T and ρ are constants, $v \propto \frac{1}{D}$ i.e., *the frequency of transverse vibration of a string is inversely proportional to the diameter of the string, when length, tension and density of the material of the string are kept constants.*

(b) Law of density : When l , T and D are constants, $v \propto \frac{1}{\sqrt{\rho}}$ i.e., *the frequency of vibration of a stretched string is inversely proportional to the square root of density of the material of the string, when length, tension and diameter are kept constants.*

These laws were first discovered by M. F. Mersenne in 1636 and can be experimentally verified by a sonometer.

11-8. Energy of Vibrating String : Energy Density

When a progressive transverse wave moves along a string, the energy and the disturbance are being transported from one place to another. In a wave motion, the particles of the string vibrate about their equilibrium position and the vibrating string possesses both potential and kinetic energy.

Consider a sine wave travelling on a stretched string along positive X -axis. The mass per unit length of the string is m and T is the tension. Let the equation of the sine wave be

$$y = a \sin \omega (t - x/v) \quad \dots(1)$$

Consider a small element of length dx and hence mass mdx at x on the string. Assume that the tension T remains the same even when this small element dx vibrates. As the element is very small, we can also assume that velocity and acceleration of its particles to be the same.

$$\text{Particle velocity at } x, \quad \frac{\partial y}{\partial t} = \omega a \cos \omega (t - x/v)$$

$$\text{Particle acceleration at } x, \quad \frac{\partial^2 y}{\partial t^2} = -\omega^2 a \sin \omega (t - x/v) = -\omega^2 y$$

Kinetic energy of the element at displacement y ,

$$K = \frac{1}{2} (mdx) \left(\frac{\partial y}{\partial t} \right)^2 = \frac{1}{2} mdx \omega^2 a^2 \cos^2 (t - x/v) \quad \dots(2)$$

$$\text{Force acting on the element,} \quad F_y = (mdx) \frac{\partial^2 y}{\partial t^2} = -mdx \omega^2 y$$

This is the restoring force acting on the element at the displacement y . Hence the total amount of work done on the element in displacing 0 to y is

$$\begin{aligned} W &= - \int_0^y F_y dy = \int_0^y mdx \omega^2 y dy \\ &= mdx \omega^2 \frac{y^2}{2} = \frac{1}{2} mdx \omega^2 a^2 \sin^2 \omega (t - x/v) \end{aligned}$$

This becomes the potential energy of the element at the displacement y i.e.,

Potential energy of the element at the displacement y ,

$$U = \frac{1}{2} mdx \omega^2 a^2 \sin^2 \omega (t - x/v) \quad \dots(3)$$

Hence the total energy of the element dx at the displacement y ,

$$dE = K + U = \frac{1}{2} m dx \omega^2 a^2 \cos^2 \omega (t - x/v) + \frac{1}{2} m dx \omega^2 a^2 \sin^2 \omega (t - x/v)$$

$$\text{or} \quad dE = \frac{1}{2} m dx \omega^2 a^2 \quad \dots(4)$$

The total energy per unit length of the string is called **energy density** (E) of the vibrating string. Hence the energy density is given by

$$E = \frac{dE}{dx} = \frac{1}{2} m \omega^2 a^2 = 2\pi^2 v^2 a^2 m \quad \dots(5)$$

where $\omega = 2\pi v$ and v is the frequency.

The **total energy per wavelength** $E(\lambda)$ in the string will be obtained by integrating the expression (4) from $x = 0$ to $x = \lambda$ i.e.,

$$E(\lambda) = \int_0^\lambda dE = \int_0^\lambda \frac{1}{2} m \omega^2 a^2 dx = \frac{1}{2} \omega^2 a^2 m \lambda = 2\pi^2 v^2 a^2 m \lambda \quad \dots(6)$$

Problem : Show that for a travelling harmonic wave on a stretched string, average kinetic energy density is equal to average potential energy density when the average is taken over a wavelength. Show that the sum of two energy densities is equal to the total energy density.

Solution : Equation of travelling harmonic wave on a string is

$$y = a \sin \omega \left(t - \frac{x}{v} \right) \quad \dots(i)$$

Potential energy of an element of length dx of the string,

$$dU = \frac{1}{2} m \omega^2 a^2 \sin^2 \omega \left(t - \frac{x}{v} \right) dx \quad \dots(ii)$$

Kinetic energy of an element of length dx of the string,

$$dK = \frac{1}{2} m \omega^2 a^2 \cos^2 \omega \left(t - \frac{x}{v} \right) dx \quad \dots(iii)$$

$$\text{Potential energy density at } x, U = \frac{dU}{dx} = m \omega^2 a^2 \sin^2 \omega \left(t - \frac{x}{v} \right) \quad \dots(iv)$$

$$\text{Kinetic energy density at } x, K = \frac{dK}{dx} = \frac{1}{2} m \omega^2 a^2 \cos^2 \omega \left(t - \frac{x}{v} \right) \quad \dots(v)$$

Average potential energy density, when average is taken over a wavelength λ ,

$$\begin{aligned} U_{av} &= \frac{1}{\lambda} \int_0^\lambda U dx = \frac{1}{\lambda} \int_0^\lambda \frac{1}{2} m \omega^2 a^2 \sin^2 \omega \left(t - \frac{x}{v} \right) dx \\ &= \frac{1}{4} \frac{m \omega^2 a^2}{\lambda} \int_0^\lambda \left[1 - \cos 2\omega \left(t - \frac{x}{v} \right) \right] dx \quad [\because \cos 2\theta = 1 - 2 \sin^2 \theta] \\ &= \frac{m \omega^2 a^2}{4\lambda} \cdot \lambda = \frac{m \omega^2 a^2}{4} = \pi^2 v^2 a^2 m \quad \dots(vi) \end{aligned}$$

Similarly average kinetic energy density

$$K_{av} = \frac{1}{\lambda} \int_0^\lambda \frac{1}{2} m \omega^2 a^2 \cos^2 \omega \left(t - \frac{x}{v} \right) dx = \pi^2 v^2 a^2 m \quad \dots(vii)$$

Thus from (vi) and (vii), $K_{av} = U_{av}$

Sum of $K_{av} + U_{av} = \pi^2 v^2 a^2 m + \pi^2 v^2 a^2 m = 2\pi^2 v^2 a^2 m = \text{total energy density.}$

11.9. Energy Transport and Rate of Energy Flow

When a travelling wave is moving along a string, energy is being transported along the direction of propagation of the wave. Let a sine wave be moving with speed v along X -axis on a string stretched with tension T . The equation for displacement in the sine wave is given by

$$y = a \sin \omega (t - x/v) \quad \dots(1)$$

where a is the amplitude and $\omega = 2\pi\nu$ is the angular frequency.

Consider a portion of the string at any time t to the right of a point at x . The string on the left of the point exerts a force T (tension) on this part. The direction of this force is along the tangent to the string at x . If the tangent is making angle ϕ with the X -axis [fig. 11.8], then the component of the force along Y -axis is

$$F = -T \sin \phi$$

But here ϕ is small so that $\sin \phi = \phi = \tan \phi = \partial y / \partial x$.

Therefore,
$$F = -T \frac{\partial y}{\partial x}.$$

The point at x moves a distance δy in δt time. In this duration, the amount of energy given by the string on the left of the point to the string on the right will be equal to the work done by the force F i.e.,

$$\begin{aligned} \delta E &= F \delta y = F \frac{\partial y}{\partial t} \delta t \\ &= \left(-T \frac{\partial y}{\partial x} \right) \frac{\partial y}{\partial t} \delta t \end{aligned} \quad \dots(2)$$

From eq. (1)

$$\frac{\partial y}{\partial x} = -a \frac{\omega}{v} \cos \omega \left(t - \frac{x}{v} \right) \text{ and } \frac{\partial y}{\partial t} = a \omega \cos \omega \left(t - \frac{x}{v} \right)$$

$$\therefore \delta E = \frac{T a^2 \omega^2}{v} \cos^2 \omega \left(t - \frac{x}{v} \right) \delta t$$

Hence the energy transmitted from left to right at x in one periodic time τ is given by

$$\begin{aligned} E(\tau) &= \int_0^\tau \frac{\omega^2 a^2 T}{v} \cos^2 \omega \left(t - \frac{x}{v} \right) dt \\ &= \frac{\omega^2 a^2 T}{v} \int_0^\tau \frac{1}{2} \left[1 + \cos 2\omega \left(t - \frac{x}{v} \right) \right] dt \quad [\because \cos 2\theta = 2\cos^2 \theta - 1] \\ &= \frac{\omega^2 a^2 T}{v} \cdot \frac{\tau}{2} \end{aligned} \quad \dots(3)$$

This expression gives the energy transferred through a point of the string in a periodic time.

But $\omega = 2\pi\nu$, ν is the frequency of the wave and $T = mv^2$ [$\because v = \sqrt{T/m}$]. Hence

$$E(\tau) = 2\pi^2 \nu^2 a^2 m v \tau \quad \dots(4)$$

But $\tau = 1/\nu$ and $\nu \tau = v/\nu = \lambda$, the wavelength. Hence

$$E(\tau) = 2\pi^2 \nu^2 a^2 m \lambda = E(\lambda) \quad \dots(5)$$

This is the same expression as we obtained for the total energy per wavelength $E(\lambda)$ of the string. Thus, the energy being transported in a periodic time through a point of the string is contained in a wavelength λ of the string.

The average rate of energy or power being transmitted through the point x of the string is obtained from (4) as

$$P_{av} = \frac{E(\tau)}{\tau} = 2\pi^2 \nu^2 a^2 m v \quad \dots(6)$$

Thus, the average rate of energy or power being transported across a point of the string in a travelling wave is proportional to the square of the frequency (ν) and square of the amplitude (a).

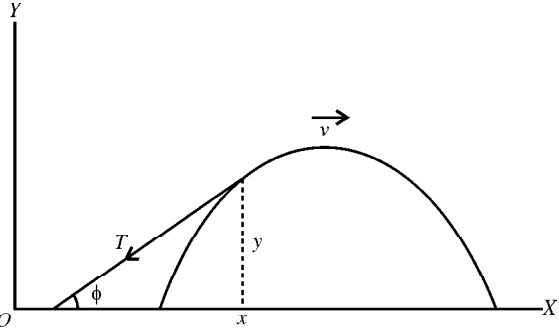


Fig. 11.8. Energy transport along a travelling wave on a string.

We know that the energy per unit length of the string or the energy density is $E = 2\pi^2 v^2 a^2 m$. Hence the average power being transported across a point of the string can be expressed as

$$P_{av} = E \times v \quad \dots(7)$$

i.e., **Average power transported = Energy per unit length of the string $\times$ Wave velocity.**

This is the energy contained in a length v of the string and passes through a point of the string per unit time. Hence expression (6) or (7) gives the rate of flow of energy i.e.,

$$\text{Rate of flow of energy} = 2\pi^2 v^2 a^2 m v \quad \dots(8)$$

This quantity is called **energy current**.

11.10. Transverse Impedance

When a progressive wave is travelling along a string, the string presents an impedance to the propagation of the wave. *The ratio of instantaneous transverse force on any particle of the string to the instantaneous transverse velocity of that particle is called transverse or characteristics impedance of the string and it is denoted by Z .* Thus,

$$Z = \frac{\text{Transverse Force}}{\text{Transverse Velocity}} = \frac{F}{u} \quad \dots(1)$$

where transverse particle velocity $u = \partial y / \partial t$.

It is in analogy with the electrical impedance

$$Z = \frac{\text{Alternating e.m.f.}}{\text{Current}} = \frac{E}{i} \quad \dots(2)$$

Consider a transverse sine wave travelling on a stretched string along X -axis. The equation for displacement (y) of a particle of the string at an instant t is given by

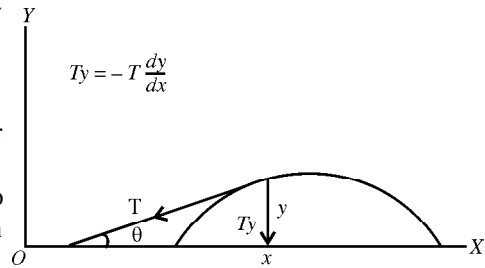
$$y = a \sin \omega (t - x/v) \quad \dots(3)$$

where a is the amplitude and ω the angular frequency. The speed of the wave (v), when it is moving on a string, is represented by

$$v = \sqrt{\frac{T}{m}} \quad \dots(4)$$

where T is the tension in the string and m is the linear density or mass per unit length of the string.

Let the tangent at a point x on the string make an angle ϕ with the X -axis at the particle displacement y [fig.11.9]. Then the transverse force (F) acting on the particle at x is



$$F = -T \sin \phi = -T \tan \phi = -T \frac{\partial y}{\partial x} \quad \dots(5)$$

because for small ϕ , $\sin \phi = \phi = \tan \phi = \partial y / \partial x$.

From eq. (3)

$$\frac{\partial y}{\partial x} = -\frac{\omega a}{v} \cos \omega (t - x/v)$$

$$\therefore F = \frac{\omega a T}{v} \cos \omega (t - x/v) \quad \dots(6)$$

Using eq. (3), particle velocity (u) is obtained to be

$$u = \frac{\partial y}{\partial t} = \omega a \cos \omega (t - x/v) \quad \dots(7)$$

Dividing eq. (6) by eq. (7), we obtain the expression for transverse impedance (Z) i.e.,

$$Z = \frac{F}{u} = \frac{T}{v} = \frac{mv^2}{v} = mv \quad \dots(8)$$

because from (4), $T = mv^2$.

Thus, the transverse impedance offered by a string to travelling transverse waves is equal to the product of linear density (m) of the string and the velocity (v) of the waves.

The transverse or characteristic impedance $Z = mv$ of the string can also be expressed as

$$Z = mv = m\sqrt{\frac{T}{m}} = \sqrt{mT}$$

$$\text{or } Z = mv = \frac{T}{v^2} \cdot v = \frac{T}{v} \quad [\because v^2 = T/m \text{ or } m = T/v^2]$$

$$\text{Thus } Z = mv = \sqrt{mT} = \frac{T}{v} \quad \dots(9)$$

Ex. 1. A wave is travelling on a long stretched string along its length taken as positive X-axis. The equation of the wave is

$$y = ae^{-\left(\frac{t}{T} - \frac{x}{\lambda}\right)}$$

where $a = 8 \text{ mm}$, $T = 1 \text{ sec}$ and $\lambda = 4 \text{ cm}$.

Find the velocity of the wave. In which direction is the wave travelling?

Sol. The equation of the travelling wave is

$$y = ae^{-\left(\frac{t}{T} - \frac{x}{\lambda}\right)} \quad \dots(i)$$

which can be written as

$$y = ae^{\frac{1}{T}\left(t - \frac{xT}{\lambda}\right)} \text{ or } y = a^{\frac{1}{T}} \left(t - \frac{x}{\lambda/T}\right) \quad \dots(ii)$$

This is the form of the general equation of the wave i.e.,

$$y = f\left(t - \frac{x}{v}\right) \quad \dots(iii)$$

Comparing (ii) and (iii), we get

$$v = \frac{\lambda}{T} = \frac{4 \text{ cm.}}{1 \text{ sec.}} = 4 \times 10^{-2} \text{ m/sec.}$$

Obviously, this wave is travelling along positive X-axis.

Ex. 2. Displacement of a wave is $y = 0.29 \cos(3x - 2t)$ in S.I. units. Find amplitude, frequency and time period. (S. K. U. 2007)

$$\text{Sol. Displacement } y = 0.29 \cos(3x - 2t) \quad \dots(i)$$

$$\text{Standard equation of a wave, } y = a \sin[(\omega t - kx) + \phi] \quad \dots(ii)$$

Eq. (i) can be written as

$$y = 0.29 \cos(2t - 3x) = 0.29 \sin[(2t - 3x) + \pi/2]$$

$\therefore$ Amplitude $a = 0.29 \text{ m.}$

$$\text{Frequency, } v = \frac{\omega}{2\pi} = \frac{2}{2\pi} = \frac{1}{\pi} = 0.31 \text{ Hz.}$$

$$\text{Time period, } T = \frac{1}{v} = \pi = 3.14 \text{ sec.}$$

Ex. 3. Standing waves are produced by the superposition of two waves, $y_1 = 10 \sin(3\pi t - 4x) \text{ cm}$ and $y_2 = 10 \sin(3\pi t + 4x) \text{ cm}$. Find the amplitude of motion at $x = 18 \text{ cm}$. (A.U. 1995)

Sol. In the standing waves, the resultant amplitude y is obtained as

$$y = y_1 + y_2 = 10 \sin(3\pi t - 4x) + 10 \sin(3\pi t + 4x)$$

$$\begin{aligned} & \left[\because \sin A + \sin B = \sin \frac{A+B}{2} \cos \frac{A-B}{2} \right] \\ & = 10 [2 \sin 3\pi t \cos 4x] \end{aligned}$$

$$= 20 \cos 4x \sin 3\pi t$$

Here, the amplitude of motion $A(x) = 20 \cos 4x$

Now at $x = 18 \text{ cm}$, $A(x = 18 \text{ cm}) = 20 \cos(4 \times 18) = \mathbf{19.5 \text{ cm}}$.

Ex. 4. A string vibrates according to the equation

$$y = 8 \sin\left(\frac{\pi x}{3}\right) \cos(20\pi t)$$

where x, y are in cm and t in sec. Find the distance between two successive nodes and the speed of particle of the string at a position $x = 1.5 \text{ cm}$ when $t = 2.25 \text{ sec}$.

Sol. At nodes, $y = 0$, i.e., $\sin\left(\frac{\pi x}{3}\right) = 0$

or $\frac{\pi x}{3} = n\pi$, where $n = 0, 1, 2, 3$

whence $x = 3n = 0, 3, 6, 9, \dots \text{ cm}$

$\therefore$ Distance between two successive nodes = 3 cm.

Velocity of particle, $\frac{dy}{dt} = -8 \sin\left(\frac{\pi x}{3}\right) 20\pi \sin(20\pi t)$

When $x = 1.5 \text{ cm}$ and $t = 2.25 \text{ sec}$, then the speed of the particle is

$$\begin{aligned} \frac{dy}{dt} &= -8 \sin\left(\frac{\pi \times 1.5}{3}\right) 20\pi \times \sin(45\pi) \\ &= -8 \sin\left(\frac{\pi}{2}\right) \times 20\pi \times \sin(45\pi) = -8 \times 20\pi \times 0 = \mathbf{0}. \end{aligned}$$

Ex. 5. The velocity of transverse wave in a 150 cm long wire of mass 5 gm subjected to a tension of 1200 N will be how much? **(K.U. 2007)**

Sol. Velocity of transverse wave, $v = \sqrt{\frac{T}{m}} = \sqrt{\frac{1200}{1.5/0.005}} \quad \left[\because m = \frac{150 \text{ cm}}{5 \text{ gm}} = \frac{1.5 \text{ m}}{0.005 \text{ kg}} \right]$

$$= \sqrt{\frac{1200}{300}} = \mathbf{2 \text{ m/sec.}}$$

Ex. 6. A steel wire 50 cm long has mass of 5 gm. It is stretched with a tension of 400 N. Find the frequency of the wire in fundamental mode of vibration. **(A.U. 1995)**

Sol. Frequency in fundamental mode, $v = \frac{1}{2l} \sqrt{\frac{T}{m}}$

Here, $l = 50 \text{ cm} = 0.5 \text{ m}$, $m = 5 \times 10^{-3} \text{ kg}/0.5 = 10^{-2} \text{ kg}$

$\therefore v = \frac{1}{2 \times 0.5} \sqrt{\frac{400}{10^{-2}}} = \frac{1}{1} \sqrt{400 \times 10^2} = \mathbf{200 \text{ Hz.}}$

Ex. 7. A flexible string of length 1 m and mass 1 gm is stretched to a tension T . The string is found to vibrate in three segments at a frequency 612 Hz. Calculate the tension in the string. **(A. U. 2002)**

Sol. As the string is vibrating in three segments, we have

$\therefore v = \frac{n}{2l} \sqrt{\frac{T}{m}} = \frac{3}{2l} \sqrt{\frac{T}{m}}$

Putting the given values, we obtain

$$612 = \frac{3}{2 \times 1} \sqrt{\frac{T}{10^{-3}}} \text{ or } \left(\frac{612 \times 2}{3}\right)^2 = \frac{T}{10^{-3}}$$

$$\therefore T = \left(\frac{612 \times 2}{3} \right)^2 \times 10^{-3} = \mathbf{0.166 \text{ N}}.$$

Ex. 8. The diameter of the iron wire is 1.2 mm. If the speed of the transverse wave in the wire be 50 m/sec, then what is the tension in the wire ? ($\rho_{\text{iron}} = 7.7 \times 10^3 \text{ kg/m}^3$) (O.U. 2008; A.N.U. 2006)

Sol. Speed of transverse waves in a string is given by

$$v = \sqrt{\frac{T}{m}} \text{ or } T = mv^2$$

where T is the tension and m is the mass of the string per unit length. Now, if r is the radius of the wire, then

$$\begin{aligned} m &= \text{mass of unit length} \\ &= \text{volume of unit length} \times \text{density of the wire} \\ &= \pi r^2 \times 1 \times \rho \quad [\rho = \text{density of the material of wire}] \end{aligned}$$

$$\begin{aligned} &= 3.14 \times \left(\frac{1.2 \times 10^{-3}}{2} \right)^2 \times 7.7 \times 10^3 \\ &= 8.7 \times 10^{-3} \text{ kg/m} \end{aligned}$$

$$\therefore T = 8.7 \times 10^{-3} \times (50)^2 = \mathbf{21.75 \text{ newton}}.$$

Ex. 9. Calculate the speed of transverse waves in a wire of 1 mm^2 cross-section under the tension produced by 0.1 kg weight (specific gravity of material of wire = 9.81 gm/cm³ and $g = 9.81 \text{ m/sec}^2$).

(A.N.U. 1991)

Sol.
$$v = \sqrt{\frac{T}{m}}$$

Here, $T = Mg = 0.1 \times 9.81 = 0.981 \text{ N}$, area of cross-section $\pi r^2 = 1 \text{ mm}^2 = (10^{-3} \text{ m})^2 = 10^{-6} \text{ m}^2$,

$$\therefore m = \pi r^2 \rho = 10^{-6} \text{ m}^2 \times (9.81 \times 1000 \text{ kg/m}^3) = 9.81 \times 10^{-3} \text{ kg/m}^3.$$

Hence,
$$v = \sqrt{\frac{0.981}{9.81 \times 10^{-3}}} = \sqrt{10^2} = \mathbf{10 \text{ m/sec}}.$$

Ex. 10. The linear density of vibrating string is $1.3 \times 10^{-4} \text{ kg/m}$. A transverse wave propagating along the length of the string is described by the equation $y = 0.03 \sin(x + 30t)$, where x and y are in meters and t is in seconds. Determine the tension in the string.

Sol. Equation of the transverse wave is

$$y = 0.03 \sin(x + 30t) \quad \dots(i)$$

Standard sine wave equation is

$$y = a \sin \omega(t - x/v) \quad \dots(ii)$$

Eq. (i) can be written as

$$y = 0.03 \sin 30(t + x/30) \quad \dots(iii)$$

Comparing (ii) and (iii), we get

$$v = -30 \text{ m/sec}$$

This wave is travelling along negative X -axis with speed $v = 30 \text{ m/sec}$.

If T is the tension in the string, then speed of the wave is given by

$$v = \sqrt{\frac{T}{m}}, \text{ whence } T = mv^2$$

Here, $m = 1.3 \times 10^{-4} \text{ kg/m}$ and $v = 30 \text{ m/sec}$

$$\therefore T = 1.3 \times 10^{-4} \times 30^2 = \mathbf{0.117 \text{ N}}.$$

Ex. 11. Two identical guitar strings are tuned to the same frequency 300 Hz. The tension of one of the strings is increased by 2%. How many beats per second will be heard when the two strings are sounded together ? (A.U. 2003)

Sol. Let the initial tension in the two identical strings be T . The tension of one of the strings is increased by 2%. Hence the new tension in that string will be

$$T' = T + \frac{2}{100} \times T = T + 0.02 T = 1.02 T$$

Frequency,
$$v = \frac{1}{2l} \sqrt{\frac{T}{m}}$$

Initial frequency of each string,
$$300 = \frac{1}{2l} \sqrt{\frac{T}{m}} \quad \dots(i)$$

Let the changed frequency of one of the string with increased tension (1.02T) be v' . Hence

$$v' = \frac{1}{2l} \sqrt{\frac{1.02T}{m}} \quad \dots(ii)$$

Dividing eq. (ii) by eq. (i), we get

$$\frac{v'}{300} = \sqrt{\frac{1.02T}{T}} = \sqrt{1.02}$$

whence

$$v' = 300 \times \sqrt{1.02} = 303 \text{ Hz}$$

$$\therefore \text{Number of beats} = v' - v = 303 - 300 = \mathbf{3 \text{ per sec.}}$$

Ex. 12. The fundamental frequency of a sonometer wire increases by 5 Hz, if its tension is increased by 21%. How will the frequency be affected if its length is increased by 25% ? (O.U. 2007)

Sol. Fundamental frequency,
$$v = \frac{1}{2l} \sqrt{\frac{T}{m}} \quad \dots(i)$$

When the tension is increased by 21%, the new tension becomes $T' = T + 0.21T = 1.21T$

$$\therefore v + 5 = \frac{1}{2l} \sqrt{\frac{1.21T}{m}} \quad \dots(ii)$$

Dividing eq. (ii) by (i), we obtain

$$\frac{v+5}{v} = \sqrt{1.21} = 1.1 \text{ or } v+5 = 1.1v \text{ or } 0.1v = 5$$

$$\therefore v = 50 \text{ Hz.}$$

When the length is increased by 25%, the new frequency v' is given by

$$v' = \frac{1}{2 \times 1.25l} \sqrt{\frac{T}{m}} = \frac{v}{1.25} = \frac{50}{1.25} = \mathbf{40 \text{ Hz.}}$$

Ex. 13. A string of length 0.5 m and linear density 0.0001 kg/m is kept under tension of 1 N. Find the frequency of the third overtone when it is plucked at its mid point. (A.N.U. 2007)

Sol. frequency string Fundamental

$$v = \frac{1}{2l} \sqrt{\frac{T}{m}} = \frac{1}{2 \times 0.5} \sqrt{\frac{1}{0.001}} = 100 \text{ Hz.}$$

When the string is plucked at mid point, even harmonics of frequencies $2v, 4v, 6v \dots$ are absent. Hence the string vibrates with the frequencies $v, 3v, 5v, 7v, \dots$. Hence the frequency of the third overtone is

$$7v = 7 \times 100 = \mathbf{700 \text{ Hz.}}$$

Ex. 14. A string of length 2.6 m and mass 0.001 kg kept under a tension 1 N. Find the fundamental frequency of the string. If the string is plucked at 0.5 m from one end, what frequencies would be stopped ? (S.V.U. 2007)

Sol. The fundamental frequency of the string

$$v = \frac{1}{2l} \sqrt{\frac{T}{m}}$$

Here, $T = 1$ N, length $l = 2.5$ m and $m = \frac{0.001}{2.5}$ kg/m

$$\therefore v = \frac{1}{2 \times 2.5} \sqrt{\frac{2.5}{0.001}} = \frac{1}{5} \sqrt{2500} = \frac{50}{5} = 10 \text{ Hz}$$

If the string is plucked at 0.5 m from one end, the point of plucking will be $0.5/2.5 = 1/5$ of the length of the string (i.e., $a = l/5$). Thus harmonics of frequencies $5v, 10v, 15v, \dots$ i.e., **50 Hz, 100 Hz, 150 Hz, ...** would be absent or stopped.

Ex. 15. A metallic wire of density 8×10^3 kg/m³ is stretched between two clamps 1 m apart and is subjected to an extension of 4.9×10^{-4} m. What will be the lowest frequency of transverse vibration in the wire? [Young's modulus of material is 8×10^{10} N/m²]

Sol. The wire will have lowest frequency in the fundamental mode of vibration i.e.,

$$v = \frac{1}{2l} \sqrt{\frac{T}{m}} = \frac{1}{2l} \sqrt{\frac{T}{A\rho}} \quad [\because m = \pi r^2 \rho = A\rho]$$

where A is the area of cross section and ρ is its density.

$$\text{Young's modulus, } Y = \frac{\text{stress}}{\text{strain}} = \frac{T/A}{\Delta l/l}, \therefore T = YA \frac{\Delta l}{l}$$

where l is the length and Δl is elongation.

$$\text{In terms of Young's modulus, } v = \frac{1}{2l} \sqrt{\frac{YA\Delta l}{lA\rho}} = \frac{1}{2l} \sqrt{\frac{Y\Delta l}{l\rho}}$$

Here, $l = 1$ m; $\Delta l = 4.9 \times 10^{-4}$ m; $Y = 8 \times 10^{10}$ N/m² and $\rho = 8 \times 10^3$ kg/m³

$$\therefore v = \frac{1}{2 \times 1} \sqrt{\frac{(8 \times 10^{10})(4.9 \times 10^{-4})}{1 \times (8 \times 10^3)}} = \mathbf{35 \text{ Hz.}}$$

Ex. 16. When the length of the stretched wire on the sonometer is 25 cm, the note emitted by it is in unison with that of a tuning fork. When the length of the wire is changed to 25.5 cm without changing its tension, 2 beats per sec are heard on sounding the two together, what is the frequency of the fork?

Sol. The fundamental frequency v of the string of length l under tension T is given by

$$v = \frac{1}{2l} \sqrt{\frac{T}{m}}$$

Since the length of wire is 25 cm = 0.25 m and it is vibrating in unison with a tuning fork. Hence the frequency of the fork is

$$v = \frac{1}{2 \times 0.25} \sqrt{\frac{T}{m}} \quad \dots(i)$$

Keeping the tension to be the same, when the length of the wire is changed to 25.5 cm = 0.255 m, the number of beats between the vibrating wire and the tuning fork are heard two. As the length of the wire is increasing, the frequency will decrease. Hence

$$v - 2 = \frac{1}{2 \times 0.255} \sqrt{\frac{T}{m}} \quad \dots(ii)$$

Dividing equation (i) by (ii), we get

$$\frac{v}{v-2} = \frac{0.255}{0.250} = \frac{51}{50}$$

or,

$$50v = 51v - 102 \text{ or, } v = 102$$

Thus, the frequency of the fork is **102**.

510 | Mechanics and Waves and Oscillations

Ex. 17. A wire of mass 1 gm and length 75 cm is making 256 vib./sec. under a tension supplied by a brass weight hanging vertically. On immersion the weight into water, the vibrating length of the wire is shortened by 5 cm to regain its original pitch. What is the density of brass ?

Sol.
$$n = \frac{1}{2l} \sqrt{\frac{T}{m}} \text{ or } T = 4l^2 mn^2$$

Here, $n = 256 \text{ vib./sec.}$, $l = 0.75 \text{ m}$, $m = 10^{-3}/0.75 = 1/750 \text{ kg/m}$

$$\therefore T = 4 \times 0.75 \times 0.75 \times (1/750) \times 256 \times 256$$

$$= \mathbf{196.61 \text{ newton.}}$$

Let on immersion in water the tension be reduced to T' . To regain the same value of n , the length of wire $l = 75 - 5 = 70 \text{ cm} = 0.7 \text{ m}$.

$$\therefore T' = 5 \times (0.7)^2 \times (1/750) \times (256)^2 = \mathbf{171 \text{ newton.}}$$

$$\therefore \text{Loss in weight due to immersion is } = (196.61 - 171) = 25.61 \text{ newtons}$$

$$\text{The relative density of brass} = \frac{\text{Wt. in air}}{\text{Loss in wt. due to immersion}}$$

$$= \frac{T}{T - T'} = \frac{196.1}{25.61} = 7.657$$

$$\therefore \text{Density of brass} = \mathbf{7.657 \text{ kg/m}^3}.$$

Ex. 18. Two wires of the same material and of the same cross-section are suspended on a sonometer. One is loaded with 16 kg, and the other with 4 kg. The first wire is tuned to the second harmonic of the second wire. If the second wire is 1 m in length, what is the length of the first wire ?

Sol. The fundamental frequency n_1 of the first wire is given by

$$n_1 = \frac{1}{2l_1} \sqrt{\frac{T_1}{m}} \quad \dots(i)$$

The two wires are made of the same material and of same cross-section, therefore mass per unit length m of the two wires is the same. Now, the second harmonic n_2 of the second wire is

$$n_2 = \frac{2}{2l_2} \sqrt{\frac{T_2}{m}} \quad \dots(ii)$$

Since first wire is tuned to the second harmonic of the second wire,

$$\therefore n_1 = n_2 \text{ i.e., } \frac{1}{2l_1} \sqrt{\frac{T_1}{m}} = \frac{2}{2l_2} \sqrt{\frac{T_2}{m}}$$

Given : $T_1 = 16 \text{ kg-wt}$, $T_2 = 4 \text{ kg-wt}$, $l_1 = ?$, $l_2 = 100 \text{ cm} = 1 \text{ m}$.

$$\therefore \frac{2}{2 \times 1} \sqrt{\frac{4}{m}} = \frac{1}{2l_1} \sqrt{\frac{16}{m}} \text{ or } l_1 = \mathbf{1 \text{ m.}}$$

Ex. 19. The frequency of the fourth harmonic in a stretched string of length 20 cm is 600 per sec. What is the velocity of the wave in the string ? If now the tension of the string is doubled, what will be the final velocity of the waves ?

Sol. Frequency of the 4th harmonic = 600 Hz

As it is four times the frequency of the fundamental,

$$\therefore \text{Frequency of the fundamental} = 600/4 = 150 \text{ Hz}$$

$$\text{Velocity } v = \sqrt{T/m} = n\lambda = n \times 2l$$

$$= 150 \times 2 \times 0.2 = 60 \text{ ms}^{-1}$$

Finally tension is doubled, then

$$v' = \sqrt{2T/m} = \sqrt{2} \times \sqrt{T/m} = \sqrt{2} \times 60 = \mathbf{85 \text{ ms}^{-1}}.$$

Ex. 20. A very long string of linear density 0.01 kg/m is stretched with a tension of 400 N. It is driven at one end in harmonic motion of amplitude 2 cm and frequency 100 Hz. Calculate the average rate of energy transported.

Sol. The average rate of energy transported is given by

$$\therefore P = 2\pi^2 v^2 a^2 m v, \text{ where } v = \sqrt{T/m}$$

$$P = 2\pi^2 v^2 a^2 \sqrt{mT}$$

Here, $v = 100$ Hz, $a = 2$ cm = 0.02 m, $m = 0.01$ kg/m and $T = 400$ N

$$\therefore P = 2 \times (3.14)^2 \times (100)^2 \times (0.02)^2 \sqrt{0.01 \times 400} = \mathbf{157.75 \text{ watt.}}$$

Ex. 21. The characteristic impedance of a string is 3.2 kg/sec and speed of the wave along the string is 32 m/sec. Find the tension in the stretched string and mass per unit length.

Sol. Tension in the string, $T = Z v$

where Z is the impedance and v is the speed of transverse wave.

$$\therefore T = 3.2 \times 32 = \mathbf{102.4 \text{ N}}$$

$$\text{Linear density, } m = \frac{Z}{v} = \frac{3.2}{32} = \mathbf{0.1 \text{ kg/m.}} \quad [\because Z = mv]$$

QUESTIONS

Long Answer Questions

1. Distinguish between transverse waves and longitudinal waves. Give examples. (O.U. 2002)
2. What are transverse waves ? Derive the expression for the velocity of a transverse wave along a stretched string and discuss the solution of wave equation. (K.U. 1997)
3. Set-up the differential equation for a transverse wave on a stretched string. Obtain the frequencies of vibration for a string of length ' l ' rigidly supported at $x = 0$ and $x = l$. (O.U. 2007)
4. What is a transverse wave ? What is the differential form of the wave equation ? Find out a solution for the wave equation.
5. Obtain the general solution of wave equation and mention its significance. (A.U. 2007)
6. Obtain general solution of wave equation on strings and derive an expression for energy density. (N.U. 2001; S.K.U. 2001)
7. Obtain an expression for velocity of transverse waves along stretched string. (O.U. 2009; 08 A.N.U. 2007; S.V.U. 2007)
8. What are transverse waves ? Obtain an expression for the velocity of transverse wave in a stretched string. What are fundamental frequency and harmonics ? (K.U. 2005)
9. What is a transverse wave ? Obtain the equation for the velocity of a transverse wave in a stretched string. Write the laws of transverse vibrations of the string. (A.U. 2007, 03)
10. Write the necessary theory giving corresponding diagrams that explain different modes of vibration of stretched string clamped at both the ends. (O.U. 2005)
11. Discuss the modes of vibration of a stretched string clamped at both ends ? What are overtones ? (A.U. 2007, 05)
12. Derive an expression for the displacement of a plucked string. What is the displacement corresponding to the first overtone when the string is plucked at the centre ? (A.N.U. 2006)
13. Discuss briefly the mathematical theory of the modes of vibration of a plucked string. How does the quality of sound depend on the position at which a string under tension is plucked ?
14. What are transverse waves ? Obtain an equation for the frequency of vibration in a stretched string. Derive an expression for the energy of vibration. (O.U. 2007; A.N.U. 2002)
15. What are the transverse waves ? Obtain an expression for the total energy per wavelength in a string when transverse wave travels on it. (A.U. 1997)
16. Define energy density in the wave motion along a string and obtain an expression for it. (O.U. 1995)
17. Obtain an expression for energy density of a wave travelling along a string. (S.V.U. 2003; O.U. 2002)

18. Derive an equation for the total energy per unit length in a string when a transverse wave travels in it. (S.K.U. 2007)
19. Discuss the energy transport in the strings and obtain an expression for energy density. (K.U. 2007)
20. Show that the characteristic impedance offered by a string to the travelling waves is $Z = \sqrt{\rho T}$, where ρ is the linear density and T is the tension with which it is stretched. (O.U. 2008)

Short Answer Questions

- What is a pulse on a string ?
- We represent a transverse wave on a string by

$$y = a \sin (bx + ct),$$
 where a , b and c are positive. In which direction, the wave is moving ?
- Write the differential form of the wave equation. What is its solution ?
- Obtain the relation between the particle velocity and the slope of the displacement curve. (A. N.U. 2003)
- Explain the formation of stationary waves in a stretched string.
- Write equations of motion of first three normal modes of vibration of a stretched string, fixed at both ends.
- Show that a stretched string fixed between two points produces all possible frequencies. Explain how ? (K.U. 1990)
- State the laws of transverse vibrations of stretched string. (A.N.U. 2008; O.U. 2007, 01; S.V.U. 2007, 03; S.K.U. 2007)
- Explain the formation of harmonics and overtones. (A.U. 2007)
- Explain fundamental frequency, overtones and harmonics. (A.N.U. 2008, 07)
- Distinguish between harmonics and overtones. (O.U. 2007, 05; A.N.U. 2006)
- Explain the energy transport in vibrating strings. (S.V.U. 2007, 05; A.U. 2005; A.N.U. 2005)
- Distinguish between wave velocity and group velocity of waves along strings. (K.U. 2005)

[Hint. Equation of a transverse wave along a string is

$$y = a \sin (\omega t - kx)$$

The relation between ω and k for transverse waves on a uniform string is

$$\omega = kv \text{ or } \omega = k \sqrt{T/m}$$

$$\therefore \text{Wave velocity, } v_p = \frac{\omega}{k} = \sqrt{\frac{T}{m}} \text{ and Group velocity, } v_g = \frac{d\omega}{dk} = \sqrt{\frac{T}{m}}$$

$$\text{Hence } v_p = v_g = \sqrt{\frac{T}{m}}.$$

Hence for transverse waves along a string the wave velocity is equal to group velocity and these waves are non-dispersive.]

14. Explain transverse impedance. (A.U. 2007)

PROBLEMS

1. The equation of a wave travelling on a stretched string along the X -axis is given by

$$y = a e^{-\left(\frac{x}{\alpha} + \frac{t}{T}\right)}.$$

Find the speed of the wave and the direction in which the wave is travelling.

[Ans. α/T ; along negative X -axis]

2. A travelling wave propagates according to the expression

$$y = 0.03 \sin (3x - 2t),$$

where y is the displacement at position x at time t . Taking the units to be in SI, determine (a) the amplitude, (b) the wavelength, (c) the frequency and (d) the period of the wave.

[Ans. (a) 0.03 cm; (b) 2.09 cm; (c) 0.31 Hz; (d) 3.14 sec]

3. Find the velocity of propagating of a transverse wave, the displacement of which is given by in S.I. system as

$$y = 2 \sin \pi (100t - x/60)$$

[Ans. 3000 m/sec]

4. A string vibrates according to the equation $y = 5 \sin (\pi x/3) \cos (40\pi t)$, where x, y are in cm and t in sec. Find the distance between two successive nodes and the speed of particle of the string at a position $x = 1.5$ cm, when $t = 9/8$ sec and $9/16$ sec.

[Ans. 3 cm; 0, 628 cm/sec]

5. The equation for the vibration of a string, fixed at both ends, vibrating in its third harmonic, is given by

$$y = a \sin kx \cos \omega t$$

where $a = 0.5$ cm, $k = 0.314 \text{ cm}^{-1}$, $\omega = 600\pi \text{ sec}^{-1}$.

Find the following :

(a) Frequency of vibration, (b) Position of nodes, (c) Length of the string, and (d) The wavelength and the speed of two travelling waves, producing the above vibration.

[Ans. (a) 300 Hz; (b) 0, 0.1 m, 0.2 m, 0.3 m; (c) 0.3 m; (d) 0.2 m, 60 m/sec]

[Hints. $\lambda = 2\pi/k$, $l = n\lambda/2$ and $v = v\lambda = \frac{\omega}{2\pi}\lambda$]

6. Standing waves are produced by superposition of two waves :

$$y_1 = 10 \sin (3\pi t - 4x) \text{ and } y_2 = 10 \sin (3\pi t + 4x).$$

Find the amplitude of motion at $x = 16$.

(K.U. 2005)

[Ans. 7.06 length unit]

[Hint. $y = y_1 + y_2 = 10 \sin (3\pi t - 4x) + 10 \sin (3\pi t + 4x)$ or $y = 20 \cos 4x \sin 3\pi t$; Amplitude $A(x) = 20 \cos 4x$; Amplitude at $x = 16$, $A(x = 16) = 20 \cos 4 \times 16 = 20 \times 0.353 = 7.06$]

7. If the frequency of stretched string of length 1 m is 256 Hz, find the frequency of a string of half the length under identical condition.

(K.U. 2005; A.N.U. 2008, 07)

[Ans. 512 Hz]

[Hint. $256 = \frac{1}{2l} \sqrt{\frac{T}{m}}$, $v = \frac{2}{2l} \sqrt{\frac{T}{m}}$ $\therefore v = 2 \times 256 = 512 \text{ Hz}$]

8. A steel wire 50 cm long has mass of 5 gm. It is stretched with a tension of 400 N. Find the frequency of the wire in fundamental mode of vibration.

[Ans. 200 Hz]

9. A flexible string of length 1 m and mass 1 gm is stretched by a tension T . The string is found to vibrate in three segments of frequency of 512 Hz. Calculate the tension.

(A.N.U. 1995)

[Ans. 116.5 N]

10. Calculate the speed of transverse waves in a wire of radius 0.05 cm under the tension produced by 1.0 kg weight. The density of the material of wire is $9.8 \times 10^3 \text{ kg/m}^3$.

[Ans. 35.7 m/sec]

[Hint. 1 kg wt. = $1 \times 9.8 \text{ N} = 9.8 \text{ N}$; $m = \pi r^2 \rho$]

11. Calculate the speed of the transverse waves in a wire of 1 mm^2 cross-section under the tension produced by 0.1 kg wt. (specific gravity of material of wire = 9.81 gm/cm^3 and $g = 9.81 \text{ m/s}^2$) (A.N.U. 2003, 02)

[Ans. 10 m/sec]

12. A string of length 8 m fixed at both ends has a tension of 49 N and a mass of 0.04 kg. Find the speed of transverse waves on this string.

(A. U. 2003)

[Ans. 98.99 m/sec]

13. Two strings vibrating side by side in a guitar have the same fundamental frequency of 256 Hz. If the tension of one of the strings is increased by 3%, find the number of beats heard.

(A.U. 2001)

[Ans. 4 beats per sec]

14. The speed of a transverse wave on a stretched string is 500 m/sec when it is stretched under a tension of

19.6 N. If the tension is changed to a value of 78.4 N, find the speed of the wave. (A.N.U. 2005)

[Ans. 1000 m/sec]

[Hint. $v_1 = \sqrt{\frac{T_1}{m_1}}$ and $v_2 = \sqrt{\frac{T_2}{m_2}} \therefore \frac{v_2}{v_1} = \sqrt{\frac{T_2}{T_1}}$]

15. A metallic wire of radius 1 cm has a length of 1 metre. It is fixed at both ends and is subjected to a tension of 10^4 N. Calculate the fundamental frequency and the frequencies first and second overtones and the corresponding wavelength. (Density of copper is 8920 kg/m^3 .)

[Ans. 298.7 Hz, 597.4 Hz, 896.1 Hz; 2 m, 1 m, 0.67 m]

16. The fundamental frequency of a sonometer wire increases by 5 Hz, its tension is increased by 21%. How will the frequency be affected if its length is increased by 10% ?

[Ans. Frequency = 45.45 Hz]

17. Two tuning forks produce 4 beats per sec when sounded together. One of them is in unison with a length of 120 cm. of a sonometer wire and other with 125 cm of the same wire under the same tension. Calculate the frequencies of the forks.

[Ans. 96 Hz and 100 Hz]

18. A fork and a string of length 100 cm give 4 beats per sec. The string is made shorter without any change of tension until it is in unison with the fork. If its length is now 99 cm, what is the frequency of the fork ?

[Ans. 400 Hz]

19. A metal wire of diameter 1 mm is held on two knife-edges separated by a distance of 0.5 m. The tension in the wire is 100 N. The wire, when vibrating with its fundamental frequency, produces 5 beats per second with a vibrating tuning fork. The tension in the wire is then reduced to 81 N. When the two are again excited, beats are heard again at the same rate. Calculate : (i) the frequency of the fork, (ii) the density of the material of the wire.

[Ans. (i) 95 second^{-1} ; (ii) $12.7 \times 10^3 \text{ kg/m}^3$]

[Hint. $(v + 5) = \frac{1}{2 \times 0.5} \sqrt{\frac{100}{\pi(0.5 \times 10^{-3})^2 \times \rho}}$

$(v - 5) = \frac{1}{2 \times 0.5} \sqrt{\frac{81}{\pi(0.5 \times 10^{-3})^2 \times \rho}}$, where v = frequency of fork]

20. The density of the material of the wire is $7.5 \times 10^3 \text{ kg/m}^3$. If the stress on the wire be $3.0 \times 10^8 \text{ N/m}^2$, then what will be the speed of the transverse wave in the wire ?

[Ans. 200 m/s]

21. P and Q are two wires whose fundamental frequencies are 256 Hz and 382 Hz respectively. How many beats in two seconds will be heard by the third harmonic of P and the second harmonic of Q .

[Ans. 8]

22. A string of length 0.5 m and linear density 0.0001 kg m^{-1} kept under a tension 1 N. Find the first three overtones of the string when it is plucked at its mid-point.

[Ans. 300 Hz, 500 Hz, 700 Hz]

23. A string of mass 2 gm/m carries progressive waves of amplitude 1.5 cm, frequency 60 Hz and speed 200 m/sec. Calculate (a) the energy per metre length of the wire and (b) the rate of energy transport in the wave.

[Ans. (a) 0.032 J; (b) 0.64 watt]

24. A steel wire is stretched by a force of 100 N. If the mass per unit length of the wire is 2.5 gm/m, find the transverse impedance.

[Ans. $0.5 \text{ N m}^{-1} - \text{sec.}$]

Ultrasonics

12-1. Introduction

Sound waves can be broadly classified into three classes on the basis of their frequencies :

(1) Sonics or Audible Waves : The sound waves whose frequencies are in the range 20 Hz to 20 kHz and which can be heard by human ear are called **sonics** or **audible waves**. These waves are produced by the vibrating bodies, e.g., by stretched strings-violin, guitar, vocal cords - animal and human voice, vibrating membranes and plates - drum, loudspeaker and vibrating air columns - clarinet, organ pipe.

(2) Subsonic or Infrasonic Waves : The sound waves, which have frequencies less than the lower limit of the audible range i.e., 20 Hz, are called subsonic or infrasonic waves. Our ear does not respond to these waves. Such waves are produced by earthquake and vibrating sources of low frequencies (< 20 Hz).

(3) Ultrasonic Waves : The sound waves, which have frequencies greater than the upper limit of audible range i.e., 20 kHz, are called ultrasonics or ultrasonic waves. As the frequencies of the ultrasonic waves are very high, these waves do not stimulate our ear and are not heard. However some animals like bats and dogs respond to the ultrasonic waves. The wavelength of these waves in air is of the order of 1 cm or less ($\lambda = v/\nu = 340/20,000 \text{ m} = 1.7 \text{ cm}$). At present, ultrasonic waves of frequencies of the order of 10^9 Hz are conveniently produced.

Hypersonic Waves : The frequency of these waves is more than 5×10^8 Hz.

Supersonic Speed : A body moving with a speed higher than the speed of sound is said to be moving with **supersonic speed**.

Shock Waves : Shock waves are produced when a sudden and abnormal increase of pressure and density occurs at any place in the medium. For example, when a jet aircraft flies in the atmosphere at supersonic speeds (i.e., at speeds greater than the speed of sound), a strong shock wave is generated and radiated from the aircraft so that a loud booming sound is heard. If the speed of the aircraft or any body is v and v_s in the speed of sound, then the ratio v/v_s is called **Mach number**. If the mach number exceeds 1, the body is moving at a supersonic speed.

Because of several important applications in the fields of metallurgy, medical, biology etc., ultrasonics need to be discussed specially. In the present chapter, we shall deal with the production, detection, measurement and applications of these waves.

12-2. Production of Ultrasonic Waves

Generally ultrasonic waves are produced by two methods : Magnetostriction method and Piezo-electric method.

(1) Magnetostriction Method : Ultrasonic waves can be generated by using the principle of magnetostriction. If a rod of ferromagnetic material (e.g., iron, nickel) is placed along an alternating magnetic field, its length is periodically increased and decreased. This phenomenon is called **magnetostriction effect**. If the natural frequency of longitudinal vibrations of the rod is equal to that of the alternating current producing the alternating magnetic field, a resonance is obtained and the amplitude of oscillations of the rod becomes maximum. Ultrasonic waves of the same frequency are produced by the oscillating rod. If the rod has been clamped in the middle, then this is the case of longitudinal vibrations of a rod fixed at the middle and free at both ends. The frequency of longitudinal vibrations of such a rod in the fundamental mode of vibration

is given by

$$v = \frac{v}{\lambda} = \frac{v}{2l} \quad [\because \lambda = 2l] \dots(1)$$

where v is the velocity of ultrasonic waves and l is the length of the rod.

The velocity of ultrasonic waves (sound) in a rod, vibrating longitudinally, is given by

$$v = \sqrt{\frac{Y}{\rho}} \dots(2)$$

where Y is the Young's modulus and ρ the density of the material of the bar.

Hence the frequency in the fundamental mode is obtained to be

$$v = \frac{1}{2l} \sqrt{\frac{Y}{\rho}} \dots(3)$$

If the rod is vibrating longitudinally in the n th harmonic, then

$$v = \frac{n}{2l} \sqrt{\frac{Y}{\rho}} \dots(4)$$

where $v = 1, 3, 5, \dots$

The high frequency alternating current required to generate the ultrasonic waves is obtained from an oscillatory circuit. Here, the iron or nickel rod serves as a *transducer* to convert the electrical energy into sound waves.

Working of Magnetostriction Vibrator : In the magnetostriction method, AB a rod of nickel, cobalt or iron, is clamped at the middle. L_1 and L_2 are two coils surrounding the AB rod. The diameter of the turns of the coil is greater than that of rod, thus leaving the rod free to move lengthwise, inside the coils [fig. 12.1]. L_1 is forming a tuned circuit with the variable condenser C which is connected to the plate P of a triode valve. The coil L_2 is connected between the grid G and the cathode F . The values of L_1 and C give the frequency of the oscillatory circuit. If this frequency of the LC circuit is equal to the frequency of the rod, then the longitudinal oscillations are maintained and ultrasonics are produced in the surrounding medium.

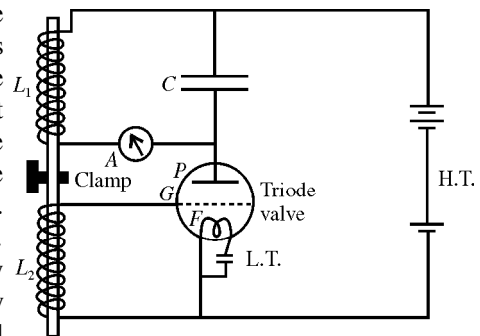


Fig. 12.1. Magnetostriction vibrator.

In the beginning, the bar is magnetised by passing direct current. The periodic variation in the anode current, passing in the coil L_1 , causes a variation in magnetisation and resulting in a variation in the length of the rod. This variation in length causes a change in the magnetic flux through the grid coil L_2 . Hence, by converse magnetostriction effect, an induced emf is produced in L_2 . This emf acts on the grid and causes an amplified current variation in L_1 . Thus the oscillations of the rod are maintained.

Initially the milliammeter A shows alteration when the oscillations are set up. The capacitor C is adjusted so that these alterations are maximum. In this position, frequency of the oscillating anode current coincides with the natural frequency of the rod. Now the vibrations are most intense and ultrasonic waves are produced.

Ultrasonics of frequencies in the range 5 to 60 kHz are produced by the magnetostrictive vibrator. The ultrasonic waves of high frequencies are produced by making use of Piezo-electric effect.

(2) Piezo-electric Method : If a plate, cut in a special way from a crystal of quartz, tourmaline, rockelle salt etc., is compressed, the other faces of the plate acquire opposite electric charges. When the

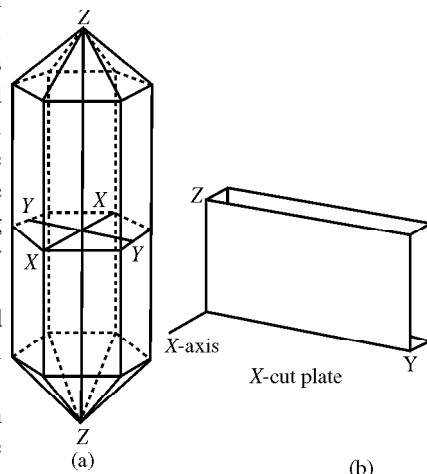


Fig. 12.2. Shape of quartz crystal plate.

plate is stretched, the signs of the charges are changed. This phenomenon is called **Piezo-electric effect**. Converse of Piezo-electric effect is also true. Thus, if the plate is placed in an alternating electric field, it contracts and expands periodically resulting in the mechanical vibrations. The frequency of vibrations of the crystal plate is in the ultrasonic range.

Shape of the Crystal Plate : In the Piezo-electric vibrator, generally quartz crystal in the form of hexagonal pyramid is used to obtain the crystal plate. The line ZZ along the long side of the crystal is called **optic axis** or **Z-axis**. XX line, which is perpendicular to Z -axis and connecting any two opposite corners of the hexagon, is called **electric axis** or **X-axis** [fig. 12.2(a)]. They are three in number. Now, if a line YY is drawn in such a way that it is perpendicular to both X - and Z - axes and normal to the two opposite faces, this line is called **mechanic axis** or **Y-axis**. They are also three in number. When a plate is cut from a crystal, so that both of its principal faces are perpendicular to X -axis, the plate is called **X-cut crystal**. Similarly, if the plate is cut from a crystal having both of its faces perpendicular to Y -axis, the plate is called **Y-cut crystal**. The distance between the two principal faces is the thickness of the plate in case of X -cut or Y -cut crystal [fig. 12.3].

Resonant Vibrations of Crystal Plate : To produce ultrasonic waves (longitudinal), X -cut crystal is frequently used. When an alternating electric field is applied along X -axis of the X -cut plate, the plate either expands along the Y -axis (length) and contracts along the X -axis (thickness) or shows contraction along the Y -axis and expansion along the X -axis, depending on the direction of the electric field. Thus the crystal plate will execute forced vibrations. If the frequency of the alternating electric field is equal to the frequency of the crystal plate, then the phenomenon of **resonance** takes place and the amplitude of vibration becomes large. The vibrational mode along the Y -axis (length) is called the **length vibration** whereas that along the X -axis (thickness) is known to be **thickness vibration**. These vibrations can be considered longitudinal, whether they are due to the strain in the length or due to the strain in the thickness. The two possible frequencies in the fundamental mode are given by

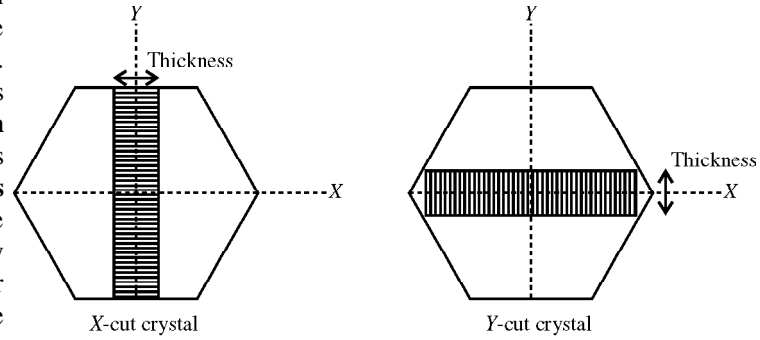


Fig. 12.3. Transverse section of hexagonal quartz crystal.

$$v_l = \frac{v}{\lambda} = \frac{v}{2l} \quad [\because \lambda = 2l] \text{ for length vibration} \quad \dots(1)$$

and

$$v_t = \frac{v}{\lambda} = \frac{v}{2t} \quad [\because \lambda = 2t] \text{ for thickness vibration} \quad \dots(2)$$

Since

$$v = \sqrt{\frac{Y}{\rho}} \quad [Y = \text{Young's modulus and } \rho = \text{density of the crystal}],$$

$$v_l = \frac{1}{2l} \sqrt{\frac{Y}{\rho}} \text{ for length vibration} \quad \dots(3)$$

and

$$v_t = \frac{1}{2t} \sqrt{\frac{Y}{\rho}} \text{ for thickness vibration} \quad \dots(4)$$

When the crystal is vibrating in the n th mode, the two frequencies are given by

$$v_l = \frac{v}{\lambda} = \frac{n}{2l} \sqrt{\frac{Y}{\rho}} \quad [\because \lambda = \frac{2l}{n}] \text{ for length vibration} \quad \dots(5)$$

and

$$v_t = \frac{v}{\lambda} = \frac{n}{2t} \sqrt{\frac{Y}{\rho}} \quad [\because \lambda = \frac{2t}{n}] \text{ for thickness vibration} \quad \dots(6)$$

where $n = 1, 2, 3, \dots$

For quartz crystal, if the average value of Young's modulus Y is $7.7 \times 10^{10} \text{ N/m}^2$ and $\rho = 2650 \text{ kg/m}^3$, then $\sqrt{Y/\rho} = \sqrt{7.7 \times 10^{10}/2650} = 5.39 \times 10^3 \text{ m/sec}$. Hence if $l = 0.05 \text{ m}$ and $t = 0.005 \text{ m}$, the fundamental frequencies of the plate are

$$v_l = \frac{1}{2 \times 0.005} \times 5.39 \times 10^3 = 5.39 \times 10^4 \text{ Hz}$$

and

$$v_t = \frac{1}{2 \times 0.005} \times 5.39 \times 10^3 = 5.39 \times 10^5 \text{ Hz}.$$

Thus, the length vibration produces low frequency and the thickness vibration generates high frequency. The thickness vibration generally used to generate longitudinal ultrasonic waves in the medium *because of much greater radiating surface*. The frequencies of other modes of the plate are an integral multiple of the fundamental frequency. Hence, if a X -cut quartz plate in thickness is vibrated by an alternating electric field of a proper frequency, longitudinal ultrasonic waves are produced. Piezo-electric vibrator can produce the frequencies upto the order of $5 \times 10^8 \text{ Hz}$. Y -cut quartz plate is mostly used to generate shear waves in the medium.

Working of Piezo-electric Vibrator : In fig. 12.4, a Piezo-electric vibrator has been shown to produce ultrasonic waves. This consists of a X -cut quartz plate Q , placed in between the metallic plates A and B . This arrangement, using the crystal plate as a dielectric, constitutes a parallel plate condenser. The plates A and B are connected to the terminals of a coil L_1 . The coil L_2 is connected to the plate circuit of a triode valve. The coil L with capacitor C constitutes the oscillating tank circuit and is connected between the grid G and filament (cathode) F . When the frequency of oscillations of the oscillating circuit is made equal to the natural frequency of the crystal mode, *resonance* occurs and the crystal undergoes linear expansions and contractions at the same rate with relatively large amplitude. In other words, the crystal vibrates vigorously, producing ultrasonic waves in the surrounding medium.

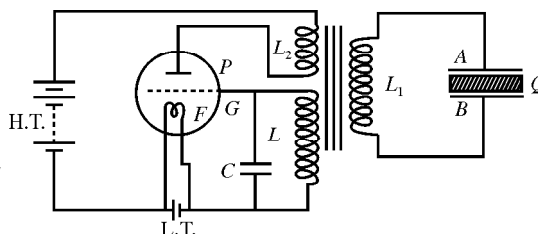


Fig. 12.4. Piezo-electric vibrator.

Remember that the frequency of the oscillating circuit with inductance L and capacitance C is given by

$$v = \frac{1}{2\pi\sqrt{LC}}.$$

By adjusting the variable capacitance C of the tank circuit, the quartz crystal is made to oscillate at its natural frequency to produce resonance.

12.3. Detection of Ultrasonics

Few methods to detect ultrasonic waves are given below :

(1) Piezo-electric Detector : Ultrasonics can be detected by using the phenomenon of Piezo-electric effect. When ultrasonic waves are incident along the mechanical axis (Y -axis) of a X -cut quartz crystal, these waves produce alternating mechanical stresses along the Y -axis. This develops small alternating e.m.f. along its electrical axis (X -axis). This small emf is amplified and detected by an electronic circuit. Thus the Piezo-electric vibrator with high gain amplifiers and an electronic circuit detects the ultrasonic waves.

(2) Kundt's Tube Method : In a Kundt's tube lycopodium power is spread and then the ultrasonic waves are passed in the tube. It is seen that the powder collects at nodal points. Thus we get $\lambda/2$, being the distance between two nodes. If we know the frequency of the wave, velocity can be calculated by the relation $v = v\lambda$. In case of liquid medium, powdered coke is used in place of lycopodium. This method can be used only when the wavelength of ultrasonic wave is not very small i.e., not less than few millimeters.

(3) Thermal Detectors : In this method a thin platinum wire is moved along the direction of propagation of ultrasonic wave. As compressions and rarefactions occur at nodal points, consequently adiabatic change in

temperature takes place. Here the resistance of platinum varies with temperature. This variation in temperature can be detected by wheatstone bridge. At the points of antinodes, the temperature and resistance remain constant. This can also be seen by the balanced position of the bridge. This method is very simple and used generally for this purpose.

(4) Sensitive Flame Method : By moving a flame along the medium, we observe that the flame is steady at the positions of antinodes and it flickers at nodal points due to the change of pressure here. Thus the distance between two adjacent nodes ($\lambda/2$) can be found. If we know the frequency of ultrasonic waves, its velocity can be determined.

12.4. Determination of Wavelength and Velocity of Ultrasonic Waves

(1) Ultrasonic Interferometer Method or Stationary Wave Method

Method : The velocity of ultrasonic waves in a gas or liquid can be determined by acoustic interferometer method. In this method, stationary ultrasonic waves are formed in the gas or liquid which enable us to determine the wavelength (λ) and velocity (v) of the waves. This is why this method is also called *stationary wave method*.

In the experiment, quartz crystal Q is mounted in front of a smooth reflecting surface R and the liquid is filled in the tank T , as shown in **fig. 12.5**. The linear dimension of the crystal Q are large in comparison to the wavelength of ultrasonic waves and hence almost plane longitudinal waves travel towards the reflector R and are reflected back to the crystal Q . The incident and reflected waves form a system of stationary waves. The position of the reflector R can be changed by means of a micrometer screw M . When the distance of the reflector from the face of the vibrating crystal is varied, the current in the micro ammeter in the anode circuit of the valve represented by A , shows maxima and minima, depending on the phase of the oncoming waves incident on the crystal. The maxima correspond to the positions of nodal planes and are determined by moving the reflecting surface towards or away the crystal vibrator. The distance between two successive maxima (or minima) gives half the wavelength i.e., $\lambda/2$ (**fig. 12.6**). Hence one can know the wavelength λ of the ultrasonic waves, produced in the experiment. The frequency ν of the crystal vibrator is determined by a standard electrical wave meter method. The velocity of ultrasonic waves in the medium (liquid or gas) can then be calculated from the formula :

$$v = \nu \lambda$$

(2) Acoustic Grating Method : In order to find the wavelength and hence the velocity of ultrasonic waves, an alternative method known as acoustic grating method, was first suggested by **Brillouin** in 1925. According to the suggestion, when ultrasonic waves are propagated in a liquid, the density varies from layer to layer due to periodic variation of pressure. Now, if monochromatic light is passed through the liquid at right angles to the waves, the liquid acts as a diffraction grating, the lines of compression being lines of greater density. Such a grating is called **acoustic grating**. This grating behaves similar to a ruled grating in optics. In 1932, **Debye and Sears** have used this technique with a stationary wave system of ultrasonics in a cell.

In the experimental arrangement, a quartz crystal plate Q is placed in a glass cell, containing the liquid and is connected to an oscillating circuit, producing plane ultrasonic waves. These waves are reflected by a

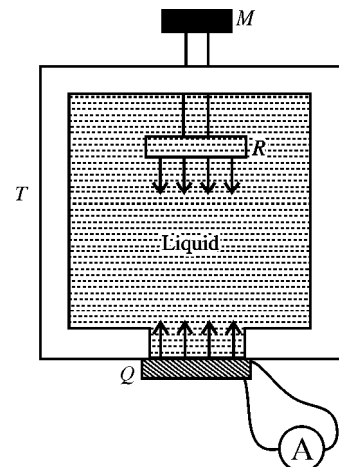


Fig. 12.5. Experimental set up for the determination of the velocity of ultrasonic waves in a liquid.

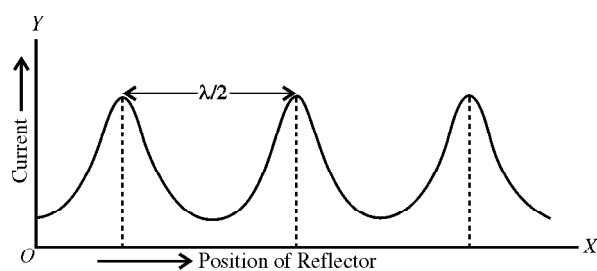


Fig. 12.6. Variation of the micrometer current with the position of the reflector.

reflector R [fig. 12.7]. The position of the reflector R can be adjusted so that the incident and reflected waves produce longitudinal stationary waves in the liquid. In these stationary waves, nodal and antinodal planes are formed at fixed positions. At the nodal planes, the density is maximum due to compression and the antinodal planes, the density is minimum due to rarefaction. These (equally spaced) nodal planes and the inter-nodal spaces in the liquid act like the grating elements. Thus, the liquid with periodic variations in density in space constitutes the acoustic grating.

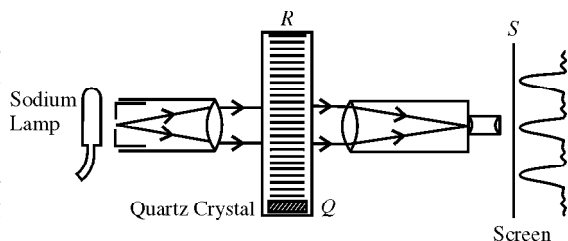


Fig. 12.7. Acoustic grating.

The acoustic grating is mounted on the prism table of a spectrometer. A parallel beam of sodium light, obtained from the collimator, is incident normally on the acoustic grating QR [fig. 12.7]. The light is diffracted by the grating and the diffracted beam of light is seen through the telescope. The diffraction pattern consists of a central maximum and a number of principal maxima on either side of it.

Let θ_n be the angle of diffraction of n th principal maximum. If d be the distance (grating element) in between two consecutive nodal planes, then the grating equation from physical optics is

$$d \sin \theta_n = n\lambda \quad \dots(1)$$

where λ is the wavelength of sodium light.

If λ_a is the wavelength of the ultrasonic waves through the liquid, then

$$d = \frac{\lambda_a}{2} \text{ or } \lambda_a = 2d \quad \dots(2)$$

Determining d from eq. (1) and substituting it in eq. (2), we can find the **wavelength (λ_a) of the ultrasonic waves**.

Hence the **velocity of ultrasonic waves** in the given medium (liquid) can be obtained as

$$v = v\lambda_a \text{ or } v = 2vd \quad \dots(3)$$

where v is the frequency of ultrasonic waves.

(3) Pulse Amplitude Method : In this method, a pulse of short duration of ultrasonic waves sent out by the quartz crystal towards a reflector. The reflected pulse received by the crystal is observed on a synchroscope. The time interval t between the emission of the pulse and its reception by the crystal is noted. As the ultrasonic waves travel twice the distance d between the crystal and the reflector in the duration t , hence one can find the velocity of the waves by using the relation

$$v = \frac{2d}{t}$$

12.5. Applications of Ultrasonic Waves

(1) Underwater Acoustics : Recently underwater acoustics have become an important tool for oceanographers. Because of the extremely short wavelength of the ultrasonic waves, these waves can be obtained in a very narrow beam. Ultrasonic waves of high frequency are used to find the depth of the sea by **echo-sounding method**. In this method, ultrasonic waves, produced by a Piezo-electric quartz oscillator, are directed towards the bed of sea. These waves are reflected back from the bed of the sea and the echo is detected by the crystal itself. Let t be the time interval between the emitted ultrasonic pulse and its echo. If v be the velocity of sound in the sea water then the depth (d) of the sea is obtained from the relation :

$$d = v \times \frac{t}{2} = \frac{vt}{2}.$$

The echo-sounding method is also used to detect the presence and depth of submarines, sea rocks, icebergs, sunken ships and different objects in ocean. The whole system, used to detect an object inside the sea, is called **sonar**, which means '**sound navigation and ranging**'. By the use of sonar, one ship can converse with the another.

(2) Application in Mining : Ultrasonic techniques are used in mining to get information about rocks of the crust of the earth. The echo-sounding method enables us to detect voids, fractures and ore bodies in the

rocks.

(3) Metallurgical Applications : The principle of sending out ultrasonic waves and measuring the reflection interval is used in industry to find the location of the flaw inside a metal block or steel bar. If a pulse of ultrasonic waves reflects back from a flaw at a distance d in a time interval t_1 and from the other end of a bar of length l in a time interval t_2 , then

$$d = \frac{vt_1}{2} \text{ and } l = \frac{vt_2}{2}$$

where v is the velocity of sound.

Therefore,
$$\frac{d}{l} = \frac{t_1}{t_2} \text{ or } d = \frac{lt_1}{t_2}$$

Hence by using echo-sounding method, one can measure the distance of flaw in a steel bar.

(4) Medical Applications : (i) Abnormal growth in the brain, tumours etc. can be detected by ultrasonic waves. They cannot be detected by X-rays.

(ii) Ultrasonic message (vibrations) are used to relieve the pain due to arthritis.

(iii) Ultrasonic waves are used in bloodless surgery. Here, ultrasonic waves of high frequency are focussed on the unwanted tissue or part of the body and the tissue is destroyed.

(5) Chemical Applications : (i) High polymers and molecular chains can be broken by high frequency vibrations.

(ii) Metals such as copper-lead, zinc-lead, etc. aluminium-cadmium etc. are alloyed with uniform composition by passing ultrasonic waves of high intensity.

(iii) Ultrasonic waves accelerate the process of crystallization.

(iv) When ultrasonic waves are passed in potassium iodide, iodine is produced.

(6) Heating Effect : When ultrasonic waves are passed in a substance, it is heated. Water can be boiled by these waves.

(7) Uses of Ultrasonics in Nature : (i) The echo-sounding method is used by bats to detect obstacles in their path. Actually bats have natural power of emitting and detecting, ultrasonic waves. During the flight in darkness, they continuously emit these waves and detect echos due to reflection from an obstacle ahead and change their path to avoid the obstacles.

(ii) The dog can hear the sound of more than 20 kHz. Hence a dog whistle, using ultrasonic waves, can be used to call a dog, which will not be heard by the neighbours, without annoying them.

Ex. 1. A magnetostriction oscillator has a frequency 20 kHz. If it produces sound wave of velocity 6.2×10^3 m/sec, find the length of the ferrite rod. **(S.V.U. 2007)**

Sol. In a magnetostriction oscillator the frequency of vibration is given by

$$v = \frac{v}{2l} \quad \dots(i)$$

where v is the velocity of sound in the rod of length l .

Therefore,
$$l = \frac{v}{2v} \quad \dots(ii)$$

Here, $v = 6.2 \times 10^3$ m/sec and $v = 20 \times 10^3$ Hz.

Hence,
$$l = \frac{6.2 \times 10^3}{2 \times 20 \times 10^3} = \mathbf{0.155 \text{ m.}}$$

Ex. 2. A quartz crystal thickness 0.005 metre is vibrating at resonance. Calculate the fundamental frequency. Given Y for quartz = 7.9×10^{10} newton/m² and ρ for quartz = 2650 kg/m³.

(A.N.U. 2005; S.K.U. 2002)

Sol. The expression for fundamental frequency of a quartz crystal of thickness t is given by

$$v_t = \frac{1}{2t} \sqrt{\frac{Y}{\rho}}$$

522 | Mechanics and Waves and Oscillations

Here, $t = 0.005$ m, $Y = 7.9 \times 10^{10}$ N/m² and $\rho = 2650$ kg/m³.

$$\therefore v_t = \frac{1}{2 \times 0.005} \sqrt{\frac{7.9 \times 10^{10}}{2650}} = 5.46 \times 10^5 \text{ Hz.}$$

Ex. 3. A Piezoelectric crystal has a thickness 0.002 m. If the velocity of sound wave in crystal is 5750 m/s, calculate the fundamental frequency of crystal. (A.N.U. 1997, 96)

Sol. Fundamental frequency for the thickness vibration is given by

$$v = \frac{v}{2t} = \frac{5750}{2 \times 0.002} = 1.4375 \times 10^6 \text{ Hz} = 1.4375 \text{ MHz.}$$

Ex. 4. A Piezoelectric crystal with vibrating length 3×10^{-3} m has density 3.5×10^3 kg/m³. If it is made of a material of Young's modulus 8×10^{10} N/m², what is its fundamental frequency? (A.N.U. 2008, 07; O.U. 1995)

Sol. The fundamental frequency for length vibration is given by

$$v = \frac{1}{2l} \sqrt{\frac{Y}{\rho}}$$

Here, $l = 3 \times 10^{-3}$ m, $Y = 8 \times 10^{10}$ N/m² and $\rho = 3.5 \times 10^3$ kg/m³

$$\therefore v = \frac{1}{2 \times (3 \times 10^{-3})} \sqrt{\frac{8 \times 10^{10}}{3.5 \times 10^3}} = 0.7967 \times 10^6 \text{ Hz} = 0.7967 \text{ MHz.}$$

Ex. 5. The velocity of sound in a Piezoelectric X-cut crystal of thickness 0.003 m is 5750 ms⁻¹. Calculate the fundamental frequency of the ultrasonic waves produced. (A.U. 2001)

Sol. Fundamental frequency of X-cut crystal is given by

$$v = \frac{v}{2t}$$

Here, $v = 5750$ m/sec and $t = 0.003$ m

$$\therefore v = \frac{5750}{2 \times 0.003} = 0.958 \times 10^6 \text{ Hz} = 0.958 \text{ MHz.}$$

Ex. 6. Calculate the capacitance to produce ultrasonic waves of 10^6 Hz with an inductance of 1 henry. (K.U. 2002)

Sol. Frequency of LC circuit $v = \frac{1}{2\pi} \sqrt{\frac{1}{LC}}$

whence $C = \frac{1}{4\pi^2 v^2 L}$

Here, $L = 1$ henry and $v = 10^6$ Hz

$$\therefore C = \frac{1}{4 \times (3.14)^2 \times (10^6)^2 \times 1} = 0.025 \times 10^{-12} = 0.025 \text{ pF.}$$

Ex. 7. Ultrasonic pulse echo method is employed to detect possible defect in a steel bar of thickness 40 cm. If the pulse arrival times are 30 and 80 microseconds, determine the distance of the defect.

Sol. Suppose x be the distance of the defect from the end of bar at which the pulse enters. The pulse, arriving first, comes from defect after reflection and the second pulse arriving later on, comes from the other end of the bar after reflection. This means that 30 microseconds are taken to move a distance of $2x$ and 80 microseconds are taken to cover a distance of 80 cm or 0.8 m. Thus

$$30 \mu\text{s} = \frac{2x}{v} \quad \dots(i)$$

$$80 \mu\text{s} = \frac{0.8}{v} \quad \dots(ii)$$

Dividing (i) equation by (ii), we get

$$\frac{2x}{v} \times \frac{v}{0.8} = \frac{30}{80}, \therefore v = \frac{3}{8} \times \frac{0.8}{2} = \mathbf{0.15 \text{ m or } 15 \text{ cm.}}$$

Ex. 8. Determine the frequency of the first and second modes of vibration for a quartz crystal of Piezoelectric oscillator. The velocity of longitudinal acoustic waves in a crystal is 5.5×10^3 m/sec. Thickness of quartz crystal is 2.5 cm.

Sol. For n th mode, the frequency of vibration of the crystal is given by

$$v_n = \frac{v}{\lambda} = \frac{nv}{2l}$$

$$\text{For the first or fundamental mode, } v_1 = \frac{v}{2l} = \frac{5.5 \times 10^3}{2 \times 0.025} = \mathbf{1.1 \times 10^5 \text{ Hz.}}$$

$$\text{For the second mode, } v_2 = \frac{2v}{2l} = \frac{v}{l} = \frac{5.5 \times 10^3}{0.025} = \mathbf{2.2 \times 10^5 \text{ Hz.}}$$

Ex. 9. An ultrasonic beam is used to determine the thickness of a steel plate. It was noticed that the difference in two adjacent harmonic frequencies is 50 kHz. The velocity of sound in steel is 5000 m/s. Calculate the thickness of the steel plate.

Sol. Frequency of the n th harmonic, $v_n = \frac{nv}{2l}$

Difference of two adjacent i.e., n th and $(n-1)$ th harmonics Frequencies

$$= \frac{nv}{2l} - \frac{(n-1)v}{2l} = \frac{v}{2l} = \text{fundamental frequency}$$

$$\text{Thus } \frac{v}{2l} = 50 \text{ kHz or } \frac{5000}{2l} = 50,000; \therefore l = \frac{1}{20} = \mathbf{0.05 \text{ m.}}$$

This is the thickness of the steel plate.

Ex. 10. Bats emit ultrasonic waves. The shortest wavelength in air by a bat is about 0.33 cm. What is the highest frequency a bat can emit ? (velocity of sound in air = 330 m/sec) **(K.U. 2007)**

Sol. The highest frequency emitted by a bat, will correspond to the shortest wavelength. If v be the velocity of sound in air, then the highest frequency is

$$v = \frac{v}{\lambda} = \frac{330 \text{ m/sec}}{0.33 \times 10^{-2} \text{ m}} = 10^5 \text{ Hz} = \mathbf{100 \text{ kHz.}}$$

QUESTIONS

Long Answer Questions

1. What are ultrasonics ? Explain the terms magnetostriction and Piezoelectric effect. Describe a method to determine the velocity and wavelength of ultrasonic waves. **(K.U. 2007)**
2. Describe the production of ultrasonics by magnetostriction method. Mention any two applications of ultrasonics. **(S.K.U. 2005)**
3. Describe how ultrasonic waves are produced by magnetostriction method with diagram. Write any five applications. **(S.K.U. 2002)**
4. What are properties of ultrasonics ? Write about the production of ultrasonics using the magnetostriction principle. **(A.N.U. 2007)**
5. What are X-cut and Y-cut crystals ? Describe the magnetostriction method of producing ultrasonics. **(O.U. 2008)**
6. Distinguish between X-cut and Y-cut crystal slabs. Which is used in Piezoelectric method and why ? **(O.U. 2001)**
7. What are ultrasonics ? Explain the terms magnetostriction and Piezoelectric effect ? Describe how ultrasonics are produced by the method of Piezoelectric method ? **(O.U. 1997)**
8. What are ultrasonics ? How ultrasonics are produced using Piezoelectric effect. **(A.N.U. 2007; A.U. 2005)**

9. Write in detail the production and detection of ultrasonic waves. What are its applications ? (A.N.U. 2008, 06)
10. What are the various methods used in the detection of ultrasonic waves ? (O.U. 2001)
11. Define ultrasonics. Explain how the ultrasonic waves are detected. (S.K.U. 2007, 05)
12. How do you detect ultrasonics ? What are its three important applications ? (K.U. 1992; S.K.U. 1988)
13. Write an essay on the production, detection and applications of ultrasonic waves. (A.U. 2007)
14. Describe ultrasonic interferometric method for the determination of wavelength and velocity of ultrasonics in liquids. (O.U. 2009)

Short Answer Questions

1. What are ultrasonics and infrasonics ?
2. Explain magnetostriction. (A.U. 2007)
3. What are ultrasonics ? Explain the terms magnetostriction and Piezoelectric effect. (K.U. 2005)
4. What is Piezo-electric effect ? Mention two examples for Piezoelectric material. (O.U. 2003)
5. What are X-cut and Y-cut crystals ? Explain. (K.U. 2005)
6. Explain the concept of Piezoelectricity. (A.U. 2008)
7. Write short notes on Piezoelectric effect. (A.U. 2007; S.V.U. 2002)
8. Explain properties of ultrasonic waves. (S.V.U. 2007)
9. Give two properties of ultrasonic waves. (A.U. 2005)
10. What are uses of ultrasonics ? (O.U. 2002)
11. Explain any five applications of ultrasonic waves. (O.U. 2007, 02; S.V.U. 2002; A.U. 2007)
12. What are shock waves ? What is Mach number. (O.U. 2009)

PROBLEMS

1. A magnetostriction oscillator has a frequency of 10 kHz. If it produces sound wave of velocity 6.2×10^3 m/s. Find the length of ferrite rod. (S.K.U. 2005)
[Ans. 0.31 m]
2. A magnetostriction oscillator has a frequency of 30 kHz. If it produces sound wave of velocity 6.2×10^3 m/sec, calculate the length of the ferrite rod.
[Ans. 10.3 cm]
3. Find out the fundamental frequency of a quartz crystal 1 mm thick and of density 2.65 gm/cm^3 . $C_{11} = 8.516 \times 10^{11} \text{ dyne/cm}^2$.
[Ans. 2.855 MHz]
4. A Piezoelectric X-cut quartz crystal plate has a thickness of 1.6 mm. If the velocity of propagation of sound waves along the X-direction is 5760 meter/sec., calculate the fundamental frequency of the crystal.
[Ans. 1.8 MHz]
5. Calculate the fundamental frequency of a quartz crystal of thickness 0.001 m. $Y = 7.9 \times 10^{10} \text{ N/m}^2$ and $\rho = 2650 \text{ kg/m}^3$. (S.K.U. 2001; A.N.U. 2002)
[Ans. 2.73 MHz]
6. A quartz crystal of thickness 3 mm is vibrating at its natural frequency. If the Young's modulus and density of quartz are $7.9 \times 10^{10} \text{ N/m}^2$ and $2,650 \text{ kg/m}^3$ respectively, calculate the crystal frequency for resonant vibration. (A.N.U. 2002)
[Ans. 0.91 MHz]
7. Calculate the frequency of fundamental note emitted by a Piezoelectric crystal. Use the following data : $Y = 8 \times 10^{11} \text{ N/m}^2$, $\rho = 2.5 \times 10^3 \text{ kg/m}^3$ and vibrating length = 3 mm. (S.K.U. 2002; A.N.U. 2001)
[Ans. 0.943 MHz]
8. Calculate the capacitance to produce ultrasonic waves of 1 MHz with inductance of 2 henry.
[Ans. 0.0125 pF]





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